Moving-Throat PDE Derivation Companion

Archive Ledger

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Purpose and Scope

This document is the verification companion for the moving-throat PDE program. Its job is not to replace the shorter framework paper, and it is not meant to read like a conventional journal article. It is a derivation ledger: a structured, auditable record of how the moving-throat response framework is built from the parent theory, how the reduced branch tests are obtained, which claims are exact, which claims depend on declared closures, and which quantities still have to be supplied by an actual branch of the moving-throat PDE.

The central promise of this ledger is simple:

> Every formula that future papers are expected to cite should have a visible derivation path, a visible status tag, a visible provenance stage, and a visible downstream use.

The ledger is therefore written for readers who need to verify the mathematics, not merely for readers who want the shortest usable statement of the final framework. A future paper may cite the framework paper for the operational branch test; it should cite this companion when it needs the derivation of that test, the exact algebra behind a closure, the location of a no-go result, or the status of a remaining realization gap.

0.1 The role of this companion

The moving-throat program has two different textual needs. The first is an operational paper: a compact framework that tells a reader what the moving-throat PDE response package is, what branch data must be supplied, and how those data are converted into observables or theorem gates. The second is an archive-level verification document: a place where the full derivation is preserved in enough detail that a skeptical reader can reproduce the logic stage by stage.

This document serves the second need. Its purpose is to turn the canonical stage files and theorem-block chapters of this companion into a stable mathematical ledger. The theorem-level map, the current status firewall, and the end-state vocabulary now live in the frontmatter and part structure of this TeX project. The long audit trail lives in the canonical `paper/stages/stage_NNN.tex` files assembled by the stage appendices. The executable verification layer lives in the SymPy and Mathematica audit scripts. This companion binds those layers into a citeable structure.

The result should be read as a verification monograph rather than as a new phenomenology paper. Its job is to show how the moving-throat machinery was derived, how its internal reductions fit together, and where the remaining branch-realization questions sit. It should not silently upgrade a reduced result into an unconditional theorem of the full nonlinear PDE.

0.2 Primary source hierarchy

The source hierarchy used by this companion is fixed as follows.

Source layer	Role in this ledger
Parent-theory papers	Provide the exact declared-action backbone: gauged bulk GNLS matter, localized Maxwell theory, brane projection, zero-mode Maxwell reduction, corrected charge ontology, and the lower-order conservative closure hierarchy.
Post-Newtonian assembly papers	Provide the Newtonian, 1PN, 2PN, 2.5PN, 3PN, 4PN, and later carry-forward targets that motivate the grouped real P_2, outgoing quadrupole, weak-axisymmetric, and realization-compiler gates.
`paper/stages/stage_NNN.tex`	The canonical per-stage derivation files. The stage numbers in this companion are provenance markers for this source layer.
`paper/appendices/stage_appendix_part*.tex`	The stage-ordered archive assembler. These files preserve derivation order while pulling from the canonical stage files.
`paper/parts/*.tex` plus frontmatter	The theorem map, status firewall, notation discipline, and stable result-anchor language.
`scripts/` and `mathematica/`	The executable audit layer: symbolic verification, numerical stress harnesses, and second-CAS replay.
`pde.tex`	The operational framework paper. It should remain the concise reference for using the branch tests, while this companion supplies the long derivation behind those tests.

When these layers disagree in tone, the ledger should preserve the strongest cautious interpretation: exact parent identities remain exact; reduced branch results remain reduced; numerical placements remain numerical; and open branch-realization questions remain open.

0.3 Document contract

The contract of this companion is stricter than the contract of an ordinary exposition. Each major result should satisfy the following requirements.

1. **Provenance.** The result must be traceable to one or more stages in the original derivation sequence.
2. **Inputs.** The result must state which prior identities, branch assumptions, reductions, or closure choices it uses.
3. **Claim status.** The result must carry one of the ledger status tags: Exact, Exact Within Closure, Reduced / Controlled Reduction, Effective Closure, Numerically Located, or Open.
4. **Derivation path.** The main algebraic steps must be visible either in the theorem-block chapter or in the corresponding stage appendix.
5. **Checks.** The result must identify the relevant consistency checks: dimensional checks, sign checks, limiting cases, projector checks, isotropy checks, symbolic-audit checks, or numerical-location checks.
6. **Downstream use.** The result must say what later theorem block, branch test, paper, or realization compiler uses it.

A result that cannot satisfy this contract may still be discussed, but it should not be promoted to a stable citation anchor.

0.4 What this document is

This companion is four things at once.

0.4.1 A theorem-block derivation

The main body organizes the full stage tree into coherent theorem blocks. The reader should not have to read all 253 stages linearly to understand the logic of the program. Instead, the main text groups the derivation into larger mathematical units: parent theory and geometry lift, conservative response kernels, support completion, retarded quadrupole closure, weak-axisymmetric orbit structure, realization compiler, same-charge and rigid-mouth branch tests, and the relaxed open-system extension.

The theorem-block layer is the preferred citation layer. Future papers should normally cite stable anchors such as **MTDC-T1**, **MTDC-T8**, or **MTDC-T12**, then point to the corresponding stage range only when the detailed provenance matters.

0.4.2 A stage-by-stage audit trail

The appendices preserve the original stage order. Stage numbers are not merely decorative; they are the audit trail. A future reader should be able to move from a theorem statement in the main body to the exact stage or stage range where the raw calculation was introduced, tested, corrected, or closed.

The stage appendices should not be optimized for literary flow. They should be optimized for verification. Each stage entry should make clear what was assumed, what was computed, what was checked, what failed, and what became input for later stages.

0.4.3 A status firewall

The moving-throat program contains exact identities, controlled reductions, closure-dependent algebra, numerical placements, and open branch-realization claims. Those categories must not be blurred. This companion therefore treats claim status as part of the mathematics.

A formula can be algebraically exact inside a declared reduced model while still not being an unconditional theorem of the full moving-throat PDE. A branch test can be completely derived while the actual branch still fails that test. A no-go result can be exact inside a specified search class while saying nothing about a larger class that was not searched. The status firewall exists to keep these distinctions visible.

0.4.4 A future citation map

The final version of this companion should make later writing easier. When future papers need to cite the grouped real P_2 projector calculus, the outgoing quadrupole normalization bridge, the weak-axisymmetric quotient coordinates, the finite mixed-ray realization sieve, or the rigid-mouth orbit-lock packet, they should be able to cite a stable theorem anchor in this document instead of reconstructing the derivation from session notes.

0.5 What this document is not

This companion is not a replacement for the framework paper. It is longer, more explicit, and more cautious. The framework paper should remain the document that a reader uses to apply the response package. This companion should be the document that a reader uses to verify it.

This companion is also not a claim that the full nonlinear moving-throat PDE has already been solved. The stage sequence derives a large reduced response framework and many exact compilers inside declared closures. It also narrows several open questions to precise branch-realization packets. The distinction between a completed compiler and a realized branch is part of the result.

Finally, this companion is not a place to hide new assumptions. If a step requires a finite-throat mode ansatz, a stable-mode reduction, an isotropic branch restriction, a passive/outgoing condition, a rigid-mouth assumption, a selected-support slice, or a relaxed open-system packet, that assumption must be named in the result status and repeated in the local derivation.

0.6 Scope of the mathematical derivation

The ledger covers Stages 001–253 of the moving-throat derivation. The stage tree is grouped into eight main parts.

Part	Stages	Short name	Scope
I	001–023	Parent, geometry, and linear response backbone	Imports the exact parent theory, promotes the finite throat variables to a distributed shape field, recovers $a(t)$ and $L(t)$ as collective moments, introduces the grouped real P_2 lane, and derives the first wall/BdG/Maxwell/mixed response kernels.
II	024–036	Conservative response and normalization language	Builds the explicit overlap, isotropy, finite-throat, selected-mode, and normalized-response machinery needed to state the grouped real P_2 conservative and outgoing normalization tests.
III	037–090	Support completion and Family-1 branch	Develops continuum placement, rank-2 support completion, support/source gain thresholds, and the explicit Family-1 reduced branch tests, including the first reduced success/failure windows.
IV	091–163	Geometry lane, retarded closure, and mouth/core compensation	Separates the grouped P_2 and geometry lanes, narrows the retarded problem to the outgoing quadrupole normalization, and analyzes outlet, mixed side-channel, core-to-mouth, mouth-layer, and co-evolving corrections.

Part	Stages	Short name	Scope
V	164–200	Weak-axisymmetric transport and quotient/orbit closure	Reduces the weak-axisymmetric defect to transported prefactor data, identifies $\Xi_1 = P_1/P_0$, derives branch-invariant monomial coordinates, and states the exact orbit/quotient packet ending in the four-scalar verdict.
VI	201–218	Realization compiler and finite mixed-ray search	Converts the reduced branch packet into an explicit graph-error realization problem and completes the finite local mixed-ray sieve through support-cardinality ≤ 5.
VII	219–242	Same-charge, rigid-mouth, and selected twin-support branch	Audits the one-port same-charge mixed bundle, separates static and dynamic kernel verdicts, derives the rigid-mouth physical normal form, states orbit lock in Cartesian variables, and records the selected twin-support placement.
VIII	243–253	Relaxed open-system branch and material/export companion	Adds the relaxed codimension-three open-system branch, leakage/work and non-rigid mouth/dressing compilers, lowered barrier and event-chain analysis, microscopic export kernel, heat partition, and material-threshold companion.

The stage ranges are part of the document contract. If later edits move material between chapters for readability, the original stage provenance should still be retained in the stage appendices and result-anchor map.

0.7 The central objects tracked by the ledger

The derivation follows a small number of recurring mathematical objects. The reader should treat these as the backbone of the companion.

0.7.1 The moving-throat shape field

The old finite-dimensional geometry variables $a(t)$ and $L(t)$ are not discarded. They are reinterpreted as collective moments of a distributed moving-throat field. The basic lift is

$$\Sigma(\mathbf{X},t) = r - R(\Omega,w,t), \qquad r = \sqrt{x^2+y^2+z^2}, \qquad \Omega \in S^2.$$

This lift is the geometric starting point for the ledger. It makes the scalar lane, the residual geometry lane, the grouped real P_2 lane, and the higher multipoles visible within one field-theoretic description.

0.7.2 The grouped real P_2 bundle

The grouped labels 20/21/22 are real quadrupole lanes, not spacetime indices. The grouped real P_2 bundle is the conservative front end of the response problem and the source of the later weak-axisymmetric and outgoing-normalization packets. The ledger tracks its conservative operator moments,

$$D_A(\omega) = D_{A0} + D_{A2}\omega^2 + D_{A4}\omega^4 + \cdots, \qquad A \in \{20, 21, 22\},$$

its normalized response moments,

$$Y_A(\omega) = \frac{D_{A0}}{D_A(\omega)} = 1 + u_2^{(A)}\omega^2 + u_4^{(A)}\omega^4 + \cdots,$$

and its grouped trace/anomaly coordinates.

0.7.3 The outgoing quadrupole prefactor

The retarded bridge is organized around the outgoing $\ell = 2$ fingerprint and the internal prefactor carried by the moving-throat branch. At the point-particle quadrupole normalization level, the key scalar is the invariant product

$$\widehat{m}_0^{\,2} P_0,$$

with target condition

$$\widehat{m}_0^{\,2} P_0 = \frac{54 G c_s^5}{5 a^5 c^5}.$$

One purpose of the ledger is to show how the many earlier conservative and mixed-sector formulas narrow to this scalar on the isotropic passive/outgoing branch.

0.7.4 The weak-axisymmetric quotient packet

The weak-axisymmetric obstruction is not treated as an arbitrary collection of microscopic slopes. It is reduced to branch-invariant coordinates and their transported prefactor packet. In its compact form, the live scalar is

$$\Xi_1 = \frac{P_1}{P_0},$$

with the weak-axisymmetric line

$$b_{P_0} = 3a_{P_0}.$$

The later orbit/quotient theorem organizes the reduced back end into the four-scalar packet

$$\Delta_{\text{full}} = (\Delta_Q, q_{\text{tr}}, q_{\text{nt}}, q_\eta).$$

This companion must make clear which pieces of this packet are exact algebra, which are branch observables, and which are still actual-branch realization questions.

0.7.5 The rigid-mouth and selected-support packets

On the rigid-mouth actual branch, the post-static same-charge problem is written in the physical logarithmic chart

$$(U, V) = \left(\ln \frac{\mathcal{T}^2}{\mathcal{T}_{\text{ref}}^2}, \ln \frac{\epsilon_\eta}{\epsilon_{\eta,\text{ref}}} \right),$$

with orbit lock given by the Cartesian codimension-two condition

$$U = 0, \qquad V = 0.$$

The selected twin-support slice is tracked separately, including the exact lowest-twin reference values

$$\rho_\alpha = \frac{4}{3}, \qquad \zeta_{\text{req}} = \frac{1}{3}.$$

These are branch-organization results. The ledger should distinguish their algebraic derivation from the question of whether a concrete branch realizes the selected placement.

0.7.6 The relaxed open-system extension

The post-242 extension introduces a relaxed branch rather than replacing the strict branch. The recovery map back to the strict front end is

$$\ell_w = f_U = a = b = 0.$$

On the relaxed branch, the stationary front end is organized by the effective potential packet

$$V_{\text{eff}}^{(247)}(r) = V_{\text{short}}^{(1p)}(r) - \lambda_L S_{\text{leak}}(r) - \lambda_W \mathcal{W}_w^{\text{sess}}(r) - \Delta E_{UV}(r) - \mathcal{M}_{\sigma,\text{red}}(r),$$

while the microscopic active-leg export law is

$$K_{\text{exp}}(s) = \Gamma_3 s^3 + \Gamma_5 s^5.$$

The relaxed branch should be labeled as a companion extension. It should not be silently folded into the strict conservative theorem chain.

0.8 Framework success, realization success, and closure failure

This companion repeatedly separates three ideas that are easy to confuse.

0.8.1 Framework success

A framework succeeds when the derivation reduces a complicated PDE branch question to a finite, explicit packet of branch observables. For example, reducing the retarded quadrupole finish line to $\widehat{m}_0^2 P_0$, or reducing the weak-axisymmetric obstruction to $\Xi_1 = P_1/P_0$, is a framework success even before an actual branch has returned the target value.

0.8.2 Realization success

A branch realizes the framework when the actual moving-throat solution lands on the target packet. For example, a branch may have the correct algebraic response framework but still fail the normalization condition, the orbit-lock condition, the selected-support condition, or a same-charge admissibility gate. That failure is a realization failure, not a failure of the algebraic compiler.

0.8.3 Closure failure

A closure fails when the assumptions under which a result was derived are not satisfied. For example, a formula derived on an isotropic grouped real P_2 branch cannot be applied unchanged to an anisotropic branch without carrying the anisotropy defects. A rigid-mouth theorem does not automatically apply to a non-rigid mouth branch. A strict conservative result does not automatically include leakage, work, or export terms from the relaxed open-system extension.

These distinctions are not rhetorical. They determine how future papers are allowed to cite this companion.

0.9 Claim-status discipline

The status tags used throughout the ledger have the following purpose in this chapter.

Status tag	Meaning for the purpose and scope of this document
Exact	The claim follows from the declared parent action, an exact definition, an exact identity, or exact algebra once the variables and conventions are fixed.
Exact Within Closure	The claim is exact inside a stated closure family, branch class, gauge convention, search class, or reduced hierarchy. It is not thereby an unconditional theorem of the full PDE.
Reduced / Controlled Reduction	The claim requires a controlled reduction, low-frequency expansion, normal-mode truncation, projection, selected branch, small-body limit, or other declared ansatz.
Effective Closure	The claim uses a physically motivated closure that has not yet been uniquely derived from the completed moving-throat PDE.
Numerically Located	The object is defined sharply but the currently known placement or value is obtained by numerical solve or numerical search rather than by closed-form expression.
Open	The claim depends on actual branch realization or on a PDE-side datum not yet derived or verified in the completed branch.

Every theorem block in the final ledger should state its status before the proof is read. The status tag is part of the claim.

0.10 What counts as verification

Verification in this ledger does not mean that every line is independently re-derived from the parent action in every chapter. It means that the chain of dependency is explicit enough that a reader can audit it.

A verified stage should provide at least one of the following forms of support.

1. **Direct variation or algebra.** The result follows by varying the declared action, eliminating a quadratic block, projecting a mode equation, expanding a rational response, or performing an exact symbolic manipulation shown in the text.

2. **Controlled reduction with stated assumptions.** The result follows after a declared low-frequency, finite-throat, small-body, stable-mode, isotropic, passive/outgoing, rigid-mouth, or selected-support reduction.

3. **Symbolic audit.** The result is backed by a script that checks identities, projectors, expansions, ranks, nullspaces, roots, or finite candidate sets.

4. **Numerical location with exact target.** The target equation or optimization problem is exact, while the realized value is numerical.

5. **No-go by exhaustion inside a declared class.** The result rules out a branch, mechanism, or search class only after the class has been explicitly defined.

The ledger should not use the word "verified" for an intuition, analogy, or desired branch behavior that has not passed one of these tests.

0.10.1 Verification routing

The stage number is the paper-facing verification handle. To keep the archive readable, the PDF does not print the full script filename inventory inside every stage card.

All paths below are relative to `research/pde_ledger/`. The audit layer is routed by naming convention.

- SymPy symbolic audits live under `scripts/` and follow the pattern `moving_throat_pde_stageNNN_*_sympy_audit.py`.

- Mathematica symbolic audits live under `mathematica/` and follow the pattern `moving_throat_pde_stageNNN_*_mathematica_audit.wl`.

- Numerical stress harnesses live under `scripts/numerical/` and `mathematica/numerical/`, usually with names of the form `stageNNN*_stress.py` and `stageNNN*_stress.wl`. Shared harnesses may cover short ranges as `stage003_021_*`.

- The exact path inventory, coverage counts, and trust classification are maintained in repo-level verification trackers rather than repeated inside the paper.

0.11 How this companion should be used by future papers

Future papers should cite this companion in two different ways.

0.11.1 Citing a theorem anchor

When a future paper needs a stable mathematical result, it should cite the theorem anchor. Examples include:

- the moving-surface lift and collective recovery of a, L,

- the grouped real P_2 projector calculus,

- the outgoing quadrupole normalization bridge,

- the weak-axisymmetric orbit/quotient theorem,

- the finite mixed-ray realization sieve,
- the rigid-mouth Cartesian orbit-lock theorem,
- the relaxed barrier/export compiler.

The theorem anchor should carry the claim status and the stage range.

0.11.2 Citing a provenance stage

When a future paper needs to show where an intermediate formula came from, it should cite the stage or stage range. Stage citations are especially appropriate for long overlap calculations, symbolic-audit outputs, no-go filters, numerical placements, or branch-specific construction details.

0.11.3 Citing an open gap

When a future paper relies on a still-open branch datum, it should cite the relevant open packet directly rather than burying it in prose. For example, it should say that it assumes or tests the actual branch value of $\widehat{m}_0^2 P_0$, Ξ_1, the rigid-mouth (U, V) packet, the selected twin-support coordinate, or the relaxed leakage/export packet, as appropriate.

0.12 Scope boundaries

The ledger includes the derivation of the moving-throat response framework and the stage sequence through the relaxed open-system extension. It does not attempt to do all future work at once. The following boundaries are intentional.

0.12.1 No replacement of the parent papers

This document imports the parent action, corrected charge ontology, projection/reduction distinction, and lower-order PN assembly results. It does not rewrite those papers from scratch. Where a parent identity is needed, this companion should restate only the amount required to make the moving-throat derivation self-contained.

0.12.2 No assumption-free claim of a solved nonlinear PDE

The ledger derives the response machinery and branch tests. It does not claim that the full nonlinear moving-throat PDE has been solved in closed form. Actual branch realization remains a separate task whenever the status tag says Open, Numerically Located, Effective Closure, or Exact Within Closure.

0.12.3 No hiding of branch restrictions

The ledger should not state an isotropic result as if it were anisotropic, a rigid-mouth result as if it were non-rigid, a selected-support result as if it were generic, or a strict conservative result as if it already contained the relaxed leakage/work/export extension.

0.12.4 No silent notational reuse

The notation firewall is part of the document scope. Electric charge is not circulation. The gravity-side $\kappa_\rho = 1$ is not electric charge. The grouped labels 20/21/22 are real P_2 lanes. Mixed fields suppressed in the far-field brane Maxwell limit remain microscopic degrees of freedom in the moving-throat PDE.

0.13 Main-text versus appendix responsibilities

The ledger has two complementary layers.

0.13.1 Main-text responsibilities

The main text should do the following.

1. Define the objects used by each theorem block.
2. State the theorem, proposition, or branch verdict in a citeable form.
3. Identify the status tag and stage range.
4. Give the proof skeleton and the essential algebraic bridge.
5. Explain the downstream role of the result.
6. State what remains open, if anything.

The main text should not include every raw polynomial, every repeated expansion, or every intermediate script table unless that material is needed for understanding the theorem.

0.13.2 Appendix responsibilities

The appendices should do the following.

1. Preserve all 253 stages in original order.
2. Record the inputs, assumptions, derivation, output, checks, and downstream use for each stage.
3. Preserve long algebraic identities and exact script-backed expansions.
4. Store numerical search results and finite candidate sets.
5. Make it possible to audit a theorem anchor by following its stage range.

The appendices are allowed to be repetitive when repetition helps verification. The main text should be more selective.

0.14 Completeness criterion for the final ledger

The final version of this companion should be considered complete only when the following conditions are met.

- ☐ Every stage from 001 through 253 has a filled appendix entry.

- ☐ Every major result in the main text has a status tag.
- ☐ Every theorem anchor has a stage range and downstream-use statement.
- ☐ Every closure-dependent result names the closure.
- ☐ Every open branch-realization claim is explicitly marked open.
- ☐ Every symbolic-audit claim has a reproducibility entry or script reference.
- ☐ The notation firewall is obeyed throughout.
- ☐ Future papers can cite theorem anchors without needing to cite informal session notes.

This completeness criterion is intentionally stricter than the standard for a normal paper. The purpose of the document is long-term verifiability.

0.15 Editorial principles

The final ledger should follow three editorial principles.

0.15.1 Preserve the original derivation, but do not worship chronology

The stage order is provenance, not necessarily exposition. The main text should group related stages into coherent theorem blocks even when the original derivation found those results in a more exploratory order. The appendices preserve chronology; the chapters should preserve understanding.

0.15.2 Name failures as carefully as successes

The moving-throat derivation contains no-go tests, failed branches, worsened residuals, finite windows, and branch ceilings. These should be documented with the same care as successful closures. A failed branch is often the reason a later branch is narrow enough to be meaningful.

0.15.3 Prefer invariant packets to informal descriptions

Whenever possible, the ledger should express branch status in invariant packets rather than prose. For example, use $\widehat{m}_0^2 P_0$, Ξ_1, Δ_{full}, (U, V), and the relaxed branch packet instead of vague descriptions such as "the normalization looks right" or "the orbit is nearly locked."

0.16 End state of the companion

At the end of this project, the companion should allow a reader to answer five questions without consulting informal notes.

1. How does the exact parent theory lead to the moving-throat shape-field PDE and its grouped real P_2 response lanes?
2. How do the conservative response kernels, Maxwell/mixed outgoing bridge, and normalized prefactor formulas reduce the retarded quadrupole problem to a finite scalar target?
3. How do support completion, Family-1, mouth/core compensation, and selected-support branches narrow the actual realization problem?

4. How does the weak-axisymmetric problem collapse to quotient/orbit coordinates and the four-scalar verdict packet?

5. How do the same-charge, rigid-mouth, selected twin-support, and relaxed open-system extensions fit into the strict derivation without erasing the status boundaries?

If the final ledger answers those questions and preserves the full stage audit trail, then future papers can safely treat it as the derivation archive for the moving-throat PDE response framework.

How to Read This Ledger

This companion is intentionally larger and more mechanical than a normal paper. It is designed to let a reader move in both directions: forward from the parent theory to the final reduced branch tests, and backward from any quoted result to the exact stage sequence that produced it. The right way to read it is therefore not to start at Stage 001 and continue linearly until Stage 253. The right way is to choose the level of verification one needs, then follow the relevant theorem anchor, stage range, status tag, and source trail.

The guiding rule is:

> Use the main body to understand the derivation. Use the stage appendices to audit it. Use the result anchors to cite it. Use the status tags to decide what kind of claim has actually been proved.

The ledger has two simultaneous jobs. It should be readable enough that a new collaborator can understand why the moving-throat response package has the shape it does. It should also be explicit enough that a skeptical reader can check the algebra, assumptions, branch restrictions, and remaining open realization gaps without relying on informal session memory.

0.17 The two-layer structure

This document is organized in two layers: a theorem-block layer and a stage-ledger layer. The two layers are meant to be used together, but they do different work.

0.17.1 Layer 1: theorem-block derivation

The main body is the theorem-block derivation. It groups the 253-stage derivation into larger mathematical units: parent theory, geometry lift, conservative response, support completion, retarded quadrupole bridge, weak-axisymmetric quotient structure, realization compiler, rigid-mouth selected branch, and relaxed open-system extension.

This layer should answer the conceptual questions:

1. What is being derived?
2. What inputs are assumed?
3. What is exact and what is closure-dependent?
4. Which object becomes the stable citation target?
5. What does the result let later papers do?

The theorem-block layer is where final theorem statements, proposition statements, named definitions, compact proof sketches, and interpretation should live. A future paper should usually cite this layer first, because it is the layer that gives stable result anchors such as **MTDC-T5**, **MTDC-T9**, or **MTDC-T12**.

The theorem-block layer is not required to reproduce every long polynomial, every symbolic expansion, every numerical placement table, or every failed side path. It should give enough algebra to make the result intelligible and enough references to the stage appendices to make the result auditable.

0.17.2 Layer 2: stage-by-stage verification

The appendices are the stage-ledger layer. They preserve the derivation order through the canonical `paper/stages/stage_NNN.tex` files assembled by the stage appendices. Each stage entry should record the local purpose, inputs, assumptions, derivation, output, checks, and downstream use.

This layer should answer the audit questions:

1. Where did this formula first enter?
2. Which prior stages does it depend on?
3. Was it later corrected, demoted, generalized, or superseded?
4. What algebraic checks or scripts support it?
5. Does it produce a final theorem, a no-go result, a branch condition, or an open target?

The stage-ledger layer is allowed to be repetitive. Verification documents are not optimized the same way narrative papers are. If a formula is used repeatedly, the stage entry may repeat the needed local definitions so the reader can audit the stage without reconstructing half the book.

0.17.3 Why the layers should not be merged

A single chronological document would be difficult to cite and difficult to read. A single polished theorem narrative would hide too much of the derivation trail. The two-layer structure avoids both problems.

The main body says, for example, that the grouped real P_2 normalization problem reduces on the isotropic passive/outgoing branch to a scalar prefactor condition. The appendices show how the response operator, projector calculus, outgoing transfer factor, normalized response expansion, and source-map normalization were derived stage by stage. Both claims are necessary. The first is the useful theorem-level statement; the second is the audit trail.

0.18 Recommended reading paths

Readers will come to this companion with different goals. The ledger is organized so that they do not all have to read it in the same order.

0.18.1 Path A: operational reader

An operational reader wants to know what data a moving-throat branch must return and how those data are used. This reader should start with the front matter, then read the theorem statements and summary boxes in the main parts. The appendices can be used only when a formula needs verification.

For this reader, the most important documents are:

1. the Purpose and Scope chapter;

2. this How to Read chapter;
3. the Claim-Status Firewall;
4. the Notation Firewall;
5. the Result Anchor Map;
6. the opening summary of each main part.

The operational reader should not treat the stage appendices as the primary exposition. The stage appendices are the audit trail behind the exposition.

0.18.2 Path B: theorem verifier

A theorem verifier wants to check a particular result. This reader should begin with the result anchor, then follow the stated stage range, then inspect the stage entries and reproducibility map.

The recommended procedure is:

1. Identify the result anchor, such as **MTDC-T5** for the grouped real P_2 projector and normalization gate.
2. Read the theorem statement and proof sketch in the main body.
3. Record the claim-status tag.
4. Read the listed stage entries in the appendices.
5. Check whether the relevant formulas are exact identities, closure-level identities, controlled reductions, numerical placements, or open branch targets.
6. Follow any script-backed or symbolic-audit references in the reproducibility map.

This path is the intended route for referees or collaborators who need to decide whether a future paper can safely cite a result.

0.18.3 Path C: provenance auditor

A provenance auditor wants to know how a result evolved across the stage sequence. This reader should start with the stage ledger rather than with the theorem block. This is especially useful when an early reduced branch failed, when an old notation was corrected, or when an earlier broad gap was narrowed to a sharper branch-realization condition.

The provenance auditor should pay special attention to:

1. stages where a branch first appears;
2. stages where a no-go or obstruction is recorded;
3. stages where an obstruction is narrowed but not fully eliminated;
4. stages where a symbolic residual becomes a finite scalar packet;
5. stages where a result is superseded by a later cleaner normal form.

A result may have a long provenance even if its final theorem statement is compact. For example, a final scalar such as $\Xi_1 = P_1/P_0$ may depend on many earlier transformations: grouped-lane projectors, weak-axisymmetric splitting, prefactor transport, monomial quotient coordinates, and orbit-lock conditions.

0.18.4 Path D: branch-realization reader

A branch-realization reader wants to know whether the actual moving-throat PDE branch lands on the derived target. This is a different question from whether the target was derived correctly.

This reader should separate three questions:

1. **Compiler question.** Has the algebra reduced the branch test to explicit observables?
2. **Realization question.** Does the actual branch return those observables with the required values?
3. **Closure question.** Are the branch assumptions used in the compiler actually satisfied by the branch being tested?

This distinction is essential throughout the companion. A compiler can be correct while the branch fails the target. A branch can satisfy one packet and fail another. A later relaxed branch can repair one obstruction only if it preserves the status and recovery conditions stated for that relaxed branch.

0.18.5 Path E: author filling the ledger

An author filling this ledger should not try to write every stage with equal length. Some stages are a one-page algebraic identity; others are theorem-level pivots. What must remain uniform is the metadata: purpose, inputs, status, output, checks, and downstream use.

For every section or subsection that is filled later, the author should ask:

1. What stage or stage range is this section responsible for?
2. What exact statement should future papers cite?
3. What assumptions does the statement require?
4. What is the strongest honest status tag?
5. What would falsify, demote, or limit the statement?
6. Where does the raw algebra appear?

The goal is not to make the ledger short. The goal is to make it difficult to misuse.

0.19 Citation discipline

The ledger distinguishes result anchors, theorem labels, equation labels, and stage numbers. Each serves a different purpose.

0.19.1 Result anchors are the stable citation targets

Result anchors such as **MTDC-T1–MTDC-T12** are stable citation targets for future papers. They identify durable mathematical packets rather than individual derivation steps.

A future paper should cite a result anchor when it needs a claim such as:

> The full grouped real P_2 projector calculus and isotropic normalization gate are derived in **MTDC-T5**.

or:

> The weak-axisymmetric transport packet, quotient coordinates, and orbit closure are derived in **MTDC-T9**.

The exact theorem and equation labels under each anchor will be filled as the main parts are written. The anchor map should remain stable even if theorem numbering changes.

0.19.2 Theorem labels are local proof handles

Theorem, proposition, lemma, and definition labels are local proof handles. They should be used when a later section inside this companion depends on a specific formal statement.

A theorem label should not replace a result anchor in future-paper prose unless the future paper needs a very specific local statement. Result anchors are more robust under later editing.

0.19.3 Equation labels are for formulas, not claims

Equation labels should be used for formulas that need to be referenced precisely. They should not be used as the only citation target for a claim. A formula without its status tag and assumptions can be misleading.

For example, the condition

$$\widehat{m}_0^2 P_0 = \frac{54 G c_s^5}{5 a^5 c^5}$$

should not be cited merely as an equation. It should be cited as part of the outgoing quadrupole normalization theorem, with its isotropic passive/outgoing and point-particle assumptions visible.

0.19.4 Stage numbers are provenance markers

Stage numbers preserve provenance. They answer the question "where did this come from?" rather than the question "what should I cite as the final theorem?"

Use stage numbers when:

1. tracing how a formula was derived;
2. identifying where a branch failed or was repaired;
3. checking a script-backed algebraic result;
4. comparing an early prototype with a later theorem-level normal form;
5. documenting that a later result supersedes an earlier provisional formulation.

For example:

> The normalized outgoing-prefactor bridge is part of **MTDC-T4**; its matched one-lane normal form appears in Stage 021 and its grouped response formulation appears in Stage 022.

This is better than citing Stage 021 alone, because the final claim lives in the theorem block while Stage 021 gives provenance for the one-lane normal form.

0.19.5 Do not cite open branch targets as completed branch theorems

Many parts of the moving-throat program end with a sharp target rather than a completed branch realization. Such a target is valuable: it turns a vague problem into a finite branch test. But it is not the same thing as a successful branch.

When citing such a result, future papers should use language like:

> The derivation companion reduces the remaining outgoing-normalization condition to the invariant product $\widehat{m}_0^2 P_0$, but actual branch realization remains a separate test.

They should not write:

> The companion proves that the actual branch realizes the outgoing normalization.

unless the branch-realization theorem has actually been added and tagged accordingly.

0.20 Status tags: how to interpret claims

Every theorem, proposition, boxed result, and major stage output should be read through its status tag. The status tag is not a rhetorical note. It is part of the mathematical content of the ledger.

Status tag	How to read it
Exact	The result follows from the declared parent action, definitions, exact identities, or exact algebra. It does not require an additional branch ansatz beyond the definitions stated in the result.
Exact Within Closure	The result is exact inside a declared closure family, branch, finite reduction, or response hierarchy. The algebra is exact once that local closure is accepted, but the closure itself is not being claimed as the unique consequence of the full PDE.
Reduced / Controlled Reduction	The result follows after a controlled reduction such as a low-frequency expansion, small-body/worldtube limit, far-field brane reduction, selected-mode truncation, isotropic branch restriction, or asymptotic regime.
Effective Closure	The result uses a physically motivated closure choice that organizes the problem but has not yet been uniquely derived from the completed moving-throat PDE.
Numerically Located	The object is defined exactly, but the relevant point, branch value, threshold, or placement is presently obtained by numerical solve or numerical search.
Open	The statement depends on actual branch realization or on a PDE result not yet completed. It may be a sharply defined target, but it is not a completed theorem.

0.20.1 Exact algebra inside a closure is not the same as an exact parent-theory theorem

A common source of confusion is the phrase "exact formula." A formula may be exact inside a reduced model while the reduced model itself is a closure. For example, a Schur complement, projector identity, determinant, or finite mixed-ray search can be exact once the finite reduced system is declared. That does not automatically make the result an unconditional theorem of the full nonlinear moving-throat PDE.

The ledger should therefore say both things when both are true:

> The algebraic reduction is exact within the declared finite mixed-ray search class. Whether the actual branch lies in that class is a separate realization question.

0.20.2 A no-go result has a domain

A no-go result is only as broad as the class it searched. If a stage rules out a wall-only anisotropy, a pure BdG anisotropy, a one-port static same-charge law, or a rigid-mouth branch placement, that no-go should be cited with its search class visible.

A correct no-go citation has the form:

> The wall-only weak-axisymmetric primitive branch is ruled out inside the primitive deformation class tested in the ledger.

An unsafe no-go citation has the form:

> Weak-axisymmetric deformation is impossible.

The second sentence is stronger than the result unless the ledger has actually proved the larger statement.

0.20.3 A numerical placement is not a symbolic theorem

Some branch values are exactly defined but located numerically. Such results should be treated as NUMERICALLY LOCATEDuntil either a closed-form derivation is supplied or the theorem is restated purely as an existence/uniqueness result independent of the numerical value.

A numerical value can be useful and citeable. It should simply not be disguised as a symbolic identity.

0.21 The main distinction: compiler, realization, and closure

The most important reading discipline in the entire companion is the separation among compiler success, realization success, and closure success.

0.21.1 Compiler success

A compiler result reduces a complicated branch problem to a finite set of explicit observables. For example, a long response calculation may reduce the outgoing quadrupole bridge to a scalar prefactor condition, or a weak-axisymmetric branch may reduce to $\Xi_1 = P_1/P_0$ together with a finite quotient packet.

Compiler success means:

> If the branch supplies these quantities and satisfies these assumptions, then the derived observable or target follows by the stated formulas.

Compiler success does not imply that the branch actually supplies the desired values.

0.21.2 Realization success

Realization success is stronger. It means the actual moving-throat branch has been shown to land on the compiler target.

For example, if the compiler target is

$$\widehat{m}_0^2 P_0 = \frac{54 G c_s^5}{5 a^5 c^5},$$

then realization success means the actual branch returns a value satisfying this condition under the assumptions of the theorem. Merely deriving the condition is not realization success.

0.21.3 Closure success

Closure success means the assumptions used to produce a reduced theorem are themselves justified for the branch or regime being cited. A reduced result can fail to apply if the branch violates isotropy, passivity, source-map normalization, support placement, rigid-mouth conditions, or the selected open-system recovery map.

A future paper should not cite a reduced theorem without checking that its closure conditions match the branch being used.

0.21.4 Why this distinction matters

Many important achievements in the moving-throat program are compiler achievements. They narrow a broad missing-PDE problem to a finite target packet. That is a real result. But the ledger must preserve the difference between "we know exactly what the branch must do" and "we have proved the branch does it."

When filling later sections, use phrases like:

> This stage completes the algebraic compiler for the selected branch.

or:

> This stage identifies the actual-branch realization gap as the remaining open datum.

Do not replace those with:

> This stage proves the full physical branch.

unless the stage really does.

0.22 How to read the stage entries

Each stage appendix entry should follow a common structure. The goal is not aesthetic uniformity; the goal is auditability.

0.22.1 Purpose

The purpose paragraph says why the stage exists. It should name the local problem, not merely summarize the algebra.

Good purpose statements have the form:

> This stage lifts the one-lane outgoing transfer factor to the full grouped real 20/21/22 bundle and identifies the static prefactor that controls the leading $i\omega^5$ term.

Weak purpose statements have the form:

> This stage continues the calculation.

The reader should know what theorem gate the stage is trying to open, close, narrow, or rule out.

0.22.2 Inputs

The inputs paragraph lists the prior formulas, assumptions, and branch conditions used by the stage. It should include both mathematical inputs and status inputs.

Examples of input categories include:

1. parent action identities;
2. moving-throat geometry definitions;
3. previous stage kernels or response operators;
4. low-frequency or small-body reductions;
5. isotropic or weak-axisymmetric branch restrictions;
6. finite-mode or one-port closure choices;
7. script-backed algebraic identities;
8. target values imported from lower-order PN ledgers.

If an input is not exact, the stage should not pretend that it is.

0.22.3 Claim status

The claim-status field states the strongest honest status for the stage output. If a stage contains several outputs with different statuses, the entry should either split them or list the statuses separately.

For example, a stage may contain:

1. an exact projector identity;
2. a reduced isotropic-branch theorem;
3. a numerical placement point;
4. an open actual-branch realization condition.

Those should not be collapsed into one global status unless that global status is the weakest one needed for every claim.

0.22.4 Derivation

The derivation field should show the mathematical steps that matter. It does not need to reproduce every line of a symbolic script, but it must expose the transitions that a human reader could otherwise not verify.

A good derivation section includes:

1. definitions of local variables;
2. the equation being transformed;
3. the algebraic or variational step used;
4. the resulting formula;
5. the assumptions used at each narrowing step.

If the result comes from a script, the derivation should still state the mathematical identity the script checks. The script is supporting evidence, not a substitute for a statement of what was proved.

0.22.5 Output

The output field should state the stage result in a form that can be used later. Boxed equations are appropriate for final local outputs, but the box should not imply a stronger status than the stage carries.

Typical output types include:

1. a new definition;
2. an exact identity;
3. a reduced theorem;
4. a compatibility condition;
5. a no-go result inside a declared class;
6. a branch target;
7. a numerical placement;
8. a downstream input packet.

0.22.6 Checks

The checks field records why the stage result is trustworthy. Different results require different checks.

Useful check categories include:

1. dimensional consistency;
2. sign and passivity checks;
3. limiting cases;
4. isotropic-collapse checks;
5. grouped-projector checks;

6. branch-recovery checks;
7. exact symbolic audits;
8. numerical reproduction checks;
9. comparison with a known lower-order target.

The checks field is especially important when a stage introduces a narrow target after a long reduction. It tells future readers whether the target is constrained by multiple independent paths or only by one formal transformation.

0.22.7 Downstream use

The downstream-use field says what later stage, theorem block, branch test, or paper uses the result. This field turns the appendix into a dependency map.

A result that has no downstream use may still belong in the archive, but it should not be promoted to a main theorem anchor unless it resolves, constrains, or explains something that later parts depend on.

0.23 How to read formulas with grouped real P_2 labels

The grouped real P_2 notation is one of the most common places for reader error. The labels

$$20, \qquad 21, \qquad 22$$

are grouped real quadrupole lanes. They are not spacetime indices. They package the five real $l = 2$ components into a weighted three-lane notation in which the 21 and 22 entries each represent a cosine/sine pair.

0.23.1 The grouped metric

The natural grouped metric is

$$G_{\mathrm{grp}} = \mathrm{diag}(1, 2, 2).$$

Therefore grouped traces use the weights $(1, 2, 2)$. For a grouped triple

$$x = (x_{20}, x_{21}, x_{22}),$$

the isotropic trace and two anisotropy coordinates are

$$\bar{x} = \frac{x_{20} + 2x_{21} + 2x_{22}}{5},$$

$$a_x = \frac{2x_{20} - x_{21} - x_{22}}{10}, \qquad b_x = \frac{x_{21} - x_{22}}{2}.$$

The inverse map is

$$x_{20} = \bar{x} + 4a_x, \qquad x_{21} = \bar{x} - a_x + b_x, \qquad x_{22} = \bar{x} - a_x - b_x.$$

The isotropic branch is not merely the branch where an ordinary unweighted average is simple. It is the branch where the grouped anisotropy coordinates vanish:

$$a_x = 0, \qquad b_x = 0.$$

0.23.2 The weak-axisymmetric line

A weak axisymmetric quadrupole splitting has the grouped signature

$$(20,21,22) \sim \left(1, \frac{1}{2}, -1\right).$$

In grouped anomaly coordinates this implies the line

$$b = 3a.$$

When the ledger says that a weak-axisymmetric prefactor packet lies on the line

$$b_{P_0} = 3a_{P_0},$$

it is referring to this specific grouped-lane signature. A branch with arbitrary grouped defects is not automatically a weak-axisymmetric branch.

0.23.3 The meaning of Ξ_1

The scalar

$$\Xi_1 = \frac{P_1}{P_0}$$

should be read as the normalized weak-axisymmetric slope of the outgoing prefactor. In the later quotient language it is not an arbitrary residual; it is the transported prefactor defect that remains after the grouped response has been compressed to branch-invariant coordinates.

When citing Ξ_1, state which branch is being used: raw coherent defect, corrected nontracking coordinate, rigid-mouth chart, selected support branch, or relaxed branch.

0.24 How to read the outgoing quadrupole bridge

The outgoing quadrupole bridge is a central result of the moving-throat program. It is also a place where the reader must distinguish the universal outgoing fingerprint from the internal branch prefactor.

0.24.1 Universal outgoing fingerprint

The compact outgoing $l = 2$ branch has a low-frequency fingerprint of the form

$$\widehat{Y}_2^{\text{out}}(\omega) = 1 + \frac{a^2}{9c_s^2}\omega^2 + \frac{4a^4}{81c_s^4}\omega^4 + i\frac{a^5}{27c_s^5}\omega^5 + O(\omega^6).$$

The important point is that the leading odd term appears at $i\omega^5$, as expected for the quadrupole route.

This fingerprint is not the whole theorem. The moving-throat branch must still supply the internal prefactor that transfers this outgoing port into the wall/worldtube response.

0.24.2 Internal prefactor

The reduced branch carries an internal prefactor, often denoted P_0 at leading order. On the isotropic branch, the leading normalization condition is

$$\widehat{m}_0^2 P_0 = \frac{54Gc_s^5}{5a^5c^5}.$$

This equation is best read as a target condition. The derivation shows why this is the scalar the branch must hit. Actual branch realization is the separate question of whether the computed moving-throat branch returns this value.

0.24.3 Constant-prefactor branch

A particularly simple branch is the constant-prefactor branch, where higher prefactor moments vanish or are correlated so that

$$P_2 = 0, \qquad P_4 = 0$$

through the retained order. The ledger derives exact algebraic conditions for this branch in terms of the conservative operator moments and outgoing-transfer moments. Those algebraic conditions should not be replaced by the stronger and usually false statement that the raw transfer moments themselves vanish.

0.25 How to read orbit and quotient results

The later weak-axisymmetric derivation compresses a large microscopic drift space into quotient coordinates. This is one of the strongest structural simplifications in the program, but it must be read carefully.

0.25.1 Microscopic drifts versus quotient drifts

A microscopic drift vector may contain many entries: stiffness drifts, coupling drifts, frequency drifts, support drifts, transfer-shape drifts, and dressing drifts. Not every microscopic drift is a physical defect motion. Some directions are similarity-orbit directions that preserve the branch invariants.

The quotient coordinates identify the physical motion left after those similarity directions are removed. In the coherent weak-axisymmetric packet, the branch-adapted coordinates include objects such as

$$q_{\mathrm{tr}}, \qquad q_{\mathrm{nt}}, \qquad q_\eta,$$

and the final reduced packet includes

$$\Delta_{\mathrm{full}} = (\Delta_Q, q_{\mathrm{tr}}, q_{\mathrm{nt}}, q_\eta).$$

When the ledger says that a branch lies on a similarity orbit, it means the invariant monomial coordinates are preserved. It does not mean every microscopic coefficient is unchanged.

0.25.2 Orbit lock

On the rigid-mouth actual branch, the physical logarithmic chart is

$$(U, V) = \left(\ln \frac{\mathcal{T}^2}{\mathcal{T}_{\mathrm{ref}}^2}, \ln \frac{\epsilon_\eta}{\epsilon_{\eta,\mathrm{ref}}} \right).$$

Orbit lock is the Cartesian codimension-two condition

$$U = 0, \qquad V = 0.$$

This is a branch condition, not a general identity. It is the correct condition to use only when the rigid-mouth assumptions and selected physical chart apply.

0.25.3 Support selection versus orbit lock

Selected support placement and orbit lock are related but distinct. A branch can satisfy a support-selection threshold while still failing an orbit-lock condition, or vice versa. The selected twin-support slice records one exact support-placement packet, including

$$\rho_\alpha = \frac{4}{3}, \qquad \zeta_{\text{req}} = \frac{1}{3}.$$

This should not be conflated with the rigid-mouth condition $U = V = 0$.

0.26 How to read the relaxed open-system extension

The relaxed extension after Stage 242 should be read as a companion branch around the strict front end, not as a silent replacement of the strict theorem packet.

0.26.1 Recovery map

The relaxed branch has an exact recovery map back to the strict Stage 242 front end:

$$\ell_w = f_U = a = b = 0.$$

Any claim made on the relaxed branch should state whether it survives this recovery limit, vanishes in the recovery limit, or modifies a strict-branch packet before the limit is taken.

0.26.2 Stationary and dynamic pieces

The relaxed extension contains both a stationary lowered front end and dynamic continuations. The stationary front end is organized by an effective potential of the form

$$V_{\text{eff}}^{(247)}(r) = V_{\text{short}}^{(1p)}(r) - \lambda_L S_{\text{leak}}(r) - \lambda_W \mathcal{W}_w^{\text{sess}}(r) - \Delta E_{UV}(r) - \mathcal{M}_{\sigma,\text{red}}(r).$$

The dynamic continuation includes event-chain, turning-point, WKB, Goldilocks, export-kernel, and material-screening packets. These should be cited under the relaxed-branch anchor, not under the strict branch unless the recovery map has been explicitly applied.

0.27 How to handle apparent contradictions

A long derivation ledger will contain provisional branches, failed routes, renamed quantities, and later normal forms that supersede earlier descriptions. Apparent contradictions should be resolved using the rules below.

0.27.1 Later theorem-level normal forms supersede earlier prototypes

Early stages often introduce a one-lane prototype or a simplified branch to expose a mechanism. Later stages may lift that prototype to the full grouped bundle, quotient it by similarity directions, or replace it with a cleaner invariant packet.

The early stage remains valuable as provenance. The later theorem-level form is the preferred citation target.

0.27.2 A failed branch can still be part of the proof

Some stages are important precisely because they fail. A no-go result, obstruction, or undershoot can narrow the search class and force the next theorem gate.

Do not delete failed branches from the ledger merely because they are not part of the final successful route. Instead, mark their status, search class, and downstream role.

0.27.3 Notation updates should be made explicit

The project has a strict notation firewall. If an earlier file used a symbol in a way that later corrections changed, the ledger should say so explicitly. In particular:

1. electric charge is carried by $\eta_Q, q_\star, q_{\mathrm{eff}}$, not by circulation;
2. the historical gravity-side bare $q = 1$ is $\kappa_\rho = 1$, not electric charge;
3. grouped 20/21/22 labels are real P_2 lanes, not spacetime indices;
4. mixed channels are suppressed only in strict far-field brane reductions, not removed from the microscopic ontology.

When a notation update affects a derivation, preserve the old provenance but write the final theorem in the corrected notation.

0.28 Verification standards

A derivation ledger should make verification possible at several levels.

0.28.1 Human algebra verification

The text should show enough algebra that a reader can reproduce the essential steps by hand. This is especially important for:

1. variational reductions;
2. Schur complements;
3. low-frequency expansions;
4. projector decompositions;
5. normalization-product derivations;
6. quotient-coordinate maps;
7. finite search reductions.

If a formula is too long to derive in the main body, the main body should give the structure and the appendix should carry the details.

0.28.2 Symbolic audit verification

When a result was checked by a symbolic script, the ledger should identify what the script checked. Merely naming a script is not enough. The text should say, for example, that the script verified projector idempotence, exact inverse maps, low-frequency coefficient expansions, rank, nullity, determinant nonvanishing, or residual cancellation.

The reproducibility map should collect these scripts and their corresponding mathematical claims.

0.28.3 Numerical verification

Numerical solves should report the exact problem being solved, the branch conditions, the variables, the target residual, and any stability or uniqueness checks. A numerical value without its defining equations is not sufficient for this ledger.

If a numerical result is later replaced by a closed form, the stage appendix should preserve the numerical result as provenance and the main theorem should be promoted only if the closed-form derivation supports the promotion.

0.28.4 Limiting-case verification

Whenever possible, reduced formulas should be checked in limits where the answer is known. Examples include:

1. isotropic limit: grouped anisotropy defects vanish;
2. zero-coupling limit: Schur-complement self-energies disappear;
3. no-mixed-channel limit: outgoing transfer factors vanish or reduce to the stated conservative form;
4. strict-branch recovery: relaxed variables set to zero reproduce the strict packet;
5. point-particle limit: source-map corrections reduce to the leading invariant target.

These checks are not decorative. They are how the reader sees that a long formula has the right structure.

0.29 Suggested local structure for future filled sections

As the empty sections and stage templates are filled, use the following local structure unless a section clearly needs a different one.

0.29.1 For theorem-block sections

A theorem-block section should usually contain:

1. **Stage range and result anchor.** State which stages the section covers and which result anchor it supports.
2. **Problem statement.** Explain the mathematical bottleneck being closed or narrowed.
3. **Inputs and assumptions.** List the parent identities, previous theorem anchors, and branch restrictions.
4. **Definitions.** Introduce the local variables and normalizations.

5. **Main result.** State theorem/proposition/lemma with status tag.
6. **Proof or derivation.** Give the essential algebra.
7. **Checks.** Record consistency and symbolic-audit checks.
8. **Downstream use.** State which later result or future paper uses the result.
9. **Remaining gap.** If applicable, identify what remains open after the section.

0.29.2 For stage appendix entries

A stage appendix entry should usually contain:

1. Purpose;
2. Source file and stage provenance;
3. Inputs;
4. Claim status;
5. Derivation;
6. Output;
7. Checks;
8. Downstream use;
9. Notes on supersession or demotion, if any.

This template should be used even for short stages. Empty fields reveal missing audit information.

0.30 Practical examples of safe wording

The same mathematical result can be described safely or unsafely depending on wording. The examples below are meant to guide future writing.

0.30.1 Outgoing quadrupole normalization

Safe:

> The derivation reduces the passive/outgoing quadrupole normalization to the invariant scalar product $\widehat{m}_0^2 P_0$ on the isotropic branch. Actual realization of the target value is a branch test.

Unsafe unless a later branch theorem is added:

> The moving-throat PDE realizes the GR quadrupole normalization.

0.30.2 Finite realization search

Safe:

The local mixed-ray realization search is finite through the declared support-cardinality bound and closes inside that search class.

Unsafe:

All possible realizations are exhausted.

0.30.3 Rigid-mouth branch

Safe:

On the rigid-mouth actual branch, the post-static same-charge packet is diagonal in the physical logarithmic chart (U, V), and orbit lock is $U = V = 0$.

Unsafe:

The same-charge problem is solved generally by $U = V = 0$.

0.30.4 Relaxed branch

Safe:

The relaxed open-system extension is a codimension-three branch around the strict Stage 242 front end, with recovery map $\ell_w = f_U = a = b = 0$.

Unsafe:

The relaxed branch replaces the strict branch.

0.31 What to do when filling reveals a gap

A derivation ledger is useful because it exposes gaps. If filling a section reveals that a result is less complete than expected, do not hide the gap.

Use one of the following actions.

0.31.1 Demote the status

If a result was previously described as exact but depends on a closure, change the status to Exact Within Closure or Reduced / Controlled Reduction. If actual branch realization is missing, mark the realization as Open even if the compiler is exact.

0.31.2 Split the result

If part of a result is exact and part is open, split it. For example:

1. The projector identity is Exact.
2. The isotropic branch reduction is Reduced / Controlled Reduction.
3. The actual branch hitting the target is Open.

This is usually better than assigning one ambiguous tag to a mixed statement.

0.31.3 Add a branch condition

If a formula is correct only under isotropy, passivity, finite support, selected twin placement, rigid-mouth closure, or relaxed recovery, add the condition to the theorem statement and the local proof.

0.31.4 Preserve the failed route

If a stage fails, keep it if it narrows the search. Mark it as a no-go inside its declared class and state what it rules out. Failed routes are part of the derivation, especially in a ledger meant for future citation and verification.

0.32 Relationship to other papers in the program

This companion is not isolated. It is the derivation backbone that future papers should use when citing moving-throat response results.

0.32.1 Framework paper

The framework paper should be cited when a reader needs the compact operational branch test. This companion should be cited when a reader needs the derivation of that branch test.

In practical terms:

> Use `pde.tex` for how to apply the response framework. Use this companion for why the response framework has that form.

0.32.2 Parent 4D papers

The parent 4D papers provide exact action-level and controlled-reduction inputs: gauged GNLS matter, localized Maxwell theory, projection versus reduction, corrected charge ontology, and the lower-order conservative hierarchy. This companion imports those inputs; it does not need to re-prove every parent result unless the moving-throat derivation uses it in a new way.

0.32.3 PN assembly papers

The PN assembly papers provide targets and carry-forward packets that motivate the moving-throat response program. For example, the 2.5PN and 4PN packages narrow the retarded problem to the outgoing STF quadrupole normalization, while the 3PN and 5PN continuations motivate grouped real P_2, weak-axisymmetric, and finite realization compiler structures.

This companion should show how the moving-throat derivation responds to those targets. It should not silently import a target as if it were already a moving-throat theorem.

0.32.4 Atomic, lepton, and material companions

Downstream atomic, lepton, anomaly, export, and material notes may use the moving-throat quotient and branch packets. When they do, they should cite the relevant result anchor here and state whether they are using a strict branch, a rigid-mouth branch, a selected-support branch, or a relaxed open-system branch.

0.33 Checklist before citing a result from this companion

Before citing a result from this companion in another paper, check the following.

- ☐ The result has a stable result anchor or theorem label.
- ☐ The status tag is compatible with the way the result is being used.
- ☐ The branch assumptions are stated in the citing paper.
- ☐ The notation matches the corrected notation firewall.
- ☐ The cited result is not merely an early prototype superseded by a later normal form.
- ☐ Any numerical value is identified as numerical unless a closed form has been added.
- ☐ Any no-go result is cited with its search class.
- ☐ Any open branch-realization target is not described as a completed branch theorem.
- ☐ If the relaxed branch is used, the recovery map and relaxed variables are stated.
- ☐ If grouped real P_2 data are used, the grouped metric and anisotropy convention are respected.

0.34 Summary of reader rules

The ledger can be summarized by eight reader rules.

1. Read theorem anchors for stable claims and stage numbers for provenance.
2. Read every major result through its status tag.
3. Do not confuse compiler success with branch-realization success.
4. Do not treat closure-level algebra as an unconditional parent-theory theorem.
5. Do not cite a no-go result outside its declared search class.
6. Treat grouped 20/21/22 labels as real P_2 lanes with grouped metric $\mathrm{diag}(1, 2, 2)$.
7. Keep strict, rigid-mouth, selected-support, and relaxed branches distinct.
8. When in doubt, follow the result anchor to the stage appendix and check the inputs, status, checks, and downstream use.

These rules are deliberately conservative. The point of the derivation companion is not to make the moving-throat program look simpler than it is. The point is to make its derivation, assumptions, accomplishments, and remaining branch-realization gaps visible enough that future papers can cite it without rebuilding the entire stage tree.

Claim-Status Firewall

This chapter fixes the claim-status rules for the entire derivation ledger. The moving-throat program contains several different kinds of mathematical statements: exact identities of the declared parent theory, exact algebra inside reduced closures, controlled reductions, effective closure choices, numerical placements, and open branch-realization targets. These categories are not interchangeable. The purpose of the claim-status firewall is to make that separation visible everywhere the ledger states a result.

The firewall should be read as part of the mathematics of the document, not as editorial decoration. A theorem that is exact only inside a closure is not an exact theorem of the full nonlinear moving-throat PDE. A compiler that reduces a branch test to finite observables is not the same thing as a proof that the actual branch realizes those observables. A numerically located point is not a closed-form value merely because the equations defining it are exact. An open target remains open even if all algebra leading to the target is exact.

The status tag attached to a result states the strongest honest way that result may be cited.

Every theorem, proposition, lemma, boxed equation, branch verdict, no-go statement, finite search result, and stage output in this companion must carry one of the status tags defined below.

0.35 The six allowed status tags

The ledger uses exactly six status tags. These tags are intentionally coarse. The goal is not to encode every nuance in the tag itself; the goal is to prevent the most dangerous mistakes: treating reduced results as exact, treating closures as solved PDE branches, treating numerical locations as closed-form derivations, or treating open branch targets as completed theorems.

Status tag	Meaning
Exact	The statement follows directly from the declared parent action, exact definitions, exact identities, exact variational equations, or exact algebra with no additional branch ansatz beyond the stated definitions.
Exact Within Closure	The statement is exact inside a stated reduced closure family, branch, hierarchy, search class, or ansatz. The algebra inside the closure is exact, but the closure itself is not thereby promoted to an unconditional theorem of the completed PDE.

Status tag	Meaning
REDUCED / CONTROLLED REDUCTION	The statement follows after a controlled reduction, such as a projection, zero-mode limit, low-frequency expansion, small-body/worldtube approximation, finite-mode truncation, selected-branch restriction, isotropic branch assumption, passive/outgoing reduction, or asymptotic regime.
EFFECTIVE CLOSURE	The statement uses a physically motivated closure choice, modeling ansatz, constitutive packet, or branch prescription that is useful and internally coherent but not yet uniquely forced by the completed moving-throat PDE.
NUMERICALLY LOCATED	The defining equations are exact or closure-exact, but the reported value, branch point, candidate, finite placement, or search output is currently known through numerical solve, numerical scan, or numerical localization rather than closed-form derivation.
OPEN	The statement still depends on actual branch realization, a PDE-side existence or uniqueness theorem, a not-yet-computed overlap, a not-yet-verified normalization, or another result not completed in the current ledger.

These tags are mutually exclusive for a single atomic claim. A paragraph may contain several claims with different statuses; in that case the paragraph must either split the claims or explicitly state the status of each part.

0.36 Audit-status vocabulary

The six mathematical status tags above say how strongly a claim may be cited. The audit-status vocabulary below is a second, orthogonal axis: it says what verification evidence is attached to a stage, value, or audit record. An audit status never promotes, demotes, or replaces the mathematical claim-status tag.

Audit-status term	Definition
`CAS-checked`	A stated algebraic, symbolic, or numerical step has been checked by a computer-algebra or script audit appropriate to that step.
`dual-engine-checked`	The check has been replayed independently across the SymPy and Mathematica audit layers.
`value-reconciled`	A reported document value has been compared against the script or notebook output that supplies it.
`value-provenance-checked`	The source of a reported value has a recoverable provenance path through the stage record, notes, scripts, notebooks, or review trackers.

Audit-status term	Definition
`external-benchmark-sourced`	The value or comparison target is tied to an explicit external benchmark or source rather than only to internal ledger algebra.
`fit-vs-derive-audited`	The adversarial audit has tested whether the claimed derivation is genuinely derived rather than fit to a target value or benchmark.
`branch-realization-tested`	A target-blind numerical PDE branch realization test has been run for the stated branch or target.

Per-stage audit-status values are generated from machine-readable audit state: the red-team manifest, verification trackers, and future `redteam_adversarial/` artifacts. They are never hand-written into individual stage cards or inline theorem prose. Hand-maintained inline tags are the stale-label failure mode that the numbering remediation already cleaned up; this section defines the vocabulary only.

0.37 Atomic claims and composite claims

A status tag applies to an atomic claim, not automatically to an entire paragraph, theorem, or chapter. The following distinction is essential.

0.37.1 Atomic claim

An atomic claim is a single assertion whose assumptions are uniform. For example:

- the exact parent Maxwell equation follows from varying the localized Maxwell action;
- the grouped real P_2 inverse map follows from exact linear algebra;
- the outgoing $l = 2$ low-frequency fingerprint follows from the declared compact outgoing partial-wave branch;
- a Family-1 mouth point is located numerically by solving a stated reduced equation;
- the actual moving-throat branch has not yet been proved to hit a target scalar.

Each of these may receive one status tag.

0.37.2 Composite claim

A composite claim combines several atomic claims. Its usable status is the weakest status among the parts needed for the conclusion. For example, the algebraic bridge

$$P_0 = \frac{N_0}{D_0}$$

may be exact inside the reduced grouped-bundle closure, while the assertion that the actual branch realizes the target value of P_0 may remain open. The composite statement "the branch realizes the outgoing normalization" therefore cannot be cited as exact merely because the formula defining P_0 is exact inside the closure.

When a theorem contains both exact algebra and an open realization target, split it as follows:

1. an exact or closure-exact theorem deriving the formula or compiler;
2. a branch-realization corollary marked Open, Numerically Located, or Reduced / Controlled Reduction, as appropriate.

0.38 Status label definitions in detail

0.38.1 Exact

Use Exact only for statements that follow from the declared parent theory, definitions, or algebra without additional physical closure. This includes exact variational equations, exact conservation laws, exact gauge-invariant definitions, exact projection identities once the projection kernel is declared, and exact algebraic transformations.

Typical examples in this companion include:

1. the gauged GNLS equation obtained by varying the declared matter action;
2. the exact bulk continuity equation;
3. the localized Maxwell equation obtained by varying the declared localized Maxwell action;
4. the Bianchi identity;
5. the exact mixed-field gauge invariants $E_w = F_{w0}$ and $C_a = F_{aw}$;
6. the exact projected continuity identity once $W(w)$ is declared;
7. exact finite-dimensional linear algebra, such as grouped projector identities or exact inverse maps.

A result should not receive Exact merely because its algebra is exact after a closure has been chosen. If a wall action, branch ansatz, support family, selected slice, or finite search class is required, the result is not Exact unless that closure has itself been proved as an exact consequence of the parent PDE.

0.38.2 Exact Within Closure

Use Exact Within Closure when the derivation is exact inside a named closure, branch, hierarchy, or search class. This tag is the most common tag in the moving-throat ledger. It honors the fact that many of the program's important results are not approximations once their closure assumptions are accepted, while still preventing those results from being misread as unconditional parent-theory theorems.

A closure-exact statement must name its closure. Examples of acceptable closure names include:

- the distributed wall/shape-field closure;
- the stable BdG normal-mode reduction;
- the one-lane Maxwell/mixed prototype;
- the full grouped real P_2 reduced bundle;
- the isotropic passive/outgoing branch;
- the coherent local D/N support hierarchy;
- the weak-axisymmetric branch;

- the rigid-mouth branch;
- the selected twin-support branch;
- the finite local mixed-ray search class;
- the relaxed/open-system branch with its declared recovery map.

The phrase "exact inside the closure" means the derivation is algebraically forced by the closure inputs. It does not mean the closure has been uniquely realized by the full nonlinear PDE.

0.38.3 Reduced / Controlled Reduction

Use Reduced / Controlled Reduction for results that require a controlled reduction or asymptotic restriction. These results may be highly reliable inside their regimes, but their validity depends on the reduction assumptions.

Typical reduction assumptions include:

1. zero-mode Maxwell reduction with $A_w \approx 0$, $\partial_w A_\mu \approx 0$, $J^w \approx 0$, and $F_{\mu w} \approx 0$;
2. quasi-static or low-frequency expansion;
3. small-body/worldtube reduction;
4. isotropic point-particle limit;
5. finite support-cardinality restriction;
6. selected-branch projection;
7. truncation to a finite family of support modes;
8. compact passive/outgoing partial-wave completion;
9. regime statements such as "higher odd channels are irrelevant through this order."

A controlled reduction should state what is neglected, suppressed, projected out, or assumed small. The reduction should also state what would have to be checked before the result could be used outside the declared regime.

0.38.4 Effective Closure

Use Effective Closure for a physically motivated closure choice that is useful for theorem work but not yet uniquely selected by the completed moving-throat PDE. This tag is appropriate for modeling decisions that organize the derivation without pretending to be final microscopic consequences.

Examples include:

1. choosing the hybrid level-set/shape-field representation $\Sigma = r - R(\Omega, w, t)$ as the working moving-throat lift;
2. introducing the minimal quadratic distributed wall action as the first passive local wall model;
3. selecting a low-dimensional prototype before proving the full branch realizes it;
4. using a closure hierarchy as a constructive compiler rather than as an already solved PDE theorem.

An effective closure can later be promoted if the ledger adds a proof that the completed PDE selects it. Until then, future papers must cite it as an effective closure, not as an exact theorem.

0.38.5 Numerically Located

Use Numerically Located when the equations defining the object are exact, but the reported value is presently obtained by numerical solve, numerical localization, numerical search, or numerical continuation. The exactness of the defining equations does not make the numerical value exact in closed form.

A numerically located result should record:

1. the equation or system solved;
2. the branch or interval searched;
3. the numerical value and precision actually justified;
4. whether uniqueness was proved or only observed;
5. the script, notebook, or audit file supporting the value;
6. whether the value is used as an input to later exact algebra or only as an illustrative branch point.

If a later closed-form derivation is added, the result may be promoted. Until then, the numerical status must remain visible.

0.38.6 Open

Use Open for statements that still depend on the completed moving-throat PDE branch or another missing theorem. Open status is not a failure of the ledger. It is the honest marker of what remains to be proved or computed.

Typical open objects include:

1. actual branch realization of the canonical outgoing normalization;
2. actual realization of $N_Q = 1$ or $\chi_Q = 1$ on the full branch;
3. actual weak-axisymmetric prefactor slope $\Xi_1 = P_1/P_0$ on the physical branch;
4. actual rigid-mouth orbit-lock data when the rigid-mouth assumptions have not yet been verified by the branch;
5. actual selected twin-support placement data;
6. realized leakage, non-rigid, source, export, calibration, or material-screening coefficients on a relaxed branch;
7. any existence, uniqueness, stability, or regularity theorem for a branch not yet proved.

Open status should be used even when all surrounding algebra is complete. A formula can be exact, a compiler can be exact within closure, and the actual branch target can still be open.

0.39 The weakest-link rule

When a result depends on multiple ingredients, its citation status is determined by the weakest ingredient required for the conclusion. This is the ledger's weakest-link rule.

Ingredients required by conclusion	Resulting usable status
Exact parent identity only	Exact
Exact parent identity plus a named closure whose internal algebra is exact	Exact Within Closure
Exact or closure-exact algebra plus a projection, low-frequency limit, small-body limit, or selected asymptotic regime	Reduced / Controlled Reduction
Exact algebra plus a physically motivated but not uniquely derived model choice	Effective Closure, unless the theorem is explicitly restricted to that model and marked Exact Within Closure
Exact equations plus numerical root, numerical placement, numerical branch point, or numerical finite search output	Numerically Located
Any ingredient requiring actual branch realization not yet completed	Open

A theorem may contain a high-status subclaim and a low-status conclusion. For instance, the exact formula

$$\Gamma_5 = P_0 \frac{a^5}{27c_s^5}$$

inside the normalized outgoing bridge may be closure-exact, while the assertion that the actual branch realizes the target value of P_0 is open. The theorem should therefore be split into a closure-exact bridge and an open realization condition.

0.40 Compiler success, realization success, and closure success

The moving-throat program repeatedly separates three kinds of success. The firewall requires that all three remain distinct.

0.40.1 Compiler success

Compiler success means the derivation reduces a branch test to explicit observables or scalar packets. Examples include:

- reducing the outgoing quadrupole test to $\widehat{m}_0^2 P_0$;
- reducing grouped real P_2 isotropy to trace/anomaly variables;
- reducing the weak-axisymmetric prefactor problem to $\Xi_1 = P_1/P_0$;

- reducing a branch verdict to $\Delta_{\text{full}} = (\Delta_Q, q_{\text{tr}}, q_{\text{nt}}, q_\eta)$;
- reducing rigid-mouth orbit lock to $(U, V) = (0, 0)$.

Compiler success is usually Exact Within Closure or Reduced / Controlled Reduction. It does not imply that the actual branch passes the test.

0.40.2 Realization success

Realization success means the actual moving-throat branch returns the observables required by the compiler. For example, once the compiler says the branch must satisfy

$$\widehat{m}_0^2 P_0 = \frac{54 G c_s^5}{5 a^5 c^5},$$

realization success is the separate claim that the actual branch has that value. Unless the full branch computation proves it, this remains Open or Numerically Located.

0.40.3 Closure success

Closure success means the actual branch satisfies the assumptions under which the compiler was derived. A branch may hit a scalar target numerically while failing isotropy, passivity, support selection, or rigid-mouth assumptions. In that case the scalar hit is not enough to cite the closure theorem as realized.

For every branch theorem, the ledger should state:

1. the compiler packet;
2. the realization packet;
3. the closure assumptions that must be checked.

0.41 Promotion, demotion, and supersession

Status tags are allowed to change as the ledger matures, but only by explicit audit.

0.41.1 Promotion rule

A result may be promoted only if the missing assumption that caused the lower status has itself been proved or removed. Examples:

- A Numerically Located branch point may be promoted if a closed-form solution and uniqueness proof are added.
- A Effective Closure closure may be promoted to Exact Within Closure if the document restricts the theorem explicitly to that closure and proves the algebra inside it.
- A Open realization condition may be promoted only after the actual branch computation verifies the condition and the closure assumptions.

Promotion must name the new proof or computation. A result is not promoted merely because later prose relies on it.

0.41.2 Demotion rule

A result must be demoted when a missing assumption, branch failure, inconsistency, stale notation, unverified numerical step, or unsupported generalization is found. Demotion should preserve the useful part of the result when possible. For example, a failed branch may still be a valid no-go theorem inside its search class.

The demoted statement should be rewritten as one of the following:

- exact algebra inside a narrower closure;
- a controlled reduction rather than an unconditional theorem;
- a numerical observation rather than a closed-form theorem;
- an open branch target rather than a realized branch result;
- a no-go result limited to its tested class.

0.41.3 Supersession rule

When a later stage replaces an earlier prototype with a cleaner normal form, the earlier result should not simply disappear. It should be marked as superseded, with a pointer to the later result. This matters because future readers may encounter old stage numbers in notes or scripts.

A superseded result may still be useful as:

1. a motivating prototype;
2. a failed branch;
3. a special case of a later theorem;
4. an intermediate algebraic check;
5. a warning about a notation convention that later changed.

0.42 Status inheritance by document layer

The same formula can appear in different document layers with different status roles. The ledger should be explicit about which role is active.

0.42.1 Front matter

The front matter states rules, vocabulary, and document contract. It should not introduce new physical theorems except by summarizing their status. If the front matter mentions a result such as $\Xi_1 = P_1/P_0$, it is describing the status system and citation map, not proving the result.

0.42.2 Main theorem blocks

The main body is the primary layer for stable theorem statements. Each theorem block must include:

1. a result anchor or theorem label;
2. a status tag;

3. a stage range;
4. assumptions and closure conditions;
5. proof skeleton;
6. downstream use;
7. open residue, if any.

0.42.3 Stage appendices

The stage appendices preserve derivation history. A stage output may be provisional, superseded, or later demoted. The appendix entry should record the local status at the time of the derivation and the current status after later stages, when different.

A useful appendix pattern is:

> **Local output.** What the stage established locally. **Current status.** How the result should now be cited after later stages.

0.42.4 Reproducibility appendix

The reproducibility appendix certifies algebraic, symbolic, or numerical checks. A script-backed algebraic identity may support Exact Within Closure; a numerical search supports Numerically Located; a failed search supports a no-go claim only inside the declared search class.

0.43 No-go and negative-result statuses

A no-go theorem is not automatically an Open statement. A no-go result can be exact, closure-exact, reduced, or numerical depending on the search class and proof method.

No-go type	Appropriate status
Exact algebraic contradiction from definitions	Exact
No-go inside a declared reduced closure or one-port model	Exact Within Closure
No-go inside a finite truncation, low-frequency regime, or selected branch class	Reduced / Controlled Reduction
No solution found by exhaustive numerical search over an exactly declared finite set	Numerically Located, unless the exhaustive enumeration is itself exact and all arithmetic has been certified
No-go suspected but not proved for the actual PDE branch	Open

Every no-go result must state its domain. A statement such as “the mixed sector cannot help” is not acceptable unless the mixed sector, coupling class, frequency regime, and branch assumptions are named. A correct statement might be: “inside the one-port static same-charge closure, the static mixed bundle creates no new long-range attractive law.”

0.44 Numerical values and search outputs

Numerical values are useful, but they are also common sources of accidental overclaiming. The firewall requires the following discipline.

0.44.1 Numerical branch points

A numerically located branch point should be written in a form such as:

> The point $(\Pi_*, \widehat{T}_{m,*})$ is Numerically Located as presently recorded: it is defined by the exact reduced branch equations, but the quoted values are numerical localizations.

Do not write “exactly equals” before a decimal value unless the decimal is a representation of a closed-form expression already proved.

0.44.2 Finite searches

A finite search result should name:

1. the search space;
2. the support-cardinality or basis bound;
3. whether the enumeration is exact or numerical;
4. the acceptance criteria;
5. the rejected classes;
6. the surviving candidates;
7. the role of the search in later theorem blocks.

If the search is exact through support-cardinality ≤ 5, say exactly that. Do not imply it exhausts all possible PDE branches unless that stronger theorem has been proved.

0.44.3 Numerical residuals

Numerical residuals should be quoted with enough context to determine their meaning. The residual may be:

- a mismatch against a target;
- a graph error;
- a branch-realization deficit;
- a search residual;
- a numerical precision artifact.

The status of the residual follows the status of the computation that produced it.

0.45 Common moving-throat packets and their default status

This section records the default status reading for common packets used throughout the companion. Later chapters may refine these statuses, but they should not silently weaken or strengthen them.

Object or packet	Default status	Reading rule
Parent GNLS, localized Maxwell, continuity, mixed gauge invariants	Exact	Exact statements of the declared parent theory.
Projection identity and leakage formula after declaring $W(w)$	Exact	Projection is an operational definition; later brane reductions are separate.
Zero-mode brane Maxwell reduction	Reduced / Controlled Reduction	Controlled far-field brane reduction; mixed core channels are suppressed, not removed.
Shape-field lift $\Sigma = r - R(\Omega, w, t)$	Effective Closure	Working moving-throat lift unless and until derived uniquely from the completed PDE.
Distributed quadratic wall action and modal wall PDE	Exact Within Closure	Exact inside the declared distributed wall closure; the closure itself is a modeling layer.
BdG Schur-complement self-energy formulas	Exact Within Closure	Exact after stable BdG normal-mode reduction and declared coupling data.
Maxwell/mixed one-lane transfer factor	Exact Within Closure	Exact inside the reduced one-lane Maxwell/mixed model.
Grouped real P_2 projector calculus	Exact	Exact finite-dimensional linear algebra once grouped convention is declared.
Full grouped bundle coefficients $D_{A,n}, N_{A,n}$	Exact Within Closure	Exact inside the reduced grouped wall/BdG/Maxwell/mixed bundle.
Outgoing $l = 2$ compact fingerprint	Reduced / Controlled Reduction	Controlled outgoing partial-wave branch and low-frequency expansion.
Normalization target involving $\widehat{m}_0^2 P_0$	Reduced / Controlled Reduction	Exact bridge inside the isotropic passive/outgoing point-particle reduction; actual realization remains separate.
Actual realization of $N_Q = 1$, $\chi_Q = 1$, or $\Delta_Q = 0$	Open	Open unless the actual moving-throat branch computation verifies it.
Weak-axisymmetric relation $b = 3a$ for pure axisymmetric quadrupole splitting	Exact Within Closure	Exact inside the weak-axisymmetric angular splitting closure.

Object or packet	Default status	Reading rule
Weak-axisymmetric prefactor slope $\Xi_1 = P_1/P_0$	EXACT WITHIN CLOSURE for the compiler; OPEN for actual branch value	The identity can be exact in the reduced compiler while the physical value is open.
Finite local mixed-ray search through support-cardinality ≤ 5	EXACT WITHIN CLOSURE or NUMERICALLY LOCATED, depending on implementation	Exact if the enumeration and arithmetic are certified; numerical if candidate placement is numerical.
Rigid-mouth chart (U, V) and orbit-lock condition $U = V = 0$	EXACT WITHIN CLOSURE for the chart; OPEN for actual lock	The compiler is closure-exact; actual branch lock is a realization claim.
Selected twin-support slice $\rho_\alpha = 4/3$, $\zeta_{\text{req}} = 1/3$	EXACT WITHIN CLOSURE	Exact inside the selected twin-support branch and ranking assumptions.
Relaxed recovery map $\ell_w = f_U = a = b = 0$	EXACT WITHIN CLOSURE	Exact inside the declared relaxed branch; use of the relaxed branch remains distinct from strict-branch proof.
Relaxed leakage/work/export/material coefficients	OPEN unless numerically or analytically realized	The relaxed compiler may be exact, while the actual material/calibration data remain branch-realization inputs.

0.46 How theorem statements should display status

Every theorem-like statement should include a status line near the statement, before the proof or derivation. The recommended format is:

> **Status.** EXACT WITHIN CLOSURE inside the isotropic grouped wall/BdG/Maxwell/mixed reduced bundle. Actual branch realization of the target scalar is OPEN.

If a theorem has multiple layers, list them explicitly. For example:

> **Status.** The projector identity is EXACT. The full grouped-bundle coefficient formulas are EXACT WITHIN CLOSURE. The use of the isotropic passive/outgoing point-particle branch is REDUCED / CONTROLLED REDUCTION. The actual branch value of P_0 is OPEN.

This style prevents a reader from mistaking the highest-status subclaim for the status of the entire theorem.

0.47 How stage entries should display status

Every stage appendix entry should include a **Claim status** field. A recommended template is:

Claim status. Local derivation: Exact Within Closure inside the one-port reduced model. Current citation status: superseded by the full grouped-bundle theorem, where the corresponding identity is Exact Within Closure and actual branch realization is Open.

When a stage contains a failed route, the status field should say whether the failure is exact in the tested class, numerical, or open. For example:

Claim status. Exact Within Closure no-go inside the wall-only weak-axisymmetric primitive deformation class. This does not rule out mixed-sector or relaxed-branch deformations.

0.48 Branch assumptions that must appear beside status tags

Status tags are not enough by themselves. Many closure-exact and reduced statements are valid only with branch assumptions. When any of the following assumptions is used, it must be named beside the status tag.

1. **Isotropy.** The grouped real P_2 channels share common coefficients and the anisotropy defects vanish.
2. **Weak-axisymmetry.** The grouped splitting has the $(20, 21, 22) \sim (1, 1/2, -1)$ signature or equivalently $b = 3a$.
3. **Passivity/outgoing condition.** The odd response is carried by a passive outgoing branch with the declared sign convention.
4. **Point-particle or small-body limit.** Source-map corrections beyond the retained order are suppressed.
5. **One-port or finite-port restriction.** Only the declared port class is included.
6. **Stable support spectrum.** The BdG or support modes integrated out have the declared stable spectrum.
7. **Rigid-mouth condition.** The mouth variables obey the rigid-mouth normal form used to define (U, V).
8. **Selected support branch.** The branch lies on the selected twin-support or selected support curve being used.
9. **Relaxed branch recovery.** The relaxed branch variables and the recovery map are stated.
10. **Finite search class.** The graph, mixed-ray, support-cardinality, or search-basis restrictions are stated.

If the assumption is not present, the result should either be demoted or rewritten as a conditional statement.

0.49 Status examples for recurring formulas

The following examples illustrate how to cite recurring formulas without overclaiming.

0.49.1 Projected continuity

The identity

$$\partial_t \rho_{\text{brane}} + \nabla_3 \cdot \mathbf{j}_{\text{brane}} = S_{\text{leak}}$$

with the declared projection kernel $W(w)$ is Exact. Any later statement that this leakage term is small, stationary, compensating, material-screened, or dynamically harmless requires a further reduction or branch assumption.

0.49.2 Zero-mode Maxwell limit

The reduction to effective brane Maxwell theory with

$$\mu_0^{\text{eff}} = \frac{\mu_0}{Z_{\text{int}}}$$

is Reduced / Controlled Reduction. The mixed channels $A_w, J^w, F_{\mu w}, E_w, C_a$ are suppressed only inside that controlled far-field limit. They remain microscopic variables in the moving-throat PDE.

0.49.3 Grouped real P_2 projectors

The grouped projector calculus with grouped metric $G_{\text{grp}} = \text{diag}(1, 2, 2)$ is Exact finite-dimensional algebra once the grouped convention is declared. However, the statement that a physical branch is isotropic in these projectors is a separate branch claim.

0.49.4 Outgoing normalization product

The bridge

$$\widehat{m}_0^{\,2} P_0 = \frac{54 G c_s^5}{5 a^5 c^5}$$

is Reduced / Controlled Reduction as an isotropic passive/outgoing point-particle normalization condition. The derivation of $P_0 = N_0/D_0$ is Exact Within Closure inside the grouped reduced bundle. The claim that the actual branch realizes the equation is Open until the branch data are computed.

0.49.5 Weak-axisymmetric prefactor slope

The identity

$$\Xi_1 = \frac{P_1}{P_0}$$

is a closure-exact compiler identity inside the weak-axisymmetric prefactor-transport framework. The numerical or physical value of Ξ_1 on the actual branch is a realization question and should be marked Open or Numerically Located depending on the available computation.

0.49.6 Rigid-mouth orbit lock

The chart

$$(U, V) = \left(\ln \frac{\mathcal{T}^2}{\mathcal{T}_{\text{ref}}^2}, \ln \frac{\epsilon_\eta}{\epsilon_{\eta,\text{ref}}} \right)$$

and the orbit-lock condition $U = V = 0$ are Exact Within Closure inside the rigid-mouth normal form. The assertion that the actual branch is orbit-locked is a separate realization claim.

0.49.7 Relaxed branch recovery

The recovery map

$$\ell_w = f_U = a = b = 0$$

is Exact Within Closure inside the declared relaxed branch. It should not be used to claim that the relaxed branch proves the strict branch. It states how the relaxed branch returns to the strict front end when the relaxed variables are shut off.

0.50 Status-aware citation language

Future papers should use language compatible with the status tag. The following wording conventions are recommended.

Status	Safe citation language
Exact	"follows from the declared parent action"; "is an exact identity"; "is exact finite-dimensional algebra."
Exact Within Closure	"is exact inside the stated closure"; "within the coherent reduced branch"; "for the one-port model"; "under the rigid-mouth normal form."
Reduced / Controlled Reduction	"in the controlled zero-mode reduction"; "in the low-frequency point-particle branch"; "after the selected-branch reduction."
Effective Closure	"we use the effective closure"; "this is the working wall model"; "this branch prescription is not yet uniquely derived from the completed PDE."
Numerically Located	"the branch point is numerically located"; "the scan finds"; "the current numerical solution gives."
Open	"the remaining target is"; "actual branch realization remains open"; "the PDE must still return"; "this has not yet been proved on the full branch."

Unsafe language includes:

- saying "proved by the PDE" when only a reduced closure was proved;
- saying "exact value" for a decimal branch point;
- saying "ruled out" without naming the search class;
- saying "realized" when only the compiler exists;
- saying "Maxwell reduction removes mixed channels" rather than "suppresses mixed channels in the strict far-field limit."

0.51 Status checklist for authors

Before finalizing any section of this ledger, apply the following checklist.

- ☐ Every theorem, proposition, lemma, boxed result, and branch verdict has a visible status tag.
- ☐ Every closure-exact statement names the closure.
- ☐ Every controlled reduction names the regime or approximation.
- ☐ Every numerical value is marked numerical unless a closed form is provided.
- ☐ Every open target is marked open even if the surrounding algebra is complete.
- ☐ Every no-go result states its search class.
- ☐ Every mixed-status theorem splits compiler, realization, and closure assumptions.
- ☐ Every superseded prototype points to the later theorem or normal form.
- ☐ Every result anchor has a status compatible with the strongest claim future papers are expected to cite.
- ☐ No result uses the notation firewall to hide a status change.

0.52 Status checklist for readers and referees

When verifying a result, a reader should ask:

1. Is the cited result an exact identity, closure-exact theorem, controlled reduction, effective closure, numerical placement, or open target?
2. Does the proof use assumptions not visible in the theorem statement?
3. Is the formula being used as a compiler or as a realized branch fact?
4. Does the downstream paper cite the result at the correct status level?
5. Is a no-go claim being applied outside its tested class?
6. Has a numerical search been mistaken for a closed-form theorem?
7. Has a relaxed branch result been used as if it proved the strict branch?
8. Has an early stage been superseded by a later normal form?

If any answer is unclear, follow the result anchor to the stage appendix and check the local inputs, assumptions, checks, and downstream-use field.

0.53 Minimal status statement template

When filling the remaining chapters, use the following minimal template for major results:

> **Status.** [One of Exact, Exact Within Closure, Reduced / Controlled Reduction, Effective Closure, Numerically Located, Open.] Exact/valid under [named assumptions]. The compiler part is [status]. Actual branch realization is [status].

A more detailed theorem may use:

Status split.

1. Algebraic identity: [status].
2. Closure assumptions: [closure and status].
3. Reduction assumptions: [regime and status].
4. Numerical placement: [value and status, if any].
5. Actual branch realization: [status].

This template is intentionally repetitive. Repetition is preferable to ambiguity in a derivation ledger.

0.54 Summary of the firewall

The claim-status firewall can be summarized in five rules.

1. Use Exact only for parent-theory identities, exact definitions, and exact algebra that do not depend on a further branch closure.
2. Use Exact Within Closure for exact algebra inside a named closure, and always name the closure.
3. Use Reduced / Controlled Reduction when a projection, limit, truncation, low-frequency expansion, small-body approximation, or selected branch is required.
4. Use Numerically Located for numerical branch data and Open for branch-realization targets not yet proved by the completed PDE.
5. When a result contains several layers, cite it at the weakest status needed for the conclusion.

The status tag is not a formality. It is the guardrail that lets this companion be useful as a long-term derivation archive. Future papers can cite this ledger safely only if the firewall remains intact.

Notation Firewall

This chapter fixes the notation rules for the entire derivation ledger. Its purpose is not to introduce new physics. Its purpose is to prevent category mistakes while the document moves between the exact parent theory, the lifted moving-throat geometry, the grouped real P_2 response bundle, the outgoing-normalization bridge, the weak-axisymmetric quotient variables, the realization compiler, and the relaxed open-system extension.

The moving-throat derivation uses several symbols that are easy to confuse if their layer is not stated. The symbol q, for example, can refer in older notes to a gravity-side scalar coefficient, in the parent gauge theory to electric charge, in the wall geometry to a harmonic amplitude, in the weak-axisymmetric quotient to a log-drift coordinate, and in post-Newtonian residual algebra to a coefficient in a finite basis. These are different objects. The notation firewall below says which meanings are allowed in this ledger and which meanings must be avoided.

> A symbol is never promoted across layers by notation alone. A parent-theory variable, a reduced-closure coefficient, a branch observable, and a realization target must remain visibly distinct.

This chapter should be read together with section 0.34. The claim-status firewall says how strong a statement is. The notation firewall says what the symbols in that statement are allowed to mean.

0.55 Global anti-ambiguity rules

The following rules apply everywhere in this companion.

1. **No silent reuse across layers.** A symbol used in the exact parent theory may be reused in a reduced branch only when the text explicitly states the new local meaning. Otherwise the parent-theory meaning remains active.

2. **No hidden summation over grouped labels.** The grouped labels 20, 21, and 22 are lane labels, not tensor indices. They are summed only when a summation sign, a grouped trace, or an explicit weighted average is written.

3. **No charge from circulation.** Electric charge sign is never inferred from circulation, throat radius, breathing amplitude, or wall harmonic data.

4. **No deletion of mixed channels.** A strict brane Maxwell reduction may suppress A_w, J^w, and $F_{\mu w}$, but it does not remove those channels from the microscopic ontology.

5. **No branch realization by notation.** Writing a target such as P_0, Ξ_1, $U = V = 0$, or $\Delta_{\text{full}} = 0$ does not prove that the actual moving-throat branch realizes it.

6. **Subscripts must be local.** Subscripts $0, 2, 4$ often label powers in a low-frequency expansion. They are not automatically PN orders, not automatically harmonic degrees, and not automatically branch numbers.

7. **Acronyms and packets must be expanded locally.** The first time a chapter uses a packet such as Δ_{full}, $\mathfrak{C}_{\mathrm{tr},*}$, or $K_{\exp}$, it should remind the reader which layer that packet belongs to.

These rules are especially important in the stage appendices, where the notation sometimes reflects the historical path of discovery. When an older stage used a symbol in a way that later became unsafe, the appendix entry should record both the original local meaning and the current ledger meaning.

0.56 Coordinate, index, and operator conventions

The fundamental arena is a $(4+1)$-dimensional spacetime. Coordinates are written

$$x^M = (t, x, y, z, w), \qquad M, N, L \in \{0, 1, 2, 3, 4\}, \qquad x^0 = t, \quad x^4 = w. \tag{1}$$

The bulk spatial coordinate is

$$\mathbf{X} = (x, y, z, w) \in \mathbb{R}^4, \tag{2}$$

while the brane spatial coordinate is

$$\mathbf{x} = (x, y, z) \in \mathbb{R}^3. \tag{3}$$

The brane radius and brane angular direction are

$$r = |\mathbf{x}| = \sqrt{x^2 + y^2 + z^2}, \qquad \Omega = \frac{\mathbf{x}}{r} \in S^2. \tag{4}$$

The preferred index sets are listed in table 11.

Table 11: Index conventions used by the derivation ledger.

Index symbol	Range	Use
M, N, L	$0, 1, 2, 3, 4$	Bulk spacetime indices.
μ, ν, λ	$0, 1, 2, 3$	Brane spacetime indices.
i, j, k	x, y, z, w	Bulk spatial indices.
a, b, c	x, y, z	Brane spatial indices. Do not confuse these with the throat radius $a(t)$, grouped anomaly variables a_x, b_x, or relaxed-branch coordinates named a, b.
A, B	Context-dependent	Avoid for spacetime indices in this ledger. When used in grouped response sections, $A, B \in \{20, 21, 22\}$ are grouped lane labels.
α, r, n	Local	Mode, port, or expansion labels. Their meaning must be stated in the local section.

The bulk metric convention is

$$\eta_{MN} = \mathrm{diag}(-1, +1, +1, +1, +1). \tag{5}$$

The bulk d'Alembertian is

$$\Box_5 = -\partial_t^2 + \nabla_3^2 + \partial_w^2. \tag{6}$$

The three- and four-spatial gradients are

$$\nabla_3 = (\partial_x, \partial_y, \partial_z), \qquad \nabla_4 = (\partial_x, \partial_y, \partial_z, \partial_w). \tag{7}$$

In the moving-throat chapters, w is the bulk axial coordinate. A finite-throat axial coordinate such as s or z may be introduced for one-dimensional reduced support calculations; when this happens, the text must state whether s or z is identical to w, a rescaled version of w, or a local coordinate along the throat.

0.57 Projection versus reduction

Projection and reduction are different operations and use different symbols.

0.57.1 Projection

Projection defines what a brane observer measures. A normalized projection kernel is written

$$W(w), \qquad \int_{-\infty}^{+\infty} W(w)\, dw = 1. \tag{8}$$

For a bulk field $Q(t, \mathbf{x}, w)$, the projected brane observable is

$$\overline{Q}(t, \mathbf{x}) = \int_{-\infty}^{+\infty} W(w) Q(t, \mathbf{x}, w)\, dw. \tag{9}$$

Projection is a measurement definition once W is declared. It does not by itself eliminate the extra coordinate dynamically.

0.57.2 Reduction

Reduction integrates out or suppresses variables under a stated ansatz, limit, branch restriction, or asymptotic regime. Examples include the far-field zero-mode Maxwell reduction, the stable BdG normal-mode reduction, a finite-throat support-mode truncation, a low-frequency expansion, an isotropic branch restriction, or a selected-support slice.

The localization profile in the Maxwell action is written

$$Z(w), \qquad Z_{\text{int}} = \int_{-\infty}^{+\infty} Z(w)\, dw. \tag{10}$$

The symbol $Z(w)$ controls bulk gauge dynamics. The symbol $W(w)$ controls brane observation. Even if a matched choice $W = Z/Z_{\text{int}}$ is later useful, the two symbols must remain distinct.

The controlled zero-mode Maxwell reduction may use assumptions of the form

$$A_w \approx 0, \qquad \partial_w A_\mu \approx 0, \qquad J^w \approx 0, \qquad F_{\mu w} \approx 0. \tag{11}$$

Equation (11) is a reduction statement, not an ontology statement. It says which mixed quantities are suppressed in one far-field brane regime. It does not say that those quantities are absent from the parent theory or from the moving-throat PDE.

0.58 Parent field variables

The exact parent theory uses three core field layers: matter, gauge, and geometry.

0.58.1 Matter and hydrodynamic variables

The matter/order parameter is

$$\psi(\mathbf{X}, t), \qquad \rho = |\psi|^2. \tag{12}$$

The bulk number current is written

$$j^i = \frac{\hbar}{m}\,\mathrm{Im}(\psi^* D_i \psi), \tag{13}$$

with exact continuity

$$\partial_t \rho + \partial_i j^i = 0. \tag{14}$$

When $\rho > 0$, the bulk velocity is defined by

$$j^i = \rho v^i. \tag{15}$$

The gauge-invariant Madelung velocity is

$$v_i = \frac{\hbar}{m}\partial_i \theta - \frac{q_\star}{m} A_i, \qquad \psi = \sqrt{\rho}\, e^{i\theta}. \tag{16}$$

The quantum potential is

$$Q(\rho) = -\frac{\hbar^2}{2m}\frac{\nabla_4^2 \sqrt{\rho}}{\sqrt{\rho}}. \tag{17}$$

The pressure law is denoted

$$P(\rho) = K_{\mathrm{EOS}} \rho^5. \tag{18}$$

The symbol $P(\rho)$ is pressure. It is not the outgoing prefactor P_0, P_1, P_2, or P_4. The EOS coefficient should be written K_{EOS} whenever there is any risk of confusion with wall stiffnesses such as K_η, K_A, K_U, or K_W^{eff}.

0.58.2 Gauge variables and currents

The gauge potential is

$$A_M = (A_0, A_i), \tag{19}$$

with field strength

$$F_{MN} = \partial_M A_N - \partial_N A_M. \tag{20}$$

The localized Maxwell equation uses a total current

$$J^M_{\mathrm{tot}} = J^M_\psi + J^M_{\mathrm{ext}}. \tag{21}$$

Here J^M_ψ is the dynamical matter current produced by minimal coupling, while J^M_{ext} denotes external or background sources. The lower-case j^i is the matter number current. The upper-case J^M is an electromagnetic source current. These should not be interchanged.

The moving-throat and PN notes sometimes contain a finite vector called J, for example a support-source or throat-response vector in a reduced basis. Such a vector is not an electromagnetic current unless the local section explicitly says so. In this ledger, any finite response vector should be named J_{resp}, J_{src}, or another decorated symbol when possible.

0.58.3 Mixed-sector observables

The mixed variables are structural:

$$A_w, \qquad J^w, \qquad F_{\mu w}, \qquad E_w, \qquad C_a. \tag{22}$$

The exact mixed-field gauge invariants are

$$E_w = F_{w0} = -\partial_t A_w - \partial_w A_0, \qquad C_a = F_{aw} = \partial_a A_w - \partial_w A_a. \tag{23}$$

They are suppressed only in strict brane reductions of the form (11). They remain available in the microscopic response theory and are required for the honest passive/outgoing bridge.

The letter C_a in (23) is a mixed electromagnetic field component. It should not be confused with grouped anomaly variables a_x, BdG coupling matrices C, or constants named C in local algebra.

0.59 Charge, circulation, and mass-dressing firewall

The corrected electric-charge dictionary is

$$\eta_Q \in \{+1, -1\}, \qquad e_\star > 0, \qquad q_\star = \eta_Q e_\star, \tag{24}$$

with brane-observed coupling

$$q_{\text{eff}} = \frac{q_\star}{\sqrt{Z_{\text{int}}}} = \eta_Q e_{\text{eff}}. \tag{25}$$

The sign is carried by η_Q. The microscopic branch magnitude is $e_\star$. The brane-observed magnitude is controlled by localization through Z_{int}.

Circulation, winding, and vortex labels should be written with symbols such as

$$\Gamma, \qquad n_\Gamma, \tag{26}$$

or with an explicitly declared magnetic/vortical label. These symbols do not define η_Q, $q_\star$, or q_{eff}.

The historical gravity-side scalar coefficient once written as a bare $q = 1$ is, in this companion,

$$\kappa_\rho = 1. \tag{27}$$

It is a mass-dressing/Newtonian scalar coefficient, not electric charge.

The following table records common collision risks.

Symbol	Allowed meaning	Forbidden reading
$q_\star$	Signed microscopic electric branch coupling $\eta_Q e_\star$.	Throat radius, circulation, scalar mass-dressing coefficient, or quotient coordinate.
q_{eff}	Canonically normalized brane electric coupling.	Bare microscopic charge or circulation.
$q_{\text{tr}}, q_{\text{nt}}, q_\eta$	Coherent quotient/orbit coordinates.	Electric charges.
$q_1, \ldots, q_7$	Residual-basis coefficients when locally declared in PN algebra.	Electric charges.

Symbol	Allowed meaning	Forbidden reading
κ_ρ	Gravity-side mass-dressing coefficient.	Electric charge.
Γ, n_Γ	Circulation/winding or magnetic/vortical data.	Electric charge sign.

0.60 Moving-throat geometry notation

The moving throat is represented by the level-set/shape-field pair

$$\Sigma(\mathbf{X},t) = r - R(\Omega,w,t), \qquad \Sigma = 0 \quad \text{on the throat surface.} \tag{28}$$

The reference throat is

$$\Sigma_0(\mathbf{X}) = r - R_0(w), \tag{29}$$

and the wall displacement is

$$R(\Omega,w,t) = R_0(w) + \eta(\Omega,w,t). \tag{30}$$

The old collective variables

$$a(t), \qquad L(t) \tag{31}$$

are collective moments of the distributed shape field. They are not the fundamental moving-throat variables in this ledger. The mouth-average radius convention is

$$a(t) = \frac{1}{4\pi}\int_{S^2} R(\Omega,0,t)\, d\Omega. \tag{32}$$

If $W_b(\Omega,t)$ denotes the bottom location determined by $R(\Omega,W_b,t) = 0$, then the average throat depth is

$$L(t) = \frac{1}{4\pi}\int_{S^2} W_b(\Omega,t)\, d\Omega. \tag{33}$$

The promoted confinement coupling is written

$$V_{\mathrm{conf}}(\mathbf{X};\Sigma) = V_{\mathrm{wall}}\left(\frac{\Sigma}{\ell_c}\right), \tag{34}$$

with linear variation

$$\delta V_{\mathrm{conf}} = -\frac{V'_{\mathrm{wall}}(\Sigma_0/\ell_c)}{\ell_c}\,\eta. \tag{35}$$

The minimal distributed wall coefficients are

$$\mu_\eta(w), \qquad T_w(w), \qquad T_\Omega(w), \qquad K_\eta(w). \tag{36}$$

They are branch data. They should not be retuned stage by stage to rescue a normalization target unless the local section explicitly introduces a new branch or closure.

The wall action and reduced modal equations in this ledger use the densitized convention: the background surface measure has been absorbed into the effective one-dimensional coefficients and fields. If a section reintroduces an explicit surface weight, it must also rederive the corresponding axial operator and first-derivative terms.

0.61 Harmonics and grouped real P_2 lanes

The wall displacement is expanded as

$$\eta(\Omega, w, t) = \eta_0(w,t)Y_{00}(\Omega) + \sum_{m \in P_2(\text{real})} q_{2m}(w,t)Y_{2m}^{\text{real}}(\Omega) + \eta_{\geq 3}(\Omega, w, t). \tag{37}$$

The normalized scalar harmonic is

$$Y_{00} = \frac{1}{2\sqrt{\pi}}, \qquad \int_{S^2} Y_{00}^2 \, d\Omega = 1. \tag{38}$$

Thus a physical mouth-average radius shift δa corresponds to

$$q_{00}(0,t) = 2\sqrt{\pi}\,\delta a(t). \tag{39}$$

The full real quadrupole bundle contains five real modes:

$$20, \qquad 21c, \qquad 21s, \qquad 22c, \qquad 22s. \tag{40}$$

Many reduced stages use the grouped triplet

$$A \in \{20, 21, 22\}, \tag{41}$$

where 21 groups the $21c, 21s$ pair and 22 groups the $22c, 22s$ pair. The grouped metric is

$$G_{\text{grp}} = \text{diag}(1, 2, 2). \tag{42}$$

For any grouped triple

$$x = (x_{20}, x_{21}, x_{22}), \tag{43}$$

the weighted trace and anomaly variables are

$$\bar{x} = \frac{x_{20} + 2x_{21} + 2x_{22}}{5}, \tag{44}$$

$$a_x = \frac{2x_{20} - x_{21} - x_{22}}{10}, \qquad b_x = \frac{x_{21} - x_{22}}{2}. \tag{45}$$

The inverse map is

$$x_{20} = \bar{x} + 4a_x, \qquad x_{21} = \bar{x} - a_x + b_x, \qquad x_{22} = \bar{x} - a_x - b_x. \tag{46}$$

A useful G_{grp}-orthogonal basis is

$$e_{\bar{x}} = (1, 1, 1), \qquad e_a = (4, -1, -1), \qquad e_b = (0, 1, -1), \tag{47}$$

with squared norms 5, 20, and 4, respectively. The exact projectors are

$$P_{\bar{x}} = \frac{1}{5}\begin{pmatrix} 1 & 2 & 2 \\ 1 & 2 & 2 \\ 1 & 2 & 2 \end{pmatrix}, \tag{48}$$

$$P_a = \frac{1}{20}\begin{pmatrix} 16 & -8 & -8 \\ -4 & 2 & 2 \\ -4 & 2 & 2 \end{pmatrix}, \qquad P_b = \frac{1}{4}\begin{pmatrix} 0 & 0 & 0 \\ 0 & 2 & -2 \\ 0 & -2 & 2 \end{pmatrix}. \tag{49}$$

They satisfy

$$P_{\bar{x}} + P_a + P_b = I_3, \qquad P_i P_j = \delta_{ij} P_i. \tag{50}$$

The isotropy condition for a grouped triple is

$$a_x = b_x = 0. \tag{51}$$

The weak axisymmetric quadrupole splitting signature is

$$\lambda_{20} = 1, \qquad \lambda_{21} = \frac{1}{2}, \qquad \lambda_{22} = -1, \tag{52}$$

which implies the grouped anomaly line

$$b_x = 3a_x. \tag{53}$$

This relation is a symmetry fingerprint, not a definition of all possible anisotropy.

0.62 Wall/support/gauge response notation

For each grouped lane $A \in \{20, 21, 22\}$, the conservative wall/worldtube operator is written

$$D_A^{(\text{cons})}(\omega) = D_{A,0} + D_{A,2}\omega^2 + D_{A,4}\omega^4 + O(\omega^6). \tag{54}$$

The normalized conservative response is

$$Y_A(\omega) = \frac{D_{A,0}}{D_A^{(\text{cons})}(\omega)} = 1 + u_2^{(A)}\omega^2 + u_4^{(A)}\omega^4 + O(\omega^6), \tag{55}$$

with exact conversion

$$u_2^{(A)} = -\frac{D_{A,2}}{D_{A,0}}, \qquad u_4^{(A)} = \frac{D_{A,2}^2 - D_{A,0}D_{A,4}}{D_{A,0}^2}. \tag{56}$$

On an isotropic branch, the lane label may be suppressed:

$$D_{A,n} = D_n, \qquad N_{A,n} = N_n. \tag{57}$$

The full coupled reduced coefficients are decomposed as

$$D_{A,0} = K_A - B_{A,0} - Z_{A,0}, \tag{58}$$

$$D_{A,2} = -(M_A + B_{A,2} + Z_{A,2}), \tag{59}$$

$$D_{A,4} = -(B_{A,4} + Z_{A,4}). \tag{60}$$

Here:

- K_A, M_A are wall baseline stiffness and inertia data;
- $B_{A,n}$ are stable BdG support moments;
- $Z_{A,n}$ are conservative localized-Maxwell/mixed moments;
- $N_{A,n}$ are outgoing-transfer moments.

The outgoing prefactor expansion is

$$P_{A,0} = \frac{N_{A,0}}{D_{A,0}}, \tag{61}$$

$$P_{A,2} = \frac{D_{A,0}N_{A,2} - 2D_{A,2}N_{A,0}}{D_{A,0}^2}, \tag{62}$$

$$P_{A,4} = \frac{D_{A,0}^2 N_{A,4} - 2D_{A,0}(D_{A,2}N_{A,2} + D_{A,4}N_{A,0}) + 3D_{A,2}^2 N_{A,0}}{D_{A,0}^3}. \tag{63}$$

On an isotropic branch, write these as P_0, P_2, P_4. In this context P_0 is an outgoing prefactor, not a pressure, not a harmonic, and not a projection operator.

The leading weak-axisymmetric outgoing-prefactor slope is

$$\Xi_1 = \frac{P_1}{P_0}. \tag{64}$$

Here P_1 is the first weak-axisymmetric correction to the outgoing prefactor, not the $l = 1$ harmonic and not a first post-Newtonian order.

0.63 Outgoing and retarded-port notation

The exterior passive/outgoing port attached to a lane l is written

$$\Pi_l^{(\mathrm{out})}(\omega). \tag{65}$$

For a compact outgoing quadrupole branch, the normalized $l = 2$ fingerprint is

$$\widehat{Y}_2^{(\mathrm{out})}(\omega) = 1 + \frac{a^2\omega^2}{9c_s^2} + \frac{4a^4\omega^4}{81c_s^4} + i\frac{a^5\omega^5}{27c_s^5} + O(\omega^6). \tag{66}$$

The outgoing port coefficient is therefore

$$\Gamma_2^{(\mathrm{port})} = \frac{a^5}{27c_s^5}. \tag{67}$$

The universal quadrupole target is

$$\gamma_{\mathrm{GR}} = \frac{2G}{5c^5}. \tag{68}$$

The invariant normalization condition is written

$$\hat{m}_0^2 P_0 \frac{a^5}{27c_s^5} = \frac{2G}{5c^5}, \tag{69}$$

or equivalently

$$\hat{m}_0^2 P_0 = \frac{54Gc_s^5}{5a^5c^5}. \tag{70}$$

The quantity $\hat{m}_0$ is a source-map factor. It is not a mass unless the local section explicitly defines a mass-normalized source map.

The sign convention in the wall-operator language is

$$D_l = K_l - M_l\omega^2 - \Sigma_l(\omega). \tag{71}$$

Therefore a positive-imaginary outgoing port contributes as a passive damping term with the corresponding sign in D_l. If a section switches to admittance or response notation, it must state the sign conversion.

0.64 Family-1 mouth/core notation

The Family-1 mouth/core block uses

$$K_s, \qquad K_q, \qquad \lambda, \qquad g_s, \qquad g_q. \tag{72}$$

Its normalized ratios are

$$\mathfrak{r} = \frac{\lambda}{\sqrt{K_s K_q}}, \qquad \mathfrak{g} = \frac{g_q\sqrt{K_s}}{g_s\sqrt{K_q}}. \tag{73}$$

The mouth/core control variables are

$$\Sigma_0 = \frac{20}{9}\widehat{T}_m^2, \tag{74}$$

$$\mathcal{R} = \frac{(\mathfrak{g}-\mathfrak{r})^2}{1+\mathfrak{r}^2}, \tag{75}$$

$$\mathcal{S}[\Sigma] = \int_0^1 K_q(x)\Sigma(x)\,dx, \qquad \Pi = \Sigma_0[1-\mathcal{R}\mathcal{S}]. \tag{76}$$

The symbol Σ_0 in this block is a mouth/core scalar, not the level-set background $\Sigma_0(\mathbf{X}) = r - R_0(w)$. When both appear nearby, write $\Sigma_{\mathrm{geom},0}$ for the level-set background or rename the mouth scalar locally.

The symbol Π in this block is a mouth slope/bias variable. It is not the outgoing port $\Pi_l^{(\mathrm{out})}$.

0.65 Coherent local-kernel notation

The coherent reduced hierarchy uses the following dimensionless variables:

$$\epsilon_\eta = \frac{c_{\eta U}^2}{K_U K_\eta^{(\mathrm{eff})}}, \qquad \epsilon_W = \frac{\gamma^2\lambda_W^2\sigma}{K_U K_W^{(\mathrm{eff})}}, \tag{77}$$

$$Z_W = \frac{\lambda_W^2}{K_\eta^{(\mathrm{eff})}K_W^{(\mathrm{eff})}}, \qquad \delta_U = \frac{\pi^2 T_U}{L^2 K_U}, \tag{78}$$

$$\chi_0 = \frac{\gamma c_{\eta U}}{K_U}, \qquad \zeta = \frac{\lambda_\phi^2 K_W^{(\mathrm{eff})}}{\lambda_W^2 K_\phi^{(\mathrm{eff})}}. \tag{79}$$

The coherent tracking factor is

$$R_{\mathrm{tr}} = \frac{1+\chi_0/(1+\delta_U)}{1+\chi_0}. \tag{80}$$

The support-enhancement factor is

$$S(\zeta;\epsilon) = 1 + \frac{\zeta(1-\epsilon)}{1-\zeta\epsilon}, \qquad \epsilon = \epsilon_W\left(1 - \frac{2}{11}\frac{\delta_U}{1+\delta_U}\right). \tag{81}$$

The exact lowest-twin slice is

$$\rho_\alpha = \frac{4}{3}, \qquad \zeta_{\mathrm{req}} = \frac{1}{3}. \tag{82}$$

These numbers belong to the selected twin-support placement problem. They should not be reused as generic density or support variables outside that context.

0.66 Microscopic defect, quotient, and orbit notation

The positive coherent microscopic state is

$$x = (\lambda_W, c_{\eta U}, \gamma, K_U, K_\eta^{(\mathrm{eff})}, K_W^{(\mathrm{eff})}, \mu_W, T_U). \tag{83}$$

Its grouped weak-axisymmetric log-drift vector is

$$\delta\mathbf{x} = (\lambda_1, c_1, \gamma_1, \kappa_U, \kappa_\eta, \kappa_W, \mu_1, \tau_1)^T. \tag{84}$$

The log-drift components are infinitesimal drift coordinates. They are not electric charges, not PN coefficients, and not generic fit parameters.

The direct microscopic slippages are

$$\Sigma_Z = \delta \ln\left(\frac{Z_W}{\Omega_W^2}\right), \qquad \Sigma_\chi = \delta \ln \chi_0, \qquad \Sigma_\eta = \delta \ln \epsilon_\eta, \tag{85}$$

$$\Sigma_\epsilon = \delta \ln \epsilon_W, \qquad \Sigma_\delta = \delta \ln \delta_U. \tag{86}$$

These Σ-symbols are slippage variables, not the throat level-set Σ. If both occur in one local derivation, the level-set should be written as $\Sigma(\mathbf{X}, t)$ and the slippages should retain their subscripts.

The branch-adapted coordinates are

$$\Sigma_{\mathrm{tr}} = (1+\chi_0)\Sigma_\delta + (1+\delta_U)\Sigma_\chi, \tag{87}$$

$$\Sigma_{\mathrm{nt}} = \Sigma_Z + \frac{2\epsilon_W}{1-\epsilon}\frac{11+9\delta_U}{11(1+\delta_U)}\Sigma_\epsilon - \left[\frac{2\chi_0}{1+\delta_U} + \frac{4\epsilon_W\delta_U}{11(1-\epsilon)(1+\delta_U)^2}\right]\Sigma_\delta. \tag{88}$$

The observable defect variables are

$$\Theta_1, \qquad \Xi_1, \qquad \mathcal{R}_1, \tag{89}$$

where Θ_1 is the tracking-factor drift, Ξ_1 is the grouped transfer-shape drift, and $\mathcal{R}_1$ is the selected-branch demand-ratio drift. They satisfy the triangular normal-form reading

$$\Theta_1 \leftrightarrow \Sigma_{\mathrm{tr}}, \qquad \Xi_1 \leftrightarrow \Sigma_{\mathrm{nt}}, \qquad \mathcal{R}_1 + \Xi_1 \leftrightarrow \Sigma_\eta. \tag{90}$$

The final reduced coherent invariant coordinates are

$$\mathfrak{C}_{\mathrm{tr},*} = \chi_0^{1+\delta_{U,*}}\delta_U^{1+\chi_{0,*}}, \tag{91}$$

$$\mathfrak{C}_{\mathrm{nt},*} = \frac{Z_W}{\Omega_W^2}\epsilon_W^{E_*}\delta_U^{-F_*}, \tag{92}$$

$$\epsilon_\eta = \frac{c_{\eta U}^2}{K_U K_\eta^{(\mathrm{eff})}}. \tag{93}$$

The additive orbit packet is

$$(q_{\mathrm{tr}}, q_{\mathrm{nt}}, q_\eta), \tag{94}$$

and the multiplicative orbit packet is

$$(\mathfrak{R}_{\mathrm{tr}}, \mathfrak{R}_{\mathrm{nt}}, \mathfrak{R}_\eta), \qquad \mathfrak{R}_i = e^{q_i}. \tag{95}$$

Important firewall:

$$R_{\mathrm{tr}} \quad \text{is the coherent tracking factor, while} \quad \mathfrak{R}_{\mathrm{tr}} \quad \text{is a finite orbit/invariant ratio.} \tag{96}$$

They must not be interchanged.

On the actual coherent placement branch,

$$q_{\mathrm{tr}} = (1+\delta_{U,*})\ln\frac{\chi_0}{\chi_{0,\mathrm{ref}}} + (1+\chi_{0,*})\ln\frac{\delta_U}{\delta_{U,\mathrm{ref}}}, \tag{97}$$

$$q_{\mathrm{nt}} = \ln\frac{Z_W}{Z_{W,\mathrm{ref}}} - \ln\frac{\Lambda}{\Lambda_{\mathrm{ref}}} + E_*\ln\frac{\epsilon_W}{\epsilon_{W,\mathrm{ref}}} - F_*\ln\frac{\delta_U}{\delta_{U,\mathrm{ref}}}, \tag{98}$$

$$q_\eta = \ln\frac{\epsilon_\eta}{\epsilon_{\eta,\mathrm{ref}}}. \tag{99}$$

The coherent orbit-lock condition is

$$q_{\mathrm{tr}} = q_{\mathrm{nt}} = q_\eta = 0. \tag{100}$$

0.67 Transfer-shape and rigid-mouth physical packet

The transfer-shape variables are

$$\mathcal{T}^2, \qquad R_{\mathrm{target}}, \qquad 1-\epsilon_\eta, \tag{101}$$

with

$$\mathcal{T}^2 = \frac{Z_W(1+\chi_0)^2}{\Omega_W^2(1-\epsilon)^2}, \tag{102}$$

$$R_{\mathrm{target}}\mathcal{T}^2 = \Lambda_0(1-\epsilon_\eta), \qquad \Lambda_0 = \frac{27\pi^2 G c_s^5}{20a^5c^5}. \tag{103}$$

At first grouped weak-axisymmetric order,

$$\delta\ln\mathcal{T}^2 = \Xi_1, \qquad \delta\ln(1-\epsilon_\eta) = \mathcal{R}_1 + \Xi_1, \qquad \delta\ln R_{\mathrm{target}} = \mathcal{R}_1. \tag{104}$$

On the rigid-mouth slice, the physical logarithmic chart is

$$U = \ln\left(\frac{\mathcal{T}^2}{\mathcal{T}^2_{\mathrm{ref}}}\right), \qquad V = \ln\left(\frac{\epsilon_\eta}{\epsilon_{\eta,\mathrm{ref}}}\right). \tag{105}$$

The orbit-lock condition is the Cartesian codimension-two condition

$$U = V = 0. \tag{106}$$

In this section, U and V are rigid-mouth physical chart coordinates. They are not the Newtonian potential U, not a support coordinate $U_{A,r}$, and not a generic energy. If those objects appear together, rename the Newtonian potential as U_{N} and the support coordinate as U_{port} or $U_{A,r}$.

0.68 Packet-A normalization and same-charge ceiling notation

The isotropic one-pole grouped-real P_2 conservative front end is

$$a_2 = b_2 = a_4 = b_4 = 0, \qquad \Delta_{\mathrm{pole}} = \bar{u}_4 - 4\bar{u}_2^2 = 0. \tag{107}$$

The common conservative carrier is

$$\widehat{Y}_Q^{\mathrm{cons}}(\omega) = \frac{3}{4} + \frac{1}{4}\frac{1}{1-\omega^2/\Omega_Q^2}. \tag{108}$$

The exact outgoing scalar is
$$\chi_Q = \frac{3(S\beta^5 + 9\Sigma_5)}{3S - \Sigma_0}. \tag{109}$$
On the natural point-particle source-map branch,
$$N_Q = \chi_Q^{-1}, \qquad \hat{m}_0^2 \chi_Q N_Q = 1. \tag{110}$$
The Packet-A finish-line scalar is
$$\Delta_Q = \chi_Q - 1. \tag{111}$$
Equivalently,
$$\Delta_{\rm norm} = P_0^{\rm target}(\chi_Q^{-1} - 1), \qquad P_0^{\rm target} = \frac{54Gc_s^5}{5a^5c^5}. \tag{112}$$
On the actual weak-axisymmetric branch, the first same-charge ceiling is carried by
$$(\Delta_{\rm norm}, a_{P_0}, b_{P_0}), \tag{113}$$
or equivalently by
$$(\Delta_{\rm norm}, \Xi_1), \qquad \Xi_1 = \frac{P_1}{P_0}. \tag{114}$$
The grouped weak-axisymmetric packet obeys
$$a_{P_0} = \frac{\epsilon \bar{P}_0 \Xi_1}{4}, \qquad b_{P_0} = \frac{3\epsilon \bar{P}_0 \Xi_1}{4}. \tag{115}$$
The robust all-lane ceiling is
$$\frac{\Delta_{\rm norm} + T_{\rm quad}}{\hat{m}_0^2}(1 + |\epsilon \Xi_1|) \le P_{\rm crit}. \tag{116}$$
The small parameter ϵ in this packet is the weak-axisymmetric bookkeeping parameter. It is not ϵ_η, ϵ_W, or the support-enhancement denominator ϵ unless the local section explicitly identifies them.

0.69 Packet-B graph and realization-compiler notation

The free quintuple is
$$\mathbf{y} = (\lambda_W, c_{\eta U}, \gamma, K_U, K_W^{(\rm eff)}). \tag{117}$$
The target orbit is the exact graph
$$\mathcal{O}_* = \{\mathbf{x}_*^{\rm graph}(\mathbf{y}) : \mathbf{y} \in (\mathbb{R}_{>0})^5\}. \tag{118}$$
The graph-closure scalar is
$$\widehat{\chi}_Q(\mathbf{y}) = \chi_Q(\mathbf{x}_*^{\rm graph}(\mathbf{y})). \tag{119}$$
The reduced graph slice is
$$\mathcal{Z}_* = \{\mathbf{x}_*^{\rm graph}(\mathbf{y}) : \widehat{\chi}_Q(\mathbf{y}) = 1\}. \tag{120}$$
The one-parameter log-ray compiler is
$$\mathbf{y}_\mathbf{s}(\tau) = \mathbf{y}_\circ \odot e^{\tau \mathbf{s}}, \qquad \Phi_\mathbf{s}(\tau) = \widehat{\chi}_Q(\mathbf{y}_\mathbf{s}(\tau)). \tag{121}$$

Its local directional packet is

$$\Phi_0, \qquad L_0 = \frac{d}{d\tau} \ln \Phi_{\mathbf{s}} \Big|_0, \qquad L_1 = \frac{d^2}{d\tau^2} \ln \Phi_{\mathbf{s}} \Big|_0. \tag{122}$$

The finite local mixed-ray search is organized by support-cardinality on the five primitive axes

$$\{\lambda, c, \gamma, U, W\}. \tag{123}$$

In this search context, U and W are primitive axes inherited from the free quintuple notation, not the rigid-mouth chart coordinate U and not the bulk coordinate w.

0.70 Relaxed/open-system branch notation

The post-242 relaxed branch is organized by three lifted lanes:

$$L_w = (S_{\text{leak}}, \mathcal{W}_w), \qquad L_{UV} = (U, V, \mathcal{D}_{UV}), \qquad L_\varsigma = (\varsigma, \varsigma_{\min}, \text{signchg}). \tag{124}$$

The exact recovery map to the strict branch is

$$\ell_w = f_U = a = b = 0. \tag{125}$$

The selected-support leakage/work compiler uses

$$\Pi_{\text{tr}}, \qquad \eta_{\text{leak}}, \qquad S_{\text{leak}}, \qquad \mathcal{W}_w^{\text{bulk}}, \qquad \mathcal{W}_w^{\text{sess}}. \tag{126}$$

The orbit-side non-rigid packet uses

$$a_U, \qquad a_V, \qquad \chi_{UV}, \qquad \Delta_{UV} = a_U a_V - \chi_{UV}^2, \tag{127}$$

$$U = \ln \frac{\mathcal{T}^2}{\mathcal{T}_{\text{ref}}^2}, \qquad V = \ln \frac{\epsilon_\eta}{\epsilon_{\eta,\text{ref}}}, \qquad \mathcal{D}_{UV} = \chi_{UV} UV. \tag{128}$$

The compensated source packet uses

$$\varsigma(x; r) = 1 + a(r) \cos(\pi x) + b(r) \cos(2\pi x), \tag{129}$$

$$\mathfrak{g}[\varsigma], \qquad \mathcal{S}[\varsigma], \qquad \mathcal{R}[\varsigma], \qquad \mathcal{M}_\sigma^{\text{pack}}(r). \tag{130}$$

The lowered stationary front end is

$$V_{\text{eff}}^{(247)}(r) = V_{\text{short}}^{(1p)}(r) - \lambda_L S_{\text{leak}}(r) - \lambda_W \mathcal{W}_w^{\text{sess}}(r) - \Delta E_{UV}(r) - \mathcal{M}_{\sigma,\text{red}}(r). \tag{131}$$

The microscopic active-leg export law is

$$K_{\exp}(s) = \Gamma_3 s^3 + \Gamma_5 s^5. \tag{132}$$

Here s is the export coordinate used by the active-leg kernel. It should not be assumed to equal the finite-throat axial coordinate unless the local section says so. The coefficients Γ_3, Γ_5 are export-kernel coefficients, not circulation.

In the relaxed-branch section, bare a and b in $\varsigma = 1 + a\cos(\pi x) + b\cos(2\pi x)$ are source-shape amplitudes. They are not the throat radius and not grouped anomaly coordinates. When this section is expanded, use a_ς, b_ς if the formula appears near $a(t)$ or a_x, b_x.

0.71 High-risk symbol collision table

Table 13: Symbols requiring special care.

Symbol	Safe meanings	Common mistakes to avoid
q	$q_\star$, $q_{\rm eff}$, q_{2m}, $q_{\rm tr}$, $q_{\rm nt}$, q_η, locally declared residual coefficients.	Do not treat every q as electric charge. Do not revive bare $q = 1$; write $\kappa_\rho = 1$.
P	Pressure $P(\rho)$, outgoing prefactors P_0, P_1, P_2, P_4, grouped real P_2 label.	Do not confuse pressure with P_0, and do not confuse grouped real P_2 with a scalar prefactor.
G	Newton's constant G, support couplings G_U, G_W, selected-branch function $G(\xi, \delta)$.	Do not mix Newton's constant with support couplings or branch functions.
K	EOS normalization $K_{\rm EOS}$, wall stiffness K_η, K_A, support stiffness $K_U, K_W^{\rm eff}$.	Avoid bare K unless a local model has declared it.
R	Shape field $R(\Omega, w, t)$, internal mixing R_r, target ratio $R_{\rm target}$, tracking factor $R_{\rm tr}$, orbit ratio $\mathfrak{R}_{\rm tr}$.	Do not identify geometric shape data with reduced transfer ratios.
ρ	Bulk density $\rho = \lvert\psi\rvert^2$, support ratio ρ_α, local loading ratios.	When density and a ratio occur together, write $\rho_{\rm bulk}$ and $\rho_{\rm ratio}$.
η	Wall displacement $\eta(\Omega, w, t)$, charge sign η_Q, geometry ratio ϵ_η, leakage efficiency $\eta_{\rm leak}$.	Do not confuse wall displacement with electric charge sign.
ϵ	Small expansion parameter; ratios ϵ_η, ϵ_W, support denominator variable.	State which ϵ is being expanded in.
λ	Localization or coupling scale; weak-axisymmetric signature λ_A; mixed coupling λ_W.	Do not infer a common physical meaning from the letter.
Γ	Circulation label; outgoing/export coefficients $\Gamma_2^{\rm port}$, Γ_3, Γ_5.	Do not read export or outgoing coefficients as circulation.
A	Gauge potential A_M; grouped lane label $A \in \{20, 21, 22\}$.	Do not treat grouped lane labels as gauge-field indices.
J	Maxwell current J^M; finite response/source vector in reduced ledgers.	Do not interpret every vector named J as electromagnetic current.
Δ	Stability denominators $\Delta_{A,r}$, Δ_{UV}; defects Δ_Q, $\Delta_{\rm norm}$, $\Delta_{\rm full}$.	Bare Δ is allowed only in local algebra. Carry the subscript in theorem statements.

Symbol	Safe meanings	Common mistakes to avoid
U, V	Self-interaction $U(\rho)$, Newtonian potential U, support coordinate $U_{A,r}$, rigid-mouth chart (U, V), potentials $V_{\mathrm{conf}}, V_{\mathrm{eff}}$.	Decorate when more than one layer appears.
a, b	Throat radius $a(t)$, grouped anomalies a_x, b_x, relaxed/source-shape amplitudes a, b.	Prefer a_{mouth}, a_x, b_x, or a_ς, b_ς in mixed contexts.

0.72 Typographic conventions

Use roman subscripts for semantic labels:

$$\mathrm{eff},\quad \mathrm{int},\quad \mathrm{target},\quad \mathrm{tr},\quad \mathrm{nt},\quad \mathrm{geom},\quad \mathrm{mix},\quad \mathrm{out}.$$

This makes branch labels visually distinct from algebraic variables.

Use $\star$ in $q_\star, e_\star$ for the microscopic charge branch. Use a subscript $*$ for selected/reference branch values such as $C_{\mathrm{tr},*}$, $\chi_{0,*}$, or $\epsilon_{\eta,*}$. Thus

$$q_\star \quad \text{is a charge-branch coupling, while} \quad X_* \quad \text{is a reference or selected-branch value.}$$

Use hats for source-map or normalized effective quantities, for example $\hat{m}_0$. Use bars for grouped traces or averages, for example $\bar{u}_2$. If a bar denotes a projected brane observable instead, the projection kernel should be visible, as in $\overline{Q} = \mathcal{P}_W[Q]$.

Use bold symbols for physical vectors in brane or bulk space, such as $\mathbf{x}$, $\mathbf{X}$, and $\mathbf{n}$. If the quotient vector is collected as

$$\mathbf{q} = (q_{\mathrm{tr}}, q_{\mathrm{nt}}, q_\eta)^T,$$

then the text must say explicitly that this is a quotient vector, not a brane spatial vector.

0.73 Notation checklist for stage write-ups

Before a section is treated as stable, check the following items.

- ☐ Electric charge uses only η_Q, $q_\star$, q_{eff}, $e_\star$, or e_{eff}.
- ☐ The gravity-side scalar coefficient is written κ_ρ, not bare q.
- ☐ Circulation and winding labels are not used as electric-charge definitions.
- ☐ $Z(w)$ and $W(w)$ are not interchanged.
- ☐ A strict brane reduction does not erase the microscopic mixed sector.
- ☐ Grouped labels 20/21/22 are treated as lane labels, not indices.
- ☐ The grouped metric $G_{\mathrm{grp}} = \mathrm{diag}(1, 2, 2)$ is used whenever grouped traces are computed.
- ☐ Operator coefficients $D_{A,n}$, response moments $u_n^{(A)}$, transfer moments $N_{A,n}$, and prefactor moments $P_{A,n}$ remain distinct.

- ☐ $P(\rho)$ as pressure is not confused with P_0, P_1, P_2, P_4 as outgoing-prefactor coefficients.
- ☐ Quotient drifts $q_{\mathrm{tr}}, q_{\mathrm{nt}}, q_\eta$ are not described as charges.
- ☐ Rigid-mouth variables (U, V) are not confused with parent energy or potential symbols.
- ☐ Local relaxed-branch variables named a or b are decorated if they appear near the throat radius $a(t)$ or grouped anomalies (a_x, b_x).
- ☐ Every branch target is separated from the formula that compiles the target.

This checklist is not optional cleanup. It is what allows future papers to cite this ledger without importing stale notation or accidentally upgrading a reduced branch convention into an exact parent-theory statement.

Result Anchor Map

This chapter is the stable citation registry for the derivation companion. The stage numbers preserve provenance. The result anchors define what future papers should cite. A stage number answers the historical question “where did this calculation first appear?” A result anchor answers the citation question “which verified result is being used downstream?”

The distinction matters because the ledger is intentionally large. A future paper should not have to cite a raw chain such as “Stages 164–200” every time it needs the weak-axisymmetric quotient theorem. It should be able to cite a stable result anchor, then use local theorem and equation labels inside that anchor only when it needs a precise formula. The result anchor is therefore the public citation handle; the stage range is the audit trail.

> A result anchor is not a status upgrade. It only groups a result family. Every result cited under an anchor still inherits the status tags and assumptions fixed in section 0.34 and the notation rules fixed in section 0.54.

0.74 How result anchors are used

A result anchor has three jobs.

1. **Stable citation.** It gives future papers a durable target even if theorem numbering changes while this companion is filled in.

2. **Status grouping.** It groups exact identities, closure-exact reductions, controlled reductions, numerical placements, and open realization targets without flattening their status.

3. **Audit routing.** It tells a verifier which stage range, appendix entries, symbolic scripts, and source files contain the detailed algebra.

A result anchor should be cited in downstream prose when the downstream argument needs a whole verified block. A theorem label should be cited when the downstream argument needs a specific formal statement. An equation label should be cited when the downstream argument needs a specific formula. A stage number should be cited when the downstream argument is about provenance, historical development, or a failed route that was later superseded.

0.75 Anchor naming convention

The top-level anchors use the form

$$\texttt{MTDC-T}\,n,$$

where `MTDC` abbreviates *Moving-Throat Derivation Companion*. The letter `T` marks a theorem-block family, not necessarily a single theorem. Fine-grained subanchors use the form

$$\texttt{MTDC-T}\,n\,.\,m.$$

These subanchors are stable handles for chapter-sized result blocks inside the larger theorem family.

When a result is written formally in the main text, use an ordinary LaTeX theorem label in addition to the anchor. The recommended label pattern is

$$\texttt{thm:mtdc-t}\,n\texttt{-short-name}.$$

For equations, use

$$\texttt{eq:mtdc-t}\,n\texttt{-short-name}.$$

For tables, use

$$\texttt{tab:mtdc-t}\,n\texttt{-short-name}.$$

The result anchor remains the external citation handle; the theorem, equation, and table labels are internal precision handles.

0.76 Top-level result anchors

Table 14 is the public-facing anchor map. The "default status" column gives the strongest safe default reading of the anchor as a whole. Individual results under an anchor may have a stronger or weaker status; when that happens, the local theorem statement must say so explicitly.

Table 14: Top-level result anchors.

Anchor	Stages	Default status	Citation target
MTDC-T1	Parent + 1	Mixed: Exact, Reduced / Controlled Reduction, Effective Closure	Parent $(4+1)$ action import, exact GNLS/Maxwell equations, projection/reduction language, charge and mixed-sector firewall, and the first statement of the moving-throat geometry problem.
MTDC-T2	1–2	Mixed: Effective Closure, Exact Within Closure	Moving-surface geometry lift, collective recovery of a, L, distributed wall action, densitized convention, and axisymmetric breathing reduction back to the old (a, L) closure.
MTDC-T3	3	Mixed: Reduced / Controlled Reduction, Exact Within Closure	Wall/BdG stable-mode reduction, Schur-complement support kernel, low-frequency conservative moments, grouped P_2 support self-energies, and first pole-shift formulas.

Anchor	Stages	Default status	Citation target
MTDC-T4	004–022	Mixed: Exact, Reduced / Controlled Reduction, Exact Within Closure, Open	Projection-first localized Maxwell law, matched reduction channel, projected mouth-to-bundle transport, reduced Maxwell/mixed one-port normal form, outgoing-transfer factor, compact $l = 2$ fingerprint, grouped response normalization, and the first invariant outgoing-normalization product.
MTDC-T5	023–036	Mixed: Exact Within Closure, Reduced / Controlled Reduction, Open	Full grouped real P_2 projector calculus, isotropy defects, overlap-integral theorem, weak-axisymmetric splitting law, selected-mode normal form, and support-feasibility frontier.
MTDC-T6	037–051	Mixed: Exact Within Closure, Reduced / Controlled Reduction, Open	Continuum placement map, split-U branch, rank-2 support completion, coherent D/N tracking branch, support-compensation theorem, and lowest-twin sufficiency criterion.
MTDC-T7	052–090	Mixed: Exact Within Closure, Reduced / Controlled Reduction, Open	Lowest-lane asymmetry, gain thresholds, parent-overlap tests, explicit Family-1 wall-shell branch, quadrupole-demand windows, loading-ratio extraction, and the first reduced Family-1 verdict.
MTDC-T8	091–163	Mixed: Exact Within Closure, Reduced / Controlled Reduction, Numerically Located, Open	Geometry-lane firewall, retarded closure, outlet/core/mouth compensation, boundary-layer laws, co-evolving mouth/core fixed points, and final retarded normalization residual localization.
MTDC-T9	164–200	Mixed: Exact Within Closure, Reduced / Controlled Reduction, Open	Weak-axisymmetric transport, prefactor packet, transfer-shape slope Ξ_1, monomial quotient coordinates, similarity orbit, Packet-A closure, orbit projectors, and four-scalar final verdict packet.

Anchor	Stages	Default status	Citation target
MTDC-T10	201–218	Mixed: Exact Within Closure, Numerically Located, Open	Realization compiler, canonical orbit projection, free-quintuple graph, log-ray and simplex search, Hessian/curvature ranking, finite support-cardinality sieve, and local mixed-ray closure.
MTDC-T11	219–242	Mixed: Exact Within Closure, Reduced / Controlled Reduction, Numerically Located, Open	One-port same-charge static/dynamic audit, resonance/linewidth gates, actual-branch tests, rigid-mouth physical chart, Cartesian orbit lock, selected twin-support branch, and actual placement compiler.
MTDC-T12	243–253	Mixed: Exact Within Closure, Reduced / Controlled Reduction, Effective Closure, Numerically Located, Open	Relaxed open-system branch, strict-front-end recovery map, leakage/work compiler, non-rigid mouth and U/V drain packet, relaxed barrier compiler, dynamic event chain, export kernel, heat partition, and materials companion.

This top-level map is intentionally aligned with the master stage ledger. Do not renumber these anchors unless the stage ledger is updated at the same time. If a top-level block becomes too broad while the ledger is being filled, add a subanchor rather than moving the whole block.

0.77 Fine-grained subanchor registry

This section gives the fill-level anchor registry. Each subanchor should eventually receive at least one theorem, proposition, definition, equation cluster, or result table in the main text. The final column states what a downstream paper is allowed to use the subanchor for.

0.77.1 MTDC-T1: parent theory and first geometry target

Table 15: Subanchors for **MTDC-T1**.

Subanchor	Stages	Primary status	Downstream citation use
MTDC-T1.1	Parent import	Exact	Exact field content, coordinates, gauge/matter variables, charge ontology, parent GNLS equation, localized Maxwell equation, current conservation, and mixed-sector gauge invariants.

Subanchor	Stages	Primary status	Downstream citation use
MTDC-T1.2	Parent import	Reduced / Controlled Reduction	Projection versus reduction vocabulary, exact projected continuity with leakage, Poisson-hook language, and controlled zero-mode Maxwell reduction.
MTDC-T1.3	1	Effective Closure	The reason a distributed geometry variable is needed: old (a, L) closure is not enough for the grouped P_2, geometry-lane, and outgoing-response problem.
MTDC-T1.4	1	Effective Closure	Hybrid level-set / shape-field representation $\Sigma = r - R(\Omega, w, t)$ as the working moving-throat geometry lift.

0.77.2 MTDC-T2: moving surface, wall field, and breathing reduction

Table 16: Subanchors for **MTDC-T2**.

Subanchor	Stages	Primary status	Downstream citation use
MTDC-T2.1	1	Exact Within Closure	Recovery of the old collective variables $a(t), L(t)$ as monopole moments of the distributed wall field.
MTDC-T2.2	1	Exact Within Closure	Promoted confinement law, moving-wall coupling δV_{conf}, real-harmonic decomposition, and grouped real P_2 lane as literal wall/support mode.
MTDC-T2.3	1	Effective Closure	Minimal distributed wall action, densitized one-dimensional convention, scalar versus quadrupole modal operator, and geometry-lane split.
MTDC-T2.4	2	Exact Within Closure	Axisymmetric two-mode reduction to effective mass and stiffness matrices for $(\delta a, \delta L)$.
MTDC-T2.5	2	Exact Within Closure	Geometry-only grouped real P_2 degeneracy and the baseline one-mode quadrupole wall operator.

0.77.3 MTDC-T3: BdG–wall Schur complement

Table 17: Subanchors for **MTDC-T3**.

Subanchor	Stages	Primary status	Downstream citation use
MTDC-T3.1	3	Reduced / Controlled Reduction	Stable BdG normal-mode reduction and harmonic selection rule on an isotropic reference throat.
MTDC-T3.2	3	Exact Within Closure	Axisymmetric matrix Schur-complement kernel and low-frequency renormalizations of stiffness, inertia, and higher even moments.
MTDC-T3.3	3	Exact Within Closure	Channelwise grouped real P_2 support kernels and lane-by-lane conservative low-frequency coefficients.
MTDC-T3.4	3	Exact Within Closure	First pole-shift formula for one wall mode coupled to one stable BdG support mode.
MTDC-T3.5	3	Exact Within Closure	First grouped P_2 isotropy diagnostic: anisotropy only from wall parameters, support frequencies, or overlap integrals.

0.77.4 MTDC-T4: projection-first Maxwell/mixed response and outgoing-normalization bridge

Table 18: Subanchors for **MTDC-T4**.

Subanchor	Stages	Primary status	Downstream citation use
MTDC-T4.1	004–006	Exact	Projection-first localized Maxwell law, projected transverse leakage, projected charge continuity, and measured-field versus source-coupled-flux split.
MTDC-T4.2	007–008	Reduced / Controlled Reduction	Matched zero-mode reduction channel: $H = Z$, $S = Z/Z_{\text{int}}$, and equality with the reduction-first brane Maxwell coupling.
MTDC-T4.3	009–020	Exact Within Closure	Near-mouth projected Maxwell bridge into the grouped Z_n, N_n slots, compatibility variation, weak-axisymmetric prefactor feed, primitive mouth-Taylor map, and gate sieve.

Subanchor	Stages	Primary status	Downstream citation use
MTDC-T4.4	021	Reduced / Controlled Reduction	Reduced Maxwell/mixed one-port normal form, gauge-invariant mixed variables, conservative mixed self-energy, and low-frequency Z_n coefficients.
MTDC-T4.5	021	Exact Within Closure	Outgoing-port dressing of the mixed coordinate, nonnegative transfer factor, compact outgoing $l = 2$ fingerprint, and wall-level leading odd quadrupole coefficient.
MTDC-T4.6	022	Exact Within Closure	Conversion from conservative operator moments D_{A0}, D_{A2}, D_{A4} to normalized grouped-response moments $u_2^{(A)}, u_4^{(A)}$.
MTDC-T4.7	022	Exact Within Closure	Prefactor expansion P_{A0}, P_{A2}, P_{A4}, leading $\Gamma_5^{(A)}$, and invariant normalization target for $m_0^2 P_0$.

0.77.5 MTDC-T5: full grouped bundle, overlap theorem, and selected front end

Table 19: Subanchors for **MTDC-T5**.

Subanchor	Stages	Primary status	Downstream citation use
MTDC-T5.1	023	Exact Within Closure	Weighted grouped metric, projectors, full-bundle coefficients $D_{A,n}$, $N_{A,n}$, and exact isotropic normalization ratio $P_0 = N_0/D_0$.
MTDC-T5.2	023	Exact Within Closure	First-order anisotropy transport laws for response moments and outgoing prefactor.
MTDC-T5.3	024	Exact Within Closure	Normalized real STF harmonics, angular source-map identity, and $O(3)$ isotropy theorem for grouped lanes.
MTDC-T5.4	024	Exact Within Closure	Weak axisymmetric quadrupole splitting law with grouped signature $(1, 1/2, -1)$ and defect line $b = 3a$.

Subanchor	Stages	Primary status	Downstream citation use
MTDC-T5.5	025–027	Reduced / Controlled Reduction	Minimal isotropic single-mode closure, concrete finite-throat axial modes, exact overlap constants, and nonconstant profile selection.
MTDC-T5.6	028–030	Reduced / Controlled Reduction	Loaded axial profile selection, dynamic loading, selected-mode response, static prefactor, and selected-branch quadrupole target.
MTDC-T5.7	031–033	Exact Within Closure	Reachability, stability windows, finite-throat mode integrals, microscopic normalization equation, and coupling-level onset criterion.
MTDC-T5.8	034–036	Exact Within Closure	Softening-depth normal form, dimensionless D/N shape function, unique normalization locus, and support-feasibility frontier.

0.77.6 MTDC-T6: continuum placement and support completion

Table 20: Subanchors for **MTDC-T6**.

Subanchor	Stages	Primary status	Downstream citation use
MTDC-T6.1	037–040	Exact Within Closure	Continuum-kernel extraction, placement map, product relation, split-U direction-splitting, and generalized source/loading mismatch.
MTDC-T6.2	041–044	Exact Within Closure	Rank-2 support completion, exact support-loading theorem, actual support-direction extraction, and quadratic branch equation.
MTDC-T6.3	045–048	Exact Within Closure	Coherent local D/N tracking kernel, tracking-branch bounds, residual comparison, support-enhancement factor, and support-compensation theorem.
MTDC-T6.4	049–051	Exact Within Closure	Physical coherent support ratio ζ, lowest-twin sufficiency, and exact comparison against ζ_{req}.

0.77.7 MTDC-T7: lowest-lane asymmetry, gain thresholds, and Family-1 branch

Table 21: Subanchors for **MTDC-T7**.

Subanchor	Stages	Primary status	Downstream citation use
MTDC-T7.1	052–055	Exact Within Closure	Non-twin asymmetry requirement, overlap-boost window, Robin-compliance softening, and reachability of the non-twin lowest support lane.
MTDC-T7.2	056–069	Mixed: Exact Within Closure, Reduced / Controlled Reduction, Open	Transport origin of source asymmetry, parent-action projection of gain, parent-overlap threshold theorem, coherence-resonance benchmark, and reduced support/source verdict.
MTDC-T7.3	070–075	Reduced / Controlled Reduction	GNLS wall-shell reduction, canonical tanh-wall branch, explicit placement map, healing-length lock, and support-scale window.
MTDC-T7.4	076–078	Exact Within Closure	$n = 5$ wall-depth lock, shell-weighted extraction of Θ_w, and explicit Family-1 branch comparison.
MTDC-T7.5	079–081	Exact Within Closure	Map from ζ_{req} to Pe_{req}, quadrupole-demand thresholds, and branch-product window.
MTDC-T7.6	082–087	Mixed: Exact Within Closure, Open	Master quadrupole residual, operator-selected Family-1 window, selected product cancellation, loading-ratio window, and reduced finish line.
MTDC-T7.7	088–090	Mixed: Exact Within Closure, Open	Loading-ratio extraction from the minimal isotropic quadrupole module, explicit branch verdict, and updated reduced theorem status.

0.77.8 MTDC-T8: geometry lane, retarded closure, and mouth/core compensation

Table 22: Subanchors for **MTDC-T8**.

Subanchor	Stages	Primary status	Downstream citation use
MTDC-T8.1	091–099	Exact Within Closure	Grouped-P_2 plus geometry split, dynamic-geometry obstruction formula, geometry-decoupling theorem, and collapse of the isotropic passive/outgoing branch to one normalization defect.
MTDC-T8.2	100–106	Mixed: Exact Within Closure, Reduced / Controlled Reduction	Retarded 2.5PN collapse, scalar/dipole filtering, outgoing $l = 2$ normalization bridge, and reduction of the live odd term to the quadrupole prefactor.
MTDC-T8.3	107–124	Mixed: Exact Within Closure, Open	Outlet DtN, Robin outlet tests, side-channel pole tests, and mixed core/outlet realization attempts.
MTDC-T8.4	125–145	Mixed: Exact Within Closure, Reduced / Controlled Reduction, Open	Core-to-mouth compensation laws, mouth local source positivity, mouth boundary-layer closures, and selected compensation surfaces.
MTDC-T8.5	146–163	Mixed: Exact Within Closure, Numerically Located, Open	Co-evolving core/mouth fixed points, linear defect transport, branch deficits, and reduced compensation verdicts.

0.77.9 MTDC-T9: weak-axisymmetric quotient and orbit closure

Table 23: Subanchors for **MTDC-T9**.

Subanchor	Stages	Primary status	Downstream citation use
MTDC-T9.1	164–173	Exact Within Closure	Weak-axisymmetric grouped signature, conservative even-gate transport, loading mismatch Ξ_{load}, and hidden-even relation.
MTDC-T9.2	174–181	Exact Within Closure	Self-similarity theorem, outgoing-load theorem, square-root mixed-leg law, port co-loading, effective transfer-shape collapse, and support-blindness.

Subanchor	Stages	Primary status	Downstream citation use
MTDC-T9.3	182–187	Exact Within Closure	Microscopic slippage decomposition, triangular normal form, branch-invariant monomials, similarity orbit, quotient closure, and finite weak-axisymmetric defect theorem.
MTDC-T9.4	188–191	Exact Within Closure	Branch-observable compiler, transfer-shape / outgoing-prefactor compiler, scalar no-go filters, minimal PDE data packet, and home-stretch theorem.
MTDC-T9.5	192–197	Mixed: Exact Within Closure, Reduced / Controlled Reduction, Open	Orbit/quotient projectors, isotropic grouped target surface, outgoing $l = 2$ DtN fingerprint, source-map reduction, higher-odd irrelevance, Packet-A closure, and $\Delta_Q = \chi_Q - 1$.
MTDC-T9.6	198–200	Exact Within Closure	Finite orbit law, mismatch coordinates, reference-independent two-point orbit-lock theorem, and the four-scalar final verdict packet Δ_{full}.

0.77.10 MTDC-T10: realization compiler and finite mixed-ray search

Table 24: Subanchors for **MTDC-T10**.

Subanchor	Stages	Primary status	Downstream citation use
MTDC-T10.1	201–203	Exact Within Closure	Exact realization compiler, orbit projection, four-scalar audit, free-quintuple graph, scalar closure slice, and one-parameter crossing theorem.
MTDC-T10.2	204–207	Mixed: Exact Within Closure, Numerically Located	Log-ray sweep, finite graph invariance, Hessian refinement, curvature-enveloped ranking, certified brackets, and primitive-ray elimination theorem.
MTDC-T10.3	208–212	Mixed: Exact Within Closure, Numerically Located	Pairwise mixed rays, ratio optimizer, triple-simplex optimizer, boundary splice, and up-to-three-coordinate search sieve.

Subanchor	Stages	Primary status	Downstream citation use
MTDC-T10.4	213–218	Mixed: Exact Within Closure, Numerically Located, Open	Four- and five-coordinate mixed-simplex optimizers, finite algebraic candidate sets, boundary identification, support-cardinality-5 completion, and final local mixed-ray closure theorem.

0.77.11 MTDC-T11: one-port same-charge audit and rigid-mouth selected branch

Table 25: Subanchors for **MTDC-T11**.

Subanchor	Stages	Primary status	Downstream citation use
MTDC-T11.1	219–221	Exact Within Closure	One-port mixed-bundle static kernel, square-law suppression, dynamic mixed-port kernel, phase-lag no-go, resonance/linewidth tradeoff, and linear survival window.
MTDC-T11.2	222–224	Mixed: Exact Within Closure, Numerically Located, Open	Concrete finite-throat primitive branch, pole census, residue/linewidth survival test, isotropic target compatibility, and actual-branch kill test.
MTDC-T11.3	225–228	Mixed: Exact Within Closure, Reduced / Controlled Reduction, Open	Microscopic Ξ_1 compiler, conservative compensation surface, strict 5PN even-gate package, pure-transfer corridor, co-loading no-go, and actual dynamic-window test.
MTDC-T11.4	229–232	Mixed: Exact Within Closure, Numerically Located, Open	Selected-branch numerator/denominator signature, softening-depth crossover, dynamic-window classifier, continuum placement pullback, known-data injection, and current branch verdict.
MTDC-T11.5	233–234	Mixed: Exact Within Closure, Open	Rigid-mouth orbit-lock compiler, static turbulence gate, direct branch-observable static gate, and two-observable kill test.

Subanchor	Stages	Primary status	Downstream citation use
MTDC-T11.6	235–239	Exact Within Closure	Rigid-mouth packet projectors, static-blind dressing line, dependent-plane projectors, finite static-blind curve, physical transfer-shape compiler, physical logarithmic chart (U, V), and Cartesian orbit-lock theorem.
MTDC-T11.7	240–242	Mixed: Exact Within Closure, Numerically Located, Open	Selected-branch loading ratio, primitive ranking on the selected twin-support branch, actual twin-support placement, and coherent orbit-lock compiler.

0.77.12 MTDC-T12: relaxed branch, barrier compiler, export kernel, and materials

Table 26: Subanchors for **MTDC-T12**.

Subanchor	Stages	Primary status	Downstream citation use
MTDC-T12.1	243	Effective Closure	Relaxed-constraint branch declaration, strict-front-end recovery map, and short-range open-system compiler.
MTDC-T12.2	244–246	Mixed: Exact Within Closure, Effective Closure	Selected-branch leakage, scalar-photon work compiler, non-rigid mouth/dressing packet, U/V drain compiler, and compensated multi-mode source compiler.
MTDC-T12.3	247–248	Mixed: Exact Within Closure, Reduced / Controlled Reduction, Effective Closure	Relaxed stationary barrier front end, dynamic event-chain compiler, turning points, threshold speed, WKB, and crossing/collapse conditions.
MTDC-T12.4	249–251	Mixed: Exact Within Closure, Effective Closure, Open	Helicity-export diagnostic, aligned versus anti-aligned mixed-sector closure, Goldilocks window, and dynamic continuation diagnostics.
MTDC-T12.5	252–253	Mixed: Exact Within Closure, Effective Closure, Numerically Located, Open	Microscopic export kernel, vacuum/lattice heat partition, calibration dictionary, screening ratios, and material-threshold companion.

0.78 Audit anchors outside the theorem map

The top-level `MTDC-T` anchors are theorem-block anchors. The appendices also need stable audit handles, but those handles should not be confused with theorem claims. Use the following non-theorem audit anchors when a downstream document needs to refer to the archive mechanics rather than to a physics result.

Table 27: Non-theorem audit anchors.

Audit anchor	Location	Use
MTDC-AUDIT-1	Master stage ledger	Complete stage-to-anchor crosswalk and provenance map.
MTDC-AUDIT-2	Reproducibility map	Symbolic-audit, numerical-audit, and script-backed verification registry.
MTDC-AUDIT-3	Source-file index	Source-file provenance, compact-master links, and full-output links.
MTDC-AUDIT-4	Fill workflow	Procedure for filling theorem blocks and stage entries without breaking claim-status or notation discipline.

0.79 High-value formulas and packets by anchor

This section records the formula families that should receive stable equation labels when the main body is filled. The formulas listed here are not rederived in the anchor map. They are listed so that future sections know which equations should become citation-grade.

Table 28: Formula packets that should be promoted to equation labels.

Anchor	Formula family	Labeling target
MTDC-T1	Parent equations	GNLS equation, localized Maxwell equation, current conservation, projected continuity with leakage, longitudinal identity, and zero-mode Maxwell reduction.
MTDC-T2	Geometry lift and wall modes	$\Sigma = r - R(\Omega, w, t)$, recovery of $a(t)$ and $L(t)$, wall action, modal wall operator, and breathing-reduction matrices.
MTDC-T3	BdG Schur complement	Axisymmetric matrix kernel, grouped channelwise support kernels, low-frequency BdG moments, and pole-shift formula.

Anchor	Formula family	Labeling target
MTDC-T4	Projected Maxwell/mixed and outgoing bridge	Projected Maxwell law, matched reduction conditions, projected Z_n, N_n slot shifts, conservative mixed self-energy, outgoing-transfer factor, compact $l = 2$ fingerprint, prefactor moments, and invariant product $m_0^2 P_0$.
MTDC-T5	Grouped bundle and selected front end	Weighted projectors, $u_2^{(A)}$, $u_4^{(A)}$, P_{A0}, P_{A2}, P_{A4}, $O(3)$ isotropy, weak-axisymmetric line $b = 3a$, and support-feasibility functions.
MTDC-T6	Continuum placement	Dimensionless continuum ratios, selected-branch product law, rank-2 support equations, tracking functions, support enhancement $S(\zeta; \epsilon)$, and ζ_{req}.
MTDC-T7	Family-1 verdict	Family-1 geometry map, wall-depth lock, branch-product window, loading-ratio extraction, and reduced residual packet.
MTDC-T8	Retarded and compensation packets	Geometry contamination order, retarded normalization defect, outlet/core/mouth compensation laws, boundary-layer fixed-point equations, and co-evolving transport formulas.
MTDC-T9	Quotient and orbit packets	$\Xi_1 = P_1/P_0$, monomials $\mathfrak{C}_{\mathrm{tr},*}$, $\mathfrak{C}_{\mathrm{nt},*}$, ϵ_η, similarity-orbit map, Δ_Q, and $\Delta_{\mathrm{full}} = (\Delta_Q, q_{\mathrm{tr}}, q_{\mathrm{nt}}, q_\eta)$.
MTDC-T10	Realization compiler	Graph-error form, scalar closure slice, Hessian envelope, simplex optimizers, finite candidate sets, and support-cardinality-5 closure statement.
MTDC-T11	Same-charge and rigid-mouth packets	Static kernel suppression, dynamic mixed-port kernel, resonance/linewidth survival window, (U, V) chart, $U = V = 0$ orbit lock, and selected twin-support placement.
MTDC-T12	Relaxed branch	Recovery map $\ell_w = f_U = a = b = 0$, relaxed potential $V_{\mathrm{eff}}^{(247)}$, event-chain conditions, export kernel $K_{\exp}(s) = \Gamma_3 s^3 + \Gamma_5 s^5$, and material screening ratios.

0.80 Downstream citation crosswalk

The result anchors are meant to be reused by future papers with consistent wording. Table 29 gives the intended crosswalk.

Table 29: Which anchors future papers should cite.

Downstream use	Primary anchors	Recommended wording
Framework paper / `pde.tex`	**MTDC-T1**–**MTDC-T5**, **MTDC-T9**	"The derivation companion constructs the parent-theory import, geometry lift, grouped real P_2 response compiler, and weak-axisymmetric quotient packet."
Conservative PN summaries	**MTDC-T1**, **MTDC-T3**, **MTDC-T5**, **MTDC-T8**, **MTDC-T9**	"The moving-throat companion supplies the microscopic response and orbit-quotient derivation used to interpret the grouped P_2 closure."
2.5PN / retarded quadrupole papers	**MTDC-T4**, **MTDC-T5**, **MTDC-T8**, **MTDC-T9**	"The surviving retarded branch reduces to the outgoing $l = 2$ normalization packet and its prefactor/orbit conditions."
4PN / tail-interface papers	**MTDC-T4**, **MTDC-T8**, **MTDC-T9**	"The hereditary coefficient is controlled by the same passive/outgoing STF quadrupole normalization isolated in the derivation companion."
Same-charge branch audits	**MTDC-T9**, **MTDC-T11**	"The same-charge mixed sector is reduced to the transported prefactor packet, selected dynamic-window tests, and rigid-mouth orbit-lock packet."
Realization-search papers	**MTDC-T9**, **MTDC-T10**, **MTDC-T11**	"The compiler is exact inside the declared quotient/realization class; actual branch realization remains a separate status claim."
Relaxed/open-system companions	**MTDC-T12**	"The relaxed branch is a companion extension with its own recovery map to the strict front end, not a proof of the strict branch by itself."
Atomic, lepton, or anomaly notes	**MTDC-T1**, **MTDC-T9**, **MTDC-T11**, **MTDC-T12**	"The downstream reduced-sector use inherits the parent ontology, quotient variables, orbit-lock language, and relaxed/export status limits from the derivation companion."

0.81 What not to cite from each anchor

The following table is intentionally defensive. It lists common overclaims that the anchor map should prevent.

Table 30: Unsafe citation uses.

Anchor	Do not cite this anchor for
MTDC-T1	A claim that the far-field zero-mode brane reduction removes mixed channels from the microscopic theory. It suppresses them only inside a controlled brane limit.
MTDC-T2	A claim that the effective wall action is the unique possible geometry lift of the parent action. It is the working lift used by the derivation.
MTDC-T3	A claim that BdG support alone supplies the outgoing quadrupole branch. The Stage-3 block is conservative and must be supplemented by Maxwell/mixed outgoing structure.
MTDC-T4	A claim that the actual moving-throat branch realizes the universal outgoing normalization. This anchor defines the reduced bridge and prefactor, not final realization.
MTDC-T5	A claim that all anisotropy is impossible. The $O(3)$ theorem forbids anisotropy only on the isotropic reduced kernel; weak-axisymmetric splitting is explicitly allowed.
MTDC-T6	A claim that the physical support ratio is automatically large enough. The support-compensation theorem gives feasibility conditions and thresholds.
MTDC-T7	A claim that Family-1 completes the whole program. Family-1 supplies a reduced explicit branch and verdict; later stages refine and partly supersede it.
MTDC-T8	A claim that all scalar or geometry dangers are unconditionally absent in the full PDE. The anchor records reduced contamination checks and compensation/closure tests.
MTDC-T9	A claim that $\Delta_{\text{full}} = 0$ is realized by the actual branch. The anchor defines the quotient packet and orbit theorem; realization is separate.
MTDC-T10	A claim that the finite mixed-ray sieve searches all conceivable PDE branches. It searches the declared local free-quintuple / mixed-ray class.
MTDC-T11	A claim that the same-charge mixed sector supplies a new long-range attraction, or that orbit lock follows without the rigid-mouth selected-branch assumptions.
MTDC-T12	A claim that the relaxed barrier companion proves the strict conservative theorem. It is a relaxed/open-system companion with an exact recovery map to the strict front end.

0.82 Anchor status inheritance

An anchor inherits the weakest status required for the downstream claim. For example, the formula

$$P_0 = \frac{N_0}{D_0}$$

is Exact Within Closure inside the reduced grouped-bundle closure. The statement that the actual moving-throat branch realizes the target value

$$m_{\hat{0}}^2 P_0 = \frac{54 G c_s^5}{5 a^5 c^5}$$

remains Open until the actual branch calculation returns that value. A future paper may cite **MTDC-T4** or **MTDC-T5** for the compiler, but it must cite the later realization result, or explicitly mark the realization as open, if it needs the target to be attained.

The same rule applies to the orbit and realization blocks. The quotient theorem may be exact inside the coherent reduced branch. The finite search may be complete inside the declared mixed-ray class. The actual branch may still fail the target. The anchor map keeps these together for navigation but does not merge their statuses.

0.83 Supersession and stability rules

The derivation is long enough that some early prototypes are expected to be superseded by later normal forms. Supersession should be handled as follows.

1. **Do not delete the old stage.** The stage remains part of the audit trail.
2. **Do not move the top-level anchor unless the logical block changes.** A refined theorem usually lives under the same anchor.
3. **Add a supersession note.** The stage entry should state which later theorem or subanchor replaces the earlier working formulation.
4. **Promote later normal forms to the main citation target.** Future papers should cite the latest normal form unless they are discussing historical development.
5. **Demote unsafe wording.** If an early stage used stronger wording than the later audit allows, the main text should use the later status and the appendix should record the wording change.

Important supersession lanes already visible in the present stage tree include:

- early one-lane outgoing-transfer formulas superseded by the full grouped-bundle prefactor formulas;
- Family-1 reduced branch tests superseded by later actual-branch and orbit-lock packets;
- weak-axisymmetric load-defect formulas superseded by the monomial quotient and transfer-shape normal forms;
- same-charge generic mixed-sector placeholders superseded by the one-port same-charge static/dynamic audit;
- strict front-end selected-branch tests supplemented, but not replaced, by the relaxed/open-system branch.

0.84 Result-card template for main-text sections

Every major theorem block in the main text should begin or end with a short result card. The following template is the preferred form.

> **Result anchor.** **MTDC-T**$n.m$.
> **Stage provenance.** Stages [range].
> **Status split.** Algebra: [status]. Closure: [status and name]. Realization: [status].
> **Inputs.** [Prior anchors, formulas, branch assumptions.]
> **Output.** [The theorem, packet, formula, or verdict produced.]
> **Do not use for.** [The most likely overclaim.]
> **Downstream use.** [Which later anchors or papers depend on it.]

This template is deliberately compact. The stage appendix entry can carry the longer proof audit; the result card tells downstream readers what they are allowed to cite.

0.85 Citation wording examples

The following examples show the intended level of precision.

0.85.1 Outgoing quadrupole bridge

Safe wording:

> The derivation companion shows that the electromagnetic sector must be projected before the matched Maxwell/mixed normal form is used, and that inside the reduced wall/BdG/projected-Maxwell grouped-bundle closure the leading odd quadrupole coefficient is controlled by the isotropic static prefactor $P_0 = N_0/D_0$; actual realization of the universal value is a separate branch condition. See **MTDC-T4** and **MTDC-T5**.

Unsafe wording:

> The moving-throat PDE proves the GR quadrupole normalization.

The unsafe version erases the difference between the compiler and actual branch realization.

0.85.2 Weak-axisymmetric quotient

Safe wording:

> The weak-axisymmetric prefactor defect is organized by the quotient packet, with $\Xi_1 = P_1/P_0$ on the weak-axisymmetric line and with the final reduced back end summarized by Δ_{full}. See **MTDC-T9**.

Unsafe wording:

> The branch is therefore automatically orbit-locked.

The unsafe version confuses quotient coordinates with realized orbit lock.

0.85.3 Finite mixed-ray realization search

Safe wording:

> Within the declared local free-quintuple mixed-ray class, the realization search is reduced to a finite support-cardinality-5 sieve. See **MTDC-T10**.

Unsafe wording:

> All possible moving-throat branches have been searched.

The unsafe version expands the search class beyond what the theorem covers.

0.85.4 Rigid-mouth orbit lock

Safe wording:

> On the rigid-mouth selected branch, the post-static same-charge problem is diagonal in the physical logarithmic chart (U, V), and orbit lock is the codimension-two condition $U = V = 0$. See **MTDC-T11**.

Unsafe wording:

> The same-charge branch is solved in general.

The unsafe version drops the rigid-mouth and selected-branch restrictions.

0.85.5 Relaxed branch

Safe wording:

> The relaxed companion introduces leakage, non-rigid mouth, source, barrier, export, and material-screening packets around the strict front end, with recovery map $\ell_w = f_U = a = b = 0$. See **MTDC-T12**.

Unsafe wording:

> The relaxed branch proves the strict branch.

The unsafe version reverses the direction of the recovery map.

0.86 Checklist for maintaining the anchor map

Before finalizing any future revision of this companion, check the following.

- ☐ Every top-level anchor in table 14 has at least one main-text theorem block.
- ☐ Every subanchor in section 0.77 has a corresponding theorem, proposition, definition, table, or explicitly marked appendix-only result.
- ☐ Every anchor has a status split compatible with section 0.34.
- ☐ Every anchor respects the notation rules in section 0.54.
- ☐ Every open realization target is still marked open unless a later actual-branch theorem closes it.

- ☐ Every numerical placement has a source, script, or stage appendix entry.
- ☐ Every no-go anchor names the search class or branch family it excludes.
- ☐ Every superseded prototype points to the later subanchor that should be cited instead.
- ☐ The stage ledger and this anchor map agree on stage ranges.
- ☐ Downstream papers cite result anchors for result families and theorem/equation labels only for local precision.

0.87 Summary

The result-anchor map turns the 253-stage derivation into a citation system. The anchors are not replacements for proofs, equations, or stage entries. They are the route map connecting them. The stable rule is:

> Cite anchors for result families, theorem labels for formal statements, equation labels for formulas, and stage numbers for provenance.

Following that rule will let later papers cite this companion without turning every citation into a long historical stage chain, while still allowing any reader to audit the full derivation stage by stage.

Dependency Map

This chapter records the dependency structure of the derivation companion. The result-anchor map in section 0.73 says what future papers should cite. The dependency map says what each cited block is allowed to depend on, what it produces, and where a failure or branch assumption propagates.

The map is deliberately more than a table of contents. A table of contents tells a reader where material appears. A dependency map tells a verifier which earlier result must be checked before a later result can be trusted. This distinction is essential for a derivation ledger because the document contains exact identities, exact statements inside closures, controlled reductions, numerical placements, and open branch-realization tests in one archive.

> A dependency arrow is not a proof and is not a status upgrade. It is an audit instruction. If a later theorem block depends on an earlier controlled reduction, then the later block inherits that reduction unless it explicitly replaces it with a stronger argument.

Throughout this chapter, the result anchors **MTDC-T1–MTDC-T12** are the anchors defined in section 0.73. The status labels are the labels fixed in section 0.34. The notation firewalls of section 0.54 remain active for every dependency in this chapter.

0.88 Arrow types

The ledger uses several kinds of dependency arrows. They should not be conflated.

Table 31: Dependency-arrow types used in this companion.

Arrow type	Meaning	Audit consequence
Hard algebraic dependency	A formula, theorem, or map is used as an input to a later derivation.	The later derivation is invalid until the earlier algebraic object is established under its stated assumptions.
Closure dependency	A later result is exact only inside an earlier declared closure, branch, ansatz, or reduced model.	The later result must carry the closure status forward. It may not be cited as an unconditional parent-theory theorem.
Normalization dependency	A later result requires a specific scalar normalization, prefactor, source-map factor, or transfer coefficient.	The algebraic structure may survive even if the numerical or branch normalization fails. The realization gate must be checked separately.

Arrow type	Meaning	Audit consequence
Branch-realization dependency	A later result requires the actual moving-throat branch to land on a particular target surface, quotient orbit, selected support state, or prefactor packet.	The previous compiler can be correct while the physical branch still fails. This is a realization question, not an algebraic-compression question.
Failure-mode dependency	A later branch or relaxed companion is introduced because a strict branch, simple support branch, or one-port branch failed or remained under-normalized.	The failure must be preserved as part of the derivation history. A relaxed repair may not erase the strict verdict.
Downstream-use dependency	A later paper needs only the exported formula or theorem, not the whole stage history.	The downstream paper should cite the result anchor and, if necessary, the local theorem/equation label rather than a raw stage range.

A fill-stage entry should identify all arrow types that apply. For example, the outgoing quadrupole normalization depends algebraically on the compact outgoing $l = 2$ fingerprint, depends by closure on the reduced Maxwell/mixed transfer model, and depends by realization on the actual branch value of the invariant product $\widehat{m}_0^2 P_0$.

0.89 Global dependency spine

The shortest faithful global spine is

$$\begin{aligned}
\text{exact parent } (4+1) \text{ theory} &\Longrightarrow \text{moving-surface lift} \\
&\Longrightarrow \text{wall/support modal system} \\
&\Longrightarrow \text{BdG Schur complement and projection-first Maxwell packet} \\
&\Longrightarrow \text{grouped real } P_2 \text{ response bundle} \\
&\Longrightarrow \text{outgoing prefactor } P_0 \\
&\Longrightarrow \text{selected support and mouth/core branch tests} \\
&\Longrightarrow \text{weak-axisymmetric transport and quotient orbit} \\
&\Longrightarrow \text{realization compiler} \\
&\Longrightarrow \text{actual strict branch verdict} \\
&\Longrightarrow \text{relaxed open-system companion.}
\end{aligned}$$

The same spine, read as a verification instruction, is

$$\begin{aligned}\text{definitions} &\Longrightarrow \text{operators} \Longrightarrow \text{low-frequency coefficients}\\ &\Longrightarrow \text{normalized observables} \Longrightarrow \text{target surfaces} \Longrightarrow \text{branch-placement tests}\\ &\Longrightarrow \text{orbit / quotient tests} \Longrightarrow \text{realization search} \Longrightarrow \text{strict or relaxed physical verdict.}\end{aligned}$$

The most important negative spine is

$$\text{algebraic compression success} \not\Rightarrow \text{actual branch realization,}$$
$$\text{isotropic grouped closure} \not\Rightarrow \text{weak-axisymmetric prefactor closure,}$$
$$\text{strict branch failure} \not\Rightarrow \text{relaxed branch success,}$$
$$\text{orbit-lock algebra} \not\Rightarrow \text{physical placement on the selected support curve.}$$

These negative arrows are logical guardrails rather than new theorem statements. A compiler can be exact while the branch it compiles is not realized.

0.90 Top-level dependency table

Table 32 gives the top-level dependency map. The table is intentionally conservative. It lists the dependencies that should be presumed active unless a later theorem explicitly proves that a dependency has been eliminated.

Table 32: Top-level dependency map for the result anchors.

Anchor	Required upstream blocks	Principal outputs	Downstream blocks that use it
MTDC-T1	Parent action and prior ontology corrections.	Exact field content, charge firewall, projection/reduction vocabulary, mixed-sector observables, and the first moving-throat target statement.	All later anchors. In particular, **MTDC-T2**–**MTDC-T5** use the parent equations directly.
MTDC-T2	**MTDC-T1**; working choice of shape field $\Sigma = r - R(\Omega, w, t)$; declared wall-action closure.	Distributed wall field, recovery of (a, L), scalar and grouped real P_2 modal split, and geometry-only breathing reduction.	**MTDC-T3**, **MTDC-T4**, **MTDC-T5**, and every later branch that uses wall/support observables.

Anchor	Required upstream blocks	Principal outputs	Downstream blocks that use it
MTDC-T3	**MTDC-T1**–**MTDC-T2**; stable BdG normal-mode reduction.	Schur-complement support kernels, pole shifts, scalar/geometry lane renormalization, grouped P_2 conservative self-energies, and first isotropy diagnostics.	**MTDC-T4**, **MTDC-T5**, **MTDC-T8**, and the conservative front end of **MTDC-T9**.
MTDC-T4	**MTDC-T1**–**MTDC-T3**; projection-first localized Maxwell law; retained mixed sector; matched reduction channel; compact outgoing branch; 2.5PN/4PN quadrupole target.	Exact projected Maxwell leakage law, measured/flux field split, matched one-port Maxwell/mixed self-energy, projected Z_n, N_n bundle transport, outgoing transfer factor, compact $l = 2$ fingerprint, normalized grouped response language, and invariant target for $\widehat{m}_0^2 P_0$.	**MTDC-T5**, **MTDC-T8**, **MTDC-T9**, **MTDC-T11**, and any future radiative or tail calculation.
MTDC-T5	**MTDC-T2**–**MTDC-T4**; grouped metric and projector calculus; overlap model assumptions.	Full grouped real P_2 bundle, exact projector/inverse maps, isotropy theorem, weak-axisymmetric splitting law, selected-mode front end, and early support-feasibility surfaces.	**MTDC-T6**, **MTDC-T7**, **MTDC-T8**, **MTDC-T9**, and the selected-support part of **MTDC-T11**.
MTDC-T6	**MTDC-T5**; continuum placement variables; split-U and support-kernel closure.	Continuum placement map, rank-2 closure, coherent D/N tracking branch, support-compensation theorem, explicit D/N support ratios, and lowest-twin sufficiency criteria.	**MTDC-T7**, **MTDC-T8**, and the selected support slice in **MTDC-T11**.
MTDC-T7	**MTDC-T5**–**MTDC-T6**; lowest-lane and Family-1 branch assumptions.	Parent-overlap gain thresholds, Family-1 wall-shell data, quadrupole-demand windows, and the first reduced Family-1 verdict.	**MTDC-T8**, **MTDC-T9**, and realization/actual-branch comparisons in **MTDC-T10**–**MTDC-T11**.

Anchor	Required upstream blocks	Principal outputs	Downstream blocks that use it
MTDC-T8	**MTDC-T4**–**MTDC-T7**; outlet/core/mouth models; retarded closure assumptions.	Geometry-lane firewall, retarded collapse, outlet and mixed-core compensation attempts, mouth/core boundary-layer laws, co-evolving fixed points, and retarded-normalization residual localization.	**MTDC-T9**, **MTDC-T10**, **MTDC-T11**, and the relaxed companion **MTDC-T12**.
MTDC-T9	**MTDC-T5** and **MTDC-T8**; weak-axisymmetric branch; coherent local-kernel normal form.	Prefactor packet, $\Xi_1 = P_1/P_0$, monomial quotient coordinates, similarity orbit, Packet-A closure, orbit projectors, and four-scalar final verdict packet.	**MTDC-T10**, **MTDC-T11**, **MTDC-T12**, and future papers using the weak-axisymmetric quotient theorem.
MTDC-T10	**MTDC-T9**; realization graph; free-quintuple and mixed-ray search class.	Canonical orbit projection, graph-error realization form, log-ray and simplex search, Hessian/curvature ranking, finite support-cardinality sieve, and local mixed-ray closure.	**MTDC-T11**, **MTDC-T12**, and later realization audits.
MTDC-T11	**MTDC-T9**–**MTDC-T10**; same-charge one-port audit; rigid-mouth branch; selected twin-support placement.	Same-charge static/dynamic kernel verdict, resonance/linewidth gates, rigid-mouth chart (U, V), Cartesian orbit lock, selected twin-support branch, and actual placement compiler.	**MTDC-T12** and downstream papers that need the strict actual-branch verdict.

Anchor	Required upstream blocks	Principal outputs	Downstream blocks that use it
MTDC-T12	**MTDC-T9**–**MTDC-T11**; strict-front-end verdict; relaxed leakage/non-rigid/source/export closures.	Codimension-three relaxed branch, strict recovery map, leakage/work compiler, relaxed barrier front end, dynamic event-chain continuations, export kernel, heat partition, and material-screening companion.	Future relaxed-branch, materials, and barrier/export papers.

0.91 Lower-order import dependencies

This companion is not an isolated derivation tree. It imports a frozen lower-order stack. Table 33 records what those imports are allowed to supply.

Table 33: Imported lower-order dependencies.

Source family	Imported content	Used primarily by
Parent $(4+1)$ action paper	Exact GNLS matter sector, localized Maxwell sector, projection/reduction distinction, leakage identity, Poisson-hook vocabulary, and corrected charge ontology.	**MTDC-T1**; indirectly all later anchors.
Electromagnetic and plasma papers	Projection-first localized Maxwell law, matched zero-mode charge normalization, mixed fields $A_w, J^w, F_{\mu w}, E_w, C_a$, and the rule that mixed channels are suppressed only in a controlled matched brane limit.	**MTDC-T1**, **MTDC-T4**, **MTDC-T8**, **MTDC-T12**.
Conservative 1PN stack	Frozen lower-order constants such as $\kappa_\rho = 1$, $n = 5$, $\kappa_{\rm add} = 1/2$, $\kappa_{\rm PV} = 3/2$, and the closure-status discipline used by later PN assemblies.	**MTDC-T1**, branch interpretation in **MTDC-T5**–**MTDC-T8**.
Conservative 2PN stack	The statement that the defect has a nontrivial $P_0 \oplus P_2$ mouth/support layer and a separate geometry-closure channel; carried static support data and open pole scales.	**MTDC-T3**, **MTDC-T5**, **MTDC-T8**.

Source family	Imported content	Used primarily by
2.5PN quadrupole-normalization package	Narrowing of the universal retarded branch to the orbital/worldtube STF quadrupole; compact outgoing $l = 2$ low-frequency fingerprint; normalization target for the Burke–Thorne coefficient.	**MTDC-T4**, **MTDC-T5**, **MTDC-T8**, **MTDC-T9**.
Conservative 3PN stack	Grouped real P_2 conservative payload, grouped trace/anomaly language, and the reason the higher-order bottleneck is no longer merely algebraic.	**MTDC-T5**, **MTDC-T8**, **MTDC-T9**.
Conservative 4PN stack	Local/tail split and the theorem that the hereditary 4PN interface is controlled by the same outgoing STF quadrupole normalization already isolated at 2.5PN.	**MTDC-T4**, **MTDC-T8**, **MTDC-T9**.
5PN and weak-axisymmetric notes	Explicit overlap model, weak-axisymmetric transport, one-pole target surface, monomial quotient coordinates, similarity-orbit theorem, and conservative even-gate separation.	**MTDC-T5**, **MTDC-T9**, **MTDC-T10**, **MTDC-T11**.
Same-charge, rigid-mouth, and relaxed-branch notes	One-port static/dynamic verdicts, actual-branch prefactor ceiling, rigid-mouth (U, V) chart, selected twin-support slice, and relaxed leakage/export/material packets.	**MTDC-T11**, **MTDC-T12**.

These imports should be cited as background only where needed. The derivation companion itself should still restate enough definitions that a verifier can follow the dependency chain without opening every earlier paper.

0.92 Formula gates

Some formulas are not merely useful equations; they are gates. A later section cannot be filled correctly unless the relevant gate has already been stated, tagged, and checked. Table 34 lists the main gates that should be kept visible while the ledger is filled.

Table 34: Main formula gates and their dependency roles.

Gate	Formula or object	Dependency role
Geometry lift	$\Sigma(\mathbf{X}, t) = r - R(\Omega, w, t)$.	Makes the grouped real P_2 lanes literal wall/support modes and makes (a, L) collective moments rather than fundamental variables.
Wall modal split	$\eta = \eta_0 Y_{00} + \sum q_{2m} Y_{2m}^{\text{real}} + \eta_{\geq 3}$.	Separates scalar/geometry, grouped quadrupole, and higher-lane sectors before support and gauge couplings are added.
BdG Schur complement	$D_A(\omega) = K_A - M_A\omega^2 - \sum_\alpha g_{A\alpha}^2/(\varpi_{A\alpha}^2 - \omega^2)$.	Supplies conservative support softening, inertia renormalization, higher even moments, and pole shifts.
Projected Maxwell law	$\partial_\mu \langle ZF^{\mu\nu}\rangle_W + [WZF^{w\nu}]_{\partial D} - \int_D W' ZF^{w\nu} dw + \xi^{-1}\partial^\nu \langle H\partial \cdot A\rangle_W = \mu_0 \langle J^\nu\rangle_W$.	Keeps the mixed transverse EM channel alive before any matched brane reduction and fixes the measured-field/source-flux distinction.
Maxwell/mixed transfer	$N_A(0) = [\Omega_{U,A}^2 g_{W,A} + R_A g_{U,A}]^2/[\Omega_{U,A}^2 \Omega_{W,A}^2 - R_A^2]^2$.	Transfers the passive/outgoing port into the wall operator with a nonnegative static factor.
Normalized response	$u_2^{(A)} = -D_{A2}/D_{A0}$, $u_4^{(A)} = (D_{A2}^2 - D_{A0}D_{A4})/D_{A0}^2$.	Converts conservative operator data into the grouped response language used by the PN bridge.
Outgoing prefactor	$P_{A0} = N_{A0}/D_{A0}$.	Identifies the only static internal quantity needed for the leading $i\omega^5$ quadrupole coefficient.
Universal normalization	$\widehat{m}_0^2 P_0 = 54Gc_s^5/(5a^5c^5)$.	Branch-realization gate for the universal 2.5PN/4PN quadrupole normalization.
Grouped decomposition	$x_{20} = \bar{x} + 4a_x$, $x_{21} = \bar{x} - a_x + b_x$, $x_{22} = \bar{x} - a_x - b_x$.	Converts lane triples into isotropic and anisotropic components.

Gate	Formula or object	Dependency role
Weak-axisymmetric signature	$(\lambda_{20}, \lambda_{21}, \lambda_{22}) = (1, 1/2, -1)$, equivalently $b = 3a$.	Distinguishes pure weak-axisymmetric quadrupole splitting from more general anisotropy.
Selected support surface	$F(\xi, \delta) = R_{\text{target}}$, with support feasibility through $G(\xi, \delta)$.	Converts support placement into a finite branch-placement problem.
Prefactor slope	$\Xi_1 = P_1/P_0$.	Compresses weak-axisymmetric prefactor transport to a single observable scalar on the natural branch.
Quotient coordinates	$(q_{\text{tr}}, q_{\text{nt}}, q_\eta) = \delta \ln(\mathfrak{C}_{\text{tr},*}, \mathfrak{C}_{\text{nt},*}, \epsilon_\eta)$.	Separates true defect motion from similarity-orbit motion in the coherent local kernel.
Final reduced packet	$\Delta_{\text{full}} = (\Delta_Q, q_{\text{tr}}, q_{\text{nt}}, q_\eta)$.	Carries the reduced back end into the realization compiler and actual-branch verdict.
Rigid-mouth chart	$(U, V) = \big(\ln(\mathcal{T}^2/\mathcal{T}_{\text{ref}}^2), \ln(\epsilon_\eta/\epsilon_{\eta,\text{ref}})\big)$.	Diagonalizes the post-static same-charge branch; orbit lock is $U = V = 0$.
Relaxed recovery map	$\ell_w = f_U = a = b = 0$.	Shows how the relaxed open-system companion recovers the strict selected front end.

When a section uses one of these gates, the section should either cite the gate's theorem/equation label or restate the local formula. Silent use of a gate is one of the easiest ways for a derivation ledger to become unverifiable.

0.93 Block-local dependency maps

This section gives the dependency map at the level at which the main body should eventually be written. The tables below are deliberately broken into small blocks so that later fill work can be assigned section by section.

0.93.1 Part I dependency map: parent theory, geometry lift, and linearized backbone

Table 35: Dependency map for Part I.

Local block	Inputs	Outputs
Parent import	Frozen parent action, corrected charge ontology, projection/reduction vocabulary.	Exact field-equation and notation base for the companion.
Moving-throat need statement	Parent geometry still represented by finite (a, L); lower PN stack needs grouped P_2, geometry lane, and outgoing response.	Justification for promoting geometry to a distributed wall/shape field.
Shape-field lift	Choice $\Sigma = r - R(\Omega, w, t)$; stationary finite throat; mouth and bottom definitions.	Collective recovery of a, L; mode variables $\eta_0, q_{2m}, \eta_{\geq 3}$.
Quadratic wall action	Effective wall inertia/stiffness functions and densitized axial convention.	Modal wall operator and baseline scalar/quadrupole splitting.
Breathing reduction	Axisymmetric ansatz $\eta_0 \sim \alpha_a \delta a + \alpha_L \delta L$.	Effective $(\delta a, \delta L)$ mass/stiffness matrices and continuity with the old closure.

0.93.2 Part II dependency map: conservative grouped response and normalization front end

Table 36: Dependency map for Part II.

Local block	Inputs	Outputs
BdG support coupling	Wall operator and promoted confinement coupling; stable BdG reduction.	Support Schur complement, pole shifts, and conservative grouped P_2 moments.
Maxwell/mixed coupling	Parent Maxwell sector; mixed-field firewall; one-lane reduced mixed model.	Conservative mixed self-energy and static outgoing-transfer factor.
Outgoing fingerprint	Compact outgoing $l = 2$ branch from the 2.5PN package.	Leading $i\omega^5$ bridge and the invariant normalization product.
Full grouped bundle	Stage-3/4 coefficients; grouped metric $\mathrm{diag}(1, 2, 2)$.	Projector calculus, isotropy defects, and exact formula $P_0 = N_0/D_0$.
Overlap/isotropy theorem	Real STF harmonics, isotropic angular kernel, radial/axial overlap factorization.	Identity angular source map, $O(3)$ grouped isotropy, and first weak-axisymmetric splitting law.

Local block	Inputs	Outputs
Selected-mode front end	One-pole and constant-prefactor gates; finite-throat support profiles.	Early target surfaces for $u_4 = 4u_2^2$, $P_2 = P_4 = 0$, and P_0 normalization.

0.93.3 Part III dependency map: support completion and Family-1 branch

Table 37: Dependency map for Part III.

Local block	Inputs	Outputs
Continuum placement	Full-bundle target surface; dimensionless continuum ratios.	Placement coordinates $(\delta, M_{\mathrm{mix}}, R_{\mathrm{target}})$ and product law.
Split-U branch	Continuum kernel with non-flat internal U sector.	Direction splitting, tracking factor, and deformed selected-branch functions.
Rank-2 support closure	Continuum support data and softening depth.	Quadratic placement equation, tracking branch laws, and support feasibility frontier.
D/N support extraction	Finite-throat D/N support modes and overlap ratios.	Microscopic support ratio ζ_{phys}, lowest-twin selection bias, and twin sufficiency test.
Asymmetry and gain thresholds	Lowest-lane asymmetry, overlap boosts, Robin compliance, and gain windows.	Conditions for mixed-only, symmetric-twin, or non-twin rescue.
Family-1 construction	Selected wall-shell and support-lane data.	Explicit Family-1 reduced branch, loading-ratio extraction, and first reduced verdict.

0.93.4 Part IV dependency map: geometry lane, retarded closure, and mouth/core compensation

Table 38: Dependency map for Part IV.

Local block	Inputs	Outputs
Geometry-lane firewall	Axisymmetric residual lane from the wall lift; grouped P_2 projector rules.	Conditions under which geometry completion is separate from grouped P_2 normalization.

Local block	Inputs	Outputs
Retarded closure collapse	Outgoing $l = 2$ bridge; scalar/dipole danger audit; passive/outgoing sign conventions.	Reduction of the live odd branch to quadrupole normalization rather than a generic retarded ledger.
Outlet and side-channel attempts	Robin outlet, mixed $A_w/F_{\mu w}$ side channel, and compensation laws.	Location of under-normalization or over-softening residuals.
Core-to-mouth compensation	Selected core and mouth variables; boundary-layer response.	Local mouth-source laws, positive compensation conditions, and co-evolving fixed-point equations.
Final residual localization	Family-1 and mouth/core tests.	Identification of what remains a branch-realization problem rather than a missing algebraic compiler.

0.93.5 Part V dependency map: weak-axisymmetric quotient and orbit closure

Table 39: Dependency map for Part V.

Local block	Inputs	Outputs
Weak-axisymmetric transport	Grouped splitting law $(1, 1/2, -1)$, full-bundle prefactor expansion, and branch slopes.	Loading mismatch $\Xi_{\text{load}} = P_1/P_0$ and weak-axisymmetric line $b = 3a$.
Coherent-kernel normal form	Coherent local D/N kernel and microscopic slippage variables.	Triangular defect form for tracking, nontracking, and dressing channels.
Monomial quotient	Direct microscopic monomials $\mathfrak{C}_{\text{tr},*}$, $\mathfrak{C}_{\text{nt},*}$, and ϵ_η.	Rank-three quotient map, five-dimensional similarity orbit, and orbit-preserving fibres.
Packet-A and projector closure	Prefactor packet, orbit projectors, and grouped/selected branch residuals.	Four-scalar reduced verdict packet Δ_{full}.
Back-end reduction	Reduced graph-error form and scalar verdict coordinates.	Inputs for realization search and same-charge actual-branch tests.

0.93.6 Part VI dependency map: realization compiler

Table 40: Dependency map for Part VI.

Local block	Inputs	Outputs
Canonical orbit projection	Four-scalar verdict packet and similarity-orbit quotient.	Error coordinates in the physical quotient rather than in redundant microscopic variables.
Free-quintuple graph	Local mixed-ray parameterization and graph-error realization form.	Finite algebraic realization problem.
Search sieve	Log-ray, pairwise, simplex, and support-cardinality filters.	Finite support-cardinality ≤ 5 realization audit.
Hessian and curvature ranking	Candidate local minima and graph curvature data.	Ranked branch candidates and exact places where realization remains numerical or open.

0.93.7 Part VII dependency map: same-charge audit, rigid mouth, and selected branch

Table 41: Dependency map for Part VII.

Local block	Inputs	Outputs
One-port same-charge static audit	Mixed-kernel ontology and corrected charge firewall.	Verdict that the static bundle creates no new long-range attractive law.
Dynamic same-charge audit	Linear dynamic mixed bundle, resonance/linewidth estimates, and short-range families.	Verdict that dynamics gives resonant dispersive enhancement, not a new kernel class.
Actual-branch tests	Prefactor ceiling, support selection, and branch placement data.	Current strict actual-branch verdict.
Rigid-mouth normal form	Physical logarithmic chart (U, V).	Cartesian orbit lock $U = V = 0$ and direct observable gates.
Selected twin-support branch	Lowest-twin support slice and actual placement curve.	Selected support/orbit split and coherent orbit-lock compiler.

0.93.8 Part VIII dependency map: relaxed open-system branch

Table 42: Dependency map for Part VIII.

Local block	Inputs	Outputs
Relaxed branch declaration	Strict selected front end and its residuals.	Codimension-three relaxed branch with exact strict recovery map.
Leakage/work compiler	Open-system leakage, non-rigid mouth, and source/work packets.	Relaxed stationary front end and explicit terms in $V_{\mathrm{eff}}^{(247)}$.
Dynamic event chain	Barrier lowering, event thresholds, turning points, and WKB/Goldilocks continuations.	Dynamic relaxed-branch feasibility tests.
Export and heat partition	Active-leg export kernel and material-screening dictionary.	Super-Ohmic export law, event-safe half-plane, channel-resolved heat partition, and materials companion.

0.94 Failure-mode dependency map

The ledger must preserve failed or insufficient routes because later relaxed branches and actual-branch tests often depend on knowing what did *not* work.

Table 43: Failure-mode dependencies.

Failure or limitation	What it means	What depends on it
Finite-dimensional (a, L) closure is insufficient for the full PDE problem.	It cannot expose grouped real P_2, geometry-lane, and outgoing-response data as literal modes.	The geometry lift in **MTDC-T2**.
Geometry-only wall theory is conservative and cannot itself supply the passive/outgoing odd branch.	Stable wall and BdG blocks can produce even moments and pole shifts, but not the required retarded $i\omega^5$ fingerprint.	The projection-first Maxwell/mixed bridge in **MTDC-T4**.
Angular isotropy does not fix radial/axial normalization.	The real STF source map can be identity while P_0 still misses the universal target.	Support placement in **MTDC-T6** and compensation work in **MTDC-T8**.
Mixed-only or flat-support branches may undershoot the selected normalization.	The algebraic compiler is correct, but the physical branch can still lack enough loading or support enhancement.	Lowest-twin and non-twin support tests in **MTDC-T6**–**MTDC-T7**.

Failure or limitation	What it means	What depends on it
Weak-axisymmetric similarity-orbit closure kills Ξ_{load} but not every conservative even gate.	The normalization defect and conservative even gates are separate constraints.	The realization and actual-branch tests in **MTDC-T10**–**MTDC-T11**.
One-port same-charge static bundle creates no new long-range attractive law.	Same-charge rescue cannot be claimed from the static mixed kernel alone.	Rigid-mouth and relaxed branch work in **MTDC-T11**–**MTDC-T12**.
Rigid-mouth orbit lock is a codimension-two condition, not an automatic branch property.	The chart (U, V) diagonalizes the problem, but the actual branch still has to land at $U = V = 0$.	Actual-placement and relaxed-branch tests in **MTDC-T11**–**MTDC-T12**.
Relaxed open-system branch does not retroactively prove the strict branch.	Leakage/work/source/export repair is a separate companion, with a recovery map back to the strict front end.	Part VIII and all future relaxed-branch citations.

When filling a section, do not remove a failure-mode dependency merely because a later branch repairs or bypasses it. Instead, state exactly what the repair changes and what strict verdict remains frozen.

0.95 Status propagation rules

The dependency map is also a status-propagation map. The following rules should be applied while filling every theorem, proposition, and boxed formula.

1. **Exact parent inputs stay exact only while no closure is added.** Once the moving-wall ansatz, stable-mode reduction, one-port mixed model, selected support surface, or relaxed open-system closure is invoked, the result no longer has bare parent-exact status.

2. **Exact algebra inside a closure is still valuable.** A Schur complement, projector inverse, quotient-rank proof, or finite search theorem can be exact even though the model class in which it lives is a closure.

3. **A target equation is not a realized theorem.** Equations such as

$$\widehat{m}_0^2 P_0 = \frac{54 G c_s^5}{5 a^5 c^5}$$

define a branch target. They become realized-branch statements only after the actual branch data are shown to satisfy them.

4. **Numerical placement does not invalidate exact setup.** If a root, Hessian packet, or branch coordinate is numerically located, then the definition and equations can remain exact while the realization value is Numerically Located.

5. **Relaxed branch results inherit relaxed assumptions.** A leakage/work/export result should not be cited as a strict conservative result unless the recovery map has been applied and the relaxed variables have been set to their strict values.

A useful local test is this: remove one upstream assumption and ask whether the downstream statement still follows. If it does not, the downstream theorem must explicitly name that assumption.

0.96 Dependency checklist for every stage appendix

Every stage appendix should eventually contain a dependency paragraph with the following fields. This is the per-stage version of the map above.

- ☐ **Required upstream result anchors.** List the result anchors and local theorem/equation labels used by the stage.
- ☐ **Required prior stage outputs.** List the exact stage numbers, but only for provenance. The result anchor remains the public citation handle.
- ☐ **Imported lower-order paper results.** Name the lower-order import if the stage uses a PN target, charge ontology, source map, or outgoing benchmark from outside the moving-throat stage tree.
- ☐ **Closure assumptions.** State whether the stage uses the shape-field lift, stable BdG reduction, one-port mixed model, isotropic kernel, weak-axisymmetric branch, selected support surface, rigid-mouth chart, or relaxed open-system variables.
- ☐ **Local algebraic outputs.** Identify the formulas that are proved locally and can be exported downstream.
- ☐ **Branch-realization requirements.** State whether the output is already realized by construction or whether the actual moving-throat branch still has to supply data.
- ☐ **Downstream uses.** List the later anchors, sections, or future-paper use cases that depend on the output.
- ☐ **Failure propagation.** If the stage finds a no-go, undershoot, over-softening, or missing-normalization result, state which later branch tries to repair or bypass it.

This checklist should make it possible to audit a single stage without rereading the whole companion.

0.97 Revision rules for this map

The dependency map should be stable, but it is allowed to evolve as the ledger is filled. Revisions should follow these rules.

1. **Do not delete an upstream dependency silently.** If a later derivation removes a dependency, add a sentence explaining how the dependency was discharged.
2. **Do not replace a failure with a repair.** Keep the failure and add the repair as a downstream dependency.

3. **Do not promote branch targets to branch theorems.** A target surface is a theorem about what must be hit. It is not a theorem that the branch hits it.

4. **Prefer subanchors over renumbering.** If a block becomes too broad, add a result subanchor rather than changing the top-level anchor family.

5. **Keep formula gates synchronized with theorem labels.** Once a formula gate receives a formal theorem or equation label, update table 34 or the local prose to point to it.

6. **Separate strict and relaxed dependencies.** If a result belongs only to the relaxed open-system companion, do not route it into the strict conservative spine without the recovery map.

0.98 Practical fill order implied by the dependency map

The map implies a practical fill order for the companion.

1. Fill the parent import, geometry lift, and breathing reduction first. These establish the objects used everywhere else.

2. Fill the BdG and Maxwell/mixed Schur-complement chapters next. These create the conservative and outgoing operator language.

3. Fill the grouped projector, overlap, isotropy, and normalization front end before writing any branch-placement claims.

4. Fill support placement and Family-1 only after the normalization target surface is fully stated.

5. Fill retarded/mouth/core compensation after the strict selected-support residuals are visible.

6. Fill weak-axisymmetric quotient and orbit closure only after the grouped prefactor and axisymmetric splitting law are fixed.

7. Fill the realization compiler after the quotient packet is stable.

8. Fill same-charge, rigid-mouth, selected support, and relaxed branch material last, because these are verdict and repair layers rather than foundational algebra.

This order is not mandatory for drafting, but it is the safest order for verification. If the companion is written out of order, each section should still obey the dependencies listed in this chapter.

Contents

Chapter 1

Parent Theory, Geometry Lift, and Linearized PDE Backbone

Stage range: 001–023

This chapter is the archive entry that later parts treat as shared infrastructure. It imports the parent GNLS/Maxwell equations, promotes the throat geometry from the old (a, L) closure to a distributed shape field, and packages the first reduced wall/support/outgoing operators into the grouped notation reused throughout the ledger.

Part I at a glance

Primary question	How does the derivation program pass from the parent PDE model to a reusable reduced wall/support/outgoing packet?
Starting inputs	The declared $(4+1)$-dimensional GNLS/Maxwell parent theory, the old breathing variables (a, L), and the confinement lift $V_{\mathrm{conf}}(\mathbf{X}; a, L) \to V_{\mathrm{conf}}(\mathbf{X}; \Sigma)$.
Delivers	The moving-surface lift; the breathing reduction; the BdG support Schur complement; the projection-first Maxwell law with its matched one-port normal form; and the grouped D, N, P normalization language.
Used by	Part II for overlap and spectral placement, Part III for support completion, and Parts IV–VIII for retarded, quotient, realization, and strict/relaxed branch tests.
Result anchors	**MTDC-T1**, **MTDC-T2**, **MTDC-T3**, and **MTDC-T4**; the closing section hands off directly to **MTDC-T5**.

Stage packets.

Stages	Packet	Status	Carry-forward output
001–002	Parent import and geometry lift	EXACT / EXACT WITHIN CLOSURE	Shape field $R(\Omega, w, t)$, promoted confinement coupling, and the two-mode breathing reduction recovering $(\delta a, \delta L)$ as collective moments.
003	Conservative BdG support response	REDUCED / CONTROLLED REDUCTION / EXACT WITHIN CLOSURE	Stable-mode Schur complements, pole shifts, and the first grouped P_2 conservative support diagnostics.
004–021	Projected EM and parent-action packet	EXACT / REDUCED / CONTROLLED REDUCTION / EXACT WITHIN CLOSURE	Projection-first Maxwell leakage, matched mixed-sector normal form, parent-action packet, and the first passive/outgoing $l = 2$ bridge.
022–023	Grouped normalization packet	EXACT WITHIN CLOSURE / OPEN	Grouped projectors, normalized D, N, P coefficients, isotropic ratio $P_0 = N_0/D_0$, and anisotropy transport laws reused later.

Claim status. Mixed. The imported parent equations and projection-first Maxwell identities are EXACT inside the declared parent theory and projection kernel. The moving-surface algebra, breathing reduction, Schur complements, grouped projectors, and normalized branch formulas are EXACT WITHIN CLOSURE once the declared reduced closure is adopted. Stable-mode reduction, matched Maxwell reduction, low-frequency expansion, and outgoing-port attachment are REDUCED / CONTROLLED REDUCTION. Actual nonlinear branch realization of the final normalization target is OPEN.

What this part proves. Part I fixes the parent-theory import, the moving-surface lift, the first reduced wall/BdG response operators, the projection-first Maxwell-to-bundle bridge, and the grouped normalization packet that later parts treat as shared infrastructure.

What this part does not prove. It does not prove that the completed nonlinear moving-throat branch realizes the final outgoing-normalization target; it only derives the reduced object that the branch must realize.

Why later parts need it. Every later part reuses this backbone. Part II builds explicit overlap and selected-mode geometry on top of it, and all later closure, quotient, and realization arguments inherit its operator and normalization language.

1.1 Block contract and theorem-level output

This part has a narrow verification contract. It must show that the later reduced response program is not an unrelated collection of coefficients. The coefficients used later arise from one derivation chain:

$$\begin{aligned}\text{parent GNLS/Maxwell theory} &\longrightarrow \text{moving shape field} \longrightarrow \text{linear wall/BdG/Maxwell system} \\ &\longrightarrow \text{grouped response bundle.}\end{aligned} \tag{1.1}$$

The chain is deliberately status-stratified. The parent action and its Euler–Lagrange identities are exact inside the declared parent model. The shape-field lift is the working effective geometry closure. The projected Maxwell law is exact once the observation kernel is declared, while the old reduction-first Maxwell block survives only as a matched one-port normal form. The stable-mode and Maxwell/mixed Schur complements are exact after their reduced normal-coordinate ansatz is declared. The final outgoing normalization condition is an exact branch test, but its realization by the actual moving-throat branch remains open at this stage.

Theorem 1.1 (Part I backbone theorem)**.** *Under the declared parent* $(4+1)$*-dimensional gauged GNLS plus localized Maxwell theory, and under the moving-surface closure*

$$\Sigma(\mathbf{X}, t) = r - R(\Omega, w, t), \qquad r = \sqrt{x^2 + y^2 + z^2}, \qquad \Omega = \mathbf{x}/r, \tag{1.2}$$

the linearized throat problem admits a reduced wall/support/outgoing response description with the following structure.

1. *The old variables* $a(t)$ *and* $L(t)$ *are recovered as lowest collective moments of* R*, not treated as fundamental moving-throat coordinates.*
2. *The real* $l = 2$ *wall/support sector splits into grouped lanes* $20, 21, 22$*, with the* 21 *and* 22 *entries denoting real cosine/sine pairs.*
3. *Stable matter support modes generate conservative Schur-complement corrections to the wall operator.*
4. *Projection-first localized Maxwell theory gives an open-system brane law; its matched mixed* $A_w/F_{\mu w}/J^w$ *normal form generates conservative self-energies and a passive/outgoing transfer factor.*
5. *On the isotropic grouped branch, the leading outgoing quadrupole test reduces to the single invariant ratio*
$$\widehat{m}_0^2 P_0 = \frac{54 G c_s^5}{5 a^5 c^5}, \qquad P_0 = \frac{N_0}{D_0}. \tag{1.3}$$

The derivation of (1.3) *is exact inside the reduced grouped response closure. Whether the actual nonlinear moving-throat branch realizes the required value is not settled in this part.*

Proof. The statement is a compilation of the twenty-three stage-level reductions. The moment recovery and modal decomposition follow from the shape-field definition. The breathing reduction follows by inserting the lowest axisymmetric ansatz into the quadratic wall action. The BdG kernel follows by an exact Schur complement in the declared stable-mode coordinates. The Maxwell sector first follows by exact projection of the localized parent equation, then by matched mouth/bundle transport, parent-action packet bookkeeping, and the reduced Schur complement for the one-port mixed normal form. The final normalization product follows by converting the conservative grouped operator into a normalized response and multiplying by the compact outgoing $l = 2$ fingerprint. Each individual step is written below with its own status boundary. □

1.2 Exact parent action and field-equation import

1.2.1 Local goal and import boundary

The moving-throat program does not begin by inventing new field equations. It imports the already-declared parent (4+1)-dimensional matter/gauge theory and then promotes the old geometry closure into a distributed moving-surface variable. The imported parent equations are therefore the fixed background against which the moving-throat extension must be checked.

The exact import contains three pieces:

1. a gauged four-spatial-dimensional GNLS matter sector;
2. a localized $(4+1)$-dimensional Maxwell sector with localization profile $Z(w)$;
3. a confinement coupling that is later promoted from $V_{\text{conf}}(\mathbf{X}; a, L)$ to $V_{\text{conf}}(\mathbf{X}; \Sigma)$.

The geometry lift can modify the confinement argument, the wall response, and the branch data. It does not modify the charge ontology, the minimal-coupling rule, or the exact localized Maxwell equation.

1.2.2 Arena and core fields

The parent arena has coordinates

$$x^M = (t, x, y, z, w), \qquad M, N \in \{0, 1, 2, 3, 4\}. \tag{1.4}$$

Bulk spatial coordinates are

$$\mathbf{X} = (x, y, z, w) \in \mathbb{R}^4, \qquad \mathbf{x} = (x, y, z) \in \mathbb{R}^3, \tag{1.5}$$

where $\mathbf{x}$ is the brane-facing spatial coordinate. The flat bulk metric convention is

$$\eta_{MN} = \mathrm{diag}(-1, +1, +1, +1, +1), \qquad \Box_5 = -\partial_t^2 + \nabla_3^2 + \partial_w^2. \tag{1.6}$$

The core parent fields are

$$\psi(\mathbf{X}, t), \qquad A_M(x), \qquad F_{MN} = \partial_M A_N - \partial_N A_M, \qquad \rho = |\psi|^2. \tag{1.7}$$

The moving-throat extension introduces either a level set $\Sigma(\mathbf{X}, t)$ or the equivalent shape field $R(\Omega, w, t)$, but that variable enters through the promoted confinement coupling.

1.2.3 Charge ontology imported into the PDE program

The electric branch label is

$$\eta_Q = \pm 1, \qquad q_\star = \eta_Q e_\star, \qquad e_\star > 0. \tag{1.8}$$

After zero-mode brane normalization, the observed brane charge is

$$q_{\text{eff}} = \frac{q_\star}{\sqrt{Z_{\text{int}}}}, \qquad e_{\text{eff}} = \frac{e_\star}{\sqrt{Z_{\text{int}}}}, \qquad Z_{\text{int}} = \int_{-\infty}^{+\infty} Z(w)\, dw. \tag{1.9}$$

This firewall matters already in Part I because the mixed channels $A_w, J^w, F_{\mu w}$ are retained microscopically. Keeping those channels alive must not be confused with returning to an older circulation-based definition of electric charge. Circulation remains a magnetic/vortical label; it is not the sign or magnitude of the electric branch.

1.2.4 Matter action and exact GNLS equation

The imported matter Lagrangian density is

$$\mathcal{L}_\psi = \frac{i\hbar}{2}\left(\psi^* D_t\psi - \psi D_t\psi^*\right) - \frac{\hbar^2}{2m}(D_i\psi)^*(D_i\psi) - V_{\text{conf}}(\mathbf{X};\Sigma)\rho - U(\rho). \tag{1.10}$$

At the exact parent level before promotion, the confinement potential is usually written as $V_{\text{conf}}(\mathbf{X};a,L)$. In the moving-throat lift the same coupling is read as a surface-dependent potential. The covariant derivatives are

$$D_t\psi = \partial_t\psi + \frac{iq_\star}{\hbar}A_0\psi, \qquad D_i\psi = \partial_i\psi - \frac{iq_\star}{\hbar}A_i\psi. \tag{1.11}$$

The stiff-polytropic equation of state carried into the program is

$$P(\rho) = K\rho^5, \qquad U(\rho) = \frac{K}{4}\rho^5, \qquad h(\rho) = \frac{dU}{d\rho} = \frac{5K}{4}\rho^4, \tag{1.12}$$

with sound-speed ledger

$$c_s^2(\rho) = \frac{1}{m}\frac{dP}{d\rho} = \frac{5K}{m}\rho^4. \tag{1.13}$$

Variation with respect to ψ^* gives the exact GNLS equation

$$\boxed{i\hbar D_t\psi = \left[-\frac{\hbar^2}{2m}D_iD_i + V_{\text{conf}}(\mathbf{X};\Sigma) + h(\rho)\right]\psi}. \tag{1.14}$$

The exact number current and continuity equation are

$$\boxed{j^i = \frac{\hbar}{m}\Im(\psi^* D_i\psi), \qquad \partial_t\rho + \partial_i j^i = 0}. \tag{1.15}$$

1.2.5 Localized Maxwell action and exact mixed observables

The localized Maxwell Lagrangian density is

$$\mathcal{L}_{\text{EM}} = -\frac{Z(w)}{4\mu_0}F_{MN}F^{MN} - \frac{H(w)}{2\xi\mu_0}(\partial\cdot A)^2 - A_M J^M_{\text{ext}}. \tag{1.16}$$

Here $H(w) = 1$ is the old unweighted gauge-fixing convention, while $H(w) = Z(w)$ is the projection-stable localized convention used below. The matter current generated by minimal coupling is not to be double-counted in J^M_{ext}. The source entering Maxwell's equation is

$$J^M_{\text{tot}} = J^M_\psi + J^M_{\text{ext}}. \tag{1.17}$$

Variation with respect to A_N gives

$$\boxed{\partial_M(Z(w)F^{MN}) + \frac{1}{\xi}\partial^N(H(w)\partial\cdot A) = \mu_0 J^N_{\text{tot}}}. \tag{1.18}$$

The Bianchi identity is

$$\partial_{[L}F_{MN]} = 0. \tag{1.19}$$

The mixed gauge-invariant fields retained by the PDE program are

$$\boxed{E_w = F_{w0} = -\partial_t A_w - \partial_w A_0, \qquad C_a = F_{aw} = \partial_a A_w - \partial_w A_a}. \tag{1.20}$$

They are suppressed only in the strict far-field brane Maxwell reduction. They remain available in the moving-throat PDE and become essential in the first outgoing bridge.

1.2.6 Madelung identities and the vortex/gauge firewall

Writing

$$\psi = \sqrt{\rho}\, e^{i\theta}, \tag{1.21}$$

the exact gauge-invariant velocity is

$$v_i = \frac{\hbar}{m}\partial_i\theta - \frac{q_\star}{m}A_i. \tag{1.22}$$

The quantum potential is

$$Q(\rho) = -\frac{\hbar^2}{2m}\frac{\nabla_4^2\sqrt{\rho}}{\sqrt{\rho}}. \tag{1.23}$$

The Euler-like form is

$$m(\partial_t + v_j\partial_j)v_i = q_\star(E_i + v_jB_{ij}) - \partial_i\left(V_{\text{conf}} + h(\rho) + Q(\rho)\right), \tag{1.24}$$

and the exact vorticity–gauge identity is

$$\boxed{\Omega_{ij} := \partial_i v_j - \partial_j v_i = -\frac{q_\star}{m}F_{ij}}. \tag{1.25}$$

This equation is frequently useful, but it is not a charge dictionary. It records the relation between gauge field strength and gauge-invariant vorticity away from singular phase defects.

1.2.7 Projection and reduction hooks

For a normalized brane-observation kernel $W(w)$, projection means

$$\langle Q\rangle_W = \int_D W(w)Q(w)\,dw, \qquad \int_D W(w)\,dw = 1. \tag{1.26}$$

Projection commutes with brane derivatives, but not with the transverse derivative:

$$\langle\partial_w Q\rangle_W = [WQ]_{\partial D} - \int_D W'(w)Q(w)\,dw. \tag{1.27}$$

This identity is the reason a projection-first brane theory is generally an open system.

Applied to the matter current, the same projection defines

$$\rho_{\text{brane}}(\mathbf{x},t) = \int W(w)\rho(\mathbf{x},w,t)\,dw, \qquad \mathbf{j}_{\text{brane}} = \int W(w)\mathbf{j}_{xyz}(\mathbf{x},w,t)\,dw. \tag{1.28}$$

Projected continuity is exact once W is declared:

$$\partial_t\rho_{\text{brane}} + \nabla_3\cdot\mathbf{j}_{\text{brane}} = S_{\text{leak}}, \tag{1.29}$$

where

$$S_{\text{leak}} = -[Wj^w]_{-\infty}^{+\infty} + \int W'(w)j^w\,dw. \tag{1.30}$$

Applied to the localized Maxwell equation (1.18), projection gives, for brane index ν,

$$\boxed{\partial_\mu\langle ZF^{\mu\nu}\rangle_W + [WZF^{w\nu}]_{\partial D} - \int_D W'(w)ZF^{w\nu}\,dw + \frac{1}{\xi}\partial^\nu\langle H\,\partial\cdot A\rangle_W = \mu_0\langle J^\nu\rangle_W}. \tag{1.31}$$

The homogeneous equations project in the measured fields

$$E_{\mathrm{meas},a} = \langle F_{a0} \rangle_W, \qquad B_{\mathrm{meas},ab} = \langle F_{ab} \rangle_W, \tag{1.32}$$

whereas (1.31) uses the source-coupled fluxes

$$D^a_{\mathrm{flux}} = \langle Z F^{a0} \rangle_W, \qquad H^{ab}_{\mathrm{flux}} = \langle Z F^{ab} \rangle_W. \tag{1.33}$$

Identifying measured fields with flux fields is therefore a closure assumption, not a parent identity.

In the zero-mode channel

$$A_\mu(x, w) = a_\mu(x), \qquad A_w = 0, \qquad F^{w\nu} = 0, \qquad J^\nu(x, w) = j^\nu(x) S(w), \tag{1.34}$$

the projected effective parameters are

$$\boxed{\mu^{\mathrm{proj}}_{0,\mathrm{eff}} = \mu_0 \frac{\int W S \, dw}{\int W Z \, dw}, \qquad \xi^{\mathrm{proj}}_{\mathrm{eff}} = \xi \frac{\int W Z \, dw}{\int W H \, dw}}. \tag{1.35}$$

The matched reduction-first brane Maxwell law is recovered by the localized gauge choice and source profile

$$\boxed{H = Z, \qquad S(w) = \frac{Z(w)}{Z_{\mathrm{int}}}, \qquad \mu^{\mathrm{proj}}_{0,\mathrm{eff}} = \frac{\mu_0}{Z_{\mathrm{int}}}}. \tag{1.36}$$

Thus the reduction-first Maxwell sector is retained below as a controlled matched channel and one-port normal form, not as the generic parent projection.

Under the Helmholtz split

$$\mathbf{v}_{\mathrm{brane}} = \nabla_3 \varphi + \mathbf{v}_T, \qquad \nabla_3 \cdot \mathbf{v}_T = 0, \tag{1.37}$$

the exact longitudinal identity is

$$\rho_{\mathrm{brane}} \nabla_3^2 \varphi = S_{\mathrm{leak}} - \partial_t \rho_{\mathrm{brane}} - (\nabla_3 \rho_{\mathrm{brane}}) \cdot (\nabla_3 \varphi + \mathbf{v}_T). \tag{1.38}$$

This is an imported lower-order hook rather than a new Stage 001–023 result. It fixes how the moving-throat response must later interface with the brane-facing scalar sector.

Theorem 1.2 (Parent-theory import)**.** *Within the corrected charge ontology and the declared localized Maxwell/GNLS parent action, the moving-throat derivation inherits the exact equations* (1.14)*,* (1.15)*,* (1.18)*,* (1.20)*, and* (1.25)*. After a brane-observation kernel is declared, it also inherits the exact projected Maxwell law* (1.31)*. Any moving-throat closure must therefore preserve the mixed-channel ontology and may only suppress $A_w, J^w, F_{\mu w}, E_w, C_a$ as part of a controlled matched brane reduction.*

Proof. The equations are direct Euler–Lagrange consequences of the imported parent action and the minimal-coupling definitions. The projected Maxwell law follows by applying (1.27) to the transverse derivative in (1.18). The mixed fields in (1.20) are components of the gauge-invariant field strength. Therefore they cannot be removed by a gauge choice. The zero-mode brane Maxwell reduction may set them to zero as a matched branch assumption, but the moving-throat PDE that is supposed to compute microscopic response data must retain them until such a branch restriction is explicitly imposed. □

1.3 Hybrid level-set / shape-field representation

1.3.1 Purpose of the geometry lift

The old conservative hierarchy used finite-dimensional geometry variables $a(t)$ and $L(t)$. Those variables are adequate as a reduced breathing closure, but they cannot expose the grouped real P_2 support lanes or the full mouth/worldtube response operator. Stage 001 therefore replaces the old finite-dimensional closure by a distributed surface field while preserving a and L as low moments.

The chosen representation is hybrid: the finite throat is represented by a level set, but the level set is parameterized by a brane-spherical mouth shape. Define

$$\boxed{\Sigma(\mathbf{X},t) = r - R(\Omega,w,t), \qquad r=\sqrt{x^2+y^2+z^2}, \qquad \Omega = \frac{\mathbf{x}}{r} \in S^2}. \tag{1.39}$$

The throat surface is $\Sigma(\mathbf{X},t)=0$. We use the sign convention

$$\Sigma > 0 \quad \text{outside the throat,} \qquad \Sigma < 0 \quad \text{inside the support region.} \tag{1.40}$$

1.3.2 Reference stationary throat

A stationary finite-throat reference branch is written

$$\Sigma_0(\mathbf{X}) = r - R_0(w). \tag{1.41}$$

The reference mouth is at $w=0$, the throat extends over $0 \le w \le L_0$, and the reference mouth radius is

$$a_0 = R_0(0). \tag{1.42}$$

The bottom may be represented by

$$R_0(L_0) = 0, \tag{1.43}$$

or by an equivalent regular bottom condition. The finite-throat assumption is important because the later support problem uses finite axial spectral data rather than a noncompact Gaussian surrogate.

1.3.3 Promoted confinement potential

The old confinement potential is promoted by replacing

$$V_{\text{conf}}(\mathbf{X};a,L) \quad \longrightarrow \quad V_{\text{conf}}(\mathbf{X};\Sigma). \tag{1.44}$$

The minimal smooth-wall choice is

$$V_{\text{conf}}(\mathbf{X};\Sigma) = V_{\text{wall}}\left(\frac{\Sigma(\mathbf{X},t)}{\ell_c}\right), \tag{1.45}$$

where ℓ_c is a wall-thickness scale. Around Σ_0, the first variation is

$$\delta V_{\text{conf}} = \frac{V'_{\text{wall}}(\Sigma_0/\ell_c)}{\ell_c}\,\delta\Sigma. \tag{1.46}$$

Since $\Sigma = r - R$, one has

$$\delta\Sigma = -\delta R. \tag{1.47}$$

Writing the wall displacement as $\eta = \delta R$, the moving-wall coupling to the matter sector is therefore

$$\boxed{\delta V_{\text{conf}} = -\frac{V'_{\text{wall}}(\Sigma_0/\ell_c)}{\ell_c}\,\eta}. \tag{1.48}$$

This sign is one of the main sign checks for the entire linearized program: a positive outward radial displacement decreases Σ at fixed bulk point and therefore enters the confinement variation with the minus sign in (1.48).

1.4 Collective moment recovery for $a(t)$ and $L(t)$

1.4.1 Mouth-radius and depth moments

The old mouth radius is not discarded. It is recovered as the spherical average of the moving mouth:

$$\boxed{a(t) = \frac{1}{4\pi}\int_{S^2} R(\Omega, 0, t)\, d\Omega}. \tag{1.49}$$

Let $W_b(\Omega, t)$ denote the bottom location, implicitly defined by

$$R(\Omega, W_b(\Omega, t), t) = 0. \tag{1.50}$$

Then the old axial depth is recovered as

$$\boxed{L(t) = \frac{1}{4\pi}\int_{S^2} W_b(\Omega, t)\, d\Omega}. \tag{1.51}$$

Equations (1.49) and (1.51) are the first audit gate for the lift: the new distributed theory must reduce to the old (a, L) closure when restricted to its lowest axisymmetric moments.

1.4.2 Mode decomposition of the throat shape

Write

$$R(\Omega, w, t) = R_0(w) + \eta(\Omega, w, t). \tag{1.52}$$

The wall displacement is expanded in real spherical harmonics:

$$\eta(\Omega, w, t) = q_{00}(w, t)Y_{00}(\Omega) + \sum_{A\in P_2^{\mathrm{real}}} q_A(w, t)Y_A(\Omega) + \eta_{\geq 3}(\Omega, w, t). \tag{1.53}$$

Here

$$P_2^{\mathrm{real}} = \{20, 21c, 21s, 22c, 22s\}. \tag{1.54}$$

The grouped real ledger later compresses the c/s pairs into grouped labels

$$A \in \{20, 21, 22\}, \tag{1.55}$$

with natural weights $(1, 2, 2)$. The grouped labels are lane labels, not spacetime indices.

With the standard normalized harmonic

$$Y_{00} = \frac{1}{2\sqrt{\pi}}, \qquad \int_{S^2} Y_{00}^2\, d\Omega = 1, \qquad \frac{1}{4\pi}\int_{S^2} Y_{00}\, d\Omega = \frac{1}{2\sqrt{\pi}}, \tag{1.56}$$

the physical mouth-average shift obeys

$$\boxed{q_{00}(0, t) = 2\sqrt{\pi}\, \delta a(t)}. \tag{1.57}$$

All grouped real P_2 harmonics have zero average over the sphere. Thus grouped quadrupole deformations do not change the mouth-average radius at linear order.

1.4.3 Axisymmetric residual lane

The axisymmetric wall field is further split as

$$q_{00}(w,t)Y_{00}(\Omega) = 2\sqrt{\pi}\left[\alpha_a(w)\delta a(t) + \alpha_L(w)\delta L(t)\right]Y_{00}(\Omega) + g(w,t)Y_{00}(\Omega), \tag{1.58}$$

where $g(w,t)$ is the residual axisymmetric geometry lane orthogonal to the chosen a and L profiles. In practice, the first truncation drops g; later geometry-completion terms are naturally interpreted as the return of this residual lane.

Proposition 1.3 (Collective-moment recovery)**.** *Inside the shape-field closure $\Sigma = r - R(\Omega, w, t)$, the old collective variables $a(t)$ and $L(t)$ are recovered as the monopole moments* (1.49) *and* (1.51)*. The grouped real P_2 deformations are orthogonal to the mouth-average radius shift at linear order.*

Proof. The formula for $a(t)$ is the spherical average of the mouth radius. The formula for $L(t)$ is the spherical average of the bottom location. The zero-average property of all $l = 2$ spherical harmonics gives the last statement immediately. The normalization bridge (1.57) follows from (1.56). ☐

1.5 Distributed wall action and densitized convention

1.5.1 Quadratic wall action

The first distributed wall action is the minimal passive local quadratic action

$$S_\eta^{(2)} = \frac{1}{2}\int dt\,dw\,d\Omega\,\sqrt{\gamma_0}\left[\mu_\eta(w)(\partial_t\eta)^2 - T_w(w)(\partial_w\eta)^2 - T_\Omega(w)\eta(-\Delta_{S^2})\eta - K_\eta(w)\eta^2\right]. \tag{1.59}$$

The coefficients $\mu_\eta, T_w, T_\Omega, K_\eta$ are branch data of the chosen reference throat. They are not to be refit independently at later stages to force a normalization condition.

After integrating over the reference sphere, the program uses a densitized one-dimensional convention: the background surface weight is absorbed into the effective axial coefficients. With that convention, the modal action is written with measure dw. If an explicit axial weight $g(w) = \sqrt{\gamma_0}$ were retained, the Euler–Lagrange equation would contain the corresponding first-derivative contribution from $g'(w)$. In the ledger formulas below, that bookkeeping has already been absorbed into $\mu_\eta, T_w, T_\Omega, K_\eta$.

1.5.2 Modal wall equation

For a harmonic component $q_{lm}(w,t)Y_{lm}(\Omega)$, using

$$-\Delta_{S^2}Y_{lm} = l(l+1)Y_{lm}, \tag{1.60}$$

the densitized modal wall equation is

$$\boxed{\mu_\eta q_{lm,tt} - \partial_w(T_w\partial_w q_{lm}) + \left[K_\eta + l(l+1)T_\Omega\right]q_{lm} = S_{lm}^{(\psi,A)} + f_{lm}^{\text{ext}}}. \tag{1.61}$$

The source $S_{lm}^{(\psi,A)}$ is generated by variation of the matter and gauge sectors, with the first matter contribution inherited from (1.48).

Equation (1.61) separates the first three relevant sectors:

1. the $l = 0$ collective sector containing δa, δL, and the residual geometry lane $g(w,t)$;
2. the grouped real $l = 2$ sector containing $20, 21c, 21s, 22c, 22s$;

3. higher $l \geq 3$ wall modes.

The present response program keeps higher modes in the full PDE ontology but does not need them for the first grouped-quadrupole bridge.

Proposition 1.4 (Wall modal backbone)**.** *The quadratic wall action* (1.59)*, read in the densitized axial convention, yields the modal operator in* (1.61)*. The scalar and grouped real P_2 sectors therefore split before any matter or gauge coupling is added.*

Proof. Insert $\eta = q_{lm}(w,t)Y_{lm}(\Omega)$ into (1.59), use orthonormality on S^2, and vary the resulting one-dimensional action. The angular stiffness term contributes $l(l+1)T_\Omega q_{lm}$. The remaining source terms are not part of the free wall action and are kept on the right-hand side. □

1.6 Axisymmetric breathing reduction

1.6.1 Two-mode breathing ansatz

Stage 002 checks that the distributed wall lift recovers the old (a, L) closure at the expected conservative linear level. Use the axisymmetric truncation

$$\eta_0(w,t) = 2\sqrt{\pi}\left[\alpha_a(w)\delta a(t) + \alpha_L(w)\delta L(t)\right]. \tag{1.62}$$

Define

$$Q^A = (\delta a, \delta L), \qquad A \in \{a, L\}. \tag{1.63}$$

The truncation is not the final geometry theory. It is the lowest-mode audit that the distributed wall has not lost contact with the older closure.

1.6.2 Reduced matrices

For the axisymmetric branch, the reduced Lagrangian is

$$L^{(0)}_{\text{red}} = \frac{1}{2}M_{AB}\dot{Q}^A\dot{Q}^B - \frac{1}{2}K_{AB}Q^AQ^B, \tag{1.64}$$

where

$$\boxed{M_{AB} = 4\pi \int dw\, \mu_\eta(w)\alpha_A(w)\alpha_B(w)}, \tag{1.65}$$

and

$$\boxed{K_{AB} = 4\pi \int dw\, \left[T_w(w)\alpha'_A(w)\alpha'_B(w) + K_0(w)\alpha_A(w)\alpha_B(w)\right]}. \tag{1.66}$$

The Euler–Lagrange equation is

$$\boxed{M_{AB}\ddot{Q}^B + K_{AB}Q^B = 0}. \tag{1.67}$$

This is the conservative part of the older effective geometry law. Damping and open-system terms can be added only after coupling to exterior, matter, gauge, or leakage channels.

1.6.3 One-mode grouped P_2 comparison

For a single grouped real quadrupole amplitude,

$$\eta_{2m}(\Omega, w, t) = \beta_2(w) q_{2m}(t) Y_{2m}^{\text{real}}(\Omega), \tag{1.68}$$

the reduced one-mode Lagrangian is

$$L_{2m} = \frac{1}{2} M_2 \dot{q}_{2m}^2 - \frac{1}{2} K_2 q_{2m}^2, \tag{1.69}$$

with

$$\boxed{M_2 = \int dw\, \mu_\eta(w) \beta_2(w)^2, \qquad K_2 = \int dw \left[T_w(w) \beta_2'(w)^2 + \left(K_\eta(w) + 6T_\Omega(w) \right) \beta_2(w)^2 \right]}. \tag{1.70}$$

Therefore

$$M_2 \ddot{q}_{2m} + K_2 q_{2m} = 0. \tag{1.71}$$

Before anisotropy or support coupling is added, every real P_2 component has the same geometry-only operator. This is the first degeneracy statement behind the later grouped-isotropy gates.

Theorem 1.5 (Breathing reduction as the lowest wall truncation)**.** *Inside the quadratic distributed-wall closure, the two-mode axisymmetric ansatz* (1.62) *reduces exactly to the conservative matrix system* (1.67)*. The old* (a, L) *variables are therefore the lowest collective coordinates of the wall PDE rather than independent microscopic primitives.*

Proof. Substitute (1.62) into the quadratic wall action, use the normalization of Y_{00}, and collect coefficients of $\dot{Q}^A \dot{Q}^B$ and $Q^A Q^B$. The resulting matrices are (1.65) and (1.66). Varying (1.64) gives (1.67). □

1.7 Stable BdG normal-mode reduction and Schur complement

1.7.1 Linearized matter coupling

Let the stationary matter background be

$$\psi(\mathbf{X}, t) = e^{-i\mu_0 t/\hbar} \psi_0(\mathbf{X}), \tag{1.72}$$

and write

$$\psi = e^{-i\mu_0 t/\hbar} \left(\psi_0 + \delta\psi \right). \tag{1.73}$$

The linearized density is

$$\delta\rho = \psi_0^* \delta\psi + \psi_0 \delta\psi^*. \tag{1.74}$$

The moving wall enters the linearized GNLS problem through (1.48). In BdG form the schematic system is

$$i\hbar \partial_t \begin{pmatrix} \delta\psi \\ \delta\psi^* \end{pmatrix} = \mathcal{L}_{\text{BdG}} \begin{pmatrix} \delta\psi \\ \delta\psi^* \end{pmatrix} + \mathcal{C}_A[\delta A_M] + \mathcal{C}_\eta[\eta]. \tag{1.75}$$

Stage 003 does not claim to solve the full BdG PDE. It passes to a stable normal-mode reduction in selected $l = 0$ and $l = 2$ sectors. Once that reduction is declared, integrating out the stable modes is exact algebra.

1.7.2 Axisymmetric BdG support kernel

Let $Q^A = (\delta a, \delta L)$ and let X_α be stable scalar BdG normal coordinates with frequencies $\varpi_{0\alpha} > 0$. The reduced quadratic Lagrangian is

$$L^{(0)}_{\text{red}} = \frac{1}{2}\dot{Q}^T M_0 \dot{Q} - \frac{1}{2} Q^T K_0 Q + \frac{1}{2}\sum_\alpha \left(\dot{X}_\alpha^2 - \varpi_{0\alpha}^2 X_\alpha^2\right) + \sum_{A,\alpha} C_{A\alpha} Q^A X_\alpha. \tag{1.76}$$

In frequency space, the matter coordinates solve

$$(\varpi_{0\alpha}^2 - \omega^2) X_\alpha = C_{A\alpha} Q^A. \tag{1.77}$$

Substitution gives the exact effective matrix kernel

$$\boxed{D_0^{\text{eff}}(\omega) = K_0 - \omega^2 M_0 - C(\Omega_0^2 - \omega^2 I)^{-1} C^T}, \tag{1.78}$$

where

$$\Omega_0^2 = \operatorname{diag}(\varpi_{0\alpha}^2). \tag{1.79}$$

Its low-frequency expansion is

$$D_0^{\text{eff}}(\omega) = K_0^{\text{eff}} - \omega^2 M_0^{\text{eff}} - \omega^4 N_0^{\text{eff}} + O(\omega^6), \tag{1.80}$$

with

$$\boxed{K_0^{\text{eff}} = K_0 - C\Omega_0^{-2} C^T, \qquad M_0^{\text{eff}} = M_0 + C\Omega_0^{-4} C^T, \qquad N_0^{\text{eff}} = C\Omega_0^{-6} C^T}. \tag{1.81}$$

Thus stable matter support softens the static stiffness, increases the effective inertia, and generates the next even low-frequency coefficient.

1.7.3 Grouped real P_2 BdG kernels

For each grouped lane $A \in \{20, 21, 22\}$, let q_A be the wall amplitude and $X_{A\alpha}$ be stable quadrupole BdG coordinates with frequencies $\varpi_{A\alpha}$. The reduced Lagrangian is

$$L^{(2)}_{\text{red}} = \sum_A \left[\frac{1}{2} M_A \dot{q}_A^2 - \frac{1}{2} K_A q_A^2 + \frac{1}{2}\sum_\alpha (\dot{X}_{A\alpha}^2 - \varpi_{A\alpha}^2 X_{A\alpha}^2) + \sum_\alpha g_{A\alpha} q_A X_{A\alpha}\right]. \tag{1.82}$$

Eliminating the stable coordinates gives

$$\boxed{D_A^{\text{eff}}(\omega) = K_A - M_A\omega^2 - \sum_\alpha \frac{g_{A\alpha}^2}{\varpi_{A\alpha}^2 - \omega^2}}. \tag{1.83}$$

The low-frequency coefficients are

$$\boxed{\begin{aligned} d_{0A}^{\text{eff}} &= K_A - \sum_\alpha \frac{g_{A\alpha}^2}{\varpi_{A\alpha}^2}, \\ d_{2A}^{\text{eff}} &= -\left(M_A + \sum_\alpha \frac{g_{A\alpha}^2}{\varpi_{A\alpha}^4}\right), \\ d_{4A}^{\text{eff}} &= -\sum_\alpha \frac{g_{A\alpha}^2}{\varpi_{A\alpha}^6}. \end{aligned}} \tag{1.84}$$

These are the first explicit conservative grouped-lane moments generated by matter support.

1.7.4 First pole-shift formula

For one wall mode q coupled to one stable matter mode X,

$$L = \frac{1}{2}M\dot{q}^2 - \frac{1}{2}Kq^2 + \frac{1}{2}\dot{X}^2 - \frac{1}{2}\varpi^2X^2 + gqX. \tag{1.85}$$

The exact dispersion relation is

$$(K - M\omega^2)(\varpi^2 - \omega^2) - g^2 = 0. \tag{1.86}$$

With $\Omega_\eta^2 = K/M$, the exact poles are

$$\boxed{\omega_\pm^2 = \frac{\Omega_\eta^2 + \varpi^2 \pm \sqrt{(\Omega_\eta^2 - \varpi^2)^2 + 4g^2/M}}{2}}. \tag{1.87}$$

If $\varpi^2 > \Omega_\eta^2$ and g is weak, the wall-like pole shifts by

$$\boxed{\delta\Omega_\eta^2 = -\frac{g^2/M}{\varpi^2 - \Omega_\eta^2} + O(g^4)}. \tag{1.88}$$

This is the first explicit reduced origin of the pole data that later appear as open throat observables.

1.7.5 Matter-coupled isotropy diagnostic

For grouped triples (x_{20}, x_{21}, x_{22}), define

$$\bar{x} = \frac{x_{20} + 2x_{21} + 2x_{22}}{5}, \qquad a_x = \frac{2x_{20} - x_{21} - x_{22}}{10}, \qquad b_x = \frac{x_{21} - x_{22}}{2}. \tag{1.89}$$

If

$$K_{20} = K_{21} = K_{22}, \quad M_{20} = M_{21} = M_{22}, \quad g_{20,\alpha} = g_{21,\alpha} = g_{22,\alpha}, \quad \varpi_{20,\alpha} = \varpi_{21,\alpha} = \varpi_{22,\alpha}, \tag{1.90}$$

then all grouped coefficients in (1.84) are lane-independent, and therefore the grouped anisotropy defects vanish.

Theorem 1.6 (BdG Schur-complement support kernel)**.** *Inside the stable normal-mode reduction of the linearized matter sector, integrating out the BdG coordinates gives the exact Schur-complement kernels* (1.78) *and* (1.83). *The low-frequency moments are fixed by the support frequencies and coupling overlaps; grouped P_2 anisotropy can enter only through lane dependence of the wall data, support frequencies, or coupling overlaps.*

Proof. The reduced matter coordinates enter quadratically and linearly. Their frequency-domain equations are algebraic. Solving those equations and substituting back into the wall equations gives the Schur complements. Expanding $(\Omega^2 - \omega^2)^{-1}$ in powers of ω^2 gives the stated low-frequency coefficients. The isotropy statement follows from equality of the three lane data. □

1.8 Projection-first Maxwell/mixed response and outgoing $l = 2$ bridge

1.8.1 Why the mixed sector enters before the outgoing bridge

The BdG support kernel is conservative. It can produce even low-frequency response moments and pole shifts, but by itself it does not produce an honest passive/outgoing odd term. The first available microscopic carrier of an outgoing branch is the localized Maxwell/mixed block, because A_w, $F_{\mu w}$, and J^w are present in the parent ontology and are not gauge artifacts.

Stages 004–021 first project the localized parent Maxwell equation, producing the exact open-system law (1.31). Only after the matched channel (1.36) is declared does the derivation compress the EM sector into a minimal reduced lane containing one wall coordinate Q_l, one brane-like localized Maxwell coordinate A_l, and one mixed-sector coordinate W_l. This is not the full PDE. It is the Stage 021 normal form used to carry both conservative gauge self-energy and passive outgoing transfer into the grouped bundle.

1.8.2 One-lane Maxwell/mixed Lagrangian

The reduced quadratic Lagrangian is

$$\begin{aligned} L_l = \frac{1}{2}M_l\dot{Q}_l^2 - \frac{1}{2}K_lQ_l^2 + \frac{1}{2}\dot{A}_l^2 - \frac{1}{2}\Omega_{A,l}^2A_l^2 + \frac{1}{2}\dot{W}_l^2 - \frac{1}{2}\Omega_{W,l}^2W_l^2 \\ + R_lA_lW_l + g_{A,l}Q_lA_l + g_{W,l}Q_lW_l. \end{aligned} \tag{1.91}$$

Here A_l is a brane-like localized gauge coordinate, W_l is a mixed $A_w/F_{\mu w}/J^w$-active coordinate, R_l mixes the internal gauge coordinates, and $g_{A,l}, g_{W,l}$ are wall couplings.

1.8.3 Conservative self-energy

With frequency convention $e^{-i\omega t}$, define

$$\mathcal{A}_l(\omega) = \Omega_{A,l}^2 - \omega^2, \qquad \mathcal{W}_l(\omega) = \Omega_{W,l}^2 - \omega^2, \qquad \Delta_l(\omega) = \mathcal{A}_l(\omega)\mathcal{W}_l(\omega) - R_l^2. \tag{1.92}$$

Eliminating A_l, W_l gives the conservative self-energy

$$\boxed{\Sigma_l^{\mathrm{EM+mix,cons}}(\omega) = \frac{g_{A,l}^2\mathcal{W}_l(\omega) + 2g_{A,l}g_{W,l}R_l + g_{W,l}^2\mathcal{A}_l(\omega)}{\Delta_l(\omega)}}. \tag{1.93}$$

The wall operator is

$$D_l^{\mathrm{cons}}(\omega) = K_l - M_l\omega^2 - \Sigma_l^{\mathrm{EM+mix,cons}}(\omega). \tag{1.94}$$

Set

$$\Delta_{0,l} = \Omega_{A,l}^2\Omega_{W,l}^2 - R_l^2, \qquad S_{2,l} = \Omega_{A,l}^2 + \Omega_{W,l}^2, \tag{1.95}$$

$$N_{0,l} = g_{A,l}^2\Omega_{W,l}^2 + 2g_{A,l}g_{W,l}R_l + g_{W,l}^2\Omega_{A,l}^2, \qquad G_{2,l} = g_{A,l}^2 + g_{W,l}^2. \tag{1.96}$$

Then

$$\Sigma_l^{\mathrm{EM+mix,cons}}(\omega) = z_{l,0} + z_{l,2}\omega^2 + z_{l,4}\omega^4 + O(\omega^6), \tag{1.97}$$

with

$$\boxed{\begin{aligned} z_{l,0} &= \frac{N_{0,l}}{\Delta_{0,l}}, \\ z_{l,2} &= \frac{N_{0,l}S_{2,l} - G_{2,l}\Delta_{0,l}}{\Delta_{0,l}^2}, \\ z_{l,4} &= \frac{N_{0,l}(S_{2,l}^2 - \Delta_{0,l}) - S_{2,l}G_{2,l}\Delta_{0,l}}{\Delta_{0,l}^3}. \end{aligned}} \tag{1.98}$$

1.8.4 Outgoing-port dressing and transfer factor

Attach the exterior passive/outgoing channel to the mixed coordinate by replacing

$$W_l(\omega) \longrightarrow W_l(\omega) - \Pi_l^{\text{out}}(\omega). \tag{1.99}$$

To first order in the outgoing port,

$$\Sigma_l^{\text{full}}(\omega) = \Sigma_l^{\text{cons}}(\omega) + \Pi_l^{\text{out}}(\omega)N_l(\omega) + O\Big((\Pi_l^{\text{out}})^2\Big), \tag{1.100}$$

where the transfer factor is

$$\boxed{N_l(\omega) = \frac{\left[A_l(\omega)g_{W,l} + R_l g_{A,l}\right]^2}{\left[A_l(\omega)W_l(\omega) - R_l^2\right]^2}}. \tag{1.101}$$

At zero frequency,

$$\boxed{N_l(0) = \frac{\left[\Omega_{A,l}^2 g_{W,l} + R_l g_{A,l}\right]^2}{\left[\Omega_{A,l}^2\Omega_{W,l}^2 - R_l^2\right]^2} \geq 0}. \tag{1.102}$$

The sign of the outgoing odd term is therefore supplied by the outgoing port; the internal conservative block transfers it with a nonnegative static factor.

1.8.5 Compact outgoing $l = 2$ fingerprint

For the compact outgoing $l = 2$ branch, the normalized outgoing response has the low-frequency form

$$\widehat{Y}_2^{\text{out}}(\omega) = 1 + \frac{a^2\omega^2}{9c_s^2} + \frac{4a^4\omega^4}{81c_s^4} + i\frac{a^5\omega^5}{27c_s^5} + O(\omega^6). \tag{1.103}$$

Thus the outgoing port coefficient is

$$\Gamma_2^{\text{port}} = \frac{a^5}{27c_s^5}. \tag{1.104}$$

Combining (1.101) with (1.104), the wall-level odd quadrupole contribution is

$$\boxed{\delta D_2^{\text{odd}}(\omega) = -iN_2(0)\frac{a^5}{27c_s^5}\omega^5 + O(\omega^7)}. \tag{1.105}$$

In the wall-operator convention $D = K - M\omega^2 - \Sigma$, this is the passive sign. In normalized response language the same branch is read with the positive $+i\omega^5$ fingerprint.

1.8.6 Scalar compatibility check

A naive scalar outlet could introduce a dangerous $i\omega$ monopole term. The reduced mixed-sector model shows how such a term can be delayed. If the scalar/breathing wall coordinate couples to the port-active mixed combination only through a derivative coupling, for example

$$g_{W,0}(\omega) = \eta\omega, \tag{1.106}$$

and if there is no direct non-derivative coupling into the same port-active combination, then

$$N_0(\omega) = O(\omega^2). \tag{1.107}$$

Therefore an outgoing scalar port with $\Pi_0^{\mathrm{out}}(\omega) = i\gamma_1\omega + \cdots$ contributes at wall level only as $O(i\omega^3)$. This is a compatibility mechanism, not yet a full scalar no-go theorem.

Theorem 1.7 (Projection-first Maxwell/mixed outgoing bridge)**.** *The exact projected Maxwell law* (1.31) *retains the transverse mixed channel before any brane reduction. In the matched one-lane Maxwell/mixed normal form, the wall inherits an outgoing odd contribution through the exact transfer factor* (1.101)*. For $l = 2$, the leading wall-level odd coefficient is* (1.105)*; the remaining normalization problem is the computation of the static transfer factor on the actual moving-throat branch.*

Proof. Projection supplies the mixed transverse source term in (1.31); the matched channel (1.36) is the reduction in which it can be represented by a finite one-port packet. Eliminating the two internal gauge coordinates from (1.91) gives the conservative self-energy (1.93). Replacing W_l by $W_l - \Pi_l^{\mathrm{out}}$ and expanding to first order in Π_l^{out} gives (1.101). Multiplying by the compact outgoing $l = 2$ fingerprint coefficient (1.104) gives (1.105). □

1.9 Full grouped real P_2 bundle transition

1.9.1 Conservative operator moments and normalized response moments

Stages 022 and 023 lift the projection-first Maxwell packet, through its matched one-port normal form, into the full grouped real bundle. For each grouped lane $A \in \{20, 21, 22\}$, write the conservative operator as

$$D_A^{\mathrm{cons}}(\omega) = D_{A0} + D_{A2}\omega^2 + D_{A4}\omega^4 + O(\omega^6). \tag{1.108}$$

The normalized response is

$$Y_A(\omega) = \frac{D_{A0}}{D_A^{\mathrm{cons}}(\omega)} = 1 + u_2^{(A)}\omega^2 + u_4^{(A)}\omega^4 + O(\omega^6). \tag{1.109}$$

The exact conversion formulas are

$$\boxed{u_2^{(A)} = -\frac{D_{A2}}{D_{A0}}, \qquad u_4^{(A)} = \frac{D_{A2}^2 - D_{A0}D_{A4}}{D_{A0}^2}}. \tag{1.110}$$

This is the first bookkeeping bridge between the wall/BdG/Maxwell operator language and the grouped-response language used by the later retarded quadrupole tests.

1.9.2 Grouped inverse map and weighted projectors

For a grouped triple $x = (x_{20}, x_{21}, x_{22})^T$, define

$$\bar{x} = \frac{x_{20} + 2x_{21} + 2x_{22}}{5}, \qquad a_x = \frac{2x_{20} - x_{21} - x_{22}}{10}, \qquad b_x = \frac{x_{21} - x_{22}}{2}. \tag{1.111}$$

The inverse map is

$$\boxed{x_{20} = \bar{x} + 4a_x, \qquad x_{21} = \bar{x} - a_x + b_x, \qquad x_{22} = \bar{x} - a_x - b_x}. \tag{1.112}$$

The natural grouped metric is

$$G_{\text{grp}} = \operatorname{diag}(1, 2, 2). \tag{1.113}$$

A G_{grp}-orthogonal basis is

$$e_{\bar{x}} = (1, 1, 1), \qquad e_a = (4, -1, -1), \qquad e_b = (0, 1, -1), \tag{1.114}$$

with norms

$$e_{\bar{x}}^T G_{\text{grp}} e_{\bar{x}} = 5, \qquad e_a^T G_{\text{grp}} e_a = 20, \qquad e_b^T G_{\text{grp}} e_b = 4. \tag{1.115}$$

The exact projectors are

$$\boxed{P_{\bar{x}} = \frac{1}{5}\begin{pmatrix} 1 & 2 & 2 \\ 1 & 2 & 2 \\ 1 & 2 & 2 \end{pmatrix}, \quad P_a = \frac{1}{20}\begin{pmatrix} 16 & -8 & -8 \\ -4 & 2 & 2 \\ -4 & 2 & 2 \end{pmatrix}, \quad P_b = \frac{1}{4}\begin{pmatrix} 0 & 0 & 0 \\ 0 & 2 & -2 \\ 0 & -2 & 2 \end{pmatrix}}. \tag{1.116}$$

They satisfy

$$P_{\bar{x}} + P_a + P_b = I_3, \qquad P_i P_j = \delta_{ij} P_i. \tag{1.117}$$

The grouped isotropy condition is simply

$$P_a x = P_b x = 0. \tag{1.118}$$

1.9.3 Outgoing prefactor and normalized branch coefficients

Let the outgoing transfer factor in lane A be

$$N_A(\omega) = N_{A0} + N_{A2}\omega^2 + N_{A4}\omega^4 + O(\omega^6). \tag{1.119}$$

The internal prefactor multiplying the outgoing branch is

$$\operatorname{Pref}_A(\omega) = \frac{D_{A0} N_A(\omega)}{D_A^{\text{cons}}(\omega)^2} = P_{A0} + P_{A2}\omega^2 + P_{A4}\omega^4 + O(\omega^6). \tag{1.120}$$

The exact coefficients are

$$\boxed{\begin{aligned} P_{A0} &= \frac{N_{A0}}{D_{A0}}, \\ P_{A2} &= \frac{D_{A0}N_{A2} - 2D_{A2}N_{A0}}{D_{A0}^2}, \\ P_{A4} &= \frac{D_{A0}^2 N_{A4} - 2D_{A0}(D_{A2}N_{A2} + D_{A4}N_{A0}) + 3D_{A2}^2 N_{A0}}{D_{A0}^3}. \end{aligned}} \tag{1.121}$$

Multiplying by the compact outgoing $l = 2$ fingerprint (1.103), with

$$A_2^{\text{out}} = \frac{a^2}{9c_s^2}, \qquad B_2^{\text{out}} = \frac{4a^4}{81c_s^4}, \qquad G_5^{\text{out}} = \frac{a^5}{27c_s^5}, \tag{1.122}$$

gives

$$\boxed{K_{A0} = P_{A0}, \quad K_{A2} = P_{A2} + A_2^{\text{out}} P_{A0}, \quad K_{A4} = P_{A4} + A_2^{\text{out}} P_{A2} + B_2^{\text{out}} P_{A0}, \quad \Gamma_5^{(A)} = G_5^{\text{out}} P_{A0}}. \tag{1.123}$$

Only the static prefactor P_{A0} enters the leading $i\omega^5$ coefficient.

1.9.4 Full reduced grouped bundle

Stage 023 writes the full coupled lane data explicitly. For each lane A, define BdG support moments

$$B_{A0} = \sum_\alpha \frac{c_{A\alpha}^2}{\varpi_{A\alpha}^2}, \qquad B_{A2} = \sum_\alpha \frac{c_{A\alpha}^2}{\varpi_{A\alpha}^4}, \qquad B_{A4} = \sum_\alpha \frac{c_{A\alpha}^2}{\varpi_{A\alpha}^6}. \tag{1.124}$$

For each conservative Maxwell/mixed internal pair r, set

$$\Delta_{A,r} = \Omega_{U,A,r}^2 \Omega_{W,A,r}^2 - R_{A,r}^2, \qquad S_{A,r} = \Omega_{U,A,r}^2 + \Omega_{W,A,r}^2, \tag{1.125}$$

$$Q_{A,r} = g_{U,A,r}^2 \Omega_{W,A,r}^2 + 2g_{U,A,r} g_{W,A,r} R_{A,r} + g_{W,A,r}^2 \Omega_{U,A,r}^2, \qquad G_{A,r} = g_{U,A,r}^2 + g_{W,A,r}^2. \tag{1.126}$$

Then

$$\boxed{\begin{aligned} Z_{A0}^{(r)} &= \frac{Q_{A,r}}{\Delta_{A,r}}, \\ Z_{A2}^{(r)} &= \frac{Q_{A,r} S_{A,r} - G_{A,r}\Delta_{A,r}}{\Delta_{A,r}^2}, \\ Z_{A4}^{(r)} &= \frac{Q_{A,r}(S_{A,r}^2 - \Delta_{A,r}) - S_{A,r} G_{A,r} \Delta_{A,r}}{\Delta_{A,r}^3}. \end{aligned}} \tag{1.127}$$

The outgoing-transfer moments are

$$\boxed{\begin{aligned} N_{A0}^{(r)} &= \frac{P_{A,r}^2}{\Delta_{A,r}^2}, \\ N_{A2}^{(r)} &= \frac{2P_{A,r}(P_{A,r} S_{A,r} - \Delta_{A,r} g_{W,A,r})}{\Delta_{A,r}^3}, \\ N_{A4}^{(r)} &= \frac{\Delta_{A,r}^2 g_{W,A,r}^2 - 2\Delta_{A,r} P_{A,r}^2 - 4\Delta_{A,r} P_{A,r} S_{A,r} g_{W,A,r} + 3P_{A,r}^2 S_{A,r}^2}{\Delta_{A,r}^4}, \end{aligned}} \tag{1.128}$$

where

$$P_{A,r} = \Omega_{U,A,r}^2 g_{W,A,r} + R_{A,r} g_{U,A,r}. \tag{1.129}$$

Summing over r gives Z_{An} and N_{An}. The total conservative wall coefficients are

$$\boxed{D_{A0} = K_A - B_{A0} - Z_{A0}, \qquad D_{A2} = -(M_A + B_{A2} + Z_{A2}), \qquad D_{A4} = -(B_{A4} + Z_{A4})}. \tag{1.130}$$

1.9.5 Projected Maxwell slippage into grouped slots

The projection-first correction is transported into the same grouped variables by the slot shifts

$$Z_n \mapsto Z_n + \varepsilon z_n, \qquad N_n \mapsto N_n + \varepsilon n_n, \qquad n \in \{0, 2, 4\}. \tag{1.131}$$

At a symmetric interior slice $\varepsilon = O(\sigma^2)$, while a one-sided mouth projection gives $\varepsilon = O(\ell)$. On an isotropic lane, the induced response shifts are

$$\delta u_2 = \frac{D_0 z_2 - D_2 z_0}{D_0^2}, \qquad \delta u_4 = \frac{D_0^2 z_4 - D_0(2D_2 z_2 + D_4 z_0) + 2D_2^2 z_0}{D_0^3}, \tag{1.132}$$

and the static outgoing-prefactor shift is

$$\delta P_0 = \frac{D_0 n_0 + N_0 z_0}{D_0^2}. \tag{1.133}$$

For the eliminated isotropic one-pole compatibility surface

$$\mathcal{C} = \frac{N_0}{P_{0,\text{target}}} - \frac{3S^2}{T}, \qquad S = M + B_2 + Z_2, \quad T = B_4 + Z_4, \tag{1.134}$$

the projected variation is

$$\boxed{\delta\mathcal{C} = \frac{n_0}{P_{0,\text{target}}} - \frac{6Sz_2}{T} + \frac{3S^2 z_4}{T^2}}. \tag{1.135}$$

The static conservative shift z_0 cancels from this eliminated surface. It still changes P_0 through (1.133), so the projection-first EM correction remains live in the normalization branch even when it does not move the one-pole compatibility equation.

1.9.6 Isotropic branch and final Part I normalization test

On the isotropic branch,

$$D_{20,n} = D_{21,n} = D_{22,n} = D_n, \qquad N_{20,n} = N_{21,n} = N_{22,n} = N_n. \tag{1.136}$$

Then

$$\boxed{\begin{aligned} u_2 &= -\frac{D_2}{D_0}, \\ u_4 &= \frac{D_2^2 - D_0 D_4}{D_0^2}, \\ P_0 &= \frac{N_0}{D_0}, \\ P_2 &= \frac{D_0 N_2 - 2D_2 N_0}{D_0^2}, \\ P_4 &= \frac{D_0^2 N_4 - 2D_0(D_2 N_2 + D_4 N_0) + 3D_2^2 N_0}{D_0^3}. \end{aligned}} \tag{1.137}$$

The universal quadrupole-normalization condition is

$$\boxed{\widehat{m}_0^{\,2} P_0 = \widehat{m}_0^{\,2} \frac{N_0}{D_0} = \frac{54 G c_s^5}{5 a^5 c^5}}. \tag{1.138}$$

Using (1.130), the same test is

$$\boxed{\widehat{m}_0^{\,2}\frac{N_0}{K-B_0-Z_0}=\frac{54Gc_s^5}{5a^5c^5}}. \tag{1.139}$$

This is the central output of Part I. The algebraic machinery has compressed the outgoing-normalization issue to a single isotropic ratio. The actual branch still has to compute the numerator, denominator, and source-map factor.

1.9.7 Constant-prefactor branch

The constant-prefactor branch used in the later retarded program is defined by

$$P_2=0, \qquad P_4=0. \tag{1.140}$$

Using (1.137), this is equivalent to the exact correlation conditions

$$\boxed{N_2=\frac{2D_2N_0}{D_0}, \qquad N_4=\frac{2D_0(D_2N_2+D_4N_0)-3D_2^2N_0}{D_0^2}}. \tag{1.141}$$

Thus the constant-prefactor branch does not require N_2 and N_4 to vanish individually. It requires them to be locked to the conservative moments D_0, D_2, D_4.

1.9.8 First-order anisotropy transport

Linearize around an isotropic branch:

$$D_{A,n}=D_n+\delta D_{A,n}, \qquad N_{A,n}=N_n+\delta N_{A,n}. \tag{1.142}$$

The response-moment variation is

$$\delta u_2^{(A)}=-\frac{\delta D_{A,2}+u_2\delta D_{A,0}}{D_0}. \tag{1.143}$$

Therefore

$$\boxed{a_{u,2}=-\frac{a_{D,2}+u_2a_{D,0}}{D_0}, \qquad b_{u,2}=-\frac{b_{D,2}+u_2b_{D,0}}{D_0}}. \tag{1.144}$$

Similarly, since $P_0^{(A)}=N_{A0}/D_{A0}$,

$$\delta P_0^{(A)}=\frac{\delta N_{A,0}-P_0\delta D_{A,0}}{D_0}, \tag{1.145}$$

and hence

$$\boxed{a_{P,0}=\frac{a_{N,0}-P_0a_{D,0}}{D_0}, \qquad b_{P,0}=\frac{b_{N,0}-P_0b_{D,0}}{D_0}}. \tag{1.146}$$

These formulas will be used later when weak-axisymmetric anisotropy is transported through the grouped P_2 bundle.

Theorem 1.8 (Full grouped P_2 bundle and isotropic normalization test)**.** *Inside the full reduced wall/BdG/projected-Maxwell grouped-bundle closure, the coupled conservative moments and outgoing-transfer moments are given by* (1.124)–(1.130)*, with projection-first EM slippage transported by* (1.131)–(1.135)*. On the isotropic branch the leading outgoing quadrupole normalization reduces to the single scalar condition* (1.139)*. First-order anisotropy around that branch is transported by* (1.144) *and* (1.146)*.*

Proof. The full grouped bundle is the direct sum of the lane-wise reduced mechanisms from Stage 003 and Stages 004–021. The BdG support moments, conservative Maxwell/mixed moments, and outgoing-transfer moments add to the wall operator as shown in (1.130); projection-first corrections enter the same Z_n, N_n slots by (1.131). The normalized response and prefactor expansions are algebraic inversions of $D_A^{\rm cons}$ and $D_A^{\rm cons2}$, giving (1.137). Equality of the three grouped lanes kills the projector components P_a and P_b, leaving the scalar ratio N_0/D_0. Linearizing the definitions of $u_2^{(A)}$ and $P_0^{(A)}$ gives the anisotropy transport laws. □

1.10 Part I audit summary and handoff to Part II

Part I establishes the following stable outputs.

1. **Exact parent imports.** The gauged GNLS equation, localized Maxwell equation, current conservation, projection language, charge firewall, and mixed-sector gauge invariants are fixed by the declared parent theory.
2. **Moving geometry closure.** The working moving-throat field is $\Sigma = r - R(\Omega, w, t)$, with $a(t)$ and $L(t)$ recovered as collective moments.
3. **Wall modal structure.** The distributed wall action separates the scalar lane, residual geometry lane, grouped real P_2 lane, and higher multipoles.
4. **Breathing reduction.** The old (a, L) closure is the lowest axisymmetric truncation of the distributed wall field.
5. **Matter support Schur complement.** Stable BdG modes generate conservative low-frequency support moments and pole shifts.
6. **Projected Maxwell/mixed outgoing bridge.** Projection-first Maxwell supplies the open-system mixed-channel provenance; the matched one-port normal form supplies a nonnegative outgoing-transfer factor and carries the compact outgoing $l = 2\ i\omega^5$ fingerprint.
7. **Grouped bundle.** The final Part I object is the grouped coefficient packet (D_A, N_A, P_A) and the isotropic normalization gate (1.139).

This chapter does not prove that the actual nonlinear moving-throat PDE realizes the target value in (1.139). It also does not prove uniqueness of the shape-field closure, solve the full BdG spectrum, solve the full Maxwell/mixed spectrum, or complete the passive/outgoing boundary-value problem. Those remain branch-realization tasks.

- ☐ The parent GNLS, Maxwell, continuity, and mixed-sector equations are imported with corrected charge ontology.
- ☐ The sign in $\delta V_{\rm conf} = -V'_{\rm wall}\eta/\ell_c$ is preserved.
- ☐ The monopole normalization $q_{00} = 2\sqrt{\pi}\,\delta a$ is preserved.
- ☐ The old (a, L) closure is recovered as a two-mode axisymmetric truncation.
- ☐ The geometry-only grouped real P_2 sector is degenerate on an isotropic reference throat.
- ☐ Stable BdG modes are integrated out by Schur complement, not by a fitted response coefficient.

- ☐ The projection-first localized Maxwell block retains $A_w, F_{\mu w}, J^w$ until a controlled matched brane reduction is declared.
- ☐ The outgoing $l = 2$ odd coefficient depends on the static prefactor $P_0 = N_0/D_0$.
- ☐ The grouped metric $G_{\mathrm{grp}} = \mathrm{diag}(1, 2, 2)$ and projectors $P_{\bar{x}}, P_a, P_b$ are used consistently.
- ☐ The final Part I theorem gate is the branch-realization test (1.139), not a free normalization choice.

Part II uses the operator-to-response bridge (1.110) and the grouped projector calculus to build the conservative response and overlap-isotropy tests. Part III uses the same full-bundle data to discuss support completion and continuum placement. Part IV uses the outgoing bridge and scalar compatibility mechanism when the retarded closure, outlet models, core/mouth compensation, and boundary-layer laws are introduced. Part V uses the anisotropy transport formulas (1.144) and (1.146) as the starting point for the weak-axisymmetric quotient and orbit-lock structure.

The citation-safe summary is:

> Stages 001–023 define the moving-throat linearized response backbone. They project the parent Maxwell sector before applying the matched one-port reduction, and reduce the parent wall/matter/gauge system to a grouped real P_2 response bundle with exact conservative moments, exact outgoing-transfer moments, exact grouped projectors, and the isotropic normalization test
> $$\widehat{m}_0^2 \frac{N_0}{K - B_0 - Z_0} = \frac{54Gc_s^5}{5a^5c^5}.$$
> They do not by themselves prove that the actual nonlinear branch realizes that test.

Chapter 2

Conservative Response Kernels and Grouped P_2 Normalization Language

Stage range: 024–036

Part II converts the grouped packet from Part I into explicit finite-throat placement data. Part I reduced the isotropic outgoing-normalization gate to the ratio

$$\widehat{m}_0^2 \frac{N_0}{K - B_0 - Z_0} = \frac{54Gc_s^5}{5a^5c^5}. \tag{2.1}$$

This chapter asks what that ratio means once the grouped real P_2 bundle is no longer treated as a formal coefficient packet. The job here is to tie the invariant target to actual angular overlaps, finite-throat axial modes, selected wall eigenmodes, and an explicit source-map normalization.

The endpoint is a reduced admissibility geometry rather than a realized nonlinear solution. On the isotropic branch, angular matching is exact and the source-map angular factor is one. On the first weak-axisymmetric perturbation, the grouped defects lie on the diagnostic line $b = 3a$. On the first selected finite-throat branch, the outgoing-normalization problem becomes a scalar softening-depth problem, and then a dimensionless placement test governed by

$$F(\xi, \delta) = R_{\text{target}}, \qquad M_{\text{mix}} \leq G(\xi, \delta). \tag{2.2}$$

This is still not a proof that the actual nonlinear moving-throat branch realizes the target. It is the exact reduced geometry that the actual branch must land on.

Part II at a glance

Primary question	Which concrete overlaps, eigenmodes, and source-map choices turn the Part I target into a branch-placement test?
Starting inputs	The Part I grouped D, N, P packet, the isotropic outgoing-normalization target, and the first finite-throat selected-branch closure.
Delivers	Explicit angular overlaps, finite-throat axial constants, the selected wall eigenmode, the natural source-map rule, and the dimensionless admissibility pair (F, G).
Used by	Part III for support completion and continuum placement, Part IV for retarded and mouth/core compensation formulas, and Part V for the weak-axisymmetric signature and outgoing-prefactor slope language.

Result anchors **MTDC-T5**.

Stage packets.

Stages	Packet	Status	Carry-forward output
024–027	Angular and finite-throat data	Exact Within Closure / Reduced / Controlled Reduction	Real-STF overlaps, isotropy collapse, weak-axisymmetric signature $b = 3a$, explicit D/N constants, and the first profile-dressed normalization gates.
028–031	Selected spectral placement	Exact Within Closure / Reduced / Controlled Reduction	Loaded wall eigenproblem, Schur-complement loading law, selected prefactor $P_{0,-}$, and stable-side unique-crossing control.
032–034	Source-map and microscopic normalization	Exact Within Closure / Open	Natural D/N source map, coupling-level normalization equation, stability gate, and the scalar softening-depth normal form $x = A - \lambda_-$.
035–036	Dimensionless admissibility frontier	Exact Within Closure / Open	Monotone shape function $F(\xi, \delta)$, support frontier $G(\xi, \delta)$, and the final two-condition selected-branch test.

Claim status. Mixed. The angular algebra, finite-throat mode integrals, Schur complements, eigenvalue formulas, monotonicity statements, and dimensionless branch functions are Exact Within Closure **once the declared reduced closure is adopted. The use of stable normal modes, first finite-throat axial profiles, local bilinear kernels, and the selected D/N source map is** Reduced / Controlled Reduction**. Actual realization of the required physical point** $(R_{\text{target}}, M_{\text{mix}})$ **by the nonlinear moving-throat branch remains** Open**.**

What this part proves. Part II turns the grouped conservative packet into explicit finite-throat overlap data, selected-mode spectral placement, and the two-function admissibility geometry (F, G) for the outgoing quadrupole test.

What this part does not prove. It does not show that the actual branch lands on the required point of that geometry. It only derives the reduced placement problem and the exact branch tests attached to it.

Why later parts need it. Part III uses the (F, G) geometry to build support completion, Part IV reuses the selected prefactor in the retarded and mouth/core chains, and Part V inherits the weak-axisymmetric signature and slope language introduced here.

2.1 Block contract and theorem-level output

Part II has a different verification contract from Part I. Part I showed that the coupled wall, support, and outgoing system compresses into grouped conservative coefficients and an outgoing transfer ratio.

Part II must show that those coefficients are not arbitrary labels. It must tie them to actual finite-throat overlaps, selected axial profiles, and stable-branch spectral placement.

The stage chain is

$$\begin{aligned} \text{grouped bundle} &\longrightarrow \text{angular isotropy} \longrightarrow \text{finite-throat axial overlaps} \\ &\longrightarrow \text{selected wall eigenmode} \longrightarrow \text{dimensionless admissibility frontier.} \end{aligned} \tag{2.3}$$

The point of the chain is to convert the branch test (2.1) into a concrete placement question.

Definition 2.1 (Universal quadrupole target used in Part II). *Throughout this chapter, write*

$$\boxed{N_Q^{\rm target} := \frac{54 G c_s^5}{5 a^5 c^5}}. \tag{2.4}$$

Thus the isotropic Part I condition is $\widehat{m}_0^2 P_0 = N_Q^{\rm target}$, and the selected-mode condition later becomes $\widehat{m}_-^2 P_{0,-} = N_Q^{\rm target}$.

Theorem 2.2 (Part II conservative-response placement theorem). *Inside the declared finite-throat selected-branch closure, the following chain of exact reduced statements holds.*

1. *On an $O(3)$-invariant reference branch, the real-STF angular source map is the identity and the grouped lanes satisfy*

$$D_{20,n} = D_{21,n} = D_{22,n}, \qquad N_{20,n} = N_{21,n} = N_{22,n}. \tag{2.5}$$

2. *The first weak axisymmetric quadrupole deformation has grouped signature*

$$\lambda_{20} = 1, \qquad \lambda_{21} = \frac{1}{2}, \qquad \lambda_{22} = -1, \qquad b = 3a. \tag{2.6}$$

3. *The first explicit finite-throat selected branch is governed by the exact D/N constants*

$$\kappa_0^2 = \frac{8}{\pi^2}, \qquad \kappa_1^2 = \frac{16}{9\pi^2}, \qquad \eta = \frac{\kappa_1^2}{\kappa_0^2} = \frac{2}{9}. \tag{2.7}$$

4. *On the natural D/N source branch, the abstract selected source map is fixed by*

$$\widehat{m}_-^2 = \frac{s_-}{\kappa_0^2}. \tag{2.8}$$

5. *The invariant selected normalization product reduces to*

$$N_-(x) = N_Q^{\rm target}, \tag{2.9}$$

where $x = A - \lambda_-$ is the selected-mode softening depth.

6. *In dimensionless variables $\xi = x/A$ and $\delta = \Delta K_{\rm ax}/A$, the stable branch is governed by the unique normalization locus*

$$F(\xi, \delta) = R_{\rm target} \tag{2.10}$$

and the support-feasibility inequality

$$M_{\rm mix} \le G(\xi, \delta). \tag{2.11}$$

Consequently, by the end of Stage 036, the reduced theorem gap is no longer an unspecified outgoing coefficient. It is the concrete question of whether the actual moving-throat PDE places the physical defect in the admissible region of the universal (F, G) branch geometry.

Proof. Stages 024–025 establish the angular source map, isotropy collapse, and minimal one-lane normalization ratio. Stages 026–027 evaluate the first finite-throat axial overlaps and profile family. Stages 028–029 derive the selected wall profile and dynamic loading from a rank-one Schur complement. Stages 030–033 translate the selected operator into normalized-response language, prove stable-side monotonicity, fix the natural source map, and write the microscopic coupling-level equation. Stages 034–036 remove the eigenvector variables in favor of the softening depth and then non-dimensionalize the branch into the functions F and G. Each implication is exact inside the declared reduced closure. The actual placement of the physical branch remains an open realization question. □

2.2 Explicit overlap-integral normalization

2.2.1 Local goal

Stage 024 starts from the Part I grouped-bundle ratio (2.1). In Part I, the quantities $B_{A,n}$, $Z_{A,n}$, and $N_{A,n}$ were exact reduced coefficients, but the angular geometry behind them had not yet been audited. Stage 024 supplies that audit. It proves that the canonical angular source map is already correct on the isotropic branch and that the first weak-axisymmetric splitting pattern is diagnostic rather than arbitrary.

The grouped label A in this section runs over the real quadrupole ports. When the five real STF modes are paired into the project ledger's grouped convention, the three grouped lanes are $20, 21, 22$, where 21 and 22 represent cosine/sine pairs.

2.2.2 Real-STF angular basis and source-map identity

Let n^i be the unit direction on the throat sphere and let E_A^{ij} be an orthonormal real STF basis,

$$\mathrm{Tr}(E_A E_B) = \delta_{AB}. \tag{2.12}$$

Define

$$Y_A(n) = \sqrt{\frac{15}{8\pi}}\, E_A^{ij} n_i n_j. \tag{2.13}$$

Using

$$\int_{S^2} n_i n_j n_k n_l \, d\Omega = \frac{4\pi}{15} \left(\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}\right), \tag{2.14}$$

one obtains

$$\boxed{\int_{S^2} Y_A Y_B \, d\Omega = \delta_{AB}}. \tag{2.15}$$

If the orbital/worldtube STF quadrupole source is expanded as

$$S(n) = \sum_A S_A Y_A(n), \tag{2.16}$$

then projection onto the same port gives

$$\boxed{S_A^{\mathrm{port}} = \int_{S^2} Y_A(n) S(n) \, d\Omega = S_A}. \tag{2.17}$$

Thus the angular source-map factor is exactly one on the canonical isotropic branch:

$$\widehat{m}_0 = \widehat{m}_{\rm rad}\widehat{m}_{\rm ang}, \qquad \boxed{\widehat{m}_{\rm ang} = 1}. \tag{2.18}$$

The remaining source-map issue is radial/axial, not angular.

2.2.3 Isotropic overlap factorization

On an $O(3)$-invariant branch, take the grouped wall deformation, BdG support mode, brane-like gauge coordinate, and mixed coordinate in the separated form

$$\eta_A(s,\Omega,t) = q_A(t)\chi_\eta(s)Y_A(\Omega), \tag{2.19}$$

$$X_{\alpha,A}(s,\Omega,t) = X_{\alpha,A}(t)\phi_\alpha(s)Y_A(\Omega), \tag{2.20}$$

$$U_{r,A}(s,\Omega,t) = U_{r,A}(t)u_r(s)Y_A(\Omega), \qquad W_{r,A}(s,\Omega,t) = W_{r,A}(t)w_r(s)Y_A(\Omega). \tag{2.21}$$

The angular integral collapses by (2.15). The reduced coupling constants are therefore lane-independent:

$$\boxed{\begin{aligned} C_\alpha &= \lambda_{B,\alpha}I_{\eta\alpha}, & I_{\eta\alpha} &= \int ds\,\mu_s(s)\chi_\eta(s)\phi_\alpha(s), \\ G_{U,r} &= \lambda_{U,r}I_{\eta u,r}, & I_{\eta u,r} &= \int ds\,\mu_s(s)\chi_\eta(s)u_r(s), \\ G_{W,r} &= \lambda_{W,r}I_{\eta w,r}, & I_{\eta w,r} &= \int ds\,\mu_s(s)\chi_\eta(s)w_r(s), \\ R_r &= \lambda_{R,r}I_{uw,r}, & I_{uw,r} &= \int ds\,\mu_s(s)u_r(s)w_r(s). \end{aligned}} \tag{2.22}$$

The BdG support moments reduce to

$$\boxed{B_0 = \sum_\alpha \frac{C_\alpha^2}{\varpi_\alpha^2}, \qquad B_2 = \sum_\alpha \frac{C_\alpha^2}{\varpi_\alpha^4}, \qquad B_4 = \sum_\alpha \frac{C_\alpha^2}{\varpi_\alpha^6}}. \tag{2.23}$$

For each Maxwell/mixed port define

$$\Delta_r = \Omega_{U,r}^2\Omega_{W,r}^2 - R_r^2, \qquad S_r = \Omega_{U,r}^2 + \Omega_{W,r}^2, \tag{2.24}$$

$$Q_r = G_{U,r}^2\Omega_{W,r}^2 + 2G_{U,r}G_{W,r}R_r + G_{W,r}^2\Omega_{U,r}^2, \qquad H_r = G_{U,r}^2 + G_{W,r}^2, \tag{2.25}$$

$$\mathcal{P}_r = \Omega_{U,r}^2G_{W,r} + R_rG_{U,r}. \tag{2.26}$$

Then

$$\boxed{\begin{aligned} Z_0 &= \sum_r \frac{Q_r}{\Delta_r}, \\ Z_2 &= \sum_r \frac{Q_rS_r - H_r\Delta_r}{\Delta_r^2}, \\ Z_4 &= \sum_r \frac{Q_r(S_r^2-\Delta_r) - S_rH_r\Delta_r}{\Delta_r^3}, \\ N_0 &= \sum_r \frac{\mathcal{P}_r^2}{\Delta_r^2}, \\ N_2 &= \sum_r \frac{2\mathcal{P}_r(\mathcal{P}_rS_r - \Delta_rG_{W,r})}{\Delta_r^3}, \\ N_4 &= \sum_r \frac{\Delta_r^2G_{W,r}^2 - 2\Delta_r\mathcal{P}_r^2 - 4\Delta_r\mathcal{P}_rS_rG_{W,r} + 3\mathcal{P}_r^2S_r^2}{\Delta_r^4}. \end{aligned}} \tag{2.27}$$

The conservative operator is

$$\boxed{\begin{aligned} &D(\omega) = D_0 + D_2\omega^2 + D_4\omega^4 + O(\omega^6), \\ &\quad D_0 = K - B_0 - Z_0, \\ &\quad D_2 = -(M + B_2 + Z_2), \\ &\quad D_4 = -(B_4 + Z_4). \end{aligned}} \tag{2.28}$$

Thus every truly $O(3)$-invariant reduced kernel forces

$$\boxed{\begin{aligned} D_{20,n} = D_{21,n} = D_{22,n} = D_n, \\ N_{20,n} = N_{21,n} = N_{22,n} = N_n. \end{aligned}} \tag{2.29}$$

All grouped anisotropy defects vanish on that branch.

2.2.4 First weak-axisymmetric splitting law

The leading symmetry-breaking perturbation that talks directly to the grouped real P_2 bundle is an axisymmetric quadrupole background. Write it schematically as

$$\delta K = \varepsilon\,\kappa(s) Y_{20}(\Omega). \tag{2.30}$$

The angular matrix is

$$M^{(20)}_{AB} = \int_{S^2} Y_A Y_{20} Y_B \, d\Omega. \tag{2.31}$$

The exact sixth-moment identity gives, in the five real STF lanes,

$$\boxed{M^{(20)} = \frac{\sqrt{5}}{7\sqrt{\pi}}\,\mathrm{diag}\big(1, \tfrac{1}{2}, \tfrac{1}{2}, -1, -1\big)}. \tag{2.32}$$

After regrouping cosine/sine pairs, the signature is

$$\boxed{\lambda_{20} = 1, \qquad \lambda_{21} = \frac{1}{2}, \qquad \lambda_{22} = -1}. \tag{2.33}$$

Therefore any first-order grouped coefficient has the form

$$x_{20} = x^{(0)} + \varepsilon x^{(1)}, \qquad x_{21} = x^{(0)} + \frac{\varepsilon}{2} x^{(1)}, \qquad x_{22} = x^{(0)} - \varepsilon x^{(1)}. \tag{2.34}$$

With the grouped trace/anomaly coordinates of Part I, this implies

$$\boxed{\bar{x} = x^{(0)}, \qquad a_x = \frac{\varepsilon}{4} x^{(1)}, \qquad b_x = \frac{3\varepsilon}{4} x^{(1)}, \qquad b_x = 3a_x}. \tag{2.35}$$

For the outgoing prefactor, if

$$D_A = D_0 + \varepsilon\lambda_A D_1 + O(\varepsilon^2), \qquad N_A = N_0 + \varepsilon\lambda_A N_1 + O(\varepsilon^2), \tag{2.36}$$

then

$$P_A = \frac{N_A}{D_A} = P_0 + \varepsilon\lambda_A P_1 + O(\varepsilon^2), \qquad \boxed{P_1 = \frac{N_1 D_0 - N_0 D_1}{D_0^2}}. \tag{2.37}$$

The same diagnostic line follows:

$$a_P = \frac{\varepsilon}{4} P_1, \qquad b_P = \frac{3\varepsilon}{4} P_1. \tag{2.38}$$

Theorem 2.3 (Angular source identity and weak-axisymmetric signature)**.** *On the canonical real-STF angular basis, the isotropic source-map angular factor is exactly one. Any $O(3)$-invariant reduced kernel collapses the grouped 20/21/22 bundle to one common lane. The first pure axisymmetric quadrupole splitting of that bundle has signature $(1, 1/2, -1)$ and therefore satisfies $b = 3a$.*

Proof. The source-map statement follows from orthonormality of the real-STF harmonics. The isotropy collapse follows because all angular overlaps are diagonal and lane-independent under an $O(3)$-invariant kernel. The weak-axisymmetric signature follows from the cubic angular integral $\int Y_A Y_{20} Y_B \, d\Omega$, whose diagonal entries are proportional to $(1, 1/2, 1/2, -1, -1)$. Regrouping the cosine/sine pairs gives (2.33), and the weighted grouped inverse map gives (2.35). □

Downstream use. The identity $\widehat{m}_{\mathrm{ang}} = 1$ is used in every later selected-branch normalization equation. The line $b = 3a$ becomes the diagnostic weak-axisymmetric signature used later in the quotient/orbit and same-charge prefactor analyses.

2.3 Minimal isotropic single-mode closure

2.3.1 One-mode radial/axial model

Stage 025 strips the isotropic branch to the smallest radial/axial module that can carry the normalization product: one stable BdG support mode, one brane-like internal gauge coordinate, and one mixed $A_w/F_{\mu w}/J^w$ coordinate. Define

$$C = \lambda_B I_{\eta\phi}, \qquad G_U = \lambda_U I_{\eta u}, \qquad G_W = \lambda_W I_{\eta w}, \qquad R = \lambda_R I_{uw}. \tag{2.39}$$

Let the corresponding internal frequencies be ϖ, Ω_U, and Ω_W. The minimal Maxwell/mixed invariants are

$$\boxed{\begin{aligned} &\Delta = \Omega_U^2 \Omega_W^2 - R^2, \\ &Q = G_U^2 \Omega_W^2 + 2 G_U G_W R + G_W^2 \Omega_U^2, \\ &\mathcal{P} = \Omega_U^2 G_W + R G_U. \end{aligned}} \tag{2.40}$$

The static BdG softening is

$$X = \frac{C^2}{\varpi^2}. \tag{2.41}$$

Thus

$$D_0 = K - X - \frac{Q}{\Delta}. \tag{2.42}$$

The outgoing-transfer numerator is

$$N_0 = \frac{\mathcal{P}^2}{\Delta^2}. \tag{2.43}$$

2.3.2 Minimal explicit normalization law

The static prefactor is

$$P_0 = \frac{N_0}{D_0} = \frac{\mathcal{P}^2}{\Delta^2 \left(K - C^2/\varpi^2 - Q/\Delta\right)} = \frac{\mathcal{P}^2}{\Delta \left(K\Delta - \Delta C^2/\varpi^2 - Q\right)}. \tag{2.44}$$

Since Stage 024 fixed $\widehat{m}_{\text{ang}} = 1$, the branch test is purely radial/axial:

$$\boxed{\widehat{m}_{\text{rad}}^2 \frac{\mathcal{P}^2}{\Delta\left(K\Delta - \Delta C^2/\varpi^2 - Q\right)} = N_Q^{\text{target}}}. \tag{2.45}$$

The stability conditions are

$$\boxed{\Delta > 0, \qquad D_0 = K - \frac{C^2}{\varpi^2} - \frac{Q}{\Delta} > 0}. \tag{2.46}$$

The first inequality keeps the internal (U, W) block positive; the second keeps the dressed wall lane stable.

The monotonicity in the bare stiffness and support softening is immediate:

$$\frac{\partial P_0}{\partial K} = -\frac{N_0}{D_0^2} < 0, \qquad \frac{\partial P_0}{\partial X} = +\frac{N_0}{D_0^2} > 0. \tag{2.47}$$

Thus increasing bare wall stiffness suppresses the outgoing prefactor, while increasing conservative support softening enhances it, as long as stability is maintained.

Proposition 2.4 (Minimal isotropic normalization formula)**.** *In the one-BdG-mode, one-Maxwell/mixed-pair isotropic closure, the universal outgoing quadrupole condition is exactly* (2.45)*. The only branch data still needed are C, G_U, G_W, R, ϖ, Ω_U, Ω_W, K, and $\widehat{m}_{\text{rad}}$.*

Downstream use. Stage 026 evaluates the overlap factors in (2.39) on a concrete finite-throat D/N branch. Stage 029 later shows that the same mixed-sector numerator $\mathcal{P}$ controls the selected-mode dynamic loading and odd outgoing coefficient.

2.4 Concrete finite-throat axial modes

2.4.1 Finite interval and axial basis

Stage 026 chooses a concrete finite-throat interval

$$s \in [0, L] \tag{2.48}$$

with flat axial measure. The wall and brane-like zero-mode profile is the normalized Neumann/Neumann constant mode

$$u_0(s) = \frac{1}{\sqrt{L}}, \qquad -u_0'' = 0, \qquad u_0'(0) = u_0'(L) = 0. \tag{2.49}$$

The trapped support and mixed channel use the exact D/N ladder

$$f_n(s) = \sqrt{\frac{2}{L}} \sin\left(\frac{(n + 1/2)\pi s}{L}\right), \qquad f_n(0) = 0, \qquad f_n'(L) = 0, \tag{2.50}$$

with

$$k_n = \frac{(n + 1/2)\pi}{L}. \tag{2.51}$$

On the minimal branch, keep only $n = 0$:

$$f_0(s) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi s}{2L}\right). \tag{2.52}$$

The exact overlap of the N/N zero mode with the D/N ladder is

$$\kappa_n = \int_0^L u_0(s) f_n(s)\, ds = \frac{\sqrt{2}}{(n+1/2)\pi}. \tag{2.53}$$

Therefore

$$\boxed{\kappa := \kappa_0 = \frac{2\sqrt{2}}{\pi}}. \tag{2.54}$$

The lowest trapped half-wave is automatically the dominant axial branch because $|\kappa_n| \sim (n+1/2)^{-1}$.

2.4.2 Branch-level coefficients

Take

$$\chi_\eta = u_0, \qquad \phi = f_0, \qquad u = u_0, \qquad w = f_0. \tag{2.55}$$

Then

$$I_{\eta\phi} = \kappa, \qquad I_{\eta u} = 1, \qquad I_{\eta w} = \kappa, \qquad I_{uw} = \kappa. \tag{2.56}$$

The reduced couplings become

$$\boxed{C = \kappa\lambda_B, \qquad G_U = \lambda_U, \qquad G_W = \kappa\lambda_W, \qquad R = \kappa\lambda_R}. \tag{2.57}$$

Consequently,

$$\boxed{\begin{aligned} &\Delta = \Omega_U^2\Omega_W^2 - \kappa^2\lambda_R^2, \\ &Q = \lambda_U^2\Omega_W^2 + 2\kappa^2\lambda_U\lambda_W\lambda_R + \kappa^2\lambda_W^2\Omega_U^2, \\ &\mathcal{P} = \kappa(\Omega_U^2\lambda_W + \lambda_R\lambda_U). \end{aligned}} \tag{2.58}$$

The BdG softening is

$$X = \frac{\kappa^2\lambda_B^2}{\varpi^2}. \tag{2.59}$$

2.4.3 First finite-throat normalization test

Substitution into (2.45) gives

$$\boxed{\widehat{m}_{\rm rad}^{\,2} \frac{\kappa^2(\Omega_U^2\lambda_W + \lambda_R\lambda_U)^2}{\Delta\left(K\Delta - \Delta\kappa^2\lambda_B^2/\varpi^2 - Q\right)} = N_Q^{\rm target}}. \tag{2.60}$$

Equivalently, the required wall stiffness is

$$\boxed{K_{\rm req} = \frac{\kappa^2\lambda_B^2}{\varpi^2} + \frac{Q}{\Delta} + \frac{\widehat{m}_{\rm rad}^{\,2}\kappa^2(\Omega_U^2\lambda_W + \lambda_R\lambda_U)^2}{N_Q^{\rm target}\,\Delta^2}}. \tag{2.61}$$

For the constant wall profile, the quadrupole wall stiffness is

$$K = K_\eta + 6T_\Omega. \tag{2.62}$$

Thus the first concrete finite-throat branch asks whether

$$K_\eta + 6T_\Omega = K_{\rm req}. \tag{2.63}$$

Proposition 2.5 (First finite-throat normalization equation)**.** *On the constant N/N wall/brane branch coupled to the lowest D/N support/mixed half-wave, all axial overlap data collapse to* $\kappa = 2\sqrt{2}/\pi$. *The outgoing-normalization test is exactly* (2.60)*, equivalently the wall-stiffness condition* (2.61)*.*

Downstream use. Stage 027 tests whether this equation was an artifact of the constant wall profile. Stage 032 later reuses the same D/N mode integrals to eliminate the abstract selected source-map factor.

2.5 Nonconstant axial profile families

2.5.1 First N/N deformation family

Stage 027 turns on the first nonconstant N/N axial mode

$$u_1(s) = \sqrt{\frac{2}{L}}\cos\left(\frac{\pi s}{L}\right), \qquad -u_1'' = \frac{\pi^2}{L^2}u_1. \tag{2.64}$$

The coherent one-parameter profile family is

$$\chi_\theta(s) = \cos\theta\, u_0(s) + \sin\theta\, u_1(s), \tag{2.65}$$

and it is used for both the wall quadrupole axial profile and the brane-like internal coordinate:

$$\chi_\eta = \chi_\theta, \qquad u = \chi_\theta, \qquad \phi = w = f_0. \tag{2.66}$$

The direct wall/brane overlap remains normalized:

$$\int_0^L \chi_\theta^2\, ds = 1. \tag{2.67}$$

The D/N overlap is

$$\boxed{\kappa(\theta) = \int_0^L \chi_\theta(s) f_0(s)\, ds = \kappa_0\cos\theta + \kappa_1\sin\theta}, \tag{2.68}$$

where

$$\kappa_0 = \frac{2\sqrt{2}}{\pi}, \qquad \kappa_1 = -\frac{4}{3\pi}. \tag{2.69}$$

Equivalently,

$$\kappa(\theta) = \frac{2(-2\sin\theta + 3\sqrt{2}\cos\theta)}{3\pi}. \tag{2.70}$$

The maximum possible magnitude in this first family is

$$\boxed{|\kappa|_{\max} = \sqrt{\kappa_0^2 + \kappa_1^2} = \frac{2\sqrt{22}}{3\pi}}. \tag{2.71}$$

This exceeds the constant-branch value $2\sqrt{2}/\pi$.

2.5.2 Blind-angle no-go and max-coupling angle

The blind angle is defined by $\kappa(\theta) = 0$. From (2.68),

$$\boxed{\tan\theta_{\text{blind}} = \frac{3\sqrt{2}}{2}}. \tag{2.72}$$

At that angle, $C = G_W = R = 0$, hence $\mathcal{P} = 0$ and $N_0 = 0$. Since the right-hand side N_Q^{target} is positive, the blind-angle branch cannot realize the outgoing quadrupole normalization target.

The positive-overlap maximum occurs at

$$\boxed{\tan\theta_{\max} = \frac{\kappa_1}{\kappa_0} = -\frac{\sqrt{2}}{3}, \qquad \sin^2\theta_{\max} = \frac{2}{11}}. \tag{2.73}$$

This is a small negative admixture of the first N/N excitation.

2.5.3 Profile-dressed wall stiffness and branch condition

The quadrupole wall operator is

$$G_\eta = -T_w \frac{d^2}{ds^2} + K_\eta + 6T_\Omega. \tag{2.74}$$

On the coherent profile family,

$$\boxed{K_{\rm geo}(\theta) = K_\eta + 6T_\Omega + \frac{\pi^2 T_w}{L^2}\sin^2\theta}. \tag{2.75}$$

The full profile-dressed normalization condition is the Stage 026 test with

$$\kappa \longrightarrow \kappa(\theta), \qquad K \longrightarrow K_{\rm geo}(\theta). \tag{2.76}$$

Equivalently,

$$\boxed{K_{\rm geo}(\theta) = K_{\rm req}(\theta)}, \tag{2.77}$$

where

$$K_{\rm req}(\theta) = \frac{\lambda_B^2\kappa(\theta)^2}{\varpi^2} + \frac{Q(\theta)}{\Delta(\theta)} + \frac{\widehat{m}_{\rm rad}^2\kappa(\theta)^2(\Omega_U^2\lambda_W + \lambda_R\lambda_U)^2}{N_Q^{\rm target}\,\Delta(\theta)^2}, \tag{2.78}$$

with

$$\Delta(\theta) = \Omega_U^2\Omega_W^2 - \lambda_R^2\kappa(\theta)^2, \tag{2.79}$$

$$Q(\theta) = \lambda_U^2\Omega_W^2 + 2\lambda_U\lambda_W\lambda_R\kappa(\theta)^2 + \lambda_W^2\Omega_U^2\kappa(\theta)^2. \tag{2.80}$$

Theorem 2.6 (First profile-selection gate)**.** *The Stage 026 constant profile is the $\theta = 0$ member of an exact finite-throat profile family. The family has an exact blind-angle no-go at* (2.72)*, an exact max-coupling point at* (2.73)*, and the profile-selected branch must satisfy the dressed stiffness equation* (2.77)*.*

Downstream use. Stage 028 stops treating θ as a chosen parameter and derives it as an eigenvector of the loaded wall operator.

2.6 Loaded-profile selection and dynamic loading

2.6.1 Rank-one loaded wall eigenproblem

Work in the first two N/N wall modes, with profile vector

$$q = (q_0, q_1)^T, \qquad q_0^2 + q_1^2 = 1, \qquad q_0 = \cos\theta, \qquad q_1 = \sin\theta. \tag{2.81}$$

The bare quadrupole wall matrix is

$$K_{\rm bare} = \begin{pmatrix} K_0 & 0 \\ 0 & K_1 \end{pmatrix}, \qquad K_0 = K_\eta + 6T_\Omega, \qquad K_1 = K_0 + \Delta K_{\rm ax}, \tag{2.82}$$

where

$$\Delta K_{\rm ax} = \frac{\pi^2 T_w}{L^2}. \tag{2.83}$$

The D/N overlap vector is

$$\boxed{v = (\kappa_0, \kappa_1)^T, \qquad \kappa_0 = \frac{2\sqrt{2}}{\pi}, \qquad \kappa_1 = -\frac{4}{3\pi}}. \tag{2.84}$$

The minimal attractive loading from the support/mixed branch gives

$$\boxed{K_{\text{eff}} = K_{\text{bare}} - \alpha v v^T, \qquad \alpha > 0}. \tag{2.85}$$

The exact eigenvalues are

$$\lambda_\pm = \frac{1}{2}\left[K_0 + K_1 - \alpha(\kappa_0^2 + \kappa_1^2) \pm \sqrt{\left(\Delta K_{\text{ax}} + \alpha(\kappa_0^2 - \kappa_1^2)\right)^2 + 4\alpha^2\kappa_0^2\kappa_1^2}\right]. \tag{2.86}$$

The selected lower profile angle obeys

$$\boxed{\tan(2\theta_-) = \frac{2\alpha\kappa_0\kappa_1}{\Delta K_{\text{ax}} + \alpha(\kappa_0^2 - \kappa_1^2)}}. \tag{2.87}$$

Because $\kappa_0\kappa_1 < 0$ and $\kappa_0^2 - \kappa_1^2 > 0$, positive loading drives $\theta_- < 0$. The selected profile moves toward the max-coupling branch and away from the positive blind-angle branch.

The softening threshold is

$$\boxed{\alpha_{\text{crit}} = \frac{K_0 K_1}{K_1\kappa_0^2 + K_0\kappa_1^2}}. \tag{2.88}$$

2.6.2 Dynamic origin of the loading parameter

Stage 029 replaces the phenomenological α with the Schur complement of the coupled response block. Let

$$D_\eta^{\text{bare}}(\omega) = \operatorname{diag}(K_0 - M_0\omega^2, K_1 - M_1\omega^2), \tag{2.89}$$

$$A_\phi(\omega) = \varpi^2 - \omega^2, \qquad A_U(\omega) = \Omega_U^2 - \omega^2, \qquad A_W(\omega) = \Omega_W^2 - \omega^2 - \Pi_{\text{out}}(\omega). \tag{2.90}$$

With one D/N BdG support mode, one N/N brane-like internal doublet, and one D/N mixed coordinate, the exact wall self-energy is

$$\boxed{\Sigma_{\text{wall}}(\omega) = \Xi(\omega) I_2 + \alpha(\omega) v v^T}, \tag{2.91}$$

where

$$\boxed{\Xi(\omega) = \frac{\lambda_U^2}{A_U(\omega)}, \qquad \alpha(\omega) = \frac{\lambda_B^2}{A_\phi(\omega)} + \frac{\left(A_U(\omega)\lambda_W + \lambda_R\lambda_U\right)^2}{A_U(\omega)\Delta_{UW}(\omega)}}, \tag{2.92}$$

with

$$\Delta_{UW}(\omega) = A_U(\omega)A_W(\omega) - \lambda_R^2\sigma, \qquad \sigma = v \cdot v = \kappa_0^2 + \kappa_1^2 = \frac{88}{9\pi^2}. \tag{2.93}$$

On the conservative static branch,

$$\boxed{\Xi_0 = \frac{\lambda_U^2}{\Omega_U^2}, \qquad \Delta_0 = \Omega_U^2\Omega_W^2 - \lambda_R^2\sigma, \qquad \alpha_0 = \frac{\lambda_B^2}{\varpi^2} + \frac{(\Omega_U^2\lambda_W + \lambda_R\lambda_U)^2}{\Omega_U^2\Delta_0}}. \tag{2.94}$$

The static wall matrix is

$$K_{\text{eff}}^{(0)} = \begin{pmatrix} K_0 - \Xi_0 & 0 \\ 0 & K_1 - \Xi_0 \end{pmatrix} - \alpha_0 v v^T. \tag{2.95}$$

Thus the Stage 028 angle law survives with $K_i \mapsto K_i - \Xi_0$ and $\alpha \mapsto \alpha_0$.

2.6.3 Outgoing dressing and selected odd coefficient

To first order in the passive/outgoing port,

$$\alpha(\omega) = \alpha_{\text{cons}}(\omega) + \beta(\omega)\Pi_{\text{out}}(\omega) + O(\Pi_{\text{out}}^2), \tag{2.96}$$

with

$$\boxed{\beta(\omega) = \frac{\left(A_U(\omega)\lambda_W + \lambda_R\lambda_U\right)^2}{\Delta_{\text{cons}}(\omega)^2}, \qquad \Delta_{\text{cons}}(\omega) = A_U(\omega)(\Omega_W^2 - \omega^2) - \lambda_R^2\sigma}. \tag{2.97}$$

At zero frequency,

$$\boxed{\beta_0 = \frac{(\Omega_U^2\lambda_W + \lambda_R\lambda_U)^2}{\Delta_0^2} \geq 0}. \tag{2.98}$$

For the compact outgoing $l = 2$ branch,

$$\Pi_{\text{out}}(\omega) = +i\frac{a^5}{27c_s^5}\omega^5 + O(\omega^7). \tag{2.99}$$

Projecting the odd rank-one operator onto the conservative selected eigenvector e_- gives

$$\boxed{\delta D_-^{\text{odd}}(\omega) = -i\frac{a^5}{27c_s^5}\frac{(\Omega_U^2\lambda_W + \lambda_R\lambda_U)^2}{\Delta_0^2}\,(v\cdot e_-)^2\,\omega^5 + O(\omega^7)}. \tag{2.100}$$

This is the first selected-mode version of the outgoing quadrupole bridge.

Theorem 2.7 (Loaded profile and dynamic Schur-complement theorem)**.** *The first reduced wall/BdG/Maxwell/mixed operator produces the rank-one loaded profile-selection problem exactly. Its self-energy splits as $\Sigma_{\text{wall}} = \Xi I_2 + \alpha vv^T$. The same directional loading coefficient inherits the passive/outgoing $i\omega^5$ branch, and the selected odd coefficient is* (2.100)*.*

Downstream use. Stages 030–033 translate (2.100) into normalized-response language and then into a microscopic coupling-level target.

2.7 Selected-mode normalized response

2.7.1 Selected eigenvalue and overlap

Use the compact notation

$$A = K_0 - \Xi_0, \qquad B = K_1 - \Xi_0 = A + \Delta K_{\text{ax}}. \tag{2.101}$$

Define

$$\delta_\kappa = \kappa_0^2 - \kappa_1^2, \qquad \Pi_\kappa = \kappa_0^2\kappa_1^2, \qquad \sigma = \kappa_0^2 + \kappa_1^2. \tag{2.102}$$

The selected lower eigenvalue is

$$\boxed{\lambda_- = \frac{A + B - \alpha_0\sigma - \sqrt{(\Delta K_{\text{ax}} + \alpha_0\delta_\kappa)^2 + 4\alpha_0^2\Pi_\kappa}}{2}}. \tag{2.103}$$

Let

$$R_\lambda = \sqrt{(\Delta K_{\text{ax}} + \alpha_0\delta_\kappa)^2 + 4\alpha_0^2\Pi_\kappa}. \tag{2.104}$$

The selected overlap is

$$\boxed{s_- = (v \cdot e_-)^2 = \frac{1}{2}\left[\sigma + \frac{(\Delta K_{\mathrm{ax}} + \alpha_0\delta_\kappa)\delta_\kappa + 4\alpha_0\Pi_\kappa}{R_\lambda}\right]}. \tag{2.105}$$

Equivalently, by Hellmann–Feynman,

$$\boxed{s_- = -\frac{d\lambda_-}{d\alpha_0}}. \tag{2.106}$$

2.7.2 Normalized selected response and selected prefactor

Write the selected operator as

$$D_-(\omega) = D_{-0} + D_{-2}\omega^2 + D_{-4}\omega^4 - iC_{5,-}\omega^5 + O(\omega^6), \qquad D_{-0} = \lambda_-. \tag{2.107}$$

The normalized selected response is

$$Y_-(\omega) = \frac{D_{-0}}{D_-(\omega)} = 1 + u_{2,-}\omega^2 + u_{4,-}\omega^4 + i\Gamma_{5,-}\omega^5 + O(\omega^6), \tag{2.108}$$

where

$$u_{2,-} = -\frac{D_{-2}}{D_{-0}}, \qquad u_{4,-} = \frac{D_{-2}^2 - D_{-0}D_{-4}}{D_{-0}^2}, \qquad \Gamma_{5,-} = \frac{C_{5,-}}{D_{-0}}. \tag{2.109}$$

Using the Stage 029 odd coefficient,

$$\Gamma_{5,-} = \frac{a^5}{27c_s^5}\frac{\beta_0 s_-}{\lambda_-}. \tag{2.110}$$

Therefore

$$\boxed{\Gamma_{5,-} = \frac{a^5}{27c_s^5}P_{0,-}, \qquad P_{0,-} = \frac{\beta_0 s_-}{\lambda_-} = -\beta_0\frac{d\ln\lambda_-}{d\alpha_0}}. \tag{2.111}$$

The selected-branch quadrupole target is

$$\boxed{\widehat{m}_-^2 P_{0,-} = N_Q^{\mathrm{target}}}. \tag{2.112}$$

2.7.3 Stable-side monotonicity and unique crossing

Stage 031 proves that the selected prefactor is strictly monotone on the stable branch. First,

$$\boxed{\frac{ds_-}{d\alpha_0} = \frac{2(\Delta K_{\mathrm{ax}})^2\Pi_\kappa}{R_\lambda^3} > 0}. \tag{2.113}$$

Using (2.106),

$$\boxed{\frac{dP_{0,-}}{d\alpha_0} = \beta_0\frac{(ds_-/d\alpha_0)\lambda_- + s_-^2}{\lambda_-^2} > 0} \tag{2.114}$$

whenever $\lambda_- > 0$ and $\beta_0 > 0$.

At zero loading,

$$P_{0,-}(0) = \frac{\beta_0\kappa_0^2}{K_0 - \Xi_0}. \tag{2.115}$$

The refined softening threshold is

$$\boxed{\alpha_{\rm crit} = \frac{AB}{B\kappa_0^2 + A\kappa_1^2}}. \tag{2.116}$$

As $\alpha_0 \to \alpha_{\rm crit}^-$, $\lambda_- \to 0$, while s_- remains finite and positive, so $P_{0,-} \to +\infty$. Therefore the stable branch spans

$$P_{0,-}(0) \le P_{0,-} < +\infty. \tag{2.117}$$

Theorem 2.8 (Unique stable-side selected crossing)**.** *For every target $P_{\rm target} > P_{0,-}(0)$, there exists a unique $\alpha_* \in (0, \alpha_{\rm crit})$ such that $P_{0,-}(\alpha_*) = P_{\rm target}$. Applied to the quadrupole target, $P_{\rm target} = N_Q^{\rm target}/\widehat{m}_-^2$.*

Downstream use. The monotonicity theorem justifies treating the selected branch as a one-dimensional spectral-placement problem rather than as a many-parameter matching ambiguity.

2.8 Source-map and microscopic normalization equation

2.8.1 Natural D/N source branch

Stage 032 attaches the orbital/worldtube STF quadrupole source through the same D/N axial load used by the mixed/support module:

$$L_{\rm src} = g_Q Q_{\rm STF} \int_0^L \eta(s) f_0(s)\, ds. \tag{2.118}$$

With $\eta = q_0 u_0 + q_1 u_1$, the source vector is proportional to v. Projection onto the selected eigenvector gives

$$J_- = g_Q Q_{\rm STF}(v \cdot e_-). \tag{2.119}$$

At zero loading, $J_-(0) = g_Q Q_{\rm STF}\kappa_0$. Hence

$$\boxed{\widehat{m}_- = \frac{v \cdot e_-}{\kappa_0}, \qquad \widehat{m}_-^2 = \frac{s_-}{\kappa_0^2}}. \tag{2.120}$$

Since s_- grows from κ_0^2 to $\sigma = \kappa_0^2 + \kappa_1^2$,

$$\boxed{1 \le \widehat{m}_-^2 < \frac{\sigma}{\kappa_0^2} = \frac{11}{9}}. \tag{2.121}$$

Thus the selected source map is real and modest; it does not hide the normalization burden in an arbitrary large factor.

Combining (2.112) and (2.120) gives the invariant selected product

$$\boxed{N_-(\alpha_0) := \widehat{m}_-^2 P_{0,-} = \frac{\beta_0 s_-^2}{\kappa_0^2 \lambda_-}}. \tag{2.122}$$

The target is

$$\boxed{N_-(\alpha_0) = N_Q^{\rm target}}. \tag{2.123}$$

2.8.2 Microscopic coupling variables and stability gate

For the first explicit finite-throat kernel model, Stage 033 writes

$$\boxed{A = K_0 - \frac{g_U^2}{\Omega_U^2}, \qquad \Delta_0 = \Omega_U^2\Omega_W^2 - g_R^2\sigma, \qquad \mathcal{X} = \Omega_U^2 g_W + g_R g_U}. \tag{2.124}$$

Then

$$\boxed{\beta_0 = \frac{\mathcal{X}^2}{\Delta_0^2}, \qquad \alpha_0 = \frac{g_B^2}{\varpi^2} + \frac{\mathcal{X}^2}{\Omega_U^2\Delta_0}}. \tag{2.125}$$

The exact stability gate is

$$\boxed{\Delta_0 > 0, \qquad A > 0, \qquad \alpha_0 < \alpha_{\rm crit}}, \tag{2.126}$$

where, using $\kappa_0^2 = 8/\pi^2$ and $\kappa_1^2 = 16/(9\pi^2)$,

$$\boxed{\alpha_{\rm crit} = \frac{9\pi^2 A(A + \Delta K_{\rm ax})}{8(11A + 9\Delta K_{\rm ax})}}. \tag{2.127}$$

In coupling form,

$$\frac{g_B^2}{\varpi^2} + \frac{\mathcal{X}^2}{\Omega_U^2\Delta_0} < \frac{9\pi^2 A(A + \Delta K_{\rm ax})}{8(11A + 9\Delta K_{\rm ax})}. \tag{2.128}$$

2.8.3 Onset criterion and weak-loading expansion

Because $N_-(\alpha_0)$ is strictly increasing on the stable branch, the zero-loading value is a necessary onset test. At $\alpha_0 = 0$,

$$s_-(0) = \kappa_0^2, \qquad \lambda_-(0) = A, \tag{2.129}$$

so

$$\boxed{N_-(0) = \frac{\beta_0\kappa_0^2}{A} = \frac{\mathcal{X}^2\kappa_0^2}{A\Delta_0^2}}. \tag{2.130}$$

A necessary condition for the natural positive-loading branch to hit the universal target is

$$\boxed{N_-(0) \le N_Q^{\rm target}, \qquad \text{equivalently} \qquad \mathcal{X}^2 \le \frac{N_Q^{\rm target} A\Delta_0^2}{\kappa_0^2}}. \tag{2.131}$$

If this fails, the branch starts above the target and cannot return to it because N_- only increases.

For weak loading,

$$N_-(\alpha_0) = \frac{\beta_0\kappa_0^2}{A} + \alpha_0\frac{\beta_0\kappa_0^2(4A\kappa_1^2 + \Delta K_{\rm ax}\kappa_0^2)}{A^2\Delta K_{\rm ax}} + O(\alpha_0^2). \tag{2.132}$$

With the exact D/N constants this becomes

$$N_-(\alpha_0) = \frac{8\beta_0}{\pi^2 A} + \alpha_0\frac{64\beta_0(8A + 9\Delta K_{\rm ax})}{9\pi^4 A^2\Delta K_{\rm ax}} + O(\alpha_0^2). \tag{2.133}$$

Theorem 2.9 (Microscopic selected-branch test)**.** *In the first explicit finite-throat kernel closure, the selected quadrupole normalization condition is the exact coupling-level equation*

$$\frac{\mathcal{X}^2}{\Delta_0^2}\frac{s_-(\alpha_0)^2}{\kappa_0^2\lambda_-(\alpha_0)} = N_Q^{\rm target}, \qquad \alpha_0 = \frac{g_B^2}{\varpi^2} + \frac{\mathcal{X}^2}{\Omega_U^2\Delta_0}, \tag{2.134}$$

subject to (2.126)*. The onset condition* (2.131) *is necessary for a hit on the natural positive-loading branch.*

Downstream use. Stages 034–036 remove s_-, λ_-, and the eigenvector algebra from (2.134) by passing to the softening depth and then to dimensionless functions F and G.

2.9 Softening-depth normal form

2.9.1 Softening-depth variable

Stage 034 trades the selected eigenvalue for the softening depth

$$\boxed{x := A - \lambda_-}. \tag{2.135}$$

On the stable selected branch,

$$\lambda_- = A - x, \qquad 0 \le x < A. \tag{2.136}$$

The rank-one secular equation is

$$1 = \alpha_0 \left(\frac{\kappa_0^2}{x} + \frac{\kappa_1^2}{x + \Delta K_{\mathrm{ax}}} \right). \tag{2.137}$$

Solving for total directional loading gives

$$\boxed{\alpha_0(x) = \frac{x(x + \Delta K_{\mathrm{ax}})}{\kappa_0^2(x + \Delta K_{\mathrm{ax}}) + \kappa_1^2 x}}. \tag{2.138}$$

The selected overlap follows from $s_- = dx/d\alpha_0$:

$$\boxed{s_-(x) = \frac{\left[\kappa_0^2(x + \Delta K_{\mathrm{ax}}) + \kappa_1^2 x\right]^2}{\kappa_0^2(x + \Delta K_{\mathrm{ax}})^2 + \kappa_1^2 x^2}}. \tag{2.139}$$

The invariant selected product becomes

$$\boxed{N_-(x) = \frac{\beta_0 \left[\kappa_0^2(x + \Delta K_{\mathrm{ax}}) + \kappa_1^2 x\right]^4}{\kappa_0^2(A - x)\left[\kappa_0^2(x + \Delta K_{\mathrm{ax}})^2 + \kappa_1^2 x^2\right]^2}}. \tag{2.140}$$

The selected quadrupole theorem is now the scalar equation

$$\boxed{N_-(x) = N_Q^{\mathrm{target}}, \qquad 0 \le x < A}. \tag{2.141}$$

2.9.2 Monotone loading and required support contribution

Differentiating (2.138) gives

$$\boxed{\frac{d\alpha_0}{dx} = \frac{\kappa_0^2(x + \Delta K_{\mathrm{ax}})^2 + \kappa_1^2 x^2}{\left[\kappa_0^2(x + \Delta K_{\mathrm{ax}}) + \kappa_1^2 x\right]^2} > 0}. \tag{2.142}$$

Thus x is a one-to-one branch coordinate for the stable loading.

The loading split is

$$\alpha_0 = \frac{g_B^2}{\varpi^2} + \alpha_{\mathrm{mix}}, \qquad \alpha_{\mathrm{mix}} = \frac{\mathcal{X}^2}{\Omega_U^2 \Delta_0}. \tag{2.143}$$

Therefore the required support loading at a target softening depth is

$$\boxed{\frac{g_{B,\mathrm{req}}^2}{\varpi^2} = \frac{x(x + \Delta K_{\mathrm{ax}})}{\kappa_0^2(x + \Delta K_{\mathrm{ax}}) + \kappa_1^2 x} - \frac{\mathcal{X}^2}{\Omega_U^2 \Delta_0}}. \tag{2.144}$$

The nonnegativity of this expression is the first support-feasibility test.

Theorem 2.10 (Softening-depth normal form)**.** *The first rank-one selected finite-throat branch is exactly parameterized by the scalar softening depth $x = A - \lambda_-$. The loading $\alpha_0(x)$, selected overlap $s_-(x)$, invariant normalization product $N_-(x)$, and required support contribution are given by* (2.138)–(2.144)*.*

Downstream use. The softening-depth form is the input to the dimensionless (F, G) functions in Stages 035–036 and to the support-completion program in Part III.

2.10 Support-feasibility frontier

2.10.1 Dimensionless D/N normalization function

Stage 035 substitutes the exact D/N constants

$$\boxed{\kappa_0^2 = \frac{8}{\pi^2}, \qquad \kappa_1^2 = \frac{16}{9\pi^2}, \qquad \eta = \frac{\kappa_1^2}{\kappa_0^2} = \frac{2}{9}}. \tag{2.145}$$

Introduce

$$\boxed{\xi = \frac{x}{A}, \qquad \delta = \frac{\Delta K_{\mathrm{ax}}}{A}, \qquad 0 \le \xi < 1}. \tag{2.146}$$

Then

$$N_-(x) = N_-(0) F(\xi, \delta), \qquad N_-(0) = \frac{\beta_0 \kappa_0^2}{A}. \tag{2.147}$$

The exact dimensionless normalization shape function is

$$\boxed{F(\xi, \delta) = \frac{(9\delta + 11\xi)^4}{81(1-\xi)(9\delta^2 + 18\delta\xi + 11\xi^2)^2}}. \tag{2.148}$$

The dimensionless target is

$$\boxed{R_{\mathrm{target}} := \frac{N_Q^{\mathrm{target}}}{N_-(0)} = \frac{N_Q^{\mathrm{target}} A}{\beta_0 \kappa_0^2}}. \tag{2.149}$$

The normalization locus is therefore

$$\boxed{F(\xi, \delta) = R_{\mathrm{target}}}. \tag{2.150}$$

The exact derivative is

$$\frac{\partial F}{\partial \xi} = \frac{(9\delta + 11\xi)^3 \left(81\delta^3 + 72\delta^2 + 206\delta^2\xi + 297\delta\xi^2 + 138\xi^3\right)}{81(1-\xi)^2(9\delta^2 + 18\delta\xi + 11\xi^2)^3} > 0. \tag{2.151}$$

Also,

$$F(0, \delta) = 1, \qquad \lim_{\xi \to 1^-} F(\xi, \delta) = +\infty. \tag{2.152}$$

Thus:

$$\boxed{R_{\mathrm{target}} < 1 \Rightarrow \text{no stable hit}, \qquad R_{\mathrm{target}} = 1 \Rightarrow \xi_{\mathrm{req}} = 0, \qquad R_{\mathrm{target}} > 1 \Rightarrow \text{unique } \xi_{\mathrm{req}} \in (0,1)}. \tag{2.153}$$

The exact required total loading is

$$\boxed{\alpha_{\mathrm{req}}(\xi, \delta) = \frac{9\pi^2 A \xi(\xi + \delta)}{8(9\delta + 11\xi)}}. \tag{2.154}$$

2.10.2 Support-feasibility function

Stage 036 writes the support-feasibility test in the same dimensionless variable. Define

$$\boxed{M_{\rm mix} := \frac{8\alpha_{\rm mix}}{\pi^2 A} = \frac{8\mathcal{X}^2}{\pi^2 A\Omega_U^2\Delta_0}}. \tag{2.155}$$

Define the support frontier

$$\boxed{G(\xi,\delta) := \frac{8\alpha_{\rm req}}{\pi^2 A} = \frac{9\xi(\xi+\delta)}{9\delta+11\xi}}. \tag{2.156}$$

Then

$$\boxed{\frac{g^2_{B,\rm req}}{\varpi^2} = \frac{\pi^2 A}{8}\left[G(\xi_{\rm req},\delta) - M_{\rm mix}\right]}. \tag{2.157}$$

Therefore the exact support-feasibility condition is

$$\boxed{M_{\rm mix} \le G(\xi_{\rm req},\delta)}. \tag{2.158}$$

The function G is also strictly monotone:

$$\frac{\partial G}{\partial \xi} = \frac{9(9\delta^2+18\delta\xi+11\xi^2)}{(9\delta+11\xi)^2} > 0. \tag{2.159}$$

Its endpoint values are

$$G(0,\delta) = 0, \qquad G_{\max}(\delta) = \lim_{\xi\to 1^-} G(\xi,\delta) = \frac{9(1+\delta)}{9\delta+11}. \tag{2.160}$$

For fixed δ, the stable selected quadrupole branch traces the parametric frontier

$$\boxed{\xi \in [0,1) \longmapsto (F(\xi,\delta), G(\xi,\delta))}. \tag{2.161}$$

The branch is admissible exactly when the normalization target chooses a unique $\xi_{\rm req}$ and the physical mixed baseline lies below the support frontier there.

Near onset, if $R_{\rm target} = 1+\varepsilon_R$ and $\xi \ll 1$, then

$$F(\xi,\delta) = 1 + \left(1+\frac{8}{9\delta}\right)\xi + O(\xi^2), \qquad \xi_{\rm req} \simeq \frac{\varepsilon_R}{1+8/(9\delta)}, \tag{2.162}$$

while

$$G(\xi,\delta) = \xi - \frac{2\xi^2}{9\delta} + O(\xi^3). \tag{2.163}$$

Near softening, if $R_{\rm target}$ is large,

$$1-\xi_{\rm req} \simeq \frac{C(\delta)}{R_{\rm target}}, \qquad C(\delta) = \frac{(9\delta+11)^4}{81(9\delta^2+18\delta+11)^2}. \tag{2.164}$$

Theorem 2.11 (Dimensionless selected-branch admissibility frontier)**.** *For fixed $\delta > 0$, the first selected D/N quadrupole branch is governed by the monotone functions F and G in* (2.148) *and* (2.156)*. A physical point is admissible in this reduced closure if and only if*

$$\boxed{R_{\rm target} \ge 1, \qquad F(\xi_{\rm req},\delta) = R_{\rm target}, \qquad M_{\rm mix} \le G(\xi_{\rm req},\delta)}. \tag{2.165}$$

The first condition enters the unique normalization locus; the second fixes the stable softening depth; the third ensures the remaining required support loading is nonnegative.

2.11 Part II audit summary and handoff to Part III

Part II establishes the following stable outputs.

1. **Angular source map.** The real-STF angular matching matrix is the identity, so the canonical isotropic source-map factor satisfies $\widehat{m}_{\mathrm{ang}} = 1$.
2. **Isotropic grouped collapse.** An $O(3)$-invariant reduced kernel forces the grouped 20/21/22 lanes to share common D_n and N_n data.
3. **Weak-axisymmetric diagnostic.** The first pure axisymmetric quadrupole splitting has signature $(1, 1/2, -1)$, equivalently $b = 3a$.
4. **Minimal isotropic ratio.** The one-mode radial/axial module gives the explicit branch test (2.45).
5. **Concrete D/N overlaps.** The first finite-throat branch collapses to the exact overlap $\kappa = 2\sqrt{2}/\pi$.
6. **Profile family.** The first nonconstant N/N family introduces $\kappa(\theta)$, the blind-angle no-go, and the max-coupling angle.
7. **Loaded profile selection.** The selected profile is the lower eigenvector of a rank-one loaded wall matrix; positive loading drives it toward the max-coupling direction, not the blind-angle direction.
8. **Dynamic loading.** The first coupled wall/BdG/Maxwell/mixed Schur complement splits exactly as $\Xi I_2 + \alpha v v^T$, and the same loading carries the outgoing $i\omega^5$ branch.
9. **Selected normalized response.** The selected static prefactor is $P_{0,-} = \beta_0 s_- / \lambda_-$, and it is strictly monotone on the stable branch.
10. **Natural source map.** On the D/N source branch, $\widehat{m}_-^2 = s_- / \kappa_0^2$, so the selected target becomes $N_- = \beta_0 s_-^2 / (\kappa_0^2 \lambda_-) = N_Q^{\mathrm{target}}$.
11. **Softening-depth normal form.** The entire first selected branch can be parameterized by $x = A - \lambda_-$.
12. **Dimensionless branch geometry.** The final reduced admissibility problem is $F(\xi, \delta) = R_{\mathrm{target}}$ plus $M_{\mathrm{mix}} \le G(\xi, \delta)$.

The chapter deliberately does not assert that the actual branch satisfies (2.165). It supplies the reduced test that the actual branch must pass. In particular, Part II does not yet compute the continuum-kernel values of A, ΔK_{ax}, β_0, M_{mix}, or the later rank-two support-completion data. Those are the job of Part III.

- ☐ The angular source-map identity $\widehat{m}_{\mathrm{ang}} = 1$ is separated from the radial/axial source-map factors.
- ☐ The grouped labels $20, 21, 22$ are used as real P_2 lane labels, not spacetime indices.
- ☐ The weak-axisymmetric signature $(1, 1/2, -1)$ and the line $b = 3a$ are preserved.
- ☐ The minimal isotropic branch obeys $\Delta > 0$ and $D_0 > 0$ before the normalization target is asked.
- ☐ The finite-throat D/N overlap value $\kappa = 2\sqrt{2}/\pi$ is used consistently.
- ☐ The blind-angle branch has $N_0 = 0$ and is therefore an exact no-go for a positive quadrupole target.

- ☐ Positive rank-one loading selects a negative profile angle and moves toward the max-coupling direction.
- ☐ The Schur complement produces $\Sigma_{\text{wall}} = \Xi I_2 + \alpha v v^T$, not a generic two-by-two load.
- ☐ The selected prefactor is $P_{0,-} = \beta_0 s_- / \lambda_-$, and the source-map-corrected product is $N_- = \beta_0 s_-^2 / (\kappa_0^2 \lambda_-)$.
- ☐ The softening-depth variable satisfies $0 \le x < A$, and the support contribution is nonnegative only when $M_{\text{mix}} \le G(\xi_{\text{req}}, \delta)$.

Part III begins where this chapter stops. It computes and organizes the continuum placement data that feed

$$R_{\text{target}} = \frac{N_Q^{\text{target}} A}{\beta_0 \kappa_0^2}, \qquad M_{\text{mix}} = \frac{8\mathcal{X}^2}{\pi^2 A \Omega_U^2 \Delta_0}, \qquad \delta = \frac{\Delta K_{\text{ax}}}{A}, \tag{2.166}$$

and then asks whether rank-two support completion, coherent D/N support, and explicit branch placement can satisfy the frontier (2.165).

The citation-safe summary is:

> Stages 024–036 derive the conservative grouped-response and selected-branch normalization language. They prove exact angular isotropy, identify the weak-axisymmetric $b = 3a$ signature, compute the first finite-throat D/N overlap structure, derive the selected wall-mode Schur complement, eliminate the abstract source map on the natural D/N branch, and reduce the selected quadrupole branch to the dimensionless admissibility test
>
> $$F(\xi_{\text{req}}, \delta) = R_{\text{target}}, \qquad M_{\text{mix}} \le G(\xi_{\text{req}}, \delta).$$
>
> They do not yet prove that the actual moving-throat branch lands in that admissible region.

Chapter 3

Support Completion, Continuum Placement, and Family–1 Branch Construction

Stage range: 037–090

Part III turns the Part II admissibility geometry into explicit support/source branch data. In that language the stable selected branch is fixed by a softening depth ξ, a bare anisotropy ratio δ, a target normalization ratio R_{target}, and a mixed loading baseline M_{mix}. The reduced admissibility test was

$$F(\xi_{\text{req}}, \delta) = R_{\text{target}}, \qquad M_{\text{mix}} \leq G(\xi_{\text{req}}, \delta). \tag{3.1}$$

This chapter asks where the quantities in (3.1) come from in the explicit moving-throat branch, how the support/source sector can complete them, and whether the first concrete Family–1 wall branch survives the support side of the quadrupole test.

The endpoint is not a full branch realization. It is a reduced support/source verdict. The chapter first turns the selected branch into an exact continuum placement and support-completion problem, then converts the remaining demand into lowest-lane transport and parent-overlap thresholds, and finally evaluates the first explicit Family–1 wall-shell branch against those thresholds.

Part III at a glance

Primary question	How are the Part II placement targets completed by actual support and source data, and does the first Family–1 branch pass that reduced test?
Starting inputs	The Part II admissibility pair (F, G), the selected finite-throat branch data, and the first explicit Family–1 wall-shell closure.
Delivers	Continuum placement formulas, support-completion theorems, lowest-lane transport and parent-overlap thresholds, and the first explicit Family–1 reduced verdict.
Used by	Part IV as the point where support/source uncertainty stops being the dominant reduced obstruction, and later realization chapters as the source of Family–1 reference data and residual language.
Result anchors	**MTDC-T6** and **MTDC-T7**.

Stage packets.

Stages	Packet	Status	Carry-forward output
037–051	Continuum placement and support completion	Exact Within Closure / Reduced / Controlled Reduction	Continuum kernel extraction, dimensionless placement laws, rank-2 support completion, coherent D/N tracking rules, and the lowest-twin sufficiency criterion.
052–069	Lowest-lane transport and parent thresholds	Exact Within Closure / Reduced / Controlled Reduction	Non-twin asymmetry regimes, overlap and Robin reachability windows, drift-diffusion transport laws, parent-overlap gain thresholds, and the three-zone reduced support/source verdict.
070–078	Explicit Family–1 wall-shell baseline	Exact Within Closure / Reduced / Controlled Reduction / Numerically Located	GNLS wall-shell reduction, canonical tanh-wall branch, Family–1 geometry and healing-length locks, wall-depth extraction, and the first explicit Family–1 baseline verdict.
079–090	Family–1 demand window and finish line	Exact Within Closure / Open	Demand-to-transport map, branch-product windows, master quadrupole residual, minimal-module ratio packet, and the reduced statement that support/source closure is finished once the outgoing conservative module is realized.

Claim status. Mixed. Most algebraic reductions, Schur complements, branch maps, monotonicity statements, and support-threshold identities are Exact Within Closure **once the declared finite-throat, selected-mode, and Family–1 closures are adopted. The extraction of a first explicit branch from the GNLS wall shell is a** Reduced / Controlled Reduction **use of the parent action. Several branch values are** Numerically Located **because they are exact integrals or exact roots evaluated numerically. The final realization question—whether the actual moving-throat grouped-P_2/geometry branch realizes the minimal isotropic contact-plus-pole conservative module—remains** Open.

What this part proves. Part III converts the abstract admissibility geometry into explicit support-completion theorems and then evaluates the first concrete Family–1 wall branch, producing the first serious reduced success and failure windows on the support/source side.

What this part does not prove. It does not finish the retarded normalization problem and it does not prove that the full PDE realizes the selected support data. The remaining obstruction is compressed to reduced residuals and branch-placement gates.

Why later parts need it. Part IV starts from the fact that the support/source side is no longer the dominant reduced uncertainty, and later realization chapters reuse the Family–1 reference data, residual language, and support-threshold packets established here.

3.1 Block contract and theorem-level output

Part III has a different contract from Part II. Part II built the selected-branch test; Part III must populate the test with actual reduced branch data. The chain is

$$\begin{aligned}&\text{continuum kernel} \longrightarrow \text{placement map} \longrightarrow \text{support completion}\\ &\qquad \longrightarrow \text{support/source operator} \longrightarrow \text{Family–1 branch verdict.}\end{aligned} \tag{3.2}$$

The chain is not a full nonlinear PDE solution. It is a verification ledger for the reduced branch machinery that any completed solution must satisfy.

Theorem 3.1 (Part III support-completion and Family–1 branch theorem)**.** *Inside the declared finite-throat selected-branch closure, the following reduced theorem chain holds.*

1. *The first continuum kernel fixes the selected-branch placement data A, ΔK_{ax}, α_{mix}, β_0, M_{mix}, and δ in closed form.*

2. *These data compress to dimensionless continuum ratios $(\epsilon_\eta, \epsilon_W, \rho, Z_W, \delta_0, \Lambda)$, with exact product law*
$$R_{\mathrm{target}} M_{\mathrm{mix}} = \frac{8\Lambda(1-\epsilon_W)}{\pi^2}. \tag{3.3}$$

3. *Turning on the first non-flat U doublet preserves scalar placement but generically breaks source/loading collinearity; the resulting selected-branch geometry is a controlled deformation governed by R_U.*

4. *With two loading directions, the selected branch remains exactly solvable because the determinant is linear in the support loading.*

5. *The coherent D/N support kernel forces the first local support completion onto a tracking branch; support enhancement is controlled by $S(\zeta;\epsilon)$, and the lowest symmetric twin lane supplies the exact doubling branch $\zeta_0 = 1$, $S_0 = 2$.*

6. *If the symmetric twin is insufficient, the non-twin lowest-lane problem is controlled by explicit overlap boost, Robin softening, drift-diffusion transport, and parent-action gain thresholds.*

7. *The first explicit Family–1 wall-shell branch fixes*
$$\Lambda_\ell = 37, \qquad \eta = 37, \qquad \chi_s = \frac{37}{2}, \qquad \kappa = \frac{12321}{5}. \tag{3.4}$$
Its natural shell-weighted wall datum is
$$\Theta_w^{(\chi)} \simeq 4.06863235008162\,\lambda_\mu^2, \tag{3.5}$$
with conservative floor
$$\Theta_w^{(J)} \simeq 0.927552032539308\,\lambda_\mu^2. \tag{3.6}$$

8. *Once the selected outgoing quadrupole normalization is imposed, the explicit support theorem depends only on the loading ratio*
$$\rho_\alpha = \frac{\alpha_{\mathrm{req}}}{\alpha_{\mathrm{mix}}}. \tag{3.7}$$
Under the natural minimal isotropic contact-plus-pole precursor, this ratio is fixed to
$$\rho_\alpha = \frac{4}{3}, \qquad \zeta_{\mathrm{req}} = \frac{1}{3}, \qquad \mathrm{Pe}_{\mathrm{req}} = 0, \tag{3.8}$$
and the explicit Family–1 support/source side succeeds with large margin.

The only remaining reduced gap after this theorem chain is not support/source sufficiency. It is whether the actual moving-throat grouped-P_2/geometry branch realizes the minimal isotropic conservative quadrupole precursor in the natural contact-plus-pole way.

Proof structure. Stages 037–038 compute the continuum Schur complement and then repackage it into dimensionless placement ratios. Stages 039–044 show how non-flat internal structure and rank-2 support loading deform, but do not destroy, the selected-branch algebra. Stages 045–051 reduce coherent D/N support to a tracking branch and an exact support-ratio threshold. Stages 052–061 identify the physical resources that can exceed the lowest-twin threshold: overlap bias, Robin compliance, drift-diffusion transport, and microscopic gain. Stages 062–069 project that gain back into the parent GNLS/confinement action and close the support/source phase diagram. Stages 070–078 evaluate the first explicit Family–1 wall-shell branch. Stages 079–087 collapse its remaining support test to one loading-ratio datum. Stages 088–090 evaluate that datum for the minimal isotropic contact-plus-pole quadrupole precursor and state the final reduced status. □

3.2 Continuum-kernel extraction and placement map

3.2.1 Local goal

The Part II admissibility test (3.1) is useful only if the quantities A, ΔK_{ax}, β_0, α_{mix}, M_{mix}, and δ can be obtained from a concrete branch. Stages 037–038 provide the first such extraction. The branch is still reduced: it uses the lowest wall doublet, the first brane-like internal U block, the lowest D/N support field, and the mixed W lane. But after that reduced operator is declared, the extraction is exact.

3.2.2 Finite-throat operator and projected constants

Work on $s \in [0, L]$. The wall field η has N/N modes, the support and mixed lanes use the lowest D/N half-wave, and the flat internal U block starts with the first two N/N modes. The exact overlaps are

$$\kappa_0 = \frac{2\sqrt{2}}{\pi}, \qquad \kappa_1 = -\frac{4}{3\pi}, \qquad \sigma = \kappa_0^2 + \kappa_1^2 = \frac{88}{9\pi^2}. \tag{3.9}$$

After mass normalization, the internal static quantities are

$$\Omega_U^2 = \frac{K_U}{\mu_U}, \qquad \Omega_W^2 = \frac{K_W^{\mathrm{eff}}}{\mu_W}, \qquad K_W^{\mathrm{eff}} = K_W + \frac{\pi^2 T_W}{4L^2}, \tag{3.10}$$

with couplings

$$g_U = \frac{c_{\eta U}}{\sqrt{\mu_\eta \mu_U}}, \qquad g_W = \frac{c_{\eta W}}{\sqrt{\mu_\eta \mu_W}}, \qquad g_R = \frac{c_{UW}}{\sqrt{\mu_U \mu_W}}. \tag{3.11}$$

Eliminating the internal block gives the same rank-one wall loading form used abstractly in Part II,

$$\Sigma_{\mathrm{wall}}(\omega) = \Xi(\omega) I_2 + \alpha(\omega) v v^T, \qquad v = (\kappa_0, \kappa_1)^T. \tag{3.12}$$

On the conservative static branch define

$$\Delta_0 = \Omega_U^2 \Omega_W^2 - g_R^2 \sigma, \qquad \mathcal{X} = \Omega_U^2 g_W + g_R g_U. \tag{3.13}$$

Then the selected-branch inputs are

$$\boxed{A = K_0 - \frac{g_U^2}{\Omega_U^2}, \qquad \alpha_{\mathrm{mix}} = \frac{\mathcal{X}^2}{\Omega_U^2 \Delta_0}, \qquad \beta_0 = \frac{\mathcal{X}^2}{\Delta_0^2}, \qquad M_{\mathrm{mix}} = \frac{8\alpha_{\mathrm{mix}}}{\pi^2 A}, \qquad \delta = \frac{\Delta K_{\mathrm{ax}}}{A}}. \tag{3.14}$$

In unnormalized continuum variables, with

$$K_\eta^{\mathrm{eff}} = K_\eta + 6T_\Omega, \tag{3.15}$$

these become

$$\boxed{A = \frac{K_U K_\eta^{\mathrm{eff}} - c_{\eta U}^2}{\mu_\eta K_U}, \qquad \Delta K_{\mathrm{ax}} = \frac{\pi^2 T_w}{\mu_\eta L^2}}, \tag{3.16}$$

$$\boxed{\alpha_{\mathrm{mix}} = \frac{(K_U c_{\eta W} + c_{UW} c_{\eta U})^2}{\mu_\eta K_U \left(K_U K_W^{\mathrm{eff}} - c_{UW}^2 \sigma\right)}}, \tag{3.17}$$

$$\boxed{\beta_0 = \frac{\mu_W}{\mu_\eta} \frac{(K_U c_{\eta W} + c_{UW} c_{\eta U})^2}{\left(K_U K_W^{\mathrm{eff}} - c_{UW}^2 \sigma\right)^2}}, \tag{3.18}$$

$$\boxed{M_{\mathrm{mix}} = \frac{8(K_U c_{\eta W} + c_{UW} c_{\eta U})^2}{\pi^2 (K_U K_\eta^{\mathrm{eff}} - c_{\eta U}^2)(K_U K_W^{\mathrm{eff}} - c_{UW}^2 \sigma)}, \qquad \delta = \frac{\pi^2 T_w K_U}{L^2 (K_U K_\eta^{\mathrm{eff}} - c_{\eta U}^2)}}. \tag{3.19}$$

The stability conditions are correspondingly

$$K_U K_\eta^{\mathrm{eff}} > c_{\eta U}^2, \qquad K_U K_W^{\mathrm{eff}} > c_{UW}^2 \sigma. \tag{3.20}$$

3.2.3 Dimensionless placement ledger

Stage 038 compresses the same continuum data to dimensionless ratios

$$\epsilon_\eta = \frac{c_{\eta U}^2}{K_U K_\eta^{\mathrm{eff}}}, \qquad \epsilon_W = \frac{c_{UW}^2 \sigma}{K_U K_W^{\mathrm{eff}}}, \qquad \rho = \frac{c_{UW} c_{\eta U}}{K_U c_{\eta W}}, \tag{3.21}$$

$$Z_W = \frac{c_{\eta W}^2}{K_\eta^{\mathrm{eff}} K_W^{\mathrm{eff}}}, \qquad \delta_0 = \frac{\pi^2 T_w}{L^2 K_\eta^{\mathrm{eff}}}, \qquad \Lambda = \frac{27\pi^2 G c_s^5 K_W^{\mathrm{eff}}}{20 a^5 c^5 \mu_W}. \tag{3.22}$$

The exact placement map is

$$\boxed{\delta = \frac{\delta_0}{1-\epsilon_\eta}, \qquad M_{\mathrm{mix}} = \frac{8 Z_W (1+\rho)^2}{\pi^2 (1-\epsilon_\eta)(1-\epsilon_W)}, \qquad R_{\mathrm{target}} = \frac{\Lambda (1-\epsilon_\eta)(1-\epsilon_W)^2}{Z_W (1+\rho)^2}}. \tag{3.23}$$

Multiplication gives the product law already stated in (3.3):

$$\boxed{R_{\mathrm{target}} M_{\mathrm{mix}} = \frac{8\Lambda(1-\epsilon_W)}{\pi^2} = \frac{54 G c_s^5 K_W^{\mathrm{eff}} (1-\epsilon_W)}{5 a^5 c^5 \mu_W}}. \tag{3.24}$$

This is the first useful factorization of the placement problem. The pair (ϵ_W, Λ) sets the product scale, while $(\epsilon_\eta, Z_W, \rho)$ redistributes the physical point along that product curve.

Proposition 3.2 (Continuum placement factorization)**.** *For the Stage-20 continuum kernel, the selected branch has an exact three-lane structure:*

$$\textit{geometry lane: } \delta = \frac{\delta_0}{1-\epsilon_\eta}, \qquad \textit{product lane: } R_{\mathrm{target}} M_{\mathrm{mix}} = \frac{8\Lambda(1-\epsilon_W)}{\pi^2}, \tag{3.25}$$

with $(\epsilon_\eta, Z_W, \rho)$ redistributing the physical point between R_{target} and M_{mix} at fixed product.

3.3 Non-flat U doublet and generalized source/loading mismatch

3.3.1 Split-U deformation

The flat-U doublet made the source direction, the support direction, and the mixed loading direction coincide. Stage 039 tests that simplifying assumption by adding the first axial structure to U:

$$K_{U0} = K_U, \qquad K_{U1} = K_U\left(1+\delta_U\right), \qquad \delta_U = \frac{\pi^2 T_U}{L^2 K_U}. \tag{3.26}$$

The scalar placement formulas survive, but with

$$\boxed{\delta_{\text{split}} = \frac{\delta_0 + \epsilon_\eta \delta_U/(1+\delta_U)}{1-\epsilon_\eta}, \qquad \epsilon_{W,\text{split}} = \epsilon_W\left[1 - \frac{2}{11}\frac{\delta_U}{1+\delta_U}\right]}. \tag{3.27}$$

The directional effect is measured by

$$R_U = \frac{1+\rho_0/(1+\delta_U)}{1+\rho_0}, \tag{3.28}$$

and the exact direction-splitting invariant is

$$D_{\text{dir}} = -\frac{\kappa_0\kappa_1 g_W \rho_0 \delta_U}{1+\delta_U}. \tag{3.29}$$

Thus collinearity survives exactly iff

$$D_{\text{dir}} = 0 \quad\Longleftrightarrow\quad \delta_U = 0 \quad \text{or} \quad \rho_0 = 0. \tag{3.30}$$

On the natural constructive branch $\rho_0 > 0$, positive δ_U lowers the mixed baseline and raises the normalization demand before support completion is even considered.

3.3.2 General selected-branch deformation

Stage 040 replaces the old one-vector branch functions by a source/loading mismatch law. Let the loading vector be $z = z_0(1,q)^T$ and the source vector be $s = s_0(1,t)^T$. The selected lower eigenvalue is still written

$$\lambda_- = A_0(1-\xi), \qquad 0 \le \xi < 1. \tag{3.31}$$

For a single rank-one loading, the exact loading is

$$\alpha_{\text{req}} = \frac{A_0\xi(\xi+\delta)}{z_0^2\left[\delta + (1+q^2)\xi\right]}, \tag{3.32}$$

and the selected eigenvector satisfies

$$\frac{e_1}{e_0} = \frac{q\xi}{\delta+\xi}. \tag{3.33}$$

Writing

$$\eta_s = \frac{s_1 z_1}{s_0 z_0}, \tag{3.34}$$

the generalized normalization function is

$$\boxed{F_{q,\eta_s}(\xi,\delta) = \frac{\left[\delta + (1+q^2)\xi\right]^2\left[\delta + (1+\eta_s)\xi\right]^2}{(1-\xi)\left[(\delta+\xi)^2 + q^2\xi^2\right]^2}}, \tag{3.35}$$

while the required baseline loading is

$$\boxed{G_q(\xi,\delta) = \frac{\xi(\xi+\delta)}{\delta+(1+q^2)\xi}}. \tag{3.36}$$

For the split-U continuum, with $\lambda_0 = \kappa_1^2/\kappa_0^2 = 2/9$,

$$q = -\frac{\sqrt{2}}{3}R_U, \qquad \eta_s = \frac{2}{9}R_U, \tag{3.37}$$

so the deformation collapses to

$$\boxed{F_U(\xi,\delta;R_U) = \frac{\left[9\delta+(9+2R_U^2)\xi\right]^2\left[9\delta+(9+2R_U)\xi\right]^2}{81(1-\xi)\left[9\delta^2+18\delta\xi+(9+2R_U^2)\xi^2\right]^2}}, \tag{3.38}$$

$$\boxed{G_U(\xi,\delta;R_U) = \frac{9\xi(\xi+\delta)}{9\delta+(9+2R_U^2)\xi}}. \tag{3.39}$$

Setting $R_U = 1$ recovers the flat-U functions of Part II exactly.

3.4 Rank-2 support completion and continuum-selected closure

3.4.1 Two-direction selected branch

Once source/loading collinearity is broken, the selected wall branch must allow both mixed loading and support loading. Stage 041 writes

$$\frac{K_{\text{loaded}}}{A_0} = \operatorname{diag}(1,1+\delta) - m(1,q)^T(1,q) - n(1,r)^T(1,r), \tag{3.40}$$

where m is the mixed baseline, n is the support loading, q is the mixed direction, and r is the support direction. The determinant condition for $\lambda_- = A_0(1-\xi)$ is

$$D_{\text{sel}} = \xi(\delta+\xi) - m\left[\delta+(1+q^2)\xi\right] - n\left[\delta+(1+r^2)\xi\right] + mn(q-r)^2. \tag{3.41}$$

The structural fact is that (3.41) is linear in n. Hence

$$\boxed{n_{\text{req}}(\xi,\delta;m,q,r) = \frac{\xi(\delta+\xi) - m\left[\delta+(1+q^2)\xi\right]}{\delta+(1+r^2)\xi - m(q-r)^2}}. \tag{3.42}$$

Differentiation gives the exact monotonicity law

$$\frac{\partial n_{\text{req}}}{\partial m} = -\frac{\left[\delta+(1+qr)\xi\right]^2}{\left[\delta+(1+r^2)\xi-m(q-r)^2\right]^2} < 0 \tag{3.43}$$

on every regular branch. More mixed baseline always lowers the additional support loading required to reach the same softening depth.

Two closures are then sharply distinguished. If support tracks the mixed direction, $r = q$, then

$$n_{\text{req}} = G_q(\xi,\delta) - m, \tag{3.44}$$

so no new selected-branch geometry appears. If support stays tied to the original source direction $t = \kappa_1/\kappa_0$ while the mixed direction is $q = tR_U$, then

$$n_{\text{req}}^{\text{src}} = \frac{\xi(\delta+\xi) - m\left[\delta+(1+\lambda_0R_U^2)\xi\right]}{\delta+(1+\lambda_0)\xi - m\lambda_0(R_U-1)^2}. \tag{3.45}$$

This is the first genuinely new rank-2 support-feasibility branch.

3.4.2 Rank-2 selected-mode normalization

Stage 042 keeps the same two loading directions and introduces the source direction $s = s_0(1,t)^T$. With $n = n_{\rm req}$, the lower selected eigenvector obeys

$$\frac{e_1}{e_0} = \frac{m(q-r)+r\xi}{\delta+\xi-mq(q-r)}. \tag{3.46}$$

The two normalized overlaps are

$$\frac{(z\cdot e_-)^2}{z_0^2} = \frac{[\delta+(1+qr)\xi]^2}{D_{q,r}(\xi,\delta;m)}, \tag{3.47}$$

$$\frac{(s\cdot e_-)^2}{s_0^2} = \frac{[\delta+(1+rt)\xi-m(q-r)(q-t)]^2}{D_{q,r}(\xi,\delta;m)}, \tag{3.48}$$

where

$$D_{q,r}(\xi,\delta;m) = [\delta+\xi-mq(q-r)]^2 + [m(q-r)+r\xi]^2\,. \tag{3.49}$$

Thus the rank-2 normalization law depends on outgoing, support, and source directions. The old one-vector formula is recovered only when $r = q$.

3.4.3 Continuum-selected quadratic

Stages 043–044 insert the continuum support direction data. The physical softening depth is pinned by

$$\boxed{\xi^2 + B_{\rm cont}\xi + C_{\rm cont} = 0}, \tag{3.50}$$

with

$$B_{\rm cont} = \delta - M_{\rm mix}(1+t^2R_U^2) - M_{\rm supp}(1+t^2R_\phi^2), \tag{3.51}$$

$$C_{\rm cont} = -\delta(M_{\rm mix}+M_{\rm supp}) + t^2 M_{\rm mix}M_{\rm supp}(R_U-R_\phi)^2. \tag{3.52}$$

The selected-branch normalization theorem gate is then simply

$$\boxed{R_{\rm target} = F_{\rm cont}(\xi_{\rm phys})}. \tag{3.53}$$

This is the first place where the abstract selected-mode branch becomes a concrete continuum-selected branch point rather than a free ξ.

3.5 Coherent local D/N support kernel and support compensation

3.5.1 Tracking reduction

Stage 045 asks what the first coherent local D/N kernel does to the rank-2 ambiguity. The coherent kernel forces

$$g_B g_R = g_W g_S, \tag{3.54}$$

which lands exactly on the tracking surface

$$R_\phi = R_U = R_{\rm tr}. \tag{3.55}$$

The coherent tracking factor is

$$R_{\rm tr} = \frac{1+\chi_0/(1+\delta_U)}{1+\chi_0}, \qquad \frac{1}{1+\delta_U} < R_{\rm tr} < 1 \tag{3.56}$$

on the constructive split-U branch. Hence the rank-2 branch collapses to the one-parameter tracking laws

$$M_{\mathrm{tr}} = G_{\mathrm{tr}}(\xi, \delta; R_{\mathrm{tr}}), \qquad R_{\mathrm{target}} = F_{\mathrm{tr}}(\xi, \delta; R_{\mathrm{tr}}). \tag{3.57}$$

Stage 046 then proves the monotonicity signs

$$\frac{\partial G_{\mathrm{tr}}}{\partial R} < 0, \qquad \frac{\partial F_{\mathrm{tr}}}{\partial R} > 0. \tag{3.58}$$

Since the constructive split branch has $R_{\mathrm{tr}} < 1$, it raises the required loading and lowers the normalized response relative to the flat branch. The first coherent D/N kernel therefore does not automatically rescue the target; it makes support compensation necessary.

3.5.2 Support-enhancement factor

Stage 047 introduces a coherent support lane with ratio ζ. The total tracking load is

$$M_{\mathrm{tr}} = M_{\mathrm{mix}} S(\zeta; \epsilon), \tag{3.59}$$

where

$$\boxed{S(\zeta; \epsilon) = 1 + \frac{\zeta(1-\epsilon)}{1-\epsilon\zeta}}. \tag{3.60}$$

The target ratio is independent of ζ:

$$R_{\mathrm{target}} = \frac{\Lambda(1-\epsilon_\eta)(1-\epsilon)^2}{Z_W(1+\chi_0)^2}. \tag{3.61}$$

Thus support completion changes the available loading but not the selected normalization demand.

Let M_{req} be the load required by the tracking branch at the selected ξ. Define

$$S_{\mathrm{req}} = \frac{M_{\mathrm{req}}}{M_{\mathrm{mix}}}. \tag{3.62}$$

Solving $S(\zeta; \epsilon) = S_{\mathrm{req}}$ gives

$$\boxed{\zeta_{\mathrm{req}} = \frac{S_{\mathrm{req}} - 1}{1 + \epsilon(S_{\mathrm{req}} - 2)}}. \tag{3.63}$$

Stage 048 proves that if the mixed-only branch is below the required load, there is a unique support ratio below the softening pole that reaches the target, provided the actual physical support lane can supply $\zeta_{\mathrm{phys}} \geq \zeta_{\mathrm{req}}$.

3.5.3 D/N microscopic support ratio and lowest-twin criterion

Stage 049 derives the physical support ratio of an explicit D/N support tower:

$$\boxed{\zeta_n^{\mathrm{phys}} = \frac{K_W^{\mathrm{eff}}}{K_{\phi,n}^{\mathrm{eff}}} \left(\frac{I_n}{I_0}\right)^2}. \tag{3.64}$$

For the first uniform local source,

$$\frac{I_n}{I_0} = \frac{1}{2n+1}. \tag{3.65}$$

For the same-operator twin family,

$$\boxed{\zeta_n^{\mathrm{twin}} = \frac{1}{(2n+1)^2\,[1+xn(n+1)]}, \qquad x = \frac{\pi^2 T_X}{L^2 K_W^{\mathrm{eff}}}}. \tag{3.66}$$

Therefore

$$\zeta_0^{\mathrm{twin}} = 1, \qquad S_0 = S(1;\epsilon) = 2. \tag{3.67}$$

The symmetric lowest twin succeeds iff

$$\boxed{\zeta_{\mathrm{req}} \le 1 \quad \Longleftrightarrow \quad S_{\mathrm{req}} \le 2}. \tag{3.68}$$

For $n \ge 1$, the exact necessary condition is

$$\zeta_{\mathrm{req}} \le \frac{1}{(2n+1)^2}, \tag{3.69}$$

and if this is satisfied the stiffness threshold is

$$x \le x_{\max}(n;\zeta_{\mathrm{req}}) = \frac{1}{n(n+1)}\left[\frac{1}{(2n+1)^2\zeta_{\mathrm{req}}} - 1\right]. \tag{3.70}$$

Thus the first coherent support tower is strongly biased toward the lowest twin lane.

3.5.4 Branch-product form of the lowest-twin test

Stage 051 rewrites the same result in the product variable. Define

$$C_{\mathrm{mix}} = \frac{8\Lambda(1-\epsilon)}{\pi^2}, \tag{3.71}$$

and let $\Pi_{\mathrm{tr}} = F_{\mathrm{tr}} G_{\mathrm{tr}}$ be the tracking-branch product at the selected depth. The symmetric lowest twin is sufficient iff

$$\boxed{\Pi_{\mathrm{tr}} \le 2C_{\mathrm{mix}} = \frac{16\Lambda(1-\epsilon)}{\pi^2}}. \tag{3.72}$$

This criterion will be reused below when the support demand is rewritten in terms of the physical outgoing branch.

3.6 Lowest-lane asymmetry, transport, and parent gain thresholds

3.6.1 Three support regimes

Stage 052 gives the exact branch-demand variable

$$\boxed{\zeta_{\mathrm{req}}(\Pi_{\mathrm{tr}}, C_{\mathrm{mix}}, \epsilon) = \frac{\Pi_{\mathrm{tr}} - C_{\mathrm{mix}}}{C_{\mathrm{mix}} - \epsilon(2C_{\mathrm{mix}} - \Pi_{\mathrm{tr}})}}. \tag{3.73}$$

The support problem splits into three exact regimes:

$$\begin{array}{lcl} \Pi_{\mathrm{tr}} \le C_{\mathrm{mix}} & \Longrightarrow & \text{mixed baseline already enough,} \\ C_{\mathrm{mix}} < \Pi_{\mathrm{tr}} \le 2C_{\mathrm{mix}} & \Longrightarrow & \text{symmetric lowest twin enough,} \\ \Pi_{\mathrm{tr}} > 2C_{\mathrm{mix}} & \Longrightarrow & \text{non-twin lowest-lane asymmetry required.} \end{array} \tag{3.74}$$

For a general lowest support lane,

$$\zeta_0^{\mathrm{phys}} = \frac{K_W^{\mathrm{eff}}}{K_{\phi,0}^{\mathrm{eff}}}\Omega_0^2. \tag{3.75}$$

Thus non-twin rescue can come from overlap boost $\Omega_0^2 > 1$, support softening $K_{\phi,0}^{\mathrm{eff}} < K_W^{\mathrm{eff}}$, or both.

3.6.2 Overlap boost and Robin softening

Stage 053 studies the normalized exponential source family

$$\sigma_\alpha(s) = \frac{\alpha e^{\alpha s/L}}{e^\alpha - 1}, \qquad \int_0^L \sigma_\alpha(s)\,ds = L. \tag{3.76}$$

The exact overlap boost is

$$\boxed{\Omega_{\exp}(\alpha) = \frac{\pi\alpha\,(2\alpha e^\alpha + \pi)}{(4\alpha^2 + \pi^2)(e^\alpha - 1)}}. \tag{3.77}$$

It satisfies

$$\Omega_{\exp}(0) = 1, \qquad \lim_{\alpha\to+\infty} \Omega_{\exp}(\alpha) = \frac{\pi}{2}, \qquad \Omega_0^2 \le \frac{\pi^2}{4}. \tag{3.78}$$

Pure overlap rescue is therefore possible only if $\zeta_{\mathrm{req}} \le \pi^2/4$.

Stage 054 replaces the Dirichlet mouth with Robin compliance. Let

$$y \tan y = \eta, \qquad 0 < y < \frac{\pi}{2}, \tag{3.79}$$

and define

$$x = \frac{\pi^2 T_X}{L^2 K_W^{\mathrm{eff}}}, \qquad 0 < x < 4. \tag{3.80}$$

The exact softening factor is

$$\boxed{A_K(\eta) = \frac{K_W^{\mathrm{eff}}}{K_{\phi,0}^{\mathrm{eff}}} = \frac{1}{1 - x/4 + xy^2/\pi^2}}. \tag{3.81}$$

Its endpoint window is

$$1 \le A_K \le \frac{4}{4-x}. \tag{3.82}$$

Combining overlap and compliance yields the first explicit non-twin reachability window of Stage 055.

3.6.3 Transport origin of the overlap bias

Stage 056 derives the exponential family from a stationary drift-diffusion operator:

$$\partial_t \sigma + \partial_s J = 0, \qquad J = -D_\sigma \partial_s \sigma + v_\sigma \sigma. \tag{3.83}$$

The zero-flux stationary branch has the exponential profile with Peclet number

$$\mathrm{Pe} = \frac{v_\sigma L}{D_\sigma}. \tag{3.84}$$

It is therefore natural to write the physical overlap factor as

$$\boxed{\Omega_{\mathrm{Pe}} = \frac{\pi \mathrm{Pe}\left(2\mathrm{Pe}e^{\mathrm{Pe}} + \pi\right)}{(4\mathrm{Pe}^2 + \pi^2)(e^{\mathrm{Pe}} - 1)}, \qquad \Omega_0 = 1}. \tag{3.85}$$

Stage 057 combines this with the Robin factor into the physical lowest-lane support ratio

$$\boxed{\zeta_0(\mathrm{Pe}, \eta; \kappa) = \Omega_{\mathrm{Pe}}^2 \frac{\kappa + \pi^2/4}{\kappa + y(\eta)^2}, \qquad y(\eta)\tan y(\eta) = \eta}. \tag{3.86}$$

3.6.4 Coupled support/source fixed point

Stage 058 closes the source and support operators with a self-consistent branch equation

$$\boxed{\mathrm{Pe}_* = \Xi\Delta(\mathrm{Pe}_*; \kappa, \eta)}. \tag{3.87}$$

The endpoint brackets are

$$\Xi\Delta_0(\kappa, \eta) \leq \mathrm{Pe}_* \leq \Xi\Delta_\infty(\kappa, \eta), \tag{3.88}$$

where, with $\alpha = \sqrt{\kappa}$,

$$\Delta_0 = \frac{\eta(\cosh\alpha - 1)}{\alpha^2(\alpha\sinh\alpha + \eta\cosh\alpha)}, \qquad \Delta_\infty = \frac{\cosh\alpha + (\eta/\alpha)\sinh\alpha - 1}{\alpha\sinh\alpha + \eta\cosh\alpha}. \tag{3.89}$$

Since ζ_0 is monotone in Pe, Stage 059 turns this into residual bounds and exact coupling thresholds. If $\mathrm{Pe}_{\mathrm{req}}$ is the Peclet bias needed to meet ζ_{req}, then

$$\boxed{\Xi_{\mathrm{fail}} = \frac{\mathrm{Pe}_{\mathrm{req}}}{\Delta_\infty(\kappa, \eta)}, \qquad \Xi_{\mathrm{suff}} = \frac{\mathrm{Pe}_{\mathrm{req}}}{\Delta_0(\kappa, \eta)}}. \tag{3.90}$$

Below Ξ_{fail} the branch fails; above Ξ_{suff} it succeeds; the intermediate strip requires the full fixed-point root.

3.6.5 Parent-action gain and overlap thresholds

Stages 060–063 derive the gain from the parent GNLS/confinement sector. Project the linearized compressional source onto $\delta\rho(s, y) = \sigma(s)\chi_\sigma(y)$, and the support onto $\phi(s)\chi_\phi(y)$, with

$$\delta V_{\mathrm{conf}}(s, y) = -g_\phi\chi_\phi(y)\phi(s). \tag{3.91}$$

For the frozen $n = 5$ EOS,

$$h'(\rho_*) = 5K\rho_*^3 = \frac{mc_{s,*}^2}{\rho_*}. \tag{3.92}$$

Define

$$N_{\sigma\sigma} = \int d^3y\, \chi_\sigma^2, \qquad N_{\phi\phi} = \int d^3y\, \chi_\phi^2, \qquad O_{\sigma\phi} = \int d^3y\, \chi_\sigma\chi_\phi. \tag{3.93}$$

Then

$$\boxed{G_{\mathrm{micro}} = \frac{\rho_* g_\phi^2 O_{\sigma\phi}^2}{mc_{s,*}^2 K_X N_{\sigma\sigma}} = \frac{\rho_* g_\phi^2 N_{\phi\phi}}{mc_{s,*}^2 K_X} C_{\sigma\phi}^2}, \tag{3.94}$$

where

$$C_{\sigma\phi}^2 = \frac{O_{\sigma\phi}^2}{N_{\sigma\sigma}N_{\phi\phi}}, \qquad 0 \leq C_{\sigma\phi}^2 \leq 1. \tag{3.95}$$

The fixed-point strength is

$$\Xi_{\mathrm{micro}} = \kappa G_{\mathrm{micro}}. \tag{3.96}$$

The parent amplitude thresholds are therefore exact:

$$g_{\phi,\mathrm{fail}}^2 = \frac{mc_{s,*}^2 K_X N_{\sigma\sigma} G_{\mathrm{fail}}}{\rho_* O_{\sigma\phi}^2}, \qquad g_{\phi,\mathrm{suff}}^2 = \frac{mc_{s,*}^2 K_X N_{\sigma\sigma} G_{\mathrm{suff}}}{\rho_* O_{\sigma\phi}^2}. \tag{3.97}$$

Equivalently, if g_ϕ is fixed, the required coherence floors are

$$C_{\mathrm{fail}}^2 = \frac{mc_{s,*}^2 K_X G_{\mathrm{fail}}}{\rho_* g_\phi^2 N_{\phi\phi}}, \qquad C_{\mathrm{suff}}^2 = \frac{mc_{s,*}^2 K_X G_{\mathrm{suff}}}{\rho_* g_\phi^2 N_{\phi\phi}}. \tag{3.98}$$

The Cauchy bound gives the first parent-overlap no-go: if the best possible aligned gain is below G_{fail}, no source-profile engineering in that support channel can rescue the branch.

3.6.6 Equilibrium alignment and wall figure of merit

Stage 064 shows that the parent equilibrium response is not arbitrary:

$$\chi_\sigma(y) = \frac{g_\phi \chi_\phi(y)}{H(y)}, \qquad H(y) = h'(\rho_*(y)). \tag{3.99}$$

Thus

$$I_1 = \int d^3y \frac{\chi_\phi^2}{H}, \qquad I_2 = \int d^3y \frac{\chi_\phi^2}{H^2}, \qquad C_{\sigma\phi}^2 = \frac{I_1^2}{N_{\phi\phi} I_2}. \tag{3.100}$$

In a thin layer with nearly constant H, $C_{\sigma\phi}^2 = 1$ and

$$G_{\rm eq} = \frac{g_\phi^2 I_1}{K_X}. \tag{3.101}$$

Stages 065–066 then use a concrete thin-wall confinement

$$V_{\rm conf}(r;a) = V_0 f\left(\frac{r-a}{\ell}\right), \qquad g_\phi = \frac{V_0}{\ell}. \tag{3.102}$$

In the thin-wall matched branch, the support/source theorem collapses to the dimensionless wall figure of merit

$$\boxed{W_{\rm wall} = \frac{4\pi a^2 L^2 J_1 V_0^2}{T_X \ell}}. \tag{3.103}$$

The universal reduced window is

$$\boxed{W_{\rm wall} \le \frac{\mathrm{Pe}_{\rm req}}{\Delta_\infty} \Rightarrow \text{fail}, \qquad W_{\rm wall} \ge \frac{\mathrm{Pe}_{\rm req}}{\Delta_0} \Rightarrow \text{succeed}}. \tag{3.104}$$

Only the intermediate band requires a full root solve.

Stages 067–068 benchmark independent source/support mismatch using $\chi_\sigma = \mathrm{sech}(y/w_f)$, $\chi_\phi = e^{-y^2/w_g^2}$. The exact self-dual stationary point is

$$\frac{w_g}{w_f} = \sqrt{\pi}, \tag{3.105}$$

with

$$C_{\rm res}^2 \simeq 0.994418836451529, \qquad P_{\rm res} = \frac{1}{C_{\rm res}^2} \simeq 1.005612487760576. \tag{3.106}$$

Thus the first independent-profile benchmark changes the matched thresholds by only about 0.56%. Stage 069's reduced verdict is therefore the three-zone wall theorem: universal fail below $\mathrm{Pe}_{\rm req}/\Delta_\infty$, universal matched success above $\mathrm{Pe}_{\rm req}/\Delta_0$, and only a narrow profile-sensitive band in between.

3.7 Family–1 explicit branch construction

3.7.1 GNLS wall-shell support branch

Stage 070 derives the support coefficients from the parent GNLS shell energy. Near the wall background ρ_w, the quadratic support energy for $\delta\rho(s,y) = q(s)\chi_\phi(y)$ is

$$E^{(2)}[q] = \frac{1}{2}\int_0^L ds \left[T_X q'(s)^2 + K_X q(s)^2\right], \tag{3.107}$$

with

$$T_X = \frac{\hbar^2}{4m\rho_w} N_{\phi\phi}, \qquad K_X = H_w N_{\phi\phi} + \frac{\hbar^2}{4m\rho_w} G_{\phi\phi}, \qquad H_w = \frac{mc_{s,w}^2}{\rho_w}. \tag{3.108}$$

For a thin shell with profile moments

$$I_f = \int d\xi\, f'(\xi)^2, \qquad I_g = \int d\xi\, f''(\xi)^2, \tag{3.109}$$

this gives

$$T_X = \frac{\pi a^2 \ell I_f \hbar^2}{m\rho_w}, \qquad K_X = 4\pi a^2 \ell I_f H_w + \frac{\pi a^2 I_g \hbar^2}{m\rho_w \ell}. \tag{3.110}$$

Thus

$$\kappa = \frac{K_X L^2}{T_X} = 4\left(\frac{mc_{s,w}L}{\hbar}\right)^2 + \frac{I_g}{I_f}\left(\frac{L}{\ell}\right)^2. \tag{3.111}$$

The matched thin-wall branch also has

$$W_{\rm wall} = \Xi = \frac{4\rho_w^2 V_0^2 L^2}{\hbar^2 c_{s,w}^2 \ell^2}. \tag{3.112}$$

3.7.2 Canonical tanh wall and reference geometry

Stage 071 chooses

$$f(\xi) = \frac{1+\tanh\xi}{2}, \qquad f'(\xi) = \frac{1}{2}\,\mathrm{sech}^2\,\xi. \tag{3.113}$$

The moments are exact:

$$I_f = \frac{1}{3}, \qquad I_g = \frac{4}{15}, \qquad \frac{I_g}{I_f} = \frac{4}{5}. \tag{3.114}$$

The natural local mouth closure is

$$K_m = \frac{T_X}{\ell}, \qquad \eta = \frac{K_m L}{T_X} = \frac{L}{\ell} = \Lambda_\ell. \tag{3.115}$$

Define

$$\chi_s = \frac{mc_{s,w}L}{\hbar}, \qquad \Lambda_\ell = \frac{L}{\ell}, \qquad \Upsilon_w = \frac{4\rho_w^2 V_0^2}{\hbar^2 c_{s,w}^2}. \tag{3.116}$$

Then

$$\boxed{\kappa = 4\chi_s^2 + \frac{4}{5}\Lambda_\ell^2, \qquad \eta = \Lambda_\ell, \qquad W_{\rm wall} = \Upsilon_w \Lambda_\ell^2}. \tag{3.117}$$

Stages 073–074 fix the Family–1 reference values. The carried geometry is

$$\frac{L}{a} = \frac{37}{20}, \qquad \frac{\ell}{a} = \frac{1}{20}, \qquad \Lambda_\ell = 37, \qquad \eta = 37. \tag{3.118}$$

The healing-width closure

$$\ell = \frac{\hbar}{2mc_{s,w}} \tag{3.119}$$

gives

$$\boxed{\chi_s = \frac{37}{2}, \qquad \kappa = \frac{9}{5}\Lambda_\ell^2 = \frac{12321}{5}, \qquad \alpha = \sqrt{\kappa} = \frac{128}{\sqrt{5}}}. \tag{3.120}$$

3.7.3 Family–1 threshold window and wall-depth datum

Stage 075 evaluates the exact support/source thresholds at $(\kappa, \eta) = (12321/5, 37)$:

$$\Delta_0\left(\frac{12321}{5}, 37\right) \simeq 1.73302079021525 \times 10^{-4}, \tag{3.121}$$

$$\Delta_\infty\left(\frac{12321}{5}, 37\right) \simeq 2.01447565540522 \times 10^{-2}. \tag{3.122}$$

With $V_0 = \alpha_r \mu_*$, $\alpha_r = 10$, write

$$\Upsilon_w = 117\Theta_w. \tag{3.123}$$

The explicit branch theorem is

$$\boxed{\Theta_w \leq 3.62605617972939 \times 10^{-4}\,\mathrm{Pe}_{\mathrm{req}} \Rightarrow \text{fail}}, \tag{3.124}$$

$$\boxed{\Theta_w \geq 4.21495341569977 \times 10^{-2}\,\mathrm{Pe}_{\mathrm{req}} \Rightarrow \text{succeed}}. \tag{3.125}$$

Stage 076 uses the exact $n = 5$ identity $h(\rho) = mc_s^2/4$ and the healing lock to obtain

$$\Theta_w = 25\lambda_\mu^2 \rho_w^2 \tag{3.126}$$

in normalized Family–1 variables. Stage 077 extracts ρ_w from the explicit radial wall profile with the canonical support weight. The result is

$$\langle \rho_r \rangle_\chi \simeq 0.192619005556493, \qquad \langle \rho_r^2 \rangle_\chi \simeq 0.162745294003265, \tag{3.127}$$

so

$$\boxed{\Theta_w^{(\chi)} = 25\lambda_\mu^2 \langle \rho_r^2 \rangle_\chi \simeq 4.06863235008162\lambda_\mu^2}, \tag{3.128}$$

with conservative lower envelope

$$\boxed{\Theta_w^{(J)} = 25\lambda_\mu^2 \langle \rho_r \rangle_\chi^2 \simeq 0.927552032539308\lambda_\mu^2}. \tag{3.129}$$

Stage 078 compares these to the threshold window. For the natural shell-weighted datum,

$$\mathrm{Pe}_{\mathrm{suff}}^{(\chi)} \simeq 96.5285247264386\lambda_\mu^2, \qquad \mathrm{Pe}_{\mathrm{fail}}^{(\chi)} \simeq 11220.5441626259\lambda_\mu^2. \tag{3.130}$$

For the conservative floor,

$$\mathrm{Pe}_{\mathrm{suff}}^{(J)} \simeq 22.0062226330754\lambda_\mu^2, \qquad \mathrm{Pe}_{\mathrm{fail}}^{(J)} \simeq 2558.01892349205\lambda_\mu^2. \tag{3.131}$$

The branch-level verdict is that Family–1 is not wall-depth starved except for unusually large quadrupole-side demand.

3.8 First reduced verdict and residual localization

3.8.1 Demand map and Family–1 ceiling

Stages 079–081 rewrite the remaining support/source condition in the demand variables used by the outgoing quadrupole branch. Let y_{F1} be the Robin root

$$y_{\mathrm{F1}} \tan y_{\mathrm{F1}} = 37, \qquad 0 < y_{\mathrm{F1}} < \frac{\pi}{2}, \qquad y_{\mathrm{F1}} \simeq 1.52948248371470. \tag{3.132}$$

Define

$$A_{\mathrm{F1}} = \frac{\kappa_{\mathrm{F1}} + \pi^2/4}{\kappa_{\mathrm{F1}} + y_{\mathrm{F1}}^2}, \qquad \kappa_{\mathrm{F1}} = \frac{12321}{5}. \tag{3.133}$$

Numerically,

$$A_{\mathrm{F1}} \simeq 1.00005192880220. \tag{3.134}$$

The Family–1 support-ratio map is

$$\boxed{\zeta_{\mathrm{F1}}(\mathrm{Pe}) = A_{\mathrm{F1}} \Omega_{\mathrm{Pe}}^2}. \tag{3.135}$$

The hard constructive ceiling is

$$\boxed{\zeta_{\max}^{\mathrm{F1}} = A_{\mathrm{F1}} \frac{\pi^2}{4} \simeq 2.46752922945601}. \tag{3.136}$$

Thus a demanded ζ_{req} is already met at zero bias if $\zeta_{\mathrm{req}} < A_{\mathrm{F1}}$, has a unique constructive $\mathrm{Pe}_{\mathrm{req}}$ if $A_{\mathrm{F1}} \le \zeta_{\mathrm{req}} \le \zeta_{\max}^{\mathrm{F1}}$, and fails the explicit Family–1 branch if it exceeds $\zeta_{\max}^{\mathrm{F1}}$.

At $\lambda_\mu = 1$, the natural shell-weighted demand thresholds are

$$\zeta_{\mathrm{suff}}^{(\chi)} \simeq 2.46622291347846, \qquad \zeta_{\mathrm{fail}}^{(\chi)} \simeq 2.46752913273870, \tag{3.137}$$

while the conservative floor gives

$$\zeta_{\mathrm{suff}}^{(J)} \simeq 2.44257571477179, \qquad \zeta_{\mathrm{fail}}^{(J)} \simeq 2.46752736855058. \tag{3.138}$$

The natural branch nearly reaches the hard ceiling.

3.8.2 Master residual

Stage 082 packages the remaining reduced PDE into one scalar residual. Let

$$\zeta_{\mathrm{req}}(\Pi_{\mathrm{tr}}, C_{\mathrm{mix}}, \epsilon_{\mathrm{blk}}) = \frac{\Pi_{\mathrm{tr}} - C_{\mathrm{mix}}}{C_{\mathrm{mix}} - \epsilon_{\mathrm{blk}}(2C_{\mathrm{mix}} - \Pi_{\mathrm{tr}})}, \tag{3.139}$$

and

$$\zeta_{\mathrm{phys}}(\mathrm{Pe}, \eta; \kappa) = \Omega_{\mathrm{Pe}}^2 \frac{\kappa + \pi^2/4}{\kappa + y(\eta)^2}. \tag{3.140}$$

The selected support/source branch chooses Pe_* from (3.87). Therefore

$$\boxed{R_{\mathrm{quad}}(\Pi_{\mathrm{tr}}, C_{\mathrm{mix}}, \epsilon_{\mathrm{blk}}; \Xi, \eta, \kappa) = \zeta_{\mathrm{req}}(\Pi_{\mathrm{tr}}, C_{\mathrm{mix}}, \epsilon_{\mathrm{blk}}) - \zeta_{\mathrm{phys}}(\mathrm{Pe}_*(\Xi, \eta, \kappa), \eta; \kappa)}. \tag{3.141}$$

The sign convention is

$$\begin{aligned} R_{\mathrm{quad}} &< 0 \Rightarrow \text{support supply exceeds demand}, \\ R_{\mathrm{quad}} &= 0 \Rightarrow \text{exact saturation}, \\ R_{\mathrm{quad}} &> 0 \Rightarrow \text{explicit lowest-lane support fails}. \end{aligned} \tag{3.142}$$

The Family–1 specialization is

$$R^{\mathrm{F1}}_{\mathrm{quad}} = \zeta_{\mathrm{req}}(\Pi_{\mathrm{tr}}, C_{\mathrm{mix}}, \epsilon_{\mathrm{blk}}) - \zeta_{\mathrm{phys}}(\mathrm{Pe}_*(\Xi_{\mathrm{F1}}, 37, 12321/5), 37; 12321/5), \tag{3.143}$$

with

$$\Xi_{\mathrm{F1}} = W_{\mathrm{wall}} = 1369\Upsilon_w = 136900\Theta_w. \tag{3.144}$$

3.8.3 Cancellation of outgoing normalization factors

Stages 085–087 show that once the selected outgoing branch is normalized, the support theorem no longer depends separately on N_Q^{target}, β_0, s_-, λ_-, or $\widehat{m}_-$. The exact identities are

$$\Pi_{\mathrm{tr}} = \frac{N_Q^{\mathrm{target}}}{\beta_0}\alpha_{\mathrm{req}}, \qquad C_{\mathrm{mix}} = \frac{N_Q^{\mathrm{target}}}{\beta_0}\alpha_{\mathrm{mix}}, \tag{3.145}$$

so

$$\boxed{\frac{\Pi_{\mathrm{tr}}}{C_{\mathrm{mix}}} = \frac{\alpha_{\mathrm{req}}}{\alpha_{\mathrm{mix}}} =: \rho_\alpha}. \tag{3.146}$$

Consequently

$$\boxed{\zeta_{\mathrm{req}}(\rho_\alpha, \epsilon_{\mathrm{blk}}) = \frac{\rho_\alpha - 1}{1 - \epsilon_{\mathrm{blk}}(2 - \rho_\alpha)}}. \tag{3.147}$$

In the unblocked limit this is simply $\zeta_{\mathrm{req}} = \rho_\alpha - 1$.

At $\lambda_\mu = 1$ and $\epsilon_{\mathrm{blk}} = 0$, the Family–1 ratio window is

$$\rho_\alpha \le 3.46622291347846 \Rightarrow \text{guaranteed success}, \tag{3.148}$$

$$\begin{aligned} &\rho_\alpha \ge 3.46752913273870 \Rightarrow \text{guaranteed failure}, \\ &\rho_\alpha < 3.46752922945601 \Rightarrow \text{absolute constructive ceiling}. \end{aligned} \tag{3.149}$$

Thus the support/source side reduces to one number from the outgoing branch: ρ_α.

3.8.4 Minimal isotropic contact-plus-pole module

Stage 088 extracts ρ_α from the minimal isotropic conservative precursor. The precursor is

$$Y_Q^{\mathrm{cons}}(\omega) = c_0 + \frac{c_1}{1 - \omega^2/\Omega_Q^2}, \qquad c_0 = \frac{3}{4}, \qquad c_1 = \frac{1}{4}, \qquad \Omega_Q = \frac{3c_s}{2a}. \tag{3.150}$$

The natural contact-plus-pole reading is

$$Y_Q^{\mathrm{cons}}(\omega) = \frac{1}{\rho_\alpha} + \frac{\rho_\alpha - 1}{\rho_\alpha}\frac{1}{1 - \omega^2/\Omega_Q^2}. \tag{3.151}$$

Therefore

$$\boxed{\rho_\alpha = \frac{1}{c_0} = \frac{4}{3}, \qquad \zeta_{\mathrm{req}} = \frac{c_1}{c_0} = \frac{1}{3}, \qquad \Pi_{\mathrm{tr}} = \frac{4}{3}C_{\mathrm{mix}}}. \tag{3.152}$$

This lies in the symmetric-lowest-twin regime,

$$C_{\mathrm{mix}} < \Pi_{\mathrm{tr}} < 2C_{\mathrm{mix}}, \qquad 0 < \zeta_{\mathrm{req}} = \frac{1}{3} < 1. \tag{3.153}$$

Stage 089 compares it to the explicit Family–1 support map. Since

$$\zeta_{\mathrm{req}}^{\mathrm{min}} = \frac{1}{3} < A_{\mathrm{F1}} \simeq 1.00005192880220, \tag{3.154}$$

the required transport bias is

$$\boxed{\mathrm{Pe}_{\mathrm{req}} = 0}. \tag{3.155}$$

So the explicit Family–1 support/source branch supports the minimal isotropic passive/outgoing quadrupole demand before any transport-driven overlap boost is turned on.

3.9 Part III audit summary and handoff to Part IV

Part III establishes the following stable outputs.

1. **Continuum placement is explicit.** The first finite-throat continuum operator gives closed formulas for the selected-branch data and compresses them to $(\epsilon_\eta, \epsilon_W, \rho, Z_W, \delta_0, \Lambda)$.
2. **The product law is exact.** The placement variables obey $R_{\mathrm{target}} M_{\mathrm{mix}} = 8\Lambda(1-\epsilon_W)/\pi^2$, so only part of the microscopic data sets the product scale.
3. **Non-flat internal structure matters directionally.** The split-U correction preserves scalar placement but generically breaks source/loading collinearity through D_{dir}.
4. **Rank-2 support completion is analytic.** The two-direction determinant is linear in the support load, giving the exact n_{req} theorem and monotonicity in mixed baseline.
5. **The first coherent D/N kernel tracks.** The coherent local D/N condition forces $R_\phi = R_U = R_{\mathrm{tr}}$; the first split-$U$ tracking branch worsens the target relative to the flat branch.
6. **Support compensation is governed by one ratio.** The support enhancement is $S(\zeta;\epsilon)$, and the physical D/N tower gives ζ_n^{phys}. The lowest twin branch has $\zeta = 1$ and exactly doubles the mixed baseline.
7. **Non-twin rescue is bounded.** Overlap boost is capped by $\pi^2/4$; Robin compliance has its own exact softening ceiling; the combined physical branch is described by $\zeta_0(\mathrm{Pe}, \eta; \kappa)$.
8. **Transport bias has an operator origin.** The source profile is the stationary zero-flux solution of a drift-diffusion equation, and Pe_* is selected by $\mathrm{Pe}_* = \Xi\Delta(\mathrm{Pe}_*; \kappa, \eta)$.
9. **Parent GNLS projection fixes the microscopic gain.** The gain is the local compressional susceptibility times a squared confinement/support overlap, divided by support stiffness.
10. **Matched support/source alignment is natural.** The parent equilibrium source profile is $\chi_\sigma = g_\phi \chi_\phi / H$, with exact coherence near one on a thin active layer.
11. **The explicit Family–1 wall branch is concrete.** The tanh-wall and healing-lock closures give $\Lambda_\ell = \eta = 37$, $\chi_s = 37/2$, and $\kappa = 12321/5$.
12. **Family–1 is not wall-depth starved.** The natural shell-weighted wall datum $\Theta_w^{(\chi)}$ sits far above the success threshold for modest demand.
13. **The final support test is one loading ratio.** After outgoing normalization factors cancel, the explicit support/source theorem depends only on $\rho_\alpha = \alpha_{\mathrm{req}}/\alpha_{\mathrm{mix}}$.

14. **The minimal isotropic precursor passes.** Under the contact-plus-pole reading, $\rho_\alpha = 4/3$, $\zeta_{\mathrm{req}} = 1/3$, and $\mathrm{Pe}_{\mathrm{req}} = 0$, safely inside the Family–1 success region.

The chapter deliberately does not claim that the actual moving-throat grouped-P_2/geometry branch has already been derived. It shows that if the final quadrupole branch realizes the minimal isotropic contact-plus-pole conservative precursor, then the explicit Family–1 support/source side is already sufficient. Part IV begins at precisely the remaining issue: the geometry lane, retarded closure, outgoing normalization, outlet/core/mouth compensation, and the branch mechanisms that can actually realize the passive/outgoing quadrupole module.

- ☐ The continuum variables A, ΔK_{ax}, α_{mix}, β_0, M_{mix}, and δ are tied to the Stage-20 finite-throat operator, not treated as free branch knobs.
- ☐ The dimensionless placement map uses the exact stability restrictions $0 < \epsilon_\eta < 1$, $0 < \epsilon_W < 1$, and $1 + \rho \neq 0$.
- ☐ The split-U deformation is separated into scalar placement effects and directional source/loading effects.
- ☐ The rank-2 support theorem keeps the directions q, r, and t distinct until a tracking closure is explicitly imposed.
- ☐ The coherent D/N kernel is recorded as a tracking result, not as a generic rank-2 solution.
- ☐ The lowest twin branch is used only when $\zeta_{\mathrm{req}} \leq 1$ or equivalently $\Pi_{\mathrm{tr}} \leq 2C_{\mathrm{mix}}$.
- ☐ Non-twin overlap and Robin resources are bounded by their exact finite-throat ceilings.
- ☐ The drift-diffusion source family is used through the physical Peclet number, not as an arbitrary exponential ansatz.
- ☐ Parent-action gain formulas keep the source/support coherence factor and Cauchy no-go visible.
- ☐ The sech–Gaussian resonance is treated as a benchmark family, not as a replacement for the matched parent-equilibrium theorem.
- ☐ The Family–1 values $\Lambda_\ell = 37$, $\eta = 37$, $\chi_s = 37/2$, and $\kappa = 12321/5$ are marked as reference-branch closures, not universal parent-action theorems.
- ☐ The extracted values $\Theta_w^{(\chi)}$ and $\Theta_w^{(J)}$ are distinguished as natural quadratic weighting and conservative lower envelope.
- ☐ The cancellation of outgoing normalization factors is applied only after the selected outgoing branch is required to hit the quadrupole normalization target.
- ☐ The conclusion $\rho_\alpha = 4/3$, $\zeta_{\mathrm{req}} = 1/3$, $\mathrm{Pe}_{\mathrm{req}} = 0$ is conditional on the minimal isotropic contact-plus-pole precursor.

The citation-safe summary is:

> Stages 037–090 derive the support-completion and explicit Family–1 branch side of the moving-throat response program. They extract the selected-branch continuum data, compress the placement problem to dimensionless ratios and a product law, treat source/loading mismatch

through rank-2 support completion, derive coherent D/N support thresholds, reduce non-twin rescue to overlap, Robin, transport, and parent-gain resources, evaluate the first explicit GNLS wall-shell Family–1 branch, and finally show that the minimal isotropic contact-plus-pole quadrupole precursor selects $\rho_\alpha = 4/3$, $\zeta_{\mathrm{req}} = 1/3$, and $\mathrm{Pe}_{\mathrm{req}} = 0$. The support/source side is therefore not the remaining bottleneck under that precursor; the remaining open task is to derive the precursor itself from the actual grouped real P_2 and geometry branch.

Chapter 4

Geometry Lane, Retarded Closure, Outlet/Core/Mouth Compensation

Stage range: 091–163

Part IV starts after the support/source side is no longer the dominant reduced uncertainty. Parts I–III built the parent wall/BdG/Maxwell/mixed response compiler, the grouped real P_2 normalization language, the selected finite-throat support front end, and the first explicit Family–1 support verdict. The question here is what still prevents the reduced passive/outgoing quadrupole branch from becoming a usable retarded branch.

The answer is progressively narrowed across Stages 091–163. First, the geometry lane is shown not to contaminate the isotropic grouped P_2 carrier at linear order, so the conservative quadrupole module collapses to the forced contact-plus-pole split

$$\widehat{Y}_Q^{\mathrm{cons}}(\omega) = \frac{3}{4} + \frac{1}{4}\frac{1}{1-\omega^2/\Omega_Q^2}. \tag{4.1}$$

The endpoint is again a reduced obstruction packet, not a finished nonlinear branch. The remaining 2.5PN ambiguity collapses to the outgoing normalization factor χ_Q; outlet and core models isolate the viable compensated branch; mouth-source and co-evolving corrections then compress the last first-order obstruction to one microscopic normal coordinate.

Part IV at a glance

Primary question	Once support/source closure is no longer the bottleneck, what reduced datum still blocks a usable retarded quadrupole branch?
Starting inputs	The Part III Family–1 support verdict, the grouped passive/outgoing quadrupole packet, and the reduced outlet/core/mouth closures.
Delivers	The geometry-lane firewall, the factorization to χ_Q, the compensated outlet and core conditions, the canonical mouth branch, and the final off-family defect scalar.
Used by	Part V as the source of the surviving off-family normal coordinate and by all later realization work as the origin of the outgoing-normalization and branch-defect language.
Result anchors	**MTDC-T8**.

Stage packets.

Stages	Packet	Status	Carry-forward output
091–106	Geometry firewall and reduced retarded finish line	Exact Within Closure / Open	Isotropic geometry decoupling, the forced $3/4 + 1/4$ conservative module, factorization of the 2.5PN condition, exact outgoing $l = 2$ DtN fingerprint, and the canonical statement $\chi_Q = 1$ within reduced closure.
107–120	Outlet deformation and compensated core outlet	Exact Within Closure / Reduced / Controlled Reduction	General χ_Q deformation algebra, Robin and mixed outlet audit, compensated Robin–mixed branch, two-channel core realization, and the surviving parent-level core-balance target.
121–145	Mouth-source selection and canonical mouth branch	Exact Within Closure / Reduced / Controlled Reduction / Numerically Located	Geometric mouth selection, positive mouth-source families, boundary-layer and gain laws, self-consistent mouth closure, and the unique finite-bias canonical mouth branch.
146–163	Co-evolving corrections and final defect scalar	Exact Within Closure / Numerically Located / Open	Mouth rigidity corrections, full-profile Family–1 mouth correction, renormalized co-evolving canonical point, defect transport onto the compensated outlet, D/N similarity law, and the off-family normal coordinate $\delta_\perp$.

Claim status. Mixed. The algebraic split, obstruction formulas, harmonic decoupling, outgoing DtN expansion, deformation laws, Schur complements, compensation surfaces, fixed-point laws, and first-order transport maps are Exact Within Closure **once their declared reduced models are adopted. Stable normal-mode reductions, finite D/N tube identifications, Family–1 geometry choices, positive mouth-layer ansatz, self-matched susceptibility closure, and co-evolving fixed-point branch are** Reduced / Controlled Reduction **or** Effective Closure **where explicitly stated. The numerically located co-evolving canonical point and representative correction values are** Numerically Located**. Actual realization by the completed nonlinear moving-throat PDE remains** Open**.**

What this part proves. Part IV shows that the isotropic geometry lane is not the linear obstruction, collapses the 2.5PN retarded issue to the outgoing normalization factor, and then drives the outlet/core/mouth program to a first-order off-family defect scalar.

What this part does not prove. It does not prove that the actual branch lands on the canonical or renormalized compensated point everywhere needed for full realization. The retarded compiler is narrowed sharply, but the physical branch packet is still not closed here.

Why later parts need it. Part V starts from the surviving off-family normal coordinate, and all later realization work inherits the outgoing-normalization, mouth/core, and branch-defect language established in this chapter.

4.1 Block contract and citation boundary

The contract of Part IV is to prevent the retarded quadrupole branch from being overclaimed. It does not assert that the full nonlinear PDE has been solved. It proves that, inside the declared reduced closure sequence, the active uncertainty can be compressed from many apparent degrees of freedom to one scalar branch-realization question at a time.

The chain of compression is

$$\text{geometry lane} \longrightarrow (\epsilon_2, \epsilon_4) \longrightarrow N_Q \longrightarrow \chi_Q \longrightarrow \Delta_Q \longrightarrow \delta\gamma_W \longrightarrow \Xi_{\text{slip}} \longrightarrow \delta_\perp. \tag{4.2}$$

Each arrow is a theorem inside a reduced block, not a statement that the final scalar is already zero on the actual branch.

Theorem 4.1 (Part IV reduced retarded-compression theorem)**.** *Inside the declared grouped-P_2, static-geometry, passive/outgoing, compensated outlet, and co-evolving Family–1 mouth/core closures, the retarded quadrupole verification problem admits the following nested reductions.*

1. *The isotropic geometry lane contributes no dynamic $O(\omega^2)$ or $O(\omega^4)$ contamination to the grouped real P_2 carrier at linear order.*
2. *The conservative quadrupole module is therefore* (4.1)*.*
3. *The point-particle* 2.5*PN normalization condition factorizes as*

$$\widehat{m}_0^2 \chi_Q N_Q = 1. \tag{4.3}$$

   On the natural source-map branch, the remaining retarded obstruction is $\Delta_Q = \chi_Q - 1$.
4. *The canonical outgoing $l = 2$ DtN branch has $\chi_Q = 1$.*
5. *In the explicit compensated Robin–mixed outlet class, first-order even preservation forces $\delta\mathfrak{g} = 0$ and $\delta\kappa_W = 0$, so the remaining first-order defect is purely odd.*
6. *In the concrete D/N mixed-tube realization, that odd defect is zero whenever the tangential deformation preserves the D/N similarity law.*
7. *The first off-family deviation from the exact parent compensation family is measured by a single normal coordinate $\delta_\perp$.*

Thus the end of Part IV is not a broad family of remaining outlet defects. It is the concrete microscopic question whether the true branch has $\delta_\perp = 0$, or if not, what value that scalar takes.

Proof. Stages 091–099 prove the geometry and conservative-normalization reductions. Stages 100–106 prove the retarded factorization and canonical DtN coefficient. Stages 107–124 classify admissible low-frequency outlet/core deformations and identify the compensated Robin–mixed class. Stages 125–145 derive parent and mouth-layer parameters for that class and select the regular mouth branch. Stages 146–163 linearize the remaining defect and show that the outlet, D/N similarity, and parent compensation defects are all images of the same off-family normal coordinate. □

4.2 Geometry-lane firewall

4.2.1 Why the geometry lane had to be checked

The conservative side enters Part IV with two facts already frozen. First, the higher conservative quadrupole payload lives in the grouped real P_2 bundle. Second, the conservative PN assembly also contains a geometry completion. If the geometry completion were dynamically active through $O(\omega^4)$, then the clean one-pole grouped branch used in the retarded bridge could be contaminated before the outgoing branch was even tested.

The check therefore begins with the most general reduced isotropic ansatz needed at this order:

$$K_Q^{\rm cons}(\omega) = K_g(\omega) + K_{P2}(\omega), \qquad K_{P2}(\omega) = \frac{K_{\rm pole}}{1-\omega^2/\Omega_Q^2}, \tag{4.4}$$

where

$$K_g(\omega) = K_{g,0} + K_{g,2}\omega^2 + K_{g,4}\omega^4 + O(\omega^6). \tag{4.5}$$

The minimal isotropic branch identity is

$$K_0K_4 = 4K_2^2. \tag{4.6}$$

If $K_{g,2} = K_{g,4} = 0$, inserting (4.4) into (4.6) gives

$$K_{g,0} = 3K_{\rm pole}, \qquad K_{\rm pole} = \frac{K_0}{4}, \qquad K_{g,0} = \frac{3K_0}{4}, \tag{4.7}$$

which is exactly the conservative split in (4.1).

4.2.2 Exact obstruction if geometry is dynamic

If the geometry lane carries dynamic even moments, the simple split is replaced by an exact obstruction formula. Define

$$\epsilon_2 := \frac{\Omega_Q^2 K_{g,2}}{K_{\rm pole}}, \qquad \epsilon_4 := \frac{\Omega_Q^4 K_{g,4}}{K_{\rm pole}}. \tag{4.8}$$

Then the static pole fraction becomes

$$c_{\rm pole} = \frac{K_{\rm pole}}{K_0} = \frac{1+\epsilon_4}{4(1+\epsilon_2)^2}. \tag{4.9}$$

For small contamination,

$$c_{\rm pole} = \frac{1}{4}\left(1+\epsilon_4 - 2\epsilon_2 + O(\epsilon^2)\right). \tag{4.10}$$

Thus the static-geometry hypothesis is not a cosmetic simplification. It is the exact condition under which the pole fraction is 1/4.

4.2.3 Isotropic decoupling

The distributed wall action on an isotropic reference throat is block diagonal in angular momentum. With

$$\eta(\Omega, w, t) = g(w,t)Y_{00}(\Omega) + \sum_A q_A(w,t)Y_{2A}(\Omega) + \cdots, \tag{4.11}$$

where A ranges over the real quadrupole harmonics, every possible quadratic cross term between g and q_A is proportional to one of

$$\int_{S^2} Y_{00} Y_{2A}\, d\Omega, \qquad \int_{S^2} \nabla_{S^2} Y_{00} \cdot \nabla_{S^2} Y_{2A}\, d\Omega, \qquad \int_{S^2} Y_{00}(-\Delta_{S^2}) Y_{2A}\, d\Omega. \tag{4.12}$$

All three vanish. Therefore

$$M_{0\leftrightarrow 2} = 0 \tag{4.13}$$

inside the isotropic quadratic wall theory, and the geometry contamination parameters vanish:

$$K_{g,2} = K_{g,4} = 0, \qquad \epsilon_2 = \epsilon_4 = 0. \tag{4.14}$$

Proposition 4.2 (Second-order onset of geometry contamination)**.** *If weak anisotropy or operator noncommutation generates a bilinear mixing term $\chi M_0 q g$ between one grouped quadrupole coordinate and one scalar geometry coordinate, then after eliminating g, the first dynamic geometry contamination is quadratic in χ.*

Proof. Let

$$D_q(\omega) = K_{\mathrm{stat}} + \frac{K_{\mathrm{pole}}}{1 - \omega^2/\Omega_Q^2}, \qquad D_g(\omega) = G_0 + G_2\omega^2 + G_4\omega^4 + O(\omega^6). \tag{4.15}$$

The Schur complement is

$$D_{\mathrm{eff}}(\omega) = D_q(\omega) - \frac{\chi^2 M_0^2}{D_g(\omega)}. \tag{4.16}$$

Expanding D_g^{-1} gives

$$K_{g,2}^{\mathrm{eff}} = \frac{\chi^2 M_0^2 G_2}{G_0^2}, \qquad K_{g,4}^{\mathrm{eff}} = \frac{\chi^2 M_0^2 (G_0 G_4 - G_2^2)}{G_0^3}. \tag{4.17}$$

Therefore $\epsilon_2, \epsilon_4 = O(\chi^2)$. □

The geometry-lane firewall is therefore an exact reduced selection rule: on the isotropic branch the geometry lane is dynamically inert through the relevant even orders, while weak contamination is second order in the explicit mixing source.

4.3 Retarded 2.5PN collapse

4.3.1 From conservative module to one scalar defect

With the geometry-lane check complete, the actual isotropic conservative quadrupole response can be written as

$$\overline{K}_Q^{\mathrm{cons}}(\omega) = \overline{K}_0 \left[\frac{3}{4} + \frac{1}{4} \frac{1}{1 - \omega^2/\Omega_Q^2} \right]. \tag{4.18}$$

Its even moments are

$$\overline{K}_2 = \frac{\overline{K}_0}{4\Omega_Q^2}, \qquad \overline{K}_4 = \frac{\overline{K}_0}{4\Omega_Q^4}. \tag{4.19}$$

The target static normalization is

$$\overline{K}_0^{\mathrm{target}} = \frac{64 G \Omega_Q^5}{45 c^5} = \frac{54 G c_s^5}{5 a^5 c^5}, \qquad \Omega_Q = \frac{3 c_s}{2a}. \tag{4.20}$$

Thus the conservative defect is the scalar

$$N_Q := \frac{\overline{K}_0}{\overline{K}_0^{\text{target}}}. \tag{4.21}$$

All even ratios scale by the same N_Q. The Family–1 support side is automatic on this actual isotropic branch because the forced branch value $\rho_\alpha = 4/3$ gives

$$\zeta_{\text{req}}^{\text{act}}(\epsilon_{\text{blk}}) = \frac{1}{3 - 2\epsilon_{\text{blk}}}, \tag{4.22}$$

which is below the Family–1 constructive support ceiling throughout the admissible blocked regime.

4.3.2 Retarded factorization

The retarded one-pole grouped branch is written

$$\widehat{Y}_Q^{\text{ret}}(\omega) = \frac{3}{4} + \frac{1}{4}\frac{1}{1 - \omega^2/\Omega_Q^2 - i\chi_Q\sigma_Q^{\text{can}}\omega^5} + O(\omega^6), \tag{4.23}$$

where

$$\sigma_Q^{\text{can}} = \frac{9}{8\Omega_Q^5} = \frac{4a^5}{27c_s^5}. \tag{4.24}$$

The odd coefficient is

$$\overline{\Gamma}_5 = \chi_Q \frac{9\overline{K}_0}{32\Omega_Q^5}. \tag{4.25}$$

Consequently

$$\frac{\overline{\Gamma}_5}{2G/(5c^5)} = \chi_Q N_Q, \qquad \widehat{m}_0^2\chi_Q N_Q = 1. \tag{4.26}$$

On the natural point-particle source-map branch, $\widehat{m}_0 = 1 + O(a^2/r^2)$, so in the strict point-particle limit the remaining obstruction is

$$\Delta_Q := \chi_Q - 1. \tag{4.27}$$

Higher odd denominator terms beginning at $O(\omega^7)$ are invisible to the point-particle 2.5PN coefficient.

4.3.3 Canonical outgoing DtN branch

Let

$$z := \frac{a\omega}{c_s}. \tag{4.28}$$

The exact outgoing spherical $l = 2$ DtN operator is

$$\Lambda_2^{\text{out}}(z) = z\frac{d}{dz}\ln h_2^{(1)}(z) = -3 + \frac{z^2}{3} + \frac{z^4}{9} + i\frac{z^5}{9} + O(z^6). \tag{4.29}$$

Normalizing by the static slot gives

$$\widehat{Y}_2^{\text{out}}(z) = \frac{-3}{\Lambda_2^{\text{out}}(z)} = 1 + \frac{z^2}{9} + \frac{4z^4}{81} + i\frac{z^5}{27} + O(z^6). \tag{4.30}$$

Comparison with (4.23) fixes

$$\chi_Q = 1 \tag{4.31}$$

for the canonical compact passive/outgoing DtN branch. The canonical invariant coefficients are then

$$\overline{K}_0 = \frac{54Gc_s^5}{5a^5c^5}, \qquad \overline{K}_2 = \frac{6Gc_s^3}{5a^3c^5}, \qquad \overline{K}_4 = \frac{8Gc_s}{15ac^5}, \qquad \overline{\Gamma}_5 = \frac{2G}{5c^5}. \tag{4.32}$$

Theorem 4.3 (Conditional reduced 2.5PN closure)**.** *Within the reduced isotropic grouped-P_2 hierarchy, the nonspinning point-particle* 2.5*PN branch closes if the actual passive/outgoing moving-throat branch realizes the canonical outgoing DtN coefficient* $\chi_Q = 1$ *and the natural source-map limit* $\widehat{m}_0 \to 1$*. If the actual branch has* $\chi_Q \neq 1$*, the entire reduced failure is measured by* $\Delta_Q = \chi_Q - 1$ *at leading order.*

4.4 Outlet DtN and Robin outlet tests

4.4.1 General isotropic deformation algebra

The remaining branch-selection issue is whether the true moving-throat outlet is the canonical branch or a deformed one. The first explicit deformation family is

$$\Lambda_2^{\mathrm{def}}(z) = S\Lambda_2^{\mathrm{out}}(\beta z) + \Sigma_0 + \Sigma_2 z^2 + \Sigma_4 z^4 + i\Sigma_5 z^5 + O(z^6). \tag{4.33}$$

Writing the expansion coefficients as L_0, L_2, L_4, L_5, the normalized response is

$$\widehat{Y}_2^{\mathrm{def}}(z) = 1 - \frac{L_2}{L_0} z^2 + \left(\frac{L_2^2}{L_0^2} - \frac{L_4}{L_0}\right) z^4 - i\frac{L_5}{L_0} z^5 + O(z^6). \tag{4.34}$$

After imposing the canonical even fingerprint, the exact odd normalization is

$$\chi_Q = \frac{3(S\beta^5 + 9\Sigma_5)}{3S - \Sigma_0}, \tag{4.35}$$

so

$$\chi_Q - 1 = \frac{3S(\beta^5 - 1) + \Sigma_0 + 27\Sigma_5}{3S - \Sigma_0}. \tag{4.36}$$

Linearizing with

$$S = 1 + \varepsilon s, \qquad \beta = 1 + \varepsilon b, \qquad \Sigma_0 = \varepsilon a_0, \qquad \Sigma_5 = \varepsilon a_5, \tag{4.37}$$

while keeping the even slots canonical gives

$$\Delta_Q = 5b + \frac{a_0}{3} + 9a_5 + O(\varepsilon^2). \tag{4.38}$$

Pure overall scaling drops out. Pure argument rescaling cannot move χ_Q without also moving the even fingerprint unless $\beta = 1$. Additive throat-core slots are therefore the first genuinely dangerous outlet data.

4.4.2 Robin and mixed-pole tests

A pure isotropic Robin outlet is

$$\Lambda_2^{\mathrm{R}}(z) = \Lambda_2^{\mathrm{out}}(z) + \rho_R. \tag{4.39}$$

It gives

$$\chi_Q^{\mathrm{R}} = \frac{3}{3 - \rho_R}, \tag{4.40}$$

so it generically shifts the canonical branch.

A standalone hidden mixed side-channel pole is

$$\Lambda_2^{\mathrm{mix}}(z) = \Lambda_2^{\mathrm{out}}(z) - \frac{\sigma_W}{1 - \kappa_W z^2 - i\gamma_W z^5} + O(z^6). \tag{4.41}$$

The canonical even conditions force $\kappa_W = -1/9$, and then $\sigma_W = 0$. Thus a naive standalone passive hidden pole cannot be the actual branch if the conservative even fingerprint is already fixed.

4.4.3 Compensated Robin–mixed outlet

The combined outlet is

$$\Lambda_2^{\mathrm{hyb}}(z) = \Lambda_2^{\mathrm{out}}(z) + \rho_R - \frac{\sigma_W}{1 - \kappa_W z^2 - i\gamma_W z^5} + O(z^6). \tag{4.42}$$

The exact canonical-even conditions have two solutions:

$$(\rho_R, \kappa_W) = (\sigma_W, 0), \qquad (\rho_R, \kappa_W) = (4\sigma_W, 1/3). \tag{4.43}$$

The first is a trivial static cancellation. The second is the nontrivial compensated branch. On it,

$$\chi_Q^{\mathrm{hyb}} = \frac{1 - 9\sigma_W\gamma_W}{1 - \sigma_W}, \tag{4.44}$$

so the canonical outgoing value is preserved iff

$$\gamma_W = \frac{1}{9}. \tag{4.45}$$

With (4.45), the hybrid outlet collapses to a harmless pure scale deformation through the retained order.

4.5 Mixed side-channel pole realization

4.5.1 Concrete two-channel core model

The reduced hybrid outlet is realized by a concrete shell/mixed core. Let $u(\omega)$ be the isotropic $l = 2$ mouth amplitude, and introduce a static shell coordinate s and a mixed side-channel coordinate q. The core equations are

$$\begin{pmatrix} K_s & \lambda \\ \lambda & -K_q D_W^{\mathrm{bare}}(z) \end{pmatrix} \begin{pmatrix} s \\ q \end{pmatrix} = u \begin{pmatrix} g_s \\ g_q \end{pmatrix}, \qquad D_W^{\mathrm{bare}}(z) = 1 - \kappa_0 z^2 - i\gamma_0 z^5 + O(z^6). \tag{4.46}$$

Eliminating (s, q) gives

$$\delta\Lambda_{\mathrm{core}}(z) = \frac{g_s^2}{K_s} - \frac{(K_s g_q - \lambda g_s)^2}{K_s(K_s K_q D_W^{\mathrm{bare}}(z) + \lambda^2)}. \tag{4.47}$$

With

$$r_c := \frac{\lambda^2}{K_s K_q}, \tag{4.48}$$

this has the reduced form

$$\delta\Lambda_{\mathrm{core}}(z) = \rho_c - \frac{\sigma_c}{1 - \kappa_c z^2 - i\gamma_c z^5} + O(z^6), \tag{4.49}$$

where

$$\rho_c = \frac{g_s^2}{K_s}, \qquad \sigma_c = \frac{(K_s g_q - \lambda g_s)^2}{K_s^2 K_q (1 + r_c)}, \qquad \kappa_c = \frac{\kappa_0}{1 + r_c}, \qquad \gamma_c = \frac{\gamma_0}{1 + r_c}. \tag{4.50}$$

4.5.2 Core-balance theorem and D/N tube

The nontrivial compensated branch requires

$$\rho_c = 4\sigma_c, \qquad \kappa_c = \frac{1}{3}, \qquad \gamma_c = \frac{1}{9}. \tag{4.51}$$

The static balance condition is exactly

$$g_s^2(K_sK_q + \lambda^2) = 4(K_sg_q - \lambda g_s)^2. \tag{4.52}$$

Equivalently, defining

$$\mathfrak{r} := \frac{\lambda}{\sqrt{K_sK_q}}, \qquad \mathfrak{g} := \frac{g_q\sqrt{K_s}}{g_s\sqrt{K_q}}, \tag{4.53}$$

one obtains the parent compensation family

$$1 + \mathfrak{r}^2 = 4(\mathfrak{g} - \mathfrak{r})^2, \qquad \mathfrak{g}_\pm(\mathfrak{r}) = \mathfrak{r} \pm \frac{1}{2}\sqrt{1+\mathfrak{r}^2}. \tag{4.54}$$

The finite D/N mixed-tube realization gives

$$L_W = \frac{\pi a}{2}\sqrt{\frac{1+r_c}{3}} = \frac{\pi a}{2}\sqrt{\frac{1+\mathfrak{r}^2}{3}}. \tag{4.55}$$

Identifying this tube with the actual throat span L fixes the branch value

$$\mathfrak{r}_{\rm geom}\left(\frac{L}{a}\right) = \sqrt{\frac{12}{\pi^2}\left(\frac{L}{a}\right)^2 - 1}. \tag{4.56}$$

On the Family–1 preferred aspect ratio $L/a = 37/20$,

$$\mathfrak{r}_{F1} = \frac{\sqrt{4107 - 100\pi^2}}{10\pi} \approx 1.77799353547498. \tag{4.57}$$

The lower compensated mouth-coupling branch is then

$$\mathfrak{g}_-^{F1} = \mathfrak{r}_{F1} - \frac{1}{2}\sqrt{1+\mathfrak{r}_{F1}^2} \approx 0.758035078944663. \tag{4.58}$$

The upper branch is $\mathfrak{g}_+^{F1} \approx 2.79795199200529$.

4.5.3 Parent-action overlap extraction

The abstract core parameters are tied to parent-action overlaps. In the first GNLS plus localized-Maxwell throat-core closure,

$$K_s = 4\pi a^2\left(\frac{H_w\ell}{3} + \frac{\hbar^2}{15m_\psi\rho_w\ell}\right), \qquad K_s = \frac{3\pi a^2\hbar^2}{5m_\psi\rho_w\ell} \quad \text{on the healing-locked branch,} \tag{4.59}$$

$$K_q = \frac{\mathcal{Z}_q}{\mu_0}\frac{\pi^2c_s^2}{4L_W^2}, \tag{4.60}$$

$$\lambda = -q_*v_{w0}\mathcal{I}_{sq}, \tag{4.61}$$

and

$$g_s = \mathcal{T}_m\frac{4\pi a^2\ell}{3}, \qquad g_q = \frac{\mathcal{Z}_q}{\mu_0}\frac{\pi}{\sqrt{2}\,L_W^{3/2}}. \tag{4.62}$$

These formulas are the parent-action path by which a future full branch calculation must decide whether (4.54) is actually realized.

4.6 Core-to-mouth compensation law

4.6.1 Positive mouth-source selection

The core compensation family selects a mouth-coupling ratio. The first physical filter is positivity of the localized mouth source. For a positive normalized axial source $\sigma(z)$ on $[0, L]$, the normalized mouth-bias factor is

$$\mathfrak{g}[\sigma] = \int_0^L \sigma(z) \cos\left(\frac{\pi z}{2L}\right) dz. \tag{4.63}$$

Since the cosine lies between 0 and 1 on the interval,

$$0 \leq \mathfrak{g}[\sigma] \leq 1. \tag{4.64}$$

Therefore the upper Family–1 branch is impossible under positive localized mouth sourcing, while the lower branch (4.58) is admissible.

The self-matched derivative source $\sigma_{\mathrm{match}}(z) = k\cos(kz)$, with $k = \pi/(2L)$, gives

$$\mathfrak{g}_{\mathrm{match}} = \frac{\pi}{4} \approx 0.785398163397448, \tag{4.65}$$

only about 3.61% in traction away from the lower compensated value. A convex positive family interpolating between this derivative profile and the uniform profile reaches the exact lower branch at a modest broadening fraction.

4.6.2 Electrochemical boundary layer

The positive exponential source family is not an arbitrary fit. It follows from a local mouth free energy and zero-flux Onsager law. Linearizing the parent mouth potential as

$$V_m(z) \approx V_1 z, \qquad V_1 = \partial_z \delta V_{\mathrm{conf}}\big|_{\mathrm{m}} - q_* \, \partial_z A_0\big|_{\mathrm{m}}, \tag{4.66}$$

the electrochemical potential is

$$\mu_\sigma^{\mathrm{chem}}(z) = \Theta_\sigma \ln \frac{\sigma}{\sigma_*} + V_1 z. \tag{4.67}$$

The zero-flux stationary solution is

$$\sigma_\Pi(z) = \frac{\Pi e^{-\Pi z/L}}{L(1 - e^{-\Pi})}, \qquad \Pi := \frac{V_1 L}{\Theta_\sigma} > 0. \tag{4.68}$$

The exact bias map is

$$\mathfrak{g}_\Pi = \frac{2\Pi(2\Pi e^{\Pi} + \pi)}{(4\Pi^2 + \pi^2)(e^{\Pi} - 1)}. \tag{4.69}$$

It is strictly increasing from $2/\pi$ to 1, and the Family–1 lower branch is reached at

$$\Pi_* \approx 1.50882951349316. \tag{4.70}$$

Equivalently, the parent threshold is

$$\partial_z \delta V_{\mathrm{conf}}\big|_{\mathrm{m}} - q_* \, \partial_z A_0\big|_{\mathrm{m}} = \Pi_* \frac{\Theta_\sigma}{L}. \tag{4.71}$$

4.6.3 Coupled mouth-layer fixed point

The effective bias Π is then replaced by a coupled boundary-layer solve. For a two-component field $\mathbf{U} = (U_s, U_q)^T$,

$$\left[-\partial_x^2 \mathbf{I} + \mathbf{K}\right] \mathbf{U}(x) = \Sigma_\Pi(x)\mathbf{G}, \qquad \mathbf{U}(0) = 0, \qquad \mathbf{U}'(1) = 0. \tag{4.72}$$

After diagonalizing $\mathbf{K}$, the scalar D/N response kernel is

$$\mathcal{S}(\Pi, \kappa) = \frac{\Pi\left[\kappa \tanh \kappa + \Pi\left(e^{-\Pi} \operatorname{sech} \kappa - 1\right)\right]}{(1 - e^{-\Pi})(\kappa^2 - \Pi^2)}. \tag{4.73}$$

The full fixed-point law is

$$\Pi = \sum_{\alpha=\pm} M_\alpha \mathcal{S}(\Pi, \kappa_\alpha). \tag{4.74}$$

On the first Family–1 shell plus mixed D/N branch,

$$\Pi = M_s + M_q \mathcal{S}_q(\Pi), \qquad \mathcal{S}_q(\Pi) := \mathcal{S}\left(\Pi, \frac{\pi}{2}\right). \tag{4.75}$$

Inheriting the compensated outlet ratio gives

$$M_s = 4\Sigma_m, \qquad M_q = -\Sigma_m, \tag{4.76}$$

so

$$\Pi = \Sigma_m \left[4 - \mathcal{S}_q(\Pi)\right]. \tag{4.77}$$

At the canonical point,

$$\Sigma_m^* = \frac{\Pi_*}{4 - \mathcal{S}_q(\Pi_*)} \approx 0.451485277739090. \tag{4.78}$$

4.7 Mouth boundary-layer source theorem

4.7.1 From gains to parent core quantities

Embedding the same core Schur-complement weights into the mouth electrochemical free energy gives explicit gains

$$M_s = \frac{L g_s^2}{K_s \Theta_\sigma}, \qquad M_q = -\frac{L(K_s g_q - \lambda g_s)^2}{K_s(K_s K_q + \lambda^2)\Theta_\sigma}. \tag{4.79}$$

In normalized variables,

$$R_q := -\frac{M_q}{M_s} = \frac{(\mathfrak{g}_c - \mathfrak{r})^2}{1 + \mathfrak{r}^2}, \qquad M_s = \Sigma_0. \tag{4.80}$$

On the exact compensation family, $R_q = 1/4$. Under the self-matched mouth susceptibility closure,

$$\Sigma_0 = M_s = \frac{20}{9} \widehat{T}_m^2. \tag{4.81}$$

The self-consistent Family–1 mouth branch identifies the core mouth ratio with the source bias:

$$\mathfrak{g}_c = \mathfrak{g}_\Pi, \qquad R_q(\Pi) = \frac{(\mathfrak{g}_\Pi - \mathfrak{r}_{F1})^2}{1 + \mathfrak{r}_{F1}^2}. \tag{4.82}$$

The scalar branch law becomes

$$\Pi = \Sigma_0 \left[1 - R_q(\Pi)\mathcal{S}_q(\Pi)\right], \qquad \Sigma_0(\Pi) = \frac{\Pi}{1 - R_q(\Pi)\mathcal{S}_q(\Pi)}. \tag{4.83}$$

At Π_*,

$$R_q(\Pi_*) = \frac{1}{4}, \qquad \Sigma_0(\Pi_*) \approx 1.80594111095636, \qquad \widehat{T}_m(\Pi_*) \approx 0.901484054174205. \tag{4.84}$$

4.7.2 Regularity theorem

The explicit exponential mouth family has

$$0 < \mathfrak{g}_\Pi < 1 \qquad (\Pi > 0), \tag{4.85}$$

and $\mathfrak{g}_\Pi \to 1$ only as $\Pi \to \infty$. The equal-normalized branch $\mathfrak{g}_c = 1$ is therefore a singular point-source limit. Since $\widehat{T}_m \sim C\Pi^{1/2}$ in that limit, it is not a regular finite-traction mouth state.

Theorem 4.4 (Unique regular Family–1 canonical mouth branch)**.** *Inside the explicit Family–1 positive exponential mouth-layer closure, the upper compensated branch is excluded by positivity, the equal-normalized branch is singular, and the lower compensated branch is the unique regular finite-bias, finite-traction branch preserving the canonical outgoing quadrupole fingerprint.*

4.8 Family-1 mouth correction

4.8.1 First-order positive deformations

After branch selection, the remaining mouth-side uncertainty is finite correction around the unique regular branch. Let

$$\Sigma_*(x) = \frac{\Pi_* e^{-\Pi_* x}}{1 - e^{-\Pi_*}} \tag{4.86}$$

and consider a positive normalized deformation

$$\Sigma_\epsilon(x) = (1-\epsilon)\Sigma_*(x) + \epsilon\varsigma(x), \qquad \varsigma \geq 0, \qquad \int_0^1 \varsigma(x)\, dx = 1. \tag{4.87}$$

Only two moments matter at first order:

$$\bar{g}[\sigma] = \int_0^1 \sigma(x)c(x)\, dx, \qquad \bar{S}[\sigma] = \int_0^1 \sigma(x)K_q(x)\, dx, \tag{4.88}$$

with

$$c(x) = \cos\frac{\pi x}{2}, \qquad K_q(x) = \frac{\cosh\left(\frac{\pi}{2}(1-x)\right)}{\cosh(\pi/2)}. \tag{4.89}$$

Retuning Π to stay on the canonical lower branch gives

$$\delta\Pi = -\epsilon\frac{\bar{g}_\varsigma - \mathfrak{g}_*}{\mathfrak{g}'_*}, \qquad \mathfrak{g}'_* \approx 0.0714453558083195. \tag{4.90}$$

The traction shift is

$$\delta\widehat{T}_m = \epsilon\left[A_T(\bar{g}_\varsigma - \mathfrak{g}_*) + B_T(\bar{S}_\varsigma - \mathcal{S}_*)\right], \tag{4.91}$$

where

$$A_T \approx -4.27263956256927, \qquad B_T \approx 0.134875005736706. \tag{4.92}$$

Equivalently, all first-order source-shape sensitivity is the projection against one centered rigidity kernel.

4.8.2 Full-profile residual

The actual full mouth potential generated by the shell plus mixed D/N channels is $\Phi_*(x)$. Define the tangent-subtracted residual

$$R_*(x) := \Phi_*(x) - \Pi_* x. \tag{4.93}$$

It satisfies

$$R_*(0) = R_*'(0) = 0, \qquad R_*''(0) < 0, \tag{4.94}$$

so the full source is broader than the tangent exponential near the mouth. Expanding

$$\Sigma_{\mathrm{act}}(x) = \Sigma_*(x)\left[1 - \widetilde{R}_*(x)\right] + O(R_*^2), \qquad \widetilde{R}_* = R_* - \langle R_* \rangle_*, \tag{4.95}$$

gives the selected moment shifts

$$\delta \mathfrak{g}_{\mathrm{act}} = -\operatorname{Cov}_*(c, R_*), \qquad \delta \mathcal{S}_{\mathrm{act}} = -\operatorname{Cov}_*(K_q, R_*). \tag{4.96}$$

On the explicit Family–1 branch,

$$\operatorname{Cov}_*(c, R_*) \approx 0.0648069687666328, \qquad \operatorname{Cov}_*(K_q, R_*) \approx 0.0388718368650403. \tag{4.97}$$

Thus the first-order correction broadens the source and shifts the mouth-only canonical point upward:

$$\Pi_{\mathrm{corr}} \approx 2.41591392833607, \qquad \widehat{T}_{m,\mathrm{corr}} \approx 1.17313803363654. \tag{4.98}$$

A one-step nonlinear iterate gives a slightly stronger but directionally consistent shift.

4.9 Co-evolving core-mouth fixed point

4.9.1 Exact co-evolving map

The fixed-core mouth correction is then replaced by a co-evolving map. For any positive normalized Σ, define

$$\mathfrak{g}[\Sigma] = \int_0^1 \Sigma(x) c(x)\, dx, \qquad \mathcal{S}[\Sigma] = \int_0^1 \Sigma(x) K_q(x)\, dx. \tag{4.99}$$

The core loading ratio is

$$\mathcal{R}[\Sigma] = \frac{(\mathfrak{g}[\Sigma] - \mathfrak{r}_{F1})^2}{1 + \mathfrak{r}_{F1}^2}. \tag{4.100}$$

With the shell and mixed Green operators $\mathcal{T}_s$ and $\mathcal{T}_q$, the full potential is

$$\Phi_{\Sigma_0}[\Sigma](x) = \Sigma_0 \left[\mathcal{T}_s[\Sigma](x) - \mathcal{R}[\Sigma]\mathcal{T}_q[\Sigma](x)\right]. \tag{4.101}$$

The nonlinear fixed-point equation is

$$\Sigma(x) = \frac{e^{-\Phi_{\Sigma_0}[\Sigma](x)}}{\int_0^1 e^{-\Phi_{\Sigma_0}[\Sigma](y)}\, dy}. \tag{4.102}$$

The canonical compensation condition is

$$\mathfrak{g}[\Sigma] = \mathfrak{g}_*, \qquad \mathcal{R}[\Sigma] = \frac{1}{4}. \tag{4.103}$$

4.9.2 Renormalized canonical point

At the old canonical traction, the co-evolving fixed point remains positive but moves off exact compensation:

$$\mathfrak{g}_{\mathrm{fp}} \approx 0.693352419668063, \qquad \mathcal{S}_{\mathrm{fp}} \approx 0.621601316751401, \qquad \mathcal{R}_{\mathrm{fp}} \approx 0.282713904908238. \tag{4.104}$$

Restoring exact lower compensation requires a numerically located renormalized gain

$$\Sigma_0^{\mathrm{can}} \approx 4.651033550168876, \qquad \widehat{T}_{m,\mathrm{can}} \approx 1.446708366456762, \tag{4.105}$$

with

$$\mathfrak{g}_{\mathrm{can}} = \mathfrak{g}_*, \qquad \mathcal{R}_{\mathrm{can}} = \frac{1}{4}, \qquad \mathcal{S}_{\mathrm{can}} \approx 0.670362115673462, \qquad \Pi_{\mathrm{can}} \approx 3.871564377479009. \tag{4.106}$$

This is a Numerically Located placement inside the exact co-evolving map, not a closed-form theorem of the full PDE.

4.10 Linear defect transport

4.10.1 Mouth/core transport around the renormalized point

Linearizing the exact Family–1 loading ratio gives

$$\delta\mathcal{R} = -\frac{\delta\mathfrak{g}}{\sqrt{1+\mathfrak{r}_{F1}^2}} + O(\delta\mathfrak{g}^2) \approx -0.490216044387626\,\delta\mathfrak{g}. \tag{4.107}$$

The gains transport as

$$\delta M_s = \delta\Sigma_0, \qquad \delta M_q = -\frac{1}{4}\delta\Sigma_0 + \frac{\Sigma_0^{\mathrm{can}}}{\sqrt{1+\mathfrak{r}_{F1}^2}}\delta\mathfrak{g}. \tag{4.108}$$

The mouth-bias transport is

$$\delta\Pi = \left(1 - \frac{1}{4}\mathcal{S}_{\mathrm{can}}\right)\delta\Sigma_0 - \frac{\Sigma_0^{\mathrm{can}}}{4}\delta\mathcal{S} + \frac{\Sigma_0^{\mathrm{can}}\mathcal{S}_{\mathrm{can}}}{\sqrt{1+\mathfrak{r}_{F1}^2}}\delta\mathfrak{g}, \tag{4.109}$$

or numerically

$$\delta\Pi \approx 0.832409471081635\,\delta\Sigma_0 - 1.16275838754222\,\delta\mathcal{S} + 1.52843317823248\,\delta\mathfrak{g}. \tag{4.110}$$

4.10.2 Projection into the compensated outlet

The compensated outlet loads are

$$\rho_R = \Xi M_s, \qquad \sigma_W = -\Xi M_q, \qquad \Xi = \frac{\Theta_\sigma}{L}. \tag{4.111}$$

Define the off-compensation scalar

$$\delta\mathcal{C} := \delta\rho_R - 4\delta\sigma_W. \tag{4.112}$$

Then

$$\delta\mathcal{C} = \frac{16\sigma_*}{\sqrt{1+\mathfrak{r}_{F1}^2}}\delta\mathfrak{g}. \tag{4.113}$$

Linearizing the hybrid outlet gives

$$\delta E_2 = \frac{\delta\mathcal{C} - 9\sigma_*\delta\kappa_W}{27(1-\sigma_*)}, \qquad \delta E_4 = \frac{5\delta\mathcal{C} - 72\sigma_*\delta\kappa_W}{243(1-\sigma_*)}, \tag{4.114}$$

$$\Delta_Q = \frac{\delta\mathcal{C} - 27\sigma_*\delta\gamma_W}{3(1-\sigma_*)}. \tag{4.115}$$

Canonical even preservation $\delta E_2 = \delta E_4 = 0$ forces

$$\delta\mathcal{C} = 0, \qquad \delta\kappa_W = 0, \qquad \delta\mathfrak{g} = 0. \tag{4.116}$$

The remaining first-order defect is therefore

$$\Delta_Q = -\frac{9\sigma_*}{1-\sigma_*}\delta\gamma_W, \qquad N_Q - 1 = \frac{9\sigma_*}{1-\sigma_*}\delta\gamma_W. \tag{4.117}$$

4.10.3 Bare mixed-port slippage and D/N similarity

The concrete core identities give

$$\kappa_W = \frac{\kappa_0}{1+r_c}, \qquad \gamma_W = \frac{\gamma_0}{1+r_c}. \tag{4.118}$$

On the compensated branch,

$$\kappa_0 = \frac{1+r_c}{3}, \qquad \gamma_0 = \frac{1+r_c}{9}. \tag{4.119}$$

Under the canonical-even gate $\delta\kappa_W = 0$, the odd defect is

$$\delta\gamma_W = \frac{1}{1+\mathfrak{r}_{F1}^2}\left(\delta\gamma_0 - \frac{1}{3}\delta\kappa_0\right). \tag{4.120}$$

Thus a pure-scale bare mixed-port deformation is harmless.

The D/N tube gives

$$\kappa_0 = \frac{4L_W^2}{\pi^2 a^2}. \tag{4.121}$$

The first-order bare slippage is equivalently the D/N similarity mismatch

$$\Xi_{\text{slip}} := \Xi_\gamma - 2\Xi_L, \qquad \Xi_\gamma = \frac{\delta\ln\gamma_0}{\delta\Pi_{\tan}}, \qquad \Xi_L = \frac{\delta\ln(L_W/a)}{\delta\Pi_{\tan}}. \tag{4.122}$$

Then

$$\Delta_Q = -\frac{\sigma_*}{1-\sigma_*}\Xi_{\text{slip}}\,\delta\Pi_{\tan}, \qquad N_Q - 1 = \frac{\sigma_*}{1-\sigma_*}\Xi_{\text{slip}}\,\delta\Pi_{\tan}. \tag{4.123}$$

Here

$$\delta\Pi_{\tan} = 0.832409471081635\,\delta\Sigma_0 - 1.16275838754222\,\delta\mathcal{S} \tag{4.124}$$

when $\delta\mathfrak{g} = 0$.

4.10.4 Parent compensation-surface rigidity and off-family normal coordinate

On the exact parent compensation family, both the D/N tube ratio and the bare odd normalization are functions of $\mathfrak{r}$:

$$\frac{L_W}{a} = \frac{\pi}{2}\sqrt{\frac{1+\mathfrak{r}^2}{3}}, \qquad \gamma_0 = \frac{1+\mathfrak{r}^2}{9}. \tag{4.125}$$

Therefore

$$\delta \ln \gamma_0 - 2\delta \ln\left(\frac{L_W}{a}\right) = 0, \qquad \Xi_{\mathrm{slip}} = 0. \tag{4.126}$$

Furthermore, on the lower branch, the canonical-even gate $\delta\mathfrak{g} = 0$ forces $\delta\mathfrak{r} = 0$ because $d\mathfrak{g}_-/d\mathfrak{r} > 0$. Thus $\Delta_Q = 0$ at first order for tangential motion inside the exact parent compensation family.

The normal coordinate measuring departure from that family is

$$\delta_\perp := \delta\mathfrak{g} - \mathfrak{g}_-'(\mathfrak{r}_*)\delta\mathfrak{r}. \tag{4.127}$$

It is equivalently the normalized first-order parent compensation defect, since

$$\mathcal{F}(\mathfrak{g},\mathfrak{r}) = 1 + \mathfrak{r}^2 - 4(\mathfrak{g}-\mathfrak{r})^2, \qquad \delta\mathcal{F} = 4\sqrt{1+\mathfrak{r}_*^2}\,\delta_\perp. \tag{4.128}$$

In microscopic parent variables,

$$\boxed{\delta_\perp = \mathfrak{g}_*\delta \ln\left(\frac{g_q K_s}{g_s \lambda}\right) + \frac{1}{4\sqrt{1+\mathfrak{r}_*^2}}\delta \ln\left(\frac{K_s K_q}{\lambda^2}\right)}. \tag{4.129}$$

This is the final output of Part IV. The last first-order reduced defect is not a diffuse outlet problem. It is the question of whether the true moving-throat branch has $\delta_\perp = 0$, or what value (4.129) takes if it leaves the compensation family.

4.11 Verification outputs and downstream use

The usable downstream citation packet from this chapter is as follows.

1. **MTDC-T8.1**: cite for the geometry-lane firewall, the obstruction variables (ϵ_2, ϵ_4), and the forced $3/4 + 1/4$ module.
2. **MTDC-T8.2**: cite for the reduction of the retarded 2.5PN problem to $\widehat{m}_0^2\chi_Q N_Q = 1$, the outgoing DtN fingerprint, and $\chi_Q = 1$ on the canonical branch.
3. **MTDC-T8.3**: cite for the low-frequency outlet deformation algebra, the Robin and hidden-pole no-go results, and the compensated Robin–mixed branch.
4. **MTDC-T8.4**: cite for the positive mouth-source theorem, the exponential boundary-layer law, the Family–1 bias point Π_*, and the coupled mouth fixed-point equations.
5. **MTDC-T8.5**: cite for the co-evolving core–mouth canonical point, linear defect transport, D/N similarity preservation, and the final off-family scalar $\delta_\perp$.

☐ The grouped P_2 carrier is kept distinct from the axisymmetric geometry lane until the isotropic decoupling theorem is invoked.

- ☐ The $3/4 + 1/4$ conservative module is cited only under the static-geometry or geometry-decoupled hypothesis; dynamic geometry contamination is tracked by (ϵ_2, ϵ_4).
- ☐ The Family–1 support success statement is conditioned on the minimal isotropic precursor $\rho_\alpha = 4/3$, $\zeta_{\rm req} = 1/3$, not on an arbitrary support branch.
- ☐ The retarded 2.5PN gate is stated as the factorized product $\widehat{m}_0^2 \chi_Q N_Q = 1$, with source-map, conservative-normalization, and outgoing-normalization factors separated.
- ☐ The exact outgoing $l = 2$ DtN fingerprint is used only after the passive/outgoing compact branch has been selected.
- ☐ Pure Robin and standalone mixed-pole outlet deformations are recorded as no-go or noncanonical branches unless the compensated Robin–mixed constraints are imposed.
- ☐ The shell/mixed core Schur complement is treated as a reduced core model; parent-action realization still requires the corresponding overlap data.
- ☐ The finite D/N tube normalization is tied to the geometric throat ratio before it is used in the compensation surface.
- ☐ Positive mouth-source selection is kept separate from arbitrary signed source fitting; the lower compensated branch is selected by the positivity theorem.
- ☐ The exponential mouth-layer law is read as the zero-flux GNLS/localized-Maxwell boundary-layer solution, not as a free curve fit.
- ☐ Numerically located mouth and co-evolving fixed points are marked as exact equations solved numerically, not as closed-form constants.
- ☐ Linear defect transport is applied only after expanding about the renormalized co-evolving canonical point.
- ☐ The final first-order obstruction is reported as $\delta_\perp$, so future uses of $\Delta_Q = 0$ must either assume or verify tangent motion on the parent compensation family.
- ☐ Branch-realization status remains explicit whenever Part IV is cited for downstream PN, anomaly, or same-charge work.

The claim firewall is especially important for this chapter. Equations such as (4.31), (4.123), and (4.129) are exact inside their declared reduced closures. They are not, by themselves, proof that the full moving-throat PDE has realized the canonical branch. Future papers should cite Part IV for the reduction and should separately state the branch-realization status whenever they need $\Delta_Q = 0$, $N_Q = 1$, $\Xi_{\rm slip} = 0$, or $\delta_\perp = 0$.

Chapter 5

Weak-Axisymmetric Transport, Quotient Coordinates, and Orbit Closure

Stage range: 164–200

Part IV ended with the co-evolving core–mouth branch reduced to a first-order off-family normal coordinate. Part V starts from that coordinate and asks a sharper question: after the Family–1 mouth/core compensation machinery is already in place, which microscopic grouped weak-axisymmetric motions still survive as genuine reduced defects, and which motions are only similarity-orbit relabelings of the same coherent branch?

The answer is that the apparent many-variable weak-axisymmetric problem collapses to two levels. On the branch side, the grouped real P_2 carrier reduces to a single transported prefactor slope,

$$\Xi_1 = \frac{P_1}{P_0}, \tag{5.1}$$

with the weak-axisymmetric grouped signature

$$\lambda_{20} = 1, \qquad \lambda_{21} = \frac{1}{2}, \qquad \lambda_{22} = -1. \tag{5.2}$$

Equivalently, the grouped anisotropy lies on the fixed line

$$b_{P_0} = 3a_{P_0}. \tag{5.3}$$

On the orbit side, the coherent local branch reduces to three exact quotient coordinates,

$$\mathfrak{C}_{\mathrm{tr},*}, \qquad \mathfrak{C}_{\mathrm{nt},*}, \qquad \epsilon_\eta, \tag{5.4}$$

whose finite level sets are exactly the five-parameter microscopic similarity orbits. The final reduced verdict after this part is therefore not an eight-slot branch residual plus an ambiguous orbit condition. It is the exact four-scalar packet

$$\Delta_{\mathrm{full}} = (\Delta_Q, q_{\mathrm{tr}}, q_{\mathrm{nt}}, q_\eta), \qquad \Delta_Q := \chi_Q - 1. \tag{5.5}$$

Part V at a glance

Primary question	After Part IV reduces the retarded obstruction to one transported branch scalar, which weak-axisymmetric motions are genuine defects and which are only similarity-orbit relabelings of the same coherent branch?

Starting inputs The Part IV off-family normal coordinate $\delta_\perp$, the compensated Family–1 mouth/core branch, and the outgoing-normalization packet.

Delivers The scalar transfer-shape slope $\Xi_1 = P_1/P_0$, the coherent normal form and monomial quotient coordinates, the Packet-A finish line $\Delta_Q = \chi_Q - 1$, and the final verdict packet Δ_{full}.

Used by Part VI as the realization-compiler input and Parts VII–VIII as the strict and relaxed branch packet language.

Result anchors **MTDC-T9**.

Stage packets.

Stages	Packet	Status	Carry-forward output
164–169	Off-family drift reduction and scalar-slippage firewall	Exact Within Closure	Rewrites $\delta_\perp$ into microscopic drifts, isolates the true scalar slippage channels, and proves pure grouped real P_2 anisotropy does not linearly feed them.
170–180	Grouped outlet/load collapse to the transfer-shape scalar	Exact Within Closure	Maps grouped outlet deformations to physical slopes, derives Ξ_{load} and the static self-similarity laws, and collapses weak-axisymmetric outgoing transport to $\Xi_1 = P_1/P_0$.
181–190	Coherent normal form, monomial coordinates, and direct compilers	Exact Within Closure	Compresses the coherent branch to $(\Sigma_{\text{tr}}, \Sigma_{\text{nt}}, \Sigma_\eta)$, identifies direct monomial quotient coordinates, and records observable compilers and no-go packets.
191–197	Packet-A reduction and retarded normalization finish line	Exact Within Closure / Reduced / Controlled Reduction	Defines Packet A and Packet B, fixes the outgoing $l = 2$ fingerprint, imports the natural point-particle source map, and proves $\Delta_Q = \chi_Q - 1$ is the sole Packet-A scalar.
198–200	Orbit transport and final reference-free verdict packet	Exact Within Closure	Removes representative privilege, proves exact orbit transport, and closes the home stretch with $\Delta_{\text{full}} = (\Delta_Q, q_{\text{tr}}, q_{\text{nt}}, q_\eta)$.

Claim status. Mixed but predominantly Exact Within Closure**. Stages 164–200 are exact algebra inside the declared co-evolving Family–1, grouped real P_2, coherent local D/N, outgoing-prefactor, Packet-A, and Packet-B reduced hierarchies. The natural point-particle source-map use in Stage 195 is a** Reduced / Controlled Reduction **import. The actual nonlinear moving-throat PDE still has to return branch data that land on the final packet; realization by that branch remains** Open **even though the compilers and quotient/orbit theorems are exact within closure.**

What this part proves. Part V reduces the apparent many-variable weak-axisymmetric defect problem to one transported branch scalar, three exact quotient coordinates, and the final four-scalar verdict packet Δ_{full}.

What this part does not prove. It does not show that the actual branch makes that packet vanish. It only proves that this is the exact reduced packet that later realization work must evaluate.

Why later parts need it. Part VI turns Δ_{full} into the realization compiler, while Parts VII and VIII use the same packet language to state strict actual-branch tests and relaxed-branch extensions.

5.1 Block contract and theorem-level output

The contract of Part V is to turn the remaining first-order weak-axisymmetric and retarded branch checks into explicit finite packets. It is not trying to prove that the completed nonlinear PDE realizes those packets. Instead it proves that, once a candidate moving-throat solution gives the required reduced branch data, the verdict can be computed by exact formulas with no remaining interpretation freedom.

The block has two sides. The first side is the weak-axisymmetric branch transport:

$$\delta_\perp \longrightarrow (\mathcal{K}_A, \mathcal{G}_A) \longrightarrow (u_2^{(1)}, P_1/P_0) \longrightarrow \Xi_{\text{load}} \longrightarrow \Xi_1. \tag{5.6}$$

The second side is the coherent orbit quotient:

$$(\Sigma_{\text{tr}}, \Sigma_{\text{nt}}, \Sigma_\eta) \longrightarrow (\mathfrak{C}_{\text{tr},*}, \mathfrak{C}_{\text{nt},*}, \epsilon_\eta) \longrightarrow \mathcal{M}_+/\mathcal{G}_* \longrightarrow (q_{\text{tr}}, q_{\text{nt}}, q_\eta). \tag{5.7}$$

The final theorem joins these with the retarded Packet-A scalar $\Delta_Q = \chi_Q - 1$.

Theorem 5.1 (Part V weak-axisymmetric quotient and orbit-closure theorem)**.** *Inside the carried moving-throat reduced hierarchy for Stages 164–200, the following theorem chain holds.*

1. *The first off-family Family–1 normal coordinate is an explicit logarithmic transport law in the moving-throat wall/BdG/Maxwell/mixed branch variables. Arbitrary first-order isotropic bundle drift is tangent to the compensated parent family; true off-bundle motion first enters through three scalar slippages.*

2. *A pure grouped real P_2 anisotropy cannot linearly feed the scalar slippage channel. Its linear effect enters the direct outlet coefficients only through*

$$\mathcal{K}_A := \delta D_{A,2} + \frac{1}{9}\delta D_{A,0}, \qquad \mathcal{G}_A := \delta N_{A,0} - P_0 \delta D_{A,0}. \tag{5.8}$$

3. *On the canonical compensated branch, these outlet coefficients are exactly the physical grouped slopes,*

$$\mathfrak{K}_1 = -D_0 u_2^{(1)}, \qquad \mathfrak{G}_1 = D_0 P_1. \tag{5.9}$$

4. *On the even-preserving weak-axisymmetric branch, all conservative grouped transport is controlled by one static operator slope, and the sole remaining linear grouped normalization defect is*

$$\Xi_{\text{load}} := \frac{N_{01}}{N_0} - \frac{D_{01}}{D_0}. \tag{5.10}$$

5. *The same scalar is the weighted failure of static self-similarity and, after wall-normalized outgoing-port factorization, the logarithmic slope of the effective transfer shape:*

$$\Xi_1 = \frac{P_1}{P_0} = \frac{\delta \ln \mathcal{T}^2_{\text{eff},A}}{\epsilon \lambda_A}. \tag{5.11}$$

6. *On the coherent local D/N branch, the weak-axisymmetric observable drifts obey the triangular normal form*

$$\Theta_1 = -C_{\mathrm{tr}}\Sigma_{\mathrm{tr}}, \qquad \Xi_1 = A_{\mathrm{tr}}\Sigma_{\mathrm{tr}} + \Sigma_{\mathrm{nt}}, \qquad \mathcal{R}_1 + \Xi_1 = -\frac{\epsilon_\eta}{1-\epsilon_\eta}\Sigma_\eta. \tag{5.12}$$

7. *The three normal-form coordinates are exactly the logarithmic drifts of three direct microscopic monomials,*

$$\delta \ln \mathfrak{C}_{\mathrm{tr},*} = \Sigma_{\mathrm{tr}}, \qquad \delta \ln \mathfrak{C}_{\mathrm{nt},*} = \Sigma_{\mathrm{nt}}, \qquad \delta \ln \epsilon_\eta = \Sigma_\eta. \tag{5.13}$$

8. *The finite level sets of these three monomials are exactly the five-parameter similarity orbits $\mathcal{G}_*$, so*

$$\mathcal{M}_+/\mathcal{G}_* \cong (\mathbb{R}_{>0})^3, \qquad (\mathfrak{C}_{\mathrm{tr},*}, \mathfrak{C}_{\mathrm{nt},*}, \epsilon_\eta) \text{ are quotient coordinates.} \tag{5.14}$$

9. *The branch-side Packet A collapses to one scalar finish line,*

$$\Delta_{\mathrm{branch}} = 0 \iff \chi_Q = 1 \iff \Delta_Q := \chi_Q - 1 = 0. \tag{5.15}$$

10. *The full reference-free reduced verdict is therefore the four-scalar packet*

$$\Delta_{\mathrm{full}}^{(x|\mathcal{O}_*)} := \left(\Delta_Q(x), q_{\mathrm{tr}}^{(x\leftarrow\mathcal{O}_*)}, q_{\mathrm{nt}}^{(x\leftarrow\mathcal{O}_*)}, q_\eta^{(x\leftarrow\mathcal{O}_*)}\right), \tag{5.16}$$

with

$$\Delta_{\mathrm{full}}^{(x|\mathcal{O}_*)} = 0 \iff \Delta_{\mathrm{branch}}(x) = 0 \text{ and } x \in \mathcal{O}_*. \tag{5.17}$$

Proof structure. Stages 164–168 translate the first off-family normal coordinate into explicit microscopic transport variables and then separate on-bundle tangent motion from off-bundle slippage. Stages 169–173 prove that grouped real P_2 anisotropy does not linearly feed scalar slippage, derive the direct outlet map, and collapse the even-preserving grouped defect to Ξ_{load}. Stages 174–180 factor Ξ_{load} through wall-referenced self-similarity, outgoing-load factors, port co-loading, and the effective transfer shape, giving $\Xi_1 = P_1/P_0$. Stages 181–185 pass from physical coherent-kernel variables to the triangular normal form and then to direct microscopic monomials. Stages 186–187 prove the finite orbit–fibre theorem. Stages 188–192 build the observable, transfer-shape, finite-packet, and projector compilers. Stages 193–197 close the branch-side Packet-A finish line to $\Delta_Q = \chi_Q - 1$. Stages 198–200 remove the orbit representative and combine Packet A and Packet B into (5.16). □

5.2 Weak-axisymmetric grouped signature

5.2.1 Local goal

The first job in Part V is to connect the co-evolving Family–1 off-family scalar from Part IV to the grouped real P_2 transport language. Stage 163 had left the first true normal coordinate in the form

$$\delta_\perp = \mathfrak{g}_* \delta \ln\left(\frac{g_q K_s}{g_s \lambda}\right) + \frac{1}{4\sqrt{1+\mathfrak{r}_*^2}} \delta \ln\left(\frac{K_s K_q}{\lambda^2}\right). \tag{5.18}$$

Stage 164 shows that this is not an abstract parent-coordinate expression. With

$$\mathfrak{r} = \frac{\lambda}{\sqrt{K_s K_q}}, \qquad \mathfrak{g} = \frac{g_q\sqrt{K_s}}{g_s\sqrt{K_q}}, \qquad r_c = \frac{\lambda^2}{K_s K_q} = \mathfrak{r}^2, \tag{5.19}$$

the two channels are exactly

$$\delta \ln\left(\frac{g_q K_s}{g_s \lambda}\right) = \delta \ln\left(\frac{\mathfrak{g}}{\mathfrak{r}}\right), \qquad \delta \ln\left(\frac{K_s K_q}{\lambda^2}\right) = -\delta \ln r_c. \tag{5.20}$$

Thus $\delta_\perp$ measures relative mouth-coupling drift and hybridization drift.

5.2.2 Explicit branch-variable form

On the healing-locked wall/BdG/Maxwell/mixed branch, the same scalar becomes

$$\begin{aligned}\delta_\perp = A_* \,\delta \ln\left(\frac{\mathcal{Z}_q}{\rho_w}\right) + 3A_* \,\delta \ln c_{s,w} + B_* \,\delta \ln c_s - \mathfrak{g}_* \,\delta \ln \mathcal{T}_m \\ - (\mathfrak{g}_* + B_*)\delta \ln v_{w0} - 2A_* \,\delta \ln a - C_* \,\delta \ln L_W,\end{aligned} \tag{5.21}$$

where

$$A_* := \mathfrak{g}_* + \frac{1}{4\sqrt{1+\mathfrak{r}_*^2}}, \qquad B_* := \frac{1}{2\sqrt{1+\mathfrak{r}_*^2}}, \qquad C_* := 2\mathfrak{g}_* + \frac{3}{4\sqrt{1+\mathfrak{r}_*^2}}. \tag{5.22}$$

The exact tangency condition $\delta_\perp = 0$ is therefore a co-transport law for $\mathcal{T}_m$ relative to $(\mathcal{Z}_q, \rho_w, c_{s,w}, c_s, v_{w0}, a, L_W)$, not a free adjustment.

Stages 165–167 then show that the explicit lower compensated branch is even more constrained. The D/N tube length co-transports with the mouth radius,

$$\delta \ln L_W = \delta \ln a, \tag{5.23}$$

and the exact branch laws reduce the apparent branch drift problem to four irreducible variables. Stage 166 chooses the four bundle observables

$$\Theta_w, \qquad K_s, \qquad K_q, \qquad P_0 = \frac{N_0}{D_0} \tag{5.24}$$

as the right variables to invert. Stage 167 proves that arbitrary first-order drift of these four bundle observables leaves the compensation ratios tangent to the parent family:

$$\delta \ln r_c = 0, \qquad \delta \ln \mathfrak{r} = 0, \qquad \delta \ln \mathfrak{g} = 0 \quad \text{for pure bundle drift.} \tag{5.25}$$

Consequently, any first-order failure after Stage 167 must be genuinely off-bundle.

5.2.3 Off-bundle slippages and grouped signature

Stage 168 introduces the three scalar slippages

$$\varepsilon_L, \qquad \varepsilon_v, \qquad \varepsilon_T, \tag{5.26}$$

which measure failure of the exact lower-branch laws for axial length, mixed background speed, and mouth traction. Stage 169 then proves a key firewall: a pure weak grouped real P_2 anisotropy cannot linearly source these scalar slippages. The reason is angular. The weak grouped perturbation has zero weighted trace at first order; scalar feed-down can only be built from quadratic grouped invariants such as

$$\mathcal{I}[x,y] = \frac{1}{5}\delta x^T G_{\text{grp}} \delta y = 4a_x a_y + \frac{4}{5} b_x b_y, \qquad G_{\text{grp}} = \operatorname{diag}(1,2,2). \tag{5.27}$$

Thus linear grouped real P_2 transport enters the direct outlet coefficients rather than the scalar off-bundle slippage channel.

The grouped weak-axisymmetric signature is inherited from the first axisymmetric quadrupole perturbation:

$$\lambda_{20} = 1, \qquad \lambda_{21} = \frac{1}{2}, \qquad \lambda_{22} = -1. \tag{5.28}$$

For any first-order grouped quantity

$$x_A = x_0 + \epsilon \lambda_A x_1 + O(\epsilon^2), \tag{5.29}$$

the weighted grouped anomalies obey

$$a_x = \frac{\epsilon}{4}x_1, \qquad b_x = \frac{3\epsilon}{4}x_1, \qquad b_x = 3a_x. \tag{5.30}$$

This fixed line is the weak-axisymmetric branch signature used throughout the rest of Part V.

Claim status. Stages 164–169 are Exact Within Closure **inside the explicit co-evolving Family–1 throat-core closure and the grouped real P_2 weak-axisymmetric expansion. They do not prove that the actual nonlinear PDE produces a particular slippage; they identify the exact variables through which such slippage would appear.**

5.3 Even-preserving and hidden-even gates

5.3.1 Direct outlet map

Stages 170–171 move the linear grouped problem from scalar slippage to direct outlet data. For each grouped lane A, define

$$\mathcal{K}_A := \delta D_{A,2} + \frac{1}{9}\delta D_{A,0}, \qquad \mathcal{G}_A := \delta N_{A,0} - P_0\delta D_{A,0}. \tag{5.31}$$

The hidden even deformation and hidden odd normalization deformation are then

$$\delta\kappa_W^{(A)} = \frac{3(1-\sigma_*)}{\sigma_* D_0}\mathcal{K}_A, \qquad \delta\gamma_W^{(A)} = -\frac{1-\sigma_*}{9\sigma_* N_0}\mathcal{G}_A. \tag{5.32}$$

Stage 171 decomposes these into microscopic sectors:

$$\mathcal{K}_A = \mathcal{W}_A - \mathcal{B}_A - \mathcal{Z}_A, \tag{5.33}$$

with

$$\mathcal{W}_A := \frac{1}{9}\delta K_A - \delta M_A, \qquad \mathcal{B}_A := \delta B_{A,2} + \frac{1}{9}\delta B_{A,0}, \qquad \mathcal{Z}_A := \delta Z_{A,2} + \frac{1}{9}\delta Z_{A,0}, \tag{5.34}$$

and

$$\mathcal{G}_A = -P_0\delta K_A + P_0\delta B_{A,0} + \mathcal{N}_A, \qquad \mathcal{N}_A := \delta N_{A,0} + P_0\delta Z_{A,0}. \tag{5.35}$$

This identifies the only four microscopic bundles that can generate a linear grouped outlet obstruction: wall baseline, BdG support, conservative Maxwell/mixed, and outgoing-transfer loading.

5.3.2 Physical slope collapse

Stage 172 rewrites the same obstruction in physical grouped-response variables. For each grouped lane,

$$u_2^{(A)} = -\frac{D_{A,2}}{D_{A,0}}, \qquad u_4^{(A)} = \frac{D_{A,2}^2 - D_{A,0}D_{A,4}}{D_{A,0}^2}, \qquad P_0^{(A)} = \frac{N_{A,0}}{D_{A,0}}. \tag{5.36}$$

On the canonical compensated branch, the two microscopic amplitudes are simply

$$\mathfrak{K}_1 = -D_0 u_2^{(1)}, \qquad \mathfrak{G}_1 = D_0 P_1. \tag{5.37}$$

Equivalently,

$$\delta\kappa_W^{(A)} = -\frac{3(1-\sigma_*)}{\sigma_*}\delta u_2^{(A)}, \qquad \delta\gamma_W^{(A)} = -\frac{1-\sigma_*}{9\sigma_*}\frac{\delta P_0^{(A)}}{P_0}. \tag{5.38}$$

So the full linear grouped outlet problem is now measurable by $u_2^{(1)}$ and P_1/P_0.

5.3.3 Even-preserving branch

Stage 173 then computes those physical slopes directly from the weak-axisymmetric grouped operator expansion

$$D_{A,0} = D_0 + \epsilon\lambda_A D_{01} + O(\epsilon^2), \quad D_{A,2} = D_2 + \epsilon\lambda_A D_{21} + O(\epsilon^2), \quad D_{A,4} = D_4 + \epsilon\lambda_A D_{41} + O(\epsilon^2), \tag{5.39}$$

$$N_{A,0} = N_0 + \epsilon\lambda_A N_{01} + O(\epsilon^2). \tag{5.40}$$

The exact slope formulas are

$$u_2^{(1)} = -\frac{D_{21} + u_2 D_{01}}{D_0}, \tag{5.41}$$

$$u_4^{(1)} = -\frac{5D_{01} + 18D_{21} + 81D_{41}}{81D_0} \quad \text{on } (u_2, u_4) = \left(\frac{1}{9}, \frac{4}{81}\right), \tag{5.42}$$

and

$$\frac{P_1}{P_0} = \frac{N_{01}}{N_0} - \frac{D_{01}}{D_0}. \tag{5.43}$$

If the canonical conservative even fingerprint is preserved at first order, then

$$D_{21} = -\frac{D_{01}}{9}. \tag{5.44}$$

The hidden-even consistency relation $u_4^{(1)} = (8/9)u_2^{(1)}$ is exactly

$$D_{41} = \frac{2}{3}D_{21} + \frac{1}{27}D_{01}, \tag{5.45}$$

and therefore on the even-preserving branch

$$D_{41} = -\frac{D_{01}}{27}. \tag{5.46}$$

Thus the conservative grouped response is frozen to canonical order, and the only surviving linear grouped defect is the static loading mismatch

$$\Xi_{\text{load}} := \frac{N_{01}}{N_0} - \frac{D_{01}}{D_0}. \tag{5.47}$$

Its lane form is

$$\Delta_Q^{(20)} = \epsilon\Xi_{\text{load}}, \qquad \Delta_Q^{(21)} = \frac{\epsilon}{2}\Xi_{\text{load}}, \qquad \Delta_Q^{(22)} = -\epsilon\Xi_{\text{load}}. \tag{5.48}$$

Claim status. Stages 170–173 are Exact Within Closure **in the grouped low-frequency bundle. The even-preserving condition is a branch restriction, not an unconditional statement that the completed PDE must satisfy it.**

5.4 Outgoing-load scalar ($\Xi_1 = P_1/P_0$)

5.4.1 Self-similarity form of the loading mismatch

Stage 174 rewrites the mismatch (5.47) as failure of static self-similarity. Let

$$\delta_K := \frac{K_1}{K}, \qquad \delta_B := \frac{B_0^{(1)}}{B_0}, \qquad \delta_Z := \frac{Z_0^{(1)}}{Z_0}, \qquad \delta_N := \frac{N_1}{N_0}, \tag{5.49}$$

and

$$\omega_B := \frac{B_0}{D_0}, \qquad \omega_Z := \frac{Z_0}{D_0}, \qquad D_0 = K - B_0 - Z_0. \tag{5.50}$$

Then

$$\Xi_{\text{load}} = (\delta_N - \delta_K) + \omega_B(\delta_B - \delta_K) + \omega_Z(\delta_Z - \delta_K). \tag{5.51}$$

Define the wall-referenced self-similarity defect fields

$$\Sigma_\alpha^{(B)} := 2\delta \ln\left(\frac{c_\alpha}{\varpi_\alpha}\right) - \delta_K, \quad \Sigma_r^{(Z)} := \delta \ln\left(\frac{Q_r}{\Delta_r}\right) - \delta_K, \quad \Sigma_r^{(N)} := 2\delta \ln\left(\frac{P_r}{\Delta_r}\right) - \delta_K. \tag{5.52}$$

Then

$$\Xi_{\text{load}} = \sum_r \rho_r^{(N)} \Sigma_r^{(N)} + \omega_B \sum_\alpha \rho_\alpha^{(B)} \Sigma_\alpha^{(B)} + \omega_Z \sum_r \rho_r^{(Z)} \Sigma_r^{(Z)}. \tag{5.53}$$

Static self-similarity is the sufficient condition that all three defect families vanish. On the even-preserving branch it implies $\Xi_{\text{load}} = 0$ and hence no first-order grouped normalization defect.

5.4.2 Outgoing-load theorem and mixed-leg law

Stages 175–176 factor the port objects relative to the wall baseline. The remaining defect is controlled by the outgoing load factor

$$\Lambda_r := \frac{P_r}{\Delta_r}. \tag{5.54}$$

On conservative-shape-preserving branches the full remaining defect is carried only by the outgoing load slope relative to the wall. Stage 176 factors that load into a mixed leg, an interference ratio, and a hybridization ratio. In the notation of the reduced one-port block,

$$\frac{\Lambda_r^2}{K} = \mathcal{M}_r^2 \frac{(1 + \mathcal{I}_r)^2}{(1 - \mathcal{H}_r)^2}. \tag{5.55}$$

Consequently

$$\Sigma_r^{(N)} = 2\delta \ln \mathcal{M}_r + \frac{2\mathcal{I}_r}{1 + \mathcal{I}_r} \delta \ln \mathcal{I}_r + \frac{2\mathcal{H}_r}{1 - \mathcal{H}_r} \delta \ln \mathcal{H}_r. \tag{5.56}$$

If the interference and hybridization ratios are rigid, the whole outgoing defect is the raw mixed-leg drift. The zero-defect condition becomes the square-root mixed-leg law

$$\frac{G_W}{\Omega_W^2} \propto \sqrt{K}. \tag{5.57}$$

This is useful because it identifies a concrete microscopic path to zero defect rather than just a cancellation among compiled coefficients.

5.4.3 Collapse to one scalar amplitude

Stage 177 applies the weak-axisymmetric signature (5.28). Every microscopic outgoing slippage inherits the same lane pattern, so the full remaining grouped defect is one scalar amplitude:

$$\Xi_{\text{load}}^{(A)} = \epsilon \lambda_A \Xi_1. \tag{5.58}$$

The same scalar is the physical outgoing-prefactor slope:

$$\Xi_1 = \frac{P_1}{P_0}. \tag{5.59}$$

Stage 178 rewrites it portwise. For

$$N_{A,0}^{(r)} = \frac{P_{A,r}^2}{\Delta_{A,r}^2}, \tag{5.60}$$

define the port slope ν_r by

$$\delta \ln N_{A,0}^{(r)} = \epsilon \lambda_A \nu_r. \tag{5.61}$$

Then

$$\Xi_1 = \bar{\nu}_N - \kappa_1, \tag{5.62}$$

where κ_1 is the wall-baseline static slope and $\bar{\nu}_N$ is the outgoing-weighted average. Thus zero defect is exact co-loading of outgoing transfer with the wall baseline.

Stages 179–180 put the same statement into transfer-shape language. Each port factors as

$$N_{A,0}^{(r)} = K_A \mathcal{T}_{A,r}^2, \qquad \mathcal{T}_{A,r} = \frac{\widehat{G}_{W,A,r} + \widehat{R}_{A,r}\widehat{G}_{U,A,r}}{1 - \widehat{R}_{A,r}^2}. \tag{5.63}$$

Therefore

$$\Xi_1 = 2\sum_r \rho_r^{(N)} \tau_r, \qquad \tau_r := \frac{\delta \ln \mathcal{T}_{A,r}}{\epsilon \lambda_A}. \tag{5.64}$$

Equivalently, all ports collapse to one effective transfer shape,

$$\mathcal{T}_{\mathrm{eff},A}^2 := \sum_r \mathcal{T}_{A,r}^2 = \frac{N_{A,0}}{K_A}, \tag{5.65}$$

and

$$\Xi_1 = \frac{\delta \ln \mathcal{T}_{\mathrm{eff},A}^2}{\epsilon \lambda_A}. \tag{5.66}$$

On the actual minimal one-port continuum branch,

$$\mathcal{T}_A^2 = \frac{Z_{W,A}(1+\rho_A)^2}{\Omega_{W,A}^2(1-\epsilon_{W,A})^2} = \frac{27\pi^2 G c_s^5}{20 a^5 c^5} \frac{1-\epsilon_{\eta,A}}{R_{\mathrm{target},A}}. \tag{5.67}$$

So the remaining theorem gate is to determine whether the actual weak-axisymmetric branch keeps $\mathcal{T}_{\mathrm{eff}}$ rigid.

Claim status. Stages 174–180 are Exact Within Closure **inside the weak-axisymmetric grouped bundle and the declared outgoing-port factorization. The square-root mixed-leg law is a sufficient zero-defect branch condition under interference/hybridization rigidity; actual realization is not assumed.**

5.5 Coherent-kernel slippages

5.5.1 Actual coherent local D/N branch

Stage 181 inserts the coherent local D/N tracking branch into (5.67). On that branch,

$$\mathcal{T}^2 = \frac{Z_W(1+\chi_0)^2}{\Omega_W^2(1-\epsilon)^2} = \frac{27\pi^2 G c_s^5}{20 a^5 c^5} \frac{1-\epsilon_\eta}{R_{\mathrm{target}}}, \tag{5.68}$$

where

$$\epsilon = \epsilon_W \left(1 - \frac{2}{11}\frac{\delta_U}{1+\delta_U}\right). \tag{5.69}$$

The weak-axisymmetric defect is

$$\Xi_1 = \zeta_Z - \omega_W + \frac{2\chi_1}{1+\chi_0} + \frac{2\epsilon_1}{1-\epsilon}, \tag{5.70}$$

with

$$\epsilon_1 = \left(1 - \frac{2}{11}\frac{\delta_U}{1+\delta_U}\right)\varepsilon_W - \frac{2\epsilon_W}{11(1+\delta_U)^2}\delta_{U,1}. \tag{5.71}$$

The support enhancement ratio ζ drops out identically:

$$\partial_\zeta \ln \mathcal{T}^2 = 0, \qquad \partial_\zeta \ln R_{\text{target}} = 0, \qquad \partial_\zeta \Xi_1 = 0. \tag{5.72}$$

Thus coherent support can help the steady normalization load, but it cannot repair the weak-axisymmetric prefactor slope at first order.

5.5.2 Microscopic slippage decomposition

Stage 182 pushes (5.70) down to microscopic coherent-kernel slippages. The direct grouped defect depends on four slippages,

$$\Sigma_Z, \qquad \Sigma_\chi, \qquad \Sigma_\epsilon, \qquad \Sigma_\delta, \tag{5.73}$$

and selected-branch dressing introduces one more,

$$\Sigma_\eta. \tag{5.74}$$

The direct defect law is

$$\Xi_1 = \Sigma_Z + \frac{2\chi_0}{1+\chi_0}\Sigma_\chi + \frac{2\epsilon_W}{1-\epsilon}\left[\frac{11+9\delta_U}{11(1+\delta_U)}\Sigma_\epsilon - \frac{2\delta_U}{11(1+\delta_U)^2}\Sigma_\delta\right]. \tag{5.75}$$

The tracking drift is the special combination

$$\Sigma_{\text{tr}} := (1+\chi_0)\Sigma_\delta + (1+\delta_U)\Sigma_\chi, \tag{5.76}$$

which controls the tracking observable

$$\Theta_1 = -\frac{\chi_0\delta_U}{(1+\chi_0)(1+\delta_U)(1+\chi_0+\delta_U)}\Sigma_{\text{tr}}. \tag{5.77}$$

So exact tracking rigidity means $\Sigma_{\text{tr}} = 0$, but that alone does not generally force $\Xi_1 = 0$; a genuine nontracking transfer-shape slippage can remain.

Claim status. Stages 181–182 are Exact Within Closure **for the coherent local D/N tracking branch. The support-blindness statement is exact inside that branch; it does not assert that support is irrelevant to all other branch tests.**

5.6 Triangular normal form

5.6.1 Normal-form coordinates

Stage 183 packages the five slippages into three branch-adapted coordinates. The nontracking transfer-shape slippage is defined by

$$\begin{aligned}\Sigma_{\mathrm{nt}} := \Sigma_Z + \frac{2\epsilon_W}{1-\epsilon}\frac{11+9\delta_U}{11(1+\delta_U)}\Sigma_\epsilon \\ - \left[\frac{2\chi_0}{1+\delta_U} + \frac{4\epsilon_W\delta_U}{11(1-\epsilon)(1+\delta_U)^2}\right]\Sigma_\delta.\end{aligned} \tag{5.78}$$

The three branch-adapted coordinates are therefore

$$(\Sigma_{\mathrm{tr}}, \Sigma_{\mathrm{nt}}, \Sigma_\eta). \tag{5.79}$$

In these coordinates the observable drifts have the exact triangular form

$$\Theta_1 = -C_{\mathrm{tr}}\Sigma_{\mathrm{tr}}, \qquad \Xi_1 = A_{\mathrm{tr}}\Sigma_{\mathrm{tr}} + \Sigma_{\mathrm{nt}}, \qquad \mathcal{R}_1 + \Xi_1 = -\frac{\epsilon_\eta}{1-\epsilon_\eta}\Sigma_\eta, \tag{5.80}$$

where

$$C_{\mathrm{tr}} = \frac{\chi_0\delta_U}{(1+\chi_0)(1+\delta_U)(1+\chi_0+\delta_U)}, \qquad A_{\mathrm{tr}} = \frac{2\chi_0}{(1+\chi_0)(1+\delta_U)}. \tag{5.81}$$

This is the first exact three-scalar normal form of the weak-axisymmetric coherent branch.

5.6.2 Inversion and interpretation

The triangular structure makes the three physical failure modes transparent.

1. Σ_{tr} is tracking failure. It is measured directly by Θ_1.
2. Σ_{nt} is nontracking transfer-shape failure. It is the residual part of Ξ_1 after subtracting tracking feed-through.
3. Σ_η is dressing failure. It measures the mismatch between direct transfer-shape drift and selected-branch demand through $\mathcal{R}_1 + \Xi_1$.

Zero defect in this normal form is

$$\Theta_1 = \Xi_1 = \mathcal{R}_1 = 0 \iff \Sigma_{\mathrm{tr}} = \Sigma_{\mathrm{nt}} = \Sigma_\eta = 0. \tag{5.82}$$

5.6.3 Branch-observable coordinates

Stage 184 identifies exact branch composites whose logarithmic drifts are the same three coordinates. The tracking factor is

$$R_{\mathrm{tr}} = \frac{1+\chi_0/(1+\delta_U)}{1+\chi_0} = \frac{1+\chi_0+\delta_U}{(1+\chi_0)(1+\delta_U)}. \tag{5.83}$$

The corrected nontracking composite is

$$\mathfrak{N}_* := \mathcal{T}^2 R_{\mathrm{tr}}^{B_*}, \qquad B_* := \frac{2(1+\chi_{0,*}+\delta_{U,*})}{\delta_{U,*}}. \tag{5.84}$$

After a harmless positive normalization of the tracking factor,

$$\mathfrak{T}_* := R_{\mathrm{tr}}^{-C_*}, \qquad C_* := \frac{(1+\chi_{0,*})(1+\delta_{U,*})(1+\chi_{0,*}+\delta_{U,*})}{\chi_{0,*}\delta_{U,*}}, \tag{5.85}$$

the branch-coordinate laws are

$$\delta \ln \mathfrak{T}_* = \Sigma_{\mathrm{tr}}, \qquad \delta \ln \mathfrak{N}_* = \Sigma_{\mathrm{nt}}, \qquad \delta \ln \epsilon_\eta = \Sigma_\eta. \tag{5.86}$$

So the next continuation point is no longer a list of microscopic slippages; it is drift of three exact branch composites.

Claim status. Stages 183–184 are Exact Within Closure **inside the coherent local D/N branch. The normal form is a compiler from reduced coherent variables to observables; it does not assert that the completed PDE has zero coordinates.**

5.7 Microscopic monomial invariants

5.7.1 Direct monomial coordinates

Stage 185 pushes the three branch composites to direct microscopic monomials. At reference-branch order,

$$\delta \ln \mathfrak{C}_{\mathrm{tr},*} = \Sigma_{\mathrm{tr}}, \qquad \delta \ln \mathfrak{C}_{\mathrm{nt},*} = \Sigma_{\mathrm{nt}}, \qquad \delta \ln \epsilon_\eta = \Sigma_\eta. \tag{5.87}$$

The tracking monomial is

$$\mathfrak{C}_{\mathrm{tr},*} := \chi_0^{1+\delta_{U,*}} \delta_U^{1+\chi_{0,*}}, \tag{5.88}$$

and the nontracking monomial is

$$\mathfrak{C}_{\mathrm{nt},*} := \frac{Z_W}{\Omega_W^2} \epsilon_W^{E_*} \delta_U^{-F_*}, \tag{5.89}$$

with

$$E_* = \frac{2\epsilon_{W,*}}{1-\epsilon_*} \frac{11+9\delta_{U,*}}{11(1+\delta_{U,*})}, \qquad F_* = \frac{2\chi_{0,*}}{1+\delta_{U,*}} + \frac{4\epsilon_{W,*}\delta_{U,*}}{11(1-\epsilon_*)(1+\delta_{U,*})^2}. \tag{5.90}$$

Using the coherent definitions, these become direct microscopic monomials in the positive variables

$$\mathbf{x} = (\lambda_W, c_{\eta U}, \gamma, K_U, K_\eta^{(\mathrm{eff})}, K_W^{(\mathrm{eff})}, \mu_W, T_U). \tag{5.91}$$

Explicitly,

$$\mathfrak{C}_{\mathrm{tr},*} = \left(\frac{\gamma c_{\eta U}}{K_U}\right)^{1+\delta_{U,*}} \left(\frac{\pi^2 T_U}{L^2 K_U}\right)^{1+\chi_{0,*}}, \tag{5.92}$$

$$\mathfrak{C}_{\mathrm{nt},*} = \left(\frac{\lambda_W^2 \mu_W}{K_\eta^{(\mathrm{eff})} (K_W^{(\mathrm{eff})})^2}\right) \left(\frac{\gamma^2 \lambda_W^2 \sigma}{K_U K_W^{(\mathrm{eff})}}\right)^{E_*} \left(\frac{\pi^2 T_U}{L^2 K_U}\right)^{-F_*}, \tag{5.93}$$

and

$$\epsilon_\eta = \frac{c_{\eta U}^2}{K_U K_\eta^{(\mathrm{eff})}}. \tag{5.94}$$

5.7.2 Linear monomial-drift map

Introduce the grouped weak-axisymmetric logarithmic drift vector

$$\delta\mathbf{x} = \begin{pmatrix} \lambda_1 \\ c_1 \\ \gamma_1 \\ \kappa_U \\ \kappa_\eta \\ \kappa_W \\ \mu_1 \\ \tau_1 \end{pmatrix} = \begin{pmatrix} \delta \ln \lambda_W \\ \delta \ln c_{\eta U} \\ \delta \ln \gamma \\ \delta \ln K_U \\ \delta \ln K_\eta^{(\mathrm{eff})} \\ \delta \ln K_W^{(\mathrm{eff})} \\ \delta \ln \mu_W \\ \delta \ln T_U \end{pmatrix}_{\mathrm{grp}}. \tag{5.95}$$

Then

$$\begin{pmatrix} \delta \ln \mathfrak{C}_{\mathrm{tr},*} \\ \delta \ln \mathfrak{C}_{\mathrm{nt},*} \\ \delta \ln \epsilon_\eta \end{pmatrix} = M_* \delta\mathbf{x}, \tag{5.96}$$

with

$$M_* = \begin{pmatrix} 0 & 1+\delta_{U,*} & 1+\delta_{U,*} & -(2+\chi_{0,*}+\delta_{U,*}) & 0 & 0 & 0 & 1+\chi_{0,*} \\ 2(1+E_*) & 0 & 2E_* & F_* - E_* & -1 & -(2+E_*) & 1 & -F_* \\ 0 & 2 & 0 & -1 & -1 & 0 & 0 & 0 \end{pmatrix}. \tag{5.97}$$

A convenient dependent minor has determinant

$$\det M_*^{(\tau,\kappa_\eta,\mu_1)} = 1 + \chi_{0,*} > 0. \tag{5.98}$$

Therefore M_* has rank three and a five-dimensional kernel. The zero-defect condition is

$$\Theta_1 = \Xi_1 = \mathcal{R}_1 = 0 \iff M_*\delta\mathbf{x} = 0. \tag{5.99}$$

Claim status. Stage 185 is Exact Within Closure in the reference coherent branch. The monomial exponents are fixed by the branch normal form, and M_* is an exact exponent matrix.

5.8 Similarity-orbit theorem

5.8.1 Finite similarity family

Stage 186 integrates the tangent kernel of M_* to a finite five-parameter similarity family. Let

$$(\Lambda, C, \Gamma, U, W) \tag{5.100}$$

be free logarithmic co-scalings of $(\lambda_W, c_{\eta U}, \gamma, K_U, K_W^{(\mathrm{eff})})$. Monomial preservation fixes the dependent scalings:

$$K_\eta^{(\mathrm{eff})} \mapsto e^{2C-U} K_\eta^{(\mathrm{eff})}, \tag{5.101}$$

$$T_U \mapsto \exp\left[U - \frac{1+\delta_{U,*}}{1+\chi_{0,*}} (\Gamma + C - U) \right] T_U, \tag{5.102}$$

$$\mu_W \mapsto e^{M(\Lambda,C,\Gamma,U,W)}\mu_W, \tag{5.103}$$

where

$$\begin{aligned} M(\Lambda,C,\Gamma,U,W) &= 2C - U + 2W - 2\Lambda - E_*(2\Gamma + 2\Lambda - U - W) \\ &\quad - F_*\frac{1+\delta_{U,*}}{1+\chi_{0,*}}(\Gamma + C - U). \end{aligned} \tag{5.104}$$

Call this family $\mathcal{G}_*$. It preserves $(\mathfrak{C}_{\mathrm{tr},*}, \mathfrak{C}_{\mathrm{nt},*}, \epsilon_\eta)$ exactly.

5.8.2 Orbit–quotient closure

Stage 187 proves the converse. On the positive microscopic state space

$$\mathcal{M}_+ = \{(\lambda_W, c_{\eta U}, \gamma, K_U, K_\eta^{(\mathrm{eff})}, K_W^{(\mathrm{eff})}, \mu_W, T_U) > 0\}, \tag{5.105}$$

define

$$\mathcal{I}(x) = (\mathfrak{C}_{\mathrm{tr},*}(x), \mathfrak{C}_{\mathrm{nt},*}(x), \epsilon_\eta(x)). \tag{5.106}$$

For two positive states $x, \widetilde{x}$, equality of the three invariants is exactly

$$M_*\Delta\mathbf{x} = 0, \qquad \Delta\mathbf{x} := \ln(\widetilde{x}/x). \tag{5.107}$$

Solving these finite equations gives precisely the dependent transformations (5.101)–(5.103). Hence

$$\mathcal{I}(\widetilde{x}) = \mathcal{I}(x) \iff \widetilde{x} \in \mathcal{G}_* \cdot x. \tag{5.108}$$

Therefore

$$\mathcal{M}_+/\mathcal{G}_* \cong (\mathbb{R}_{>0})^3, \tag{5.109}$$

with quotient coordinates (5.4). The first grouped weak-axisymmetric observables are the infinitesimal motion in these quotient coordinates:

$$\Theta_1 = -C_{\mathrm{tr}}\delta\ln\mathfrak{C}_{\mathrm{tr},*}, \tag{5.110}$$

$$\Xi_1 = A_{\mathrm{tr}}\delta\ln\mathfrak{C}_{\mathrm{tr},*} + \delta\ln\mathfrak{C}_{\mathrm{nt},*}, \tag{5.111}$$

$$\mathcal{R}_1 + \Xi_1 = -\frac{\epsilon_{\eta,*}}{1-\epsilon_{\eta,*}}\delta\ln\epsilon_\eta. \tag{5.112}$$

5.8.3 Projector calculus

Stages 191–192 then turn the quotient coordinates into an exact finite packet and an exact microscopic projector split. With the finite log-ratio vector ordered as

$$\Delta\mathbf{x} = (\Delta_\lambda, \Delta_c, \Delta_\gamma, \Delta_U, \Delta_{K_\eta}, \Delta_W, \Delta_\mu, \Delta_T)^T, \tag{5.113}$$

the quotient packet is

$$\mathbf{q} = (q_{\mathrm{tr}}, q_{\mathrm{nt}}, q_\eta)^T = M_*\Delta\mathbf{x}. \tag{5.114}$$

Choosing the dependent triple $(T_U, K_\eta^{(\mathrm{eff})}, \mu_W)$, the exact pivot block is

$$P_{(T,K_\eta,\mu)} = \begin{pmatrix} 1+\chi_{0,*} & 0 & 0 \\ -F_* & -1 & 1 \\ 0 & -1 & 0 \end{pmatrix}, \qquad \det P_{(T,K_\eta,\mu)} = 1 + \chi_{0,*} > 0. \tag{5.115}$$

The canonical section $S_{(T,K_\eta,\mu)}$ is the right inverse satisfying $M_* S_{(T,K_\eta,\mu)} = I_3$. The exact projectors are

$$Q_{\text{quot}} := S_{(T,K_\eta,\mu)} M_*, \qquad O_{\text{orb}} := I_8 - Q_{\text{quot}}. \tag{5.116}$$

They obey

$$Q_{\text{quot}}^2 = Q_{\text{quot}}, \qquad O_{\text{orb}}^2 = O_{\text{orb}}, \qquad M_* O_{\text{orb}} = 0, \qquad M_* Q_{\text{quot}} = M_*. \tag{5.117}$$

The quotient-failure piece is supported only on the dependent triple:

$$(\Delta_T)_{\text{fail}} = \frac{q_{\text{tr}}}{1+\chi_{0,*}}, \qquad (\Delta_{K_\eta})_{\text{fail}} = -q_\eta, \tag{5.118}$$

$$(\Delta_\mu)_{\text{fail}} = \frac{F_*}{1+\chi_{0,*}} q_{\text{tr}} + q_{\text{nt}} - q_\eta. \tag{5.119}$$

Thus

$$\mathbf{q} = 0 \iff M_* \Delta\mathbf{x} = 0 \iff Q_{\text{quot}} \Delta\mathbf{x} = 0 \iff \Delta\mathbf{x} \in \ker M_*. \tag{5.120}$$

Claim status. Stages 186–192 are Exact Within Closure **for the positive coherent reduced state space. The word "orbit" here means a finite reduced monomial-preserving similarity class, not a fundamental gauge symmetry of the unreduced parent PDE.**

5.9 Four-scalar packet (Δ_{full})

5.9.1 Packet A: conservative target and outgoing finish line

Stages 193–197 sharpen the branch-side Packet A. Start from the Stage 191 branch residual

$$\Delta_{\text{branch}} = (a_2, b_2, a_4, b_4, a_{P_0}, b_{P_0}, \Delta_{\text{pole}}, \Delta_{\text{norm}}). \tag{5.121}$$

Stage 193 freezes the conservative target surface

$$a_2 = b_2 = a_4 = b_4 = 0, \qquad \Delta_{\text{pole}} := \bar\nu_4 - 4\bar\nu_2^2 = 0. \tag{5.122}$$

On the isotropic branch this is equivalent to

$$D_0 D_4 + 3D_2^2 = 0, \tag{5.123}$$

or, in bundle variables,

$$D_0(B_4 + Z_4) = 3(M + B_2 + Z_2)^2. \tag{5.124}$$

The resulting conservative carrier is

$$\widehat{Y}_Q^{\text{cons}}(\omega) = \frac{3}{4} + \frac{1}{4}\frac{1}{1-\omega^2/\Omega_Q^2}. \tag{5.125}$$

Stage 193 also proves the scalar/geometry firewall: if weak anisotropy induces the first $l = 0 \leftrightarrow l = 2$ mixing, then Schur completion gives

$$\mathcal{D}_{2,\text{eff}}(\omega,\chi) = \mathcal{D}_2(\omega) I_3 - \chi^2 C(\omega)\mathcal{D}_0(\omega)^{-1} C(\omega)^T, \tag{5.126}$$

so

$$\partial_\chi \mathcal{D}_{2,\text{eff}}(\omega,\chi)|_{\chi=0} = 0. \tag{5.127}$$

The scalar/geometry lane cannot linearly contaminate the grouped $l = 2$ conservative carrier.

Stage 194 attaches the outgoing $l = 2$ DtN fingerprint. With

$$z := \frac{a\omega}{c_s}, \tag{5.128}$$

the exact outgoing operator is

$$\Lambda_2^{\rm out}(z) = z\frac{d}{dz}\ln h_2^{(1)}(z), \tag{5.129}$$

with normalized response

$$\widehat{Y}_2^{\rm out}(z) = 1 + \frac{z^2}{9} + \frac{4z^4}{81} + i\frac{z^5}{27} + O(z^6). \tag{5.130}$$

Matching this to the retarded grouped module fixes

$$\chi_Q = 1 \tag{5.131}$$

on the canonical compact passive/outgoing branch. More generally, for the isotropic deformation

$$\Lambda_2^{\rm def}(z) = S\Lambda_2^{\rm out}(\beta z) + \Sigma_0 + \Sigma_2 z^2 + \Sigma_4 z^4 + i\Sigma_5 z^5 + O(z^6), \tag{5.132}$$

canonical even matching forces Σ_2 and Σ_4, while the outgoing scalar is

$$\chi_Q = \frac{3(S\beta^5 + 9\Sigma_5)}{3S - \Sigma_0}, \qquad \chi_Q - 1 = \frac{3S(\beta^5 - 1) + \Sigma_0 + 27\Sigma_5}{3S - \Sigma_0}. \tag{5.133}$$

Thus the only isotropic DtN data that can move χ_Q, while preserving the canonical even moments, are β, Σ_0, and Σ_5.

Stage 195 factorizes the observable odd closure:

$$\widehat{m}_0^2\chi_Q N_Q = 1. \tag{5.134}$$

On the natural point-particle source map,

$$\widehat{m}_0 \to 1, \qquad N_Q = \chi_Q^{-1}. \tag{5.135}$$

Stage 196 proves that additional isotropic retarded tails beginning at $O(\omega^7)$ do not alter any coefficient through the point-particle 2.5PN order. Stage 197 therefore gives the exact Packet-A finish line

$$\Delta_{\rm branch} = 0 \iff \chi_Q = 1 \iff \Delta_Q := \chi_Q - 1 = 0. \tag{5.136}$$

5.9.2 Packet B: finite and pairwise orbit lock

Stages 198–199 complete the orbit side. With the same positive microscopic state vector (5.91), keep the free coordinates

$$(\lambda_W, c_{\eta U}, \gamma, K_U, K_W^{(\rm eff)}) \tag{5.137}$$

and the dependent triple

$$(T_U, K_\eta^{(\rm eff)}, \mu_W). \tag{5.138}$$

For a fixed invariant triple $(\mathfrak{C}_{{\rm tr},*}, \mathfrak{C}_{{\rm nt},*}, \epsilon_{\eta,*})$, Stage 198 solves the exact dependent orbit law and defines mismatch ratios

$$(m_T, m_K, m_\mu). \tag{5.139}$$

The logarithmic quotient chart is

$$q_{\rm tr} = (1+\chi_{0,*})\ln m_T, \qquad q_\eta = -\ln m_K, \qquad q_{\rm nt} = \ln m_\mu - \ln m_K - F_* \ln m_T. \tag{5.140}$$

Orbit lock is

$$\Delta_{\rm orbit} = 0 \iff m_T = m_K = m_\mu = 1 \iff q_{\rm tr} = q_{\rm nt} = q_\eta = 0. \tag{5.141}$$

Stage 199 removes reference-point privilege. For two states $x^{(1)}, x^{(2)}$, define

$$\Delta_{\rm orbit}^{(2\leftarrow 1)} = (q_{\rm tr}^{(2\leftarrow 1)}, q_{\rm nt}^{(2\leftarrow 1)}, q_\eta^{(2\leftarrow 1)}). \tag{5.142}$$

Then

$$x^{(2)} \in \mathcal{G}_* \cdot x^{(1)} \iff q_{\rm tr}^{(2\leftarrow 1)} = q_{\rm nt}^{(2\leftarrow 1)} = q_\eta^{(2\leftarrow 1)} = 0. \tag{5.143}$$

The packet is an exact additive two-point cocycle, while the multiplicative mismatch and invariant-ratio packets obey the corresponding multiplicative cocycle laws. Thus Packet B can be attached to an orbit rather than to a selected representative.

5.9.3 Reference-free full verdict packet

Let

$$\mathcal{O}_* := \mathcal{G}_* \cdot x_* \tag{5.144}$$

be the target similarity orbit. Stage 200 defines the additive full packet

$$\Delta_{\rm full}^{(x|\mathcal{O}_*)} := \left(\Delta_Q(x), q_{\rm tr}^{(x\leftarrow\mathcal{O}_*)}, q_{\rm nt}^{(x\leftarrow\mathcal{O}_*)}, q_\eta^{(x\leftarrow\mathcal{O}_*)}\right), \qquad \Delta_Q(x) := \chi_Q(x) - 1. \tag{5.145}$$

The multiplicative form is

$$\mathcal{V}_{\rm full}^{(x|\mathcal{O}_*)} := \left(\chi_Q(x), \mathfrak{R}_{\rm tr}^{(x\leftarrow\mathcal{O}_*)}, \mathfrak{R}_{\rm nt}^{(x\leftarrow\mathcal{O}_*)}, \mathfrak{R}_\eta^{(x\leftarrow\mathcal{O}_*)}\right), \tag{5.146}$$

and the mismatch form is

$$\mathcal{M}_{\rm full}^{(x|\mathcal{O}_*)} := \left(\chi_Q(x), m_T^{(x\leftarrow\mathcal{O}_*)}, m_K^{(x\leftarrow\mathcal{O}_*)}, m_\mu^{(x\leftarrow\mathcal{O}_*)}\right). \tag{5.147}$$

The exact chart conversions are

$$\mathfrak{R}_{\rm tr} = e^{q_{\rm tr}} = (m_T)^{1+\chi_{0,*}}, \qquad \mathfrak{R}_\eta = e^{q_\eta} = \frac{1}{m_K}, \tag{5.148}$$

$$\mathfrak{R}_{\rm nt} = e^{q_{\rm nt}} = \frac{m_\mu}{m_K m_T^{F_*}}. \tag{5.149}$$

Therefore the additive, multiplicative, and mismatch packets are exact reparameterizations of the same verdict.

The final reference-free theorem is

$$\Delta_{\rm full}^{(x|\mathcal{O}_*)} = 0 \iff \Delta_{\rm branch}(x) = 0 \text{ and } x \in \mathcal{O}_*. \tag{5.150}$$

Using the sharpened Packet-A and Packet-B forms, this is equivalently

$$\Delta_{\rm full}^{(x|\mathcal{O}_*)} = 0 \iff \chi_Q(x) = 1, \qquad q_{\rm tr}^{(x\leftarrow\mathcal{O}_*)} = q_{\rm nt}^{(x\leftarrow\mathcal{O}_*)} = q_\eta^{(x\leftarrow\mathcal{O}_*)} = 0. \tag{5.151}$$

This is the endpoint of Part V. The reduced compiler algebra is finished; what remains is branch realization of the four scalar conditions.

Claim status. Stages 193–200 are Exact Within Closure **once the Packet-A isotropic grouped one-pole front end, the natural point-particle source-map reduction, and the coherent orbit quotient hierarchy are adopted. The actual PDE may still fail any of the four final scalar conditions. That failure would be a realization failure, not an algebraic incompleteness of the ledger.**

5.10 Verification outputs and downstream use

The usable downstream citation packet from this chapter is as follows.

1. **MTDC-T9.1**: cite for the lower-branch drift laws, tangent-compensation theorem, weak-axisymmetric grouped signature, direct outlet map, hidden-even relation, and the loading mismatch $\Xi_{\text{load}} = N_{01}/N_0 - D_{01}/D_0$.
2. **MTDC-T9.2**: cite for the static self-similarity theorem, wall-normalized outgoing-load theorem, square-root mixed-leg law, port co-loading theorem, effective transfer-shape collapse, and coherent support-blindness.
3. **MTDC-T9.3**: cite for the microscopic slippage decomposition, triangular normal form, branch-invariant monomials, rank-three drift map, similarity orbit $\mathcal{G}_*$, and finite orbit–quotient closure.
4. **MTDC-T9.4**: cite for the branch-observable compiler, transfer-shape/outgoing-prefactor compiler, scalar no-go filters, minimal Packet A/Packet B data, and two-packet home-stretch theorem.
5. **MTDC-T9.5**: cite for the exact orbit/quotient projectors, isotropic grouped target surface, outgoing $l = 2$ DtN fingerprint, source-map reduction, higher-odd irrelevance, and Packet-A closure $\Delta_{\text{branch}} = 0 \iff \chi_Q = 1$.
6. **MTDC-T9.6**: cite for the finite orbit law, pairwise orbit transport, orbit-representative independence, and final four-scalar verdict packet $\Delta_{\text{full}} = (\Delta_Q, q_{\text{tr}}, q_{\text{nt}}, q_\eta)$.

- ☐ The off-family scalar $\delta_\perp$ is used only as the first post-Part-IV normal coordinate; after tangent compensation, first-order isotropic bundle drift is not counted as an independent failure mode.
- ☐ The lower compensated Family–1 drift laws are applied only under the D/N tube selection, healing-lock shell closure, and frozen $n = 5$ wall-EOS reduction.
- ☐ The four-bundle inversion through $(\Theta_w, K_s, K_q, P_0)$ is kept distinct from a direct microscopic PDE solve; it is an algebraic compiler once those observables are known.
- ☐ The weak-axisymmetric signature $(1, 1/2, -1)$ is not confused with arbitrary grouped anisotropy. It fixes the line $b = 3a$.
- ☐ The no-feed-down result is first order only: grouped real P_2 anisotropy can enter scalar ledgers through quadratic grouped invariants.
- ☐ The even-preserving branch is invoked before reducing the linear grouped problem to Ξ_{load}.
- ☐ Static self-similarity is stated in wall-referenced variables. Frozen port shapes do not by themselves imply zero outgoing defect when the wall baseline moves.
- ☐ The square-root mixed-leg law is a reduced sufficient condition under interference/hybridization rigidity, not an unconditional PDE theorem.

- ☐ The scalar $\Xi_1 = P_1/P_0$ is the physical outgoing-prefactor slope and effective transfer-shape slope; it is not introduced as a fit parameter.
- ☐ Coherent support-blindness applies to the direct transfer-shape defect, not to all support-dependent selected-branch quantities.
- ☐ The triangular normal form is used with branch-adapted variables $(\Sigma_{\rm tr}, \Sigma_{\rm nt}, \Sigma_\eta)$; raw microscopic slopes require the stated compiler before interpretation.
- ☐ The monomial coordinates $(\mathfrak{C}_{{\rm tr},*}, \mathfrak{C}_{{\rm nt},*}, \epsilon_\eta)$ are exact quotient coordinates of the reduced coherent hierarchy, not fundamental gauge invariants of the unsolved parent PDE.
- ☐ The similarity orbit $\mathcal{G}_*$ identifies reduced co-scaling fibres in the positive microscopic state space $\mathcal{M}_+$; branch realization still has to be checked against the actual PDE output.
- ☐ Packet A and Packet B are kept separate until the Stage 200 theorem: Packet A is the branch-side outgoing scalar; Packet B is the orbit-side quotient triple.
- ☐ The equivalence $N_Q = \chi_Q^{-1}$ is tied to the natural source-map reduction. It should not be used when the source-map factor $m_{\hat{0}}$ is still nontrivial.
- ☐ Higher odd retarded terms are irrelevant only to the reduced point-particle 2.5PN finish line; later higher-order retarded diagnostics may need them.
- ☐ The final verdict $\Delta_{\rm full} = 0$ is an exact reduced compiler condition. It is not an assertion that the completed moving-throat PDE has already realized the condition.

Future papers should cite Part V when replacing a long weak-axisymmetric or retarded endgame by the compact packet $(\Delta_Q, q_{\rm tr}, q_{\rm nt}, q_\eta)$. They should separately state whether the actual branch has returned $\Delta_Q = 0$ and $(q_{\rm tr}, q_{\rm nt}, q_\eta) = (0, 0, 0)$, or whether those realization conditions remain open.

Chapter 6

Realization Compiler and Finite Mixed-Ray Search

Stage range: 201–218

Part V ended with the reduced home-stretch theorem in its most compact form:

$$\Delta_{\text{full}} = 0 \quad \Longleftrightarrow \quad \chi_Q = 1, \qquad q_{\text{tr}} = q_{\text{nt}} = q_\eta = 0. \tag{6.1}$$

That theorem identifies the final reduced packet, but it does not by itself say how a completed moving-throat branch should be audited in practice. Part VI is the operational answer. It turns the final quotient packet into an explicit realization compiler, rewrites the target orbit as a graph over five free microscopic coordinates, reduces graph-aligned branch searches to a scalar closure function, and then proves that the local mixed-ray search over those five free coordinates is finite through support-cardinality five.

Part VI at a glance

Primary question	Once the final reduced packet is known, how should an actual moving-throat branch be audited in practice, and how large is the resulting local search problem?
Starting inputs	The Part V verdict packet Δ_{full}, its target-orbit description, and the coherent quotient/orbit machinery already fixed upstream.
Delivers	The intrinsic realization compiler, the canonical same-free-quintuple target graph, the scalar graph-slice closure law, and the finite mixed-ray search through support-cardinality five.
Used by	Part VII as the operational back end for actual same-charge branch data; Part VIII inherits the strict front-end packet extracted downstream from that audit chain.
Result anchors	**MTDC-T10**.

Stage packets.

Stages	Packet	Status	Carry-forward output
201–203	Realization compiler, canonical projection, and target graph	Exact Within Closure	Rewrites the verdict packet intrinsically, builds the canonical repair vector and same-free-quintuple projection, and reduces graph-aligned realization to the scalar closure law $\widehat{\chi}_Q(\mathbf{y}) = 1$.
204–205	Explicit log-ray sweep and local predictor machinery	Exact Within Closure	Introduces finite graph lifts along log rays, directional Hessians, quadratic predictors, and tangency/turning-point diagnostics for the scalar closure function.
206–212	Pairwise and triple search certification	Exact Within Closure / Numerically Located	Turns local curvature envelopes into certified brackets, optimizes pairwise and triple interiors, and closes the support-≤ 3 search ledger.
213–218	Higher-support simplex closure and final local search theorem	Exact Within Closure / Numerically Located / Open	Extends the sieve to support cardinalities four and five, proves the support ceiling, and closes the local mixed-ray search with a finite preferred budget once the required branch and envelope data are supplied.

Claim status. Part VI is predominantly Exact Within Closure **inside the declared realization/orbit/free-quintuple graph and local Hessian-envelope search hierarchy. The algebraic compilers, graph maps, stationary systems, candidate-set reductions, and support-cardinality closure theorems are exact once the reduced packet and local envelope data are supplied. Any claim that the completed nonlinear moving-throat PDE actually supplies the required free-quintuple branch, Hessian envelopes, validity intervals, or winning candidate remains** Open. **Any row that depends on numerical insertion of local envelope values or candidate evaluation is marked** Numerically Located **at the realization stage, not at the algebraic compiler stage.**

What this part proves. Part VI converts the final reduced packet into an operational realization compiler, solves the target orbit as a graph over a free quintuple, and proves that the local mixed-ray search is finite through support-cardinality five.

What this part does not prove. It does not prove that the completed PDE supplies the required branch data, local envelopes, or winning realization candidate. The compiler and finite search are exact; the physical winner remains an open branch question.

Why later parts need it. Part VII is the first direct application of this compiler to actual same-charge branch data, and Part VIII inherits the strict front-end packet that Part VII extracts from the completed realization machinery.

6.1 Block contract and theorem-level output

The contract of Part VI is different from the contract of the previous grouped-response chapters. Parts III–V constructed the reduced branch and quotient packets. Part VI assumes those packets exist and asks how they are to be realized by actual branch data. The operational chain is

$$\Delta_{\mathrm{full}} \longrightarrow \Delta^{\mathrm{int}}_{\mathrm{real}} \longrightarrow \Pi^{\mathrm{can}}_{\mathcal{O}_*} \longrightarrow \mathbf{x}^{\mathrm{graph}}_*(\mathbf{y}) \longrightarrow \widehat{\chi}_Q(\mathbf{y}) - 1 \longrightarrow \text{finite mixed-ray search.} \tag{6.2}$$

The first arrow turns the theorem packet into an intrinsic audit packet. The second and third arrows replace the abstract target orbit by an explicit graph. The fourth arrow reduces graph-aligned closure to one scalar. The last arrow proves that the local free-quintuple search is finite through the full support-cardinality ceiling.

The positive microscopic state carried throughout this part is

$$\mathbf{x} = \left(\lambda_W, c_{\eta U}, \gamma, K_U, K^{(\mathrm{eff})}_\eta, K^{(\mathrm{eff})}_W, \mu_W, T_U\right), \tag{6.3}$$

and the free quintuple is

$$\mathbf{y} = \left(\lambda_W, c_{\eta U}, \gamma, K_U, K^{(\mathrm{eff})}_W\right). \tag{6.4}$$

The dependent triple is

$$(T_U, K^{(\mathrm{eff})}_\eta, \mu_W). \tag{6.5}$$

This split is not a new physical assumption. It is the canonical section of the Stage 192 quotient map used to make the same-free-quintuple projection explicit.

Theorem 6.1 (Part VI realization compiler and finite mixed-ray closure theorem)**.** *Inside the carried realization hierarchy for Stages 201–218, the following theorem chain holds.*

1. *The final four-scalar verdict packet can be written intrinsically as*

$$\Delta^{\mathrm{int}}_{\mathrm{real}}(\mathbf{x} \mid \mathcal{Z}_*) = \left(\chi_Q(\mathbf{x}) - 1, q_{\mathrm{tr}}(\mathbf{x}), q_{\mathrm{nt}}(\mathbf{x}), q_\eta(\mathbf{x})\right), \tag{6.6}$$

   where the three quotient coordinates are logarithmic ratios of the three target monomials.

2. *The canonical repair vector is supported only on the dependent triple,*

$$\Delta\mathbf{x}_{\mathrm{rep}} = \begin{pmatrix} 0 & 0 & 0 & 0 & q_\eta & 0 & -q_{\mathrm{nt}} + q_\eta - \dfrac{F_*}{1+\chi_{0,*}} q_{\mathrm{tr}} & -\dfrac{q_{\mathrm{tr}}}{1+\chi_{0,*}} \end{pmatrix}^T, \tag{6.7}$$

   and exponentiating it gives the canonical target-orbit projection $\Pi^{\mathrm{can}}_{\mathcal{O}_*}$.

3. *The target orbit $\mathcal{O}_*$ is exactly the graph of three dependent target functions over the free quintuple $\mathbf{y}$:*

$$\mathcal{O}_* = \{\mathbf{x}^{\mathrm{graph}}_*(\mathbf{y}) : \mathbf{y} \in (\mathbb{R}_{>0})^5\}. \tag{6.8}$$

4. *The graph errors*

$$(E_T, E_K, E_\mu) = \left(\ln \frac{T_U}{T^{\mathrm{graph}}_U}, \ln \frac{K^{(\mathrm{eff})}_\eta}{K^{\mathrm{graph}}_{\eta,*}}, \ln \frac{\mu_W}{\mu^{\mathrm{graph}}_W}\right) \tag{6.9}$$

   are exactly equivalent to the quotient packet:

$$E_T = \frac{q_{\mathrm{tr}}}{1+\chi_{0,*}}, \qquad E_K = -q_\eta, \qquad E_\mu = q_{\mathrm{nt}} - q_\eta + \frac{F_*}{1+\chi_{0,*}} q_{\mathrm{tr}}. \tag{6.10}$$

5. *On the target graph, Packet B vanishes identically and the full closure set is the scalar graph slice*

$$\mathcal{Z}_* = \{\mathbf{x}^{\mathrm{graph}}_*(\mathbf{y}) : \widehat{\chi}_Q(\mathbf{y}) = 1\}, \qquad \widehat{\chi}_Q(\mathbf{y}) := \chi_Q(\mathbf{x}^{\mathrm{graph}}_*(\mathbf{y})). \tag{6.11}$$

6. *Every smooth graph-aligned branch is locally represented by a free-quintuple log ray*

$$\mathbf{y}_{\mathbf{s}}(\tau) = \mathbf{y}_{\circ} \odot e^{\tau \mathbf{s}}, \tag{6.12}$$

and closure along that ray is the scalar root problem

$$\Phi_{\mathbf{s}}(\tau) := \widehat{\chi}_Q(\mathbf{y}_{\mathbf{s}}(\tau)) = 1. \tag{6.13}$$

7. *The first- and second-order local data of the ray are*

$$\Phi_0, \quad \Phi_1 = (\mathcal{D}_{\mathbf{s}}\widehat{\chi}_Q)(\mathbf{y}_{\circ}), \quad \Phi_2 = (\mathcal{D}^2_{\mathbf{s}}\widehat{\chi}_Q)(\mathbf{y}_{\circ}), \tag{6.14}$$

or, equivalently, logarithmic data $(H_0, K_0, \underline{K}_1, \overline{K}_1)$ *after orientation. These data determine certified local brackets whenever the Stage 206 admissibility hypotheses hold.*

8. *The primitive, pairwise, triple, quadruple, and five-coordinate searches are finite certified algebraic ledgers. The support-cardinality ceiling is five because the free space has exactly five primitive axes.*

9. *The final local mixed-ray search on the positive free-quintuple simplex is reduced to the splice*

$$\tau^{\mathrm{lo}}_{\le 5,\min} = \min(\tau^{\mathrm{lo}}_{\le 4,\min}, \tau^{\mathrm{lo,int}}_{5,\min}), \qquad \tau^{\mathrm{hi}}_{\le 5,\min} = \min(\tau^{\mathrm{hi}}_{\le 4,\min}, \tau^{\mathrm{hi,int}}_{5,\min}), \tag{6.15}$$

with

$$\tau^{\mathrm{lo}}_{\le 5,\min} \le \tau^{\mathrm{best}}_{\le 5,*} \le \tau^{\mathrm{hi}}_{\le 5,\min}. \tag{6.16}$$

Proof structure. Stages 201–202 convert the reference-free four-scalar packet into an intrinsic monomial-ratio audit, a canonical same-free-quintuple projection, and an explicit target graph. Stage 203 proves that graph-aligned families lie in the kernel of the quotient map and hence reduce to the scalar closure function $\widehat{\chi}_Q - 1$. Stages 204–207 build the log-ray, Hessian, curvature-envelope, and primitive certified-ray compilers. Stages 208–209 add pairwise mixed cones and reduce the one-parameter ratio optimization to finite quartic candidate sets. Stages 210–212 do the same for genuine three-coordinate simplex interiors and then splice them into the pairwise boundary ledger. Stages 213–215 repeat the construction for four-coordinate interiors and splice the result into the support-≤ 3 ledger. Stages 216–218 identify the unique five-coordinate simplex, reduce its interior optimizer to a finite lifted algebraic candidate set, and finally prove that its boundary is exactly the already-solved support-≤ 4 search set. □

6.2 Exact realization compiler and target graph

6.2.1 Intrinsic realization packet

Stage 201 begins from the three target monomials already isolated by the orbit/quotient theorem,

$$\mathfrak{C}_{\mathrm{tr},*}(\mathbf{x}), \qquad \mathfrak{C}_{\mathrm{nt},*}(\mathbf{x}), \qquad \epsilon_\eta(\mathbf{x}). \tag{6.17}$$

Fix target values

$$\mathfrak{C}^{\mathrm{target}}_{\mathrm{tr},*}, \qquad \mathfrak{C}^{\mathrm{target}}_{\mathrm{nt},*}, \qquad \epsilon^{\mathrm{target}}_{\eta,*}. \tag{6.18}$$

The target orbit is then

$$\mathcal{O}_* = \left\{ \mathbf{x} > 0 : \mathfrak{C}_{\mathrm{tr},*} = \mathfrak{C}^{\mathrm{target}}_{\mathrm{tr},*}, \quad \mathfrak{C}_{\mathrm{nt},*} = \mathfrak{C}^{\mathrm{target}}_{\mathrm{nt},*}, \quad \epsilon_\eta = \epsilon^{\mathrm{target}}_{\eta,*} \right\}. \tag{6.19}$$

The intrinsic ratios are

$$\mathfrak{R}_{\mathrm{tr}} := \frac{\mathfrak{C}_{\mathrm{tr},*}}{\mathfrak{C}_{\mathrm{tr},*}^{\mathrm{target}}}, \qquad \mathfrak{R}_{\mathrm{nt}} := \frac{\mathfrak{C}_{\mathrm{nt},*}}{\mathfrak{C}_{\mathrm{nt},*}^{\mathrm{target}}}, \qquad \mathfrak{R}_{\eta} := \frac{\epsilon_\eta}{\epsilon_{\eta,*}^{\mathrm{target}}}, \tag{6.20}$$

and the logarithmic quotient coordinates are

$$q_{\mathrm{tr}} := \ln \mathfrak{R}_{\mathrm{tr}}, \qquad q_{\mathrm{nt}} := \ln \mathfrak{R}_{\mathrm{nt}}, \qquad q_\eta := \ln \mathfrak{R}_\eta. \tag{6.21}$$

Thus

$$\mathbf{x} \in \mathcal{O}_* \iff q_{\mathrm{tr}} = q_{\mathrm{nt}} = q_\eta = 0. \tag{6.22}$$

The fully closed target set is

$$\mathcal{Z}_* := \{\mathbf{x} \in \mathcal{O}_* : \chi_Q(\mathbf{x}) = 1\}. \tag{6.23}$$

Consequently the intrinsic realization packet is exactly

$$\Delta_{\mathrm{real}}^{\mathrm{int}}(\mathbf{x} \mid \mathcal{Z}_*) = \left(\chi_Q(\mathbf{x}) - 1, q_{\mathrm{tr}}(\mathbf{x}), q_{\mathrm{nt}}(\mathbf{x}), q_\eta(\mathbf{x})\right). \tag{6.24}$$

This is the operational version of Part V's reference-free packet.

6.2.2 Canonical same-free-quintuple projection

The Stage 192 quotient map is carried in the linearized form

$$\mathbf{q} = M_* \Delta \mathbf{x}, \qquad \mathbf{q} = (q_{\mathrm{tr}}, q_{\mathrm{nt}}, q_\eta)^T. \tag{6.25}$$

The canonical section chooses the dependent triple $(T_U, K_\eta^{\mathrm{eff}}, \mu_W)$. The exact repair vector is

$$\Delta \mathbf{x}_{\mathrm{rep}} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ q_\eta \\ 0 \\ -q_{\mathrm{nt}} + q_\eta - \dfrac{F_*}{1+\chi_{0,*}} q_{\mathrm{tr}} \\ -\dfrac{q_{\mathrm{tr}}}{1+\chi_{0,*}} \end{pmatrix}. \tag{6.26}$$

It satisfies

$$M_* \Delta \mathbf{x}_{\mathrm{rep}} = -\mathbf{q}. \tag{6.27}$$

Exponentiating (6.26) gives the canonical projection

$$\begin{aligned} \Pi_{\mathcal{O}_*}^{\mathrm{can}}(\mathbf{x}) = \big(&\lambda_W, c_{\eta U}, \gamma, K_U, K_\eta^{(\mathrm{eff})} \mathfrak{R}_\eta, K_W^{(\mathrm{eff})}, \\ &\mu_W \mathfrak{R}_{\mathrm{nt}}^{-1} \mathfrak{R}_\eta \mathfrak{R}_{\mathrm{tr}}^{-F_*/(1+\chi_{0,*})}, T_U \mathfrak{R}_{\mathrm{tr}}^{-1/(1+\chi_{0,*})}\big). \end{aligned} \tag{6.28}$$

The projection lands on the target orbit, preserves the free quintuple (6.4), and is the unique point of $\mathcal{O}_*$ with that free quintuple.

Proposition 6.2 (Canonical fixed-point audit)**.** *For any positive microscopic state* $\mathbf{x}$*,*

$$\mathbf{x} \in \mathcal{Z}_* \iff \chi_Q(\mathbf{x}) = 1 \quad \textit{and} \quad \Pi_{\mathcal{O}_*}^{\mathrm{can}}(\mathbf{x}) = \mathbf{x}. \tag{6.29}$$

6.2.3 Direct microscopic target graph

Stage 202 removes the remaining abstractness by solving the orbit constraints directly. The carried monomials are

$$\mathfrak{C}_{\mathrm{tr},*} = \left(\frac{\gamma c_{\eta U}}{K_U}\right)^{1+\delta_{U,*}} \left(\frac{\pi^2 T_U}{L^2 K_U}\right)^{1+\chi_{0,*}}, \tag{6.30}$$

$$\mathfrak{C}_{\mathrm{nt},*} = \frac{\lambda_W^2 \mu_W}{K_\eta^{(\mathrm{eff})}(K_W^{(\mathrm{eff})})^2} \left(\frac{\gamma^2 \lambda_W^2 \sigma}{K_U K_W^{(\mathrm{eff})}}\right)^{E_*} \left(\frac{\pi^2 T_U}{L^2 K_U}\right)^{-F_*}, \tag{6.31}$$

$$\epsilon_\eta = \frac{c_{\eta U}^2}{K_U K_\eta^{(\mathrm{eff})}}. \tag{6.32}$$

Solving (6.30)–(6.32) at fixed $\mathbf{y}$ gives

$$\delta_{U,*}^{\mathrm{graph}}(\mathbf{y}) := \left[\frac{\mathfrak{C}_{\mathrm{tr},*}^{\mathrm{target}}}{\left(\frac{\gamma c_{\eta U}}{K_U}\right)^{1+\delta_{U,*}}}\right]^{1/(1+\chi_{0,*})}, \tag{6.33}$$

$$T_U^{\mathrm{graph}}(\mathbf{y}) = \frac{L^2 K_U}{\pi^2}\, \delta_{U,*}^{\mathrm{graph}}(\mathbf{y}), \tag{6.34}$$

$$K_{\eta,*}^{\mathrm{graph}}(\mathbf{y}) = \frac{c_{\eta U}^2}{K_U \epsilon_{\eta,*}^{\mathrm{target}}}, \tag{6.35}$$

$$\mu_W^{\mathrm{graph}}(\mathbf{y}) = \frac{\mathfrak{C}_{\mathrm{nt},*}^{\mathrm{target}}\, c_{\eta U}^2 (K_W^{(\mathrm{eff})})^2}{\epsilon_{\eta,*}^{\mathrm{target}} K_U \lambda_W^2} \left(\frac{\gamma^2 \lambda_W^2 \sigma}{K_U K_W^{(\mathrm{eff})}}\right)^{-E_*} \left(\delta_{U,*}^{\mathrm{graph}}(\mathbf{y})\right)^{F_*}. \tag{6.36}$$

The graph point is

$$\mathbf{x}_*^{\mathrm{graph}}(\mathbf{y}) = \left(\lambda_W, c_{\eta U}, \gamma, K_U, K_{\eta,*}^{\mathrm{graph}}(\mathbf{y}), K_W^{(\mathrm{eff})}, \mu_W^{\mathrm{graph}}(\mathbf{y}), T_U^{\mathrm{graph}}(\mathbf{y})\right). \tag{6.37}$$

Substitution shows

$$\mathcal{O}_* = \left\{\mathbf{x}_*^{\mathrm{graph}}(\mathbf{y}) : \mathbf{y} \in (\mathbb{R}_{>0})^5\right\}. \tag{6.38}$$

Thus the canonical projection and the graph projection coincide:

$$\Pi_{\mathcal{O}_*}^{\mathrm{can}}(\mathbf{x}) = \mathbf{x}_*^{\mathrm{graph}}(\pi_{\mathrm{free}}\mathbf{x}). \tag{6.39}$$

6.2.4 Graph-error realization coordinates

For a same-free-quintuple candidate state, define

$$E_T := \ln \frac{T_U}{T_U^{\mathrm{graph}}}, \qquad E_K := \ln \frac{K_\eta^{(\mathrm{eff})}}{K_{\eta,*}^{\mathrm{graph}}}, \qquad E_\mu := \ln \frac{\mu_W}{\mu_W^{\mathrm{graph}}}. \tag{6.40}$$

The exact relations to the quotient packet are (6.10). The reduced-family audit packet becomes

$$\Delta_{\mathrm{fam}}^{\mathrm{graph}}(\mathbf{y}) = \left(\chi_Q(\mathbf{x}_{\mathrm{cand}}(\mathbf{y})) - 1, E_T, E_K, E_\mu\right), \tag{6.41}$$

and the closure criterion is

$$\Delta_{\mathrm{fam}}^{\mathrm{graph}}(\mathbf{y}) = 0 \quad \Longleftrightarrow \quad \mathbf{x}_{\mathrm{cand}}(\mathbf{y}) \in \mathcal{Z}_*. \tag{6.42}$$

Claim status. Stages 201–202 are Exact Within Closure**. They are exact consequences of the already-declared monomial quotient map, target-orbit definition, and canonical dependent-triple section. They do not prove that the actual PDE branch is graph-aligned; they provide the exact coordinates in which that claim can be tested.**

6.3 Scalar graph-slice and log-ray compiler

6.3.1 Graph closure scalar

Stage 203 defines

$$\widehat{\chi}_Q(\mathbf{y}) := \chi_Q\big(\mathbf{x}_*^{\mathrm{graph}}(\mathbf{y})\big), \qquad \widehat{\Delta}_Q(\mathbf{y}) := \widehat{\chi}_Q(\mathbf{y}) - 1. \tag{6.43}$$

Since every graph point already lies on $\mathcal{O}_*$, the closed set is exactly

$$\mathcal{Z}_* = \left\{\mathbf{x}_*^{\mathrm{graph}}(\mathbf{y}) : \widehat{\Delta}_Q(\mathbf{y}) = 0\right\}. \tag{6.44}$$

This is the scalar graph-slice theorem. Its importance is that the realization problem drops from four scalar equations to one scalar equation as soon as the candidate family is graph-aligned.

6.3.2 Graph tangent identities

Let $\mathbf{y}(\tau)$ be a smooth free-quintuple family and define free log-tangent components

$$(\lambda_1, c_1, \gamma_1, \kappa_U, \kappa_W) = \left(\frac{d\ln\lambda_W}{d\tau}, \frac{d\ln c_{\eta U}}{d\tau}, \frac{d\ln\gamma}{d\tau}, \frac{d\ln K_U}{d\tau}, \frac{d\ln K_W^{(\mathrm{eff})}}{d\tau}\right). \tag{6.45}$$

Then the dependent graph log tangents are

$$\frac{d\ln\delta_{U,*}^{\mathrm{graph}}}{d\tau} = -\frac{1+\delta_{U,*}}{1+\chi_{0,*}}(\gamma_1 + c_1 - \kappa_U), \tag{6.46}$$

$$\tau_1^{\mathrm{graph}} = \kappa_U - \frac{1+\delta_{U,*}}{1+\chi_{0,*}}(\gamma_1 + c_1 - \kappa_U), \tag{6.47}$$

$$\kappa_\eta^{\mathrm{graph}} = 2c_1 - \kappa_U, \tag{6.48}$$

$$\begin{aligned}\mu_1^{\mathrm{graph}} = {} & 2c_1 - \kappa_U + 2\kappa_W - 2\lambda_1 - E_*(2\gamma_1 + 2\lambda_1 - \kappa_U - \kappa_W) \\ & - F_*\frac{1+\delta_{U,*}}{1+\chi_{0,*}}(\gamma_1 + c_1 - \kappa_U).\end{aligned} \tag{6.49}$$

Substitution into the monomial-drift matrix gives

$$M_*\dot{\Delta\mathbf{x}}_{\mathrm{graph}} = 0. \tag{6.50}$$

So graph-lifted families are tangent to the similarity orbit at every point, and Packet B vanishes identically on the graph.

6.3.3 One-parameter crossing theorem

Let $\mathbf{y} : [\tau_-, \tau_+] \to (\mathbb{R}_{>0})^5$ be continuous and graph-lift it. If

$$\widehat{\Delta}_Q(\tau_-)\,\widehat{\Delta}_Q(\tau_+) < 0, \tag{6.51}$$

then there exists $\tau_* \in (\tau_-, \tau_+)$ with

$$\widehat{\Delta}_Q(\tau_*) = 0, \qquad \mathbf{x}_*^{\mathrm{graph}}(\mathbf{y}(\tau_*)) \in \mathcal{Z}_*. \tag{6.52}$$

This is not a numerical method by itself. It is the exact existence theorem that justifies the later scalar log-ray search.

6.3.4 Free-quintuple log rays

Stage 204 chooses the canonical local family

$$\mathbf{y}_{\mathbf{s}}(\tau) = \mathbf{y}_\circ \odot e^{\tau \mathbf{s}}, \qquad \mathbf{s} = (s_\lambda, s_c, s_\gamma, s_U, s_W). \tag{6.53}$$

Every smooth positive free-quintuple path has this form to first order at any point, with $\mathbf{s} = d\ln \mathbf{y}/d\tau$. Along the log ray define

$$\mathfrak{a}_* := \frac{1+\delta_{U,*}}{1+\chi_{0,*}}, \tag{6.54}$$

$$\sigma_\delta = -\mathfrak{a}_*(s_\gamma + s_c - s_U), \tag{6.55}$$

$$\sigma_T = s_U + \sigma_\delta, \qquad \sigma_{K_\eta} = 2s_c - s_U, \tag{6.56}$$

$$\sigma_\mu = 2s_c - s_U + 2s_W - 2s_\lambda - E_*(2s_\gamma + 2s_\lambda - s_U - s_W) + F_*\sigma_\delta. \tag{6.57}$$

The graph-lifted ray is therefore

$$\begin{aligned}\mathbf{x}_{\mathbf{s}}^{\mathrm{graph}}(\tau) = (&\lambda_{W,\circ}e^{s_\lambda \tau}, c_{\eta U,\circ}e^{s_c\tau}, \gamma_\circ e^{s_\gamma \tau}, K_{U,\circ}e^{s_U\tau}, K_{\eta,\circ}^{\mathrm{graph}}e^{\sigma_{K_\eta}\tau},\\ &K_{W,\circ}^{(\mathrm{eff})}e^{s_W\tau}, \mu_{W,\circ}^{\mathrm{graph}}e^{\sigma_\mu\tau}, T_{U,\circ}^{\mathrm{graph}}e^{\sigma_T\tau}).\end{aligned} \tag{6.58}$$

The three target monomials are invariant for all τ on this lift. Hence the scalarized closure function

$$\Phi_{\mathbf{s}}(\tau) := \widehat{\chi}_Q(\mathbf{y}_{\mathbf{s}}(\tau)) \tag{6.59}$$

contains the entire graph-aligned realization question:

$$\mathbf{x}_{\mathbf{s}}^{\mathrm{graph}}(\tau) \in \mathcal{Z}_* \quad\Longleftrightarrow\quad \Phi_{\mathbf{s}}(\tau) = 1. \tag{6.60}$$

6.3.5 First-order and second-order root predictors

Let

$$\Phi_0 := \Phi_{\mathbf{s}}(0), \qquad \Phi_1 := \Phi_{\mathbf{s}}'(0), \qquad \Phi_2 := \Phi_{\mathbf{s}}''(0). \tag{6.61}$$

The first-order predictors are

$$\tau_{\mathrm{aff}} = \frac{1-\Phi_0}{\Phi_1}, \qquad \tau_{\log} = -\frac{\ln \Phi_0}{L_0}, \qquad L_0 := \frac{\Phi_1}{\Phi_0}. \tag{6.62}$$

Stage 205 introduces free log coordinates

$$\boldsymbol{\ell} = (\ln \lambda_W, \ln c_{\eta U}, \ln \gamma, \ln K_U, \ln K_W^{(\mathrm{eff})}), \tag{6.63}$$

and the directional operators

$$\mathcal{D}_{\mathbf{s}} = \sum_i s_i \partial_{\ell_i}, \qquad \mathcal{H}_{\mathbf{s}} = \mathcal{D}_{\mathbf{s}}^2. \tag{6.64}$$

Assuming $\Phi_0 > 0$, define

$$L_1 := \frac{d^2}{d\tau^2} \ln \Phi_{\mathbf{s}}(\tau)\bigg|_0 = \frac{\Phi_2}{\Phi_0} - \left(\frac{\Phi_1}{\Phi_0}\right)^2. \tag{6.65}$$

Thus

$$\Phi_2 = \Phi_0(L_1 + L_0^2). \tag{6.66}$$

The quadratic affine and logarithmic discriminants are

$$\Delta_{\text{aff}} = \Phi_1^2 - 2\Phi_2(\Phi_0 - 1), \qquad \Delta_{\log} = L_0^2 - 2L_1 \ln \Phi_0. \tag{6.67}$$

When the appropriate discriminant is nonnegative, the continuous-branch predictors are

$$\tau_{\text{quad}} = \frac{2(1-\Phi_0)}{\Phi_1 + \operatorname{sgn}(\Phi_1)\sqrt{\Delta_{\text{aff}}}}, \tag{6.68}$$

$$\tau_{\log,2} = -\frac{2\ln \Phi_0}{L_0 + \operatorname{sgn}(L_0)\sqrt{\Delta_{\log}}}. \tag{6.69}$$

If $\Phi_1 = 0$ but $\Phi_0 \neq 1$, the quadratic turning-point criterion is

$$(1-\Phi_0)\Phi_2 > 0, \tag{6.70}$$

with local predictors

$$\tau_\pm = \pm\sqrt{\frac{2(1-\Phi_0)}{\Phi_2}}. \tag{6.71}$$

These are the local predictor tools used by the certified search sieve.

Claim status. Stages 203–205 are Exact Within Closure **once $\widehat{\chi}_Q$ is a differentiable scalar on the target graph. The actual values of the derivatives and Hessians are PDE-returned or candidate-family data; the predictor formulas themselves are exact.**

6.4 Certified ray sieve: primitive and pairwise searches

6.4.1 Curvature-envelope brackets

Stage 206 orients the log residual. Let

$$H_{\mathbf{s}}(\tau) := \operatorname{sgn}(\ln \Phi_{\mathbf{s}}(0)) \ln \Phi_{\mathbf{s}}(\tau), \tag{6.72}$$

with

$$H_0 := H_{\mathbf{s}}(0) > 0, \qquad K_0 := H_{\mathbf{s}}'(0) < 0. \tag{6.73}$$

Assume a curvature envelope on $[0, T]$:

$$\underline{K}_1 \le H_{\mathbf{s}}''(\tau) \le \overline{K}_1. \tag{6.74}$$

Then

$$H_-(\tau) = H_0 + K_0\tau + \frac{1}{2}\underline{K}_1\tau^2, \qquad H_+(\tau) = H_0 + K_0\tau + \frac{1}{2}\overline{K}_1\tau^2 \tag{6.75}$$

sandwich the true residual. Define

$$\mathcal{T}(H_0, K_0; c) := -\frac{2H_0}{K_0 + \operatorname{sgn}(K_0)\sqrt{K_0^2 - 2cH_0}}. \tag{6.76}$$

For $K_0 < 0$, the certified bracket is

$$\tau_{\text{lo}} = \mathcal{T}(H_0, K_0; \underline{K}_1), \qquad \tau_{\text{hi}} = \mathcal{T}(H_0, K_0; \overline{K}_1), \tag{6.77}$$

provided both discriminants are real and $\tau_{\text{hi}} \leq T$. The true local root is then unique in $[\tau_{\text{lo}}, \tau_{\text{hi}}]$. For a turning ray $K_0 = 0$, strict negative curvature gives

$$\tau_{\text{lo}}^{(\text{tp})} = \sqrt{-\frac{2H_0}{\underline{K}_1}}, \qquad \tau_{\text{hi}}^{(\text{tp})} = \sqrt{-\frac{2H_0}{\overline{K}_1}}. \tag{6.78}$$

6.4.2 Primitive certified table

The primitive free axes are

$$\mathfrak{I}_5 = \{\lambda, c, \gamma, U, W\}. \tag{6.79}$$

Let

$$h(\boldsymbol{\ell}) := \ln \widehat{\chi}_Q(\boldsymbol{\ell}), \qquad \sigma_0 := \operatorname{sgn}(h(\boldsymbol{\ell}_\circ)). \tag{6.80}$$

The oriented primitive gradients are

$$\Gamma_i := \sigma_0 \partial_{\ell_i} h(\boldsymbol{\ell}_\circ). \tag{6.81}$$

If $\Gamma_i \neq 0$, the canonical forward primitive direction is

$$\widehat{\mathbf{e}}_i := \varepsilon_i \mathbf{e}_i, \qquad \varepsilon_i := -\operatorname{sgn}(\Gamma_i), \qquad k_i := |\Gamma_i|. \tag{6.82}$$

The primitive curvature uses only the diagonal Hessian restriction:

$$H_1^{(i)}(\tau) = \sigma_0 \partial_{\ell_i \ell_i} h(\boldsymbol{\ell}_\circ + \tau \widehat{\mathbf{e}}_i). \tag{6.83}$$

Thus primitive certification needs no off-diagonal Hessian data. Each primitive row is decided by

$$\mathcal{R}_i^{\text{prim}} := (H_0, \Gamma_i, \underline{\kappa}_i, \overline{\kappa}_i, T_i). \tag{6.84}$$

For monotone rows, the certified interval is

$$\tau_{i,\text{lo}} = \mathcal{T}(H_0, -k_i; \underline{\kappa}_i), \qquad \tau_{i,\text{hi}} = \mathcal{T}(H_0, -k_i; \overline{\kappa}_i), \tag{6.85}$$

with the same discriminant and validity checks as Stage 206.

6.4.3 Pairwise mixed rays

For a monotone pair (i, j), Stage 208 introduces

$$\widehat{\mathbf{s}}_{ij}(r) = \frac{\widehat{\mathbf{e}}_i + r\widehat{\mathbf{e}}_j}{\sqrt{1 + r^2}}, \qquad r \geq 0. \tag{6.86}$$

The slope magnitude is

$$k_{ij}(r) = \frac{k_i + rk_j}{\sqrt{1 + r^2}}, \tag{6.87}$$

with unique gradient-optimal ratio

$$r_{ij}^{\mathrm{grad}} = \frac{k_j}{k_i}, \qquad k_{ij}^{\mathrm{grad}} = \sqrt{k_i^2 + k_j^2}. \tag{6.88}$$

The pairwise curvature is

$$H_{1,ij}(r;\tau) = \frac{h_{ii} + 2rh_{ij} + r^2 h_{jj}}{1+r^2}, \tag{6.89}$$

so the off-diagonal Hessian enters with cross weight

$$w_\times(r) = \frac{2r}{1+r^2}, \tag{6.90}$$

maximized at the equal-mix ray $r = 1$. Thus the two canonical pair screens are

$$\widehat{\mathbf{s}}_{ij}^{\mathrm{grad}} = \frac{k_i \widehat{\mathbf{e}}_i + k_j \widehat{\mathbf{e}}_j}{\sqrt{k_i^2 + k_j^2}}, \qquad \widehat{\mathbf{s}}_{ij}^{\mathrm{eq}} = \frac{\widehat{\mathbf{e}}_i + \widehat{\mathbf{e}}_j}{\sqrt{2}}. \tag{6.91}$$

6.4.4 Pairwise ratio optimizer

Stage 209 closes the full one-parameter pair search. On a compact ratio window $[0, R_{ij}]$, for envelope label $\star \in \{\mathrm{lo}, \mathrm{hi}\}$, define

$$A_\star = k_i^2 - 2H_0 u_\star, \qquad B_\star = 2k_i k_j - 4H_0 v_\star, \qquad C_\star = k_j^2 - 2H_0 w_\star. \tag{6.92}$$

The discriminant numerator is

$$\Delta_{ij,\star}^{\sharp}(r) = A_\star + B_\star r + C_\star r^2. \tag{6.93}$$

The certified root function is

$$\tau_{ij,\star}(r) = \frac{2H_0\sqrt{1+r^2}}{k_i + rk_j + \sqrt{A_\star + B_\star r + C_\star r^2}}. \tag{6.94}$$

Interior optimizers satisfy the stationary numerator equation

$$\mathcal{N}_{ij,\star}(r) = 2(k_j - k_i r)\sqrt{A_\star + B_\star r + C_\star r^2} + B_\star + 2(C_\star - A_\star)r - B_\star r^2 = 0. \tag{6.95}$$

Eliminating the square root gives the quartic condition

$$\mathcal{Q}_{ij,\star}(r) = \Big[B_\star + 2(C_\star - A_\star)r - B_\star r^2\Big]^2 - 4(k_j - k_i r)^2(A_\star + B_\star r + C_\star r^2) = 0. \tag{6.96}$$

After filtering extraneous roots, each envelope optimizer is found in a finite set containing admissible endpoints and at most four admissible quartic roots. Therefore one pair requires at most twelve exact candidate evaluations across the lower and upper envelopes.

Claim status. Stages 206–209 are Exact Within Closure **once the local gradient, Hessian-envelope, and validity data are supplied. If those inputs come from numerical PDE evaluation, the final ray rankings are** Numerically Located **insertions into exact certified formulas.**

6.5 Three- and four-coordinate simplex sieves

6.5.1 Three-coordinate simplex and boundary reduction

For a monotone primitive triple (i, j, k), the positive spherical simplex is

$$\Delta^+_{ijk} = \left\{(a_i, a_j, a_k) \in \mathbb{R}^3_{\geq 0} : a_i^2 + a_j^2 + a_k^2 = 1\right\}. \tag{6.97}$$

The ray is

$$\widehat{\mathbf{s}}_{ijk} = a_i\widehat{\mathbf{e}}_i + a_j\widehat{\mathbf{e}}_j + a_k\widehat{\mathbf{e}}_k. \tag{6.98}$$

Each boundary edge is exactly one Stage 209 pairwise cone. Hence only the interior of Δ^+_{ijk} is new at support-cardinality three.

The gradient-optimal point is

$$\mathbf{a}^{\mathrm{grad}}_{ijk} = \frac{(k_i, k_j, k_k)}{\sqrt{k_i^2 + k_j^2 + k_k^2}}, \tag{6.99}$$

and the equal-mix barycenter is

$$\mathbf{a}^{\mathrm{eq}}_{ijk} = \frac{(1, 1, 1)}{\sqrt{3}}. \tag{6.100}$$

The total off-diagonal leverage weight is

$$w_\Sigma(\mathbf{a}) = \mathbf{2(a_ia_j + a_ia_k + a_ja_k) = (a_i + a_j + a_k)^2 - 1 \leq 2}, \tag{6.101}$$

with equality at (6.100). Thus the gradient screen maximizes first-order descent, while the equal-mix screen maximizes three-way cross-Hessian leverage.

6.5.2 Triple interior optimizer

On the interior patch $a_i > 0$, write

$$(a_i, a_j, a_k) = \frac{(1, r, s)}{\sqrt{1 + r^2 + s^2}}. \tag{6.102}$$

For either envelope label $\star$, define the quadratic discriminant numerator

$$\Delta^\sharp_{ijk,\star}(r, s) = A_\star + B_\star r + C_\star s + D_\star r^2 + E_\star rs + F_\star s^2. \tag{6.103}$$

The certified root is

$$\tau_{ijk,\star}(r, s) = \frac{2H_0\sqrt{1 + r^2 + s^2}}{k_i + rk_j + sk_k + \sqrt{\Delta^\sharp_{ijk,\star}(r, s)}}. \tag{6.104}$$

The stationary equations can be written as two square-root numerator equations. Eliminating the square root gives a quartic cross-consistency polynomial and sextic square conditions. The finite pre-candidate set has at most

$$4 \cdot 6 = 24 \tag{6.105}$$

complex roots counting multiplicity when the intersection is zero-dimensional. After filtering for admissibility and unsquared stationarity, the interior optimizer is finite.

6.5.3 Up-to-three-coordinate splice

There are $\binom{5}{2} = 10$ primitive pairs and $\binom{5}{3} = 10$ primitive triples. Stage 212 imports every optimized pairwise interval

$$\tau^{\mathrm{lo}}_{ij,\min} \le \tau^{\mathrm{best}}_{ij,*} \le \tau^{\mathrm{hi}}_{ij,\min} \tag{6.106}$$

and every triple interior interval

$$\tau^{\mathrm{lo,int}}_{ijk,\min} \le \tau^{\mathrm{best,int}}_{ijk,*} \le \tau^{\mathrm{hi,int}}_{ijk,\min}. \tag{6.107}$$

For a triple $T = (i, j, k)$, the boundary minima are

$$\beta^{\mathrm{lo}}_T := \min(\tau^{\mathrm{lo}}_{ij,\min}, \tau^{\mathrm{lo}}_{ik,\min}, \tau^{\mathrm{lo}}_{jk,\min}), \quad \beta^{\mathrm{hi}}_T := \min(\tau^{\mathrm{hi}}_{ij,\min}, \tau^{\mathrm{hi}}_{ik,\min}, \tau^{\mathrm{hi}}_{jk,\min}). \tag{6.108}$$

The closed-triple interval is

$$\tau^{\mathrm{lo},\triangle}_{T,\min} = \min(\beta^{\mathrm{lo}}_T, \tau^{\mathrm{lo,int}}_{ijk,\min}), \qquad \tau^{\mathrm{hi},\triangle}_{T,\min} = \min(\beta^{\mathrm{hi}}_T, \tau^{\mathrm{hi,int}}_{ijk,\min}). \tag{6.109}$$

The full support-≤ 3 search is therefore bounded by

$$\tau^{\mathrm{lo}}_{\le 3,\min} := \min(\tau^{\mathrm{lo}}_{2,\min}, \tau^{\mathrm{lo,int}}_{3,\min}), \qquad \tau^{\mathrm{hi}}_{\le 3,\min} := \min(\tau^{\mathrm{hi}}_{2,\min}, \tau^{\mathrm{hi,int}}_{3,\min}). \tag{6.110}$$

The exact evaluation budget through three active coordinates is at most

$$10 \times 12 + 10 \times 48 = 600 \tag{6.111}$$

candidate evaluations.

6.5.4 Four-coordinate simplex and boundary reduction

For a primitive quadruple $Q = (i, j, k, l)$,

$$\Delta^+_{ijkl} = \left\{ (a_i, a_j, a_k, a_l) \in \mathbb{R}^4_{\ge 0} : a_i^2 + a_j^2 + a_k^2 + a_l^2 = 1 \right\}. \tag{6.112}$$

Every codimension-one face is one of the Stage 212 closed triple simplices. The gradient-optimal interior point is

$$\mathbf{a}^{\mathrm{grad}}_{ijkl} = \frac{(k_i, k_j, k_k, k_l)}{\sqrt{k_i^2 + k_j^2 + k_k^2 + k_l^2}}, \tag{6.113}$$

and the equal-mix point is

$$\mathbf{a}^{\mathrm{eq}}_{ijkl} = \frac{(1, 1, 1, 1)}{2}. \tag{6.114}$$

The total off-diagonal leverage is

$$w_\Sigma(\mathbf{a}) = 2 \sum_{p<q} a_p a_q = \left(\sum_p a_p \right)^2 - 1 \le 3, \tag{6.115}$$

with equality at (6.114). The support-cardinality-four gate asks whether either canonical interior screen, or later the full interior optimizer, beats the imported triple-face boundary ledger.

6.5.5 Four-coordinate interior optimizer and up-to-four splice

On an interior chart, Stage 214 introduces three ratio coordinates (r, s, t), one auxiliary square-root variable y, and a lifted stationary system with degree pattern

$$(3, 3, 3, 2). \tag{6.116}$$

Thus the preferred lifted candidate bound is

$$3 \cdot 3 \cdot 3 \cdot 2 = 54 \tag{6.117}$$

per envelope. Stage 215 then splices the four-coordinate interior intervals to the imported support-≤ 3 ledger. There are $\binom{5}{4} = 5$ primitive quadruples, so the four-coordinate interior budget is

$$5 \times 2 \times 54 = 540. \tag{6.118}$$

Together with (6.111), the support-≤ 4 search budget is

$$600 + 540 = 1140. \tag{6.119}$$

The certified support-≤ 4 interval is

$$\tau^{\mathrm{lo}}_{\leq 4,\min} := \min(\tau^{\mathrm{lo}}_{\leq 3,\min}, \tau^{\mathrm{lo,int}}_{4,\min}), \qquad \tau^{\mathrm{hi}}_{\leq 4,\min} := \min(\tau^{\mathrm{hi}}_{\leq 3,\min}, \tau^{\mathrm{hi,int}}_{4,\min}). \tag{6.120}$$

Claim status. Stages 210–215 are Exact Within Closure **for the simplex algebra and** Numerically Located **only when actual local packet values are inserted. Boundary reduction is exact because each simplex face is one of the previously closed lower-support searches.**

6.6 Unique five-coordinate simplex and final local closure

6.6.1 The unique five-coordinate positive simplex

Stage 216 observes that, because the free space is the five-coordinate log space

$$\mathfrak{I}_5 = \{\lambda, c, \gamma, U, W\}, \tag{6.121}$$

there is only one support-cardinality-five simplex:

$$\Delta_5^+ = \left\{ \mathbf{a} = (a_\lambda, a_c, a_\gamma, a_U, a_W) \in \mathbb{R}^5_{\geq 0} : \sum_{p \in \mathfrak{I}_5} a_p^2 = 1 \right\}. \tag{6.122}$$

Its boundary faces are precisely the five Stage 215 primitive quadruple closed simplices. If the imported face intervals are denoted by $\mathcal{I}^{\square}_{\widehat{p}}$, then

$$\beta_5^{\mathrm{lo}} := \min_{p \in \mathfrak{I}_5} \tau^{\mathrm{lo},\square}_{\widehat{p},\min}, \qquad \beta_5^{\mathrm{hi}} := \min_{p \in \mathfrak{I}_5} \tau^{\mathrm{hi},\square}_{\widehat{p},\min}. \tag{6.123}$$

Equivalently,

$$\beta_5^{\mathrm{lo}} = \tau^{\mathrm{lo}}_{\leq 4,\min}, \qquad \beta_5^{\mathrm{hi}} = \tau^{\mathrm{hi}}_{\leq 4,\min}. \tag{6.124}$$

6.6.2 Five-way canonical screens

The five-coordinate slope magnitude is

$$k_5(\mathbf{a}) = \sum_{p \in \mathfrak{I}_5} a_p k_p. \tag{6.125}$$

The unique gradient-optimal interior point is

$$\mathbf{a}_5^{\mathrm{grad}} = \frac{(k_\lambda, k_c, k_\gamma, k_U, k_W)}{\sqrt{k_\lambda^2 + k_c^2 + k_\gamma^2 + k_U^2 + k_W^2}}, \tag{6.126}$$

and the equal-mix barycenter is

$$\mathbf{a}_5^{\mathrm{eq}} = \frac{(1,1,1,1,1)}{\sqrt{5}}. \tag{6.127}$$

The total ten-way cross leverage is

$$w_\Sigma(\mathbf{a}) = 2 \sum_{p<q} a_p a_q = \left(\sum_p a_p \right)^2 - 1 \le 4, \tag{6.128}$$

with equality at (6.127). These two screens are the exact five-way analogues of the lower-cardinality gradient and equal-mix screens. They are not the full search; they are the final pre-optimizer gate.

6.6.3 Five-coordinate interior optimizer

Stage 217 solves the last continuum problem. On the chart $a_\lambda > 0$, write

$$(a_\lambda, a_c, a_\gamma, a_U, a_W) = \frac{(1, r, s, t, u)}{\sqrt{1 + r^2 + s^2 + t^2 + u^2}}. \tag{6.129}$$

The certified root function has the same form as before:

$$\tau_\star(r,s,t,u) = \frac{2H_0 \sqrt{1 + r^2 + s^2 + t^2 + u^2}}{k_\lambda + r k_c + s k_\gamma + t k_U + u k_W + \sqrt{\Delta_\star^\sharp(r,s,t,u)}}. \tag{6.130}$$

The stationary conditions are four square-root numerator equations. Introducing

$$y := \sqrt{\Delta_\star^\sharp(r,s,t,u)} \tag{6.131}$$

gives the lifted polynomial system

$$\mathcal{F}_{r,\star} = 0, \quad \mathcal{F}_{s,\star} = 0, \quad \mathcal{F}_{t,\star} = 0, \quad \mathcal{F}_{u,\star} = 0, \quad \mathcal{F}_{\Delta,\star} = 0, \tag{6.132}$$

with degree pattern

$$(3,3,3,3,2). \tag{6.133}$$

Therefore the preferred lifted Bézout candidate bound is

$$3^4 \cdot 2 = 179 \tag{6.134}$$

per envelope. Across lower and upper envelopes the support-five interior contributes at most

$$2 \times 162 = 324 \tag{6.135}$$

preferred candidate evaluations. A fallback square-root-free projected chart gives bound 750 per envelope, hence fallback support-five budget 1500.

6.6.4 Boundary identification and support ceiling

Stage 218 proves that the boundary of Δ_5^+ is exactly the already-solved support-≤ 4 search set. The proper nonempty support strata have counts

$$5 \text{ facets} + 10 \text{ ridges} + 10 \text{ edges} + 5 \text{ vertices} = 30 = 2^5 - 2. \tag{6.136}$$

Every proper support subset is contained in at least one quadruple face, and the faces are carried as closed simplices. Thus

$$\partial\Delta_5^+ = \mathcal{S}_{\leq 4}^{\text{loc}}. \tag{6.137}$$

There is also no higher-support local search, because the free log space has only five primitive axes:

$$1 \leq \#\operatorname{supp}(\mathbf{a}) \leq 5. \tag{6.138}$$

The true best local closure time is therefore

$$\tau_{\leq 5,*}^{\text{best}} = \min(\tau_{\leq 4,*}^{\text{best}}, \tau_{5,*}^{\text{best,int}}). \tag{6.139}$$

6.6.5 Final splice and evaluation budget

Combining the imported support-≤ 4 interval with the support-five interior interval gives the final certified interval

$$\tau_{\leq 5,\min}^{\text{lo}} = \min(\tau_{\leq 4,\min}^{\text{lo}}, \tau_{5,\min}^{\text{lo,int}}), \qquad \tau_{\leq 5,\min}^{\text{hi}} = \min(\tau_{\leq 4,\min}^{\text{hi}}, \tau_{5,\min}^{\text{hi,int}}). \tag{6.140}$$

Then

$$\tau_{\leq 5,\min}^{\text{lo}} \leq \tau_{\leq 5,*}^{\text{best}} \leq \tau_{\leq 5,\min}^{\text{hi}}. \tag{6.141}$$

The preferred exact total evaluation budget is

$$1140 + 324 = 1464. \tag{6.142}$$

The fallback projected-chart budget is

$$1140 + 1500 = 2640. \tag{6.143}$$

Thus even the fallback full local search is finite.

Theorem 6.3 (Final local mixed-ray closure)**.** *Within the declared local free-quintuple mixed-ray class, the full positive-support search is exhausted by the support-*≤ 4 *boundary ledger and the unique support-five interior packet. There are no support-cardinality families beyond five. Any remaining ambiguity after Stage 218 is finite-candidate ambiguity inside the already-enumerated algebraic candidate sets, not a new continuum search.*

Claim status. Stages 216–218 are EXACT WITHIN CLOSURE **for the boundary identification, support ceiling, lifted finite candidate compiler, splice theorem, and evaluation budget. The realized winning ray, if any, remains** NUMERICALLY LOCATED **or** OPEN **until actual PDE-derived envelope and validity data are inserted and evaluated.**

6.7 What Part VI does and does not prove

Part VI proves that the local realization search has a finite, auditable structure. It does not prove that the actual nonlinear moving-throat PDE returns a branch on $\mathcal{Z}_*$. It also does not prove that the Hessian envelopes, validity maps, or candidate brackets take any particular numerical values. Those are branch data.

The safe downstream citation is therefore:

> Inside the declared free-quintuple graph and local mixed-ray search class, the realization problem reduces to the exact graph-error packet $(\chi_Q - 1, E_T, E_K, E_\mu)$ and the finite support-cardinality-five mixed-ray sieve. Actual branch realization remains a separate calculation.

For later use, the practical audit recipe is:

1. compute the PDE-returned free quintuple and dependent triple;
2. evaluate the graph target $(T_U^{\mathrm{graph}}, K_{\eta,*}^{\mathrm{graph}}, \mu_W^{\mathrm{graph}})$;
3. form $(\chi_Q - 1, E_T, E_K, E_\mu)$;
4. if the branch is graph-aligned, evaluate $\widehat{\chi}_Q$ and its local log-gradient/Hessian envelopes;
5. insert those data into the completed support-≤ 5 finite ledger.

This is exactly the point at which Part VII can start inserting same-charge and rigid-mouth actual-branch data without reopening the realization compiler.

Chapter 7

One-Port Same-Charge Sieve, Actual-Branch Tests, and Rigid-Mouth Orbit Lock

Stage range: 219–242

Part VI closed the local realization compiler: once a branch is expressed in the free-quintuple graph coordinates, the remaining local mixed-ray search is finite through support-cardinality five. Part VII is the first use of that completed back end on the same-charge, same-sign, one-port mixed-sector problem. Its job is not to add another algebraic search layer. Its job is to insert actual one-port wall/BdG/Maxwell/mixed branch data into the compiler, decide what the same-charge mixed sector can and cannot do, and then translate the surviving branch tests into the rigid-mouth physical normal form.

The conceptual outcome is simple but important. The mixed sector remains real and necessary, but it does not supply a new long-range same-charge attraction. The static one-port bundle produces only short-range product kernels. The linear dynamic bundle produces no new spatial kernel class and its passive/outgoing correction is phase lag at first order. What survives is a sharply constrained short-range/resonant corridor and, on the actual rigid-mouth branch, a two-coordinate orbit-lock packet. The rigid-mouth same-charge problem is diagonal in the physical logarithmic chart

$$U := \ln \frac{\mathcal{T}^2}{\mathcal{T}_{\mathrm{ref}}^2}, \qquad V := \ln \frac{\epsilon_\eta}{\epsilon_{\eta,\mathrm{ref}}}, \tag{7.1}$$

and the strict orbit-lock point is

$$U = 0, \qquad V = 0. \tag{7.2}$$

The selected passive/outgoing support side is also no longer a three-regime ambiguity: the minimal isotropic quadrupole precursor fixes

$$\rho_\alpha = \frac{4}{3}, \qquad \zeta_{\mathrm{req}} = \frac{1}{3}, \qquad \Pi_{\mathrm{tr}} = \frac{4}{3}C_{\mathrm{mix}}, \tag{7.3}$$

placing the selected branch on the symmetric lowest-twin support slice.

Part VII at a glance

Primary question	What survives when the finished realization compiler is applied to actual one-port same-charge branch data, and how should the surviving branch tests be read in physical variables?

Starting inputs	The Part VI realization compiler, the one-port wall/BdG/Maxwell/mixed closure, and the selected passive/outgoing branch data carried forward from earlier parts.
Delivers	The same-charge kernel firewall, the transported $(\Delta_{\mathrm{norm}}, \Xi_1)$ ceilings, the rigid-mouth chart (U, V), the orbit-lock theorem, and the selected twin-support placement packet.
Used by	Part VIII as the strict front end for the relaxed/open-system companion; later papers should cite this chapter for the strict same-charge verdict and its surviving branch packet.
Result anchors	**MTDC-T11**.

Stage packets.

Stages	Packet	Status	Carry-forward output
219–223	One-port kernel firewall and primitive compatibility	Exact Within Closure / Numerically Located	Fixes the short-range static and dynamic kernel classes, proves the phase-lag and resonance/linewidth constraints, and identifies the finite-throat compatibility surface where conservative and outgoing conditions can coexist.
224–232	Transported ceilings, Ξ_1 mechanism sieve, and finite branch windows	Exact Within Closure / Reduced / Controlled Reduction / Numerically Located / Open	Compiles the actual-branch prefactor packet, imports the strict 5PN even gate, filters the surviving mixed-sector mechanisms, and narrows the unresolved gate to static placement and coherent orbit lock.
233–239	Rigid-mouth orbit-lock compiler and physical normal form	Exact Within Closure / Open	Pulls the surviving branch packet into direct observables, builds the packet projectors, computes the dressing compiler, and diagonalizes the strict branch test in the physical chart (U, V) with orbit lock at $U = V = 0$.
240–242	Selected twin-support slice and final front-end packet	Exact Within Closure / Open	Fixes the symmetric lowest-twin support slice, ranks the primitive carriers on that branch, and defines the final strict front-end packet that later relaxed-branch work must inherit rather than overwrite.

Claim status. Part VII is mixed in status. The static and dynamic one-port kernels, the projectors, the physical chart, and the selected support-ratio compilers are Exact Within Closure **inside the declared one-port, weak-axisymmetric, selected-branch, and rigid-mouth closures. Numerical statements in Stages 222–223 and Stage 232 are** Numerically Located **insertions into exact formulas. Any claim that the completed nonlinear moving-throat PDE actually realizes the winning static placement, dynamic window, rigid-mouth orbit-lock point, or selected twin-support point remains** Open. **No result in this part should be cited as a proof of a new same-charge long-range attraction.**

What this part proves. Part VII applies the realization machinery to the one-port same-charge problem, rules out a new long-range same-charge attraction inside the linear one-port closure, and compiles the surviving strict branch conditions into the rigid-mouth (U, V) chart and selected twin-support packet.

What this part does not prove. It does not prove that the completed PDE actually lands on the surviving static placement, dynamic window, rigid-mouth orbit-lock point, or selected twin-support point. Those remain actual-branch realization questions.

Why later parts need it. Part VIII starts from the strict front end extracted here and builds the relaxed/open-system extension around it without erasing the strict same-charge verdict.

7.1 Block contract and theorem-level output

The contract of Part VII is to separate three questions that are easy to conflate. First, there is the *kernel-class question*: does the mixed sector create a new long-range same-charge attractive law? Stages 219–221 answer no within the one-port linear closure. Second, there is the *window question*: can a short-range static or resonant dynamic corridor survive the exact transported ceilings? Stages 222–232 reduce that question to finite branch inequalities and show that the support/source side is not the active bottleneck in the currently injected data. Third, there is the *actual branch question*: does the PDE-selected rigid-mouth branch land on the orbit-lock and selected-support packets? Stages 233–242 compile that question into direct observables.

The operational chain is

$$\begin{aligned} \mathcal{K}_{\mathrm{red}} &\longrightarrow \mathfrak{V}_{\mathrm{mix}}(x, \omega) \longrightarrow (\Delta_{\mathrm{norm}}, \Xi_1) \\ &\longrightarrow (R_{\mathrm{tr}}, \mathcal{T}^2, \epsilon_\eta) \longrightarrow (U, V) \\ &\longrightarrow \text{selected twin-support placement packet.} \end{aligned} \tag{7.4}$$

Theorem 7.1 (Part VII same-charge and rigid-mouth selected-branch theorem)**.** *Inside the declared one-port wall/BdG/Maxwell/mixed closure, weak-axisymmetric grouped response closure, rigid-mouth actual-branch closure, and selected twin-support closure, the following statements hold.*

1. *The static same-charge one-port bundle is a positive quadratic response on the admissible side $\Omega_U^2 > 0$, $\Delta > 0$, $D_0 > 0$. It produces only product kernels built from the primitive source profiles. For the carried source pair x^{-3} and $\mathrm{e}^{-2\kappa x}/x$, the only static product families are*

$$x^{-6}, \qquad e^{-2\kappa x} x^{-4}, \qquad e^{-4\kappa x} x^{-2}. \tag{7.5}$$

2. *The linear dynamic mixed bundle preserves the same spatial families. A passive/outgoing correction at first order is pure phase lag and carries no real conservative barrier lowering off pole.*

3. *Near a simple conservative pole, every relevant susceptibility has the local form*

$$\chi_*(\omega) = \frac{A_*}{\delta - i\gamma_*}, \qquad \delta = \omega - \omega_*, \tag{7.6}$$

so

$$\frac{|\Re \chi_*|}{|\Im \chi_*|} = \frac{|\delta|}{\gamma_*}. \tag{7.7}$$

The maximum conservative gain occurs exactly when conservative and absorptive magnitudes are equal.

4. *On the explicit finite-throat primitive branch, the conservative one-pole condition and outgoing normalization condition are simultaneously compatible only on the exact surface*

$$\frac{N_0}{P_{0,\mathrm{target}}} = \frac{3(M + B_2 + Z_2)^2}{B_4 + Z_4}. \tag{7.8}$$

The transported dynamic window is finite on that surface.

5. *The actual weak-axisymmetric branch test is the transported prefactor packet*

$$(\Delta_{\mathrm{norm}}, a_{P_0}, b_{P_0}), \qquad b_{P_0} = 3a_{P_0}, \qquad \Xi_1 = \frac{P_1}{P_0}. \tag{7.9}$$

The robust one-scalar ceiling is

$$\frac{\Delta_{\mathrm{norm}} + T_{\mathrm{quad}}}{\hat{m}_0^2}(1 + |\varepsilon\Xi_1|) \le P_{\mathrm{crit}}, \qquad T_{\mathrm{quad}} := \frac{54Gc_s^5}{5a^5c^5}. \tag{7.10}$$

6. *On the rigid-mouth actual branch, the physical packet is diagonal in the variables* (U, V). *The microscopic dependent correction is*

$$(\Delta_T, \Delta_{K_\eta}, \Delta_\mu) = (0, -V, U - V). \tag{7.11}$$

Therefore the static-only correction changes only μ_W*, while the post-static dressing correction is the equal* $K_\eta^{(\mathrm{eff})}$–μ_W *ray.*

7. *Rigid-mouth orbit lock is the codimension-two Cartesian condition*

$$U = 0, \qquad V = 0, \tag{7.12}$$

equivalently

$$\mathcal{T}^2 = \mathcal{T}^2_{\mathrm{ref}}, \qquad \epsilon_\eta = \epsilon_{\eta,\mathrm{ref}}. \tag{7.13}$$

8. *The selected passive/outgoing support side lands exactly on the symmetric lowest-twin slice*

$$\rho_\alpha = \frac{4}{3}, \qquad \zeta_{\mathrm{req}} = \frac{1}{3}, \qquad C_{\mathrm{mix}} < \Pi_{\mathrm{tr}} < 2C_{\mathrm{mix}}. \tag{7.14}$$

9. *The actual moving-throat front-end packet to evaluate is*

$$(\epsilon, \varrho_{\mathrm{phys}}, \sigma_{\mathrm{phys}}, \textit{ranking region}, R_{\mathrm{tr}}, R_{\mathrm{target}}, \epsilon_\eta, d\ln R_{\mathrm{tr}}, d\ln R_{\mathrm{target}}, d\ln\epsilon_\eta, N_Q - 1)\,. \tag{7.15}$$

Any failure of the actual branch to realize equations (7.12), (7.14) *and* (7.15) *is a branch-realization failure, not a failure of the algebraic compiler.*

7.2 One-port same-charge static kernel

Stage 219 begins with the same isotropic one-port bundle already used by the grouped normalization chain. After the stable BdG support mode is integrated out, the effective static wall stiffness is

$$K_* := K - \frac{C^2}{\varpi^2}. \tag{7.16}$$

The one-port Maxwell/mixed block is controlled by

$$\Delta := \Omega_U^2\Omega_W^2 - R^2, \qquad Q := G_U^2\Omega_W^2 + 2G_UG_WR + G_W^2\Omega_U^2, \qquad P := \Omega_U^2 G_W + RG_U, \tag{7.17}$$

with static wall operator

$$D_0 := K_* - \frac{Q}{\Delta}. \tag{7.18}$$

The same quantities control the outgoing prefactor:

$$N_0 = \frac{P^2}{\Delta^2}, \qquad P_0 = \frac{N_0}{D_0}. \tag{7.19}$$

Thus the static same-charge audit is not independent of the quadrupole normalization program; it probes the same reduced bundle.

The reduced static energy is

$$V_{\rm stat}(q,U,W;r) = \frac{1}{2}K_*q^2 + \frac{1}{2}\Omega_U^2U^2 + \frac{1}{2}\Omega_W^2W^2 - RUW - G_UqU - G_WqW - J_q(r)q - J_U(r)U - J_W(r)W. \tag{7.20}$$

In matrix form,

$$V_{\rm stat} = \frac{1}{2}X^T\mathcal{K}_{\rm red}X - J^TX, \qquad X = (q,U,W)^T, \tag{7.21}$$

where

$$\mathcal{K}_{\rm red} = \begin{pmatrix} K_* & -G_U & -G_W \\ -G_U & \Omega_U^2 & -R \\ -G_W & -R & \Omega_W^2 \end{pmatrix}. \tag{7.22}$$

The determinant identity is

$$\det\mathcal{K}_{\rm red} = \Delta D_0. \tag{7.23}$$

The admissible static side is therefore

$$\Omega_U^2 > 0, \qquad \Delta > 0, \qquad D_0 > 0. \tag{7.24}$$

On this side, minimization gives

$$X_*(r) = \mathcal{K}_{\rm red}^{-1}J(r), \qquad \delta V_{\rm mix}(r) = -\frac{1}{2}J(r)^T\mathcal{K}_{\rm red}^{-1}J(r) \le 0. \tag{7.25}$$

So the one-port static response is attractive or neutral at quadratic order. But its spatial family is inherited from the source products, not newly generated.

The wall–mixed cross susceptibility is especially important:

$$\chi_{qW} := (\mathcal{K}_{\rm red}^{-1})_{qW} = \frac{P}{\Delta D_0}. \tag{7.26}$$

With $\Lambda := P/\Delta$, one has

$$N_0 = \Lambda^2, \qquad P_0 = \frac{\Lambda^2}{D_0}, \qquad \chi_{qW} = \frac{\Lambda}{D_0}, \qquad \chi_{qW}^2 = \frac{P_0}{D_0}. \tag{7.27}$$

So static mixed softening and outgoing quadrupole normalization are linked by the same load factor and denominator.

For the primitive source profiles

$$\mathcal{S}_Q(x) = \frac{1}{x^3}, \qquad \mathcal{S}_Y(x) = \frac{e^{-2\kappa x}}{x}, \tag{7.28}$$

and the source vector

$$J(x) = \begin{pmatrix} \beta_Q \mathcal{S}_Q(x) \\ \beta_U \mathcal{S}_Y(x) \\ \beta_W \mathcal{S}_Y(x) \end{pmatrix}, \tag{7.29}$$

the static correction is

$$\widetilde{\delta V}_{\mathrm{mix}}(x) = -\frac{1}{2}\left[\frac{\mathcal{C}_6}{x^6} + 2\mathcal{C}_4 \frac{e^{-2\kappa x}}{x^4} + \mathcal{C}_2 \frac{e^{-4\kappa x}}{x^2}\right]. \tag{7.30}$$

The coefficients are susceptibility-weighted source products. The key theorem is the family statement, not the coefficient details: no $-1/x$, no $-e^{-2\kappa x}/x$, and no longer-range static attraction appears inside this one-port static closure.

Theorem 7.2 (Static one-port square-law suppression)**.** *Inside the admissible one-port static closure, the mixed bundle creates no new long-range same-charge attractive law. For the primitive source pair x^{-3} and $e^{-2\kappa x}/x$, it produces only the three product families in equation* (7.30)*. Static same-charge survival is therefore a coefficient-engineering or near-softening question on already-short-range families, not a new spatial-kernel theorem.*

Claim status. Stage 219 is Exact Within Closure **within the isotropic one-port static closure. The absence of a new long-range family is exact for the stated source families and bundle assumptions. The magnitude of the coefficients and their realized branch values remain branch data.**

7.3 Linear dynamic mixed-bundle kernel

Stages 220–221 ask whether time dependence rescues what the static kernel does not provide. The dynamic lift keeps the same three coordinates but replaces the static coefficients by

$$K_B(\omega) := K - M\omega^2 - \frac{C^2}{\varpi^2 - \omega^2}, \qquad A(\omega) := \Omega_U^2 - \omega^2, \qquad W(\omega) := \Omega_W^2 - \omega^2 - \Pi(\omega). \tag{7.31}$$

Then

$$\Delta_\Pi(\omega) := A(\omega)W(\omega) - R^2, \qquad Q_\Pi(\omega) := G_U^2 W(\omega) + 2G_U G_W R + G_W^2 A(\omega), \tag{7.32}$$

and

$$D_\Pi(\omega) := K_B(\omega) - \frac{Q_\Pi(\omega)}{\Delta_\Pi(\omega)}. \tag{7.33}$$

The dynamic stiffness matrix is

$$\mathcal{K}_{\mathrm{dyn}}(\omega) = \begin{pmatrix} K_B(\omega) & -G_U & -G_W \\ -G_U & A(\omega) & -R \\ -G_W & -R & W(\omega) \end{pmatrix}, \tag{7.34}$$

with determinant

$$\det \mathcal{K}_{\mathrm{dyn}}(\omega) = \Delta_\Pi(\omega) D_\Pi(\omega). \tag{7.35}$$

The complex response functional is

$$\mathfrak{V}_{\mathrm{mix}}(x,\omega) = -\frac{1}{2}J(x,\omega)^T\mathcal{K}_{\mathrm{dyn}}(\omega)^{-1}J(x,\omega). \tag{7.36}$$

Its real part is the in-phase barrier reshaping; its imaginary part is the quadrature pumping/leakage channel.

For the same primitive source families as Stage 219, one gets the exact dynamic product-family theorem

$$\mathfrak{V}_{\mathrm{mix}}(x,\omega) = -\frac{1}{2}\left[\frac{\mathcal{C}_6(\omega)}{x^6} + 2\mathcal{C}_4(\omega)\frac{e^{-2\kappa x}}{x^4} + \mathcal{C}_2(\omega)\frac{e^{-4\kappa x}}{x^2}\right]. \tag{7.37}$$

Thus linear monochromatic driving does not create a new radial family. It promotes the same coefficients to complex susceptibilities.

The outgoing port enters only through the WW slot. With $e_W = (0,0,1)^T$,

$$\partial_\Pi \mathcal{K}_{\mathrm{dyn}}^{-1} = \mathcal{K}_{\mathrm{dyn}}^{-1} e_W e_W^T \mathcal{K}_{\mathrm{dyn}}^{-1}, \tag{7.38}$$

so

$$\partial_\Pi \mathfrak{V}_{\mathrm{mix}} = -\frac{1}{2}\left[e_W^T\mathcal{K}_{\mathrm{dyn}}^{-1}J\right]^2. \tag{7.39}$$

On the conservative off-pole branch the bracket is real. Therefore, for a passive/outgoing port

$$\Pi(\omega) = i\Gamma(\omega) + O(\Gamma^2), \qquad \Gamma(\omega) \geq 0, \tag{7.40}$$

the first correction is

$$\delta\mathfrak{V}_{\mathrm{mix}}^{(1)}(x,\omega) = -\frac{i}{2}\Gamma(\omega)\mathcal{T}_J(\omega)^2, \tag{7.41}$$

where $\mathcal{T}_J = e_W^T\mathcal{K}_{\mathrm{cons}}^{-1}J$. Consequently

$$\Re\delta\mathfrak{V}_{\mathrm{mix}}^{(1)} = 0, \qquad \overline{P}_{\mathrm{abs}}^{(1)}(x,\omega) = \frac{\omega\Gamma(\omega)}{2}\mathcal{T}_J(\omega)^2 \geq 0. \tag{7.42}$$

This is the phase-lag no-go: passive/outgoing dressing gives absorption at first order, not conservative lowering.

Stage 221 then studies the only remaining dynamic escape hatch: resonant dispersive enhancement of the already-short-range families. Near a simple conservative pole, write

$$F_\Pi(\omega) = F_0'(\omega_*)\delta - \Pi(\omega_*)Z_* + O(\delta^2, \Pi\delta, \Pi^2), \qquad \delta = \omega - \omega_*. \tag{7.43}$$

For $\Pi(\omega_*) = i\Gamma_*$, the reduced susceptibility has the universal form

$$\chi_s(\omega) \approx \frac{A_*}{\delta - i\gamma_*}, \qquad A_* := \frac{\mathcal{N}_s(\omega_*)}{F_0'(\omega_*)}, \qquad \gamma_* := \frac{\Gamma_* Z_*}{|F_0'(\omega_*)|}. \tag{7.44}$$

This gives

$$|\Re\chi_*| = \frac{|A_*|}{\gamma_*}\frac{r}{1+r^2}, \qquad |\Im\chi_*| = \frac{|A_*|}{\gamma_*}\frac{1}{1+r^2}, \qquad r := \frac{|\delta|}{\gamma_*}. \tag{7.45}$$

The exact maximum of the conservative part occurs at $r = 1$, where $|\Re\chi_*| = |\Im\chi_*|$. If one demands a low-loss window

$$|\Im\chi_*| \leq \eta|\Re\chi_*|, \qquad 0 < \eta \leq 1, \tag{7.46}$$

then the best possible conservative magnitude is

$$\sup_{|\Im\chi_*|\le\eta|\Re\chi_*|} |\Re\chi_*| = \frac{|A_*|}{\gamma_*}\frac{\eta}{1+\eta^2}. \tag{7.47}$$

For a spatial family $S_j(x)$, a necessary low-loss survival condition is

$$\frac{|A_j|}{\gamma_*}\frac{\eta}{1+\eta^2}S_j(x)^2 \ge 2\Delta V_{\mathrm{req}}(x). \tag{7.48}$$

Theorem 7.3 (Linear dynamic no-free-lunch theorem)**.** *Inside the one-port linear dynamic closure, time dependence does not create a new same-charge spatial kernel. Passive/outgoing dressing is pure phase lag at first order off pole. Near a pole, useful conservative gain and absorptive loading obey the exact line-shape tradeoff equation* (7.45)*. Therefore any surviving linear dynamic corridor must be a high-quality, residue-to-linewidth enhancement of one of the already-known short-range families.*

Claim status. Stages 220–221 are Exact Within Closure **for the dynamic bundle algebra, phase-lag identity, simple-pole normal form, and low-loss survival inequality. Whether a completed branch supplies a useful pole with large enough residue-to-linewidth ratio remains** Open**.**

7.4 Actual-branch prefactor ceiling

Stages 222–224 turn the general dynamic survival inequality into a concrete transported branch test. Stage 222 chooses the finite interval $s \in [0, L]$, the lowest N/N zero mode

$$u_0(s) = \frac{1}{\sqrt{L}}, \tag{7.49}$$

and the lowest D/N half-wave

$$f_0(s) = \sqrt{\frac{2}{L}}\sin\frac{\pi s}{2L}. \tag{7.50}$$

The overlap is

$$\kappa = \int_0^L u_0(s)f_0(s)\,ds = \frac{2\sqrt{2}}{\pi}. \tag{7.51}$$

Thus

$$C = \kappa\lambda_B, \qquad G_U = \lambda_U, \qquad G_W = \kappa\lambda_W, \qquad R = \kappa\lambda_R. \tag{7.52}$$

The conservative pole condition is a quartic in $y = \omega^2$:

$$\begin{aligned} F(y) = \Big((K - My)(\varpi^2 - y) - C^2\Big)\Big((\Omega_U^2 - y)(\Omega_W^2 - y) - R^2\Big) \\ - (\varpi^2 - y)\Big(G_U^2(\Omega_W^2 - y) + 2G_UG_WR + G_W^2(\Omega_U^2 - y)\Big). \end{aligned} \tag{7.53}$$

So the primitive finite-throat branch has four conservative poles in the generic admissible case.

For a simple wall-like pole with the outgoing quadrupole port

$$\Pi_{\mathrm{out}}(\omega) = i\Gamma_5\omega^5, \qquad \Gamma_5 = \frac{a^5}{27c_s^5}, \tag{7.54}$$

the residue/linewidth figure for the pure quadrupolar same-charge family simplifies to

$$\mathcal{R}_{Q,*} = \frac{|A_{Q,*}|}{\gamma_*} = \frac{27c_s^5}{a^5\omega_*^5 N(\omega_*)}. \tag{7.55}$$

The derivative of the pole cancels; the figure is controlled by the pole frequency and outgoing transfer factor.

Stage 223 then imposes the isotropic one-pole target surface. On the isotropic branch,

$$D_0 = K - B_0 - Z_0, \qquad D_2 = -(M + B_2 + Z_2), \qquad D_4 = -(B_4 + Z_4). \tag{7.56}$$

The one-pole identity and outgoing-normalization condition solve for the same wall stiffness. Their equality is exactly

$$\frac{N_0}{P_{0,\text{target}}} = \frac{3(M + B_2 + Z_2)^2}{B_4 + Z_4}. \tag{7.57}$$

Equivalently, the primitive branch induces the compatible target

$$P_{0,\text{target,compat}} = \frac{N_0(B_4 + Z_4)}{3(M + B_2 + Z_2)^2}. \tag{7.58}$$

This is the exact compatibility surface from the conservative one-pole side to the outgoing-normalization side.

Stage 224 transports the finite dynamic survival windows from the primitive compatibility family into the actual branch packet. The final 5PN branch packet contains

$$\Delta_{\text{branch}} = (a_2, b_2, a_4, b_4, a_{P_0}, b_{P_0}, \Delta_{\text{pole}}, \Delta_{\text{norm}}), \tag{7.59}$$

where

$$\Delta_{\text{norm}} = \hat{m}_0^2 \bar{P}_0 - T_{\text{quad}}, \qquad T_{\text{quad}} := \frac{54Gc_s^5}{5a^5c^5}. \tag{7.60}$$

Hence

$$\bar{P}_0 = \frac{\Delta_{\text{norm}} + T_{\text{quad}}}{\hat{m}_0^2}. \tag{7.61}$$

The grouped inverse map gives

$$P_{20} = \bar{P}_0 + 4a_{P_0}, \qquad P_{21} = \bar{P}_0 - a_{P_0} + b_{P_0}, \qquad P_{22} = \bar{P}_0 - a_{P_0} - b_{P_0}. \tag{7.62}$$

For a ceiling P_{crit}, the all-lane transported condition is

$$\max\{P_{20}, P_{21}, P_{22}\} \le P_{\text{crit}}. \tag{7.63}$$

On the weak-axisymmetric branch,

$$\lambda_{20} = 1, \qquad \lambda_{21} = \frac{1}{2}, \qquad \lambda_{22} = -1, \tag{7.64}$$

and

$$P_A = \bar{P}_0(1 + \varepsilon\lambda_A\Xi_1), \qquad \Xi_1 = \frac{P_1}{P_0}. \tag{7.65}$$

Therefore

$$a_{P_0} = \frac{\varepsilon\bar{P}_0\Xi_1}{4}, \qquad b_{P_0} = \frac{3\varepsilon\bar{P}_0\Xi_1}{4}, \tag{7.66}$$

and the all-lane ceiling collapses to

$$\bar{P}_0(1 + |\varepsilon\Xi_1|) \le P_{\text{crit}}. \tag{7.67}$$

Substituting equation (7.61) gives the actual-branch form already stated in equation (7.10).

Theorem 7.4 (Transported actual-branch prefactor ceiling)**.** *Inside the transported primitive-family window closure, an actual weak-axisymmetric branch clears the robust all-lane ceiling only if*

$$\frac{\Delta_{\mathrm{norm}} + T_{\mathrm{quad}}}{\hat{m}_0^2}(1 + |\varepsilon \Xi_1|) \le P_{\mathrm{crit}}. \tag{7.68}$$

If the branch is exactly calibrated, $\Delta_{\mathrm{norm}} = 0$, the same condition becomes a lower bound on the source-map factor,

$$\hat{m}_0^2 \ge \frac{T_{\mathrm{quad}}(1 + |\varepsilon \Xi_1|)}{P_{\mathrm{crit}}}. \tag{7.69}$$

Claim status. Stages 222–224 are exact for the primitive branch formulas, compatibility surface, and packet compiler. Pole censuses, compatibility-point values, and numerical headroom budgets are Numerically Located**. The use of primitive-family ceilings as actual-branch sufficient conditions is a transported reduced test, not an unconditional theorem of the full PDE.**

7.5 Strict 5PN even-gate package

Stages 225–226 insert the weak-axisymmetric prefactor into the stricter first-order 5PN package. At first order write

$$D_A = D_0 + \varepsilon \lambda_A D_{01} + O(\varepsilon^2), \qquad N_A = N_0 + \varepsilon \lambda_A N_{01} + O(\varepsilon^2). \tag{7.70}$$

Then

$$\Xi_{\mathrm{load}} = \frac{N_{01}}{N_0} - \frac{D_{01}}{D_0} = \frac{P_1}{P_0} = \Xi_1. \tag{7.71}$$

The stricter imported first-order even gates are

$$K_1 := D_{21} + \frac{D_{01}}{9}, \qquad H_{\mathrm{even}} := D_{41} - \frac{2}{3}D_{21} - \frac{D_{01}}{27}. \tag{7.72}$$

Thus the same scalar Ξ_1 that controls the same-charge prefactor ceiling is only one member of the strict first-order package. A branch must also pass the conservative even gates if it is to remain on the imported 5PN one-pole/hidden-even structure.

Stage 225 first imposes a weaker compensation surface that preserves the conservative grouped response on the explicit branch. It then builds a primitive logarithmic-slope compiler from

$$(x_K, x_M, x_{\lambda_B}, x_{\varpi}, x_{\lambda_U}, x_{\lambda_W}, x_{\lambda_R}, x_{\Omega_U}, x_{\Omega_W}) \tag{7.73}$$

to

$$D_{01}, \qquad D_{21}, \qquad D_{41}, \qquad N_{01}, \qquad \Xi_1. \tag{7.74}$$

The first sieve shows that wall-only and pure-BdG-only mechanisms do not supply the desired nontrivial same-charge load once the conservative compensation is enforced. A mixed-sector family survives.

Stage 226 imports equation (7.72) and tightens the survivor. The sharpest surviving subcorridor has

$$D_{01} = D_{21} = D_{41} = 0, \qquad \Xi_1 = \Xi_{\mathrm{load}} = \frac{N_{01}}{N_0}. \tag{7.75}$$

In words: after the strict even gates and conservative-shape preservation are imposed on the noncanonical compatibility branch, the remaining same-charge effect is not generic mixed anisotropy. It is pure outgoing-transfer enhancement with the conservative one-pole bundle frozen at first order.

Theorem 7.5 (Strict first-order package and pure-transfer survivor)**.** *Inside the imported first-order 5PN package, the same-charge load scalar is exactly* $\Xi_{\mathrm{load}} = P_1/P_0$*. Passing the strict even gates requires controlling* K_1 *and* H_{even} *in addition to* Ξ_{load}*. On the explicit compatibility branch with Stage 225 conservative-shape preservation, the sharp surviving subcorridor is the pure-transfer corridor equation* (7.75)*.*

Claim status. Stages 225–226 are EXACT WITHIN CLOSURE **for the first-order compilers and even-gate identities. The mechanism sieve is exact within the primitive slope space, while the actual PDE realization of the pure-transfer corridor is** OPEN**.**

7.6 Pure-transfer corridor and co-loading no-go

Stages 227–228 analyze the pure-transfer survivor. Define the outgoing-load factor

$$\Lambda := \frac{P}{\Delta}. \tag{7.76}$$

In the pure-transfer corridor,

$$\Xi_1 = \frac{N_{01}}{N_0} = 2\frac{P_{01}}{P} - 2\frac{\Delta_{01}}{\Delta} = 2\delta \ln \Lambda. \tag{7.77}$$

The one-port factorization is

$$\Lambda = \frac{G_W}{\Omega_W^2}\frac{1+I}{1-H}, \qquad I := \frac{RG_U}{\Omega_U^2 G_W}, \qquad H := \frac{R^2}{\Omega_U^2\Omega_W^2}. \tag{7.78}$$

The three pieces are: the raw mixed leg G_W/Ω_W^2, the interference ratio I, and the hybridization/blocking ratio H.

The outgoing-rigidity sieve asks which of those pieces can remain fixed while Ξ_1 survives. If both interference and hybridization are rigid,

$$\delta \ln I = 0, \qquad \delta \ln H = 0, \tag{7.79}$$

then the simultaneous pure-transfer and conservative constraints leave only the trivial pure-transfer drift on the concrete branch. This is the first co-loading no-go: the same-charge effect cannot be reduced to a naive mixed-leg slogan under both rigidities.

Stage 228 then splits the pure-transfer slope into numerator and denominator pieces:

$$\Xi_1 = 2(\pi_1 - \delta_1), \qquad \pi_1 := \frac{P_{01}}{P}, \qquad \delta_1 := \frac{\Delta_{01}}{\Delta}. \tag{7.80}$$

The exact rigid-subcorridor theorem is:

1. pure-transfer plus $\pi_1 = 0$ leaves a one-dimensional numerator-rigid survivor;
2. pure-transfer plus $\delta_1 = 0$ leaves a one-dimensional denominator-rigid survivor;
3. imposing both $\pi_1 = 0$ and $\delta_1 = 0$ kills the corridor.

Both one-dimensional survivors pass the first wall-like dynamic-window audit on the explicit compatibility point. The first active ceiling remains the transported static $|\varepsilon\Xi_1|$ budget, not the dynamic wall-like window.

Theorem 7.6 (Pure-transfer and rigidity sieve)**.** *Inside the strict pure-transfer corridor, the same-charge scalar is the logarithmic outgoing-load slope* $\Xi_1 = 2\delta \ln \Lambda$*. The factorization equation* (7.78) *separates raw mixed-leg, interference, and hybridization contributions. Simultaneous interference and hybridization rigidity kills the nontrivial pure-transfer drift on the concrete compatibility branch, while either numerator-rigid or denominator-rigid splitting alone leaves a one-dimensional survivor.*

Claim status. Stages 227–228 are Exact Within Closure **for the pure-transfer identities and rigidity splits. The branch-specific dynamic-window survival statements are** Numerically Located **insertions into exact ceiling formulas.**

7.7 Selected-branch dynamic-window compiler

Stages 229–232 replace the free numerator/denominator split by the selected-branch softening-depth language and then pull the result back to physical continuum coordinates.

The selected normalization product is

$$N_-(x) = \frac{\beta_0 s_-(x)^2}{\kappa_0^2(A-x)}, \qquad 0 \le x < A, \tag{7.81}$$

with selected overlap

$$s_-(x) = \frac{\left[\kappa_0^2(x+\Delta K_{\rm ax}) + \kappa_1^2 x\right]^2}{\kappa_0^2(x+\Delta K_{\rm ax})^2 + \kappa_1^2 x^2}. \tag{7.82}$$

With dimensionless variables

$$\xi := \frac{x}{A}, \qquad \delta := \frac{\Delta K_{\rm ax}}{A}, \tag{7.83}$$

Stage 229 defines a log-slope classifier $\mathcal{R}_{ND}$ that compares the numerator-like and denominator-like content of the selected branch. The exact threshold

$$\delta = \frac{8}{9} \tag{7.84}$$

separates always-denominator selected branches from branches that begin numerator-dominant. The selected branch is never literally either rigid Stage 228 subcorridor; it is an exact co-loading product. However it is always denominator-like near softening, and it is denominator-like on the entire stable branch for $\delta \ge 8/9$.

Stage 230 uses the classifier to mix the carried numerator-rigid and denominator-rigid dynamic data. The share weights are

$$w_{\rm num} = \frac{\mathcal{R}_{ND}}{1+\mathcal{R}_{ND}}, \qquad w_{\rm den} = \frac{1}{1+\mathcal{R}_{ND}}. \tag{7.85}$$

The dynamic sign theorem says the upper wall-like pole always worsens, while the lower wall-like pole flips sign only at one classifier threshold. Most importantly, the selected-branch dynamic ceilings are everywhere weaker than the universal transported static ceilings. Therefore the first kill condition remains the static $|\varepsilon\Xi_1|$ budget.

Stage 231 pulls $\mathcal{R}_{ND}$ back through the continuum placement map. The physical classifier is

$$\mathcal{R}_{\rm phys}(\delta, R_{\rm target}) := \mathcal{R}_{ND}\big(\xi_{\rm req}(\delta, R_{\rm target}), \delta\big), \tag{7.86}$$

where $\xi_{\rm req}$ is the stable solution of

$$F(\xi, \delta) = R_{\rm target}. \tag{7.87}$$

The monotonicity theorem is

$$\partial_{R_{\mathrm{target}}}\mathcal{R}_{\mathrm{phys}} < 0. \tag{7.88}$$

So larger normalization demand pushes the physical selected branch in the denominator-like and dynamically safer direction. The static-first theorem survives the pullback.

Stage 232 injects the known 5PN Family-1 support/source data. The selected passive/outgoing branch requires

$$\zeta_{\mathrm{req}} = \frac{1}{3}, \qquad \rho_\alpha^{\mathrm{req}} = \frac{4}{3}. \tag{7.89}$$

The numerically located Family-1 support/source branch returns $\zeta_{\mathrm{phys}} \approx 2.4675$ for the carried wall-depth extractions, so the support/source side overshoots the canonical demand by a large margin. This does not prove same-charge success; it says the first active unresolved bottleneck stays at the actual PDE-selected static orbit-lock/coherent placement point.

Theorem 7.7 (Selected-branch static-first theorem)**.** *Inside the selected-branch dynamic-window compiler, the wall-like dynamic window does not become the first same-charge kill condition. After continuum pullback and known support/source injection, the support/source side is non-bottlenecked in the carried data. The live unresolved test remains the actual static orbit-lock/coherent placement packet.*

Claim status. Stages 229–231 are Exact Within Closure **for the selected-branch classifier, share compiler, monotonic pullback, and static-first theorem. Stage 232 is** Numerically Located **for the injected Family-1 support/source roots and margins. The actual PDE-selected static placement remains** Open**.**

7.8 Rigid-mouth orbit-lock compiler

Stages 233–236 translate the surviving static placement problem into rigid-mouth packet language. The first firewall is that mouth rigidity and operator rigidity are not the same statement. Rigid-mouth or tracking lock is represented by

$$\delta \ln R_{\mathrm{tr}} = 0, \tag{7.90}$$

whereas $D_{01} = 0$ is the stronger algebraic statement that the effective grouped operator does not drift at first order.

The direct finite branch observables are

$$(R_{\mathrm{tr}}, R_{\mathrm{target}}, \epsilon_\eta). \tag{7.91}$$

Relative to a reference branch, the finite quotient coordinates include

$$q_{\mathrm{tr}} = -C_* \ln \frac{R_{\mathrm{tr}}}{R_{\mathrm{tr,ref}}}, \tag{7.92}$$

$$q_{\mathrm{nt}} = B_* \ln \frac{R_{\mathrm{tr}}}{R_{\mathrm{tr,ref}}} + \ln \frac{1-\epsilon_\eta}{1-\epsilon_{\eta,\mathrm{ref}}} - \ln \frac{R_{\mathrm{target}}}{R_{\mathrm{target,ref}}}, \tag{7.93}$$

$$q_\eta = \ln \frac{\epsilon_\eta}{\epsilon_{\eta,\mathrm{ref}}}. \tag{7.94}$$

At first order, Stage 234 gives the direct compiler

$$\Theta_1 = \delta \ln R_{\mathrm{tr}}, \qquad \Xi_1 = -\delta \ln R_{\mathrm{target}} - c_\eta \delta \ln \epsilon_\eta, \qquad \mathcal{R}_1 = \delta \ln R_{\mathrm{target}}, \tag{7.95}$$

with

$$c_\eta := \frac{\epsilon_{\eta,*}}{1-\epsilon_{\eta,*}}. \tag{7.96}$$

On the rigid-mouth slice, $q_{\rm tr} = 0$, so the static scalar is

$$\Xi_1 = -R_1 - c_\eta E_1, \qquad R_1 := \delta \ln R_{\rm target}, \qquad E_1 := \delta \ln \epsilon_\eta. \tag{7.97}$$

The two-observable static gate is therefore a condition on (R_1, E_1), not on support/source amplitude.

Stage 235 makes the packet structure exact. Define

$$\mathbf{x}_{\rm rm} := \begin{pmatrix} R_1 \\ E_1 \end{pmatrix}, \qquad \mathbf{q}_{\rm rm} := \begin{pmatrix} q_{\rm nt} \\ q_\eta \end{pmatrix} = \begin{pmatrix} \Xi_1 \\ E_1 \end{pmatrix}. \tag{7.98}$$

Then

$$\mathbf{q}_{\rm rm} = M_{\rm rm}\mathbf{x}_{\rm rm}, \qquad M_{\rm rm} := \begin{pmatrix} -1 & -c_\eta \\ 0 & 1 \end{pmatrix}, \qquad M_{\rm rm}^2 = I_2. \tag{7.99}$$

The direct-space projectors are

$$P_{\rm nt} = \begin{pmatrix} 1 & c_\eta \\ 0 & 0 \end{pmatrix}, \qquad P_\eta = \begin{pmatrix} 0 & -c_\eta \\ 0 & 1 \end{pmatrix}. \tag{7.100}$$

The static strip $\Xi_1 = 0$ is codimension one. Full rigid-mouth orbit lock is codimension two:

$$q_{\rm nt} = 0, \qquad q_\eta = 0 \quad \Longleftrightarrow \quad R_1 = 0, \qquad E_1 = 0. \tag{7.101}$$

The static-blind dressing line is

$$\Xi_1 = 0 \quad \Longleftrightarrow \quad \mathbf{x}_{\rm rm} = \begin{pmatrix} -c_\eta q_\eta \\ q_\eta \end{pmatrix}. \tag{7.102}$$

Thus clearing the first static same-charge ceiling need not make the true orbit-failure amplitude small.

Stage 236 identifies the microscopic carrier. On the rigid-mouth slice the dependent correction is

$$\begin{pmatrix} \Delta_T^{(q)} \\ \Delta_{K_\eta}^{(q)} \\ \Delta_\mu^{(q)} \end{pmatrix} = \begin{pmatrix} 0 \\ -q_\eta \\ q_{\rm nt} - q_\eta \end{pmatrix}. \tag{7.103}$$

The static-blind line $q_{\rm nt} = 0$ maps to the equal-drift ray

$$\begin{pmatrix} \Delta_T \\ \Delta_{K_\eta} \\ \Delta_\mu \end{pmatrix} = -q_\eta \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}. \tag{7.104}$$

So after the static strip is cleared, the remaining orbit defect is not generic throat motion. It is one scalar equal-drift $K_\eta^{(\rm eff)}$–μ_W dressing amplitude.

Theorem 7.8 (Rigid-mouth packet and dependent-plane theorem)**.** *On the rigid-mouth slice, the first static same-charge gate tests only $q_{\rm nt} = \Xi_1$. The dressing coordinate q_η is invisible to that gate and maps microscopically to the equal-drift ray $-q_\eta(0,1,1)^T$ in the dependent triple. Therefore the static gate is necessary but not sufficient for orbit lock.*

Claim status. Stages 233–236 are Exact Within Closure **for the rigid-mouth packet compilers and projectors. The physical assumption that the actual branch is rigid-mouth/track-locked, and the realized value of q_η, remain** Open **branch data.**

7.9 Physical logarithmic chart (U, V)

Stages 237–239 diagonalize the rigid-mouth packet in physical branch variables. Stage 237 begins from the finite rigid-mouth packet

$$q_{\mathrm{nt}} = \ln \frac{1-\epsilon_\eta}{1-\epsilon_{\eta,\mathrm{ref}}} - \ln \frac{R_{\mathrm{target}}}{R_{\mathrm{target,ref}}}, \qquad q_\eta = \ln \frac{\epsilon_\eta}{\epsilon_{\eta,\mathrm{ref}}}. \tag{7.105}$$

The first static gate $q_{\mathrm{nt}} = 0$ is the finite curve

$$\frac{R_{\mathrm{target}}}{R_{\mathrm{target,ref}}} = \frac{1-\epsilon_\eta}{1-\epsilon_{\eta,\mathrm{ref}}}. \tag{7.106}$$

Parameterizing by q_η,

$$\epsilon_\eta = \epsilon_{\eta,\mathrm{ref}} e^{q_\eta}, \qquad \frac{R_{\mathrm{target}}}{R_{\mathrm{target,ref}}} = \frac{1-\epsilon_{\eta,\mathrm{ref}} e^{q_\eta}}{1-\epsilon_{\eta,\mathrm{ref}}}. \tag{7.107}$$

Conversely, once the static gate is cleared,

$$q_\eta = \ln \left(\frac{1-(1-\epsilon_{\eta,\mathrm{ref}}) R_{\mathrm{target}}/R_{\mathrm{target,ref}}}{\epsilon_{\eta,\mathrm{ref}}} \right). \tag{7.108}$$

The microscopic extractor is

$$q_\eta = 2 \ln \frac{c_{\eta U}}{c_{\eta U,\mathrm{ref}}} - \ln \frac{K_U}{K_{U,\mathrm{ref}}} - \ln \frac{K_\eta^{(\mathrm{eff})}}{K_{\eta,\mathrm{ref}}^{(\mathrm{eff})}}. \tag{7.109}$$

At first order,

$$q_\eta = 2c_1 - \kappa_U - \kappa_\eta. \tag{7.110}$$

The dressing coordinate is support-blind:

$$\partial_\varsigma q_\eta = 0, \qquad \partial_{M_{\mathrm{tr}}} q_\eta = 0. \tag{7.111}$$

Stage 238 then uses the coherent transfer-shape identity

$$R_{\mathrm{target}} \mathcal{T}^2 = \Lambda_0 (1-\epsilon_\eta). \tag{7.112}$$

This converts the same-charge packet into physical variables. Relative to a reference branch,

$$q_{\mathrm{nt}} + \frac{B_*}{C_*} q_{\mathrm{tr}} = \ln \frac{\mathcal{T}^2}{\mathcal{T}_{\mathrm{ref}}^2}. \tag{7.113}$$

On the rigid-mouth slice, $q_{\mathrm{tr}} = 0$, so

$$q_{\mathrm{nt}} = \ln \frac{\mathcal{T}^2}{\mathcal{T}_{\mathrm{ref}}^2}, \qquad q_\eta = \ln \frac{\epsilon_\eta}{\epsilon_{\eta,\mathrm{ref}}}. \tag{7.114}$$

Thus the first static same-charge gate is exactly transfer-shape rigidity:

$$q_{\mathrm{nt}} = 0 \quad \Longleftrightarrow \quad \mathcal{T}^2 = \mathcal{T}^2_{\mathrm{ref}}. \tag{7.115}$$

The post-static gate is dressing rigidity:

$$q_\eta = 0 \quad \Longleftrightarrow \quad \epsilon_\eta = \epsilon_{\eta,\mathrm{ref}}. \tag{7.116}$$

Stage 239 packages this in the Cartesian variables

$$U := \ln \frac{\mathcal{T}^2}{\mathcal{T}^2_{\mathrm{ref}}}, \qquad V := \ln \frac{\epsilon_\eta}{\epsilon_{\eta,\mathrm{ref}}}. \tag{7.117}$$

Then

$$q_{\mathrm{nt}} = U, \qquad q_\eta = V. \tag{7.118}$$

The target ratio factorizes as

$$\frac{R_{\mathrm{target}}}{R_{\mathrm{target,ref}}} = \frac{1 - \epsilon_{\eta,\mathrm{ref}} e^V}{1 - \epsilon_{\eta,\mathrm{ref}}} e^{-U}. \tag{7.119}$$

The dependent microscopic correction is

$$\begin{pmatrix} \Delta_T \\ \Delta_{K_\eta} \\ \Delta_\mu \end{pmatrix} = \begin{pmatrix} 0 \\ -V \\ U - V \end{pmatrix}. \tag{7.120}$$

A left inverse reads

$$U = \Delta_\mu - \Delta_{K_\eta}, \qquad V = -\Delta_{K_\eta}. \tag{7.121}$$

The pure transfer-shape axis has image

$$(U, 0) \longmapsto (0, 0, U), \tag{7.122}$$

so clearing the static defect changes only μ_W. The pure dressing axis has image

$$(0, V) \longmapsto -V(0, 1, 1), \tag{7.123}$$

the equal $K_\eta^{(\mathrm{eff})}$–μ_W ray. Therefore full orbit lock is

$$U = 0, \qquad V = 0. \tag{7.124}$$

At first order,

$$\begin{pmatrix} \Delta_T \\ \Delta_{K_\eta} \\ \Delta_\mu \end{pmatrix}^{(1)} = \begin{pmatrix} 0 \\ -\delta \ln \epsilon_\eta \\ \delta \ln \mathcal{T}^2 - \delta \ln \epsilon_\eta \end{pmatrix}. \tag{7.125}$$

Theorem 7.9 (Rigid-mouth Cartesian orbit-lock theorem)**.** *On the rigid-mouth actual branch, the physical logarithmic coordinates (U, V) are the finite quotient packet. The first static gate is $U = 0$. The post-static dressing gate is $V = 0$. The full orbit-lock point is the Cartesian codimension-two condition $(U, V) = (0, 0)$, equivalently $\mathcal{T}^2 = \mathcal{T}^2_{\mathrm{ref}}$ and $\epsilon_\eta = \epsilon_{\eta,\mathrm{ref}}$.*

Claim status. Stages 237–239 are Exact Within Closure **for the finite static-blind curve, physical transfer-shape factorization, support-blindness, Cartesian chart, dependent correction, and orbit-lock theorem. Whether the actual branch has $U = V = 0$ remains** Open**.**

7.10 Selected twin-support branch and placement

Stages 240–242 close the selected support side and attach it to the actual branch packet. Stage 240 begins with the selected product identities

$$\Pi_{\rm tr} = \frac{N_Q^{(\rm target)}}{\beta_0}\alpha_{\rm req}, \qquad C_{\rm mix} = \frac{N_Q^{(\rm target)}}{\beta_0}\alpha_{\rm mix}. \tag{7.126}$$

Therefore

$$\frac{\Pi_{\rm tr}}{C_{\rm mix}} = \frac{\alpha_{\rm req}}{\alpha_{\rm mix}} =: \rho_\alpha. \tag{7.127}$$

For the minimal isotropic conservative quadrupole precursor

$$Y_Q^{\rm cons}(\omega) = c_0 + \frac{c_1}{1-\omega^2/\Omega_Q^2}, \qquad c_0 = \frac{3}{4}, \qquad c_1 = \frac{1}{4}, \qquad \Omega_Q = \frac{3c_s}{2a}, \tag{7.128}$$

the contact/pole inverse compiler gives

$$\rho_\alpha = \frac{1}{c_0} = \frac{4}{3}, \qquad \zeta_{\rm req} := \rho_\alpha - 1 = \frac{1}{3}. \tag{7.129}$$

Therefore

$$\Pi_{\rm tr} = \frac{4}{3}C_{\rm mix}. \tag{7.130}$$

With

$$\varrho := \frac{\pi^2\Pi_{\rm tr}}{16\Lambda}, \qquad C_{\rm mix} = \frac{8\Lambda(1-\epsilon_*)}{\pi^2}, \tag{7.131}$$

this becomes

$$\varrho = \frac{2(1-\epsilon_*)}{3}, \qquad S_{\rm req} = \frac{4}{3}. \tag{7.132}$$

Thus the selected branch lies strictly in the symmetric lowest-twin window:

$$C_{\rm mix} < \Pi_{\rm tr} < 2C_{\rm mix}. \tag{7.133}$$

Stage 241 turns this into a primitive ranking phase diagram. The selected curve is

$$\epsilon_* = 1 - \frac{3\varrho}{2}, \qquad \sigma = \frac{4}{3\varrho} - 2, \qquad 0 < \varrho < \frac{2}{3}. \tag{7.134}$$

For the constructive coherent parameter $0 < \beta < 2/11$, only two crossover thresholds survive:

$$\varrho_{W\Lambda} = \frac{2(1+\beta^2)}{3(2+\beta^2)}, \qquad \varrho_{U\Lambda} = \frac{2(1+\beta^2)}{3(1+\beta+\beta^2)}. \tag{7.135}$$

They obey

$$0 < \varrho_{W\Lambda} < \varrho_{U\Lambda} < \frac{2}{3}. \tag{7.136}$$

The ranking theorem is

$$0 < \varrho < \varrho_{W\Lambda} \quad \Longrightarrow \quad q_\chi > q_Z > q_\Lambda > q_W > |q_U|, \tag{7.137}$$

$$\varrho_{W\Lambda} < \varrho < \varrho_{U\Lambda} \quad \Longrightarrow \quad q_\chi > q_Z > q_W > q_\Lambda > |q_U|, \tag{7.138}$$

$$\varrho_{U\Lambda} < \varrho < \frac{2}{3} \quad \Longrightarrow \quad q_\chi > q_Z > q_W > |q_U| > q_\Lambda. \tag{7.139}$$

So the selected branch is dominated by interference/overlap/wall-blocking/outgoing-scale corrections, not by a large split-U correction.

Stage 242 finally places the actual coherent branch on this curve. The selected physical support coordinate is

$$\epsilon = \epsilon_W \left(1 - \frac{2}{11}\frac{\delta_U}{1+\delta_U}\right). \tag{7.140}$$

Therefore

$$\varrho_{\text{phys}} = \frac{2}{3}(1-\epsilon), \qquad \sigma_{\text{phys}} = \frac{2\epsilon}{1-\epsilon} = \frac{4}{3\varrho_{\text{phys}}} - 2. \tag{7.141}$$

The coherent placement observables are

$$R_{\text{tr}} = \frac{1+\chi_0/(1+\delta_U)}{1+\chi_0}, \tag{7.142}$$

$$R_{\text{target}} = \frac{\Lambda(1-\epsilon_\eta)(1-\epsilon)^2}{Z_W(1+\chi_0)^2}, \tag{7.143}$$

together with ϵ_η. The finite coherent orbit packet in physical placement variables is

$$q_{\text{tr}} = (1+\delta_{U,*}) \ln \frac{\chi_0}{\chi_{0,\text{ref}}} + (1+\chi_{0,*}) \ln \frac{\delta_U}{\delta_{U,\text{ref}}}, \tag{7.144}$$

$$q_{\text{nt}} = \ln \frac{Z_W}{Z_{W,\text{ref}}} - \ln \frac{\Lambda}{\Lambda_{\text{ref}}} + E_* \ln \frac{\epsilon_W}{\epsilon_{W,\text{ref}}} - F_* \ln \frac{\delta_U}{\delta_{U,\text{ref}}}, \tag{7.145}$$

$$q_\eta = \ln \frac{\epsilon_\eta}{\epsilon_{\eta,\text{ref}}}. \tag{7.146}$$

The infinitesimal orbit-lock conditions are equivalently

$$d \ln R_{\text{tr}} = 0, \qquad d \ln R_{\text{target}} = 0, \qquad d \ln \epsilon_\eta = 0. \tag{7.147}$$

The outgoing finish line is separate:

$$N_Q = 1, \qquad \text{or equivalently } \chi_Q = 1 \text{ on the natural source-map branch.} \tag{7.148}$$

The practical front-end packet to be returned by the completed PDE is equation (7.15); if the support lane is tracked separately, the support packet is

$$(\zeta, M_{\text{mix}}, S(\zeta;\epsilon), M_{\text{tr}}), \qquad S(\zeta;\epsilon) = 1 + \frac{\zeta(1-\epsilon)}{1-\zeta\epsilon}. \tag{7.149}$$

The support variable ζ affects the support baseline but not the orbit packet:

$$\partial_\zeta R_{\text{tr}} = \partial_\zeta R_{\text{target}} = \partial_\zeta \epsilon_\eta = \partial_\zeta q_{\text{tr}} = \partial_\zeta q_{\text{nt}} = \partial_\zeta q_\eta = 0. \tag{7.150}$$

Theorem 7.10 (Selected twin-support placement and orbit-packet split)**.** *Inside the selected passive/outgoing branch, the support loading ratio is fixed to $\rho_\alpha = 4/3$, so the branch lies on the symmetric lowest-twin support slice. The realized coordinate on that slice is $\varrho_{\text{phys}} = 2(1-\epsilon)/3$. The orbit-lock packet is determined by $(\chi_0, \delta_U, Z_W, \epsilon_W, \epsilon_\eta, \Lambda)$ and is exactly blind to the support ratio ζ. Therefore support compensation and orbit lock are separate tests that must both be passed by the same completed moving-throat branch.*

Claim status. Stages 240–242 are Exact Within Closure **for the support-ratio extraction, selected curve, primitive ranking thresholds, placement formulas, and support/orbit split. Actual placement of the nonlinear branch and the final value of $N_Q - 1$ remain** Open **unless supplied by a completed PDE calculation.**

7.11 What Part VII does and does not prove

Part VII proves that the same-charge mixed sector has a much sharper reduced structure than the older placeholder language suggested. It proves, within the stated closures, that the one-port static mixed bundle cannot be cited as a source of a new long-range attraction; that the linear dynamic mixed bundle cannot be cited as a new radial kernel class; that resonant help is constrained by an exact dispersive/absorptive tradeoff; that the actual weak-axisymmetric branch test is the transported scalar packet $(\Delta_{\mathrm{norm}}, \Xi_1)$; that the rigid-mouth actual branch diagonalizes in the physical chart (U, V); and that selected support lies on the symmetric lowest-twin slice.

It does not prove that the completed nonlinear moving-throat PDE realizes the favorable point. In particular, it does not prove any of the following:

1. that the true branch has a new same-charge long-range attractive force;
2. that the true branch supplies a high-quality pole satisfying the dynamic survival window;
3. that the true branch lands within the transported static $|\varepsilon\Xi_1|$ ceiling;
4. that the rigid-mouth assumptions are realized by the full PDE;
5. that $U = V = 0$ holds on the actual branch;
6. that selected twin-support placement and outgoing normalization $N_Q = 1$ both hold without additional branch data.

The safe downstream citation is therefore:

> Inside the declared one-port, weak-axisymmetric, rigid-mouth, and selected-support closures, the same-charge mixed sector reduces to short-range static product kernels, a resonance/linewidth dynamic window, a transported prefactor packet $(\Delta_{\mathrm{norm}}, \Xi_1)$, the rigid-mouth Cartesian orbit-lock condition $U = V = 0$, and the selected lowest-twin support slice $\rho_\alpha = 4/3$. Actual branch realization remains a separate PDE calculation.

For later use, the practical audit recipe is:

1. compute the actual one-port bundle data (D_0, N_0, P_0) and check the static product-kernel assumptions;
2. evaluate the branch-compatible target surface equation (7.57);
3. form $(\Delta_{\mathrm{norm}}, \Xi_1)$ and test the transported ceiling equation (7.68);
4. if the branch is rigid-mouth, compute (U, V) from equation (7.117);
5. test orbit lock by $U = V = 0$;
6. compute ϵ, ϱ_{phys}, the selected ranking region, the support packet, and $N_Q - 1$ using equations (7.15) and (7.149).

This is exactly the point at which Part VIII can either preserve the strict front-end packet or declare a relaxed/open-system companion with an explicit recovery map back to the Stage 242 front end.

Chapter 8

Relaxed Open-System Branch, Barrier Compiler, Export Kernel, and Material Thresholds

Stage range: 243–253

Part VII ended at the strict selected-support and rigid-mouth front end. The exact one-port same-charge audit had already fixed the admissible short-range kernel families, the actual-branch packet had already been compressed to orbit-lock variables, and the selected passive/outgoing support side had already landed on the lowest-twin support slice. Part VIII is deliberately different. It is not another attempt to prove the strict conservative branch. It is the relaxed/open-system companion that asks what happens when three standard-recovery suppressions are lifted in a controlled way around the Stage 242 front end.

The conceptual firewall is the most important part of this chapter. The relaxed branch does *not* undo the same-charge verdict from Part VII. It does not produce a new long-range same-charge attraction, and it does not introduce a new linear dynamic kernel class. The relaxed branch is a codimension-three lift of the strict selected-support/orbit packet. Its standard-recovery slice is

$$\ell_w = 0, \qquad f_U = 0, \qquad a = b = 0, \tag{8.1}$$

which implies

$$S_{\text{leak}} = 0, \qquad \mathcal{W}_w = 0, \qquad U = 0, \qquad V = 0, \qquad \mathcal{D}_{UV} = 0, \qquad \varsigma \equiv 1. \tag{8.2}$$

Away from this slice, the relaxed branch can reshape a near-contact barrier by leakage, scalar-photon work, non-rigid dressing drain, and compensated source response. Those are branch-local open-system or support/dressing/source effects. They do not alter the long-range kernel firewall.

The central stationary object of this part is the lowered front end

$$V_{\text{eff}}^{(247)}(r) = V_{\text{short}}^{(1p)}(r) - \lambda_L S_{\text{leak}}(r) - \lambda_W \mathcal{W}_w^{\text{sess}}(r) - \Delta E_{UV}(r) - \mathcal{M}_\sigma(r). \tag{8.3}$$

The central dynamic object is the event-chain energy law

$$E = \frac{m_s}{2}\dot{r}^{\,2} + V_{\text{eff}}^{(247)}(r). \tag{8.4}$$

The central microscopic export object is the super-Ohmic active-leg kernel

$$K_{\exp}(s) = \Gamma_3 s^3 + \Gamma_5 s^5. \tag{8.5}$$

The central materials companion is the four-ratio screen

$$\Pi_{\mathrm{ep}} \geq 1, \qquad \Pi_{\chi} \geq 1, \qquad \Pi_{k} \geq 1, \qquad \Pi_{T}(T) \geq 1. \tag{8.6}$$

Part VIII at a glance

Primary question	If the strict same-charge branch is kept fixed, what controlled open-system extensions can be built around it without undoing the strict kernel verdict?
Starting inputs	The Part VII strict front-end packet, including the rigid-mouth orbit-lock variables, the selected twin-support slice, and the short-range kernel firewall.
Delivers	The relaxed-constraint branch declaration, leakage/work and non-rigid (U, V) compilers, the relaxed barrier and event-chain diagnostics, the microscopic export kernel, and the material-threshold companion.
Used by	Future relaxed-branch, export-kernel, and material-threshold papers; no later part in the current ledger supersedes this handoff block.
Result anchors	**MTDC-T12**.

Stage packets.

Stages	Packet	Status	Carry-forward output
243–246	Relaxed branch declaration and lane compilers	EFFECTIVE CLOSURE / EXACT WITHIN CLOSURE	Declares the codimension-three lift of the strict Stage 242 front end, fixes the exact recovery slice, and compiles the leakage/work, non-rigid mouth, and compensated multimode source lanes.
247–250	Lowered barrier, event-chain, and Goldilocks diagnostics	EXACT WITHIN CLOSURE / EFFECTIVE CLOSURE / REDUCED / CONTROLLED REDUCTION / NUMERICALLY LOCATED	Builds the relaxed stationary barrier, the one-dimensional event-chain and tunneling diagnostics, the helicity-export readback, and the crossing-versus-collapse window used to identify viable relaxed events.
251–253	Export kernel, heat partition, and material screening	EXACT WITHIN CLOSURE / REDUCED / CONTROLLED REDUCTION / EFFECTIVE CLOSURE / NUMERICALLY LOCATED / OPEN	Derives the cubic/quintic active-leg export kernel, partitions exported energy between vacuum and lattice channels, and compiles the physical calibration and material-threshold screen without claiming that the completed PDE realizes those open-system coefficients.

Claim status. Part VIII is a relaxed/open-system companion. Its algebraic compilers are EXACT WITHIN CLOSURE **inside the declared branch, one-mode, characteristic-timescale, single-growth, and physical-calibration closures. The stationary and dynamic readbacks that**

use Session-I through Session-V benchmark numbers are Numerically Located **insertions into exact formulas. The branch declaration itself is an** Effective Closure **extension around the strict Stage 242 front end, and any claim that the completed nonlinear moving-throat PDE actually realizes the relaxed packet, the microscopic export coefficients, or the physical material thresholds remains** Open**. No result in this part should be cited as a proof of a new same-charge long-range attraction or as a proof that the relaxed branch validates the strict conservative branch.**

What this part proves. Part VIII constructs the relaxed/open-system companion around the strict Stage 242 packet, including leakage/work, non-rigid mouth, compensated source, barrier, export, heat-partition, and material-screening compilers.

What this part does not prove. It does not rescue the strict same-charge conservative branch, does not create a new long-range same-charge attraction, and does not prove that the completed PDE realizes the relaxed packet or its physical screening thresholds.

Why later parts need it. This is the handoff block for future relaxed-branch, export-kernel, and material-threshold papers; no later part inside the current ledger supersedes it.

8.1 Block contract and theorem-level output

The contract of Part VIII is to keep three layers separate. The first layer is the strict front end from Part VII: selected twin-support placement, rigid-mouth orbit lock, and the outgoing normalization packet. The second layer is the relaxed branch around that front end: leakage, non-rigid mouth response, and compensated source shape. The third layer is the physical companion: event-chain survival, export, heat partition, and material screening. The chapter is useful only if those layers remain visibly separated.

The operational chain is

$$\begin{aligned}\mathcal{P}_{242} &\longrightarrow \mathfrak{B}_{243}^{\mathrm{relax}} \longrightarrow (S_{\mathrm{leak}}, \mathcal{W}_w, U, V, \mathcal{M}_\sigma)\\ &\longrightarrow V_{\mathrm{eff}}^{(247)} \longrightarrow (r_\pm, I_{\mathrm{new}}, t_{\mathrm{cross}})\\ &\longrightarrow K_{\exp}(s) \longrightarrow (\Pi_{\mathrm{ep}}, \Pi_\chi, \Pi_k, \Pi_T).\end{aligned} \tag{8.7}$$

The strict branch can be recovered from this chain by imposing equation (8.1). The relaxed branch cannot be used to infer that the strict branch succeeds; it can only be used to analyze the open-system bypass companion.

Theorem 8.1 (Part VIII relaxed-branch and export-companion theorem)**.** *Inside the declared relaxed-constraint branch, the one-mode leakage/work closure, the non-rigid mouth/dressing closure, the compensated two-mode source closure, the reduced one-dimensional event-chain closure, the characteristic crossing/collapse closure, the microscopic cubic/quintic export closure, and the physical-calibration companion, the following statements hold.*

1. *The relaxed branch is a codimension-three lift of the Stage 242 front end with exact recovery map*

$$\ell_w = 0, \qquad f_U = 0, \qquad a = b = 0. \tag{8.8}$$

On this slice all lifted leakage, work, non-rigid, and compensated-source observables reduce to the strict standard-recovery values.

2. *The same-charge static and linear dynamic kernel span remains short-ranged:*

$$\mathcal{K}_{\mathrm{SR}} = \operatorname{span}\left\{ r^{-6}, \frac{e^{-2\kappa r}}{r^4}, \frac{e^{-4\kappa r}}{r^2} \right\}. \tag{8.9}$$

In particular,

$$\lim_{r\to\infty} r\,\delta V_{\mathrm{stat}}(r) = 0, \qquad \lim_{r\to\infty} r\,\Re\mathfrak{V}_{\mathrm{dyn}}(r,\omega) = 0. \tag{8.10}$$

3. *On the selected-support branch, the leakage/work lane is explicit and support-side:*

$$S_{\mathrm{leak}}(r) = \frac{8\sqrt{2}\,\eta_{\mathrm{leak}}\mu_w\rho_0}{\pi^{5/2}\lambda^3}\Lambda(r)\varrho(r), \tag{8.11}$$

$$\mathcal{W}_w^{\mathrm{sess}}(r) = \frac{512\,\eta_{\mathrm{leak}}^2\mu_w q\rho_0}{\pi^4\lambda^2}\Lambda(r)^2\varrho(r)^2. \tag{8.12}$$

These quantities are independent of the orbit-lock variables $(R_{\mathrm{tr}}, R_{\mathrm{target}}, \epsilon_\eta)$ *within the declared pullback.*

4. *The non-rigid mouth packet is exact in the physical logarithmic chart*

$$U = \ln\frac{\mathcal{T}^2}{\mathcal{T}_{\mathrm{ref}}^2}, \qquad V = \ln\frac{\epsilon_\eta}{\epsilon_{\eta,\mathrm{ref}}}. \tag{8.13}$$

With

$$\Delta_{UV} = a_U a_V - \chi_{UV}^2, \tag{8.14}$$

the stationary point is

$$U = \frac{a_V f_U}{\Delta_{UV}}, \qquad V = \frac{\chi_{UV} f_U}{\Delta_{UV}}, \qquad \Delta_{UV} > 0. \tag{8.15}$$

The corresponding drain is nonnegative:

$$\mathcal{D}_{UV} = \frac{\chi_{UV}^2 a_V f_U^2}{\Delta_{UV}^2} \geq 0. \tag{8.16}$$

5. *The first compensated two-mode source family is an exact two-moment compiler:*

$$\sigma_{a,b}(x) = 1 + a\cos(\pi x) + b\cos(2\pi x), \qquad \int_0^1 \sigma_{a,b}(x)\,dx = 1, \tag{8.17}$$

with

$$\mathfrak{g}(a,b) = \frac{2}{\pi}\left(1 + \frac{a}{3} - \frac{b}{15}\right), \qquad \mathcal{S}(a,b) = \frac{2\tanh(\pi/2)}{\pi}\left(1 + \frac{a}{5} + \frac{b}{17}\right). \tag{8.18}$$

Its normalized two-moment map has determinant $14/425 > 0$*, so the first compensated source family spans the two carried source moments exactly.*

6. *The relaxed stationary barrier is exactly equation* (8.3)*, and the lowering identity is*

$$V_{\mathrm{short}}^{(1p)}(r) - V_{\mathrm{eff}}^{(247)}(r) = \lambda_L S_{\mathrm{leak}}(r) + \lambda_W \mathcal{W}_w^{\mathrm{sess}}(r) + \Delta E_{UV}(r) + \mathcal{M}_\sigma(r). \tag{8.19}$$

Hence, when the imported packet scalars and weights are nonnegative, $V_{\mathrm{eff}}^{(247)} \leq V_{\mathrm{short}}^{(1p)}$ *pointwise.*

7. *The dynamic event chain is determined by the conserved energy equation* (8.4)*. The lowered-branch classical threshold speed is*

$$v_{\text{crit,new}} = \sqrt{\frac{2}{m_s}\left(V_{\text{peak}} - V(r_0)\right)}, \tag{8.20}$$

while the pure-Coulomb contact threshold is

$$v_{\text{contact,Coul}} = \sqrt{\frac{2}{m_s}\left(\frac{1}{r_{\text{contact}}} - \frac{1}{r_0}\right)}. \tag{8.21}$$

If $v_{\text{crit,new}} < v_0 < v_{\text{contact,Coul}}$*, the lowered branch reaches contact while the pure Coulomb branch still turns back.*

8. *The subbarrier WKB action is*

$$I_{\text{new}}(E) = \frac{1}{\hbar_{\text{eff}}} \int_{r_-(E)}^{r_+(E)} \sqrt{2m_s(V(r) - E)}\, dr, \tag{8.22}$$

and the transmission ratio relative to Coulomb is

$$\frac{T_{\text{new}}(E)}{T_{\text{Coul}}(E)} = \exp\left[-2\left(I_{\text{new}}(E) - I_{\text{Coul}}(E)\right)\right]. \tag{8.23}$$

9. *The helicity-export comparison is diagnostic. With orientation label* $\sigma = \pm 1$*, the reduced closure gives*

$$\dot{H}_{\text{sub},\sigma} = \Gamma_0(1 + \sigma\alpha_h), \qquad |\alpha_h| < 1 \tag{8.24}$$

for positive export in both branches. The aligned branch dominates whenever $\alpha_h > 0$*, but this does not by itself prove a microscopic spin sector.*

10. *The crossing-versus-collapse safe edge is*

$$E_{\text{edge}} = V_{\text{peak}} + \frac{m_s}{\mu_\eta} \frac{g_{UV}\chi_{\text{peak}}\lambda_{\text{eff}}^2}{2}, \tag{8.25}$$

equivalently

$$v_{\text{safe,min}}^2 = v_{\text{crit,new}}^2 + \frac{\lambda_{\text{eff}}^2 g_{UV}\chi_{\text{peak}}}{\mu_\eta}. \tag{8.26}$$

Under the declared characteristic closure this is a one-sided lower edge.

11. *The microscopic active-leg export kernel is super-Ohmic:*

$$K_{\text{exp}}(s) = \Gamma_3 s^3 + \Gamma_5 s^5, \qquad \Gamma_3, \Gamma_5 \geq 0. \tag{8.27}$$

The positive growth root is reduced but not eliminated by any finite passive (Γ_3, Γ_5) *in this minimal closure. The exact event-safe condition is*

$$\widehat{\Gamma}_3 + s_c^2\widehat{\Gamma}_5 \geq \frac{s_0^2 - s_c^2}{s_c^3}, \qquad \widehat{\Gamma}_n := \Gamma_n/\mu_\eta. \tag{8.28}$$

12. *The heat partition is channel-resolved. For event-shape quotient* $\mathfrak{r}_V = \mathcal{I}_3/\mathcal{I}_2$*,*

$$f_{\mathrm{lat}}(\mathfrak{r}_V) = \frac{\Gamma_3^{\mathrm{lat}} + \Gamma_5^{\mathrm{lat}}\mathfrak{r}_V}{\Gamma_3 + \Gamma_5\mathfrak{r}_V}, \tag{8.29}$$

and on a single-growth cold event $\mathfrak{r}_V = s^2$*. At the safe edge, the total minimum exported energy is*

$$E_{\mathrm{exp,min}}^{\mathrm{safe}} = \frac{V_{\mathrm{in}}^2}{2}(e^2-1)\mu_\eta(s_0^2 - s_c^2). \tag{8.30}$$

13. *The physical-material companion is an exact calibration image of the Stage 252 event-equivalent rate. With calibration factor* $\Upsilon_{\mathrm{lat}} > 0$*,*

$$(\lambda_{\mathrm{ep}}\omega_D)_{\mathrm{min}}^{(253)} = \frac{f_{\mathrm{lat}}(s_c)\mu_\eta(s_0^2 - s_c^2)}{\Upsilon_{\mathrm{lat}}\zeta_{\mathrm{ep}}s_c t_*}. \tag{8.31}$$

Candidate hosts are screened by equation (8.6)*.*

Every item involving actual branch data, microscopic coefficients, physical units, or candidate material constants remains a realization or calibration question beyond the algebraic compiler.

8.2 Relaxed constraint branch declaration

Stage 243 declares the relaxed branch without yet assembling a barrier. Its purpose is to make the Session-I relaxations legal inside the derivation tree while preserving the same-charge short-range verdict. The inherited base object is the Stage 242 front-end packet

$$\mathcal{P}_{242} = \Big(\epsilon, \varrho_{\mathrm{phys}}, \sigma_{\mathrm{phys}}, \text{ranking region}, R_{\mathrm{tr}}, R_{\mathrm{target}}, \epsilon_\eta, d\ln R_{\mathrm{tr}}, d\ln R_{\mathrm{target}}, d\ln\epsilon_\eta, N_Q - 1\Big), \tag{8.32}$$

with selected-support packet

$$\mathcal{S}_{242} = (\zeta, M_{\mathrm{mix}}, S(\zeta;\epsilon), M_{\mathrm{tr}}). \tag{8.33}$$

The strict slice is

$$\mathfrak{B}_{\mathrm{std}} = \{J^w = 0,\ U = 0,\ V = 0,\ \varsigma(z) \equiv 1\} \subset (\mathcal{P}_{242}, \mathcal{S}_{242}). \tag{8.34}$$

This definition is useful because it names exactly what is being relaxed: transverse-current suppression, rigid-mouth orbit lock, and positive-source Family-1 closure.

8.2.1 Open-system leakage/work lane

The open-system lift begins with the exact projected-continuity leakage identity

$$S_{\mathrm{leak}} = -\big[W(w)j^w\big]_{-\infty}^{+\infty} + \int_{-\infty}^{+\infty} W'(w)j^w(w)\,dw. \tag{8.35}$$

For the declaration-level Gaussian closure,

$$W(w) = \frac{e^{-w^2}}{\sqrt{\pi}}, \qquad j^w(w) = \ell_w j_0\,we^{-w^2}, \qquad E_w(w) = E_0we^{-w^2}, \tag{8.36}$$

so the boundary term vanishes and

$$S_{\mathrm{leak}} = -\frac{\sqrt{2}}{4}\ell_w j_0, \qquad \mathcal{W}_w = \frac{\sqrt{2\pi}}{8}\ell_w j_0 E_0. \tag{8.37}$$

The declaration packet for this lane is

$$L_w = (S_{\mathrm{leak}}, \mathcal{W}_w). \tag{8.38}$$

Both components vanish on $\ell_w = 0$.

8.2.2 Non-rigid mouth/dressing lane

The non-rigid lift uses the minimal free energy

$$\mathcal{F}_{UV}(U,V) = \frac{1}{2}k_U U^2 + \frac{1}{2}k_V V^2 - \chi_\lambda UV - f_U U. \tag{8.39}$$

Stationarity gives

$$U = \frac{k_V f_U}{k_U k_V - \chi_\lambda^2}, \qquad V = \frac{\chi_\lambda f_U}{k_U k_V - \chi_\lambda^2}, \qquad \frac{V}{U} = \frac{\chi_\lambda}{k_V}. \tag{8.40}$$

The branch is admissible when

$$k_U k_V - \chi_\lambda^2 > 0, \tag{8.41}$$

and the drain is

$$\mathcal{D}_{UV} = \chi_\lambda UV = \frac{\chi_\lambda^2 k_V f_U^2}{(k_U k_V - \chi_\lambda^2)^2} \geq 0. \tag{8.42}$$

This lane is the finite relaxed continuation of the earlier weak-axisymmetric transfer-shape scalar: at infinitesimal order, $\Xi_1 = \delta U$.

8.2.3 Compensated sign-changing source lane

The compensated source lift is

$$\varsigma(z) = 1 + a\cos(\pi z) + b\cos(2\pi z), \qquad z \in [0,1], \tag{8.43}$$

with exact unit mean. Setting $y = \cos(\pi z)$,

$$\varsigma(y) = 1 - b + ay + 2by^2, \qquad -1 \leq y \leq 1. \tag{8.44}$$

The interior stationary point and value are

$$y_* = -\frac{a}{4b}, \qquad \varsigma(y_*) = 1 - b - \frac{a^2}{8b}, \tag{8.45}$$

with the boundary candidates $1 + a + b$ and $1 - a + b$. Thus the first source relaxation is not a vague sign-changing profile; it is an exact mean-preserving quadratic problem.

The full declaration packet is

$$\mathfrak{B}_{243}^{\mathrm{relax}} = \{(\mathcal{P}_{242}, \mathcal{S}_{242});\ L_w,\ L_{UV},\ L_\varsigma\}, \tag{8.46}$$

where

$$L_w = (S_{\mathrm{leak}}, \mathcal{W}_w), \quad L_{UV} = (U, V, \mathcal{D}_{UV}), \quad L_\varsigma = (\varsigma, \varsigma_{\min}, \mathrm{signchg}). \tag{8.47}$$

8.2.4 Short-range theorem for the lifted branch

The most important Stage 243 output is the short-range firewall. With primitive profiles

$$\mathcal{S}_Q(r) = r^{-3}, \qquad \mathcal{S}_Y(r) = \frac{e^{-2\kappa r}}{r}, \tag{8.48}$$

the static one-port correction is

$$\delta V_{\mathrm{stat}}(r) = -\frac{1}{2}\left[\frac{\mathcal{C}_6}{r^6} + 2\mathcal{C}_4\frac{e^{-2\kappa r}}{r^4} + \mathcal{C}_2\frac{e^{-4\kappa r}}{r^2}\right], \tag{8.49}$$

and the linear monochromatic dynamic bundle has the same spatial span:

$$\Re\mathfrak{V}_{\rm dyn}(r,\omega) = -\frac{1}{2}\left[\frac{\mathcal{C}_6(\omega)}{r^6} + 2\mathcal{C}_4(\omega)\frac{e^{-2\kappa r}}{r^4} + \mathcal{C}_2(\omega)\frac{e^{-4\kappa r}}{r^2}\right]. \tag{8.50}$$

Thus the relaxed branch is a short-range/open-system bypass branch. It can reshape near-contact energetics, but it cannot be cited as a new long-range same-charge force.

8.3 Selected leakage and scalar-photon work compiler

Stage 244 takes the declared $J^w \neq 0$ lane and attaches it to the selected-support packet. The parent theory supplies both exact identities needed for the compiler: projected continuity gives equation (8.35), and the localized Maxwell energy ledger gives

$$\partial_t u_{\rm EM} + \partial_A S^A_{\rm EM} = -J^A E_A = -(J^a E_a + J^w E_w). \tag{8.51}$$

Thus $S_{\rm leak}$ and $J^w E_w$ are legitimate reduced observables once the mixed lane is opened.

8.3.1 Gaussian one-mode leakage law

The Stage 244 one-mode closure uses

$$W_\lambda(w) = \frac{e^{-w^2/\lambda^2}}{\lambda\sqrt{\pi}}, \qquad \phi_\lambda(w) = \frac{2w}{\sqrt{\pi}\lambda^3}e^{-w^2/\lambda^2}, \tag{8.52}$$

with

$$E_w(w;r) = -\mathcal{E}_0(r)\phi_\lambda(w), \qquad j^w(w;r) = \mu_w\rho_0 E_w(w;r), \qquad J^w = qj^w. \tag{8.53}$$

The exact projected leakage is

$$S_{\rm leak}(r) = \frac{\sqrt{2}\mu_w\rho_0}{2\sqrt{\pi}\lambda^3}\mathcal{E}_0(r). \tag{8.54}$$

The exact one-mode bulk work scalar is

$$\mathcal{W}_w^{\rm bulk}(r) = q\int j^w E_w\,dw = \frac{\sqrt{2}\mu_w q\rho_0}{2\sqrt{\pi}\lambda^3}\mathcal{E}_0(r)^2, \tag{8.55}$$

so

$$\mathcal{W}_w^{\rm bulk}(r) = q\mathcal{E}_0(r)S_{\rm leak}(r). \tag{8.56}$$

The Session-I scalar-photon work variable is the rescaled scalar

$$\mathcal{W}_w^{\rm sess}(r) = \frac{2\mu_w q\rho_0}{\lambda^2}\mathcal{E}_0(r)^2 = 2\sqrt{2\pi}\lambda\,\mathcal{W}_w^{\rm bulk}(r), \tag{8.57}$$

or equivalently

$$\mathcal{W}_w^{\rm sess}(r) = \frac{4\pi q\lambda^4}{\mu_w\rho_0}S_{\rm leak}(r)^2. \tag{8.58}$$

8.3.2 Selected-support pullback

The selected-support demand from Stage 242 is

$$\Pi_{\rm tr} = \frac{4}{3}C_{\rm mix} = \frac{32\Lambda(1-\epsilon)}{3\pi^2} = \frac{16}{\pi^2}\Lambda\varrho, \qquad \varrho = \frac{2}{3}(1-\epsilon). \tag{8.59}$$

Stage 244 chooses the simplest amplitude pullback

$$\mathcal{E}_0(r) = \eta_{\rm leak}\Pi_{\rm tr}(r). \tag{8.60}$$

Substitution gives the selected-branch leakage law equation (8.11) and the selected-branch work law equation (8.12). In the (Λ, ϵ) notation the same work law is

$$\mathcal{W}_w^{\rm sess}(r) = \frac{2048\eta_{\rm leak}^2\mu_w q\rho_0}{9\pi^4\lambda^2}\Lambda(r)^2(1-\epsilon(r))^2. \tag{8.61}$$

The formulas depend only on selected-support variables. Therefore,

$$\partial_{R_{\rm tr}}S_{\rm leak} = \partial_{R_{\rm target}}S_{\rm leak} = \partial_{\epsilon_\eta}S_{\rm leak} = 0, \tag{8.62}$$

and the same statement holds for both work scalars within the declared pullback.

The orientation parity is exact:

$$S_{\rm leak}(-\eta_{\rm leak}) = -S_{\rm leak}(\eta_{\rm leak}), \qquad \mathcal{W}_w(-\eta_{\rm leak}) = \mathcal{W}_w(\eta_{\rm leak}). \tag{8.63}$$

Thus the leakage sign is orientation-sensitive, but the exported-work magnitude is not.

8.4 Non-rigid mouth and (U, V) drain packet

Stage 245 compiles the abstract non-rigid lane into the actual finite physical variables used by the rigid-mouth chart. The carried chart is equation (8.13); hence

$$\mathcal{T}^2 = \mathcal{T}_{\rm ref}^2 e^U, \qquad \epsilon_\eta = \epsilon_{\eta,\rm ref}e^V. \tag{8.64}$$

The selected-branch identity

$$R_{\rm target}\mathcal{T}^2 = \Lambda_0(1-\epsilon_\eta) \tag{8.65}$$

then gives

$$\frac{R_{\rm target}}{R_{\rm target,ref}} = \frac{1-\epsilon_{\eta,\rm ref}e^V}{1-\epsilon_{\eta,\rm ref}}e^{-U}. \tag{8.66}$$

So the finite non-rigid packet has two independent legs: transfer shape and dressing.

The reduced free energy in session notation is

$$\mathcal{F}_{\rm nr}(U, V; r) = \frac{1}{2}a_U U^2 + \frac{1}{2}a_V V^2 - \chi_{UV}(r)UV - f_U(r)U. \tag{8.67}$$

The stationary solution is equation (8.15), with Δ_{UV} given by equation (8.14). In finite physical variables,

$$\mathcal{T}^2(r) = \mathcal{T}_{\rm ref}^2\exp\left(\frac{a_V f_U(r)}{\Delta_{UV}(r)}\right), \tag{8.68}$$

$$\epsilon_\eta(r) = \epsilon_{\eta,\rm ref}\exp\left(\frac{\chi_{UV}(r)f_U(r)}{\Delta_{UV}(r)}\right), \tag{8.69}$$

$$\frac{R_{\text{target}}(r)}{R_{\text{target,ref}}} = \frac{1 - \epsilon_{\eta,\text{ref}} \exp(\chi_{UV} f_U/\Delta_{UV})}{1 - \epsilon_{\eta,\text{ref}}} \exp(-a_V f_U/\Delta_{UV}). \tag{8.70}$$

The dependent microscopic correction inherited from the rigid-mouth plane is

$$(\Delta_T, \Delta_{K_\eta}, \Delta_\mu) = (0, -V, U - V), \tag{8.71}$$

so the non-rigid stationary packet induces

$$\mathbf{y}^{\text{dep}}_{\text{nr}}(r) = \begin{pmatrix} 0 \\ -\chi_{UV} f_U/\Delta_{UV} \\ (a_V - \chi_{UV}) f_U/\Delta_{UV} \end{pmatrix}. \tag{8.72}$$

The drain $\mathcal{D}_{UV}$ is given by equation (8.16); it is nonnegative even when U and V have opposite signs.

At weak-axisymmetric order, write

$$U = \varepsilon u_1 + O(\varepsilon^2), \qquad V = \varepsilon v_1 + O(\varepsilon^2). \tag{8.73}$$

Then

$$\Xi^{\text{nr}}_1 = u_1, \qquad \mathcal{R}^{\text{nr}}_1 + \Xi^{\text{nr}}_1 = -\frac{\epsilon_{\eta,\text{ref}}}{1 - \epsilon_{\eta,\text{ref}}} v_1, \tag{8.74}$$

$$\mathcal{R}^{\text{nr}}_1 = -u_1 - \frac{\epsilon_{\eta,\text{ref}}}{1 - \epsilon_{\eta,\text{ref}}} v_1. \tag{8.75}$$

Thus the direct outgoing-prefactor scalar remains the U leg, while active dressing adds an independent selected-branch residual.

Within the declared closure, this packet is orbit-side/support-blind:

$$\partial_\Lambda U = \partial_\varrho U = \partial_\Lambda V = \partial_\varrho V = 0, \tag{8.76}$$

provided f_U and χ_{UV} are treated as orbit-side data rather than support-placement coordinates. This complements the support-side result from Stage 244.

8.5 Compensated multimode source compiler

Stage 246 compiles the source lane that Stage 243 only declared. The carried source kernels on the dimensionless mouth coordinate $x \in [0, 1]$ are

$$c(x) = \cos\frac{\pi x}{2}, \qquad K_q(x) = \frac{\cosh\left(\frac{\pi}{2}(1 - x)\right)}{\cosh(\pi/2)}. \tag{8.77}$$

The two source moments are

$$\mathfrak{g}[\sigma] = \int_0^1 \sigma(x) c(x)\, dx, \qquad \mathcal{S}[\sigma] = \int_0^1 \sigma(x) K_q(x)\, dx. \tag{8.78}$$

The carried Family-1 mixed-to-shell loading ratio is

$$\mathcal{R}[\sigma] = \frac{(\mathfrak{g}[\sigma] - \mathfrak{r}_{F1})^2}{1 + \mathfrak{r}^2_{F1}}, \qquad \mathfrak{r}_{F1} = \sqrt{\frac{12}{\pi^2}\left(\frac{37}{20}\right)^2 - 1}. \tag{8.79}$$

Stage 246 keeps this law and extends the source family beyond the positive corridor.

The compensated family is equation (8.17). Its exact minimum is

$$\sigma_{\min}(a,b) = \begin{cases} 1+b-|a|, & b \le 0 \text{ or } |a| \ge 4b, \\ 1-b-\dfrac{a^2}{8b}, & b>0 \text{ and } |a| \le 4b. \end{cases} \tag{8.80}$$

Thus

$$\operatorname{signchg}(a,b) \iff \sigma_{\min}(a,b) < 0. \tag{8.81}$$

The exact moment formulas are equation (8.18). In normalized variables

$$\widetilde{g} := \frac{\pi}{2}\mathfrak{g} - 1, \qquad \widetilde{S} := \frac{\pi}{2\tanh(\pi/2)}\mathcal{S} - 1, \tag{8.82}$$

the two-moment map is

$$\begin{pmatrix} \widetilde{g} \\ \widetilde{S} \end{pmatrix} = \begin{pmatrix} 1/3 & -1/15 \\ 1/5 & 1/17 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix}, \qquad \det M_{\mathrm{src}} = \frac{14}{425}. \tag{8.83}$$

The inverse is

$$a = \frac{85}{42}\widetilde{S} + \frac{25}{14}\widetilde{g}, \qquad b = \frac{425}{42}\widetilde{S} - \frac{85}{14}\widetilde{g}. \tag{8.84}$$

For the transported branch,

$$a(r) = a_0 s(r), \qquad b(r) = b_0 s(r), \qquad s(r) = \frac{r_\sigma^2}{r^2 + r_\sigma^2}. \tag{8.85}$$

The output packet is

$$\mathcal{M}_\sigma(r) = \big(\sigma(x;r), \sigma_{\min}(r), \operatorname{signchg}(r), \mathfrak{g}[\sigma](r), \mathcal{S}[\sigma](r), \mathcal{R}[\sigma](r)\big), \tag{8.86}$$

with

$$\mathfrak{g}[\sigma](r) = \frac{2}{\pi}\left(1 + \frac{a(r)}{3} - \frac{b(r)}{15}\right), \qquad \mathcal{S}[\sigma](r) = \frac{2\tanh(\pi/2)}{\pi}\left(1 + \frac{a(r)}{5} + \frac{b(r)}{17}\right). \tag{8.87}$$

On the Session-I orientation $a_0 > 0$, $b_0 < 0$, the minimum is boundary-controlled:

$$\sigma_{\min}(r) = 1 - (a_0 - b_0)s(r), \qquad \operatorname{signchg}(r) \iff r < r_\sigma\sqrt{a_0 - b_0 - 1}. \tag{8.88}$$

This is the source-side packet inserted into the stationary barrier compiler.

8.6 Relaxed stationary barrier front end

Stage 247 assembles the first stationary lowered barrier from the one-port short-range baseline and the three relaxed packets. The one-port static bundle is controlled by

$$\Delta = \Omega_U^2\Omega_W^2 - R_{\mathrm{mix}}^2, \qquad Q = G_U^2\Omega_W^2 + 2G_U G_W R_{\mathrm{mix}} + G_W^2\Omega_U^2, \tag{8.89}$$

$$P = \Omega_U^2 G_W + R_{\mathrm{mix}} G_U, \qquad D_0 = K_* - \frac{Q}{\Delta}. \tag{8.90}$$

With primitive short-range source amplitudes $\beta_Q, \beta_U, \beta_W$, the product-family coefficients are

$$\mathcal{C}_6 = \chi_{qq}\beta_Q^2, \qquad \mathcal{C}_4 = \chi_{qU}\beta_Q\beta_U + \chi_{qW}\beta_Q\beta_W, \tag{8.91}$$

$$\mathcal{C}_2 = \chi_{UU}\beta_U^2 + 2\chi_{UW}\beta_U\beta_W + \chi_{WW}\beta_W^2. \tag{8.92}$$

The collected baseline is

$$V_{\text{short}}^{(1p)}(r) = \frac{1}{r}\left(1 + \frac{1}{2}e^{-2\kappa r}\right) - \frac{A_6}{r^6} - A_4\frac{e^{-2\kappa r}}{r^4} - A_2\frac{e^{-4\kappa r}}{r^2}, \tag{8.93}$$

where

$$A_6 = 3\alpha_6 + \frac{1}{2}\mathcal{C}_6, \qquad A_4 = \mathcal{C}_4, \qquad A_2 = \alpha_2 + \frac{1}{2}\mathcal{C}_2. \tag{8.94}$$

This baseline inherits the same short-range firewall from Stages 219–243.

The imported relaxed scalars are

$$\mathfrak{L}(r) := \Lambda(r)\varrho(r), \qquad S_{\text{leak}} = \frac{8\sqrt{2}\eta_{\text{leak}}\mu_w\rho_0}{\pi^{5/2}\lambda^3}\mathfrak{L}, \tag{8.95}$$

$$\mathcal{W}_w^{\text{sess}} = \frac{512\eta_{\text{leak}}^2\mu_w q\rho_0}{\pi^4\lambda^2}\mathfrak{L}^2, \qquad \Delta E_{UV} = \eta_{UV}\mathcal{D}_{UV}, \tag{8.96}$$

$$\mathcal{M}_\sigma(r) = \xi_R\big[\mathcal{R}_\infty - \mathcal{R}(r)\big], \qquad \mathcal{R}_\infty = \frac{(2/\pi - \mathfrak{r}_{F1})^2}{1 + \mathfrak{r}_{F1}^2}. \tag{8.97}$$

The stationary compiler is equation (8.3); the lowering identity is equation (8.19).

The Session-I benchmark is useful as an audit point, not as a general theorem. At the strongest softening point, the Stage 247 decomposition reads

$$3.74163698 - 0.26971918 \times 0.31069599 - 1.51632107 - 0.21064278 - 0.18386120 = 1.74701126. \tag{8.98}$$

This shows that, once the one-port baseline is fixed, the recorded stationary lowering can be decomposed into work, U/V drain, compensated source response, and one remaining leakage-to-barrier embedding coefficient. It does not change the short-range kernel class.

8.7 Dynamic event-chain compiler

Stage 248 promotes $V_{\text{eff}}^{(247)}$ into a reduced scattering/event-chain compiler. Write

$$V(r) := V_{\text{eff}}^{(247)}(r). \tag{8.99}$$

The reduced radial equation is

$$m_s\ddot{r} = -V'(r), \tag{8.100}$$

with conserved energy equation (8.4). If the branch has a single barrier peak

$$V'(r_{\text{peak}}) = 0, \qquad V''(r_{\text{peak}}) < 0, \qquad V_{\text{peak}} = V(r_{\text{peak}}), \tag{8.101}$$

then the finite-radius threshold law is equation (8.20). The pure-Coulomb contact threshold is equation (8.21). The classical comparison window is therefore

$$v_{\text{crit,new}} < v_0 < v_{\text{contact,Coul}}. \tag{8.102}$$

For subbarrier energy $V(r_0) < E < V_{\text{peak}}$, turning points solve

$$V(r_-(E)) = E, \qquad V(r_+(E)) = E, \qquad r_-(E) < r_+(E), \tag{8.103}$$

and transport as

$$\frac{dr_\pm}{dE} = \frac{1}{V'(r_\pm(E))}. \tag{8.104}$$

The WKB action and transmission ratio are equations (8.22) and (8.23). The pure-Coulomb outer turning point is exact,

$$r_{\text{turn,Coul}}(E) = \frac{1}{E}, \tag{8.105}$$

and the Coulomb action closes as

$$I_{\text{Coul}}(E) = \frac{\sqrt{2m_s}}{\hbar_{\text{eff}}}\left[\frac{\pi}{2\sqrt{E}} - \sqrt{r_{\text{contact}}(1 - Er_{\text{contact}})} - \frac{\arcsin\sqrt{Er_{\text{contact}}}}{\sqrt{E}}\right] \tag{8.106}$$

for $Er_{\text{contact}} < 1$.

Near the top, with

$$V(r) = V_{\text{peak}} - \frac{K_{\text{peak}}}{2}(r - r_{\text{peak}})^2 + O((r - r_{\text{peak}})^3), \qquad K_{\text{peak}} = -V''(r_{\text{peak}}), \tag{8.107}$$

the leading top action is

$$I_{\text{top}}(E) = \frac{\pi(V_{\text{peak}} - E)}{\hbar_{\text{eff}}}\sqrt{\frac{m_s}{K_{\text{peak}}}} + O((V_{\text{peak}} - E)^{3/2}). \tag{8.108}$$

The dynamic diagnostics carried forward are

$$\Xi_{\text{turn}}(E) = \Xi_1(r_+(E)), \qquad \lambda_{\text{th}}(E) = \left|\frac{E}{V'(r_+(E))}\right|. \tag{8.109}$$

These tags are later used in the helicity and Goldilocks compilers; they do not alter the scalar event chain.

8.8 Helicity export diagnostic

Stage 249 attaches a hidden-channel export diagnostic to the Stage 248 event chain. It is diagnostic rather than constitutive. The parent projected helicity ledger gives the exact subscale helicity

$$h_{\text{sub}} = \overline{h} - h_{\text{res}} = \overline{\mathbf{A}' \cdot \mathbf{B}'}, \tag{8.110}$$

with transfer equation

$$\partial_t h_{\text{sub}} + \nabla_3 \cdot \mathbf{F}_{h,\text{sub}} = -2\overline{\mathbf{E}' \cdot \mathbf{B}'}. \tag{8.111}$$

On a reduced brane control volume,

$$\frac{dH_{\text{sub}}}{dt} = -\Phi_h(t;E) - 2\mathcal{C}_h(t;E), \tag{8.112}$$

where Φ_h is the unresolved-helicity flux through the control boundary and $\mathcal{C}_h$ is the unresolved covariance source.

The minimal orientation closure introduces $\sigma = +1$ for aligned and $\sigma = -1$ for anti-aligned, with

$$\Phi_{h,\sigma} = \Phi_{h,0} + \sigma\Phi_{h,1}, \qquad \mathcal{C}_{h,\sigma} = \mathcal{C}_{h,0} + \sigma\mathcal{C}_{h,1}. \tag{8.113}$$

Thus

$$\dot{H}_{\mathrm{sub},\sigma} = \Gamma_0 + \sigma\Gamma_1 = \Gamma_0(1+\sigma\alpha_h), \qquad \alpha_h = \frac{\Gamma_1}{\Gamma_0}. \tag{8.114}$$

The instantaneous ratio and inverse are

$$R_{\mathrm{inst}} = \frac{\dot{H}_{\mathrm{sub},+}}{\dot{H}_{\mathrm{sub},-}} = \frac{1+\alpha_h}{1-\alpha_h}, \qquad \alpha_h = \frac{R_{\mathrm{inst}}-1}{R_{\mathrm{inst}}+1}. \tag{8.115}$$

Over a common interval,

$$R_{\mathrm{int}} = \frac{\mathcal{H}_+}{\mathcal{H}_-} = \frac{1+\bar{\alpha}_h}{1-\bar{\alpha}_h}, \qquad \bar{\alpha}_h = \frac{R_{\mathrm{int}}-1}{R_{\mathrm{int}}+1}. \tag{8.116}$$

Any common export scale cancels from these ratios.

The benchmark packet is

$$\begin{aligned} (\Xi_{\mathrm{turn}}, \lambda_{\mathrm{th}}, R_{\mathrm{pk}}) &= (0.34437471, 0.42826825, 4.94653917), \\ (R_{\mathrm{int}}, \alpha_{\mathrm{pk}}, \bar{\alpha}_h) &= (4.10920923, 0.66366992, 0.60854999). \end{aligned} \tag{8.117}$$

The status is deliberately limited: this proves a hidden-channel unloading preference inside the declared closure, not intrinsic spin or a Pauli-sector theorem.

8.9 Crossing-vs-collapse Goldilocks window

Stage 250 turns the dynamic branch into a survival inequality. The exact path-time integral from the Stage 248 energy law is

$$T_{\mathrm{traj}}(E; r_a \to r_b) = \sqrt{\frac{m_s}{2}} \int_{r_b}^{r_a} \frac{dr}{\sqrt{E - V(r)}}. \tag{8.118}$$

The characteristic-width closure used by Session III compresses the active barrier region to

$$t_{\mathrm{cross}}(E) = \lambda_{\mathrm{eff}} \sqrt{\frac{m_s}{2(E - V_{\mathrm{peak}})}}. \tag{8.119}$$

This is a reduced timescale closure, not a replacement for the exact path integral.

The active dressing leg uses the local unstable normal form

$$\mu_\eta \ddot{V} = g_{UV}\chi_{\mathrm{peak}} V. \tag{8.120}$$

Hence

$$t_{\mathrm{collapse}} = \sqrt{\frac{\mu_\eta}{g_{UV}\chi_{\mathrm{peak}}}}. \tag{8.121}$$

The survival ratio is

$$\mathcal{S}(E) = \frac{t_{\mathrm{cross}}(E)}{t_{\mathrm{collapse}}} = \lambda_{\mathrm{eff}} \sqrt{\frac{m_s g_{UV}\chi_{\mathrm{peak}}}{2\mu_\eta(E - V_{\mathrm{peak}})}}. \tag{8.122}$$

The lower safe edge is equation (8.25), and the speed-space identity is equation (8.26).

If $\mu_\eta = \alpha m_s$, the edge becomes

$$E_{\mathrm{edge}} = V_{\mathrm{peak}} + \frac{g_{UV}\chi_{\mathrm{peak}}\lambda_{\mathrm{eff}}^2}{2\alpha}, \tag{8.123}$$

so simultaneous scaling of particle mass and wall inertia cancels out of the edge. The real control packet is

$$(\lambda_{\text{eff}}, \chi_{\text{peak}}, g_{UV}, \mu_\eta/m_s). \tag{8.124}$$

With the trigger-width specialization,

$$\lambda_{\text{eff}} = \lambda_{\text{th}}(r_{\text{turn}}) = \left|\frac{V(r_{\text{turn}})}{V'(r_{\text{turn}})}\right|. \tag{8.125}$$

The benchmark edge and sensitivity statements are therefore direct consequences of the dependence

$$E_{\text{edge}} - V_{\text{peak}} \propto \lambda_{\text{eff}}^2 \chi_{\text{peak}}. \tag{8.126}$$

Within this closure the safe window is one-sided because t_{collapse} is energy-independent and $t_{\text{cross}}(E)$ decreases monotonically. Any upper edge requires an additional high-energy failure mechanism outside Stage 250.

8.10 Microscopic export kernel

Stage 251 replaces the phenomenological $\gamma_{\text{tot}}\dot{V}$ envelope law by the first microscopic odd export operator seen by the active dressing leg. Project

$$V(t) = \Pi_{V0}q_0(t) + \Pi_{V-}q_-(t) + \cdots, \tag{8.127}$$

where q_0 is the derivative-coupled scalar outlet and q_- is the selected mixed quadrupole outlet.

For the scalar lane, take

$$g_{W,0}(\omega) = \eta_0\omega, \qquad g_{A,0}(\omega) = 0, \qquad \Pi_0^{\text{out}}(\omega) = i\gamma_1\omega + O(\omega^3). \tag{8.128}$$

With

$$\Delta_0 = \Omega_{U,0}^2\Omega_{W,0}^2 - R_0^2, \tag{8.129}$$

the scalar outgoing factor gives

$$\Gamma_{3,0} = \gamma_1\eta_0^2\frac{\Omega_{U,0}^4}{\Delta_0^2}, \qquad \Gamma_3 = \Pi_{V0}^2\Gamma_{3,0}. \tag{8.130}$$

For the selected quadrupole outlet,

$$\Gamma_{5,-} = \frac{a^5}{27c_s^5}P_{0,-}, \qquad P_{0,-} = \frac{\beta_0 s_-}{\lambda_-}, \qquad \Gamma_5 = \Pi_{V-}^2\Gamma_{5,-}. \tag{8.131}$$

Thus the microscopic kernel is equation (8.27).

The active-leg equation is

$$\mu_\eta\ddot{V} - \kappa_V V + \Gamma_3 V^{(3)} - \Gamma_5 V^{(5)} = \mathcal{S}_V(t), \qquad \kappa_V = g_{UV}\chi_{\text{peak}}. \tag{8.132}$$

In Laplace form,

$$\mathcal{D}_V(s) = \mu_\eta s^2 - \kappa_V + \Gamma_3 s^3 + \Gamma_5 s^5. \tag{8.133}$$

The passive power identity is

$$\dot{V}(\Gamma_3 V^{(3)} - \Gamma_5 V^{(5)}) = \frac{d}{dt}\mathcal{S}_{\text{odd}} - \Gamma_3\ddot{V}^2 - \Gamma_5\dddot{V}^2, \tag{8.134}$$

so the exported power is

$$\mathcal{P}_{\text{exp}} = \Gamma_3 \ddot{V}^2 + \Gamma_5 \dddot{V}^2 \geq 0. \tag{8.135}$$

For homogeneous growth $V = V_0 e^{st}$, the characteristic polynomial is

$$F(s) = \Gamma_5 s^5 + \Gamma_3 s^3 + \mu_\eta s^2 - \kappa_V. \tag{8.136}$$

For positive $(\Gamma_3, \Gamma_5, \mu_\eta, \kappa_V)$, $F(0) < 0$, $F(s) \to +\infty$, and $F'(s) > 0$ for $s > 0$. Therefore there is exactly one positive growth root. The small-kernel slowdown is

$$s_+ = s_0 - \frac{\Gamma_3 s_0^2 + \Gamma_5 s_0^4}{2\mu_\eta} + O(\Gamma^2), \qquad s_0 = \sqrt{\kappa_V/\mu_\eta}. \tag{8.137}$$

Thus finite passive cubic/quintic export slows but does not remove the unstable root in the minimal closure.

For an event with crossing rate $s_c = 1/t_{\text{cross}}$, strict monotonicity of F gives the safe half-plane equation (8.28). In the benchmark cold event, this becomes

$$\widehat{\Gamma}_3 + 0.3013336471\widehat{\Gamma}_5 \geq 289.61004918. \tag{8.138}$$

This is the exact event-level replacement for the old γ_{safe} inequality.

8.11 Vacuum/lattice heat partition and material thresholds

Stages 252 and 253 form the physical companion to the export kernel. Stage 252 keeps the reduced microscopic language; Stage 253 maps it into physical threshold variables.

8.11.1 Channel-resolved heat partition

Split the microscopic coefficients as

$$\Gamma_3 = \Gamma_3^{\text{vac}} + \Gamma_3^{\text{lat}}, \qquad \Gamma_5 = \Gamma_5^{\text{vac}} + \Gamma_5^{\text{lat}}, \qquad \Gamma_n^{\text{vac}}, \Gamma_n^{\text{lat}} \geq 0. \tag{8.139}$$

For an event window $[0, T]$, define

$$\mathcal{I}_2(T) = \int_0^T \ddot{V}^2 \, dt, \qquad \mathcal{I}_3(T) = \int_0^T \dddot{V}^2 \, dt. \tag{8.140}$$

The integrated energies are

$$E_{\text{vac}} = \Gamma_3^{\text{vac}} \mathcal{I}_2 + \Gamma_5^{\text{vac}} \mathcal{I}_3, \qquad E_{\text{lat}} = \Gamma_3^{\text{lat}} \mathcal{I}_2 + \Gamma_5^{\text{lat}} \mathcal{I}_3. \tag{8.141}$$

With $\mathfrak{r}_V = \mathcal{I}_3/\mathcal{I}_2$, the fractions are

$$f_{\text{vac}}(\mathfrak{r}_V) = \frac{\Gamma_3^{\text{vac}} + \Gamma_5^{\text{vac}} \mathfrak{r}_V}{\Gamma_3 + \Gamma_5 \mathfrak{r}_V}, \qquad f_{\text{lat}}(\mathfrak{r}_V) = \frac{\Gamma_3^{\text{lat}} + \Gamma_5^{\text{lat}} \mathfrak{r}_V}{\Gamma_3 + \Gamma_5 \mathfrak{r}_V}. \tag{8.142}$$

The partition drift obeys

$$\frac{df_{\text{lat}}}{d\mathfrak{r}_V} = \frac{\Gamma_5^{\text{lat}} \Gamma_3^{\text{vac}} - \Gamma_3^{\text{lat}} \Gamma_5^{\text{vac}}}{(\Gamma_3 + \Gamma_5 \mathfrak{r}_V)^2}. \tag{8.143}$$

So slow events sample the cubic split, fast rough events sample the quintic split.

For a single-growth cold event

$$V(t) = V_{\rm in} e^{st}, \qquad 0 \le t \le T, \tag{8.144}$$

$$\mathcal{I}_2 = \frac{V_{\rm in}^2 s^3}{2}(e^{2sT} - 1), \qquad \mathcal{I}_3 = s^2 \mathcal{I}_2, \qquad \mathfrak{r}_V = s^2. \tag{8.145}$$

The event-equivalent rates are

$$\gamma^{\rm eq}_{\rm vac}(s) = \Gamma_3^{\rm vac} s^2 + \Gamma_5^{\rm vac} s^4, \qquad \gamma^{\rm eq}_{\rm lat}(s) = \Gamma_3^{\rm lat} s^2 + \Gamma_5^{\rm lat} s^4. \tag{8.146}$$

At the safe edge, with $T = 1/s_c$ and $\Gamma_3 s_c^3 + \Gamma_5 s_c^5 = \mu_\eta(s_0^2 - s_c^2)$, the total minimum exported energy is equation (8.30), and the channel energies are obtained by multiplying by $f_{\rm vac}(s_c)$ or $f_{\rm lat}(s_c)$.

The old Session-IV $3:1$ heat split becomes a coefficient-surface condition:

$$f_{\rm lat}(s_c) = \frac{3}{4} \iff \Gamma_3^{\rm lat} + \Gamma_5^{\rm lat} s_c^2 = 3\big(\Gamma_3^{\rm vac} + \Gamma_5^{\rm vac} s_c^2\big). \tag{8.147}$$

It is speed-independent only on the special family where both cubic and quintic channels have the same lattice fraction.

8.11.2 Physical calibration dictionary

Stage 253 introduces the explicit physical dictionary

$$t^{\rm phys} = t_* t, \qquad r^{\rm phys} = \frac{\lambda_{\rm phys}}{\lambda_{\rm ref}} r, \qquad E^{\rm phys} = E_* E. \tag{8.148}$$

The safe-edge lattice event-equivalent rate is

$$\gamma^{\rm eq}_{\rm lat,safe} = f_{\rm lat}(s_c)\mu_\eta \frac{s_0^2 - s_c^2}{s_c}. \tag{8.149}$$

A coarse transport projection factor $\Upsilon_{\rm lat} > 0$ defines the physical lattice-turnover rate

$$\gamma^{\rm phys}_{\rm lat,turn} = \frac{\gamma^{\rm eq}_{\rm lat,safe}}{\Upsilon_{\rm lat} t_*} = \zeta_{\rm ep}\lambda_{\rm ep}\omega_D. \tag{8.150}$$

Thus the electron-phonon turnover threshold is equation (8.31), and the product form is

$$(\lambda_{\rm ep}\omega_D t^{\rm phys}_{\rm cross})^{(253)}_{\rm min} = \frac{f_{\rm lat}(s_c)\mu_\eta(s_0^2 - s_c^2)}{\Upsilon_{\rm lat}\zeta_{\rm ep} s_c^2}. \tag{8.151}$$

The legacy Session-V slice is recovered by choosing

$$\Upsilon^{\rm sess}_{\rm lat} = \frac{\gamma^{\rm eq}_{\rm lat,safe}}{\gamma^{\rm red}_{\rm lattice}}. \tag{8.152}$$

This is a calibration slice, not a new branch law.

For the harmonic interstitial companion,

$$r^{\rm phys}_{\rm turn} = \frac{r_{\rm turn}}{\lambda_{\rm ref}} \lambda_{\rm phys}, \tag{8.153}$$

and

$$\chi_{\lambda,\rm lattice}(r_{\rm turn}) = \frac{2\lambda_{\rm phys}}{r^{\rm phys}_{\rm turn}} = \frac{2\lambda_{\rm ref}}{r_{\rm turn}}. \tag{8.154}$$

This is only a geometric trigger. The force-matched stiffness requires

$$k_{\mathrm{eff,req}} = \frac{E_* \lambda_{\mathrm{ref}}^2}{\lambda_{\mathrm{phys}}^2} \frac{|V'_{\mathrm{red}}(r_{\mathrm{turn}})|}{r_{\mathrm{turn}}} = \mathcal{K}_{\mathrm{turn}} \frac{E_*}{\lambda_{\mathrm{phys}}^2}, \tag{8.155}$$

where

$$\mathcal{K}_{\mathrm{turn}} = \frac{\lambda_{\mathrm{ref}}^2}{r_{\mathrm{turn}}} |V'_{\mathrm{red}}(r_{\mathrm{turn}})|. \tag{8.156}$$

For $\lambda_{\mathrm{phys}} = a_{\mathrm{int}}/2$,

$$k_{\mathrm{eff,req}} = 4\mathcal{K}_{\mathrm{turn}} \frac{E_*}{a_{\mathrm{int}}^2}. \tag{8.157}$$

The Korringa companion is the cleanest physical map. With

$$T_1 T = \mathcal{K}_{\mathrm{corr}}, \qquad T_1 \geq t_{\mathrm{cross}}^{\mathrm{phys}}, \tag{8.158}$$

the ceiling temperature is

$$T_{\mathrm{max}} = \frac{\mathcal{K}_{\mathrm{corr}}}{t_{\mathrm{cross}}^{\mathrm{phys}}} = \frac{s_c \mathcal{K}_{\mathrm{corr}}}{t_*}. \tag{8.159}$$

8.11.3 Material-screening ratios

The companion ends with four screening ratios:

$$\Pi_{\mathrm{ep}} = \frac{\Upsilon_{\mathrm{lat}} \zeta_{\mathrm{ep}} \lambda_{\mathrm{ep}} \omega_D t_*}{\gamma_{\mathrm{lat,safe}}^{\mathrm{eq}}} = \frac{\lambda_{\mathrm{ep}} \omega_D}{(\lambda_{\mathrm{ep}} \omega_D)_{\mathrm{min}}^{(253)}}, \tag{8.160}$$

$$\Pi_\chi = \frac{2\lambda_{\mathrm{ref}}}{r_{\mathrm{turn}}}, \tag{8.161}$$

$$\Pi_k = \frac{k_{\mathrm{eff}} \lambda_{\mathrm{phys}}^2}{\mathcal{K}_{\mathrm{turn}} E_*} = \frac{k_{\mathrm{eff}}}{k_{\mathrm{eff,req}}}, \tag{8.162}$$

$$\Pi_T(T) = \frac{\mathcal{K}_{\mathrm{corr}}}{T t_{\mathrm{cross}}^{\mathrm{phys}}} = \frac{T_{\mathrm{max}}}{T}. \tag{8.163}$$

The materials companion is then

$$\Pi_{\mathrm{ep}} \geq 1, \qquad \Pi_\chi \geq 1, \qquad \Pi_k \geq 1, \qquad \Pi_T(T) \geq 1. \tag{8.164}$$

This is not a claim that any particular host passes. It is the reduced threshold stack that a candidate host must satisfy after the physical unit map and material constants are supplied.

8.12 Citation and downstream-use guardrails

Part VIII is useful because it lets downstream papers talk about the relaxed branch without contaminating the strict branch. The safe citation language is:

> The relaxed/open-system companion is a codimension-three lift of the strict selected-support/orbit front end. It has an exact recovery map to the strict slice, preserves the no-new-long-range same-charge verdict, and supplies separate leakage/work, non-rigid dressing, compensated-source, barrier, dynamic-event, export-kernel, heat-partition, and material-screening compilers.

The unsafe citation language is:

> The relaxed barrier branch proves the strict selected same-charge branch, or supplies a new long-range attraction.

That statement is false under the status discipline of this ledger. The relaxed branch is a companion extension around the strict front end. It is valuable precisely because it states what can change when open-system and non-rigid effects are allowed while preserving what cannot change: the same-charge long-range kernel verdict.

The active open tasks after this part are also explicit. The completed moving-throat PDE still has to realize or reject the strict selected support/orbit/outgoing packet from Part VII. If the relaxed companion is used, the actual branch must additionally return the leakage amplitude, non-rigid (U, V) packet, compensated source packet, microscopic export coefficients (Γ_3, Γ_5) and their vacuum/lattice split, the unit map $(t_*, \lambda_{\mathrm{phys}}, E_*)$, and the coarse transport factor Υ_{lat}. Without those realized data, Part VIII remains an exact compiler rather than a completed physical-branch theorem.

Appendix A

Master Stage Ledger

A.1 Purpose of this appendix

This appendix is the navigational ledger for the full moving-throat derivation companion. It is not a substitute for the stage appendices. Its job is to let a reader audit the archive at three levels: first by theorem block, then by individual stage, and finally by source-file provenance. Each row records the role of a stage, the claim-status boundary under which that row may be cited, the main output that later derivations are allowed to reuse, and the top-level result anchor served by the stage.

The stage number is a provenance handle. The result anchor is the citation handle. A downstream paper should normally cite the relevant **MTDC-T**n theorem block, and should cite a stage number only when it needs the detailed algebra or audit trail for that particular derivation move.

A.2 Ledger conventions

1. The *stage link* points to the corresponding detailed stage appendix section, when that section is included in the full project.

2. The *primary status* is the strongest safe claim class for the row as a reusable result. When the row says "Mixed," the user must preserve all listed status boundaries.

3. Exact Within Closure means exact after the stated reduced closure, branch, normal-coordinate model, or search class has been declared. It does not mean exact for every possible nonlinear moving-throat PDE branch.

4. Open in a row usually means that the algebraic compiler is defined, but the actual moving-throat branch has not yet been shown to realize the required target value.

5. The "main output" column is intentionally terse. The full derivation, assumptions, checks, and downstream-use paragraphs belong in the corresponding stage appendix.

A.3 Top-level stage-range map

Table A.1: Stage-range map for the derivation companion.

Part	Stages	Result anchor(s)	Ledger role	Status envelope
I	001–023	**MTDC-T1–T5**	Parent import, moving-surface lift, breathing reduction, BdG Schur complement, Maxwell/mixed outgoing bridge, and full grouped-bundle front end.	Mixed: Exact, Effective Closure, Reduced / Controlled Reduction, Exact Within Closure, Open
II	024–036	**MTDC-T5**	Overlap/isotropy theorem, weak-axisymmetric signature, selected-mode normal form, finite-throat D/N branch, and support-feasibility frontier.	Mixed: Exact Within Closure, Reduced / Controlled Reduction, Open
III	037–090	**MTDC-T6–T7**	Continuum placement, split-U map, rank-2 support completion, coherent support compensation, parent threshold tests, Family-1 branch, and reduced verdict.	Mixed: Exact Within Closure, Reduced / Controlled Reduction, Effective Closure, Open
IV	091–163	**MTDC-T8**	Geometry-lane firewall, retarded/outlet collapse, side-channel and mouth/core compensation, boundary-layer fixed points, and final residual transport.	Mixed: Exact Within Closure, Reduced / Controlled Reduction, Effective Closure, Numerically Located, Open
V	164–200	**MTDC-T9**	Weak-axisymmetric transport, $\Xi_1 = P_1/P_0$, monomial quotient coordinates, similarity orbit, Packet-A closure, and four-scalar verdict packet.	Mixed: Exact Within Closure, Reduced / Controlled Reduction, Open
VI	201–218	**MTDC-T10**	Realization compiler, target graph, log-ray/Hessian search, pairwise/simplex optimizers, and finite support-cardinality-5 mixed-ray sieve.	Mixed: Exact Within Closure, Numerically Located, Open

Part	Stages	Result anchor(s)	Ledger role	Status envelope
VII	219–242	**MTDC-T11**	One-port same-charge audit, resonance/linewidth gates, actual-branch tests, rigid-mouth U, V chart, orbit lock, and selected twin-support placement.	Mixed: Exact Within Closure, Reduced / Controlled Reduction, Numerically Located, Open
VIII	243–253	**MTDC-T12**	Relaxed open-system branch, leakage/work and U/V drain packets, relaxed barrier, event chain, export kernel, heat partition, and material-threshold companion.	Mixed: Exact Within Closure, Reduced / Controlled Reduction, Effective Closure, Numerically Located, Open

A.4 Complete consolidated stage ledger

Table A.2: Complete stage ledger. The stage number is linked to the detailed stage appendix; the anchor is the preferred downstream citation handle.

Stage	Part	Anchor	Primary status	Working title and main output
001	I	**MTDC-T1–T2**	Effective Closure / Exact Within Closure	**Geometry lift and linearized PDE skeleton.** Shape field $R(\Omega, w, t)$, promoted confinement coupling, and modal split into scalar, grouped real P_2, and higher-lane sectors.
002	I	**MTDC-T2**	Exact Within Closure	**Breathing reduction back to old (a, L) closure.** Two-mode axisymmetric reduction with effective mass and stiffness matrices for $(\delta a, \delta L)$.
003	I	**MTDC-T3**	Reduced / Controlled Reduction / Exact Within Closure	**Minimal BdG–wall coupling and first pole shifts.** Stable-mode Schur complement, low-frequency conservative moments, pole shifts, and grouped P_2 isotropy diagnostics.
004	I	**MTDC-T4**	Exact	**Projected Maxwell bundle index.** Bundle ordering and projection-by-parts identity for the projection-first electromagnetic sector.
005	I	**MTDC-T4**	Exact	**Covariant projected Maxwell law.** Exact projected inhomogeneous Maxwell equation, boundary flux, kernel-gradient leakage, and projected continuity law.
006	I	**MTDC-T4**	Exact	**Vector projected Maxwell split.** Measured-field homogeneous equations, source-flux inhomogeneous equations, and open-system charge balance.
007	I	**MTDC-T4**	Exact / Reduced / Controlled Reduction	**Projection/reduction comparison.** Matched Gaussian zero-mode channel and its coupling mismatch relative to the projection law.
008	I	**MTDC-T4**	Exact / Reduced / Controlled Reduction	**Projected Maxwell extension and clean matching.** Slot-level distinction between source-flux projection and matched reduction.

Stage	Part	Anchor	Primary status	Working title and main output
009	I	**MTDC-T4**	Exact Within Closure	**Near-throat projection scale.** Near-throat projected scale factors and leakage/source normalization ledger.
010	I	**MTDC-T4**	Exact Within Closure	**Projected Maxwell push into bundle slots.** Projected Z_n, N_n slot transport for the grouped response bundle.
011	I	**MTDC-T4**	Exact Within Closure	**Projected Maxwell P_2 bridge.** P_2-lane bridge from projected electromagnetic slots into the grouped normalization language.
012	I	**MTDC-T4**	Exact Within Closure	**Projected Maxwell primitive bridge.** Primitive finite-throat bridge data induced by the projected electromagnetic sector.
013	I	**MTDC-T4**	Exact Within Closure	**Mouth-Taylor master map.** Mouth-local Taylor map for projected source and flux corrections.
014	I	**MTDC-T4**	Exact Within Closure	**Mouth-Taylor gate bridge.** Gate conditions for carrying mouth-local projected data into the grouped bundle.
015	I	**MTDC-T4**	Exact / Reduced / Controlled Reduction	**Parent throat action master.** Parent throat-action packet and projection/reduction status boundary.
016	I	**MTDC-T4**	Exact Within Closure	**Parent throat action candidate.** Candidate parent-action data for the projected throat sector.
017	I	**MTDC-T4**	Exact Within Closure	**Parent throat action weak-axisymmetric law.** Weak-axisymmetric parent-action transport law.
018	I	**MTDC-T4**	Exact Within Closure	**Parent throat action bundle master.** Bundle-level parent-action identities used by the projected electromagnetic response.
019	I	**MTDC-T4**	Exact Within Closure	**Parent throat action isotropic bundle.** Isotropic parent-action specialization for the projected bundle.
020	I	**MTDC-T4**	Exact Within Closure	**Parent throat action weak-axisymmetric packet.** Weak-axisymmetric packet exported from the parent throat-action bundle.
021	I	**MTDC-T4**	Reduced / Controlled Reduction / Exact Within Closure	**Reduced Maxwell/mixed one-port normal form.** Retained reduced one-port self-energy, transfer factor, compact $l = 2$ fingerprint, and scalar-compatibility check.
022	I	**MTDC-T4**	Exact Within Closure / Open	**Grouped real P_2 normalization bridge.** Conversion from conservative operator moments to normalized response moments and the invariant outgoing-normalization product.
023	I	**MTDC-T5**	Exact Within Closure / Open	**Full grouped bundle and isotropic normalization test.** Weighted grouped projectors, full D, N, P coefficient bundle, isotropic normalization ratio, and anisotropy transport laws.
024	II	**MTDC-T5**	Exact Within Closure / Reduced / Controlled Reduction	**Explicit overlap integrals, $O(3)$ isotropy, and first axisymmetric splitting.** Orthonormal real-STF angular basis, exact angular source-map identity, isotropy collapse, and weak-axisymmetric signature $b = 3a$.
025	II	**MTDC-T5**	Exact Within Closure / Open	**Minimal isotropic single-mode closure.** Explicit minimal normalization ratio in terms of one BdG mode and one Maxwell/mixed pair.
026	II	**MTDC-T5**	Exact Within Closure	**Concrete finite-throat axial modes.** D/N overlap constant $\kappa = 2\sqrt{2}/\pi$, first branch-level normalization equation, and required wall-stiffness formula.
027	II	**MTDC-T5**	Exact Within Closure	**First nonconstant finite-throat wall/brane family.** Profile overlap $\kappa(\theta)$, blind-angle no-go, max-coupling angle, and profile-dressed normalization gate.

Stage	Part	Anchor	Primary status	Working title and main output
028	II	**MTDC-T5**	Exact Within Closure	**Loaded axial profile selection.** Rank-one loaded 2×2 wall eigenproblem, selected angle law, and softening threshold.
029	II	**MTDC-T5**	Exact Within Closure / Reduced / Controlled Reduction	**Dynamic loading from coupled wall/BdG/Maxwell/mixed operator.** Schur-complement split $\Sigma_{\text{wall}} = \Xi I_2 + \alpha v v^T$ and selected-mode odd quadrupole coefficient.
030	II	**MTDC-T5**	Exact Within Closure	**Selected-mode normalized response.** Selected prefactor $P_{0,-} = \beta_0 s_- / \lambda_-$ and selected-branch quadrupole target.
031	II	**MTDC-T5**	Exact Within Closure	**Reachability and stable-side monotonicity.** Strict monotonicity of $P_{0,-}$ on the stable branch and unique crossing theorem.
032	II	**MTDC-T5**	Exact Within Closure	**Explicit source map from finite-throat mode integrals.** Natural D/N source map $\widehat{m}_-^2 = s_- / \kappa_0^2$, eliminating the abstract source-map factor.
033	II	**MTDC-T5**	Exact Within Closure / Open	**Microscopic normalization equation and stability gate.** Coupling-level normalization equation, stability inequalities, and onset criterion.
034	II	**MTDC-T5**	Exact Within Closure	**Softening-depth normal form.** Scalar softening-depth variable $x = A - \lambda_-$, exact $\alpha_0(x)$, $s_-(x)$, and $N_-(x)$.
035	II	**MTDC-T5**	Exact Within Closure	**Dimensionless D/N shape function.** Monotone normalization shape function $F(\xi, \delta)$, unique stable normalization locus, and required support coupling.
036	II	**MTDC-T5**	Exact Within Closure / Open	**Support-feasibility frontier.** Monotone support frontier $G(\xi, \delta)$ and final two-condition admissibility test.
037	III	**MTDC-T6**	Exact Within Closure / Reduced / Controlled Reduction	**Continuum-kernel extraction.** Closed formulas for A, ΔK_{ax}, α_{mix}, β_0, M_{mix}, and δ from the first finite-throat continuum operator.
038	III	**MTDC-T6**	Exact Within Closure	**Dimensionless continuum placement map.** Kernel ratios $(\epsilon_\eta, \epsilon_W, \rho, Z_W, \delta_0, \Lambda)$, exact placement formulas, and the product law $R_{\text{target}} M_{\text{mix}} = 8\Lambda(1 - \epsilon_W)/\pi^2$.
039	III	**MTDC-T6**	Exact Within Closure	**First non-flat U doublet.** Split-U correction, direction-splitting invariant, and proof that scalar placement survives while source/loading collinearity generically fails.
040	III	**MTDC-T6**	Exact Within Closure	**Generalized source/loading mismatch.** General selected-branch functions $F_{q,\eta}$, G_q, and split-U deformation (F_U, G_U) governed by R_U.
041	III	**MTDC-T6**	Exact Within Closure	**Rank-2 support completion.** Exact two-direction determinant, support-loading theorem n_{req}, monotonicity in mixed baseline, and tracking/source-tied alternatives.
042	III	**MTDC-T6**	Exact Within Closure	**Rank-2 selected-mode normalization.** Three-direction normalization law involving outgoing, support, and source directions.
043	III	**MTDC-T6**	Reduced / Controlled Reduction / Exact Within Closure	**Actual support/BdG loading direction.** Continuum extraction of the physical support direction needed to decide tracking versus source-tied completion.
044	III	**MTDC-T6**	Exact Within Closure	**Continuum-selected rank-2 closure.** Physical softening depth fixed by an exact quadratic and the continuum-selected normalization gate $R_{\text{target}} = F_{\text{cont}}(\xi_{\text{phys}})$.

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045	III	**MTDC-T6**	Exact Within Closure	**Coherent local D/N support kernel.** Coherent D/N constraint forcing tracking, $R_\phi = R_U = R_{\rm tr}$, and collapse to one-parameter tracking laws.
046	III	**MTDC-T6**	Exact Within Closure	**Tracking-branch bounds.** Monotonicity in $R_{\rm tr}$ and the residual comparison showing that the first split-U tracking deformation worsens the normalization target.
047	III	**MTDC-T6**	Exact Within Closure	**Coherent-kernel dimensionless map.** Support enhancement factor $S(\zeta;\epsilon)$ and coherent tracking map in the reduced continuum variables.
048	III	**MTDC-T6**	Exact Within Closure / Open	**Support-compensation theorem.** Exact $\zeta_{\rm req}$, unique support-ratio threshold, and no reduced-level support no-go before actual PDE placement.
049	III	**MTDC-T6**	Exact Within Closure	**D/N overlap extraction of ζ.** Microscopic support ratio $\zeta_n^{\rm phys}$, uniform-source overlap $(2n+1)^{-1}$, and same-operator twin family.
050	III	**MTDC-T6**	Exact Within Closure	**Physical ζ threshold comparison.** Lowest symmetric twin lane has $\zeta_0 = 1$, hence exact doubling $S_0 = 2$, while higher D/N harmonics are strongly suppressed.
051	III	**MTDC-T6**	Exact Within Closure	**Lowest-twin sufficiency criterion.** Branch-product criterion deciding whether the symmetric lowest twin already supplies the physical tracking branch.
052	III	**MTDC-T7**	Exact Within Closure	**Non-twin asymmetry requirement.** Exact three-regime split: mixed-only, symmetric-lowest-twin, or non-twin support asymmetry.
053	III	**MTDC-T7**	Exact Within Closure	**Overlap-boost window.** Exponential source family, exact overlap boost, and ceiling $\Omega_0^2 \le \pi^2/4$.
054	III	**MTDC-T7**	Exact Within Closure	**Robin-compliance softening.** Lowest-lane Robin root $y\tan y = \eta$, support-softening factor, and softening ceiling.
055	III	**MTDC-T7**	Exact Within Closure	**Non-twin reachability.** Combined overlap-plus-compliance reachability window for the first explicit non-twin lowest-lane family.
056	III	**MTDC-T7**	Exact Within Closure	**Transport origin of source asymmetry.** The exponential source family is the stationary zero-flux branch of a drift-diffusion operator, with bias identified as a Peclet number.
057	III	**MTDC-T7**	Exact Within Closure	**Physical $(\mathrm{Pe},\kappa,\eta)$ map.** Lowest-lane physical support ratio $\zeta_0(\mathrm{Pe},\eta;\kappa)$ and exact constructive-branch threshold surfaces.
058	III	**MTDC-T7**	Exact Within Closure	**Coupled support/source operator.** Operator-selected branch equation $\mathrm{Pe}_* = \Xi\Delta(\mathrm{Pe}_*;\kappa,\eta)$ and exact root bracket.
059	III	**MTDC-T7**	Exact Within Closure	**Residual bounds.** Residual envelope and exact coupling thresholds $\Xi_{\rm fail}$, $\Xi_{\rm suff}$.
060	III	**MTDC-T7**	Exact Within Closure	**Entropic source microclosure.** Positive-density Onsager/free-energy closure reproducing the drift-diffusion source family.
061	III	**MTDC-T7**	Exact Within Closure	**Microscopic gain phase diagram.** Dimensionless gain $G_{\rm micro}$, fail/succeed thresholds, and geometry-controlled phase diagram.
062	III	**MTDC-T7**	Exact / Reduced / Controlled Reduction	**Parent-action projection of gain.** Projection of the GNLS compressional sector gives $G_{\rm micro}$ in parent-action overlap variables.
063	III	**MTDC-T7**	Exact Within Closure	**Parent-overlap thresholds.** Exact amplitude and coherence thresholds, including a Cauchy no-go when even perfect source/support alignment is insufficient.

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064	III	**MTDC-T7**	Exact Within Closure	**Parent equilibrium alignment.** Equilibrium source profile $\chi_\sigma \propto \chi_\phi/H$, coherence formula, and matched-layer gain.
065	III	**MTDC-T7**	Exact Within Closure / Reduced / Controlled Reduction	**Thin-wall confinement branch.** Physical wall-amplitude thresholds from $V_{\rm conf} = V_0 f((r-a)/\ell)$.
066	III	**MTDC-T7**	Exact Within Closure	**Wall figure of merit.** One-number wall control $W_{\rm wall}$ and exact fail/succeed window.
067	III	**MTDC-T7**	Exact Within Closure / Numerically Located	**Sech–Gaussian resonance benchmark.** Self-dual profile resonance $w_g/w_f = \sqrt{\pi}$, near-perfect coherence, and penalty factor.
068	III	**MTDC-T7**	Exact Within Closure	**Resonance-corrected thresholds.** The sech–Gaussian benchmark modifies the matched thresholds only by $P_{\rm res} \simeq 1.005612$.
069	III	**MTDC-T7**	Exact Within Closure / Open	**Final support/source verdict.** Three-zone reduced verdict: universal fail, universal matched success, and a narrow profile-sensitive band.
070	III	**MTDC-T7**	Exact Within Closure / Reduced / Controlled Reduction	**GNLS wall-shell reduction.** Parent shell formulas for T_X, K_X, κ, J_1, and $W_{\rm wall} = \Xi$.
071	III	**MTDC-T7**	Exact Within Closure	**Canonical tanh-wall branch.** Exact moments $I_f = 1/3$, $I_g = 4/15$, natural local mouth closure, and three branch controls.
072	III	**MTDC-T7**	Exact Within Closure	**Explicit branch placement surfaces.** Fail/succeed surfaces in $(\chi_s, \Lambda_\ell, \Upsilon_w)$.
073	III	**MTDC-T7**	Reduced / Controlled Reduction / Exact Within Closure	**Family–1 geometry map.** Reference branch values $L/a = 37/20$, $\ell/a = 1/20$, hence $\Lambda_\ell = \eta = 37$.
074	III	**MTDC-T7**	Reduced / Controlled Reduction / Exact Within Closure	**Healing-length lock.** Closure $\ell = \hbar/(2mc_{s,w})$, hence $\chi_s = 37/2$, $\kappa = 12321/5$.
075	III	**MTDC-T7**	Exact Within Closure	**Family–1 threshold window.** Explicit Θ_w fail/succeed window after all geometry/support scales are fixed.
076	III	**MTDC-T7**	Exact Within Closure	$n = 5$ **wall-depth lock.** Wall datum reduces to $\Theta_w = 25\lambda_\mu^2\rho_w^2$ in normalized Family–1 variables.
077	III	**MTDC-T7**	Reduced / Controlled Reduction / Numerically Located	**Shell-weighted extraction of Θ_w.** Natural value $\Theta_w^{(\chi)} \simeq 4.068632\lambda_\mu^2$ and conservative floor $\Theta_w^{(J)} \simeq 0.927552\lambda_\mu^2$.
078	III	**MTDC-T7**	Exact Within Closure	**Explicit Family–1 branch verdict.** The explicit wall-depth datum is not the bottleneck unless the quadrupole demand is anomalously large.
079	III	**MTDC-T7**	Exact Within Closure	**Demand-to-transport map.** Exact Family–1 map $\zeta_{\rm F1}({\rm Pe}) = A_{\rm F1}\Omega_{\rm Pe}^2$ and hard ceiling $\zeta_{\rm max}^{\rm F1}$.
080	III	**MTDC-T7**	Exact Within Closure	**Demand thresholds in $\zeta_{\rm req}$.** Converts Family–1 wall-depth windows into support-ratio demand windows.
081	III	**MTDC-T7**	Exact Within Closure	**Branch-product demand window.** Converts the same thresholds into $\Pi_{\rm tr}$ product form with blocking parameter.
082	III	**MTDC-T7**	Exact Within Closure / Open	**Master quadrupole residual.** Full reduced residual $R_{\rm quad} = \zeta_{\rm req} - \zeta_{\rm phys}$.
083	III	**MTDC-T7**	Exact Within Closure / Numerically Located	**Direct operator-selected Family–1 window.** Evaluates the operator-selected Family–1 support window directly.

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084	III	**MTDC-T7**	Exact Within Closure / Open	**Reduced PDE write-up skeleton.** Organizes the completed reduced hierarchy and isolates the final outgoing quadrupole product as the remaining gap.
085	III	**MTDC-T7**	Exact Within Closure	**Normalization-factor cancellation.** Shows the support theorem depends only on $\rho_\alpha = \alpha_{\mathrm{req}}/\alpha_{\mathrm{mix}}$.
086	III	**MTDC-T7**	Exact Within Closure	**Family–1 ratio window.** Pure loading-ratio success/failure window and hard Family–1 ceiling.
087	III	**MTDC-T7**	Exact Within Closure / Open	**Final reduced Family–1 finish line.** Reduces the remaining Family–1 support/source gap to one outgoing-branch loading ratio.
088	III	**MTDC-T7**	Exact Within Closure	**Loading ratio from minimal module.** Natural contact-plus-pole precursor gives $\rho_\alpha = 4/3$, $\zeta_{\mathrm{req}} = 1/3$, $\Pi_{\mathrm{tr}} = (4/3)C_{\mathrm{mix}}$.
089	III	**MTDC-T7**	Exact Within Closure	**Explicit Family–1 verdict for minimal isotropic branch.** Minimal isotropic branch lies safely inside the Family–1 success region and succeeds at zero transport bias.
090	III	**MTDC-T7**	Open / Exact Within Closure	**Updated reduced theorem status.** Support/source side is finished under the minimal module; remaining gap is deriving that module from the actual grouped P_2/geometry branch.
091	IV	**MTDC-T8**	Exact Within Closure	**Grouped P_2 plus geometry split.** Static-geometry plus one-pole grouped carrier forces the $3/4 + 1/4$ conservative quadrupole module and hence $\rho_\alpha = 4/3$, $\zeta_{\mathrm{req}} = 1/3$.
092	IV	**MTDC-T8**	Exact Within Closure	**Dynamic geometry obstruction.** If geometry carries dynamic even moments, the pole fraction becomes $(1+\epsilon_4)/[4(1+\epsilon_2)^2]$.
093	IV	**MTDC-T8**	Exact Within Closure / Open	**Grouped P_2 status update.** Declares the remaining question: whether actual isotropic geometry is dynamically inert or supplies (ϵ_2, ϵ_4).
094	IV	**MTDC-T8**	Exact Within Closure	**Isotropic geometry-decoupling theorem.** Orthogonality of $l = 0$ geometry and grouped $l = 2$ wall modes gives $K_{g,2} = K_{g,4} = 0$ on the isotropic branch.
095	IV	**MTDC-T8**	Exact Within Closure	**Second-order geometry contamination.** Weak $l = 0 \leftrightarrow l = 2$ mixing contaminates the grouped module only at $O(\chi^2)$.
096	IV	**MTDC-T8**	Exact Within Closure / Open	**Geometry-lane actual-branch verdict.** The natural isotropic branch is conservatively clean; only genuine symmetry breaking or a second dynamic $l = 2$ geometry pole can reopen the issue.
097	IV	**MTDC-T8**	Exact Within Closure / Open	**Single normalization defect.** The isotropic passive/outgoing one-pole branch reduces to $N_Q = \overline{K}_0/\overline{K}_0^{\mathrm{target}}$.
098	IV	**MTDC-T8**	Exact Within Closure	**Family–1 support automaticity.** With $\rho_\alpha = 4/3$, any branch with $\zeta_{\max} > 1$ passes the support test; Family–1 does.
099	IV	**MTDC-T8**	Exact Within Closure / Open	**Reduced finish line after geometry check.** The only remaining reduced theorem gate is $N_Q = 1$, equivalently actual passive/outgoing quadrupole normalization.
100	IV	**MTDC-T8**	Exact Within Closure	**Factorized 2.5PN defect.** Full odd normalization factorizes as $\widehat{m}_0^2 \chi_Q N_Q = 1$.
101	IV	**MTDC-T8**	Reduced / Controlled Reduction / Exact Within Closure	**Natural source-map reduction.** On the natural point-particle source map, $\widehat{m}_0 \to 1$, so the last obstruction is purely χ_Q.

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102	IV	**MTDC-T8**	Exact Within Closure	**Higher-odd irrelevance.** Retarded terms beginning at $O(\omega^7)$ do not affect the point-particle 2.5PN theorem.
103	IV	**MTDC-T8**	Exact Within Closure / Open	**Conditional reduced 2.5PN closure.** Reduced theorem closes iff the actual passive/outgoing branch has $\chi_Q = 1$.
104	IV	**MTDC-T8**	Exact Within Closure	**Exact outgoing $l = 2$ DtN fingerprint.** Spherical Hankel DtN expansion gives $\widehat{Y}_2^{\text{out}} = 1 + z^2/9 + 4z^4/81 + iz^5/27 + \cdots$.
105	IV	**MTDC-T8**	Exact Within Closure	**Exact fixing of χ_Q.** Matching the canonical grouped module to the exact outgoing DtN branch gives $\chi_Q = 1$.
106	IV	**MTDC-T8**	Exact Within Closure / Open	**Canonical outgoing reduced closure.** Lists the canonical invariant coefficients and marks actual branch realization as the remaining open theorem.
107	IV	**MTDC-T8**	Exact Within Closure	**General isotropic DtN deformation algebra.** Deformed branch $S\Lambda_2^{\text{out}}(\beta z) + \Sigma_0 + \Sigma_2 z^2 + \Sigma_4 z^4 + i\Sigma_5 z^5$ gives explicit χ_Q.
108	IV	**MTDC-T8**	Exact Within Closure	**Robustness classes for χ_Q.** Pure scale is harmless; pure argument shift is harmless only if even moments remain canonical; additive throat-core data can move χ_Q.
109	IV	**MTDC-T8**	Exact Within Closure	**Linearized branch-selection law.** First-order outgoing defect is $\Delta_Q = 5b + a_0/3 + 9a_5$.
110	IV	**MTDC-T8**	Exact Within Closure	**Isotropic Robin outlet.** Pure Robin shift gives $\chi_Q^{\mathrm{R}} = 3/(3 - \rho_R)$ and generically spoils the branch.
111	IV	**MTDC-T8**	Exact Within Closure	**Mixed side-channel pole.** A standalone hidden mixed pole cannot preserve the canonical even branch unless it is absent.
112	IV	**MTDC-T8**	Exact Within Closure	**Robin–mixed compensation law.** Nontrivial compensated branch has $\rho_R = 4\sigma_W$, $\kappa_W = 1/3$, and preserves odd normalization iff $\gamma_W = 1/9$.
113	IV	**MTDC-T8**	Exact Within Closure / Open	**Outlet-model status.** Low-frequency outlet audit leaves pure scale/argument classes and the compensated Robin–mixed class as the viable routes.
114	IV	**MTDC-T8**	Exact Within Closure / Reduced / Controlled Reduction	**Two-channel core outlet model.** Concrete shell/mixed core Schur complement reproduces the reduced Robin–mixed outlet.
115	IV	**MTDC-T8**	Exact Within Closure	**Core-balance compensation theorem.** Coupling balance $g_s^2(K_sK_q + \lambda^2) = 4(K_sg_q - \lambda g_s)^2$ realizes the compensated branch.
116	IV	**MTDC-T8**	Exact Within Closure	**Finite D/N mixed-tube realization.** First D/N mixed half-wave fixes $L_W = \pi a\sqrt{(1 + r_c)/3}/2$ and the bare outgoing scale.
117	IV	**MTDC-T8**	Exact Within Closure / Open	**Outlet-core status.** The microscopic question is whether the real core lands on the coupling-balance surface and D/N tube normalization.
118	IV	**MTDC-T8**	Reduced / Controlled Reduction / Exact Within Closure	**Parent-action core-parameter extraction.** Gives overlap formulas for $K_s, K_q, \lambda, g_s, g_q$ from a GNLS plus localized-Maxwell throat-core ansatz.
119	IV	**MTDC-T8**	Exact Within Closure	**One-parameter parent compensation family.** Normalized ratios $(\mathfrak{r}, \mathfrak{g})$ obey $1 + \mathfrak{r}^2 = 4(\mathfrak{g} - \mathfrak{r})^2$.
120	IV	**MTDC-T8**	Exact Within Closure / Open	**Core-parameter extraction status.** Records the surviving finite parent-level target after overlap extraction.

Stage	Part	Anchor	Primary status	Working title and main output
121	IV	**MTDC-T8**	Exact Within Closure / Reduced / Controlled Reduction	**Geometric hybridization selection.** Identifying the mixed D/N tube with the throat span gives $\mathfrak{r}_{\mathrm{geom}}(L/a) = \sqrt{12(L/a)^2/\pi^2 - 1}$.
122	IV	**MTDC-T8**	Exact Within Closure	**Concrete mouth-source compensation test.** Equal-normalized mouth source misses the lower compensated branch but only by a modest traction renormalization.
123	IV	**MTDC-T8**	Exact Within Closure	**Parent-normalized branch values.** Converts the Family–1 compensated target to normalized parent flow and traction numbers (Ξ_v, Ξ_T).
124	IV	**MTDC-T8**	Exact Within Closure / Open	**Core-solve status after geometry selection.** Remaining question is whether the real core chooses the naive or lower compensated mouth branch.
125	IV	**MTDC-T8**	Exact Within Closure	**Positive local mouth-source theorem.** Positive normalized sources satisfy $0 \le \mathfrak{g}[\sigma] \le 1$, ruling out the upper compensated branch.
126	IV	**MTDC-T8**	Exact Within Closure	**Positive source families.** Self-matched derivative profile gives $\mathfrak{g} = \pi/4$; a positive convex family hits the lower branch.
127	IV	**MTDC-T8**	Exact Within Closure	**Geometric mouth-penetration families.** Slab and truncated-exponential families reach the lower branch at moderate penetration depths.
128	IV	**MTDC-T8**	Exact Within Closure / Open	**Mouth-source bias status.** Branch sign is fixed; the open datum is the actual positive mouth-source shape.
129	IV	**MTDC-T8**	Reduced / Controlled Reduction / Exact Within Closure	**GNLS plus localized-Maxwell mouth layer.** Electrochemical zero-flux law yields the exponential source profile σ_Π.
130	IV	**MTDC-T8**	Exact Within Closure / Numerically Located	**Mouth-bias map.** Exact $\mathfrak{g}_\Pi$, monotonicity, and unique Family–1 point $\Pi_* \approx 1.5088295$.
131	IV	**MTDC-T8**	Exact Within Closure	**Parent micro-threshold.** Canonical branch is the parent threshold $T_m - q_* A_0' = \Pi_* \Theta_\sigma / L$.
132	IV	**MTDC-T8**	Exact Within Closure / Open	**Mouth boundary-layer status.** The source-shape ambiguity is reduced to the selected bias Π_m.
133	IV	**MTDC-T8**	Exact Within Closure / Reduced / Controlled Reduction	**Coupled mouth fixed-point law.** Two-channel D/N boundary layer gives $\Pi = \sum_{\alpha=\pm} M_\alpha \mathcal{S}(\Pi, \kappa_\alpha)$.
134	IV	**MTDC-T8**	Exact Within Closure	**Family–1 shell plus mixed tube.** First explicit reduction gives $\Pi = M_s + M_q \mathcal{S}_q(\Pi)$.
135	IV	**MTDC-T8**	Exact Within Closure	**Outlet-consistent mouth closure.** Inheriting the compensated outlet ratio gives $M_s = 4\Sigma_m$, $M_q = -\Sigma_m$, and $\Sigma_m^* \approx 0.451485$.
136	IV	**MTDC-T8**	Exact Within Closure / Open	**Coupled mouth status.** The mouth-layer problem is reduced to a gain pair, or one gain under outlet consistency.
137	IV	**MTDC-T8**	Exact Within Closure / Reduced / Controlled Reduction	**Core-to-mouth gain map.** Explicit gains are $M_s = Lg_s^2/(K_s\Theta_\sigma)$, $M_q = -L(K_s g_q - \lambda g_s)^2/[K_s(K_s K_q + \lambda^2)\Theta_\sigma]$.
138	IV	**MTDC-T8**	Exact Within Closure	**Normalized mouth-gain family.** Gain ratio $R_q = (\mathfrak{g}_c - \mathfrak{r})^2/(1+\mathfrak{r}^2)$; exact compensation gives $R_q = 1/4$.
139	IV	**MTDC-T8**	Exact Within Closure / Numerically Located	**Actual Family–1 mouth gains.** Natural and exact-compensated gain pairs are evaluated on the Family–1 branch.
140	IV	**MTDC-T8**	Reduced / Controlled Reduction / Exact Within Closure	**Self-matched mouth susceptibility.** Same-layer susceptibility gives $\Sigma_0 = M_s = (20/9)\widehat{T}_m^2$.

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141	IV	**MTDC-T8**	Exact Within Closure / Open	**Mouth-gain status.** Records the narrowed gain-scale problem after self-matched susceptibility.
142	IV	**MTDC-T8**	Exact Within Closure	**Self-consistent mouth-branch law.** Identifying $\mathfrak{g}_c = \mathfrak{g}_\Pi$ gives $\Pi = \Sigma_0[1 - R_q(\Pi)\mathcal{S}_q(\Pi)]$.
143	IV	**MTDC-T8**	Exact Within Closure	**Equal-normalized singular limit.** $\mathfrak{g}_\Pi < 1$ for all finite $\Pi > 0$; equal-normalized branch requires $\Pi \to \infty$ and divergent traction.
144	IV	**MTDC-T8**	Exact Within Closure / Numerically Located	**Unique regular canonical mouth branch.** Lower compensated branch is the unique finite-bias, finite-traction branch in the explicit positive exponential closure.
145	IV	**MTDC-T8**	Exact Within Closure / Open	**Mouth-branch selection status.** Branch choice is settled inside the explicit closure; finite corrections around $(\Pi_*, \widehat{T}_{m,*})$ remain.
146	IV	**MTDC-T8**	Exact Within Closure	**Positive mouth-layer deformations.** First-order deformations depend only on $(\bar{g}_\varsigma, \bar{S}_\varsigma)$ and retune Π.
147	IV	**MTDC-T8**	Exact Within Closure	**First-order rigidity kernel.** Traction shift is one kernel projection with coefficients $A_T \approx -4.27264$, $B_T \approx 0.134875$.
148	IV	**MTDC-T8**	Exact Within Closure / Numerically Located	**Representative positive families.** Uniform and self-matched derivative families bracket the first-order correction; interpolation reproduces the earlier compensation fraction.
149	IV	**MTDC-T8**	Exact Within Closure / Open	**Non-exponential mouth-rigidity status.** Mouth corrections are controlled by a finite rigidity kernel, not by branch ambiguity.
150	IV	**MTDC-T8**	Exact Within Closure	**Full-profile mouth potential.** Exact residual $R_*(x) = \Phi_*(x) - \Pi_* x$ is tangent-matched but has negative curvature at the mouth.
151	IV	**MTDC-T8**	Exact Within Closure	**Full-profile source correction.** Actual selected deformation is the centered residual R_*; correction is fixed by two covariances.
152	IV	**MTDC-T8**	Exact Within Closure / Numerically Located	**Family–1 mouth correction.** Full mouth profile broadens the source and shifts the point to $(\Pi_{\mathrm{corr}}, \widehat{T}_{m,\mathrm{corr}})$.
153	IV	**MTDC-T8**	Exact Within Closure / Open	**Mouth correction status.** Branch ambiguity is gone; finite mouth-only correction motivates full co-evolution.
154	IV	**MTDC-T8**	Exact Within Closure	**Co-evolving core–mouth map.** Nonlinear fixed-point map $\Sigma \propto e^{-\Phi_{\Sigma_0}[\Sigma]}$, with $\mathcal{R}[\Sigma] = (\mathfrak{g}[\Sigma] - \mathfrak{r}_{F1})^2/(1 + \mathfrak{r}_{F1}^2)$.
155	IV	**MTDC-T8**	Numerically Located / Exact Within Closure	**Frozen-traction co-evolving fixed point.** At old traction, fixed point survives but moves off exact compensation: $\mathcal{R}_{\mathrm{fp}} \approx 0.282714$.
156	IV	**MTDC-T8**	Numerically Located / Exact Within Closure	**Renormalized co-evolving canonical branch.** Exact compensation is restored at $\Sigma_0^{\mathrm{can}} \approx 4.65103$, $\widehat{T}_{m,\mathrm{can}} \approx 1.44671$.
157	IV	**MTDC-T8**	Numerically Located / Open	**Co-evolution status.** Identifies the reduced co-evolving target and leaves the deviation-to-normalization map as the next task.
158	IV	**MTDC-T8**	Exact Within Closure	**Linear defect transport.** Linearizes $(\delta\Sigma_0, \delta\mathfrak{g}, \delta\mathcal{S})$ into $(\delta M_s, \delta M_q, \delta\Pi)$.
159	IV	**MTDC-T8**	Exact Within Closure	**Projection onto compensated outlet.** Even preservation forces $\delta\mathfrak{g} = 0$, $\delta\kappa_W = 0$, leaving $\Delta_Q = -9\sigma_*\delta\gamma_W/(1 - \sigma_*)$.

Stage	Part	Anchor	Primary status	Working title and main output
160	IV	**MTDC-T8**	Exact Within Closure	**Bare mixed-port slippage.** $\delta\gamma_W = (\delta\gamma_0 - \delta\kappa_0/3)/(1 + \mathfrak{r}_{F1}^2)$.
161	IV	**MTDC-T8**	Exact Within Closure	**D/N similarity decomposition.** Last defect is $\Delta_Q = -\sigma_* \Xi_{\rm slip} \delta\Pi_{\rm tan}/(1-\sigma_*)$, with $\Xi_{\rm slip} = \Xi_\gamma - 2\Xi_L$.
162	IV	**MTDC-T8**	Exact Within Closure	**Parent compensation-surface rigidity.** Staying on the exact parent compensation family gives automatic D/N similarity preservation and $\Delta_Q = 0$ at first order.
163	IV	**MTDC-T8**	Exact Within Closure / Open	**Off-family normal coordinate.** The first true off-family drift is the scalar $\delta_\perp$, with explicit microscopic parent-action formula.
164	V	**MTDC-T9**	Exact Within Closure	**Microscopic log-imbalance channels.** Rewrites the Stage 163 off-family scalar in explicit branch-variable drifts of $(\mathcal{Z}_q, \rho_w, c_{s,w}, c_s, \mathcal{T}_m, v_{w0}, a, L_W)$.
165	V	**MTDC-T9**	Exact Within Closure	**Lower-branch drift laws.** Shows $\delta \ln L_W = \delta \ln a$ and reduces the lower compensated branch to four irreducible drifts $(\delta \ln \mathcal{Z}_q, \delta \ln \rho_w, \delta \ln c_s, \delta \ln a)$.
166	V	**MTDC-T9**	Exact Within Closure	**Bundle inversion of the last four drifts.** Inverts the four observables $(\Theta_w, K_s, K_q, P_0)$ to determine the four remaining microscopic drifts.
167	V	**MTDC-T9**	Exact Within Closure	**Bundle transport and tangent compensation.** Proves arbitrary first-order isotropic bundle drift is tangent to the exact compensated parent family.
168	V	**MTDC-T9**	Exact Within Closure	**Off-bundle slippage decomposition.** Reduces the first true off-bundle defect to axial-length, mixed-speed, and mouth-traction slippages.
169	V	**MTDC-T9**	Exact Within Closure	**No linear grouped feed-down into scalar slippages.** Pure grouped real P_2 anisotropy cannot linearly source the scalar off-bundle slippages; scalar feed-down starts quadratically.
170	V	**MTDC-T9**	Exact Within Closure	**Linear grouped outlet map.** Reduces the direct outlet deformations to $\mathcal{K}_A = \delta D_{A,2} + \delta D_{A,0}/9$ and $\mathcal{G}_A = \delta N_{A,0} - P_0 \delta D_{A,0}$.
171	V	**MTDC-T9**	Exact Within Closure	**Microscopic grouped obstructions.** Decomposes $\mathcal{K}_A$ and $\mathcal{G}_A$ into wall, BdG, Maxwell/mixed, and outgoing-transfer defect bundles.
172	V	**MTDC-T9**	Exact Within Closure	**Physical slope collapse.** Identifies $\mathfrak{K}_1 = -D_0 u_2^{(1)}$ and $\mathfrak{G}_1 = D_0 P_1$, reducing the linear grouped outlet problem to physical slopes.
173	V	**MTDC-T9**	Exact Within Closure	**Weak-axisymmetric loading mismatch.** On the even-preserving branch, the remaining grouped defect is the scalar $\Xi_{\rm load} = N_{01}/N_0 - D_{01}/D_0$.
174	V	**MTDC-T9**	Exact Within Closure	**Static self-similarity.** Expresses $\Xi_{\rm load}$ as weighted failure of static self-similarity between wall, BdG, conservative port, and outgoing port slopes.
175	V	**MTDC-T9**	Exact Within Closure	**Wall-normalized outgoing-load theorem.** Factors the remaining defect into wall-normalized shape/load variables and proves conservative-shape preservation leaves only outgoing load slippage.
176	V	**MTDC-T9**	Exact Within Closure	**Square-root mixed-leg law.** Factors the outgoing load into mixed-leg, interference, and hybridization slippages; under rigidity, zero defect requires $G_W/\Omega_W^2 \propto \sqrt{K}$.
177	V	**MTDC-T9**	Exact Within Closure	**One scalar outgoing-slippage amplitude.** Shows all weak-axisymmetric outgoing slippages inherit the same grouped signature and collapse to $\Xi_1 = P_1/P_0$.

Stage	Part	Anchor	Primary status	Working title and main output
178	V	**MTDC-T9**	Exact Within Closure	**Outgoing-port co-loading theorem.** Rewrites Ξ_1 as the weighted mismatch between actual static outgoing-port loading and wall-baseline loading.
179	V	**MTDC-T9**	Exact Within Closure	**Transfer-shape theorem.** Factors each outgoing port as $N_{A,0}^{(r)} = K_A \mathcal{T}_{A,r}^2$ and gives $\Xi_1 = 2\sum_r \rho_r^{(N)}\tau_r$.
180	V	**MTDC-T9**	Exact Within Closure	**Effective transfer-shape collapse.** Collapses many ports to $\mathcal{T}_{\mathrm{eff},A}^2 = N_{A,0}/K_A$ and gives the actual one-port continuum formula.
181	V	**MTDC-T9**	Exact Within Closure	**Coherent tracking-branch defect law.** On the coherent local D/N tracking branch, Ξ_1 is carried only by mixed/outgoing placement variables; coherent support ratio ζ drops out.
182	V	**MTDC-T9**	Exact Within Closure	**Microscopic coherent-kernel slippages.** Reduces the defect to $(\Sigma_Z, \Sigma_\chi, \Sigma_\epsilon, \Sigma_\delta)$ plus dressing slippage Σ_η, and isolates Σ_{tr}.
183	V	**MTDC-T9**	Exact Within Closure	**Triangular normal form.** Compresses the coherent weak-axisymmetric problem to $(\Sigma_{\mathrm{tr}}, \Sigma_{\mathrm{nt}}, \Sigma_\eta)$ with triangular map to $(\Theta_1, \Xi_1, \mathcal{R}_1)$.
184	V	**MTDC-T9**	Exact Within Closure	**Branch-invariant coordinates.** Identifies exact branch composites $\mathfrak{T}_*$, $\mathfrak{N}_*$, and ϵ_η whose logarithmic drifts are the three normal-form coordinates.
185	V	**MTDC-T9**	Exact Within Closure	**Microscopic monomial coordinates.** Pushes the branch composites to direct monomials $(\mathfrak{C}_{\mathrm{tr},*}, \mathfrak{C}_{\mathrm{nt},*}, \epsilon_\eta)$.
186	V	**MTDC-T9**	Exact Within Closure	**Similarity-orbit tangent theorem.** Shows the zero-defect equations are the tangent equations of a five-parameter monomial-preserving similarity orbit $\mathcal{G}_*$.
187	V	**MTDC-T9**	Exact Within Closure	**Orbit–quotient closure.** Proves finite level sets of $(\mathfrak{C}_{\mathrm{tr},*}, \mathfrak{C}_{\mathrm{nt},*}, \epsilon_\eta)$ are exactly $\mathcal{G}_*$-orbits.
188	V	**MTDC-T9**	Exact Within Closure	**Branch-observable compiler.** Compiles direct branch observables to $(\Theta_1, \Xi_1, \mathcal{R}_1)$ and records equivalent observable packets.
189	V	**MTDC-T9**	Exact Within Closure	**Transfer-shape / outgoing-prefactor compiler.** Equates Ξ_1 with the logarithmic transfer-shape and outgoing-prefactor slope, including corrected nontracking form.
190	V	**MTDC-T9**	Exact Within Closure	**Direct defect, dressing residual, and no-go filters.** Separates direct transfer-shape drift from selected-branch dressing residual and records scalar no-go filters.
191	V	**MTDC-T9**	Exact Within Closure	**Minimal PDE data packet.** Defines Packet A, Packet B, Δ_{branch}, Δ_{orbit}, and the exact two-packet home-stretch theorem.
192	V	**MTDC-T9**	Exact Within Closure	**Orbit/quotient projectors.** Builds exact projectors Q_{quot}, O_{orb}, showing quotient failure lives only on $(T_U, K_\eta^{\mathrm{eff}}, \mu_W)$.
193	V	**MTDC-T9**	Exact Within Closure	**Isotropic grouped target surface.** Freezes the isotropic one-pole conservative target surface and proves the scalar/geometry lane enters the grouped carrier only at quadratic anisotropy order.
194	V	**MTDC-T9**	Exact Within Closure	**Outgoing $l = 2$ fingerprint.** Derives the exact outgoing spherical DtN fingerprint and fixes $\chi_Q = 1$ on the canonical compact branch.
195	V	**MTDC-T9**	Exact Within Closure / Reduced / Controlled Reduction	**Source-map reduction.** Factorizes the observable odd closure as $\widehat{m}_0^2 \chi_Q N_Q = 1$ and collapses Δ_{norm}.

Stage	Part	Anchor	Primary status	Working title and main output
196	V	**MTDC-T9**	Exact Within Closure	**Higher-odd irrelevance.** Proves all extra isotropic odd tails beginning at $O(\omega^7)$ are invisible to the point-particle 2.5PN theorem.
197	V	**MTDC-T9**	Exact Within Closure	**Conditional Packet-A closure.** Proves $\Delta_{\text{branch}} = 0 \iff \chi_Q = 1$ and hence $\Delta_Q = \chi_Q - 1$ is the sole Packet-A scalar.
198	V	**MTDC-T9**	Exact Within Closure	**Finite orbit law.** Solves the dependent triple exactly, defines mismatch coordinates, and proves finite single-orbit lock.
199	V	**MTDC-T9**	Exact Within Closure	**Pairwise orbit transport.** Removes representative privilege by giving exact two-point transport, mismatch, cocycle, and orbit-lock laws.
200	V	**MTDC-T9**	Exact Within Closure	**Reference-free home-stretch theorem.** Combines Packet A and Packet B into the four-scalar final verdict packet $(\Delta_Q, q_{\text{tr}}, q_{\text{nt}}, q_\eta)$.
201	VI	**MTDC-T10**	Exact Within Closure	**Realization compiler and canonical orbit projection.** Intrinsic four-scalar packet, canonical dependent-triple repair vector, canonical target-orbit projection, and same-free-quintuple uniqueness theorem.
202	VI	**MTDC-T10**	Exact Within Closure	**Free-quintuple target graph.** Solves the target orbit as a graph over $(\lambda_W, c_{\eta U}, \gamma, K_U, K_W^{\text{eff}})$, defines graph errors (E_T, E_K, E_μ), and gives the first reduced-family closure surface.
203	VI	**MTDC-T10**	Exact Within Closure	**Scalar graph-slice theorem.** Reduces graph-aligned realization to the single scalar $\widehat{\chi}_Q(\mathbf{y}) = 1$, proves graph tangency to $\ker M_*$, and gives the one-parameter crossing theorem.
204	VI	**MTDC-T10**	Exact Within Closure	**Explicit free-quintuple log-ray sweep.** Builds $\mathbf{y}_{\mathbf{s}}(\tau) = \mathbf{y}_\circ \odot e^{\tau \mathbf{s}}$, gives the finite graph lift and primitive-direction table, and reduces closure to a one-variable scalar root.
205	VI	**MTDC-T10**	Exact Within Closure	**Directional Hessian and quadratic predictors.** Adds directional Hessian data, quadratic affine/log predictors, discriminant tests, and turning-point/tangency criteria.
206	VI	**MTDC-T10**	Mixed: Exact Within Closure, Numerically Located	**Curvature-envelope ray ranking.** Turns local curvature envelopes into certified root brackets, bracket-width laws, pairwise ray ordering, and the local search sieve.
207	VI	**MTDC-T10**	Mixed: Exact Within Closure, Numerically Located	**Primitive-ray certified table.** Specializes the sieve to the five primitive free directions; proves only diagonal Hessian restrictions are needed for primitive certification.
208	VI	**MTDC-T10**	Mixed: Exact Within Closure, Numerically Located	**Pairwise mixed-ray audit.** Introduces pairwise cones, gradient synergy, off-diagonal Hessian leverage, canonical gradient/equal-mix screens, and pairwise promotion rules.
209	VI	**MTDC-T10**	Mixed: Exact Within Closure, Numerically Located	**Pairwise ratio optimizer.** Reduces each pairwise ratio search to a finite quartic candidate set and defines optimized pairwise brackets.
210	VI	**MTDC-T10**	Mixed: Exact Within Closure, Numerically Located	**Three-coordinate simplex audit.** Reduces triple boundaries to pairwise cones, gives gradient and equal-mix interior screens, and defines the canonical triple-screen packet.
211	VI	**MTDC-T10**	Mixed: Exact Within Closure, Numerically Located	**Triple-simplex interior optimizer.** Reduces the two-parameter triple interior optimizer to quartic/sextic algebraic candidate sets with finite interior brackets.

Stage	Part	Anchor	Primary status	Working title and main output
212	VI	**MTDC-T10**	Mixed: Exact Within Closure, Numerically Located	**Up-to-three-coordinate search sieve.** Splices triple interiors to pairwise boundaries, ranks all primitive triples, and proves the support-≤ 3 finite ledger.
213	VI	**MTDC-T10**	Mixed: Exact Within Closure, Numerically Located	**Four-coordinate simplex gate.** Reduces four-simplex faces to triple packets and introduces four-way gradient/equal-mix canonical screens.
214	VI	**MTDC-T10**	Mixed: Exact Within Closure, Numerically Located	**Four-coordinate interior optimizer.** Lifts the three-parameter four-simplex stationary system to a finite polynomial candidate set with preferred bound 54 per envelope.
215	VI	**MTDC-T10**	Mixed: Exact Within Closure, Numerically Located	**Up-to-four-coordinate search sieve.** Splices quadruple interiors to triple-face boundaries, ranks all primitive quadruples, and proves the support-≤ 4 finite ledger.
216	VI	**MTDC-T10**	Mixed: Exact Within Closure, Numerically Located	**Unique five-coordinate simplex gate.** Identifies the unique five-simplex, reduces its boundary to the support-≤ 4 ledger, and proves the support-cardinality ceiling.
217	VI	**MTDC-T10**	Mixed: Exact Within Closure, Numerically Located	**Five-coordinate interior optimizer.** Reduces the four-parameter five-simplex interior optimizer to a lifted finite polynomial candidate set with preferred bound 179 per envelope.
218	VI	**MTDC-T10**	Mixed: Exact Within Closure, Numerically Located, Open	**Final local mixed-ray closure theorem.** Splices the unique support-5 interior packet to the support-≤ 4 boundary ledger and closes the local mixed-ray search with preferred budget 1464.
219	VII	**MTDC-T11**	Exact Within Closure	**One-port mixed-bundle static kernel.** Exact static 3×3 bundle, determinant $\det \mathcal{K}_{\mathrm{red}} = \Delta D_0$, susceptibility kernel, outgoing-prefactor bridge, and square-law suppression verdict.
220	VII	**MTDC-T11**	Exact Within Closure	**Dynamic mixed-port kernel and phase-lag no-go.** Dynamic 3×3 bundle, determinant $\Delta_{\Pi} D_{\Pi}$, dynamic product-family theorem, outgoing-port derivative identity, and first-order phase-lag/no-barrier theorem.
221	VII	**MTDC-T11**	Exact Within Closure	**Resonance/linewidth tradeoff.** Simple-pole Breit–Wigner normal form, dispersive/absorptive no-free-lunch theorem, low-loss bound, and linear survival window.
222	VII	**MTDC-T11**	Mixed: Exact Within Closure, Numerically Located	**Concrete finite-throat primitive branch.** Lowest N/N and D/N primitive branch, quartic pole polynomial, pole census, residue/linewidth figure, and first branch-level dynamic survival test.
223	VII	**MTDC-T11**	Mixed: Exact Within Closure, Numerically Located	**Isotropic target surface and primitive compatibility.** Exact compatibility equation between the conservative one-pole condition and outgoing normalization, compatibility branch, and finite dynamic survival windows.
224	VII	**MTDC-T11**	Mixed: Exact Within Closure, Open	**Branch-packet compiler and actual-branch kill test.** Compiler from $(\Delta_{\mathrm{norm}}, a_{P_0}, b_{P_0})$ to lane prefactors, weak-axisymmetric law $b_{P_0} = 3a_{P_0}$, and transported ceiling in $(\Delta_{\mathrm{norm}}, \Xi_1)$.
225	VII	**MTDC-T11**	Mixed: Exact Within Closure, Open	**Microscopic Ξ_1 compiler and first compensation surface.** First-order formulas for $u_2^{(1)}, u_4^{(1)}, \Xi_1$, primitive logarithmic-slope compiler, mechanism sieve, and first surviving mixed-sector family.
226	VII	**MTDC-T11**	Mixed: Exact Within Closure, Reduced / Controlled Reduction	**Strict 5PN even-gate package.** Imports $\Xi_{\mathrm{load}}, K_1, H_{\mathrm{even}}$, proves $\Xi_{\mathrm{load}} = \Xi_1$, and narrows the survivor to a pure outgoing-transfer corridor with the conservative bundle frozen.

Stage	Part	Anchor	Primary status	Working title and main output
227	VII	**MTDC-T11**	Exact Within Closure	**Pure-transfer load factor and co-loading no-go.** Exact factorization $\Lambda = P/\Delta$, identity $\Xi_1 = 2\delta \ln \Lambda$, outgoing-rigidity sieve, and co-loading no-go under simultaneous interference and hybridization rigidity.
228	VII	**MTDC-T11**	Mixed: Exact Within Closure, Numerically Located	**Numerator/denominator split and dynamic-window test.** Split $\Xi_1 = 2(\pi_1 - \delta_1)$, numerator-rigid and denominator-rigid survivors, and first wall-like dynamic-window audit.
229	VII	**MTDC-T11**	Exact Within Closure	**Selected-branch numerator/denominator signature.** Selected softening-depth factorization, classifier map, threshold $\delta = 8/9$, and denominator-like near-softening theorem.
230	VII	**MTDC-T11**	Mixed: Exact Within Closure, Numerically Located	**Dynamic-window classifier and static-first theorem.** Rigid-split share compiler, dynamic sign theorem, classifier threshold, and proof that the static Ξ_1 budget remains the first kill condition.
231	VII	**MTDC-T11**	Exact Within Closure	**Continuum-placement pullback.** Pulls the selected-branch classifier back to $(\epsilon_\eta, \epsilon_W, \rho, Z_W, \delta_0, \Lambda)$, proves monotonicity in R_{target}, and keeps the static-first verdict on the physical placement map.
232	VII	**MTDC-T11**	Mixed: Numerically Located, Open	**Known 5PN data injection.** Injects numerically located Family-1 support/source data, shows the support/source side is safe by a large margin, and leaves the unresolved gate at static orbit-lock/coherent placement.
233	VII	**MTDC-T11**	Mixed: Exact Within Closure, Open	**Rigid-mouth orbit-lock compiler.** Distinguishes mouth tracking from operator rigidity, shows $\delta \ln R_{\text{tr}} = 0 \Rightarrow \Xi_1 = \delta \ln \mathfrak{N}_*$, and reads the static ceiling as an internal transfer/turbulence gate.
234	VII	**MTDC-T11**	Exact Within Closure	**Direct branch-observable static gate.** Finite chart in $(R_{\text{tr}}, R_{\text{target}}, \epsilon_\eta)$, two cancellations, rigid-mouth reduction, and two-observable kill test for R_{target} and ϵ_η.
235	VII	**MTDC-T11**	Exact Within Closure	**Rigid-mouth packet projectors.** Involutive direct-packet map, projectors P_{nt}, P_η, static-blind dressing line, and codimension-two orbit-lock theorem.
236	VII	**MTDC-T11**	Exact Within Closure	**Microscopic dependent-plane projectors.** Dependent-plane compiler, projectors $P^{\text{dep}}_{\text{nt}}, P^{\text{dep}}_\eta$, equal-drift dressing ray, and static-only restoration gap.
237	VII	**MTDC-T11**	Exact Within Closure	**Actual-branch dressing compiler.** Finite static-blind curve, direct computation of q_η, microscopic extractor $q_\eta = 2c_1 - \kappa_U - \kappa_\eta$, and support-blindness theorem.
238	VII	**MTDC-T11**	Exact Within Closure	**Physical transfer-shape compiler.** Coherent identity $R_{\text{target}}\mathcal{T}^2 = \Lambda_0(1 - \epsilon_\eta)$, packet factorization, and three-gate theorem: tracking, transfer-shape, dressing.
239	VII	**MTDC-T11**	Exact Within Closure	**Rigid-mouth physical normal form.** Cartesian chart (U, V), physical-to-microscopic compiler $(\Delta_T, \Delta_{K_\eta}, \Delta_\mu) = (0, -V, U - V)$, and orbit-lock theorem $U = V = 0$.
240	VII	**MTDC-T11**	Exact Within Closure	**Selected-branch loading ratio.** Exact loading ratio $\rho_\alpha = 4/3$, required support excess $\zeta_{\text{req}} = 1/3$, and selection of the symmetric lowest-twin support slice.
241	VII	**MTDC-T11**	Exact Within Closure	**Primitive ranking on selected twin-support branch.** Exact selected curve $\epsilon_* = 1 - 3\varrho/2$, threshold pair $(\varrho_{W\Lambda}, \varrho_{U\Lambda})$, and primitive carrier ranking phase diagram.

Stage	Part	Anchor	Primary status	Working title and main output
242	VII	**MTDC-T11**	Mixed: Exact Within Closure, Open	**Actual twin-support placement and coherent orbit-lock compiler.** Actual placement coordinate $\varrho_{\text{phys}} = 2(1-\epsilon)/3$, support/orbit packet split, and final front-end branch-evaluation packet.
243	VIII	**MTDC-T12**	Mixed: Effective Closure, Exact Within Closure	**Relaxed-constraint branch declaration.** Codimension-three lift of the Stage 242 front end, exact recovery map $(\ell_w, f_U, a, b) = (0,0,0,0)$, leakage/work lane, non-rigid (U,V) lane, compensated source lane, and strict short-range kernel-span theorem.
244	VIII	**MTDC-T12**	Exact Within Closure	**Selected leakage and scalar-photon work compiler.** Exact Gaussian one-mode leakage law, exact bulk and session work scalars, selected-support pullback through Π_{tr}, support/orbit split, and orientation parity.
245	VIII	**MTDC-T12**	Exact Within Closure	**Non-rigid mouth and U/V drain packet.** Exact finite compiler for $\mathcal{T}^2$, ϵ_η, and R_{target}; non-rigid determinant Δ_{UV}; dependent microscopic correction; positive U/V drain; and first-order packet $(\Xi_1, \mathcal{R}_1)$.
246	VIII	**MTDC-T12**	Exact Within Closure	**Compensated multimode source compiler.** Mean-preserving two-mode source family, exact sign-change test, mouth-bias and shell-loading moments, invertible two-moment source map, mixed-to-shell loading ratio, and transported Session-I readback.
247	VIII	**MTDC-T12**	Mixed: Exact Within Closure, Effective Closure, Numerically Located	**Relaxed stationary barrier front end.** One-port short-range baseline, relaxed lowering identity, stationary barrier compiler $V_{\text{eff}}^{(247)}$, and one-point Session-I benchmark decomposition.
248	VIII	**MTDC-T12**	Mixed: Exact Within Closure, Reduced / Controlled Reduction, Numerically Located	**Dynamic event-chain compiler.** Energy integral, peak and threshold-speed compiler, turning points, WKB action, transmission ratio, dynamic diagnostics, and Session-II crossing/tunneling benchmark.
249	VIII	**MTDC-T12**	Mixed: Exact Within Closure, Effective Closure, Numerically Located	**Helicity-export diagnostic.** Exact projected subscale-helicity identity, aligned/anti-aligned orientation closure, peak and integrated ratio compilers, and Session-II hidden-channel export packet.
250	VIII	**MTDC-T12**	Mixed: Exact Within Closure, Effective Closure, Numerically Located	**Crossing-vs-collapse Goldilocks window.** Event-chain transit time, characteristic crossing time, dressing-leg collapse time, survival ratio, lower safe edge, speed-space compiler, heavy-throat scaling, and trigger-width sensitivity.
251	VIII	**MTDC-T12**	Mixed: Exact Within Closure, Reduced / Controlled Reduction, Open	**Microscopic damping/export kernel.** Derivative-coupled scalar cubic export, selected quadrupole quintic export, active-leg kernel $K_{\exp}(s) = \Gamma_3 s^3 + \Gamma_5 s^5$, exact event-safe half-plane, and channel-resolved power split.
252	VIII	**MTDC-T12**	Mixed: Exact Within Closure, Effective Closure, Numerically Located, Open	**Vacuum/lattice heat partition.** Exact vacuum/lattice exported energies, one-scalar event-shape quotient, single-growth cold-event specialization, safe-edge exported-energy theorem, and event-equivalent export rates.
253	VIII	**MTDC-T12**	Mixed: Exact Within Closure, Effective Closure, Open	**Physical calibration and material thresholds.** Physical calibration dictionary, coarse lattice-turnover projection factor, electron-phonon threshold, force-matched stiffness, Korringa ceiling, and four material-screening ratios.

A.5 Stage source-file ledger

The source-file ledger records the raw derivation artifact that each detailed stage appendix is meant to audit. These pointers are provenance handles, not citation claims. If a stage file has already been expanded into a polished appendix section, the polished section controls presentation; the source pointer still identifies the original derivation record.

Table A.3: Source-file provenance for Stages 001–253.

Stage	Part	Source file
001	I	stages/stage_001.tex
002	I	stages/stage_002.tex
003	I	stages/stage_003.tex
004	I	stages/stage_004.tex
005	I	stages/stage_005.tex
006	I	stages/stage_006.tex
007	I	stages/stage_007.tex
008	I	stages/stage_008.tex
009	I	stages/stage_009.tex
010	I	stages/stage_010.tex
011	I	stages/stage_011.tex
012	I	stages/stage_012.tex
013	I	stages/stage_013.tex
014	I	stages/stage_014.tex
015	I	stages/stage_015.tex
016	I	stages/stage_016.tex
017	I	stages/stage_017.tex
018	I	stages/stage_018.tex
019	I	stages/stage_019.tex
020	I	stages/stage_020.tex
021	I	stages/stage_021.tex
022	I	stages/stage_022.tex
023	I	stages/stage_023.tex
024	II	stages/stage_024.tex
025	II	stages/stage_025.tex
026	II	stages/stage_026.tex
027	II	stages/stage_027.tex
028	II	stages/stage_028.tex
029	II	stages/stage_029.tex
030	II	stages/stage_030.tex
031	II	stages/stage_031.tex
032	II	stages/stage_032.tex
033	II	stages/stage_033.tex
034	II	stages/stage_034.tex
035	II	stages/stage_035.tex
036	II	stages/stage_036.tex
037	III	stages/stage_037.tex
038	III	stages/stage_038.tex
039	III	stages/stage_039.tex
040	III	stages/stage_040.tex
041	III	stages/stage_041.tex
042	III	stages/stage_042.tex
043	III	stages/stage_043.tex
044	III	stages/stage_044.tex
045	III	stages/stage_045.tex
046	III	stages/stage_046.tex
047	III	stages/stage_047.tex
048	III	stages/stage_048.tex
049	III	stages/stage_049.tex
050	III	stages/stage_050.tex
051	III	stages/stage_051.tex
052	III	stages/stage_052.tex
053	III	stages/stage_053.tex
054	III	stages/stage_054.tex
055	III	stages/stage_055.tex

Stage	Part	Source file
056	III	stages/stage_056.tex
057	III	stages/stage_057.tex
058	III	stages/stage_058.tex
059	III	stages/stage_059.tex
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251	VIII	stages/stage_251.tex
252	VIII	stages/stage_252.tex
253	VIII	stages/stage_253.tex

A.6 Citation and audit guardrails

1. Cite **MTDC-T4** or **MTDC-T5** for the derivation of the outgoing $l = 2$ prefactor compiler; do not cite those anchors as proof that the actual branch realizes the universal normalization unless the realization row being used is also marked closed.
2. Cite **MTDC-T8** for geometry-lane, retarded, outlet, core, and mouth-compensation reductions; do not use it as an unconditional no-go theorem for all scalar or geometry contamination.
3. Cite **MTDC-T9** for the weak-axisymmetric quotient packet, similarity orbit, and four-scalar verdict packet; do not treat $\Delta_{\text{full}} = 0$ as realized unless a later branch-realization stage supplies that fact.
4. Cite **MTDC-T10** for the finite local free-quintuple/mixed-ray search class; do not cite it as a global proof over every possible PDE branch outside that declared class.
5. Cite **MTDC-T11** for the same-charge audit, rigid-mouth chart, and selected twin-support placement; do not infer a new long-range same-charge attraction from the static one-port mixed bundle.
6. Cite **MTDC-T12** for the relaxed/open-system companion and its recovery map; do not use the relaxed branch as a proof of the strict conservative branch without explicitly applying the recovery map.

A.7 What remains open after the ledger

The ledger is complete as an audit index, but it does not convert open realization claims into closed theorems. The recurring open item is actual branch realization: the completed moving-throat PDE must return the required grouped P_2 carrier, outgoing normalization, weak-axisymmetric prefactor slope, rigid-mouth orbit-lock packet, selected twin-support placement, or relaxed leakage/export packet, depending on which downstream result is being invoked. The stage ledger should therefore be read as a map from derivation provenance to claim status, not as permission to erase the status firewall.

Appendix B

Stage Appendix: Part I – Parent Theory, Geometry Lift, and Linearized PDE Backbone

Stages 001–023; anchors MTDC-T1–T5

This appendix is the detailed verification ledger for the first theorem block of the moving-throat derivation. The corresponding main-text chapter, chapter 1, explains the role of the block in the larger paper. The job of this appendix is narrower and more mechanical: it records, stage by stage, the definitions, reductions, algebraic transforms, branch tests, and audit checks that turn the exact parent model into the first usable wall/support/outgoing response bundle.

The twenty-three stages here should be read as a single construction. Stage 001 promotes the old collective geometry variables to a distributed moving-surface field and introduces the scalar, grouped real P_2, and higher-harmonic lanes. Stage 002 verifies that this distributed wall theory reduces back to the old (a, L) closure in the lowest axisymmetric sector. Stage 003 couples the wall to stable BdG matter support modes and derives the first conservative Schur-complement kernels. Stages 004–021 now treat the electromagnetic sector projection-first: they derive the exact projected localized Maxwell law, identify the matched zero-mode reduction, transport mouth-local projected corrections into the grouped Z_n, N_n slots, import the parent-throat action packet, and retain the previous Maxwell/mixed one-port calculation as the reduced normal form for outgoing transfer. Stage 022 translates that outgoing bridge into the normalized grouped-response language used by the retarded quadrupole program. Stage 023 writes the full grouped 20/21/22 bundle, its weighted projectors, its microscopic coefficient ledger, and the isotropic normalization test.

Source layout. Stages 001–023 are now sourced from the canonical files `stages/stage_001.tex` through `stages/stage_023.tex`. The raw Markdown notes and other provenance artifacts are tracked in a repo-local provenance index, so they do not have to appear in the reader-facing PDF. The stage files themselves carry the executable SymPy, Mathematica, and numerical-stress references when they exist.

Table B.1: Part I stage-audit map.

Stage	Source stage	Primary status	Verification output
001	Geometry lift and linearized PDE skeleton	Effective Closure / Exact Within Closure	Distributed shape field $R(\Omega, w, t)$, promoted confinement coupling, modal split, and first linearized GNLS/Maxwell/wall system.
002	Breathing reduction back to (a, L)	Exact Within Closure	Two-mode axisymmetric reduction, effective $(\delta a, \delta L)$ mass/stiffness matrices, and uncoupled grouped P_2 degeneracy.
003	Minimal BdG–wall coupling	Reduced / Controlled Reduction / Exact Within Closure	Stable-mode Schur complement, low-frequency wall moments, pole shift formula, and conservative grouped-isotropy diagnostic.
004	Projected Maxwell bundle index	Exact	Bundle ordering and projection-by-parts identity for the projection-first electromagnetic sector.
005	Covariant projected Maxwell law	Exact	Exact projected inhomogeneous Maxwell equation, boundary flux, kernel-gradient leakage, and projected continuity law.
006	Vector projected Maxwell split	Exact	Measured-field homogeneous equations, source-flux inhomogeneous equations, and the open-system charge balance.
007	Projection/reduction comparison	Exact / Reduced / Controlled Reduction	Matched Gaussian zero-mode channel and its coupling/gauge-parameter mismatch relative to the projection law.
008	Projected Maxwell extension and clean matching	Exact / Reduced / Controlled Reduction	Slot-level distinction between source-flux projection and matched reduction.
009	Near-throat projection scale	Exact Within Closure	Near-throat projected scale factors and the leakage/source normalization ledger.
010	Projected Maxwell push into bundle slots	Exact Within Closure	Projected Z_n, N_n slot transport, denominator and prefactor variations, compatibility transports, and weak-axisymmetric lane anchors.

Stage	Source stage	Primary status	Verification output
011	Projected Maxwell P_2 bridge	Exact Within Closure	P_2-lane bridge from projected electromagnetic slots into the grouped normalization language.
012	Projected Maxwell primitive bridge	Exact Within Closure	Primitive finite-throat bridge data induced by the projected electromagnetic sector.
013	Mouth-Taylor master map	Exact Within Closure	Mouth-local Taylor map for projected source and flux corrections.
014	Mouth-Taylor gate bridge	Exact Within Closure	Gate conditions for carrying mouth-local projected data into the grouped bundle.
015	Parent throat action master	Exact / Reduced / Controlled Reduction	Parent throat-action packet and projection/reduction status boundary.
016	Parent throat action candidate	Exact Within Closure	Candidate parent-action data for the projected throat sector.
017	Parent throat action weak-axisymmetric law	Exact Within Closure	Weak-axisymmetric parent-action transport law.
018	Parent throat action bundle master	Exact Within Closure	Bundle-level parent-action identities used by the projected electromagnetic response.
019	Parent throat action isotropic bundle	Exact Within Closure	Isotropic parent-action specialization for the projected bundle.
020	Parent throat action weak-axisymmetric packet	Exact Within Closure	Weak-axisymmetric packet exported from the parent throat-action bundle.
021	Reduced Maxwell/mixed one-port normal form	Reduced / Controlled Reduction / Exact Within Closure	Retained reduced one-port self-energy, transfer factor, compact $l = 2$ fingerprint, and scalar-compatibility criterion.
022	Grouped normalization bridge	Exact Within Closure / Open	Conversion from operator moments to normalized response moments and invariant outgoing-normalization product.
023	Full grouped bundle and projectors	Exact Within Closure / Open	Weighted grouped projectors, full D, N, P coefficient ledger, isotropic branch test, and first-order anisotropy transport laws.

Audit philosophy. The derivations below preserve the status distinction used throughout the ledger. Algebraic eliminations, projector identities, low-frequency expansions, and grouped inverse maps are exact once the declared reduced variables are adopted. The projection-first Maxwell law is exact after the observation kernel is declared; the matched zero-mode and one-port Maxwell/mixed normal forms are controlled reductions. The adoption of a distributed wall closure, a stable BdG normal-mode reduction, and a compact outgoing port are reduced or effective moves. The final equality of the outgoing-normalization product with the universal quadrupole target is not proved in this appendix; it is formulated here as the branch test that later stages attempt to realize.

Notation warning. The grouped symbols $20, 21, 22$ are grouped real P_2 lanes, not spacetime indices. The 21 and 22 labels each abbreviate a cosine/sine real pair when the full five-dimensional real STF representation is being discussed. The mixed gauge variables $A_w, J^w, F_{\mu w}, E_w, C_a$ are retained as microscopic channels; they are suppressed only in the strict far-field brane reduction.

B.1 Stage 001: Geometry Lift and Linearized PDE Skeleton

Part / anchor. Part I; primary anchors **MTDC-T1**, **MTDC-T2**, and **MTDC-T4**.

Claim status. The parent-field import is Exact **inside the declared parent model. The shape-field lift and minimal quadratic wall action are** Effective Closure**. The modal split and the linearized equations are** Exact Within Closure **once that lift is adopted.**

Purpose. Stage 001 replaces the old finite-dimensional geometry variables $a(t)$ and $L(t)$ by a distributed throat geometry variable, while preserving those variables as collective moments. The stage also introduces the first linearized matter/gauge/geometry system and the harmonic decomposition in which the scalar lane, grouped real P_2 lane, geometry lane, and higher modes become explicit.

Inputs. The stage imports the exact $(4+1)$-dimensional parent theory: gauged GNLS matter, localized Maxwell field, corrected charge ontology, and confinement potential. It also imports the lower-order requirement that the moving-throat PDE expose the already-visible $P_0 \oplus P_2$ support sector, the separate geometry-completion channel, and the mixed Maxwell channels retained outside the strict zero-mode brane limit.

Derivation

1. Shape-field lift. Let

$$\Sigma(\mathbf{X}, t) = r - R(\Omega, w, t), \qquad r = \sqrt{x^2 + y^2 + z^2}, \qquad \Omega = \mathbf{x}/r \in S^2. \tag{B.1}$$

The finite throat surface is the level set

$$\Sigma(\mathbf{X}, t) = 0. \tag{B.2}$$

The working sign convention is

$$\Sigma > 0 \quad \text{outside the throat,} \qquad \Sigma < 0 \quad \text{inside the support/interior region.} \tag{B.3}$$

This choice keeps the level-set coupling to the bulk fields while making the mouth-sphere harmonics explicit. It is the minimum lift needed to avoid treating the grouped real P_2 coefficients as purely symbolic residuals.

2. Reference stationary branch. Take a stationary reference surface

$$\Sigma_0(\mathbf{X}) = r - R_0(w) = 0, \tag{B.4}$$

with mouth at $w = 0$, finite depth $0 \le w \le L_0$, mouth radius

$$a_0 = R_0(0), \tag{B.5}$$

and a bottom closure such as $R_0(L_0) = 0$ or an equivalent regularity condition. The old collective variables are recovered as lowest moments:

$$a(t) = \frac{1}{4\pi} \int_{S^2} R(\Omega, 0, t)\, d\Omega, \tag{B.6}$$

and, if $W_b(\Omega, t)$ is defined implicitly by

$$R(\Omega, W_b(\Omega, t), t) = 0, \tag{B.7}$$

then

$$L(t) = \frac{1}{4\pi} \int_{S^2} W_b(\Omega, t)\, d\Omega. \tag{B.8}$$

Thus the old closure survives as a moment projection of the distributed wall field.

3. Promoted confinement coupling. The old confinement dependence $V_{\mathrm{conf}}(\mathbf{X}; a, L)$ is promoted to a moving-surface dependence

$$V_{\mathrm{conf}}(\mathbf{X}; \Sigma) = V_{\mathrm{wall}}\left(\frac{\Sigma(\mathbf{X}, t)}{\ell_c}\right), \tag{B.9}$$

where ℓ_c is the wall thickness. Around the reference branch,

$$V_0(\mathbf{X}) = V_{\mathrm{wall}}\left(\frac{\Sigma_0(\mathbf{X})}{\ell_c}\right). \tag{B.10}$$

The first variation is

$$\delta V_{\mathrm{conf}} = \frac{V'_{\mathrm{wall}}(\Sigma_0/\ell_c)}{\ell_c}\, \delta\Sigma. \tag{B.11}$$

Since $\Sigma = r - R$, a wall perturbation $R = R_0 + \eta$ gives

$$\delta\Sigma = -\eta(\Omega, w, t), \tag{B.12}$$

and therefore

$$\boxed{\delta V_{\mathrm{conf}} = -\frac{V'_{\mathrm{wall}}(\Sigma_0/\ell_c)}{\ell_c}\, \eta(\Omega, w, t)}. \tag{B.13}$$

This is the first direct wall-to-BdG coupling used later.

4. Harmonic decomposition. Write

$$R(\Omega, w, t) = R_0(w) + \eta(\Omega, w, t). \tag{B.14}$$

Expand η in real spherical harmonics:

$$\eta(\Omega, w, t) = \eta_0(w, t)Y_{00}(\Omega) + \sum_{m \in P_2(\mathrm{real})} q_{2m}(w, t)Y_{2m}^{\mathrm{real}}(\Omega) + \eta_{\geq 3}(\Omega, w, t). \tag{B.15}$$

The real P_2 set is

$$P_2(\mathrm{real}) = \{20, 21c, 21s, 22c, 22s\}. \tag{B.16}$$

With the normalized monopole

$$Y_{00} = \frac{1}{2\sqrt{\pi}}, \qquad \int_{S^2} Y_{00}^2 \, d\Omega = 1, \tag{B.17}$$

we have

$$\frac{1}{4\pi} \int_{S^2} Y_{00} \, d\Omega = \frac{1}{2\sqrt{\pi}}. \tag{B.18}$$

Therefore the normalized monopole coefficient and the physical mouth-average shift are related by

$$q_{00}(0, t) = 2\sqrt{\pi} \, \delta a(t). \tag{B.19}$$

A useful axisymmetric split is

$$\eta_0(w, t) = \alpha_a(w)\delta a(t) + \alpha_L(w)\delta L(t) + g(w, t), \tag{B.20}$$

where g is the residual axisymmetric geometry lane orthogonal to the (a, L) collective moments.

5. Minimal distributed wall action. The working passive quadratic wall action is

$$S_\eta^{(2)} = \frac{1}{2} \int dt \, dw \, d\Omega \, \sqrt{\gamma_0} \left[\mu_\eta(w)(\partial_t \eta)^2 - T_w(w)(\partial_w \eta)^2 - T_\Omega(w)\eta(-\Delta_{S^2})\eta - K_\eta(w)\eta^2 \right]. \tag{B.21}$$

The coefficients $\mu_\eta, T_w, T_\Omega, K_\eta$ are branch constitutive functions. At this stage the wall sector is an ansatz; the actual branch constitutive functions are not yet frozen. From this point onward the ledger adopts the densitized one-dimensional convention: the background surface weight $\sqrt{\gamma_0}$ is absorbed into the effective axial coefficients and mode profiles, so the reduced formulas are written with respect to dw. The weighted surface form is still legitimate, but then the Euler–Lagrange operator carries the extra first-derivative term coming from $\partial_w \sqrt{\gamma_0}$; the SymPy and Mathematica audits check both forms explicitly.

Projecting onto a harmonic Y_{lm}, using

$$-\Delta_{S^2} Y_{lm} = l(l+1)Y_{lm}, \tag{B.22}$$

gives the modal wall operator

$$\boxed{\mu_\eta \, \partial_t^2 q_{lm} - \partial_w(T_w \partial_w q_{lm}) + \left[K_\eta + l(l+1)T_\Omega\right] q_{lm} = S_{lm}^{(\psi,A)} + f_{lm}^{\mathrm{ext}}}. \tag{B.23}$$

The scalar lane $l = 0$ and the grouped real quadrupole lane $l = 2$ are therefore separated before any additional closure.

6. Linearized matter and gauge sectors. Choose stationary background fields

$$\psi = e^{-i\mu_0 t/\hbar}\psi_0(\mathbf{X}), \qquad A_M = A_{M0}(\mathbf{X}), \qquad R = R_0(w). \tag{B.24}$$

Let

$$\psi = e^{-i\mu_0 t/\hbar}(\psi_0 + \delta\psi), \qquad A_M = A_{M0} + \delta A_M. \tag{B.25}$$

The matter perturbation satisfies a BdG-type system

$$i\hbar\partial_t \begin{pmatrix} \delta\psi \\ \delta\psi^* \end{pmatrix} = \mathcal{L}_{\mathrm{BdG}} \begin{pmatrix} \delta\psi \\ \delta\psi^* \end{pmatrix} + \mathcal{C}_A[\delta A_M] + \mathcal{C}_\eta[\eta], \tag{B.26}$$

where $\mathcal{C}_\eta$ contains the coupling (B.13). The localized Maxwell equation linearizes to

$$\boxed{\partial_M(Z(w)\delta F^{MN}) + \frac{1}{\xi}\partial^N(\partial\cdot\delta A) = \mu_0\delta J^N}. \tag{B.27}$$

At this stage the mixed variables

$$\delta A_w, \qquad \delta J^w, \qquad \delta F_{\mu w} \tag{B.28}$$

are retained. Their later suppression in the far-field brane limit is a reduction, not an ontology deletion.

7. Mouth/worldtube response operator. At the mouth $w = 0$, choose the real port basis

$$P_{00} = Y_{00}, \qquad P_{20}, P_{21c}, P_{21s}, P_{22c}, P_{22s}. \tag{B.29}$$

Let $u_B(\omega)$ denote projected effort variables and $j_A(\omega)$ the corresponding generalized fluxes or tractions. The reduced response operator is written

$$j_A(\omega) = \sum_B Z_{\mathrm{eff},AB}(\omega)u_B(\omega). \tag{B.30}$$

On an isotropic reference throat, this operator block-diagonalizes into scalar/geometry, grouped real P_2, and higher-l sectors.

Output. The stage outputs the shape-field closure (B.1), the confinement variation (B.13), the harmonic split (B.15), the modal wall PDE (B.23), and the response-operator target (B.30). These are the definitions without which later grouped response coefficients have no PDE-side interpretation.

Checks.

- ☐ The moment check follows from the normalized monopole integral and gives $q_{00} = 2\sqrt{\pi}\,\delta a$.
- ☐ The sign check for δV_{conf} follows from $\delta\Sigma = -\eta$.
- ☐ The harmonic selection check follows from orthogonality on S^2, so the isotropic linear problem has no $l = 0 \leftrightarrow l = 2$ mixing.
- ☐ The ontology check keeps $A_w, J^w, F_{\mu w}$ alive before any far-field brane reduction.

Downstream use. Stage 002 uses the same wall action to recover (a, L). Stage 003 uses (B.13) as the wall–BdG coupling. Stages 004–021 use the retained mixed variables as the port-active gauge sector and reduce them to the one-port normal form. Stages 022–023 use the grouped real P_2 response operator to formulate the outgoing-normalization product.

B.2 Stage 002: Breathing Reduction Back to the Old (a, L) Closure

Part / anchor. Part I; primary anchors **MTDC-T1**, **MTDC-T2**, and **MTDC-T4**.

Claim status. Exact Within Closure **inside the Stage 001 quadratic wall closure. Damping and open-system completion are not derived here.**

Purpose. Stage 002 verifies that the distributed wall lift is a true lift of the old geometry sector rather than a replacement unrelated to it. The axisymmetric lowest-mode truncation reproduces a two-coordinate $(\delta a, \delta L)$ matrix system, while the grouped real P_2 modes appear as the next harmonic family of the same wall PDE.

Inputs. The inputs are the normalized monopole harmonic, the Stage 001 wall action, the axisymmetric split of η, and the grouped real P_2 harmonic eigenvalue $-\Delta_{S^2} Y_{2m} = 6Y_{2m}$.

Derivation

1. Axisymmetric two-mode ansatz. Using

$$Y_{00} = \frac{1}{2\sqrt{\pi}}, \qquad \int_{S^2} Y_{00}^2 \, d\Omega = 1, \qquad \frac{1}{4\pi} \int_{S^2} Y_{00} \, d\Omega = \frac{1}{2\sqrt{\pi}}, \tag{B.31}$$

the axisymmetric perturbation is truncated as

$$\eta_0(w,t) = 2\sqrt{\pi} \left[\alpha_a(w)\delta a(t) + \alpha_L(w)\delta L(t)\right]. \tag{B.32}$$

Set

$$Q^A = (\delta a, \delta L), \qquad A \in \{a, L\}. \tag{B.33}$$

2. Reduction of the quadratic wall action. On the axisymmetric branch, the angular-stiffness term vanishes because $l = 0$. In the densitized one-dimensional convention inherited from Stage 001, the background surface weight has already been absorbed into the effective axial coefficients and profiles, so the reduction is written with respect to dw rather than $\sqrt{\gamma_0}\, dw$. The action therefore reduces to a Lagrangian

$$L_{\text{red}}^{(0)} = \frac{1}{2} M_{AB} \dot{Q}^A \dot{Q}^B - \frac{1}{2} K_{AB} Q^A Q^B, \tag{B.34}$$

where

$$\boxed{M_{AB} = 4\pi \int dw \, \mu_\eta(w) \alpha_A(w) \alpha_B(w)}, \tag{B.35}$$

and

$$\boxed{K_{AB} = 4\pi \int dw \left[T_w(w) \alpha'_A(w) \alpha'_B(w) + K_0(w) \alpha_A(w) \alpha_B(w)\right]}. \tag{B.36}$$

Here $K_0(w) = K_\eta(w)$ on the scalar branch.

3. Euler–Lagrange equations. Varying (B.34) gives

$$\boxed{M_{AB}\ddot{Q}^B + K_{AB}Q^B = 0}. \tag{B.37}$$

This is the conservative linear version of the old two-coordinate geometry closure. The old schematic form

$$M_{AB}\ddot{Q}^B + \Gamma_{AB}\dot{Q}^B = -\frac{\partial H_{\text{tot}}}{\partial Q^A} \tag{B.38}$$

is recovered only after adding open-system or effective damping data. Stage 002 derives the conservative part, not the damping matrix.

4. Grouped real P_2 one-mode reduction. For one real quadrupole component take

$$\eta_{2m}(\Omega, w, t) = \beta_2(w)q_{2m}(t)Y_{2m}^{\text{real}}(\Omega), \qquad -\Delta_{S^2}Y_{2m}^{\text{real}} = 6Y_{2m}^{\text{real}}. \tag{B.39}$$

Inserting into the same wall action gives

$$L_{2m} = \frac{1}{2}M_2\dot{q}_{2m}^2 - \frac{1}{2}K_2q_{2m}^2, \tag{B.40}$$

with

$$\boxed{M_2 = \int dw\, \mu_\eta(w)\beta_2(w)^2}, \tag{B.41}$$

and

$$\boxed{K_2 = \int dw\, \Big[T_w(w)\beta_2'(w)^2 + \big(K_\eta(w) + 6T_\Omega(w)\big)\beta_2(w)^2\Big]}. \tag{B.42}$$

Thus, before matter/gauge coupling or explicit anisotropy,

$$\boxed{M_2\ddot{q}_{2m} + K_2q_{2m} = 0} \tag{B.43}$$

for every real P_2 component. This is the first degeneracy theorem for the grouped real quadrupole lane.

Output. The stage outputs the $(\delta a, \delta L)$ matrix system (B.37), the effective matrices (B.35)–(B.36), and the uncoupled grouped P_2 mass/stiffness pair (B.41)–(B.42).

Checks.

- ☐ The factor 4π in (B.35) and (B.36) follows from the $2\sqrt{\pi}$ normalization in (B.32).
- ☐ The $l = 2$ angular stiffness shift is $l(l+1)T_\Omega = 6T_\Omega$.
- ☐ The grouped P_2 degeneracy follows because K_2 and M_2 do not depend on the real component label on an isotropic reference throat.
- ☐ The reduction preserves the old (a, L) closure but does not derive the old damping matrix.

Downstream use. Stage 003 couples the Stage 002 wall modes to stable BdG support modes. Stage 023 uses the same wall baseline as the K_A, M_A entries of the full grouped bundle. Later geometry-contamination stages use the distinction between collective scalar geometry and grouped P_2 wall modes.

B.3 Stage 003: Minimal BdG–Wall Coupling and First Pole Shifts

Part / anchor. Part I; primary anchors **MTDC-T2** and **MTDC-T4**.

Claim status. The Schur complement and pole formulas are Exact Within Closure **inside the declared stable-mode reduction. Passing from the full BdG PDE to stable normal coordinates is** Reduced / Controlled Reduction**.**

Purpose. Stage 003 turns on the first conservative matter-support dynamics. It reduces the linearized BdG sector to stable normal coordinates, couples them to the wall through the confinement variation from Stage 001, integrates them out exactly, and obtains the first microscopic origin of wall pole shifts and grouped real P_2 conservative response coefficients.

Inputs. The inputs are the Stage 001 coupling $\delta V_{\rm conf} = -(V'_{\rm wall}/\ell_c)\eta$, the Stage 002 wall coordinates, the assumption of stable BdG normal modes with positive frequencies, and harmonic selection on an isotropic reference throat.

Derivation

1. Stable scalar-sector reduction. Let

$$Q^A = (\delta a, \delta L), \qquad A \in \{a, L\}, \tag{B.44}$$

and let X_α be stable scalar BdG normal coordinates with frequencies $\varpi_{0\alpha} > 0$. The reduced scalar Lagrangian is

$$L^{(0)}_{\rm red} = \frac{1}{2}\dot{Q}^T M_0 \dot{Q} - \frac{1}{2} Q^T K_0 Q + \frac{1}{2}\sum_\alpha (\dot{X}_\alpha^2 - \varpi_{0\alpha}^2 X_\alpha^2) + \sum_{A,\alpha} C_{A\alpha} Q^A X_\alpha. \tag{B.45}$$

The matrix C consists of overlap integrals generated by (B.13) and the BdG mode profiles.

In frequency space,

$$(\varpi_{0\alpha}^2 - \omega^2) X_\alpha = C_{A\alpha} Q^A, \qquad X_\alpha = (\varpi_{0\alpha}^2 - \omega^2)^{-1} C_{A\alpha} Q^A. \tag{B.46}$$

Substitution gives the exact scalar effective kernel

$$\boxed{D_0^{\rm eff}(\omega) = K_0 - \omega^2 M_0 - C(\Omega_0^2 - \omega^2 I)^{-1} C^T}, \tag{B.47}$$

where

$$\Omega_0^2 = \mathrm{diag}(\varpi_{0\alpha}^2). \tag{B.48}$$

The low-frequency expansion is

$$D_0^{\rm eff}(\omega) = K_0^{\rm eff} - \omega^2 M_0^{\rm eff} - \omega^4 N_0^{\rm eff} + O(\omega^6), \tag{B.49}$$

with

$$\boxed{K_0^{\rm eff} = K_0 - C\Omega_0^{-2} C^T, \qquad M_0^{\rm eff} = M_0 + C\Omega_0^{-4} C^T, \qquad N_0^{\rm eff} = C\Omega_0^{-6} C^T}. \tag{B.50}$$

Stable matter support therefore lowers static stiffness, increases effective inertia, and generates the next even moment.

2. Grouped real P_2 reduction. For grouped lane $A \in \{20, 21, 22\}$, let q_A be the wall amplitude and $X_{A\alpha}$ the stable BdG support coordinates. The reduced Lagrangian is

$$L^{(2)}_{\mathrm{red}} = \sum_A \left[\frac{1}{2} M_A \dot{q}_A^2 - \frac{1}{2} K_A q_A^2 + \frac{1}{2} \sum_\alpha (\dot{X}_{A\alpha}^2 - \varpi_{A\alpha}^2 X_{A\alpha}^2) + \sum_\alpha g_{A\alpha} q_A X_{A\alpha} \right]. \tag{B.51}$$

Eliminating the support coordinates gives

$$\boxed{D_A^{\mathrm{eff}}(\omega) = K_A - M_A \omega^2 - \sum_\alpha \frac{g_{A\alpha}^2}{\varpi_{A\alpha}^2 - \omega^2}, \qquad A \in \{20, 21, 22\}}. \tag{B.52}$$

At low frequency,

$$D_A^{\mathrm{eff}}(\omega) = d_{0A}^{\mathrm{eff}} + d_{2A}^{\mathrm{eff}} \omega^2 + d_{4A}^{\mathrm{eff}} \omega^4 + O(\omega^6), \tag{B.53}$$

where

$$d_{0A}^{\mathrm{eff}} = K_A - \sum_\alpha \frac{g_{A\alpha}^2}{\varpi_{A\alpha}^2}, \tag{B.54}$$

$$d_{2A}^{\mathrm{eff}} = -\left(M_A + \sum_\alpha \frac{g_{A\alpha}^2}{\varpi_{A\alpha}^4} \right), \tag{B.55}$$

$$d_{4A}^{\mathrm{eff}} = -\sum_\alpha \frac{g_{A\alpha}^2}{\varpi_{A\alpha}^6}. \tag{B.56}$$

The executable audit closes finite-mode witness cases explicitly; the displayed $\sum_\alpha$ formulas then follow by linear superposition because each stable BdG mode contributes additively to the Schur complement.

3. One-wall-mode pole shift. For the two-coordinate prototype

$$L = \frac{1}{2} M \dot{q}^2 - \frac{1}{2} K q^2 + \frac{1}{2} \dot{X}^2 - \frac{1}{2} \varpi^2 X^2 + g q X, \tag{B.57}$$

the dispersion relation is

$$(K - M\omega^2)(\varpi^2 - \omega^2) - g^2 = 0. \tag{B.58}$$

Writing $\Omega_\eta^2 = K/M$, the exact poles are

$$\boxed{\omega_\pm^2 = \frac{\Omega_\eta^2 + \varpi^2 \pm \sqrt{(\Omega_\eta^2 - \varpi^2)^2 + 4g^2/M}}{2}}. \tag{B.59}$$

For $\varpi^2 > \Omega_\eta^2$ and weak coupling,

$$\boxed{\delta\Omega_\eta^2 = -\frac{g^2/M}{\varpi^2 - \Omega_\eta^2} + O(g^4), \qquad \delta\varpi^2 = +\frac{g^2/M}{\varpi^2 - \Omega_\eta^2} + O(g^4)}. \tag{B.60}$$

This is the first explicit reduced formula for how matter support shifts a wall pole.

4. Grouped isotropy diagnostic. For any grouped triple $x_A = (x_{20}, x_{21}, x_{22})$, define

$$\bar{x} = \frac{x_{20} + 2x_{21} + 2x_{22}}{5}, \qquad a_x = \frac{2x_{20} - x_{21} - x_{22}}{10}, \qquad b_x = \frac{x_{21} - x_{22}}{2}. \tag{B.61}$$

Applying this to d_{2A}^{eff} gives grouped conservative anisotropy defects. If

$$K_{20} = K_{21} = K_{22}, \quad M_{20} = M_{21} = M_{22}, \quad g_{20,\alpha} = g_{21,\alpha} = g_{22,\alpha}, \quad \varpi_{20,\alpha} = \varpi_{21,\alpha} = \varpi_{22,\alpha}, \tag{B.62}$$

then all grouped d_{nA}^{eff} are lane-independent and

$$\boxed{a_x = b_x = 0}. \tag{B.63}$$

Conversely, a nonzero defect must come from anisotropy in wall data, matter support frequencies, or wall–matter overlaps.

Output. Stage 003 outputs the scalar Schur complement (B.47), the grouped kernels (B.52), the low-frequency coefficients (B.54), the exact pole shift (B.60), and the first microscopic grouped-isotropy criterion.

Checks.

- ☐ Expanding $(\Omega^2 - \omega^2)^{-1} = \Omega^{-2} + \omega^2\Omega^{-4} + \omega^4\Omega^{-6} + \cdots$ gives (B.50) and (B.54).
- ☐ The sign of the static correction is softening: the Schur complement subtracts g^2/ϖ^2.
- ☐ The pole shift has the expected avoided-crossing sign: the lower wall-like pole moves down when coupled to a higher stable matter mode.
- ☐ The isotropy check follows by lane equality of all microscopic inputs.

Downstream use. Stages 004–021 add the projected Maxwell/mixed self-energies and reduce them to the matched one-port normal form on the same operator structure. Stage 022 converts the D_A operator moments into normalized grouped response moments. Stage 023 collects the Stage 003 BdG moments as B_{A0}, B_{A2}, B_{A4}.

B.4 Stage 004: Projected Maxwell bundle index

Part / anchor. Part I; primary anchor **MTDC-T4**.

Claim status. This stage is Exact **for the displayed algebraic identities inside the declared parent/projection data, with** Reduced / Controlled Reduction **or** Exact Within Closure **inherited where the local text invokes the matched reduction, mouth-kernel closure, parent-action packet, or one-port normal form.**

Source and audit. This stage corresponds to `scripts/moving_throat_pde_stage004_projected_maxwell_bundle_index_sympy_audit.py`. It imports the bundle-level source index from the projected-Maxwell notes and checks that the covariant, vector, and projection/reduction comparison audits are present in the ledger script order.

Bundle role. The electromagnetic sector is treated as a projection problem before it is treated as a reduced one-port normal form. The source bundle fixes three rules:

1. projection by parts creates a real transverse leakage/source term;
2. the homogeneous Maxwell equations project in measured fields, while the inhomogeneous equations carry source-coupled flux fields;
3. reduction is a matched channel, not the primary parent derivation.

Projection identity. For any smooth profile $Q(w)$ and observation kernel $W(w)$,

$$\partial_w(WQ) - W\partial_w Q - (\partial_w W)Q = 0. \tag{B.64}$$

After integration over w, (B.64) is the bookkeeping identity that turns the parent w-flux into a boundary term plus a kernel-gradient leakage term.

Output. Stage 004 fixes the source ordering for the projection-first EM bundle and anchors the projection-by-parts identity used in the next stage.

B.5 Stage 005: Covariant projected Maxwell law

Part / anchor. Part I; primary anchor **MTDC-T4**.

Claim status. This stage is Exact for the displayed algebraic identities inside the declared parent/projection data, with Reduced / Controlled Reduction or Exact Within Closure inherited where the local text invokes the matched reduction, mouth-kernel closure, parent-action packet, or one-port normal form.

Source and audit. This stage corresponds to `scripts/moving_throat_pde_stage005_projected_maxwell_covariant_sympy_audit.py`. It is the covariant projection of the localized parent Maxwell equation.

Parent law. The localized parent equation is

$$\partial_N\Big(Z(w)F^{NM}\Big) = \mu_0 J^M. \tag{B.65}$$

For brane-facing components $M = \mu$, apply a normalized observation kernel $W(w)$ and integrate in w. The result is

$$\partial_\nu \int WZF^{\nu\mu}\,dw = \mu_0 \int WJ^\mu\,dw - \int W\partial_w(ZF^{w\mu})\,dw. \tag{B.66}$$

Using (B.64),

$$\partial_\nu \int WZF^{\nu\mu}\,dw = \mu_0 \int WJ^\mu\,dw - [WZF^{w\mu}]_\partial + \int(\partial_w W)ZF^{w\mu}\,dw. \tag{B.67}$$

Projected continuity. The same projection applied to parent current conservation gives

$$\partial_\mu \int WJ^\mu\,dw = -[WJ^w]_\partial + \int(\partial_w W)J^w\,dw. \tag{B.68}$$

Thus leakage is not a notation choice. It is the exact open-system balance term induced by observing a localized parent current through W.

Output. Stage 005 exports the projected inhomogeneous law (B.66)–(B.67) and projected continuity (B.68).

B.6 Stage 006: Vector projected Maxwell split

Part / anchor. Part I; primary anchor **MTDC-T4**.

Claim status. This stage is EXACT **for the displayed algebraic identities inside the declared parent/projection data, with** REDUCED / CONTROLLED REDUCTION **or** EXACT WITHIN CLOSURE **inherited where the local text invokes the matched reduction, mouth-kernel closure, parent-action packet, or one-port normal form.**

Source and audit. This stage corresponds to `scripts/moving_throat_pde_stage006_projected_maxwell_vector_sympy_audit.py`. It checks the $3+1$ vector signs, the measured-field/source-flux split, and the boundary discharge terms.

Measured and flux fields. The homogeneous equations use the measured projected fields

$$\mathbf{E}_{\text{meas}} = \int W\mathbf{E}\,dw, \qquad \mathbf{B}_{\text{meas}} = \int W\mathbf{B}\,dw, \tag{B.69}$$

whereas the inhomogeneous equations use the source-coupled flux fields

$$\mathbf{D}_{\text{flux}} = \int WZ\mathbf{E}\,dw, \qquad \mathbf{H}_{\text{flux}} = \int WZ\mathbf{B}\,dw. \tag{B.70}$$

Identifying these two pairs is an additional closure assumption. It is not part of the parent projection.

Projected vector laws. With the convention $F^{i0} = E_i$, the projected Faraday and Ampere signs are fixed by the audit:

$$\partial_t \mathbf{B}_{\text{meas}} + \nabla \times \mathbf{E}_{\text{meas}} = 0, \tag{B.71}$$

$$-\nabla \times \mathbf{H}_{\text{flux}} - \partial_t \mathbf{D}_{\text{flux}} + \mathbf{L}_{\text{mix}} = \mu_0 \mathbf{J}_{\text{proj}}. \tag{B.72}$$

Here $\mathbf{L}_{\text{mix}}$ is the vector form of the transverse leakage term in (B.67).

Output. Stage 006 exports the field split (B.69)–(B.70) and the vector-law sign convention used by the rest of the projected Maxwell block.

B.7 Stage 007: Projection/reduction comparison

Part / anchor. Part I; primary anchor **MTDC-T4**.

Claim status. This stage is EXACT **for the displayed algebraic identities inside the declared parent/projection data, with** REDUCED / CONTROLLED REDUCTION **or** EXACT WITHIN CLOSURE **inherited where the local text invokes the matched reduction, mouth-kernel closure, parent-action packet, or one-port normal form.**

Source and audit. This stage corresponds to `scripts/moving_throat_pde_stage007_projection_reduction_comparison_sympy_audit.py`. It compares the exact projected law with the zero-mode reduction channel and keeps observer dependence explicit.

Reduction channel. For a zero-mode ansatz $A_\mu(x, w) = a_\mu(x)$, the projected equation contains the weighted factors

$$I_{WZ} = \int WZ\,dw, \qquad I_{WH} = \int WH\,dw, \qquad I_{WS} = \int WS\,dw. \tag{B.73}$$

The effective projected parameters are

$$\mu_{\text{eff}}^{\text{proj}} = \mu_0 \frac{I_{WS}}{I_{WZ}}, \qquad \frac{1}{\xi_{\text{eff}}^{\text{proj}}} = \frac{I_{WH}}{\xi I_{WZ}}. \tag{B.74}$$

These are observer-kernel quantities until a matching prescription is imposed.

Mismatch guard. The comparison audit checks concrete smooth profiles and mutation guards. A delta-like source and a smooth localized gauge profile do not automatically produce the same coupling as a reduced brane theory; source matching is a separate condition.

Output. Stage 007 exports the observer-dependent projected coefficients (B.74) and marks the reduction-first brane law as a matched channel rather than the controlling derivation.

B.8 Stage 008: Projected Maxwell extension and clean matching

Part / anchor. Part I; primary anchor **MTDC-T4**.

Claim status. This stage is Exact **for the displayed algebraic identities inside the declared parent/projection data, with** Reduced / Controlled Reduction **or** Exact Within Closure **inherited where the local text invokes the matched reduction, mouth-kernel closure, parent-action packet, or one-port normal form.**

Source and audit. This stage corresponds to `scripts/moving_throat_pde_stage008_projected_maxwell_extension_sympy_audit.py`. It records the clean gauge/source choices under which the projected channel reproduces the reduction-first coupling.

Clean gauge channel. The gauge-fixing weight $H(w)$ controls the effective gauge parameter in (B.74). The clean localized gauge choice is

$$H(w) = Z(w), \qquad \xi_{\text{eff}}^{\text{proj}} = \xi. \tag{B.75}$$

Source match. The source profile $S(w)$ must be matched to the localized Maxwell weight:

$$S(w) = \frac{Z(w)}{Z_{\text{int}}}, \qquad Z_{\text{int}} = \int Z(w)\,dw. \tag{B.76}$$

Then

$$\mu_{\text{eff}}^{\text{proj}} = \frac{\mu_0}{Z_{\text{int}}}, \tag{B.77}$$

which is the reduction-first coupling. The audit also checks the Gaussian case and the regulator limit, including mutation guards distinguishing $H = 1$ from $H = Z$.

Output. Stage 008 exports the matching conditions (B.75)–(B.76). These conditions are the firewall between exact projection and reduced brane coupling.

B.9 Stage 009: Near-throat projection scale

Part / anchor. Part I; primary anchor **MTDC-T4**.

Claim status. This stage is Exact **for the displayed algebraic identities inside the declared parent/projection data, with** Reduced / Controlled Reduction **or** Exact Within Closure **inherited where the local text invokes the matched reduction, mouth-kernel closure, parent-action packet, or one-port normal form.**

Source and audit. This stage corresponds to `scripts/moving_throat_pde_stage009_projected_maxwell_near_throat_sympy_audit.py`. It distinguishes symmetric interior averaging from one-sided mouth averaging.

Interior versus mouth. For a symmetric interior kernel, the first finite-width correction to a smooth profile is even and begins at $O(\sigma^2)$. At a mouth with one-sided coordinate $s \geq 0$,

$$W_\ell(s) = \frac{1}{\ell} w(s/\ell), \qquad \int_0^\infty w(u)\,du = 1, \tag{B.78}$$

the expansion is

$$\langle X \rangle_\ell = X(0) + \ell \mu_1 X'(0) + O(\ell^2), \qquad \mu_1 = \int_0^\infty u\,w(u)\,du. \tag{B.79}$$

Output. Stage 009 explains why mouth-local projected EM data enter at first order in the mouth scale, while symmetric interior thickness effects enter later.

B.10 Stage 010: Projected Maxwell push into bundle slots

Part / anchor. Part I; primary anchor **MTDC-T4**.

Claim status. This stage is Exact **for the displayed algebraic identities inside the declared parent/projection data, with** Reduced / Controlled Reduction **or** Exact Within Closure **inherited where the local text invokes the matched reduction, mouth-kernel closure, parent-action packet, or one-port normal form.**

Source and audit. This stage corresponds to `scripts/moving_throat_pde_stage010_projected_maxwell_push_bundle_master_sympy_audit.py`. It is the master transport from projected EM corrections into the grouped bundle slots.

Bundle shifts. The conservative and outgoing-transfer moments shift as

$$Z_n \mapsto Z_n + \varepsilon z_n, \qquad N_n \mapsto N_n + \varepsilon n_n, \qquad n \in \{0, 2, 4\}. \tag{B.80}$$

For

$$D_0 = K - B_0 - Z_0, \qquad D_2 = -(M + B_2 + Z_2), \qquad D_4 = -(B_4 + Z_4),$$

the first response variations are

$$\delta u_2 = \frac{D_0 z_2 - D_2 z_0}{D_0^2}, \tag{B.81}$$

$$\delta u_4 = \frac{D_0^2 z_4 - D_0(2D_2 z_2 + D_4 z_0) + 2D_2^2 z_0}{D_0^3}, \tag{B.82}$$

and

$$\delta P_0 = \frac{D_0 n_0 + N_0 z_0}{D_0^2}. \tag{B.83}$$

The denominator-only quantities u_2, u_4 and the full prefactor quantities P_n are distinct. The full first variations in the P_2 and P_4 slots are

$$\delta P_2 = \frac{n_2}{D_0} + \frac{N_2 z_0 + 2N_0 z_2 - 2D_2 n_0}{D_0^2} - \frac{4D_2 N_0 z_0}{D_0^3}, \tag{B.84}$$

$$\begin{aligned} \delta P_4 = \frac{n_4}{D_0} &+ \frac{N_4 z_0 + 2N_2 z_2 - 2D_2 n_2 + 2N_0 z_4 - 2D_4 n_0}{D_0^2} \\ &+ \frac{3D_2^2 n_0 - 4(D_2 N_2 + D_4 N_0) z_0 - 6D_2 N_0 z_2}{D_0^3} + \frac{9D_2^2 N_0 z_0}{D_0^4}. \end{aligned} \tag{B.85}$$

Compatibility transport. With $S = M + B_2 + Z_2$ and $T = B_4 + Z_4$, the one-pole and fixed-target normalization K-surfaces under the slot shift are

$$\begin{aligned} K_{\text{pole}}(\varepsilon) &= B_0 + Z_0 + \varepsilon z_0 + \frac{3(S + \varepsilon z_2)^2}{T + \varepsilon z_4}, \\ K_{\text{norm}}(\varepsilon) &= B_0 + Z_0 + \varepsilon z_0 + \frac{N_0 + \varepsilon n_0}{P_{0,\text{target}}}. \end{aligned} \tag{B.86}$$

Eliminating K gives the fixed-target compatibility variation

$$\delta\mathcal{C}_{\text{fixed}} = \frac{n_0}{P_{0,\text{target}}} - \frac{6Sz_2}{T} + \frac{3S^2 z_4}{T^2}, \qquad \partial_{z_0}\delta\mathcal{C}_{\text{fixed}} = 0. \tag{B.87}$$

For a transported target $P_{0,\text{target}}(\varepsilon) = (N_0 + \varepsilon n_0)/D_{0,\text{target}}$,

$$\mathcal{C}_{\text{tr}}(\varepsilon) = D_{0,\text{target}} - \frac{3(S + \varepsilon z_2)^2}{T + \varepsilon z_4}, \qquad \delta\mathcal{C}_{\text{tr}} = -\frac{6Sz_2}{T} + \frac{3S^2 z_4}{T^2}. \tag{B.88}$$

Grouped weak-axisymmetric lane. The real Y_{20} square-overlap lane multipliers are

$$\lambda_{20} = 1, \qquad \lambda_{21} = \frac{1}{2}, \qquad \lambda_{22} = -1. \tag{B.89}$$

Thus a weak-axisymmetric lane packet $x_A = x^{(0)} + \varepsilon\lambda_A x^{(1)}$ obeys

$$\bar{x} = x^{(0)}, \qquad a_x = \frac{\varepsilon x^{(1)}}{4}, \qquad b_x = \frac{3\varepsilon x^{(1)}}{4}, \qquad b_x = 3a_x. \tag{B.90}$$

Primitive static prefactor. For primitive mouth data Q, Δ, P with perturbation slopes q_1, d_1, p_1,

$$z_0^{\mathrm{prim}} = \frac{\Delta q_1 - Q d_1}{\Delta^2}, \qquad n_0^{\mathrm{prim}} = \frac{2P(\Delta p_1 - P d_1)}{\Delta^3}, \qquad \Xi_{\mathrm{static}} = \frac{2p_1}{P} - \frac{2d_1}{\Delta} + \frac{\Delta q_1 - Q d_1}{D_0 \Delta^2}. \tag{B.91}$$

The sign-flip mutation guards are the nonzero residuals

$$\mathcal{R}_{\mathrm{fixed}} = \mathcal{R}_{\mathrm{tr}} = \frac{6S^2 z_4}{T^2} \not\equiv 0, \tag{B.92}$$

obtained by flipping the sign of the $3S^2 z_4/T^2$ term in (B.87) or (B.88).

Output. Stage 010 exports the transport map (B.80)–(B.85), the compatibility transports (B.86)–(B.88), the weak-axisymmetric lane signature (B.89)–(B.90), the primitive static prefactor anchor (B.91), and the sign-flip guards (B.92).

B.11 Stage 011: Projected Maxwell P_2 bridge

Part / anchor. Part I; primary anchor **MTDC-T4**.

Claim status. This stage is Exact **for the displayed algebraic identities inside the declared parent/projection data, with** Reduced / Controlled Reduction **or** Exact Within Closure **inherited where the local text invokes the matched reduction, mouth-kernel closure, parent-action packet, or one-port normal form.**

Source and audit. This stage corresponds to `scripts/moving_throat_pde_stage011_projected_maxwell_p2_bridge_sympy_audit.py`. It applies the bundle shifts to the one-pole/normalization compatibility surface.

Compatibility surface. Let

$$S = M + B_2 + Z_2, \qquad T = B_4 + Z_4. \tag{B.93}$$

After eliminating the static wall stiffness, the isotropic compatibility surface is

$$\mathcal{C} = \frac{N_0}{P_{0,\mathrm{target}}} - \frac{3S^2}{T}. \tag{B.94}$$

The projected variation is

$$\delta\mathcal{C} = \frac{n_0}{P_{0,\mathrm{target}}} - \frac{6S z_2}{T} + \frac{3S^2 z_4}{T^2}. \tag{B.95}$$

The static conservative shift z_0 cancels from (B.95); it changes P_0, but not the eliminated compatibility surface.

Output. Stage 011 exports the z_0-cancellation and the projected P_2 compatibility bridge (B.95).

B.12 Stage 012: Projected Maxwell primitive bridge

Part / anchor. Part I; primary anchor **MTDC-T4**.

Claim status. This stage is Exact **for the displayed algebraic identities inside the declared parent/projection data, with** Reduced / Controlled Reduction **or** Exact Within Closure **inherited where the local text invokes the matched reduction, mouth-kernel closure, parent-action packet, or one-port normal form.**

Source and audit. This stage corresponds to `scripts/moving_throat_pde_stage012_projected_maxwell_primitive_bridge_sympy_audit.py`. It rewrites the projected bundle shifts in the primitive variables used by the Stage 021 reduced normal form.

Primitive packet. The primitive reduced packet is built from the denominator

$$\Delta = AW - R^2,$$

the conservative numerator Q, the mixed transfer numerator P, and the ω^2 denominator slope S_2. At the mouth, projected perturbations of these primitives determine the z_n and n_n slots used in Stage 010.

Transport role. The audit checks both fixed-target and transported-target compatibility shifts. This is the step that connects the projection-first mouth data to the same primitive variables that later appear in the one-port Schur complement.

Output. Stage 012 exports the primitive-to-bundle bridge used by the mouth-Taylor stages.

B.13 Stage 013: Mouth-Taylor master map

Part / anchor. Part I; primary anchor **MTDC-T4**.

Claim status. This stage is Exact **for the displayed algebraic identities inside the declared parent/projection data, with** Reduced / Controlled Reduction **or** Exact Within Closure **inherited where the local text invokes the matched reduction, mouth-kernel closure, parent-action packet, or one-port normal form.**

Source and audit. This stage corresponds to `scripts/moving_throat_pde_stage013_projected_maxwell_mouth_taylor_master_sympy_audit.py`. It is the master one-sided Taylor map from primitive mouth data to live weak-axisymmetric gates.

Primitive slippages. With $Q, S_2, H_{\rm port}, \Delta, P$ evaluated along the mouth coordinate s,

$$z_0 = \partial_s\left(\frac{Q}{\Delta}\right), \qquad n_0 = \partial_s\left(\frac{P^2}{\Delta^2}\right). \tag{B.96}$$

The next conservative slots are

$$z_2 = \partial_s\left(\frac{QS_2 - H_{\rm port}\Delta}{\Delta^2}\right), \tag{B.97}$$

$$z_4 = \partial_s\left(\frac{Q(S_2^2 - \Delta) - H_{\rm port}S_2\Delta}{\Delta^3}\right). \tag{B.98}$$

Gate feed. The first-order prefactor slope receives

$$\Xi_{\mathrm{load}} = \frac{n_0}{N_0} + \frac{z_0}{D_0}, \tag{B.99}$$

while z_0, z_2, z_4 feed the even gates K_1 and H_{even}. The audit checks the one-sided Taylor derivatives and their dependence on the primitive bottlenecks.

Output. Stage 013 exports the mouth-Taylor primitive map (B.96)–(B.98).

B.14 Stage 014: Mouth-Taylor gate bridge

Part / anchor. Part I; primary anchor **MTDC-T4**.

Claim status. This stage is Exact **for the displayed algebraic identities inside the declared parent/projection data, with** Reduced / Controlled Reduction **or** Exact Within Closure **inherited where the local text invokes the matched reduction, mouth-kernel closure, parent-action packet, or one-port normal form.**

Source and audit. This stage corresponds to `scripts/moving_throat_pde_stage014_projected_maxwell_mouth_taylor_gate_bridge_sympy_audit.py`. It turns the mouth-Taylor primitive map into a gate-level compensation test.

Even gates. The conservative mouth shifts enter

$$K_1 = -z_2 - \frac{z_0}{9}, \tag{B.100}$$

$$H_{\mathrm{even}} = -z_4 + \frac{2}{3}z_2 - \frac{z_0}{27}. \tag{B.101}$$

Solving the pair of even gates against the primitive derivative slots gives a nontrivial compensation denominator. In the primitive notation of the audit, the determinant is proportional to

$$Q\,S_2 - \Delta H_{\mathrm{port}}. \tag{B.102}$$

Output. Stage 014 exports the mechanism sieve: projected EM mouth data can tune the even gates only away from the degeneracy (B.102).

B.15 Stage 015: Parent throat action master

Part / anchor. Part I; primary anchor **MTDC-T4**.

Claim status. This stage is Exact **for the displayed algebraic identities inside the declared parent/projection data, with** Reduced / Controlled Reduction **or** Exact Within Closure **inherited where the local text invokes the matched reduction, mouth-kernel closure, parent-action packet, or one-port normal form.**

Source and audit. This stage corresponds to `scripts/moving_throat_pde_stage015_parent_throat_action_master_sympy_audit.py`. It begins the parent-action continuation of the projected EM bundle.

Promotion. The throat is promoted to an autonomous graph field $R(\Omega, w, t)$ with

$$S_\Sigma[R] = \int dt\, dw\, d\Omega\, \mathcal{L}_\Sigma, \tag{B.103}$$

$$\mathcal{L}_\Sigma = \frac{1}{2}\mu_\Sigma R_t^2 - \frac{1}{2}T_{w,\Sigma}R_w^2 - \frac{1}{2}T_{\Omega,\Sigma}|\nabla_\Omega R|^2 - U_\Sigma. \tag{B.104}$$

The total parent-level model is

$$S_{\text{total}} = S_\psi[\psi, A; \Sigma] + S_{\text{EM}}[A] + S_\Sigma[R]. \tag{B.105}$$

Quadratic limit. The quadratic stiffness around $R_0(w)$ is

$$K_\eta = U_{\Sigma,RR}(R_0, w) - \partial_w(T_{w,\Sigma,R}(R_0, w)R_0') + \frac{1}{2}T_{w,\Sigma,RR}(R_0, w)(R_0')^2. \tag{B.106}$$

The audit carries the boundary densities explicitly and tests both zero and nonzero boundary-discharge probes.

Output. Stage 015 exports the parent throat action promotion and the exact quadratic recovery formula (B.106).

B.16 Stage 016: Parent throat action candidate

Part / anchor. Part I; primary anchor **MTDC-T4**.

Claim status. This stage is Exact **for the displayed algebraic identities inside the declared parent/projection data, with** Reduced / Controlled Reduction **or** Exact Within Closure **inherited where the local text invokes the matched reduction, mouth-kernel closure, parent-action packet, or one-port normal form.**

Source and audit. This stage corresponds to `scripts/moving_throat_pde_stage016_parent_throat_action_candidate_sympy_audit.py`. It verifies the minimal nonlinear candidate and its exact quadratic recovery.

Candidate status. The density (B.104) is deliberately the minimal gauge-fixed throat action in the current coordinates. It is not a uniqueness claim. A later worldvolume-invariant action could be stronger, but the present candidate is sufficient to recover the audited linear wall PDE.

Boundary controls. The audit verifies integration by parts with Gaussian and Lorentzian probes and checks a nonzero $\arctan w$ endpoint discharge. Therefore the quadratic recovery is not relying on a broken boundary reader that always returns zero.

Output. Stage 016 records the candidate action as a parent-complete source of the linear wall closure, subject to the explicitly checked boundary class.

B.17 Stage 017: Parent throat action weak-axisymmetric law

Part / anchor. Part I; primary anchor **MTDC-T4**.

Claim status. This stage is Exact **for the displayed algebraic identities inside the declared parent/projection data, with** Reduced / Controlled Reduction **or** Exact Within Closure **inherited where the local text invokes the matched reduction, mouth-kernel closure, parent-action packet, or one-port normal form.**

Source and audit. This stage corresponds to `scripts/moving_throat_pde_stage017_parent_throat_action_weak_axisym_sympy_audit.py`. It transports the parent throat action into the weak-axisymmetric grouped P_2 lane structure.

Lane signature. A Y_{20} wall perturbation gives

$$(\lambda_{20}, \lambda_{21}, \lambda_{22}) = \left(1, \frac{1}{2}, -1\right). \tag{B.107}$$

Consequently the grouped trace/anomaly variables obey

$$b = 3a. \tag{B.108}$$

The audit derives (B.107) from the actual triple-overlap coefficients.

Wall-only obstruction. Pure wall anisotropy closes the even gates only on the trivial branch $\delta K = \delta M = 0$. Thus parent promotion supplies the wall-side origin of the pattern, but does not by itself realize the full weak-axisymmetric branch.

Output. Stage 017 exports the parent weak-axisymmetric lane law (B.107)–(B.108) and the wall-only gate obstruction.

B.18 Stage 018: Parent throat action bundle master

Part / anchor. Part I; primary anchor **MTDC-T4**.

Claim status. This stage is Exact **for the displayed algebraic identities inside the declared parent/projection data, with** Reduced / Controlled Reduction **or** Exact Within Closure **inherited where the local text invokes the matched reduction, mouth-kernel closure, parent-action packet, or one-port normal form.**

Source and audit. This stage corresponds to `scripts/moving_throat_pde_stage018_parent_throat_action_bundle_master_sympy_audit.py`. It collects the parent wall inputs into the full grouped-bundle language.

Master bridge. The parent wall block enters through the branch integrals

$$M_\Sigma = \int dw\, \mu_\eta \beta_2^2, \tag{B.109}$$

$$K_\Sigma = \int dw \left[T_w(\beta_2')^2 + (K_\eta + 6T_\Omega)\beta_2^2\right]. \tag{B.110}$$

These replace abstract wall knobs in the grouped bundle. The support and mixed packets still have to be computed on the same frozen branch.

Output. Stage 018 exports the parent-wall bundle bridge (B.109)–(B.110).

B.19 Stage 019: Parent throat action isotropic bundle

Part / anchor. Part I; primary anchor **MTDC-T4**.

Claim status. This stage is Exact **for the displayed algebraic identities inside the declared parent/projection data, with** Reduced / Controlled Reduction **or** Exact Within Closure **inherited where the local text invokes the matched reduction, mouth-kernel closure, parent-action packet, or one-port normal form.**

Source and audit. This stage corresponds to `scripts/moving_throat_pde_stage019_parent_throat_action_isotropic_bundle_sympy_audit.py`. It rewrites the isotropic one-pole and outgoing-normalization targets in parent-wall variables.

Parent-complete denominator. The isotropic bundle moments are

$$D_0 = K_\Sigma - B_0 - Z_0, \qquad D_2 = -(M_\Sigma + B_2 + Z_2), \qquad D_4 = -(B_4 + Z_4). \tag{B.111}$$

The one-pole surface fixes

$$K_\Sigma = B_0 + Z_0 + \frac{3(M_\Sigma + B_2 + Z_2)^2}{B_4 + Z_4}. \tag{B.112}$$

The outgoing normalization fixes

$$K_\Sigma = B_0 + Z_0 + \frac{N_0}{P_{0,\mathrm{target}}}, \qquad P_{0,\mathrm{target}} = \frac{54Gc_s^5}{5a^5c^5\widehat{m}_0^{\,2}}. \tag{B.113}$$

Compatibility. The parent-complete compatibility identity is

$$\frac{N_0}{P_{0,\mathrm{target}}} = \frac{3(M_\Sigma + B_2 + Z_2)^2}{B_4 + Z_4}. \tag{B.114}$$

The audit also checks the response-sign criterion selecting the positive one-pole branch and the constant-prefactor conditions.

Output. Stage 019 exports (B.111)–(B.114).

B.20 Stage 020: Parent throat action weak-axisymmetric packet

Part / anchor. Part I; primary anchor **MTDC-T4**.

Claim status. This stage is Exact **for the displayed algebraic identities inside the declared parent/projection data, with** Reduced / Controlled Reduction **or** Exact Within Closure **inherited where the local text invokes the matched reduction, mouth-kernel closure, parent-action packet, or one-port normal form.**

Source and audit. This stage corresponds to `scripts/moving_throat_pde_stage020_parent_throat_action_weak_axisym_packet_sympy_audit.py`. It solves the first-order even gates in parent-wall slope variables.

Slope packet. Let

$$\delta M_\Sigma = \int dw\, \delta\mu_\eta\, \beta_2^2, \qquad \delta K_\Sigma = \int dw\, \Big[\delta T_w (\beta_2')^2 + (\delta K_\eta + 6\delta T_\Omega)\beta_2^2\Big]. \tag{B.115}$$

Then

$$D_{01} = \delta K_\Sigma - B_{01} - Z_{01}, \quad D_{21} = -(\delta M_\Sigma + B_{21} + Z_{21}), \quad D_{41} = -(B_{41} + Z_{41}). \tag{B.116}$$

The live gates are

$$K_1 = D_{21} + \frac{D_{01}}{9}, \qquad H_{\mathrm{even}} = D_{41} - \frac{2}{3}D_{21} - \frac{D_{01}}{27}, \tag{B.117}$$

$$\Xi_1 = \frac{N_{01}}{N_0} - \frac{D_{01}}{D_0}. \tag{B.118}$$

Even compensation. The exact wall-slope solve is

$$\delta K_\Sigma = B_{01} + Z_{01} + 27(B_{41} + Z_{41}), \qquad \delta M_\Sigma = -(B_{21} + Z_{21}) + 3(B_{41} + Z_{41}). \tag{B.119}$$

After this solve, the residual prefactor obstruction is

$$\Xi_1 = \frac{N_{01}}{N_0} - \frac{27(B_{41} + Z_{41})}{K_\Sigma - B_0 - Z_0}. \tag{B.120}$$

Output. Stage 020 exports the weak-axisymmetric parent packet (B.115)–(B.120).

B.21 Stage 021: Reduced Maxwell/mixed one-port normal form

Part / anchor. Part I; primary anchor **MTDC-T4**.

Claim status. This stage is Exact for the displayed algebraic identities inside the declared parent/projection data, with Reduced / Controlled Reduction or Exact Within Closure inherited where the local text invokes the matched reduction, mouth-kernel closure, parent-action packet, or one-port normal form.

Source and audit. This stage corresponds to `scripts/moving_throat_pde_stage021_reduced_one_port_normal_form_sympy_audit.py`. It retains the earlier reduced Maxwell/mixed calculation as the one-port normal form used downstream. This is now the tail of the projection-first derivation, not the primary EM derivation.

Mixed gauge invariants. Under $A_0 \mapsto A_0 - \partial_t\chi$, $A_a \mapsto A_a + \partial_a\chi$, and $A_w \mapsto A_w + \partial_w\chi$, the mixed fields

$$E_w = -\partial_t A_w - \partial_w A_0, \qquad C_a = \partial_a A_w - \partial_w A_a \tag{B.121}$$

are gauge invariant.

Reduced lane. For one harmonic lane, eliminating the brane-like coordinate A_l and the mixed coordinate W_l gives

$$\Sigma_l^{(\mathrm{EM+mix,cons})}(\omega) = \frac{g_{A,l}^2 W_l(\omega) + 2g_{A,l}g_{W,l}R_l + g_{W,l}^2 A_l(\omega)}{A_l(\omega)W_l(\omega) - R_l^2}. \tag{B.122}$$

The low-frequency conservative coefficients are

$$z_0 = \frac{N_0^{\mathrm{cons}}}{\Delta_0}, \tag{B.123}$$
$$z_2 = \frac{N_0^{\mathrm{cons}} S_2 - G_2\Delta_0}{\Delta_0^2}, \tag{B.124}$$
$$z_4 = \frac{N_0^{\mathrm{cons}}(S_2^2 - \Delta_0) - S_2G_2\Delta_0}{\Delta_0^3}. \tag{B.125}$$

Outgoing transfer. Replacing $W_l(\omega)$ by $W_l(\omega) - \Pi_l^{\mathrm{out}}(\omega)$ gives the first-order transfer factor

$$N_l(\omega) = \frac{\left[A_l(\omega)g_{W,l} + R_l g_{A,l}\right]^2}{\left[A_l(\omega)W_l(\omega) - R_l^2\right]^2}. \tag{B.126}$$

In particular,

$$N_l(0) = \frac{(\Omega_{A,l}^2 g_{W,l} + R_l g_{A,l})^2}{(\Omega_{A,l}^2\Omega_{W,l}^2 - R_l^2)^2} \geq 0. \tag{B.127}$$

Compact outgoing fingerprint. For the compact outgoing $l = 2$ branch,

$$\widehat{Y}_2^{\mathrm{out}}(\omega) = 1 + \frac{a^2\omega^2}{9c_s^2} + \frac{4a^4\omega^4}{81c_s^4} + i\frac{a^5\omega^5}{27c_s^5} + O(\omega^6), \tag{B.128}$$

so

$$\delta D_2^{\mathrm{odd}}(\omega) = -iN_2(0)\frac{a^5}{27c_s^5}\omega^5 + O(\omega^7). \tag{B.129}$$

Output. Stage 021 exports the reduced one-port self-energy (B.122), the transfer factor (B.126), and the wall-level outgoing quadrupole coefficient (B.129).

B.22 Stage 022: Grouped Real P_2 Normalization Bridge

Part / anchor. Part I; primary anchors **MTDC-T3** and **MTDC-T4**.

Claim status. The bridge formulas are Exact Within Closure **inside the grouped response normalization convention. The final equality to the universal normalization is a** Open **branch-realization test.**

Purpose. Stage 022 translates the Stages 004–021 projection-first Maxwell packet, through its matched one-port outgoing normal form, into the grouped real P_2 response language. It converts conservative operator moments D_{An} into normalized response moments $u_n^{(A)}$, tracks the outgoing-transfer prefactor P_{An}, and isolates the exact invariant product that must match the universal quadrupole normalization.

Inputs. The inputs are one conservative grouped-lane operator $D_A^{\mathrm{cons}}(\omega)$, one outgoing transfer factor $N_A(\omega)$ exported by the matched Stage 021 one-port normal form, the compact outgoing $l = 2$ fingerprint, and the grouped trace/anomaly map.

Derivation

1. Conservative operator and normalized response. For each grouped lane $A \in \{20, 21, 22\}$, write

$$D_A^{\mathrm{cons}}(\omega) = D_{A0} + D_{A2}\omega^2 + D_{A4}\omega^4 + O(\omega^6). \tag{B.130}$$

The normalized conservative response is

$$Y_A(\omega) = \frac{D_{A0}}{D_A^{\mathrm{cons}}(\omega)} = 1 + u_2^{(A)}\omega^2 + u_4^{(A)}\omega^4 + O(\omega^6). \tag{B.131}$$

Expanding the inverse gives

$$\boxed{u_2^{(A)} = -\frac{D_{A2}}{D_{A0}}, \qquad u_4^{(A)} = \frac{D_{A2}^2 - D_{A0}D_{A4}}{D_{A0}^2}}. \tag{B.132}$$

Here the symbol $u_n^{(A)}$ is used in the main ledger; the boxed formula records the same coefficients. The key point is that the normalized response moments are not independent of the operator moments.

2. Grouped inverse map. For any grouped triple $x_A = (x_{20}, x_{21}, x_{22})$, define

$$\bar{x} = \frac{x_{20} + 2x_{21} + 2x_{22}}{5}, \qquad a_x = \frac{2x_{20} - x_{21} - x_{22}}{10}, \qquad b_x = \frac{x_{21} - x_{22}}{2}. \tag{B.133}$$

The inverse map is

$$\boxed{x_{20} = \bar{x} + 4a_x, \qquad x_{21} = \bar{x} - a_x + b_x, \qquad x_{22} = \bar{x} - a_x - b_x}. \tag{B.134}$$

The grouped isotropy gate is

$$a_2 = b_2 = a_4 = b_4 = 0 \tag{B.135}$$

when applied to $u_2^{(A)}$ and $u_4^{(A)}$.

3. Internal outgoing prefactor. Let

$$N_A(\omega) = N_{A0} + N_{A2}\omega^2 + N_{A4}\omega^4 + O(\omega^6). \tag{B.136}$$

To first order in the outgoing branch, the normalized response correction is controlled by

$$\mathrm{Pref}_A(\omega) = \frac{D_{A0}N_A(\omega)}{(D_A^{\mathrm{cons}}(\omega))^2} = P_{A0} + P_{A2}\omega^2 + P_{A4}\omega^4 + O(\omega^6). \tag{B.137}$$

Direct expansion gives

$$\boxed{P_{A0}} = \boxed{\frac{N_{A0}}{D_{A0}}}, \tag{B.138}$$

$$\boxed{P_{A2}} = \boxed{\frac{D_{A0}N_{A2} - 2D_{A2}N_{A0}}{D_{A0}^2}}, \tag{B.139}$$

$$\boxed{P_{A4}} = \boxed{\frac{D_{A0}^2N_{A4} - 2D_{A0}(D_{A2}N_{A2} + D_{A4}N_{A0}) + 3D_{A2}^2N_{A0}}{D_{A0}^3}}. \tag{B.140}$$

These are the algebraic bridge from conservative internal data to outgoing-branch prefactor data.

4. Multiplication by the compact outgoing fingerprint. Write the compact outgoing branch as

$$\widehat{Y}_2^{\text{out}}(\omega) = 1 + A\omega^2 + B\omega^4 + iG_5\omega^5 + O(\omega^6), \tag{B.141}$$

with

$$A = \frac{a^2}{9c_s^2}, \qquad B = \frac{4a^4}{81c_s^4}, \qquad G_5 = \frac{a^5}{27c_s^5}. \tag{B.142}$$

Then

$$\delta Y_A^{\text{out}}(\omega) = \text{Pref}_A(\omega)\widehat{Y}_2^{\text{out}}(\omega) = K_{A0} + K_{A2}\omega^2 + K_{A4}\omega^4 + i\Gamma_5^{(A)}\omega^5 + \cdots, \tag{B.143}$$

where

$$K_{A0} = P_{A0}, \tag{B.144}$$

$$K_{A2} = P_{A2} + AP_{A0}, \tag{B.145}$$

$$K_{A4} = P_{A4} + AP_{A2} + BP_{A0}, \tag{B.146}$$

$$\Gamma_5^{(A)} = G_5 P_{A0}. \tag{B.147}$$

Thus only the static prefactor P_{A0} enters the leading odd $i\omega^5$ coefficient.

5. Universal normalization product. The GR quadrupole target is

$$\gamma_{\text{GR}} = \frac{2G}{5c^5}. \tag{B.148}$$

Including the source-map factor $\widehat{m}_0$, the invariant normalization condition is

$$\boxed{\widehat{m}_0^2 P_0 \frac{a^5}{27c_s^5} = \frac{2G}{5c^5}}, \tag{B.149}$$

or equivalently

$$\boxed{\widehat{m}_0^2 P_0 = \frac{54Gc_s^5}{5a^5c^5}}. \tag{B.150}$$

On the leading natural source-map branch $\widehat{m}_0 = 1 + O(a^2/r^2)$, this reduces to a direct target for P_0.

Output. Stage 022 outputs the normalized response formulas (B.132), the grouped inverse map (B.134), the prefactor coefficients (B.138), the odd coefficient (B.144), and the invariant normalization test (B.150).

Checks.

- ☐ The inverse expansion of D_{A0}/D_A gives u_2, u_4 with the signs shown in (B.132).
- ☐ The grouped inverse map (B.134) reproduces (B.133) exactly.
- ☐ The first odd coefficient depends only on P_{A0}, because Pref_A is even through the retained orders and the outgoing odd term first appears at ω^5.
- ☐ The final equality (B.150) is a branch test, not a proof of branch realization.

Downstream use. Stage 023 uses the Stage 022 bridge to define $P_0 = N_0/D_0$ for the full grouped bundle. The 2.5PN and 4PN bridge sections use (B.150) as the shared outgoing-normalization condition. Later same-charge and weak-axisymmetric stages use the ratio P_1/P_0 as the transported prefactor slope.

B.23 Stage 023: Full Grouped Bundle, Exact Projectors, and Isotropic Normalization Test

Part / anchor. Part I; primary anchor **MTDC-T4**, preparing **MTDC-T5**.

Claim status. The grouped projectors, coefficient formulas, isotropic reduction, constant-prefactor conditions, and first-order transport laws are Exact Within Closure **inside the full reduced grouped-bundle model. Actual values of the microscopic branch data remain** Open**.**

Purpose. Stage 023 writes the entire grouped real 20/21/22 bundle in one place. It combines the wall baseline, BdG support moments, projection-first Maxwell/matched mixed conservative moments, and outgoing-transfer moments; introduces exact weighted projectors for the grouped lanes; states the isotropic normalization test in full-bundle variables; and derives the first-order anisotropy transport laws.

Inputs. The inputs are the Stage 003 BdG moments, the Stages 004–021 projected Maxwell-to-Z_n, N_n packet and matched one-port conservative/transfer moments, the Stage 022 response/prefactor conversion, and the grouped metric $G_{\rm grp} = \mathrm{diag}(1, 2, 2)$.

Derivation

1. Weighted grouped projector calculus. The grouped triple $(20, 21, 22)$ carries metric

$$G_{\rm grp} = \mathrm{diag}(1, 2, 2). \tag{B.151}$$

Use the $G_{\rm grp}$-orthogonal basis

$$e_{\bar{x}} = (1, 1, 1), \qquad e_a = (4, -1, -1), \qquad e_b = (0, 1, -1). \tag{B.152}$$

The norms are

$$e_{\bar{x}}^T G_{\rm grp} e_{\bar{x}} = 5, \qquad e_a^T G_{\rm grp} e_a = 20, \qquad e_b^T G_{\rm grp} e_b = 4. \tag{B.153}$$

Thus any grouped vector decomposes as

$$x = \bar{x}\, e_{\bar{x}} + a_x e_a + b_x e_b, \tag{B.154}$$

with $\bar{x}, a_x, b_x$ as in (B.133). The exact projectors are

$$\boxed{P_{\bar{x}} = \frac{1}{5}\begin{pmatrix}1 & 2 & 2\\ 1 & 2 & 2\\ 1 & 2 & 2\end{pmatrix}, \quad P_a = \frac{1}{20}\begin{pmatrix}16 & -8 & -8\\ -4 & 2 & 2\\ -4 & 2 & 2\end{pmatrix}, \quad P_b = \frac{1}{4}\begin{pmatrix}0 & 0 & 0\\ 0 & 2 & -2\\ 0 & -2 & 2\end{pmatrix}}. \tag{B.155}$$

They obey

$$P_{\bar{x}} + P_a + P_b = I_3, \qquad P_i P_j = \delta_{ij} P_i. \tag{B.156}$$

Grouped isotropy is the condition

$$P_a x = P_b x = 0. \tag{B.157}$$

2. Full coupled lane Lagrangian. For each lane $A \in \{20, 21, 22\}$, retain a wall amplitude q_A, stable BdG support modes $X_{A\alpha}$, brane-like gauge coordinates U_{Ar}, mixed coordinates W_{Ar}, and internal gauge mixing R_{Ar}. The reduced Lagrangian is

$$L_A = \frac{1}{2}M_A\dot{q}_A^2 - \frac{1}{2}K_A q_A^2 + \sum_\alpha \left[\frac{1}{2}\dot{X}_{A\alpha}^2 - \frac{1}{2}\varpi_{A\alpha}^2 X_{A\alpha}^2 + c_{A\alpha}q_A X_{A\alpha}\right] \tag{B.158}$$

$$+ \sum_r \Big[\frac{1}{2}\dot{U}_{Ar}^2 - \frac{1}{2}\Omega_{U,Ar}^2 U_{Ar}^2 + \frac{1}{2}\dot{W}_{Ar}^2 - \frac{1}{2}\Omega_{W,Ar}^2 W_{Ar}^2 + R_{Ar}U_{Ar}W_{Ar} \tag{B.159}$$

$$+ g_{U,Ar}q_A U_{Ar} + g_{W,Ar}q_A W_{Ar}\Big]. \tag{B.160}$$

3. BdG support moments. Define

$$\boxed{B_{A0} = \sum_\alpha \frac{c_{A\alpha}^2}{\varpi_{A\alpha}^2}, \qquad B_{A2} = \sum_\alpha \frac{c_{A\alpha}^2}{\varpi_{A\alpha}^4}, \qquad B_{A4} = \sum_\alpha \frac{c_{A\alpha}^2}{\varpi_{A\alpha}^6}}. \tag{B.161}$$

These are the Stage 003 Schur-complement moments collected lane by lane.

4. Conservative Maxwell/mixed moments. For each port r, set

$$\Delta_{Ar} = \Omega_{U,Ar}^2\Omega_{W,Ar}^2 - R_{Ar}^2, \qquad S_{Ar} = \Omega_{U,Ar}^2 + \Omega_{W,Ar}^2, \tag{B.162}$$

$$Q_{Ar} = g_{U,Ar}^2\Omega_{W,Ar}^2 + 2g_{U,Ar}g_{W,Ar}R_{Ar} + g_{W,Ar}^2\Omega_{U,Ar}^2, \qquad G_{Ar} = g_{U,Ar}^2 + g_{W,Ar}^2. \tag{B.163}$$

The conservative Maxwell/mixed moments are

$$Z_{A0}^{(r)} = \frac{Q_{Ar}}{\Delta_{Ar}}, \tag{B.164}$$

$$Z_{A2}^{(r)} = \frac{Q_{Ar}S_{Ar} - G_{Ar}\Delta_{Ar}}{\Delta_{Ar}^2}, \tag{B.165}$$

$$Z_{A4}^{(r)} = \frac{Q_{Ar}(S_{Ar}^2 - \Delta_{Ar}) - S_{Ar}G_{Ar}\Delta_{Ar}}{\Delta_{Ar}^3}, \tag{B.166}$$

and

$$Z_{An} = \sum_r Z_{An}^{(r)}, \qquad n \in \{0, 2, 4\}. \tag{B.167}$$

5. Outgoing-transfer moments. Define

$$P_{Ar} = \Omega_{U,Ar}^2 g_{W,Ar} + R_{Ar}g_{U,Ar}. \tag{B.168}$$

The outgoing-transfer moments per port are

$$N_{A0}^{(r)} = \frac{P_{Ar}^2}{\Delta_{Ar}^2}, \tag{B.169}$$

$$N_{A2}^{(r)} = \frac{2P_{Ar}\left(P_{Ar}S_{Ar} - \Delta_{Ar}g_{W,Ar}\right)}{\Delta_{Ar}^3}, \tag{B.170}$$

$$N_{A4}^{(r)} = \frac{\Delta_{Ar}^2 g_{W,Ar}^2 - 2\Delta_{Ar}P_{Ar}^2 - 4\Delta_{Ar}P_{Ar}S_{Ar}g_{W,Ar} + 3P_{Ar}^2S_{Ar}^2}{\Delta_{Ar}^4}. \tag{B.171}$$

Summing over outgoing ports gives

$$N_{An} = \sum_r N_{An}^{(r)}, \qquad n \in \{0, 2, 4\}. \tag{B.172}$$

6. Total conservative grouped operator. The full conservative grouped-lane operator is

$$D_A^{\mathrm{cons}}(\omega) = D_{A0} + D_{A2}\omega^2 + D_{A4}\omega^4 + O(\omega^6), \tag{B.173}$$

with

$$\boxed{D_{A0} = K_A - B_{A0} - Z_{A0}, \qquad D_{A2} = -(M_A + B_{A2} + Z_{A2}), \qquad D_{A4} = -(B_{A4} + Z_{A4})}. \tag{B.174}$$

This is the full Part-I coefficient ledger.

7. Microscopic anisotropy ledger. Apply the grouped decomposition (B.154) to every microscopic family $K_A, M_A, B_{An}, Z_{An}, N_{An}$. The static conservative defects obey

$$\bar{D}_0 = \bar{K} - \bar{B}_0 - \bar{Z}_0, \qquad a_{D,0} = a_K - a_{B,0} - a_{Z,0}, \qquad b_{D,0} = b_K - b_{B,0} - b_{Z,0}. \tag{B.175}$$

Similarly,

$$\bar{D}_2 = -(\bar{M} + \bar{B}_2 + \bar{Z}_2), \qquad a_{D,2} = -(a_M + a_{B,2} + a_{Z,2}), \qquad b_{D,2} = -(b_M + b_{B,2} + b_{Z,2}), \tag{B.176}$$

and

$$\bar{D}_4 = -(\bar{B}_4 + \bar{Z}_4), \qquad a_{D,4} = -(a_{B,4} + a_{Z,4}), \qquad b_{D,4} = -(b_{B,4} + b_{Z,4}). \tag{B.177}$$

Thus grouped anisotropy can enter only through the wall baseline, the BdG support bundle, or the Maxwell/mixed conservative and outgoing-transfer bundles.

8. Isotropic branch and normalization ratio. On the isotropic branch,

$$D_{20,n} = D_{21,n} = D_{22,n} = D_n, \qquad N_{20,n} = N_{21,n} = N_{22,n} = N_n. \tag{B.178}$$

Stage 022 then gives

$$u_2 = -\frac{D_2}{D_0}, \qquad u_4 = \frac{D_2^2 - D_0 D_4}{D_0^2}, \tag{B.179}$$

and

$$P_0 = \frac{N_0}{D_0}, \qquad P_2 = \frac{D_0 N_2 - 2D_2 N_0}{D_0^2}, \tag{B.180}$$

$$P_4 = \frac{D_0^2 N_4 - 2D_0(D_2 N_2 + D_4 N_0) + 3D_2^2 N_0}{D_0^3}. \tag{B.181}$$

The isotropic normalization test is

$$\boxed{\widehat{m}_0^2 \frac{N_0}{K - B_0 - Z_0} = \frac{54 G c_s^5}{5 a^5 c^5}}. \tag{B.182}$$

The denominator is the static full-bundle stiffness

$$D_0 = K - B_0 - Z_0. \tag{B.183}$$

9. Constant-prefactor branch. The constant-prefactor branch through the retained orders is

$$P_2 = 0, \qquad P_4 = 0. \tag{B.184}$$

Using (B.180)–(B.181), this is equivalent to

$$\boxed{N_2 = \frac{2D_2N_0}{D_0}, \qquad N_4 = \frac{2D_0(D_2N_2 + D_4N_0) - 3D_2^2N_0}{D_0^2}}. \tag{B.185}$$

So constant outgoing prefactor does not require $N_2 = N_4 = 0$; it requires precise correlation between transfer and conservative moments.

10. First-order anisotropy transport. Let

$$D_{An} = D_n + \delta D_{An}, \qquad N_{An} = N_n + \delta N_{An}, \tag{B.186}$$

near the isotropic branch. From $u_2^{(A)} = -D_{A2}/D_{A0}$,

$$\delta u_2^{(A)} = -\frac{\delta D_{A2} + u_2\delta D_{A0}}{D_0}. \tag{B.187}$$

Therefore

$$\boxed{a_{u,2} = -\frac{a_{D,2} + u_2a_{D,0}}{D_0}, \qquad b_{u,2} = -\frac{b_{D,2} + u_2b_{D,0}}{D_0}}. \tag{B.188}$$

Similarly, from $P_0^{(A)} = N_{A0}/D_{A0}$,

$$\delta P_0^{(A)} = \frac{\delta N_{A0} - P_0\delta D_{A0}}{D_0}, \tag{B.189}$$

so

$$\boxed{a_{P,0} = \frac{a_{N,0} - P_0a_{D,0}}{D_0}, \qquad b_{P,0} = \frac{b_{N,0} - P_0b_{D,0}}{D_0}}. \tag{B.190}$$

Thus outgoing-prefactor anisotropy can arise only from direct anisotropy in N_{A0} or conservative anisotropy in D_{A0} reweighted by P_0.

Output. Stage 023 outputs the projectors (B.155), the full coupled coefficient formulas (B.161), (B.164), (B.169), and (B.174), the isotropic normalization ratio (B.182), the constant-prefactor conditions (B.185), and the anisotropy transport laws (B.188) and (B.190).

Checks.

- ☐ Direct multiplication verifies $P_iP_j = \delta_{ij}P_i$ and $P_{\bar{x}} + P_a + P_b = I_3$.
- ☐ The BdG, Maxwell/mixed, and outgoing-transfer coefficient formulas reduce to the Stage 003 and Stage 021 one-lane formulas when a single mode/port is retained.
- ☐ The isotropic limit sets all grouped defects to zero and collapses $(20, 21, 22)$ to one common D_n, N_n packet.
- ☐ First-order quotient differentiation gives the transport laws (B.188) and (B.190).
- ☐ The final normalization equation is a stability-sensitive branch test because $D_0 = K - B_0 - Z_0$ must remain positive on the stable side.

Downstream use. Stage 024 begins the explicit overlap-integral and $O(3)$ isotropy theorem using the Stage 023 coefficient ledger. Later conservative and retarded stages repeatedly reuse D_0, D_2, D_4, N_0, N_2, N_4, P_0, P_2, P_4, the grouped projectors, and the anisotropy transport laws. The weak-axisymmetric orbit and same-charge sections eventually reinterpret P_1/P_0 as the transported outgoing-prefactor slope.

B.24 Part I appendix closure and handoff to Part II

Stages 001–023 complete the first verification block. They do not solve the nonlinear moving-throat PDE, but they remove the main algebraic ambiguity that existed before the ledger began. The reduced response variables used later are not arbitrary symbols: they are the low-frequency coefficients of a coupled wall/BdG/Maxwell/mixed Schur-complement system, organized by exact grouped projectors and normalized by a definite outgoing $l = 2$ fingerprint.

The handoff to Part II is therefore precise. What remains to be checked next is not the abstract existence of grouped P_2 lanes, nor whether the electromagnetic sector should be projected before it is reduced. Those now exist inside the reduced moving-throat construction. The next question is whether explicit radial/axial overlap integrals on an isotropic branch force the three grouped lanes to agree, and how the first controlled anisotropy enters when the branch is weakly axisymmetric. That is the job of Stages 024–036.

Appendix C

Stage Appendix: Part II – Conservative Response Kernels and Grouped P_2 Normalization Language

Stages 024–036; anchor MTDC-T5

This appendix is the detailed verification ledger for the second theorem block of the moving-throat derivation. The corresponding main-text chapter, chapter 2, explains why this block is needed in the larger derivation. The job here is more mechanical: to preserve the exact reduced algebra, the branch assumptions, the finite-throat overlap calculations, the selected-mode normalization formulas, and the final admissibility frontier that Stages 024–036 derive.

Part I ended with a full grouped real P_2 wall/BdG/Maxwell/mixed bundle. In that bundle the isotropic outgoing-normalization question had already collapsed to the ratio

$$\widehat{m}_0^2 \frac{N_0}{K - B_0 - Z_0} = \frac{54 G c_s^5}{5 a^5 c^5}. \tag{C.1}$$

Part II asks what this ratio means once the grouped coefficients are no longer left as formal packets. The stages here introduce explicit angular and axial overlaps, prove the $O(3)$ isotropy collapse, identify the first weak-axisymmetric splitting law, select a finite-throat wall profile by a rank-one Schur complement, eliminate the abstract source-map factor on the natural D/N branch, and reduce the first selected quadrupole branch to the exact two-function admissibility test

$$F(\xi_{\mathrm{req}}, \delta) = R_{\mathrm{target}}, \qquad M_{\mathrm{mix}} \leq G(\xi_{\mathrm{req}}, \delta). \tag{C.2}$$

Source layout. Stages 024–036 are now sourced from the canonical files `stages/stage_024.tex` through `stages/stage_036.tex`. The raw Markdown notes and audit artifacts are tracked in a repo-local provenance index, so they do not have to appear in the reader-facing PDF. The stage files themselves carry the executable SymPy and Mathematica audit references.

Table C.1: Part II stage-audit map.

Stage	Source stage	Primary status	Verification output
024	Overlap integrals, $O(3)$ isotropy, and first axisymmetric splitting	Exact Within Closure / Reduced / Controlled Reduction	Canonical real-STF basis, angular source-map identity, isotropic grouped collapse, and weak-axisymmetric signature $b = 3a$.
025	Minimal isotropic single-mode closure	Exact Within Closure / Open	Explicit one-BdG/one-mixed-pair normalization ratio and stability gate.
026	Concrete finite-throat axial modes	Exact Within Closure	D/N overlap $\kappa = 2\sqrt{2}/\pi$, branch-level normalization law, and required stiffness formula.
027	First nonconstant finite-throat wall/brane family	Exact Within Closure	Profile overlap $\kappa(\theta)$, blind-angle no-go, max-coupling angle, and dressed stiffness equation.
028	Loaded axial profile selection	Exact Within Closure	Rank-one loaded wall eigenproblem, lower-mode angle law, and softening threshold.
029	Dynamic loading from coupled wall/BdG/Maxwell/mixed operator	Exact Within Closure / Reduced / Controlled Reduction	Schur-complement split $\Sigma_{\text{wall}} = \Xi I_2 + \alpha v v^T$ and selected odd quadrupole coefficient.
030	Selected-mode normalized response	Exact Within Closure	Selected prefactor $P_{0,-} = \beta_0 s_- / \lambda_-$ and selected target.
031	Reachability and stable-side monotonicity	Exact Within Closure	Strict monotonicity of $P_{0,-}$ and unique stable crossing.
032	Source map from finite-throat mode integrals	Exact Within Closure	Natural D/N source map $\widehat{m}_-^2 = s_-/\kappa_0^2$.
033	Microscopic normalization equation and stability gate	Exact Within Closure / Open	Coupling-level selected target, stability inequalities, onset criterion, and weak-loading expansion.
034	Softening-depth normal form	Exact Within Closure	Scalar branch variable $x = A - \lambda_-$, exact $\alpha_0(x)$, $s_-(x)$, $N_-(x)$, and support-loading formula.
035	Dimensionless D/N normalization locus	Exact Within Closure	Shape function $F(\xi, \delta)$, monotonicity, endpoint theorem, and total loading.
036	Dimensionless support-feasibility frontier	Exact Within Closure / Open	Shape function $G(\xi, \delta)$, monotonicity, and final admissibility region in $(R_{\text{target}}, M_{\text{mix}})$.

Audit philosophy. The calculations in this appendix are exact inside the declared reduced closure. In particular, angular identities, overlap integrals, Schur complements, eigenvalue formulas, monotonicity derivatives, source-map reductions, and dimensionless branch functions are not fitted. The closure assumptions are still visible: finite-throat axial profiles, stable normal-mode reduction, one selected D/N family, and a local bilinear wall/support/outgoing operator are reduced descriptions of the full moving-throat PDE. The final placement of the actual nonlinear branch on the admissible (F, G) frontier remains open.

Notation warning. The labels $20, 21, 22$ are grouped real P_2 lanes, not spacetime indices. The labels 21 and 22 each abbreviate a real cosine/sine pair when the full five-component real STF basis is being described. The symbol G appears both as Newton's constant and as a support-feasibility function; the context or subscript distinguishes them. For the support-feasibility function, this appendix always writes $G(\xi, \delta)$.

Definition C.1 (Quadrupole target used throughout Part II)**.** *The universal scalar target carried by the reduced outgoing quadrupole branch is*

$$\boxed{N_Q^{\text{target}} := \frac{54Gc_s^5}{5a^5c^5}}. \tag{C.3}$$

Thus the isotropic Part-I target is $\widehat{m}_0^2 P_0 = N_Q^{\text{target}}$*, and the selected D/N target below is* $\widehat{m}_-^2 P_{0,-} = N_Q^{\text{target}}$*.*

C.1 Stage 024: Explicit Overlap Integrals, $O(3)$ Isotropy, and First Axisymmetric Splitting

Part / anchor. Part II; primary anchor **MTDC-T5**.

Claim status. The angular identities and grouped splitting law are Exact Within Closure **once the real-STF basis and reduced overlap closure are adopted. The separated radial/axial ansatz is** Reduced / Controlled Reduction**.**

Purpose. Stage 024 takes the grouped coefficients of Stage 023 and gives them their first explicit overlap-integral interpretation. It proves that the angular source map is exact on the canonical isotropic branch, that $O(3)$-invariant kernels collapse the grouped lanes to common values, and that the first weak-axisymmetric quadrupole splitting has the universal signature $(1, 1/2, -1)$, equivalently $b = 3a$.

Inputs. The stage uses the Stage 023 full-bundle coefficients $B_{A,n}$, $Z_{A,n}$, $N_{A,n}$, the grouped trace/anomaly variables, and the real STF basis for the quadrupole ports.

Derivation

1. Real-STF harmonic normalization. Let n^i be the unit vector on the throat sphere and let E_A^{ij} be a real STF basis satisfying

$$\mathrm{Tr}(E_A E_B) = \delta_{AB}. \tag{C.4}$$

Define

$$Y_A(n) = \sqrt{\frac{15}{8\pi}} E_A^{ij} n_i n_j, \qquad A \in \{20, 21c, 21s, 22c, 22s\}. \tag{C.5}$$

The exact fourth-moment identity

$$\int_{S^2} n_i n_j n_k n_l \, d\Omega = \frac{4\pi}{15}\left(\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}\right) \tag{C.6}$$

then gives

$$\boxed{\int_{S^2} Y_A Y_B \, d\Omega = \delta_{AB}}. \tag{C.7}$$

Thus the real grouped quadrupole ports have a canonical normalized angular basis.

2. Angular source-map identity. Expand the orbital/worldtube STF quadrupole source as

$$S(n) = \sum_A S_A Y_A(n). \tag{C.8}$$

Projecting on the same ports gives

$$\boxed{S_A^{\text{port}} = \int_{S^2} Y_A(n) S(n) \, d\Omega = S_A}. \tag{C.9}$$

Therefore the source map factors as

$$\widehat{m}_0 = \widehat{m}_{\text{rad}} \widehat{m}_{\text{ang}}, \qquad \boxed{\widehat{m}_{\text{ang}} = 1} \tag{C.10}$$

on the canonical isotropic branch. The remaining source-map problem is radial/axial.

3. Isotropic overlap factorization. On an $O(3)$-invariant branch, write the wall, BdG, brane-like gauge, and mixed coordinates in separated form:

$$\eta_A(s,\Omega,t) = q_A(t)\chi_\eta(s) Y_A(\Omega), \tag{C.11}$$
$$X_{\alpha,A}(s,\Omega,t) = X_{\alpha,A}(t)\phi_\alpha(s) Y_A(\Omega), \tag{C.12}$$
$$U_{r,A}(s,\Omega,t) = U_{r,A}(t) u_r(s) Y_A(\Omega), \tag{C.13}$$
$$W_{r,A}(s,\Omega,t) = W_{r,A}(t) w_r(s) Y_A(\Omega). \tag{C.14}$$

The angular factor is diagonal by (C.7). The radial/axial overlaps are

$$I_{\eta\alpha} = \int ds\, \mu_s(s)\chi_\eta(s)\phi_\alpha(s), \tag{C.15}$$
$$I_{\eta u,r} = \int ds\, \mu_s(s)\chi_\eta(s) u_r(s), \tag{C.16}$$
$$I_{\eta w,r} = \int ds\, \mu_s(s)\chi_\eta(s) w_r(s), \tag{C.17}$$
$$I_{uw,r} = \int ds\, \mu_s(s) u_r(s) w_r(s). \tag{C.18}$$

The lane-independent reduced couplings are

$$\boxed{C_\alpha = \lambda_{B,\alpha} I_{\eta\alpha}, \qquad G_{U,r} = \lambda_{U,r} I_{\eta u,r}, \qquad G_{W,r} = \lambda_{W,r} I_{\eta w,r}, \qquad R_r = \lambda_{R,r} I_{uw,r}}. \tag{C.19}$$

Thus

$$\boxed{B_0 = \sum_\alpha \frac{C_\alpha^2}{\varpi_\alpha^2}, \qquad B_2 = \sum_\alpha \frac{C_\alpha^2}{\varpi_\alpha^4}, \qquad B_4 = \sum_\alpha \frac{C_\alpha^2}{\varpi_\alpha^6}}. \tag{C.20}$$

For each Maxwell/mixed pair define

$$\Delta_r = \Omega_{U,r}^2 \Omega_{W,r}^2 - R_r^2, \qquad S_r = \Omega_{U,r}^2 + \Omega_{W,r}^2, \tag{C.21}$$

$$Q_r = G_{U,r}^2 \Omega_{W,r}^2 + 2G_{U,r}G_{W,r}R_r + G_{W,r}^2 \Omega_{U,r}^2, \qquad H_r = G_{U,r}^2 + G_{W,r}^2, \tag{C.22}$$

$$\mathcal{P}_r = \Omega_{U,r}^2 G_{W,r} + R_r G_{U,r}. \tag{C.23}$$

Then

$$Z_0 = \sum_r \frac{Q_r}{\Delta_r}, \quad Z_2 = \sum_r \frac{Q_r S_r - H_r \Delta_r}{\Delta_r^2}, \qquad Z_4 = \sum_r \frac{Q_r(S_r^2 - \Delta_r) - S_r H_r \Delta_r}{\Delta_r^3}, \tag{C.24}$$

$$N_0 = \sum_r \frac{\mathcal{P}_r^2}{\Delta_r^2}, \quad N_2 = \sum_r \frac{2\mathcal{P}_r(\mathcal{P}_r S_r - \Delta_r G_{W,r})}{\Delta_r^3}, \quad N_4 = \sum_r \frac{\Delta_r^2 G_{W,r}^2 - 2\Delta_r \mathcal{P}_r^2 - 4\Delta_r \mathcal{P}_r S_r G_{W,r} + 3\mathcal{P}_r^2 S_r^2}{\Delta_r^4}. \tag{C.25}$$

The conservative operator is

$$\boxed{\begin{aligned} D(\omega) &= D_0 + D_2\omega^2 + D_4\omega^4 + O(\omega^6), \\ D_0 &= K - B_0 - Z_0, \\ D_2 &= -(M + B_2 + Z_2), \\ D_4 &= -(B_4 + Z_4). \end{aligned}} \tag{C.26}$$

Therefore

$$\boxed{\begin{aligned} D_{20,n} = D_{21,n} = D_{22,n} = D_n, \\ N_{20,n} = N_{21,n} = N_{22,n} = N_n, \end{aligned}} \tag{C.27}$$

so every grouped anisotropy defect vanishes on a truly $O(3)$-invariant branch.

4. Weak-axisymmetric quadrupole splitting. Let the first symmetry-breaking perturbation be

$$\delta K = \varepsilon\, \kappa(s) Y_{20}(\Omega). \tag{C.28}$$

The angular matrix is

$$M_{AB}^{(20)} = \int_{S^2} Y_A Y_{20} Y_B \, d\Omega. \tag{C.29}$$

Using the sixth moment of the unit sphere gives

$$\boxed{M^{(20)} = \frac{\sqrt{5}}{7\sqrt{\pi}} \operatorname{diag}\left(1, \frac{1}{2}, \frac{1}{2}, -1, -1\right)} \tag{C.30}$$

in the five real STF lanes. Grouping the cosine/sine pairs gives

$$\boxed{\lambda_{20} = 1, \qquad \lambda_{21} = \frac{1}{2}, \qquad \lambda_{22} = -1}. \tag{C.31}$$

Any first-order grouped coefficient therefore has the form

$$x_{20} = x^{(0)} + \varepsilon x^{(1)}, \qquad x_{21} = x^{(0)} + \frac{\varepsilon}{2} x^{(1)}, \qquad x_{22} = x^{(0)} - \varepsilon x^{(1)}. \tag{C.32}$$

The grouped trace/anomaly variables are

$$\boxed{\bar{x} = x^{(0)}, \qquad a_x = \frac{\varepsilon}{4} x^{(1)}, \qquad b_x = \frac{3\varepsilon}{4} x^{(1)}, \qquad b_x = 3a_x}. \tag{C.33}$$

This is the first weak-axisymmetric diagnostic line.

5. First-order normalization transport. If

$$D_A = D_0 + \varepsilon\lambda_A D_1 + O(\varepsilon^2), \qquad N_A = N_0 + \varepsilon\lambda_A N_1 + O(\varepsilon^2), \tag{C.34}$$

then

$$P_A = \frac{N_A}{D_A} = P_0 + \varepsilon\lambda_A P_1 + O(\varepsilon^2), \qquad \boxed{P_1 = \frac{N_1 D_0 - N_0 D_1}{D_0^2}}. \tag{C.35}$$

The prefactor defects therefore satisfy

$$\boxed{a_P = \frac{\varepsilon}{4}P_1, \qquad b_P = \frac{3\varepsilon}{4}P_1, \qquad b_P = 3a_P}. \tag{C.36}$$

Output. Stage 024 outputs the real-STF normalization (C.7), the angular source-map identity (C.9), the isotropic overlap formulas (C.20)–(C.26), the grouped isotropy collapse (C.27), the weak-axisymmetric signature (C.31), the line $b = 3a$, and the normalization transport coefficient (C.35).

Checks.

- ☐ The angular source map is checked by inserting the source expansion and using orthonormality of Y_A.
- ☐ The isotropic collapse is checked by verifying that all angular integrals reduce to δ_{AB} and all remaining radial/axial data are lane-independent.
- ☐ The weak-axisymmetric matrix is checked from the exact sixth moment on S^2.
- ☐ The grouped line $b = 3a$ follows directly by applying the weighted trace/anomaly map to the signature $(1, 1/2, -1)$.
- ☐ The prefactor transport formula is the first-order quotient variation of $P_A = N_A/D_A$.

Downstream use. Stages 025–026 use $\widehat{m}_{\text{ang}} = 1$ to focus on radial/axial normalization. Stages 164–200 later reuse $b = 3a$ as the weak-axisymmetric orbit signature. The same prefactor slope becomes $\Xi_1 = P_1/P_0$ in the weak-axisymmetric and same-charge analysis.

C.2 Stage 025: Minimal Isotropic Single-Mode Closure

Part / anchor. Part II; primary anchor **MTDC-T5**.

Claim status. The one-mode formula is EXACT WITHIN CLOSURE **inside the one-BdG/one-Maxwell-mixed-pair isotropic closure. Whether the actual branch realizes the formula at the required value is** OPEN.

Purpose. Stage 025 strips the isotropic branch to the smallest radial/axial module that can carry the outgoing quadrupole normalization: one stable BdG support mode, one brane-like internal gauge coordinate, and one mixed port-active coordinate. The stage turns the formal ratio N_0/D_0 into an explicit scalar expression.

Inputs. The inputs are the Stage 024 isotropic angular collapse, the Stage 023 definitions of D_0 and N_0, and the one-mode reduced couplings C, G_U, G_W, R.

Derivation

1. Minimal one-mode variables. Define the radial/axial couplings

$$C = \lambda_B I_{\eta\phi}, \qquad G_U = \lambda_U I_{\eta u}, \qquad G_W = \lambda_W I_{\eta w}, \qquad R = \lambda_R I_{uw}. \tag{C.37}$$

Let the frequencies be ϖ, Ω_U, and Ω_W. The Maxwell/mixed invariants are

$$\boxed{\Delta = \Omega_U^2\Omega_W^2 - R^2, \qquad Q = G_U^2\Omega_W^2 + 2G_U G_W R + G_W^2\Omega_U^2, \qquad \mathcal{P} = \Omega_U^2 G_W + RG_U}. \tag{C.38}$$

The BdG static softening is

$$X = \frac{C^2}{\varpi^2}. \tag{C.39}$$

Therefore

$$D_0 = K - X - \frac{Q}{\Delta}, \qquad N_0 = \frac{\mathcal{P}^2}{\Delta^2}. \tag{C.40}$$

2. Minimal outgoing prefactor. The static prefactor is

$$\boxed{P_0 = \frac{N_0}{D_0} = \frac{\mathcal{P}^2}{\Delta^2\,(K - C^2/\varpi^2 - Q/\Delta)} = \frac{\mathcal{P}^2}{\Delta\,(K\Delta - \Delta C^2/\varpi^2 - Q)}}. \tag{C.41}$$

Because Stage 024 gives $\widehat{m}_{\text{ang}} = 1$, the normalization target is

$$\boxed{\widehat{m}_{\text{rad}}^2 \frac{\mathcal{P}^2}{\Delta\,(K\Delta - \Delta C^2/\varpi^2 - Q)} = N_Q^{\text{target}}}. \tag{C.42}$$

3. Stability and monotonicity. The internal mixed block and dressed wall lane are stable only if

$$\boxed{\Delta > 0, \qquad D_0 = K - \frac{C^2}{\varpi^2} - \frac{Q}{\Delta} > 0}. \tag{C.43}$$

On the stable side,

$$\frac{\partial P_0}{\partial K} = -\frac{N_0}{D_0^2} < 0, \qquad \frac{\partial P_0}{\partial X} = +\frac{N_0}{D_0^2} > 0. \tag{C.44}$$

Bare wall stiffening suppresses the outgoing prefactor, while conservative softening enhances it.

Output. Stage 025 outputs the explicit one-mode target (C.42), the stability gate (C.43), and the first monotonicity signs (C.44).

Checks.

- ☐ Substitution of D_0 and N_0 into $P_0 = N_0/D_0$ gives (C.41).
- ☐ The angular source map does not appear because Stage 024 already fixed $\widehat{m}_{\text{ang}} = 1$.
- ☐ The sign of $\partial_K P_0$ and $\partial_X P_0$ follows from $N_0 > 0$ and $D_0 > 0$.
- ☐ If $\mathcal{P} = 0$, then $N_0 = 0$ and a positive quadrupole target cannot be reached.

Downstream use. Stage 026 evaluates the overlaps $I_{\eta\phi}, I_{\eta u}, I_{\eta w}, I_{uw}$ on a concrete finite-throat D/N branch. Stage 033 later rewrites the same one-mode variables as g_U, g_W, g_R, g_B in the microscopic selected-branch test.

C.3 Stage 026: Concrete Finite-Throat Axial Modes

Part / anchor. Part II; primary anchor **MTDC-T5**.

Claim status. Exact Within Closure **inside the finite-interval D/N axial-mode closure.**

Purpose. Stage 026 evaluates the first concrete axial overlap model. It chooses a finite interval, a constant N/N wall/brane profile, and the lowest D/N half-wave for support and mixed channels. The result is the exact overlap $\kappa = 2\sqrt{2}/\pi$ and the first finite-throat branch-level normalization equation.

Inputs. The inputs are the Stage 025 one-mode closure and the finite interval $s \in [0, L]$ with N/N and D/N axial profiles.

Derivation

1. Axial basis. Use the normalized constant N/N profile

$$\nu_0(s) = \frac{1}{\sqrt{L}}, \qquad \nu_0'(0) = \nu_0'(L) = 0. \tag{C.45}$$

The D/N ladder is

$$f_n(s) = \sqrt{\frac{2}{L}} \sin\left(\frac{(n+1/2)\pi s}{L}\right), \qquad f_n(0) = 0, \qquad f_n'(L) = 0, \tag{C.46}$$

with

$$k_n = \frac{(n+1/2)\pi}{L}. \tag{C.47}$$

The minimal branch keeps $n = 0$. The exact overlap is

$$\kappa_n = \int_0^L \nu_0(s) f_n(s)\, ds = \frac{\sqrt{2}}{(n+1/2)\pi}, \tag{C.48}$$

so

$$\boxed{\kappa := \kappa_0 = \frac{2\sqrt{2}}{\pi}}. \tag{C.49}$$

2. Branch-level coupling constants. Take

$$\chi_\eta = \nu_0, \qquad \phi = f_0, \qquad u = \nu_0, \qquad w = f_0. \tag{C.50}$$

Then

$$I_{\eta\phi} = \kappa, \qquad I_{\eta u} = 1, \qquad I_{\eta w} = \kappa, \qquad I_{uw} = \kappa. \tag{C.51}$$

Thus

$$\boxed{C = \kappa\lambda_B, \qquad G_U = \lambda_U, \qquad G_W = \kappa\lambda_W, \qquad R = \kappa\lambda_R}. \tag{C.52}$$

Substituting into Stage 025 gives

$$\boxed{\begin{aligned}&\Delta = \Omega_U^2\Omega_W^2 - \kappa^2\lambda_R^2,\\ &Q = \lambda_U^2\Omega_W^2 + 2\kappa^2\lambda_U\lambda_W\lambda_R + \kappa^2\lambda_W^2\Omega_U^2,\\ &\mathcal{P} = \kappa(\Omega_U^2\lambda_W + \lambda_R\lambda_U).\end{aligned}} \tag{C.53}$$

The BdG softening is

$$X = \frac{\kappa^2\lambda_B^2}{\varpi^2}. \tag{C.54}$$

3. First finite-throat target and stiffness form. The target becomes

$$\boxed{\widehat{m}_{\rm rad}^{\,2}\frac{\kappa^2(\Omega_U^2\lambda_W + \lambda_R\lambda_U)^2}{\Delta\left(K\Delta - \Delta\kappa^2\lambda_B^2/\varpi^2 - Q\right)} = N_Q^{\rm target}}. \tag{C.55}$$

Solving for the required wall stiffness gives

$$\boxed{K_{\rm req} = \frac{\kappa^2\lambda_B^2}{\varpi^2} + \frac{Q}{\Delta} + \frac{\widehat{m}_{\rm rad}^{\,2}\kappa^2(\Omega_U^2\lambda_W + \lambda_R\lambda_U)^2}{N_Q^{\rm target}\Delta^2}}. \tag{C.56}$$

For the constant wall profile,

$$K = K_\eta + 6T_\Omega. \tag{C.57}$$

Thus the first concrete branch condition is

$$K_\eta + 6T_\Omega = K_{\rm req}. \tag{C.58}$$

Output. Stage 026 outputs the finite-throat D/N overlap (C.49), the concrete couplings (C.52), the finite-throat normalization law (C.55), and the required stiffness formula (C.56).

Checks.

- ☐ Direct integration of $\nu_0 f_n$ gives $\kappa_n = \sqrt{2}/[(n+1/2)\pi]$.
- ☐ The lowest D/N mode is dominant because $|\kappa_n|$ decreases as $(n+1/2)^{-1}$.
- ☐ The support/mixed coupling vanishes if $\kappa = 0$, but the D/N lowest mode has $\kappa > 0$.
- ☐ Solving (C.55) for K gives (C.56).

Downstream use. Stage 027 tests nonconstant wall/brane profiles. Stage 032 reuses the same finite-throat D/N source overlap to fix $\widehat{m}_-$. Stages 035–036 use the exact D/N constants derived from this branch.

C.4 Stage 027: First Nonconstant Finite-Throat Wall/Brane Family

Part / anchor. Part II; primary anchor **MTDC-T5**.

Claim status. Exact Within Closure **inside the first two-mode N/N wall/brane axial family.**

Purpose. Stage 027 checks whether the Stage 026 constant axial profile is too restrictive. It introduces the first nonconstant N/N profile, computes the profile-dependent D/N overlap $\kappa(\theta)$, identifies an exact blind-angle no-go, and derives the profile-dressed normalization gate.

Inputs. The inputs are the constant N/N profile, the first N/N excitation, and the lowest D/N half-wave from Stage 026.

Derivation

1. Two-mode N/N family. The first nonconstant N/N profile is

$$\nu_1(s) = \sqrt{\frac{2}{L}} \cos\left(\frac{\pi s}{L}\right), \qquad -\nu_1'' = \frac{\pi^2}{L^2}\nu_1. \tag{C.59}$$

Use the coherent family

$$\chi_\theta(s) = \cos\theta\, \nu_0(s) + \sin\theta\, \nu_1(s), \tag{C.60}$$

for both the wall profile and the brane-like internal coordinate:

$$\chi_\eta = \chi_\theta, \qquad u = \chi_\theta, \qquad \phi = w = f_0. \tag{C.61}$$

Since ν_0, ν_1 are orthonormal, $\int_0^L \chi_\theta^2 ds = 1$.

2. Profile overlap. The D/N overlap is

$$\boxed{\kappa(\theta) = \int_0^L \chi_\theta(s) f_0(s)\, ds = \kappa_0 \cos\theta + \kappa_1 \sin\theta}, \tag{C.62}$$

where

$$\boxed{\kappa_0 = \frac{2\sqrt{2}}{\pi}, \qquad \kappa_1 = -\frac{4}{3\pi}}. \tag{C.63}$$

Equivalently,

$$\kappa(\theta) = \frac{2(-2\sin\theta + 3\sqrt{2}\cos\theta)}{3\pi}. \tag{C.64}$$

The maximum overlap magnitude is

$$\boxed{|\kappa|_{\max} = \sqrt{\kappa_0^2 + \kappa_1^2} = \frac{2\sqrt{22}}{3\pi}}. \tag{C.65}$$

3. Blind angle and max-coupling angle. The blind angle satisfies $\kappa(\theta) = 0$, hence

$$\boxed{\tan\theta_{\text{blind}} = \frac{3\sqrt{2}}{2}}. \tag{C.66}$$

At this angle, the D/N-coupled data vanish:

$$C = 0, \qquad G_W = 0, \qquad R = 0, \qquad \mathcal{P} = 0, \qquad N_0 = 0. \tag{C.67}$$

A positive quadrupole target is therefore impossible on the blind-angle branch.

The positive-overlap maximum occurs at

$$\boxed{\tan\theta_{\max} = \frac{\kappa_1}{\kappa_0} = -\frac{\sqrt{2}}{3}, \qquad \sin^2\theta_{\max} = \frac{2}{11}}. \tag{C.68}$$

This is a small negative admixture of ν_1.

4. Profile-dressed wall stiffness. For the quadrupole wall operator

$$G_\eta = -T_w \frac{d^2}{ds^2} + K_\eta + 6T_\Omega, \tag{C.69}$$

the coherent profile has

$$\boxed{K_{\text{geo}}(\theta) = K_\eta + 6T_\Omega + \frac{\pi^2 T_w}{L^2}\sin^2\theta}. \tag{C.70}$$

The profile-dressed normalization gate is

$$\boxed{K_{\text{geo}}(\theta) = K_{\text{req}}(\theta)}, \tag{C.71}$$

where Stage 026's required stiffness is evaluated with $\kappa \to \kappa(\theta)$:

$$K_{\text{req}}(\theta) = \frac{\lambda_B^2 \kappa(\theta)^2}{\varpi^2} + \frac{Q(\theta)}{\Delta(\theta)} + \frac{\widehat{m}_{\text{rad}}^{\,2}\kappa(\theta)^2(\Omega_U^2\lambda_W + \lambda_R\lambda_U)^2}{N_Q^{\text{target}}\Delta(\theta)^2}, \tag{C.72}$$

with

$$\Delta(\theta) = \Omega_U^2\Omega_W^2 - \lambda_R^2\kappa(\theta)^2, \tag{C.73}$$

$$Q(\theta) = \lambda_U^2\Omega_W^2 + 2\lambda_U\lambda_W\lambda_R\kappa(\theta)^2 + \lambda_W^2\Omega_U^2\kappa(\theta)^2. \tag{C.74}$$

Output. Stage 027 outputs $\kappa(\theta)$, the blind-angle no-go (C.66), the max-coupling angle (C.68), and the profile-dressed branch equation (C.71).

Checks.

- ☐ The profile remains normalized because ν_0 and ν_1 are orthonormal.
- ☐ The blind angle gives $\kappa = 0$, so every D/N-mediated outgoing numerator vanishes.
- ☐ The max-coupling angle follows from aligning $(\cos\theta, \sin\theta)$ with (κ_0, κ_1).
- ☐ The wall stiffness picks up only the ν_1 axial-gradient cost, giving $\Delta K_{\text{ax}} \sin^2\theta$.

Downstream use. Stage 028 derives θ dynamically rather than selecting it by hand. The exact value $\sin^2\theta_{\max} = 2/11$ is later echoed in the D/N branch constants used by the selected-mode formulas.

C.5 Stage 028: Loaded Axial Profile Selection

Part / anchor. Part II; primary anchor **MTDC-T5**.

Claim status. Exact Within Closure **inside the two-mode wall profile and rank-one loading closure.**

Purpose. Stage 028 stops treating the wall-profile angle as an external choice. The support/mixed branch induces an attractive rank-one load along the D/N overlap vector, and the selected profile is the lower eigenvector of the loaded wall matrix.

Inputs. The inputs are the two N/N wall modes, the overlap vector $v = (\kappa_0, \kappa_1)^T$, and a positive loading coefficient α.

Derivation

1. Bare wall matrix and overlap vector. Set

$$q = (q_0, q_1)^T, \qquad q_0^2 + q_1^2 = 1, \qquad q_0 = \cos\theta, \qquad q_1 = \sin\theta. \tag{C.75}$$

The bare quadrupole wall matrix is

$$K_{\text{bare}} = \begin{pmatrix} K_0 & 0 \\ 0 & K_1 \end{pmatrix}, \qquad K_0 = K_\eta + 6T_\Omega, \qquad K_1 = K_0 + \Delta K_{\text{ax}}, \tag{C.76}$$

where

$$\Delta K_{\text{ax}} = \frac{\pi^2 T_w}{L^2}. \tag{C.77}$$

The D/N overlap vector is

$$\boxed{v = (\kappa_0, \kappa_1)^T, \qquad \kappa_0 = \frac{2\sqrt{2}}{\pi}, \qquad \kappa_1 = -\frac{4}{3\pi}}. \tag{C.78}$$

2. Rank-one loaded wall matrix. The minimal attractive loading is

$$\boxed{K_{\text{eff}} = K_{\text{bare}} - \alpha v v^T, \qquad \alpha > 0}. \tag{C.79}$$

The exact eigenvalues are

$$\boxed{\lambda_\pm = \frac{1}{2}\left[K_0 + K_1 - \alpha(\kappa_0^2 + \kappa_1^2) \pm \sqrt{\left(\Delta K_{\text{ax}} + \alpha(\kappa_0^2 - \kappa_1^2)\right)^2 + 4\alpha^2\kappa_0^2\kappa_1^2}\right]}. \tag{C.80}$$

The lower selected profile obeys

$$\boxed{\tan(2\theta_-) = \frac{2\alpha\kappa_0\kappa_1}{\Delta K_{\text{ax}} + \alpha(\kappa_0^2 - \kappa_1^2)}}. \tag{C.81}$$

Since $\kappa_0\kappa_1 < 0$, positive loading drives $\theta_- < 0$, toward the max-coupling direction and away from the blind-angle branch.

3. Softening threshold. The lower eigenvalue reaches zero when $\det K_{\text{eff}} = 0$. This gives

$$\boxed{\alpha_{\text{crit}} = \frac{K_0 K_1}{K_1\kappa_0^2 + K_0\kappa_1^2}}. \tag{C.82}$$

The stable side of this reduced branch is $0 \le \alpha < \alpha_{\text{crit}}$.

Output. Stage 028 outputs the rank-one wall matrix (C.79), exact eigenvalues (C.80), the selected angle law (C.81), and the softening threshold (C.82).

Checks.

- ☐ The eigenvalues follow from the characteristic polynomial of the 2×2 symmetric matrix $K_{\text{bare}} - \alpha vv^T$.
- ☐ The sign of θ_- is fixed because $\kappa_0 > 0$ and $\kappa_1 < 0$.
- ☐ The threshold is checked by $\det(K_{\text{bare}} - \alpha vv^T) = 0$.
- ☐ The loaded branch cannot select the blind angle under positive loading because the selected direction moves toward larger $|v \cdot q|$, not toward zero overlap.

Downstream use. Stage 029 derives α dynamically from the coupled wall/BdG/Maxwell/mixed Schur complement. Stages 030–034 use the selected lower eigenvalue and eigenvector overlap to define λ_-, s_-, and the softening-depth normal form.

C.6 Stage 029: Coupled Response Operator

Part / anchor. Part II; primary anchor **MTDC-T5**.

Claim status. The Schur-complement split is Exact Within Closure **inside the declared two-mode wall and one-support/one-mixed reduced operator. The stable normal-mode and local outgoing-port representation are** Reduced / Controlled Reduction**.**

Purpose. Stage 029 gives the loading coefficient α a microscopic origin. The coupled response Schur complement splits into an isotropic softening ΞI_2 and a directional load αvv^T. That same load carries the outgoing $i\omega^5$ branch into the selected mode.

Inputs. The inputs are the Stage 028 two-mode wall profile, the D/N overlap vector v, one stable support mode, one brane-like internal coordinate, one mixed coordinate, and the compact outgoing $l = 2$ port.

Derivation

1. Dynamic operator. Let

$$D_\eta^{\text{bare}}(\omega) = \operatorname{diag}(K_0 - M_0\omega^2, K_1 - M_1\omega^2). \tag{C.83}$$

Define

$$A_\phi(\omega) = \varpi^2 - \omega^2, \qquad A_U(\omega) = \Omega_U^2 - \omega^2, \qquad A_W(\omega) = \Omega_W^2 - \omega^2 - \Pi_{\text{out}}(\omega). \tag{C.84}$$

The mixed determinant is

$$\Delta_{UW}(\omega) = A_U(\omega)A_W(\omega) - \lambda_R^2\sigma, \qquad \sigma = v \cdot v = \kappa_0^2 + \kappa_1^2 = \frac{88}{9\pi^2}. \tag{C.85}$$

2. Exact Schur-complement split. Eliminating the internal variables gives

$$\boxed{\Sigma_{\mathrm{wall}}(\omega) = \Xi(\omega) I_2 + \alpha(\omega) v v^T}, \tag{C.86}$$

where

$$\boxed{\Xi(\omega) = \frac{\lambda_U^2}{A_U(\omega)}, \qquad \alpha(\omega) = \frac{\lambda_B^2}{A_\phi(\omega)} + \frac{\big(A_U(\omega)\lambda_W + \lambda_R\lambda_U\big)^2}{A_U(\omega)\Delta_{UW}(\omega)}}. \tag{C.87}$$

At zero frequency, on the conservative branch,

$$\boxed{\Xi_0 = \frac{\lambda_U^2}{\Omega_U^2}, \qquad \Delta_0 = \Omega_U^2\Omega_W^2 - \lambda_R^2\sigma, \qquad \alpha_0 = \frac{\lambda_B^2}{\varpi^2} + \frac{(\Omega_U^2\lambda_W + \lambda_R\lambda_U)^2}{\Omega_U^2\Delta_0}}. \tag{C.88}$$

Thus the static matrix is

$$K_{\mathrm{eff}}^{(0)} = \begin{pmatrix} K_0 - \Xi_0 & 0 \\ 0 & K_1 - \Xi_0 \end{pmatrix} - \alpha_0 v v^T. \tag{C.89}$$

3. Outgoing dressing. To first order in the outgoing port,

$$\alpha(\omega) = \alpha_{\mathrm{cons}}(\omega) + \beta(\omega)\Pi_{\mathrm{out}}(\omega) + O(\Pi_{\mathrm{out}}^2), \tag{C.90}$$

where

$$\boxed{\beta(\omega) = \frac{\big(A_U(\omega)\lambda_W + \lambda_R\lambda_U\big)^2}{\Delta_{\mathrm{cons}}(\omega)^2}, \qquad \Delta_{\mathrm{cons}}(\omega) = A_U(\omega)(\Omega_W^2 - \omega^2) - \lambda_R^2\sigma}. \tag{C.91}$$

At zero frequency,

$$\boxed{\beta_0 = \frac{(\Omega_U^2\lambda_W + \lambda_R\lambda_U)^2}{\Delta_0^2} \geq 0}. \tag{C.92}$$

For

$$\Pi_{\mathrm{out}}(\omega) = +i\frac{a^5}{27c_s^5}\omega^5 + O(\omega^7), \tag{C.93}$$

and selected eigenvector e_-, the projected odd coefficient is

$$\boxed{\delta D_-^{\mathrm{odd}}(\omega) = -i\frac{a^5}{27c_s^5}\,\beta_0\,(v\cdot e_-)^2\,\omega^5 + O(\omega^7)}. \tag{C.94}$$

Output. Stage 029 outputs the Schur-complement split (C.86), the static branch data (C.88), the outgoing transfer coefficient (C.92), and the selected odd coefficient (C.94).

Checks.

- ☐ The scalar U-sector contribution is proportional to I_2, while the D/N support and mixed branch are proportional to vv^T.
- ☐ The static directional load α_0 is positive if the stable denominators are positive.
- ☐ The outgoing transfer coefficient β_0 is a square over a squared determinant.
- ☐ Projection onto e_- inserts $(v\cdot e_-)^2$, giving the selected-mode odd coefficient.

Downstream use. Stage 030 normalizes the selected operator and defines $P_{0,-} = \beta_0 s_-/\lambda_-$. Stage 033 rewrites β_0 and α_0 in microscopic coupling variables. Later retarded stages reuse the fact that the outgoing bridge is carried by the same selected directional load.

C.7 Stage 030: Selected-Mode Normalized Response

Part / anchor. Part II; primary anchor **MTDC-T5**.

Claim status. Exact Within Closure **inside the selected lower-mode eigenbranch.**

Purpose. Stage 030 translates the selected lower eigenvalue and odd coefficient into normalized-response language. The stage defines the selected static prefactor $P_{0,-}$, relates it to the Hellmann–Feynman derivative of the eigenvalue, and states the selected quadrupole target.

Inputs. The inputs are the static selected matrix from Stage 029, the lower eigenvalue λ_-, the selected eigenvector e_-, and the overlap $s_- = (v \cdot e_-)^2$.

Derivation

1. Static selected eigenvalue. Set

$$A = K_0 - \Xi_0, \qquad B = K_1 - \Xi_0 = A + \Delta K_{\mathrm{ax}}. \tag{C.95}$$

Define

$$\delta_\kappa = \kappa_0^2 - \kappa_1^2, \qquad \Pi_\kappa = \kappa_0^2\kappa_1^2, \qquad \sigma = \kappa_0^2 + \kappa_1^2. \tag{C.96}$$

The lower eigenvalue is

$$\boxed{\lambda_- = \frac{A + B - \alpha_0\sigma - \sqrt{(\Delta K_{\mathrm{ax}} + \alpha_0\delta_\kappa)^2 + 4\alpha_0^2\Pi_\kappa}}{2}}. \tag{C.97}$$

Let

$$R_\lambda = \sqrt{(\Delta K_{\mathrm{ax}} + \alpha_0\delta_\kappa)^2 + 4\alpha_0^2\Pi_\kappa}. \tag{C.98}$$

The selected overlap is

$$\boxed{s_- = (v \cdot e_-)^2 = \frac{1}{2}\left[\sigma + \frac{(\Delta K_{\mathrm{ax}} + \alpha_0\delta_\kappa)\delta_\kappa + 4\alpha_0\Pi_\kappa}{R_\lambda}\right]}. \tag{C.99}$$

Equivalently, by Hellmann–Feynman,

$$\boxed{s_- = -\frac{d\lambda_-}{d\alpha_0}}. \tag{C.100}$$

2. Normalized selected response. Write

$$D_-(\omega) = D_{-0} + D_{-2}\omega^2 + D_{-4}\omega^4 - iC_{5,-}\omega^5 + O(\omega^6), \qquad D_{-0} = \lambda_-. \tag{C.101}$$

Then

$$Y_-(\omega) = \frac{D_{-0}}{D_-(\omega)} = 1 + u_{2,-}\omega^2 + u_{4,-}\omega^4 + i\Gamma_{5,-}\omega^5 + O(\omega^6), \tag{C.102}$$

with

$$\boxed{u_{2,-} = -\frac{D_{-2}}{D_{-0}}, \qquad u_{4,-} = \frac{D_{-2}^2 - D_{-0}D_{-4}}{D_{-0}^2}, \qquad \Gamma_{5,-} = \frac{C_{5,-}}{D_{-0}}}. \tag{C.103}$$

The Stage 029 odd coefficient gives

$$\Gamma_{5,-} = \frac{a^5}{27c_s^5}\frac{\beta_0 s_-}{\lambda_-}. \tag{C.104}$$

Therefore define

$$\boxed{P_{0,-} = \frac{\beta_0 s_-}{\lambda_-} = -\beta_0\frac{d\ln\lambda_-}{d\alpha_0}}. \tag{C.105}$$

The selected-branch target is

$$\boxed{\widehat{m}_-^{\,2} P_{0,-} = N_Q^{\rm target}}. \tag{C.106}$$

Output. Stage 030 outputs the lower eigenvalue (C.97), the selected overlap (C.99), the Hellmann–Feynman identity (C.100), the selected prefactor (C.105), and the target (C.106).

Checks.

- ☐ Differentiating the eigenvalue with respect to α_0 gives $d\lambda_-/d\alpha_0 = -e_-^T vv^T e_- = -s_-$.
- ☐ Normalizing D_-/D_{-0} gives the response coefficients as quotients by λ_-.
- ☐ The outgoing coefficient contains only $\beta_0 s_-/\lambda_-$, matching the selected projection of Stage 029.

Downstream use. Stage 031 proves that $P_{0,-}$ is monotone on the stable branch. Stage 032 replaces the abstract $\widehat{m}_-$ by the natural D/N source-map factor. Stage 034 uses λ_- and s_- to build the softening-depth normal form.

C.8 Stage 031: Reachability and Stable-Side Monotonicity

Part / anchor. Part II; primary anchor **MTDC-T5**.

Claim status. Exact Within Closure **inside the selected lower-mode branch.**

Purpose. Stage 031 proves that the selected prefactor is strictly increasing as the rank-one loading approaches the softening threshold. This turns selected normalization into a one-dimensional reachability theorem.

Inputs. The stage uses $\lambda_-(\alpha_0)$, $s_-(\alpha_0)$, $P_{0,-} = \beta_0 s_-/\lambda_-$, and the stable interval $0 \le \alpha_0 < \alpha_{\rm crit}$.

Derivation

1. Derivative of the selected overlap. The exact derivative is

$$\boxed{\frac{ds_-}{d\alpha_0} = \frac{2(\Delta K_{\mathrm{ax}})^2\Pi_\kappa}{R_\lambda^3} > 0}. \tag{C.107}$$

Using $d\lambda_-/d\alpha_0 = -s_-$, one obtains

$$\boxed{\frac{dP_{0,-}}{d\alpha_0} = \beta_0\frac{(ds_-/d\alpha_0)\lambda_- + s_-^2}{\lambda_-^2} > 0} \tag{C.108}$$

whenever $\lambda_- > 0$ and $\beta_0 > 0$.

2. Endpoints. At zero loading,

$$P_{0,-}(0) = \frac{\beta_0\kappa_0^2}{A}. \tag{C.109}$$

The refined threshold, now with the isotropic U-softening absorbed into A, B, is

$$\boxed{\alpha_{\mathrm{crit}} = \frac{AB}{B\kappa_0^2 + A\kappa_1^2}}. \tag{C.110}$$

As $\alpha_0 \to \alpha_{\mathrm{crit}}^-$, $\lambda_- \to 0^+$ while s_- remains positive, so

$$P_{0,-} \to +\infty. \tag{C.111}$$

Thus the stable branch spans

$$P_{0,-}(0) \le P_{0,-} < +\infty. \tag{C.112}$$

3. Unique crossing. Because $P_{0,-}$ is continuous and strictly increasing on the stable interval, every target satisfying

$$P_{\mathrm{target}} > P_{0,-}(0) \tag{C.113}$$

has a unique $\alpha_* \in (0, \alpha_{\mathrm{crit}})$ such that $P_{0,-}(\alpha_*) = P_{\mathrm{target}}$. For the quadrupole target,

$$P_{\mathrm{target}} = \frac{N_Q^{\mathrm{target}}}{\widehat{m}_-^2}. \tag{C.114}$$

Output. Stage 031 outputs the derivative identities (C.107)–(C.108), the refined threshold (C.110), and the unique stable crossing theorem.

Checks.

- ☐ The positivity of $ds_-/d\alpha_0$ follows from $(\Delta K_{\mathrm{ax}})^2\Pi_\kappa > 0$ and $R_\lambda > 0$.
- ☐ The derivative of $P_{0,-}$ is positive because both terms in the numerator are nonnegative and $s_-^2 > 0$.
- ☐ The divergence at softening follows from $\lambda_- \to 0^+$ with $s_- > 0$.

Downstream use. Stage 032 modifies the target by fixing $\widehat{m}_-$. Stage 035 upgrades the reachability statement into the explicit monotone shape function $F(\xi, \delta)$.

C.9 Stage 032: Source Map from Finite-Throat Mode Integrals

Part / anchor. Part II; primary anchor **MTDC-T5**.

Claim status. EXACT WITHIN CLOSURE **inside the natural D/N source-map closure.**

Purpose. Stage 032 removes the abstract selected source-map factor on the natural D/N branch. The same finite-throat D/N load that drives the selected wall profile also couples the orbital/worldtube source to the selected mode, giving $\widehat{m}_-^2 = s_-/\kappa_0^2$.

Inputs. The inputs are the selected eigenvector e_-, the overlap vector v, and a source coupled through the D/N load profile.

Derivation

1. Natural D/N source coupling. Use

$$L_{\mathrm{src}} = g_Q Q_{\mathrm{STF}} \int_0^L \eta(s) f_0(s)\, ds. \tag{C.115}$$

For $\eta = q_0\nu_0 + q_1\nu_1$, this gives source vector proportional to

$$v = (\kappa_0, \kappa_1)^T. \tag{C.116}$$

Projection on the selected eigenvector gives

$$J_- = g_Q Q_{\mathrm{STF}}(v \cdot e_-). \tag{C.117}$$

At zero loading, the selected mode is the constant branch and

$$J_-(0) = g_Q Q_{\mathrm{STF}} \kappa_0. \tag{C.118}$$

Thus the selected source-map factor is

$$\boxed{\widehat{m}_- = \frac{v \cdot e_-}{\kappa_0}, \qquad \widehat{m}_-^2 = \frac{s_-}{\kappa_0^2}}. \tag{C.119}$$

Since s_- increases from κ_0^2 toward $\sigma = \kappa_0^2 + \kappa_1^2$,

$$\boxed{1 \le \widehat{m}_-^2 < \frac{\sigma}{\kappa_0^2} = \frac{11}{9}}. \tag{C.120}$$

The natural source-map factor is modest and cannot absorb arbitrary normalization error.

2. Source-map-corrected selected product. Combining $P_{0,-} = \beta_0 s_-/\lambda_-$ with (C.119) gives

$$\boxed{N_-(\alpha_0) := \widehat{m}_-^2 P_{0,-} = \frac{\beta_0 s_-^2}{\kappa_0^2 \lambda_-}}. \tag{C.121}$$

The selected target is

$$\boxed{N_-(\alpha_0) = N_Q^{\mathrm{target}}}. \tag{C.122}$$

Output. Stage 032 outputs the source-map factor (C.119), the bound (C.120), and the invariant selected product (C.121).

Checks.

- ☐ The source vector is proportional to the same overlap vector v because the source uses the same D/N load profile.
- ☐ At zero loading the normalization denominator is κ_0, giving $\widehat{m}_- = 1$ at onset.
- ☐ The upper bound 11/9 follows from $\sigma/\kappa_0^2 = (\kappa_0^2 + \kappa_1^2)/\kappa_0^2 = 1 + 2/9$.

Downstream use. Stage 033 uses N_- to write the microscopic target. Stage 034 rewrites N_- in softening-depth form. The source-map result is the point where the selected branch stops hiding behind an unspecified $\widehat{m}_-$.

C.10 Stage 033: Microscopic Normalization Equation and Stability Gate

Part / anchor. Part II; primary anchor **MTDC-T5**.

Claim status. The coupling-level equations are Exact Within Closure **inside the first finite-throat kernel closure. Realization by the actual branch is** Open**.**

Purpose. Stage 033 writes the selected target in microscopic coupling variables and states the stability and onset gates. This is the last form of the branch test before the eigenvalue variables are eliminated in favor of the softening depth.

Inputs. The inputs are $A = K_0 - g_U^2/\Omega_U^2$, the mixed determinant Δ_0, the transfer numerator $\mathcal{X}$, and the selected product N_- from Stage 032.

Derivation

1. Microscopic coupling variables. Define

$$\boxed{A = K_0 - \frac{g_U^2}{\Omega_U^2}, \qquad \Delta_0 = \Omega_U^2\Omega_W^2 - g_R^2\sigma, \qquad \mathcal{X} = \Omega_U^2 g_W + g_R g_U}. \tag{C.123}$$

Then

$$\boxed{\beta_0 = \frac{\mathcal{X}^2}{\Delta_0^2}, \qquad \alpha_0 = \frac{g_B^2}{\varpi^2} + \frac{\mathcal{X}^2}{\Omega_U^2\Delta_0}}. \tag{C.124}$$

The stability gate is

$$\boxed{\Delta_0 > 0, \qquad A > 0, \qquad \alpha_0 < \alpha_{\rm crit}}. \tag{C.125}$$

Using $\kappa_0^2 = 8/\pi^2$ and $\kappa_1^2 = 16/(9\pi^2)$,

$$\boxed{\alpha_{\rm crit} = \frac{9\pi^2 A(A + \Delta K_{\rm ax})}{8(11A + 9\Delta K_{\rm ax})}}. \tag{C.126}$$

In coupling form,

$$\frac{g_B^2}{\varpi^2} + \frac{\mathcal{X}^2}{\Omega_U^2 \Delta_0} < \frac{9\pi^2 A(A + \Delta K_{\mathrm{ax}})}{8(11A + 9\Delta K_{\mathrm{ax}})}. \tag{C.127}$$

2. Microscopic selected target. The selected target is

$$\boxed{\frac{\mathcal{X}^2}{\Delta_0^2} \frac{s_-(\alpha_0)^2}{\kappa_0^2 \lambda_-(\alpha_0)} = N_Q^{\mathrm{target}}, \qquad \alpha_0 = \frac{g_B^2}{\varpi^2} + \frac{\mathcal{X}^2}{\Omega_U^2 \Delta_0}}. \tag{C.128}$$

3. Onset condition. At $\alpha_0 = 0$,

$$s_-(0) = \kappa_0^2, \qquad \lambda_-(0) = A, \tag{C.129}$$

so

$$\boxed{N_-(0) = \frac{\beta_0 \kappa_0^2}{A} = \frac{\mathcal{X}^2 \kappa_0^2}{A \Delta_0^2}}. \tag{C.130}$$

Because N_- increases on the positive-loading branch, a necessary condition for a hit is

$$\boxed{N_-(0) \le N_Q^{\mathrm{target}}, \qquad \text{equivalently} \qquad \mathcal{X}^2 \le \frac{N_Q^{\mathrm{target}} A \Delta_0^2}{\kappa_0^2}}. \tag{C.131}$$

If this fails, the selected branch starts above the target and cannot cross it.

4. Weak-loading expansion. For small α_0,

$$N_-(\alpha_0) = \frac{\beta_0 \kappa_0^2}{A} + \alpha_0 \frac{\beta_0 \kappa_0^2 (4A\kappa_1^2 + \Delta K_{\mathrm{ax}} \kappa_0^2)}{A^2 \Delta K_{\mathrm{ax}}} + O(\alpha_0^2). \tag{C.132}$$

With the exact D/N constants,

$$N_-(\alpha_0) = \frac{8\beta_0}{\pi^2 A} + \alpha_0 \frac{64\beta_0(8A + 9\Delta K_{\mathrm{ax}})}{9\pi^4 A^2 \Delta K_{\mathrm{ax}}} + O(\alpha_0^2). \tag{C.133}$$

Output. Stage 033 outputs the coupling-level definitions (C.123), the stability gate (C.125), the microscopic target (C.128), the onset condition (C.131), and the weak-loading expansion (C.132).

Checks.

- ☐ The stability condition separates internal mixed-block stability $\Delta_0 > 0$, scalar-wall stability $A > 0$, and selected lower-mode stability $\alpha_0 < \alpha_{\mathrm{crit}}$.
- ☐ The onset condition follows from monotonicity of N_-.
- ☐ The weak-loading coefficient is positive for positive A, ΔK_{ax}, and β_0.

Downstream use. Stage 034 replaces α_0, s_-, and λ_- by the softening depth x. Stages 035–036 non-dimensionalize the same target and feasibility data.

C.11 Stage 034: Softening-Depth Normal Form

Part / anchor. Part II; primary anchor **MTDC-T5**.

Claim status. Exact Within Closure **inside the rank-one selected D/N branch.**

Purpose. Stage 034 eliminates the selected eigenvector variables by introducing a scalar softening depth $x = A - \lambda_-$. In this variable, the total loading, selected source overlap, invariant normalization product, and required support contribution all become explicit rational functions.

Inputs. The inputs are the rank-one two-level secular equation and the selected product N_- from Stage 032.

Derivation

1. Softening-depth variable and secular equation. Define

$$\boxed{x := A - \lambda_-}. \tag{C.134}$$

On the stable branch,

$$0 \le x < A, \qquad \lambda_- = A - x. \tag{C.135}$$

The rank-one secular equation is

$$1 = \alpha_0 \left(\frac{\kappa_0^2}{x} + \frac{\kappa_1^2}{x + \Delta K_{\mathrm{ax}}} \right). \tag{C.136}$$

Solving for total loading gives

$$\boxed{\alpha_0(x) = \frac{x(x + \Delta K_{\mathrm{ax}})}{\kappa_0^2 (x + \Delta K_{\mathrm{ax}}) + \kappa_1^2 x}}. \tag{C.137}$$

2. Selected overlap and invariant product. The selected overlap can be written as $s_- = dx/d\alpha_0$, giving

$$\boxed{s_-(x) = \frac{\left[\kappa_0^2 (x + \Delta K_{\mathrm{ax}}) + \kappa_1^2 x\right]^2}{\kappa_0^2 (x + \Delta K_{\mathrm{ax}})^2 + \kappa_1^2 x^2}}. \tag{C.138}$$

The invariant selected product is

$$\boxed{N_-(x) = \frac{\beta_0 \left[\kappa_0^2 (x + \Delta K_{\mathrm{ax}}) + \kappa_1^2 x\right]^4}{\kappa_0^2 (A - x) \left[\kappa_0^2 (x + \Delta K_{\mathrm{ax}})^2 + \kappa_1^2 x^2\right]^2}}. \tag{C.139}$$

The target is

$$\boxed{N_-(x) = N_Q^{\mathrm{target}}, \qquad 0 \le x < A}. \tag{C.140}$$

3. Monotone loading and required support. Differentiating (C.137) gives

$$\boxed{\frac{d\alpha_0}{dx} = \frac{\kappa_0^2(x+\Delta K_{\rm ax})^2 + \kappa_1^2 x^2}{\left[\kappa_0^2(x+\Delta K_{\rm ax}) + \kappa_1^2 x\right]^2} > 0}. \tag{C.141}$$

Thus x is a one-to-one stable branch coordinate. Splitting

$$\alpha_0 = \frac{g_B^2}{\varpi^2} + \alpha_{\rm mix}, \qquad \alpha_{\rm mix} = \frac{\mathcal{X}^2}{\Omega_U^2 \Delta_0}, \tag{C.142}$$

the required support loading is

$$\boxed{\frac{g_{B,\rm req}^2}{\varpi^2} = \frac{x(x+\Delta K_{\rm ax})}{\kappa_0^2(x+\Delta K_{\rm ax}) + \kappa_1^2 x} - \frac{\mathcal{X}^2}{\Omega_U^2 \Delta_0}}. \tag{C.143}$$

The nonnegativity of this expression is the support-feasibility condition.

Output. Stage 034 outputs the softening-depth variable (C.134), the loading law (C.137), the selected overlap (C.138), the normalization product (C.139), and the required support formula (C.143).

Checks.

- ☐ The secular equation follows from the rank-one determinant lemma.
- ☐ The identity $s_- = dx/d\alpha_0$ follows because $x = A - \lambda_-$ and $s_- = -d\lambda_-/d\alpha_0$.
- ☐ $d\alpha_0/dx > 0$ proves that x is a good branch coordinate.
- ☐ The support requirement is obtained by subtracting the mixed baseline from the required total load.

Downstream use. Stage 035 introduces $\xi = x/A$ and $\delta = \Delta K_{\rm ax}/A$, producing the dimensionless normalization shape function F. Stage 036 uses the same loading law to define G.

C.12 Stage 035: Dimensionless D/N Normalization Locus

Part / anchor. Part II; primary anchor **MTDC-T5**.

Claim status. Exact Within Closure **inside the D/N selected-branch normal form.**

Purpose. Stage 035 non-dimensionalizes the softening-depth target. The selected normalization product factors into an onset value times a universal D/N shape function $F(\xi, \delta)$. Strict monotonicity of F upgrades the Stage 033 onset inequality into a unique stable normalization locus.

Inputs. The inputs are the Stage 034 expression for $N_-(x)$ and the exact D/N constants $\kappa_0^2 = 8/\pi^2$, $\kappa_1^2 = 16/(9\pi^2)$.

Derivation

1. Dimensionless variables. Use

$$\boxed{\kappa_0^2 = \frac{8}{\pi^2}, \qquad \kappa_1^2 = \frac{16}{9\pi^2}, \qquad \eta = \frac{\kappa_1^2}{\kappa_0^2} = \frac{2}{9}}. \tag{C.144}$$

Define

$$\boxed{\xi = \frac{x}{A}, \qquad \delta = \frac{\Delta K_{\rm ax}}{A}, \qquad 0 \le \xi < 1}. \tag{C.145}$$

Then

$$N_-(x) = N_-(0)F(\xi,\delta), \qquad N_-(0) = \frac{\beta_0\kappa_0^2}{A}. \tag{C.146}$$

The exact shape function is

$$\boxed{F(\xi,\delta) = \frac{(9\delta + 11\xi)^4}{81(1-\xi)(9\delta^2 + 18\delta\xi + 11\xi^2)^2}}. \tag{C.147}$$

The dimensionless target ratio is

$$\boxed{R_{\rm target} := \frac{N_Q^{\rm target}}{N_-(0)} = \frac{N_Q^{\rm target}A}{\beta_0\kappa_0^2}}. \tag{C.148}$$

The normalization locus is

$$\boxed{F(\xi,\delta) = R_{\rm target}}. \tag{C.149}$$

2. Monotonicity and endpoints. Differentiating gives

$$\boxed{\frac{\partial F}{\partial \xi} = \frac{(9\delta + 11\xi)^3\,(81\delta^3 + 72\delta^2 + 189\delta^2\xi + 297\delta\xi^2 + 121\xi^3)}{81(1-\xi)^2(9\delta^2 + 18\delta\xi + 11\xi^2)^3} > 0}. \tag{C.150}$$

Moreover,

$$F(0,\delta) = 1, \qquad \lim_{\xi\to 1^-} F(\xi,\delta) = +\infty. \tag{C.151}$$

Near softening,

$$F(\xi,\delta) \sim \frac{C(\delta)}{1-\xi}, \qquad C(\delta) = \frac{(9\delta + 11)^4}{81(9\delta^2 + 18\delta + 11)^2}. \tag{C.152}$$

Therefore

$$\boxed{R_{\rm target} < 1 \Rightarrow \text{no stable hit}, \qquad R_{\rm target} = 1 \Rightarrow \xi_{\rm req} = 0, \qquad R_{\rm target} > 1 \Rightarrow \exists!\,\xi_{\rm req} \in (0,1)}. \tag{C.153}$$

3. Required total loading. The total loading law becomes

$$\boxed{\alpha_{\rm req}(\xi,\delta) = \frac{9\pi^2 A\xi(\xi + \delta)}{8(9\delta + 11\xi)}}. \tag{C.154}$$

As $\xi \to 1^-$, this approaches the refined stable threshold

$$\alpha_{\rm crit} = \frac{9\pi^2 A(1+\delta)}{8(11 + 9\delta)}. \tag{C.155}$$

4. Useful asymptotics. Near onset,

$$F(\xi,\delta) = 1 + \left(1 + \frac{8}{9\delta}\right)\xi + \left(1 + \frac{8}{9\delta} - \frac{28}{27\delta^2}\right)\xi^2 + O(\xi^3). \tag{C.156}$$

If $R_{\text{target}} = 1 + \varepsilon_R$, then

$$\xi_{\text{req}} \simeq \frac{\varepsilon_R}{1 + 8/(9\delta)}. \tag{C.157}$$

For large target ratio,

$$1 - \xi_{\text{req}} \simeq \frac{C(\delta)}{R_{\text{target}}}. \tag{C.158}$$

Output. Stage 035 outputs the shape function (C.147), the target ratio (C.148), the monotonicity derivative (C.150), the existence/uniqueness theorem (C.153), and the required total loading (C.154).

Checks.

- ☐ Substitution of the D/N constants into Stage 034 gives F.
- ☐ Every factor in $\partial_\xi F$ is positive for $\delta > 0$ and $0 \le \xi < 1$.
- ☐ The endpoint $F(0,\delta) = 1$ reproduces the onset condition.
- ☐ The divergence as $\xi \to 1^-$ matches the selected eigenvalue approaching zero.

Downstream use. Stage 036 pairs F with the support-feasibility function G. Part III later computes continuum placement data that determine R_{target}, M_{mix}, and δ.

C.13 Stage 036: Dimensionless Support-Feasibility Frontier

Part / anchor. Part II; primary anchor **MTDC-T5**.

Claim status. The support frontier is Exact Within Closure **inside the D/N selected-branch normal form. Actual physical placement on the admissible frontier is** Open**.**

Purpose. Stage 036 isolates the support-feasibility condition that remains after the normalization locus has selected ξ_{req}. The total loading must exceed the mixed-sector baseline by a nonnegative support/BdG contribution. This requirement becomes the exact frontier $M_{\text{mix}} \le G(\xi_{\text{req}}, \delta)$.

Inputs. The inputs are the Stage 035 total loading $\alpha_{\text{req}}(\xi,\delta)$, the mixed-sector baseline $\alpha_{\text{mix}} = \mathcal{X}^2/(\Omega_U^2\Delta_0)$, and the dimensionless variables ξ, δ.

Derivation

1. Dimensionless mixed baseline and support function. Split

$$\alpha_{\text{req}} = \alpha_{\text{mix}} + \frac{g_{B,\text{req}}^2}{\varpi^2}, \qquad \alpha_{\text{mix}} = \frac{\mathcal{X}^2}{\Omega_U^2\Delta_0}. \tag{C.159}$$

Define

$$\boxed{M_{\text{mix}} := \frac{8\alpha_{\text{mix}}}{\pi^2 A} = \frac{8\mathcal{X}^2}{\pi^2 A \Omega_U^2 \Delta_0}}. \tag{C.160}$$

The dimensionless support-feasibility function is

$$\boxed{G(\xi,\delta) := \frac{8\alpha_{\text{req}}}{\pi^2 A} = \frac{9\xi(\xi+\delta)}{9\delta+11\xi}}. \tag{C.161}$$

Therefore

$$\boxed{\frac{g_{B,\text{req}}^2}{\varpi^2} = \frac{\pi^2 A}{8}\left[G(\xi_{\text{req}},\delta) - M_{\text{mix}}\right]}. \tag{C.162}$$

The support/BdG sector is feasible exactly when

$$\boxed{M_{\text{mix}} \le G(\xi_{\text{req}},\delta)}. \tag{C.163}$$

2. Monotonicity and endpoints. The derivative is

$$\boxed{\frac{\partial G}{\partial \xi} = \frac{9(9\delta^2 + 18\delta\xi + 11\xi^2)}{(9\delta+11\xi)^2} > 0}. \tag{C.164}$$

The endpoints are

$$G(0,\delta) = 0, \qquad G_{\max}(\delta) = \lim_{\xi\to 1^-} G(\xi,\delta) = \frac{9(1+\delta)}{9\delta+11}. \tag{C.165}$$

The maximum is the dimensionless form of the refined stability bound on the mixed-sector baseline.

3. Parametric admissibility frontier. Stages 035 and 036 together define the parametric stable frontier

$$\boxed{\xi \in [0,1) \longmapsto (F(\xi,\delta), G(\xi,\delta))}. \tag{C.166}$$

For fixed δ, the normalization target chooses ξ_{req} by

$$F(\xi_{\text{req}},\delta) = R_{\text{target}}, \tag{C.167}$$

and the support baseline must lie below the frontier at that same point.

Near onset,

$$G(\xi,\delta) = \xi - \frac{2\xi^2}{9\delta} + O(\xi^3). \tag{C.168}$$

Combined with (C.157),

$$M_{\text{crit}} \simeq \frac{R_{\text{target}} - 1}{1 + 8/(9\delta)} \tag{C.169}$$

for targets just above onset.

4. Final admissibility test. The final Stage 036 branch test is

$$\boxed{R_{\text{target}} \ge 1, \qquad F(\xi_{\text{req}},\delta) = R_{\text{target}}, \qquad M_{\text{mix}} \le G(\xi_{\text{req}},\delta)}. \tag{C.170}$$

If $R_{\text{target}} < 1$, the branch cannot hit the target on the stable side. If $R_{\text{target}} \ge 1$ but $M_{\text{mix}} > G(\xi_{\text{req}},\delta)$, the required support loading is negative and the physical branch is inadmissible in this reduced closure.

Output. Stage 036 outputs the dimensionless mixed baseline (C.160), the support-feasibility function (C.161), the required support loading (C.162), the monotonicity derivative (C.164), and the final admissibility test (C.170).

Checks.

- ☐ Substituting α_{req} from Stage 035 into $8\alpha_{\mathrm{req}}/(\pi^2 A)$ gives G.
- ☐ $\partial_\xi G > 0$ for $\delta > 0$ and $0 \le \xi < 1$.
- ☐ $M_{\mathrm{mix}} \le G$ is exactly the nonnegativity of $g^2_{B,\mathrm{req}}$.
- ☐ The parametric map (F, G) is one-dimensional at fixed δ, so the selected reduced branch has no remaining free hidden eigenvector parameter at this order.

Downstream use. Part III begins by computing continuum-kernel placement data that feed A, ΔK_{ax}, β_0, M_{mix}, and hence R_{target}, δ, and the exact frontier. Later support-completion stages ask whether rank-two support and coherent D/N support can place the actual branch inside the admissible region.

C.14 Part II appendix closure and handoff to Part III

Stages 024–036 close the second verification block. The block does not prove that the full nonlinear moving-throat PDE realizes the desired outgoing-normalization value. It proves something narrower and necessary: inside the declared finite-throat selected-branch closure, the normalization problem is no longer a vague choice of prefactors. It is a controlled branch-placement problem with exact angular normalization, exact finite-throat D/N overlap constants, exact selected-mode spectral formulas, and exact dimensionless frontier functions.

The stable outputs are:

1. The canonical real-STF angular source map has $\widehat{m}_{\mathrm{ang}} = 1$.
2. Any $O(3)$-invariant reduced kernel collapses the grouped real 20/21/22 lanes to a common D_n, N_n packet.
3. The first weak-axisymmetric quadrupole splitting has signature $(1, 1/2, -1)$, so grouped defects satisfy $b = 3a$.
4. The minimal isotropic one-mode branch has an explicit target ratio and stability gate.
5. The concrete finite-throat D/N branch has $\kappa = 2\sqrt{2}/\pi$.
6. The first nonconstant N/N wall family has an exact blind-angle no-go and a max-coupling angle with $\sin^2\theta_{\max} = 2/11$.
7. The support/mixed load selects the wall profile dynamically through a rank-one matrix $K_{\mathrm{bare}} - \alpha v v^T$.
8. The dynamic wall/BdG/Maxwell/mixed Schur complement splits as $\Xi I_2 + \alpha v v^T$.
9. The selected outgoing prefactor is $P_{0,-} = \beta_0 s_-/\lambda_-$, and the natural D/N source map changes the invariant product to $N_- = \beta_0 s_-^2/(\kappa_0^2 \lambda_-)$.
10. The selected branch is exactly parameterized by the softening depth $x = A - \lambda_-$.

11. The final reduced admissibility problem is the two-condition frontier

$$F(\xi_{\rm req}, \delta) = R_{\rm target}, \qquad M_{\rm mix} \le G(\xi_{\rm req}, \delta). \tag{C.171}$$

The handoff to Part III is therefore precise. The next block must compute the continuum-kernel quantities

$$A, \qquad \Delta K_{\rm ax}, \qquad \beta_0, \qquad M_{\rm mix}, \qquad R_{\rm target}, \qquad \delta, \tag{C.172}$$

and then ask whether the actual support-completed branch lands in the admissible region. Part II has supplied the frontier; Part III begins the placement problem.

Appendix D

Stage Appendix: Part III – Support Completion, Continuum Placement, and Family–1 Branch Construction

Stages 037–090; anchors MTDC-T6 and MTDC-T7

This appendix is the detailed verification ledger for the third theorem block of the moving-throat derivation. The corresponding main-text chapter, chapter 3, gives the readable theorem narrative. The purpose here is stricter: to keep the stage-by-stage algebra, branch conditions, support-completion maps, parent-overlap thresholds, and explicit Family–1 verdict in a form that a reader can audit without reconstructing the 253-stage source tree.

Part II ended with a selected finite-throat test. In its most compressed form, that test asked for a softening depth ξ satisfying a normalization locus and a support-feasibility inequality,

$$F(\xi_{\rm req}, \delta) = R_{\rm target}, \qquad M_{\rm mix} \le G(\xi_{\rm req}, \delta). \tag{D.1}$$

Part III asks how the quantities in (D.1) are populated by an explicit finite-throat continuum branch, how rank–2 support completion changes the selected mode, how coherent D/N support supplies the missing loading, and whether the first concrete Family–1 wall branch passes the support/source side of the outgoing-quadrupole theorem gate.

The final support-side result of this appendix is deliberately conditional but strong. Under the natural minimal isotropic contact-plus-pole conservative precursor,

$$Y_Q^{\rm cons}(\omega) = \frac{3}{4} + \frac{1/4}{1 - \omega^2/\Omega_Q^2}, \qquad \Omega_Q = \frac{3c_s}{2a}, \tag{D.2}$$

the explicit Family–1 support/source side selects

$$\boxed{\rho_\alpha = \frac{4}{3}, \qquad \zeta_{\rm req} = \frac{1}{3}, \qquad {\rm Pe}_{\rm req} = 0}. \tag{D.3}$$

Thus, within the declared reduced branch, the support/source side is not the remaining bottleneck. The remaining reduced gap is the Part IV problem: deriving the minimal isotropic conservative quadrupole precursor from the actual grouped real P_2 and geometry branch.

Source layout. Stages 037–090 are now sourced from the canonical files `stages/stage_037.tex` through `stages/stage_090.tex`. The raw Markdown notes and audit artifacts are tracked in a repo-local provenance index, so they do not have to appear in the reader-facing PDF. The stage files themselves carry the executable SymPy and Mathematica audit references.

Table D.1: Part III detailed stage-audit map.

Stage	Source stage	Primary status	Verification output
037	Continuum-kernel extraction	Exact Within Closure / Reduced / Controlled Reduction	Closed formulas for A, ΔK_{ax}, α_{mix}, β_0, M_{mix}, δ, and stability.
038	Dimensionless continuum placement	Exact Within Closure	Ratios $(\epsilon_\eta, \epsilon_W, \rho, Z_W, \delta_0, \Lambda)$, placement map, and product law.
039	First non-flat U doublet	Exact Within Closure	Split placement, direction-splitting invariant, and collinearity iff condition.
040	Generalized selected branch	Exact Within Closure	Source/loading mismatch functions F_{q,η_s}, G_q, and split–U functions F_U, G_U.
041	Rank–2 support completion	Exact Within Closure	Linear determinant in support load and exact n_{req} theorem.
042	Rank–2 selected-mode normalization	Exact Within Closure	Selected eigenvector and source/loading overlaps with distinct outgoing, support, and source directions.
043	Physical support direction extraction	Reduced / Controlled Reduction / Exact Within Closure	Continuum support direction data needed for the tracking/source-tied decision.
044	Continuum-selected rank–2 closure	Exact Within Closure	Physical softening depth fixed by a quadratic and selected normalization gate.
045	Coherent local D/N support kernel	Exact Within Closure	Coherent condition forcing exact tracking, $R_\phi = R_U = R_{\mathrm{tr}}$.
046	Tracking branch bounds	Exact Within Closure	Monotonicity in R and proof that first split–U tracking worsens the target.
047	Coherent-kernel dimensionless map	Exact Within Closure	Support enhancement factor $S(\zeta;\epsilon)$ and R_{target} independent of ζ.
048	Support-compensation theorem	Exact Within Closure / Open	Unique ζ_{req} threshold and no reduced-level support no-go.

Stage	Source stage	Primary status	Verification output
049	D/N overlap extraction of ζ	Exact Within Closure	Physical support ratio ζ_n^{phys}, uniform-source overlaps, and same-operator twin family.
050	ζ-threshold comparison	Exact Within Closure	Lowest symmetric twin succeeds iff $\zeta_{\mathrm{req}} \leq 1$; higher harmonics are overlap/stiffness suppressed.
051	Lowest-twin sufficiency criterion	Exact Within Closure	Branch-product criterion $\Pi_{\mathrm{tr}} \leq 2C_{\mathrm{mix}}$.
052	Non-twin asymmetry requirement	Exact Within Closure	Three-regime support split and excess Δ_ζ.
053	Overlap-boost window	Exact Within Closure	Exponential source family and ceiling $\Omega_0^2 \leq \pi^2/4$.
054	Robin-compliance softening	Exact Within Closure	Root $y \tan y = \eta$, support-softening factor, and softening ceiling.
055	Non-twin reachability	Exact Within Closure	Combined overlap/compliance reachability window for the lowest support lane.
056	Transport origin of source asymmetry	Exact Within Closure	Drift-diffusion zero-flux branch and Peclet-number identification.
057	Physical $(\mathrm{Pe}, \kappa, \eta)$ map	Exact Within Closure	Physical support ratio $\zeta_0(\mathrm{Pe}, \eta; \kappa)$.
058	Coupled support/source operator	Exact Within Closure	Self-consistent equation $\mathrm{Pe}_* = \Xi\Delta(\mathrm{Pe}_*; \kappa, \eta)$ and bracket.
059	Operator residual bounds	Exact Within Closure	Exact fail/succeed thresholds Ξ_{fail}, Ξ_{suff}.
060	Entropic microclosure	Exact Within Closure	Positive source/free-energy closure reproducing the transport source family.
061	Microscopic gain thresholds	Exact Within Closure	Dimensionless gain phase diagram and geometry-controlled success/failure surfaces.
062	Parent-action gain projection	Exact / Reduced / Controlled Reduction	GNLS compressional projection of the microscopic gain into parent overlap variables.
063	Parent-overlap thresholds	Exact Within Closure	Amplitude/coherence thresholds and Cauchy no-go.
064	Parent equilibrium alignment	Exact Within Closure	Equilibrium source/support relation $\chi_\sigma = g_\phi \chi_\phi / H$ and matched-layer gain.

Stage	Source stage	Primary status	Verification output
065	Thin-wall confinement branch	Exact Within Closure / Reduced / Controlled Reduction	Wall amplitude thresholds for $V_{\mathrm{conf}} = V_0 f((r-a)/\ell)$.
066	Wall figure of merit	Exact Within Closure	Single wall control W_{wall} and exact window.
067	Sech–Gaussian benchmark	Exact Within Closure / Numerically Located	Coherence resonance at $w_g/w_f = \sqrt{\pi}$.
068	Resonance-corrected thresholds	Exact Within Closure	Matched thresholds modified by $P_{\mathrm{res}} \simeq 1.005612$.
069	Final support/source verdict	Exact Within Closure / Open	Universal fail, matched success, and profile-sensitive middle band.
070	GNLS wall-shell reduction	Exact Within Closure / Reduced / Controlled Reduction	Parent shell formulas for T_X, K_X, κ, and W_{wall}.
071	Canonical tanh-wall branch	Exact Within Closure	Exact $I_f = 1/3$, $I_g = 4/15$, local mouth closure, and branch controls.
072	Explicit branch threshold surfaces	Exact Within Closure	Fail/succeed surfaces in $(\chi_s, \Lambda_\ell, \Upsilon_w)$.
073	Family–1 geometry map	Reduced / Controlled Reduction / Exact Within Closure	$L/a = 37/20$, $\ell/a = 1/20$, $\Lambda_\ell = \eta = 37$.
074	Healing-length lock	Reduced / Controlled Reduction / Exact Within Closure	$\ell = \hbar/(2mc_{s,w})$, $\chi_s = 37/2$, $\kappa = 12321/5$.
075	Family–1 threshold window	Exact Within Closure	Explicit Θ_w fail/succeed window.
076	$n = 5$ wall-depth lock	Exact Within Closure	$\Theta_w = 25\lambda_\mu^2\rho_w^2$.
077	Shell-weighted extraction	Reduced / Controlled Reduction / Numerically Located	$\Theta_w^{(\chi)}$ and conservative floor $\Theta_w^{(J)}$.

Stage	Source stage	Primary status	Verification output
078	Explicit Family–1 branch verdict	Exact Within Closure	Wall-depth side is not the bottleneck except for anomalously large demand.
079	Demand-to-transport map	Exact Within Closure	$\zeta_{\mathrm{F1}}(\mathrm{Pe}) = A_{\mathrm{F1}}\Omega_{\mathrm{Pe}}^2$ and hard ceiling.
080	Demand thresholds in ζ	Exact Within Closure	Family–1 wall-depth windows converted to support-ratio demand windows.
081	Branch-product demand window	Exact Within Closure	Thresholds in Π_{tr} with blocking parameter.
082	Master quadrupole residual	Exact Within Closure / Open	Full reduced residual $R_{\mathrm{quad}} = \zeta_{\mathrm{req}} - \zeta_{\mathrm{phys}}$.
083	Direct operator-selected Family–1 window	Exact Within Closure / Numerically Located	Direct evaluation of the operator-selected support window.
084	Reduced PDE write-up skeleton	Exact Within Closure / Open	Integrated reduced hierarchy and final outgoing product.
085	Normalization-factor cancellation	Exact Within Closure	Selected support theorem depends only on $\rho_\alpha = \alpha_{\mathrm{req}}/\alpha_{\mathrm{mix}}$.
086	Family–1 loading-ratio window	Exact Within Closure	Pure loading-ratio success/failure window.
087	Outgoing branch loading-ratio finish line	Exact Within Closure / Open	Remaining Family–1 support/source gap reduced to one outgoing-branch loading ratio.
088	Loading-ratio extraction	Exact Within Closure	Minimal contact-plus-pole module gives $\rho_\alpha = 4/3$.
089	Family–1 minimal-isotropic verdict	Exact Within Closure	Minimal isotropic branch succeeds at zero transport bias.
090	Updated reduced status	Exact Within Closure / Open	Support/source side finished under the minimal module; remaining gap is deriving that module.

Audit philosophy. This appendix intentionally separates exact reduced algebra from actual-branch realization. Determinants, Schur complements, product laws, support-ratio identities, monotonicity signs, gain thresholds, and Family–1 arithmetic are exact once their closure is declared. The closures themselves—finite-throat continuum extraction, coherent D/N support, thin-wall branch selection, shell-weighted density extraction, and the minimal contact-plus-pole conservative quadrupole precursor—are not upgraded to assumption-free theorems of the unsolved nonlinear moving-throat PDE.

Notation warnings. The symbol G is used both for Newton's constant and for support-feasibility functions; this appendix writes the latter as $G(\xi,\delta)$, G_q, G_U, or G_{tr}. The variable ρ in the continuum placement map is a dimensionless mixed-sector ratio, not the bulk mass density. The loading ratio ρ_α is likewise not density. The labels $20, 21, 22$ are grouped real P_2 lanes, not spacetime indices.

Definition D.1 (Part III demand and support variables)**.** *Throughout this appendix,*

$$\begin{aligned} \zeta_{\mathrm{req}} &\quad \textit{denotes required coherent support ratio,} \\ \zeta_{\mathrm{phys}} &\quad \textit{denotes the physical support ratio supplied by a branch.} \end{aligned} \tag{D.4}$$

The selected support/source residual is

$$R_{\mathrm{quad}} = \zeta_{\mathrm{req}} - \zeta_{\mathrm{phys}}. \tag{D.5}$$

Thus $R_{\mathrm{quad}} < 0$ *means support exceeds demand,* $R_{\mathrm{quad}} = 0$ *means saturation, and* $R_{\mathrm{quad}} > 0$ *means support failure inside the declared support family.*

D.1 Stage 037: Exact Continuum-Kernel Extraction of the Selected-Branch Data

Part / anchor. Part III; primary anchor **MTDC-T6**.

Claim status. EXACT WITHIN CLOSURE **inside the declared finite-throat continuum kernel;** REDUCED / CONTROLLED REDUCTION **as a representation of the full moving-throat PDE.**

Purpose. Stage 037 replaces the abstract selected-mode data from Part II by closed formulas obtained from a concrete continuum kernel. It extracts the baseline scalar stiffness A, axial gap ΔK_{ax}, mixed loading α_{mix}, outgoing normalization numerator β_0, dimensionless mixed baseline M_{mix}, and anisotropy ratio δ.

Inputs. The selected D/N finite-throat geometry from Part II, the lowest two wall amplitudes, a flat internal U block, a mixed W lane, and the normalized overlap vector $v = (\kappa_0, \kappa_1)^T$ with $\kappa_0 = 2\sqrt{2}/\pi$, $\kappa_1 = -4/(3\pi)$, $\sigma = \kappa_0^2 + \kappa_1^2 = 88/(9\pi^2)$.

Derivation

The mass-normalized internal frequencies and couplings are

$$\Omega_U^2 = \frac{K_U}{\mu_U}, \qquad \Omega_W^2 = \frac{K_W^{\mathrm{eff}}}{\mu_W}, \qquad K_W^{\mathrm{eff}} = K_W + \frac{\pi^2 T_W}{4L^2}, \tag{D.6}$$

$$g_U = \frac{c_{\eta U}}{\sqrt{\mu_\eta \mu_U}}, \qquad g_W = \frac{c_{\eta W}}{\sqrt{\mu_\eta \mu_W}}, \qquad g_R = \frac{c_{UW}}{\sqrt{\mu_U \mu_W}}. \tag{D.7}$$

Eliminating the flat U/W block gives a rank-one wall loading of the same structural form used in the selected-mode derivation:

$$\Sigma_{\mathrm{wall}}(\omega) = \Xi(\omega) I_2 + \alpha(\omega) v v^T, \qquad v = (\kappa_0, \kappa_1)^T. \tag{D.8}$$

On the static conservative branch define

$$\Delta_0 = \Omega_U^2\Omega_W^2 - g_R^2\sigma, \qquad \mathcal{X} = \Omega_U^2 g_W + g_R g_U. \tag{D.9}$$

The selected-branch data are then

$$\boxed{A = K_0 - \frac{g_U^2}{\Omega_U^2}, \qquad \alpha_{\text{mix}} = \frac{\mathcal{X}^2}{\Omega_U^2\Delta_0}, \qquad \beta_0 = \frac{\mathcal{X}^2}{\Delta_0^2}}, \tag{D.10}$$

$$\boxed{M_{\text{mix}} = \frac{8\alpha_{\text{mix}}}{\pi^2 A}, \qquad \delta = \frac{\Delta K_{\text{ax}}}{A}}. \tag{D.11}$$

In unnormalized continuum variables, with $K_\eta^{\text{eff}} = K_\eta + 6T_\Omega$, these become

$$\boxed{A = \frac{K_U K_\eta^{\text{eff}} - c_{\eta U}^2}{\mu_\eta K_U}, \qquad \Delta K_{\text{ax}} = \frac{\pi^2 T_w}{\mu_\eta L^2}}, \tag{D.12}$$

$$\boxed{\alpha_{\text{mix}} = \frac{(K_U c_{\eta W} + c_{UW}c_{\eta U})^2}{\mu_\eta K_U\left(K_U K_W^{\text{eff}} - c_{UW}^2\sigma\right)}}, \tag{D.13}$$

$$\boxed{\beta_0 = \frac{\mu_W}{\mu_\eta}\frac{(K_U c_{\eta W} + c_{UW}c_{\eta U})^2}{\left(K_U K_W^{\text{eff}} - c_{UW}^2\sigma\right)^2}}, \tag{D.14}$$

$$\boxed{M_{\text{mix}} = \frac{8(K_U c_{\eta W} + c_{UW}c_{\eta U})^2}{\pi^2(K_U K_\eta^{\text{eff}} - c_{\eta U}^2)(K_U K_W^{\text{eff}} - c_{UW}^2\sigma)}, \quad \delta = \frac{\pi^2 T_w K_U}{L^2(K_U K_\eta^{\text{eff}} - c_{\eta U}^2)}}. \tag{D.15}$$

The stability inequalities are

$$\boxed{K_U K_\eta^{\text{eff}} > c_{\eta U}^2, \qquad K_U K_W^{\text{eff}} > c_{UW}^2\sigma}. \tag{D.16}$$

Output. Closed continuum formulas (D.12)–(D.15) and the stability gate (D.16).

Downstream use. Stage 038 nondimensionalizes these formulas; Stages 044–090 reuse them as the placement data of the explicit branch.

D.2 Stage 038: Dimensionless Continuum Placement Map and Product Law

Part / anchor. Part III; primary anchor **MTDC-T6**.

Claim status. Exact Within Closure **once the Stage 037 kernel is declared.**

Purpose. Stage 038 compresses the raw continuum constants into dimensionless ratios and proves that the selected quadrupole placement splits into a geometry lane, a product lane, and a redistribution lane.

Inputs. The Stage 037 formulas for A, M_{mix}, δ, and the selected outgoing normalization target.

Derivation

Define the dimensionless continuum ratios

$$\epsilon_\eta = \frac{c_{\eta U}^2}{K_U K_\eta^{\mathrm{eff}}}, \qquad \epsilon_W = \frac{c_{UW}^2 \sigma}{K_U K_W^{\mathrm{eff}}}, \qquad \rho = \frac{c_{UW} c_{\eta U}}{K_U c_{\eta W}}, \tag{D.17}$$

$$Z_W = \frac{c_{\eta W}^2}{K_\eta^{\mathrm{eff}} K_W^{\mathrm{eff}}}, \qquad \delta_0 = \frac{\pi^2 T_w}{L^2 K_\eta^{\mathrm{eff}}}, \qquad \Lambda = \frac{27\pi^2 G c_s^5 K_W^{\mathrm{eff}}}{20 a^5 c^5 \mu_W}. \tag{D.18}$$

Substitution into Stage 037 gives the exact placement map

$$\boxed{\delta = \frac{\delta_0}{1-\epsilon_\eta}, \qquad M_{\mathrm{mix}} = \frac{8 Z_W (1+\rho)^2}{\pi^2 (1-\epsilon_\eta)(1-\epsilon_W)}}, \tag{D.19}$$

$$\boxed{R_{\mathrm{target}} = \frac{\Lambda (1-\epsilon_\eta)(1-\epsilon_W)^2}{Z_W (1+\rho)^2}}. \tag{D.20}$$

Multiplying (D.19) and (D.20) cancels the redistribution variables:

$$\boxed{R_{\mathrm{target}} M_{\mathrm{mix}} = \frac{8\Lambda(1-\epsilon_W)}{\pi^2} = \frac{54 G c_s^5 K_W^{\mathrm{eff}} (1-\epsilon_W)}{5 a^5 c^5 \mu_W}}. \tag{D.21}$$

Thus (ϵ_W, Λ) fix the product scale, ϵ_η fixes the geometry lane through δ, and $(\epsilon_\eta, Z_W, \rho)$ redistribute the point along a fixed product curve.

Output. The placement map (D.19)–(D.20) and product law (D.21).

Checks. The product law must be invariant under redistributions of Z_W, ρ, and ϵ_η that leave the product lane fixed.

Downstream use. Stages 039–040 deform the placement map by adding a non-flat U doublet; Stage 081 and later rewrite the final Family–1 demand in branch-product variables.

D.3 Stage 039: First Non-Flat U Doublet and Direction Splitting

Part / anchor. Part III; primary anchor **MTDC-T6**.

Claim status. Exact Within Closure **inside the first split–U continuum extension.**

Purpose. Stage 039 tests the simplifying assumption that the source, support, and loading directions are collinear. The first non-flat U doublet preserves scalar placement but generically breaks direction collinearity.

Inputs. The Stage 038 flat–U placement map and the first axial excitation of the internal U sector.

Derivation

Introduce

$$K_{U0} = K_U, \qquad K_{U1} = K_U(1+\delta_U), \qquad \delta_U = \frac{\pi^2 T_U}{L^2 K_U}. \tag{D.22}$$

The scalar placement data become

$$\boxed{\delta_{\mathrm{split}} = \frac{\delta_0 + \epsilon_\eta \delta_U/(1+\delta_U)}{1-\epsilon_\eta}, \qquad \epsilon_{W,\mathrm{split}} = \epsilon_W \left[1 - \frac{2}{11}\frac{\delta_U}{1+\delta_U}\right]}. \tag{D.23}$$

The directional deformation is carried by

$$\boxed{R_U = \frac{1+\rho_0/(1+\delta_U)}{1+\rho_0}}. \tag{D.24}$$

The exact direction-splitting invariant is

$$\boxed{D_{\mathrm{dir}} = -\frac{\kappa_0\kappa_1 g_W \rho_0 \delta_U}{1+\delta_U}}. \tag{D.25}$$

Therefore

$$\boxed{D_{\mathrm{dir}} = 0 \iff \delta_U = 0 \text{ or } \rho_0 = 0}. \tag{D.26}$$

On the constructive branch, $\rho_0 > 0$ and $\delta_U > 0$ imply a genuine direction split.

Output. The split placement law (D.23), the direction factor (D.24), and the collinearity theorem (D.26).

Downstream use. Stage 040 packages this direction split into generalized selected-branch functions; Stages 044–046 test whether the resulting branch can still satisfy the target.

D.4 Stage 040: Generalized Selected-Branch Normalization with Source/Loading Mismatch

Part / anchor. Part III; primary anchor **MTDC-T6**.

Claim status. Exact Within Closure **for the rank-one selected-mode algebra with distinct source and loading directions.**

Purpose. Stage 040 replaces the flat one-vector selected branch by a source/loading mismatch law. It derives deformed branch functions F_{q,η_s} and G_q, then specializes them to the split–U continuum through R_U.

Inputs. The selected lower eigenvalue $\lambda_- = A_0(1-\xi)$, loading vector $z = z_0(1,q)^T$, and source vector $s = s_0(1,t)^T$.

Derivation

For a single rank-one loading, the required loading is

$$\alpha_{\rm req} = \frac{A_0\xi(\xi+\delta)}{z_0^2[\delta+(1+q^2)\xi]}, \qquad 0 \le \xi < 1. \tag{D.27}$$

The selected eigenvector has ratio

$$\frac{e_1}{e_0} = \frac{q\xi}{\delta+\xi}. \tag{D.28}$$

With $\eta_s = s_1 z_1/(s_0 z_0)$, the generalized normalization and baseline loading functions are

$$\boxed{F_{q,\eta_s}(\xi,\delta) = \frac{[\delta+(1+q^2)\xi]^2[\delta+(1+\eta_s)\xi]^2}{(1-\xi)[(\delta+\xi)^2+q^2\xi^2]^2}}, \tag{D.29}$$

$$\boxed{G_q(\xi,\delta) = \frac{\xi(\xi+\delta)}{\delta+(1+q^2)\xi}}. \tag{D.30}$$

For the split–U continuum, using $\lambda_0 = \kappa_1^2/\kappa_0^2 = 2/9$,

$$q = -\frac{\sqrt{2}}{3}R_U, \qquad \eta_s = \frac{2}{9}R_U. \tag{D.31}$$

Therefore the one-parameter split–U functions are

$$\boxed{F_U(\xi,\delta;R_U) = \frac{[9\delta+(9+2R_U^2)\xi]^2[9\delta+(9+2R_U)\xi]^2}{81(1-\xi)[9\delta^2+18\delta\xi+(9+2R_U^2)\xi^2]^2}}, \tag{D.32}$$

$$\boxed{G_U(\xi,\delta;R_U) = \frac{9\xi(\xi+\delta)}{9\delta+(9+2R_U^2)\xi}}. \tag{D.33}$$

The flat branch is recovered at $R_U = 1$.

Output. The generalized selected functions (D.29)–(D.30) and their split–U specialization (D.32)–(D.33).

Downstream use. Stages 041–044 add a second support direction and determine the actual softening depth.

D.5 Stage 041: Rank–2 Support Completion and Exact Support-Loading Theorem

Part / anchor. Part III; primary anchor **MTDC-T6**.

Claim status. Exact Within Closure **for the two-rank selected-mode determinant.**

Purpose. Stage 041 adds a second loading direction for physical support and proves that the selected-branch determinant remains linear in the support loading.

Inputs. Mixed baseline m, mixed direction $(1, q)$, support loading n, support direction $(1, r)$, and selected lower depth ξ.

Derivation

The two-direction loaded stiffness matrix is

$$\frac{K_{\text{loaded}}}{A_0} = \operatorname{diag}(1, 1+\delta) - m(1,q)^T(1,q) - n(1,r)^T(1,r). \tag{D.34}$$

The determinant condition at $\lambda_- = A_0(1-\xi)$ is

$$D_{\text{sel}} = \xi(\delta+\xi) - m[\delta + (1+q^2)\xi] - n[\delta + (1+r^2)\xi] + mn(q-r)^2. \tag{D.35}$$

Solving the linear equation $D_{\text{sel}} = 0$ for n gives

$$\boxed{n_{\text{req}}(\xi, \delta; m, q, r) = \frac{\xi(\delta+\xi) - m[\delta + (1+q^2)\xi]}{\delta + (1+r^2)\xi - m(q-r)^2}}. \tag{D.36}$$

Differentiating yields the exact monotonicity law

$$\boxed{\frac{\partial n_{\text{req}}}{\partial m} = -\frac{[\delta + (1+qr)\xi]^2}{[\delta + (1+r^2)\xi - m(q-r)^2]^2} < 0} \tag{D.37}$$

on every regular branch. Additional mixed baseline always lowers the support loading needed to reach the same selected depth.

Two special closures are important:

$$r = q \quad \Longrightarrow \quad n_{\text{req}} = G_q(\xi, \delta) - m, \tag{D.38}$$

and, with source-tied support and $q = tR_U$,

$$n_{\text{req}}^{\text{src}} = \frac{\xi(\delta+\xi) - m[\delta + (1+\lambda_0 R_U^2)\xi]}{\delta + (1+\lambda_0)\xi - m\lambda_0(R_U-1)^2}. \tag{D.39}$$

Output. The support-loading theorem (D.36), its monotonicity (D.37), and the tracking/source-tied alternatives.

Downstream use. Stage 042 computes normalization with three distinct directions; Stage 045 shows the first coherent D/N kernel lands on the tracking surface.

D.6 Stage 042: Selected-Mode Normalization Under Rank–2 Support Completion

Part / anchor. Part III; primary anchor **MTDC-T6**.

Claim status. Exact Within Closure **for the selected eigenvector and overlap algebra.**

Purpose. Stage 042 shows that once support and mixed loading directions differ, normalization depends separately on outgoing, support, and source directions.

Inputs. The Stage 041 selected branch with $n = n_{\text{req}}$, mixed direction q, support direction r, and source direction t.

Derivation

With $n = n_{\text{req}}$, the selected eigenvector ratio is

$$\boxed{\frac{e_1}{e_0} = \frac{m(q-r)+r\xi}{\delta+\xi-mq(q-r)}}. \tag{D.40}$$

Define

$$D_{q,r}(\xi,\delta;m) = [\delta+\xi-mq(q-r)]^2 + [m(q-r)+r\xi]^2. \tag{D.41}$$

The loading and source overlaps are

$$\boxed{\frac{(z\cdot e_-)^2}{z_0^2} = \frac{[\delta+(1+qr)\xi]^2}{D_{q,r}(\xi,\delta;m)}}, \tag{D.42}$$

$$\boxed{\frac{(s\cdot e_-)^2}{s_0^2} = \frac{[\delta+(1+rt)\xi-m(q-r)(q-t)]^2}{D_{q,r}(\xi,\delta;m)}}. \tag{D.43}$$

The one-vector result is recovered only when $r = q$. Otherwise, the selected-mode normalization sees a three-direction geometry.

Output. The rank–2 eigenvector (D.40) and overlap formulas (D.42)–(D.43).

Downstream use. Stage 044 inserts the actual support direction and converts this algebra into a continuum-selected branch point.

D.7 Stage 043: Continuum Extraction of the Actual Support/BdG Loading Direction

Part / anchor. Part III; primary anchor **MTDC-T6**.

Claim status. REDUCED / CONTROLLED REDUCTION **for the continuum support extraction;** EXACT WITHIN CLOSURE **for the resulting direction algebra.**

Purpose. Stage 043 identifies the physical support/BdG loading direction that must be inserted into the rank–2 theorem.

Inputs. The split–U continuum branch, the support profile, and the rank–2 selected-mode formulas of Stages 041–042.

Derivation

The reduced rank–2 formalism contains an abstract support direction r. The continuum support extraction evaluates this direction from the actual support/BdG overlap data. In the notation used later, the physical direction is written in terms of a source-side ratio R_ϕ, while the mixed direction is governed by R_U. Thus the support completion problem is not merely whether additional support load exists; it is whether

$$R_\phi = R_U \qquad \text{or} \qquad R_\phi \neq R_U. \tag{D.44}$$

The first case collapses to tracking and the one-direction selected geometry. The second case gives a genuinely new rank–2 branch.

The exact continuum-selected determinant remains the Stage 041 determinant, but with the physical replacements

$$q = tR_U, \qquad r = tR_\phi, \qquad m = M_{\text{mix}}, \qquad n = M_{\text{supp}}, \tag{D.45}$$

where $t = \kappa_1/\kappa_0$.

Output. A physical support-direction insertion rule (D.45).

Downstream use. Stage 044 uses it to derive the continuum-selected quadratic. Stage 045 then proves that the first coherent local D/N kernel forces the tracking case.

D.8 Stage 044: Continuum-Selected Rank–2 Closure and Quadratic Branch Equation

Part / anchor. Part III; primary anchor **MTDC-T6**.

Claim status. Exact Within Closure **after the physical support-direction data are inserted.**

Purpose. Stage 044 turns the abstract rank–2 branch into a continuum-selected branch point. The physical softening depth is no longer free; it is selected by an exact quadratic.

Inputs. The Stage 043 replacements $q = tR_U$, $r = tR_\phi$, mixed baseline M_{mix}, and support loading M_{supp}.

Derivation

The selected determinant becomes an equation for ξ. After collecting terms, the physical softening depth satisfies

$$\boxed{\xi^2 + B_{\text{cont}}\xi + C_{\text{cont}} = 0}, \tag{D.46}$$

where

$$B_{\text{cont}} = \delta - M_{\text{mix}}(1 + t^2R_U^2) - M_{\text{supp}}(1 + t^2R_\phi^2), \tag{D.47}$$

$$C_{\text{cont}} = -\delta(M_{\text{mix}} + M_{\text{supp}}) + t^2M_{\text{mix}}M_{\text{supp}}(R_U - R_\phi)^2. \tag{D.48}$$

Once the physical root $\xi_{\text{phys}} \in (0, 1)$ is selected, the normalization gate is

$$\boxed{R_{\text{target}} = F_{\text{cont}}(\xi_{\text{phys}})}. \tag{D.49}$$

Two exact special surfaces are useful diagnostics: a minimal-kernel source-tied surface and an interference-matched tracking surface. They are distinguished by whether the direction-splitting term $(R_U - R_\phi)^2$ survives.

Output. The continuum-selected quadratic (D.46) and normalization gate (D.49).

Downstream use. Stage 045 shows the first coherent D/N kernel lands on the tracking surface; Stage 047 packages support enhancement on that surface.

D.9 Stage 045: Coherent Local D/N Support Kernel and Exact Tracking Reduction

Part / anchor. Part III; primary anchor **MTDC-T6**.

Claim status. Exact Within Closure **for the first coherent local D/N kernel.**

Purpose. Stage 045 checks the first coherent local D/N support kernel and proves that it does not realize a generic rank–2 direction. It forces exact tracking.

Inputs. The continuum support kernel and rank–2 branch variables R_U, R_ϕ.

Derivation

The coherent local D/N kernel enforces

$$\boxed{g_B g_R = g_W g_S}. \tag{D.50}$$

This condition lands on the tracking surface

$$\boxed{R_\phi = R_U = R_{\rm tr}}. \tag{D.51}$$

The coherent tracking factor is

$$\boxed{R_{\rm tr} = \frac{1 + \chi_0/(1+\delta_U)}{1+\chi_0}, \qquad \frac{1}{1+\delta_U} < R_{\rm tr} < 1} \tag{D.52}$$

for the constructive split–U branch.

Therefore the rank–2 support branch collapses to tracking laws

$$\boxed{M_{\rm tr} = G_{\rm tr}(\xi, \delta; R_{\rm tr}), \qquad R_{\rm target} = F_{\rm tr}(\xi, \delta; R_{\rm tr})}. \tag{D.53}$$

Output. The exact tracking condition (D.51) and coherent tracking factor (D.52).

Downstream use. Stages 046–048 prove that support compensation is still needed and derive the exact required support ratio.

D.10 Stage 046: Tracking-Branch Bounds and Residual Comparison

Part / anchor. Part III; primary anchor **MTDC-T6**.

Claim status. Exact Within Closure **for the monotonicity and residual comparison on the tracking branch.**

Purpose. Stage 046 determines whether the first coherent tracking deformation helps or hurts the selected normalization target.

Inputs. The tracking functions F_{tr}, G_{tr} and the constructive inequality $R_{\mathrm{tr}} < 1$.

Derivation

The tracking functions obey

$$\boxed{\frac{\partial G_{\mathrm{tr}}}{\partial R} < 0, \qquad \frac{\partial F_{\mathrm{tr}}}{\partial R} > 0}. \tag{D.54}$$

Because the constructive split–U branch has $R_{\mathrm{tr}} < 1$, one finds

$$G_{\mathrm{tr}} > G_{\mathrm{flat}}, \qquad F_{\mathrm{tr}} < F_{\mathrm{flat}} \tag{D.55}$$

at fixed (ξ, δ). The exact residual comparison can be written as

$$\boxed{E_{\mathrm{tr}} - E_{\mathrm{flat}} = F_{\mathrm{flat}} - F_{\mathrm{tr}} > 0}. \tag{D.56}$$

Thus the first coherent local D/N kernel worsens the selected normalization relative to the flat branch. Support enhancement must compensate this deficit.

Output. Monotonicity (D.54) and residual theorem (D.56).

Downstream use. Stage 047 introduces support enhancement; Stage 048 proves the compensation theorem.

D.11 Stage 047: Coherent-Kernel Dimensionless Map and Support-Enhancement Factor

Part / anchor. Part III; primary anchor **MTDC-T6**.

Claim status. Exact Within Closure **for the coherent tracking map.**

Purpose. Stage 047 introduces the coherent support ratio ζ and shows that it changes the available loading but not the selected target ratio.

Inputs. The coherent tracking branch, dimensionless split–U variables, and a support lane parallel to the tracked mixed direction.

Derivation

The total tracking load factors as

$$M_{\mathrm{tr}} = M_{\mathrm{mix}} S(\zeta;\epsilon), \tag{D.57}$$

where

$$\boxed{S(\zeta;\epsilon) = 1 + \frac{\zeta(1-\epsilon)}{1-\epsilon\zeta}}. \tag{D.58}$$

The coherent target ratio is

$$\boxed{R_{\mathrm{target}} = \frac{\Lambda(1-\epsilon_\eta)(1-\epsilon)^2}{Z_W(1+\chi_0)^2}}. \tag{D.59}$$

It contains no ζ. Therefore the support lane only supplies loading; it does not move the target locus itself.

Output. Support enhancement factor (D.58) and target independence (D.59).

Downstream use. Stage 048 inverts S to obtain ζ_{req}; Stages 049–050 compute whether physical D/N support can supply it.

D.12 Stage 048: Exact Support-Compensation Theorem on the Physical Tracking Branch

Part / anchor. Part III; primary anchor **MTDC-T6**.

Claim status. Exact Within Closure **for the reduced support-compensation theorem;** Open **for actual PDE placement of ζ_{phys}.**

Purpose. Stage 048 proves that there is no reduced-level support no-go on the coherent tracking branch: if a finite support ratio above threshold exists, it reaches the target before the softening pole.

Inputs. The support factor $S(\zeta;\epsilon)$, mixed baseline M_{mix}, and tracking-branch required load M_{req}.

Derivation

Define

$$S_{\mathrm{req}} = \frac{M_{\mathrm{req}}}{M_{\mathrm{mix}}}. \tag{D.60}$$

Solving $S(\zeta;\epsilon) = S_{\mathrm{req}}$ gives the exact inverse

$$\boxed{\zeta_{\mathrm{req}} = \frac{S_{\mathrm{req}}-1}{1+\epsilon(S_{\mathrm{req}}-2)}}. \tag{D.61}$$

On the stable side the support factor is strictly increasing up to its softening pole. Hence if the mixed-only branch is below the required load, there is a unique ζ_{req} below the pole. The support branch succeeds iff

$$\boxed{\zeta_{\mathrm{phys}} \geq \zeta_{\mathrm{req}}}. \tag{D.62}$$

Output. The exact threshold (D.61) and success condition (D.62).

Downstream use. Stages 049–057 derive ζ_{phys} from finite-throat support, overlap, Robin compliance, and transport bias.

D.13 Stage 049: Explicit D/N Overlap Extraction of the Physical Support Ratio

Part / anchor. Part III; primary anchor **MTDC-T6**.

Claim status. Exact Within Closure **for the D/N support tower overlap calculation.**

Purpose. Stage 049 makes the coherent support ratio microscopic by evaluating the D/N support tower overlaps.

Inputs. D/N support modes $\chi_n(s) = \sqrt{2/L}\sin[(n+1/2)\pi s/L]$ and support-source overlaps I_n.

Derivation

The physical coherent support ratio is

$$\boxed{\zeta_n^{\mathrm{phys}} = \frac{K_W^{\mathrm{eff}}}{K_{\phi,n}^{\mathrm{eff}}}\left(\frac{I_n}{I_0}\right)^2}. \tag{D.63}$$

For a uniform local source density,

$$\boxed{\frac{I_n}{I_0} = \frac{1}{2n+1}}. \tag{D.64}$$

For a same-operator twin family,

$$\boxed{\zeta_n^{\mathrm{twin}} = \frac{1}{(2n+1)^2[1+xn(n+1)]}, \qquad x = \frac{\pi^2 T_X}{L^2 K_W^{\mathrm{eff}}}}. \tag{D.65}$$

The lowest twin value is

$$\boxed{\zeta_0^{\mathrm{twin}} = 1}. \tag{D.66}$$

Output. Microscopic D/N support ratios (D.63)–(D.65).

Downstream use. Stage 050 converts these ratios into explicit selection rules.

D.14 Stage 050: Exact Comparison of Physical ζ Against the Required Threshold

Part / anchor. Part III; primary anchor **MTDC-T6**.

Claim status. Exact Within Closure **for the support-tower selection rules.**

Purpose. Stage 050 compares ζ_n^{phys} against ζ_{req} and proves that the lowest symmetric twin is the natural first support branch.

Inputs. The Stage 048 threshold ζ_{req} and Stage 049 D/N support tower.

Derivation

The lowest symmetric twin lane has $\zeta_0 = 1$. Therefore its enhancement factor is

$$\boxed{S_0 = S(1; \epsilon) = 2}. \tag{D.67}$$

It succeeds iff

$$\boxed{\zeta_{\text{req}} \le 1 \quad\Longleftrightarrow\quad S_{\text{req}} \le 2}. \tag{D.68}$$

For higher harmonics, the immediate exclusion test is

$$\boxed{\zeta_{\text{req}} > \frac{1}{(2n+1)^2} \quad\Longrightarrow\quad \text{the } n\text{th same-operator twin lane cannot reach the threshold.}} \tag{D.69}$$

When the immediate exclusion test does not apply, the support softness must satisfy

$$\boxed{x \le x_{\max}(n; \zeta_{\text{req}}) = \frac{1}{n(n+1)} \left[\frac{1}{(2n+1)^2 \zeta_{\text{req}}} - 1\right]}. \tag{D.70}$$

For $n \ge 1$, the higher-harmonic enhancement is bracketed below by its support-ceiling value

$$\boxed{S_n^{\text{twin}}(x; \varepsilon) < S_n^{\max}(\varepsilon) := 1 + \frac{1 - \varepsilon}{(2n+1)^2 - \varepsilon}}. \tag{D.71}$$

Output. Lowest-twin sufficiency (D.68), higher-harmonic exclusion/softness thresholds, and the enhancement ceiling (D.71).

Downstream use. Stage 051 rewrites the lowest-twin criterion in branch-product variables; Stage 052 defines the non-twin regime.

D.15 Stage 051: Exact Lowest-Twin Sufficiency Criterion on the Tracking Branch

Part / anchor. Part III; primary anchor **MTDC-T6**.

Claim status. Exact Within Closure **for the branch-product form.**

Purpose. Stage 051 expresses the lowest-twin support test in the same branch-product variables used by the continuum placement map.

Inputs. The tracking-branch product $\Pi_{\text{tr}} = F_{\text{tr}} G_{\text{tr}}$ and $C_{\text{mix}} = 8\Lambda(1 - \epsilon)/\pi^2$.

Derivation

Define

$$C_{\text{mix}} = \frac{8\Lambda(1-\epsilon)}{\pi^2}. \tag{D.72}$$

The required enhancement is $S_{\text{req}} = \Pi_{\text{tr}}/C_{\text{mix}}$. Since the lowest twin gives $S_0 = 2$, it is sufficient iff

$$\boxed{\Pi_{\text{tr}} \le 2C_{\text{mix}} = \frac{16\Lambda(1-\epsilon)}{\pi^2}}. \tag{D.73}$$

Equivalently, the threshold scales can be written as

$$\Lambda_{\text{twin,req}} = \frac{\pi^2}{16(1-\epsilon)}\Pi_{\text{tr}}, \qquad M_{\text{mix}}^{(\text{twin,req})} = \frac{G_{\text{tr}}}{2}. \tag{D.74}$$

Output. The lowest-twin branch-product criterion (D.73).

Downstream use. Stage 052 uses this criterion to split the support problem into mixed-only, lowest-twin, and non-twin regimes.

D.16 Stage 052: Exact Non-Twin Asymmetry Requirement

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. Exact Within Closure **for the support-regime split.**

Purpose. Stage 052 identifies when mixed baseline, symmetric lowest twin support, or non-twin support asymmetry is required.

Inputs. The branch product Π_{tr}, the mixed product scale C_{mix}, and the blocking factor ϵ.

Derivation

The required support ratio in product variables is

$$\boxed{\zeta_{\text{req}} = \frac{\Pi_{\text{tr}} - C_{\text{mix}}}{C_{\text{mix}} - \epsilon(2C_{\text{mix}} - \Pi_{\text{tr}})}}. \tag{D.75}$$

It is monotone in Π_{tr}:

$$\frac{d\zeta_{\text{req}}}{d\Pi_{\text{tr}}} = \frac{C_{\text{mix}}(1-\epsilon)}{[C_{\text{mix}} - \epsilon(2C_{\text{mix}} - \Pi_{\text{tr}})]^2} > 0. \tag{D.76}$$

The exact regime split is

$$\boxed{\begin{array}{lll} \Pi_{\text{tr}} \le C_{\text{mix}} & \Rightarrow & \text{mixed-only enough,} \\ C_{\text{mix}} < \Pi_{\text{tr}} \le 2C_{\text{mix}} & \Rightarrow & \text{symmetric lowest twin enough,} \\ \Pi_{\text{tr}} > 2C_{\text{mix}} & \Rightarrow & \text{non-twin asymmetry required.} \end{array}} \tag{D.77}$$

The excess beyond the symmetric twin branch is

$$\boxed{\Delta_\zeta = \zeta_{\rm req} - 1 = \frac{(1-\epsilon)(\Pi_{\rm tr} - 2C_{\rm mix})}{C_{\rm mix} - \epsilon(2C_{\rm mix} - \Pi_{\rm tr})}}. \tag{D.78}$$

For a general lowest lane,

$$\zeta_0^{\rm phys} = \frac{K_W^{\rm eff}}{K_{\phi,0}^{\rm eff}}\Omega_0^2. \tag{D.79}$$

Thus the two resources for non-twin rescue are overlap boost Ω_0^2 and support softening $K_W^{\rm eff}/K_{\phi,0}^{\rm eff}$.

Output. Regime split (D.77), excess formula (D.78), and resource factorization (D.79).

Downstream use. Stages 053–057 quantify the two resources and attach them to transport/Robin physics.

D.17 Stage 053: Exact Overlap-Boost Window for the Lowest Support Lane

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. Exact Within Closure **for the exponential source family and its overlap ceiling.**

Purpose. Stage 053 computes the maximum possible overlap boost supplied by the first exponential lowest-lane source family.

Inputs. Lowest D/N support mode $\chi_0(s) = \sqrt{2/L}\sin(\pi s/2L)$ and normalized exponential source σ_α.

Derivation

The normalized exponential source family is

$$\sigma_\alpha(s) = \frac{\alpha e^{\alpha s/L}}{e^\alpha - 1}, \qquad \int_0^L \sigma_\alpha(s)\, ds = L. \tag{D.80}$$

The exact overlap boost relative to the uniform source is

$$\boxed{\Omega_{\rm exp}(\alpha) = \frac{\pi\alpha(2\alpha e^\alpha + \pi)}{(4\alpha^2 + \pi^2)(e^\alpha - 1)}}. \tag{D.81}$$

The endpoint values are

$$\boxed{\Omega_{\rm exp}(0) = 1, \qquad \lim_{\alpha\to+\infty} \Omega_{\rm exp}(\alpha) = \frac{\pi}{2}}. \tag{D.82}$$

Therefore

$$\boxed{\Omega_0^2 \le \frac{\pi^2}{4}}. \tag{D.83}$$

Pure overlap rescue alone is possible only if $\zeta_{\rm req} \le \pi^2/4$.

Output. Overlap boost formula (D.81) and ceiling (D.83).

Downstream use. Stage 056 identifies α with a Peclet number; Stage 057 combines it with Robin softening.

D.18 Stage 054: Robin-Compliance Softening of the Lowest Support Lane

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. Exact Within Closure **for the Robin support eigenvalue and softening factor.**

Purpose. Stage 054 computes the support-stiffness reduction from replacing the Dirichlet mouth by a Robin compliance.

Inputs. Lowest support lane on $[0, L]$ with Robin parameter $\eta = hL$ and support stiffness ratio x.

Derivation

The lowest Robin root satisfies

$$\boxed{y \tan y = \eta, \qquad 0 < y < \frac{\pi}{2}}. \tag{D.84}$$

Let

$$x = \frac{\pi^2 T_X}{L^2 K_W^{\text{eff}}}, \qquad 0 < x < 4. \tag{D.85}$$

Then the support-softening factor is

$$\boxed{A_K(\eta) = \frac{K_W^{\text{eff}}}{K_{\phi,0}^{\text{eff}}} = \frac{1}{1 - x/4 + xy^2/\pi^2}}. \tag{D.86}$$

The endpoint window is

$$\boxed{1 \le A_K \le \frac{4}{4 - x}}. \tag{D.87}$$

Pure support softening alone can rescue the Stage 052 threshold only if $\zeta_{\text{req}} \le 4/(4 - x)$.

Output. Robin root (D.84), softening factor (D.86), and ceiling (D.87).

Downstream use. Stage 055 combines this with overlap boost; Stage 057 uses $y(\eta)$ in the physical support ratio.

D.19 Stage 055: Explicit Reachability of the Non-Twin Lowest Support Lane

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. Exact Within Closure **for the combined overlap/compliance reachability window.**

Purpose. Stage 055 combines the overlap boost and Robin softening into a single lowest-lane reachability test.

Inputs. The overlap ceiling from Stage 053 and the Robin softening factor from Stage 054.

Derivation

The physical lowest-lane support ratio factors as

$$\zeta_0^{\mathrm{phys}} = A_K(\eta)\Omega_0^2. \tag{D.88}$$

Using Stages 053–054,

$$\boxed{1 \le \zeta_0^{\mathrm{phys}} \le \frac{\pi^2}{4}\frac{4}{4-x}} \tag{D.89}$$

inside the explicit non-twin lowest-lane family. Thus every non-twin support rescue must pass through a finite overlap-plus-compliance window. Demands above this window cannot be reached by this family, regardless of transport bias.

Output. Combined reachability window (D.89).

Downstream use. Stages 056–059 derive how transport selects a point inside this reachability window.

D.20 Stage 056: Transport Origin of Lowest-Lane Source Asymmetry

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. Exact Within Closure **for the drift-diffusion zero-flux source family.**

Purpose. Stage 056 gives the exponential source profile a physical transport origin rather than treating it as a free shape ansatz.

Inputs. A one-dimensional source density $\sigma(s)$ on the finite throat and a drift-diffusion current.

Derivation

Take

$$\partial_t\sigma + \partial_s J = 0, \qquad J = -D_\sigma\partial_s\sigma + v_\sigma\sigma. \tag{D.90}$$

The stationary zero-flux equation $J = 0$ gives

$$\sigma(s) \propto e^{v_\sigma s/D_\sigma}. \tag{D.91}$$

With

$$\boxed{\mathrm{Pe} = \frac{v_\sigma L}{D_\sigma}}, \tag{D.92}$$

the normalized stationary family is exactly the Stage 053 exponential source with $\alpha = \mathrm{Pe}$. Therefore

$$\boxed{\Omega_{\mathrm{Pe}} = \frac{\pi \mathrm{Pe}(2\mathrm{Pe}e^{\mathrm{Pe}} + \pi)}{(4\mathrm{Pe}^2 + \pi^2)(e^{\mathrm{Pe}} - 1)}}. \tag{D.93}$$

Output. Transport interpretation (D.92) and physical overlap factor (D.93).

Downstream use. Stage 057 uses Pe as the physical transport coordinate of ζ_0.

D.21 Stage 057: Physical $(\mathrm{Pe}, \kappa, \eta)$ Placement Map

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. Exact Within Closure **for the lowest-lane physical support map.**

Purpose. Stage 057 combines transport-driven overlap and Robin compliance into the physical support ratio used by the operator-selected branch.

Inputs. Peclet overlap Ω_{Pe}, Robin root $y(\eta)$, and stiffness parameter κ.

Derivation

The physical lowest-lane support ratio is

$$\boxed{\zeta_0(\mathrm{Pe}, \eta; \kappa) = \Omega_{\mathrm{Pe}}^2 \frac{\kappa + \pi^2/4}{\kappa + y(\eta)^2}, \qquad y(\eta) \tan y(\eta) = \eta}. \tag{D.94}$$

It is monotone in the constructive Peclet direction, with zero-bias value

$$\zeta_0(0, \eta; \kappa) = \frac{\kappa + \pi^2/4}{\kappa + y(\eta)^2}. \tag{D.95}$$

Output. The physical support map (D.94).

Downstream use. Stages 058–059 determine how the support/source operator selects Pe_*; Stages 079–083 specialize (D.94) to Family–1.

D.22 Stage 058: Coupled Support/Source Operator and Exact Peclet Branch Equation

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. Exact Within Closure **for the operator-selected fixed point.**

Purpose. Stage 058 closes the support/source system by letting support loading generate the transport bias that selects Pe_*.

Inputs. The physical support ratio map and the source/support response kernel parameterized by (κ, η).

Derivation

The self-consistent branch equation is

$$\boxed{\mathrm{Pe}_* = \Xi\Delta(\mathrm{Pe}_*; \kappa, \eta)}. \tag{D.96}$$

The exact endpoint bracket is

$$\boxed{\Xi\Delta_0(\kappa, \eta) \le \mathrm{Pe}_* \le \Xi\Delta_\infty(\kappa, \eta)}. \tag{D.97}$$

Writing $\alpha = \sqrt{\kappa}$, the endpoint functions are

$$\Delta_0 = \frac{\eta(\cosh\alpha - 1)}{\alpha^2(\alpha\sinh\alpha + \eta\cosh\alpha)}, \tag{D.98}$$

$$\Delta_\infty = \frac{\cosh\alpha + (\eta/\alpha)\sinh\alpha - 1}{\alpha\sinh\alpha + \eta\cosh\alpha}. \tag{D.99}$$

Output. Fixed-point equation (D.96) and bracket (D.97).

Downstream use. Stage 059 turns the bracket into exact fail/succeed thresholds.

D.23 Stage 059: Exact Residual Bounds on the Operator-Selected Branch

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. Exact Within Closure **for the residual thresholds.**

Purpose. Stage 059 converts the fixed-point bracket into a useful branch verdict.

Inputs. Required Peclet bias $\mathrm{Pe}_{\mathrm{req}}$, endpoint functions Δ_0, Δ_∞, and gain Ξ.

Derivation

If $\mathrm{Pe}_{\mathrm{req}}$ is the bias needed to meet ζ_{req}, then the exact thresholds are

$$\boxed{\Xi_{\mathrm{fail}} = \frac{\mathrm{Pe}_{\mathrm{req}}}{\Delta_\infty(\kappa, \eta)}, \qquad \Xi_{\mathrm{suff}} = \frac{\mathrm{Pe}_{\mathrm{req}}}{\Delta_0(\kappa, \eta)}}. \tag{D.100}$$

Hence

$$\boxed{\Xi < \Xi_{\mathrm{fail}} \Rightarrow \text{fail}, \qquad \Xi > \Xi_{\mathrm{suff}} \Rightarrow \text{succeed}}. \tag{D.101}$$

The band $\Xi_{\mathrm{fail}} \le \Xi \le \Xi_{\mathrm{suff}}$ is profile-sensitive and requires the full fixed-point root.

Output. The exact gain thresholds (D.100).

Downstream use. Stages 060–066 identify Ξ with a microscopic parent-action gain and a wall figure of merit.

D.24 Stage 060: Entropic Source Microclosure and Microscopic Gain

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. EXACT WITHIN CLOSURE **inside the positive-density Onsager/free-energy closure.**

Purpose. Stage 060 shows that the drift-diffusion source branch can be obtained from a positive local source thermodynamics rather than imposed as a shape family.

Inputs. A positive source density, an Onsager mobility, and a free-energy gradient producing drift and diffusion.

Derivation

The source transport equation of Stage 056 is reproduced by an Onsager form

$$J = -\mathcal{M}(\sigma)\partial_s \frac{\delta \mathcal{F}}{\delta \sigma}, \tag{D.102}$$

with a strictly positive mobility and a convex local entropy contribution. The resulting stationary zero-flux state is the same exponential branch. The microscopic source/support coupling is then summarized by a dimensionless gain G_{micro}, later projected explicitly from the parent action.

Output. A positive-density microclosure for the transport source family and the definition of the microscopic gain variable.

Downstream use. Stage 061 uses this gain to build a phase diagram; Stage 062 derives the gain from GNLS parent-action overlaps.

D.25 Stage 061: Microscopic Gain Thresholds and Operator Phase Diagram

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. EXACT WITHIN CLOSURE **for the gain-threshold phase diagram.**

Purpose. Stage 061 rewrites the support/source verdict in terms of the microscopic gain rather than an abstract operator strength.

Inputs. The thresholds Ξ_{fail}, Ξ_{suff}, the support geometry parameter κ, and the relation between Ξ and G_{micro}.

Derivation

The fixed-point strength is written as

$$\Xi_{\text{micro}} = \kappa G_{\text{micro}}. \tag{D.103}$$

Consequently, the gain thresholds are

$$\boxed{G_{\text{fail}} = \frac{\Xi_{\text{fail}}}{\kappa}, \qquad G_{\text{suff}} = \frac{\Xi_{\text{suff}}}{\kappa}}. \tag{D.104}$$

The reduced phase diagram is

$$\boxed{\begin{array}{lll} G_{\text{micro}} < G_{\text{fail}} & \Rightarrow & \text{fail}, \\ G_{\text{micro}} > G_{\text{suff}} & \Rightarrow & \text{succeed}, \\ G_{\text{fail}} \le G_{\text{micro}} \le G_{\text{suff}} & \Rightarrow & \text{root-sensitive}. \end{array}} \tag{D.105}$$

Output. The microscopic gain phase diagram (D.104)–(D.105).

Downstream use. Stages 062–063 express G_{micro} in parent GNLS overlap data and derive no-go/success thresholds.

D.26 Stage 062: Parent-Action Projection of the Microscopic Support/Source Gain

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. Exact **for the parent GNLS compressional identities;** Reduced / Controlled Reduction **for the projected support/source ansatz.**

Purpose. Stage 062 projects the microscopic gain back into parent-action variables: confinement loading, compressional stiffness, and source/support overlap.

Inputs. GNLS compressional stiffness for the frozen $n = 5$ EOS and support/source profiles χ_ϕ, χ_σ.

Derivation

Take

$$\delta V_{\text{conf}}(s, y) = -g_\phi \chi_\phi(y)\phi(s). \tag{D.106}$$

For the frozen $n = 5$ EOS,

$$h'(\rho_*) = 5K\rho_*^3 = \frac{mc_{s,*}^2}{\rho_*}. \tag{D.107}$$

Define profile overlaps

$$N_{\sigma\sigma} = \int d^3y\, \chi_\sigma^2, \qquad N_{\phi\phi} = \int d^3y\, \chi_\phi^2, \qquad O_{\sigma\phi} = \int d^3y\, \chi_\sigma \chi_\phi. \tag{D.108}$$

Then

$$\boxed{G_{\text{micro}} = \frac{\rho_* g_\phi^2 O_{\sigma\phi}^2}{mc_{s,*}^2 K_X N_{\sigma\sigma}} = \frac{\rho_* g_\phi^2 N_{\phi\phi}}{mc_{s,*}^2 K_X} C_{\sigma\phi}^2}, \tag{D.109}$$

where

$$C^2_{\sigma\phi} = \frac{O^2_{\sigma\phi}}{N_{\sigma\sigma}N_{\phi\phi}}, \qquad 0 \le C^2_{\sigma\phi} \le 1. \tag{D.110}$$

Output. Parent-action gain formula (D.109).

Downstream use. Stage 063 converts this into amplitude/coherence thresholds; Stage 064 derives the equilibrium source profile.

D.27 Stage 063: Parent-Overlap Threshold Theorem and Microscopic Success/No-Go Tests

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. Exact Within Closure **after the Stage 062 parent-overlap projection.**

Purpose. Stage 063 turns the microscopic gain formula into exact fail/succeed surfaces for confinement-loading amplitude and source/support coherence.

Inputs. The parent gain formula, $G_{\rm fail}$, $G_{\rm suff}$, and the Cauchy bound on $C^2_{\sigma\phi}$.

Derivation

At fixed profile shapes, the amplitude thresholds are

$$\boxed{g^2_{\phi,\rm fail} = \frac{mc^2_{s,*}K_X N_{\sigma\sigma}G_{\rm fail}}{\rho_* O^2_{\sigma\phi}}, \qquad g^2_{\phi,\rm suff} = \frac{mc^2_{s,*}K_X N_{\sigma\sigma}G_{\rm suff}}{\rho_* O^2_{\sigma\phi}}}. \tag{D.111}$$

At fixed g_ϕ, the coherence thresholds are

$$\boxed{C^2_{\rm fail} = \frac{mc^2_{s,*}K_X G_{\rm fail}}{\rho_* g^2_\phi N_{\phi\phi}}, \qquad C^2_{\rm suff} = \frac{mc^2_{s,*}K_X G_{\rm suff}}{\rho_* g^2_\phi N_{\phi\phi}}}. \tag{D.112}$$

The Cauchy bound yields the no-go:

$$C^2_{\rm fail} > 1 \quad \Longrightarrow \quad \text{no source profile in this support channel can reach even the fail threshold.} \tag{D.113}$$

Output. Amplitude thresholds (D.111), coherence thresholds (D.112), and Cauchy no-go (D.113).

Downstream use. Stage 064 shows that parent equilibrium tends toward matched source/support alignment; Stages 065–069 reduce the thresholds to a wall figure of merit.

D.28 Stage 064: Parent Equilibrium Source/Support Alignment

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. Exact Within Closure **inside the same local compressional linearization used in Stages 060–063.**

Purpose. Stage 064 shows that the parent equilibrium equations tie the source profile to the support profile instead of leaving $C_{\sigma\phi}$ arbitrary.

Inputs. The local compressional stiffness $H(y) = h'(\rho_*(y))$ and support profile χ_ϕ.

Derivation

The equilibrium source profile is

$$\boxed{\chi_\sigma(y) = \frac{g_\phi \chi_\phi(y)}{H(y)}, \qquad H(y) = h'(\rho_*(y))}. \tag{D.114}$$

Define

$$I_1 = \int d^3y \frac{\chi_\phi^2}{H}, \qquad I_2 = \int d^3y \frac{\chi_\phi^2}{H^2}. \tag{D.115}$$

Then

$$O_{\sigma\phi} = g_\phi I_1, \qquad N_{\sigma\sigma} = g_\phi^2 I_2, \tag{D.116}$$

and the coherence is

$$\boxed{C_{\sigma\phi}^2 = \frac{I_1^2}{N_{\phi\phi} I_2} \le 1}. \tag{D.117}$$

The eliminated-source support softening is

$$\Delta K_X^{\text{eq}} = g_\phi^2 I_1, \qquad G_{\text{eq}} = \frac{g_\phi^2 I_1}{K_X}. \tag{D.118}$$

In a thin active layer where H is nearly constant,

$$C_{\sigma\phi}^2 = 1, \qquad G_{\text{eq}} = \frac{g_\phi^2 N_{\phi\phi}}{K_X H_w}. \tag{D.119}$$

Output. Equilibrium source/support alignment (D.114) and matched-layer gain (D.119).

Downstream use. Stages 065–066 insert an explicit thin-wall confinement and collapse the test to W_{wall}.

D.29 Stage 065: Explicit Thin-Wall Confinement Branch

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. Exact Within Closure / Reduced / Controlled Reduction **for the declared thin-wall confinement branch.**

Purpose. Stage 065 translates the parent gain thresholds into wall-amplitude thresholds for a concrete confinement profile.

Inputs. Thin-wall confinement $V_{\mathrm{conf}}(r;a) = V_0 f((r-a)/\ell)$ and matched active layer.

Derivation

The wall-loading amplitude is

$$g_\phi = \frac{V_0}{\ell}. \tag{D.120}$$

Inserting this into the matched-layer gain yields thresholds directly in terms of V_0, ℓ, support stiffness, and wall geometry. Stage 066 packages those quantities into the dimensionless wall figure of merit.

Output. A physical wall-amplitude interpretation of g_ϕ.

Downstream use. Stage 066 defines W_{wall}, the scalar control of the matched support/source theorem.

D.30 Stage 066: Dimensionless Wall Figure of Merit

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. Exact Within Closure **for the matched thin-wall theorem.**

Purpose. Stage 066 collapses the matched support/source threshold to one dimensionless wall control.

Inputs. The thin-wall branch of Stage 065 and the operator thresholds of Stage 059.

Derivation

The dimensionless wall figure of merit is

$$\boxed{W_{\mathrm{wall}} = \frac{4\pi a^2 L^2 J_1 V_0^2}{T_X \ell}}. \tag{D.121}$$

The exact matched-branch verdict is

$$\boxed{W_{\mathrm{wall}} \le \frac{\mathrm{Pe}_{\mathrm{req}}}{\Delta_\infty} \Rightarrow \text{fail}, \qquad W_{\mathrm{wall}} \ge \frac{\mathrm{Pe}_{\mathrm{req}}}{\Delta_0} \Rightarrow \text{succeed}}. \tag{D.122}$$

Only the intermediate band requires solving the full operator branch.

Output. The wall control (D.121) and exact window (D.122).

Downstream use. Stages 067–069 quantify how much profile mismatch can change this matched verdict.

D.31 Stage 067: Exact Sech–Gaussian Coherence Resonance Benchmark

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. Exact Within Closure **for the benchmark family;** Numerically Located **for the quoted coherence value.**

Purpose. Stage 067 tests how sensitive the matched theorem is to independent source/support shapes by using a sech–Gaussian benchmark family.

Inputs. Source profile $\chi_\sigma = \operatorname{sech}(y/w_f)$ and support profile $\chi_\phi = e^{-y^2/w_g^2}$.

Derivation

The exact self-dual stationary point of the shape family is

$$\boxed{\frac{w_g}{w_f} = \sqrt{\pi}}. \tag{D.123}$$

At this point the coherence is

$$\boxed{C_{\mathrm{res}}^2 \simeq 0.994418836451529}. \tag{D.124}$$

The corresponding penalty factor is

$$\boxed{P_{\mathrm{res}} = \frac{1}{C_{\mathrm{res}}^2} \simeq 1.005612487760576}. \tag{D.125}$$

Output. The near-perfect benchmark coherence (D.124) and penalty (D.125).

Downstream use. Stage 068 applies P_{res} to the matched thresholds; Stage 069 gives the final support/source verdict.

D.32 Stage 068: Resonance-Corrected Thresholds for the Sech–Gaussian Benchmark

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. Exact Within Closure **after inserting the benchmark coherence penalty.**

Purpose. Stage 068 shows that the first independent-profile benchmark changes the matched thresholds only mildly.

Inputs. The matched window (D.122) and penalty factor P_{res}.

Derivation

The resonance-corrected matched thresholds are multiplied by

$$P_{\mathrm{res}} \simeq 1.005612487760576. \tag{D.126}$$

Thus the independent-profile benchmark raises the required wall figure of merit by about 0.56%, not by an order-one factor.

Output. Threshold correction factor P_{res}.

Downstream use. Stage 069 uses this to state a three-zone support/source verdict.

D.33 Stage 069: Final Reduced Verdict for the Support/Source Program

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. Exact Within Closure **for the reduced verdict;** Open **for actual branch placement outside the matched/prototyped families.**

Purpose. Stage 069 summarizes the support/source program before inserting the explicit Family–1 wall branch.

Inputs. The matched wall theorem, profile penalty benchmark, and operator-selected support/source thresholds.

Derivation

The reduced support/source side has three regimes:

$$\boxed{\begin{array}{lcl} W_{\text{wall}} < \text{Pe}_{\text{req}}/\Delta_\infty & \Rightarrow & \text{universal fail,} \\ W_{\text{wall}} > \text{Pe}_{\text{req}}/\Delta_0 & \Rightarrow & \text{universal matched success,} \\ \text{Pe}_{\text{req}}/\Delta_\infty \le W_{\text{wall}} \le \text{Pe}_{\text{req}}/\Delta_0 & \Rightarrow & \text{profile-sensitive band.} \end{array}} \tag{D.127}$$

The sech–Gaussian benchmark shows that realistic smooth mismatch need not significantly enlarge the middle band, but the actual PDE branch still has to supply its own profiles.

Output. The final pre-Family–1 support/source verdict (D.127).

Downstream use. Stages 070–078 evaluate W_{wall}, κ, and η for the explicit Family–1 branch.

D.34 Stage 070: Explicit GNLS Wall-Shell Reduction for the First Support Branch

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. Exact Within Closure / Reduced / Controlled Reduction **for the thin GNLS shell support reduction.**

Purpose. Stage 070 begins the explicit Family–1 construction by deriving support stiffness and gradient energy from the parent GNLS shell.

Inputs. A support perturbation $\delta\rho(s, y) = q(s)\chi_\phi(y)$, wall density ρ_w, and frozen $n = 5$ compressional stiffness.

Derivation

The quadratic support energy is

$$E^{(2)}[q] = \frac{1}{2}\int_0^L ds\,[T_X q'(s)^2 + K_X q(s)^2]. \tag{D.128}$$

The coefficients are

$$\boxed{T_X = \frac{\hbar^2}{4m\rho_w}N_{\phi\phi}, \qquad K_X = H_w N_{\phi\phi} + \frac{\hbar^2}{4m\rho_w}G_{\phi\phi}, \qquad H_w = \frac{mc_{s,w}^2}{\rho_w}}. \tag{D.129}$$

For a thin shell with moments $I_f = \int f'^2 d\xi$ and $I_g = \int f''^2 d\xi$,

$$\boxed{T_X = \frac{\pi a^2 \ell I_f \hbar^2}{m\rho_w}, \qquad K_X = 4\pi a^2 \ell I_f H_w + \frac{\pi a^2 I_g \hbar^2}{m\rho_w \ell}}. \tag{D.130}$$

Thus

$$\boxed{\kappa = \frac{K_X L^2}{T_X} = 4\left(\frac{mc_{s,w}L}{\hbar}\right)^2 + \frac{I_g}{I_f}\left(\frac{L}{\ell}\right)^2}. \tag{D.131}$$

The matched wall control is

$$\boxed{W_{\text{wall}} = \Xi = \frac{4\rho_w^2 V_0^2 L^2}{\hbar^2 c_{s,w}^2 \ell^2}}. \tag{D.132}$$

Output. The parent-shell coefficients (D.129)–(D.132).

Downstream use. Stage 071 evaluates these for the canonical tanh wall; Stages 073–074 fix the Family–1 geometry and healing lock.

D.35 Stage 071: Canonical Tanh-Wall Branch and Natural Local Mouth Closure

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. Exact Within Closure **for the canonical wall-profile moments.**

Purpose. Stage 071 chooses the canonical tanh-wall shell and records the resulting branch controls.

Inputs. Profile $f(\xi) = (1 + \tanh \xi)/2$.

Derivation

For

$$f(\xi) = \frac{1 + \tanh \xi}{2}, \qquad f'(\xi) = \frac{1}{2}\operatorname{sech}^2 \xi, \tag{D.133}$$

the exact moments are

$$\boxed{I_f = \frac{1}{3}, \qquad I_g = \frac{4}{15}, \qquad \frac{I_g}{I_f} = \frac{4}{5}}. \tag{D.134}$$

The natural local mouth closure is

$$K_m = \frac{T_X}{\ell}, \qquad \eta = \frac{K_m L}{T_X} = \frac{L}{\ell} = \Lambda_\ell. \tag{D.135}$$

Define

$$\chi_s = \frac{m c_{s,w} L}{\hbar}, \qquad \Lambda_\ell = \frac{L}{\ell}, \qquad \Upsilon_w = \frac{4\rho_w^2 V_0^2}{\hbar^2 c_{s,w}^2}. \tag{D.136}$$

Then

$$\boxed{\kappa = 4\chi_s^2 + \frac{4}{5}\Lambda_\ell^2, \qquad \eta = \Lambda_\ell, \qquad W_{\rm wall} = \Upsilon_w \Lambda_\ell^2}. \tag{D.137}$$

Output. Canonical branch controls (D.137).

Downstream use. Stages 073–074 fix Λ_ℓ, η, χ_s, and κ for Family–1.

D.36 Stage 072: Explicit Branch Placement Map and Threshold Surfaces

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. Exact Within Closure **for the branch-threshold map in $(\chi_s, \Lambda_\ell, \Upsilon_w)$.**

Purpose. Stage 072 expresses the support/source fail/succeed surfaces directly in terms of the canonical branch controls.

Inputs. The canonical branch controls χ_s, Λ_ℓ, Υ_w and the exact thresholds from Stage 069.

Derivation

Using (D.137), the support/source thresholds become surfaces in the three branch controls:

$$\Upsilon_w \Lambda_\ell^2 \lesseqgtr \frac{\mathrm{Pe}_{\rm req}}{\Delta_{\infty,0}(\kappa(\chi_s, \Lambda_\ell), \eta = \Lambda_\ell)}. \tag{D.138}$$

This is the branch-level version of the wall figure of merit. It is exact once $(\chi_s, \Lambda_\ell, \Upsilon_w)$ are selected.

Output. A threshold-surface map in the canonical branch variables.

Downstream use. Stages 073–078 insert the Family–1 values and compare actual wall-depth data against the thresholds.

D.37 Stage 073: Family–1 Reference-Branch Geometry Map

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. Reduced / Controlled Reduction / Exact Within Closure **for the declared Family–1 reference geometry.**

Purpose. Stage 073 fixes the geometry ratios of the first explicit Family–1 branch.

Inputs. The reference finite-throat geometry selected earlier in the program.

Reference-branch freeze

The selected Family–1 reference ratios, posited for this branch, are

$$\boxed{\frac{L}{a} = \frac{37}{20} \quad \text{(reference freeze)}, \qquad \frac{\ell}{a} = \frac{1}{20} \quad \text{(posited)}}. \tag{D.139}$$

Therefore

$$\boxed{\Lambda_\ell = \frac{L}{\ell} = 37, \qquad \eta = 37}. \tag{D.140}$$

Output. The Family–1 geometry values (D.140).

Downstream use. Stage 074 uses the healing lock to fix χ_s and κ.

D.38 Stage 074: Healing-Length Lock and Reference-Branch Support Scale

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. Reduced / Controlled Reduction / Exact Within Closure **for the declared healing-width closure.**

Purpose. Stage 074 fixes the support scale of Family–1 through a healing-length lock.

Inputs. The Family–1 geometry $L/\ell = 37$ and closure $\ell = \hbar/(2mc_{s,w})$.

Derivation

The healing lock is

$$\boxed{\ell = \frac{\hbar}{2mc_{s,w}}}. \tag{D.141}$$

Thus

$$\boxed{\chi_s = \frac{mc_{s,w}L}{\hbar} = \frac{37}{2}}. \tag{D.142}$$

Using $\kappa = 4\chi_s^2 + (4/5)\Lambda_\ell^2$ gives

$$\boxed{\kappa = \frac{9}{5}\Lambda_\ell^2 = \frac{12321}{5}, \qquad \alpha = \sqrt{\kappa} = \frac{111}{\sqrt{5}}}. \tag{D.143}$$

Output. Family–1 support values $\chi_s = 37/2$, $\kappa = 12321/5$, $\eta = 37$.

Downstream use. Stage 075 evaluates the threshold window at these values.

D.39 Stage 075: Explicit Family–1 Threshold Window

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. EXACT WITHIN CLOSURE **for the numerical evaluation of the exact threshold functions.**

Purpose. Stage 075 evaluates the operator thresholds for the fixed Family–1 geometry and support scale.

Inputs. $\kappa = 12321/5$, $\eta = 37$, and $\Upsilon_w = 100\Theta_w$.

Derivation

The endpoint functions evaluate to

$$\Delta_0\left(\frac{12321}{5}, 37\right) \simeq 1.73302079021525 \times 10^{-4}, \tag{D.144}$$

$$\Delta_\infty\left(\frac{12321}{5}, 37\right) \simeq 2.01447565540522 \times 10^{-2}. \tag{D.145}$$

With $V_0 = \alpha_r \mu_*$, $\alpha_r = 10$, and $\Upsilon_w = 100\Theta_w$, the Family–1 threshold window is

$$\boxed{\Theta_w \leq 3.62605617972939 \times 10^{-4}\,\mathrm{Pe}_{\mathrm{req}} \Rightarrow \text{fail}}, \tag{D.146}$$

$$\boxed{\Theta_w \geq 4.21495341569977 \times 10^{-2}\,\mathrm{Pe}_{\mathrm{req}} \Rightarrow \text{succeed}}. \tag{D.147}$$

Output. The explicit Family–1 wall-depth window (D.146)–(D.147).

Downstream use. Stages 076–077 compute Θ_w from the wall profile; Stage 078 compares it to the window.

D.40 Stage 076: Exact $n = 5$ Wall-Depth Lock

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. EXACT WITHIN CLOSURE **inside the Family–1 normalization conventions and frozen $n = 5$ EOS.**

Purpose. Stage 076 expresses the remaining wall-depth datum Θ_w in normalized Family–1 density variables.

Inputs. The $n = 5$ identity $h(\rho) = mc_s^2/4$, healing lock, and wall-density normalization.

Derivation

The exact $n = 5$ relation and healing lock reduce the wall-depth datum to

$$\boxed{\Theta_w = 25\lambda_\mu^2 \rho_w^2}. \tag{D.148}$$

Thus only the effective wall density weighting remains to be extracted from the explicit wall profile.

Output. The wall-depth lock (D.148).

Downstream use. Stage 077 evaluates ρ_w^2 with shell weights; Stage 078 uses it in the threshold comparison.

D.41 Stage 077: Shell-Weighted Extraction of Θ_w on Family–1

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. Reduced / Controlled Reduction / Numerically Located **for the shell-weighted profile extraction.**

Purpose. Stage 077 evaluates the natural wall-density weighting and a conservative lower envelope.

Inputs. The explicit radial wall profile and the canonical support weighting.

Derivation

The shell-weighted profile extraction gives

$$\langle \rho_r \rangle_\chi \simeq 0.192619005556493, \qquad \langle \rho_r^2 \rangle_\chi \simeq 0.162745294003265. \tag{D.149}$$

Using (D.148), the natural shell-weighted datum is

$$\boxed{\Theta_w^{(\chi)} = 25\lambda_\mu^2 \langle \rho_r^2 \rangle_\chi \simeq 4.06863235008162\lambda_\mu^2}. \tag{D.150}$$

The conservative lower envelope is

$$\boxed{\Theta_w^{(J)} = 25\lambda_\mu^2 \langle \rho_r \rangle_\chi^2 \simeq 0.927552032539308\lambda_\mu^2}. \tag{D.151}$$

Output. Natural and conservative Family–1 wall-depth data (D.150)–(D.151).

Downstream use. Stage 078 compares these against the exact threshold window.

D.42 Stage 078: Explicit Family–1 Branch Comparison and Closing Verdict for This Subprogram

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. Exact Within Closure **for the comparison of extracted Family–1 data against the threshold window.**

Purpose. Stage 078 decides whether the explicit Family–1 wall-depth datum is the bottleneck.

Inputs. The threshold window of Stage 075 and the extracted Θ_w data from Stage 077.

Derivation

For the natural shell-weighted datum,

$$\boxed{\mathrm{Pe}_{\mathrm{suff}}^{(\chi)} \simeq 96.5285247264386\lambda_\mu^2, \qquad \mathrm{Pe}_{\mathrm{fail}}^{(\chi)} \simeq 11220.5441626259\lambda_\mu^2}. \tag{D.152}$$

For the conservative floor,

$$\boxed{\mathrm{Pe}_{\mathrm{suff}}^{(J)} \simeq 22.0062226330754\lambda_\mu^2, \qquad \mathrm{Pe}_{\mathrm{fail}}^{(J)} \simeq 2558.01892349205\lambda_\mu^2}. \tag{D.153}$$

The Family–1 branch is therefore not wall-depth starved except if the quadrupole-side demand requires an anomalously large transport bias.

Output. The first explicit Family–1 support/source verdict: wall-depth is not the leading bottleneck for moderate demand.

Downstream use. Stages 079–087 translate quadrupole demand into the remaining loading ratio.

D.43 Stage 079: Exact Family–1 Map from Quadrupole Demand to Required Transport Bias

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. Exact Within Closure **for the Family–1 demand-to-transport map.**

Purpose. Stage 079 converts a required support ratio into the transport bias required on the explicit Family–1 branch.

Inputs. The Family–1 parameters $\kappa_{\mathrm{F1}} = 12321/5$, $\eta = 37$, and the physical support map of Stage 057.

Derivation

Let y_{F1} solve

$$y_{\mathrm{F1}} \tan y_{\mathrm{F1}} = 37, \qquad 0 < y_{\mathrm{F1}} < \frac{\pi}{2}, \qquad y_{\mathrm{F1}} \simeq 1.52948248371470. \tag{D.154}$$

Define

$$A_{\mathrm{F1}} = \frac{\kappa_{\mathrm{F1}} + \pi^2/4}{\kappa_{\mathrm{F1}} + y_{\mathrm{F1}}^2}, \qquad \kappa_{\mathrm{F1}} = \frac{12321}{5}. \tag{D.155}$$

Then

$$\boxed{A_{\mathrm{F1}} \simeq 1.00005192880220}. \tag{D.156}$$

The Family–1 support-ratio map is

$$\boxed{\zeta_{\mathrm{F1}}(\mathrm{Pe}) = A_{\mathrm{F1}}\Omega_{\mathrm{Pe}}^2}. \tag{D.157}$$

The hard constructive ceiling is

$$\boxed{\zeta_{\max}^{\mathrm{F1}} = A_{\mathrm{F1}}\frac{\pi^2}{4} \simeq 2.46752922945601}. \tag{D.158}$$

Output. Family–1 demand map (D.157) and ceiling (D.158).

Downstream use. Stages 080–081 convert wall-depth thresholds into ζ and product windows.

D.44 Stage 080: Family–1 Demand Thresholds in ζ_{req}

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. Exact Within Closure **for the conversion of** $\mathrm{Pe}_{\mathrm{req}}$ **windows to** ζ_{req} **windows.**

Purpose. Stage 080 expresses the explicit Family–1 wall-depth windows in the support-ratio demand variable.

Inputs. The demand map $\zeta_{\mathrm{F1}} = A_{\mathrm{F1}}\Omega_{\mathrm{Pe}}^2$ and the Stage 078 Peclet windows.

Derivation

At $\lambda_\mu = 1$, the natural shell-weighted thresholds are

$$\boxed{\zeta_{\mathrm{suff}}^{(\chi)} \simeq 2.46622291347846, \qquad \zeta_{\mathrm{fail}}^{(\chi)} \simeq 2.46752913273870}. \tag{D.159}$$

The conservative floor gives

$$\boxed{\zeta_{\mathrm{suff}}^{(J)} \simeq 2.44257571477179, \qquad \zeta_{\mathrm{fail}}^{(J)} \simeq 2.46752736855058}. \tag{D.160}$$

Both are close to the hard Family–1 ceiling (D.158); the natural branch nearly reaches the largest possible constructive support ratio in this family.

Output. Demand thresholds (D.159)–(D.160).

Downstream use. Stage 081 rewrites the same window in Π_{tr}.

D.45 Stage 081: Family–1 Demand Window in the Branch-Product Variable

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. Exact Within Closure **for the branch-product translation.**

Purpose. Stage 081 expresses the Family–1 demand thresholds in the branch-product variable $\Pi_{\rm tr}$, including blocking.

Inputs. The support-ratio threshold formula and blocking parameter $\epsilon_{\rm blk}$.

Derivation

The branch-product demand is related to $\zeta_{\rm req}$ by

$$\zeta_{\rm req} = \frac{\Pi_{\rm tr} - C_{\rm mix}}{C_{\rm mix} - \epsilon_{\rm blk}(2C_{\rm mix} - \Pi_{\rm tr})}. \tag{D.161}$$

Solving for $\Pi_{\rm tr}$ gives the equivalent product threshold. In the unblocked limit, $\epsilon_{\rm blk} = 0$, this reduces to

$$\frac{\Pi_{\rm tr}}{C_{\rm mix}} = 1 + \zeta_{\rm req}. \tag{D.162}$$

Thus the ζ-windows of Stage 080 can be read directly as product-ratio windows.

Output. Product-variable conversion (D.161)–(D.162).

Downstream use. Stage 082 packages the full reduced branch into a master residual.

D.46 Stage 082: Master Quadrupole Residual of the Full Reduced Moving-Throat PDE

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. Exact Within Closure **for the residual definition;** Open **for the actual PDE branch values inserted into it.**

Purpose. Stage 082 defines the master scalar residual that compares support demand against physical support supply.

Inputs. The support-demand function $\zeta_{\rm req}$, physical support map $\zeta_{\rm phys}$, and operator-selected bias Pe_*.

Derivation

The demand is

$$\zeta_{\rm req}(\Pi_{\rm tr}, C_{\rm mix}, \epsilon_{\rm blk}) = \frac{\Pi_{\rm tr} - C_{\rm mix}}{C_{\rm mix} - \epsilon_{\rm blk}(2C_{\rm mix} - \Pi_{\rm tr})}. \tag{D.163}$$

The physical supply is

$$\zeta_{\rm phys}(\mathrm{Pe}, \eta; \kappa) = \Omega_{\rm Pe}^2 \frac{\kappa + \pi^2/4}{\kappa + y(\eta)^2}. \tag{D.164}$$

With Pe_* selected by Stage 058, define

$$\boxed{R_{\rm quad} = \zeta_{\rm req}(\Pi_{\rm tr}, C_{\rm mix}, \epsilon_{\rm blk}) - \zeta_{\rm phys}(\mathrm{Pe}_*(\Xi, \eta, \kappa), \eta; \kappa)}. \tag{D.165}$$

The sign convention is

$$R_{\text{quad}} < 0 \Rightarrow \text{support exceeds demand}, \quad R_{\text{quad}} = 0 \Rightarrow \text{saturation}, \quad R_{\text{quad}} > 0 \Rightarrow \text{support failure}. \tag{D.166}$$

The Family–1 specialization is obtained by setting $(\eta, \kappa) = (37, 12321/5)$ and

$$\Xi_{\text{F1}} = W_{\text{wall}} = 1369\Upsilon_w = 136900\Theta_w. \tag{D.167}$$

Output. The master residual (D.165).

Downstream use. Stages 083–087 narrow the Family–1 residual to one loading ratio from the outgoing branch.

D.47 Stage 083: Direct Operator-Selected Family–1 Window

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. Exact Within Closure / Numerically Located **for direct evaluation of the operator-selected Family–1 branch.**

Purpose. Stage 083 evaluates the Family–1 support window directly through the operator-selected branch rather than only through endpoint bounds.

Inputs. The Family–1 parameters, Ξ_{F1}, and the fixed-point equation for Pe_*.

Derivation

The direct operator-selected support supply is

$$\zeta_{\text{F1}}^{\text{op}} = \zeta_{\text{phys}}(\text{Pe}_*(\Xi_{\text{F1}}, 37, 12321/5), 37; 12321/5). \tag{D.168}$$

The direct window agrees with the endpoint logic: the branch has a hard constructive ceiling and a narrow endpoint-sensitive band near that ceiling. This calculation is a numerical placement of exact functions, not an independent fit.

Output. Direct operator-selected support window for Family–1.

Downstream use. Stage 084 records the reduced write-up skeleton and final remaining theorem gap.

D.48 Stage 084: Full Reduced Moving-Throat PDE Write-Up Skeleton and Remaining Theorem Gap

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. Exact Within Closure **for the reduced hierarchy organization;** Open **for final actual-branch realization.**

Purpose. Stage 084 organizes the completed reduced hierarchy into a write-up-ready theorem chain and isolates the final outgoing quadrupole product as the remaining gap.

Inputs. Stages 037–083.

Derivation

The reduced PDE chain is now

$$\begin{aligned}\text{continuum placement} &\to \text{support completion} \to \text{transport/parent gain}\\ &\to \text{Family–1 support map} \to R_{\text{quad}}.\end{aligned} \tag{D.169}$$

The remaining open datum is the outgoing branch's demand side, i.e. the conservative quadrupole module and its loading ratio. The support/source machinery can be cited as completed within the declared reduced Family–1 closure once that ratio is supplied.

Output. A structured reduced-PDE theorem chain and handoff to the outgoing-branch loading-ratio problem.

Downstream use. Stages 085–090 finish the support side by showing that the natural minimal module supplies a small ratio.

D.49 Stage 085: Exact Cancellation of Outgoing-Normalization Factors in the Selected Demand Product

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. Exact Within Closure **for the selected-branch cancellation.**

Purpose. Stage 085 proves that after the selected outgoing quadrupole branch is normalized, the support theorem depends only on a loading ratio, not on the absolute outgoing normalization factors.

Inputs. Selected outgoing normalization, N_Q^{target}, β_0, s_-, λ_-, and selected support loading.

Derivation

The selected product and mixed product are

$$\boxed{\Pi_{\text{tr}} = \frac{N_Q^{\text{target}}}{\beta_0}\alpha_{\text{req}}, \qquad C_{\text{mix}} = \frac{N_Q^{\text{target}}}{\beta_0}\alpha_{\text{mix}}}. \tag{D.170}$$

Therefore the absolute factors cancel:

$$\boxed{\frac{\Pi_{\text{tr}}}{C_{\text{mix}}} = \frac{\alpha_{\text{req}}}{\alpha_{\text{mix}}} =: \rho_\alpha}. \tag{D.171}$$

The support demand becomes

$$\boxed{\zeta_{\text{req}}(\rho_\alpha, \epsilon_{\text{blk}}) = \frac{\rho_\alpha - 1}{1 - \epsilon_{\text{blk}}(2 - \rho_\alpha)}}. \tag{D.172}$$

In the unblocked limit, $\zeta_{\text{req}} = \rho_\alpha - 1$.

Output. The cancellation theorem (D.171) and reduced demand formula (D.172).

Downstream use. Stages 086–087 express the Family–1 support window entirely in ρ_α; Stage 088 computes ρ_α from the minimal module.

D.50 Stage 086: Exact Family–1 Success/Failure Window in the Pure Loading-Ratio Variable

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. Exact Within Closure **for the Family–1 ratio window.**

Purpose. Stage 086 converts the explicit Family–1 support/source verdict into a pure window for ρ_α.

Inputs. The cancellation theorem of Stage 085 and the Family–1 demand thresholds.

Derivation

In the unblocked branch at $\lambda_\mu = 1$, the Family–1 success/failure window is

$$\boxed{\rho_\alpha \le 3.46622291347846 \Rightarrow \text{guaranteed success}}, \tag{D.173}$$

$$\boxed{\begin{aligned} &\rho_\alpha \ge 3.46752913273870 \Rightarrow \text{guaranteed failure}, \\ &\rho_\alpha < 3.46752922945601 \Rightarrow \text{absolute constructive ceiling}. \end{aligned}} \tag{D.174}$$

Thus the entire explicit support/source side waits for one number from the outgoing conservative precursor: ρ_α.

Output. The pure Family–1 loading-ratio window (D.173)–(D.174).

Downstream use. Stage 087 states the finish line; Stage 088 supplies ρ_α for the minimal module.

D.51 Stage 087: Final Reduced Finish Line for the Explicit Family–1 Outgoing Quadrupole Branch

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. Exact Within Closure **for the reduction to a single ratio;** Open **until the outgoing conservative module supplies that ratio.**

Purpose. Stage 087 records that the explicit Family–1 support/source side has been reduced to a single outgoing-branch loading ratio.

Inputs. Stages 085–086.

Derivation

The explicit support theorem now reads:

$$\boxed{\rho_\alpha = \frac{\alpha_{\text{req}}}{\alpha_{\text{mix}}} \quad \text{is the only remaining support-side input from the outgoing branch.}} \tag{D.175}$$

Given ρ_α, the Family–1 support/source verdict follows from Stage 086. No separate dependence on N_Q^{target}, β_0, s_-, λ_-, or $\widehat{m}_-$ remains.

Output. The one-ratio finish line (D.175).

Downstream use. Stage 088 evaluates ρ_α for the minimal isotropic contact-plus-pole quadrupole module.

D.52 Stage 088: Loading-Ratio Extraction from the Minimal Isotropic Quadrupole Module

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. Exact Within Closure **under the natural contact-plus-pole interpretation of the minimal isotropic conservative quadrupole precursor.**

Purpose. Stage 088 supplies the one missing loading ratio by reading the minimal isotropic conservative quadrupole module as contact plus pole.

Inputs. The minimal isotropic conservative precursor $Y_Q^{\text{cons}} = 3/4 + (1/4)/(1 - \omega^2/\Omega_Q^2)$.

Derivation

The minimal precursor is

$$\boxed{Y_Q^{\text{cons}}(\omega) = c_0 + \frac{c_1}{1 - \omega^2/\Omega_Q^2}, \qquad c_0 = \frac{3}{4}, \qquad c_1 = \frac{1}{4}, \qquad \Omega_Q = \frac{3c_s}{2a}}. \tag{D.176}$$

The natural contact-plus-pole reading writes

$$Y_Q^{\text{cons}}(\omega) = \frac{1}{\rho_\alpha} + \frac{\rho_\alpha - 1}{\rho_\alpha} \frac{1}{1 - \omega^2/\Omega_Q^2}. \tag{D.177}$$

Matching coefficients gives

$$\boxed{\rho_\alpha = \frac{1}{c_0} = \frac{4}{3}, \qquad \zeta_{\text{req}} = \frac{c_1}{c_0} = \frac{1}{3}, \qquad \Pi_{\text{tr}} = \frac{4}{3} C_{\text{mix}}}. \tag{D.178}$$

This lies in the symmetric-lowest-twin regime:

$$C_{\text{mix}} < \Pi_{\text{tr}} < 2C_{\text{mix}}, \qquad 0 < \zeta_{\text{req}} = \frac{1}{3} < 1. \tag{D.179}$$

Output. The minimal-module ratio (D.178).

Downstream use. Stage 089 compares $\zeta_{\rm req} = 1/3$ against the explicit Family–1 support map.

D.53 Stage 089: Explicit Family–1 Verdict for the Minimal Isotropic Passive/Outgoing Branch

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. Exact Within Closure **given the Stage 088 minimal isotropic precursor.**

Purpose. Stage 089 evaluates the minimal isotropic demand on the explicit Family–1 support/source branch.

Inputs. $\zeta_{\rm req} = 1/3$ and the zero-bias Family–1 value $A_{\rm F1} \simeq 1.00005192880220$.

Derivation

At zero transport bias, $\Omega_{\rm Pe=0} = 1$, so Family–1 supplies

$$\zeta_{\rm F1}(0) = A_{\rm F1} \simeq 1.00005192880220. \tag{D.180}$$

Since

$$\boxed{\zeta_{\rm req}^{\rm min} = \frac{1}{3} < A_{\rm F1}}, \tag{D.181}$$

the required transport bias is

$$\boxed{\mathrm{Pe}_{\rm req} = 0}. \tag{D.182}$$

The explicit Family–1 support/source branch therefore supports the minimal isotropic passive/outgoing quadrupole demand before any transport-driven overlap boost is turned on.

Output. The zero-bias Family–1 success theorem (D.182).

Downstream use. Stage 090 records the updated reduced theorem status and hands the problem to the grouped P_2/geometry derivation of Part IV.

D.54 Stage 090: Updated Reduced Theorem Status After Loading-Ratio Extraction

Part / anchor. Part III; secondary anchor **MTDC-T7**.

Claim status. Exact Within Closure **for the support/source verdict under the minimal module;** Open **for derivation of that minimal module from the actual moving-throat grouped P_2/geometry branch.**

Purpose. Stage 090 records the clean status update after Stages 088–089.

Inputs. The explicit Family–1 support/source side and the minimal isotropic contact-plus-pole conservative precursor.

Derivation

Inside the reduced Family–1 branch, the following are settled:

1. the explicit Family–1 support/source branch has a hard constructive ceiling;
2. outgoing normalization factors cancel from the support theorem;
3. the support theorem depends only on $\rho_\alpha = \alpha_{\mathrm{req}}/\alpha_{\mathrm{mix}}$;
4. the natural contact-plus-pole reading gives $\rho_\alpha = 4/3$;
5. this lies strictly inside the Family–1 success region;
6. the explicit branch succeeds at zero transport bias.

The remaining reduced question is not whether the Family–1 support/source side can meet the minimal module. It is whether the actual grouped real P_2/geometry throat branch realizes that minimal isotropic conservative quadrupole module.

Output. The support/source side is finished under the minimal module. The next theorem gate is derivation of the module itself.

Downstream use. Part IV begins with the grouped P_2+geometry derivation of the $3/4 + 1/4$ conservative quadrupole module.

D.55 Part III verification summary

The Part III verification ledger establishes a chain of exact reduced implications:

$$\begin{aligned}\text{Stage 037 continuum data} &\Rightarrow \text{Stage 038 placement product}\\ &\Rightarrow \text{Stages 039–044 rank–2 support closure}\\ &\Rightarrow \text{Stages 045–051 coherent D/N support threshold}\\ &\Rightarrow \text{Stages 052–069 parent support/source gain theorem}\\ &\Rightarrow \text{Stages 070–078 explicit Family–1 wall data}\\ &\Rightarrow \text{Stages 085–090 one-ratio finish line.}\end{aligned} \tag{D.183}$$

The most important audit points are:

1. The selected branch variables are not free: they are extracted from a finite-throat continuum Schur complement.
2. The dimensionless placement map has an exact product law, so the target and mixed load cannot be varied independently once (ϵ_W, Λ) are fixed.
3. The first non-flat U correction preserves scalar placement but breaks source/loading collinearity unless $\delta_U = 0$ or $\rho_0 = 0$.

4. Rank–2 support completion is analytic because the determinant is linear in support loading.
5. The first coherent local D/N support kernel forces tracking rather than an arbitrary rank–2 rescue.
6. Support enhancement is controlled by $S(\zeta;\epsilon)$, and the lowest symmetric twin gives exact doubling.
7. If non-twin support is needed, overlap boost, Robin softening, and transport bias are all bounded by explicit finite-throat formulas.
8. The source transport bias has a drift-diffusion origin and is selected by a coupled support/source fixed point.
9. The microscopic gain projects back to parent GNLS/confinement overlaps, so insufficient parent confinement loading or poor source/support coherence produces an exact no-go in the declared channel.
10. The canonical tanh-wall Family–1 branch fixes $\Lambda_\ell = \eta = 37$, $\chi_s = 37/2$, and $\kappa = 12321/5$.
11. The explicit Family–1 wall-depth datum is not starved for the minimal branch; its remaining support-side input is only ρ_α.
12. The minimal contact-plus-pole quadrupole precursor gives $\rho_\alpha = 4/3$, $\zeta_{\mathrm{req}} = 1/3$, and $\mathrm{Pe}_{\mathrm{req}} = 0$, safely inside the Family–1 success region.

- ☐ Check that every use of M_{mix}, R_{target}, and δ traces back to the Stage 037 continuum extraction and Stage 038 dimensionless placement map.
- ☐ Keep ρ, ρ_0, and ρ_α distinct: only ρ_α is the outgoing-branch loading ratio.
- ☐ Do not use the split–U deformation as if it were a generic rank–2 support rescue; Stages 045–046 show the first coherent D/N kernel tracks.
- ☐ Apply the lowest-twin criterion only when $\Pi_{\mathrm{tr}} \le 2C_{\mathrm{mix}}$ or $\zeta_{\mathrm{req}} \le 1$.
- ☐ Apply non-twin overlap and Robin softening only within the finite ceilings $\Omega_0^2 \le \pi^2/4$ and $A_K \le 4/(4-x)$.
- ☐ Treat Pe as a transport-selected physical bias, not as an arbitrary fitting variable.
- ☐ Preserve the parent-overlap Cauchy no-go: if the aligned gain is below the fail threshold, source-profile engineering cannot rescue the branch in the declared support channel.
- ☐ Distinguish the natural shell-weighted $\Theta_w^{(\chi)}$ from the conservative lower envelope $\Theta_w^{(J)}$.
- ☐ Apply the cancellation of outgoing normalization factors only after the selected outgoing branch has been normalized.
- ☐ Mark the final Family–1 success as conditional on the minimal isotropic contact-plus-pole conservative precursor; deriving that precursor is the next block, not an output of Part III.

The citation-safe summary is:

Stages 037–090 derive the support-completion and explicit Family–1 side of the moving-throat response program. The stages extract the finite-throat continuum placement data, prove an exact product law, treat split–U source/loading mismatch by rank–2 support completion, reduce coherent local D/N support to a tracking branch with an explicit support-ratio threshold, bound non-twin rescue through overlap, Robin, transport, and parent-action gain, evaluate the first GNLS wall-shell Family–1 branch, and finally show that the minimal isotropic contact-plus-pole quadrupole precursor gives $\rho_\alpha = 4/3$, $\zeta_{\mathrm{req}} = 1/3$, and $\mathrm{Pe}_{\mathrm{req}} = 0$. Under that precursor, the explicit Family–1 support/source side succeeds; the remaining reduced gap is to derive the precursor itself from the grouped real P_2/geometry branch.

Appendix E

Stage Appendix: Part IV – Geometry Lane, Retarded Closure, Outlet/Core/Mouth Compensation

Stage range: 091–163

This appendix records the fourth theorem block of the moving-throat derivation ledger. Parts I–III built the parent wall/BdG/Maxwell/mixed response compiler, the grouped real P_2 normalization language, the selected finite-throat support front end, and the first explicit Family–1 support verdict. Part IV asks a different question. Once the support/source side is no longer the dominant reduced uncertainty, what prevents the reduced passive/outgoing quadrupole branch from becoming a full usable retarded branch?

The answer is progressively narrowed across Stages 091–163. First, the geometry lane is shown not to contaminate the isotropic grouped P_2 carrier at linear order, so the conservative quadrupole module collapses to the forced contact-plus-pole split

$$\widehat{Y}_Q^{\mathrm{cons}}(\omega) = \frac{3}{4} + \frac{1}{4}\frac{1}{1-\omega^2/\Omega_Q^2}. \tag{E.1}$$

Second, the remaining 2.5PN retarded ambiguity collapses to the leading outgoing ω^5 normalization factor χ_Q, with all higher odd data pushed above the point-particle 2.5PN order. Third, explicit outlet models show which low-frequency deformations preserve the canonical outgoing fingerprint and which do not. Pure Robin loading and a standalone hidden mixed pole fail; a compensated Robin–mixed outlet survives. Fourth, that compensated outlet is embedded in a concrete shell/mixed throat-core Schur complement, and then into a positive mouth-source boundary layer. Finally, the fully co-evolving core–mouth branch is linearized around its renormalized canonical point, where the last first-order obstruction is reduced to a single microscopic normal coordinate.

The result anchor served by this appendix is **MTDC-T8**. The block should be cited when a downstream paper needs the derivation of the geometry-lane firewall, the outgoing $l = 2$ DtN fingerprint, the factorization of the reduced 2.5PN defect, the compensated Robin–mixed outlet, the mouth boundary-layer fixed point, or the final first-order off-family defect scalar.

Audit-path map. For readability, this appendix organizes the seventy-three stages into eight audit paths:

1. Stages 091–099: geometry-lane firewall and the conservative $3/4 + 1/4$ module.

2. Stages 100–106: factorization of the reduced 2.5PN obstruction and canonical outgoing $l = 2$ DtN closure.

3. Stages 107–113: low-frequency outlet deformations and the compensated Robin–mixed branch.

4. Stages 114–124: two-channel core realization, parent extraction, and geometric core selection.

5. Stages 125–132: positive mouth-source theorem and explicit GNLS/localized-Maxwell mouth boundary layer.

6. Stages 133–145: coupled mouth fixed point, core-to-mouth gain map, and canonical regular branch selection.

7. Stages 146–153: first-order mouth-profile rigidity and finite mouth-only correction.

8. Stages 154–163: co-evolving core–mouth branch, linear defect transport, D/N similarity, and the final off-family normal coordinate.

The full stage-by-stage audit cards appear in appendix E.12.

Claim status. Mixed. The algebraic split, obstruction formulas, harmonic decoupling, outgoing DtN expansion, deformation laws, Schur complements, compensation surfaces, fixed-point laws, and first-order transport maps are Exact Within Closure **once their declared reduced models are adopted. Stable normal-mode reductions, finite D/N tube identifications, Family–1 geometry choices, positive mouth-layer ansatz, self-matched susceptibility closure, and co-evolving fixed-point branch are** Reduced / Controlled Reduction **or** Effective Closure **where explicitly stated. The numerically located co-evolving canonical point and representative correction values are** Numerically Located**. Actual realization by the completed nonlinear moving-throat PDE remains** Open**.**

E.1 Block contract and citation boundary

The contract of Part IV is to prevent the retarded quadrupole branch from being overclaimed. It does not assert that the full nonlinear PDE has been solved. It proves that, inside the declared reduced closure sequence, the active uncertainty can be compressed from many apparent degrees of freedom to one scalar branch-realization question at a time.

The chain of compression is

$$\text{geometry lane} \longrightarrow (\epsilon_2, \epsilon_4) \longrightarrow N_Q \longrightarrow \chi_Q \longrightarrow \Delta_Q \longrightarrow \delta\gamma_W \longrightarrow \Xi_{\text{slip}} \longrightarrow \delta_\perp. \tag{E.2}$$

Each arrow is a theorem inside a reduced block, not a statement that the final scalar is already zero on the actual branch.

Theorem E.1 (Part IV reduced retarded-compression theorem)**.** *Inside the declared grouped-P_2, static-geometry, passive/outgoing, compensated outlet, and co-evolving Family–1 mouth/core closures, the retarded quadrupole verification problem admits the following nested reductions.*

1. *The isotropic geometry lane contributes no dynamic $O(\omega^2)$ or $O(\omega^4)$ contamination to the grouped real P_2 carrier at linear order.*

2. *The conservative quadrupole module is therefore* (E.1)*.*

3. *The point-particle* 2.5*PN normalization condition factorizes as*

$$\widehat{m}_0^2 \chi_Q N_Q = 1. \tag{E.3}$$

On the natural source-map branch, the remaining retarded obstruction is $\Delta_Q = \chi_Q - 1$.

4. *The canonical outgoing* $l = 2$ *DtN branch has* $\chi_Q = 1$.

5. *In the explicit compensated Robin–mixed outlet class, first-order even preservation forces* $\delta\mathfrak{g} = 0$ *and* $\delta\kappa_W = 0$, *so the remaining first-order defect is purely odd.*

6. *In the concrete D/N mixed-tube realization, that odd defect is zero whenever the tangential deformation preserves the D/N similarity law.*

7. *The first off-family deviation from the exact parent compensation family is measured by a single normal coordinate* $\delta_\perp$.

Thus the end of Part IV is not a broad family of remaining outlet defects. It is the concrete microscopic question whether the true branch has $\delta_\perp = 0$, *or if not, what value that scalar takes.*

Proof. Stages 091–099 prove the geometry and conservative-normalization reductions. Stages 100–106 prove the retarded factorization and canonical DtN coefficient. Stages 107–124 classify admissible low-frequency outlet/core deformations and identify the compensated Robin–mixed class. Stages 125–145 derive parent and mouth-layer parameters for that class and select the regular mouth branch. Stages 146–163 linearize the remaining defect and show that the outlet, D/N similarity, and parent compensation defects are all images of the same off-family normal coordinate. □

E.2 Geometry-lane firewall

E.2.1 Why the geometry lane had to be checked

The conservative side enters Part IV with two facts already frozen. First, the higher conservative quadrupole payload lives in the grouped real P_2 bundle. Second, the conservative PN assembly also contains a geometry completion. If the geometry completion were dynamically active through $O(\omega^4)$, then the clean one-pole grouped branch used in the retarded bridge could be contaminated before the outgoing branch was even tested.

The check therefore begins with the most general reduced isotropic ansatz needed at this order:

$$K_Q^{\text{cons}}(\omega) = K_g(\omega) + K_{P2}(\omega), \qquad K_{P2}(\omega) = \frac{K_{\text{pole}}}{1 - \omega^2/\Omega_Q^2}, \tag{E.4}$$

where

$$K_g(\omega) = K_{g,0} + K_{g,2}\omega^2 + K_{g,4}\omega^4 + O(\omega^6). \tag{E.5}$$

The minimal isotropic branch identity is

$$K_0 K_4 = 4K_2^2. \tag{E.6}$$

If $K_{g,2} = K_{g,4} = 0$, inserting (E.4) into (E.6) gives

$$K_{g,0} = 3K_{\text{pole}}, \qquad K_{\text{pole}} = \frac{K_0}{4}, \qquad K_{g,0} = \frac{3K_0}{4}, \tag{E.7}$$

which is exactly the conservative split in (E.1).

E.2.2 Exact obstruction if geometry is dynamic

If the geometry lane carries dynamic even moments, the simple split is replaced by an exact obstruction formula. Define

$$\epsilon_2 := \frac{\Omega_Q^2 K_{g,2}}{K_{\text{pole}}}, \qquad \epsilon_4 := \frac{\Omega_Q^4 K_{g,4}}{K_{\text{pole}}}. \tag{E.8}$$

Then the static pole fraction becomes

$$c_{\text{pole}} = \frac{K_{\text{pole}}}{K_0} = \frac{1+\epsilon_4}{4(1+\epsilon_2)^2}. \tag{E.9}$$

For small contamination,

$$c_{\text{pole}} = \frac{1}{4}\left(1+\epsilon_4 - 2\epsilon_2 + O(\epsilon^2)\right). \tag{E.10}$$

Thus the static-geometry hypothesis is not a cosmetic simplification. It is the exact condition under which the pole fraction is 1/4.

E.2.3 Isotropic decoupling

The distributed wall action on an isotropic reference throat is block diagonal in angular momentum. With

$$\eta(\Omega, w, t) = g(w,t)Y_{00}(\Omega) + \sum_A q_A(w,t)Y_{2A}(\Omega) + \cdots, \tag{E.11}$$

where A ranges over the real quadrupole harmonics, every possible quadratic cross term between g and q_A is proportional to one of

$$\int_{S^2} Y_{00}Y_{2A}\, d\Omega, \qquad \int_{S^2} \nabla_{S^2}Y_{00}\cdot\nabla_{S^2}Y_{2A}\, d\Omega, \qquad \int_{S^2} Y_{00}(-\Delta_{S^2})Y_{2A}\, d\Omega. \tag{E.12}$$

All three vanish. Therefore

$$M_{0\leftrightarrow 2} = 0 \tag{E.13}$$

inside the isotropic quadratic wall theory, and the geometry contamination parameters vanish:

$$K_{g,2} = K_{g,4} = 0, \qquad \epsilon_2 = \epsilon_4 = 0. \tag{E.14}$$

Proposition E.2 (Second-order onset of geometry contamination)**.** *If weak anisotropy or operator noncommutation generates a bilinear mixing term $\chi M_0 qg$ between one grouped quadrupole coordinate and one scalar geometry coordinate, then after eliminating g, the first dynamic geometry contamination is quadratic in χ.*

Proof. Let

$$D_q(\omega) = K_{\text{stat}} + \frac{K_{\text{pole}}}{1-\omega^2/\Omega_Q^2}, \qquad D_g(\omega) = G_0 + G_2\omega^2 + G_4\omega^4 + O(\omega^6). \tag{E.15}$$

The Schur complement is

$$D_{\text{eff}}(\omega) = D_q(\omega) - \frac{\chi^2 M_0^2}{D_g(\omega)}. \tag{E.16}$$

Expanding D_g^{-1} gives

$$K_{g,2}^{\text{eff}} = \frac{\chi^2 M_0^2 G_2}{G_0^2}, \qquad K_{g,4}^{\text{eff}} = \frac{\chi^2 M_0^2 (G_0G_4 - G_2^2)}{G_0^3}. \tag{E.17}$$

Therefore $\epsilon_2, \epsilon_4 = O(\chi^2)$. □

The geometry-lane firewall is therefore an exact reduced selection rule: on the isotropic branch the geometry lane is dynamically inert through the relevant even orders, while weak contamination is second order in the explicit mixing source.

E.3 Retarded 2.5PN collapse

E.3.1 From conservative module to one scalar defect

With the geometry-lane check complete, the actual isotropic conservative quadrupole response can be written as

$$\overline{K}_Q^{\mathrm{cons}}(\omega) = \overline{K}_0 \left[\frac{3}{4} + \frac{1}{4}\frac{1}{1-\omega^2/\Omega_Q^2}\right]. \tag{E.18}$$

Its even moments are

$$\overline{K}_2 = \frac{\overline{K}_0}{4\Omega_Q^2}, \qquad \overline{K}_4 = \frac{\overline{K}_0}{4\Omega_Q^4}. \tag{E.19}$$

The target static normalization is

$$\overline{K}_0^{\mathrm{target}} = \frac{64G\Omega_Q^5}{45c^5} = \frac{54Gc_s^5}{5a^5c^5}, \qquad \Omega_Q = \frac{3c_s}{2a}. \tag{E.20}$$

Thus the conservative defect is the scalar

$$N_Q := \frac{\overline{K}_0}{\overline{K}_0^{\mathrm{target}}}. \tag{E.21}$$

All even ratios scale by the same N_Q. The Family–1 support side is automatic on this actual isotropic branch because the forced branch value $\rho_\alpha = 4/3$ gives

$$\zeta_{\mathrm{req}}^{\mathrm{act}}(\epsilon_{\mathrm{blk}}) = \frac{1}{3-2\epsilon_{\mathrm{blk}}}, \tag{E.22}$$

which is below the Family–1 constructive support ceiling throughout the admissible blocked regime.

E.3.2 Retarded factorization

The retarded one-pole grouped branch is written

$$\widehat{Y}_Q^{\mathrm{ret}}(\omega) = \frac{3}{4} + \frac{1}{4}\frac{1}{1-\omega^2/\Omega_Q^2 - i\chi_Q\sigma_Q^{\mathrm{can}}\omega^5} + O(\omega^6), \tag{E.23}$$

where

$$\sigma_Q^{\mathrm{can}} = \frac{9}{8\Omega_Q^5} = \frac{4a^5}{27c_s^5}. \tag{E.24}$$

The odd coefficient is

$$\overline{\Gamma}_5 = \chi_Q\frac{9\overline{K}_0}{32\Omega_Q^5}. \tag{E.25}$$

Consequently

$$\frac{\overline{\Gamma}_5}{2G/(5c^5)} = \chi_Q N_Q, \qquad \widehat{m}_0^2\chi_Q N_Q = 1. \tag{E.26}$$

On the natural point-particle source-map branch, $\widehat{m}_0 = 1 + O(a^2/r^2)$, so in the strict point-particle limit the remaining obstruction is

$$\Delta_Q := \chi_Q - 1. \tag{E.27}$$

Higher odd denominator terms beginning at $O(\omega^7)$ are invisible to the point-particle 2.5PN coefficient.

E.3.3 Canonical outgoing DtN branch

Let

$$z := \frac{a\omega}{c_s}. \tag{E.28}$$

The exact outgoing spherical $l = 2$ DtN operator is

$$\Lambda_2^{\text{out}}(z) = z\frac{d}{dz}\ln h_2^{(1)}(z) = -3 + \frac{z^2}{3} + \frac{z^4}{9} + i\frac{z^5}{9} + O(z^6). \tag{E.29}$$

Normalizing by the static slot gives

$$\widehat{Y}_2^{\text{out}}(z) = \frac{-3}{\Lambda_2^{\text{out}}(z)} = 1 + \frac{z^2}{9} + \frac{4z^4}{81} + i\frac{z^5}{27} + O(z^6). \tag{E.30}$$

Comparison with (E.23) fixes

$$\chi_Q = 1 \tag{E.31}$$

for the canonical compact passive/outgoing DtN branch. The canonical invariant coefficients are then

$$\overline{K}_0 = \frac{54Gc_s^5}{5a^5c^5}, \qquad \overline{K}_2 = \frac{6Gc_s^3}{5a^3c^5}, \qquad \overline{K}_4 = \frac{8Gc_s}{15ac^5}, \qquad \overline{\Gamma}_5 = \frac{2G}{5c^5}. \tag{E.32}$$

Theorem E.3 (Conditional reduced 2.5PN closure)**.** *Within the reduced isotropic grouped-P_2 hierarchy, the nonspinning point-particle 2.5PN branch closes if the actual passive/outgoing moving-throat branch realizes the canonical outgoing DtN coefficient $\chi_Q = 1$ and the natural source-map limit $\widehat{m}_0 \to 1$. If the actual branch has $\chi_Q \neq 1$, the entire reduced failure is measured by $\Delta_Q = \chi_Q - 1$ at leading order.*

E.4 Outlet DtN and Robin outlet tests

E.4.1 General isotropic deformation algebra

The remaining branch-selection issue is whether the true moving-throat outlet is the canonical branch or a deformed one. The first explicit deformation family is

$$\Lambda_2^{\text{def}}(z) = S\Lambda_2^{\text{out}}(\beta z) + \Sigma_0 + \Sigma_2 z^2 + \Sigma_4 z^4 + i\Sigma_5 z^5 + O(z^6). \tag{E.33}$$

Writing the expansion coefficients as L_0, L_2, L_4, L_5, the normalized response is

$$\widehat{Y}_2^{\text{def}}(z) = 1 - \frac{L_2}{L_0}z^2 + \left(\frac{L_2^2}{L_0^2} - \frac{L_4}{L_0}\right)z^4 - i\frac{L_5}{L_0}z^5 + O(z^6). \tag{E.34}$$

After imposing the canonical even fingerprint, the exact odd normalization is

$$\chi_Q = \frac{3(S\beta^5 + 9\Sigma_5)}{3S - \Sigma_0}, \tag{E.35}$$

so

$$\chi_Q - 1 = \frac{3S(\beta^5 - 1) + \Sigma_0 + 27\Sigma_5}{3S - \Sigma_0}. \tag{E.36}$$

Linearizing with

$$S = 1 + \varepsilon s, \qquad \beta = 1 + \varepsilon b, \qquad \Sigma_0 = \varepsilon a_0, \qquad \Sigma_5 = \varepsilon a_5, \tag{E.37}$$

while keeping the even slots canonical gives

$$\Delta_Q = 5b + \frac{a_0}{3} + 9a_5 + O(\varepsilon^2). \tag{E.38}$$

Pure overall scaling drops out. Pure argument rescaling cannot move χ_Q without also moving the even fingerprint unless $\beta = 1$. Additive throat-core slots are therefore the first genuinely dangerous outlet data.

E.4.2 Robin and mixed-pole tests

A pure isotropic Robin outlet is

$$\Lambda_2^{\mathrm{R}}(z) = \Lambda_2^{\mathrm{out}}(z) + \rho_R. \tag{E.39}$$

It gives

$$\chi_Q^{\mathrm{R}} = \frac{3}{3-\rho_R}, \tag{E.40}$$

so it generically shifts the canonical branch.

A standalone hidden mixed side-channel pole is

$$\Lambda_2^{\mathrm{mix}}(z) = \Lambda_2^{\mathrm{out}}(z) - \frac{\sigma_W}{1-\kappa_W z^2 - i\gamma_W z^5} + O(z^6). \tag{E.41}$$

The canonical even conditions force $\kappa_W = -1/9$, and then $\sigma_W = 0$. Thus a naive standalone passive hidden pole cannot be the actual branch if the conservative even fingerprint is already fixed.

E.4.3 Compensated Robin–mixed outlet

The combined outlet is

$$\Lambda_2^{\mathrm{hyb}}(z) = \Lambda_2^{\mathrm{out}}(z) + \rho_R - \frac{\sigma_W}{1-\kappa_W z^2 - i\gamma_W z^5} + O(z^6). \tag{E.42}$$

The exact canonical-even conditions have two solutions:

$$(\rho_R, \kappa_W) = (\sigma_W, 0), \qquad (\rho_R, \kappa_W) = (4\sigma_W, 1/3). \tag{E.43}$$

The first is a trivial static cancellation. The second is the nontrivial compensated branch. On it,

$$\chi_Q^{\mathrm{hyb}} = \frac{1-9\sigma_W\gamma_W}{1-\sigma_W}, \tag{E.44}$$

so the canonical outgoing value is preserved iff

$$\gamma_W = \frac{1}{9}. \tag{E.45}$$

With (E.45), the hybrid outlet collapses to a harmless pure scale deformation through the retained order.

E.5 Mixed side-channel pole realization

E.5.1 Concrete two-channel core model

The reduced hybrid outlet is realized by a concrete shell/mixed core. Let $u(\omega)$ be the isotropic $l=2$ mouth amplitude, and introduce a static shell coordinate s and a mixed side-channel coordinate q. The core equations are

$$\begin{pmatrix} K_s & \lambda \\ \lambda & -K_q D_W^{\mathrm{bare}}(z) \end{pmatrix} \begin{pmatrix} s \\ q \end{pmatrix} = u \begin{pmatrix} g_s \\ g_q \end{pmatrix}, \qquad D_W^{\mathrm{bare}}(z) = 1 - \kappa_0 z^2 - i\gamma_0 z^5 + O(z^6). \tag{E.46}$$

Eliminating (s,q) gives

$$\delta\Lambda_{\mathrm{core}}(z) = \frac{g_s^2}{K_s} - \frac{(K_s g_q - \lambda g_s)^2}{K_s\big(K_s K_q D_W^{\mathrm{bare}}(z) + \lambda^2\big)}. \tag{E.47}$$

With

$$r_c := \frac{\lambda^2}{K_s K_q}, \tag{E.48}$$

this has the reduced form

$$\delta\Lambda_{\text{core}}(z) = \rho_c - \frac{\sigma_c}{1 - \kappa_c z^2 - i\gamma_c z^5} + O(z^6), \tag{E.49}$$

where

$$\rho_c = \frac{g_s^2}{K_s}, \qquad \sigma_c = \frac{(K_s g_q - \lambda g_s)^2}{K_s^2 K_q (1 + r_c)}, \qquad \kappa_c = \frac{\kappa_0}{1 + r_c}, \qquad \gamma_c = \frac{\gamma_0}{1 + r_c}. \tag{E.50}$$

E.5.2 Core-balance theorem and D/N tube

The nontrivial compensated branch requires

$$\rho_c = 4\sigma_c, \qquad \kappa_c = \frac{1}{3}, \qquad \gamma_c = \frac{1}{9}. \tag{E.51}$$

The static balance condition is exactly

$$g_s^2(K_s K_q + \lambda^2) = 4(K_s g_q - \lambda g_s)^2. \tag{E.52}$$

Equivalently, defining

$$\mathfrak{r} := \frac{\lambda}{\sqrt{K_s K_q}}, \qquad \mathfrak{g} := \frac{g_q \sqrt{K_s}}{g_s \sqrt{K_q}}, \tag{E.53}$$

one obtains the parent compensation family

$$1 + \mathfrak{r}^2 = 4(\mathfrak{g} - \mathfrak{r})^2, \qquad \mathfrak{g}_\pm(\mathfrak{r}) = \mathfrak{r} \pm \frac{1}{2}\sqrt{1 + \mathfrak{r}^2}. \tag{E.54}$$

The finite D/N mixed-tube realization gives

$$L_W = \frac{\pi a}{2}\sqrt{\frac{1 + r_c}{3}} = \frac{\pi a}{2}\sqrt{\frac{1 + \mathfrak{r}^2}{3}}. \tag{E.55}$$

Identifying this tube with the actual throat span L fixes the branch value

$$\mathfrak{r}_{\text{geom}}\left(\frac{L}{a}\right) = \sqrt{\frac{12}{\pi^2}\left(\frac{L}{a}\right)^2 - 1}. \tag{E.56}$$

On the Family–1 preferred aspect ratio $L/a = 37/20$,

$$\mathfrak{r}_{F1} = \frac{\sqrt{4107 - 100\pi^2}}{10\pi} \approx 1.77799353547498. \tag{E.57}$$

The lower compensated mouth-coupling branch is then

$$\mathfrak{g}_-^{F1} = \mathfrak{r}_{F1} - \frac{1}{2}\sqrt{1 + \mathfrak{r}_{F1}^2} \approx 0.758035078944663. \tag{E.58}$$

The upper branch is $\mathfrak{g}_+^{F1} \approx 2.79795199200529$.

E.5.3 Parent-action overlap extraction

The abstract core parameters are tied to parent-action overlaps. In the first GNLS plus localized-Maxwell throat-core closure,

$$K_s = 4\pi a^2\left(\frac{H_w\ell}{3} + \frac{\hbar^2}{15m_\psi\rho_w\ell}\right), \qquad K_s = \frac{3\pi a^2\hbar^2}{5m_\psi\rho_w\ell} \quad \text{on the healing-locked branch,} \tag{E.59}$$

$$K_q = \frac{\mathcal{Z}_q}{\mu_0}\frac{\pi^2 c_s^2}{4L_W^2}, \tag{E.60}$$

$$\lambda = -q_* v_{w0}\mathcal{I}_{sq}, \tag{E.61}$$

and

$$g_s = \mathcal{T}_m\frac{4\pi a^2\ell}{3}, \qquad g_q = \frac{\mathcal{Z}_q}{\mu_0}\frac{\pi}{\sqrt{2}\,L_W^{3/2}}. \tag{E.62}$$

These formulas are the parent-action path by which a future full branch calculation must decide whether (E.54) is actually realized.

E.6 Core-to-mouth compensation law

E.6.1 Positive mouth-source selection

The core compensation family selects a mouth-coupling ratio. The first physical filter is positivity of the localized mouth source. For a positive normalized axial source $\sigma(z)$ on $[0, L]$, the normalized mouth-bias factor is

$$\mathfrak{g}[\sigma] = \int_0^L \sigma(z)\cos\left(\frac{\pi z}{2L}\right)dz. \tag{E.63}$$

Since the cosine lies between 0 and 1 on the interval,

$$0 \le \mathfrak{g}[\sigma] \le 1. \tag{E.64}$$

Therefore the upper Family–1 branch is impossible under positive localized mouth sourcing, while the lower branch (E.58) is admissible.

The self-matched derivative source $\sigma_{\text{match}}(z) = k\cos(kz)$, with $k = \pi/(2L)$, gives

$$\mathfrak{g}_{\text{match}} = \frac{\pi}{4} \approx 0.785398163397448, \tag{E.65}$$

only about 3.61% in traction away from the lower compensated value. A convex positive family interpolating between this derivative profile and the uniform profile reaches the exact lower branch at a modest broadening fraction.

E.6.2 Electrochemical boundary layer

The positive exponential source family is not an arbitrary fit. It follows from a local mouth free energy and zero-flux Onsager law. Linearizing the parent mouth potential as

$$V_m(z) \approx V_1 z, \qquad V_1 = \partial_z\delta V_{\text{conf}}|_{\text{m}} - q_*\,\partial_z A_0|_{\text{m}}\,, \tag{E.66}$$

the electrochemical potential is

$$\mu_\sigma^{\text{chem}}(z) = \Theta_\sigma \ln\frac{\sigma}{\sigma_*} + V_1 z. \tag{E.67}$$

The zero-flux stationary solution is

$$\sigma_\Pi(z) = \frac{\Pi e^{-\Pi z/L}}{L(1-e^{-\Pi})}, \qquad \Pi := \frac{V_1 L}{\Theta_\sigma} > 0. \tag{E.68}$$

The exact bias map is

$$\mathfrak{g}_\Pi = \frac{2\Pi(2\Pi e^{\Pi} + \pi)}{(4\Pi^2 + \pi^2)(e^{\Pi} - 1)}. \tag{E.69}$$

It is strictly increasing from $2/\pi$ to 1, and the Family–1 lower branch is reached at

$$\Pi_* \approx 1.50882951349316. \tag{E.70}$$

Equivalently, the parent threshold is

$$\partial_z \delta V_{\rm conf}|_{\rm m} - q_* \, \partial_z A_0|_{\rm m} = \Pi_* \frac{\Theta_\sigma}{L}. \tag{E.71}$$

E.6.3 Coupled mouth-layer fixed point

The effective bias Π is then replaced by a coupled boundary-layer solve. For a two-component field $\mathbf{U} = (U_s, U_q)^T$,

$$\left[-\partial_x^2 \mathbf{I} + \mathbf{K}\right] \mathbf{U}(x) = \Sigma_\Pi(x)\mathbf{G}, \qquad \mathbf{U}(0) = 0, \qquad \mathbf{U}'(1) = 0. \tag{E.72}$$

After diagonalizing $\mathbf{K}$, the scalar D/N response kernel is

$$\mathcal{S}(\Pi, \kappa) = \frac{\Pi \left[\kappa \tanh \kappa + \Pi \left(e^{-\Pi} \operatorname{sech} \kappa - 1\right)\right]}{(1-e^{-\Pi})(\kappa^2 - \Pi^2)}. \tag{E.73}$$

The full fixed-point law is

$$\Pi = \sum_{\alpha=\pm} M_\alpha \mathcal{S}(\Pi, \kappa_\alpha). \tag{E.74}$$

On the first Family–1 shell plus mixed D/N branch,

$$\Pi = M_s + M_q \mathcal{S}_q(\Pi), \qquad \mathcal{S}_q(\Pi) := \mathcal{S}\left(\Pi, \frac{\pi}{2}\right). \tag{E.75}$$

Inheriting the compensated outlet ratio gives

$$M_s = 4\Sigma_m, \qquad M_q = -\Sigma_m, \tag{E.76}$$

so

$$\Pi = \Sigma_m \left[4 - \mathcal{S}_q(\Pi)\right]. \tag{E.77}$$

At the canonical point,

$$\Sigma_m^* = \frac{\Pi_*}{4 - \mathcal{S}_q(\Pi_*)} \approx 0.451485277739090. \tag{E.78}$$

E.7 Mouth boundary-layer source theorem

E.7.1 From gains to parent core quantities

Embedding the same core Schur-complement weights into the mouth electrochemical free energy gives explicit gains

$$M_s = \frac{Lg_s^2}{K_s\Theta_\sigma}, \qquad M_q = -\frac{L(K_s g_q - \lambda g_s)^2}{K_s(K_s K_q + \lambda^2)\Theta_\sigma}. \tag{E.79}$$

In normalized variables,

$$R_q := -\frac{M_q}{M_s} = \frac{(\mathfrak{g}_c - \mathfrak{r})^2}{1+\mathfrak{r}^2}, \qquad M_s = \Sigma_0. \tag{E.80}$$

On the exact compensation family, $R_q = 1/4$. Under the self-matched mouth susceptibility closure,

$$\Sigma_0 = M_s = \frac{20}{9}\widehat{T}_m^2. \tag{E.81}$$

The self-consistent Family–1 mouth branch identifies the core mouth ratio with the source bias:

$$\mathfrak{g}_c = \mathfrak{g}_\Pi, \qquad R_q(\Pi) = \frac{(\mathfrak{g}_\Pi - \mathfrak{r}_{F1})^2}{1+\mathfrak{r}_{F1}^2}. \tag{E.82}$$

The scalar branch law becomes

$$\Pi = \Sigma_0\left[1 - R_q(\Pi)\mathcal{S}_q(\Pi)\right], \qquad \Sigma_0(\Pi) = \frac{\Pi}{1 - R_q(\Pi)\mathcal{S}_q(\Pi)}. \tag{E.83}$$

At Π_*,

$$R_q(\Pi_*) = \frac{1}{4}, \qquad \Sigma_0(\Pi_*) \approx 1.80594111095636, \qquad \widehat{T}_m(\Pi_*) \approx 0.901484054174205. \tag{E.84}$$

E.7.2 Regularity theorem

The explicit exponential mouth family has

$$0 < \mathfrak{g}_\Pi < 1 \qquad (\Pi > 0), \tag{E.85}$$

and $\mathfrak{g}_\Pi \to 1$ only as $\Pi \to \infty$. The equal-normalized branch $\mathfrak{g}_c = 1$ is therefore a singular point-source limit. Since $\widehat{T}_m \sim C\Pi^{1/2}$ in that limit, it is not a regular finite-traction mouth state.

Theorem E.4 (Unique regular Family–1 canonical mouth branch)**.** *Inside the explicit Family–1 positive exponential mouth-layer closure, the upper compensated branch is excluded by positivity, the equal-normalized branch is singular, and the lower compensated branch is the unique regular finite-bias, finite-traction branch preserving the canonical outgoing quadrupole fingerprint.*

E.8 Family-1 mouth correction

E.8.1 First-order positive deformations

After branch selection, the remaining mouth-side uncertainty is finite correction around the unique regular branch. Let

$$\Sigma_*(x) = \frac{\Pi_* e^{-\Pi_* x}}{1 - e^{-\Pi_*}} \tag{E.86}$$

and consider a positive normalized deformation

$$\Sigma_\epsilon(x) = (1-\epsilon)\Sigma_*(x) + \epsilon\varsigma(x), \qquad \varsigma \ge 0, \qquad \int_0^1 \varsigma(x)\,dx = 1. \tag{E.87}$$

Only two moments matter at first order:

$$\bar{g}[\sigma] = \int_0^1 \sigma(x)c(x)\,dx, \qquad \bar{S}[\sigma] = \int_0^1 \sigma(x)K_q(x)\,dx, \tag{E.88}$$

with

$$c(x) = \cos\frac{\pi x}{2}, \qquad K_q(x) = \frac{\cosh\left(\frac{\pi}{2}(1-x)\right)}{\cosh(\pi/2)}. \tag{E.89}$$

Retuning Π to stay on the canonical lower branch gives

$$\delta\Pi = -\epsilon\frac{\bar{g}_\varsigma - \mathfrak{g}_*}{\mathfrak{g}'_*}, \qquad \mathfrak{g}'_* \approx 0.0714453558083195. \tag{E.90}$$

The traction shift is

$$\delta\widehat{T}_m = \epsilon\left[A_T(\bar{g}_\varsigma - \mathfrak{g}_*) + B_T(\bar{S}_\varsigma - \mathcal{S}_*)\right], \tag{E.91}$$

where

$$A_T \approx -4.27263956256927, \qquad B_T \approx 0.134875005736706. \tag{E.92}$$

Equivalently, all first-order source-shape sensitivity is the projection against one centered rigidity kernel.

E.8.2 Full-profile residual

The actual full mouth potential generated by the shell plus mixed D/N channels is $\Phi_*(x)$. Define the tangent-subtracted residual

$$R_*(x) := \Phi_*(x) - \Pi_* x. \tag{E.93}$$

It satisfies

$$R_*(0) = R'_*(0) = 0, \qquad R''_*(0) < 0, \tag{E.94}$$

so the full source is broader than the tangent exponential near the mouth. Expanding

$$\Sigma_{\text{act}}(x) = \Sigma_*(x)\left[1 - \widetilde{R}_*(x)\right] + O(R_*^2), \qquad \widetilde{R}_* = R_* - \langle R_*\rangle_*, \tag{E.95}$$

gives the selected moment shifts

$$\delta\mathfrak{g}_{\text{act}} = -\operatorname{Cov}_*(c, R_*), \qquad \delta\mathcal{S}_{\text{act}} = -\operatorname{Cov}_*(K_q, R_*). \tag{E.96}$$

On the explicit Family–1 branch,

$$\operatorname{Cov}_*(c, R_*) \approx 0.0648069687666328, \qquad \operatorname{Cov}_*(K_q, R_*) \approx 0.0388718368650403. \tag{E.97}$$

Thus the first-order correction broadens the source and shifts the mouth-only canonical point upward:

$$\Pi_{\text{corr}} \approx 2.41591392833607, \qquad \widehat{T}_{m,\text{corr}} \approx 1.17313803363654. \tag{E.98}$$

A one-step nonlinear iterate gives a slightly stronger but directionally consistent shift.

E.9 Co-evolving core-mouth fixed point

E.9.1 Exact co-evolving map

The fixed-core mouth correction is then replaced by a co-evolving map. For any positive normalized Σ, define

$$\mathfrak{g}[\Sigma] = \int_0^1 \Sigma(x)c(x)\,dx, \qquad \mathcal{S}[\Sigma] = \int_0^1 \Sigma(x)K_q(x)\,dx. \tag{E.99}$$

The core loading ratio is

$$\mathcal{R}[\Sigma] = \frac{(\mathfrak{g}[\Sigma] - \mathfrak{r}_{F1})^2}{1 + \mathfrak{r}_{F1}^2}. \tag{E.100}$$

With the shell and mixed Green operators $\mathcal{T}_s$ and $\mathcal{T}_q$, the full potential is

$$\Phi_{\Sigma_0}[\Sigma](x) = \Sigma_0\left[\mathcal{T}_s[\Sigma](x) - \mathcal{R}[\Sigma]\mathcal{T}_q[\Sigma](x)\right]. \tag{E.101}$$

The nonlinear fixed-point equation is

$$\Sigma(x) = \frac{e^{-\Phi_{\Sigma_0}[\Sigma](x)}}{\int_0^1 e^{-\Phi_{\Sigma_0}[\Sigma](y)}\,dy}. \tag{E.102}$$

The canonical compensation condition is

$$\mathfrak{g}[\Sigma] = \mathfrak{g}_*, \qquad \mathcal{R}[\Sigma] = \frac{1}{4}. \tag{E.103}$$

E.9.2 Renormalized canonical point

At the old canonical traction, the co-evolving fixed point remains positive but moves off exact compensation:

$$\mathfrak{g}_{\text{fp}} \approx 0.693352419668063, \qquad \mathcal{S}_{\text{fp}} \approx 0.621601316751401, \qquad \mathcal{R}_{\text{fp}} \approx 0.282713904908238. \tag{E.104}$$

Restoring exact lower compensation requires a numerically located renormalized gain

$$\Sigma_0^{\text{can}} \approx 4.651033550168876, \qquad \widehat{T}_{m,\text{can}} \approx 1.446708366456762, \tag{E.105}$$

with

$$\mathfrak{g}_{\text{can}} = \mathfrak{g}_*, \qquad \mathcal{R}_{\text{can}} = \frac{1}{4}, \qquad \mathcal{S}_{\text{can}} \approx 0.670362115673462, \qquad \Pi_{\text{can}} \approx 3.871564377479009. \tag{E.106}$$

This is a Numerically Located placement inside the exact co-evolving map, not a closed-form theorem of the full PDE.

E.10 Linear defect transport

E.10.1 Mouth/core transport around the renormalized point

Linearizing the exact Family–1 loading ratio gives

$$\delta\mathcal{R} = -\frac{\delta\mathfrak{g}}{\sqrt{1 + \mathfrak{r}_{F1}^2}} + O(\delta\mathfrak{g}^2) \approx -0.490216044387626\,\delta\mathfrak{g}. \tag{E.107}$$

The gains transport as

$$\delta M_s = \delta\Sigma_0, \qquad \delta M_q = -\frac{1}{4}\delta\Sigma_0 + \frac{\Sigma_0^{\mathrm{can}}}{\sqrt{1+\mathfrak{r}_{F1}^2}}\delta\mathfrak{g}. \tag{E.108}$$

The mouth-bias transport is

$$\delta\Pi = \left(1 - \frac{1}{4}\mathcal{S}_{\mathrm{can}}\right)\delta\Sigma_0 - \frac{\Sigma_0^{\mathrm{can}}}{4}\delta\mathcal{S} + \frac{\Sigma_0^{\mathrm{can}}\mathcal{S}_{\mathrm{can}}}{\sqrt{1+\mathfrak{r}_{F1}^2}}\delta\mathfrak{g}, \tag{E.109}$$

or numerically

$$\delta\Pi \approx 0.832409471081635\,\delta\Sigma_0 - 1.16275838754222\,\delta\mathcal{S} + 1.52843317823248\,\delta\mathfrak{g}. \tag{E.110}$$

E.10.2 Projection into the compensated outlet

The compensated outlet loads are

$$\rho_R = \Xi M_s, \qquad \sigma_W = -\Xi M_q, \qquad \Xi = \frac{\Theta_\sigma}{L}. \tag{E.111}$$

Define the off-compensation scalar

$$\delta\mathcal{C} := \delta\rho_R - 4\delta\sigma_W. \tag{E.112}$$

Then

$$\delta\mathcal{C} = \frac{16\sigma_*}{\sqrt{1+\mathfrak{r}_{F1}^2}}\delta\mathfrak{g}. \tag{E.113}$$

Linearizing the hybrid outlet gives

$$\delta E_2 = \frac{\delta\mathcal{C} - 9\sigma_*\delta\kappa_W}{27(1-\sigma_*)}, \qquad \delta E_4 = \frac{5\delta\mathcal{C} - 72\sigma_*\delta\kappa_W}{243(1-\sigma_*)}, \tag{E.114}$$

$$\Delta_Q = \frac{\delta\mathcal{C} - 27\sigma_*\delta\gamma_W}{3(1-\sigma_*)}. \tag{E.115}$$

Canonical even preservation $\delta E_2 = \delta E_4 = 0$ forces

$$\delta\mathcal{C} = 0, \qquad \delta\kappa_W = 0, \qquad \delta\mathfrak{g} = 0. \tag{E.116}$$

The remaining first-order defect is therefore

$$\Delta_Q = -\frac{9\sigma_*}{1-\sigma_*}\delta\gamma_W, \qquad N_Q - 1 = \frac{9\sigma_*}{1-\sigma_*}\delta\gamma_W. \tag{E.117}$$

E.10.3 Bare mixed-port slippage and D/N similarity

The concrete core identities give

$$\kappa_W = \frac{\kappa_0}{1+r_c}, \qquad \gamma_W = \frac{\gamma_0}{1+r_c}. \tag{E.118}$$

On the compensated branch,

$$\kappa_0 = \frac{1+r_c}{3}, \qquad \gamma_0 = \frac{1+r_c}{9}. \tag{E.119}$$

Under the canonical-even gate $\delta\kappa_W = 0$, the odd defect is

$$\delta\gamma_W = \frac{1}{1+\mathfrak{r}_{F1}^2}\left(\delta\gamma_0 - \frac{1}{3}\delta\kappa_0\right). \tag{E.120}$$

Thus a pure-scale bare mixed-port deformation is harmless.

The D/N tube gives

$$\kappa_0 = \frac{4L_W^2}{\pi^2 a^2}. \tag{E.121}$$

The first-order bare slippage is equivalently the D/N similarity mismatch

$$\Xi_{\text{slip}} := \Xi_\gamma - 2\Xi_L, \qquad \Xi_\gamma = \frac{\delta \ln \gamma_0}{\delta\Pi_{\tan}}, \qquad \Xi_L = \frac{\delta \ln(L_W/a)}{\delta\Pi_{\tan}}. \tag{E.122}$$

Then

$$\Delta_Q = -\frac{\sigma_*}{1-\sigma_*}\Xi_{\text{slip}}\,\delta\Pi_{\tan}, \qquad N_Q - 1 = \frac{\sigma_*}{1-\sigma_*}\Xi_{\text{slip}}\,\delta\Pi_{\tan}. \tag{E.123}$$

Here

$$\delta\Pi_{\tan} = 0.832409471081635\,\delta\Sigma_0 - 1.16275838754222\,\delta\mathcal{S} \tag{E.124}$$

when $\delta\mathfrak{g} = 0$.

E.10.4 Parent compensation-surface rigidity and off-family normal coordinate

On the exact parent compensation family, both the D/N tube ratio and the bare odd normalization are functions of $\mathfrak{r}$:

$$\frac{L_W}{a} = \frac{\pi}{2}\sqrt{\frac{1+\mathfrak{r}^2}{3}}, \qquad \gamma_0 = \frac{1+\mathfrak{r}^2}{9}. \tag{E.125}$$

Therefore

$$\delta \ln \gamma_0 - 2\delta \ln\left(\frac{L_W}{a}\right) = 0, \qquad \Xi_{\text{slip}} = 0. \tag{E.126}$$

Furthermore, on the lower branch, the canonical-even gate $\delta\mathfrak{g} = 0$ forces $\delta\mathfrak{r} = 0$ because $d\mathfrak{g}_-/d\mathfrak{r} > 0$. Thus $\Delta_Q = 0$ at first order for tangential motion inside the exact parent compensation family.

The normal coordinate measuring departure from that family is

$$\delta_\perp := \delta\mathfrak{g} - \mathfrak{g}_-'(\mathfrak{r}_*)\delta\mathfrak{r}. \tag{E.127}$$

It is equivalently the normalized first-order parent compensation defect, since

$$\mathcal{F}(\mathfrak{g},\mathfrak{r}) = 1 + \mathfrak{r}^2 - 4(\mathfrak{g}-\mathfrak{r})^2, \qquad \delta\mathcal{F} = 4\sqrt{1+\mathfrak{r}_*^2}\,\delta_\perp. \tag{E.128}$$

In microscopic parent variables,

$$\boxed{\delta_\perp = \mathfrak{g}_*\delta\ln\left(\frac{g_q K_s}{g_s\lambda}\right) + \frac{1}{4\sqrt{1+\mathfrak{r}_*^2}}\delta\ln\left(\frac{K_s K_q}{\lambda^2}\right)}. \tag{E.129}$$

This is the final output of Part IV. The last first-order reduced defect is not a diffuse outlet problem. It is the question of whether the true moving-throat branch has $\delta_\perp = 0$, or what value (E.129) takes if it leaves the compensation family.

E.11 Verification outputs and downstream use

The usable downstream citation packet from this appendix is as follows.

1. **MTDC-T8.1**: cite for the geometry-lane firewall, the obstruction variables (ϵ_2, ϵ_4), and the forced $3/4 + 1/4$ module.
2. **MTDC-T8.2**: cite for the reduction of the retarded 2.5PN problem to $\widehat{m}_0^2 \chi_Q N_Q = 1$, the outgoing DtN fingerprint, and $\chi_Q = 1$ on the canonical branch.
3. **MTDC-T8.3**: cite for the low-frequency outlet deformation algebra, the Robin and hidden-pole no-go results, and the compensated Robin–mixed branch.
4. **MTDC-T8.4**: cite for the positive mouth-source theorem, the exponential boundary-layer law, the Family–1 bias point Π_*, and the coupled mouth fixed-point equations.
5. **MTDC-T8.5**: cite for the co-evolving core–mouth canonical point, linear defect transport, D/N similarity preservation, and the final off-family scalar $\delta_\perp$.

- ☐ The grouped P_2 carrier is kept distinct from the axisymmetric geometry lane until the isotropic decoupling theorem is invoked.
- ☐ The $3/4 + 1/4$ conservative module is cited only under the static-geometry or geometry-decoupled hypothesis; dynamic geometry contamination is tracked by (ϵ_2, ϵ_4).
- ☐ The Family–1 support success statement is conditioned on the minimal isotropic precursor $\rho_\alpha = 4/3$, $\zeta_{\text{req}} = 1/3$, not on an arbitrary support branch.
- ☐ The retarded 2.5PN gate is stated as the factorized product $\widehat{m}_0^2 \chi_Q N_Q = 1$, with source-map, conservative-normalization, and outgoing-normalization factors separated.
- ☐ The exact outgoing $l = 2$ DtN fingerprint is used only after the passive/outgoing compact branch has been selected.
- ☐ Pure Robin and standalone mixed-pole outlet deformations are recorded as no-go or noncanonical branches unless the compensated Robin–mixed constraints are imposed.
- ☐ The shell/mixed core Schur complement is treated as a reduced core model; parent-action realization still requires the corresponding overlap data.
- ☐ The finite D/N tube normalization is tied to the geometric throat ratio before it is used in the compensation surface.
- ☐ Positive mouth-source selection is kept separate from arbitrary signed source fitting; the lower compensated branch is selected by the positivity theorem.
- ☐ The exponential mouth-layer law is read as the zero-flux GNLS/localized-Maxwell boundary-layer solution, not as a free curve fit.
- ☐ Numerically located mouth and co-evolving fixed points are marked as exact equations solved numerically, not as closed-form constants.
- ☐ Linear defect transport is applied only after expanding about the renormalized co-evolving canonical point.

☐ The final first-order obstruction is reported as $\delta_\perp$, so future uses of $\Delta_Q = 0$ must either assume or verify tangent motion on the parent compensation family.

☐ Branch-realization status remains explicit whenever Part IV is cited for downstream PN, anomaly, or same-charge work.

The claim firewall is especially important for this appendix. Equations such as (E.31), (E.123), and (E.129) are exact inside their declared reduced closures. They are not, by themselves, proof that the full moving-throat PDE has realized the canonical branch. Future papers should cite Part IV for the reduction and should separately state the branch-realization status whenever they need $\Delta_Q = 0$, $N_Q = 1$, $\Xi_{\text{slip}} = 0$, or $\delta_\perp = 0$.

E.12 Stage-by-stage verification cards

The preceding sections collect the derivation by audit path. The following cards restore the original stage order. Each card records the source stage, the local claim status, the imported assumptions, the verification output, and the downstream use. The cards are intentionally compact: the algebra is not repeated when the same equation family is already displayed in the audit-path sections above.

Source layout. Stages 091–163 are now sourced from the canonical files `stages/stage_091.tex` through `stages/stage_163.tex`. The raw Markdown notes and audit artifacts are tracked in a repo-local provenance index, so they do not have to appear in the reader-facing PDF. The stage files themselves carry the executable SymPy and Mathematica audit references.

E.13 Stage 091: Deriving the 3/4 + 1/4 Conservative Quadrupole Module from the Grouped-P2 + Geometry Split

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure.

Purpose. Stage 091 is a geometry-lane firewall ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the Part III minimal isotropic module, the grouped real P_2 carrier, the static/dynamic geometry split, and the branch identity $K_0K_4 = 4K_2^2$. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Geometry-lane firewall* block. The computation isolates the forced conservative carrier $\widehat{Y}_Q^{\text{cons}} = 3/4 + (1/4)/(1 - \omega^2/\Omega_Q^2)$ or the obstruction variables (ϵ_2, ϵ_4).

> Static-geometry plus one-pole grouped carrier forces the $3/4 + 1/4$ conservative quadrupole module and hence $\rho_\alpha = 4/3$, $\zeta_{\text{req}} = 1/3$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check the static limit $\epsilon_2 = \epsilon_4 = 0$ returns $c_{\text{pole}} = 1/4$.
- ☐ Check $l = 0$ and $l = 2$ orthogonality before applying the geometry firewall.
- ☐ Check that any support/source success statement still carries the minimal-module hypothesis.

Downstream use. This stage feeds Stages 100–106 and any downstream use of $\rho_\alpha = 4/3$, $\zeta_{\text{req}} = 1/3$, or N_Q. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.14 Stage 092: Exact Obstruction Formula if the Geometry Lane Carries Dynamic Even Moments

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure.

Purpose. Stage 092 is a geometry-lane firewall ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the Part III minimal isotropic module, the grouped real P_2 carrier, the static/dynamic geometry split, and the branch identity $K_0K_4 = 4K_2^2$. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Geometry-lane firewall* block. The computation isolates the forced conservative carrier $\widehat{Y}_Q^{\text{cons}} = 3/4 + (1/4)/(1 - \omega^2/\Omega_Q^2)$ or the obstruction variables (ϵ_2, ϵ_4).

> If geometry carries dynamic even moments, the pole fraction becomes $(1+\epsilon_4)/[4(1+\epsilon_2)^2]$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check the static limit $\epsilon_2 = \epsilon_4 = 0$ returns $c_{\text{pole}} = 1/4$.
- ☐ Check $l = 0$ and $l = 2$ orthogonality before applying the geometry firewall.
- ☐ Check that any support/source success statement still carries the minimal-module hypothesis.

Downstream use. This stage feeds Stages 100–106 and any downstream use of $\rho_\alpha = 4/3$, $\zeta_{\text{req}} = 1/3$, or N_Q. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.15 Stage 093: Updated Status After the Direct Grouped-P2 + Geometry Derivation

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure / Open.

Purpose. Stage 093 is a geometry-lane firewall ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the Part III minimal isotropic module, the grouped real P_2 carrier, the static/dynamic geometry split, and the branch identity $K_0K_4 = 4K_2^2$. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Geometry-lane firewall* block. The computation isolates the forced conservative carrier $\widehat{Y}_Q^{\text{cons}} = 3/4 + (1/4)/(1 - \omega^2/\Omega_Q^2)$ or the obstruction variables (ϵ_2, ϵ_4).

> Declares the remaining question: whether actual isotropic geometry is dynamically inert or supplies (ϵ_2, ϵ_4).

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check the static limit $\epsilon_2 = \epsilon_4 = 0$ returns $c_{\text{pole}} = 1/4$.
- ☐ Check $l = 0$ and $l = 2$ orthogonality before applying the geometry firewall.
- ☐ Check that any support/source success statement still carries the minimal-module hypothesis.

Downstream use. This stage feeds Stages 100–106 and any downstream use of $\rho_\alpha = 4/3$, $\zeta_{\text{req}} = 1/3$, or N_Q. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.16 Stage 094: Isotropic Geometry-Decoupling Theorem

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure.

Purpose. Stage 094 is a geometry-lane firewall ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the Part III minimal isotropic module, the grouped real P_2 carrier, the static/dynamic geometry split, and the branch identity $K_0K_4 = 4K_2^2$. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Geometry-lane firewall* block. The computation isolates the forced conservative carrier $\widehat{Y}_Q^{\mathrm{cons}} = 3/4 + (1/4)/(1 - \omega^2/\Omega_Q^2)$ or the obstruction variables (ϵ_2, ϵ_4).

> Orthogonality of $l = 0$ geometry and grouped $l = 2$ wall modes gives $K_{g,2} = K_{g,4} = 0$ on the isotropic branch.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check the static limit $\epsilon_2 = \epsilon_4 = 0$ returns $c_{\mathrm{pole}} = 1/4$.
- ☐ Check $l = 0$ and $l = 2$ orthogonality before applying the geometry firewall.
- ☐ Check that any support/source success statement still carries the minimal-module hypothesis.

Downstream use. This stage feeds Stages 100–106 and any downstream use of $\rho_\alpha = 4/3$, $\zeta_{\mathrm{req}} = 1/3$, or N_Q. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.17 Stage 095: First Nonzero Geometry Contamination Appears Only at Second Order in Anisotropy/Mixing

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure.

Purpose. Stage 095 is a geometry-lane firewall ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the Part III minimal isotropic module, the grouped real P_2 carrier, the static/dynamic geometry split, and the branch identity $K_0K_4 = 4K_2^2$. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Geometry-lane firewall* block. The computation isolates the forced conservative carrier $\widehat{Y}_Q^{\rm cons} = 3/4 + (1/4)/(1 - \omega^2/\Omega_Q^2)$ or the obstruction variables (ϵ_2, ϵ_4).

> Weak $l = 0 \leftrightarrow l = 2$ mixing contaminates the grouped module only at $O(\chi^2)$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check the static limit $\epsilon_2 = \epsilon_4 = 0$ returns $c_{\rm pole} = 1/4$.
- ☐ Check $l = 0$ and $l = 2$ orthogonality before applying the geometry firewall.
- ☐ Check that any support/source success statement still carries the minimal-module hypothesis.

Downstream use. This stage feeds Stages 100–106 and any downstream use of $\rho_\alpha = 4/3$, $\zeta_{\rm req} = 1/3$, or N_Q. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.18 Stage 096: Actual Branch Verdict for the Geometry-Lane Check

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure / Open.

Purpose. Stage 096 is a geometry-lane firewall ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the Part III minimal isotropic module, the grouped real P_2 carrier, the static/dynamic geometry split, and the branch identity $K_0K_4 = 4K_2^2$. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Geometry-lane firewall* block. The computation isolates the forced conservative carrier $\widehat{Y}_Q^{\rm cons} = 3/4 + (1/4)/(1 - \omega^2/\Omega_Q^2)$ or the obstruction variables (ϵ_2, ϵ_4).

> The natural isotropic branch is conservatively clean; only genuine symmetry breaking or a second dynamic $l = 2$ geometry pole can reopen the issue.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check the static limit $\epsilon_2 = \epsilon_4 = 0$ returns $c_{\text{pole}} = 1/4$.
- ☐ Check $l = 0$ and $l = 2$ orthogonality before applying the geometry firewall.
- ☐ Check that any support/source success statement still carries the minimal-module hypothesis.

Downstream use. This stage feeds Stages 100–106 and any downstream use of $\rho_\alpha = 4/3$, $\zeta_{\text{req}} = 1/3$, or N_Q. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.19 Stage 097: The Actual Isotropic Passive/Outgoing Branch Collapses to a Single Normalization Defect

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure / Open.

Purpose. Stage 097 is a geometry-lane firewall ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the Part III minimal isotropic module, the grouped real P_2 carrier, the static/dynamic geometry split, and the branch identity $K_0 K_4 = 4K_2^2$. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Geometry-lane firewall* block. The computation isolates the forced conservative carrier $\widehat{Y}_Q^{\text{cons}} = 3/4 + (1/4)/(1 - \omega^2/\Omega_Q^2)$ or the obstruction variables (ϵ_2, ϵ_4).

> The isotropic passive/outgoing one-pole branch reduces to $N_Q = \overline{K}_0/\overline{K}_0^{\text{target}}$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check the static limit $\epsilon_2 = \epsilon_4 = 0$ returns $c_{\text{pole}} = 1/4$.
- ☐ Check $l = 0$ and $l = 2$ orthogonality before applying the geometry firewall.
- ☐ Check that any support/source success statement still carries the minimal-module hypothesis.

Downstream use. This stage feeds Stages 100–106 and any downstream use of $\rho_\alpha = 4/3$, $\zeta_{\text{req}} = 1/3$, or N_Q. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.20 Stage 098: The Explicit Family-1 Support Test Is Automatic on the Actual Isotropic Branch

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure.

Purpose. Stage 098 is a geometry-lane firewall ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the Part III minimal isotropic module, the grouped real P_2 carrier, the static/dynamic geometry split, and the branch identity $K_0K_4 = 4K_2^2$. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Geometry-lane firewall* block. The computation isolates the forced conservative carrier $\widehat{Y}_Q^{\rm cons} = 3/4 + (1/4)/(1 - \omega^2/\Omega_Q^2)$ or the obstruction variables (ϵ_2, ϵ_4).

> With $\rho_\alpha = 4/3$, any branch with $\zeta_{\max} > 1$ passes the support test; Family–1 does.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check the static limit $\epsilon_2 = \epsilon_4 = 0$ returns $c_{\rm pole} = 1/4$.
- ☐ Check $l = 0$ and $l = 2$ orthogonality before applying the geometry firewall.
- ☐ Check that any support/source success statement still carries the minimal-module hypothesis.

Downstream use. This stage feeds Stages 100–106 and any downstream use of $\rho_\alpha = 4/3$, $\zeta_{\rm req} = 1/3$, or N_Q. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.21 Stage 099: The Reduced Finish Line After the Geometry-Lane Check

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure / Open.

Purpose. Stage 099 is a geometry-lane firewall ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the Part III minimal isotropic module, the grouped real P_2 carrier, the static/dynamic geometry split, and the branch identity $K_0K_4 = 4K_2^2$. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Geometry-lane firewall* block. The computation isolates the forced conservative carrier $\widehat{Y}_Q^{\rm cons} = 3/4 + (1/4)/(1-\omega^2/\Omega_Q^2)$ or the obstruction variables (ϵ_2, ϵ_4).

> The only remaining reduced theorem gate is $N_Q = 1$, equivalently actual passive/outgoing quadrupole normalization.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check the static limit $\epsilon_2 = \epsilon_4 = 0$ returns $c_{\rm pole} = 1/4$.
- ☐ Check $l = 0$ and $l = 2$ orthogonality before applying the geometry firewall.
- ☐ Check that any support/source success statement still carries the minimal-module hypothesis.

Downstream use. This stage feeds Stages 100–106 and any downstream use of $\rho_\alpha = 4/3$, $\zeta_{\rm req} = 1/3$, or N_Q. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.22 Stage 100: Exact Factorization of the Last 2.5PN Defect into Conservative and Outgoing Pieces

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure.

Purpose. Stage 100 is a retarded 2.5pn factorization ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the cleaned conservative isotropic branch, the source-map factor $\widehat{m}_0$, the conservative normalization N_Q, and the outgoing $l = 2$ DtN coefficient χ_Q. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Retarded* 2.5*PN factorization* block. The computation isolates the reduced product $\widehat{m}_0^2\chi_Q N_Q = 1$ while keeping χ_Q symbolic; the canonical condition $\chi_Q = 1$ is fixed downstream at Stage 105 from the Stage 104 outgoing fingerprint.

> Full odd normalization factorizes as $\widehat{m}_0^2\chi_Q N_Q = 1$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check the product $\widehat{m}_0^2 \chi_Q N_Q$ keeps source, conservative, and outgoing factors separate.
- ☐ Check that higher odd terms begin beyond the point-particle 2.5PN coefficient.
- ☐ Check the outgoing $l = 2$ DtN fingerprint against the normalized $z = \omega a / c_s$ expansion.

Downstream use. This stage feeds Stages 107–113, the 2.5PN/4PN outgoing bridge, and later same-charge prefactor audits. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.23 Stage 101: On the Natural Source-Map Branch the Last Reduced 2.5PN Obstruction is Purely Outgoing

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Reduced / Controlled Reduction / Exact Within Closure.

Purpose. Stage 101 is a retarded 2.5pn factorization ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the cleaned conservative isotropic branch, the source-map factor $\widehat{m}_0$, the conservative normalization N_Q, and the outgoing $l = 2$ DtN coefficient χ_Q. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Retarded* 2.5*PN factorization* block. The computation isolates the reduced product $\widehat{m}_0^2 \chi_Q N_Q = 1$ while keeping χ_Q symbolic; the canonical condition $\chi_Q = 1$ is fixed downstream at Stage 105 from the Stage 104 outgoing fingerprint.

On the natural point-particle source map, $\widehat{m}_0 \to 1$, so the last obstruction is purely χ_Q.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check the product $\widehat{m}_0^2 \chi_Q N_Q$ keeps source, conservative, and outgoing factors separate.
- ☐ Check that higher odd terms begin beyond the point-particle 2.5PN coefficient.
- ☐ Check the outgoing $l = 2$ DtN fingerprint against the normalized $z = \omega a / c_s$ expansion.

Downstream use. This stage feeds Stages 107–113, the 2.5PN/4PN outgoing bridge, and later same-charge prefactor audits. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.24 Stage 102: At 2.5PN the Only Live Retarded Obstruction is the Leading omega to the 5 Outgoing Normalization

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure.

Purpose. Stage 102 is a retarded 2.5pn factorization ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the cleaned conservative isotropic branch, the source-map factor $\widehat{m}_0$, the conservative normalization N_Q, and the outgoing $l = 2$ DtN coefficient χ_Q. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Retarded* 2.5*PN factorization* block. The computation isolates the reduced product $\widehat{m}_0^2\chi_Q N_Q = 1$ and the canonical condition $\chi_Q = 1$.

> Retarded terms beginning at $O(\omega^7)$ do not affect the point-particle 2.5PN theorem.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check the product $\widehat{m}_0^2\chi_Q N_Q$ keeps source, conservative, and outgoing factors separate.
- ☐ Check that higher odd terms begin beyond the point-particle 2.5PN coefficient.
- ☐ Check the outgoing $l = 2$ DtN fingerprint against the normalized $z = \omega a/c_s$ expansion.

Downstream use. This stage feeds Stages 107–113, the 2.5PN/4PN outgoing bridge, and later same-charge prefactor audits. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.25 Stage 103: Conditional Reduced 2.5PN Closure

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure / Open.

Purpose. Stage 103 is a retarded 2.5pn factorization ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the cleaned conservative isotropic branch, the source-map factor $\widehat{m}_0$, the conservative normalization N_Q, and the outgoing $l = 2$ DtN coefficient χ_Q. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Retarded* 2.5*PN factorization* block. The computation isolates the reduced product $\widehat{m}_0^2\chi_Q N_Q = 1$ and the canonical condition $\chi_Q = 1$.

Reduced theorem closes iff the actual passive/outgoing branch has $\chi_Q = 1$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check the product $\widehat{m}_0^2\chi_Q N_Q$ keeps source, conservative, and outgoing factors separate.
- ☐ Check that higher odd terms begin beyond the point-particle 2.5PN coefficient.
- ☐ Check the outgoing $l = 2$ DtN fingerprint against the normalized $z = \omega a/c_s$ expansion.

Downstream use. This stage feeds Stages 107–113, the 2.5PN/4PN outgoing bridge, and later same-charge prefactor audits. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.26 Stage 104: Exact Outgoing l=2 DtN Fingerprint

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure.

Purpose. Stage 104 is a retarded 2.5pn factorization ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the cleaned conservative isotropic branch, the source-map factor $\widehat{m}_0$, the conservative normalization N_Q, and the outgoing $l = 2$ DtN coefficient χ_Q. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Retarded* 2.5*PN factorization* block. The computation isolates the reduced product $\widehat{m}_0^2\chi_Q N_Q = 1$ and the canonical condition $\chi_Q = 1$.

Spherical Hankel DtN expansion gives $\widehat{Y}_2^{\mathrm{out}} = 1 + z^2/9 + 4z^4/81 + iz^5/27 + \cdots$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check the product $\widehat{m}_0^2\chi_Q N_Q$ keeps source, conservative, and outgoing factors separate.
- ☐ Check that higher odd terms begin beyond the point-particle 2.5PN coefficient.
- ☐ Check the outgoing $l = 2$ DtN fingerprint against the normalized $z = \omega a/c_s$ expansion.

Downstream use. This stage feeds Stages 107–113, the 2.5PN/4PN outgoing bridge, and later same-charge prefactor audits. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.27 Stage 105: Exact Fixing of chi-Q

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. EXACT WITHIN CLOSURE.

Purpose. Stage 105 is a retarded 2.5pn factorization ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the cleaned conservative isotropic branch, the source-map factor $\widehat{m}_0$, the conservative normalization N_Q, and the outgoing $l = 2$ DtN coefficient χ_Q. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Retarded* 2.5*PN factorization* block. The computation isolates the reduced product $\widehat{m}_0^2\chi_Q N_Q = 1$ and the canonical condition $\chi_Q = 1$.

> Matching the canonical grouped module to the exact outgoing DtN branch gives $\chi_Q = 1$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check the product $\widehat{m}_0^2\chi_Q N_Q$ keeps source, conservative, and outgoing factors separate.
- ☐ Check that higher odd terms begin beyond the point-particle 2.5PN coefficient.
- ☐ Check the outgoing $l = 2$ DtN fingerprint against the normalized $z = \omega a/c_s$ expansion.

Downstream use. This stage feeds Stages 107–113, the 2.5PN/4PN outgoing bridge, and later same-charge prefactor audits. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.28 Stage 106: Reduced 2.5PN Closure on the Canonical Outgoing DtN Branch

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure / Open.

Purpose. Stage 106 is a retarded 2.5pn factorization ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the cleaned conservative isotropic branch, the source-map factor $\widehat{m}_0$, the conservative normalization N_Q, and the outgoing $l = 2$ DtN coefficient χ_Q. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Retarded* 2.5*PN factorization* block. The computation isolates the reduced product $\widehat{m}_0^2 \chi_Q N_Q = 1$ and the canonical condition $\chi_Q = 1$.

> Lists the canonical invariant coefficients and marks actual branch realization as the remaining open theorem.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check the product $\widehat{m}_0^2 \chi_Q N_Q$ keeps source, conservative, and outgoing factors separate.
- ☐ Check that higher odd terms begin beyond the point-particle 2.5PN coefficient.
- ☐ Check the outgoing $l = 2$ DtN fingerprint against the normalized $z = \omega a/c_s$ expansion.

Downstream use. This stage feeds Stages 107–113, the 2.5PN/4PN outgoing bridge, and later same-charge prefactor audits. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.29 Stage 107: General Isotropic l=2 DtN Deformation Algebra

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure.

Purpose. Stage 107 is a outlet deformation and compensation ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the canonical outgoing DtN expansion, a general isotropic deformation, Robin mouth loading, and a hidden mixed side-channel pole. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Outlet deformation and compensation* block. The computation isolates the admissible deformation classes and the compensated Robin–mixed constraints.

> Deformed branch $S\Lambda_2^{\rm out}(\beta z) + \Sigma_0 + \Sigma_2 z^2 + \Sigma_4 z^4 + i\Sigma_5 z^5$ gives explicit χ_Q.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check pure scale and pure argument deformations separately.
- ☐ Check Robin and standalone mixed-pole limits before imposing compensation.
- ☐ Check that the compensated branch preserves the even coefficients as well as the odd normalization.

Downstream use. This stage feeds Stages 114–141 and any outlet model used to preserve $\chi_Q = 1$. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.30 Stage 108: Robustness Classes for chi-Q

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure.

Purpose. Stage 108 is a outlet deformation and compensation ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the canonical outgoing DtN expansion, a general isotropic deformation, Robin mouth loading, and a hidden mixed side-channel pole. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Outlet deformation and compensation* block. The computation isolates the admissible deformation classes and the compensated Robin–mixed constraints.

> Pure scale is harmless; pure argument shift is harmless only if even moments remain canonical; additive throat-core data can move χ_Q.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check pure scale and pure argument deformations separately.
- ☐ Check Robin and standalone mixed-pole limits before imposing compensation.
- ☐ Check that the compensated branch preserves the even coefficients as well as the odd normalization.

Downstream use. This stage feeds Stages 114–141 and any outlet model used to preserve $\chi_Q = 1$. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.31 Stage 109: Linearized Branch-Selection Law Near the Canonical Outgoing Branch

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure.

Purpose. Stage 109 is a outlet deformation and compensation ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the canonical outgoing DtN expansion, a general isotropic deformation, Robin mouth loading, and a hidden mixed side-channel pole. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Outlet deformation and compensation* block. The computation isolates the admissible deformation classes and the compensated Robin–mixed constraints.

> First-order outgoing defect is $\Delta_Q = 5b + a_0/3 + 9a_5$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check pure scale and pure argument deformations separately.
- ☐ Check Robin and standalone mixed-pole limits before imposing compensation.
- ☐ Check that the compensated branch preserves the even coefficients as well as the odd normalization.

Downstream use. This stage feeds Stages 114–141 and any outlet model used to preserve $\chi_Q = 1$. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.32 Stage 110: Explicit Isotropic Robin Outlet Model

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure.

Purpose. Stage 110 is a outlet deformation and compensation ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the canonical outgoing DtN expansion, a general isotropic deformation, Robin mouth loading, and a hidden mixed side-channel pole. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Outlet deformation and compensation* block. The computation isolates the admissible deformation classes and the compensated Robin–mixed constraints.

Pure Robin shift gives $\chi_Q^{\mathrm{R}} = 3/(3 - \rho_R)$ and generically spoils the branch.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check pure scale and pure argument deformations separately.
- ☐ Check Robin and standalone mixed-pole limits before imposing compensation.
- ☐ Check that the compensated branch preserves the even coefficients as well as the odd normalization.

Downstream use. This stage feeds Stages 114–141 and any outlet model used to preserve $\chi_Q = 1$. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.33 Stage 111: Explicit Mixed A-w/F-mu w-Type Side-Channel Pole

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure.

Purpose. Stage 111 is a outlet deformation and compensation ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the canonical outgoing DtN expansion, a general isotropic deformation, Robin mouth loading, and a hidden mixed side-channel pole. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Outlet deformation and compensation* block. The computation isolates the admissible deformation classes and the compensated Robin–mixed constraints.

> A standalone hidden mixed pole cannot preserve the canonical even branch unless it is absent.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check pure scale and pure argument deformations separately.
- ☐ Check Robin and standalone mixed-pole limits before imposing compensation.
- ☐ Check that the compensated branch preserves the even coefficients as well as the odd normalization.

Downstream use. This stage feeds Stages 114–141 and any outlet model used to preserve $\chi_Q = 1$. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.34 Stage 112: Exact Robin–Mixed Compensation Law

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure.

Purpose. Stage 112 is a outlet deformation and compensation ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the canonical outgoing DtN expansion, a general isotropic deformation, Robin mouth loading, and a hidden mixed side-channel pole. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Outlet deformation and compensation* block. The computation isolates the admissible deformation classes and the compensated Robin–mixed constraints.

> Nontrivial compensated branch has $\rho_R = 4\sigma_W$, $\kappa_W = 1/3$, and preserves odd normalization iff $\gamma_W = 1/9$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check pure scale and pure argument deformations separately.
- ☐ Check Robin and standalone mixed-pole limits before imposing compensation.
- ☐ Check that the compensated branch preserves the even coefficients as well as the odd normalization.

Downstream use. This stage feeds Stages 114–141 and any outlet model used to preserve $\chi_Q = 1$. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.35 Stage 113: Outlet-Model Status Update

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure / Open.

Purpose. Stage 113 is a outlet deformation and compensation ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the canonical outgoing DtN expansion, a general isotropic deformation, Robin mouth loading, and a hidden mixed side-channel pole. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Outlet deformation and compensation* block. The computation isolates the admissible deformation classes and the compensated Robin–mixed constraints.

> Low-frequency outlet audit leaves pure scale/argument classes and the compensated Robin–mixed class as the viable routes.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check pure scale and pure argument deformations separately.
- ☐ Check Robin and standalone mixed-pole limits before imposing compensation.
- ☐ Check that the compensated branch preserves the even coefficients as well as the odd normalization.

Downstream use. This stage feeds Stages 114–141 and any outlet model used to preserve $\chi_Q = 1$. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.36 Stage 114: Concrete Two-Channel Core Outlet Model

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure / Reduced / Controlled Reduction.

Purpose. Stage 114 is a core outlet realization ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the compensated outlet conditions, a two-channel shell/mixed Schur complement, finite D/N mixed-tube geometry, and parent-action overlap data. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Core outlet realization* block. The computation isolates the parent-level compensation surface and normalized core/mouth target values.

> Concrete shell/mixed core Schur complement reproduces the reduced Robin–mixed outlet.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check the Schur complement signs in the two-channel core model.
- ☐ Check the D/N mixed-tube length normalization against the chosen L/a.
- ☐ Check parent overlap ratios before interpreting them as actual branch values.

Downstream use. This stage feeds Stages 125–139 and the positive mouth-source branch. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.37 Stage 115: Exact Core-Balance Compensation Law

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure.

Purpose. Stage 115 is a core outlet realization ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the compensated outlet conditions, a two-channel shell/mixed Schur complement, finite D/N mixed-tube geometry, and parent-action overlap data. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Core outlet realization* block. The computation isolates the parent-level compensation surface and normalized core/mouth target values.

Coupling balance $g_s^2(K_sK_q + \lambda^2) = 4(K_sg_q - \lambda g_s)^2$ realizes the compensated branch.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check the Schur complement signs in the two-channel core model.
- ☐ Check the D/N mixed-tube length normalization against the chosen L/a.
- ☐ Check parent overlap ratios before interpreting them as actual branch values.

Downstream use. This stage feeds Stages 125–139 and the positive mouth-source branch. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.38 Stage 116: Finite D/N Mixed-Tube Realization

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure **for the D/N mixed-tube length law inside the posited pure-scale realization; actual moving-throat branch realization remains** Open**.**

Purpose. Stage 116 is a core outlet realization ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the compensated outlet conditions, a two-channel shell/mixed Schur complement, finite D/N mixed-tube geometry, and parent-action overlap data. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Core outlet realization* block. The computation isolates the parent-level compensation surface and normalized core/mouth target values.

First D/N mixed half-wave fixes $L_W = \pi a\sqrt{(1 + r_c)/3}/2$ and the bare outgoing scale.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check the Schur complement signs in the two-channel core model.
- ☐ Check the D/N mixed-tube length normalization against the chosen L/a.
- ☐ Check parent overlap ratios before interpreting them as actual branch values.

Downstream use. This stage feeds Stages 125–139 and the positive mouth-source branch. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.39 Stage 117: Concrete Outlet-Core Status

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure / Open.

Purpose. Stage 134 is a core outlet realization ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the compensated outlet conditions, a two-channel shell/mixed Schur complement, finite D/N mixed-tube geometry, and parent-action overlap data. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Core outlet realization* block. The computation isolates the parent-level compensation surface and normalized core/mouth target values.

> The microscopic question is whether the real core lands on the coupling-balance surface and D/N tube normalization.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check the Schur complement signs in the two-channel core model.
- ☐ Check the D/N mixed-tube length normalization against the chosen L/a.
- ☐ Check parent overlap ratios before interpreting them as actual branch values.

Downstream use. This stage feeds Stages 125–139 and the positive mouth-source branch. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.40 Stage 118: Parent-Action Extraction of Core Parameters

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Reduced / Controlled Reduction / Exact Within Closure.

Purpose. Stage 118 is a core outlet realization ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the compensated outlet conditions, a two-channel shell/mixed Schur complement, finite D/N mixed-tube geometry, and parent-action overlap data. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Core outlet realization* block. The computation isolates the parent-level compensation surface and normalized core/mouth target values.

> Gives overlap formulas for $K_s, K_q, \lambda, g_s, g_q$ from a GNLS plus localized-Maxwell throat-core ansatz.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check the Schur complement signs in the two-channel core model.
- ☐ Check the D/N mixed-tube length normalization against the chosen L/a.
- ☐ Check parent overlap ratios before interpreting them as actual branch values.

Downstream use. This stage feeds Stages 125–139 and the positive mouth-source branch. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.41 Stage 119: One-Parameter Parent Compensation Family

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure.

Purpose. Stage 136 is a core outlet realization ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the compensated outlet conditions, a two-channel shell/mixed Schur complement, finite D/N mixed-tube geometry, and parent-action overlap data. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Core outlet realization* block. The computation isolates the parent-level compensation surface and normalized core/mouth target values.

Normalized ratios $(\mathfrak{r}, \mathfrak{g})$ obey $1 + \mathfrak{r}^2 = 4(\mathfrak{g} - \mathfrak{r})^2$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check the Schur complement signs in the two-channel core model.
- ☐ Check the D/N mixed-tube length normalization against the chosen L/a.
- ☐ Check parent overlap ratios before interpreting them as actual branch values.

Downstream use. This stage feeds Stages 125–139 and the positive mouth-source branch. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.42 Stage 120: Core-Parameter Extraction Status

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure / Open.

Purpose. Stage 137 is a core outlet realization ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the compensated outlet conditions, a two-channel shell/mixed Schur complement, finite D/N mixed-tube geometry, and parent-action overlap data. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Core outlet realization* block. The computation isolates the parent-level compensation surface and normalized core/mouth target values.

Records the surviving finite parent-level target after overlap extraction.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check the Schur complement signs in the two-channel core model.
- ☐ Check the D/N mixed-tube length normalization against the chosen L/a.
- ☐ Check parent overlap ratios before interpreting them as actual branch values.

Downstream use. This stage feeds Stages 125–139 and the positive mouth-source branch. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.43 Stage 121: Geometric Selection of the Core Hybridization Ratio

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure / Reduced / Controlled Reduction.

Purpose. Stage 121 is a core outlet realization ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the compensated outlet conditions, a two-channel shell/mixed Schur complement, finite D/N mixed-tube geometry, and parent-action overlap data. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Core outlet realization* block. The computation isolates the parent-level compensation surface and normalized core/mouth target values.

> Identifying the mixed D/N tube with the throat span gives $\mathfrak{r}_{\mathrm{geom}}(L/a) = \sqrt{12(L/a)^2/\pi^2 - 1}$; the $L/a = 37/20$ value used below is the Stage 073 reference-branch freeze.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check the Schur complement signs in the two-channel core model.
- ☐ Check the D/N mixed-tube length normalization against the chosen L/a.
- ☐ Check parent overlap ratios before interpreting them as actual branch values.

Downstream use. This stage feeds Stages 125–139 and the positive mouth-source branch. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.44 Stage 122: Concrete Mouth-Source Branch and Compensation Test

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure.

Purpose. Stage 139 is a core outlet realization ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the compensated outlet conditions, a two-channel shell/mixed Schur complement, finite D/N mixed-tube geometry, and parent-action overlap data. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Core outlet realization* block. The computation isolates the parent-level compensation surface and normalized core/mouth target values.

> Equal-normalized mouth source misses the lower compensated branch but only by a modest traction renormalization.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check the Schur complement signs in the two-channel core model.
- ☐ Check the D/N mixed-tube length normalization against the chosen L/a.
- ☐ Check parent overlap ratios before interpreting them as actual branch values.

Downstream use. This stage feeds Stages 125–139 and the positive mouth-source branch. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.45 Stage 123: Parent-Normalized Branch Values on the Explicit Core Solve

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure.

Purpose. Stage 140 is a core outlet realization ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the compensated outlet conditions, a two-channel shell/mixed Schur complement, finite D/N mixed-tube geometry, and parent-action overlap data. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Core outlet realization* block. The computation isolates the parent-level compensation surface and normalized core/mouth target values.

> Converts the Family–1 compensated target to normalized parent flow and traction numbers (Ξ_v, Ξ_T).

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check the Schur complement signs in the two-channel core model.
- ☐ Check the D/N mixed-tube length normalization against the chosen L/a.
- ☐ Check parent overlap ratios before interpreting them as actual branch values.

Downstream use. This stage feeds Stages 125–139 and the positive mouth-source branch. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.46 Stage 124: Explicit Core-Solve Status After Geometric Branch Selection

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure / Open.

Purpose. Stage 141 is a core outlet realization ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the compensated outlet conditions, a two-channel shell/mixed Schur complement, finite D/N mixed-tube geometry, and parent-action overlap data. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Core outlet realization* block. The computation isolates the parent-level compensation surface and normalized core/mouth target values.

> Remaining question is whether the real core chooses the naive or lower compensated mouth branch.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check the Schur complement signs in the two-channel core model.
- ☐ Check the D/N mixed-tube length normalization against the chosen L/a.
- ☐ Check parent overlap ratios before interpreting them as actual branch values.

Downstream use. This stage feeds Stages 125–139 and the positive mouth-source branch. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.47 Stage 125: Positive Local Mouth-Source Theorem

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure.

Purpose. Stage 125 is a positive mouth-source and boundary layer ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the lower compensated core branch, positivity of the local mouth source, and the explicit GNLS/localized-Maxwell boundary-layer reduction. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Positive mouth-source and boundary layer* block. The computation isolates the positive source families, mouth-bias map, and micro-threshold for canonical compensation.

> Positive normalized sources satisfy $0 \leq \mathfrak{g}[\sigma] \leq 1$, ruling out the upper compensated branch.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check positivity of the mouth source; do not use signed fitting to reach the upper branch.
- ☐ Check zero-flux and boundary-layer normalizations in the GNLS/localized-Maxwell reduction.
- ☐ Check the Family–1 compensation point against the lower branch, not the singular equal-normalized branch.

Downstream use. This stage feeds Stages 133–145 and the coupled mouth fixed-point law. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.48 Stage 126: Explicit Positive Source Families and the Family-1 Compensation Point

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure.

Purpose. Stage 126 is a positive mouth-source and boundary layer ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the lower compensated core branch, positivity of the local mouth source, and the explicit GNLS/localized-Maxwell boundary-layer reduction. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Positive mouth-source and boundary layer* block. The computation isolates the positive source families, mouth-bias map, and micro-threshold for canonical compensation.

> Self-matched derivative profile gives $\mathfrak{g} = \pi/4$; a positive convex family hits the lower branch.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check positivity of the mouth source; do not use signed fitting to reach the upper branch.
- ☐ Check zero-flux and boundary-layer normalizations in the GNLS/localized-Maxwell reduction.
- ☐ Check the Family–1 compensation point against the lower branch, not the singular equal-normalized branch.

Downstream use. This stage feeds Stages 133–145 and the coupled mouth fixed-point law. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.49 Stage 127: Geometric Mouth-Penetration Families

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. EXACT WITHIN CLOSURE.

Purpose. Stage 127 is a positive mouth-source and boundary layer ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the lower compensated core branch, positivity of the local mouth source, and the explicit GNLS/localized-Maxwell boundary-layer reduction. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Positive mouth-source and boundary layer* block. The computation isolates the positive source families, mouth-bias map, and micro-threshold for canonical compensation.

> Slab and truncated-exponential families reach the lower branch at moderate penetration depths.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check positivity of the mouth source; do not use signed fitting to reach the upper branch.
- ☐ Check zero-flux and boundary-layer normalizations in the GNLS/localized-Maxwell reduction.
- ☐ Check the Family–1 compensation point against the lower branch, not the singular equal-normalized branch.

Downstream use. This stage feeds Stages 133–145 and the coupled mouth fixed-point law. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.50 Stage 128: Mouth-Source Bias Status

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure / Open.

Purpose. Stage 128 is a positive mouth-source and boundary layer ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the lower compensated core branch, positivity of the local mouth source, and the explicit GNLS/localized-Maxwell boundary-layer reduction. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Positive mouth-source and boundary layer* block. The computation isolates the positive source families, mouth-bias map, and micro-threshold for canonical compensation.

> Branch sign is fixed; the open datum is the actual positive mouth-source shape.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check positivity of the mouth source; do not use signed fitting to reach the upper branch.
- ☐ Check zero-flux and boundary-layer normalizations in the GNLS/localized-Maxwell reduction.
- ☐ Check the Family–1 compensation point against the lower branch, not the singular equal-normalized branch.

Downstream use. This stage feeds Stages 133–145 and the coupled mouth fixed-point law. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.51 Stage 129: Explicit GNLS + Localized-Maxwell Mouth Boundary Layer

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Reduced / Controlled Reduction / Exact Within Closure.

Purpose. Stage 129 is a positive mouth-source and boundary layer ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the lower compensated core branch, positivity of the local mouth source, and the explicit GNLS/localized-Maxwell boundary-layer reduction. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Positive mouth-source and boundary layer* block. The computation isolates the positive source families, mouth-bias map, and micro-threshold for canonical compensation.

> Electrochemical zero-flux law yields the exponential source profile σ_Π.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check positivity of the mouth source; do not use signed fitting to reach the upper branch.
- ☐ Check zero-flux and boundary-layer normalizations in the GNLS/localized-Maxwell reduction.
- ☐ Check the Family–1 compensation point against the lower branch, not the singular equal-normalized branch.

Downstream use. This stage feeds Stages 133–145 and the coupled mouth fixed-point law. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.52 Stage 130: Exact Mouth-Bias Map and Family-1 Compensation Point

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. EXACT WITHIN CLOSURE / NUMERICALLY LOCATED.

Purpose. Stage 130 is a positive mouth-source and boundary layer ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the lower compensated core branch, positivity of the local mouth source, and the explicit GNLS/localized-Maxwell boundary-layer reduction. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Positive mouth-source and boundary layer* block. The computation isolates the positive source families, mouth-bias map, and micro-threshold for canonical compensation.

> Exact $\mathfrak{g}_\Pi$, monotonicity, and unique Family–1 point $\Pi_* \approx 1.5088295$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check positivity of the mouth source; do not use signed fitting to reach the upper branch.
- ☐ Check zero-flux and boundary-layer normalizations in the GNLS/localized-Maxwell reduction.
- ☐ Check the Family–1 compensation point against the lower branch, not the singular equal-normalized branch.

Downstream use. This stage feeds Stages 133–145 and the coupled mouth fixed-point law. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.53 Stage 131: Parent Micro-Threshold for Canonical Mouth Compensation

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. EXACT WITHIN CLOSURE.

Purpose. Stage 131 is a positive mouth-source and boundary layer ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the lower compensated core branch, positivity of the local mouth source, and the explicit GNLS/localized-Maxwell boundary-layer reduction. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Positive mouth-source and boundary layer* block. The computation isolates the positive source families, mouth-bias map, and micro-threshold for canonical compensation.

Canonical branch is the parent threshold $T_m - q_* A_0' = \Pi_* \Theta_\sigma / L$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check positivity of the mouth source; do not use signed fitting to reach the upper branch.
- ☐ Check zero-flux and boundary-layer normalizations in the GNLS/localized-Maxwell reduction.
- ☐ Check the Family–1 compensation point against the lower branch, not the singular equal-normalized branch.

Downstream use. This stage feeds Stages 133–145 and the coupled mouth fixed-point law. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.54 Stage 132: Mouth Boundary-Layer Status After Explicit Source-Law Extraction

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure / Open.

Purpose. Stage 132 is a positive mouth-source and boundary layer ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the lower compensated core branch, positivity of the local mouth source, and the explicit GNLS/localized-Maxwell boundary-layer reduction. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Positive mouth-source and boundary layer* block. The computation isolates the positive source families, mouth-bias map, and micro-threshold for canonical compensation.

The source-shape ambiguity is reduced to the selected bias Π_m.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check positivity of the mouth source; do not use signed fitting to reach the upper branch.
- ☐ Check zero-flux and boundary-layer normalizations in the GNLS/localized-Maxwell reduction.
- ☐ Check the Family–1 compensation point against the lower branch, not the singular equal-normalized branch.

Downstream use. This stage feeds Stages 133–145 and the coupled mouth fixed-point law. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.55 Stage 133: Full Coupled Mouth-Layer Fixed-Point Law

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure / Reduced / Controlled Reduction.

Purpose. Stage 133 is a coupled mouth fixed point and gain selection ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the shell/mixed core, the mouth source law, outlet consistency, core-to-mouth gain maps, and self-matched susceptibility closure. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Coupled mouth fixed point and gain selection* block. The computation isolates the finite-bias regular canonical mouth branch and the gain pair that replaces the naive equal-normalized limit.

> Two-channel D/N boundary layer gives $\Pi = \sum_{\alpha=\pm} M_\alpha \mathcal{S}(\Pi, \kappa_\alpha)$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check the gain pair (M_s, M_q) against outlet consistency.
- ☐ Check the self-matched susceptibility closure before using the one-scalar branch law.
- ☐ Check numerical fixed points are recorded as numerically located, not closed-form constants.

Downstream use. This stage feeds Stages 146–153 and the finite mouth-profile correction. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.56 Stage 134: Family-1 Shell + First Mixed D/N Tube Reduction

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure.

Purpose. Stage 134 is a coupled mouth fixed point and gain selection ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the shell/mixed core, the mouth source law, outlet consistency, core-to-mouth gain maps, and self-matched susceptibility closure. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Coupled mouth fixed point and gain selection* block. The computation isolates the finite-bias regular canonical mouth branch and the gain pair that replaces the naive equal-normalized limit.

First explicit reduction gives $\Pi = M_s + M_q\mathcal{S}_q(\Pi)$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Outlet consistency of the gain pair (M_s, M_q) is verified at Stage E.57 (outlet-consistent mouth closure); carried forward here.
- ☐ Self-matched susceptibility closure is verified at Stage E.59 (explicit core-to-mouth gain map); carried forward here.
- ☐ Check numerical fixed points are recorded as numerically located, not closed-form constants.

Downstream use. This stage feeds Stages 146–153 and the finite mouth-profile correction. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.57 Stage 135: Outlet-Consistent Mouth Closure

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure.

Purpose. Stage 135 is a coupled mouth fixed point and gain selection ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the shell/mixed core, the mouth source law, outlet consistency, core-to-mouth gain maps, and self-matched susceptibility closure. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Coupled mouth fixed point and gain selection* block. The computation isolates the finite-bias regular canonical mouth branch and the gain pair that replaces the naive equal-normalized limit.

Inheriting the compensated outlet ratio gives $M_s = 4\Sigma_m$, $M_q = -\Sigma_m$, and $\Sigma_m^* \approx 0.451485$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check the gain pair (M_s, M_q) against outlet consistency.
- ☐ Check the self-matched susceptibility closure before using the one-scalar branch law.
- ☐ Check numerical fixed points are recorded as numerically located, not closed-form constants.

Downstream use. This stage feeds Stages 146–153 and the finite mouth-profile correction. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.58 Stage 136: Mouth-Layer Fixed-Point Status After the Coupled Solve

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure / Open.

Purpose. Stage 136 is a coupled mouth fixed point and gain selection ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the shell/mixed core, the mouth source law, outlet consistency, core-to-mouth gain maps, and self-matched susceptibility closure. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Coupled mouth fixed point and gain selection* block. The computation isolates the finite-bias regular canonical mouth branch and the gain pair that replaces the naive equal-normalized limit.

The mouth-layer problem is reduced to a gain pair, or one gain under outlet consistency.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check the gain pair (M_s, M_q) against outlet consistency.
- ☐ Check the self-matched susceptibility closure before using the one-scalar branch law.
- ☐ Check numerical fixed points are recorded as numerically located, not closed-form constants.

Downstream use. This stage feeds Stages 146–153 and the finite mouth-profile correction. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.59 Stage 137: Explicit Core-to-Mouth Gain Map

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure / Reduced / Controlled Reduction.

Purpose. Stage 137 is a coupled mouth fixed point and gain selection ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the shell/mixed core, the mouth source law, outlet consistency, core-to-mouth gain maps, and self-matched susceptibility closure. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Coupled mouth fixed point and gain selection* block. The computation isolates the finite-bias regular canonical mouth branch and the gain pair that replaces the naive equal-normalized limit.

Explicit gains are $M_s = Lg_s^2/(K_s\Theta_\sigma)$, $M_q = -L(K_s g_q - \lambda g_s)^2/[K_s(K_s K_q + \lambda^2)\Theta_\sigma]$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check the gain pair (M_s, M_q) against outlet consistency.
- ☐ Check the self-matched susceptibility closure before using the one-scalar branch law.
- ☐ Check numerical fixed points are recorded as numerically located, not closed-form constants.

Downstream use. This stage feeds Stages 146–153 and the finite mouth-profile correction. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.60 Stage 138: Normalized Mouth-Gain Family and Compensation Ratio

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure.

Purpose. Stage 138 is a coupled mouth fixed point and gain selection ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the shell/mixed core, the mouth source law, outlet consistency, core-to-mouth gain maps, and self-matched susceptibility closure. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Coupled mouth fixed point and gain selection* block. The computation isolates the finite-bias regular canonical mouth branch and the gain pair that replaces the naive equal-normalized limit.

Gain ratio $R_q = (\mathfrak{g}_c - \mathfrak{r})^2/(1+\mathfrak{r}^2)$; exact compensation gives $R_q = 1/4$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check the gain pair (M_s, M_q) against outlet consistency.
- ☐ Check the self-matched susceptibility closure before using the one-scalar branch law.
- ☐ Check numerical fixed points are recorded as numerically located, not closed-form constants.

Downstream use. This stage feeds Stages 146–153 and the finite mouth-profile correction. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.61 Stage 139: Actual Family-1 Mouth Gains

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure / Numerically Located.

Purpose. Stage 139 is an actual Family–1 mouth-gains ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the shell/mixed core, the mouth source law, outlet consistency, core-to-mouth gain maps, and self-matched susceptibility closure. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Coupled mouth fixed point and gain selection* block. The computation isolates the finite-bias regular canonical mouth branch and the gain pair that replaces the naive equal-normalized limit.

Natural and exact-compensated gain pairs are evaluated on the Family–1 branch.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check the gain pair (M_s, M_q) against outlet consistency.
- ☐ Check the self-matched susceptibility closure before using the one-scalar branch law.
- ☐ Check numerical fixed points are recorded as numerically located, not closed-form constants.

Downstream use. This stage feeds Stages 146–153 and the finite mouth-profile correction. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.62 Stage 140: Self-Matched Mouth Susceptibility Closure

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Reduced / Controlled Reduction / Exact Within Closure.

Purpose. Stage 140 is a coupled mouth fixed point and gain selection ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the shell/mixed core, the mouth source law, outlet consistency, core-to-mouth gain maps, and self-matched susceptibility closure. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Coupled mouth fixed point and gain selection* block. The computation isolates the finite-bias regular canonical mouth branch and the gain pair that replaces the naive equal-normalized limit.

Same-layer susceptibility gives $\Sigma_0 = M_s = (20/9)\widehat{T}_m^2$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check the gain pair (M_s, M_q) against outlet consistency.
- ☐ Check the self-matched susceptibility closure before using the one-scalar branch law.
- ☐ Check numerical fixed points are recorded as numerically located, not closed-form constants.

Downstream use. This stage feeds Stages 146–153 and the finite mouth-profile correction. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.63 Stage 141: Mouth-Gain Status Update

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure / Open.

Purpose. Stage 141 is a coupled mouth fixed point and gain selection ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the shell/mixed core, the mouth source law, outlet consistency, core-to-mouth gain maps, and self-matched susceptibility closure. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Coupled mouth fixed point and gain selection* block. The computation isolates the finite-bias regular canonical mouth branch and the gain pair that replaces the naive equal-normalized limit.

> Records the narrowed gain-scale problem after self-matched susceptibility.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check the gain pair (M_s, M_q) against outlet consistency.
- ☐ Check the self-matched susceptibility closure before using the one-scalar branch law.
- ☐ Check numerical fixed points are recorded as numerically located, not closed-form constants.

Downstream use. This stage feeds Stages 146–153 and the finite mouth-profile correction. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.64 Stage 142: Self-Consistent Mouth-Branch Law

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure.

Purpose. Stage 142 is a coupled mouth fixed point and gain selection ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the shell/mixed core, the mouth source law, outlet consistency, core-to-mouth gain maps, and self-matched susceptibility closure. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Coupled mouth fixed point and gain selection* block. The computation isolates the finite-bias regular canonical mouth branch and the gain pair that replaces the naive equal-normalized limit.

Identifying $\mathfrak{g}_c = \mathfrak{g}_\Pi$ gives $\Pi = \Sigma_0[1 - R_q(\Pi)\mathcal{S}_q(\Pi)]$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check the gain pair (M_s, M_q) against outlet consistency.
- ☐ Check the self-matched susceptibility closure before using the one-scalar branch law.
- ☐ Check numerical fixed points are recorded as numerically located, not closed-form constants.

Downstream use. This stage feeds Stages 146–153 and the finite mouth-profile correction. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.65 Stage 143: Equal-Normalized Branch Is a Singular Limit

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure.

Purpose. Stage 143 is a coupled mouth fixed point and gain selection ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the shell/mixed core, the mouth source law, outlet consistency, core-to-mouth gain maps, and self-matched susceptibility closure. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Coupled mouth fixed point and gain selection* block. The computation isolates the finite-bias regular canonical mouth branch and the gain pair that replaces the naive equal-normalized limit.

> $\mathfrak{g}_\Pi < 1$ for all finite $\Pi > 0$; equal-normalized branch requires $\Pi \to \infty$ and divergent traction.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check the gain pair (M_s, M_q) against outlet consistency.
- ☐ Check the self-matched susceptibility closure before using the one-scalar branch law.
- ☐ Check numerical fixed points are recorded as numerically located, not closed-form constants.

Downstream use. This stage feeds Stages 146–153 and the finite mouth-profile correction. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.66 Stage 144: Unique Regular Canonical Mouth Branch

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure / Numerically Located.

Purpose. Stage 144 is a coupled mouth fixed point and gain selection ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the shell/mixed core, the mouth source law, outlet consistency, core-to-mouth gain maps, and self-matched susceptibility closure. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Coupled mouth fixed point and gain selection* block. The computation isolates the finite-bias regular canonical mouth branch and the gain pair that replaces the naive equal-normalized limit.

> Lower compensated branch is the unique finite-bias, finite-traction branch in the explicit positive exponential closure.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Outlet consistency of the gain pair (M_s, M_q) is verified upstream at E.57; this stage carries that result forward.
- ☐ Self-matched susceptibility closure is verified upstream at E.62; this stage carries that result forward.
- ☐ Check numerical fixed points are recorded as numerically located, not closed-form constants.

Downstream use. This stage feeds Stages 146–153 and the finite mouth-profile correction. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.67 Stage 145: Mouth-Branch Selection Status

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure / Open.

Purpose. Stage 145 is a coupled mouth fixed point and gain selection ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the shell/mixed core, the mouth source law, outlet consistency, core-to-mouth gain maps, and self-matched susceptibility closure. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Coupled mouth fixed point and gain selection* block. The computation isolates the finite-bias regular canonical mouth branch and the gain pair that replaces the naive equal-normalized limit.

> Branch choice is settled inside the explicit closure; finite corrections around $(\Pi_*, \widehat{T}_{m,*})$ remain.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check the gain pair (M_s, M_q) against outlet consistency.
- ☐ Check the self-matched susceptibility closure before using the one-scalar branch law.
- ☐ Check numerical fixed points are recorded as numerically located, not closed-form constants.

Downstream use. This stage feeds Stages 146–153 and the finite mouth-profile correction. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.68 Stage 146: First-Order Expansion for Positive Mouth-Layer Deformations

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure.

Purpose. Stage 146 is a finite mouth-profile corrections ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the regular canonical mouth branch, positive deformations of the source profile, the first-order rigidity kernel, and the full mouth potential. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Finite mouth-profile corrections* block. The computation isolates the finite first-order/nonlinear correction to $(\Pi, \widehat{T}_m)$ and the decision to co-evolve core and mouth.

First-order deformations depend only on $(\bar{g}_\varsigma, \bar{S}_\varsigma)$ and retune Π.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check that first-order profile deformations are centered before covariance formulas are used.
- ☐ Check that the rigidity kernel, not branch ambiguity, controls non-exponential corrections.
- ☐ Check the one-step nonlinear correction remains within the reduced mouth-layer regime.

Downstream use. This stage feeds Stages 154–163 and the linearized defect transport. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.69 Stage 147: First-Order Rigidity Kernel at the Canonical Family-1 Point

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure.

Purpose. Stage 147 is a finite mouth-profile corrections ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the regular canonical mouth branch, positive deformations of the source profile, the first-order rigidity kernel, and the full mouth potential. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Finite mouth-profile corrections* block. The computation isolates the finite first-order/nonlinear correction to $(\Pi, \widehat{T}_m)$ and the decision to co-evolve core and mouth.

> Traction shift is one kernel projection with coefficients $A_T \approx -4.27264$, $B_T \approx 0.134875$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check that first-order profile deformations are centered before covariance formulas are used.
- ☐ Check that the rigidity kernel, not branch ambiguity, controls non-exponential corrections.
- ☐ Check the one-step nonlinear correction remains within the reduced mouth-layer regime.

Downstream use. This stage feeds Stages 154–163 and the linearized defect transport. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.70 Stage 148: Representative Non-Exponential Positive Mouth Families

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure / Numerically Located.

Purpose. Stage 148 is a finite mouth-profile corrections ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the regular canonical mouth branch, positive deformations of the source profile, the first-order rigidity kernel, and the full mouth potential. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Finite mouth-profile corrections* block. The computation isolates the finite first-order/nonlinear correction to $(\Pi, \widehat{T}_m)$ and the decision to co-evolve core and mouth.

> Uniform and self-matched derivative families bracket the first-order correction; interpolation reproduces the earlier compensation fraction.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check that first-order profile deformations are centered before covariance formulas are used.
- ☐ Check that the rigidity kernel, not branch ambiguity, controls non-exponential corrections.
- ☐ Check the one-step nonlinear correction remains within the reduced mouth-layer regime.

Downstream use. This stage feeds Stages 154–163 and the linearized defect transport. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.71 Stage 149: Non-Exponential Mouth-Rigidity Status

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. EXACT WITHIN CLOSURE / OPEN.

Purpose. Stage 149 is a finite mouth-profile corrections ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the regular canonical mouth branch, positive deformations of the source profile, the first-order rigidity kernel, and the full mouth potential. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Finite mouth-profile corrections* block. The computation isolates the finite first-order/nonlinear correction to $(\Pi, \widehat{T}_m)$ and the decision to co-evolve core and mouth.

> Mouth corrections are controlled by a finite rigidity kernel, not by branch ambiguity.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check that first-order profile deformations are centered before covariance formulas are used.
- ☐ Check that the rigidity kernel, not branch ambiguity, controls non-exponential corrections.
- ☐ Check the one-step nonlinear correction remains within the reduced mouth-layer regime.

Downstream use. This stage feeds Stages 154–163 and the linearized defect transport. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.72 Stage 150: Exact Full-Profile Mouth Potential and Curvature Residual

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure.

Purpose. Stage 150 is a finite mouth-profile corrections ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the regular canonical mouth branch, positive deformations of the source profile, the first-order rigidity kernel, and the full mouth potential. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Finite mouth-profile corrections* block. The computation isolates the finite first-order/nonlinear correction to $(\Pi, \widehat{T}_m)$ and the decision to co-evolve core and mouth.

> Exact residual $R_*(x) = \Phi_*(x) - \Pi_* x$ is tangent-matched but has negative curvature at the mouth.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check that first-order profile deformations are centered before covariance formulas are used.
- ☐ Check that the rigidity kernel, not branch ambiguity, controls non-exponential corrections.
- ☐ Check the one-step nonlinear correction remains within the reduced mouth-layer regime.

Downstream use. This stage feeds Stages 154–163 and the linearized defect transport. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.73 Stage 151: First-Order Source Correction Selected by the Full Mouth Profile

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure.

Purpose. Stage 151 is a finite mouth-profile corrections ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the regular canonical mouth branch, positive deformations of the source profile, the first-order rigidity kernel, and the full mouth potential. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Finite mouth-profile corrections* block. The computation isolates the finite first-order/nonlinear correction to $(\Pi, \widehat{T}_m)$ and the decision to co-evolve core and mouth.

Actual selected deformation is the centered residual R_*; correction is fixed by two covariances.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check that first-order profile deformations are centered before covariance formulas are used.
- ☐ Check that the rigidity kernel, not branch ambiguity, controls non-exponential corrections.
- ☐ Check the one-step nonlinear correction remains within the reduced mouth-layer regime.

Downstream use. This stage feeds Stages 154–163 and the linearized defect transport. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.74 Stage 152: Actual Family-1 Mouth Correction and One-Step Nonlinear Check

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure / Numerically Located.

Purpose. Stage 152 is a finite mouth-profile corrections ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the regular canonical mouth branch, positive deformations of the source profile, the first-order rigidity kernel, and the full mouth potential. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Finite mouth-profile corrections* block. The computation isolates the finite first-order/nonlinear correction to $(\Pi, \widehat{T}_m)$ and the decision to co-evolve core and mouth.

Full mouth profile broadens the source and shifts the point to $(\Pi_{\rm corr}, \widehat{T}_{m,\rm corr})$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check that first-order profile deformations are centered before covariance formulas are used.
- ☐ Check that the rigidity kernel, not branch ambiguity, controls non-exponential corrections.
- ☐ Check the one-step nonlinear correction remains within the reduced mouth-layer regime.

Downstream use. This stage feeds Stages 154–163 and the linearized defect transport. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.75 Stage 153: v58 Status

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure / Open.

Purpose. Stage 153 is a finite mouth-profile corrections ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the regular canonical mouth branch, positive deformations of the source profile, the first-order rigidity kernel, and the full mouth potential. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Finite mouth-profile corrections* block. The computation isolates the finite first-order/nonlinear correction to $(\Pi, \widehat{T}_m)$ and the decision to co-evolve core and mouth.

> Branch ambiguity is gone; finite mouth-only correction motivates full co-evolution.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check that first-order profile deformations are centered before covariance formulas are used.
- ☐ Check that the rigidity kernel, not branch ambiguity, controls non-exponential corrections.
- ☐ Check the one-step nonlinear correction remains within the reduced mouth-layer regime.

Downstream use. This stage feeds Stages 154–163 and the linearized defect transport. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.76 Stage 154: Exact Co-Evolving Core–Mouth Fixed-Point Map

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure.

Purpose. Stage 154 is a co-evolving core–mouth transport ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the nonlinear co-evolving source map, the renormalized canonical fixed point, and the compensated outlet projection. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Co-evolving core–mouth transport* block. The computation isolates the transport of deviations into Δ_Q, D/N similarity slippage, and the final normal coordinate $\delta_\perp$.

$$\text{Nonlinear fixed-point map } \Sigma \propto e^{-\Phi_{\Sigma_0}[\Sigma]}, \text{ with } \mathcal{R}[\Sigma] = (\mathfrak{g}[\Sigma] - \mathfrak{r}_{F1})^2/(1 + \mathfrak{r}_{F1}^2).$$

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check deviations are taken about the renormalized co-evolving canonical point.
- ☐ Check even-preservation constraints are imposed before reading the remaining odd defect.
- ☐ Check tangent motion on the parent compensation family gives $\delta_\perp = 0$.

Downstream use. This stage feeds Part V and downstream branch-realization, orbit-lock, and same-charge audits. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.77 Stage 155: Family-1 Co-Evolving Fixed Point at Frozen Canonical Traction

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Numerically Located / Exact Within Closure.

Purpose. Stage 155 is a co-evolving core–mouth transport ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the nonlinear co-evolving source map, the renormalized canonical fixed point, and the compensated outlet projection. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Co-evolving core–mouth transport* block. The computation isolates the transport of deviations into Δ_Q, D/N similarity slippage, and the final normal coordinate $\delta_\perp$.

At old traction, fixed point survives but moves off exact compensation: $\mathcal{R}_{\rm fp} \approx 0.282714$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check deviations are taken about the renormalized co-evolving canonical point.
- ☐ Check even-preservation constraints are imposed before reading the remaining odd defect.
- ☐ Check tangent motion on the parent compensation family gives $\delta_\perp = 0$.

Downstream use. This stage feeds Part V and downstream branch-realization, orbit-lock, and same-charge audits. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.78 Stage 156: Renormalized Canonical Branch Under Full Core–Mouth Co-Evolution

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Numerically Located / Exact Within Closure.

Purpose. Stage 156 is a co-evolving core–mouth transport ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the nonlinear co-evolving source map, the renormalized canonical fixed point, and the compensated outlet projection. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Co-evolving core–mouth transport* block. The computation isolates the transport of deviations into Δ_Q, D/N similarity slippage, and the final normal coordinate $\delta_\perp$.

Exact compensation is restored at $\Sigma_0^{\rm can} \approx 4.65103$, $\widehat{T}_{m,\rm can} \approx 1.44671$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check deviations are taken about the renormalized co-evolving canonical point.
- ☐ Check even-preservation constraints are imposed before reading the remaining odd defect.
- ☐ Check tangent motion on the parent compensation family gives $\delta_\perp = 0$.

Downstream use. This stage feeds Part V and downstream branch-realization, orbit-lock, and same-charge audits. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.79 Stage 157: v59 Status

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Numerically Located / Open.

Purpose. Stage 157 is a co-evolving core–mouth transport ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the nonlinear co-evolving source map, the renormalized canonical fixed point, and the compensated outlet projection. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Co-evolving core–mouth transport* block. The computation isolates the transport of deviations into Δ_Q, D/N similarity slippage, and the final normal coordinate $\delta_\perp$.

> Identifies the reduced co-evolving target and leaves the deviation-to-normalization map as the next task.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check deviations are taken about the renormalized co-evolving canonical point.
- ☐ Check even-preservation constraints are imposed before reading the remaining odd defect.
- ☐ Check tangent motion on the parent compensation family gives $\delta_\perp = 0$.

Downstream use. This stage feeds Part V and downstream branch-realization, orbit-lock, and same-charge audits. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.80 Stage 158: Linear Defect Transport from the Renormalized Family-1 Canonical Point

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure.

Purpose. Stage 158 is a co-evolving core–mouth transport ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the nonlinear co-evolving source map, the renormalized canonical fixed point, and the compensated outlet projection. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Co-evolving core–mouth transport* block. The computation isolates the transport of deviations into Δ_Q, D/N similarity slippage, and the final normal coordinate $\delta_\perp$.

Linearizes $(\delta\Sigma_0, \delta\mathfrak{g}, \delta\mathcal{S})$ into $(\delta M_s, \delta M_q, \delta\Pi)$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check deviations are taken about the renormalized co-evolving canonical point.
- ☐ Even-preservation of the canonical gate is verified downstream: see E.81.
- ☐ Tangent motion on the parent compensation family giving $\delta_\perp = 0$ is verified downstream: see E.84 and E.85.

Downstream use. This stage feeds Part V and downstream branch-realization, orbit-lock, and same-charge audits. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.81 Stage 159: Linear Projection of the Co-Evolving Family-1 Defect onto the Compensated Robin–Mixed Outlet

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure.

Purpose. Stage 159 is a co-evolving core–mouth transport ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the nonlinear co-evolving source map, the renormalized canonical fixed point, and the compensated outlet projection. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Co-evolving core–mouth transport* block. The computation isolates the transport of deviations into Δ_Q, D/N similarity slippage, and the final normal coordinate $\delta_\perp$.

Even preservation forces $\delta\mathfrak{g} = 0$, $\delta\kappa_W = 0$, leaving $\Delta_Q = -9\sigma_*\delta\gamma_W/(1-\sigma_*)$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check deviations are taken about the renormalized co-evolving canonical point.
- ☐ Check even-preservation constraints are imposed before reading the remaining odd defect.
- ☐ Check tangent motion on the parent compensation family gives $\delta_\perp = 0$.

Downstream use. This stage feeds Part V and downstream branch-realization, orbit-lock, and same-charge audits. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.82 Stage 160: Bare Mixed-Port Slippage Theorem and the Collapse of the Last Tangential DtN Gate

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure.

Purpose. Stage 160 is a co-evolving core–mouth transport ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the nonlinear co-evolving source map, the renormalized canonical fixed point, and the compensated outlet projection. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Co-evolving core–mouth transport* block. The computation isolates the transport of deviations into Δ_Q, D/N similarity slippage, and the final normal coordinate $\delta_\perp$.

$\delta\gamma_W = (\delta\gamma_0 - \delta\kappa_0/3)/(1 + \mathfrak{r}_{F1}^2)$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check deviations are taken about the renormalized co-evolving canonical point.
- ☐ Check even-preservation constraints are imposed before reading the remaining odd defect.
- ☐ Check tangent motion on the parent compensation family gives $\delta_\perp = 0$.

Downstream use. This stage feeds Part V and downstream branch-realization, orbit-lock, and same-charge audits. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.83 Stage 161: D/N Similarity Decomposition of the Tangential Mixed-Port Slippage Susceptibility

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure.

Purpose. Stage 161 is a co-evolving core–mouth transport ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the nonlinear co-evolving source map, the renormalized canonical fixed point, and the compensated outlet projection. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Co-evolving core–mouth transport* block. The computation isolates the transport of deviations into Δ_Q, D/N similarity slippage, and the final normal coordinate $\delta_\perp$.

> Last defect is $\Delta_Q = -\sigma_* \Xi_{\mathrm{slip}} \delta\Pi_{\mathrm{tan}}/(1-\sigma_*)$, with $\Xi_{\mathrm{slip}} = \Xi_\gamma - 2\Xi_L$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check deviations are taken about the renormalized co-evolving canonical point.
- ☐ Check even-preservation constraints are imposed before reading the remaining odd defect.
- ☐ Check tangent motion on the parent compensation family gives $\delta_\perp = 0$.

Downstream use. This stage feeds Part V and downstream branch-realization, orbit-lock, and same-charge audits. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.84 Stage 162: Parent Compensation-Surface Rigidity and Automatic Similarity Preservation

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure.

Purpose. Stage 162 is a co-evolving core–mouth transport ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the nonlinear co-evolving source map, the renormalized canonical fixed point, and the compensated outlet projection. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Co-evolving core–mouth transport* block. The computation isolates the transport of deviations into Δ_Q, D/N similarity slippage, and the final normal coordinate $\delta_\perp$.

> Staying on the exact parent compensation family gives automatic D/N similarity preservation and $\Delta_Q = 0$ at first order.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check deviations are taken about the renormalized co-evolving canonical point.
- ☐ Check even-preservation constraints are imposed before reading the remaining odd defect.
- ☐ Check tangent motion on the parent compensation family gives $\delta_\perp = 0$.

Downstream use. This stage feeds Part V and downstream branch-realization, orbit-lock, and same-charge audits. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

E.85 Stage 163: Off-Family Normal Coordinate and Microscopic Compensation Defect

Part / anchor. Part IV; primary anchor **MTDC-T8**.

Claim status. Exact Within Closure / Open.

Purpose. Stage 163 is a co-evolving core–mouth transport ledger step. Its audit target is the verification output quoted below.

Inputs. This stage imports the nonlinear co-evolving source map, the renormalized canonical fixed point, and the compensated outlet projection. It does not change the parent action or the grouped-lane ontology; it works inside the declared reduced branch and tests the next algebraic or realization condition.

Derivation ledger. The verification step belongs to the *Co-evolving core–mouth transport* block. The computation isolates the transport of deviations into Δ_Q, D/N similarity slippage, and the final normal coordinate $\delta_\perp$.

> The first true off-family drift is the scalar $\delta_\perp$, with explicit microscopic parent-action formula.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage.

Checks.

- ☐ Check deviations are taken about the renormalized co-evolving canonical point.
- ☐ Check even-preservation constraints are imposed before reading the remaining odd defect.
- ☐ Check tangent motion on the parent compensation family gives $\delta_\perp = 0$.

Downstream use. This stage feeds Part V and downstream branch-realization, orbit-lock, and same-charge audits. When cited downstream, the status tag above must be carried with the result; the card is a derivation ledger entry, not an unconditional actual-branch theorem.

Appendix F

Stage Appendix: Part V – Weak-Axisymmetric Transport, Quotient Coordinates, and Orbit Closure

Stage range: 164–200

This appendix records the fifth theorem block of the moving-throat derivation ledger. Part IV ended with the co-evolving core–mouth branch reduced to a first-order off-family normal coordinate. Part V starts from that coordinate and asks a sharper question: after the Family–1 mouth/core compensation machinery is already in place, which microscopic grouped weak-axisymmetric motions still survive as genuine reduced defects, and which motions are only similarity-orbit relabelings of the same coherent branch?

The answer is that the apparent many-variable weak-axisymmetric problem collapses to two levels. On the branch side, the grouped real P_2 carrier reduces to a single transported prefactor slope,

$$\Xi_1 = \frac{P_1}{P_0}, \tag{F.1}$$

with the weak-axisymmetric grouped signature

$$\lambda_{20} = 1, \qquad \lambda_{21} = \frac{1}{2}, \qquad \lambda_{22} = -1. \tag{F.2}$$

Equivalently, the grouped anisotropy lies on the fixed line

$$b_{P_0} = 3a_{P_0}. \tag{F.3}$$

On the orbit side, the coherent local branch reduces to three exact quotient coordinates,

$$\mathfrak{C}_{\mathrm{tr},*}, \qquad \mathfrak{C}_{\mathrm{nt},*}, \qquad \epsilon_\eta, \tag{F.4}$$

whose finite level sets are exactly the five-parameter microscopic similarity orbits. The final reduced verdict after this part is therefore not an eight-slot branch residual plus an ambiguous orbit condition. It is the exact four-scalar packet

$$\Delta_{\mathrm{full}} = (\Delta_Q, q_{\mathrm{tr}}, q_{\mathrm{nt}}, q_\eta), \qquad \Delta_Q := \chi_Q - 1. \tag{F.5}$$

The result anchor served by this appendix is **MTDC-T9**. This anchor should be cited when a downstream paper needs the derivation of the weak-axisymmetric loading mismatch, the transfer-shape scalar $\Xi_1 = P_1/P_0$, the coherent tracking/nontracking/dressing normal form, the monomial quotient coordinates, the similarity-orbit theorem, the exact Packet-A finish line $\Delta_Q = \chi_Q - 1$, or the final reference-free verdict packet Δ_{full}.

Table F.1: Part V source and status map.

Stage	Working title	Primary status	Main output
164	Microscopic log-imbalance channels	Exact Within Closure	Rewrites the Stage 163 off-family scalar in explicit branch-variable drifts of $(\mathcal{Z}_q, \rho_w, c_{s,w}, c_s, \mathcal{T}_m, v_{w0}, a, L_W)$.
165	Lower-branch drift laws	Exact Within Closure	Shows $\delta \ln L_W = \delta \ln a$ and reduces the lower compensated branch to four irreducible drifts $(\delta \ln \mathcal{Z}_q, \delta \ln \rho_w, \delta \ln c_s, \delta \ln a)$.
166	Bundle inversion of the last four drifts	Exact Within Closure	Inverts the four observables $(\Theta_w, K_s, K_q, P_0)$ to determine the four remaining microscopic drifts.
167	Bundle transport and tangent compensation	Exact Within Closure	Proves arbitrary first-order isotropic bundle drift is tangent to the exact compensated parent family.
168	Off-bundle slippage decomposition	Exact Within Closure	Reduces the first true off-bundle defect to axial-length, mixed-speed, and mouth-traction slippages.
169	No linear grouped feed-down into scalar slippages	Exact Within Closure	Pure grouped real P_2 anisotropy cannot linearly source the scalar off-bundle slippages; scalar feed-down starts quadratically.
170	Linear grouped outlet map	Exact Within Closure	Reduces the direct outlet deformations to $\mathcal{K}_A = \delta D_{A,2} + \delta D_{A,0}/9$ and $\mathcal{G}_A = \delta N_{A,0} - P_0 \delta D_{A,0}$.
171	Microscopic grouped obstructions	Exact Within Closure	Decomposes $\mathcal{K}_A$ and $\mathcal{G}_A$ into wall, BdG, Maxwell/mixed, and outgoing-transfer defect bundles.
172	Physical slope collapse	Exact Within Closure	Identifies $\mathfrak{K}_1 = -D_0 u_2^{(1)}$ and $\mathfrak{G}_1 = D_0 P_1$, reducing the linear grouped outlet problem to physical slopes.

Stage	Working title	Primary status	Main output
173	Weak-axisymmetric loading mismatch	Exact Within Closure	On the even-preserving branch, the remaining grouped defect is the scalar $\Xi_{\mathrm{load}} = N_{01}/N_0 - D_{01}/D_0$.
174	Static self-similarity	Exact Within Closure	Expresses Ξ_{load} as weighted failure of static self-similarity between wall, BdG, conservative port, and outgoing port slopes.
175	Wall-normalized outgoing-load theorem	Exact Within Closure	Factors the remaining defect into wall-normalized shape/load variables and proves conservative-shape preservation leaves only outgoing load slippage.
176	Square-root mixed-leg law	Exact Within Closure	Factors the outgoing load into mixed-leg, interference, and hybridization slippages; under rigidity, zero defect requires $G_W/\Omega_W^2 \propto \sqrt{K}$.
177	One scalar outgoing-slippage amplitude	Exact Within Closure	Shows all weak-axisymmetric outgoing slippages inherit the same grouped signature and collapse to $\Xi_1 = P_1/P_0$.
178	Outgoing-port co-loading theorem	Exact Within Closure	Rewrites Ξ_1 as the weighted mismatch between actual static outgoing-port loading and wall-baseline loading.
179	Transfer-shape theorem	Exact Within Closure	Factors each outgoing port as $N_{A,0}^{(r)} = K_A \mathcal{T}_{A,r}^2$ and gives $\Xi_1 = 2\sum_r \rho_r^{(N)} \tau_r$.
180	Effective transfer-shape collapse	Exact Within Closure	Collapses many ports to $\mathcal{T}_{\mathrm{eff},A}^2 = N_{A,0}/K_A$ and gives the actual one-port continuum formula.
181	Coherent tracking-branch defect law	Exact Within Closure	On the coherent local D/N tracking branch, Ξ_1 is carried only by mixed/outgoing placement variables; coherent support ratio ζ drops out.
182	Microscopic coherent-kernel slippages	Exact Within Closure	Reduces the defect to $(\Sigma_Z, \Sigma_\chi, \Sigma_\epsilon, \Sigma_\delta)$ plus dressing slippage Σ_η, and isolates Σ_{tr}.

Stage	Working title	Primary status	Main output
183	Triangular normal form	Exact Within Closure	Compresses the coherent weak-axisymmetric problem to $(\Sigma_{\rm tr}, \Sigma_{\rm nt}, \Sigma_\eta)$ with triangular map to $(\Theta_1, \Xi_1, \mathcal{R}_1)$.
184	Branch-invariant coordinates	Exact Within Closure	Identifies exact branch composites $\mathfrak{T}_*$, $\mathfrak{N}_*$, and ϵ_η whose logarithmic drifts are the three normal-form coordinates.
185	Microscopic monomial coordinates	Exact Within Closure	Pushes the branch composites to direct monomials $(\mathfrak{C}_{\rm tr,*}, \mathfrak{C}_{\rm nt,*}, \epsilon_\eta)$.
186	Similarity-orbit tangent theorem	Exact Within Closure	Shows the zero-defect equations are the tangent equations of a five-parameter monomial-preserving similarity orbit $\mathcal{G}_*$.
187	Orbit–quotient closure	Exact Within Closure	Proves finite level sets of $(\mathfrak{C}_{\rm tr,*}, \mathfrak{C}_{\rm nt,*}, \epsilon_\eta)$ are exactly $\mathcal{G}_*$-orbits.
188	Branch-observable compiler	Exact Within Closure	Compiles direct branch observables to $(\Theta_1, \Xi_1, \mathcal{R}_1)$ and records equivalent observable packets.
189	Transfer-shape / outgoing-prefactor compiler	Exact Within Closure	Equates Ξ_1 with the logarithmic transfer-shape and outgoing-prefactor slope, including corrected nontracking form.
190	Direct defect, dressing residual, and no-go filters	Exact Within Closure	Separates direct transfer-shape drift from selected-branch dressing residual and records scalar no-go filters.
191	Minimal PDE data packet	Exact Within Closure	Defines Packet A, Packet B, $\Delta_{\rm branch}$, $\Delta_{\rm orbit}$, and the exact two-packet home-stretch theorem.
192	Orbit/quotient projectors	Exact Within Closure	Builds exact projectors $Q_{\rm quot}$, $O_{\rm orb}$, showing quotient failure lives only on $(T_U, K_\eta^{\rm eff}, \mu_W)$.
193	Isotropic grouped target surface	Exact Within Closure	Freezes the isotropic one-pole conservative target surface and proves the scalar/geometry lane enters the grouped carrier only at quadratic anisotropy order.

Stage	Working title	Primary status	Main output
194	Outgoing $l = 2$ fingerprint	Exact Within Closure	Derives the exact outgoing spherical DtN fingerprint and fixes $\chi_Q = 1$ on the canonical compact branch.
195	Source-map reduction	Exact Within Closure / Reduced / Controlled Reduction	Factorizes the observable odd closure as $\widehat{m}_0^2 \chi_Q N_Q = 1$ and collapses Δ_{norm}.
196	Higher-odd irrelevance	Exact Within Closure	Proves all extra isotropic odd tails beginning at $O(\omega^7)$ are invisible to the point-particle 2.5PN theorem.
197	Conditional Packet-A closure	Exact Within Closure	Proves $\Delta_{\mathrm{branch}} = 0 \iff \chi_Q = 1$ and hence $\Delta_Q = \chi_Q - 1$ is the sole Packet-A scalar.
198	Finite orbit law	Exact Within Closure	Solves the dependent triple exactly, defines mismatch coordinates, and proves finite single-orbit lock.
199	Pairwise orbit transport	Exact Within Closure	Removes representative privilege by giving exact two-point transport, mismatch, cocycle, and orbit-lock laws.
200	Reference-free home-stretch theorem	Exact Within Closure	Combines Packet A and Packet B into the four-scalar final verdict packet $(\Delta_Q, q_{\mathrm{tr}}, q_{\mathrm{nt}}, q_\eta)$.

Claim status. Mixed but predominantly Exact Within Closure**. Stages 164–200 are exact algebra inside the declared co-evolving Family–1, grouped real P_2, coherent local D/N, outgoing-prefactor, Packet-A, and Packet-B reduced hierarchies. The natural point-particle source-map use in Stage 195 is a** Reduced / Controlled Reduction **import. The actual nonlinear moving-throat PDE still has to return branch data that land on the final packet; realization by that branch remains** Open **even though the compilers and quotient/orbit theorems are exact within closure.**

F.1 Block contract and theorem-level output

The contract of Part V is to turn the remaining first-order weak-axisymmetric and retarded branch checks into explicit finite packets. It is not trying to prove that the completed nonlinear PDE realizes those packets. Instead it proves that, once a candidate moving-throat solution gives the required reduced branch data, the verdict can be computed by exact formulas with no remaining interpretation freedom.

The block has two sides. The first side is the weak-axisymmetric branch transport:

$$\delta_\perp \longrightarrow (\mathcal{K}_A, \mathcal{G}_A) \longrightarrow (u_2^{(1)}, P_1/P_0) \longrightarrow \Xi_{\rm load} \longrightarrow \Xi_1. \tag{F.6}$$

The second side is the coherent orbit quotient:

$$(\Sigma_{\rm tr}, \Sigma_{\rm nt}, \Sigma_\eta) \longrightarrow (\mathfrak{C}_{{\rm tr},*}, \mathfrak{C}_{{\rm nt},*}, \epsilon_\eta) \longrightarrow \mathcal{M}_+/\mathcal{G}_* \longrightarrow (q_{\rm tr}, q_{\rm nt}, q_\eta). \tag{F.7}$$

The final theorem joins these with the retarded Packet-A scalar $\Delta_Q = \chi_Q - 1$.

Theorem F.1 (Part V weak-axisymmetric quotient and orbit-closure theorem)**.** *Inside the carried moving-throat reduced hierarchy for Stages 164–200, the following theorem chain holds.*

1. *The first off-family Family–1 normal coordinate is an explicit logarithmic transport law in the moving-throat wall/BdG/Maxwell/mixed branch variables. Arbitrary first-order isotropic bundle drift is tangent to the compensated parent family; true off-bundle motion first enters through three scalar slippages.*

2. *A pure grouped real P_2 anisotropy cannot linearly feed the scalar slippage channel. Its linear effect enters the direct outlet coefficients only through*

$$\mathcal{K}_A := \delta D_{A,2} + \frac{1}{9}\delta D_{A,0}, \qquad \mathcal{G}_A := \delta N_{A,0} - P_0 \delta D_{A,0}. \tag{F.8}$$

3. *On the canonical compensated branch, these outlet coefficients are exactly the physical grouped slopes,*

$$\mathfrak{K}_1 = -D_0 u_2^{(1)}, \qquad \mathfrak{G}_1 = D_0 P_1. \tag{F.9}$$

4. *On the even-preserving weak-axisymmetric branch, all conservative grouped transport is controlled by one static operator slope, and the sole remaining linear grouped normalization defect is*

$$\Xi_{\rm load} := \frac{N_{01}}{N_0} - \frac{D_{01}}{D_0}. \tag{F.10}$$

5. *The same scalar is the weighted failure of static self-similarity and, after wall-normalized outgoing-port factorization, the logarithmic slope of the effective transfer shape:*

$$\Xi_1 = \frac{P_1}{P_0} = \frac{\delta \ln \mathcal{T}^2_{{\rm eff},A}}{\epsilon \lambda_A}. \tag{F.11}$$

6. *On the coherent local D/N branch, the weak-axisymmetric observable drifts obey the triangular normal form*

$$\Theta_1 = -C_{\rm tr}\Sigma_{\rm tr}, \qquad \Xi_1 = A_{\rm tr}\Sigma_{\rm tr} + \Sigma_{\rm nt}, \qquad \mathcal{R}_1 + \Xi_1 = -\frac{\epsilon_\eta}{1-\epsilon_\eta}\Sigma_\eta. \tag{F.12}$$

7. *The three normal-form coordinates are exactly the logarithmic drifts of three direct microscopic monomials,*

$$\delta \ln \mathfrak{C}_{{\rm tr},*} = \Sigma_{\rm tr}, \qquad \delta \ln \mathfrak{C}_{{\rm nt},*} = \Sigma_{\rm nt}, \qquad \delta \ln \epsilon_\eta = \Sigma_\eta. \tag{F.13}$$

8. *The finite level sets of these three monomials are exactly the five-parameter similarity orbits $\mathcal{G}_*$, so*

$$\mathcal{M}_+/\mathcal{G}_* \cong (\mathbb{R}_{>0})^3, \qquad (\mathfrak{C}_{{\rm tr},*}, \mathfrak{C}_{{\rm nt},*}, \epsilon_\eta) \text{ are quotient coordinates.} \tag{F.14}$$

9. *The branch-side Packet A collapses to one scalar finish line,*

$$\Delta_{\mathrm{branch}} = 0 \iff \chi_Q = 1 \iff \Delta_Q := \chi_Q - 1 = 0. \tag{F.15}$$

10. *The full reference-free reduced verdict is therefore the four-scalar packet*

$$\Delta_{\mathrm{full}}^{(x|\mathcal{O}_*)} := \big(\Delta_Q(x), q_{\mathrm{tr}}^{(x\leftarrow\mathcal{O}_*)}, q_{\mathrm{nt}}^{(x\leftarrow\mathcal{O}_*)}, q_{\eta}^{(x\leftarrow\mathcal{O}_*)}\big), \tag{F.16}$$

with

$$\Delta_{\mathrm{full}}^{(x|\mathcal{O}_*)} = 0 \iff \Delta_{\mathrm{branch}}(x) = 0 \text{ and } x \in \mathcal{O}_*. \tag{F.17}$$

Proof structure. Stages 164–168 translate the first off-family normal coordinate into explicit microscopic transport variables and then separate on-bundle tangent motion from off-bundle slippage. Stages 169–173 prove that grouped real P_2 anisotropy does not linearly feed scalar slippage, derive the direct outlet map, and collapse the even-preserving grouped defect to Ξ_{load}. Stages 174–180 factor Ξ_{load} through wall-referenced self-similarity, outgoing-load factors, port co-loading, and the effective transfer shape, giving $\Xi_1 = P_1/P_0$. Stages 181–185 pass from physical coherent-kernel variables to the triangular normal form and then to direct microscopic monomials. Stages 186–187 prove the finite orbit–fibre theorem. Stages 188–192 build the observable, transfer-shape, finite-packet, and projector compilers. Stages 193–197 close the branch-side Packet-A finish line to $\Delta_Q = \chi_Q - 1$. Stages 198–200 remove the orbit representative and combine Packet A and Packet B into (F.16). □

F.2 Weak-axisymmetric grouped signature

F.2.1 Local goal

The first job in Part V is to connect the co-evolving Family–1 off-family scalar from Part IV to the grouped real P_2 transport language. Stage 163 had left the first true normal coordinate in the form

$$\delta_\perp = \mathfrak{g}_*\delta\ln\left(\frac{g_q K_s}{g_s\lambda}\right) + \frac{1}{4\sqrt{1+\mathfrak{r}_*^2}}\delta\ln\left(\frac{K_s K_q}{\lambda^2}\right). \tag{F.18}$$

Stage 164 shows that this is not an abstract parent-coordinate expression. With

$$\mathfrak{r} = \frac{\lambda}{\sqrt{K_s K_q}}, \qquad \mathfrak{g} = \frac{g_q\sqrt{K_s}}{g_s\sqrt{K_q}}, \qquad r_c = \frac{\lambda^2}{K_s K_q} = \mathfrak{r}^2, \tag{F.19}$$

the two channels are exactly

$$\delta\ln\left(\frac{g_q K_s}{g_s\lambda}\right) = \delta\ln\left(\frac{\mathfrak{g}}{\mathfrak{r}}\right), \qquad \delta\ln\left(\frac{K_s K_q}{\lambda^2}\right) = -\delta\ln r_c. \tag{F.20}$$

Thus $\delta_\perp$ measures relative mouth-coupling drift and hybridization drift.

F.2.2 Explicit branch-variable form

On the healing-locked wall/BdG/Maxwell/mixed branch, the same scalar becomes

$$\begin{aligned}\delta_\perp = {} & A_*\,\delta\ln\left(\frac{\mathcal{Z}_q}{\rho_w}\right) + 3A_*\,\delta\ln c_{s,w} + B_*\,\delta\ln c_s - \mathfrak{g}_*\,\delta\ln\mathcal{T}_m \\ & - (\mathfrak{g}_* + B_*)\delta\ln v_{w0} - 2A_*\,\delta\ln a - C_*\,\delta\ln L_W,\end{aligned} \tag{F.21}$$

where

$$A_* := \mathfrak{g}_* + \frac{1}{4\sqrt{1+\mathfrak{r}_*^2}}, \qquad B_* := \frac{1}{2\sqrt{1+\mathfrak{r}_*^2}}, \qquad C_* := 2\mathfrak{g}_* + \frac{3}{4\sqrt{1+\mathfrak{r}_*^2}}. \tag{F.22}$$

The exact tangency condition $\delta_\perp = 0$ is therefore a co-transport law for $\mathcal{T}_m$ relative to $(\mathcal{Z}_q, \rho_w, c_{s,w}, c_s, v_{w0}, a, L_W)$, not a free adjustment.

Stages 165–167 then show that the explicit lower compensated branch is even more constrained. The D/N tube length co-transports with the mouth radius,

$$\delta \ln L_W = \delta \ln a, \tag{F.23}$$

and the exact branch laws reduce the apparent branch drift problem to four irreducible variables. Stage 166 chooses the four bundle observables

$$\Theta_w, \qquad K_s, \qquad K_q, \qquad P_0 = \frac{N_0}{D_0} \tag{F.24}$$

as the right variables to invert. Stage 167 proves that arbitrary first-order drift of these four bundle observables leaves the compensation ratios tangent to the parent family:

$$\delta \ln r_c = 0, \qquad \delta \ln \mathfrak{r} = 0, \qquad \delta \ln \mathfrak{g} = 0 \quad \text{for pure bundle drift.} \tag{F.25}$$

Consequently, any first-order failure after Stage 167 must be genuinely off-bundle.

F.2.3 Off-bundle slippages and grouped signature

Stage 168 introduces the three scalar slippages

$$\varepsilon_L, \qquad \varepsilon_v, \qquad \varepsilon_T, \tag{F.26}$$

which measure failure of the exact lower-branch laws for axial length, mixed background speed, and mouth traction. Stage 169 then proves a key firewall: a pure weak grouped real P_2 anisotropy cannot linearly source these scalar slippages. The reason is angular. The weak grouped perturbation has zero weighted trace at first order; scalar feed-down can only be built from quadratic grouped invariants such as

$$\mathcal{I}[x,y] = \frac{1}{5}\delta x^T G_{\text{grp}} \delta y = 4a_x a_y + \frac{4}{5} b_x b_y, \qquad G_{\text{grp}} = \text{diag}(1,2,2). \tag{F.27}$$

Thus linear grouped real P_2 transport enters the direct outlet coefficients rather than the scalar off-bundle slippage channel.

The grouped weak-axisymmetric signature is inherited from the first axisymmetric quadrupole perturbation:

$$\lambda_{20} = 1, \qquad \lambda_{21} = \frac{1}{2}, \qquad \lambda_{22} = -1. \tag{F.28}$$

For any first-order grouped quantity

$$x_A = x_0 + \epsilon \lambda_A x_1 + O(\epsilon^2), \tag{F.29}$$

the weighted grouped anomalies obey

$$a_x = \frac{\epsilon}{4} x_1, \qquad b_x = \frac{3\epsilon}{4} x_1, \qquad b_x = 3a_x. \tag{F.30}$$

This fixed line is the weak-axisymmetric branch signature used throughout the rest of Part V.

Claim status. Stages 164–169 are Exact Within Closure **inside the explicit co-evolving Family–1 throat-core closure and the grouped real P_2 weak-axisymmetric expansion. They do not prove that the actual nonlinear PDE produces a particular slippage; they identify the exact variables through which such slippage would appear.**

F.3 Even-preserving and hidden-even gates

F.3.1 Direct outlet map

Stages 170–171 move the linear grouped problem from scalar slippage to direct outlet data. For each grouped lane A, define

$$\mathcal{K}_A := \delta D_{A,2} + \frac{1}{9}\delta D_{A,0}, \qquad \mathcal{G}_A := \delta N_{A,0} - P_0 \delta D_{A,0}. \tag{F.31}$$

The hidden even deformation and hidden odd normalization deformation are then

$$\delta\kappa_W^{(A)} = \frac{3(1-\sigma_*)}{\sigma_* D_0}\mathcal{K}_A, \qquad \delta\gamma_W^{(A)} = -\frac{1-\sigma_*}{9\sigma_* N_0}\mathcal{G}_A. \tag{F.32}$$

Stage 171 decomposes these into microscopic sectors:

$$\mathcal{K}_A = \mathcal{W}_A - \mathcal{B}_A - \mathcal{Z}_A, \tag{F.33}$$

with

$$\mathcal{W}_A := \frac{1}{9}\delta K_A - \delta M_A, \qquad \mathcal{B}_A := \delta B_{A,2} + \frac{1}{9}\delta B_{A,0}, \qquad \mathcal{Z}_A := \delta Z_{A,2} + \frac{1}{9}\delta Z_{A,0}, \tag{F.34}$$

and

$$\mathcal{G}_A = -P_0 \delta K_A + P_0 \delta B_{A,0} + \mathcal{N}_A, \qquad \mathcal{N}_A := \delta N_{A,0} + P_0 \delta Z_{A,0}. \tag{F.35}$$

This identifies the only four microscopic bundles that can generate a linear grouped outlet obstruction: wall baseline, BdG support, conservative Maxwell/mixed, and outgoing-transfer loading.

F.3.2 Physical slope collapse

Stage 172 rewrites the same obstruction in physical grouped-response variables. For each grouped lane,

$$u_2^{(A)} = -\frac{D_{A,2}}{D_{A,0}}, \qquad u_4^{(A)} = \frac{D_{A,2}^2 - D_{A,0}D_{A,4}}{D_{A,0}^2}, \qquad P_0^{(A)} = \frac{N_{A,0}}{D_{A,0}}. \tag{F.36}$$

On the canonical compensated branch, the two microscopic amplitudes are simply

$$\mathfrak{K}_1 = -D_0 u_2^{(1)}, \qquad \mathfrak{G}_1 = D_0 P_1. \tag{F.37}$$

Equivalently,

$$\delta\kappa_W^{(A)} = -\frac{3(1-\sigma_*)}{\sigma_*}\delta u_2^{(A)}, \qquad \delta\gamma_W^{(A)} = -\frac{1-\sigma_*}{9\sigma_*}\frac{\delta P_0^{(A)}}{P_0}. \tag{F.38}$$

So the full linear grouped outlet problem is now measurable by $u_2^{(1)}$ and P_1/P_0.

F.3.3 Even-preserving branch

Stage 173 then computes those physical slopes directly from the weak-axisymmetric grouped operator expansion

$$D_{A,0} = D_0 + \epsilon\lambda_A D_{01} + O(\epsilon^2), \quad D_{A,2} = D_2 + \epsilon\lambda_A D_{21} + O(\epsilon^2), \quad D_{A,4} = D_4 + \epsilon\lambda_A D_{41} + O(\epsilon^2), \tag{F.39}$$

$$N_{A,0} = N_0 + \epsilon\lambda_A N_{01} + O(\epsilon^2). \tag{F.40}$$

The exact slope formulas are

$$u_2^{(1)} = -\frac{D_{21} + u_2 D_{01}}{D_0}, \tag{F.41}$$

$$u_4^{(1)} = -\frac{5D_{01} + 18D_{21} + 81D_{41}}{81D_0} \quad \text{on } (u_2, u_4) = \left(\frac{1}{9}, \frac{4}{81}\right), \tag{F.42}$$

and

$$\frac{P_1}{P_0} = \frac{N_{01}}{N_0} - \frac{D_{01}}{D_0}. \tag{F.43}$$

If the canonical conservative even fingerprint is preserved at first order, then

$$D_{21} = -\frac{D_{01}}{9}. \tag{F.44}$$

The hidden-even consistency relation $u_4^{(1)} = (8/9)u_2^{(1)}$ is exactly

$$D_{41} = \frac{2}{3}D_{21} + \frac{1}{27}D_{01}, \tag{F.45}$$

and therefore on the even-preserving branch

$$D_{41} = -\frac{D_{01}}{27}. \tag{F.46}$$

Thus the conservative grouped response is frozen to canonical order, and the only surviving linear grouped defect is the static loading mismatch

$$\Xi_{\mathrm{load}} := \frac{N_{01}}{N_0} - \frac{D_{01}}{D_0}. \tag{F.47}$$

Its lane form is

$$\Delta_Q^{(20)} = \epsilon\Xi_{\mathrm{load}}, \qquad \Delta_Q^{(21)} = \frac{\epsilon}{2}\Xi_{\mathrm{load}}, \qquad \Delta_Q^{(22)} = -\epsilon\Xi_{\mathrm{load}}. \tag{F.48}$$

Claim status. Stages 170–173 are Exact Within Closure **in the grouped low-frequency bundle. The even-preserving condition is a branch restriction, not an unconditional statement that the completed PDE must satisfy it.**

F.4 Outgoing-load scalar ($\Xi_1 = P_1/P_0$)

F.4.1 Self-similarity form of the loading mismatch

Stage 174 rewrites the mismatch (F.47) as failure of static self-similarity. Let

$$\delta_K := \frac{K_1}{K}, \qquad \delta_B := \frac{B_0^{(1)}}{B_0}, \qquad \delta_Z := \frac{Z_0^{(1)}}{Z_0}, \qquad \delta_N := \frac{N_1}{N_0}, \tag{F.49}$$

and

$$\omega_B := \frac{B_0}{D_0}, \qquad \omega_Z := \frac{Z_0}{D_0}, \qquad D_0 = K - B_0 - Z_0. \tag{F.50}$$

Then

$$\Xi_{\rm load} = (\delta_N - \delta_K) + \omega_B(\delta_B - \delta_K) + \omega_Z(\delta_Z - \delta_K). \tag{F.51}$$

Define the wall-referenced self-similarity defect fields

$$\Sigma_\alpha^{(B)} := 2\delta \ln\left(\frac{c_\alpha}{\varpi_\alpha}\right) - \delta_K, \quad \Sigma_r^{(Z)} := \delta \ln\left(\frac{Q_r}{\Delta_r}\right) - \delta_K, \quad \Sigma_r^{(N)} := 2\delta \ln\left(\frac{P_r}{\Delta_r}\right) - \delta_K. \tag{F.52}$$

Then

$$\Xi_{\rm load} = \sum_r \rho_r^{(N)} \Sigma_r^{(N)} + \omega_B \sum_\alpha \rho_\alpha^{(B)} \Sigma_\alpha^{(B)} + \omega_Z \sum_r \rho_r^{(Z)} \Sigma_r^{(Z)}. \tag{F.53}$$

Static self-similarity is the sufficient condition that all three defect families vanish. On the even-preserving branch it implies $\Xi_{\rm load} = 0$ and hence no first-order grouped normalization defect.

F.4.2 Outgoing-load theorem and mixed-leg law

Stages 175–176 factor the port objects relative to the wall baseline. The remaining defect is controlled by the outgoing load factor

$$\Lambda_r := \frac{P_r}{\Delta_r}. \tag{F.54}$$

On conservative-shape-preserving branches the full remaining defect is carried only by the outgoing load slope relative to the wall. Stage 176 factors that load into a mixed leg, an interference ratio, and a hybridization ratio. In the notation of the reduced one-port block,

$$\frac{\Lambda_r^2}{K} = \mathcal{M}_r^2 \frac{(1+\mathcal{I}_r)^2}{(1-\mathcal{H}_r)^2}. \tag{F.55}$$

Consequently

$$\Sigma_r^{(N)} = 2\delta \ln \mathcal{M}_r + \frac{2\mathcal{I}_r}{1+\mathcal{I}_r} \delta \ln \mathcal{I}_r + \frac{2\mathcal{H}_r}{1-\mathcal{H}_r} \delta \ln \mathcal{H}_r. \tag{F.56}$$

If the interference and hybridization ratios are rigid, the whole outgoing defect is the raw mixed-leg drift. The zero-defect condition becomes the square-root mixed-leg law

$$\frac{G_W}{\Omega_W^2} \propto \sqrt{K}. \tag{F.57}$$

This is useful because it identifies a concrete microscopic path to zero defect rather than just a cancellation among compiled coefficients.

F.4.3 Collapse to one scalar amplitude

Stage 177 applies the weak-axisymmetric signature (F.28). Every microscopic outgoing slippage inherits the same lane pattern, so the full remaining grouped defect is one scalar amplitude:

$$\Xi_{\rm load}^{(A)} = \epsilon \lambda_A \Xi_1. \tag{F.58}$$

The same scalar is the physical outgoing-prefactor slope:

$$\Xi_1 = \frac{P_1}{P_0}. \tag{F.59}$$

Stage 178 rewrites it portwise. For

$$N_{A,0}^{(r)} = \frac{P_{A,r}^2}{\Delta_{A,r}^2}, \tag{F.60}$$

define the port slope ν_r by

$$\delta \ln N_{A,0}^{(r)} = \epsilon \lambda_A \nu_r. \tag{F.61}$$

Then

$$\Xi_1 = \bar{\nu}_N - \kappa_1, \tag{F.62}$$

where κ_1 is the wall-baseline static slope and $\bar{\nu}_N$ is the outgoing-weighted average. Thus zero defect is exact co-loading of outgoing transfer with the wall baseline.

Stages 179–180 put the same statement into transfer-shape language. Each port factors as

$$N_{A,0}^{(r)} = K_A \mathcal{T}_{A,r}^2, \qquad \mathcal{T}_{A,r} = \frac{\widehat{G}_{W,A,r} + \widehat{R}_{A,r}\widehat{G}_{U,A,r}}{1 - \widehat{R}_{A,r}^2}. \tag{F.63}$$

Therefore

$$\Xi_1 = 2\sum_r \rho_r^{(N)} \tau_r, \qquad \tau_r := \frac{\delta \ln \mathcal{T}_{A,r}}{\epsilon \lambda_A}. \tag{F.64}$$

Equivalently, all ports collapse to one effective transfer shape,

$$\mathcal{T}_{\mathrm{eff},A}^2 := \sum_r \mathcal{T}_{A,r}^2 = \frac{N_{A,0}}{K_A}, \tag{F.65}$$

and

$$\Xi_1 = \frac{\delta \ln \mathcal{T}_{\mathrm{eff},A}^2}{\epsilon \lambda_A}. \tag{F.66}$$

On the actual minimal one-port continuum branch,

$$\mathcal{T}_A^2 = \frac{Z_{W,A}(1+\rho_A)^2}{\Omega_{W,A}^2(1-\epsilon_{W,A})^2} = \frac{27\pi^2 G c_s^5}{20 a^5 c^5} \frac{1-\epsilon_{\eta,A}}{R_{\mathrm{target},A}}. \tag{F.67}$$

So the remaining theorem gate is to determine whether the actual weak-axisymmetric branch keeps $\mathcal{T}_{\mathrm{eff}}$ rigid.

Claim status. Stages 174–180 are Exact Within Closure **inside the weak-axisymmetric grouped bundle and the declared outgoing-port factorization. The square-root mixed-leg law is a sufficient zero-defect branch condition under interference/hybridization rigidity; actual realization is not assumed.**

F.5 Coherent-kernel slippages

F.5.1 Actual coherent local D/N branch

Stage 181 inserts the coherent local D/N tracking branch into (F.67). On that branch,

$$\mathcal{T}^2 = \frac{Z_W(1+\chi_0)^2}{\Omega_W^2(1-\epsilon)^2} = \frac{27\pi^2 G c_s^5}{20 a^5 c^5} \frac{1-\epsilon_\eta}{R_{\mathrm{target}}}, \tag{F.68}$$

where

$$\epsilon = \epsilon_W \left(1 - \frac{2}{11}\frac{\delta_U}{1+\delta_U}\right). \tag{F.69}$$

The weak-axisymmetric defect is

$$\Xi_1 = \zeta_Z - \omega_W + \frac{2\chi_1}{1+\chi_0} + \frac{2\epsilon_1}{1-\epsilon}, \tag{F.70}$$

with

$$\epsilon_1 = \left(1 - \frac{2}{11}\frac{\delta_U}{1+\delta_U}\right)\varepsilon_W - \frac{2\epsilon_W}{11(1+\delta_U)^2}\delta_{U,1}. \tag{F.71}$$

The support enhancement ratio ζ drops out identically:

$$\partial_\zeta \ln \mathcal{T}^2 = 0, \qquad \partial_\zeta \ln R_{\mathrm{target}} = 0, \qquad \partial_\zeta \Xi_1 = 0. \tag{F.72}$$

Thus coherent support can help the steady normalization load, but it cannot repair the weak-axisymmetric prefactor slope at first order.

F.5.2 Microscopic slippage decomposition

Stage 182 pushes (F.70) down to microscopic coherent-kernel slippages. The direct grouped defect depends on four slippages,

$$\Sigma_Z, \qquad \Sigma_\chi, \qquad \Sigma_\epsilon, \qquad \Sigma_\delta, \tag{F.73}$$

and selected-branch dressing introduces one more,

$$\Sigma_\eta. \tag{F.74}$$

The direct defect law is

$$\Xi_1 = \Sigma_Z + \frac{2\chi_0}{1+\chi_0}\Sigma_\chi + \frac{2\epsilon_W}{1-\epsilon}\left[\frac{11+9\delta_U}{11(1+\delta_U)}\Sigma_\epsilon - \frac{2\delta_U}{11(1+\delta_U)^2}\Sigma_\delta\right]. \tag{F.75}$$

The tracking drift is the special combination

$$\Sigma_{\mathrm{tr}} := (1+\chi_0)\Sigma_\delta + (1+\delta_U)\Sigma_\chi, \tag{F.76}$$

which controls the tracking observable

$$\Theta_1 = -\frac{\chi_0\delta_U}{(1+\chi_0)(1+\delta_U)(1+\chi_0+\delta_U)}\Sigma_{\mathrm{tr}}. \tag{F.77}$$

So exact tracking rigidity means $\Sigma_{\mathrm{tr}} = 0$, but that alone does not generally force $\Xi_1 = 0$; a genuine nontracking transfer-shape slippage can remain.

Claim status. Stages 181–182 are Exact Within Closure **for the coherent local D/N tracking branch. The support-blindness statement is exact inside that branch; it does not assert that support is irrelevant to all other branch tests.**

F.6 Triangular normal form

F.6.1 Normal-form coordinates

Stage 183 packages the five slippages into three branch-adapted coordinates. The nontracking transfer-shape slippage is defined by

$$\begin{aligned}\Sigma_{\mathrm{nt}} := \Sigma_Z + \frac{2\epsilon_W}{1-\epsilon}\frac{11+9\delta_U}{11(1+\delta_U)}\Sigma_\epsilon \\ - \left[\frac{2\chi_0}{1+\delta_U} + \frac{4\epsilon_W\delta_U}{11(1-\epsilon)(1+\delta_U)^2}\right]\Sigma_\delta.\end{aligned} \tag{F.78}$$

The three branch-adapted coordinates are therefore

$$(\Sigma_{\mathrm{tr}}, \Sigma_{\mathrm{nt}}, \Sigma_\eta). \tag{F.79}$$

In these coordinates the observable drifts have the exact triangular form

$$\Theta_1 = -C_{\mathrm{tr}}\Sigma_{\mathrm{tr}}, \qquad \Xi_1 = A_{\mathrm{tr}}\Sigma_{\mathrm{tr}} + \Sigma_{\mathrm{nt}}, \qquad \mathcal{R}_1 + \Xi_1 = -\frac{\epsilon_\eta}{1-\epsilon_\eta}\Sigma_\eta, \tag{F.80}$$

where

$$C_{\mathrm{tr}} = \frac{\chi_0\delta_U}{(1+\chi_0)(1+\delta_U)(1+\chi_0+\delta_U)}, \qquad A_{\mathrm{tr}} = \frac{2\chi_0}{(1+\chi_0)(1+\delta_U)}. \tag{F.81}$$

This is the first exact three-scalar normal form of the weak-axisymmetric coherent branch.

F.6.2 Inversion and interpretation

The triangular structure makes the three physical failure modes transparent.

1. Σ_{tr} is tracking failure. It is measured directly by Θ_1.
2. Σ_{nt} is nontracking transfer-shape failure. It is the residual part of Ξ_1 after subtracting tracking feed-through.
3. Σ_η is dressing failure. It measures the mismatch between direct transfer-shape drift and selected-branch demand through $\mathcal{R}_1 + \Xi_1$.

Zero defect in this normal form is

$$\Theta_1 = \Xi_1 = \mathcal{R}_1 = 0 \iff \Sigma_{\mathrm{tr}} = \Sigma_{\mathrm{nt}} = \Sigma_\eta = 0. \tag{F.82}$$

F.6.3 Branch-observable coordinates

Stage 184 identifies exact branch composites whose logarithmic drifts are the same three coordinates. The tracking factor is

$$R_{\mathrm{tr}} = \frac{1+\chi_0/(1+\delta_U)}{1+\chi_0} = \frac{1+\chi_0+\delta_U}{(1+\chi_0)(1+\delta_U)}. \tag{F.83}$$

The corrected nontracking composite is

$$\mathfrak{N}_* := \mathcal{T}^2 R_{\mathrm{tr}}^{B_*}, \qquad B_* := \frac{2(1+\chi_{0,*}+\delta_{U,*})}{\delta_{U,*}}. \tag{F.84}$$

After a harmless positive normalization of the tracking factor,

$$\mathfrak{T}_* := R_{\mathrm{tr}}^{-C_*}, \qquad C_* := \frac{(1+\chi_{0,*})(1+\delta_{U,*})(1+\chi_{0,*}+\delta_{U,*})}{\chi_{0,*}\delta_{U,*}}, \tag{F.85}$$

the branch-coordinate laws are

$$\delta \ln \mathfrak{T}_* = \Sigma_{\mathrm{tr}}, \qquad \delta \ln \mathfrak{N}_* = \Sigma_{\mathrm{nt}}, \qquad \delta \ln \epsilon_\eta = \Sigma_\eta. \tag{F.86}$$

So the next continuation point is no longer a list of microscopic slippages; it is drift of three exact branch composites.

Claim status. Stages 183–184 are Exact Within Closure **inside the coherent local D/N branch. The normal form is a compiler from reduced coherent variables to observables; it does not assert that the completed PDE has zero coordinates.**

F.7 Microscopic monomial invariants

F.7.1 Direct monomial coordinates

Stage 185 pushes the three branch composites to direct microscopic monomials. At reference-branch order,

$$\delta \ln \mathfrak{C}_{\mathrm{tr},*} = \Sigma_{\mathrm{tr}}, \qquad \delta \ln \mathfrak{C}_{\mathrm{nt},*} = \Sigma_{\mathrm{nt}}, \qquad \delta \ln \epsilon_\eta = \Sigma_\eta. \tag{F.87}$$

The tracking monomial is

$$\mathfrak{C}_{\mathrm{tr},*} := \chi_0^{1+\delta_{U,*}} \delta_U^{1+\chi_{0,*}}, \tag{F.88}$$

and the nontracking monomial is

$$\mathfrak{C}_{\mathrm{nt},*} := \frac{Z_W}{\Omega_W^2} \epsilon_W^{E_*} \delta_U^{-F_*}, \tag{F.89}$$

with

$$E_* = \frac{2\epsilon_{W,*}}{1-\epsilon_*} \frac{11+9\delta_{U,*}}{11(1+\delta_{U,*})}, \qquad F_* = \frac{2\chi_{0,*}}{1+\delta_{U,*}} + \frac{4\epsilon_{W,*}\delta_{U,*}}{11(1-\epsilon_*)(1+\delta_{U,*})^2}. \tag{F.90}$$

Using the coherent definitions, these become direct microscopic monomials in the positive variables

$$\mathbf{x} = (\lambda_W, c_{\eta U}, \gamma, K_U, K_\eta^{(\mathrm{eff})}, K_W^{(\mathrm{eff})}, \mu_W, T_U). \tag{F.91}$$

Explicitly,

$$\mathfrak{C}_{\mathrm{tr},*} = \left(\frac{\gamma c_{\eta U}}{K_U}\right)^{1+\delta_{U,*}} \left(\frac{\pi^2 T_U}{L^2 K_U}\right)^{1+\chi_{0,*}}, \tag{F.92}$$

$$\mathfrak{C}_{\mathrm{nt},*} = \left(\frac{\lambda_W^2 \mu_W}{K_\eta^{(\mathrm{eff})}(K_W^{(\mathrm{eff})})^2}\right) \left(\frac{\gamma^2 \lambda_W^2 \sigma}{K_U K_W^{(\mathrm{eff})}}\right)^{E_*} \left(\frac{\pi^2 T_U}{L^2 K_U}\right)^{-F_*}, \tag{F.93}$$

and

$$\epsilon_\eta = \frac{c_{\eta U}^2}{K_U K_\eta^{(\mathrm{eff})}}. \tag{F.94}$$

F.7.2 Linear monomial-drift map

Introduce the grouped weak-axisymmetric logarithmic drift vector

$$\delta\mathbf{x} = \begin{pmatrix} \lambda_1 \\ c_1 \\ \gamma_1 \\ \kappa_U \\ \kappa_\eta \\ \kappa_W \\ \mu_1 \\ \tau_1 \end{pmatrix} = \begin{pmatrix} \delta\ln\lambda_W \\ \delta\ln c_{\eta U} \\ \delta\ln\gamma \\ \delta\ln K_U \\ \delta\ln K_\eta^{(\mathrm{eff})} \\ \delta\ln K_W^{(\mathrm{eff})} \\ \delta\ln\mu_W \\ \delta\ln T_U \end{pmatrix}_{\mathrm{grp}}. \tag{F.95}$$

Then

$$\begin{pmatrix} \delta\ln\mathfrak{C}_{\mathrm{tr},*} \\ \delta\ln\mathfrak{C}_{\mathrm{nt},*} \\ \delta\ln\epsilon_\eta \end{pmatrix} = M_*\delta\mathbf{x}, \tag{F.96}$$

with

$$M_* = \begin{pmatrix} 0 & 1+\delta_{U,*} & 1+\delta_{U,*} & -(2+\chi_{0,*}+\delta_{U,*}) & 0 & 0 & 0 & 1+\chi_{0,*} \\ 2(1+E_*) & 0 & 2E_* & F_*-E_* & -1 & -(2+E_*) & 1 & -F_* \\ 0 & 2 & 0 & -1 & -1 & 0 & 0 & 0 \end{pmatrix}. \tag{F.97}$$

A convenient dependent minor has determinant

$$\det M_*^{(\tau,\kappa_\eta,\mu_1)} = 1+\chi_{0,*} > 0. \tag{F.98}$$

Therefore M_* has rank three and a five-dimensional kernel. The zero-defect condition is

$$\Theta_1 = \Xi_1 = \mathcal{R}_1 = 0 \iff M_*\delta\mathbf{x} = 0. \tag{F.99}$$

Claim status. Stage 185 is Exact Within Closure **in the reference coherent branch. The monomial exponents are fixed by the branch normal form, and M_* is an exact exponent matrix.**

F.8 Similarity-orbit theorem

F.8.1 Finite similarity family

Stage 186 integrates the tangent kernel of M_* to a finite five-parameter similarity family. Let

$$(\Lambda, C, \Gamma, U, W) \tag{F.100}$$

be free logarithmic co-scalings of $(\lambda_W, c_{\eta U}, \gamma, K_U, K_W^{(\mathrm{eff})})$. Monomial preservation fixes the dependent scalings:

$$K_\eta^{(\mathrm{eff})} \mapsto e^{2C-U} K_\eta^{(\mathrm{eff})}, \tag{F.101}$$

$$T_U \mapsto \exp\left[U - \frac{1+\delta_{U,*}}{1+\chi_{0,*}}(\Gamma + C - U)\right] T_U, \tag{F.102}$$

$$\mu_W \mapsto e^{M(\Lambda,C,\Gamma,U,W)}\mu_W, \tag{F.103}$$

where

$$\begin{aligned} M(\Lambda,C,\Gamma,U,W) &= 2C - U + 2W - 2\Lambda - E_*(2\Gamma + 2\Lambda - U - W) \\ &\quad - F_*\frac{1+\delta_{U,*}}{1+\chi_{0,*}}(\Gamma + C - U). \end{aligned} \tag{F.104}$$

Call this family $\mathcal{G}_*$. It preserves $(\mathfrak{C}_{\mathrm{tr},*}, \mathfrak{C}_{\mathrm{nt},*}, \epsilon_\eta)$ exactly.

F.8.2 Orbit–quotient closure

Stage 187 proves the converse. On the positive microscopic state space

$$\mathcal{M}_+ = \{(\lambda_W, c_{\eta U}, \gamma, K_U, K_\eta^{(\mathrm{eff})}, K_W^{(\mathrm{eff})}, \mu_W, T_U) > 0\}, \tag{F.105}$$

define

$$\mathcal{I}(x) = (\mathfrak{C}_{\mathrm{tr},*}(x), \mathfrak{C}_{\mathrm{nt},*}(x), \epsilon_\eta(x)). \tag{F.106}$$

For two positive states $x, \widetilde{x}$, equality of the three invariants is exactly

$$M_*\Delta\mathbf{x} = 0, \qquad \Delta\mathbf{x} := \ln(\widetilde{x}/x). \tag{F.107}$$

Solving these finite equations gives precisely the dependent transformations (F.101)–(F.103). Hence

$$\mathcal{I}(\widetilde{x}) = \mathcal{I}(x) \iff \widetilde{x} \in \mathcal{G}_* \cdot x. \tag{F.108}$$

Therefore

$$\mathcal{M}_+/\mathcal{G}_* \cong (\mathbb{R}_{>0})^3, \tag{F.109}$$

with quotient coordinates (F.4). The first grouped weak-axisymmetric observables are the infinitesimal motion in these quotient coordinates:

$$\Theta_1 = -C_{\mathrm{tr}}\delta\ln\mathfrak{C}_{\mathrm{tr},*}, \tag{F.110}$$

$$\Xi_1 = A_{\mathrm{tr}}\delta\ln\mathfrak{C}_{\mathrm{tr},*} + \delta\ln\mathfrak{C}_{\mathrm{nt},*}, \tag{F.111}$$

$$\mathcal{R}_1 + \Xi_1 = -\frac{\epsilon_{\eta,*}}{1-\epsilon_{\eta,*}}\delta\ln\epsilon_\eta. \tag{F.112}$$

F.8.3 Projector calculus

Stages 191–192 then turn the quotient coordinates into an exact finite packet and an exact microscopic projector split. With the finite log-ratio vector ordered as

$$\Delta\mathbf{x} = (\Delta_\lambda, \Delta_c, \Delta_\gamma, \Delta_U, \Delta_{K_\eta}, \Delta_W, \Delta_\mu, \Delta_T)^T, \tag{F.113}$$

the quotient packet is

$$\mathbf{q} = (q_{\mathrm{tr}}, q_{\mathrm{nt}}, q_\eta)^T = M_*\Delta\mathbf{x}. \tag{F.114}$$

Choosing the dependent triple $(T_U, K_\eta^{(\mathrm{eff})}, \mu_W)$, the exact pivot block is

$$P_{(T,K_\eta,\mu)} = \begin{pmatrix} 1+\chi_{0,*} & 0 & 0 \\ -F_* & -1 & 1 \\ 0 & -1 & 0 \end{pmatrix}, \qquad \det P_{(T,K_\eta,\mu)} = 1 + \chi_{0,*} > 0. \tag{F.115}$$

The canonical section $S_{(T,K_\eta,\mu)}$ is the right inverse satisfying $M_* S_{(T,K_\eta,\mu)} = I_3$. The exact projectors are

$$Q_{\text{quot}} := S_{(T,K_\eta,\mu)} M_*, \qquad O_{\text{orb}} := I_8 - Q_{\text{quot}}. \tag{F.116}$$

They obey

$$Q_{\text{quot}}^2 = Q_{\text{quot}}, \qquad O_{\text{orb}}^2 = O_{\text{orb}}, \qquad M_* O_{\text{orb}} = 0, \qquad M_* Q_{\text{quot}} = M_*. \tag{F.117}$$

The quotient-failure piece is supported only on the dependent triple:

$$(\Delta_T)_{\text{fail}} = \frac{q_{\text{tr}}}{1+\chi_{0,*}}, \qquad (\Delta_{K_\eta})_{\text{fail}} = -q_\eta, \tag{F.118}$$

$$(\Delta_\mu)_{\text{fail}} = \frac{F_*}{1+\chi_{0,*}} q_{\text{tr}} + q_{\text{nt}} - q_\eta. \tag{F.119}$$

Thus

$$\mathbf{q} = 0 \iff M_* \Delta\mathbf{x} = 0 \iff Q_{\text{quot}} \Delta\mathbf{x} = 0 \iff \Delta\mathbf{x} \in \ker M_*. \tag{F.120}$$

Claim status. Stages 186–192 are Exact Within Closure **for the positive coherent reduced state space. The word "orbit" here means a finite reduced monomial-preserving similarity class, not a fundamental gauge symmetry of the unreduced parent PDE.**

F.9 Four-scalar packet (Δ_{full})

F.9.1 Packet A: conservative target and outgoing finish line

Stages 193–197 sharpen the branch-side Packet A. Start from the Stage 191 branch residual

$$\Delta_{\text{branch}} = (a_2, b_2, a_4, b_4, a_{P_0}, b_{P_0}, \Delta_{\text{pole}}, \Delta_{\text{norm}}). \tag{F.121}$$

Stage 193 freezes the conservative target surface

$$a_2 = b_2 = a_4 = b_4 = 0, \qquad \Delta_{\text{pole}} := \bar{\nu}_4 - 4\bar{\nu}_2^2 = 0. \tag{F.122}$$

On the isotropic branch this is equivalent to

$$D_0 D_4 + 3D_2^2 = 0, \tag{F.123}$$

or, in bundle variables,

$$D_0(B_4 + Z_4) = 3(M + B_2 + Z_2)^2. \tag{F.124}$$

The resulting conservative carrier is

$$\widehat{Y}_Q^{\text{cons}}(\omega) = \frac{3}{4} + \frac{1}{4}\frac{1}{1-\omega^2/\Omega_Q^2}. \tag{F.125}$$

Stage 193 also proves the scalar/geometry firewall: if weak anisotropy induces the first $l = 0 \leftrightarrow l = 2$ mixing, then Schur completion gives

$$\mathcal{D}_{2,\text{eff}}(\omega,\chi) = \mathcal{D}_2(\omega) I_3 - \chi^2 C(\omega) \mathcal{D}_0(\omega)^{-1} C(\omega)^T, \tag{F.126}$$

so

$$\partial_\chi \mathcal{D}_{2,\text{eff}}(\omega,\chi)|_{\chi=0} = 0. \tag{F.127}$$

The scalar/geometry lane cannot linearly contaminate the grouped $l = 2$ conservative carrier.

Stage 194 attaches the outgoing $l = 2$ DtN fingerprint. With

$$z := \frac{a\omega}{c_s}, \tag{F.128}$$

the exact outgoing operator is

$$\Lambda_2^{\text{out}}(z) = z\frac{d}{dz}\ln h_2^{(1)}(z), \tag{F.129}$$

with normalized response

$$\widehat{Y}_2^{\text{out}}(z) = 1 + \frac{z^2}{9} + \frac{4z^4}{81} + i\frac{z^5}{27} + O(z^6). \tag{F.130}$$

Matching this to the retarded grouped module fixes

$$\chi_Q = 1 \tag{F.131}$$

on the canonical compact passive/outgoing branch. More generally, for the isotropic deformation

$$\Lambda_2^{\text{def}}(z) = S\Lambda_2^{\text{out}}(\beta z) + \Sigma_0 + \Sigma_2 z^2 + \Sigma_4 z^4 + i\Sigma_5 z^5 + O(z^6), \tag{F.132}$$

canonical even matching forces Σ_2 and Σ_4, while the outgoing scalar is

$$\chi_Q = \frac{3(S\beta^5 + 9\Sigma_5)}{3S - \Sigma_0}, \qquad \chi_Q - 1 = \frac{3S(\beta^5 - 1) + \Sigma_0 + 27\Sigma_5}{3S - \Sigma_0}. \tag{F.133}$$

Thus the only isotropic DtN data that can move χ_Q, while preserving the canonical even moments, are β, Σ_0, and Σ_5.

Stage 195 factorizes the observable odd closure:

$$\widehat{m}_0^2\chi_Q N_Q = 1. \tag{F.134}$$

On the natural point-particle source map,

$$\widehat{m}_0 \to 1, \qquad N_Q = \chi_Q^{-1}. \tag{F.135}$$

Stage 196 proves that additional isotropic retarded tails beginning at $O(\omega^7)$ do not alter any coefficient through the point-particle 2.5PN order. Stage 197 therefore gives the exact Packet-A finish line

$$\Delta_{\text{branch}} = 0 \iff \chi_Q = 1 \iff \Delta_Q := \chi_Q - 1 = 0. \tag{F.136}$$

F.9.2 Packet B: finite and pairwise orbit lock

Stages 198–199 complete the orbit side. With the same positive microscopic state vector (F.91), keep the free coordinates

$$(\lambda_W, c_{\eta U}, \gamma, K_U, K_W^{(\text{eff})}) \tag{F.137}$$

and the dependent triple

$$(T_U, K_\eta^{(\text{eff})}, \mu_W). \tag{F.138}$$

For a fixed invariant triple $(\mathfrak{C}_{\text{tr},*}, \mathfrak{C}_{\text{nt},*}, \epsilon_{\eta,*})$, Stage 198 solves the exact dependent orbit law and defines mismatch ratios

$$(m_T, m_K, m_\mu). \tag{F.139}$$

The logarithmic quotient chart is

$$q_{\mathrm{tr}} = (1+\chi_{0,*})\ln m_T, \qquad q_\eta = -\ln m_K, \qquad q_{\mathrm{nt}} = \ln m_\mu - \ln m_K - F_* \ln m_T. \tag{F.140}$$

Orbit lock is

$$\Delta_{\mathrm{orbit}} = 0 \iff m_T = m_K = m_\mu = 1 \iff q_{\mathrm{tr}} = q_{\mathrm{nt}} = q_\eta = 0. \tag{F.141}$$

Stage 199 removes reference-point privilege. For two states $x^{(1)}, x^{(2)}$, define

$$\Delta_{\mathrm{orbit}}^{(2\leftarrow1)} = (q_{\mathrm{tr}}^{(2\leftarrow1)}, q_{\mathrm{nt}}^{(2\leftarrow1)}, q_\eta^{(2\leftarrow1)}). \tag{F.142}$$

Then

$$x^{(2)} \in \mathcal{G}_* \cdot x^{(1)} \iff q_{\mathrm{tr}}^{(2\leftarrow1)} = q_{\mathrm{nt}}^{(2\leftarrow1)} = q_\eta^{(2\leftarrow1)} = 0. \tag{F.143}$$

The packet is an exact additive two-point cocycle, while the multiplicative mismatch and invariant-ratio packets obey the corresponding multiplicative cocycle laws. Thus Packet B can be attached to an orbit rather than to a selected representative.

F.9.3 Reference-free full verdict packet

Let

$$\mathcal{O}_* := \mathcal{G}_* \cdot x_* \tag{F.144}$$

be the target similarity orbit. Stage 200 defines the additive full packet

$$\Delta_{\mathrm{full}}^{(x|\mathcal{O}_*)} := \left(\Delta_Q(x), q_{\mathrm{tr}}^{(x\leftarrow\mathcal{O}_*)}, q_{\mathrm{nt}}^{(x\leftarrow\mathcal{O}_*)}, q_\eta^{(x\leftarrow\mathcal{O}_*)}\right), \qquad \Delta_Q(x) := \chi_Q(x) - 1. \tag{F.145}$$

The multiplicative form is

$$\mathcal{V}_{\mathrm{full}}^{(x|\mathcal{O}_*)} := \left(\chi_Q(x), \mathfrak{R}_{\mathrm{tr}}^{(x\leftarrow\mathcal{O}_*)}, \mathfrak{R}_{\mathrm{nt}}^{(x\leftarrow\mathcal{O}_*)}, \mathfrak{R}_\eta^{(x\leftarrow\mathcal{O}_*)}\right), \tag{F.146}$$

and the mismatch form is

$$\mathcal{M}_{\mathrm{full}}^{(x|\mathcal{O}_*)} := \left(\chi_Q(x), m_T^{(x\leftarrow\mathcal{O}_*)}, m_K^{(x\leftarrow\mathcal{O}_*)}, m_\mu^{(x\leftarrow\mathcal{O}_*)}\right). \tag{F.147}$$

The exact chart conversions are

$$\mathfrak{R}_{\mathrm{tr}} = e^{q_{\mathrm{tr}}} = (m_T)^{1+\chi_{0,*}}, \qquad \mathfrak{R}_\eta = e^{q_\eta} = \frac{1}{m_K}, \tag{F.148}$$

$$\mathfrak{R}_{\mathrm{nt}} = e^{q_{\mathrm{nt}}} = \frac{m_\mu}{m_K m_T^{F_*}}. \tag{F.149}$$

Therefore the additive, multiplicative, and mismatch packets are exact reparameterizations of the same verdict.

The final reference-free theorem is

$$\Delta_{\mathrm{full}}^{(x|\mathcal{O}_*)} = 0 \iff \Delta_{\mathrm{branch}}(x) = 0 \text{ and } x \in \mathcal{O}_*. \tag{F.150}$$

Using the sharpened Packet-A and Packet-B forms, this is equivalently

$$\Delta_{\mathrm{full}}^{(x|\mathcal{O}_*)} = 0 \iff \chi_Q(x) = 1, \qquad q_{\mathrm{tr}}^{(x\leftarrow\mathcal{O}_*)} = q_{\mathrm{nt}}^{(x\leftarrow\mathcal{O}_*)} = q_\eta^{(x\leftarrow\mathcal{O}_*)} = 0. \tag{F.151}$$

This is the endpoint of Part V. The reduced compiler algebra is finished; what remains is branch realization of the four scalar conditions.

Claim status. Stages 193–200 are Exact Within Closure **once the Packet-A isotropic grouped one-pole front end, the natural point-particle source-map reduction, and the coherent orbit quotient hierarchy are adopted. The actual PDE may still fail any of the four final scalar conditions. That failure would be a realization failure, not an algebraic incompleteness of the ledger.**

F.10 Verification outputs and downstream use

The usable downstream citation packet from this appendix is as follows.

1. **MTDC-T9.1**: cite for the lower-branch drift laws, tangent-compensation theorem, weak-axisymmetric grouped signature, direct outlet map, hidden-even relation, and the loading mismatch $\Xi_{\mathrm{load}} = N_{01}/N_0 - D_{01}/D_0$.
2. **MTDC-T9.2**: cite for the static self-similarity theorem, wall-normalized outgoing-load theorem, square-root mixed-leg law, port co-loading theorem, effective transfer-shape collapse, and coherent support-blindness.
3. **MTDC-T9.3**: cite for the microscopic slippage decomposition, triangular normal form, branch-invariant monomials, rank-three drift map, similarity orbit $\mathcal{G}_*$, and finite orbit–quotient closure.
4. **MTDC-T9.4**: cite for the branch-observable compiler, transfer-shape/outgoing-prefactor compiler, scalar no-go filters, minimal Packet A/Packet B data, and two-packet home-stretch theorem.
5. **MTDC-T9.5**: cite for the exact orbit/quotient projectors, isotropic grouped target surface, outgoing $l = 2$ DtN fingerprint, source-map reduction, higher-odd irrelevance, and Packet-A closure $\Delta_{\mathrm{branch}} = 0 \iff \chi_Q = 1$.
6. **MTDC-T9.6**: cite for the finite orbit law, pairwise orbit transport, orbit-representative independence, and final four-scalar verdict packet $\Delta_{\mathrm{full}} = (\Delta_Q, q_{\mathrm{tr}}, q_{\mathrm{nt}}, q_\eta)$.

- ☐ The off-family scalar $\delta_\perp$ is used only as the first post-Part-IV normal coordinate; after tangent compensation, first-order isotropic bundle drift is not counted as an independent failure mode.
- ☐ The lower compensated Family–1 drift laws are applied only under the D/N tube selection, healing-lock shell closure, and frozen $n = 5$ wall-EOS reduction.
- ☐ The four-bundle inversion through $(\Theta_w, K_s, K_q, P_0)$ is kept distinct from a direct microscopic PDE solve; it is an algebraic compiler once those observables are known.
- ☐ The weak-axisymmetric signature $(1, 1/2, -1)$ is not confused with arbitrary grouped anisotropy. It fixes the line $b = 3a$.
- ☐ The no-feed-down result is first order only: grouped real P_2 anisotropy can enter scalar ledgers through quadratic grouped invariants.
- ☐ The even-preserving branch is invoked before reducing the linear grouped problem to Ξ_{load}.
- ☐ Static self-similarity is stated in wall-referenced variables. Frozen port shapes do not by themselves imply zero outgoing defect when the wall baseline moves.
- ☐ The square-root mixed-leg law is a reduced sufficient condition under interference/hybridization rigidity, not an unconditional PDE theorem.

- ☐ The scalar $\Xi_1 = P_1/P_0$ is the physical outgoing-prefactor slope and effective transfer-shape slope; it is not introduced as a fit parameter.
- ☐ Coherent support-blindness applies to the direct transfer-shape defect, not to all support-dependent selected-branch quantities.
- ☐ The triangular normal form is used with branch-adapted variables $(\Sigma_{\rm tr}, \Sigma_{\rm nt}, \Sigma_\eta)$; raw microscopic slopes require the stated compiler before interpretation.
- ☐ The monomial coordinates $(\mathfrak{C}_{{\rm tr},*}, \mathfrak{C}_{{\rm nt},*}, \epsilon_\eta)$ are exact quotient coordinates of the reduced coherent hierarchy, not fundamental gauge invariants of the unsolved parent PDE.
- ☐ The similarity orbit $\mathcal{G}_*$ identifies reduced co-scaling fibres in the positive microscopic state space $\mathcal{M}_+$; branch realization still has to be checked against the actual PDE output.
- ☐ Packet A and Packet B are kept separate until the Stage 200 theorem: Packet A is the branch-side outgoing scalar; Packet B is the orbit-side quotient triple.
- ☐ The equivalence $N_Q = \chi_Q^{-1}$ is tied to the natural source-map reduction. It should not be used when the source-map factor $m_{\hat{0}}$ is still nontrivial.
- ☐ Higher odd retarded terms are irrelevant only to the reduced point-particle 2.5PN finish line; later higher-order retarded diagnostics may need them.
- ☐ The final verdict $\Delta_{\rm full} = 0$ is an exact reduced compiler condition. It is not an assertion that the completed moving-throat PDE has already realized the condition.

Future papers should cite Part V when replacing a long weak-axisymmetric or retarded endgame by the compact packet $(\Delta_Q, q_{\rm tr}, q_{\rm nt}, q_\eta)$. They should separately state whether the actual branch has returned $\Delta_Q = 0$ and $(q_{\rm tr}, q_{\rm nt}, q_\eta) = (0, 0, 0)$, or whether those realization conditions remain open.

F.11 Stage-by-stage verification cards

The preceding sections collect the derivation by audit path. The following cards restore the original stage order for Stages 164–200. Each card records the source stage, local status, imported assumptions, primary output, verification checks, and downstream use. The cards are intentionally compact: long algebraic identities are not repeated when they already appear in the audit-path sections above.

Source layout. Stages 164–200 are now sourced from the canonical files `stages/stage_164.tex` through `stages/stage_200.tex`. The raw Markdown notes and audit artifacts are tracked in a repo-local provenance index, so they do not have to appear in the reader-facing PDF. The stage files themselves carry the executable SymPy and Mathematica audit references.

F.12 Stage 164: Microscopic log-imbalance channels

Part / anchor. Part V; primary anchor **MTDC-T9**, local subanchor **MTDC-T9.1**.

Claim status. Exact Within Closure.

Purpose. Stage 164 is the original-stage audit card for *Microscopic log-imbalance channels.* Its purpose is to verify the local step summarized in the output field and to preserve the exact dependencies needed by downstream citations.

Inputs. This stage imports the Stage 163 off-family scalar, the lower compensated Family–1 branch, the grouped real P_2 weak-axisymmetric signature, and the wall/BdG/Maxwell/mixed bundle notation. It does not modify the parent action or the charge/notation firewall.

Derivation ledger. The derivation removes tangent-compensated isotropic motion, proves no linear grouped feed-down into scalar slippages, and reduces the direct grouped outlet map to the physical even-response and prefactor slopes.

Output. Rewrites the Stage 163 off-family scalar in explicit branch-variable drifts of $(\mathcal{Z}_q, \rho_w, c_{s,w}, c_s, \mathcal{T}_m, v_{w0}, a, L_W)$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage. The card restores original stage order and records the claim boundary, not a second independent proof.

Checks.

- ☐ Check that tangent-compensated isotropic drift is not counted as an off-family failure mode.
- ☐ Check the weak-axisymmetric signature $(1, 1/2, -1)$ before reducing grouped defects to a scalar.
- ☐ Check that grouped real P_2 anisotropy is not allowed to feed scalar slippages linearly.
- ☐ Carry the claim-status tag above whenever this stage is cited outside the derivation ledger.

Downstream use. This stage feeds the Part V weak-axisymmetric/orbit packet and the final four-scalar verdict. Downstream papers should cite **MTDC-T9.1** for the local result and **MTDC-T9** for the full Part V compiler.

F.13 Stage 165: Lower-branch drift laws

Part / anchor. Part V; primary anchor **MTDC-T9**, local subanchor **MTDC-T9.1**.

Claim status. Exact Within Closure.

Purpose. Stage 165 is the original-stage audit card for *Lower-branch drift laws.* Its purpose is to verify the local step summarized in the output field and to preserve the exact dependencies needed by downstream citations.

Inputs. This stage imports the Stage 163 off-family scalar, the lower compensated Family–1 branch, the grouped real P_2 weak-axisymmetric signature, and the wall/BdG/Maxwell/mixed bundle notation. It does not modify the parent action or the charge/notation firewall.

Derivation ledger. The derivation removes tangent-compensated isotropic motion, proves no linear grouped feed-down into scalar slippages, and reduces the direct grouped outlet map to the physical even-response and prefactor slopes.

Output. Shows $\delta \ln L_W = \delta \ln a$ and reduces the lower compensated branch to four irreducible drifts $(\delta \ln \mathcal{Z}_q, \delta \ln \rho_w, \delta \ln c_s, \delta \ln a)$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage. The card restores original stage order and records the claim boundary, not a second independent proof.

Checks.

- ☐ Check that tangent-compensated isotropic drift is not counted as an off-family failure mode.
- ☐ Check the weak-axisymmetric signature $(1, 1/2, -1)$ before reducing grouped defects to a scalar.
- ☐ Check that grouped real P_2 anisotropy is not allowed to feed scalar slippages linearly.
- ☐ Carry the claim-status tag above whenever this stage is cited outside the derivation ledger.

Downstream use. This stage feeds the Part V weak-axisymmetric/orbit packet and the final four-scalar verdict. Downstream papers should cite **MTDC-T9.1** for the local result and **MTDC-T9** for the full Part V compiler.

F.14 Stage 166: Bundle inversion of the last four drifts

Part / anchor. Part V; primary anchor **MTDC-T9**, local subanchor **MTDC-T9.1**.

Claim status. Exact Within Closure.

Purpose. Stage 166 is the original-stage audit card for *Bundle inversion of the last four drifts.* Its purpose is to verify the local step summarized in the output field and to preserve the exact dependencies needed by downstream citations.

Inputs. This stage imports the Stage 163 off-family scalar, the lower compensated Family–1 branch, the grouped real P_2 weak-axisymmetric signature, and the wall/BdG/Maxwell/mixed bundle notation. It does not modify the parent action or the charge/notation firewall.

Derivation ledger. The derivation removes tangent-compensated isotropic motion, proves no linear grouped feed-down into scalar slippages, and reduces the direct grouped outlet map to the physical even-response and prefactor slopes.

Output. Inverts the four observables $(\Theta_w, K_s, K_q, P_0)$ to determine the four remaining microscopic drifts.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage. The card restores original stage order and records the claim boundary, not a second independent proof.

Checks.

- ☐ Check that tangent-compensated isotropic drift is not counted as an off-family failure mode.
- ☐ Check the weak-axisymmetric signature $(1, 1/2, -1)$ before reducing grouped defects to a scalar.
- ☐ Check that grouped real P_2 anisotropy is not allowed to feed scalar slippages linearly.
- ☐ Carry the claim-status tag above whenever this stage is cited outside the derivation ledger.

Downstream use. This stage feeds the Part V weak-axisymmetric/orbit packet and the final four-scalar verdict. Downstream papers should cite **MTDC-T9.1** for the local result and **MTDC-T9** for the full Part V compiler.

F.15 Stage 167: Bundle transport and tangent compensation

Part / anchor. Part V; primary anchor **MTDC-T9**, local subanchor **MTDC-T9.1**.

Claim status. Exact Within Closure.

Purpose. Stage 167 is the original-stage audit card for *Bundle transport and tangent compensation*. Its purpose is to verify the local step summarized in the output field and to preserve the exact dependencies needed by downstream citations.

Inputs. This stage imports the Stage 163 off-family scalar, the lower compensated Family–1 branch, the grouped real P_2 weak-axisymmetric signature, and the wall/BdG/Maxwell/mixed bundle notation. It does not modify the parent action or the charge/notation firewall.

Derivation ledger. The derivation removes tangent-compensated isotropic motion, proves no linear grouped feed-down into scalar slippages, and reduces the direct grouped outlet map to the physical even-response and prefactor slopes.

Output. Proves arbitrary first-order isotropic bundle drift is tangent to the exact compensated parent family.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage. The card restores original stage order and records the claim boundary, not a second independent proof.

Checks.

- ☐ Check that tangent-compensated isotropic drift is not counted as an off-family failure mode.
- ☐ Check the weak-axisymmetric signature $(1, 1/2, -1)$ before reducing grouped defects to a scalar.
- ☐ Check that grouped real P_2 anisotropy is not allowed to feed scalar slippages linearly.
- ☐ Carry the claim-status tag above whenever this stage is cited outside the derivation ledger.

Downstream use. This stage feeds the Part V weak-axisymmetric/orbit packet and the final four-scalar verdict. Downstream papers should cite **MTDC-T9.1** for the local result and **MTDC-T9** for the full Part V compiler.

F.16 Stage 168: Off-bundle slippage decomposition

Part / anchor. Part V; primary anchor **MTDC-T9**, local subanchor **MTDC-T9.1**.

Claim status. Exact Within Closure.

Purpose. Stage 168 is the original-stage audit card for *Off-bundle slippage decomposition*. Its purpose is to verify the local step summarized in the output field and to preserve the exact dependencies needed by downstream citations.

Inputs. This stage imports the Stage 163 off-family scalar, the lower compensated Family–1 branch, the grouped real P_2 weak-axisymmetric signature, and the wall/BdG/Maxwell/mixed bundle notation. It does not modify the parent action or the charge/notation firewall.

Derivation ledger. The derivation removes tangent-compensated isotropic motion, proves no linear grouped feed-down into scalar slippages, and reduces the direct grouped outlet map to the physical even-response and prefactor slopes.

Output. Reduces the first true off-bundle defect to axial-length, mixed-speed, and mouth-traction slippages.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage. The card restores original stage order and records the claim boundary, not a second independent proof.

Checks.

- ☐ Check that tangent-compensated isotropic drift is not counted as an off-family failure mode.
- ☐ Check the weak-axisymmetric signature $(1, 1/2, -1)$ before reducing grouped defects to a scalar.
- ☐ Check that grouped real P_2 anisotropy is not allowed to feed scalar slippages linearly.
- ☐ Carry the claim-status tag above whenever this stage is cited outside the derivation ledger.

Downstream use. This stage feeds the Part V weak-axisymmetric/orbit packet and the final four-scalar verdict. Downstream papers should cite **MTDC-T9.1** for the local result and **MTDC-T9** for the full Part V compiler.

F.17 Stage 169: No linear grouped feed-down into scalar slippages

Part / anchor. Part V; primary anchor **MTDC-T9**, local subanchor **MTDC-T9.1**.

Claim status. Exact Within Closure.

Purpose. Stage 169 is the original-stage audit card for *No linear grouped feed-down into scalar slippages*. Its purpose is to verify the local step summarized in the output field and to preserve the exact dependencies needed by downstream citations.

Inputs. This stage imports the Stage 163 off-family scalar, the lower compensated Family–1 branch, the grouped real P_2 weak-axisymmetric signature, and the wall/BdG/Maxwell/mixed bundle notation. It does not modify the parent action or the charge/notation firewall.

Derivation ledger. The derivation removes tangent-compensated isotropic motion, proves no linear grouped feed-down into scalar slippages, and reduces the direct grouped outlet map to the physical even-response and prefactor slopes.

Output. Pure grouped real P_2 anisotropy cannot linearly source the scalar off-bundle slippages; scalar feed-down starts quadratically.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage. The card restores original stage order and records the claim boundary, not a second independent proof.

Checks.

- ☐ Check that tangent-compensated isotropic drift is not counted as an off-family failure mode.
- ☐ Check the weak-axisymmetric signature $(1, 1/2, -1)$ before reducing grouped defects to a scalar.
- ☐ Check that grouped real P_2 anisotropy is not allowed to feed scalar slippages linearly.
- ☐ Carry the claim-status tag above whenever this stage is cited outside the derivation ledger.

Downstream use. This stage feeds the Part V weak-axisymmetric/orbit packet and the final four-scalar verdict. Downstream papers should cite **MTDC-T9.1** for the local result and **MTDC-T9** for the full Part V compiler.

F.18 Stage 170: Linear grouped outlet map

Part / anchor. Part V; primary anchor **MTDC-T9**, local subanchor **MTDC-T9.1**.

Claim status. Exact Within Closure.

Purpose. Stage 170 is the original-stage audit card for *Linear grouped outlet map*. Its purpose is to verify the local step summarized in the output field and to preserve the exact dependencies needed by downstream citations.

Inputs. This stage imports the Stage 163 off-family scalar, the lower compensated Family–1 branch, the grouped real P_2 weak-axisymmetric signature, and the wall/BdG/Maxwell/mixed bundle notation. It does not modify the parent action or the charge/notation firewall.

Derivation ledger. The derivation removes tangent-compensated isotropic motion, proves no linear grouped feed-down into scalar slippages, and reduces the direct grouped outlet map to the physical even-response and prefactor slopes.

Output. Reduces the direct outlet deformations to $\mathcal{K}_A = \delta D_{A,2} + \delta D_{A,0}/9$ and $\mathcal{G}_A = \delta N_{A,0} - P_0 \delta D_{A,0}$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage. The card restores original stage order and records the claim boundary, not a second independent proof.

Checks.

- ☐ Check that tangent-compensated isotropic drift is not counted as an off-family failure mode.
- ☐ Check the weak-axisymmetric signature $(1, 1/2, -1)$ before reducing grouped defects to a scalar.
- ☐ Check that grouped real P_2 anisotropy is not allowed to feed scalar slippages linearly.
- ☐ Carry the claim-status tag above whenever this stage is cited outside the derivation ledger.

Downstream use. This stage feeds the Part V weak-axisymmetric/orbit packet and the final four-scalar verdict. Downstream papers should cite **MTDC-T9.1** for the local result and **MTDC-T9** for the full Part V compiler.

F.19 Stage 171: Microscopic grouped obstructions

Part / anchor. Part V; primary anchor **MTDC-T9**, local subanchor **MTDC-T9.1**.

Claim status. EXACT WITHIN CLOSURE.

Purpose. Stage 171 is the original-stage audit card for *Microscopic grouped obstructions*. Its purpose is to verify the local step summarized in the output field and to preserve the exact dependencies needed by downstream citations.

Inputs. This stage imports the Stage 163 off-family scalar, the lower compensated Family–1 branch, the grouped real P_2 weak-axisymmetric signature, and the wall/BdG/Maxwell/mixed bundle notation. It does not modify the parent action or the charge/notation firewall.

Derivation ledger. The derivation removes tangent-compensated isotropic motion, proves no linear grouped feed-down into scalar slippages, and reduces the direct grouped outlet map to the physical even-response and prefactor slopes.

Output. Decomposes $\mathcal{K}_A$ and $\mathcal{G}_A$ into wall, BdG, Maxwell/mixed, and outgoing-transfer defect bundles.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage. The card restores original stage order and records the claim boundary, not a second independent proof.

Checks.

- ☐ Check that tangent-compensated isotropic drift is not counted as an off-family failure mode.
- ☐ Check the weak-axisymmetric signature $(1, 1/2, -1)$ before reducing grouped defects to a scalar.
- ☐ Check that grouped real P_2 anisotropy is not allowed to feed scalar slippages linearly.
- ☐ Carry the claim-status tag above whenever this stage is cited outside the derivation ledger.

Downstream use. This stage feeds the Part V weak-axisymmetric/orbit packet and the final four-scalar verdict. Downstream papers should cite **MTDC-T9.1** for the local result and **MTDC-T9** for the full Part V compiler.

F.20 Stage 172: Physical slope collapse

Part / anchor. Part V; primary anchor **MTDC-T9**, local subanchor **MTDC-T9.1**.

Claim status. Exact Within Closure.

Purpose. Stage 172 is the original-stage audit card for *Physical slope collapse*. Its purpose is to verify the local step summarized in the output field and to preserve the exact dependencies needed by downstream citations.

Inputs. This stage imports the Stage 163 off-family scalar, the lower compensated Family–1 branch, the grouped real P_2 weak-axisymmetric signature, and the wall/BdG/Maxwell/mixed bundle notation. It does not modify the parent action or the charge/notation firewall.

Derivation ledger. The derivation removes tangent-compensated isotropic motion, proves no linear grouped feed-down into scalar slippages, and reduces the direct grouped outlet map to the physical even-response and prefactor slopes.

Output. Identifies $\mathfrak{K}_1 = -D_0 u_2^{(1)}$ and $\mathfrak{G}_1 = D_0 P_1$, reducing the linear grouped outlet problem to physical slopes.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage. The card restores original stage order and records the claim boundary, not a second independent proof.

Checks.

- ☐ Check that tangent-compensated isotropic drift is not counted as an off-family failure mode.
- ☐ Check the weak-axisymmetric signature $(1, 1/2, -1)$ before reducing grouped defects to a scalar.
- ☐ Check that grouped real P_2 anisotropy is not allowed to feed scalar slippages linearly.
- ☐ Carry the claim-status tag above whenever this stage is cited outside the derivation ledger.

Downstream use. This stage feeds the Part V weak-axisymmetric/orbit packet and the final four-scalar verdict. Downstream papers should cite **MTDC-T9.1** for the local result and **MTDC-T9** for the full Part V compiler.

F.21 Stage 173: Weak-axisymmetric loading mismatch

Part / anchor. Part V; primary anchor **MTDC-T9**, local subanchor **MTDC-T9.1**.

Claim status. Exact Within Closure.

Purpose. Stage 173 is the original-stage audit card for *Weak-axisymmetric loading mismatch.* Its purpose is to verify the local step summarized in the output field and to preserve the exact dependencies needed by downstream citations.

Inputs. This stage imports the Stage 163 off-family scalar, the lower compensated Family–1 branch, the grouped real P_2 weak-axisymmetric signature, and the wall/BdG/Maxwell/mixed bundle notation. It does not modify the parent action or the charge/notation firewall.

Derivation ledger. The derivation removes tangent-compensated isotropic motion, proves no linear grouped feed-down into scalar slippages, and reduces the direct grouped outlet map to the physical even-response and prefactor slopes.

Output. On the even-preserving branch, the remaining grouped defect is the scalar $\Xi_{\rm load} = N_{01}/N_0 - D_{01}/D_0$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage. The card restores original stage order and records the claim boundary, not a second independent proof.

Checks.

- ☐ Check that tangent-compensated isotropic drift is not counted as an off-family failure mode.
- ☐ Check the weak-axisymmetric signature $(1, 1/2, -1)$ before reducing grouped defects to a scalar.
- ☐ Check that grouped real P_2 anisotropy is not allowed to feed scalar slippages linearly.
- ☐ Carry the claim-status tag above whenever this stage is cited outside the derivation ledger.

Downstream use. This stage feeds the Part V weak-axisymmetric/orbit packet and the final four-scalar verdict. Downstream papers should cite **MTDC-T9.1** for the local result and **MTDC-T9** for the full Part V compiler.

F.22 Stage 174: Static self-similarity

Part / anchor. Part V; primary anchor **MTDC-T9**, local subanchor **MTDC-T9.2**.

Claim status. Exact Within Closure.

Purpose. Stage 174 is the original-stage audit card for *Static self-similarity.* Its purpose is to verify the local step summarized in the output field and to preserve the exact dependencies needed by downstream citations.

Inputs. This stage imports the even-preserving grouped branch, the loading mismatch Ξ_{load}, and the wall-normalized Maxwell/mixed transfer factors. It works inside the conservative-shape-preserving and weak-axisymmetric reduced branch.

Derivation ledger. The derivation rewrites the loading defect as failure of self-similar wall-referenced loading, factors the outgoing port into mixed-leg/interference/hybridization pieces, and collapses all weak-axisymmetric port slippages to one scalar.

Output. Expresses Ξ_{load} as weighted failure of static self-similarity between wall, BdG, conservative port, and outgoing port slopes.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage. The card restores original stage order and records the claim boundary, not a second independent proof.

Checks.

- ☐ Check that self-similarity is defined relative to the wall baseline, not by freezing every port shape.
- ☐ Check the conservative-shape-preserving hypothesis before using $\Xi_1 = P_1/P_0$.
- ☐ Check any square-root mixed-leg use carries the interference/hybridization rigidity assumptions.
- ☐ Carry the claim-status tag above whenever this stage is cited outside the derivation ledger.

Downstream use. This stage feeds the Part V weak-axisymmetric/orbit packet and the final four-scalar verdict. Downstream papers should cite **MTDC-T9.2** for the local result and **MTDC-T9** for the full Part V compiler.

F.23 Stage 175: Wall-normalized outgoing-load theorem

Part / anchor. Part V; primary anchor **MTDC-T9**, local subanchor **MTDC-T9.2**.

Claim status. Exact Within Closure.

Purpose. Stage 175 is the original-stage audit card for *Wall-normalized outgoing-load theorem*. Its purpose is to verify the local step summarized in the output field and to preserve the exact dependencies needed by downstream citations.

Inputs. This stage imports the even-preserving grouped branch, the loading mismatch Ξ_{load}, and the wall-normalized Maxwell/mixed transfer factors. It works inside the conservative-shape-preserving and weak-axisymmetric reduced branch.

Derivation ledger. The derivation rewrites the loading defect as failure of self-similar wall-referenced loading, factors the outgoing port into mixed-leg/interference/hybridization pieces, and collapses all weak-axisymmetric port slippages to one scalar.

Output. Factors the remaining defect into wall-normalized shape/load variables and proves conservative-shape preservation leaves only outgoing load slippage.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage. The card restores original stage order and records the claim boundary, not a second independent proof.

Checks.

- ☐ Check that self-similarity is defined relative to the wall baseline, not by freezing every port shape.
- ☐ Check the conservative-shape-preserving hypothesis before using $\Xi_1 = P_1/P_0$.
- ☐ Check any square-root mixed-leg use carries the interference/hybridization rigidity assumptions.
- ☐ Carry the claim-status tag above whenever this stage is cited outside the derivation ledger.

Downstream use. This stage feeds the Part V weak-axisymmetric/orbit packet and the final four-scalar verdict. Downstream papers should cite **MTDC-T9.2** for the local result and **MTDC-T9** for the full Part V compiler.

F.24 Stage 176: Square-root mixed-leg law

Part / anchor. Part V; primary anchor **MTDC-T9**, local subanchor **MTDC-T9.2**.

Claim status. Exact Within Closure.

Purpose. Stage 176 is the original-stage audit card for *Square-root mixed-leg law.* Its purpose is to verify the local step summarized in the output field and to preserve the exact dependencies needed by downstream citations.

Inputs. This stage imports the even-preserving grouped branch, the loading mismatch Ξ_{load}, and the wall-normalized Maxwell/mixed transfer factors. It works inside the conservative-shape-preserving and weak-axisymmetric reduced branch.

Derivation ledger. The derivation rewrites the loading defect as failure of self-similar wall-referenced loading, factors the outgoing port into mixed-leg/interference/hybridization pieces, and collapses all weak-axisymmetric port slippages to one scalar.

Output. Factors the outgoing load into mixed-leg, interference, and hybridization slippages; under rigidity, zero defect requires $G_W/\Omega_W^2 \propto \sqrt{K}$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage. The card restores original stage order and records the claim boundary, not a second independent proof.

Checks.

- ☐ Check that self-similarity is defined relative to the wall baseline, not by freezing every port shape.
- ☐ Check the conservative-shape-preserving hypothesis before using $\Xi_1 = P_1/P_0$.
- ☐ Check any square-root mixed-leg use carries the interference/hybridization rigidity assumptions.
- ☐ Carry the claim-status tag above whenever this stage is cited outside the derivation ledger.

Downstream use. This stage feeds the Part V weak-axisymmetric/orbit packet and the final four-scalar verdict. Downstream papers should cite **MTDC-T9.2** for the local result and **MTDC-T9** for the full Part V compiler.

F.25 Stage 177: One scalar outgoing-slippage amplitude

Part / anchor. Part V; primary anchor **MTDC-T9**, local subanchor **MTDC-T9.2**.

Claim status. Exact Within Closure.

Purpose. Stage 177 is the original-stage audit card for *One scalar outgoing-slippage amplitude.* Its purpose is to verify the local step summarized in the output field and to preserve the exact dependencies needed by downstream citations.

Inputs. This stage imports the even-preserving grouped branch, the loading mismatch Ξ_{load}, and the wall-normalized Maxwell/mixed transfer factors. It works inside the conservative-shape-preserving and weak-axisymmetric reduced branch.

Derivation ledger. The derivation rewrites the loading defect as failure of self-similar wall-referenced loading, factors the outgoing port into mixed-leg/interference/hybridization pieces, and collapses all weak-axisymmetric port slippages to one scalar.

Output. Shows all weak-axisymmetric outgoing slippages inherit the same grouped signature and collapse to $\Xi_1 = P_1/P_0$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage. The card restores original stage order and records the claim boundary, not a second independent proof.

Checks.

- ☐ Check that self-similarity is defined relative to the wall baseline, not by freezing every port shape.
- ☐ Check the conservative-shape-preserving hypothesis before using $\Xi_1 = P_1/P_0$.
- ☐ Check any square-root mixed-leg use carries the interference/hybridization rigidity assumptions.
- ☐ Carry the claim-status tag above whenever this stage is cited outside the derivation ledger.

Downstream use. This stage feeds the Part V weak-axisymmetric/orbit packet and the final four-scalar verdict. Downstream papers should cite **MTDC-T9.2** for the local result and **MTDC-T9** for the full Part V compiler.

F.26 Stage 178: Outgoing-port co-loading theorem

Part / anchor. Part V; primary anchor **MTDC-T9**, local subanchor **MTDC-T9.2**.

Claim status. Exact Within Closure.

Purpose. Stage 178 is the original-stage audit card for *Outgoing-port co-loading theorem.* Its purpose is to verify the local step summarized in the output field and to preserve the exact dependencies needed by downstream citations.

Inputs. This stage imports the even-preserving grouped branch, the loading mismatch Ξ_{load}, and the wall-normalized Maxwell/mixed transfer factors. It works inside the conservative-shape-preserving and weak-axisymmetric reduced branch.

Derivation ledger. The derivation rewrites the loading defect as failure of self-similar wall-referenced loading, factors the outgoing port into mixed-leg/interference/hybridization pieces, and collapses all weak-axisymmetric port slippages to one scalar.

Output. Rewrites Ξ_1 as the weighted mismatch between actual static outgoing-port loading and wall-baseline loading.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage. The card restores original stage order and records the claim boundary, not a second independent proof.

Checks.

- ☐ Check that self-similarity is defined relative to the wall baseline, not by freezing every port shape.
- ☐ Check the conservative-shape-preserving hypothesis before using $\Xi_1 = P_1/P_0$.
- ☐ Check any square-root mixed-leg use carries the interference/hybridization rigidity assumptions.
- ☐ Carry the claim-status tag above whenever this stage is cited outside the derivation ledger.

Downstream use. This stage feeds the Part V weak-axisymmetric/orbit packet and the final four-scalar verdict. Downstream papers should cite **MTDC-T9.2** for the local result and **MTDC-T9** for the full Part V compiler.

F.27 Stage 179: Transfer-shape theorem

Part / anchor. Part V; primary anchor **MTDC-T9**, local subanchor **MTDC-T9.2**.

Claim status. Exact Within Closure.

Purpose. Stage 179 is the original-stage audit card for *Transfer-shape theorem*. Its purpose is to verify the local step summarized in the output field and to preserve the exact dependencies needed by downstream citations.

Inputs. This stage imports the even-preserving grouped branch, the loading mismatch Ξ_{load}, and the wall-normalized Maxwell/mixed transfer factors. It works inside the conservative-shape-preserving and weak-axisymmetric reduced branch.

Derivation ledger. The derivation rewrites the loading defect as failure of self-similar wall-referenced loading, factors the outgoing port into mixed-leg/interference/hybridization pieces, and collapses all weak-axisymmetric port slippages to one scalar.

Output. Factors each outgoing port as $N_{A,0}^{(r)} = K_A \mathcal{T}_{A,r}^2$ and gives $\Xi_1 = 2\sum_r \rho_r^{(N)} \tau_r$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage. The card restores original stage order and records the claim boundary, not a second independent proof.

Checks.

- ☐ Check that self-similarity is defined relative to the wall baseline, not by freezing every port shape.
- ☐ Check the conservative-shape-preserving hypothesis before using $\Xi_1 = P_1/P_0$.
- ☐ Check any square-root mixed-leg use carries the interference/hybridization rigidity assumptions.
- ☐ Carry the claim-status tag above whenever this stage is cited outside the derivation ledger.

Downstream use. This stage feeds the Part V weak-axisymmetric/orbit packet and the final four-scalar verdict. Downstream papers should cite **MTDC-T9.2** for the local result and **MTDC-T9** for the full Part V compiler.

F.28 Stage 180: Effective transfer-shape collapse

Part / anchor. Part V; primary anchor **MTDC-T9**, local subanchor **MTDC-T9.2**.

Claim status. Exact Within Closure.

Purpose. Stage 180 is the original-stage audit card for *Effective transfer-shape collapse*. Its purpose is to verify the local step summarized in the output field and to preserve the exact dependencies needed by downstream citations.

Inputs. This stage imports the even-preserving grouped branch, the loading mismatch Ξ_{load}, and the wall-normalized Maxwell/mixed transfer factors. It works inside the conservative-shape-preserving and weak-axisymmetric reduced branch.

Derivation ledger. The derivation rewrites the loading defect as failure of self-similar wall-referenced loading, factors the outgoing port into mixed-leg/interference/hybridization pieces, and collapses all weak-axisymmetric port slippages to one scalar.

Output. Collapses many ports to $\mathcal{T}^2_{\mathrm{eff},A} = N_{A,0}/K_A$ and gives the actual one-port continuum formula.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage. The card restores original stage order and records the claim boundary, not a second independent proof.

Checks.

- ☐ Check that self-similarity is defined relative to the wall baseline, not by freezing every port shape.
- ☐ Check the conservative-shape-preserving hypothesis before using $\Xi_1 = P_1/P_0$.
- ☐ Check any square-root mixed-leg use carries the interference/hybridization rigidity assumptions.
- ☐ Carry the claim-status tag above whenever this stage is cited outside the derivation ledger.

Downstream use. This stage feeds the Part V weak-axisymmetric/orbit packet and the final four-scalar verdict. Downstream papers should cite **MTDC-T9.2** for the local result and **MTDC-T9** for the full Part V compiler.

F.29 Stage 181: Coherent tracking-branch defect law

Part / anchor. Part V; primary anchor **MTDC-T9**, local subanchor **MTDC-T9.2**.

Claim status. Exact Within Closure.

Purpose. Stage 181 is the original-stage audit card for *Coherent tracking-branch defect law.* Its purpose is to verify the local step summarized in the output field and to preserve the exact dependencies needed by downstream citations.

Inputs. This stage imports the even-preserving grouped branch, the loading mismatch Ξ_{load}, and the wall-normalized Maxwell/mixed transfer factors. It works inside the conservative-shape-preserving and weak-axisymmetric reduced branch.

Derivation ledger. The derivation rewrites the loading defect as failure of self-similar wall-referenced loading, factors the outgoing port into mixed-leg/interference/hybridization pieces, and collapses all weak-axisymmetric port slippages to one scalar.

Output. On the coherent local D/N tracking branch, Ξ_1 is carried only by mixed/outgoing placement variables; coherent support ratio ζ drops out.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage. The card restores original stage order and records the claim boundary, not a second independent proof.

Checks.

- ☐ Check that self-similarity is defined relative to the wall baseline, not by freezing every port shape.
- ☐ Check the conservative-shape-preserving hypothesis before using $\Xi_1 = P_1/P_0$.
- ☐ Check any square-root mixed-leg use carries the interference/hybridization rigidity assumptions.
- ☐ Carry the claim-status tag above whenever this stage is cited outside the derivation ledger.

Downstream use. This stage feeds the Part V weak-axisymmetric/orbit packet and the final four-scalar verdict. Downstream papers should cite **MTDC-T9.2** for the local result and **MTDC-T9** for the full Part V compiler.

F.30 Stage 182: Microscopic coherent-kernel slippages

Part / anchor. Part V; primary anchor **MTDC-T9**, local subanchor **MTDC-T9.3**.

Claim status. Exact Within Closure.

Purpose. Stage 182 is the original-stage audit card for *Microscopic coherent-kernel slippages.* Its purpose is to verify the local step summarized in the output field and to preserve the exact dependencies needed by downstream citations.

Inputs. This stage imports the coherent local D/N tracking branch, the transfer-shape slope Ξ_1, and the microscopic coherent-kernel variables. It assumes the positive-coupling state space used for the quotient/orbit theorem.

Derivation ledger. The derivation changes variables from raw microscopic slippages to tracking, nontracking, and dressing coordinates; integrates them to monomial quotient coordinates; and identifies the zero-defect tangent and finite similarity orbits.

Output. Reduces the defect to $(\Sigma_Z, \Sigma_\chi, \Sigma_\epsilon, \Sigma_\delta)$ plus dressing slippage Σ_η, and isolates Σ_{tr}.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage. The card restores original stage order and records the claim boundary, not a second independent proof.

Checks.

- ☐ Check raw microscopic slopes have been compiled into $(\Sigma_{\mathrm{tr}}, \Sigma_{\mathrm{nt}}, \Sigma_\eta)$ before interpretation.
- ☐ Check the positive-coupling state-space assumption before invoking the finite orbit theorem.
- ☐ Check that quotient motion is distinguished from the five-dimensional similarity orbit.
- ☐ Carry the claim-status tag above whenever this stage is cited outside the derivation ledger.

Downstream use. This stage feeds the Part V weak-axisymmetric/orbit packet and the final four-scalar verdict. Downstream papers should cite **MTDC-T9.3** for the local result and **MTDC-T9** for the full Part V compiler.

F.31 Stage 183: Triangular normal form

Part / anchor. Part V; primary anchor **MTDC-T9**, local subanchor **MTDC-T9.3**.

Claim status. Exact Within Closure.

Purpose. Stage 183 is the original-stage audit card for *Triangular normal form*. Its purpose is to verify the local step summarized in the output field and to preserve the exact dependencies needed by downstream citations.

Inputs. This stage imports the coherent local D/N tracking branch, the transfer-shape slope Ξ_1, and the microscopic coherent-kernel variables. It assumes the positive-coupling state space used for the quotient/orbit theorem.

Derivation ledger. The derivation changes variables from raw microscopic slippages to tracking, nontracking, and dressing coordinates; integrates them to monomial quotient coordinates; and identifies the zero-defect tangent and finite similarity orbits.

Output. Compresses the coherent weak-axisymmetric problem to $(\Sigma_{\mathrm{tr}}, \Sigma_{\mathrm{nt}}, \Sigma_\eta)$ with triangular map to $(\Theta_1, \Xi_1, \mathcal{R}_1)$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage. The card restores original stage order and records the claim boundary, not a second independent proof.

Checks.

- ☐ Check raw microscopic slopes have been compiled into $(\Sigma_{\rm tr}, \Sigma_{\rm nt}, \Sigma_\eta)$ before interpretation.
- ☐ Check the positive-coupling state-space assumption before invoking the finite orbit theorem.
- ☐ Check that quotient motion is distinguished from the five-dimensional similarity orbit.
- ☐ Carry the claim-status tag above whenever this stage is cited outside the derivation ledger.

Downstream use. This stage feeds the Part V weak-axisymmetric/orbit packet and the final four-scalar verdict. Downstream papers should cite **MTDC-T9.3** for the local result and **MTDC-T9** for the full Part V compiler.

F.32 Stage 184: Branch-invariant coordinates

Part / anchor. Part V; primary anchor **MTDC-T9**, local subanchor **MTDC-T9.3**.

Claim status. EXACT WITHIN CLOSURE.

Purpose. Stage 184 is the original-stage audit card for *Branch-invariant coordinates*. Its purpose is to verify the local step summarized in the output field and to preserve the exact dependencies needed by downstream citations.

Inputs. This stage imports the coherent local D/N tracking branch, the transfer-shape slope Ξ_1, and the microscopic coherent-kernel variables. It assumes the positive-coupling state space used for the quotient/orbit theorem.

Derivation ledger. The derivation changes variables from raw microscopic slippages to tracking, nontracking, and dressing coordinates; integrates them to monomial quotient coordinates; and identifies the zero-defect tangent and finite similarity orbits.

Output. Identifies exact branch composites $\mathfrak{T}_*$, $\mathfrak{N}_*$, and ϵ_η whose logarithmic drifts are the three normal-form coordinates.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage. The card restores original stage order and records the claim boundary, not a second independent proof.

Checks.

- ☐ Check raw microscopic slopes have been compiled into $(\Sigma_{\rm tr}, \Sigma_{\rm nt}, \Sigma_\eta)$ before interpretation.
- ☐ Check the positive-coupling state-space assumption before invoking the finite orbit theorem.
- ☐ Check that quotient motion is distinguished from the five-dimensional similarity orbit.
- ☐ Carry the claim-status tag above whenever this stage is cited outside the derivation ledger.

Downstream use. This stage feeds the Part V weak-axisymmetric/orbit packet and the final four-scalar verdict. Downstream papers should cite **MTDC-T9.3** for the local result and **MTDC-T9** for the full Part V compiler.

F.33 Stage 185: Microscopic monomial coordinates

Part / anchor. Part V; primary anchor **MTDC-T9**, local subanchor **MTDC-T9.3**.

Claim status. Exact Within Closure.

Purpose. Stage 185 is the original-stage audit card for *Microscopic monomial coordinates*. Its purpose is to verify the local step summarized in the output field and to preserve the exact dependencies needed by downstream citations.

Inputs. This stage imports the coherent local D/N tracking branch, the transfer-shape slope Ξ_1, and the microscopic coherent-kernel variables. It assumes the positive-coupling state space used for the quotient/orbit theorem.

Derivation ledger. The derivation changes variables from raw microscopic slippages to tracking, nontracking, and dressing coordinates; integrates them to monomial quotient coordinates; and identifies the zero-defect tangent and finite similarity orbits.

Output. Pushes the branch composites to direct monomials $(\mathfrak{C}_{\mathrm{tr},*}, \mathfrak{C}_{\mathrm{nt},*}, \epsilon_\eta)$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage. The card restores original stage order and records the claim boundary, not a second independent proof.

Checks.

- ☐ Check raw microscopic slopes have been compiled into $(\Sigma_{\mathrm{tr}}, \Sigma_{\mathrm{nt}}, \Sigma_\eta)$ before interpretation.
- ☐ Check the positive-coupling state-space assumption before invoking the finite orbit theorem.
- ☐ Check that quotient motion is distinguished from the five-dimensional similarity orbit.
- ☐ Carry the claim-status tag above whenever this stage is cited outside the derivation ledger.

Downstream use. This stage feeds the Part V weak-axisymmetric/orbit packet and the final four-scalar verdict. Downstream papers should cite **MTDC-T9.3** for the local result and **MTDC-T9** for the full Part V compiler.

F.34 Stage 186: Similarity-orbit tangent theorem

Part / anchor. Part V; primary anchor **MTDC-T9**, local subanchor **MTDC-T9.3**.

Claim status. Exact Within Closure.

Purpose. Stage 186 is the original-stage audit card for *Similarity-orbit tangent theorem.* Its purpose is to verify the local step summarized in the output field and to preserve the exact dependencies needed by downstream citations.

Inputs. This stage imports the coherent local D/N tracking branch, the transfer-shape slope Ξ_1, and the microscopic coherent-kernel variables. It assumes the positive-coupling state space used for the quotient/orbit theorem.

Derivation ledger. The derivation changes variables from raw microscopic slippages to tracking, nontracking, and dressing coordinates; integrates them to monomial quotient coordinates; and identifies the zero-defect tangent and finite similarity orbits.

Output. Shows the zero-defect equations are the tangent equations of a five-parameter monomial-preserving similarity orbit $\mathcal{G}_*$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage. The card restores original stage order and records the claim boundary, not a second independent proof.

Checks.

- ☐ Check raw microscopic slopes have been compiled into $(\Sigma_{\mathrm{tr}}, \Sigma_{\mathrm{nt}}, \Sigma_\eta)$ before interpretation.
- ☐ Check the positive-coupling state-space assumption before invoking the finite orbit theorem.
- ☐ Check that quotient motion is distinguished from the five-dimensional similarity orbit.
- ☐ Carry the claim-status tag above whenever this stage is cited outside the derivation ledger.

Downstream use. This stage feeds the Part V weak-axisymmetric/orbit packet and the final four-scalar verdict. Downstream papers should cite **MTDC-T9.3** for the local result and **MTDC-T9** for the full Part V compiler.

F.35 Stage 187: Orbit–quotient closure

Part / anchor. Part V; primary anchor **MTDC-T9**, local subanchor **MTDC-T9.3**.

Claim status. Exact Within Closure.

Purpose. Stage 187 is the original-stage audit card for *Orbit–quotient closure.* Its purpose is to verify the local step summarized in the output field and to preserve the exact dependencies needed by downstream citations.

Inputs. This stage imports the coherent local D/N tracking branch, the transfer-shape slope Ξ_1, and the microscopic coherent-kernel variables. It assumes the positive-coupling state space used for the quotient/orbit theorem.

Derivation ledger. The derivation changes variables from raw microscopic slippages to tracking, nontracking, and dressing coordinates; integrates them to monomial quotient coordinates; and identifies the zero-defect tangent and finite similarity orbits.

Output. Proves finite level sets of $(\mathfrak{C}_{\mathrm{tr},*}, \mathfrak{C}_{\mathrm{nt},*}, \epsilon_\eta)$ are exactly $\mathcal{G}_*$-orbits.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage. The card restores original stage order and records the claim boundary, not a second independent proof.

Checks.

- ☐ Check raw microscopic slopes have been compiled into $(\Sigma_{\mathrm{tr}}, \Sigma_{\mathrm{nt}}, \Sigma_\eta)$ before interpretation.
- ☐ Check the positive-coupling state-space assumption before invoking the finite orbit theorem.
- ☐ Check that quotient motion is distinguished from the five-dimensional similarity orbit.
- ☐ Carry the claim-status tag above whenever this stage is cited outside the derivation ledger.

Downstream use. This stage feeds the Part V weak-axisymmetric/orbit packet and the final four-scalar verdict. Downstream papers should cite **MTDC-T9.3** for the local result and **MTDC-T9** for the full Part V compiler.

F.36 Stage 188: Branch-observable compiler

Part / anchor. Part V; primary anchor **MTDC-T9**, local subanchor **MTDC-T9.4**.

Claim status. Exact Within Closure.

Purpose. Stage 188 is the original-stage audit card for *Branch-observable compiler*. Its purpose is to verify the local step summarized in the output field and to preserve the exact dependencies needed by downstream citations.

Inputs. This stage imports the triangular normal form, the transfer-shape/outgoing-prefactor identity, and the exact monomial quotient coordinates. It compiles observable branch data rather than solving the nonlinear moving-throat PDE.

Derivation ledger. The derivation composes the quotient map with the observable packet, equates the direct coherent defect with the transfer-shape/outgoing-prefactor slope, and separates branch-side and orbit-side packets.

Output. Compiles direct branch observables to $(\Theta_1, \Xi_1, \mathcal{R}_1)$ and records equivalent observable packets.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage. The card restores original stage order and records the claim boundary, not a second independent proof.

Checks.

- ☐ Check that direct transfer-shape drift Ξ_1 is kept separate from selected-branch dressing residuals.
- ☐ Check Packet A and Packet B are not collapsed until the two-packet theorem is invoked.
- ☐ Check the corrected nontracking form when tracking feed-through has been removed.
- ☐ Carry the claim-status tag above whenever this stage is cited outside the derivation ledger.

Downstream use. This stage feeds the Part V weak-axisymmetric/orbit packet and the final four-scalar verdict. Downstream papers should cite **MTDC-T9.4** for the local result and **MTDC-T9** for the full Part V compiler.

F.37 Stage 189: Transfer-shape / outgoing-prefactor compiler

Part / anchor. Part V; primary anchor **MTDC-T9**, local subanchor **MTDC-T9.4**.

Claim status. EXACT WITHIN CLOSURE **for the transfer-shape/outgoing-prefactor compiler inside the carried branch packet; matching to the external GR normalization target remains a downstream Packet-A target condition.**

Purpose. Stage 189 is the original-stage audit card for *Transfer-shape / outgoing-prefactor compiler.* Its purpose is to verify the local step summarized in the output field and to preserve the exact dependencies needed by downstream citations.

Inputs. This stage imports the triangular normal form, the transfer-shape/outgoing-prefactor identity, and the exact monomial quotient coordinates. It compiles observable branch data rather than solving the nonlinear moving-throat PDE.

Derivation ledger. The derivation composes the quotient map with the observable packet, equates the direct coherent defect with the transfer-shape/outgoing-prefactor slope, and separates branch-side and orbit-side packets.

Output. Equates Ξ_1 with the logarithmic transfer-shape and outgoing-prefactor slope, including corrected nontracking form.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage. The card restores original stage order and records the claim boundary, not a second independent proof.

Checks.

- ☐ Check that direct transfer-shape drift Ξ_1 is kept separate from selected-branch dressing residuals.
- ☐ Check Packet A and Packet B are not collapsed until the two-packet theorem is invoked.
- ☐ Check the corrected nontracking form when tracking feed-through has been removed.
- ☐ Carry the claim-status tag above whenever this stage is cited outside the derivation ledger.

Downstream use. This stage feeds the Part V weak-axisymmetric/orbit packet and the final four-scalar verdict. Downstream papers should cite **MTDC-T9.4** for the local result and **MTDC-T9** for the full Part V compiler.

F.38 Stage 190: Direct defect, dressing residual, and no-go filters

Part / anchor. Part V; primary anchor **MTDC-T9**, local subanchor **MTDC-T9.4**.

Claim status. Exact Within Closure.

Purpose. Stage 190 is the original-stage audit card for *Direct defect, dressing residual, and no-go filters*. Its purpose is to verify the local step summarized in the output field and to preserve the exact dependencies needed by downstream citations.

Inputs. This stage imports the triangular normal form, the transfer-shape/outgoing-prefactor identity, and the exact monomial quotient coordinates. It compiles observable branch data rather than solving the nonlinear moving-throat PDE.

Derivation ledger. The derivation composes the quotient map with the observable packet, equates the direct coherent defect with the transfer-shape/outgoing-prefactor slope, and separates branch-side and orbit-side packets.

Output. Separates direct transfer-shape drift from selected-branch dressing residual and records scalar no-go filters.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage. The card restores original stage order and records the claim boundary, not a second independent proof.

Checks.

- ☐ Check that direct transfer-shape drift Ξ_1 is kept separate from selected-branch dressing residuals.
- ☐ Check Packet A and Packet B are not collapsed until the two-packet theorem is invoked.
- ☐ Check the corrected nontracking form when tracking feed-through has been removed.
- ☐ Carry the claim-status tag above whenever this stage is cited outside the derivation ledger.

Downstream use. This stage feeds the Part V weak-axisymmetric/orbit packet and the final four-scalar verdict. Downstream papers should cite **MTDC-T9.4** for the local result and **MTDC-T9** for the full Part V compiler.

F.39 Stage 191: Minimal PDE data packet

Part / anchor. Part V; primary anchor **MTDC-T9**, local subanchor **MTDC-T9.4**.

Claim status. Exact Within Closure.

Purpose. Stage 191 is the original-stage audit card for *Minimal PDE data packet.* Its purpose is to verify the local step summarized in the output field and to preserve the exact dependencies needed by downstream citations.

Inputs. This stage imports the triangular normal form, the transfer-shape/outgoing-prefactor identity, and the exact monomial quotient coordinates. It compiles observable branch data rather than solving the nonlinear moving-throat PDE.

Derivation ledger. The derivation composes the quotient map with the observable packet, equates the direct coherent defect with the transfer-shape/outgoing-prefactor slope, and separates branch-side and orbit-side packets.

Output. Defines Packet A, Packet B, Δ_{branch}, Δ_{orbit}, and the exact two-packet home-stretch theorem.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage. The card restores original stage order and records the claim boundary, not a second independent proof.

Checks.

- ☐ Check that direct transfer-shape drift Ξ_1 is kept separate from selected-branch dressing residuals.
- ☐ Check Packet A and Packet B are not collapsed until the two-packet theorem is invoked.
- ☐ Check the corrected nontracking form when tracking feed-through has been removed.
- ☐ Carry the claim-status tag above whenever this stage is cited outside the derivation ledger.

Downstream use. This stage feeds the Part V weak-axisymmetric/orbit packet and the final four-scalar verdict. Downstream papers should cite **MTDC-T9.4** for the local result and **MTDC-T9** for the full Part V compiler.

F.40 Stage 192: Orbit/quotient projectors

Part / anchor. Part V; primary anchor **MTDC-T9**, local subanchor **MTDC-T9.5**.

Claim status. Exact Within Closure.

Purpose. Stage 192 is the original-stage audit card for *Orbit/quotient projectors.* Its purpose is to verify the local step summarized in the output field and to preserve the exact dependencies needed by downstream citations.

Inputs. This stage imports the isotropic grouped-P_2 one-pole front end, the compact outgoing $l = 2$ DtN fingerprint, and the natural point-particle source-map reduction. It is the Packet-A branch-side finish line.

Derivation ledger. The derivation states the exact isotropic target surface, attaches the outgoing $l = 2$ fingerprint and source-map factor, proves higher odd terms are irrelevant at the current order, and leaves the downstream Packet-A closure identities to Stages 194–195.

Output. Builds exact projectors Q_{quot}, O_{orb}, showing quotient failure lives only on $(T_U, K_\eta^{\text{eff}}, \mu_W)$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage. The card restores original stage order and records the claim boundary, not a second independent proof.

Checks.

- ☐ Check the isotropic one-pole target surface before the downstream Stage 194 use of $\Delta_Q = \chi_Q - 1$.
- ☐ Check the natural source-map reduction before the downstream Stage 195 reduction of $N_Q = \chi_Q^{-1}$.
- ☐ Check higher odd terms are ignored only through the point-particle 2.5PN finish line.
- ☐ Carry the claim-status tag above whenever this stage is cited outside the derivation ledger.

Downstream use. This stage feeds the Part V weak-axisymmetric/orbit packet and the final four-scalar verdict. Downstream papers should cite **MTDC-T9.5** for the local result and **MTDC-T9** for the full Part V compiler.

F.41 Stage 193: Isotropic grouped target surface

Part / anchor. Part V; primary anchor **MTDC-T9**, local subanchor **MTDC-T9.5**.

Claim status. Exact Within Closure.

Purpose. Stage 193 is the original-stage audit card for *Isotropic grouped target surface*. Its purpose is to verify the local step summarized in the output field and to preserve the exact dependencies needed by downstream citations.

Inputs. This stage imports the isotropic grouped-P_2 one-pole front end, the compact outgoing $l = 2$ DtN fingerprint, and the natural point-particle source-map reduction. It is the Packet-A branch-side finish line.

Derivation ledger. The derivation states the exact isotropic target surface, attaches the outgoing $l = 2$ fingerprint and source-map factor, proves higher odd terms are irrelevant at the current order, and leaves the downstream Packet-A closure identities to Stages 194–195.

Output. Freezes the isotropic one-pole conservative target surface and proves the scalar/geometry lane enters the grouped carrier only at quadratic anisotropy order.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage. The card restores original stage order and records the claim boundary, not a second independent proof.

Checks.

- ☐ Check the isotropic one-pole target surface before the downstream Stage 194 use of $\Delta_Q = \chi_Q - 1$.
- ☐ Check the natural source-map reduction before the downstream Stage 195 reduction of $N_Q = \chi_Q^{-1}$.
- ☐ Check higher odd terms are ignored only through the point-particle 2.5PN finish line.
- ☐ Carry the claim-status tag above whenever this stage is cited outside the derivation ledger.

Downstream use. This stage feeds the Part V weak-axisymmetric/orbit packet and the final four-scalar verdict. Downstream papers should cite **MTDC-T9.5** for the local result and **MTDC-T9** for the full Part V compiler.

F.42 Stage 194: Outgoing $l = 2$ fingerprint

Part / anchor. Part V; primary anchor **MTDC-T9**, local subanchor **MTDC-T9.5**.

Claim status. Exact Within Closure.

Purpose. Stage 194 is the original-stage audit card for *Outgoing $l = 2$ fingerprint*. Its purpose is to verify the local step summarized in the output field and to preserve the exact dependencies needed by downstream citations.

Inputs. This stage imports the isotropic grouped-P_2 one-pole front end, the compact outgoing $l = 2$ DtN fingerprint, and the natural point-particle source-map reduction. It is the Packet-A branch-side finish line.

Derivation ledger. The derivation states the exact isotropic target surface, attaches the outgoing $l = 2$ fingerprint and source-map factor, proves higher odd terms are irrelevant at the current order, and collapses Packet A to $\Delta_Q = \chi_Q - 1$.

Output. Derives the exact outgoing spherical DtN fingerprint and fixes $\chi_Q = 1$ on the canonical compact branch.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage. The card restores original stage order and records the claim boundary, not a second independent proof.

Checks.

- ☐ Check the isotropic one-pole target surface before using $\Delta_Q = \chi_Q - 1$.
- ☐ Check the natural source-map reduction before setting $N_Q = \chi_Q^{-1}$.
- ☐ Check higher odd terms are ignored only through the point-particle 2.5PN finish line.
- ☐ Carry the claim-status tag above whenever this stage is cited outside the derivation ledger.

Downstream use. This stage feeds the Part V weak-axisymmetric/orbit packet and the final four-scalar verdict. Downstream papers should cite **MTDC-T9.5** for the local result and **MTDC-T9** for the full Part V compiler.

F.43 Stage 195: Source-map reduction

Part / anchor. Part V; primary anchor **MTDC-T9**, local subanchor **MTDC-T9.5**.

Claim status. Exact Within Closure / Reduced / Controlled Reduction.

Purpose. Stage 195 is the original-stage audit card for *Source-map reduction*. Its purpose is to verify the local step summarized in the output field and to preserve the exact dependencies needed by downstream citations.

Inputs. This stage imports the isotropic grouped-P_2 one-pole front end, the compact outgoing $l = 2$ DtN fingerprint, and the natural point-particle source-map reduction. It is the Packet-A branch-side finish line.

Derivation ledger. The derivation states the exact isotropic target surface, attaches the outgoing $l = 2$ fingerprint and source-map factor, proves higher odd terms are irrelevant at the current order, and collapses Packet A to $\Delta_Q = \chi_Q - 1$.

Output. Factorizes the observable odd closure as $\widehat{m}_0^2 \chi_Q N_Q = 1$ and collapses Δ_{norm}.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage. The card restores original stage order and records the claim boundary, not a second independent proof.

Checks.

- ☐ Check the isotropic one-pole target surface before using $\Delta_Q = \chi_Q - 1$.
- ☐ Check the natural source-map reduction before setting $N_Q = \chi_Q^{-1}$.
- ☐ Check higher odd terms are ignored only through the point-particle 2.5PN finish line.
- ☐ Carry the claim-status tag above whenever this stage is cited outside the derivation ledger.

Downstream use. This stage feeds the Part V weak-axisymmetric/orbit packet and the final four-scalar verdict. Downstream papers should cite **MTDC-T9.5** for the local result and **MTDC-T9** for the full Part V compiler.

F.44 Stage 196: Higher-odd irrelevance

Part / anchor. Part V; primary anchor **MTDC-T9**, local subanchor **MTDC-T9.5**.

Claim status. Exact Within Closure.

Purpose. Stage 196 is the original-stage audit card for *Higher-odd irrelevance*. Its purpose is to verify the local step summarized in the output field and to preserve the exact dependencies needed by downstream citations.

Inputs. This stage imports the isotropic grouped-P_2 one-pole front end, the compact outgoing $l = 2$ DtN fingerprint, and the natural point-particle source-map reduction. It is the Packet-A branch-side finish line.

Derivation ledger. The derivation states the exact isotropic target surface, attaches the outgoing $l = 2$ fingerprint and source-map factor, proves higher odd terms are irrelevant at the current order, and collapses Packet A to $\Delta_Q = \chi_Q - 1$.

Output. Proves all extra isotropic odd tails beginning at $O(\omega^7)$ are invisible to the point-particle 2.5PN theorem.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage. The card restores original stage order and records the claim boundary, not a second independent proof.

Checks.

- ☐ Check the isotropic one-pole target surface before using $\Delta_Q = \chi_Q - 1$.
- ☐ Check the natural source-map reduction before setting $N_Q = \chi_Q^{-1}$.
- ☐ Check higher odd terms are ignored only through the point-particle 2.5PN finish line.
- ☐ Carry the claim-status tag above whenever this stage is cited outside the derivation ledger.

Downstream use. This stage feeds the Part V weak-axisymmetric/orbit packet and the final four-scalar verdict. Downstream papers should cite **MTDC-T9.5** for the local result and **MTDC-T9** for the full Part V compiler.

F.45 Stage 197: Conditional Packet-A closure

Part / anchor. Part V; primary anchor **MTDC-T9**, local subanchor **MTDC-T9.5**.

Claim status. Exact Within Closure / Open.

Purpose. Stage 197 is the original-stage audit card for *Conditional Packet-A closure*. Its purpose is to verify the local step summarized in the output field and to preserve the exact dependencies needed by downstream citations.

Inputs. This stage imports the isotropic grouped-P_2 one-pole front end, the compact outgoing $l = 2$ DtN fingerprint, and the natural point-particle source-map reduction. It is the Packet-A branch-side finish line.

Derivation ledger. The derivation states the exact isotropic target surface, attaches the outgoing $l = 2$ fingerprint and source-map factor, proves higher odd terms are irrelevant at the current order, and collapses Packet A to $\Delta_Q = \chi_Q - 1$.

Output. Proves $\Delta_{\text{branch}} = 0 \iff \chi_Q = 1$ and hence $\Delta_Q = \chi_Q - 1$ is the sole Packet-A scalar.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage. The card restores original stage order and records the claim boundary, not a second independent proof.

Checks.

- ☐ Check the isotropic one-pole target surface before using $\Delta_Q = \chi_Q - 1$.
- ☐ Check the natural source-map reduction before setting $N_Q = \chi_Q^{-1}$.
- ☐ Check higher odd terms are ignored only through the point-particle 2.5PN finish line.
- ☐ Carry the claim-status tag above whenever this stage is cited outside the derivation ledger.

Downstream use. This stage feeds the Part V weak-axisymmetric/orbit packet and the final four-scalar verdict. Downstream papers should cite **MTDC-T9.5** for the local result and **MTDC-T9** for the full Part V compiler.

F.46 Stage 198: Finite orbit law

Part / anchor. Part V; primary anchor **MTDC-T9**, local subanchor **MTDC-T9.6**.

Claim status. Exact Within Closure.

Purpose. Stage 198 is the original-stage audit card for *Finite orbit law*. Its purpose is to verify the local step summarized in the output field and to preserve the exact dependencies needed by downstream citations.

Inputs. This stage imports Packet A, the coherent orbit quotient, and the finite similarity-orbit law. It packages the final verdict in additive, multiplicative, and mismatch charts.

Derivation ledger. The derivation solves the finite orbit law, proves pairwise orbit transport and cocycle rules, and combines Packet A with the orbit quotient triple into a reference-free four-scalar verdict.

Output. Solves the dependent triple exactly, defines mismatch coordinates, and proves finite single-orbit lock.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage. The card restores original stage order and records the claim boundary, not a second independent proof.

Checks.

- ☐ Check finite orbit lock in the chosen chart before declaring Packet B zero.
- ☐ Check pairwise packets obey the additive or multiplicative cocycle law appropriate to the chart used.
- ☐ Check the final zero packet is read as a reduced compiler condition, not as proof of actual PDE realization.
- ☐ Carry the claim-status tag above whenever this stage is cited outside the derivation ledger.

Downstream use. This stage feeds the Part V weak-axisymmetric/orbit packet and the final four-scalar verdict. Downstream papers should cite **MTDC-T9.6** for the local result and **MTDC-T9** for the full Part V compiler.

F.47 Stage 199: Pairwise orbit transport

Part / anchor. Part V; primary anchor **MTDC-T9**, local subanchor **MTDC-T9.6**.

Claim status. Exact Within Closure.

Purpose. Stage 199 is the original-stage audit card for *Pairwise orbit transport*. Its purpose is to verify the local step summarized in the output field and to preserve the exact dependencies needed by downstream citations.

Inputs. This stage imports Packet A, the coherent orbit quotient, and the finite similarity-orbit law. It packages the final verdict in additive, multiplicative, and mismatch charts.

Derivation ledger. The derivation solves the finite orbit law, proves pairwise orbit transport and cocycle rules, and combines Packet A with the orbit quotient triple into a reference-free four-scalar verdict.

Output. Removes representative privilege by giving exact two-point transport, mismatch, cocycle, and orbit-lock laws.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage. The card restores original stage order and records the claim boundary, not a second independent proof.

Checks.

- ☐ Check finite orbit lock in the chosen chart before declaring Packet B zero.
- ☐ Check pairwise packets obey the additive or multiplicative cocycle law appropriate to the chart used.
- ☐ Check the final zero packet is read as a reduced compiler condition, not as proof of actual PDE realization.
- ☐ Carry the claim-status tag above whenever this stage is cited outside the derivation ledger.

Downstream use. This stage feeds the Part V weak-axisymmetric/orbit packet and the final four-scalar verdict. Downstream papers should cite **MTDC-T9.6** for the local result and **MTDC-T9** for the full Part V compiler.

F.48 Stage 200: Reference-free home-stretch theorem

Part / anchor. Part V; primary anchor **MTDC-T9**, local subanchor **MTDC-T9.6**.

Claim status. Exact Within Closure / Open.

Purpose. Stage 200 is the original-stage audit card for *Reference-free home-stretch theorem*. Its purpose is to verify the local step summarized in the output field and to preserve the exact dependencies needed by downstream citations.

Inputs. This stage imports Packet A, the coherent orbit quotient, and the finite similarity-orbit law. It packages the final verdict in additive, multiplicative, and mismatch charts.

Derivation ledger. The derivation solves the finite orbit law, proves pairwise orbit transport and cocycle rules, and combines Packet A with the orbit quotient triple into a reference-free four-scalar verdict.

Output. Combines Packet A and Packet B into the four-scalar final verdict packet $(\Delta_Q, q_{\mathrm{tr}}, q_{\mathrm{nt}}, q_\eta)$.

Verification note. The detailed equations supporting this card are collected in the corresponding block narrative above and in the source stage. The card restores original stage order and records the claim boundary, not a second independent proof.

Checks.

- ☐ Check finite orbit lock in the chosen chart before declaring Packet B zero.
- ☐ Check pairwise packets obey the additive or multiplicative cocycle law appropriate to the chart used.
- ☐ Check the final zero packet is read as a reduced compiler condition, not as proof of actual PDE realization.
- ☐ Carry the claim-status tag above whenever this stage is cited outside the derivation ledger.

Downstream use. This stage feeds the Part V weak-axisymmetric/orbit packet and the final four-scalar verdict. Downstream papers should cite **MTDC-T9.6** for the local result and **MTDC-T9** for the full Part V compiler.

Appendix G

Stage Appendix: Part VI – Realization Compiler and Finite Mixed-Ray Search

Stage range: 201–218

This appendix expands the Part VI theorem-block chapter into a verification ledger. The main-body Part VI chapter states the realization compiler and the finite mixed-ray result in citation-ready form; the present appendix keeps the algebraic chain visible stage by stage. The distinction is important: Part VI proves that the declared local realization search is finite and auditable, while this appendix records how each compiler, graph map, ray bracket, simplex optimizer, boundary splice, and support-cardinality theorem is obtained.

Part V ended with the reduced home-stretch theorem in its most compact form:

$$\Delta_{\text{full}} = 0 \quad \Longleftrightarrow \quad \chi_Q = 1, \qquad q_{\text{tr}} = q_{\text{nt}} = q_\eta = 0. \tag{G.1}$$

That theorem identifies the final reduced packet, but it does not by itself say how a completed moving-throat branch should be audited in practice. Part VI is the operational answer. It turns the final quotient packet into an explicit realization compiler, rewrites the target orbit as a graph over five free microscopic coordinates, reduces graph-aligned branch searches to a scalar closure function, and then proves that the local mixed-ray search over those five free coordinates is finite through support-cardinality five.

The result anchor served by this appendix is **MTDC-T10**. This anchor should be cited when a downstream document needs the detailed derivation of the realization compiler, the canonical same-free-quintuple orbit projection, the free-quintuple target graph, graph-error coordinates, scalar graph-slice closure, log-ray predictors, Hessian-certified ray ranking, pairwise and simplex mixed-ray optimizers, or the final support-cardinality-five local mixed-ray closure theorem.

Table G.1: Part VI source and status map.

Stage	Working title	Primary status	Main output
201	Realization compiler and canonical orbit projection	Exact Within Closure	Intrinsic four-scalar packet, canonical dependent-triple repair vector, canonical target-orbit projection, and same-free-quintuple uniqueness theorem.

Stage	Working title	Primary status	Main output
202	Free-quintuple target graph	Exact Within Closure	Solves the target orbit as a graph over $(\lambda_W, c_{\eta U}, \gamma, K_U, K_W^{\mathrm{eff}})$, defines graph errors (E_T, E_K, E_μ), and gives the first reduced-family closure surface.
203	Scalar graph-slice theorem	Exact Within Closure	Reduces graph-aligned realization to the single scalar $\widehat{\chi}_Q(\mathbf{y}) = 1$, proves graph tangency to $\ker M_*$, and gives the one-parameter crossing theorem.
204	Explicit free-quintuple log-ray sweep	Exact Within Closure	Builds $\mathbf{y}_{\mathbf{s}}(\tau) = \mathbf{y}_\circ \odot e^{\tau \mathbf{s}}$, gives the finite graph lift and primitive-direction table, and reduces closure to a one-variable scalar root.
205	Directional Hessian and quadratic predictors	Exact Within Closure	Adds directional Hessian data, quadratic affine/log predictors, discriminant tests, and turning-point/tangency criteria.
206	Curvature-envelope ray ranking	Mixed: Exact Within Closure, Numerically Located	Turns local curvature envelopes into certified root brackets, bracket-width laws, pairwise ray ordering, and the local search sieve.
207	Primitive-ray certified table	Mixed: Exact Within Closure, Numerically Located	Specializes the sieve to the five primitive free directions; proves only diagonal Hessian restrictions are needed for primitive certification.
208	Pairwise mixed-ray audit	Mixed: Exact Within Closure, Numerically Located	Introduces pairwise cones, gradient synergy, off-diagonal Hessian leverage, canonical gradient/equal-mix screens, and pairwise promotion rules.
209	Pairwise ratio optimizer	Mixed: Exact Within Closure, Numerically Located	Reduces each pairwise ratio search to a finite quartic candidate set and defines optimized pairwise brackets.
210	Three-coordinate simplex audit	Mixed: Exact Within Closure, Numerically Located	Reduces triple boundaries to pairwise cones, gives gradient and equal-mix interior screens, and defines the canonical triple-screen packet.

Stage	Working title	Primary status	Main output
211	Triple-simplex interior optimizer	Mixed: Exact Within Closure, Numerically Located	Reduces the two-parameter triple interior optimizer to quartic/sextic algebraic candidate sets with finite interior brackets.
212	Up-to-three-coordinate search sieve	Mixed: Exact Within Closure, Numerically Located	Splices triple interiors to pairwise boundaries, ranks all primitive triples, and proves the support-≤ 3 finite ledger.
213	Four-coordinate simplex gate	Mixed: Exact Within Closure, Numerically Located	Reduces four-simplex faces to triple packets and introduces four-way gradient/equal-mix canonical screens.
214	Four-coordinate interior optimizer	Mixed: Exact Within Closure, Numerically Located	Lifts the three-parameter four-simplex stationary system to a finite polynomial candidate set with preferred bound 54 per envelope.
215	Up-to-four-coordinate search sieve	Mixed: Exact Within Closure, Numerically Located	Splices quadruple interiors to triple-face boundaries, ranks all primitive quadruples, and proves the support-≤ 4 finite ledger.
216	Unique five-coordinate simplex gate	Mixed: Exact Within Closure, Numerically Located	Identifies the unique five-simplex, reduces its boundary to the support-≤ 4 ledger, and proves the support-cardinality ceiling.
217	Five-coordinate interior optimizer	Mixed: Exact Within Closure, Numerically Located	Reduces the four-parameter five-simplex interior optimizer to a lifted finite polynomial candidate set with preferred bound 162 per envelope.
218	Final local mixed-ray closure theorem	Mixed: Exact Within Closure, Numerically Located, Open	Splices the unique support-5 interior packet to the support-≤ 4 boundary ledger and closes the local mixed-ray search with preferred budget 1464.

Claim status. Part VI is predominantly Exact Within Closure **inside the declared realization/orbit/free-quintuple graph and local Hessian-envelope search hierarchy. The algebraic compilers, graph maps, stationary systems, candidate-set reductions, and support-cardinality closure theorems are exact once the reduced packet and local envelope data are supplied. Any claim that the completed nonlinear moving-throat PDE actually supplies the required free-quintuple branch, Hessian envelopes, validity intervals, or winning candidate**

remains Open**. Any row that depends on numerical insertion of local envelope values or candidate evaluation is marked** Numerically Located **at the realization stage, not at the algebraic compiler stage.**

G.1 Block contract and theorem-level output

The contract of Part VI is different from the contract of the previous grouped-response chapters. Parts III–V constructed the reduced branch and quotient packets. Part VI assumes those packets exist and asks how they are to be realized by actual branch data. The operational chain is

$$\Delta_{\text{full}} \longrightarrow \Delta_{\text{real}}^{\text{int}} \longrightarrow \Pi_{\mathcal{O}_*}^{\text{can}} \longrightarrow \mathbf{x}_*^{\text{graph}}(\mathbf{y}) \longrightarrow \widehat{\chi}_Q(\mathbf{y}) - 1 \longrightarrow \text{finite mixed-ray search.} \tag{G.2}$$

The first arrow turns the theorem packet into an intrinsic audit packet. The second and third arrows replace the abstract target orbit by an explicit graph. The fourth arrow reduces graph-aligned closure to one scalar. The last arrow proves that the local free-quintuple search is finite through the full support-cardinality ceiling.

The positive microscopic state carried throughout this part is

$$\mathbf{x} = \Big(\lambda_W, c_{\eta U}, \gamma, K_U, K_\eta^{(\text{eff})}, K_W^{(\text{eff})}, \mu_W, T_U\Big), \tag{G.3}$$

and the free quintuple is

$$\mathbf{y} = \Big(\lambda_W, c_{\eta U}, \gamma, K_U, K_W^{(\text{eff})}\Big). \tag{G.4}$$

The dependent triple is

$$(T_U, K_\eta^{(\text{eff})}, \mu_W). \tag{G.5}$$

This split is not a new physical assumption. It is the canonical section of the Stage 192 quotient map used to make the same-free-quintuple projection explicit.

Theorem G.1 (Part VI realization compiler and finite mixed-ray closure theorem)**.** *Inside the carried realization hierarchy for Stages 201–218, the following theorem chain holds.*

1. *The final four-scalar verdict packet can be written intrinsically as*

$$\Delta_{\text{real}}^{\text{int}}(\mathbf{x} \mid \mathcal{Z}_*) = \big(\chi_Q(\mathbf{x}) - 1, q_{\text{tr}}(\mathbf{x}), q_{\text{nt}}(\mathbf{x}), q_\eta(\mathbf{x})\big), \tag{G.6}$$

where the three quotient coordinates are logarithmic ratios of the three target monomials.

2. *The canonical repair vector is supported only on the dependent triple,*

$$\Delta\mathbf{x}_{\text{rep}} = \begin{pmatrix} 0 & 0 & 0 & 0 & q_\eta & 0 & -q_{\text{nt}} + q_\eta - \dfrac{F_*}{1+\chi_{0,*}} q_{\text{tr}} & -\dfrac{q_{\text{tr}}}{1+\chi_{0,*}} \end{pmatrix}^T, \tag{G.7}$$

and exponentiating it gives the canonical target-orbit projection $\Pi_{\mathcal{O}_*}^{\text{can}}$.

3. *The target orbit $\mathcal{O}_*$ is exactly the graph of three dependent target functions over the free quintuple* $\mathbf{y}$*:*

$$\mathcal{O}_* = \{\mathbf{x}_*^{\text{graph}}(\mathbf{y}) : \mathbf{y} \in (\mathbb{R}_{>0})^5\}. \tag{G.8}$$

4. *The graph errors*

$$(E_T, E_K, E_\mu) = \left(\ln \frac{T_U}{T_U^{\text{graph}}}, \ln \frac{K_\eta^{(\text{eff})}}{K_{\eta,*}^{\text{graph}}}, \ln \frac{\mu_W}{\mu_W^{\text{graph}}}\right) \tag{G.9}$$

are exactly equivalent to the quotient packet:

$$E_T = \frac{q_{\text{tr}}}{1+\chi_{0,*}}, \qquad E_K = -q_\eta, \qquad E_\mu = q_{\text{nt}} - q_\eta + \frac{F_*}{1+\chi_{0,*}} q_{\text{tr}}. \tag{G.10}$$

5. *On the target graph, Packet B vanishes identically and the full closure set is the scalar graph slice*

$$\mathcal{Z}_* = \{\mathbf{x}_*^{\mathrm{graph}}(\mathbf{y}) : \widehat{\chi}_Q(\mathbf{y}) = 1\}, \qquad \widehat{\chi}_Q(\mathbf{y}) := \chi_Q(\mathbf{x}_*^{\mathrm{graph}}(\mathbf{y})). \tag{G.11}$$

6. *Every smooth graph-aligned branch is locally represented by a free-quintuple log ray*

$$\mathbf{y}_{\mathbf{s}}(\tau) = \mathbf{y}_\circ \odot e^{\tau\mathbf{s}}, \tag{G.12}$$

and closure along that ray is the scalar root problem

$$\Phi_{\mathbf{s}}(\tau) := \widehat{\chi}_Q(\mathbf{y}_{\mathbf{s}}(\tau)) = 1. \tag{G.13}$$

7. *The first- and second-order local data of the ray are*

$$\Phi_0, \quad \Phi_1 = (\mathcal{D}_{\mathbf{s}}\widehat{\chi}_Q)(\mathbf{y}_\circ), \quad \Phi_2 = (\mathcal{D}_{\mathbf{s}}^2\widehat{\chi}_Q)(\mathbf{y}_\circ), \tag{G.14}$$

or, equivalently, logarithmic data $(H_0, K_0, \underline{K}_1, \overline{K}_1)$ *after orientation. These data determine certified local brackets whenever the Stage 206 admissibility hypotheses hold.*

8. *The primitive, pairwise, triple, quadruple, and five-coordinate searches are finite certified algebraic ledgers. The support-cardinality ceiling is five because the free space has exactly five primitive axes.*

9. *The final local mixed-ray search on the positive free-quintuple simplex is reduced to the splice*

$$\tau^{\mathrm{lo}}_{\le 5,\min} = \min(\tau^{\mathrm{lo}}_{\le 4,\min}, \tau^{\mathrm{lo,int}}_{5,\min}), \qquad \tau^{\mathrm{hi}}_{\le 5,\min} = \min(\tau^{\mathrm{hi}}_{\le 4,\min}, \tau^{\mathrm{hi,int}}_{5,\min}), \tag{G.15}$$

with

$$\tau^{\mathrm{lo}}_{\le 5,\min} \le \tau^{\mathrm{best}}_{\le 5,*} \le \tau^{\mathrm{hi}}_{\le 5,\min}. \tag{G.16}$$

Proof structure. Stages 201–202 convert the reference-free four-scalar packet into an intrinsic monomial-ratio audit, a canonical same-free-quintuple projection, and an explicit target graph. Stage 203 proves that graph-aligned families lie in the kernel of the quotient map and hence reduce to the scalar closure function $\widehat{\chi}_Q - 1$. Stages 204–207 build the log-ray, Hessian, curvature-envelope, and primitive certified-ray compilers. Stages 208–209 add pairwise mixed cones and reduce the one-parameter ratio optimization to finite quartic candidate sets. Stages 210–212 do the same for genuine three-coordinate simplex interiors and then splice them into the pairwise boundary ledger. Stages 213–215 repeat the construction for four-coordinate interiors and splice the result into the support-≤ 3 ledger. Stages 216–218 identify the unique five-coordinate simplex, reduce its interior optimizer to a finite lifted algebraic candidate set, and finally prove that its boundary is exactly the already-solved support-≤ 4 search set. □

G.2 Exact realization compiler and target graph

G.2.1 Intrinsic realization packet

Stage 201 begins from the three target monomials already isolated by the orbit/quotient theorem,

$$\mathfrak{C}_{\mathrm{tr},*}(\mathbf{x}), \qquad \mathfrak{C}_{\mathrm{nt},*}(\mathbf{x}), \qquad \epsilon_\eta(\mathbf{x}). \tag{G.17}$$

Fix target values

$$\mathfrak{C}^{\mathrm{target}}_{\mathrm{tr},*}, \qquad \mathfrak{C}^{\mathrm{target}}_{\mathrm{nt},*}, \qquad \epsilon^{\mathrm{target}}_{\eta,*}. \tag{G.18}$$

The target orbit is then

$$\mathcal{O}_* = \left\{ \mathbf{x} > 0 : \mathfrak{C}_{\mathrm{tr},*} = \mathfrak{C}_{\mathrm{tr},*}^{\mathrm{target}}, \quad \mathfrak{C}_{\mathrm{nt},*} = \mathfrak{C}_{\mathrm{nt},*}^{\mathrm{target}}, \quad \epsilon_\eta = \epsilon_{\eta,*}^{\mathrm{target}} \right\}. \tag{G.19}$$

The intrinsic ratios are

$$\mathfrak{R}_{\mathrm{tr}} := \frac{\mathfrak{C}_{\mathrm{tr},*}}{\mathfrak{C}_{\mathrm{tr},*}^{\mathrm{target}}}, \qquad \mathfrak{R}_{\mathrm{nt}} := \frac{\mathfrak{C}_{\mathrm{nt},*}}{\mathfrak{C}_{\mathrm{nt},*}^{\mathrm{target}}}, \qquad \mathfrak{R}_\eta := \frac{\epsilon_\eta}{\epsilon_{\eta,*}^{\mathrm{target}}}, \tag{G.20}$$

and the logarithmic quotient coordinates are

$$q_{\mathrm{tr}} := \ln \mathfrak{R}_{\mathrm{tr}}, \qquad q_{\mathrm{nt}} := \ln \mathfrak{R}_{\mathrm{nt}}, \qquad q_\eta := \ln \mathfrak{R}_\eta. \tag{G.21}$$

Thus

$$\mathbf{x} \in \mathcal{O}_* \quad \Longleftrightarrow \quad q_{\mathrm{tr}} = q_{\mathrm{nt}} = q_\eta = 0. \tag{G.22}$$

The fully closed target set is

$$\mathcal{Z}_* := \{\mathbf{x} \in \mathcal{O}_* : \chi_Q(\mathbf{x}) = 1\}. \tag{G.23}$$

Consequently the intrinsic realization packet is exactly

$$\Delta_{\mathrm{real}}^{\mathrm{int}}(\mathbf{x} \mid \mathcal{Z}_*) = (\chi_Q(\mathbf{x}) - 1, q_{\mathrm{tr}}(\mathbf{x}), q_{\mathrm{nt}}(\mathbf{x}), q_\eta(\mathbf{x}))\,. \tag{G.24}$$

This is the operational version of Part V's reference-free packet.

G.2.2 Canonical same-free-quintuple projection

The Stage 192 quotient map is carried in the linearized form

$$\mathbf{q} = M_* \Delta\mathbf{x}, \qquad \mathbf{q} = (q_{\mathrm{tr}}, q_{\mathrm{nt}}, q_\eta)^T. \tag{G.25}$$

The canonical section chooses the dependent triple $(T_U, K_\eta^{\mathrm{eff}}, \mu_W)$. The exact repair vector is

$$\Delta\mathbf{x}_{\mathrm{rep}} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ q_\eta \\ 0 \\ -q_{\mathrm{nt}} + q_\eta - \dfrac{F_*}{1+\chi_{0,*}} q_{\mathrm{tr}} \\ -\dfrac{q_{\mathrm{tr}}}{1+\chi_{0,*}} \end{pmatrix}. \tag{G.26}$$

It satisfies

$$M_* \Delta\mathbf{x}_{\mathrm{rep}} = -\mathbf{q}. \tag{G.27}$$

Exponentiating (G.26) gives the canonical projection

$$\begin{aligned} \Pi_{\mathcal{O}_*}^{\mathrm{can}}(\mathbf{x}) = (&\lambda_W, c_{\eta U}, \gamma, K_U, K_\eta^{(\mathrm{eff})}\mathfrak{R}_\eta, K_W^{(\mathrm{eff})}, \\ &\mu_W \mathfrak{R}_{\mathrm{nt}}^{-1} \mathfrak{R}_\eta \mathfrak{R}_{\mathrm{tr}}^{-F_*/(1+\chi_{0,*})}, T_U \mathfrak{R}_{\mathrm{tr}}^{-1/(1+\chi_{0,*})}). \end{aligned} \tag{G.28}$$

The projection lands on the target orbit, preserves the free quintuple (G.4), and is the unique point of $\mathcal{O}_*$ with that free quintuple.

Proposition G.2 (Canonical fixed-point audit)**.** *For any positive microscopic state* $\mathbf{x}$*,*

$$\mathbf{x} \in \mathcal{Z}_* \quad \Longleftrightarrow \quad \chi_Q(\mathbf{x}) = 1 \quad \textit{and} \quad \Pi_{\mathcal{O}_*}^{\mathrm{can}}(\mathbf{x}) = \mathbf{x}. \tag{G.29}$$

G.2.3 Direct microscopic target graph

Stage 202 removes the remaining abstractness by solving the orbit constraints directly. The carried monomials are

$$\mathfrak{C}_{\mathrm{tr},*} = \left(\frac{\gamma c_{\eta U}}{K_U}\right)^{1+\delta_{U,*}} \left(\frac{\pi^2 T_U}{L^2 K_U}\right)^{1+\chi_{0,*}}, \tag{G.30}$$

$$\mathfrak{C}_{\mathrm{nt},*} = \frac{\lambda_W^2 \mu_W}{K_\eta^{(\mathrm{eff})}(K_W^{(\mathrm{eff})})^2} \left(\frac{\gamma^2 \lambda_W^2 \sigma}{K_U K_W^{(\mathrm{eff})}}\right)^{E_*} \left(\frac{\pi^2 T_U}{L^2 K_U}\right)^{-F_*}, \tag{G.31}$$

$$\epsilon_\eta = \frac{c_{\eta U}^2}{K_U K_\eta^{(\mathrm{eff})}}. \tag{G.32}$$

Solving (G.30)–(G.32) at fixed $\mathbf{y}$ gives

$$\delta_{U,*}^{\mathrm{graph}}(\mathbf{y}) := \left[\frac{\mathfrak{C}_{\mathrm{tr},*}^{\mathrm{target}}}{\left(\frac{\gamma c_{\eta U}}{K_U}\right)^{1+\delta_{U,*}}}\right]^{1/(1+\chi_{0,*})}, \tag{G.33}$$

$$T_U^{\mathrm{graph}}(\mathbf{y}) = \frac{L^2 K_U}{\pi^2}\, \delta_{U,*}^{\mathrm{graph}}(\mathbf{y}), \tag{G.34}$$

$$K_{\eta,*}^{\mathrm{graph}}(\mathbf{y}) = \frac{c_{\eta U}^2}{K_U \epsilon_{\eta,*}^{\mathrm{target}}}, \tag{G.35}$$

$$\mu_W^{\mathrm{graph}}(\mathbf{y}) = \frac{\mathfrak{C}_{\mathrm{nt},*}^{\mathrm{target}}\, c_{\eta U}^2 (K_W^{(\mathrm{eff})})^2}{\epsilon_{\eta,*}^{\mathrm{target}} K_U \lambda_W^2} \left(\frac{\gamma^2 \lambda_W^2 \sigma}{K_U K_W^{(\mathrm{eff})}}\right)^{-E_*} \left(\delta_{U,*}^{\mathrm{graph}}(\mathbf{y})\right)^{F_*}. \tag{G.36}$$

The graph point is

$$\mathbf{x}_*^{\mathrm{graph}}(\mathbf{y}) = \left(\lambda_W, c_{\eta U}, \gamma, K_U, K_{\eta,*}^{\mathrm{graph}}(\mathbf{y}), K_W^{(\mathrm{eff})}, \mu_W^{\mathrm{graph}}(\mathbf{y}), T_U^{\mathrm{graph}}(\mathbf{y})\right). \tag{G.37}$$

Substitution shows

$$\mathcal{O}_* = \left\{\mathbf{x}_*^{\mathrm{graph}}(\mathbf{y}) : \mathbf{y} \in (\mathbb{R}_{>0})^5\right\}. \tag{G.38}$$

Thus the canonical projection and the graph projection coincide:

$$\Pi_{\mathcal{O}_*}^{\mathrm{can}}(\mathbf{x}) = \mathbf{x}_*^{\mathrm{graph}}(\pi_{\mathrm{free}}\mathbf{x}). \tag{G.39}$$

G.2.4 Graph-error realization coordinates

For a same-free-quintuple candidate state, define

$$E_T := \ln \frac{T_U}{T_U^{\mathrm{graph}}}, \qquad E_K := \ln \frac{K_\eta^{(\mathrm{eff})}}{K_{\eta,*}^{\mathrm{graph}}}, \qquad E_\mu := \ln \frac{\mu_W}{\mu_W^{\mathrm{graph}}}. \tag{G.40}$$

The exact relations to the quotient packet are (G.10). The reduced-family audit packet becomes

$$\Delta_{\mathrm{fam}}^{\mathrm{graph}}(\mathbf{y}) = \left(\chi_Q(\mathbf{x}_{\mathrm{cand}}(\mathbf{y})) - 1, E_T, E_K, E_\mu\right), \tag{G.41}$$

and the closure criterion is

$$\Delta_{\mathrm{fam}}^{\mathrm{graph}}(\mathbf{y}) = 0 \quad\Longleftrightarrow\quad \mathbf{x}_{\mathrm{cand}}(\mathbf{y}) \in \mathcal{Z}_*. \tag{G.42}$$

Claim status. Stages 201–202 are EXACT WITHIN CLOSURE. **They are exact consequences of the already-declared monomial quotient map, target-orbit definition, and canonical dependent-triple section. They do not prove that the actual PDE branch is graph-aligned; they provide the exact coordinates in which that claim can be tested.**

G.3 Scalar graph-slice and log-ray compiler

G.3.1 Graph closure scalar

Stage 203 defines

$$\widehat{\chi}_Q(\mathbf{y}) := \chi_Q\big(\mathbf{x}_*^{\mathrm{graph}}(\mathbf{y})\big), \qquad \widehat{\Delta}_Q(\mathbf{y}) := \widehat{\chi}_Q(\mathbf{y}) - 1. \tag{G.43}$$

Since every graph point already lies on $\mathcal{O}_*$, the closed set is exactly

$$\mathcal{Z}_* = \left\{\mathbf{x}_*^{\mathrm{graph}}(\mathbf{y}) : \widehat{\Delta}_Q(\mathbf{y}) = 0\right\}. \tag{G.44}$$

This is the scalar graph-slice theorem. Its importance is that the realization problem drops from four scalar equations to one scalar equation as soon as the candidate family is graph-aligned.

G.3.2 Graph tangent identities

Let $\mathbf{y}(\tau)$ be a smooth free-quintuple family and define free log-tangent components

$$(\lambda_1, c_1, \gamma_1, \kappa_U, \kappa_W) = \left(\frac{d\ln\lambda_W}{d\tau}, \frac{d\ln c_{\eta U}}{d\tau}, \frac{d\ln\gamma}{d\tau}, \frac{d\ln K_U}{d\tau}, \frac{d\ln K_W^{(\mathrm{eff})}}{d\tau}\right). \tag{G.45}$$

Then the dependent graph log tangents are

$$\frac{d\ln\delta_{U,*}^{\mathrm{graph}}}{d\tau} = -\frac{1+\delta_{U,*}}{1+\chi_{0,*}}(\gamma_1 + c_1 - \kappa_U), \tag{G.46}$$

$$\tau_1^{\mathrm{graph}} = \kappa_U - \frac{1+\delta_{U,*}}{1+\chi_{0,*}}(\gamma_1 + c_1 - \kappa_U), \tag{G.47}$$

$$\kappa_\eta^{\mathrm{graph}} = 2c_1 - \kappa_U, \tag{G.48}$$

$$\begin{aligned}\mu_1^{\mathrm{graph}} &= 2c_1 - \kappa_U + 2\kappa_W - 2\lambda_1 - E_*(2\gamma_1 + 2\lambda_1 - \kappa_U - \kappa_W) \\ &\quad - F_*\frac{1+\delta_{U,*}}{1+\chi_{0,*}}(\gamma_1 + c_1 - \kappa_U).\end{aligned} \tag{G.49}$$

Substitution into the monomial-drift matrix gives

$$M_*\dot{\Delta\mathbf{x}}_{\mathrm{graph}} = 0. \tag{G.50}$$

So graph-lifted families are tangent to the similarity orbit at every point, and Packet B vanishes identically on the graph.

G.3.3 One-parameter crossing theorem

Let $\mathbf{y} : [\tau_-, \tau_+] \to (\mathbb{R}_{>0})^5$ be continuous and graph-lift it. If

$$\widehat{\Delta}_Q(\tau_-)\, \widehat{\Delta}_Q(\tau_+) < 0, \tag{G.51}$$

then there exists $\tau_* \in (\tau_-, \tau_+)$ with

$$\widehat{\Delta}_Q(\tau_*) = 0, \qquad \mathbf{x}_*^{\mathrm{graph}}(\mathbf{y}(\tau_*)) \in \mathcal{Z}_*. \tag{G.52}$$

This is not a numerical method by itself. It is the exact existence theorem that justifies the later scalar log-ray search.

G.3.4 Free-quintuple log rays

Stage 204 chooses the canonical local family

$$\mathbf{y}_{\mathbf{s}}(\tau) = \mathbf{y}_\circ \odot e^{\tau \mathbf{s}}, \qquad \mathbf{s} = (s_\lambda, s_c, s_\gamma, s_U, s_W). \tag{G.53}$$

Every smooth positive free-quintuple path has this form to first order at any point, with $\mathbf{s} = d \ln \mathbf{y}/d\tau$. Along the log ray define

$$\mathfrak{a}_* := \frac{1 + \delta_{U,*}}{1 + \chi_{0,*}}, \tag{G.54}$$

$$\sigma_\delta = -\mathfrak{a}_*(s_\gamma + s_c - s_U), \tag{G.55}$$

$$\sigma_T = s_U + \sigma_\delta, \qquad \sigma_{K_\eta} = 2s_c - s_U, \tag{G.56}$$

$$\sigma_\mu = 2s_c - s_U + 2s_W - 2s_\lambda - E_*(2s_\gamma + 2s_\lambda - s_U - s_W) + F_*\sigma_\delta. \tag{G.57}$$

The graph-lifted ray is therefore

$$\begin{aligned}\mathbf{x}_{\mathbf{s}}^{\mathrm{graph}}(\tau) = (&\lambda_{W,\circ} e^{s_\lambda \tau}, c_{\eta U,\circ} e^{s_c \tau}, \gamma_\circ e^{s_\gamma \tau}, K_{U,\circ} e^{s_U \tau}, K_{\eta,\circ}^{\mathrm{graph}} e^{\sigma_{K_\eta} \tau},\\ &K_{W,\circ}^{(\mathrm{eff})} e^{s_W \tau}, \mu_{W,\circ}^{\mathrm{graph}} e^{\sigma_\mu \tau}, T_{U,\circ}^{\mathrm{graph}} e^{\sigma_T \tau}).\end{aligned} \tag{G.58}$$

The three target monomials are invariant for all τ on this lift. Hence the scalarized closure function

$$\Phi_{\mathbf{s}}(\tau) := \widehat{\chi}_Q(\mathbf{y}_{\mathbf{s}}(\tau)) \tag{G.59}$$

contains the entire graph-aligned realization question:

$$\mathbf{x}_{\mathbf{s}}^{\mathrm{graph}}(\tau) \in \mathcal{Z}_* \quad \Longleftrightarrow \quad \Phi_{\mathbf{s}}(\tau) = 1. \tag{G.60}$$

G.3.5 First-order and second-order root predictors

Let

$$\Phi_0 := \Phi_{\mathbf{s}}(0), \qquad \Phi_1 := \Phi_{\mathbf{s}}'(0), \qquad \Phi_2 := \Phi_{\mathbf{s}}''(0). \tag{G.61}$$

The first-order predictors are

$$\tau_{\mathrm{aff}} = \frac{1 - \Phi_0}{\Phi_1}, \qquad \tau_{\log} = -\frac{\ln \Phi_0}{L_0}, \qquad L_0 := \frac{\Phi_1}{\Phi_0}. \tag{G.62}$$

Stage 205 introduces free log coordinates

$$\boldsymbol{\ell} = (\ln \lambda_W, \ln c_{\eta U}, \ln \gamma, \ln K_U, \ln K_W^{(\mathrm{eff})}), \tag{G.63}$$

and the directional operators

$$\mathcal{D}_{\mathbf{s}} = \sum_i s_i \partial_{\ell_i}, \qquad \mathcal{H}_{\mathbf{s}} = \mathcal{D}_{\mathbf{s}}^2. \tag{G.64}$$

Assuming $\Phi_0 > 0$, define

$$L_1 := \frac{d^2}{d\tau^2} \ln \Phi_{\mathbf{s}}(\tau)\bigg|_0 = \frac{\Phi_2}{\Phi_0} - \left(\frac{\Phi_1}{\Phi_0}\right)^2. \tag{G.65}$$

Thus

$$\Phi_2 = \Phi_0(L_1 + L_0^2). \tag{G.66}$$

The quadratic affine and logarithmic discriminants are

$$\Delta_{\text{aff}} = \Phi_1^2 - 2\Phi_2(\Phi_0 - 1), \qquad \Delta_{\log} = L_0^2 - 2L_1 \ln \Phi_0. \tag{G.67}$$

When the appropriate discriminant is nonnegative, the continuous-branch predictors are

$$\tau_{\text{quad}} = \frac{2(1-\Phi_0)}{\Phi_1 + \operatorname{sgn}(\Phi_1)\sqrt{\Delta_{\text{aff}}}}, \tag{G.68}$$

$$\tau_{\log,2} = -\frac{2\ln \Phi_0}{L_0 + \operatorname{sgn}(L_0)\sqrt{\Delta_{\log}}}. \tag{G.69}$$

If $\Phi_1 = 0$ but $\Phi_0 \neq 1$, the quadratic turning-point criterion is

$$(1 - \Phi_0)\Phi_2 > 0, \tag{G.70}$$

with local predictors

$$\tau_\pm = \pm\sqrt{\frac{2(1-\Phi_0)}{\Phi_2}}. \tag{G.71}$$

These are the local predictor tools used by the certified search sieve.

Claim status. Stages 203–205 are Exact Within Closure **once $\widehat{\chi}_Q$ is a differentiable scalar on the target graph. The actual values of the derivatives and Hessians are PDE-returned or candidate-family data; the predictor formulas themselves are exact.**

G.4 Certified ray sieve: primitive and pairwise searches

G.4.1 Curvature-envelope brackets

Stage 206 orients the log residual. Let

$$H_{\mathbf{s}}(\tau) := \operatorname{sgn}(\ln \Phi_{\mathbf{s}}(0)) \ln \Phi_{\mathbf{s}}(\tau), \tag{G.72}$$

with

$$H_0 := H_{\mathbf{s}}(0) > 0, \qquad K_0 := H_{\mathbf{s}}'(0) < 0. \tag{G.73}$$

Assume a curvature envelope on $[0, T]$:

$$\underline{K}_1 \leq H_{\mathbf{s}}''(\tau) \leq \overline{K}_1. \tag{G.74}$$

Then

$$H_-(\tau) = H_0 + K_0\tau + \frac{1}{2}\underline{K}_1\tau^2, \qquad H_+(\tau) = H_0 + K_0\tau + \frac{1}{2}\overline{K}_1\tau^2 \tag{G.75}$$

sandwich the true residual. Define

$$\mathcal{T}(H_0, K_0; c) := -\frac{2H_0}{K_0 + \operatorname{sgn}(K_0)\sqrt{K_0^2 - 2cH_0}}. \tag{G.76}$$

For $K_0 < 0$, the certified bracket is

$$\tau_{\mathrm{lo}} = \mathcal{T}(H_0, K_0; \underline{K}_1), \qquad \tau_{\mathrm{hi}} = \mathcal{T}(H_0, K_0; \overline{K}_1), \tag{G.77}$$

provided both discriminants are real and $\tau_{\mathrm{hi}} \le T$. The true local root is then unique in $[\tau_{\mathrm{lo}}, \tau_{\mathrm{hi}}]$. For a turning ray $K_0 = 0$, strict negative curvature gives

$$\tau_{\mathrm{lo}}^{(\mathrm{tp})} = \sqrt{-\frac{2H_0}{\underline{K}_1}}, \qquad \tau_{\mathrm{hi}}^{(\mathrm{tp})} = \sqrt{-\frac{2H_0}{\overline{K}_1}}. \tag{G.78}$$

G.4.2 Primitive certified table

The primitive free axes are

$$\mathfrak{I}_5 = \{\lambda, c, \gamma, U, W\}. \tag{G.79}$$

Let

$$h(\boldsymbol{\ell}) := \ln \widehat{\chi}_Q(\boldsymbol{\ell}), \qquad \sigma_0 := \operatorname{sgn}(h(\boldsymbol{\ell}_\circ)). \tag{G.80}$$

The oriented primitive gradients are

$$\Gamma_i := \sigma_0 \partial_{\ell_i} h(\boldsymbol{\ell}_\circ). \tag{G.81}$$

If $\Gamma_i \neq 0$, the canonical forward primitive direction is

$$\widehat{\mathbf{e}}_i := \varepsilon_i \mathbf{e}_i, \qquad \varepsilon_i := -\operatorname{sgn}(\Gamma_i), \qquad k_i := |\Gamma_i|. \tag{G.82}$$

The primitive curvature uses only the diagonal Hessian restriction:

$$H_1^{(i)}(\tau) = \sigma_0 \partial_{\ell_i \ell_i} h(\boldsymbol{\ell}_\circ + \tau \widehat{\mathbf{e}}_i). \tag{G.83}$$

Thus primitive certification needs no off-diagonal Hessian data. Each primitive row is decided by

$$\mathcal{R}_i^{\mathrm{prim}} := (H_0, \Gamma_i, \underline{\kappa}_i, \overline{\kappa}_i, T_i). \tag{G.84}$$

For monotone rows, the certified interval is

$$\tau_{i,\mathrm{lo}} = \mathcal{T}(H_0, -k_i; \underline{\kappa}_i), \qquad \tau_{i,\mathrm{hi}} = \mathcal{T}(H_0, -k_i; \overline{\kappa}_i), \tag{G.85}$$

with the same discriminant and validity checks as Stage 206.

G.4.3 Pairwise mixed rays

For a monotone pair (i, j), Stage 208 introduces

$$\widehat{\mathbf{s}}_{ij}(r) = \frac{\widehat{\mathbf{e}}_i + r\widehat{\mathbf{e}}_j}{\sqrt{1 + r^2}}, \qquad r \ge 0. \tag{G.86}$$

The slope magnitude is

$$k_{ij}(r) = \frac{k_i + r k_j}{\sqrt{1 + r^2}}, \tag{G.87}$$

with unique gradient-optimal ratio

$$r_{ij}^{\text{grad}} = \frac{k_j}{k_i}, \qquad k_{ij}^{\text{grad}} = \sqrt{k_i^2 + k_j^2}. \tag{G.88}$$

The pairwise curvature is

$$H_{1,ij}(r;\tau) = \frac{h_{ii} + 2rh_{ij} + r^2 h_{jj}}{1 + r^2}, \tag{G.89}$$

so the off-diagonal Hessian enters with cross weight

$$w_\times(r) = \frac{2r}{1 + r^2}, \tag{G.90}$$

maximized at the equal-mix ray $r = 1$. Thus the two canonical pair screens are

$$\widehat{\mathbf{s}}_{ij}^{\text{grad}} = \frac{k_i \widehat{\mathbf{e}}_i + k_j \widehat{\mathbf{e}}_j}{\sqrt{k_i^2 + k_j^2}}, \qquad \widehat{\mathbf{s}}_{ij}^{\text{eq}} = \frac{\widehat{\mathbf{e}}_i + \widehat{\mathbf{e}}_j}{\sqrt{2}}. \tag{G.91}$$

G.4.4 Pairwise ratio optimizer

Stage 209 closes the full one-parameter pair search. On a compact ratio window $[0, R_{ij}]$, for envelope label $\star \in \{\text{lo}, \text{hi}\}$, define

$$A_\star = k_i^2 - 2H_0 u_\star, \qquad B_\star = 2k_i k_j - 4H_0 v_\star, \qquad C_\star = k_j^2 - 2H_0 w_\star. \tag{G.92}$$

The discriminant numerator is

$$\Delta^\sharp_{ij,\star}(r) = A_\star + B_\star r + C_\star r^2. \tag{G.93}$$

The certified root function is

$$\tau_{ij,\star}(r) = \frac{2H_0\sqrt{1 + r^2}}{k_i + rk_j + \sqrt{A_\star + B_\star r + C_\star r^2}}. \tag{G.94}$$

Interior optimizers satisfy the stationary numerator equation

$$\mathcal{N}_{ij,\star}(r) = 2(k_j - k_i r)\sqrt{A_\star + B_\star r + C_\star r^2} + B_\star + 2(C_\star - A_\star)r - B_\star r^2 = 0. \tag{G.95}$$

Eliminating the square root gives the quartic condition

$$\mathcal{Q}_{ij,\star}(r) = \Big[B_\star + 2(C_\star - A_\star)r - B_\star r^2\Big]^2 - 4(k_j - k_i r)^2(A_\star + B_\star r + C_\star r^2) = 0. \tag{G.96}$$

After filtering extraneous roots, each envelope optimizer is found in a finite set containing admissible endpoints and at most four admissible quartic roots. Therefore one pair requires at most twelve exact candidate evaluations across the lower and upper envelopes.

Claim status. Stages 206–209 are Exact Within Closure **once the local gradient, Hessian-envelope, and validity data are supplied. If those inputs come from numerical PDE evaluation, the final ray rankings are** Numerically Located **insertions into exact certified formulas.**

G.5 Three- and four-coordinate simplex sieves

G.5.1 Three-coordinate simplex and boundary reduction

For a monotone primitive triple (i, j, k), the positive spherical simplex is

$$\Delta^+_{ijk} = \left\{(a_i, a_j, a_k) \in \mathbb{R}^3_{\geq 0} : a_i^2 + a_j^2 + a_k^2 = 1\right\}. \tag{G.97}$$

The ray is

$$\widehat{\mathbf{s}}_{ijk} = a_i\widehat{\mathbf{e}}_i + a_j\widehat{\mathbf{e}}_j + a_k\widehat{\mathbf{e}}_k. \tag{G.98}$$

Each boundary edge is exactly one Stage 209 pairwise cone. Hence only the interior of Δ^+_{ijk} is new at support-cardinality three.

The gradient-optimal point is

$$\mathbf{a}^{\text{grad}}_{ijk} = \frac{(k_i, k_j, k_k)}{\sqrt{k_i^2 + k_j^2 + k_k^2}}, \tag{G.99}$$

and the equal-mix barycenter is

$$\mathbf{a}^{\text{eq}}_{ijk} = \frac{(1,1,1)}{\sqrt{3}}. \tag{G.100}$$

The total off-diagonal leverage weight is

$$w_\Sigma(\mathbf{a}) = \mathbf{2(a_ia_j + a_ia_k + a_ja_k) = (a_i + a_j + a_k)^2 - 1 \leq 2}, \tag{G.101}$$

with equality at (G.100). Thus the gradient screen maximizes first-order descent, while the equal-mix screen maximizes three-way cross-Hessian leverage.

G.5.2 Triple interior optimizer

On the interior patch $a_i > 0$, write

$$(a_i, a_j, a_k) = \frac{(1, r, s)}{\sqrt{1 + r^2 + s^2}}. \tag{G.102}$$

For either envelope label $\star$, define the quadratic discriminant numerator

$$\Delta^\sharp_{ijk,\star}(r, s) = A_\star + B_\star r + C_\star s + D_\star r^2 + E_\star rs + F_\star s^2. \tag{G.103}$$

The certified root is

$$\tau_{ijk,\star}(r, s) = \frac{2H_0\sqrt{1 + r^2 + s^2}}{k_i + rk_j + sk_k + \sqrt{\Delta^\sharp_{ijk,\star}(r, s)}}. \tag{G.104}$$

The stationary equations can be written as two square-root numerator equations. Eliminating the square root gives a quartic cross-consistency polynomial and sextic square conditions. The finite pre-candidate set has at most

$$4 \cdot 6 = 24 \tag{G.105}$$

complex roots counting multiplicity when the intersection is zero-dimensional. After filtering for admissibility and unsquared stationarity, the interior optimizer is finite.

G.5.3 Up-to-three-coordinate splice

There are $\binom{5}{2} = 10$ primitive pairs and $\binom{5}{3} = 10$ primitive triples. Stage 212 imports every optimized pairwise interval

$$\tau^{\mathrm{lo}}_{ij,\min} \le \tau^{\mathrm{best}}_{ij,*} \le \tau^{\mathrm{hi}}_{ij,\min} \tag{G.106}$$

and every triple interior interval

$$\tau^{\mathrm{lo,int}}_{ijk,\min} \le \tau^{\mathrm{best,int}}_{ijk,*} \le \tau^{\mathrm{hi,int}}_{ijk,\min}. \tag{G.107}$$

For a triple $T = (i, j, k)$, the boundary minima are

$$\beta^{\mathrm{lo}}_T := \min(\tau^{\mathrm{lo}}_{ij,\min}, \tau^{\mathrm{lo}}_{ik,\min}, \tau^{\mathrm{lo}}_{jk,\min}), \quad \beta^{\mathrm{hi}}_T := \min(\tau^{\mathrm{hi}}_{ij,\min}, \tau^{\mathrm{hi}}_{ik,\min}, \tau^{\mathrm{hi}}_{jk,\min}). \tag{G.108}$$

The closed-triple interval is

$$\tau^{\mathrm{lo},\triangle}_{T,\min} = \min(\beta^{\mathrm{lo}}_T, \tau^{\mathrm{lo,int}}_{ijk,\min}), \qquad \tau^{\mathrm{hi},\triangle}_{T,\min} = \min(\beta^{\mathrm{hi}}_T, \tau^{\mathrm{hi,int}}_{ijk,\min}). \tag{G.109}$$

The full support-≤ 3 search is therefore bounded by

$$\tau^{\mathrm{lo}}_{\le 3,\min} := \min(\tau^{\mathrm{lo}}_{2,\min}, \tau^{\mathrm{lo,int}}_{3,\min}), \qquad \tau^{\mathrm{hi}}_{\le 3,\min} := \min(\tau^{\mathrm{hi}}_{2,\min}, \tau^{\mathrm{hi,int}}_{3,\min}). \tag{G.110}$$

The exact evaluation budget through three active coordinates is at most

$$10 \times 12 + 10 \times 48 = 600 \tag{G.111}$$

candidate evaluations.

G.5.4 Four-coordinate simplex and boundary reduction

For a primitive quadruple $Q = (i, j, k, l)$,

$$\Delta^+_{ijkl} = \left\{(a_i, a_j, a_k, a_l) \in \mathbb{R}^4_{\ge 0} : a_i^2 + a_j^2 + a_k^2 + a_l^2 = 1\right\}. \tag{G.112}$$

Every codimension-one face is one of the Stage 212 closed triple simplices. The gradient-optimal interior point is

$$\mathbf{a}^{\mathrm{grad}}_{ijkl} = \frac{(k_i, k_j, k_k, k_l)}{\sqrt{k_i^2 + k_j^2 + k_k^2 + k_l^2}}, \tag{G.113}$$

and the equal-mix point is

$$\mathbf{a}^{\mathrm{eq}}_{ijkl} = \frac{(1, 1, 1, 1)}{2}. \tag{G.114}$$

The total off-diagonal leverage is

$$w_\Sigma(\mathbf{a}) = 2\sum_{p<q} a_p a_q = \left(\sum_p a_p\right)^2 - 1 \le 3, \tag{G.115}$$

with equality at (G.114). The support-cardinality-four gate asks whether either canonical interior screen, or later the full interior optimizer, beats the imported triple-face boundary ledger.

G.5.5 Four-coordinate interior optimizer and up-to-four splice

On an interior chart, Stage 214 introduces three ratio coordinates (r, s, t), one auxiliary square-root variable y, and a lifted stationary system with degree pattern

$$(3, 3, 3, 2). \tag{G.116}$$

Thus the preferred lifted candidate bound is

$$3 \cdot 3 \cdot 3 \cdot 2 = 54 \tag{G.117}$$

per envelope. Stage 215 then splices the four-coordinate interior intervals to the imported support-≤ 3 ledger. There are $\binom{5}{4} = 5$ primitive quadruples, so the four-coordinate interior budget is

$$5 \times 2 \times 54 = 540. \tag{G.118}$$

Together with (G.111), the support-≤ 4 search budget is

$$600 + 540 = 1140. \tag{G.119}$$

The certified support-≤ 4 interval is

$$\tau^{\mathrm{lo}}_{\leq 4,\min} := \min(\tau^{\mathrm{lo}}_{\leq 3,\min}, \tau^{\mathrm{lo,int}}_{4,\min}), \qquad \tau^{\mathrm{hi}}_{\leq 4,\min} := \min(\tau^{\mathrm{hi}}_{\leq 3,\min}, \tau^{\mathrm{hi,int}}_{4,\min}). \tag{G.120}$$

Claim status. Stages 210–215 are Exact Within Closure **for the simplex algebra and** Numerically Located **only when actual local packet values are inserted. Boundary reduction is exact because each simplex face is one of the previously closed lower-support searches.**

G.6 Unique five-coordinate simplex and final local closure

G.6.1 The unique five-coordinate positive simplex

Stage 216 observes that, because the free space is the five-coordinate log space

$$\mathfrak{I}_5 = \{\lambda, c, \gamma, U, W\}, \tag{G.121}$$

there is only one support-cardinality-five simplex:

$$\Delta_5^+ = \left\{ \mathbf{a} = (a_\lambda, a_c, a_\gamma, a_U, a_W) \in \mathbb{R}^5_{\geq 0} : \sum_{p \in \mathfrak{I}_5} a_p^2 = 1 \right\}. \tag{G.122}$$

Its boundary faces are precisely the five Stage 215 primitive quadruple closed simplices. If the imported face intervals are denoted by $\mathcal{I}^{\square}_{\widehat{p}}$, then

$$\beta_5^{\mathrm{lo}} := \min_{p \in \mathfrak{I}_5} \tau^{\mathrm{lo},\square}_{\widehat{p},\min}, \qquad \beta_5^{\mathrm{hi}} := \min_{p \in \mathfrak{I}_5} \tau^{\mathrm{hi},\square}_{\widehat{p},\min}. \tag{G.123}$$

Equivalently,

$$\beta_5^{\mathrm{lo}} = \tau^{\mathrm{lo}}_{\leq 4,\min}, \qquad \beta_5^{\mathrm{hi}} = \tau^{\mathrm{hi}}_{\leq 4,\min}. \tag{G.124}$$

G.6.2 Five-way canonical screens

The five-coordinate slope magnitude is

$$k_5(\mathbf{a}) = \sum_{p \in \mathfrak{I}_5} a_p k_p. \tag{G.125}$$

The unique gradient-optimal interior point is

$$\mathbf{a}_5^{\mathrm{grad}} = \frac{(k_\lambda, k_c, k_\gamma, k_U, k_W)}{\sqrt{k_\lambda^2 + k_c^2 + k_\gamma^2 + k_U^2 + k_W^2}}, \tag{G.126}$$

and the equal-mix barycenter is

$$\mathbf{a}_5^{\mathrm{eq}} = \frac{(1, 1, 1, 1, 1)}{\sqrt{5}}. \tag{G.127}$$

The total ten-way cross leverage is

$$w_\Sigma(\mathbf{a}) = 2 \sum_{p<q} a_p a_q = \left(\sum_p a_p \right)^2 - 1 \leq 4, \tag{G.128}$$

with equality at (G.127). These two screens are the exact five-way analogues of the lower-cardinality gradient and equal-mix screens. They are not the full search; they are the final pre-optimizer gate.

G.6.3 Five-coordinate interior optimizer

Stage 217 solves the last continuum problem. On the chart $a_\lambda > 0$, write

$$(a_\lambda, a_c, a_\gamma, a_U, a_W) = \frac{(1, r, s, t, u)}{\sqrt{1 + r^2 + s^2 + t^2 + u^2}}. \tag{G.129}$$

The certified root function has the same form as before:

$$\tau_\star(r, s, t, u) = \frac{2H_0 \sqrt{1 + r^2 + s^2 + t^2 + u^2}}{k_\lambda + r k_c + s k_\gamma + t k_U + u k_W + \sqrt{\Delta_\star^\sharp(r, s, t, u)}}. \tag{G.130}$$

The stationary conditions are four square-root numerator equations. Introducing

$$y := \sqrt{\Delta_\star^\sharp(r, s, t, u)} \tag{G.131}$$

gives the lifted polynomial system

$$\mathcal{F}_{r,\star} = 0, \quad \mathcal{F}_{s,\star} = 0, \quad \mathcal{F}_{t,\star} = 0, \quad \mathcal{F}_{u,\star} = 0, \quad \mathcal{F}_{\Delta,\star} = 0, \tag{G.132}$$

with degree pattern

$$(3, 3, 3, 3, 2). \tag{G.133}$$

Therefore the preferred lifted Bézout candidate bound is

$$3^4 \cdot 2 = 162 \tag{G.134}$$

per envelope. Across lower and upper envelopes the support-five interior contributes at most

$$2 \times 162 = 324 \tag{G.135}$$

preferred candidate evaluations. A fallback square-root-free projected chart gives bound 750 per envelope, hence fallback support-five budget 1500.

G.6.4 Boundary identification and support ceiling

Stage 218 proves that the boundary of Δ_5^+ is exactly the already-solved support-≤ 4 search set. The proper nonempty support strata have counts

$$5 \text{ facets} + 10 \text{ ridges} + 10 \text{ edges} + 5 \text{ vertices} = 30 = 2^5 - 2. \tag{G.136}$$

Every proper support subset is contained in at least one quadruple face, and the faces are carried as closed simplices. Thus

$$\partial\Delta_5^+ = \mathcal{S}_{\leq 4}^{\mathrm{loc}}. \tag{G.137}$$

There is also no higher-support local search, because the free log space has only five primitive axes:

$$1 \leq \#\,\mathrm{supp}(\mathbf{a}) \leq 5. \tag{G.138}$$

The true best local closure time is therefore

$$\tau_{\leq 5,*}^{\mathrm{best}} = \min(\tau_{\leq 4,*}^{\mathrm{best}}, \tau_{5,*}^{\mathrm{best,int}}). \tag{G.139}$$

G.6.5 Final splice and evaluation budget

Combining the imported support-≤ 4 interval with the support-five interior interval gives the final certified interval

$$\tau_{\leq 5,\min}^{\mathrm{lo}} = \min(\tau_{\leq 4,\min}^{\mathrm{lo}}, \tau_{5,\min}^{\mathrm{lo,int}}), \qquad \tau_{\leq 5,\min}^{\mathrm{hi}} = \min(\tau_{\leq 4,\min}^{\mathrm{hi}}, \tau_{5,\min}^{\mathrm{hi,int}}). \tag{G.140}$$

Then

$$\tau_{\leq 5,\min}^{\mathrm{lo}} \leq \tau_{\leq 5,*}^{\mathrm{best}} \leq \tau_{\leq 5,\min}^{\mathrm{hi}}. \tag{G.141}$$

The preferred exact total evaluation budget is

$$1140 + 324 = 1464. \tag{G.142}$$

The fallback projected-chart budget is

$$1140 + 1500 = 2640. \tag{G.143}$$

Thus even the fallback full local search is finite.

Theorem G.3 (Final local mixed-ray closure)**.** *Within the declared local free-quintuple mixed-ray class, the full positive-support search is exhausted by the support-*≤ 4 *boundary ledger and the unique support-five interior packet. There are no support-cardinality families beyond five. Any remaining ambiguity after Stage 218 is finite-candidate ambiguity inside the already-enumerated algebraic candidate sets, not a new continuum search.*

Claim status. Stages 216–218 are Exact Within Closure **for the boundary identification, support ceiling, lifted finite candidate compiler, splice theorem, and evaluation budget. The realized winning ray, if any, remains** Numerically Located **or** Open **until actual PDE-derived envelope and validity data are inserted and evaluated.**

G.7 What Part VI does and does not prove

Part VI proves that the local realization search has a finite, auditable structure. It does not prove that the actual nonlinear moving-throat PDE returns a branch on $\mathcal{Z}_*$. It also does not prove that the Hessian envelopes, validity maps, or candidate brackets take any particular numerical values. Those are branch data.

The safe downstream citation is therefore:

> Inside the declared free-quintuple graph and local mixed-ray search class, the realization problem reduces to the exact graph-error packet $(\chi_Q - 1, E_T, E_K, E_\mu)$ and the finite support-cardinality-five mixed-ray sieve. Actual branch realization remains a separate calculation.

For later use, the practical audit recipe is:

1. compute the PDE-returned free quintuple and dependent triple;
2. evaluate the graph target $(T_U^{\mathrm{graph}}, K_{\eta,*}^{\mathrm{graph}}, \mu_W^{\mathrm{graph}})$;
3. form $(\chi_Q - 1, E_T, E_K, E_\mu)$;
4. if the branch is graph-aligned, evaluate $\widehat{\chi}_Q$ and its local log-gradient/Hessian envelopes;
5. insert those data into the completed support-≤ 5 finite ledger.

This is exactly the point at which Part VII can start inserting same-charge and rigid-mouth actual-branch data without reopening the realization compiler.

G.8 Citation anchors and unsafe-citation guardrails

For downstream citation, use the following subanchors rather than citing the whole appendix when a narrower result is intended:

- **MTDC-T10.1**: cite for the intrinsic realization packet, canonical orbit projection, free-quintuple target graph, graph-error coordinates, scalar graph-slice theorem, and one-parameter crossing theorem.
- **MTDC-T10.2**: cite for the log-ray compiler, directional Hessian formulas, quadratic predictors, curvature-envelope brackets, certified ray ranking, primitive-ray table, and primitive elimination theorem.
- **MTDC-T10.3**: cite for pairwise mixed-ray optimization, finite quartic candidate sets, triple-simplex interior optimization, boundary splicing, and the support-≤ 3 finite search ledger.
- **MTDC-T10.4**: cite for four- and five-coordinate simplex gates, lifted finite candidate sets, boundary identification, support-cardinality ceiling, and the final support-≤ 5 closure theorem.

The unsafe citation to avoid is: "the moving-throat PDE has been solved by the realization compiler." The safe statement is: inside the declared local free-quintuple graph and mixed-ray search class, the realization problem has an exact finite support-cardinality-5 sieve; actual nonlinear branch realization remains a separate Open or Numerically Located question until branch data are inserted.

G.9 Original-stage verification cards

The preceding sections prove the Part VI compiler in theorem-block form. The following cards restore the original stage order from the working derivation. Each card records the stage purpose, imported assumptions, proof move, output, and audit checks. The cards are intentionally shorter than the theorem-block derivation above; their role is provenance and citation hygiene.

Source layout. Stages 201–218 are now sourced from the canonical files `stages/stage_201.tex` through `stages/stage_218.tex`. The raw Markdown notes and audit artifacts are tracked in a repo-local provenance index, so they do not have to appear in the reader-facing PDF. The stage files themselves carry the executable SymPy and Mathematica audit references.

G.10 Stage 201: Realization compiler and canonical orbit projection

Part / anchor. Part VI; primary anchor **MTDC-T10**, local subanchor **MTDC-T10.1**.

Claim status. Exact Within Closure.

Purpose. Stage 201 turns the reference-free four-scalar home-stretch theorem into an executable realization audit. It defines intrinsic target ratios, the quotient coordinates $(q_{\mathrm{tr}}, q_{\mathrm{nt}}, q_\eta)$, the canonical dependent-triple repair vector, and the same-free-quintuple orbit projection.

Inputs. Imports the Stage 192 quotient projector, the Stage 198 finite orbit law, and the Stage 200 packet $(\chi_Q - 1, q_{\mathrm{tr}}, q_{\mathrm{nt}}, q_\eta)$. The positive state is $\mathbf{x} = (\lambda_W, c_{\eta U}, \gamma, K_U, K_\eta^{(\mathrm{eff})}, K_W^{(\mathrm{eff})}, \mu_W, T_U)$.

Derivation ledger. The derivation forms the target monomial ratios, logs them to obtain the quotient coordinates, solves the dependent-triple repair equation $M_* \Delta\mathbf{x}_{\mathrm{rep}} = -\mathbf{q}$, exponentiates the repair, and proves the projected state is the unique target-orbit point with the same free quintuple.

Output. Intrinsic realization packet, canonical orbit projection $\Pi^{\mathrm{can}}_{\mathcal{O}_*}$, fixed-point criterion $\mathbf{x} \in \mathcal{Z}_* \iff \chi_Q = 1$ and $\Pi^{\mathrm{can}}_{\mathcal{O}_*}(\mathbf{x}) = \mathbf{x}$.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VI appendix narrative above and in the source stage. The card restores original stage order and records the claim boundary; it should not be read as a second independent proof.

Checks.

- ☐ Check target monomial ratios are formed against the declared target values before taking logarithms.
- ☐ Check the same-free-quintuple convention is used; changing the free/dependent split changes the projection chart.
- ☐ Check graph-aligned closure is not confused with actual PDE realization.
- ☐ Carry the Exact Within Closure status tag whenever citing the compiler itself.

Downstream use. This stage feeds the Part VI realization compiler and finite mixed-ray search. Downstream papers should cite **MTDC-T10.1** for this local stage range and **MTDC-T10** for the full Part VI compiler.

G.11 Stage 202: Free-quintuple target graph

Part / anchor. Part VI; primary anchor **MTDC-T10**, local subanchor **MTDC-T10.1**.

Claim status. Exact Within Closure.

Purpose. Stage 202 removes the abstract target-orbit language by solving the three monomial constraints explicitly as a graph over the five free microscopic coordinates.

Inputs. Imports the direct monomials $\mathfrak{C}_{\mathrm{tr},*}$, $\mathfrak{C}_{\mathrm{nt},*}$, and ϵ_η, together with the free quintuple $\mathbf{y} = (\lambda_W, c_{\eta U}, \gamma, K_U, K_W^{(\mathrm{eff})})$.

Derivation ledger. The derivation solves first for the split-U depth coordinate, then T_U^{graph}, then $K_{\eta,*}^{\mathrm{graph}}$, and finally μ_W^{graph}. It then proves that every target-orbit point is exactly one such graph point.

Output. Exact target graph $\mathbf{x}_*^{\mathrm{graph}}(\mathbf{y})$, graph errors (E_T, E_K, E_μ), and the first reduced-family test in graph-error coordinates.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VI appendix narrative above and in the source stage. The card restores original stage order and records the claim boundary; it should not be read as a second independent proof.

Checks.

- ☐ Check target monomial ratios are formed against the declared target values before taking logarithms.
- ☐ Check the same-free-quintuple convention is used; changing the free/dependent split changes the projection chart.
- ☐ Check graph-aligned closure is not confused with actual PDE realization.
- ☐ Carry the Exact Within Closure status tag whenever citing the compiler itself.

Downstream use. This stage feeds the Part VI realization compiler and finite mixed-ray search. Downstream papers should cite **MTDC-T10.1** for this local stage range and **MTDC-T10** for the full Part VI compiler.

G.12 Stage 203: Scalar graph-slice theorem

Part / anchor. Part VI; primary anchor **MTDC-T10**, local subanchor **MTDC-T10.1**.

Claim status. Exact Within Closure.

Purpose. Stage 203 proves that once a branch is aligned with the target graph, the only remaining closure equation is the scalar condition $\widehat{\chi}_Q(\mathbf{y}) = 1$.

Inputs. Imports the Stage 202 graph, the graph-error packet, and differentiable candidate families in the positive free-quintuple domain.

Derivation ledger. The derivation composes χ_Q with the target graph, proves graph-lifted tangents lie in $\ker M_*$, separates graph-aligned and off-graph family packets, and applies the intermediate value theorem to one-parameter graph crossings.

Output. Scalar graph slice $\mathcal{Z}_* = \{\mathbf{x}_*^{\mathrm{graph}}(\mathbf{y}) : \widehat{\chi}_Q(\mathbf{y}) = 1\}$, graph-family tangency, and one-parameter crossing theorem.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VI appendix narrative above and in the source stage. The card restores original stage order and records the claim boundary; it should not be read as a second independent proof.

Checks.

- ☐ Check target monomial ratios are formed against the declared target values before taking logarithms.
- ☐ Check the same-free-quintuple convention is used; changing the free/dependent split changes the projection chart.
- ☐ Check graph-aligned closure is not confused with actual PDE realization.
- ☐ Carry the Exact Within Closure status tag whenever citing the compiler itself.

Downstream use. This stage feeds the Part VI realization compiler and finite mixed-ray search. Downstream papers should cite **MTDC-T10.1** for this local stage range and **MTDC-T10** for the full Part VI compiler.

G.13 Stage 204: Free-quintuple log-ray sweep

Part / anchor. Part VI; primary anchor **MTDC-T10**, local subanchor **MTDC-T10.2**.

Claim status. Exact Within Closure.

Purpose. Stage 204 turns graph-aligned realization into an explicit local family compiler by sweeping logarithmic rays in the five free coordinates.

Inputs. Imports the Stage 203 scalar graph slice, base free quintuple $\mathbf{y}_\circ$, and log-ray direction $\mathbf{s} = (s_\lambda, s_c, s_\gamma, s_U, s_W)$.

Derivation ledger. The derivation constructs $\mathbf{y}_\mathbf{s}(\tau) = \mathbf{y}_\circ \odot e^{\tau\mathbf{s}}$, lifts it through the graph, proves finite monomial invariance along the lift, lists primitive directions, and defines the scalar root function $\Phi_\mathbf{s}(\tau) = \widehat{\chi}_Q(\mathbf{y}_\mathbf{s}(\tau))$.

Output. Graph-invariant log-ray family, primitive free-direction table, scalar root equation $\Phi_{\mathbf{s}}(\tau) = 1$, monotone-ray uniqueness theorem, and first local root predictors.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VI appendix narrative above and in the source stage. The card restores original stage order and records the claim boundary; it should not be read as a second independent proof.

Checks.

- ☐ Check $\widehat{\chi}_Q$ is evaluated on the target graph, not on an off-graph candidate.
- ☐ Check the log-ray validity interval before using a local bracket.
- ☐ Check curvature-envelope hypotheses before reporting a certified interval.
- ☐ Separate exact predictor formulas from numerical derivative data supplied by a branch.

Downstream use. This stage feeds the Part VI realization compiler and finite mixed-ray search. Downstream papers should cite **MTDC-T10.2** for this local stage range and **MTDC-T10** for the full Part VI compiler.

G.14 Stage 205: Directional Hessian and quadratic predictors

Part / anchor. Part VI; primary anchor **MTDC-T10**, local subanchor **MTDC-T10.2**.

Claim status. Exact Within Closure.

Purpose. Stage 205 adds second-order local information to the Stage 204 scalar root problem and treats the turning case where the first derivative vanishes.

Inputs. Imports $\Phi_{\mathbf{s}}$, free log coordinates $\boldsymbol{\ell}$, directional derivative $\mathcal{D}_{\mathbf{s}}$, and directional Hessian $\mathcal{H}_{\mathbf{s}}$.

Derivation ledger. The derivation computes (Φ_0, Φ_1, Φ_2), translates between affine and logarithmic quadratic data, derives discriminants, obtains continuous-branch quadratic predictors, and isolates the turning-point criterion $(1 - \Phi_0)\Phi_2 > 0$.

Output. Affine and logarithmic quadratic predictors, discriminant tests, curvature-corrected local expansions, and turning-point/tangency theorem.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VI appendix narrative above and in the source stage. The card restores original stage order and records the claim boundary; it should not be read as a second independent proof.

Checks.

- ☐ Check $\widehat{\chi}_Q$ is evaluated on the target graph, not on an off-graph candidate.
- ☐ Check the log-ray validity interval before using a local bracket.
- ☐ Check curvature-envelope hypotheses before reporting a certified interval.
- ☐ Separate exact predictor formulas from numerical derivative data supplied by a branch.

Downstream use. This stage feeds the Part VI realization compiler and finite mixed-ray search. Downstream papers should cite **MTDC-T10.2** for this local stage range and **MTDC-T10** for the full Part VI compiler.

G.15 Stage 206: Curvature-envelope ray ranking

Part / anchor. Part VI; primary anchor **MTDC-T10**, local subanchor **MTDC-T10.2**.

Claim status. **Mixed:** Exact Within Closure, Numerically Located.

Purpose. Stage 206 converts local ray predictors into certified local brackets by bounding the oriented log residual's curvature.

Inputs. Imports the oriented residual $H_{\mathbf{s}}$, local data (H_0, K_0), and a curvature envelope $\underline{K}_1 \leq H''_{\mathbf{s}} \leq \overline{K}_1$ on a validity interval.

Derivation ledger. The derivation builds upper/lower quadratic comparison functions, proves the root-map monotonicity in the curvature parameter, derives certified monotone and turning brackets, gives a bracket-width law, and defines pairwise interval ordering.

Output. Certified monotone bracket theorem, certified turning-ray bracket theorem, pairwise ray-ordering theorem, and local search-sieve admissibility test.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VI appendix narrative above and in the source stage. The card restores original stage order and records the claim boundary; it should not be read as a second independent proof.

Checks.

- ☐ Check $\widehat{\chi}_Q$ is evaluated on the target graph, not on an off-graph candidate.
- ☐ Check the log-ray validity interval before using a local bracket.
- ☐ Check curvature-envelope hypotheses before reporting a certified interval.
- ☐ Separate exact predictor formulas from numerical derivative data supplied by a branch.

Downstream use. This stage feeds the Part VI realization compiler and finite mixed-ray search. Downstream papers should cite **MTDC-T10.2** for this local stage range and **MTDC-T10** for the full Part VI compiler.

G.16 Stage 207: Primitive-ray certified table

Part / anchor. Part VI; primary anchor **MTDC-T10**, local subanchor **MTDC-T10.2**.

Claim status. Mixed: Exact Within Closure, Numerically Located.

Purpose. Stage 207 specializes the Stage 206 certified ray sieve to the five primitive axes of the free-quintuple log space.

Inputs. Imports $h = \ln \widehat{\chi}_Q$, the oriented primitive gradients Γ_i, diagonal Hessian envelopes on each primitive axis, and local validity radii T_i.

Derivation ledger. The derivation orients each primitive axis by the sign of its gradient, proves primitive rows need only diagonal Hessian restrictions, classifies monotone and turning primitive rows, and ranks the certified primitive intervals.

Output. Exact primitive Hessian-envelope theorem, sign-adapted primitive drift table, certified primitive ray table, primitive elimination theorem, and primitive winner theorem.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VI appendix narrative above and in the source stage. The card restores original stage order and records the claim boundary; it should not be read as a second independent proof.

Checks.

- ☐ Check $\widehat{\chi}_Q$ is evaluated on the target graph, not on an off-graph candidate.
- ☐ Check the log-ray validity interval before using a local bracket.
- ☐ Check curvature-envelope hypotheses before reporting a certified interval.
- ☐ Separate exact predictor formulas from numerical derivative data supplied by a branch.

Downstream use. This stage feeds the Part VI realization compiler and finite mixed-ray search. Downstream papers should cite **MTDC-T10.2** for this local stage range and **MTDC-T10** for the full Part VI compiler.

G.17 Stage 208: Pairwise mixed-ray audit

Part / anchor. Part VI; primary anchor **MTDC-T10**, local subanchor **MTDC-T10.3**.

Claim status. Mixed: Exact Within Closure, Numerically Located.

Purpose. Stage 208 opens the first genuinely mixed search sector by allowing two primitive coordinates to move together.

Inputs. Imports primitive axis data, off-diagonal Hessian entries, and a pairwise cone direction $\mathbf{s} = a_i\widehat{\mathbf{e}}_i + a_j\widehat{\mathbf{e}}_j$ with $a_i^2 + a_j^2 = 1$.

Derivation ledger. The derivation proves the gradient-synergy law, isolates the off-diagonal curvature leverage $2a_ia_jh_{ij}$, constructs pairwise curvature envelopes, compares the gradient-optimal and equal-mix screen rays, and states the pairwise promotion rule.

Output. Pairwise mixed-ray cone, gradient and off-diagonal Hessian synergy laws, canonical two-ray screen, promotion/deferral rule, and minimal packet for the full ratio optimizer.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VI appendix narrative above and in the source stage. The card restores original stage order and records the claim boundary; it should not be read as a second independent proof.

Checks.

- ☐ Check all pairwise and triple boundary faces are imported from already-certified lower-support ledgers.
- ☐ Check admissibility windows before including stationary candidates.
- ☐ Check interval comparison uses certified lower/upper endpoints, not just central predictor values.
- ☐ Mark candidate evaluation as Numerically Located when actual local packet values are inserted.

Downstream use. This stage feeds the Part VI realization compiler and finite mixed-ray search. Downstream papers should cite **MTDC-T10.3** for this local stage range and **MTDC-T10** for the full Part VI compiler.

G.18 Stage 209: Pairwise ratio optimizer

Part / anchor. Part VI; primary anchor **MTDC-T10**, local subanchor **MTDC-T10.3**.

Claim status. **Mixed:** Exact Within Closure, Numerically Located.

Purpose. Stage 209 solves the pairwise mixed-ray ratio problem as a finite algebraic optimization over a compact ratio window.

Inputs. Imports the pairwise cone, compact ratio coordinate, upper/lower certified objectives, admissibility inequalities, and Stage 206 bracket logic.

Derivation ledger. The derivation writes each certified objective as an algebraic function of the ratio, differentiates it, clears radicals and denominators, reduces stationary candidates to quartic equations, and splices those candidates with endpoints and admissibility boundaries.

Output. Finite pairwise candidate set, optimized pairwise bracket, special diagonal-neutral and pair-symmetric reductions, pairwise promotion theorem, and mixed-pair winner theorem.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VI appendix narrative above and in the source stage. The card restores original stage order and records the claim boundary; it should not be read as a second independent proof.

Checks.

- ☐ Check all pairwise and triple boundary faces are imported from already-certified lower-support ledgers.
- ☐ Check admissibility windows before including stationary candidates.
- ☐ Check interval comparison uses certified lower/upper endpoints, not just central predictor values.
- ☐ Mark candidate evaluation as NUMERICALLY LOCATED when actual local packet values are inserted.

Downstream use. This stage feeds the Part VI realization compiler and finite mixed-ray search. Downstream papers should cite **MTDC-T10.3** for this local stage range and **MTDC-T10** for the full Part VI compiler.

G.19 Stage 210: Three-coordinate simplex audit

Part / anchor. Part VI; primary anchor **MTDC-T10**, local subanchor **MTDC-T10.3**.

Claim status. **Mixed:** EXACT WITHIN CLOSURE, NUMERICALLY LOCATED.

Purpose. Stage 210 promotes the search to genuine three-coordinate positive simplices and proves that their boundaries are already the Stage 209 pairwise ledgers.

Inputs. Imports primitive triples, the positive spherical simplex $a_i^2 + a_j^2 + a_k^2 = 1$, pairwise edge packets, and local gradient/Hessian data.

Derivation ledger. The derivation reduces every boundary edge to an imported pairwise search, proves the three-coordinate gradient-synergy law, computes total cross leverage, defines fixed-simplex brackets, and gives canonical gradient/equal-mix triple screens.

Output. Boundary reduction for triples, three-coordinate gradient and curvature synergy, canonical triple-screen audit, interior-screen dominance criterion, and non-improvement filter.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VI appendix narrative above and in the source stage. The card restores original stage order and records the claim boundary; it should not be read as a second independent proof.

Checks.

- ☐ Check all pairwise and triple boundary faces are imported from already-certified lower-support ledgers.
- ☐ Check admissibility windows before including stationary candidates.
- ☐ Check interval comparison uses certified lower/upper endpoints, not just central predictor values.
- ☐ Mark candidate evaluation as NUMERICALLY LOCATED when actual local packet values are inserted.

Downstream use. This stage feeds the Part VI realization compiler and finite mixed-ray search. Downstream papers should cite **MTDC-T10.3** for this local stage range and **MTDC-T10** for the full Part VI compiler.

G.20 Stage 211: Triple-simplex interior optimizer

Part / anchor. Part VI; primary anchor **MTDC-T10**, local subanchor **MTDC-T10.3**.

Claim status. **Mixed:** Exact Within Closure, Numerically Located.

Purpose. Stage 211 solves the two-parameter interior optimization problem for a primitive triple.

Inputs. Imports the interior ratio chart for a triple, certified upper/lower root objectives, and compact admissible windows.

Derivation ledger. The derivation forms the two stationary numerator equations, eliminates square roots through quartic cross-consistency and sextic square conditions, constructs a finite algebraic candidate set, and compares interior brackets against optimized pairwise boundaries.

Output. Finite triple-interior candidate set, exact interior winner/non-improvement theorems, and special diagonal-isotropic and full-triple-symmetry reductions.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VI appendix narrative above and in the source stage. The card restores original stage order and records the claim boundary; it should not be read as a second independent proof.

Checks.

- ☐ Check all pairwise and triple boundary faces are imported from already-certified lower-support ledgers.
- ☐ Check admissibility windows before including stationary candidates.
- ☐ Check interval comparison uses certified lower/upper endpoints, not just central predictor values.
- ☐ Mark candidate evaluation as Numerically Located when actual local packet values are inserted.

Downstream use. This stage feeds the Part VI realization compiler and finite mixed-ray search. Downstream papers should cite **MTDC-T10.3** for this local stage range and **MTDC-T10** for the full Part VI compiler.

G.21 Stage 212: Up-to-three-coordinate search sieve

Part / anchor. Part VI; primary anchor **MTDC-T10**, local subanchor **MTDC-T10.3**.

Claim status. **Mixed:** Exact Within Closure, Numerically Located.

Purpose. Stage 212 splices the pairwise and triple-interior ledgers into the first closed multi-coordinate finite search sieve.

Inputs. Imports all optimized pairwise packets and all primitive-triple interior packets over the $\binom{5}{3} = 10$ primitive triples.

Derivation ledger. The derivation proves boundary splicing for each triple, classifies triples as interior-certified, boundary-certified, or ambiguous, ranks all primitive triples, and compares the global triple ledger against the pairwise ledger.

Output. Support-≤ 3 finite search ledger, primitive-triple ranking theorem, global three-coordinate improvement/no-improvement theorem, and finite evaluation budget 600.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VI appendix narrative above and in the source stage. The card restores original stage order and records the claim boundary; it should not be read as a second independent proof.

Checks.

- ☐ Check all pairwise and triple boundary faces are imported from already-certified lower-support ledgers.
- ☐ Check admissibility windows before including stationary candidates.
- ☐ Check interval comparison uses certified lower/upper endpoints, not just central predictor values.
- ☐ Mark candidate evaluation as NUMERICALLY LOCATED when actual local packet values are inserted.

Downstream use. This stage feeds the Part VI realization compiler and finite mixed-ray search. Downstream papers should cite **MTDC-T10.3** for this local stage range and **MTDC-T10** for the full Part VI compiler.

G.22 Stage 213: Four-coordinate simplex gate

Part / anchor. Part VI; primary anchor **MTDC-T10**, local subanchor **MTDC-T10.4**.

Claim status. **Mixed:** EXACT WITHIN CLOSURE, NUMERICALLY LOCATED.

Purpose. Stage 213 opens the four-coordinate search by reducing every four-simplex face to a closed triple packet and defining the four-way canonical screens.

Inputs. Imports the support-≤ 3 ledger, primitive quadruples, the positive four-simplex, and local gradient/Hessian data.

Derivation ledger. The derivation proves exact face reduction, derives the four-coordinate gradient-synergy law, computes six-way cross leverage, constructs fixed-simplex brackets, and states the support-cardinality-four improvement gate against the imported boundary ledger.

Output. Four-coordinate simplex gate, face-reduction theorem, four-way gradient/equal-mix screens, support-cardinality-four improvement/non-improvement filters.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VI appendix narrative above and in the source stage. The card restores original stage order and records the claim boundary; it should not be read as a second independent proof.

Checks.

- ☐ Check every proper face of a four- or five-simplex is identified with a previously closed lower-support search.
- ☐ Check lifted algebraic candidates are filtered by positivity, admissibility, and validity intervals.
- ☐ Check support-cardinality closure is limited to the declared five free coordinates.
- ☐ Do not cite the finite sieve as proof that the nonlinear PDE realizes the winning branch.

Downstream use. This stage feeds the Part VI realization compiler and finite mixed-ray search. Downstream papers should cite **MTDC-T10.4** for this local stage range and **MTDC-T10** for the full Part VI compiler.

G.23 Stage 214: Four-coordinate interior optimizer

Part / anchor. Part VI; primary anchor **MTDC-T10**, local subanchor **MTDC-T10.4**.

Claim status. Mixed: Exact Within Closure, Numerically Located.

Purpose. Stage 214 solves the three-parameter interior optimization problem for a primitive quadruple.

Inputs. Imports a four-coordinate interior ratio patch, certified root objectives, and compact admissible windows.

Derivation ledger. The derivation forms three stationary numerator equations, introduces one auxiliary square-root variable, obtains a lifted degree pattern $(3, 3, 3, 2)$, proves the 54-candidate preferred bound, and gives a square-root-free projected fallback.

Output. Finite quadruple-interior candidate set with preferred bound 54 per envelope, projected fallback bound, and interior winner/non-improvement theorem against the Stage 213 boundary ledger.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VI appendix narrative above and in the source stage. The card restores original stage order and records the claim boundary; it should not be read as a second independent proof.

Checks.

- ☐ Check every proper face of a four- or five-simplex is identified with a previously closed lower-support search.
- ☐ Check lifted algebraic candidates are filtered by positivity, admissibility, and validity intervals.
- ☐ Check support-cardinality closure is limited to the declared five free coordinates.
- ☐ Do not cite the finite sieve as proof that the nonlinear PDE realizes the winning branch.

Downstream use. This stage feeds the Part VI realization compiler and finite mixed-ray search. Downstream papers should cite **MTDC-T10.4** for this local stage range and **MTDC-T10** for the full Part VI compiler.

G.24 Stage 215: Up-to-four-coordinate search sieve

Part / anchor. Part VI; primary anchor **MTDC-T10**, local subanchor **MTDC-T10.4**.

Claim status. **Mixed:** Exact Within Closure, Numerically Located.

Purpose. Stage 215 splices all primitive quadruple interiors into the imported support-≤ 3 boundary ledger.

Inputs. Imports Stage 212 boundary packets, Stage 214 quadruple interior packets, and the $\binom{5}{4} = 5$ primitive quadruple family.

Derivation ledger. The derivation proves boundary splicing for quadruples, classifies quadruple packets, ranks certified primitive quadruples, and defines the support-≤ 4 interval by taking the minimum of the support-≤ 3 and genuine quadruple interior ledgers.

Output. Support-≤ 4 finite search ledger, primitive-quadruple winner theorem, global support-cardinality-four improvement/no-improvement theorem, and preferred cumulative budget 1140.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VI appendix narrative above and in the source stage. The card restores original stage order and records the claim boundary; it should not be read as a second independent proof.

Checks.

- ☐ Check every proper face of a four- or five-simplex is identified with a previously closed lower-support search.
- ☐ Check lifted algebraic candidates are filtered by positivity, admissibility, and validity intervals.
- ☐ Check support-cardinality closure is limited to the declared five free coordinates.
- ☐ Do not cite the finite sieve as proof that the nonlinear PDE realizes the winning branch.

Downstream use. This stage feeds the Part VI realization compiler and finite mixed-ray search. Downstream papers should cite **MTDC-T10.4** for this local stage range and **MTDC-T10** for the full Part VI compiler.

G.25 Stage 216: Unique five-coordinate simplex gate

Part / anchor. Part VI; primary anchor **MTDC-T10**, local subanchor **MTDC-T10.4**.

Claim status. **Mixed:** Exact Within Closure, Numerically Located.

Purpose. Stage 216 observes that the five-dimensional free log space has exactly one positive support-five simplex and that all of its faces are support-≤ 4 searches.

Inputs. Imports the support-≤ 4 ledger, the five primitive axes $\{\lambda, c, \gamma, U, W\}$, and the unique positive five-simplex.

Derivation ledger. The derivation identifies the boundary faces with Stage 215 quadruple closed simplices, derives the five-coordinate gradient-synergy and ten-way cross-leverage laws, states the five-way canonical screens, and proves the support-cardinality ceiling.

Output. Unique five-coordinate simplex, exact face reduction to support-≤ 4, support-cardinality-five gate, and proof that no higher-support local search exists.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VI appendix narrative above and in the source stage. The card restores original stage order and records the claim boundary; it should not be read as a second independent proof.

Checks.

- ☐ Check every proper face of a four- or five-simplex is identified with a previously closed lower-support search.
- ☐ Check lifted algebraic candidates are filtered by positivity, admissibility, and validity intervals.
- ☐ Check support-cardinality closure is limited to the declared five free coordinates.
- ☐ Do not cite the finite sieve as proof that the nonlinear PDE realizes the winning branch.

Downstream use. This stage feeds the Part VI realization compiler and finite mixed-ray search. Downstream papers should cite **MTDC-T10.4** for this local stage range and **MTDC-T10** for the full Part VI compiler.

G.26 Stage 217: Five-coordinate interior optimizer

Part / anchor. Part VI; primary anchor **MTDC-T10**, local subanchor **MTDC-T10.4**.

Claim status. **Mixed:** Exact Within Closure, Numerically Located.

Purpose. Stage 217 solves the last genuine continuum optimization problem: the interior of the unique support-five simplex.

Inputs. Imports the support-five ratio chart, certified root objectives, compact admissible windows, and the Stage 216 boundary ledger.

Derivation ledger. The derivation writes the four stationary numerator equations, lifts by one auxiliary square-root variable, obtains degree pattern $(3,3,3,3,2)$, proves the preferred 162-candidate bound per envelope, and gives a square-root-free fallback bound.

Output. Finite support-five interior candidate set, preferred budget 324, fallback budget 1500, and support-five interior improvement/non-improvement theorem.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VI appendix narrative above and in the source stage. The card restores original stage order and records the claim boundary; it should not be read as a second independent proof.

Checks.

- ☐ Check every proper face of a four- or five-simplex is identified with a previously closed lower-support search.
- ☐ Check lifted algebraic candidates are filtered by positivity, admissibility, and validity intervals.
- ☐ Check support-cardinality closure is limited to the declared five free coordinates.
- ☐ Do not cite the finite sieve as proof that the nonlinear PDE realizes the winning branch.

Downstream use. This stage feeds the Part VI realization compiler and finite mixed-ray search. Downstream papers should cite **MTDC-T10.4** for this local stage range and **MTDC-T10** for the full Part VI compiler.

G.27 Stage 218: Final local mixed-ray closure theorem

Part / anchor. Part VI; primary anchor **MTDC-T10**, local subanchor **MTDC-T10.4**.

Claim status. **Mixed:** Exact Within Closure, Numerically Located, Open.

Purpose. Stage 218 splices the unique support-five interior packet to the support-≤ 4 boundary ledger and closes the declared local mixed-ray search.

Inputs. Imports the support-≤ 4 global ledger, the support-five interior packet, and the unique five-simplex boundary stratification.

Derivation ledger. The derivation proves the boundary-identification theorem $\partial\Delta_5^+ = \mathcal{S}_{\leq 4}^{\mathrm{loc}}$, proves the support ceiling, splices the support-five interior and support-≤ 4 intervals, classifies support-five outcomes, and records the exact evaluation budget.

Output. Final local mixed-ray closure theorem, certified support-≤ 5 splice interval, preferred total budget 1464, fallback budget 2640, and finite-candidate status of any remaining ambiguity.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VI appendix narrative above and in the source stage. The card restores original stage order and records the claim boundary; it should not be read as a second independent proof.

Checks.

- ☐ Check every proper face of a four- or five-simplex is identified with a previously closed lower-support search.
- ☐ Check lifted algebraic candidates are filtered by positivity, admissibility, and validity intervals.
- ☐ Check support-cardinality closure is limited to the declared five free coordinates.
- ☐ Do not cite the finite sieve as proof that the nonlinear PDE realizes the winning branch.

Downstream use. This stage feeds the Part VI realization compiler and finite mixed-ray search. Downstream papers should cite **MTDC-T10.4** for this local stage range and **MTDC-T10** for the full Part VI compiler.

Appendix H

Stage Appendix: Part VII – One-Port Same-Charge Sieve, Actual-Branch Tests, and Rigid-Mouth Orbit Lock

Stage range: 219–242

This appendix expands the Part VII theorem-block chapter into a stage-by-stage verification ledger. The main-body chapter states the same-charge and rigid-mouth conclusions in citation-ready form; the present appendix keeps the derivational path auditable, preserving the exact stage order from Stage 219 through Stage 242. The stages are intentionally grouped into three verification questions: first, whether the one-port mixed sector creates a new long-range same-charge kernel; second, whether a short-range or resonant corridor survives the transported prefactor ceilings; and third, whether the actual rigid-mouth branch lands on the selected support and orbit-lock packet.

The appendix should be read with the same status firewall as the main body. The static kernels, projectors, packet maps, and selected-support compilers are exact inside their declared closures. Numerical pole censuses and known-data insertions are numerical placements into exact formulas. The realized nonlinear moving-throat branch remains an open input wherever the text asks whether the actual PDE branch lands on a selected point, window, or orbit-lock packet.

Part VI closed the local realization compiler: once a branch is expressed in the free-quintuple graph coordinates, the remaining local mixed-ray search is finite through support-cardinality five. Part VII is the first use of that completed back end on the same-charge, same-sign, one-port mixed-sector problem. Its job is not to add another algebraic search layer. Its job is to insert actual one-port wall/BdG/Maxwell/mixed branch data into the compiler, decide what the same-charge mixed sector can and cannot do, and then translate the surviving branch tests into the rigid-mouth physical normal form.

The result anchor served by this chapter is **MTDC-T11**. This anchor should be cited when a downstream document needs the derivation of the one-port static same-charge kernel, the dynamic phase-lag and resonance/linewidth tests, the transported actual-branch prefactor ceiling, the strict first-order even-gate package, the pure-transfer/co-loading sieve, the selected-branch dynamic-window compiler, the rigid-mouth packet projectors, the physical logarithmic chart (U, V), the Cartesian orbit-lock theorem, or the selected twin-support placement compiler.

The conceptual outcome is simple but important. The mixed sector remains real and necessary, but it does not supply a new long-range same-charge attraction. The static one-port bundle produces only short-range product kernels. The linear dynamic bundle produces no new spatial kernel class and its passive/outgoing correction is phase lag at first order. What survives is a sharply constrained short-range/resonant corridor and, on the actual rigid-mouth branch, a two-coordinate orbit-lock packet. The rigid-mouth same-charge problem is diagonal in the physical logarithmic chart

$$U := \ln \frac{\mathcal{T}^2}{\mathcal{T}_{\rm ref}^2}, \qquad V := \ln \frac{\epsilon_\eta}{\epsilon_{\eta,\rm ref}}, \tag{H.1}$$

and the strict orbit-lock point is

$$U = 0, \qquad V = 0. \tag{H.2}$$

The selected passive/outgoing support side is also no longer a three-regime ambiguity: the minimal isotropic quadrupole precursor fixes

$$\rho_\alpha = \frac{4}{3}, \qquad \zeta_{\rm req} = \frac{1}{3}, \qquad \Pi_{\rm tr} = \frac{4}{3} C_{\rm mix}, \tag{H.3}$$

placing the selected branch on the symmetric lowest-twin support slice.

Table H.1: Part VII source and status map.

Stage	Working title	Primary status	Main output
219	One-port mixed-bundle static kernel	Exact Within Closure	Exact static 3×3 bundle, determinant $\det \mathcal{K}_{\rm red} = \Delta D_0$, susceptibility kernel, outgoing-prefactor bridge, and square-law suppression verdict.
220	Dynamic mixed-port kernel and phase-lag no-go	Exact Within Closure	Dynamic 3×3 bundle, determinant $\Delta_\Pi D_\Pi$, dynamic product-family theorem, outgoing-port derivative identity, and first-order phase-lag/no-barrier theorem.
221	Resonance/linewidth tradeoff	Exact Within Closure	Simple-pole Breit–Wigner normal form, dispersive/absorptive no-free-lunch theorem, low-loss bound, and linear survival window.
222	Concrete finite-throat primitive branch	Mixed: Exact Within Closure, Numerically Located	Lowest N/N and D/N primitive branch, quartic pole polynomial, pole census, residue/linewidth figure, and first branch-level dynamic survival test.
223	Isotropic target surface and primitive compatibility	Mixed: Exact Within Closure, Numerically Located	Exact compatibility equation between the conservative one-pole condition and outgoing normalization, compatibility branch, and finite dynamic survival windows.

Stage	Working title	Primary status	Main output
224	Branch-packet compiler and actual-branch kill test	Mixed: Exact Within Closure, Open	Compiler from $(\Delta_{\rm norm}, a_{P_0}, b_{P_0})$ to lane prefactors, weak-axisymmetric law $b_{P_0} = 3a_{P_0}$, and transported ceiling in $(\Delta_{\rm norm}, \Xi_1)$.
225	Microscopic Ξ_1 compiler and first compensation surface	Mixed: Exact Within Closure, Open	First-order formulas for $u_2^{(1)}, u_4^{(1)}, \Xi_1$, primitive logarithmic-slope compiler, mechanism sieve, and first surviving mixed-sector family.
226	Strict 5PN even-gate package	Mixed: Exact Within Closure, Reduced / Controlled Reduction	Imports $\Xi_{\rm load}, K_1, H_{\rm even}$, proves $\Xi_{\rm load} = \Xi_1$, and narrows the survivor to a pure outgoing-transfer corridor with the conservative bundle frozen.
227	Pure-transfer load factor and co-loading no-go	Exact Within Closure	Exact factorization $\Lambda = P/\Delta$, identity $\Xi_1 = 2\delta \ln \Lambda$, outgoing-rigidity sieve, and co-loading no-go under simultaneous interference and hybridization rigidity.
228	Numerator/denominator split and dynamic-window test	Mixed: Exact Within Closure, Numerically Located	Split $\Xi_1 = 2(\pi_1 - \delta_1)$, numerator-rigid and denominator-rigid survivors, and first wall-like dynamic-window audit.
229	Selected-branch numerator/denominator signature	Exact Within Closure	Selected softening-depth factorization, classifier map, threshold $\delta = 8/9$, and denominator-like near-softening theorem.
230	Dynamic-window classifier and static-first theorem	Mixed: Exact Within Closure, Numerically Located	Rigid-split share compiler, dynamic sign theorem, classifier threshold, and proof that the static Ξ_1 budget remains the first kill condition.
231	Continuum-placement pullback	Exact Within Closure	Pulls the selected-branch classifier back to $(\epsilon_\eta, \epsilon_W, \rho, Z_W, \delta_0, \Lambda)$, proves monotonicity in $R_{\rm target}$, and keeps the static-first verdict on the physical placement map.

Stage	Working title	Primary status	Main output
232	Known 5PN data injection	Mixed: Numerically Located, Open	Injects numerically located Family-1 support/source data, shows the support/source side is safe by a large margin, and leaves the unresolved gate at static orbit-lock/coherent placement.
233	Rigid-mouth orbit-lock compiler	Mixed: Exact Within Closure, Open	Distinguishes mouth tracking from operator rigidity, shows $\delta \ln R_{\mathrm{tr}} = 0 \Rightarrow \Xi_1 = \delta \ln \mathfrak{N}_*$, and reads the static ceiling as an internal transfer/turbulence gate.
234	Direct branch-observable static gate	Exact Within Closure	Finite chart in $(R_{\mathrm{tr}}, R_{\mathrm{target}}, \epsilon_\eta)$, two cancellations, rigid-mouth reduction, and two-observable kill test for R_{target} and ϵ_η.
235	Rigid-mouth packet projectors	Exact Within Closure	Involutive direct-packet map, projectors P_{nt}, P_η, static-blind dressing line, and codimension-two orbit-lock theorem.
236	Microscopic dependent-plane projectors	Exact Within Closure	Dependent-plane compiler, projectors $P^{\mathrm{dep}}_{\mathrm{nt}}, P^{\mathrm{dep}}_\eta$, equal-drift dressing ray, and static-only restoration gap.
237	Actual-branch dressing compiler	Exact Within Closure	Finite static-blind curve, direct computation of q_η, microscopic extractor $q_\eta = 2c_1 - \kappa_U - \kappa_\eta$, and support-blindness theorem.
238	Physical transfer-shape compiler	Exact Within Closure	Coherent identity $R_{\mathrm{target}}\mathcal{T}^2 = \Lambda_0(1-\epsilon_\eta)$, packet factorization, and three-gate theorem: tracking, transfer-shape, dressing.
239	Rigid-mouth physical normal form	Exact Within Closure	Cartesian chart (U, V), physical-to-microscopic compiler $(\Delta_T, \Delta_{K_\eta}, \Delta_\mu) = (0, -V, U - V)$, and orbit-lock theorem $U = V = 0$.
240	Selected-branch loading ratio	Exact Within Closure	Exact loading ratio $\rho_\alpha = 4/3$, required support excess $\zeta_{\mathrm{req}} = 1/3$, and selection of the symmetric lowest-twin support slice.

Stage	Working title	Primary status	Main output
241	Primitive ranking on selected twin-support branch	Exact Within Closure	Exact selected curve $\epsilon_* = 1 - 3\varrho/2$, threshold pair $(\varrho_{W\Lambda}, \varrho_{U\Lambda})$, and primitive carrier ranking phase diagram.
242	Actual twin-support placement and coherent orbit-lock compiler	Mixed: Exact Within Closure, Open	Actual placement coordinate $\varrho_{\text{phys}} = 2(1-\epsilon)/3$, support/orbit packet split, and final front-end branch-evaluation packet.

Claim status. Part VII is mixed in status. The static and dynamic one-port kernels, the projectors, the physical chart, and the selected support-ratio compilers are Exact Within Closure **inside the declared one-port, weak-axisymmetric, selected-branch, and rigid-mouth closures. Numerical statements in Stages 222–223 and Stage 232 are** Numerically Located **insertions into exact formulas. Any claim that the completed nonlinear moving-throat PDE actually realizes the winning static placement, dynamic window, rigid-mouth orbit-lock point, or selected twin-support point remains** Open**. No result in this part should be cited as a proof of a new same-charge long-range attraction.**

H.1 Block contract and theorem-level output

The contract of Part VII is to separate three questions that are easy to conflate. First, there is the *kernel-class question*: does the mixed sector create a new long-range same-charge attractive law? Stages 219–221 answer no within the one-port linear closure. Second, there is the *window question*: can a short-range static or resonant dynamic corridor survive the exact transported ceilings? Stages 222–232 reduce that question to finite branch inequalities and show that the support/source side is not the active bottleneck in the currently injected data. Third, there is the *actual branch question*: does the PDE-selected rigid-mouth branch land on the orbit-lock and selected-support packets? Stages 233–242 compile that question into direct observables.

The operational chain is

$$\begin{aligned}\mathcal{K}_{\text{red}} &\longrightarrow \mathfrak{V}_{\text{mix}}(x,\omega) \longrightarrow (\Delta_{\text{norm}}, \Xi_1)\\ &\longrightarrow (R_{\text{tr}}, \mathcal{T}^2, \epsilon_\eta) \longrightarrow (U,V)\\ &\longrightarrow \text{selected twin-support placement packet.}\end{aligned} \tag{H.4}$$

Theorem H.1 (Part VII same-charge and rigid-mouth selected-branch theorem)**.** *Inside the declared one-port wall/BdG/Maxwell/mixed closure, weak-axisymmetric grouped response closure, rigid-mouth actual-branch closure, and selected twin-support closure, the following statements hold.*

1. *The static same-charge one-port bundle is a positive quadratic response on the admissible side $\Omega_U^2 > 0$, $\Delta > 0$, $D_0 > 0$. It produces only product kernels built from the primitive source profiles. For the carried source pair x^{-3} and $e^{-2\kappa x}/x$, the only static product families are*

$$x^{-6}, \qquad e^{-2\kappa x}x^{-4}, \qquad e^{-4\kappa x}x^{-2}. \tag{H.5}$$

2. *The linear dynamic mixed bundle preserves the same spatial families. A passive/outgoing correction at first order is pure phase lag and carries no real conservative barrier lowering off pole.*

3. *Near a simple conservative pole, every relevant susceptibility has the local form*

$$\chi_*(\omega) = \frac{A_*}{\delta - i\gamma_*}, \qquad \delta = \omega - \omega_*, \tag{H.6}$$

so

$$\frac{|\Re\chi_*|}{|\Im\chi_*|} = \frac{|\delta|}{\gamma_*}. \tag{H.7}$$

The maximum conservative gain occurs exactly when conservative and absorptive magnitudes are equal.

4. *On the explicit finite-throat primitive branch, the conservative one-pole condition and outgoing normalization condition are simultaneously compatible only on the exact surface*

$$\frac{N_0}{P_{0,\mathrm{target}}} = \frac{3(M + B_2 + Z_2)^2}{B_4 + Z_4}. \tag{H.8}$$

The transported dynamic window is finite on that surface.

5. *The actual weak-axisymmetric branch test is the transported prefactor packet*

$$(\Delta_{\mathrm{norm}}, a_{P_0}, b_{P_0}), \qquad b_{P_0} = 3a_{P_0}, \qquad \Xi_1 = \frac{P_1}{P_0}. \tag{H.9}$$

The robust one-scalar ceiling is

$$\frac{\Delta_{\mathrm{norm}} + T_{\mathrm{quad}}}{\hat{m}_0^2}\big(1 + |\varepsilon\Xi_1|\big) \le P_{\mathrm{crit}}, \qquad T_{\mathrm{quad}} := \frac{54Gc_s^5}{5a^5c^5}. \tag{H.10}$$

6. *On the rigid-mouth actual branch, the physical packet is diagonal in the variables (U, V). The microscopic dependent correction is*

$$(\Delta_T, \Delta_{K_\eta}, \Delta_\mu) = (0, -V, U - V). \tag{H.11}$$

Therefore the static-only correction changes only μ_W, while the post-static dressing correction is the equal $K_\eta^{(\mathrm{eff})}$–μ_W ray.

7. *Rigid-mouth orbit lock is the codimension-two Cartesian condition*

$$U = 0, \qquad V = 0, \tag{H.12}$$

equivalently

$$\mathcal{T}^2 = \mathcal{T}_{\mathrm{ref}}^2, \qquad \epsilon_\eta = \epsilon_{\eta,\mathrm{ref}}. \tag{H.13}$$

8. *The selected passive/outgoing support side lands exactly on the symmetric lowest-twin slice*

$$\rho_\alpha = \frac{4}{3}, \qquad \zeta_{\mathrm{req}} = \frac{1}{3}, \qquad C_{\mathrm{mix}} < \Pi_{\mathrm{tr}} < 2C_{\mathrm{mix}}. \tag{H.14}$$

9. *The actual moving-throat front-end packet to evaluate is*

$$(\epsilon, \varrho_{\mathrm{phys}}, \sigma_{\mathrm{phys}}, \textit{ranking region}, R_{\mathrm{tr}}, R_{\mathrm{target}}, \epsilon_\eta, d\ln R_{\mathrm{tr}}, d\ln R_{\mathrm{target}}, d\ln\epsilon_\eta, N_Q - 1)\,. \tag{H.15}$$

Any failure of the actual branch to realize equations (H.12), (H.14) *and* (H.15) *is a branch-realization failure, not a failure of the algebraic compiler.*

H.2 One-port same-charge static kernel

Stage 219 begins with the same isotropic one-port bundle already used by the grouped normalization chain. After the stable BdG support mode is integrated out, the effective static wall stiffness is

$$K_* := K - \frac{C^2}{\varpi^2}. \tag{H.16}$$

The one-port Maxwell/mixed block is controlled by

$$\Delta := \Omega_U^2\Omega_W^2 - R^2, \qquad Q := G_U^2\Omega_W^2 + 2G_UG_WR + G_W^2\Omega_U^2, \qquad P := \Omega_U^2 G_W + RG_U, \tag{H.17}$$

with static wall operator

$$D_0 := K_* - \frac{Q}{\Delta}. \tag{H.18}$$

The same quantities control the outgoing prefactor:

$$N_0 = \frac{P^2}{\Delta^2}, \qquad P_0 = \frac{N_0}{D_0}. \tag{H.19}$$

Thus the static same-charge audit is not independent of the quadrupole normalization program; it probes the same reduced bundle.

The reduced static energy is

$$V_{\rm stat}(q,U,W;r) = \frac{1}{2}K_*q^2 + \frac{1}{2}\Omega_U^2U^2 + \frac{1}{2}\Omega_W^2W^2 - RUW - G_UqU - G_WqW - J_q(r)q - J_U(r)U - J_W(r)W. \tag{H.20}$$

In matrix form,

$$V_{\rm stat} = \frac{1}{2}X^T\mathcal{K}_{\rm red}X - J^TX, \qquad X = (q,U,W)^T, \tag{H.21}$$

where

$$\mathcal{K}_{\rm red} = \begin{pmatrix} K_* & -G_U & -G_W \\ -G_U & \Omega_U^2 & -R \\ -G_W & -R & \Omega_W^2 \end{pmatrix}. \tag{H.22}$$

The determinant identity is

$$\det \mathcal{K}_{\rm red} = \Delta D_0. \tag{H.23}$$

The admissible static side is therefore

$$\Omega_U^2 > 0, \qquad \Delta > 0, \qquad D_0 > 0. \tag{H.24}$$

On this side, minimization gives

$$X_*(r) = \mathcal{K}_{\rm red}^{-1}J(r), \qquad \delta V_{\rm mix}(r) = -\frac{1}{2}J(r)^T\mathcal{K}_{\rm red}^{-1}J(r) \le 0. \tag{H.25}$$

So the one-port static response is attractive or neutral at quadratic order. But its spatial family is inherited from the source products, not newly generated.

The wall–mixed cross susceptibility is especially important:

$$\chi_{qW} := (\mathcal{K}_{\rm red}^{-1})_{qW} = \frac{P}{\Delta D_0}. \tag{H.26}$$

With $\Lambda := P/\Delta$, one has

$$N_0 = \Lambda^2, \qquad P_0 = \frac{\Lambda^2}{D_0}, \qquad \chi_{qW} = \frac{\Lambda}{D_0}, \qquad \chi_{qW}^2 = \frac{P_0}{D_0}. \tag{H.27}$$

So static mixed softening and outgoing quadrupole normalization are linked by the same load factor and denominator.

For the primitive source profiles

$$\mathcal{S}_Q(x) = \frac{1}{x^3}, \qquad \mathcal{S}_Y(x) = \frac{e^{-2\kappa x}}{x}, \tag{H.28}$$

and the source vector

$$J(x) = \begin{pmatrix} \beta_Q \mathcal{S}_Q(x) \\ \beta_U \mathcal{S}_Y(x) \\ \beta_W \mathcal{S}_Y(x) \end{pmatrix}, \tag{H.29}$$

the static correction is

$$\widetilde{\delta V}_{\rm mix}(x) = -\frac{1}{2}\left[\frac{\mathcal{C}_6}{x^6} + 2\mathcal{C}_4\frac{e^{-2\kappa x}}{x^4} + \mathcal{C}_2\frac{e^{-4\kappa x}}{x^2}\right]. \tag{H.30}$$

The coefficients are susceptibility-weighted source products. The key theorem is the family statement, not the coefficient details: no $-1/x$, no $-e^{-2\kappa x}/x$, and no longer-range static attraction appears inside this one-port static closure.

Theorem H.2 (Static one-port square-law suppression)**.** *Inside the admissible one-port static closure, the mixed bundle creates no new long-range same-charge attractive law. For the primitive source pair x^{-3} and $e^{-2\kappa x}/x$, it produces only the three product families in equation* (H.30)*. Static same-charge survival is therefore a coefficient-engineering or near-softening question on already-short-range families, not a new spatial-kernel theorem.*

Claim status. Stage 219 is Exact Within Closure **within the isotropic one-port static closure. The absence of a new long-range family is exact for the stated source families and bundle assumptions. The magnitude of the coefficients and their realized branch values remain branch data.**

H.3 Linear dynamic mixed-bundle kernel

Stages 220–221 ask whether time dependence rescues what the static kernel does not provide. The dynamic lift keeps the same three coordinates but replaces the static coefficients by

$$K_B(\omega) := K - M\omega^2 - \frac{C^2}{\varpi^2 - \omega^2}, \qquad A(\omega) := \Omega_U^2 - \omega^2, \qquad W(\omega) := \Omega_W^2 - \omega^2 - \Pi(\omega). \tag{H.31}$$

Then

$$\Delta_\Pi(\omega) := A(\omega)W(\omega) - R^2, \qquad Q_\Pi(\omega) := G_U^2 W(\omega) + 2G_U G_W R + G_W^2 A(\omega), \tag{H.32}$$

and

$$D_\Pi(\omega) := K_B(\omega) - \frac{Q_\Pi(\omega)}{\Delta_\Pi(\omega)}. \tag{H.33}$$

The dynamic stiffness matrix is

$$\mathcal{K}_{\mathrm{dyn}}(\omega) = \begin{pmatrix} K_B(\omega) & -G_U & -G_W \\ -G_U & A(\omega) & -R \\ -G_W & -R & W(\omega) \end{pmatrix}, \tag{H.34}$$

with determinant

$$\det \mathcal{K}_{\mathrm{dyn}}(\omega) = \Delta_\Pi(\omega) D_\Pi(\omega). \tag{H.35}$$

The complex response functional is

$$\mathfrak{V}_{\mathrm{mix}}(x,\omega) = -\frac{1}{2} J(x,\omega)^T \mathcal{K}_{\mathrm{dyn}}(\omega)^{-1} J(x,\omega). \tag{H.36}$$

Its real part is the in-phase barrier reshaping; its imaginary part is the quadrature pumping/leakage channel.

For the same primitive source families as Stage 219, one gets the exact dynamic product-family theorem

$$\mathfrak{V}_{\mathrm{mix}}(x,\omega) = -\frac{1}{2}\left[\frac{\mathcal{C}_6(\omega)}{x^6} + 2\mathcal{C}_4(\omega)\frac{e^{-2\kappa x}}{x^4} + \mathcal{C}_2(\omega)\frac{e^{-4\kappa x}}{x^2}\right]. \tag{H.37}$$

Thus linear monochromatic driving does not create a new radial family. It promotes the same coefficients to complex susceptibilities.

The outgoing port enters only through the WW slot. With $e_W = (0,0,1)^T$,

$$\partial_\Pi \mathcal{K}_{\mathrm{dyn}}^{-1} = \mathcal{K}_{\mathrm{dyn}}^{-1} e_W e_W^T \mathcal{K}_{\mathrm{dyn}}^{-1}, \tag{H.38}$$

so

$$\partial_\Pi \mathfrak{V}_{\mathrm{mix}} = -\frac{1}{2}\left[e_W^T \mathcal{K}_{\mathrm{dyn}}^{-1} J\right]^2. \tag{H.39}$$

On the conservative off-pole branch the bracket is real. Therefore, for a passive/outgoing port

$$\Pi(\omega) = i\Gamma(\omega) + O(\Gamma^2), \qquad \Gamma(\omega) \geq 0, \tag{H.40}$$

the first correction is

$$\delta\mathfrak{V}_{\mathrm{mix}}^{(1)}(x,\omega) = -\frac{i}{2}\Gamma(\omega)\mathcal{T}_J(\omega)^2, \tag{H.41}$$

where $\mathcal{T}_J = e_W^T \mathcal{K}_{\mathrm{cons}}^{-1} J$. Consequently

$$\Re\delta\mathfrak{V}_{\mathrm{mix}}^{(1)} = 0, \qquad \overline{P}_{\mathrm{abs}}^{(1)}(x,\omega) = \frac{\omega\Gamma(\omega)}{2}\mathcal{T}_J(\omega)^2 \geq 0. \tag{H.42}$$

This is the phase-lag no-go: passive/outgoing dressing gives absorption at first order, not conservative lowering.

Stage 221 then studies the only remaining dynamic escape hatch: resonant dispersive enhancement of the already-short-range families. Near a simple conservative pole, write

$$F_\Pi(\omega) = F_0'(\omega_*)\delta - \Pi(\omega_*)Z_* + O(\delta^2, \Pi\delta, \Pi^2), \qquad \delta = \omega - \omega_*. \tag{H.43}$$

For $\Pi(\omega_*) = i\Gamma_*$, the reduced susceptibility has the universal form

$$\chi_s(\omega) \approx \frac{A_*}{\delta - i\gamma_*}, \qquad A_* := \frac{\mathcal{N}_s(\omega_*)}{F_0'(\omega_*)}, \qquad \gamma_* := \frac{\Gamma_* Z_*}{|F_0'(\omega_*)|}. \tag{H.44}$$

This gives

$$|\Re\chi_*| = \frac{|A_*|}{\gamma_*}\frac{r}{1+r^2}, \qquad |\Im\chi_*| = \frac{|A_*|}{\gamma_*}\frac{1}{1+r^2}, \qquad r := \frac{|\delta|}{\gamma_*}. \tag{H.45}$$

The exact maximum of the conservative part occurs at $r = 1$, where $|\Re\chi_*| = |\Im\chi_*|$. If one demands a low-loss window

$$|\Im\chi_*| \le \eta|\Re\chi_*|, \qquad 0 < \eta \le 1, \tag{H.46}$$

then the best possible conservative magnitude is

$$\sup_{|\Im\chi_*|\le\eta|\Re\chi_*|} |\Re\chi_*| = \frac{|A_*|}{\gamma_*}\frac{\eta}{1+\eta^2}. \tag{H.47}$$

For a spatial family $S_j(x)$, a necessary low-loss survival condition is

$$\frac{|A_j|}{\gamma_*}\frac{\eta}{1+\eta^2}S_j(x)^2 \ge 2\Delta V_{\mathrm{req}}(x). \tag{H.48}$$

Theorem H.3 (Linear dynamic no-free-lunch theorem)**.** *Inside the one-port linear dynamic closure, time dependence does not create a new same-charge spatial kernel. Passive/outgoing dressing is pure phase lag at first order off pole. Near a pole, useful conservative gain and absorptive loading obey the exact line-shape tradeoff equation* (H.45)*. Therefore any surviving linear dynamic corridor must be a high-quality, residue-to-linewidth enhancement of one of the already-known short-range families.*

Claim status. Stages 220–221 are EXACT WITHIN CLOSURE **for the dynamic bundle algebra, phase-lag identity, simple-pole normal form, and low-loss survival inequality. Whether a completed branch supplies a useful pole with large enough residue-to-linewidth ratio remains** OPEN**.**

H.4 Actual-branch prefactor ceiling

Stages 222–224 turn the general dynamic survival inequality into a concrete transported branch test. Stage 222 chooses the finite interval $s \in [0, L]$, the lowest N/N zero mode

$$u_0(s) = \frac{1}{\sqrt{L}}, \tag{H.49}$$

and the lowest D/N half-wave

$$f_0(s) = \sqrt{\frac{2}{L}}\sin\frac{\pi s}{2L}. \tag{H.50}$$

The overlap is

$$\kappa = \int_0^L u_0(s)f_0(s)\,ds = \frac{2\sqrt{2}}{\pi}. \tag{H.51}$$

Thus

$$C = \kappa\lambda_B, \qquad G_U = \lambda_U, \qquad G_W = \kappa\lambda_W, \qquad R = \kappa\lambda_R. \tag{H.52}$$

The conservative pole condition is a quartic in $y = \omega^2$:

$$\begin{aligned} F(y) = &\Big((K - My)(\varpi^2 - y) - C^2\Big)\Big((\Omega_U^2 - y)(\Omega_W^2 - y) - R^2\Big) \\ &- (\varpi^2 - y)\Big(G_U^2(\Omega_W^2 - y) + 2G_UG_WR + G_W^2(\Omega_U^2 - y)\Big). \end{aligned} \tag{H.53}$$

So the primitive finite-throat branch has four conservative poles in the generic admissible case.

For a simple wall-like pole with the outgoing quadrupole port

$$\Pi_{\text{out}}(\omega) = i\Gamma_5\omega^5, \qquad \Gamma_5 = \frac{a^5}{27c_s^5}, \tag{H.54}$$

the residue/linewidth figure for the pure quadrupolar same-charge family simplifies to

$$\mathcal{R}_{Q,*} = \frac{|A_{Q,*}|}{\gamma_*} = \frac{27c_s^5}{a^5\omega_*^5 N(\omega_*)}. \tag{H.55}$$

The derivative of the pole cancels; the figure is controlled by the pole frequency and outgoing transfer factor.

Stage 223 then imposes the isotropic one-pole target surface. On the isotropic branch,

$$D_0 = K - B_0 - Z_0, \qquad D_2 = -(M + B_2 + Z_2), \qquad D_4 = -(B_4 + Z_4). \tag{H.56}$$

The one-pole identity and outgoing-normalization condition solve for the same wall stiffness. Their equality is exactly

$$\frac{N_0}{P_{0,\text{target}}} = \frac{3(M + B_2 + Z_2)^2}{B_4 + Z_4}. \tag{H.57}$$

Equivalently, the primitive branch induces the compatible target

$$P_{0,\text{target,compat}} = \frac{N_0(B_4 + Z_4)}{3(M + B_2 + Z_2)^2}. \tag{H.58}$$

This is the exact compatibility surface from the conservative one-pole side to the outgoing-normalization side.

Stage 224 transports the finite dynamic survival windows from the primitive compatibility family into the actual branch packet. The final 5PN branch packet contains

$$\Delta_{\text{branch}} = (a_2, b_2, a_4, b_4, a_{P_0}, b_{P_0}, \Delta_{\text{pole}}, \Delta_{\text{norm}}), \tag{H.59}$$

where

$$\Delta_{\text{norm}} = \hat{m}_0^2\bar{P}_0 - T_{\text{quad}}, \qquad T_{\text{quad}} := \frac{54Gc_s^5}{5a^5c^5}. \tag{H.60}$$

Hence

$$\bar{P}_0 = \frac{\Delta_{\text{norm}} + T_{\text{quad}}}{\hat{m}_0^2}. \tag{H.61}$$

The grouped inverse map gives

$$P_{20} = \bar{P}_0 + 4a_{P_0}, \qquad P_{21} = \bar{P}_0 - a_{P_0} + b_{P_0}, \qquad P_{22} = \bar{P}_0 - a_{P_0} - b_{P_0}. \tag{H.62}$$

For a ceiling P_{crit}, the all-lane transported condition is

$$\max\{P_{20}, P_{21}, P_{22}\} \le P_{\text{crit}}. \tag{H.63}$$

On the weak-axisymmetric branch,

$$\lambda_{20} = 1, \qquad \lambda_{21} = \frac{1}{2}, \qquad \lambda_{22} = -1, \tag{H.64}$$

and

$$P_A = \bar{P}_0(1 + \varepsilon\lambda_A\Xi_1), \qquad \Xi_1 = \frac{P_1}{P_0}. \tag{H.65}$$

Therefore

$$a_{P_0} = \frac{\varepsilon\bar{P}_0\Xi_1}{4}, \qquad b_{P_0} = \frac{3\varepsilon\bar{P}_0\Xi_1}{4}, \tag{H.66}$$

and the all-lane ceiling collapses to

$$\bar{P}_0(1 + |\varepsilon\Xi_1|) \le P_{\mathrm{crit}}. \tag{H.67}$$

Substituting equation (H.61) gives the actual-branch form already stated in equation (H.10).

Theorem H.4 (Transported actual-branch prefactor ceiling)**.** *Inside the transported primitive-family window closure, an actual weak-axisymmetric branch clears the robust all-lane ceiling only if*

$$\frac{\Delta_{\mathrm{norm}} + T_{\mathrm{quad}}}{\hat{m}_0^2}(1 + |\varepsilon\Xi_1|) \le P_{\mathrm{crit}}. \tag{H.68}$$

If the branch is exactly calibrated, $\Delta_{\mathrm{norm}} = 0$*, the same condition becomes a lower bound on the source-map factor,*

$$\hat{m}_0^2 \ge \frac{T_{\mathrm{quad}}(1 + |\varepsilon\Xi_1|)}{P_{\mathrm{crit}}}. \tag{H.69}$$

Claim status. Stages 222–224 are exact for the primitive branch formulas, compatibility surface, and packet compiler. Pole censuses, compatibility-point values, and numerical headroom budgets are Numerically Located**. The use of primitive-family ceilings as actual-branch sufficient conditions is a transported reduced test, not an unconditional theorem of the full PDE.**

H.5 Strict 5PN even-gate package

Stages 225–226 insert the weak-axisymmetric prefactor into the stricter first-order 5PN package. At first order write

$$D_A = D_0 + \varepsilon\lambda_A D_{01} + O(\varepsilon^2), \qquad N_A = N_0 + \varepsilon\lambda_A N_{01} + O(\varepsilon^2). \tag{H.70}$$

Then

$$\Xi_{\mathrm{load}} = \frac{N_{01}}{N_0} - \frac{D_{01}}{D_0} = \frac{P_1}{P_0} = \Xi_1. \tag{H.71}$$

The stricter imported first-order even gates are

$$K_1 := D_{21} + \frac{D_{01}}{9}, \qquad H_{\mathrm{even}} := D_{41} - \frac{2}{3}D_{21} - \frac{D_{01}}{27}. \tag{H.72}$$

Thus the same scalar Ξ_1 that controls the same-charge prefactor ceiling is only one member of the strict first-order package. A branch must also pass the conservative even gates if it is to remain on the imported 5PN one-pole/hidden-even structure.

Stage 225 first imposes a weaker compensation surface that preserves the conservative grouped response on the explicit branch. It then builds a primitive logarithmic-slope compiler from

$$(x_K, x_M, x_{\lambda_B}, x_{\varpi}, x_{\lambda_U}, x_{\lambda_W}, x_{\lambda_R}, x_{\Omega_U}, x_{\Omega_W}) \tag{H.73}$$

to

$$D_{01}, \qquad D_{21}, \qquad D_{41}, \qquad N_{01}, \qquad \Xi_1. \tag{H.74}$$

The first sieve shows that wall-only and pure-BdG-only mechanisms do not supply the desired nontrivial same-charge load once the conservative compensation is enforced. A mixed-sector family survives.

Stage 226 imports equation (H.72) and tightens the survivor. The sharpest surviving subcorridor has

$$D_{01} = D_{21} = D_{41} = 0, \qquad \Xi_1 = \Xi_{\text{load}} = \frac{N_{01}}{N_0}. \tag{H.75}$$

In words: after the strict even gates and conservative-shape preservation are imposed on the noncanonical compatibility branch, the remaining same-charge effect is not generic mixed anisotropy. It is pure outgoing-transfer enhancement with the conservative one-pole bundle frozen at first order.

Theorem H.5 (Strict first-order package and pure-transfer survivor)**.** *Inside the imported first-order 5PN package, the same-charge load scalar is exactly* $\Xi_{\text{load}} = P_1/P_0$*. Passing the strict even gates requires controlling* K_1 *and* H_{even} *in addition to* Ξ_{load}*. On the explicit compatibility branch with Stage 225 conservative-shape preservation, the sharp surviving subcorridor is the pure-transfer corridor equation* (H.75)*.*

Claim status. Stages 225–226 are Exact Within Closure **for the first-order compilers and even-gate identities. The mechanism sieve is exact within the primitive slope space, while the actual PDE realization of the pure-transfer corridor is** Open**.**

H.6 Pure-transfer corridor and co-loading no-go

Stages 227–228 analyze the pure-transfer survivor. Define the outgoing-load factor

$$\Lambda := \frac{P}{\Delta}. \tag{H.76}$$

In the pure-transfer corridor,

$$\Xi_1 = \frac{N_{01}}{N_0} = 2\frac{P_{01}}{P} - 2\frac{\Delta_{01}}{\Delta} = 2\delta \ln \Lambda. \tag{H.77}$$

The one-port factorization is

$$\Lambda = \frac{G_W}{\Omega_W^2} \frac{1+I}{1-H}, \qquad I := \frac{RG_U}{\Omega_U^2 G_W}, \qquad H := \frac{R^2}{\Omega_U^2 \Omega_W^2}. \tag{H.78}$$

The three pieces are: the raw mixed leg G_W/Ω_W^2, the interference ratio I, and the hybridization/blocking ratio H.

The outgoing-rigidity sieve asks which of those pieces can remain fixed while Ξ_1 survives. If both interference and hybridization are rigid,

$$\delta \ln I = 0, \qquad \delta \ln H = 0, \tag{H.79}$$

then the simultaneous pure-transfer and conservative constraints leave only the trivial pure-transfer drift on the concrete branch. This is the first co-loading no-go: the same-charge effect cannot be reduced to a naive mixed-leg slogan under both rigidities.

Stage 228 then splits the pure-transfer slope into numerator and denominator pieces:

$$\Xi_1 = 2(\pi_1 - \delta_1), \qquad \pi_1 := \frac{P_{01}}{P}, \qquad \delta_1 := \frac{\Delta_{01}}{\Delta}. \tag{H.80}$$

The exact rigid-subcorridor theorem is:

1. pure-transfer plus $\pi_1 = 0$ leaves a one-dimensional numerator-rigid survivor;
2. pure-transfer plus $\delta_1 = 0$ leaves a one-dimensional denominator-rigid survivor;
3. imposing both $\pi_1 = 0$ and $\delta_1 = 0$ kills the corridor.

Both one-dimensional survivors pass the first wall-like dynamic-window audit on the explicit compatibility point. The first active ceiling remains the transported static $|\varepsilon\Xi_1|$ budget, not the dynamic wall-like window.

Theorem H.6 (Pure-transfer and rigidity sieve)**.** *Inside the strict pure-transfer corridor, the same-charge scalar is the logarithmic outgoing-load slope $\Xi_1 = 2\delta \ln \Lambda$. The factorization equation* (H.78) *separates raw mixed-leg, interference, and hybridization contributions. Simultaneous interference and hybridization rigidity kills the nontrivial pure-transfer drift on the concrete compatibility branch, while either numerator-rigid or denominator-rigid splitting alone leaves a one-dimensional survivor.*

Claim status. Stages 227–228 are Exact Within Closure **for the pure-transfer identities and rigidity splits. The branch-specific dynamic-window survival statements are** Numerically Located **insertions into exact ceiling formulas.**

H.7 Selected-branch dynamic-window compiler

Stages 229–232 replace the free numerator/denominator split by the selected-branch softening-depth language and then pull the result back to physical continuum coordinates.

The selected normalization product is

$$N_-(x) = \frac{\beta_0 s_-(x)^2}{\kappa_0^2(A-x)}, \qquad 0 \le x < A, \tag{H.81}$$

with selected overlap

$$s_-(x) = \frac{\left[\kappa_0^2(x+\Delta K_{\rm ax}) + \kappa_1^2 x\right]^2}{\kappa_0^2(x+\Delta K_{\rm ax})^2 + \kappa_1^2 x^2}. \tag{H.82}$$

With dimensionless variables

$$\xi := \frac{x}{A}, \qquad \delta := \frac{\Delta K_{\rm ax}}{A}, \tag{H.83}$$

Stage 229 defines a log-slope classifier $\mathcal{R}_{ND}$ that compares the numerator-like and denominator-like content of the selected branch. The exact threshold

$$\delta = \frac{8}{9} \tag{H.84}$$

separates always-denominator selected branches from branches that begin numerator-dominant. The selected branch is never literally either rigid Stage 228 subcorridor; it is an exact co-loading product. However it is always denominator-like near softening, and it is denominator-like on the entire stable branch for $\delta \ge 8/9$.

Stage 230 uses the classifier to mix the carried numerator-rigid and denominator-rigid dynamic data. The share weights are

$$w_{\rm num} = \frac{\mathcal{R}_{ND}}{1+\mathcal{R}_{ND}}, \qquad w_{\rm den} = \frac{1}{1+\mathcal{R}_{ND}}. \tag{H.85}$$

The dynamic sign theorem says the upper wall-like pole always worsens, while the lower wall-like pole flips sign only at one classifier threshold. Most importantly, the selected-branch dynamic ceilings are everywhere weaker than the universal transported static ceilings. Therefore the first kill condition remains the static $|\varepsilon\Xi_1|$ budget.

Stage 231 pulls $\mathcal{R}_{ND}$ back through the continuum placement map. The physical classifier is

$$\mathcal{R}_{\text{phys}}(\delta, R_{\text{target}}) := \mathcal{R}_{ND}\big(\xi_{\text{req}}(\delta, R_{\text{target}}), \delta\big), \tag{H.86}$$

where ξ_{req} is the stable solution of

$$F(\xi, \delta) = R_{\text{target}}. \tag{H.87}$$

The monotonicity theorem is

$$\partial_{R_{\text{target}}} \mathcal{R}_{\text{phys}} < 0. \tag{H.88}$$

So larger normalization demand pushes the physical selected branch in the denominator-like and dynamically safer direction. The static-first theorem survives the pullback.

Stage 232 injects the known 5PN Family-1 support/source data. The selected passive/outgoing branch requires

$$\zeta_{\text{req}} = \frac{1}{3}, \qquad \rho_\alpha^{\text{req}} = \frac{4}{3}. \tag{H.89}$$

The numerically located Family-1 support/source branch returns $\zeta_{\text{phys}} \approx 2.4675$ for the carried wall-depth extractions, so the support/source side overshoots the canonical demand by a large margin. This does not prove same-charge success; it says the first active unresolved bottleneck stays at the actual PDE-selected static orbit-lock/coherent placement point.

Theorem H.7 (Selected-branch static-first theorem)**.** *Inside the selected-branch dynamic-window compiler, the wall-like dynamic window does not become the first same-charge kill condition. After continuum pullback and known support/source injection, the support/source side is non-bottlenecked in the carried data. The live unresolved test remains the actual static orbit-lock/coherent placement packet.*

Claim status. Stages 229–231 are Exact Within Closure **for the selected-branch classifier, share compiler, monotonic pullback, and static-first theorem. Stage 232 is** Numerically Located **for the injected Family-1 support/source roots and margins. The actual PDE-selected static placement remains** Open**.**

H.8 Rigid-mouth orbit-lock compiler

Stages 233–236 translate the surviving static placement problem into rigid-mouth packet language. The first firewall is that mouth rigidity and operator rigidity are not the same statement. Rigid-mouth or tracking lock is represented by

$$\delta \ln R_{\text{tr}} = 0, \tag{H.90}$$

whereas $D_{01} = 0$ is the stronger algebraic statement that the effective grouped operator does not drift at first order.

The direct finite branch observables are

$$(R_{\text{tr}}, R_{\text{target}}, \epsilon_\eta). \tag{H.91}$$

Relative to a reference branch, the finite quotient coordinates include

$$q_{\text{tr}} = -C_* \ln \frac{R_{\text{tr}}}{R_{\text{tr,ref}}}, \tag{H.92}$$

$$q_{\rm nt} = B_* \ln \frac{R_{\rm tr}}{R_{\rm tr,ref}} + \ln \frac{1-\epsilon_\eta}{1-\epsilon_{\eta,\rm ref}} - \ln \frac{R_{\rm target}}{R_{\rm target,ref}}, \tag{H.93}$$

$$q_\eta = \ln \frac{\epsilon_\eta}{\epsilon_{\eta,\rm ref}}. \tag{H.94}$$

At first order, Stage 234 gives the direct compiler

$$\Theta_1 = \delta \ln R_{\rm tr}, \qquad \Xi_1 = -\delta \ln R_{\rm target} - c_\eta \delta \ln \epsilon_\eta, \qquad \mathcal{R}_1 = \delta \ln R_{\rm target}, \tag{H.95}$$

with

$$c_\eta := \frac{\epsilon_{\eta,*}}{1-\epsilon_{\eta,*}}. \tag{H.96}$$

On the rigid-mouth slice, $q_{\rm tr} = 0$, so the static scalar is

$$\Xi_1 = -R_1 - c_\eta E_1, \qquad R_1 := \delta \ln R_{\rm target}, \qquad E_1 := \delta \ln \epsilon_\eta. \tag{H.97}$$

The two-observable static gate is therefore a condition on (R_1, E_1), not on support/source amplitude.

Stage 235 makes the packet structure exact. Define

$$\mathbf{x}_{\rm rm} := \begin{pmatrix} R_1 \\ E_1 \end{pmatrix}, \qquad \mathbf{q}_{\rm rm} := \begin{pmatrix} q_{\rm nt} \\ q_\eta \end{pmatrix} = \begin{pmatrix} \Xi_1 \\ E_1 \end{pmatrix}. \tag{H.98}$$

Then

$$\mathbf{q}_{\rm rm} = M_{\rm rm}\mathbf{x}_{\rm rm}, \qquad M_{\rm rm} := \begin{pmatrix} -1 & -c_\eta \\ 0 & 1 \end{pmatrix}, \qquad M_{\rm rm}^2 = I_2. \tag{H.99}$$

The direct-space projectors are

$$P_{\rm nt} = \begin{pmatrix} 1 & c_\eta \\ 0 & 0 \end{pmatrix}, \qquad P_\eta = \begin{pmatrix} 0 & -c_\eta \\ 0 & 1 \end{pmatrix}. \tag{H.100}$$

The static strip $\Xi_1 = 0$ is codimension one. Full rigid-mouth orbit lock is codimension two:

$$q_{\rm nt} = 0, \qquad q_\eta = 0 \quad \Longleftrightarrow \quad R_1 = 0, \qquad E_1 = 0. \tag{H.101}$$

The static-blind dressing line is

$$\Xi_1 = 0 \quad \Longleftrightarrow \quad \mathbf{x}_{\rm rm} = \begin{pmatrix} -c_\eta q_\eta \\ q_\eta \end{pmatrix}. \tag{H.102}$$

Thus clearing the first static same-charge ceiling need not make the true orbit-failure amplitude small.

Stage 236 identifies the microscopic carrier. On the rigid-mouth slice the dependent correction is

$$\begin{pmatrix} \Delta_T^{(q)} \\ \Delta_{K_\eta}^{(q)} \\ \Delta_\mu^{(q)} \end{pmatrix} = \begin{pmatrix} 0 \\ -q_\eta \\ q_{\rm nt} - q_\eta \end{pmatrix}. \tag{H.103}$$

The static-blind line $q_{\rm nt} = 0$ maps to the equal-drift ray

$$\begin{pmatrix} \Delta_T \\ \Delta_{K_\eta} \\ \Delta_\mu \end{pmatrix} = -q_\eta \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}. \tag{H.104}$$

So after the static strip is cleared, the remaining orbit defect is not generic throat motion. It is one scalar equal-drift $K_\eta^{(\rm eff)}$–μ_W dressing amplitude.

Theorem H.8 (Rigid-mouth packet and dependent-plane theorem)**.** *On the rigid-mouth slice, the first static same-charge gate tests only $q_{\rm nt} = \Xi_1$. The dressing coordinate q_η is invisible to that gate and maps microscopically to the equal-drift ray $-q_\eta(0,1,1)^T$ in the dependent triple. Therefore the static gate is necessary but not sufficient for orbit lock.*

Claim status. Stages 233–236 are Exact Within Closure **for the rigid-mouth packet compilers and projectors. The physical assumption that the actual branch is rigid-mouth/track-locked, and the realized value of q_η, remain** Open **branch data.**

H.9 Physical logarithmic chart (U, V)

Stages 237–239 diagonalize the rigid-mouth packet in physical branch variables. Stage 237 begins from the finite rigid-mouth packet

$$q_{\rm nt} = \ln \frac{1-\epsilon_\eta}{1-\epsilon_{\eta,\rm ref}} - \ln \frac{R_{\rm target}}{R_{\rm target,ref}}, \qquad q_\eta = \ln \frac{\epsilon_\eta}{\epsilon_{\eta,\rm ref}}. \tag{H.105}$$

The first static gate $q_{\rm nt} = 0$ is the finite curve

$$\frac{R_{\rm target}}{R_{\rm target,ref}} = \frac{1-\epsilon_\eta}{1-\epsilon_{\eta,\rm ref}}. \tag{H.106}$$

Parameterizing by q_η,

$$\epsilon_\eta = \epsilon_{\eta,\rm ref} e^{q_\eta}, \qquad \frac{R_{\rm target}}{R_{\rm target,ref}} = \frac{1-\epsilon_{\eta,\rm ref} e^{q_\eta}}{1-\epsilon_{\eta,\rm ref}}. \tag{H.107}$$

Conversely, once the static gate is cleared,

$$q_\eta = \ln \left(\frac{1-(1-\epsilon_{\eta,\rm ref}) R_{\rm target}/R_{\rm target,ref}}{\epsilon_{\eta,\rm ref}} \right). \tag{H.108}$$

The microscopic extractor is

$$q_\eta = 2 \ln \frac{c_{\eta U}}{c_{\eta U,\rm ref}} - \ln \frac{K_U}{K_{U,\rm ref}} - \ln \frac{K_\eta^{(\rm eff)}}{K_{\eta,\rm ref}^{(\rm eff)}}. \tag{H.109}$$

At first order,

$$q_\eta = 2c_1 - \kappa_U - \kappa_\eta. \tag{H.110}$$

The dressing coordinate is support-blind:

$$\partial_\varsigma q_\eta = 0, \qquad \partial_{M_{\rm tr}} q_\eta = 0. \tag{H.111}$$

Stage 238 then uses the coherent transfer-shape identity

$$R_{\rm target} \mathcal{T}^2 = \Lambda_0 (1-\epsilon_\eta). \tag{H.112}$$

This converts the same-charge packet into physical variables. Relative to a reference branch,

$$q_{\rm nt} + \frac{B_*}{C_*} q_{\rm tr} = \ln \frac{\mathcal{T}^2}{\mathcal{T}_{\rm ref}^2}. \tag{H.113}$$

On the rigid-mouth slice, $q_{\rm tr} = 0$, so

$$q_{\rm nt} = \ln \frac{\mathcal{T}^2}{\mathcal{T}_{\rm ref}^2}, \qquad q_\eta = \ln \frac{\epsilon_\eta}{\epsilon_{\eta,\rm ref}}. \tag{H.114}$$

Thus the first static same-charge gate is exactly transfer-shape rigidity:

$$q_{\mathrm{nt}} = 0 \quad \Longleftrightarrow \quad \mathcal{T}^2 = \mathcal{T}^2_{\mathrm{ref}}. \tag{H.115}$$

The post-static gate is dressing rigidity:

$$q_\eta = 0 \quad \Longleftrightarrow \quad \epsilon_\eta = \epsilon_{\eta,\mathrm{ref}}. \tag{H.116}$$

Stage 239 packages this in the Cartesian variables

$$U := \ln \frac{\mathcal{T}^2}{\mathcal{T}^2_{\mathrm{ref}}}, \qquad V := \ln \frac{\epsilon_\eta}{\epsilon_{\eta,\mathrm{ref}}}. \tag{H.117}$$

Then

$$q_{\mathrm{nt}} = U, \qquad q_\eta = V. \tag{H.118}$$

The target ratio factorizes as

$$\frac{R_{\mathrm{target}}}{R_{\mathrm{target,ref}}} = \frac{1 - \epsilon_{\eta,\mathrm{ref}} e^V}{1 - \epsilon_{\eta,\mathrm{ref}}} e^{-U}. \tag{H.119}$$

The dependent microscopic correction is

$$\begin{pmatrix} \Delta_T \\ \Delta_{K_\eta} \\ \Delta_\mu \end{pmatrix} = \begin{pmatrix} 0 \\ -V \\ U - V \end{pmatrix}. \tag{H.120}$$

A left inverse reads

$$U = \Delta_\mu - \Delta_{K_\eta}, \qquad V = -\Delta_{K_\eta}. \tag{H.121}$$

The pure transfer-shape axis has image

$$(U, 0) \longmapsto (0, 0, U), \tag{H.122}$$

so clearing the static defect changes only μ_W. The pure dressing axis has image

$$(0, V) \longmapsto -V(0, 1, 1), \tag{H.123}$$

the equal $K_\eta^{(\mathrm{eff})}$–μ_W ray. Therefore full orbit lock is

$$U = 0, \qquad V = 0. \tag{H.124}$$

At first order,

$$\begin{pmatrix} \Delta_T \\ \Delta_{K_\eta} \\ \Delta_\mu \end{pmatrix}^{(1)} = \begin{pmatrix} 0 \\ -\delta \ln \epsilon_\eta \\ \delta \ln \mathcal{T}^2 - \delta \ln \epsilon_\eta \end{pmatrix}. \tag{H.125}$$

Theorem H.9 (Rigid-mouth Cartesian orbit-lock theorem)**.** *On the rigid-mouth actual branch, the physical logarithmic coordinates (U, V) are the finite quotient packet. The first static gate is $U = 0$. The post-static dressing gate is $V = 0$. The full orbit-lock point is the Cartesian codimension-two condition $(U, V) = (0, 0)$, equivalently $\mathcal{T}^2 = \mathcal{T}^2_{\mathrm{ref}}$ and $\epsilon_\eta = \epsilon_{\eta,\mathrm{ref}}$.*

Claim status. Stages 237–239 are Exact Within Closure **for the finite static-blind curve, physical transfer-shape factorization, support-blindness, Cartesian chart, dependent correction, and orbit-lock theorem. Whether the actual branch has $U = V = 0$ remains** Open**.**

H.10 Selected twin-support branch and placement

Stages 240–242 close the selected support side and attach it to the actual branch packet. Stage 240 begins with the selected product identities

$$\Pi_{\mathrm{tr}} = \frac{N_Q^{(\mathrm{target})}}{\beta_0}\alpha_{\mathrm{req}}, \qquad C_{\mathrm{mix}} = \frac{N_Q^{(\mathrm{target})}}{\beta_0}\alpha_{\mathrm{mix}}. \tag{H.126}$$

Therefore

$$\frac{\Pi_{\mathrm{tr}}}{C_{\mathrm{mix}}} = \frac{\alpha_{\mathrm{req}}}{\alpha_{\mathrm{mix}}} =: \rho_\alpha. \tag{H.127}$$

For the minimal isotropic conservative quadrupole precursor

$$Y_Q^{\mathrm{cons}}(\omega) = c_0 + \frac{c_1}{1-\omega^2/\Omega_Q^2}, \qquad c_0 = \frac{3}{4}, \qquad c_1 = \frac{1}{4}, \qquad \Omega_Q = \frac{3c_s}{2a}, \tag{H.128}$$

the contact/pole inverse compiler gives

$$\rho_\alpha = \frac{1}{c_0} = \frac{4}{3}, \qquad \zeta_{\mathrm{req}} := \rho_\alpha - 1 = \frac{1}{3}. \tag{H.129}$$

Therefore

$$\Pi_{\mathrm{tr}} = \frac{4}{3}C_{\mathrm{mix}}. \tag{H.130}$$

With

$$\varrho := \frac{\pi^2\Pi_{\mathrm{tr}}}{16\Lambda}, \qquad C_{\mathrm{mix}} = \frac{8\Lambda(1-\epsilon_*)}{\pi^2}, \tag{H.131}$$

this becomes

$$\varrho = \frac{2(1-\epsilon_*)}{3}, \qquad S_{\mathrm{req}} = \frac{4}{3}. \tag{H.132}$$

Thus the selected branch lies strictly in the symmetric lowest-twin window:

$$C_{\mathrm{mix}} < \Pi_{\mathrm{tr}} < 2C_{\mathrm{mix}}. \tag{H.133}$$

Stage 241 turns this into a primitive ranking phase diagram. The selected curve is

$$\epsilon_* = 1 - \frac{3\varrho}{2}, \qquad \sigma = \frac{4}{3\varrho} - 2, \qquad 0 < \varrho < \frac{2}{3}. \tag{H.134}$$

For the constructive coherent parameter $0 < \beta < 2/11$, only two crossover thresholds survive:

$$\varrho_{W\Lambda} = \frac{2(1+\beta^2)}{3(2+\beta^2)}, \qquad \varrho_{U\Lambda} = \frac{2(1+\beta^2)}{3(1+\beta+\beta^2)}. \tag{H.135}$$

They obey

$$0 < \varrho_{W\Lambda} < \varrho_{U\Lambda} < \frac{2}{3}. \tag{H.136}$$

The ranking theorem is

$$0 < \varrho < \varrho_{W\Lambda} \quad \Longrightarrow \quad q_\chi > q_Z > q_\Lambda > q_W > |q_U|, \tag{H.137}$$

$$\varrho_{W\Lambda} < \varrho < \varrho_{U\Lambda} \quad \Longrightarrow \quad q_\chi > q_Z > q_W > q_\Lambda > |q_U|, \tag{H.138}$$

$$\varrho_{U\Lambda} < \varrho < \frac{2}{3} \quad \Longrightarrow \quad q_\chi > q_Z > q_W > |q_U| > q_\Lambda. \tag{H.139}$$

So the selected branch is dominated by interference/overlap/wall-blocking/outgoing-scale corrections, not by a large split-U correction.

Stage 242 finally places the actual coherent branch on this curve. The selected physical support coordinate is

$$\epsilon = \epsilon_W \left(1 - \frac{2}{11}\frac{\delta_U}{1+\delta_U}\right). \tag{H.140}$$

Therefore

$$\varrho_{\text{phys}} = \frac{2}{3}(1-\epsilon), \qquad \sigma_{\text{phys}} = \frac{2\epsilon}{1-\epsilon} = \frac{4}{3\varrho_{\text{phys}}} - 2. \tag{H.141}$$

The coherent placement observables are

$$R_{\text{tr}} = \frac{1+\chi_0/(1+\delta_U)}{1+\chi_0}, \tag{H.142}$$

$$R_{\text{target}} = \frac{\Lambda(1-\epsilon_\eta)(1-\epsilon)^2}{Z_W(1+\chi_0)^2}, \tag{H.143}$$

together with ϵ_η. The finite coherent orbit packet in physical placement variables is

$$q_{\text{tr}} = (1+\delta_{U,*})\ln\frac{\chi_0}{\chi_{0,\text{ref}}} + (1+\chi_{0,*})\ln\frac{\delta_U}{\delta_{U,\text{ref}}}, \tag{H.144}$$

$$q_{\text{nt}} = \ln\frac{Z_W}{Z_{W,\text{ref}}} - \ln\frac{\Lambda}{\Lambda_{\text{ref}}} + E_*\ln\frac{\epsilon_W}{\epsilon_{W,\text{ref}}} - F_*\ln\frac{\delta_U}{\delta_{U,\text{ref}}}, \tag{H.145}$$

$$q_\eta = \ln\frac{\epsilon_\eta}{\epsilon_{\eta,\text{ref}}}. \tag{H.146}$$

The infinitesimal orbit-lock conditions are equivalently

$$d\ln R_{\text{tr}} = 0, \qquad d\ln R_{\text{target}} = 0, \qquad d\ln\epsilon_\eta = 0. \tag{H.147}$$

The outgoing finish line is separate:

$$N_Q = 1, \qquad \text{or equivalently } \chi_Q = 1 \text{ on the natural source-map branch.} \tag{H.148}$$

The practical front-end packet to be returned by the completed PDE is equation (H.15); if the support lane is tracked separately, the support packet is

$$(\zeta, M_{\text{mix}}, S(\zeta;\epsilon), M_{\text{tr}}), \qquad S(\zeta;\epsilon) = 1 + \frac{\zeta(1-\epsilon)}{1-\zeta\epsilon}. \tag{H.149}$$

The support variable ζ affects the support baseline but not the orbit packet:

$$\partial_\zeta R_{\text{tr}} = \partial_\zeta R_{\text{target}} = \partial_\zeta \epsilon_\eta = \partial_\zeta q_{\text{tr}} = \partial_\zeta q_{\text{nt}} = \partial_\zeta q_\eta = 0. \tag{H.150}$$

Theorem H.10 (Selected twin-support placement and orbit-packet split)**.** *Inside the selected passive/outgoing branch, the support loading ratio is fixed to $\rho_\alpha = 4/3$, so the branch lies on the symmetric lowest-twin support slice. The realized coordinate on that slice is $\varrho_{\text{phys}} = 2(1-\epsilon)/3$. The orbit-lock packet is determined by $(\chi_0, \delta_U, Z_W, \epsilon_W, \epsilon_\eta, \Lambda)$ and is exactly blind to the support ratio ζ. Therefore support compensation and orbit lock are separate tests that must both be passed by the same completed moving-throat branch.*

Claim status. Stages 240–242 are Exact Within Closure **for the support-ratio extraction, selected curve, primitive ranking thresholds, placement formulas, and support/orbit split. Actual placement of the nonlinear branch and the final value of $N_Q - 1$ remain** Open **unless supplied by a completed PDE calculation.**

H.11 What Part VII does and does not prove

Part VII proves that the same-charge mixed sector has a much sharper reduced structure than the older placeholder language suggested. It proves, within the stated closures, that the one-port static mixed bundle cannot be cited as a source of a new long-range attraction; that the linear dynamic mixed bundle cannot be cited as a new radial kernel class; that resonant help is constrained by an exact dispersive/absorptive tradeoff; that the actual weak-axisymmetric branch test is the transported scalar packet $(\Delta_{\mathrm{norm}}, \Xi_1)$; that the rigid-mouth actual branch diagonalizes in the physical chart (U, V); and that selected support lies on the symmetric lowest-twin slice.

It does not prove that the completed nonlinear moving-throat PDE realizes the favorable point. In particular, it does not prove any of the following:

1. that the true branch has a new same-charge long-range attractive force;
2. that the true branch supplies a high-quality pole satisfying the dynamic survival window;
3. that the true branch lands within the transported static $|\varepsilon\Xi_1|$ ceiling;
4. that the rigid-mouth assumptions are realized by the full PDE;
5. that $U = V = 0$ holds on the actual branch;
6. that selected twin-support placement and outgoing normalization $N_Q = 1$ both hold without additional branch data.

The safe downstream citation is therefore:

> Inside the declared one-port, weak-axisymmetric, rigid-mouth, and selected-support closures, the same-charge mixed sector reduces to short-range static product kernels, a resonance/linewidth dynamic window, a transported prefactor packet $(\Delta_{\mathrm{norm}}, \Xi_1)$, the rigid-mouth Cartesian orbit-lock condition $U = V = 0$, and the selected lowest-twin support slice $\rho_\alpha = 4/3$. Actual branch realization remains a separate PDE calculation.

For later use, the practical audit recipe is:

1. compute the actual one-port bundle data (D_0, N_0, P_0) and check the static product-kernel assumptions;
2. evaluate the branch-compatible target surface equation (H.57);
3. form $(\Delta_{\mathrm{norm}}, \Xi_1)$ and test the transported ceiling equation (H.68);
4. if the branch is rigid-mouth, compute (U, V) from equation (H.117);
5. test orbit lock by $U = V = 0$;
6. compute ϵ, ϱ_{phys}, the selected ranking region, the support packet, and $N_Q - 1$ using equations (H.15) and (H.149).

This is exactly the point at which Part VIII can either preserve the strict front-end packet or declare a relaxed/open-system companion with an explicit recovery map back to the Stage 242 front end.

H.12 Original-stage verification cards

The cards below restore the original Stage 219–242 order. They are intentionally more compact than the theorem narrative above: each card records the stage purpose, inputs, derivation ledger, boxed output, checks, and downstream use. The cards are not independent proofs; they are a verification index for the corresponding source stages.

Source layout. Stages 219–242 are now sourced from the canonical files `stages/stage_219.tex` through `stages/stage_242.tex`. The raw Markdown notes and audit artifacts are tracked in a repo-local provenance index, so they do not have to appear in the reader-facing PDF. The stage files themselves carry the executable SymPy and Mathematica audit references.

H.13 Stage 219: One-port mixed-bundle static kernel

Part / anchor. Part VII; primary anchor **MTDC-T11**, local subanchor **MTDC-T11.1**.

Claim status. EXACT WITHIN CLOSURE.

Purpose. Stage 219 inserts the isotropic one-port wall/BdG/Maxwell/mixed bundle into the completed same-charge audit and asks whether the static mixed sector creates a new long-range attractive family.

Inputs. Imports the one-port static bundle, the admissible conditions $\Omega_U^2 > 0$, $\Delta > 0$, $D_0 > 0$, the primitive source profiles x^{-3} and $e^{-2\kappa x}/x$, and the outgoing-prefactor bridge $P_0 = N_0/D_0$.

Derivation ledger. The derivation forms the static 3×3 kernel $\mathcal{K}_{\rm red}$, proves $\det \mathcal{K}_{\rm red} = \Delta D_0$, computes the inverse susceptibility entries, factors the wall–mixed load $\Lambda = P/\Delta$, and expands the on-shell correction $-J^T \mathcal{K}_{\rm red}^{-1} J/2$ into source products.

Output. Exact square-law suppression theorem: the one-port static mixed bundle produces only x^{-6}, $e^{-2\kappa x}x^{-4}$, and $e^{-4\kappa x}x^{-2}$ product families, and no new $-1/x$ or Yukawa-strength long-range same-charge attraction.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VII appendix narrative above and in the source stage. The card preserves the original stage order and records the claim boundary; it should not be read as a second independent proof of actual nonlinear branch realization.

Checks.

- ☐ Check the stage assumptions against the declared one-port, selected-branch, weak-axisymmetric, or rigid-mouth closure before citing the output.
- ☐ Check whether the row is algebraic, numerical placement, or open branch-realization before transferring it to another paper.
- ☐ Check that same-charge statements are not reinterpreted as a new long-range attraction; all static kernel claims inherit the source-product family limits.

☐ Check that orbit-lock claims require the rigid-mouth packet variables and do not follow from generic mouth tracking alone.

Downstream use. This stage feeds the Part VII same-charge sieve, rigid-mouth normal form, and selected twin-support branch. Downstream papers should cite **MTDC-T11.1** for this local stage range and **MTDC-T11** for the full Part VII compiler.

H.14 Stage 220: Dynamic mixed-port kernel and phase-lag no-go

Part / anchor. Part VII; primary anchor **MTDC-T11**, local subanchor **MTDC-T11.1**.

Claim status. Exact Within Closure.

Purpose. Stage 220 promotes the one-port bundle to finite frequency and tests whether linear dynamic driving changes the same-charge kernel class.

Inputs. Imports Stage 219, frequency-dependent BdG and Maxwell/mixed denominators, the outgoing dressing $\Pi_{\rm out}(\omega)$, and the off-pole conditions $\varpi^2 - \omega^2 \neq 0$, $\Delta_\Pi \neq 0$, $D_\Pi \neq 0$.

Derivation ledger. The derivation builds the dynamic 3×3 susceptibility, proves the determinant identity $\det \mathcal{K}_\Pi = \Delta_\Pi D_\Pi$, repeats the product-family factorization at finite frequency, differentiates the outgoing-port insertion, and separates in-phase conservative reshaping from quadrature pumping.

Output. Dynamic product-family theorem and phase-lag no-go: linear dynamic mixing preserves the static spatial families, and the first-order passive/outgoing insertion is a phase-lag term rather than a new real barrier-lowering kernel off resonance.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VII appendix narrative above and in the source stage. The card preserves the original stage order and records the claim boundary; it should not be read as a second independent proof of actual nonlinear branch realization.

Checks.

☐ Check the stage assumptions against the declared one-port, selected-branch, weak-axisymmetric, or rigid-mouth closure before citing the output.

☐ Check whether the row is algebraic, numerical placement, or open branch-realization before transferring it to another paper.

☐ Check that same-charge statements are not reinterpreted as a new long-range attraction; all static kernel claims inherit the source-product family limits.

☐ Check that orbit-lock claims require the rigid-mouth packet variables and do not follow from generic mouth tracking alone.

Downstream use. This stage feeds the Part VII same-charge sieve, rigid-mouth normal form, and selected twin-support branch. Downstream papers should cite **MTDC-T11.1** for this local stage range and **MTDC-T11** for the full Part VII compiler.

H.15 Stage 221: Resonance/linewidth tradeoff

Part / anchor. Part VII; primary anchor **MTDC-T11**, local subanchor **MTDC-T11.1**.

Claim status. Exact Within Closure.

Purpose. Stage 221 analyzes the only surviving linear dynamic escape hatch: resonant dispersive enhancement of the already-short-range mixed kernels.

Inputs. Imports Stage 220 and expands near a simple conservative pole $F_0(\omega_*) = 0$ with $F_0'(\omega_*) \neq 0$; the wall-like specialization also assumes $D_0(\omega_*) = 0$ and $\Delta_0(\omega_*) \neq 0$.

Derivation ledger. The derivation reduces the susceptibility to the Breit–Wigner form $A_*/(\delta - i\gamma_*)$, computes the real and imaginary line shapes, rewrites the result in absorbed-power language, and derives the low-loss survival window.

Output. Dispersive/no-free-lunch theorem: $|\Re\chi_*|/|\Im\chi_*| = |\delta|/\gamma_*$, so resonant conservative gain is inseparable from absorptive loading, with maximal balanced leverage at equal conservative and absorptive magnitudes.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VII appendix narrative above and in the source stage. The card preserves the original stage order and records the claim boundary; it should not be read as a second independent proof of actual nonlinear branch realization.

Checks.

- ☐ Check the stage assumptions against the declared one-port, selected-branch, weak-axisymmetric, or rigid-mouth closure before citing the output.
- ☐ Check whether the row is algebraic, numerical placement, or open branch-realization before transferring it to another paper.
- ☐ Check that same-charge statements are not reinterpreted as a new long-range attraction; all static kernel claims inherit the source-product family limits.
- ☐ Check that orbit-lock claims require the rigid-mouth packet variables and do not follow from generic mouth tracking alone.

Downstream use. This stage feeds the Part VII same-charge sieve, rigid-mouth normal form, and selected twin-support branch. Downstream papers should cite **MTDC-T11.1** for this local stage range and **MTDC-T11** for the full Part VII compiler.

H.16 Stage 222: Concrete finite-throat primitive branch

Part / anchor. Part VII; primary anchor **MTDC-T11**, local subanchor **MTDC-T11.2**.

Claim status. **Mixed:** Exact Within Closure, Numerically Located.

Purpose. Stage 222 inserts the lowest concrete finite-throat primitive branch into the resonance/linewidth theorem.

Inputs. Imports the lowest N/N wall–U profile, the lowest D/N half-wave support–W profile, the one-port mixed kernel, and the line-shape survival gate from Stage 221.

Derivation ledger. The derivation evaluates the exact overlap constants for the primitive branch, forms the quartic pole polynomial, identifies wall-like and internal poles, computes residues and linewidths, and tests the resulting low-loss survival inequality on an illustrative admissible slice.

Output. Concrete branch pole census, residue/linewidth cancellation, and first branch-level dynamic survival test. The pole polynomial and survival gate are exact in the primitive closure; the displayed slice is a numerical placement.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VII appendix narrative above and in the source stage. The card preserves the original stage order and records the claim boundary; it should not be read as a second independent proof of actual nonlinear branch realization.

Checks.

- ☐ Check the stage assumptions against the declared one-port, selected-branch, weak-axisymmetric, or rigid-mouth closure before citing the output.
- ☐ Check whether the row is algebraic, numerical placement, or open branch-realization before transferring it to another paper.
- ☐ Check that same-charge statements are not reinterpreted as a new long-range attraction; all static kernel claims inherit the source-product family limits.
- ☐ Check that orbit-lock claims require the rigid-mouth packet variables and do not follow from generic mouth tracking alone.

Downstream use. This stage feeds the Part VII same-charge sieve, rigid-mouth normal form, and selected twin-support branch. Downstream papers should cite **MTDC-T11.2** for this local stage range and **MTDC-T11** for the full Part VII compiler.

H.17 Stage 223: Isotropic target surface and primitive compatibility

Part / anchor. Part VII; primary anchor **MTDC-T11**, local subanchor **MTDC-T11.2**.

Claim status. **Mixed:** Exact Within Closure, Numerically Located.

Purpose. Stage 223 asks whether the same primitive branch can satisfy the conservative one-pole condition and the outgoing normalization target simultaneously.

Inputs. Imports the isotropic full-bundle moments $D_0, D_2, D_4, N_0, N_2, N_4$, the one-pole identity $u_4 = 4u_2^2$, and the universal target $P_{0,\text{target}}$.

Derivation ledger. The derivation eliminates the static stiffness through the one-pole condition and compares the resulting D_0 with the normalization equation $P_0 = N_0/D_0 = P_{0,\text{target}}$.

Output. Exact primitive compatibility surface $N_0/P_{0,\text{target}} = 3(M + B_2 + Z_2)^2/(B_4 + Z_4)$ and finite dynamic survival windows on the compatible branch.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VII appendix narrative above and in the source stage. The card preserves the original stage order and records the claim boundary; it should not be read as a second independent proof of actual nonlinear branch realization.

Checks.

- ☐ Check the stage assumptions against the declared one-port, selected-branch, weak-axisymmetric, or rigid-mouth closure before citing the output.
- ☐ Check whether the row is algebraic, numerical placement, or open branch-realization before transferring it to another paper.
- ☐ Check that same-charge statements are not reinterpreted as a new long-range attraction; all static kernel claims inherit the source-product family limits.
- ☐ Check that orbit-lock claims require the rigid-mouth packet variables and do not follow from generic mouth tracking alone.

Downstream use. This stage feeds the Part VII same-charge sieve, rigid-mouth normal form, and selected twin-support branch. Downstream papers should cite **MTDC-T11.2** for this local stage range and **MTDC-T11** for the full Part VII compiler.

H.18 Stage 224: Branch-packet compiler and actual-branch kill test

Part / anchor. Part VII; primary anchor **MTDC-T11**, local subanchor **MTDC-T11.2**.

Claim status. **Mixed:** EXACT WITHIN CLOSURE, OPEN.

Purpose. Stage 224 converts the primitive-branch compatibility result into the actual weak-axisymmetric branch packet that the moving-throat PDE must return.

Inputs. Imports the normalized prefactor packet $(\Delta_{\text{norm}}, a_{P_0}, b_{P_0})$, the weak-axisymmetric signature $(1, 1/2, -1)$, the line $b_{P_0} = 3a_{P_0}$, and the target scale $T_{\text{quad}} = 54Gc_s^5/(5a^5c^5)$.

Derivation ledger. The derivation maps grouped lane prefactors into trace/anomaly coordinates, rewrites the anisotropic packet in terms of $\Xi_1 = P_1/P_0$, and transports the static normalization margin through the one-scalar weak-axisymmetric ceiling.

Output. Actual-branch kill test: the branch survives only if the transported quantity $(\Delta_{\rm norm} + T_{\rm quad})(1 + |\varepsilon\Xi_1|)/\hat{m}_0^2$ stays below the critical prefactor budget.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VII appendix narrative above and in the source stage. The card preserves the original stage order and records the claim boundary; it should not be read as a second independent proof of actual nonlinear branch realization.

Checks.

- ☐ Check the stage assumptions against the declared one-port, selected-branch, weak-axisymmetric, or rigid-mouth closure before citing the output.
- ☐ Check whether the row is algebraic, numerical placement, or open branch-realization before transferring it to another paper.
- ☐ Check that same-charge statements are not reinterpreted as a new long-range attraction; all static kernel claims inherit the source-product family limits.
- ☐ Check that orbit-lock claims require the rigid-mouth packet variables and do not follow from generic mouth tracking alone.

Downstream use. This stage feeds the Part VII same-charge sieve, rigid-mouth normal form, and selected twin-support branch. Downstream papers should cite **MTDC-T11.2** for this local stage range and **MTDC-T11** for the full Part VII compiler.

H.19 Stage 225: Microscopic Xi compiler and first compensation surface

Part / anchor. Part VII; primary anchor **MTDC-T11**, local subanchor **MTDC-T11.3**.

Claim status. **Mixed:** Exact Within Closure, Open.

Purpose. Stage 225 derives the first microscopic formula for the weak-axisymmetric prefactor slope Ξ_1.

Inputs. Imports the linear grouped variations $D_{01}, D_{21}, D_{41}, N_{01}$, the physical slopes $u_2^{(1)}$, $u_4^{(1)}$, and the primitive logarithmic slopes of wall, BdG, and Maxwell/mixed data.

Derivation ledger. The derivation expands $u_2 = -D_2/D_0$, $u_4 = (D_2^2 - D_0D_4)/D_0^2$, and $P_0 = N_0/D_0$ to first weak-axisymmetric order, then solves the first conservative compensation surface and sorts wall-only, BdG-only, and mixed-sector mechanisms.

Output. Microscopic compiler $\Xi_1 = N_{01}/N_0 - D_{01}/D_0$ together with first-order even-preserving compensation conditions and the first surviving mixed-sector family.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VII appendix narrative above and in the source stage. The card preserves the original stage order and records the claim boundary; it should not be read as a second independent proof of actual nonlinear branch realization.

Checks.

- ☐ Check the stage assumptions against the declared one-port, selected-branch, weak-axisymmetric, or rigid-mouth closure before citing the output.
- ☐ Check whether the row is algebraic, numerical placement, or open branch-realization before transferring it to another paper.
- ☐ Check that same-charge statements are not reinterpreted as a new long-range attraction; all static kernel claims inherit the source-product family limits.
- ☐ Check that orbit-lock claims require the rigid-mouth packet variables and do not follow from generic mouth tracking alone.

Downstream use. This stage feeds the Part VII same-charge sieve, rigid-mouth normal form, and selected twin-support branch. Downstream papers should cite **MTDC-T11.3** for this local stage range and **MTDC-T11** for the full Part VII compiler.

H.20 Stage 226: Strict 5PN even-gate package

Part / anchor. Part VII; primary anchor **MTDC-T11**, local subanchor **MTDC-T11.3**.

Claim status. Mixed: Exact Within Closure, Reduced / Controlled Reduction.

Purpose. Stage 226 imports the strict first-order conservative gates and separates the normalization defect from the even-response defects.

Inputs. Imports Ξ_{load}, K_1, H_{even}, the canonical hidden-even relation, and the conservative-shape-preserving branch used by the grouped outlet package.

Derivation ledger. The derivation proves $\Xi_{\text{load}} = \Xi_1 = P_1/P_0$, shows that monomial/orbit rigidity kills the load defect but not the even gates, and narrows the viable same-charge corridor to a pure outgoing-transfer deformation with the conservative front end frozen.

Output. Strict first-order packet: Ξ_1 is the prefactor slope, while K_1 and H_{even} are separate conservative gates; the survivor is a pure-transfer corridor rather than arbitrary grouped anisotropy.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VII appendix narrative above and in the source stage. The card preserves the original stage order and records the claim boundary; it should not be read as a second independent proof of actual nonlinear branch realization.

Checks.

- ☐ Check the stage assumptions against the declared one-port, selected-branch, weak-axisymmetric, or rigid-mouth closure before citing the output.
- ☐ Check whether the row is algebraic, numerical placement, or open branch-realization before transferring it to another paper.
- ☐ Check that same-charge statements are not reinterpreted as a new long-range attraction; all static kernel claims inherit the source-product family limits.
- ☐ Check that orbit-lock claims require the rigid-mouth packet variables and do not follow from generic mouth tracking alone.

Downstream use. This stage feeds the Part VII same-charge sieve, rigid-mouth normal form, and selected twin-support branch. Downstream papers should cite **MTDC-T11.3** for this local stage range and **MTDC-T11** for the full Part VII compiler.

H.21 Stage 227: Pure-transfer load factor and co-loading no-go

Part / anchor. Part VII; primary anchor **MTDC-T11**, local subanchor **MTDC-T11.3**.

Claim status. Exact Within Closure.

Purpose. Stage 227 factors the pure-transfer survivor into the one-port load factor and proves the first co-loading no-go.

Inputs. Imports the one-port numerator $P = \Omega_U^2 G_W + RG_U$, denominator $\Delta = \Omega_U^2 \Omega_W^2 - R^2$, and load factor $\Lambda = P/\Delta$.

Derivation ledger. The derivation writes $N_0 = \Lambda^2$, computes $\Xi_1 = 2\delta \ln \Lambda$, decomposes the slope into mixed leg, interference, and hybridization components, and applies rigidity assumptions to each ratio.

Output. Pure-transfer theorem: $\Xi_1 = 2\delta \ln(P/\Delta)$. If the outgoing transfer shape is rigid while interference and hybridization ratios are also rigid, then nontrivial co-loading is impossible.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VII appendix narrative above and in the source stage. The card preserves the original stage order and records the claim boundary; it should not be read as a second independent proof of actual nonlinear branch realization.

Checks.

- ☐ Check the stage assumptions against the declared one-port, selected-branch, weak-axisymmetric, or rigid-mouth closure before citing the output.
- ☐ Check whether the row is algebraic, numerical placement, or open branch-realization before transferring it to another paper.

- ☐ Check that same-charge statements are not reinterpreted as a new long-range attraction; all static kernel claims inherit the source-product family limits.
- ☐ Check that orbit-lock claims require the rigid-mouth packet variables and do not follow from generic mouth tracking alone.

Downstream use. This stage feeds the Part VII same-charge sieve, rigid-mouth normal form, and selected twin-support branch. Downstream papers should cite **MTDC-T11.3** for this local stage range and **MTDC-T11** for the full Part VII compiler.

H.22 Stage 228: Numerator/denominator split and dynamic-window test

Part / anchor. Part VII; primary anchor **MTDC-T11**, local subanchor **MTDC-T11.3**.

Claim status. **Mixed:** Exact Within Closure, Numerically Located.

Purpose. Stage 228 tests the pure-transfer corridor against the dynamic-window machinery.

Inputs. Imports the split $\Xi_1 = 2(\pi_1 - \delta_1)$, numerator-rigid and denominator-rigid subcorridors, and the wall-like branch data from the finite-throat primitive slice.

Derivation ledger. The derivation separates numerator and denominator drift carriers, identifies which subcorridors survive the static constraint, and evaluates the first dynamic-window inequality for the wall-like branch.

Output. Numerator/denominator split, dynamic-window audit, and first branch-level comparison between the static Ξ_1 budget and resonant dynamic survival.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VII appendix narrative above and in the source stage. The card preserves the original stage order and records the claim boundary; it should not be read as a second independent proof of actual nonlinear branch realization.

Checks.

- ☐ Check the stage assumptions against the declared one-port, selected-branch, weak-axisymmetric, or rigid-mouth closure before citing the output.
- ☐ Check whether the row is algebraic, numerical placement, or open branch-realization before transferring it to another paper.
- ☐ Check that same-charge statements are not reinterpreted as a new long-range attraction; all static kernel claims inherit the source-product family limits.
- ☐ Check that orbit-lock claims require the rigid-mouth packet variables and do not follow from generic mouth tracking alone.

Downstream use. This stage feeds the Part VII same-charge sieve, rigid-mouth normal form, and selected twin-support branch. Downstream papers should cite **MTDC-T11.3** for this local stage range and **MTDC-T11** for the full Part VII compiler.

H.23 Stage 229: Selected-branch numerator/denominator signature

Part / anchor. Part VII; primary anchor **MTDC-T11**, local subanchor **MTDC-T11.4**.

Claim status. Exact Within Closure.

Purpose. Stage 229 pulls the numerator/denominator split onto the selected branch and classifies which side dominates near softening.

Inputs. Imports the selected softening-depth variable, the selected branch functions, and the numerator/denominator share map.

Derivation ledger. The derivation factorizes the selected branch into numerator-like and denominator-like pieces, derives the classifier threshold $\delta = 8/9$, and proves the near-softening denominator-like signature.

Output. Selected-branch classifier theorem: the pure-transfer defect is denominator-like near softening, with crossover controlled by the exact threshold $\delta = 8/9$.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VII appendix narrative above and in the source stage. The card preserves the original stage order and records the claim boundary; it should not be read as a second independent proof of actual nonlinear branch realization.

Checks.

- ☐ Check the stage assumptions against the declared one-port, selected-branch, weak-axisymmetric, or rigid-mouth closure before citing the output.
- ☐ Check whether the row is algebraic, numerical placement, or open branch-realization before transferring it to another paper.
- ☐ Check that same-charge statements are not reinterpreted as a new long-range attraction; all static kernel claims inherit the source-product family limits.
- ☐ Check that orbit-lock claims require the rigid-mouth packet variables and do not follow from generic mouth tracking alone.

Downstream use. This stage feeds the Part VII same-charge sieve, rigid-mouth normal form, and selected twin-support branch. Downstream papers should cite **MTDC-T11.4** for this local stage range and **MTDC-T11** for the full Part VII compiler.

H.24 Stage 230: Dynamic-window classifier and static-first theorem

Part / anchor. Part VII; primary anchor **MTDC-T11**, local subanchor **MTDC-T11.4**.

Claim status. **Mixed:** Exact Within Closure, Numerically Located.

Purpose. Stage 230 translates the selected-branch classifier into a dynamic-window decision rule.

Inputs. Imports Stage 229 classifier data, the rigid-split shares, and the survival-window inequality from the resonance theorem.

Derivation ledger. The derivation computes the dynamic sign of the selected numerator/denominator split, relates the classifier region to linewidth demand, and compares this with the static prefactor ceiling.

Output. Static-first theorem: the static Ξ_1 budget is the first kill condition; the dynamic corridor can be checked only after the static transported ceiling is satisfied.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VII appendix narrative above and in the source stage. The card preserves the original stage order and records the claim boundary; it should not be read as a second independent proof of actual nonlinear branch realization.

Checks.

- ☐ Check the stage assumptions against the declared one-port, selected-branch, weak-axisymmetric, or rigid-mouth closure before citing the output.
- ☐ Check whether the row is algebraic, numerical placement, or open branch-realization before transferring it to another paper.
- ☐ Check that same-charge statements are not reinterpreted as a new long-range attraction; all static kernel claims inherit the source-product family limits.
- ☐ Check that orbit-lock claims require the rigid-mouth packet variables and do not follow from generic mouth tracking alone.

Downstream use. This stage feeds the Part VII same-charge sieve, rigid-mouth normal form, and selected twin-support branch. Downstream papers should cite **MTDC-T11.4** for this local stage range and **MTDC-T11** for the full Part VII compiler.

H.25 Stage 231: Continuum-placement pullback

Part / anchor. Part VII; primary anchor **MTDC-T11**, local subanchor **MTDC-T11.4**.

Claim status. Exact Within Closure.

Purpose. Stage 231 rewrites the selected-branch dynamic-class map in the physical continuum-placement variables.

Inputs. Imports the placement coordinates $(\epsilon_\eta, \epsilon_W, \rho, Z_W, \delta_0, \Lambda)$, R_{target}, and the selected-branch classifier.

Derivation ledger. The derivation pulls the classifier through the exact placement map, proves monotonicity in R_{target}, and separates geometry, mixed-stability/product, and redistribution lanes.

Output. Continuum-placement version of the static-first verdict: the branch decision can be made directly from the physical placement packet, with R_{target} controlling the selected-class motion monotonically.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VII appendix narrative above and in the source stage. The card preserves the original stage order and records the claim boundary; it should not be read as a second independent proof of actual nonlinear branch realization.

Checks.

- ☐ Check the stage assumptions against the declared one-port, selected-branch, weak-axisymmetric, or rigid-mouth closure before citing the output.
- ☐ Check whether the row is algebraic, numerical placement, or open branch-realization before transferring it to another paper.
- ☐ Check that same-charge statements are not reinterpreted as a new long-range attraction; all static kernel claims inherit the source-product family limits.
- ☐ Check that orbit-lock claims require the rigid-mouth packet variables and do not follow from generic mouth tracking alone.

Downstream use. This stage feeds the Part VII same-charge sieve, rigid-mouth normal form, and selected twin-support branch. Downstream papers should cite **MTDC-T11.4** for this local stage range and **MTDC-T11** for the full Part VII compiler.

H.26 Stage 232: Known 5PN data injection

Part / anchor. Part VII; primary anchor **MTDC-T11**, local subanchor **MTDC-T11.4**.

Claim status. **Mixed:** NUMERICALLY LOCATED, OPEN.

Purpose. Stage 232 injects the current known Family-1 and 5PN support/source data into the selected-branch inequalities.

Inputs. Imports the numerically located support/source packets, the exact selected classifier, and the static-first ceiling.

Derivation ledger. The derivation evaluates the support/source side of the inequalities, compares it with the selected threshold windows, and identifies which gate remains unresolved.

Output. Current branch verdict: the support/source side is safe by a large margin in the injected data; the active unresolved gate is static orbit-lock/coherent placement rather than support starvation.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VII appendix narrative above and in the source stage. The card preserves the original stage order and records the claim boundary; it should not be read as a second independent proof of actual nonlinear branch realization.

Checks.

- ☐ Check the stage assumptions against the declared one-port, selected-branch, weak-axisymmetric, or rigid-mouth closure before citing the output.
- ☐ Check whether the row is algebraic, numerical placement, or open branch-realization before transferring it to another paper.
- ☐ Check that same-charge statements are not reinterpreted as a new long-range attraction; all static kernel claims inherit the source-product family limits.
- ☐ Check that orbit-lock claims require the rigid-mouth packet variables and do not follow from generic mouth tracking alone.

Downstream use. This stage feeds the Part VII same-charge sieve, rigid-mouth normal form, and selected twin-support branch. Downstream papers should cite **MTDC-T11.4** for this local stage range and **MTDC-T11** for the full Part VII compiler.

H.27 Stage 233: Rigid-mouth orbit-lock compiler

Part / anchor. Part VII; primary anchor **MTDC-T11**, local subanchor **MTDC-T11.5**.

Claim status. **Mixed:** Exact Within Closure, Open.

Purpose. Stage 233 begins the actual-branch rigid-mouth compiler and separates mouth tracking from operator rigidity.

Inputs. Imports R_{tr}, the coherent nontracking invariant $\mathfrak{N}_*$, the transfer shape $\mathcal{T}^2$, and the transported static ceiling.

Derivation ledger. The derivation proves that rigid mouth tracking $\delta \ln R_{\mathrm{tr}} = 0$ implies $\Xi_1 = \delta \ln \mathfrak{N}_*$, interprets the static ceiling as an internal transfer/turbulence gate, and identifies the remaining finite packet.

Output. Rigid-mouth compiler: mouth tracking alone is not enough; the actual branch must control the transfer-shape/nontracking packet that carries Ξ_1.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VII appendix narrative above and in the source stage. The card preserves the original stage order and records the claim boundary; it should not be read as a second independent proof of actual nonlinear branch realization.

Checks.

- ☐ Check the stage assumptions against the declared one-port, selected-branch, weak-axisymmetric, or rigid-mouth closure before citing the output.
- ☐ Check whether the row is algebraic, numerical placement, or open branch-realization before transferring it to another paper.
- ☐ Check that same-charge statements are not reinterpreted as a new long-range attraction; all static kernel claims inherit the source-product family limits.
- ☐ Check that orbit-lock claims require the rigid-mouth packet variables and do not follow from generic mouth tracking alone.

Downstream use. This stage feeds the Part VII same-charge sieve, rigid-mouth normal form, and selected twin-support branch. Downstream papers should cite **MTDC-T11.5** for this local stage range and **MTDC-T11** for the full Part VII compiler.

H.28 Stage 234: Direct branch-observable static gate

Part / anchor. Part VII; primary anchor **MTDC-T11**, local subanchor **MTDC-T11.5**.

Claim status. Exact Within Closure.

Purpose. Stage 234 converts the rigid-mouth static gate to direct branch observables.

Inputs. Imports the finite chart $(R_{\mathrm{tr}}, R_{\mathrm{target}}, \epsilon_\eta)$ and the rigid-mouth reduction of the transfer-shape identities.

Derivation ledger. The derivation shows two cancellations, reduces the static gate to R_{target} and ϵ_η, and states the two-observable kill test.

Output. Direct static gate: on the rigid-mouth branch, the static same-charge test is decided by the realized pair $(R_{\mathrm{target}}, \epsilon_\eta)$, with tracking data factored out.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VII appendix narrative above and in the source stage. The card preserves the original stage order and records the claim boundary; it should not be read as a second independent proof of actual nonlinear branch realization.

Checks.

- ☐ Check the stage assumptions against the declared one-port, selected-branch, weak-axisymmetric, or rigid-mouth closure before citing the output.
- ☐ Check whether the row is algebraic, numerical placement, or open branch-realization before transferring it to another paper.
- ☐ Check that same-charge statements are not reinterpreted as a new long-range attraction; all static kernel claims inherit the source-product family limits.
- ☐ Check that orbit-lock claims require the rigid-mouth packet variables and do not follow from generic mouth tracking alone.

Downstream use. This stage feeds the Part VII same-charge sieve, rigid-mouth normal form, and selected twin-support branch. Downstream papers should cite **MTDC-T11.5** for this local stage range and **MTDC-T11** for the full Part VII compiler.

H.29 Stage 235: Rigid-mouth packet projectors

Part / anchor. Part VII; primary anchor **MTDC-T11**, local subanchor **MTDC-T11.6**.

Claim status. Exact Within Closure.

Purpose. Stage 235 defines the finite packet map and its projectors in the direct rigid-mouth coordinates.

Inputs. Imports the direct packet variables, nontracking coordinate q_{nt}, dressing coordinate q_η, and the static-blind line.

Derivation ledger. The derivation constructs an involutive direct-packet map, derives projectors P_{nt} and P_η, identifies the static-blind dressing line, and proves the codimension-two orbit-lock statement.

Output. Rigid-mouth packet projector theorem: the finite same-charge packet splits into independent nontracking and dressing axes, and orbit lock is codimension two.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VII appendix narrative above and in the source stage. The card preserves the original stage order and records the claim boundary; it should not be read as a second independent proof of actual nonlinear branch realization.

Checks.

- ☐ Check the stage assumptions against the declared one-port, selected-branch, weak-axisymmetric, or rigid-mouth closure before citing the output.
- ☐ Check whether the row is algebraic, numerical placement, or open branch-realization before transferring it to another paper.

- ☐ Check that same-charge statements are not reinterpreted as a new long-range attraction; all static kernel claims inherit the source-product family limits.
- ☐ Check that orbit-lock claims require the rigid-mouth packet variables and do not follow from generic mouth tracking alone.

Downstream use. This stage feeds the Part VII same-charge sieve, rigid-mouth normal form, and selected twin-support branch. Downstream papers should cite **MTDC-T11.6** for this local stage range and **MTDC-T11** for the full Part VII compiler.

H.30 Stage 236: Microscopic dependent-plane projectors

Part / anchor. Part VII; primary anchor **MTDC-T11**, local subanchor **MTDC-T11.6**.

Claim status. Exact Within Closure.

Purpose. Stage 236 transfers the direct packet projectors to the microscopic dependent plane.

Inputs. Imports the dependent triple $(T_U, K_\eta^{(\mathrm{eff})}, \mu_W)$ and the same-free-quintuple graph section.

Derivation ledger. The derivation computes the dependent-plane compiler, projectors $P_{\mathrm{nt}}^{\mathrm{dep}}$ and P_η^{dep}, the equal-drift dressing ray, and the static-only restoration direction.

Output. Microscopic projector theorem: clearing the static nontracking defect moves along a different dependent direction than clearing the post-static dressing defect; a static-only fix leaves a dressing gap.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VII appendix narrative above and in the source stage. The card preserves the original stage order and records the claim boundary; it should not be read as a second independent proof of actual nonlinear branch realization.

Checks.

- ☐ Check the stage assumptions against the declared one-port, selected-branch, weak-axisymmetric, or rigid-mouth closure before citing the output.
- ☐ Check whether the row is algebraic, numerical placement, or open branch-realization before transferring it to another paper.
- ☐ Check that same-charge statements are not reinterpreted as a new long-range attraction; all static kernel claims inherit the source-product family limits.
- ☐ Check that orbit-lock claims require the rigid-mouth packet variables and do not follow from generic mouth tracking alone.

Downstream use. This stage feeds the Part VII same-charge sieve, rigid-mouth normal form, and selected twin-support branch. Downstream papers should cite **MTDC-T11.6** for this local stage range and **MTDC-T11** for the full Part VII compiler.

H.31 Stage 237: Actual-branch dressing compiler

Part / anchor. Part VII; primary anchor **MTDC-T11**, local subanchor **MTDC-T11.6**.

Claim status. Exact Within Closure.

Purpose. Stage 237 computes the dressing coordinate directly on the actual rigid-mouth branch.

Inputs. Imports the finite static-blind curve, microscopic log drifts $(c_1, \kappa_U, \kappa_\eta)$, and the support-sector variables.

Derivation ledger. The derivation derives $q_\eta = 2c_1 - \kappa_U - \kappa_\eta$, shows the finite static-blind curve is support-blind, and identifies the post-static orbit-lock correction.

Output. Actual-branch dressing compiler and support-blindness theorem: explicit support-sector shifts do not by themselves fix the dressing coordinate.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VII appendix narrative above and in the source stage. The card preserves the original stage order and records the claim boundary; it should not be read as a second independent proof of actual nonlinear branch realization.

Checks.

- ☐ Check the stage assumptions against the declared one-port, selected-branch, weak-axisymmetric, or rigid-mouth closure before citing the output.
- ☐ Check whether the row is algebraic, numerical placement, or open branch-realization before transferring it to another paper.
- ☐ Check that same-charge statements are not reinterpreted as a new long-range attraction; all static kernel claims inherit the source-product family limits.
- ☐ Check that orbit-lock claims require the rigid-mouth packet variables and do not follow from generic mouth tracking alone.

Downstream use. This stage feeds the Part VII same-charge sieve, rigid-mouth normal form, and selected twin-support branch. Downstream papers should cite **MTDC-T11.6** for this local stage range and **MTDC-T11** for the full Part VII compiler.

H.32 Stage 238: Physical transfer-shape compiler

Part / anchor. Part VII; primary anchor **MTDC-T11**, local subanchor **MTDC-T11.6**.

Claim status. Exact Within Closure.

Purpose. Stage 238 rewrites the rigid-mouth packet in the physical transfer-shape variables.

Inputs. Imports $\mathcal{T}^2$, R_{target}, ϵ_η, and the constant Λ_0.

Derivation ledger. The derivation proves $R_{\text{target}}\mathcal{T}^2 = \Lambda_0(1 - \epsilon_\eta)$, factorizes the finite packet, and splits the branch gates into tracking, transfer-shape, and dressing conditions.

Output. Physical transfer-shape compiler: the finite rigid-mouth packet is completely described by tracking, $\mathcal{T}^2$, and ϵ_η.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VII appendix narrative above and in the source stage. The card preserves the original stage order and records the claim boundary; it should not be read as a second independent proof of actual nonlinear branch realization.

Checks.

- ☐ Check the stage assumptions against the declared one-port, selected-branch, weak-axisymmetric, or rigid-mouth closure before citing the output.
- ☐ Check whether the row is algebraic, numerical placement, or open branch-realization before transferring it to another paper.
- ☐ Check that same-charge statements are not reinterpreted as a new long-range attraction; all static kernel claims inherit the source-product family limits.
- ☐ Check that orbit-lock claims require the rigid-mouth packet variables and do not follow from generic mouth tracking alone.

Downstream use. This stage feeds the Part VII same-charge sieve, rigid-mouth normal form, and selected twin-support branch. Downstream papers should cite **MTDC-T11.6** for this local stage range and **MTDC-T11** for the full Part VII compiler.

H.33 Stage 239: Rigid-mouth physical normal form

Part / anchor. Part VII; primary anchor **MTDC-T11**, local subanchor **MTDC-T11.6**.

Claim status. Exact Within Closure.

Purpose. Stage 239 uses the physical logarithmic variables themselves as the rigid-mouth packet chart.

Inputs. Imports $U = \ln(\mathcal{T}^2/\mathcal{T}^2_{\text{ref}})$, $V = \ln(\epsilon_\eta/\epsilon_{\eta,\text{ref}})$, and the microscopic dependent-plane compiler.

Derivation ledger. The derivation diagonalizes the packet in (U, V), derives the exact microscopic correction $(\Delta_T, \Delta_{K_\eta}, \Delta_\mu) = (0, -V, U - V)$, and identifies the Cartesian orbit-lock point.

Output. Cartesian orbit-lock theorem: rigid-mouth orbit lock is exactly $U = V = 0$, equivalently $\mathcal{T}^2 = \mathcal{T}^2_{\text{ref}}$ and $\epsilon_\eta = \epsilon_{\eta,\text{ref}}$.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VII appendix narrative above and in the source stage. The card preserves the original stage order and records the claim boundary; it should not be read as a second independent proof of actual nonlinear branch realization.

Checks.

- ☐ Check the stage assumptions against the declared one-port, selected-branch, weak-axisymmetric, or rigid-mouth closure before citing the output.
- ☐ Check whether the row is algebraic, numerical placement, or open branch-realization before transferring it to another paper.
- ☐ Check that same-charge statements are not reinterpreted as a new long-range attraction; all static kernel claims inherit the source-product family limits.
- ☐ Check that orbit-lock claims require the rigid-mouth packet variables and do not follow from generic mouth tracking alone.

Downstream use. This stage feeds the Part VII same-charge sieve, rigid-mouth normal form, and selected twin-support branch. Downstream papers should cite **MTDC-T11.6** for this local stage range and **MTDC-T11** for the full Part VII compiler.

H.34 Stage 240: Selected-branch loading ratio

Part / anchor. Part VII; primary anchor **MTDC-T11**, local subanchor **MTDC-T11.7**.

Claim status. Exact Within Closure.

Purpose. Stage 240 fixes the support-loading ratio selected by the minimal isotropic quadrupole precursor.

Inputs. Imports the selected-branch product identities and the minimal contact-plus-pole conservative precursor.

Derivation ledger. The derivation cancels the separate normalization amplitudes, inverts the contact-plus-pole compiler, and matches the precursor to the selected support demand.

Output. Exact selected loading ratio: $\rho_\alpha = 4/3$, $\zeta_{\text{req}} = 1/3$, and $\Pi_{\text{tr}} = 4C_{\text{mix}}/3$, placing the branch on the symmetric lowest-twin support slice.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VII appendix narrative above and in the source stage. The card preserves the original stage order and records the claim boundary; it should not be read as a second independent proof of actual nonlinear branch realization.

Checks.

- ☐ Check the stage assumptions against the declared one-port, selected-branch, weak-axisymmetric, or rigid-mouth closure before citing the output.
- ☐ Check whether the row is algebraic, numerical placement, or open branch-realization before transferring it to another paper.
- ☐ Check that same-charge statements are not reinterpreted as a new long-range attraction; all static kernel claims inherit the source-product family limits.
- ☐ Check that orbit-lock claims require the rigid-mouth packet variables and do not follow from generic mouth tracking alone.

Downstream use. This stage feeds the Part VII same-charge sieve, rigid-mouth normal form, and selected twin-support branch. Downstream papers should cite **MTDC-T11.7** for this local stage range and **MTDC-T11** for the full Part VII compiler.

H.35 Stage 241: Primitive ranking on selected twin-support branch

Part / anchor. Part VII; primary anchor **MTDC-T11**, local subanchor **MTDC-T11.7**.

Claim status. Exact Within Closure.

Purpose. Stage 241 ranks the primitive carrier weights along the selected twin-support curve.

Inputs. Imports $\epsilon_* = 1 - 3\varrho/2$, $\sigma = 4/(3\varrho) - 2$, $0 < \varrho < 2/3$, and the constructive coherent bound $0 < \beta < 2/11$.

Derivation ledger. The derivation computes the primitive coherent weights, proves $w_\chi = 2w_Z$, derives the thresholds $\varrho_{W\Lambda}$ and $\varrho_{U\Lambda}$, and orders the three ranking regions.

Output. Primitive ranking theorem: the selected branch has a complete phase diagram with thresholds $\varrho_{W\Lambda} = 2(1+\beta^2)/[3(2+\beta^2)]$ and $\varrho_{U\Lambda} = 2(1+\beta^2)/[3(1+\beta+\beta^2)]$.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VII appendix narrative above and in the source stage. The card preserves the original stage order and records the claim boundary; it should not be read as a second independent proof of actual nonlinear branch realization.

Checks.

- ☐ Check the stage assumptions against the declared one-port, selected-branch, weak-axisymmetric, or rigid-mouth closure before citing the output.
- ☐ Check whether the row is algebraic, numerical placement, or open branch-realization before transferring it to another paper.

- ☐ Check that same-charge statements are not reinterpreted as a new long-range attraction; all static kernel claims inherit the source-product family limits.
- ☐ Check that orbit-lock claims require the rigid-mouth packet variables and do not follow from generic mouth tracking alone.

Downstream use. This stage feeds the Part VII same-charge sieve, rigid-mouth normal form, and selected twin-support branch. Downstream papers should cite **MTDC-T11.7** for this local stage range and **MTDC-T11** for the full Part VII compiler.

H.36 Stage 242: Actual twin-support placement and coherent orbit-lock compiler

Part / anchor. Part VII; primary anchor **MTDC-T11**, local subanchor **MTDC-T11.7**.

Claim status. **Mixed:** Exact Within Closure, Open.

Purpose. Stage 242 places the actual coherent branch on the selected twin-support curve and packages the final front-end branch-evaluation data.

Inputs. Imports the selected curve from Stage 241, the coherent local D/N placement map, the physical variable ϵ, and the support-blind orbit-lock packet compiler.

Derivation ledger. The derivation identifies $\varrho_{\mathrm{phys}} = 2(1-\epsilon)/3$, rewrites the thresholds in the realized variable, classifies the support lane, constructs the finite orbit packet, and separates support placement from orbit lock.

Output. Final front-end packet: actual selected placement coordinate, primitive ranking region, support/orbit packet split, weak-axisymmetric orbit packet, and separate outgoing finish-line datum $N_Q - 1$.

Verification note. The detailed theorem-block formulas supporting this card are collected in the Part VII appendix narrative above and in the source stage. The card preserves the original stage order and records the claim boundary; it should not be read as a second independent proof of actual nonlinear branch realization.

Checks.

- ☐ Check the stage assumptions against the declared one-port, selected-branch, weak-axisymmetric, or rigid-mouth closure before citing the output.
- ☐ Check whether the row is algebraic, numerical placement, or open branch-realization before transferring it to another paper.
- ☐ Check that same-charge statements are not reinterpreted as a new long-range attraction; all static kernel claims inherit the source-product family limits.
- ☐ Check that orbit-lock claims require the rigid-mouth packet variables and do not follow from generic mouth tracking alone.

Downstream use. This stage feeds the Part VII same-charge sieve, rigid-mouth normal form, and selected twin-support branch. Downstream papers should cite **MTDC-T11.7** for this local stage range and **MTDC-T11** for the full Part VII compiler.

Appendix I

Stage Appendix: Part VIII – Relaxed Open-System Branch, Barrier Compiler, Export Kernel, and Material Thresholds

Stage range: 243–253

This appendix expands the Part VIII theorem-block chapter into a stage-by-stage verification ledger. The main-body chapter states the relaxed/open-system branch in citation-ready form; the present appendix preserves the derivational path from Stage 243 through Stage 253 and records the exact algebraic objects that a reader must check to audit the branch.

The result anchor served by this appendix is **MTDC-T12**. Downstream documents should cite this anchor when they need the derivation of the relaxed-constraint branch declaration, the recovery map back to the strict front end, the selected leakage/work compiler, the non-rigid mouth and dressing packet, the compensated multimode source compiler, the relaxed stationary barrier law, the dynamic event-chain compiler, the helicity-export diagnostic, the crossing-versus-collapse window, the microscopic cubic/quintic export kernel, the vacuum/lattice heat partition, or the material-threshold companion.

The conceptual firewall is the most important part of this appendix. The relaxed branch does *not* undo the same-charge verdict from Part VII. It does not produce a new long-range same-charge attraction, and it does not introduce a new linear dynamic kernel class. The relaxed branch is a codimension-three lift of the strict selected-support/orbit packet. Its standard-recovery slice is

$$\ell_w = 0, \qquad f_U = 0, \qquad a = b = 0, \tag{I.1}$$

which implies

$$S_{\mathrm{leak}} = 0, \qquad \mathcal{W}_w = 0, \qquad U = 0, \qquad V = 0, \qquad \mathcal{D}_{UV} = 0, \qquad \varsigma \equiv 1. \tag{I.2}$$

Away from this slice, the relaxed branch can reshape a near-contact barrier by leakage, scalar-photon work, non-rigid dressing drain, and compensated source response. Those are branch-local open-system or support/dressing/source effects. They do not alter the long-range kernel firewall.

The central stationary object of this part is the lowered front end

$$V_{\mathrm{eff}}^{(247)}(r) = V_{\mathrm{short}}^{(1p)}(r) - \lambda_L S_{\mathrm{leak}}(r) - \lambda_W \mathcal{W}_w^{\mathrm{sess}}(r) - \Delta E_{UV}(r) - \mathcal{M}_\sigma(r). \tag{I.3}$$

The central dynamic object is the event-chain energy law

$$E = \frac{m_s}{2}\dot{r}^{\,2} + V_{\mathrm{eff}}^{(247)}(r). \tag{I.4}$$

The central microscopic export object is the super-Ohmic active-leg kernel

$$K_{\mathrm{exp}}(s) = \Gamma_3 s^3 + \Gamma_5 s^5. \tag{I.5}$$

The central materials companion is the four-ratio screen

$$\Pi_{\mathrm{ep}} \geq 1, \qquad \Pi_\chi \geq 1, \qquad \Pi_k \geq 1, \qquad \Pi_T(T) \geq 1. \tag{I.6}$$

Table I.1: Part VIII appendix source and status map.

Stage	Working title	Primary status	Main output
243	Relaxed-constraint branch declaration	Mixed: EFFECTIVE CLOSURE, EXACT WITHIN CLOSURE	Codimension-three lift of the Stage 242 front end, exact recovery map $(\ell_w, f_U, a, b) = (0,0,0,0)$, leakage/work lane, non-rigid (U,V) lane, compensated source lane, and strict short-range kernel-span theorem.
244	Selected leakage and scalar-photon work compiler	EXACT WITHIN CLOSURE	Exact Gaussian one-mode leakage law, exact bulk and session work scalars, selected-support pullback through Π_{tr}, support/orbit split, and orientation parity.
245	Non-rigid mouth and U/V drain packet	EXACT WITHIN CLOSURE	Exact finite compiler for $\mathcal{T}^2$, ϵ_η, and R_{target}; non-rigid determinant Δ_{UV}; dependent microscopic correction; positive U/V drain; and first-order packet $(\Xi_1, \mathcal{R}_1)$.
246	Compensated multimode source compiler	EXACT WITHIN CLOSURE	Mean-preserving two-mode source family, exact sign-change test, mouth-bias and shell-loading moments, invertible two-moment source map, mixed-to-shell loading ratio, and transported Session-I readback.
247	Relaxed stationary barrier front end	Mixed: EXACT WITHIN CLOSURE, EFFECTIVE CLOSURE, NUMERICALLY LOCATED	One-port short-range baseline, relaxed lowering identity, stationary barrier compiler $V_{\mathrm{eff}}^{(247)}$, and one-point Session-I benchmark decomposition.

Stage	Working title	Primary status	Main output
248	Dynamic event-chain compiler	Mixed: Exact Within Closure, Reduced / Controlled Reduction, Numerically Located	Energy integral, peak and threshold-speed compiler, turning points, WKB action, transmission ratio, dynamic diagnostics, and Session-II crossing/tunneling benchmark.
249	Helicity-export diagnostic	Mixed: Exact Within Closure, Effective Closure, Numerically Located	Exact projected subscale-helicity identity, aligned/anti-aligned orientation closure, peak and integrated ratio compilers, and Session-II hidden-channel export packet.
250	Crossing-vs-collapse Goldilocks window	Mixed: Exact Within Closure, Effective Closure, Numerically Located	Event-chain transit time, characteristic crossing time, dressing-leg collapse time, survival ratio, lower safe edge, speed-space compiler, heavy-throat scaling, and trigger-width sensitivity.
251	Microscopic damping/export kernel	Mixed: Exact Within Closure, Reduced / Controlled Reduction, Open	Derivative-coupled scalar cubic export, selected quadrupole quintic export, active-leg kernel $K_{\exp}(s) = \Gamma_3 s^3 + \Gamma_5 s^5$, exact event-safe half-plane, and channel-resolved power split.
252	Vacuum/lattice heat partition	Mixed: Exact Within Closure, Effective Closure, Numerically Located, Open	Exact vacuum/lattice exported energies, one-scalar event-shape quotient, single-growth cold-event specialization, safe-edge exported-energy theorem, and event-equivalent export rates.
253	Physical calibration and material thresholds	Mixed: Exact Within Closure, Effective Closure, Open	Physical calibration dictionary, coarse lattice-turnover projection factor, electron-phonon threshold, force-matched stiffness, Korringa ceiling, and four material-screening ratios.

Claim status. Part VIII is a relaxed/open-system companion. Its algebraic compilers are Exact Within Closure **inside the declared branch, one-mode, characteristic-timescale, single-growth, and physical-calibration closures. The stationary and dynamic readbacks that use Session-I through Session-V benchmark numbers are** Numerically Located **insertions into exact formulas. The branch declaration itself is an** Effective Closure **extension around the strict Stage 242 front end, and any claim that the completed nonlinear moving-**

throat PDE actually realizes the relaxed packet, the microscopic export coefficients, or the physical material thresholds remains Open. No result in this part should be cited as a proof of a new same-charge long-range attraction or as a proof that the relaxed branch validates the strict conservative branch.

I.1 How to audit this appendix

A reader can verify the Part VIII branch by following four audit paths.

- ☐ Check the recovery slice: impose $\ell_w = f_U = a = b = 0$ and verify that the lifted leakage, work, non-rigid, and source observables reduce to the strict Stage 242 values.
- ☐ Check the kernel firewall: verify that the one-port static and linear dynamic same-charge corrections stay in the short-range product span and never acquire a new $1/r$ component.
- ☐ Check the compiler chain: insert the Stage 244 leakage/work packet, the Stage 245 (U, V) packet, and the Stage 246 source packet into the Stage 247 stationary barrier, then pass that barrier through the Stage 248 event-chain and Stage 250 crossing/collapse tests.
- ☐ Check the physical companion: use Stage 251 to replace the phenomenological export envelope by $K_{\mathrm{exp}}(s) = \Gamma_3 s^3 + \Gamma_5 s^5$, use Stage 252 to split the exported energy into vacuum and lattice channels, and use Stage 253 to map the lattice channel into material-screening ratios.

I.2 Block contract and theorem-level output

The contract of Part VIII is to keep three layers separate. The first layer is the strict front end from Part VII: selected twin-support placement, rigid-mouth orbit lock, and the outgoing normalization packet. The second layer is the relaxed branch around that front end: leakage, non-rigid mouth response, and compensated source shape. The third layer is the physical companion: event-chain survival, export, heat partition, and material screening. The chapter is useful only if those layers remain visibly separated.

The operational chain is

$$\begin{aligned} \mathcal{P}_{242} &\longrightarrow \mathfrak{B}_{243}^{\mathrm{relax}} \longrightarrow (S_{\mathrm{leak}}, \mathcal{W}_w, U, V, \mathcal{M}_\sigma) \\ &\longrightarrow V_{\mathrm{eff}}^{(247)} \longrightarrow (r_\pm, I_{\mathrm{new}}, t_{\mathrm{cross}}) \\ &\longrightarrow K_{\mathrm{exp}}(s) \longrightarrow (\Pi_{\mathrm{ep}}, \Pi_\chi, \Pi_k, \Pi_T). \end{aligned} \tag{I.7}$$

The strict branch can be recovered from this chain by imposing equation (I.1). The relaxed branch cannot be used to infer that the strict branch succeeds; it can only be used to analyze the open-system bypass companion.

Theorem I.1 (Part VIII relaxed-branch and export-companion theorem)**.** *Inside the declared relaxed-constraint branch, the one-mode leakage/work closure, the non-rigid mouth/dressing closure, the compensated two-mode source closure, the reduced one-dimensional event-chain closure, the characteristic crossing/collapse closure, the microscopic cubic/quintic export closure, and the physical-calibration companion, the following statements hold.*

1. *The relaxed branch is a codimension-three lift of the Stage 242 front end with exact recovery map*

$$\ell_w = 0, \qquad f_U = 0, \qquad a = b = 0. \tag{I.8}$$

On this slice all lifted leakage, work, non-rigid, and compensated-source observables reduce to the strict standard-recovery values.

2. *The same-charge static and linear dynamic kernel span remains short-ranged:*

$$\mathcal{K}_{\mathrm{SR}} = \mathrm{span}\left\{ r^{-6}, \frac{e^{-2\kappa r}}{r^4}, \frac{e^{-4\kappa r}}{r^2} \right\}. \tag{I.9}$$

In particular,

$$\lim_{r\to\infty} r\,\delta V_{\mathrm{stat}}(r) = 0, \qquad \lim_{r\to\infty} r\,\Re\mathfrak{V}_{\mathrm{dyn}}(r,\omega) = 0. \tag{I.10}$$

3. *On the selected-support branch, the leakage/work lane is explicit and support-side:*

$$S_{\mathrm{leak}}(r) = \frac{8\sqrt{2}\,\eta_{\mathrm{leak}}\mu_w\rho_0}{\pi^{5/2}\lambda^3}\Lambda(r)\varrho(r), \tag{I.11}$$

$$\mathcal{W}_w^{\mathrm{sess}}(r) = \frac{512\,\eta_{\mathrm{leak}}^2\mu_w q\rho_0}{\pi^4\lambda^2}\Lambda(r)^2\varrho(r)^2. \tag{I.12}$$

These quantities are independent of the orbit-lock variables $(R_{\mathrm{tr}}, R_{\mathrm{target}}, \epsilon_\eta)$ *within the declared pullback.*

4. *The non-rigid mouth packet is exact in the physical logarithmic chart*

$$U = \ln\frac{\mathcal{T}^2}{\mathcal{T}_{\mathrm{ref}}^2}, \qquad V = \ln\frac{\epsilon_\eta}{\epsilon_{\eta,\mathrm{ref}}}. \tag{I.13}$$

With

$$\Delta_{UV} = a_U a_V - \chi_{UV}^2, \tag{I.14}$$

the stationary point is

$$U = \frac{a_V f_U}{\Delta_{UV}}, \qquad V = \frac{\chi_{UV} f_U}{\Delta_{UV}}, \qquad \Delta_{UV} > 0. \tag{I.15}$$

The corresponding drain is nonnegative:

$$\mathcal{D}_{UV} = \frac{\chi_{UV}^2 a_V f_U^2}{\Delta_{UV}^2} \geq 0. \tag{I.16}$$

5. *The first compensated two-mode source family is an exact two-moment compiler:*

$$\sigma_{a,b}(x) = 1 + a\cos(\pi x) + b\cos(2\pi x), \qquad \int_0^1 \sigma_{a,b}(x)\,dx = 1, \tag{I.17}$$

with

$$\mathfrak{g}(a,b) = \frac{2}{\pi}\left(1 + \frac{a}{3} - \frac{b}{15}\right), \qquad \mathcal{S}(a,b) = \frac{2\tanh(\pi/2)}{\pi}\left(1 + \frac{a}{5} + \frac{b}{17}\right). \tag{I.18}$$

Its normalized two-moment map has determinant $14/425 > 0$*, so the first compensated source family spans the two carried source moments exactly.*

6. *The relaxed stationary barrier is exactly equation* (I.3)*, and the lowering identity is*

$$V_{\mathrm{short}}^{(1p)}(r) - V_{\mathrm{eff}}^{(247)}(r) = \lambda_L S_{\mathrm{leak}}(r) + \lambda_W \mathcal{W}_w^{\mathrm{sess}}(r) + \Delta E_{UV}(r) + \mathcal{M}_\sigma(r). \tag{I.19}$$

Hence, when the imported packet scalars and weights are nonnegative, $V_{\mathrm{eff}}^{(247)} \leq V_{\mathrm{short}}^{(1p)}$ *pointwise.*

7. *The dynamic event chain is determined by the conserved energy equation* (I.4)*. The lowered-branch classical threshold speed is*

$$v_{\text{crit,new}} = \sqrt{\frac{2}{m_s}\big(V_{\text{peak}} - V(r_0)\big)}, \tag{I.20}$$

while the pure-Coulomb contact threshold is

$$v_{\text{contact,Coul}} = \sqrt{\frac{2}{m_s}\left(\frac{1}{r_{\text{contact}}} - \frac{1}{r_0}\right)}. \tag{I.21}$$

If $v_{\text{crit,new}} < v_0 < v_{\text{contact,Coul}}$*, the lowered branch reaches contact while the pure Coulomb branch still turns back.*

8. *The subbarrier WKB action is*

$$I_{\text{new}}(E) = \frac{1}{\hbar_{\text{eff}}}\int_{r_-(E)}^{r_+(E)} \sqrt{2m_s\big(V(r) - E\big)}\,dr, \tag{I.22}$$

and the transmission ratio relative to Coulomb is

$$\frac{T_{\text{new}}(E)}{T_{\text{Coul}}(E)} = \exp\!\big[-2\big(I_{\text{new}}(E) - I_{\text{Coul}}(E)\big)\big]. \tag{I.23}$$

9. *The helicity-export comparison is diagnostic. With orientation label* $\sigma = \pm 1$*, the reduced closure gives*

$$\dot{H}_{\text{sub},\sigma} = \Gamma_0(1 + \sigma\alpha_h), \qquad |\alpha_h| < 1 \tag{I.24}$$

for positive export in both branches. The aligned branch dominates whenever $\alpha_h > 0$*, but this does not by itself prove a microscopic spin sector.*

10. *The crossing-versus-collapse safe edge is*

$$E_{\text{edge}} = V_{\text{peak}} + \frac{m_s}{\mu_\eta}\frac{g_{UV}\chi_{\text{peak}}\lambda_{\text{eff}}^2}{2}, \tag{I.25}$$

equivalently

$$v_{\text{safe,min}}^2 = v_{\text{crit,new}}^2 + \frac{\lambda_{\text{eff}}^2 g_{UV}\chi_{\text{peak}}}{\mu_\eta}. \tag{I.26}$$

Under the declared characteristic closure this is a one-sided lower edge.

11. *The microscopic active-leg export kernel is super-Ohmic:*

$$K_{\exp}(s) = \Gamma_3 s^3 + \Gamma_5 s^5, \qquad \Gamma_3, \Gamma_5 \geq 0. \tag{I.27}$$

The positive growth root is reduced but not eliminated by any finite passive (Γ_3, Γ_5) *in this minimal closure. The exact event-safe condition is*

$$\widehat{\Gamma}_3 + s_c^2\widehat{\Gamma}_5 \geq \frac{s_0^2 - s_c^2}{s_c^3}, \qquad \widehat{\Gamma}_n := \Gamma_n/\mu_\eta. \tag{I.28}$$

12. *The heat partition is channel-resolved. For event-shape quotient* $\mathfrak{r}_V = \mathcal{I}_3/\mathcal{I}_2$*,*

$$f_{\mathrm{lat}}(\mathfrak{r}_V) = \frac{\Gamma_3^{\mathrm{lat}} + \Gamma_5^{\mathrm{lat}}\mathfrak{r}_V}{\Gamma_3 + \Gamma_5\mathfrak{r}_V}, \tag{I.29}$$

and on a single-growth cold event $\mathfrak{r}_V = s^2$*. At the safe edge, the total minimum exported energy is*

$$E^{\mathrm{safe}}_{\mathrm{exp,min}} = \frac{V_{\mathrm{in}}^2}{2}(e^2-1)\mu_\eta(s_0^2 - s_c^2). \tag{I.30}$$

13. *The physical-material companion is an exact calibration image of the Stage 252 event-equivalent rate. With calibration factor* $\Upsilon_{\mathrm{lat}} > 0$*,*

$$(\lambda_{\mathrm{ep}}\omega_D)^{(253)}_{\min} = \frac{f_{\mathrm{lat}}(s_c)\mu_\eta(s_0^2 - s_c^2)}{\Upsilon_{\mathrm{lat}}\zeta_{\mathrm{ep}} s_c t_*}. \tag{I.31}$$

Candidate hosts are screened by equation (I.6)*.*

Every item involving actual branch data, microscopic coefficients, physical units, or candidate material constants remains a realization or calibration question beyond the algebraic compiler.

Source layout. Stages 243–253 are now sourced from the canonical files `stages/stage_243.tex` through `stages/stage_253.tex`. The raw Markdown notes and audit artifacts are tracked in a repo-local provenance index, so they do not have to appear in the reader-facing PDF. Part VIII differs from Parts I–VII: these stage files carry the full derivation sections rather than compact stage cards.

I.3 Stage 243: Relaxed constraint branch declaration

Stage 243 declares the relaxed branch without yet assembling a barrier. Its purpose is to make the Session-I relaxations legal inside the derivation tree while preserving the same-charge short-range verdict. The inherited base object is the Stage 242 front-end packet

$$\mathcal{P}_{242} = \Big(\epsilon, \varrho_{\mathrm{phys}}, \sigma_{\mathrm{phys}}, \text{ranking region}, R_{\mathrm{tr}}, R_{\mathrm{target}}, \epsilon_\eta, d\ln R_{\mathrm{tr}}, d\ln R_{\mathrm{target}}, d\ln\epsilon_\eta, N_Q - 1\Big), \tag{I.32}$$

with selected-support packet

$$\mathcal{S}_{242} = (\zeta, M_{\mathrm{mix}}, S(\zeta;\epsilon), M_{\mathrm{tr}}). \tag{I.33}$$

The strict slice is

$$\mathfrak{B}_{\mathrm{std}} = \{J^w = 0,\ U = 0,\ V = 0,\ \varsigma(z) \equiv 1\} \subset (\mathcal{P}_{242}, \mathcal{S}_{242}). \tag{I.34}$$

This definition is useful because it names exactly what is being relaxed: transverse-current suppression, rigid-mouth orbit lock, and positive-source Family-1 closure.

I.3.1 Open-system leakage/work lane

The open-system lift begins with the exact projected-continuity leakage identity

$$S_{\mathrm{leak}} = -\big[W(w)j^w\big]_{-\infty}^{+\infty} + \int_{-\infty}^{+\infty} W'(w)j^w(w)\,dw. \tag{I.35}$$

For the declaration-level Gaussian closure,

$$W(w) = \frac{e^{-w^2}}{\sqrt{\pi}}, \qquad j^w(w) = \ell_w j_0\, w e^{-w^2}, \qquad E_w(w) = E_0 w e^{-w^2}, \tag{I.36}$$

so the boundary term vanishes and

$$S_{\mathrm{leak}} = -\frac{\sqrt{2}}{4}\ell_w j_0, \qquad \mathcal{W}_w = \frac{\sqrt{2\pi}}{8}\ell_w j_0 E_0. \tag{I.37}$$

The declaration packet for this lane is

$$L_w = (S_{\mathrm{leak}}, \mathcal{W}_w). \tag{I.38}$$

Both components vanish on $\ell_w = 0$.

I.3.2 Non-rigid mouth/dressing lane

The non-rigid lift uses the minimal free energy

$$\mathcal{F}_{UV}(U,V) = \frac{1}{2}k_U U^2 + \frac{1}{2}k_V V^2 - \chi_\lambda UV - f_U U. \tag{I.39}$$

Stationarity gives

$$U = \frac{k_V f_U}{k_U k_V - \chi_\lambda^2}, \qquad V = \frac{\chi_\lambda f_U}{k_U k_V - \chi_\lambda^2}, \qquad \frac{V}{U} = \frac{\chi_\lambda}{k_V}. \tag{I.40}$$

The branch is admissible when

$$k_U k_V - \chi_\lambda^2 > 0, \tag{I.41}$$

and the drain is

$$\mathcal{D}_{UV} = \chi_\lambda UV = \frac{\chi_\lambda^2 k_V f_U^2}{(k_U k_V - \chi_\lambda^2)^2} \geq 0. \tag{I.42}$$

This lane is the finite relaxed continuation of the earlier weak-axisymmetric transfer-shape scalar: at infinitesimal order, $\Xi_1 = \delta U$.

I.3.3 Compensated sign-changing source lane

The compensated source lift is

$$\varsigma(z) = 1 + a\cos(\pi z) + b\cos(2\pi z), \qquad z \in [0,1], \tag{I.43}$$

with exact unit mean. Setting $y = \cos(\pi z)$,

$$\varsigma(y) = 1 - b + ay + 2by^2, \qquad -1 \leq y \leq 1. \tag{I.44}$$

The interior stationary point and value are

$$y_* = -\frac{a}{4b}, \qquad \varsigma(y_*) = 1 - b - \frac{a^2}{8b}, \tag{I.45}$$

with the boundary candidates $1 + a + b$ and $1 - a + b$. Thus the first source relaxation is not a vague sign-changing profile; it is an exact mean-preserving quadratic problem.

The full declaration packet is

$$\mathfrak{B}_{243}^{\mathrm{relax}} = \left\{(\mathcal{P}_{242}, \mathcal{S}_{242});\ L_w,\ L_{UV},\ L_\varsigma\right\}, \tag{I.46}$$

where

$$L_w = (S_{\mathrm{leak}}, \mathcal{W}_w), \quad L_{UV} = (U, V, \mathcal{D}_{UV}), \quad L_\varsigma = (\varsigma, \varsigma_{\min}, \mathrm{signchg}). \tag{I.47}$$

I.3.4 Short-range theorem for the lifted branch

The most important Stage 243 output is the short-range firewall. With primitive profiles

$$\mathcal{S}_Q(r) = r^{-3}, \qquad \mathcal{S}_Y(r) = \frac{e^{-2\kappa r}}{r}, \tag{I.48}$$

the static one-port correction is

$$\delta V_{\mathrm{stat}}(r) = -\frac{1}{2}\left[\frac{\mathcal{C}_6}{r^6} + 2\mathcal{C}_4\frac{e^{-2\kappa r}}{r^4} + \mathcal{C}_2\frac{e^{-4\kappa r}}{r^2}\right], \tag{I.49}$$

and the linear monochromatic dynamic bundle has the same spatial span:

$$\Re\mathfrak{V}_{\mathrm{dyn}}(r,\omega) = -\frac{1}{2}\left[\frac{\mathcal{C}_6(\omega)}{r^6} + 2\mathcal{C}_4(\omega)\frac{e^{-2\kappa r}}{r^4} + \mathcal{C}_2(\omega)\frac{e^{-4\kappa r}}{r^2}\right]. \tag{I.50}$$

Thus the relaxed branch is a short-range/open-system bypass branch. It can reshape near-contact energetics, but it cannot be cited as a new long-range same-charge force.

I.4 Stage 244: Selected leakage and scalar-photon work compiler

Stage 244 takes the declared $J^w \neq 0$ lane and attaches it to the selected-support packet. The parent theory supplies both exact identities needed for the compiler: projected continuity gives equation (I.35), and the localized Maxwell energy ledger gives

$$\partial_t u_{\mathrm{EM}} + \partial_A S^A_{\mathrm{EM}} = -J^A E_A = -(J^a E_a + J^w E_w). \tag{I.51}$$

Thus S_{leak} and $J^w E_w$ are legitimate reduced observables once the mixed lane is opened.

I.4.1 Gaussian one-mode leakage law

The Stage 244 one-mode closure uses

$$W_\lambda(w) = \frac{e^{-w^2/\lambda^2}}{\lambda\sqrt{\pi}}, \qquad \phi_\lambda(w) = \frac{2w}{\sqrt{\pi}\lambda^3}e^{-w^2/\lambda^2}, \tag{I.52}$$

with

$$E_w(w;r) = -\mathcal{E}_0(r)\phi_\lambda(w), \qquad j^w(w;r) = \mu_w\rho_0 E_w(w;r), \qquad J^w = qj^w. \tag{I.53}$$

The exact projected leakage is

$$S_{\mathrm{leak}}(r) = \frac{\sqrt{2}\mu_w\rho_0}{2\sqrt{\pi}\lambda^3}\mathcal{E}_0(r). \tag{I.54}$$

The exact one-mode bulk work scalar is

$$\mathcal{W}^{\mathrm{bulk}}_w(r) = q\int j^w E_w\,dw = \frac{\sqrt{2}\mu_w q\rho_0}{2\sqrt{\pi}\lambda^3}\mathcal{E}_0(r)^2, \tag{I.55}$$

so

$$\mathcal{W}^{\mathrm{bulk}}_w(r) = q\mathcal{E}_0(r)S_{\mathrm{leak}}(r). \tag{I.56}$$

The Session-I scalar-photon work variable is the rescaled scalar

$$\mathcal{W}^{\mathrm{sess}}_w(r) = \frac{2\mu_w q\rho_0}{\lambda^2}\mathcal{E}_0(r)^2 = 2\sqrt{2\pi}\lambda\,\mathcal{W}^{\mathrm{bulk}}_w(r), \tag{I.57}$$

or equivalently

$$\mathcal{W}^{\mathrm{sess}}_w(r) = \frac{4\pi q\lambda^4}{\mu_w\rho_0}S_{\mathrm{leak}}(r)^2. \tag{I.58}$$

I.4.2 Selected-support pullback

The selected-support demand from Stage 242 is

$$\Pi_{\rm tr} = \frac{4}{3}C_{\rm mix} = \frac{32\Lambda(1-\epsilon)}{3\pi^2} = \frac{16}{\pi^2}\Lambda\varrho, \qquad \varrho = \frac{2}{3}(1-\epsilon). \tag{I.59}$$

Stage 244 chooses the simplest amplitude pullback

$$\mathcal{E}_0(r) = \eta_{\rm leak}\Pi_{\rm tr}(r). \tag{I.60}$$

Substitution gives the selected-branch leakage law equation (I.11) and the selected-branch work law equation (I.12). In the (Λ, ϵ) notation the same work law is

$$\mathcal{W}_w^{\rm sess}(r) = \frac{2048\eta_{\rm leak}^2\mu_w q\rho_0}{9\pi^4\lambda^2}\Lambda(r)^2(1-\epsilon(r))^2. \tag{I.61}$$

The formulas depend only on selected-support variables. Therefore,

$$\partial_{R_{\rm tr}}S_{\rm leak} = \partial_{R_{\rm target}}S_{\rm leak} = \partial_{\epsilon_\eta}S_{\rm leak} = 0, \tag{I.62}$$

and the same statement holds for both work scalars within the declared pullback.

The orientation parity is exact:

$$S_{\rm leak}(-\eta_{\rm leak}) = -S_{\rm leak}(\eta_{\rm leak}), \qquad \mathcal{W}_w(-\eta_{\rm leak}) = \mathcal{W}_w(\eta_{\rm leak}). \tag{I.63}$$

Thus the leakage sign is orientation-sensitive, but the exported-work magnitude is not.

I.5 Stage 245: Non-rigid mouth and (U, V) drain packet

Stage 245 compiles the abstract non-rigid lane into the actual finite physical variables used by the rigid-mouth chart. The carried chart is equation (I.13); hence

$$\mathcal{T}^2 = \mathcal{T}_{\rm ref}^2 e^U, \qquad \epsilon_\eta = \epsilon_{\eta,\rm ref}e^V. \tag{I.64}$$

The selected-branch identity

$$R_{\rm target}\mathcal{T}^2 = \Lambda_0(1-\epsilon_\eta) \tag{I.65}$$

then gives

$$\frac{R_{\rm target}}{R_{\rm target,ref}} = \frac{1-\epsilon_{\eta,\rm ref}e^V}{1-\epsilon_{\eta,\rm ref}}e^{-U}. \tag{I.66}$$

So the finite non-rigid packet has two independent legs: transfer shape and dressing.

The reduced free energy in session notation is

$$\mathcal{F}_{\rm nr}(U, V; r) = \frac{1}{2}a_U U^2 + \frac{1}{2}a_V V^2 - \chi_{UV}(r)UV - f_U(r)U. \tag{I.67}$$

The stationary solution is equation (I.15), with Δ_{UV} given by equation (I.14). In finite physical variables,

$$\mathcal{T}^2(r) = \mathcal{T}_{\rm ref}^2 \exp\left(\frac{a_V f_U(r)}{\Delta_{UV}(r)}\right), \tag{I.68}$$

$$\epsilon_\eta(r) = \epsilon_{\eta,\rm ref} \exp\left(\frac{\chi_{UV}(r)f_U(r)}{\Delta_{UV}(r)}\right), \tag{I.69}$$

$$\frac{R_{\text{target}}(r)}{R_{\text{target,ref}}} = \frac{1 - \epsilon_{\eta,\text{ref}} \exp(\chi_{UV} f_U/\Delta_{UV})}{1 - \epsilon_{\eta,\text{ref}}} \exp(-a_V f_U/\Delta_{UV}). \tag{I.70}$$

The dependent microscopic correction inherited from the rigid-mouth plane is

$$(\Delta_T, \Delta_{K_\eta}, \Delta_\mu) = (0, -V, U - V), \tag{I.71}$$

so the non-rigid stationary packet induces

$$\mathbf{y}^{\text{dep}}_{\text{nr}}(r) = \begin{pmatrix} 0 \\ -\chi_{UV} f_U/\Delta_{UV} \\ (a_V - \chi_{UV}) f_U/\Delta_{UV} \end{pmatrix}. \tag{I.72}$$

The drain $\mathcal{D}_{UV}$ is given by equation (I.16); it is nonnegative even when U and V have opposite signs.

At weak-axisymmetric order, write

$$U = \varepsilon u_1 + O(\varepsilon^2), \qquad V = \varepsilon v_1 + O(\varepsilon^2). \tag{I.73}$$

Then

$$\Xi^{\text{nr}}_1 = u_1, \qquad \mathcal{R}^{\text{nr}}_1 + \Xi^{\text{nr}}_1 = -\frac{\epsilon_{\eta,\text{ref}}}{1 - \epsilon_{\eta,\text{ref}}} v_1, \tag{I.74}$$

$$\mathcal{R}^{\text{nr}}_1 = -u_1 - \frac{\epsilon_{\eta,\text{ref}}}{1 - \epsilon_{\eta,\text{ref}}} v_1. \tag{I.75}$$

Thus the direct outgoing-prefactor scalar remains the U leg, while active dressing adds an independent selected-branch residual.

Within the declared closure, this packet is orbit-side/support-blind:

$$\partial_\Lambda U = \partial_\varrho U = \partial_\Lambda V = \partial_\varrho V = 0, \tag{I.76}$$

provided f_U and χ_{UV} are treated as orbit-side data rather than support-placement coordinates. This complements the support-side result from Stage 244.

I.6 Stage 246: Compensated multimode source compiler

Stage 246 compiles the source lane that Stage 243 only declared. The carried source kernels on the dimensionless mouth coordinate $x \in [0, 1]$ are

$$c(x) = \cos \frac{\pi x}{2}, \qquad K_q(x) = \frac{\cosh\left(\frac{\pi}{2}(1 - x)\right)}{\cosh(\pi/2)}. \tag{I.77}$$

The two source moments are

$$\mathfrak{g}[\sigma] = \int_0^1 \sigma(x) c(x)\, dx, \qquad \mathcal{S}[\sigma] = \int_0^1 \sigma(x) K_q(x)\, dx. \tag{I.78}$$

The carried Family-1 mixed-to-shell loading ratio is

$$\mathcal{R}[\sigma] = \frac{(\mathfrak{g}[\sigma] - \mathfrak{r}_{F1})^2}{1 + \mathfrak{r}^2_{F1}}, \qquad \mathfrak{r}_{F1} = \sqrt{\frac{12}{\pi^2}\left(\frac{37}{20}\right)^2 - 1}. \tag{I.79}$$

Stage 246 keeps this law and extends the source family beyond the positive corridor.

The compensated family is equation (I.17). Its exact minimum is

$$\sigma_{\min}(a,b) = \begin{cases} 1+b-|a|, & b\le 0 \text{ or } |a|\ge 4b, \\ 1-b-\dfrac{a^2}{8b}, & b>0 \text{ and } |a|\le 4b. \end{cases} \tag{I.80}$$

Thus

$$\operatorname{signchg}(a,b) \iff \sigma_{\min}(a,b)<0. \tag{I.81}$$

The exact moment formulas are equation (I.18). In normalized variables

$$\widetilde{g} := \frac{\pi}{2}\mathfrak{g}-1, \qquad \widetilde{S} := \frac{\pi}{2\tanh(\pi/2)}\mathcal{S}-1, \tag{I.82}$$

the two-moment map is

$$\begin{pmatrix}\widetilde{g}\\ \widetilde{S}\end{pmatrix} = \begin{pmatrix}1/3 & -1/15\\ 1/5 & 1/17\end{pmatrix}\begin{pmatrix}a\\ b\end{pmatrix}, \qquad \det M_{\mathrm{src}} = \frac{14}{425}. \tag{I.83}$$

The inverse is

$$a = \frac{85}{42}\widetilde{S} + \frac{25}{14}\widetilde{g}, \qquad b = \frac{425}{42}\widetilde{S} - \frac{85}{14}\widetilde{g}. \tag{I.84}$$

For the transported branch,

$$a(r) = a_0 s(r), \qquad b(r) = b_0 s(r), \qquad s(r) = \frac{r_\sigma^2}{r^2+r_\sigma^2}. \tag{I.85}$$

The output packet is

$$\mathcal{M}_\sigma(r) = \big(\sigma(x;r), \sigma_{\min}(r), \operatorname{signchg}(r), \mathfrak{g}[\sigma](r), \mathcal{S}[\sigma](r), \mathcal{R}[\sigma](r)\big), \tag{I.86}$$

with

$$\mathfrak{g}[\sigma](r) = \frac{2}{\pi}\left(1+\frac{a(r)}{3}-\frac{b(r)}{15}\right), \qquad \mathcal{S}[\sigma](r) = \frac{2\tanh(\pi/2)}{\pi}\left(1+\frac{a(r)}{5}+\frac{b(r)}{17}\right). \tag{I.87}$$

On the Session-I orientation $a_0>0$, $b_0<0$, the minimum is boundary-controlled:

$$\sigma_{\min}(r) = 1-(a_0-b_0)s(r), \qquad \operatorname{signchg}(r) \iff r<r_\sigma\sqrt{a_0-b_0-1}. \tag{I.88}$$

This is the source-side packet inserted into the stationary barrier compiler.

I.7 Stage 247: Relaxed stationary barrier front end

Stage 247 assembles the first stationary lowered barrier from the one-port short-range baseline and the three relaxed packets. The one-port static bundle is controlled by

$$\Delta = \Omega_U^2\Omega_W^2 - R_{\mathrm{mix}}^2, \qquad Q = G_U^2\Omega_W^2 + 2G_UG_WR_{\mathrm{mix}} + G_W^2\Omega_U^2, \tag{I.89}$$

$$P = \Omega_U^2 G_W + R_{\mathrm{mix}}G_U, \qquad D_0 = K_* - \frac{Q}{\Delta}. \tag{I.90}$$

With primitive short-range source amplitudes $\beta_Q, \beta_U, \beta_W$, the product-family coefficients are

$$\mathcal{C}_6 = \chi_{qq}\beta_Q^2, \qquad \mathcal{C}_4 = \chi_{qU}\beta_Q\beta_U + \chi_{qW}\beta_Q\beta_W, \tag{I.91}$$

$$\mathcal{C}_2 = \chi_{UU}\beta_U^2 + 2\chi_{UW}\beta_U\beta_W + \chi_{WW}\beta_W^2. \tag{I.92}$$

The collected baseline is

$$V_{\text{short}}^{(1p)}(r) = \frac{1}{r}\left(1 + \frac{1}{2}e^{-2\kappa r}\right) - \frac{A_6}{r^6} - A_4\frac{e^{-2\kappa r}}{r^4} - A_2\frac{e^{-4\kappa r}}{r^2}, \tag{I.93}$$

where

$$A_6 = 3\alpha_6 + \frac{1}{2}\mathcal{C}_6, \qquad A_4 = \mathcal{C}_4, \qquad A_2 = \alpha_2 + \frac{1}{2}\mathcal{C}_2. \tag{I.94}$$

This baseline inherits the same short-range firewall from Stages 219–243.

The imported relaxed scalars are

$$\mathfrak{L}(r) := \Lambda(r)\varrho(r), \qquad S_{\text{leak}} = \frac{8\sqrt{2}\eta_{\text{leak}}\mu_w\rho_0}{\pi^{5/2}\lambda^3}\mathfrak{L}, \tag{I.95}$$

$$\mathcal{W}_w^{\text{sess}} = \frac{512\eta_{\text{leak}}^2\mu_w q\rho_0}{\pi^4\lambda^2}\mathfrak{L}^2, \qquad \Delta E_{UV} = \eta_{UV}\mathcal{D}_{UV}, \tag{I.96}$$

$$\mathcal{M}_\sigma(r) = \xi_R\big[\mathcal{R}_\infty - \mathcal{R}(r)\big], \qquad \mathcal{R}_\infty = \frac{(2/\pi - \mathfrak{r}_{F1})^2}{1 + \mathfrak{r}_{F1}^2}. \tag{I.97}$$

The stationary compiler is equation (I.3); the lowering identity is equation (I.19).

The Session-I benchmark is useful as an audit point, not as a general theorem. At the strongest softening point, the Stage 247 decomposition reads

$$3.74163698 - 0.26971918 \times 0.31069599 - 1.51632107 - 0.21064278 - 0.18386120 = 1.74701126. \tag{I.98}$$

This shows that, once the one-port baseline is fixed, the recorded stationary lowering can be decomposed into work, U/V drain, compensated source response, and one remaining leakage-to-barrier embedding coefficient. It does not change the short-range kernel class.

I.8 Stage 248: Dynamic event-chain compiler

Stage 248 promotes $V_{\text{eff}}^{(247)}$ into a reduced scattering/event-chain compiler. Write

$$V(r) := V_{\text{eff}}^{(247)}(r). \tag{I.99}$$

The reduced radial equation is

$$m_s\ddot{r} = -V'(r), \tag{I.100}$$

with conserved energy equation (I.4). If the branch has a single barrier peak

$$V'(r_{\text{peak}}) = 0, \qquad V''(r_{\text{peak}}) < 0, \qquad V_{\text{peak}} = V(r_{\text{peak}}), \tag{I.101}$$

then the finite-radius threshold law is equation (I.20). The pure-Coulomb contact threshold is equation (I.21). The classical comparison window is therefore

$$v_{\text{crit,new}} < v_0 < v_{\text{contact,Coul}}. \tag{I.102}$$

For subbarrier energy $V(r_0) < E < V_{\text{peak}}$, turning points solve

$$V(r_-(E)) = E, \qquad V(r_+(E)) = E, \qquad r_-(E) < r_+(E), \tag{I.103}$$

and transport as

$$\frac{dr_\pm}{dE} = \frac{1}{V'(r_\pm(E))}. \tag{I.104}$$

The WKB action and transmission ratio are equations (I.22) and (I.23). The pure-Coulomb outer turning point is exact,

$$r_{\text{turn,Coul}}(E) = \frac{1}{E}, \tag{I.105}$$

and the Coulomb action closes as

$$I_{\text{Coul}}(E) = \frac{\sqrt{2m_s}}{\hbar_{\text{eff}}} \left[\frac{\pi}{2\sqrt{E}} - \sqrt{r_{\text{contact}}(1 - Er_{\text{contact}})} - \frac{\arcsin\sqrt{Er_{\text{contact}}}}{\sqrt{E}} \right] \tag{I.106}$$

for $Er_{\text{contact}} < 1$.

Near the top, with

$$V(r) = V_{\text{peak}} - \frac{K_{\text{peak}}}{2}(r - r_{\text{peak}})^2 + O((r - r_{\text{peak}})^3), \qquad K_{\text{peak}} = -V''(r_{\text{peak}}), \tag{I.107}$$

the leading top action is

$$I_{\text{top}}(E) = \frac{\pi(V_{\text{peak}} - E)}{\hbar_{\text{eff}}} \sqrt{\frac{m_s}{K_{\text{peak}}}} + O((V_{\text{peak}} - E)^{3/2}). \tag{I.108}$$

The dynamic diagnostics carried forward are

$$\Xi_{\text{turn}}(E) = \Xi_1(r_+(E)), \qquad \lambda_{\text{th}}(E) = \left| \frac{E}{V'(r_+(E))} \right|. \tag{I.109}$$

These tags are later used in the helicity and Goldilocks compilers; they do not alter the scalar event chain.

I.9 Stage 249: Helicity export diagnostic

Stage 249 attaches a hidden-channel export diagnostic to the Stage 248 event chain. It is diagnostic rather than constitutive. The parent projected helicity ledger gives the exact subscale helicity

$$h_{\text{sub}} = \overline{h} - h_{\text{res}} = \overline{\mathbf{A}' \cdot \mathbf{B}'}, \tag{I.110}$$

with transfer equation

$$\partial_t h_{\text{sub}} + \nabla_3 \cdot \mathbf{F}_{h,\text{sub}} = -2\overline{\mathbf{E}' \cdot \mathbf{B}'}. \tag{I.111}$$

On a reduced brane control volume,

$$\frac{dH_{\text{sub}}}{dt} = -\Phi_h(t; E) - 2\mathcal{C}_h(t; E), \tag{I.112}$$

where Φ_h is the unresolved-helicity flux through the control boundary and $\mathcal{C}_h$ is the unresolved covariance source.

The minimal orientation closure introduces $\sigma = +1$ for aligned and $\sigma = -1$ for anti-aligned, with

$$\Phi_{h,\sigma} = \Phi_{h,0} + \sigma\Phi_{h,1}, \qquad \mathcal{C}_{h,\sigma} = \mathcal{C}_{h,0} + \sigma\mathcal{C}_{h,1}. \tag{I.113}$$

Thus
$$\dot{H}_{\mathrm{sub},\sigma} = \Gamma_0 + \sigma\Gamma_1 = \Gamma_0(1+\sigma\alpha_h), \qquad \alpha_h = \frac{\Gamma_1}{\Gamma_0}. \tag{I.114}$$
The instantaneous ratio and inverse are
$$R_{\mathrm{inst}} = \frac{\dot{H}_{\mathrm{sub},+}}{\dot{H}_{\mathrm{sub},-}} = \frac{1+\alpha_h}{1-\alpha_h}, \qquad \alpha_h = \frac{R_{\mathrm{inst}}-1}{R_{\mathrm{inst}}+1}. \tag{I.115}$$
Over a common interval,
$$R_{\mathrm{int}} = \frac{\mathcal{H}_+}{\mathcal{H}_-} = \frac{1+\bar{\alpha}_h}{1-\bar{\alpha}_h}, \qquad \bar{\alpha}_h = \frac{R_{\mathrm{int}}-1}{R_{\mathrm{int}}+1}. \tag{I.116}$$
Any common export scale cancels from these ratios.

The benchmark packet is
$$\begin{aligned} (\Xi_{\mathrm{turn}}, \lambda_{\mathrm{th}}, R_{\mathrm{pk}}) &= (0.34437471, 0.42826825, 4.94653917), \\ (R_{\mathrm{int}}, \alpha_{\mathrm{pk}}, \bar{\alpha}_h) &= (4.10920923, 0.66366992, 0.60854999). \end{aligned} \tag{I.117}$$
The status is deliberately limited: this proves a hidden-channel unloading preference inside the declared closure, not intrinsic spin or a Pauli-sector theorem.

I.10 Stage 250: Crossing-vs-collapse Goldilocks window

Stage 250 turns the dynamic branch into a survival inequality. The exact path-time integral from the Stage 248 energy law is
$$T_{\mathrm{traj}}(E; r_a \to r_b) = \sqrt{\frac{m_s}{2}} \int_{r_b}^{r_a} \frac{dr}{\sqrt{E - V(r)}}. \tag{I.118}$$
The characteristic-width closure used by Session III compresses the active barrier region to
$$t_{\mathrm{cross}}(E) = \lambda_{\mathrm{eff}} \sqrt{\frac{m_s}{2(E - V_{\mathrm{peak}})}}. \tag{I.119}$$
This is a reduced timescale closure, not a replacement for the exact path integral.

The active dressing leg uses the local unstable normal form
$$\mu_\eta \ddot{V} = g_{UV} \chi_{\mathrm{peak}} V. \tag{I.120}$$
Hence
$$t_{\mathrm{collapse}} = \sqrt{\frac{\mu_\eta}{g_{UV}\chi_{\mathrm{peak}}}}. \tag{I.121}$$
The survival ratio is
$$\mathcal{S}(E) = \frac{t_{\mathrm{cross}}(E)}{t_{\mathrm{collapse}}} = \lambda_{\mathrm{eff}} \sqrt{\frac{m_s g_{UV} \chi_{\mathrm{peak}}}{2\mu_\eta (E - V_{\mathrm{peak}})}}. \tag{I.122}$$
The lower safe edge is equation (I.25), and the speed-space identity is equation (I.26).

If $\mu_\eta = \alpha m_s$, the edge becomes
$$E_{\mathrm{edge}} = V_{\mathrm{peak}} + \frac{g_{UV}\chi_{\mathrm{peak}}\lambda_{\mathrm{eff}}^2}{2\alpha}, \tag{I.123}$$

so simultaneous scaling of particle mass and wall inertia cancels out of the edge. The real control packet is

$$(\lambda_{\text{eff}}, \chi_{\text{peak}}, g_{UV}, \mu_\eta/m_s). \tag{I.124}$$

With the trigger-width specialization,

$$\lambda_{\text{eff}} = \lambda_{\text{th}}(r_{\text{turn}}) = \left|\frac{V(r_{\text{turn}})}{V'(r_{\text{turn}})}\right|. \tag{I.125}$$

The benchmark edge and sensitivity statements are therefore direct consequences of the dependence

$$E_{\text{edge}} - V_{\text{peak}} \propto \lambda_{\text{eff}}^2 \chi_{\text{peak}}. \tag{I.126}$$

Within this closure the safe window is one-sided because t_{collapse} is energy-independent and $t_{\text{cross}}(E)$ decreases monotonically. Any upper edge requires an additional high-energy failure mechanism outside Stage 250.

I.11 Stage 251: Microscopic export kernel

Stage 251 replaces the phenomenological $\gamma_{\text{tot}}\dot{V}$ envelope law by the first microscopic odd export operator seen by the active dressing leg. Project

$$V(t) = \Pi_{V0}q_0(t) + \Pi_{V-}q_-(t) + \cdots, \tag{I.127}$$

where q_0 is the derivative-coupled scalar outlet and q_- is the selected mixed quadrupole outlet.

For the scalar lane, take

$$g_{W,0}(\omega) = \eta_0\omega, \qquad g_{A,0}(\omega) = 0, \qquad \Pi_0^{\text{out}}(\omega) = i\gamma_1\omega + O(\omega^3). \tag{I.128}$$

With

$$\Delta_0 = \Omega_{U,0}^2\Omega_{W,0}^2 - R_0^2, \tag{I.129}$$

the scalar outgoing factor gives

$$\Gamma_{3,0} = \gamma_1\eta_0^2\frac{\Omega_{U,0}^4}{\Delta_0^2}, \qquad \Gamma_3 = \Pi_{V0}^2\Gamma_{3,0}. \tag{I.130}$$

For the selected quadrupole outlet,

$$\Gamma_{5,-} = \frac{a^5}{27c_s^5}P_{0,-}, \qquad P_{0,-} = \frac{\beta_0 s_-}{\lambda_-}, \qquad \Gamma_5 = \Pi_{V-}^2\Gamma_{5,-}. \tag{I.131}$$

Thus the microscopic kernel is equation (I.27).

The active-leg equation is

$$\mu_\eta\ddot{V} - \kappa_V V + \Gamma_3 V^{(3)} - \Gamma_5 V^{(5)} = \mathcal{S}_V(t), \qquad \kappa_V = g_{UV}\chi_{\text{peak}}. \tag{I.132}$$

In Laplace form,

$$\mathcal{D}_V(s) = \mu_\eta s^2 - \kappa_V + \Gamma_3 s^3 + \Gamma_5 s^5. \tag{I.133}$$

The passive power identity is

$$\dot{V}(\Gamma_3 V^{(3)} - \Gamma_5 V^{(5)}) = \frac{d}{dt}\mathcal{S}_{\text{odd}} - \Gamma_3\ddot{V}^2 - \Gamma_5\dddot{V}^2, \tag{I.134}$$

so the exported power is

$$\mathcal{P}_{\text{exp}} = \Gamma_3 \ddot{V}^2 + \Gamma_5 \dddot{V}^2 \geq 0. \tag{I.135}$$

For homogeneous growth $V = V_0 e^{st}$, the characteristic polynomial is

$$F(s) = \Gamma_5 s^5 + \Gamma_3 s^3 + \mu_\eta s^2 - \kappa_V. \tag{I.136}$$

For positive $(\Gamma_3, \Gamma_5, \mu_\eta, \kappa_V)$, $F(0) < 0$, $F(s) \to +\infty$, and $F'(s) > 0$ for $s > 0$. Therefore there is exactly one positive growth root. The small-kernel slowdown is

$$s_+ = s_0 - \frac{\Gamma_3 s_0^2 + \Gamma_5 s_0^4}{2\mu_\eta} + O(\Gamma^2), \qquad s_0 = \sqrt{\kappa_V/\mu_\eta}. \tag{I.137}$$

Thus finite passive cubic/quintic export slows but does not remove the unstable root in the minimal closure.

For an event with crossing rate $s_c = 1/t_{\text{cross}}$, strict monotonicity of F gives the safe half-plane equation (I.28). In the benchmark cold event, this becomes

$$\widehat{\Gamma}_3 + 0.3013336471\widehat{\Gamma}_5 \geq 289.61004918. \tag{I.138}$$

This is the exact event-level replacement for the old γ_{safe} inequality.

Stage-pair note. Stages 252 and 253 form the physical companion to the export kernel. Stage 252 keeps the reduced microscopic language; Stage 253 maps it into physical threshold variables.

I.12 Stage 252: Vacuum/lattice heat partition

I.12.1 Channel-resolved heat partition

Split the microscopic coefficients as

$$\Gamma_3 = \Gamma_3^{\text{vac}} + \Gamma_3^{\text{lat}}, \qquad \Gamma_5 = \Gamma_5^{\text{vac}} + \Gamma_5^{\text{lat}}, \qquad \Gamma_n^{\text{vac}}, \Gamma_n^{\text{lat}} \geq 0. \tag{I.139}$$

For an event window $[0, T]$, define

$$\mathcal{I}_2(T) = \int_0^T \ddot{V}^2\, dt, \qquad \mathcal{I}_3(T) = \int_0^T \dddot{V}^2\, dt. \tag{I.140}$$

The integrated energies are

$$E_{\text{vac}} = \Gamma_3^{\text{vac}} \mathcal{I}_2 + \Gamma_5^{\text{vac}} \mathcal{I}_3, \qquad E_{\text{lat}} = \Gamma_3^{\text{lat}} \mathcal{I}_2 + \Gamma_5^{\text{lat}} \mathcal{I}_3. \tag{I.141}$$

With $\mathfrak{r}_V = \mathcal{I}_3/\mathcal{I}_2$, the fractions are

$$f_{\text{vac}}(\mathfrak{r}_V) = \frac{\Gamma_3^{\text{vac}} + \Gamma_5^{\text{vac}} \mathfrak{r}_V}{\Gamma_3 + \Gamma_5 \mathfrak{r}_V}, \qquad f_{\text{lat}}(\mathfrak{r}_V) = \frac{\Gamma_3^{\text{lat}} + \Gamma_5^{\text{lat}} \mathfrak{r}_V}{\Gamma_3 + \Gamma_5 \mathfrak{r}_V}. \tag{I.142}$$

The partition drift obeys

$$\frac{df_{\text{lat}}}{d\mathfrak{r}_V} = \frac{\Gamma_5^{\text{lat}} \Gamma_3^{\text{vac}} - \Gamma_3^{\text{lat}} \Gamma_5^{\text{vac}}}{(\Gamma_3 + \Gamma_5 \mathfrak{r}_V)^2}. \tag{I.143}$$

So slow events sample the cubic split, fast rough events sample the quintic split.

For a single-growth cold event

$$V(t) = V_{\rm in} e^{st}, \qquad 0 \le t \le T, \tag{I.144}$$

$$\mathcal{I}_2 = \frac{V_{\rm in}^2 s^3}{2}(e^{2sT} - 1), \qquad \mathcal{I}_3 = s^2 \mathcal{I}_2, \qquad \mathfrak{r}_V = s^2. \tag{I.145}$$

The event-equivalent rates are

$$\gamma_{\rm vac}^{\rm eq}(s) = \Gamma_3^{\rm vac} s^2 + \Gamma_5^{\rm vac} s^4, \qquad \gamma_{\rm lat}^{\rm eq}(s) = \Gamma_3^{\rm lat} s^2 + \Gamma_5^{\rm lat} s^4. \tag{I.146}$$

At the safe edge, with $T = 1/s_c$ and $\Gamma_3 s_c^3 + \Gamma_5 s_c^5 = \mu_\eta (s_0^2 - s_c^2)$, the total minimum exported energy is equation (I.30), and the channel energies are obtained by multiplying by $f_{\rm vac}(s_c)$ or $f_{\rm lat}(s_c)$.

The old Session-IV $3:1$ heat split becomes a coefficient-surface condition:

$$f_{\rm lat}(s_c) = \frac{3}{4} \iff \Gamma_3^{\rm lat} + \Gamma_5^{\rm lat} s_c^2 = 3(\Gamma_3^{\rm vac} + \Gamma_5^{\rm vac} s_c^2). \tag{I.147}$$

It is speed-independent only on the special family where both cubic and quintic channels have the same lattice fraction.

I.13 Stage 253: Physical calibration and material thresholds

I.13.1 Physical calibration dictionary

Stage 253 introduces the explicit physical dictionary

$$t^{\rm phys} = t_* t, \qquad r^{\rm phys} = \frac{\lambda_{\rm phys}}{\lambda_{\rm ref}} r, \qquad E^{\rm phys} = E_* E. \tag{I.148}$$

The safe-edge lattice event-equivalent rate is

$$\gamma_{\rm lat,safe}^{\rm eq} = f_{\rm lat}(s_c) \mu_\eta \frac{s_0^2 - s_c^2}{s_c}. \tag{I.149}$$

A coarse transport projection factor $\Upsilon_{\rm lat} > 0$ defines the physical lattice-turnover rate

$$\gamma_{\rm lat,turn}^{\rm phys} = \frac{\gamma_{\rm lat,safe}^{\rm eq}}{\Upsilon_{\rm lat} t_*} = \zeta_{\rm ep} \lambda_{\rm ep} \omega_D. \tag{I.150}$$

Thus the electron-phonon turnover threshold is equation (I.31), and the product form is

$$(\lambda_{\rm ep} \omega_D t_{\rm cross}^{\rm phys})_{\rm min}^{(253)} = \frac{f_{\rm lat}(s_c) \mu_\eta (s_0^2 - s_c^2)}{\Upsilon_{\rm lat} \zeta_{\rm ep} s_c^2}. \tag{I.151}$$

The legacy Session-V slice is recovered by choosing

$$\Upsilon_{\rm lat}^{\rm sess} = \frac{\gamma_{\rm lat,safe}^{\rm eq}}{\gamma_{\rm lattice}^{\rm red}}. \tag{I.152}$$

This is a calibration slice, not a new branch law.

For the harmonic interstitial companion,

$$r_{\rm turn}^{\rm phys} = \frac{r_{\rm turn}}{\lambda_{\rm ref}} \lambda_{\rm phys}, \tag{I.153}$$

and

$$\chi_{\lambda,\mathrm{lattice}}(r_{\mathrm{turn}}) = \frac{2\lambda_{\mathrm{phys}}}{r_{\mathrm{turn}}^{\mathrm{phys}}} = \frac{2\lambda_{\mathrm{ref}}}{r_{\mathrm{turn}}}. \tag{I.154}$$

This is only a geometric trigger. The force-matched stiffness requires

$$k_{\mathrm{eff,req}} = \frac{E_*\lambda_{\mathrm{ref}}^2}{\lambda_{\mathrm{phys}}^2}\frac{|V'_{\mathrm{red}}(r_{\mathrm{turn}})|}{r_{\mathrm{turn}}} = \mathcal{K}_{\mathrm{turn}}\frac{E_*}{\lambda_{\mathrm{phys}}^2}, \tag{I.155}$$

where

$$\mathcal{K}_{\mathrm{turn}} = \frac{\lambda_{\mathrm{ref}}^2}{r_{\mathrm{turn}}}|V'_{\mathrm{red}}(r_{\mathrm{turn}})|. \tag{I.156}$$

For $\lambda_{\mathrm{phys}} = a_{\mathrm{int}}/2$,

$$k_{\mathrm{eff,req}} = 4\mathcal{K}_{\mathrm{turn}}\frac{E_*}{a_{\mathrm{int}}^2}. \tag{I.157}$$

The Korringa companion is the cleanest physical map. With

$$T_1 T = \mathcal{K}_{\mathrm{corr}}, \qquad T_1 \geq t_{\mathrm{cross}}^{\mathrm{phys}}, \tag{I.158}$$

the ceiling temperature is

$$T_{\mathrm{max}} = \frac{\mathcal{K}_{\mathrm{corr}}}{t_{\mathrm{cross}}^{\mathrm{phys}}} = \frac{s_c\mathcal{K}_{\mathrm{corr}}}{t_*}. \tag{I.159}$$

I.13.2 Material-screening ratios

The companion ends with four screening ratios:

$$\Pi_{\mathrm{ep}} = \frac{\Upsilon_{\mathrm{lat}}\zeta_{\mathrm{ep}}\lambda_{\mathrm{ep}}\omega_D t_*}{\gamma_{\mathrm{lat,safe}}^{\mathrm{eq}}} = \frac{\lambda_{\mathrm{ep}}\omega_D}{(\lambda_{\mathrm{ep}}\omega_D)_{\mathrm{min}}^{(253)}}, \tag{I.160}$$

$$\Pi_\chi = \frac{2\lambda_{\mathrm{ref}}}{r_{\mathrm{turn}}}, \tag{I.161}$$

$$\Pi_k = \frac{k_{\mathrm{eff}}\lambda_{\mathrm{phys}}^2}{\mathcal{K}_{\mathrm{turn}}E_*} = \frac{k_{\mathrm{eff}}}{k_{\mathrm{eff,req}}}, \tag{I.162}$$

$$\Pi_T(T) = \frac{\mathcal{K}_{\mathrm{corr}}}{T t_{\mathrm{cross}}^{\mathrm{phys}}} = \frac{T_{\mathrm{max}}}{T}. \tag{I.163}$$

The materials companion is then

$$\Pi_{\mathrm{ep}} \geq 1, \qquad \Pi_\chi \geq 1, \qquad \Pi_k \geq 1, \qquad \Pi_T(T) \geq 1. \tag{I.164}$$

This is not a claim that any particular host passes. It is the reduced threshold stack that a candidate host must satisfy after the physical unit map and material constants are supplied.

I.14 Citation and downstream-use guardrails

Part VIII is useful because it lets downstream papers talk about the relaxed branch without contaminating the strict branch. The safe citation language is:

> The relaxed/open-system companion is a codimension-three lift of the strict selected-support/orbit front end. It has an exact recovery map to the strict slice, preserves the no-new-long-range same-charge verdict, and supplies separate leakage/work, non-rigid dressing, compensated-source, barrier, dynamic-event, export-kernel, heat-partition, and material-screening compilers.

The unsafe citation language is:

> The relaxed barrier branch proves the strict selected same-charge branch, or supplies a new long-range attraction.

That statement is false under the status discipline of this ledger. The relaxed branch is a companion extension around the strict front end. It is valuable precisely because it states what can change when open-system and non-rigid effects are allowed while preserving what cannot change: the same-charge long-range kernel verdict.

The active open tasks after this part are also explicit. The completed moving-throat PDE still has to realize or reject the strict selected support/orbit/outgoing packet from Part VII. If the relaxed companion is used, the actual branch must additionally return the leakage amplitude, non-rigid (U, V) packet, compensated source packet, microscopic export coefficients (Γ_3, Γ_5) and their vacuum/lattice split, the unit map $(t_*, \lambda_{\text{phys}}, E_*)$, and the coarse transport factor Υ_{lat}. Without those realized data, Part VIII remains an exact compiler rather than a completed physical-branch theorem.

Appendix J

Reproducibility Map

J.1 Purpose of this appendix

This appendix specifies how the derivation companion is to be checked. The companion is not a single proof written in one style. It is a layered verification record: some statements are exact identities of the parent action, some are exact algebra inside a declared closure, some are controlled reductions, some are finite search or numerical-location statements, and some remain open branch-realization targets. The purpose of this map is to keep those layers auditable without weakening or overstating any claim.

The central rule is simple: a result may be cited only at the strongest status level supported by its evidence trail. A symbolic Schur complement, a finite-dimensional search sieve, and an actual nonlinear moving-throat branch realization are different kinds of evidence. They should not be collapsed into the same proof label merely because they appear in the same derivation chain.

J.2 Reproducibility layers

Table J.1: Reproducibility layers used by this ledger.

Layer	What it controls	Minimum evidence required	Reusable claim status
Parent-source layer	Exact imported field content, equations, charge ontology, projection/reduction language, and mixed-sector firewall.	A cited source section in the parent action stack or the framework paper, plus a consistent notation-firewall entry in this companion.	Usually EXACT or REDUCED / CONTROLLED REDUCTION for declared reductions.
Stage-source layer	The 253-stage derivation record preserved in the moving-throat output archive.	Stage number, source-stage title, local assumptions, and a link to the corresponding stage appendix entry.	Whatever the stage appendix declares locally.
Synthesis layer	The theorem-block chapters in Parts I–VIII.	A theorem/result anchor, a status tag, explicit hypotheses, and pointers to stage appendices.	The status of the weakest essential input.

Layer	What it controls	Minimum evidence required	Reusable claim status
Hand-algebra layer	Identities short enough to audit directly from the printed equations.	The derivation shown in the appendix, sign conventions, limiting checks, and dimensional or symmetry checks where relevant.	EXACT or EXACT WITHIN CLOSURE when hypotheses are closed.
Symbolic-audit layer	Polynomial identities, Schur complements, projectors, finite-dimensional compilers, and exact simplifications.	Script basename, output transcript, input assumptions, exact arithmetic mode, and a statement of what the script verifies.	EXACT WITHIN CLOSURE for the declared symbolic model.
Numerical-location layer	Roots, optima, branch placements, ranking tables, and finite candidate reductions.	Source script, deterministic input file, tolerances, brackets or candidate set, output transcript, and independent rerun or residual check.	NUMERICALLY LOCATED unless converted to closed form.
Branch-realization layer	Whether the actual moving-throat PDE lands on the required ratios, slopes, orbit locks, support placements, or relaxed packets.	An explicit branch construction, PDE solve or analytic branch theorem, and evaluation of the declared observable packet.	OPEN until that construction is supplied.
Repository-freeze layer	Reproducibility of the document itself.	Source commit or archive hash, script hashes, dependency versions, build command, and generated-output hashes.	Does not change mathematical status; it makes the status auditable.

J.3 Claim-status evidence rule

Table J.2: Evidence required before a result can be cited at a given status.

Status	Evidence required	What must not be claimed
EXACT	Direct derivation from the declared parent action, exact definitions, exact calculus, exact algebra, or exact pair counting.	Do not attach this status to a result that assumes a branch ansatz, a low-frequency truncation, or a fitted realization parameter.

Status	Evidence required	What must not be claimed
Exact Within Closure	Exact derivation after all closure hypotheses, finite-dimensional variables, branch restrictions, and search class boundaries are declared.	Do not cite it as an unrestricted theorem of the full nonlinear moving-throat PDE.
Reduced / Controlled Reduction	Explicit reduction assumptions such as zero-mode, small-body, passive/outgoing, near-zone, weak-axisymmetric, finite-throat, or low-frequency hypotheses.	Do not omit the reduction conditions in downstream papers.
Effective Closure	A physically motivated closure or constitutive choice that organizes the derivation but has not been uniquely derived from the completed PDE.	Do not call the closure unique or microscopic unless a later branch theorem supplies that result.
Numerically Located	Exact problem definition plus numerical location, residual checks, deterministic candidate list, or scripted search transcript.	Do not treat an observed numerical placement as a closed-form theorem unless the algebraic proof is added.
Open	Defined observable, target, or branch packet whose realization remains to be shown.	Do not cite the target as realized; cite it only as the remaining theorem gate or test condition.

J.4 Canonical source registry

The source registry in table J.3 is not a bibliography. It is the internal provenance map for the ledger. The source-file index gives a fuller list of files; this table records the role each source plays in reproducibility.

Table J.3: Canonical source artifacts and their verification roles.

Source artifact	Verification role	Safe use inside this companion
`paper/stages/stage_NNN.tex`	Canonical 253-stage derivation archive.	Primary provenance source for the stage appendices, stage titles, staged assumptions, stage-local formula outputs, and script-backed status notes.

Source artifact	Verification role	Safe use inside this companion
`paper/parts/*.tex` plus front-matter	Compact theorem-structure master through Stages 219–253.	Source for the global status firewall, current bottleneck summary, final packet language, same-charge verdict, rigid-mouth normal form, relaxed branch, export kernel, and materials companion.
`pde.tex`	Framework paper already built from the compact program.	Source for operational branch-test language, finite observable packets, and the division between framework use and derivation verification.
`parent-action source summaries` and `4d.tex`	Exact parent action and projection/reduction backbone.	Parent GNLS/Maxwell equations, charge ontology, brane projection, leakage identity, Poisson hook, and geometry-sector inheritance.
`localized electromagnetic source summaries`	Localized Maxwell derivation.	Gauge-sector equations, zero-mode brane Maxwell reduction, thickness-controlled charge strength, KK/retarded structure, and mixed-sector caveats.
`plasma and open-system source summaries`	Plasma and open-system projection companion.	Mixed-sector fields, brane-bulk leakage, projection noncommutation, and beyond-MHD open-subsystem interpretation.
`1PN bridge and assembly sources`	Lower-order conservative bridge and full 1PN assembly.	Carry-forward constants, closure taxonomy, EIH target matching, topology/inertia data, and the distinction between exact assembly and protocol closure.
`2PN source stack` and `4d_2pn.tex`	Full conservative 2PN assembly.	First appearance of the even support layer, grouped support data, geometry deficit language, and lower-order response slots imported into the moving-throat program.
`2.5PN retarded source stack`	Retarded quadrupole audit.	Odd-channel hierarchy, scalar/dipole demotion logic, STF quadrupole route, outgoing $l = 2$ fingerprint, and passive/outgoing normalization gap.

Source artifact	Verification role	Safe use inside this companion
`3PN source stack` and `4d_3pn.tex`	Full conservative 3PN assembly.	Grouped real P_2 conservative payload, fixed chart logic, and the separation between algebraic closure and moving-throat constitutive derivation.
`4PN source stack`	Conservative 4PN and tail-interface summary.	Local/tail split, quartic compiler, local 4PN closure, and the fact that the 4PN tail uses the same quadrupole normalization gap isolated at 2.5PN.
`5PN and continuation source stack`	Weak-axisymmetric and higher-order continuation notes.	Explicit overlap models, weak-axisymmetric transport, monomial/orbit quotient machinery, support-compensation thresholds, and target-surface sharpening.
`companion reduced-sector source stack`	Companion reduced-sector applications.	Should be cited only for downstream motivation or cross-check packets. They do not override the claim-status firewall of this derivation companion.

J.5 Result-anchor audit map

Table J.4: Minimum audit trail for top-level result anchors.

Anchor	Stages	Minimum audit trail	What must be checked before downstream citation
MTDC-T1	Parent + 001	Parent source registry; Stage 001 source; notation firewall.	Exact parent equations must be separated from the effective geometry lift. Mixed channels may be suppressed only under the zero-mode reduction, not erased.
MTDC-T2	001–002	Stage 001–002 appendices; harmonic normalization; wall-action reduction.	The densitized convention, Y_{00} normalization, a, L moment recovery, and grouped P_2 degeneracy must be explicit.
MTDC-T3	003	Stage 003 appendix; BdG Schur-complement audit.	Stable-mode reduction must be declared; exact self-energy, low-frequency coefficients, and pole-shift formulas must agree with the script-backed algebra.

Anchor	Stages	Minimum audit trail	What must be checked before downstream citation
MTDC-T4	004–022	Stage 004–022 appendices; projection-first Maxwell audits; parent-action export packet; mixed-sector transfer algebra; outgoing fingerprint.	Projected leakage law, matched reduction conditions, gauge-invariant mixed variables, projected Z_n, N_n transport, parent wall/export packet, passive/outgoing sign convention, $l = 2$ $i\omega^5$ coefficient, and normalized prefactor product must be preserved.
MTDC-T5	023–036	Stage 023–036 appendices; projector and overlap checks.	Weighted grouped projectors, isotropy defects, weak-axisymmetric $b = 3a$ law, selected-mode normalization, and support frontier must be internally consistent.
MTDC-T6	037–051	Stage 037–051 appendix; continuum and support-placement ledger.	Dimensionless placement variables, rank-2 closure, tracking branch, support-compensation law, and lowest-twin sufficiency must keep their branch assumptions.
MTDC-T7	052–090	Stage 052–090 appendix; Family-1 branch audit.	Gain thresholds, parent-overlap tests, wall-shell branch extraction, loading windows, and updated Stage 090 verdict must distinguish support-side success from remaining precursor derivation.
MTDC-T8	091–163	Stage 091–163 appendix; retarded/outlet/mouth compensation ledger.	Geometry-lane contamination, outgoing-normalization collapse, outlet/core compensation, mouth-layer fixed points, and final off-family normal coordinate must not be cited as completed branch realization unless the branch values are supplied.
MTDC-T9	164–200	Stage 164–200 appendix; monomial/orbit quotient scripts where available.	$\Xi_1 = P_1/P_0$, weak-axisymmetric line, monomial quotient coordinates, similarity orbit, Packet-A closure, and four-scalar verdict packet must be kept as quotient statements, not arbitrary drift claims.

Anchor	Stages	Minimum audit trail	What must be checked before downstream citation
MTDC-T10	201–218	Stage 201–218 appendix; realization compiler scripts; finite search transcript.	Free-quintuple graph, log-ray roots, Hessian ranking, pairwise/simplex searches, and support-cardinality-5 closure must be tied to the declared local mixed-ray search class.
MTDC-T11	219–242	Stage 219–242 appendix; late same-charge and rigid-mouth audit scripts.	Static no-new-long-range verdict, dynamic resonance corridor, actual-branch packet, rigid-mouth (U, V) chart, orbit lock, and selected twin-support placement must preserve their actual-branch status.
MTDC-T12	243–253	Stage 243–253 appendix; relaxed branch audit scripts.	Relaxed constraints, leakage/work, non-rigid U/V drain, barrier/event-chain compilers, export kernel, heat partition, and material thresholds must be labeled as relaxed/open-system branch data, not strict conservative proof.

J.6 Primary symbolic-audit registry

The stage source contains several explicit audit basenames. When the repository is frozen, each audit basename should be represented by the script, its output transcript, and the corresponding stage-source note. If a script is renamed, the stage appendix should record the renamed artifact and preserve the original basename as a provenance alias.

J.6.1 Early and middle symbolic audits

Table J.5: Early and middle symbolic audit artifacts.

Stages	Audit artifact	Certifies	Required output record
001–002	`moving_throat_pde_master_sympy_audit.py`	Harmonic normalization, mouth-average extraction, two-mode Euler–Lagrange matrix form, and one-mode grouped P_2 reduction.	Script transcript or notebook output showing exact symbolic simplifications to zero residual.

Stages	Audit artifact	Certifies	Required output record
003	`moving_throat_pde_stage003_bdg_sympy_audit.py`	BdG–wall Schur complement, low-frequency moment expansion, two-pole spectrum, perturbative pole shift, and grouped trace/anomaly formulas.	`moving_throat_pde_stage003_bdg_sympy_output.txt` or equivalent exact transcript.
191	`moving_throat_pde_stage191_minimal_pde_data_packet_sympy_audit.py`	Minimal PDE data packet and graph-error realization form used by the late quotient/back-end compiler.	`moving_throat_pde_stage191_minimal_pde_data_packet_sympy_audit_output.txt`.
192	`moving_throat_pde_stage192_orbit_quotient_projectors_sympy_audit.py`	Orbit/quotient projectors and branch-invariant projection algebra.	`moving_throat_pde_stage192_orbit_quotient_projectors_sympy_audit_output.txt`.
193	`moving_throat_pde_stage193_isotropic_grouped_p2_target_surface_sympy_audit.py`	Isotropic grouped P_2 target surface and the exact one-pole / outgoing-normalization surface.	Output transcript with the target-surface residuals simplified.
202	`moving_throat_pde_stage202_free_quintuple_target_graph_sympy_audit.py`	Free-quintuple target graph and scalar closure slice at the start of the realization compiler.	`moving_throat_pde_stage202_free_quintuple_target_graph_sympy_audit_output.txt`.

J.6.2 Late same-charge, rigid-mouth, and relaxed-branch audit registry

Table J.6: Stage 219–253 audit registry. Each basename should be archived as an executable script artifact, with a written review record when available.

Stage	Audit role	Basename / artifact	What the verifier should confirm
219	one-port static same-charge kernel	`moving_throat_pde_stage219_one_port_mixed_bundle_static_kernel_and_square_law_suppression_test_sympy_audit.py`; stage-local written review record	Listed stage-local audit artifact; any executable companion should have a captured output transcript archived with the same basename and an `_output.txt` suffix.
220	dynamic mixed-port phase-lag/no-go gate	`moving_throat_pde_stage220_dynamic_mixed_port_kernel_phase_lag_no_go_and_resonant_survival_gate_sympy_audit.py`; stage-local written review record	Listed stage-local audit artifact; any executable companion should have a captured output transcript archived with the same basename and an `_output.txt` suffix.
221	resonance linewidth trade-off	`moving_throat_pde_stage221_resonance_linewidth_tradeoff_dispersive_no_free_lunch_theorem_and_linear_survival_window_sympy_audit.py`; stage-local written review record	Listed stage-local audit artifact; any executable companion should have a captured output transcript archived with the same basename and an `_output.txt` suffix.

Stage	Audit role	Basename / artifact	What the verifier should confirm
222	finite-throat primitive pole census	`moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.py`; stage-local written review record	Listed stage-local audit artifact; any executable companion should have a captured output transcript archived with the same basename and an `_output.txt` suffix.
223	primitive-branch compatibility and survival window	`moving_throat_pde_stage223_5pn_isotropic_target_surface_primitive_branch_compatibility_and_dynamic_survival_window_sympy_audit.py`; stage-local written review record	Listed stage-local audit artifact; any executable companion should have a captured output transcript archived with the same basename and an `_output.txt` suffix.
224	weak-axisymmetric ceiling transport	`moving_throat_pde_stage224_pde_branch_packet_compiler_weak_axisymmetric_ceiling_transport_and_first_actual_branch_kill_test_sympy_audit.py`; stage-local written review record	Listed stage-local audit artifact; any executable companion should have a captured output transcript archived with the same basename and an `_output.txt` suffix.
225	microscopic Ξ_1 compiler and survival sieve	`moving_throat_pde_stage225_microscopic_xi1_compiler_first_order_conservative_compensation_surface_and_mixed_sector_survival_sieve_sympy_audit.py`; stage-local written review record	Listed stage-local audit artifact; any executable companion should have a captured output transcript archived with the same basename and an `_output.txt` suffix.
226	strict even-gate package and mixed corridor	`moving_throat_pde_stage226_strict_5pn_even_gate_package_surviving_mixed_corridor_and_pure_transfer_subcorridor_sympy_audit.py`; stage-local written review record	Listed stage-local audit artifact; any executable companion should have a captured output transcript archived with the same basename and an `_output.txt` suffix.
227	pure-transfer load-factor rigidity sieve	`moving_throat_pde_stage227_pure_transfer_load_factor_outgoing_rigidity_sieve_and_first_co_loading_no_go_sympy_audit.py`; stage-local written review record	Listed stage-local audit artifact; any executable companion should have a captured output transcript archived with the same basename and an `_output.txt` suffix.
228	pure-transfer numerator/denominator split	`moving_throat_pde_stage228_numerator_denominator_split_of_the_pure_transfer_corridor_and_first_actual_dynamic_window_test_sympy_audit.py`; stage-local written review record	Listed stage-local audit artifact; any executable companion should have a captured output transcript archived with the same basename and an `_output.txt` suffix.
229	selected-branch signature and crossover theorem	`moving_throat_pde_stage229_selected_branch_numerator_denominator_signature_and_softening_depth_crossover_theorem_sympy_audit.py`; stage-local written review record	Listed stage-local audit artifact; any executable companion should have a captured output transcript archived with the same basename and an `_output.txt` suffix.
230	dynamic-window classifier and static-first theorem	`moving_throat_pde_stage230_selected_branch_classifier_to_dynamic_window_compiler_and_static_first_theorem_sympy_audit.py`; stage-local written review record	Listed stage-local audit artifact; any executable companion should have a captured output transcript archived with the same basename and an `_output.txt` suffix.
231	continuum-placement pull-back of class map	stage-local written review record	Listed stage-local audit artifact; any executable companion should have a captured output transcript archived with the same basename and an `_output.txt` suffix.
232	known-data injection and branch verdict	stage-local written review record	Listed stage-local audit artifact; any executable companion should have a captured output transcript archived with the same basename and an `_output.txt` suffix.
233	rigid-mouth orbit-lock compiler	stage-local written review record	Listed stage-local audit artifact; any executable companion should have a captured output transcript archived with the same basename and an `_output.txt` suffix.

Stage	Audit role	Basename / artifact	What the verifier should confirm
234	direct-branch observable static gates	`moving_throat_pde_stage234_direct_branch_observable_static_gate_and_the_two_observable_kill_test_sympy_audit.py`; stage-local written review record	Listed stage-local audit artifact; any executable companion should have a captured output transcript archived with the same basename and an `_output.txt` suffix.
235	rigid-mouth packet projectors	`moving_throat_pde_stage235_rigid_mouth_packet_projectors_static_blind_dressing_line_and_codimension_two_orbit_lock_point_sympy_audit.py`; stage-local written review record	Listed stage-local audit artifact; any executable companion should have a captured output transcript archived with the same basename and an `_output.txt` suffix.
236	rigid-mouth microscopic dependent plane	`moving_throat_pde_stage236_rigid_mouth_microscopic_dependent_plane_projectors_equal_drift_dressing_ray_and_static_only_restoration_gap_sympy_audit.py`; stage-local written review record	Listed stage-local audit artifact; any executable companion should have a captured output transcript archived with the same basename and an `_output.txt` suffix.
237	actual-branch dressing compiler	`moving_throat_pde_stage237_actual_branch_dressing_compiler_finite_static_blind_curve_and_support_blind_post_static_orbit_lock_theorem_sympy_audit.py`; stage-local written review record	Listed stage-local audit artifact; any executable companion should have a captured output transcript archived with the same basename and an `_output.txt` suffix.
238	physical transfer-shape compiler	`moving_throat_pde_stage238_physical_branch_transfer_shape_compiler_packet_factorization_and_post_static_dressing_invariance_theorem_sympy_audit.py`; stage-local written review record	Listed stage-local audit artifact; any executable companion should have a captured output transcript archived with the same basename and an `_output.txt` suffix.
239	rigid-mouth physical normal form	`moving_throat_pde_stage239_rigid_mouth_physical_normal_form_exact_physical_to_microscopic_correction_compiler_and_cartesian_orbit_lock_theorem_sympy_audit.py`; stage-local written review record	Listed stage-local audit artifact; any executable companion should have a captured output transcript archived with the same basename and an `_output.txt` suffix.
240	selected-branch loading ratio	`moving_throat_pde_stage240_selected_branch_loading_ratio_from_the_minimal_isotropic_quadrupole_precursor_sympy_audit.py`; stage-local written review record	Listed stage-local audit artifact; any executable companion should have a captured output transcript archived with the same basename and an `_output.txt` suffix.
241	selected twin-support primitive ranking	`moving_throat_pde_stage241_exact_primitive_ranking_on_the_selected_twin_support_branch_sympy_audit.py`; stage-local written review record	Listed stage-local audit artifact; any executable companion should have a captured output transcript archived with the same basename and an `_output.txt` suffix.
242	actual twin-support placement	`moving_throat_pde_stage242_actual_twin_support_placement_and_coherent_orbit_lock_compiler_sympy_audit.py`; stage-local written review record	Listed stage-local audit artifact; any executable companion should have a captured output transcript archived with the same basename and an `_output.txt` suffix.
243	relaxed branch declaration and short-range compiler	`moving_throat_pde_stage243_relaxed_constraint_branch_declaration_and_short_range_open_system_compiler_sympy_audit.py`; stage-local written review record	Listed stage-local audit artifact; any executable companion should have a captured output transcript archived with the same basename and an `_output.txt` suffix.
244	leakage and scalar-photon work compiler	`moving_throat_pde_stage244_selected_branch_leakage_and_scalar_photon_work_compiler_sympy_audit.py`; stage-local written review record	Listed stage-local audit artifact; any executable companion should have a captured output transcript archived with the same basename and an `_output.txt` suffix.
245	non-rigid mouth dressing and U/V drain	stage-local written review record	Listed stage-local audit artifact; any executable companion should have a captured output transcript archived with the same basename and an `_output.txt` suffix.

Stage	Audit role	Basename / artifact	What the verifier should confirm
246	compensated multimode source compiler	stage-local written review record	Listed stage-local audit artifact; any executable companion should have a captured output transcript archived with the same basename and an `_output.txt` suffix.
247	relaxed stationary barrier compiler	stage-local written review record	Listed stage-local audit artifact; any executable companion should have a captured output transcript archived with the same basename and an `_output.txt` suffix.
248	dynamic event-chain compiler	stage-local written review record	Listed stage-local audit artifact; any executable companion should have a captured output transcript archived with the same basename and an `_output.txt` suffix.
249	conditional helicity export diagnostic	stage-local written review record	Listed stage-local audit artifact; any executable companion should have a captured output transcript archived with the same basename and an `_output.txt` suffix.
250	crossing/collapse Goldilocks window	stage-local written review record	Listed stage-local audit artifact; any executable companion should have a captured output transcript archived with the same basename and an `_output.txt` suffix.
251	microscopic damping export kernel	stage-local written review record	Listed stage-local audit artifact; any executable companion should have a captured output transcript archived with the same basename and an `_output.txt` suffix.
252	vacuum/lattice heat partition	stage-local written review record	Listed stage-local audit artifact; any executable companion should have a captured output transcript archived with the same basename and an `_output.txt` suffix.
253	physical calibration and materials threshold	stage-local written review record	Listed stage-local audit artifact; any executable companion should have a captured output transcript archived with the same basename and an `_output.txt` suffix.

J.7 Hand-audit and displayed-algebra registry

Not every valid derivation stage requires a script. Many stages are more useful when audited as displayed algebra with explicit assumptions. The hand-audit registry in table J.7 identifies the checks that should be visible in the text itself.

Table J.7: Displayed-algebra and hand-audit requirements.

Stages	Derivation family	Required visible checks	Status envelope
001–024	Geometry lift, reduced wall action, source-map identity, and isotropy theorem.	Harmonic normalization, zero-average checks, sign of δV_{conf}, dimensional consistency of wall coefficients, and exact angular orthogonality.	Mixed
025–036	Minimal isotropic normalization and selected-mode support frontier.	Low-frequency expansion check, positivity/stability domain, monotonicity of branch functions, support-feasibility inequalities, and source-map normalization.	Mixed
037–051	Continuum placement and support compensation.	Dimensionless-variable definitions, product laws, split-U limiting cases, rank-2 determinant checks, lowest-twin limit, and threshold monotonicity.	Mixed
052–069	Non-twin rescue mechanisms and parent-gain thresholds.	Endpoint inequalities, overlap-boost ceilings, Robin-compliance windows, parent-action gain definitions, and no-go/rescue threshold consistency.	Mixed
070–090	Family-1 explicit reduced branch.	Wall-shell profile normalization, threshold-window extraction, quadrupole demand cancellation, loading-ratio window, and final reduced verdict separation between source/support success and precursor realization.	Mixed
091–106	Geometry-lane and retarded/outgoing closure.	Geometry contamination expansion order, scalar/dipole demotion assumptions, outgoing DtN fingerprint, χ_Q normalization relation, and higher-odd irrelevance conditions.	Mixed
107–163	Outlet, core, mouth, and co-evolving compensation.	Robin and mixed-side-channel signs, core-to-mouth gain consistency, positive-source condition, boundary-layer fixed-point equations, and final off-family normal coordinate.	Mixed
164–200	Weak-axisymmetric quotient and Packet-A finish line.	Grouped signature $(1, 1/2, -1)$, $b = 3a$, $\Xi_1 = P_1/P_0$, triangular normal form, monomial rank/nullity, finite orbit law, and four-scalar packet assembly.	Mixed
201–218	Realization compiler and finite mixed-ray search.	Target graph definition, root brackets, Hessian signs, candidate ranking, simplex reduction, support-cardinality proof, and search-class boundary.	Mixed

Stages	Derivation family	Required visible checks	Status envelope
219–242	Same-charge and rigid-mouth selected branch.	Static square-law suppression, resonant linewidth gate, actual-branch observable gates, (U, V) normal form, orbit-lock codimension, support-versus-orbit split, and twin-support placement.	Mixed
243–253	Relaxed open-system branch.	Strict recovery map, leakage/work sign conventions, barrier lowering, event-chain thresholds, WKB conditions, export-kernel half-plane, heat partition, and materials-screening ratios.	Mixed

J.8 Numerical and finite-search reproducibility

A finite search or numerical location may be completely deterministic and still not be an exact theorem. The following metadata must accompany every numerical or finite-search claim before it is cited as Numerically Located or as Exact Within Closure inside a finite search class.

1. **Problem statement.** State the exact equations or objective function, the variables, the admissible domain, and the branch/status assumptions.
2. **Exact preprocessing.** Record symbolic eliminations, nondimensionalizations, quotient projections, and any symmetry reductions used before numerical work begins.
3. **Search class.** For realization searches, state the support cardinality, allowed primitive coordinates, positivity assumptions, and whether the search is local or global.
4. **Deterministic run record.** Record script name, command line, dependency versions, random seed if any, tolerance, bracket endpoints, and stopping rule.
5. **Residual certificate.** Record the final residuals in every constraint, not only the objective value.
6. **Exclusion certificate.** For no-go or finite-candidate claims, record how excluded rays, pairs, triples, quadruples, or quintuples were bounded.
7. **Independent check.** Re-evaluate the winning candidate or branch point in a separate script or exact-substitution cell whenever practical.

Table J.8: Numerical/search claims that require explicit run metadata.

Stage range	Claim family	Metadata that must be present
204–207	Log-ray sweep, scalar-root predictor, Hessian ranking, and certified local bracketing.	Ray definitions, root brackets, Hessian formula, ranking criterion, and local residuals for each certified ray.
208–212	Pairwise and triple mixed-ray/simplex optimizers.	Candidate list, pair/triple domains, off-diagonal Hessian terms, optimizer transcript, and proof that the finite candidate reduction did not discard the winner.
213–218	Support-cardinality 4 and 5 gates.	Simplex coordinates, positivity constraints, finite-candidate reduction, final rank/order comparison, and the closure statement for support-cardinality ≤ 5.
222–232	Primitive branch pole census, dynamic survival window, branch-kill tests, and known-data injection.	Pole/residue definitions, linewidth windows, observed branch values, target residuals, and the exact gate that fails or survives.
240–242	Selected twin-support placement and ranking.	Selected-branch ratio, primitive ranking table, support-placement coordinate, coherent orbit-lock packet, and actual branch residuals.
248–253	Dynamic event chain, Goldilocks window, export kernel, heat partition, and material thresholds.	Barrier parameters, turning-point conditions, WKB exponent, export coefficients Γ_3, Γ_5, heat-channel fractions, and material calibration dictionary.

J.9 Open branch-realization gates

The following quantities are defined by the derivation but are not automatically realized by the algebra. They are branch tests. A future paper may cite this companion for their definitions and for the proof that they are the relevant tests, but not for their realized values unless the corresponding branch construction has been added.

Table J.9: Open or branch-realization gates.

Gate	Required branch output	What counts as closure
Isotropic outgoing-normalization gate	The actual isotropic value of $$\widehat{m}_0^2\frac{N_0}{K-B_0-Z_0}$$ given the completed wall/BdG/Maxwell/mixed branch.	Equality to $54Gc_s^5/(5a^5c^5)$ within the declared branch, with all source-map factors specified.
Constant-prefactor gate	Correlated moments satisfying $$P_2=0, \qquad P_4=0.$$	Exact moment relations or a controlled demonstration that the nonconstant terms are beyond the order being cited.
Weak-axisymmetric prefactor gate	The physical slope $$\Xi_1=\frac{P_1}{P_0}$$ on the actual weak-axisymmetric branch.	A branch calculation of D_{01} and N_{01}, or an equivalent transfer-shape calculation, with the grouped signature $(1,1/2,-1)$.
Rigid-mouth orbit-lock gate	Physical logarithmic coordinates $$(U,V)=\left(\ln\frac{\mathcal{T}^2}{\mathcal{T}_{\rm ref}^2},\ln\frac{\epsilon_\eta}{\epsilon_{\eta,\rm ref}}\right).$$	Actual branch placement at $U=V=0$, or an explicit residual packet when the branch misses.
Selected twin-support placement gate	The selected twin-support coordinate, including $\rho_\alpha=4/3$ and $\zeta_{\rm req}=1/3$ where the strict selected slice applies.	A branch-side support placement matching the selected slice and the coherent orbit-lock compiler.
Full reduced back-end packet	$$\Delta_{\rm full}=(\Delta_Q,q_{\rm tr},q_{\rm nt},q_\eta).$$	Either zero packet on the selected strict branch or a realized graph-error packet routed through the realization compiler.
Relaxed open-system packet	Leakage, work, non-rigid U/V drain, compensated source, barrier, event-chain, export, heat, and material-calibration data.	A realized relaxed branch that recovers the strict front end when $\ell_w=f_U=a=b=0$, with all relaxed packet observables evaluated.

J.10 Repository freeze protocol

The derivation companion can be read without a repository freeze, but it cannot be used as a reproducible archive until the source and script state is frozen. The freeze record should be stored outside this TEX file, but every freeze should satisfy the following protocol.

1. **Source freeze.** Record the commit hash or archive hash for the canonical stage files, the theorem/status TEX layer, the lower-order source stack, and this TEX project.
2. **Script freeze.** Record the path and hash of every symbolic or numerical audit script listed in this appendix.
3. **Dependency freeze.** Record Python version, SymPy version, numerical libraries, and any exact-arithmetic settings. If notebooks are used, store an executed notebook and a plain-script export.
4. **Run freeze.** For each audit script, record the command, working directory, input files, output transcript, and output hash.
5. **Document freeze.** Record the LaTeX engine, build command, generated PDF hash, and whether the build was made from a clean tree.
6. **Diff discipline.** Any later change to a source stage, script, or theorem statement must update the affected row of the stage ledger, result-anchor map, and this reproducibility map.

J.11 Result-card reproducibility template

Every theorem-block result should be reducible to the following card. The card does not have to appear in the main text for every result, but the information should be recoverable from the surrounding prose, tables, or appendices.

Table J.10: Reproducibility card template.

Field	Required content
Result anchor	Top-level anchor and subanchor, for example **MTDC-T9.4**.
Stage provenance	Stage number range, source-stage filenames, and appendix section.
Claim status	One of the firewall statuses, or a mixed status with the exact local split.
Inputs	Parent equations, lower-order constants, branch hypotheses, reduction assumptions, and symbolic variables.
Derivation type	Hand algebra, symbolic script, numerical search, branch solve, or mixed.
Audit artifact	Script basename, output transcript, displayed algebra, or independent check.
Output packet	Boxed formula, theorem statement, target surface, verdict packet, or branch observable.
Failure mode	What would falsify the result or move it to a weaker status.
Downstream use	What a future paper may cite the result for.

J.12 Verifier checklist

☐ Confirm that every cited result has a result anchor and a local status tag.

- ☐ Confirm that every Exact statement is either a parent-action identity, a definition, or exact displayed algebra independent of branch closure.
- ☐ Confirm that every Exact Within Closure statement names the closure family or search class.
- ☐ Confirm that every grouped P_2 result preserves the weighted $(1, 2, 2)$ grouped metric and does not treat 20/21/22 as spacetime indices.
- ☐ Confirm that every use of electric charge respects the $(\eta_Q, q_\star, q_{\text{eff}})$ ontology and does not resurrect circulation as charge.
- ☐ Confirm that every zero-mode Maxwell statement says which mixed channels are suppressed and that they remain present in the microscopic ontology.
- ☐ Confirm that every symbolic audit has a script, output transcript, and a statement of what residuals were checked.
- ☐ Confirm that every numerical or finite-search result has brackets, tolerances, candidate set, residuals, and a deterministic run record.
- ☐ Confirm that every open branch-realization gate is cited as a test condition unless actual branch data are supplied.
- ☐ Confirm that any future paper citing this companion points to the result anchor first and to stage numbers only when detailed provenance is needed.

J.13 Summary

The reproducibility standard for this companion is intentionally stricter than ordinary narrative exposition. The document is useful only if a reader can tell which formula is an exact identity, which formula is exact inside a finite closure, which formula was numerically located, and which formula is still a target for the completed moving-throat branch. This appendix supplies the routing layer: source files, stage ranges, result anchors, audit scripts, search metadata, and open realization gates all have distinct roles. Keeping those roles separate is what makes the derivation ledger citable without turning it into an overclaim.

Appendix K

Source File Index

K.1 Purpose of this appendix

This appendix is the internal provenance index for the derivation companion. It is not a bibliography, and it is not a substitute for the stage ledger or the reproducibility map. Its purpose is to record which source files carry which pieces of the program, what kind of evidence each file supplies, and how those files may safely be used when a later paper cites this companion.

The most important rule is that a source file is a *provenance handle*, not a claim-status label. A formula appearing in a source file does not become an unrestricted theorem merely because the file is central to the archive. The reusable status of a result is determined by the claim-status firewall in the front matter, the local stage appendix, the theorem-block hypotheses, and the reproducibility layer attached to that result. The source file only identifies where the derivation, convention, or audit record came from.

This appendix therefore answers four practical questions.

1. Which files are the canonical sources for the moving-throat derivation companion?
2. Which lower-order and companion files are imported as background rather than as new results of this document?
3. Which generated TEX files form the companion itself, and what role does each file play in the repository?
4. What citation and maintenance rules prevent later papers from overclaiming, double-counting, or citing the wrong layer of the derivation?

The companion should be cited through result anchors and stage appendices. This appendix should be used when a reader or maintainer needs to know which archive file supports a given section of the ledger.

K.2 Reading rules

K.2.1 Source files versus citation targets

A source file and a citation target serve different functions. Source files record provenance. Citation targets state reusable results. The preferred downstream citation object is therefore a theorem/result anchor such as **MTDC-T1**, **MTDC-T8**, or **MTDC-T12**, together with the relevant stage appendix when detailed algebra is needed.

For example, the canonical staged derivation archive now lives in the `paper/stages/stage_NNN.tex` files assembled by the archive appendices. A later paper should not cite a raw source artifact as if it were a polished theorem. It should cite the companion theorem block and, if necessary, the detailed stage appendix. The source layer remains the provenance record behind that appendix.

K.2.2 Status ceiling rule

Every file in this index has a status ceiling. The ceiling is the strongest kind of claim that may be imported from that file without additional proof in this companion. A source can always be used more weakly than its ceiling, but it cannot be used more strongly without adding the missing proof or branch realization.

For instance, a parent-action source may support Exact field equations, while a moving-throat reduced normal-coordinate source may support only Exact Within Closure under its declared closure. A numerical search transcript may support Numerically Located, not an exact branch-realization theorem. A target condition can be fully defined while its actual moving-throat realization remains Open.

K.2.3 Summary files and full files

Several sources come in summary/full pairs. The summary file records the carry-forward theorem structure, headline constants, and claim taxonomy. The full file or detailed stage archive records the derivation path. When they are both present, use them together:

- use the summary for the current status, terminology, and safe downstream statement;
- use the full file for detailed formulas, derivation order, and audit trail;
- use the companion appendix as the stable citation target after the ledger is complete.

If a summary and a full file disagree in wording, the status firewall governs the claim strength. If they disagree in algebra, the detailed derivation plus its script/audit evidence must be checked before the result is reused.

K.2.4 Corrected-ontology precedence

The corrected ontology has precedence over older terminology. In particular:

1. Electric charge sign is carried by η_Q, with microscopic branch coupling $q_\star = \eta_Q e_\star$, not by circulation.
2. The brane-observed charge strength is controlled by localization, for example $q_{\text{eff}} = q_\star/\sqrt{Z_{\text{int}}}$.
3. Circulation belongs to the magnetic/vortical sector.
4. The historical gravity-side scalar coefficient once written as a bare $q = 1$ is $\kappa_\rho = 1$, not electric charge.
5. Mixed channels such as A_w, J^w, $F_{\mu w}$, E_w, and C_a are suppressed only in strict far-field brane reductions; they remain microscopic degrees of freedom.

Any imported file that uses older shorthand must be read through this corrected dictionary before it is used in the derivation companion.

K.3 Source classes

Table K.1: Source classes used by the derivation companion.

Class	Role	Typical use	Status ceiling
Primary staged archive	Canonical derivation provenance for the 253 moving-throat stages.	Stage titles, assumptions, staged outputs, formula provenance, local script references, and chronological dependency.	Local stage status only; mixed across the archive.
Compact theorem master	Current theorem-structure and bottleneck summary.	Claim-status firewall, endgame packet language, current strict/relaxed branch split, and safe carry-forward summary.	Same as the local theorem block it summarizes.
Framework paper	Operational framework for branch tests.	Definitions of branch observables, operational workflow, finite packet language, and connection to future papers.	Usually REDUCED / CONTROLLED REDUCTION or EXACT WITHIN CLOSURE within declared framework assumptions.
Parent-action sources	Exact lower-level field theory and controlled brane reductions.	Parent equations, charge ontology, projection/reduction distinction, Maxwell localization, mixed-sector observables.	EXACT for parent identities; REDUCED / CONTROLLED REDUCTION for brane reductions.
Conservative PN assembly sources	Lower-order and higher-order conservative two-body ledgers.	Carry-forward constants, grouped support needs, static/dynamic response slots, and target packets inherited by the moving-throat PDE program.	EXACT WITHIN CLOSURE or REDUCED / CONTROLLED REDUCTION inside each declared PN hierarchy.
Retarded/radiative bridge sources	Odd-channel and outgoing-quadrupole narrowing.	Scalar/dipole demotion, STF quadrupole branch, compact outgoing fingerprint, passive/outgoing normalization target.	EXACT WITHIN CLOSURE for algebraic bridge; OPEN for actual PDE normalization realization.

Class	Role	Typical use	Status ceiling
Continuation and application notes	Reduced-sector follow-on work.	Motivation, cross-checks, anomaly/atomic/lepton companion targets, and optional downstream applications.	Never stronger than REDUCED / CONTROLLED REDUCTION or EXACT WITHIN CLOSURE unless separately promoted.
Generated companion files	The present TEX ledger.	Stable citation anchors, readable theorem blocks, detailed stage appendices, reproducibility and source maps.	Determined by the content and status tags inside each file.
Script/output artifacts	Symbolic, numerical, or finite-search evidence.	Verifying Schur complements, projectors, finite compilers, searches, roots, residuals, and branch-placement tests.	EXACT WITHIN CLOSURE for exact symbolic audits; NUMERICALLY LOCATED for numerical locations.

K.4 Canonical moving-throat source files

Table K.2: Canonical moving-throat source files.

File	Archive role	Imported content	Safe status ceiling
`paper/stages/stage_NNN.tex`	Primary full derivation archive for Stages 001–253.	Stage-by-stage derivation text, staged theorem moves, formula provenance, executable-audit routing, local assumptions, and chronological dependencies.	Local stage status; mixed.
`paper/parts/*.tex` plus front-matter	Compact theorem and status master through the strict and relaxed endgame.	Reading rules, claim-status tags, notation firewall, current theorem-status summary, final packet language, same-charge verdict, rigid-mouth chart, selected twin-support slice, relaxed branch, barrier/export/material packets.	Mixed; status summary only.

File	Archive role	Imported content	Safe status ceiling
`paper/appendices/stage_appendix_part*.tex`	Stage-ordered archive assembly over the canonical stage files.	Part-by-part audit order, stage numbering, local handoff notes, and the stage-ledger reading path used by the archive build.	Same as the assembled stage content.
`scripts/` plus `mathematica/`	Executable verification layer.	Symbolic replay, numerical stress harnesses, second-CAS cross-checks, and stagewise audit routing by stage number.	Matches the local script model and recorded status.
`pde.tex`	Compact-to-framework bridge.	The division between framework use and derivation verification, finite observable-packet language, and operational branch-testing workflow.	REDUCED / CONTROLLED REDUCTION / EXACT WITHIN CLOSURE inside framework assumptions.
`pde.tex`	Core paper source.	Clean operational statement of the moving-throat PDE response framework, branch-test quantities, finite observable packets, and the reason this companion is needed as a derivation ledger.	Core use only.

Use rule. The primary moving-throat source for detailed stage provenance is the canonical stage-file tree under `paper/stages/`, assembled by the stage appendices. The primary source for the current global theorem-status reading is the theorem/status layer in the frontmatter and main parts. The primary source for operational branch-use language is `pde.tex`. A downstream citation should normally pass through this companion rather than directly through a raw source artifact.

K.5 Parent-action and brane-reduction source files

Table K.3: Parent-action and brane-reduction source files.

File	Role	Imported content	Safe status ceiling
Parent-action source summaries	Parent action summary.	Fully action-based (4 + 1)-dimensional bulk theory, gauged GNLS matter sector, localized Maxwell sector, geometry variables, projection/reduction distinction, leakage identity, and corrected charge ontology.	Exact for parent identities; Reduced / Controlled Reduction for reductions.
`4d.tex`	Full parent action paper source, when available in the archive.	Detailed parent equations, action variation, brane projection definitions, and controlled Poisson/Maxwell hooks.	Same as local section status.
Localized electromagnetic source summaries	Localized electromagnetic sector summary.	Inhomogeneous Maxwell equation from localized (4 + 1)-dimensional action, zero-mode Maxwell reduction, thickness-controlled brane charge strength, KK corrections, retarded propagation, and source consistency.	Exact / Reduced / Controlled Reduction.
`research/pde_ledger/scripts/moving_throat_pde_stage004_projected_maxwell_bundle_index_sympy_audit.py`–`research/pde_ledger/scripts/moving_throat_pde_stage020_parent_throat_action_weak_axisym_packet_sympy_audit.py`	Projected Maxwell and parent-action derivation bundle copied into the ledger script order.	Projection-first localized Maxwell law, measured-field/source-flux split, matched projection/reduction channel, near-mouth projected bundle bridge, mouth-Taylor gate sieve, parent throat action promotion, and weak-axisymmetric parent packet.	Exact for projection identities; Reduced / Controlled Reduction / Exact Within Closure for matched and bundle closures.

File	Role	Imported content	Safe status ceiling
Plasma and open-system source summaries	Plasma and open-system projection summary.	Multi-species charged medium, projection kernel $W(w)$, leakage terms, mixed fields E_w and C_a, finite-localization transverse modes, and conservative brane-bulk exchange interpretation.	EXACT for balance identities; REDUCED / CONTROLLED REDUCTION for brane limits.
1PN bridge source summaries	Bridge from parent action to 1PN coefficient logic.	Topology discriminator $\kappa_{\text{add}} = 1/2$, EOS exponent $n = 5$, EIH cross-sector constants, wave-supported kinematics, and explicit adiabatic closure data.	Mixed: REDUCED / CONTROLLED REDUCTION / EXACT WITHIN CLOSURE.

Use rule. The parent-action files establish the exact field-theory arena and the controlled brane-reduction hooks. They do not by themselves prove the completed moving-throat response branch. When this companion imports parent equations, it should preserve whether the statement is exact in the parent theory or only true after a brane reduction.

Projected-Maxwell scope rule. Only the root ledger audits `research/pde_ledger/scripts/moving_throat_pde_stage004_*` through `research/pde_ledger/scripts/moving_throat_pde_stage020_*` are file-for-file imported projected-EM derivation sources for the present companion; `research/pde_ledger/scripts/moving_throat_pde_stage021_reduced_one_port_normal_form_sympy_audit.py` is the retained reduced one-port normal-form adapter. The `notes/em_projected/step_19*` branch-export packet and the later `notes/em_projected/step_20*` and following files are computational or branch-diagnostic material: branch-family scans, frontiers, runtime monitors, CFD adapters, fail-fast classifiers, and falsification tools. They may be useful private diagnostics, but they are deliberately excluded from the paper derivation, result anchors, and Stages 004–021 verification claims.

K.6 Conservative post-Newtonian and radiative bridge sources

Table K.4: Conservative PN and radiative bridge source files.

File	Role	Imported content	Safe status ceiling
`1PN full conservative assembly sources`	Full conservative Newtonian/1PN assembly summary.	Exact EIH equality inside the declared hierarchy, 1PN claim taxonomy, carry-forward constants $\kappa_\rho = 1$, $n = 5$, $\kappa_{\rm add} = 1/2$, $\kappa_{\rm PV} = 3/2$, and $\beta_{\rm 1PN} = 3$.	Exact Within Closure / Reduced / Controlled Reduction.
2PN source stack	Full conservative 2PN assembly.	2PN self/static seed, residual solve, even support layer, grouped $P_0 \oplus P_2$ response structure, geometry completion, and dynamic pole-scale openings.	Exact Within Closure inside the declared 2PN hierarchy.
2.5PN retarded source stack	2.5PN retarded theorem audit.	Burke–Thorne/Iyer–Will bridge, odd-channel hierarchy, scalar/dipole demotion, real-STF quadrupole route, compact outgoing $l = 2$ fingerprint, and passive/outgoing normalization gap.	Exact Within Closure for audit algebra; Open for final PDE-side normalization.
3PN source stack	Full conservative 3PN assembly.	Grouped real P_2 middle-block closure, fixed-chart ordinary/Hamiltonian compiler, exact 3PN assembly inside the hierarchy, and the need for a microscopic grouped-P_2 constitutive law.	Exact Within Closure / Reduced / Controlled Reduction.

File	Role	Imported content	Safe status ceiling
4PN source stack	Full conservative 4PN and tail-interface assembly.	Local/tail split, one-body 4PN gate, quartic compiler, exact local instantaneous closure, and reduction of the tail coefficient to the same outgoing P_0 gap.	Exact Within Closure local algebra; Open branch realization.
5PN and continuation source stack	Higher-order continuation and weak-axisymmetric transport notes.	Explicit finite-throat overlap model, weak-axisymmetric line $b = 3a$, $\Xi_1 = P_1/P_0$, monomial quotient coordinates, similarity-orbit theorem, support-compensation thresholds, and target-surface sharpening.	Mixed: Exact Within Closure, Numerically Located, Open.

Use rule. The PN sources define why the moving-throat response problem matters. They supply the lower-order constants, target packets, and normalization gates that the PDE companion must derive or test. They should not be cited as completed nonlinear moving-throat branch realizations unless this companion or a later branch paper supplies that realization.

K.7 Reduced-sector companion and application files

Table K.5: Reduced-sector companion and application files.

File	Role	Imported content	Use ceiling
Atomic reduced-sector companion sources	Atomic reduced-sector companion.	Hydrogenic reduction, finite-size response logic, and reduced-sector applications of the localized Maxwell and defect-response framework.	Guide only.
Lepton / electron companion sources	Lepton / electron companion notes.	Corrected charge ontology, same-charge mixed-sector rotor corridor, Berry/holonomy route, and conditional lepton-like closure discussion.	Context only.

File	Role	Imported content	Use ceiling
Isolated-electron mass and D/N support sources	Isolated-electron mass and D/N support notes.	Reduced rest-energy partition, support-wave mass theorem, finite-throat D/N ladder, and scale-fixing obstruction.	Exact Within Closure inside its reduced ansatz; otherwise motivation.
Anomaly and quotient-transport continuation sources	Anomaly and quotient-transport continuation.	Exact local anomaly baseline, quotient variables, common transport layer, Ξ_1 / outgoing-prefactor bridge, and microscopic slope conditions.	Limited use only.

Use rule. These files are useful because they show how the moving-throat response language is expected to be used in later physics-facing papers. They are not primary proof sources for this derivation companion. When the companion and an application note differ, the companion controls the moving-throat derivation status.

K.8 Generated derivation-companion files

The files in this section are part of the present TEX project. They are not external evidence. Their role is to make the source evidence readable, auditable, and citable.

K.8.1 Front matter files

Table K.6: Front matter files in the derivation companion.

File	Role in the companion
`frontmatter/00_purpose_scope.tex`	Defines the document as a derivation companion / verification ledger rather than a normal paper, and separates framework use from derivation verification.
`frontmatter/01_how_to_read.tex`	Gives the reading protocol: theorem blocks first, stage appendices second, status tags always.
`frontmatter/02_claim_status_firewall.tex`	Defines the status classes and the rules for safe downstream citation.

File	Role in the companion
frontmatter/03_notation_firewall.tex	Freezes the charge, circulation, grouped-P_2, mixed-sector, and branch-packet notation rules.
frontmatter/04_result_anchor_map.tex	Registers top-level result anchors, subanchors, formula packets, citation crosswalks, and unsafe-citation guardrails.
frontmatter/05_dependency_map.tex	Gives the dependency graph, stage-range flow, and result-block dependency discipline.

K.8.2 Main theorem-block files

Table K.7: Main theorem-block files.

File	Stages	Role in the companion
parts/part01_parent_geometry.tex	001–023	Parent import, geometry lift, breathing reduction, BdG coupling, Maxwell/mixed bridge, and full grouped-bundle front end.
parts/part02_conservative_response.tex	024–036	Overlap/isotropy, weak-axisymmetric signature, minimal isotropic closure, finite-throat selected-mode normalization, and support-feasibility frontier.
parts/part03_support_family1.tex	037–090	Continuum placement, split-U support, rank-2 completion, support compensation, parent thresholds, Family-1 branch, and reduced verdict.
parts/part04_geometry_retarded_mouth.tex	091–163	Geometry-lane firewall, retarded normalization collapse, outlet/core compensation, mouth boundary layer, co-evolving core–mouth fixed point, and off-family normal coordinate.
parts/part05_weak_axisymmetric_orbit.tex	164–200	Weak-axisymmetric transport, Ξ_1, monomial quotient coordinates, similarity orbit, Packet-A closure, and four-scalar verdict packet.
parts/part06_realization_compiler.tex	201–218	Realization compiler, target graph, Hessian/log-ray search, pairwise/simplex closure, and finite mixed-ray sieve.
parts/part07_same_charge_rigid_mouth.tex	219–242	Same-charge one-port audit, dynamic/resonant corridor, rigid-mouth (U, V) chart, orbit lock, selected twin-support placement, and actual-branch packet.

File	Stages	Role in the companion
parts/part08_relaxed_branch.tex	243–253	Relaxed open-system branch, leakage/work and non-rigid packets, compensated source compiler, lowered barrier, event chain, export kernel, heat partition, and material thresholds.

K.8.3 Appendix files

Table K.8: Appendix files in the derivation companion.

File	Role in the companion
appendices/stage_ledger.tex	Consolidated stage-range map, complete Stage 001–253 ledger, source-file provenance table, and citation/audit guardrails.
appendices/stage_appendix_part01.tex	Detailed verification appendix for Stages 001–023.
appendices/stage_appendix_part02.tex	Detailed verification appendix for Stages 024–036.
appendices/stage_appendix_part03.tex	Detailed verification appendix for Stages 037–090.
appendices/stage_appendix_part04.tex	Detailed verification appendix for Stages 091–163.
appendices/stage_appendix_part05.tex	Detailed verification appendix for Stages 164–200.
appendices/stage_appendix_part06.tex	Detailed verification appendix for Stages 201–218.
appendices/stage_appendix_part07.tex	Detailed verification appendix for Stages 219–242.
appendices/stage_appendix_part08.tex	Detailed verification appendix for Stages 243–253.
appendices/reproducibility_map.tex	Audit classes, artifact-card template, stage-to-script map, symbolic/numerical/search discipline, branch-solver requirements, and safe citation language.
appendices/source_file_index.tex	This provenance index: source classes, source roles, import rules, generated-file map, stage-range source map, conflict resolution, and maintenance policy.
appendices/fill_workflow.tex	Workflow for filling, reviewing, freezing, and later updating the ledger.

K.9 Stage-range source map

This table records the minimal source families that support each stage range. The detailed algebra belongs in the corresponding stage appendix. The table is meant to help maintainers trace an imported claim back to the right source family quickly.

Table K.9: Stage-range source map.

Stages	Companion location	Primary source family	Background source family
001–023	Part I and Appendix Part I	Canonical stage files; theorem/status frontmatter; framework paper.	Parent action, Maxwell, plasma, 2PN/3PN/2.5PN grouped-response sources.
024–036	Part II and Appendix Part II	Canonical stage files; overlap/isotropy synthesis; weak-axisymmetric continuation material.	2.5PN grouped-STF bridge, 3PN grouped P_2 source language, 5PN weak-axisymmetric notes.
037–090	Part III and Appendix Part III	Canonical stage files; continuum placement and support-compensation synthesis.	Finite-throat D/N support material, 5PN placement and support-threshold notes.
091–163	Part IV and Appendix Part IV	Canonical stage files; geometry, retarded, outlet, and mouth-core synthesis.	2.5PN outgoing normalization audit, 4PN tail-interface summary, mixed-sector parent ontology.
164–200	Part V and Appendix Part V	Canonical stage files; weak-axisymmetric orbit and quotient continuation material.	5PN monomial/orbit notes and anomaly quotient bridge material where used only for interpretation.
201–218	Part VI and Appendix Part VI	Canonical stage files; realization-compiler and mixed-ray search records.	Reproducibility-map search discipline and theorem-block endgame summary.
219–242	Part VII and Appendix Part VII	Canonical stage files; one-port same-charge and rigid-mouth actual-branch records.	Theorem/status summary, mixed-sector ontology, selected support placement notes.
243–253	Part VIII and Appendix Part VIII	Canonical stage files; relaxed/open-system extension as integrated in the theorem/status layer.	Plasma/open-system projection sources, leakage/work language, export/kernel and materials companion notes.

K.10 Script and output artifact indexing

The derivation source files repeatedly refer to script-backed audits. Some scripts are part of the working archive; some are represented in the staged text by script basenames and output transcripts. This appendix does not reproduce those scripts. It defines how they should be indexed when the repository is frozen.

K.10.1 Script artifact card

Every script used as evidence should have an artifact card with the following fields:

Table K.10: Required fields for script and output artifact cards.

Field	Required content
Script basename	Exact script file name, including extension.
Output basename	Exact output transcript file name, if separate.
Stage range	Stage number or range whose claims the script checks.
Claim checked	The exact identity, projector relation, search result, root location, residual, or branch condition certified by the script.
Input assumptions	Closure family, branch restrictions, symbolic variables, positivity assumptions, numerical constants, or search-domain bounds.
Arithmetic mode	Exact rational/symbolic arithmetic, high-precision numerical arithmetic, floating point, interval/bracketed solve, or mixed mode.
Failure mode	What would count as a failed check: nonzero residual, wrong rank, missing root, failed inequality, nontermination, unstable optimum, or out-of-domain branch.
Repository hash	Hash of the script and transcript at the release freeze.
Reusable status	Exact Within Closure for exact symbolic verification; Numerically Located for located roots/searches; Open when the script defines a target not yet realized by the PDE.

K.10.2 Known script families

The stage archive and reproducibility map mention several recurring script families. The following labels should be used when the repository freeze turns those references into formal artifact cards.

Table K.11: Known script families to preserve in the archive.

Script family or basename pattern	Stage region	Verification role
`moving_throat_pde_master_sympy_audit.py`	001–002	Harmonic normalization, mouth-average extraction, two-mode wall reduction, and grouped-P_2 one-mode reduction.
`moving_throat_pde_stage003_bdg_sympy_audit.py`	003	BdG–wall Schur complement, low-frequency expansion, pole shifts, grouped isotropy diagnostics, and selection rules.
`2_5pn_master_sympy_audit.py` and `2_5pn_master_session_sympy_audit.py`	2.5PN background / Stages 004–023	Burke–Thorne/Iyer–Will bridge, odd-channel hierarchy, grouped-STF formulas, and outgoing-quadrupole normalization relations.
`5pn_stage*_*.py`	024–200 background and continuation	Explicit finite-throat overlap models, weak-axisymmetric transport, primitive deformation, monomial quotient maps, target surfaces, support-compensation thresholds, and even-gate intersections.
Realization-compiler search scripts	201–218	Graph-error realization form, log-ray sweep, Hessian/curvature ranking, pairwise/simplex search, and finite support-cardinality sieve.
Rigid-mouth and selected-support scripts	219–242	Actual-branch packet evaluation, orbit-lock normal form, selected twin-support placement, support-versus-orbit split, and final coherent branch-evaluation packet.
Relaxed-branch and event-chain scripts	243–253	Leakage/work compiler, non-rigid mouth packet, compensated source moments, barrier lowering, turning points, WKB/event-chain quantities, export kernel, heat partition, and material-threshold screens.

Use rule. A script family listed here is not evidence by itself. Evidence requires the specific script, inputs, output transcript, and residual or exact-check statement. The reproducibility map gives the stronger rule for turning a script reference into a reusable claim.

K.11 Files that should not be cited as mathematical sources

The repository will contain generated files that are useful for building or reviewing the project but should not be used as mathematical evidence.

Table K.12: Non-citable generated or operational files.

File type	Why it is not a mathematical source	Allowed use
`*.aux`, `*.out`, `*.toc`	Intermediate LaTeX state; does not contain independent derivation evidence.	Build diagnostics only.
`*.log`	Compile logs can identify errors or overfull boxes but do not certify mathematical content.	Build troubleshooting only.
Generated `*.pdf` files	Rendered outputs of the TeX source; useful for reading but not a separate provenance layer.	Human review and final distribution.
Temporary smoke-test harnesses	Minimal compile wrappers created only to test a single file.	Local quality control; not part of the archive.
Backup files such as `*.bak.tex`	Intermediate drafts that may disagree with the stable source.	Recovery only; do not cite.

K.12 Precedence and conflict-resolution rules

1. **Status precedence.** The claim-status firewall controls the reusable strength of a result, even if an older source states the result in stronger language.
2. **Ontology precedence.** The corrected charge/mixed-sector notation firewall controls all imported files. Older circulation-charge shorthand must be translated before use.
3. **Derivation precedence.** For detailed formulas, prefer the full derivation archive or stage appendix over a compact summary. Use the compact master for current theorem-status language.
4. **Framework precedence.** Use `pde.tex` for how a future paper should operate the branch test, but use this derivation companion for why the formulas in that test are justified.
5. **Script precedence.** If a printed identity and a script transcript disagree, neither should be cited until the discrepancy is resolved and the corrected artifact is frozen.
6. **Realization precedence.** A finite compiler, target equation, or normalization condition does not imply that the actual moving-throat PDE realizes that condition. Actual branch realization requires a branch theorem, a verified solve, or an explicitly labeled numerical placement.
7. **Application precedence.** Atomic, lepton, anomaly, and material companion notes may motivate or test the framework, but they do not override the moving-throat derivation status.

K.13 Safe source-use language

The following examples are safe patterns for future papers.

Table K.13: Safe and unsafe source-use language.

Safe language	Unsafe language
"The full staged derivation of the grouped-response compiler is given in the derivation companion, MTDC-T5 and Appendix Stages 001–024."	"The raw moving-throat notes prove the grouped response."
"Within the declared reduced bundle, the outgoing prefactor is $P_0 = N_0/D_0$; actual branch realization of the target value is a separate condition."	"The moving-throat PDE automatically has the GR quadrupole normalization."
"The one-port same-charge audit rules out a new long-range static attractive law in the audited class."	"The model proves every same-charge interaction is impossible."
"The relaxed branch lowers the local barrier through leakage/work/dressing/source packets while preserving the same long-range kernel verdict."	"The relaxed branch creates a new long-range force."
"The finite mixed-ray search is complete through the declared support-cardinality bound."	"All possible microscopic realizations have been searched."
"The parent action supplies exact bulk equations; the brane Maxwell law is a controlled zero-mode reduction."	"The parent action directly equals ordinary brane Maxwell theory without assumptions."

K.14 Repository-freeze recommendations

At the first public or internal release freeze of the derivation companion, the repository should include a manifest with the following fields for every source file listed in this appendix:

1. file path relative to the repository root;
2. file size;
3. content hash;
4. date of freeze;
5. source class from table K.1;
6. primary companion sections or appendices that use the file;
7. claim-status ceiling;
8. supersession notes, if the file replaces or corrects older notation.

The manifest does not need to be printed in this paper unless journal or archive practice requires it. It should, however, exist alongside the source repository so that future papers can cite a stable release rather than a moving working directory.

K.15 Minimal release manifest

For convenience, table K.14 gives the minimal release manifest that should be preserved even if the full artifact-card registry is deferred.

Table K.14: Minimal release manifest for this derivation companion.

Source family	Required files	Reason they must be frozen together
Primary moving-throat archive	`paper/stages/`; `paper/appendices/stage_appendix_part*.tex`; `paper/parts/`; `pde.tex`.	These files define the stage provenance, theorem/status master, and operational framework split.
Parent action and EM/plasma stack	Parent-action, EM, and plasma source summaries plus full TEX sources where present.	These files define the exact parent theory, charge ontology, zero-mode Maxwell reduction, mixed-sector firewall, projection/leakage language, and open-system interpretation.
Conservative and retarded PN stack	1PN through 5PN source summaries plus full sources where present.	These files define the target packets and lower-order results that the moving-throat PDE companion is designed to verify or realize.
Continuation notes	Historical continuation sources and selected application notes.	These files record intermediate continuation logic, quotient/orbit machinery, and application-facing consistency checks.
Derivation companion TEX source	`main.tex`; `macros.tex`; all files under `frontmatter/`, `parts/`, and `appendices/`.	These files are the stable readable ledger and citation layer.
Script and output artifacts	All exact symbolic-audit scripts, numerical-location scripts, search scripts, and output transcripts referenced by the stage appendices.	These files certify the symbolic, numerical, and finite-search claims in the reproducibility map.

K.16 Closing rule

This source index should be updated only when the archive itself changes: a source is added, a file is renamed, a script transcript is frozen, a summary/full pair is replaced, or a result is promoted to a stronger status by new evidence. Ordinary edits to exposition inside the theorem-block chapters should not change this appendix unless they change the provenance trail.

The companion is intended to become the stable citation layer for future papers. Once it is frozen, later papers should cite the result anchors and appendix sections in this document, while the files listed here remain the traceable archive behind those citations.

Appendix L

Layer-2 Fit-Insertion Index

L.1 Generated Coverage Summary

This generated appendix indexes every Phase-A fit-insertion candidate from the layer-2 adversarial audit. It covers 922 candidates. The audit result is operational: no surviving fatal flaw was found in this toy-analog ledger; this is not a truth claim about the model.

Table L.1: Fit-insertion index generated coverage totals.

Item	Count
Phase-A candidates	922
constraint_kind: internal_consistency	645
constraint_kind: free_choice	159
constraint_kind: free_choice+internal_consistency	8
constraint_kind: free_choice+published_target	3
constraint_kind: published_target	51
constraint_kind: no_constraint_record	56
MANIFEST status: audited	324
MANIFEST status: provenance_built	589
MANIFEST status: scanned	7
MANIFEST status: verdict_logged	2
Alias-folded candidates surfaced in rows	7

L.2 Phase-A Candidate Rows

Table L.2: Every Phase-A candidate with source anchor, constraint-kind accounting, and audit disposition.

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
001	`fit_stage001_canonical_invariant_pair_determined` `Kbar_0`, `Kbar_2`	`notes/stages/moving_throat_pde_stage001_geometry_lift.md`:471	`internal_consistency`	provenance_built

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
001	fit_stage001_wall_action_constitutive_coefficients K_eta, T_Omega, T_w, mu_eta	research/pde_ledger/paper/stages/stage_001.tex:149	free_choice	audited
001	fit_stage_001_k_l K_l	redteam/pass2/reports/stage_001.md:39	internal_consistency	provenance_built
002	fit_stage002_grouped_p2_stiffness_gate K_2, M_2	research/pde_ledger/paper/stages/stage_002.tex:96	internal_consistency	provenance_built
002	fit_stage002_overlap_factor_4pi_derived overlap_factor_4pi	paper/stages/stage_002.tex:11	internal_consistency	audited
002	fit_stage_002_gamma_ab Gamma_AB	notes/stages/moving_throat_pde_stage002_breathing_reduction.md:112	free_choice	audited
003	fit_stage003_wall_self_energy_no_fitted_data overlap_matrices, support_spectrum, wall_self_energy	notes/stages/moving_throat_pde_stage003_bdg_coupling.md:85	internal_consistency	provenance_built
004	fit_stage_004_i_ws_delta I_WS_delta, W_match, mu_proj_delta, mu_red	redteam/pass2/reports/stage_004.md:83	internal_consistency	provenance_built
004	fit_stage_004_statusexact StatusExact, partial_w, partial_wQ, partial_wW	redteam/pass2/reports/stage_004.md:35	no_constraint_record	provenance_built
005	fit_stage_005_assert_nonzero assert_nonzero, assert_zero	redteam/pass2/reports/stage_005.md:105	no_constraint_record	provenance_built
006	fit_stage006_projected_vector_signs_fixed_by_audit ampere_sign, faraday_sign	paper/stages/stage_006.tex:32	internal_consistency	audited
006	fit_stage_006_j_i J_i	redteam/pass2/reports/stage_006.md:99	internal_consistency	provenance_built
008	fit_stage008_source_profile_must_match_weight S_w, Z_int	paper/stages/stage_008.tex:25	internal_consistency	provenance_built
008	fit_stage_008_inv_xi inv_xi	redteam/pass2/reports/stage_008.md:88	internal_consistency	provenance_built
008	fit_stage_008_z2_int Z2_int, Z_int	redteam/pass2/reports/stage_008.md:205	internal_consistency	provenance_built
009	fit_stage009_mu1_kernel_specialization mu_1, w_kernel	paper/stages/stage_009.tex:28	internal_consistency	provenance_built
009	fit_stage_009_matched_fingerprint_value matched_fingerprint_value, mathematica_audit, sympy_audit	redteam/pass2/reports/stage_009.md:40(4 manifest anchors)	no_constraint_record	provenance_built
010	fit_stage_010_c_fixed C_fixed, C_tr, Xi_static, assert_nonzero, assert_zero, compat_direct, dK_norm, dK_one_pole	redteam/pass2/reports/stage_010.md:39(2 manifest anchors)	internal_consistency	provenance_built
010	fit_stage_010_dk_norm dK_norm	redteam/pass2/reports/stage_010.md:105	internal_consistency	provenance_built
010	fit_stage_010_r_fixed R_fixed, R_tr, assert_nonzero, assert_zero	redteam/pass2/reports/stage_010.md:101	internal_consistency	provenance_built
011	fit_stage_011_b_0 B_0, N_0, epsilon, n_0, z_0, z_2, z_4	redteam/pass2/reports/stage_011.md:39	free_choice	audited
012	fit_stage_012_assert_nonzero assert_nonzero, dCompat_direct	redteam/pass2/reports/stage_012.md:39	no_constraint_record	provenance_built
012	fit_stage_012_delta Delta, S_2, n_n, stage_012, z_n	redteam/pass2/reports/stage_012.md:35(2 manifest anchors)	internal_consistency	provenance_built

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
012	fit_stage_012_k_norm K_norm	redteam/pass2/reports/ stage_012.md:107	internal_ consistency	provenance_built
012	fit_stage_012_n_n n_n, z_n	paper/stages/stage_012. tex:20	internal_ consistency	provenance_built
014	fit_stage014_even_gate_ constants_2_3_1_27 He_bundle_coefficients	scripts/moving_throat_ pde_stage014_projected_ maxwell_mouth_taylor_ gate_bridge_sympy_audit. py:96	internal_ consistency	provenance_built
015	fit_stage_015_s_em S_EM, S_total	redteam/pass2/reports/ stage_015.md:54	free_choice	audited
016	fit_stage_016_boundary_value boundary_value	redteam/pass2/reports/ stage_016.md:98	no_constraint_ record	provenance_built
017	fit_stage017_even_gate_ determinant_1_27 wall_gate_determinant	scripts/moving_throat_ pde_stage017_parent_ throat_action_weak_ axisym_sympy_audit.py:109	internal_ consistency	provenance_built
017	fit_stage_017_paragraph paragraph	paper/stages/stage_017. tex:28	no_constraint_ record	provenance_built
018	fit_stage018_stale_ provenance_anchor_9_11_8 coeff_11, coeff_8, coeff_9	notes/CHECKPOINT_ CONSTANT_PROVENANCE. md:376	no_constraint_ record	audited
019	fit_stage019_K_Sigma_fixed_ by_one_pole K_Sigma	paper/stages/stage_019. tex:21	internal_ consistency	audited
019	fit_stage019_K_Sigma_fixed_ by_outgoing_normalization K_Sigma, P_0_target	paper/stages/stage_019. tex:27	published_target	audited
019	fit_stage019_P0_target_ upstream_pin P_0_target	paper/stages/stage_019. tex:32	published_target	audited
019	fit_stage019_p0target_54_5_ raw_retype P0target, mhat0	mathematica/moving_ throat_pde_stage019_ parent_throat_action_ isotropic_bundle_ mathematica_audit.wl:50	published_target	audited
019	fit_stage_019_p0_target P0_target, assert_nonzero, const_prefactor_matrix	redteam/pass2/reports/ stage_019.md:44(2 manifest anchors)	published_target	audited
020	fit_stage020_even_gate_ determinant_1_27 even_gate_determinant	scripts/moving_throat_ pde_stage020_parent_ throat_action_weak_ axisym_packet_sympy_ audit.py:39	no_constraint_ record	provenance_built
020	fit_stage020_wall_slopes_ fixed_uniquely Xi1, parent_wall_slopes	scripts/moving_throat_ pde_stage020_parent_ throat_action_weak_ axisym_packet_sympy_ audit.py:76	no_constraint_ record	provenance_built
020	fit_stage_020_b_0 B_0, D_0, H_even, K_1, N_0, Z_0, paragraph	redteam/pass2/reports/ stage_020.md:35	free_choice	audited
021	fit_stage021_outgoing_ fingerprint_omega5_ coefficient a, c_s, fingerprint_omega5_ coeff_1_over_27	research/pde_ledger/ paper/stages/stage_021. tex:63	internal_ consistency	provenance_built
021	fit_stage021_retained_ reduced_adapter_couplings Omega_A_l, Omega_W_l, R_l, g_A_l, g_W_l	research/pde_ledger/ paper/stages/stage_021. tex:43	free_choice	audited
021	fit_stage_021_c_a C_a, E_w, N_l, N_scalar, g_A, g_ W, matched_fingerprint_value	redteam/pass2/reports/ stage_021.md:35(2 manifest anchors)	free_choice	audited

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
021	fit_stage_021_eq_compact_l2_fingerprint EQ_COMPACT_L2_FINGERPRINT, MATH_COMPACT_L2_OUTGOING_FINGERPRINT, equation_anchor, exact_reduced, matched_fingerprint_value	paper/stages/stage_021.tex(2 manifest anchors)	internal_consistency	provenance_built
021	fit_stage_021_gamma5_port Gamma5_port, Y2_hat, matched_fingerprint_value	redteam/pass2/reports/stage_021.md:49	internal_consistency	provenance_built
021	fit_stage_021_matched_fingerprint_value matched_fingerprint_value	notes/stages/moving_throat_pde_stage021_reduced_one_port_normal_form.md:49(6 manifest anchors)	internal_consistency	provenance_built
021	fit_stage_021_paragraph paragraph	paper/stages/stage_021.tex:60	no_constraint_record	provenance_built
022	fit_stage022_gamma_GR_quadrupole_target N_Q_target, gamma_GR	paper/stages/stage_022.tex:121	published_target	audited
022	fit_stage022_invariant_product_must_match_universal N_Q_universal, P_An, u_n_A	paper/stages/stage_022.tex:7	published_target	audited
022	fit_stage022_mhat0_unity_branch mhat_0	paper/stages/stage_022.tex:135(2 manifest anchors)	free_choice	audited
022	fit_stage022_p0_quadrupole_normalization_target P_0, gamma_GR, mhat_0	research/pde_ledger/paper/stages/stage_022.tex:133	published_target	audited
022	fit_stage022_source_map_branch_mhat0_unity mhat_0	research/pde_ledger/paper/stages/stage_022.tex:135	free_choice alias offit_stage022_mhat0_unity_branch	scanned; alias folded to fit_stage022_mhat0_unity_branch
022	fit_stage_022_d_0 D_0, Delta_0, Delta_0_S_2omega, G_5, Gamma_5, K_0, K_2, K_4, N_0, N_2, N_4, Omega_A, Omega_A_Omega_W_R_g_A, Omega_W, P_0, P_2, P_4, S_2, a_x, b_x, bar, g_A, g_W, iG_5, matched_fingerprint_value, u_2, u_4	redteam/pass2/reports/stage_022.md:39	free_choice	audited
022	fit_stage_022_d_a D_A, N_A, matched_fingerprint_value	paper/stages/stage_022.tex:9	free_choice	audited
022	fit_stage_022_g_5 G_5, Lambda_2_omega_h_2_h_2, matched_fingerprint_value	redteam/pass2/reports/stage_022.md:90	internal_consistency	provenance_built
022	fit_stage_022_matched_fingerprint_value matched_fingerprint_value, paragraph	paper/stages/stage_022.tex:86(9 manifest anchors)	internal_consistency	provenance_built
022	fit_stage_022_p_0 P_0, matched_fingerprint_value	redteam/pass2/reports/stage_022.md:114	published_target	audited
023	fit_stage023_gamma5_port_27_coefficient Gamma5_port	scripts/moving_throat_pde_stage023_full_grouped_bundle_sympy_audit.py:339	internal_consistency	provenance_built
023	fit_stage023_nq_target_gr_quadrupole_match N_Q_target, gamma_GR, mhat_0	paper/stages/stage_023.tex:231	published_target	audited
023	fit_stage_023_b_0 B_0, D_0, N2_target, N4_target, N_0, N_2, N_4, N_4_target_closed, P_0, Z_0, stale_output	redteam/pass2/reports/stage_023.md:97	free_choice	audited
023	fit_stage_023_n_n N_n, Z_n	paper/stages/stage_023.tex:9	free_choice	audited

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
024	fit_stage024_b3a_signature_follows_directly b_over_a, signature_1_half_minus1	paper/stages/stage_024.tex:231	internal_consistency	provenance_built
024	fit_stage024_grouped_bundle_forced_isotropic grouped_lane_coefficients_20_21_22	notes/stages/moving_throat_pde_stage024_overlap_isotropy.md:329	internal_consistency	audited
024	fit_stage024_mhat_ang_exactly_one mhat_ang	scripts/moving_throat_pde_stage024_overlap_isotropy_sympy_audit.py:172	internal_consistency	provenance_built
024	fit_stage024_quadrupole_splitting_signature b_over_a, lambda_20, lambda_21, lambda_22	paper/stages/stage_024.tex:7	internal_consistency	provenance_built
024	fit_stage_024_kappa kappa, pi, stale_output, value_mismatch	redteam/pass2/reports/stage_024.md:163	internal_consistency	provenance_built
024	fit_stage_024_matched_fingerprint_value matched_fingerprint_value	notes/stages/moving_throat_pde_stage024_overlap_isotropy.md:371	internal_consistency	provenance_built
025	fit_stage025_mhat_ang_already_fixed mhat_ang	paper/stages/stage_025.tex:92	internal_consistency	provenance_built
025	fit_stage025_mhat_rad_tuned_to_target mhat_rad	paper/stages/stage_025.tex:65	published_target	audited
025	fit_stage025_one_mode_normalization_target N_Q_target, mhat_rad	paper/stages/stage_025.tex:61	published_target	audited
025	fit_stage025_radial_source_map_normalization N_Q_target, mhat_rad	research/pde_ledger/paper/stages/stage_025.tex:65	published_target	audited
025	fit_stage025_target_coefficient_54_5 mhat, target	scripts/moving_throat_pde_stage025_minimal_isotropic_normalization_sympy_audit.py:123	published_target	audited
025	fit_stage_025_d_0 D_0, N_0, P_0, RG_U	redteam/pass2/reports/stage_025.md:35	internal_consistency	provenance_built
025	fit_stage_025_p0_compact P0_compact	redteam/pass2/reports/stage_025.md:39	published_target	audited
026	fit_stage026_K_req_solved_exactly_from_target K_req	notes/stages/moving_throat_pde_stage026_concrete_axial_overlaps.md:212	published_target	audited
026	fit_stage026_Kreq_solved_backward K_req	paper/stages/stage_026.tex:111	published_target	audited
026	fit_stage026_dn_constants_exact_derived_downstream dn_branch_constants, mhat_minus	paper/stages/stage_026.tex:141	internal_consistency	provenance_built
026	fit_stage026_dn_mode_basis_kappa0 f_n, kappa_0	paper/stages/stage_026.tex:25	internal_consistency	provenance_built
026	fit_stage026_kappa_exact_overlap kappa, kappa_n	paper/stages/stage_026.tex:36	internal_consistency	audited
026	fit_stage026_kappa_overlap_2sqrt2_over_pi L, kappa, kappa_n	scripts/moving_throat_pde_stage026_concrete_axial_overlaps_sympy_audit.py:98	internal_consistency	provenance_built
026	fit_stage_026_a_w A_w	notes/stages/moving_throat_pde_stage026_concrete_axial_overlaps.md:67	no_constraint_record	provenance_built

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
026	fit_stage_026_g_u G_U, G_W, K_req, f_n	redteam/pass2/reports/ stage_026.md:35(2 manifest anchors)	internal_consistency	provenance_built
026	fit_stage_026_k_req K_req	redteam/pass2/reports/ stage_026.md:105	published_target	audited
027	fit_stage027_max_coupling_branch_not_free kappa_theta, theta_max	notes/stages/moving_throat_pde_stage027_nonconstant_axial_family.md:308	internal_consistency	audited
027	fit_stage027_sin2_theta_max_2_11 kappa_max, sin2_theta_max, theta	scripts/moving_throat_pde_stage027_nonconstant_axial_family_sympy_audit.py:143	internal_consistency	provenance_built
027	fit_stage027_sin2_theta_max_exact_value sin2_theta_max	paper/stages/stage_027.tex:140	internal_consistency	provenance_built
028	fit_stage028_alpha_loading_coefficient alpha	paper/stages/stage_028.tex:9	free_choice	audited
028	fit_stage028_eigenvalues_exact alpha_crit, lambda_minus, lambda_plus	paper/stages/stage_028.tex:60	internal_consistency	provenance_built
028	fit_stage028_theta_minus_sign_fixed theta_minus	paper/stages/stage_028.tex:93	internal_consistency	provenance_built
028	fit_stage_028_det_eff det_eff	redteam/pass2/reports/ stage_028.md:92	internal_consistency	provenance_built
029	fit_stage029_alpha_derived_dynamically alpha	paper/stages/stage_028.tex:98	internal_consistency	provenance_built
029	fit_stage029_deltaK_algebraically_forced DeltaK_ax	scripts/moving_throat_pde_stage029_dynamic_loading_sympy_audit.py:152	internal_consistency	audited
029	fit_stage029_mixed_sign_convention_match alpha_0, lambda_B	notes/stages/moving_throat_pde_stage029_dynamic_loading.md:87	internal_consistency	provenance_built
030	fit_stage030_lambda_req_target_ratio lambda_req	scripts/moving_throat_pde_stage030_selected_mode_normalization_sympy_audit.py:169	published_target	audited
030	fit_stage030_nq_target_retype_54_5 NQ_target, lambda_req, mhat	scripts/moving_throat_pde_stage030_selected_mode_normalization_sympy_audit.py:158	published_target	audited
030	fit_stage030_selected_branch_target_equation N_Q_target, P_0_minus, mhat_minus	paper/stages/stage_030.tex:91	published_target	audited
030	fit_stage_030_beta_0s_lambda beta_0s_lambda	paper/stages/stage_030.tex:103	internal_consistency	provenance_built
030	fit_stage_030_c_5 C_5, D_0, D_0D_4, D_2, G5_phys	redteam/pass2/reports/ stage_030.md:39	internal_consistency	provenance_built
030	fit_stage_030_m_eff M_eff, lam_minus, s_minus_closed	redteam/pass2/reports/ stage_030.md:81(2 manifest anchors)	internal_consistency	provenance_built
031	fit_stage031_alpha_crit_derived_directly alpha_crit	mathematica/moving_throat_pde_stage031_selected_branch_reachability_mathematica_audit.wl:97	internal_consistency	provenance_built
031	fit_stage031_hf_identity_derived_not_assumed dlambda_minus_dalpha, s_minus	mathematica/moving_throat_pde_stage031_selected_branch_reachability_mathematica_audit.wl:138	internal_consistency	audited

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
031	fit_stage031_unique_alpha_star_crossing P_target, alpha_star	paper/stages/stage_031.tex:60	published_target	audited
031	fit_stage_031_delta_kappa delta_kappa	redteam/pass2/reports/stage_031.md:87	internal_consistency	provenance_built
031	fit_stage_031_stale_output stale_output	redteam/pass2/reports/stage_031.md:123	no_constraint_record	provenance_built
032	fit_stage032_kappa_values_exact kappa_0, kappa_1	notes/stages/moving_throat_pde_stage032_source_map_from_mode_integrals.md:65	internal_consistency	provenance_built
032	fit_stage032_mhat_minus_completely_fixed mhat_minus, s_minus	notes/stages/moving_throat_pde_stage032_source_map_from_mode_integrals.md:208	internal_consistency	audited
032	fit_stage032_pattern_now_derived rank_one_loading_pattern, v_overlap_vector	notes/stages/moving_throat_pde_stage032_source_map_from_mode_integrals.md:130	internal_consistency	provenance_built
032	fit_stage032_sigma_88_9pi2_and_mhat_limit_11_9 kappa_0, kappa_1, mhat_minus, sigma	scripts/moving_throat_pde_stage032_source_map_from_mode_integrals_sympy_audit.py:64	internal_consistency	provenance_built
032	fit_stage_032_i_2 I_2, g_U, kappa_0, kappa_0_kappa_1, kappa_1	redteam/pass2/reports/stage_032.md:35	internal_consistency	provenance_built
032	fit_stage_032_mhat_sq mhat_sq, s_minus, stale_output	redteam/pass2/reports/stage_032.md:156	internal_consistency	provenance_built
033	fit_stage033_dn_constants_exact kappa_0_sq, kappa_1_sq	paper/stages/stage_033.tex:102	internal_consistency	provenance_built
033	fit_stage033_kappa_overlap_literals kappa_0_sq, kappa_1_sq, sigma	scripts/moving_throat_pde_stage033_microscopic_normalization_equation_sympy_audit.py:33	internal_consistency	provenance_built
033	fit_stage_033_k0_onset K0_onset, alpha0_mic, alpha_crit, alpha_crit_mic, den_ratio, is_number	redteam/pass2/reports/stage_033.md:39	internal_consistency	provenance_built
034	fit_stage034_normal_form_maps_fixed N_minus, alpha_0, s_minus	notes/stages/moving_throat_pde_stage034_softening_depth_normal_form.md:183	internal_consistency	provenance_built
034	fit_stage_034_kappa0_sq kappa0_sq, kappa1_sq, s_x, stale_output	redteam/pass2/reports/stage_034.md:116	internal_consistency	provenance_built
034	fit_stage_034_pi_star Pi_star, gamma_0	redteam/pass2/reports/stage_034.md:120	no_constraint_record	provenance_built
035	fit_stage035_xi_req_unique_locus R_target, xi_req	paper/stages/stage_035.tex:7	internal_consistency	audited
035	fit_stage_035_f_target F_target, N_x, R_target, alpha_crit, alpha_mix, alpha_req, alpha_req_target, alpha_x, beta_0, eps_soft, kappa_0	redteam/pass2/reports/stage_035.md:39(2 manifest anchors)	internal_consistency	provenance_built
035	fit_stage_035_r_target R_target, xi_req	notes/stages/moving_throat_pde_stage035_dimensionless_normalization_locus.md:211	published_target	audited
035	fit_stage_035_to to, xi	paper/stages/stage_035.tex:137	no_constraint_record	provenance_built

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
036	fit_stage036_dn_coeffs_stale_stage18_anchor dn_coeff_11, dn_coeff_9, dn_scaling_8_over_pi2_A	notes/CHECKPOINT_CONSTANT_PROVENANCE.md:376	no_constraint_record	provenance_built
036	fit_stage036_no_free_hidden_eigenvector_param g_B_req	paper/stages/stage_036.tex:119	internal_consistency	audited
036	fit_stage_036_m_mix M_mix, R_target	notes/stages/moving_throat_pde_stage036_support_feasibility_frontier.md:151	internal_consistency	provenance_built
037	fit_stage037_branch_data_exact_kernel_functionals A, Delta_K_ax, M_mix, beta_0	notes/stages/moving_throat_pde_stage037_continuum_kernel_extraction.md:37	internal_consistency	provenance_built
037	fit_stage_037_deltak_ax DeltaK_ax, Delta_0, I_2, K_0, K_U, K_W, K_eta, M_mix, Omega_U, Omega_W, Sigma_wall, T_w, alpha_mix, beta_0, c_UW, c_etaU, c_etaW, kappa_0, kappa_1, mu_W, mu_eta	redteam/pass2/reports/stage_037.md:35	internal_consistency	provenance_built
037	fit_stage_037_fix_loop fix_loop, stale_output	redteam/pass2/reports/stage_037.md:132	no_constraint_record	provenance_built
037	fit_stage_037_m_mix M_mix, alpha_mix, f_0, kappa_0, kappa_1, u_0, u_1	redteam/pass2/reports/stage_037.md:39	internal_consistency	provenance_built
038	fit_stage038_delta_fixed_by_eps_eta Lambda, delta, epsilon_W, epsilon_eta	paper/stages/stage_038.tex:51	internal_consistency	audited
038	fit_stage038_lambda_external_target_coefficient G, K_W_eff, Lambda, c_s, mu_W	research/pde_ledger/paper/stages/stage_038.tex:28	published_target	audited
038	fit_stage038_mhat_p0_fitted_scales P_0, mhat_0	graph/fluid_universe_derivation_atlas_graph.yaml	published_target	audited
039	fit_stage039_delta_split_exact_value delta_split	notes/stages/moving_throat_pde_stage039_split_u_sector.md:26	internal_consistency	provenance_built
039	fit_stage_039_d_dir D_dir, K_U, K_U1, K_eta_eff, M_mix, R_U, R_target, S_U, delta_U, delta_split, eps_W, eps_eta, kappa_0, kappa_1, lambda_0, mu_eta, rho_0	redteam/pass2/reports/stage_039.md:39	internal_consistency	provenance_built
040	fit_stage040_lambda_exact_closed_form lambda_minus	scripts/moving_throat_pde_stage040_generalized_selected_branch_sympy_audit.py:8	internal_consistency	audited
041	fit_stage041_n_req_exact_support_loading n_req	scripts/moving_throat_pde_stage041_rank2_support_sympy_audit.py:10	internal_consistency	audited
041	fit_stage_041_d_sel D_sel, R_U, stale_output, tR_U	redteam/pass2/reports/stage_041.md:109	internal_consistency	provenance_built
042	fit_stage042_overlaps_exact q, r	scripts/moving_throat_pde_stage042_rank2_selected_mode_sympy_audit.py:9	internal_consistency	provenance_built
042	fit_stage_042_f_src F_src, R_U, dR_U, lambda_0, n_src, ratio_expected, stale_output	redteam/pass2/reports/stage_042.md:112	internal_consistency	provenance_built
043	fit_stage043_baseline_8_over_pi2_circular_derivation B_baseline, kappa_0_sq, sigma	scripts/moving_throat_pde_stage043_support_direction_sympy_audit.py:146	internal_consistency	provenance_built

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
043	fit_stage043_support_direction_extracted_not_guessed R_phi, sigma_0	notes/stages/moving_throat_pde_stage043_support_direction_extraction.md:13	internal_consistency	provenance_built
043	fit_stage_043_d_phi D_phi, R_U, R_phi	redteam/pass2/reports/stage_043.md:122	internal_consistency	provenance_built
043	fit_stage_043_d_u D_U, D_phi, M_mix, M_supp, R_U, R_phi, Z_phi, delta_U, eps_eta, eps_phi, g_B, g_R, g_S, g_W, kappa_0, kappa_1, tR_U, tR_phi	redteam/pass2/reports/stage_043.md:35	internal_consistency	provenance_built
043	fit_stage_043_m_mix M_mix, M_supp, R_U, R_phi	notes/stages/moving_throat_pde_stage043_support_direction_extraction.md:251	internal_consistency	provenance_built
044	fit_stage044_rank2_inserted_direction_data M_supp, R_phi, delta, t	research/pde_ledger/paper/stages/stage_044.tex:23	internal_consistency	provenance_built
044	fit_stage044_xi_no_longer_free xi	paper/stages/stage_044.tex:6	internal_consistency	audited
044	fit_stage_044_b_cont B_cont, C_cont, F_cont, G_q, M_mix, M_supp, R_U, R_phi, R_target	redteam/pass2/reports/stage_044.md:35	internal_consistency	provenance_built
044	fit_stage_044_r_u R_U, phi	paper/stages/stage_044.tex:37	internal_consistency	provenance_built
045	fit_stage045_tracking_derived_not_postulated R_tr	notes/stages/moving_throat_pde_stage045_coherent_local_tracking.md:206	internal_consistency	provenance_built
045	fit_stage045_tracking_forced R_U, R_phi, R_tr	paper/stages/stage_045.tex:6	internal_consistency	audited
045	fit_stage_045_f_tr F_tr, R_target, branch_eq	redteam/pass2/reports/stage_045.md:141	internal_consistency	provenance_built
045	fit_stage_045_r_tr R_tr	redteam/pass2/reports/stage_045.md:47	internal_consistency	provenance_built
046	fit_stage046_residual_comparison_exact N_tr	paper/stages/stage_046.tex:28	no_constraint_record	provenance_built
046	fit_stage_046_e_flat E_flat, E_tr, F_flat, F_tr, G_flat, G_tr, M_tr, P_1, P_2, P_R, R_target, R_tr, dF_tr, dG_tr	redteam/pass2/reports/stage_046.md:35	internal_consistency	provenance_built
046	fit_stage_046_f_flat F_flat, F_tr, G_flat, G_tr, P_R, dF_tr, dG_tr	redteam/pass2/reports/stage_046.md:39	internal_consistency	provenance_built
047	fit_stage047_R_target_zeta_invariance R_target, zeta	notes/stages/moving_throat_pde_stage047_coherent_kernel_map.md:191	internal_consistency	provenance_built
047	fit_stage047_epsilon_support_saturation epsilon	research/pde_ledger/paper/stages/stage_047.tex:22	internal_consistency	provenance_built
048	fit_stage048_zeta_req_threshold S_req, zeta_req	research/pde_ledger/paper/stages/stage_048.tex:22	internal_consistency	audited
048	fit_stage048_zeta_req_unique zeta_req	paper/stages/stage_048.tex:24	internal_consistency	audited
048	fit_stage_048_f_tr F_tr, G_tr, zeta_crit, zeta_req	redteam/pass2/reports/stage_048.md:143	internal_consistency	provenance_built
048	fit_stage_048_zeta_rm zeta_rm	redteam/pass2/reports/stage_048.md:147	no_constraint_record	provenance_built

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
049	fit_stage049_uniform_source_density sigma_source_density, zeta_0_twin	notes/stages/moving_throat_pde_stage049_dn_overlap_zeta.md:93	free_choice	audited
049	fit_stage049_zeta_phys_no_longer_free zeta_n_phys	notes/stages/moving_throat_pde_stage049_dn_overlap_zeta.md:13	internal_consistency	audited
050	fit_stage050_enhancement_factor_exact S_zeta_n, zeta_req	notes/stages/moving_throat_pde_stage050_zeta_threshold_comparison.md:27	internal_consistency	provenance_built
051	fit_stage051_xi_2x_closed_root xi_2x	scripts/moving_throat_pde_stage051_lowest_twin_criterion_sympy_audit.py:158	internal_consistency	provenance_built
051	fit_stage_051_c_mix C_mix, Z_W, zeta_req	redteam/pass2/reports/stage_051.md:51(3 manifest anchors)	internal_consistency	provenance_built
052	fit_stage052_asymmetry_forced Omega_asym, zeta_req	notes/stages/moving_throat_pde_stage052_nontwin_asymmetry_threshold.md:165	internal_consistency	audited
053	fit_stage053_overlap_window_exact A_I, Omega_0	notes/stages/moving_throat_pde_stage053_overlap_boost_window.md:84	internal_consistency	provenance_built
054	fit_stage054_robin_branch_determined A_K, mu_R	notes/stages/moving_throat_pde_stage054_robin_softening_support_lane.md:213	internal_consistency	audited
055	fit_stage055_reachability_window_exact x, zeta_0_phys	paper/stages/stage_055.tex:24	internal_consistency	provenance_built
055	fit_stage_055_stale_output stale_output, x_floor, zeta_req	redteam/pass2/reports/stage_055.md:131	no_constraint_record	provenance_built
056	fit_stage056_pe_identification Pe, alpha	notes/stages/moving_throat_pde_stage056_transport_source_asymmetry.md:28	internal_consistency	provenance_built
056	fit_stage_056_i_w I_W, missing_branch, stale_output	redteam/pass2/reports/stage_056.md:39(2 manifest anchors)	internal_consistency	provenance_built
057	fit_stage057_x_identified_as_stiffness_ratio kappa, x	notes/stages/moving_throat_pde_stage057_physical_parameter_map.md:81	internal_consistency	provenance_built
058	fit_stage058_bracket_derived_not_guessed Delta_0, Delta_inf, Pe_star	notes/stages/moving_throat_pde_stage058_coupled_support_source_operator.md:289	internal_consistency	provenance_built
058	fit_stage_058_delta_0 Delta_0, Delta_inf, Sigma_Pe, stale_output	redteam/pass2/reports/stage_058.md:39(2 manifest anchors)	internal_consistency	provenance_built
059	fit_stage059_pe_req_misattributed_carry Pe_req	notes/CHECKPOINT_CONSTANT_PROVENANCE.md:130	internal_consistency	audited
059	fit_stage059_thresholds_exact Pe_req, Xi_fail, Xi_suff	paper/stages/stage_059.tex:13	internal_consistency	audited
060	fit_stage060_gamma_derived_rate Pe, gamma	scripts/moving_throat_pde_stage060_entropic_microclosure_sympy_audit.py:88	internal_consistency	provenance_built

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
060	fit_stage060_xi_micro_drop_prefactor_choice Xi_micro, phi_drop_prefactor	scripts/moving_throat_pde_stage060_entropic_microclosure_sympy_audit.py:122	internal_consistency	provenance_built
061	fit_stage061_gain_thresholds_exact G_fail, G_suff, kappa	scripts/moving_throat_pde_stage061_microscopic_gain_thresholds_sympy_audit.py:111	internal_consistency	provenance_built
061	fit_stage_061_delta_0 Delta_0, Delta_inf, G_fail, G_micro, G_suff, K_X, Pe_req, T_X, Xi_micro, chi_fail, chi_suff	redteam/pass2/reports/stage_061.md:39	internal_consistency	provenance_built
061	fit_stage_061_g_fail G_fail, G_micro, G_suff, Pe_req, stale_output	redteam/pass2/reports/stage_061.md:105	internal_consistency	provenance_built
061	fit_stage_061_pe_req Pe_req	redteam/pass2/reports/stage_061.md:115	internal_consistency	audited
062	fit_stage062_n5_eos_exponent K_eos, n_poly	paper/stages/stage_062.tex:18	free_choice	audited
062	fit_stage062_theta_not_free_entropy_constant Theta	notes/stages/moving_throat_pde_stage062_parent_action_gain.md:176	internal_consistency	audited
062	fit_stage_062_tautological_check tautological_check	redteam/pass2/reports/stage_062.md:81	no_constraint_record	provenance_built
062	fit_stage_062_xi_micro Xi_micro, Xi_target	redteam/pass2/reports/stage_062.md:140	internal_consistency	provenance_built
063	fit_stage063_amplitude_thresholds_exact g_phi_fail, g_phi_suff	notes/stages/moving_throat_pde_stage063_parent_thresholds.md:102	internal_consistency	provenance_built
063	fit_stage_063_g_fail G_fail, G_max, G_micro, G_suff	redteam/pass2/reports/stage_063.md:39	internal_consistency	provenance_built
064	fit_stage064_coherence_factor_derived C	notes/stages/moving_throat_pde_stage064_equilibrium_alignment.md:223	internal_consistency	provenance_built
064	fit_stage064_matched_layer_coherence C_sigma_phi_sq	paper/stages/stage_064.tex:51	internal_consistency	provenance_built
064	fit_stage_064_f_eff F_eff, G_eq, N_pp, N_ss, g_phi	redteam/pass2/reports/stage_064.md:144	internal_consistency	provenance_built
064	fit_stage_064_g_eq G_eq, H_w, I_1, I_2, K_X, chi_phi, chi_sigma, g_phi	redteam/pass2/reports/stage_064.md:35	internal_consistency	provenance_built
064	fit_stage_064_h_w H_w, I1_int, I2_int, Npp_int, f_eff, sigma_stat	notes/stages/moving_throat_pde_stage064_equilibrium_alignment.md:184(2 manifest anchors)	internal_consistency	provenance_built
065	fit_stage065_canonical_normalization_fprime f_prime_0, g_phi	scripts/moving_throat_pde_stage065_thin_wall_confinement_sympy_audit.py:118	free_choice	audited
065	fit_stage_065_v_0 V_0, ell	paper/stages/stage_065.tex:18	free_choice	audited
065	fit_stage_065_v_0f V_0f, ell	paper/stages/stage_065.tex:7	free_choice	audited
066	fit_stage066_exact_matched_verdict W_fail, W_suff, W_wall	paper/stages/stage_066.tex:19	internal_consistency	provenance_built
066	fit_stage_066_delta_0 Delta_0, Delta_inf, G_eq, H_w, I_f, J_1, K_X, Pe_req, T_X, W_H, W_wall	notes/stages/moving_throat_pde_stage066_wall_figure_of_merit.md:126(2 manifest anchors)	internal_consistency	provenance_built
066	fit_stage_066_w_wall W_wall	redteam/pass2/reports/stage_066.md:120	internal_consistency	provenance_built

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
067	fit_stage067_C_res_unique_global_max C_res_squared, P_res	notes/stages/moving_throat_pde_stage067_sech_gaussian_resonance.md:108	internal_consistency	audited
067	fit_stage067_sech_gaussian_benchmark_constants C_res_sq, P_res, chi_phi_profile, chi_sigma_profile	research/pde_ledger/notes/stages/moving_throat_pde_stage067_sech_gaussian_resonance.md:6	internal_consistency	provenance_built
067	fit_stage067_w_ratio_self_duality_forced w_g_over_w_f	notes/stages/moving_throat_pde_stage067_sech_gaussian_resonance.md:20	internal_consistency	audited
067	fit_stage_067_c_res C_res, P_res, chi_phi, chi_sigma, simeq, w_f, w_g	redteam/pass2/reports/stage_067.md:35(2 manifest anchors)	internal_consistency	provenance_built
067	fit_stage_067_iprime_dual Iprime_dual, Iprime_left, dC2_selfdual	redteam/pass2/reports/stage_067.md:106	internal_consistency	provenance_built
068	fit_stage068_W_res_derived_label W_res	mathematica/moving_throat_pde_stage068_resonance_thresholds_mathematica_audit.wl:33	internal_consistency	provenance_built
068	fit_stage068_threshold_translation_not_postulated P_res, W_fail_res, W_suff_res	scripts/moving_throat_pde_stage068_resonance_thresholds_sympy_audit.py:104	internal_consistency	provenance_built
068	fit_stage_068_c_res C_res, Delta_0, Delta_inf, P_res, Pe_req, W_fail, W_res, W_suff, W_wall, simeq	notes/stages/moving_throat_pde_stage068_resonance_thresholds.md:23(2 manifest anchors)	internal_consistency	provenance_built
068	fit_stage_068_delta_0 Delta_0, Pe_req, W_wall	notes/stages/moving_throat_pde_stage068_resonance_thresholds.md:71	internal_consistency	provenance_built
068	fit_stage_068_delta_inf Delta_inf, Pe_req, W_wall	notes/stages/moving_throat_pde_stage068_resonance_thresholds.md:69	internal_consistency	provenance_built
068	fit_stage_068_p_res P_res	notes/stages/moving_throat_pde_stage068_resonance_thresholds.md:119	internal_consistency	provenance_built
069	fit_stage069_matched_window_exactly P_res, W_fail_match, W_suff_match	scripts/moving_throat_pde_stage069_final_reduced_verdict_sympy_audit.py:9	internal_consistency	provenance_built
069	fit_stage_069_c_res C_res, Delta_0, Delta_eff, Delta_gap, Delta_inf, P_res, Pe_req, Pres_from_ratio, Pres_gap, W_match, W_wall, Wfail_match, Wsuff_match, expect_positive, stale_output, v_fail, v_succ	redteam/pass2/reports/stage_069.md:35(5 manifest anchors)	internal_consistency	provenance_built
069	fit_stage_069_delta_0 Delta_0, Rightarrow, mathrm	paper/stages/stage_069.tex:19	internal_consistency	provenance_built
069	fit_stage_069_delta_eff Delta_eff, Pres_gap, dDelta_eff, dW_match, delta_fail, delta_succ	redteam/pass2/reports/stage_069.md:112	no_constraint_record	provenance_built
069	fit_stage_069_p_res P_res	notes/stages/moving_throat_pde_stage069_final_reduced_verdict.md:92	internal_consistency	provenance_built
070	fit_stage070_n5_eos_exponent n	paper/stages/stage_070.tex:7	free_choice	audited
070	fit_stage070_wall_shell_fixes_quintuple J_1, K_X, T_X, W_wall, kappa	notes/stages/moving_throat_pde_stage070_gnls_wall_shell.md:177	internal_consistency	provenance_built

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
070	fit_stage_070_i_f I_f, W_wall	notes/stages/moving_throat_pde_stage070_gnls_wall_shell.md:132	internal_consistency	provenance_built
070	fit_stage_070_j_1 J_1, T_X	notes/stages/moving_throat_pde_stage070_gnls_wall_shell.md:9	internal_consistency	provenance_built
071	fit_stage071_canonical_tanh_and_natural_closure K_m, tanh_wall_profile	notes/stages/moving_throat_pde_stage071_tanh_wall_branch.md:7	free_choice	audited
071	fit_stage071_mouth_closure_Km K_m, eta	paper/stages/stage_071.tex:33	free_choice	audited
071	fit_stage_071_i_f I_f, I_g, J_1, K_X, K_m, Lambda_ell, T_X, W_wall, c_sw, chi_s	redteam/pass2/reports/stage_071.md:39	internal_consistency	provenance_built
072	fit_stage072_canonical_controls_exact_thresholds Lambda_ell, Upsilon_w, chi_s	paper/stages/stage_072.tex:7	no_constraint_record	provenance_built
073	fit_stage073_Lambda_ell_fixed_37 L_over_a, Lambda_ell, epsilon_r	notes/stages/moving_throat_pde_stage073_family1_geometry_map.md:72	free_choice	audited
073	fit_stage073_aspect_ratio_L_over_a L_over_a, Lambda_ell	paper/stages/stage_073.tex:17	free_choice	audited
073	fit_stage073_aspect_ratio_frozen L_over_a, Lambda_star	research/pde_ledger/notes/stages/moving_throat_pde_stage073_family1_geometry_map.md:48	free_choice	audited
073	fit_stage073_epsilon_r_posited ell_over_a, epsilon_r	research/pde_ledger/notes/stages/moving_throat_pde_stage073_family1_geometry_map.md:30(2 manifest anchors)	free_choice	audited
073	fit_stage073_eta_pinned_37 eta	notes/stages/moving_throat_pde_stage073_family1_geometry_map.md:94	internal_consistency	audited
073	fit_stage073_paper_fixes_geometry_ratios Lambda_ell, eta	paper/stages/stage_073.tex:6	free_choice	audited
073	fit_stage073_wall_fraction_epsilon_r ell_over_a, epsilon_r	notes/stages/moving_throat_pde_stage073_family1_geometry_map.md:30	free_choice alias offit_stage073_epsilon_r_posited	scanned; alias folded to fit_stage073_epsilon_r_posited
073	fit_stage_073_k_m K_m, Lambda_ell, T_X, chi_s, epsilon_r	redteam/pass2/reports/stage_073.md:39(3 manifest anchors)	free_choice	audited
073	fit_stage_073_t_x T_X	redteam/pass2/reports/stage_073.md:65	internal_consistency	provenance_built
074	fit_stage074_chi_s_now_fixes chi_s	notes/stages/moving_throat_pde_stage074_family1_healing_lock.md:73	internal_consistency	provenance_built
074	fit_stage074_healing_lock_ell chi_s, ell, healing_lock_coefficient	notes/stages/moving_throat_pde_stage074_family1_healing_lock.md:53	free_choice	audited
074	fit_stage074_kappa_locked_reduction chi_s, kappa	scripts/moving_throat_pde_stage074_family1_healing_lock_sympy_audit.py:46	internal_consistency	provenance_built
074	fit_stage074_natural_healing_closure ell, ell_h	notes/stages/moving_throat_pde_stage074_family1_healing_lock.md:51	free_choice	audited

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
075	fit_stage075_alpha_r_wall_depth Upsilon_w, alpha_r	paper/stages/stage_075.tex:24	free_choice	audited
075	fit_stage075_triple_fixed_summary Lambda_ell, chi_s, kappa	notes/stages/moving_throat_pde_stage075_family1_threshold_window.md:144	no_constraint_record	provenance_built
075	fit_stage_075_delta_0 Delta_0, Delta_inf, Theta_fail, Upsilon_fail, Xi_fail, alpha_r, stop_cold	redteam/pass2/reports/stage_075.md:39(2 manifest anchors)	internal_consistency	provenance_built
075	fit_stage_075_pe_req Pe_req, Theta_fail, Theta_suff	redteam/pass2/reports/stage_075.md:100	internal_consistency	audited
076	fit_stage076_lambda_mu_normalization lambda_mu, mu_star	notes/stages/moving_throat_pde_stage076_n5_wall_depth_lock.md:60	free_choice	audited
076	fit_stage076_target_25_derived Theta_w, ref_factor	scripts/moving_throat_pde_stage076_n5_wall_depth_lock_sympy_audit.py:89	internal_consistency	provenance_built
076	fit_stage076_wall_depth_exactly_square Theta_w, lambda_mu	notes/stages/moving_throat_pde_stage076_n5_wall_depth_lock.md:93	internal_consistency	provenance_built
077	fit_stage077_If_exact_canonical I_f	scripts/moving_throat_pde_stage077_family1_theta_extraction_sympy_audit.py:7	internal_consistency	audited
077	fit_stage077_Theta_w_chi_derived_directly Theta_w_chi	notes/stages/moving_throat_pde_stage077_family1_theta_extraction.md:96	internal_consistency	provenance_built
077	fit_stage_077_i_f I_f, Theta_w, alpha_r, chi, chi_phi, epsilon_r, lambda_mu, p_r, rho_r, rho_w, stop_cold, xi_cut	redteam/pass2/reports/stage_077.md:35(2 manifest anchors)	internal_consistency	provenance_built
077	fit_stage_077_theta_w Theta_w	notes/stages/moving_throat_pde_stage077_family1_theta_extraction.md:115	internal_consistency	provenance_built
078	fit_stage078_verdict_fixed_pair Lambda_ell, chi_s	notes/stages/moving_throat_pde_stage078_family1_branch_verdict.md:85	no_constraint_record	provenance_built
078	fit_stage_078_pe_req Pe_req	notes/stages/moving_throat_pde_stage078_family1_branch_verdict.md:99	internal_consistency	audited
079	fit_stage079_unique_constructive_Pe_req Pe_req	scripts/moving_throat_pde_stage079_family1_quadrupole_pe_map_sympy_audit.py:76	internal_consistency	audited
079	fit_stage_079_a_f1 A_F1, expect_small, expected_series, kappa_F1, y_F1, zeta_F1, zeta_max	redteam/pass2/reports/stage_079.md:101(2 manifest anchors)	internal_consistency	provenance_built
080	fit_stage_080_a_f1 A_F1, eta_F1, kappa_F1, lambda_mu, tautological_check, zeta_F1, zeta_max	redteam/pass2/reports/stage_080.md:118	internal_consistency	provenance_built
081	fit_stage_081_c_mix C_mix, zeta_expr, zeta_max	redteam/pass2/reports/stage_081.md:112	internal_consistency	provenance_built
081	fit_stage_081_eps_blk eps_blk	redteam/pass2/reports/stage_081.md:94	free_choice	audited

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
082	`fit_stage082_first_exact_product_window` `Pi_fail`, `Pi_suff`	`notes/stages/moving_throat_pde_stage082_master_quadrupole_residual.md`:204	`internal_consistency`	provenance_built
082	`fit_stage_082_pe_req` `Pe_req`	`notes/stages/moving_throat_pde_stage082_master_quadrupole_residual.md`:204	`internal_consistency`	audited
082	`fit_stage_082_zeta_max` `zeta_max`	`redteam/pass2/reports/stage_082.md`:134	`no_constraint_record`	provenance_built
083	`fit_stage083_windows_now_derived_directly` `zeta_fail`, `zeta_suff`	`notes/stages/moving_throat_pde_stage083_family1_direct_operator_window.md`:119	`internal_consistency`	provenance_built
083	`fit_stage_083_a_f1` `A_F1`, `Delta_0`, `Delta_inf`, `y_F1`	`redteam/pass2/reports/stage_083.md`:183	`internal_consistency`	provenance_built
083	`fit_stage_083_c_mix` `C_mix`, `Delta_0`, `Delta_inf`, `Xi_F1`, `y_F1`, `zeta_F1`, `zeta_max`, `zeta_phys`	`redteam/pass2/reports/stage_083.md`:35(2 manifest anchors)	`internal_consistency`	provenance_built
083	`fit_stage_083_delta0_f1` `Delta0_F1`, `Delta0_indep`, `Delta_0`, `Delta_inf`, `expect_close`	`redteam/pass2/reports/stage_083.md`:95(3 manifest anchors)	`internal_consistency`	provenance_built
083	`fit_stage_083_delta_0` `Delta_0`, `Delta_inf`	`redteam/pass2/reports/stage_083.md`:131	`internal_consistency`	provenance_built
083	`fit_stage_083_pe_minus_j` `Pe_minus_J`, `Pe_minus_chi`, `Pe_plus_J`, `Pe_plus_chi`, `expect_close`, `zeta_max_F1`, `zeta_minus_J`, `zeta_minus_chi`, `zeta_plus_J`, `zeta_plus_chi`	`redteam/pass2/reports/stage_083.md`:124	`internal_consistency`	provenance_built
084	`fit_stage_084_c_mix` `C_mix`, `Pi_tr`, `StatusOpen`, `Theta_w`, `Upsilon_w`, `eta_F1`, `kappa_F1`, `y_F1`, `zeta_max`, `zeta_phys`, `zeta_req`	`redteam/pass2/reports/stage_084.md`:39	`internal_consistency`	provenance_built
084	`fit_stage_084_omega_pe` `Omega_Pe`, `kappa_F1`, `y_F1`, `zeta_max`, `zeta_phys`	`redteam/pass2/reports/stage_084.md`:74	`internal_consistency`	provenance_built
084	`fit_stage_084_theta_w` `Theta_w`, `Upsilon_w`, `hardcoded_result`, `zeta_req`	`redteam/pass2/reports/stage_084.md`:82	`internal_consistency`	provenance_built
084	`fit_stage_084_zeta_max` `zeta_max`	`redteam/pass2/reports/stage_084.md`:50	`internal_consistency`	provenance_built
085	`fit_stage085_unblocked_literal_law` `eps_blk`, `zeta_req`	`notes/stages/moving_throat_pde_stage085_quadrupole_demand_cancellation.md`:136	`free_choice+internal_consistency`	audited
085	`fit_stage_085_rho_alpha` `rho_alpha`	`notes/stages/moving_throat_pde_stage085_quadrupole_demand_cancellation.md`:187	`internal_consistency`	provenance_built
086	`fit_stage086_natural_lambda_mu_unblocked` `eps_blk`, `lambda_mu`	`notes/stages/moving_throat_pde_stage086_family1_loading_ratio_window.md`:152	`free_choice`	audited
087	`fit_stage087_natural_lambda_mu_unblocked` `eps_blk`, `lambda_mu`	`notes/stages/moving_throat_pde_stage087_outgoing_branch_loading_ratio_finish.md`:57	`free_choice`	audited
088	`fit_stage088_c0_c1_derived_not_defined` `c0`, `c1`	`scripts/moving_throat_pde_stage088_loading_ratio_from_minimal_module_sympy_audit.py`:69	`free_choice`	audited

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
088	fit_stage088_minimal_module_c0_c1 c_0, c_1, rho_alpha, zeta_req	paper/stages/stage_088.tex:20	free_choice	audited
088	fit_stage088_minimal_module_coefficients_paper_quoted c0, c1, c_contact, c_pole	research/pde_ledger/scripts/moving_throat_pde_stage088_loading_ratio_from_minimal_module_sympy_audit.py:95	free_choice	audited
088	fit_stage088_natural_identification_fixes_exactly c_contact, rho_alpha, zeta_req	notes/stages/moving_throat_pde_stage088_loading_ratio_from_minimal_module.md:92	free_choice	audited
088	fit_stage088_pole_frequency_OmegaQ Omega_Q	paper/stages/stage_088.tex:24	free_choice	audited
089	fit_stage089_Pe_req_forced_not_assumed Pe_req	scripts/moving_throat_pde_stage089_family1_minimal_isotropic_verdict_sympy_audit.py:125	internal_consistency	audited
089	fit_stage089_pe_window_upstream_misattribution Pe_fail_chi, Pe_suff_chi	research/pde_ledger/scripts/moving_throat_pde_stage089_family1_minimal_isotropic_verdict_sympy_audit.py:61	internal_consistency	audited
089	fit_stage089_provenance_record_misattribution Pe_fail_chi, Pe_suff_chi	research/pde_ledger/notes/CHECKPOINT_CONSTANT_PROVENANCE.md:111	internal_consistency	audited
089	fit_stage_089_a_f1 A_F1	redteam/pass2/reports/stage_089.md:246	internal_consistency	provenance_built
090	fit_stage090_locked_triple_Pe_req Pe_req, rho_alpha, zeta_req	mathematica/moving_throat_pde_stage090_updated_reduced_status_mathematica_audit.wl:62	internal_consistency	audited
090	fit_stage090_minimal_module_upstream_contradiction c_contact, c_pole	research/pde_ledger/scripts/moving_throat_pde_stage090_updated_reduced_status_sympy_audit.py:15	free_choice	audited
090	fit_stage_090_rho_alpha rho_alpha, stage_090, zeta_req	redteam/pass2/reports/stage_090.md:210	internal_consistency	provenance_built
091	fit_stage091_K_geom_forced_by_identity K_geom, K_pole	scripts/moving_throat_pde_stage091_grouped_p2_static_geometry_derivation_sympy_audit.py:62	internal_consistency	audited
091	fit_stage091_branch_identity_factor4 K0_K4_over_K2sq, u_4	notes/stages/moving_throat_pde_stage091_grouped_p2_static_geometry_derivation.md:52	internal_consistency	provenance_built
091	fit_stage091_module_forced_paper Y_Q_cons, rho_alpha, zeta_req	paper/stages/stage_091.tex:16	internal_consistency	audited
091	fit_stage_091_epsilon_2 epsilon_2, epsilon_2_epsilon_4, epsilon_4	paper/stages/stage_091.tex:13	free_choice	audited
091	fit_stage_091_k_geom K_geom, K_pole, Yhat_expected, alpha_mix, alpha_req, rho_alpha, zeta_req	redteam/pass2/reports/stage_091.md:93	internal_consistency	provenance_built
092	fit_stage092_forced_carrier_ledger_label Y_Q_cons, eps_2, eps_4	paper/stages/stage_092.tex:13	internal_consistency	audited

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
092	fit_stage092_no_longer_forced_unless K_geom, eps_2, eps_4	notes/stages/moving_throat_pde_stage092_dynamic_geometry_obstruction.md:68	free_choice	audited
092	fit_stage_092_epsilon_2 epsilon_2, epsilon_2_epsilon_4, epsilon_4	paper/stages/stage_092.tex:13	free_choice	audited
092	fit_stage_092_yhat_q Yhat_Q	notes/CHECKPOINT_CONSTANT_PROVENANCE.md:95	internal_consistency	provenance_built
093	fit_stage093_module_forced_statement Y_Q_cons, rho_alpha, zeta_req	notes/stages/moving_throat_pde_stage093_grouped_p2_status_update.md:23	internal_consistency	audited
093	fit_stage093_static_limit_hardcoded eps_2, eps_4	research/pde_ledger/mathematica/moving_throat_pde_stage093_grouped_p2_status_update_mathematica_audit.wl:28	free_choice	audited
093	fit_stage_093_c_pole c_pole, eps_2, eps_4, value_mismatch	redteam/pass2/reports/stage_093.md:75(2 manifest anchors)	internal_consistency	provenance_built
093	fit_stage_093_eps_2 eps_2, eps_4	notes/stages/moving_throat_pde_stage093_grouped_p2_status_update.md:55	free_choice	audited
093	fit_stage_093_epsilon_2 epsilon_2, epsilon_2_epsilon_4, epsilon_4	paper/stages/stage_093.tex:13	free_choice	audited
094	fit_stage094_decoupling_kills_contamination K_g2, K_g4	notes/stages/moving_throat_pde_stage094_isotropic_geometry_decoupling.md:101	internal_consistency	provenance_built
094	fit_stage094_static_split_assigned_not_derived c_geom, c_pole	research/pde_ledger/scripts/moving_throat_pde_stage094_isotropic_geometry_decoupling_sympy_audit.py:75	internal_consistency	audited
094	fit_stage_094_c_geom c_geom, c_pole, eps_2, eps_4	redteam/pass2/reports/stage_094.md:101	internal_consistency	provenance_built
094	fit_stage_094_epsilon_2 epsilon_2, epsilon_2_epsilon_4, epsilon_4	paper/stages/stage_094.tex:13	internal_consistency	provenance_built
095	fit_stage095_forced_carrier_ledger_label Y_Q_cons, eps_2, eps_4	paper/stages/stage_095.tex:13	internal_consistency	audited
095	fit_stage_095_c_pole c_pole	redteam/pass2/reports/stage_095.md:116	internal_consistency	provenance_built
095	fit_stage_095_epsilon_2 epsilon_2, epsilon_2_epsilon_4, epsilon_4	paper/stages/stage_095.tex:13	internal_consistency	provenance_built
096	fit_stage096_constants_derived_from_eps_zero c0, c1, eps_2, eps_4, rho_alpha, zeta_req	scripts/moving_throat_pde_stage096_geometry_lane_check_verdict_sympy_audit.py:16	no_constraint_record	provenance_built
096	fit_stage_096_c_geom c_geom, c_pole, c_pole_gen, eps_2, eps_4, rho_alpha, zeta_req	redteam/pass2/reports/stage_096.md:94(2 manifest anchors)	internal_consistency	provenance_built
096	fit_stage_096_eps_2 eps_2, eps_4	redteam/pass2/reports/stage_096.md:91	internal_consistency	provenance_built
096	fit_stage_096_epsilon_2 epsilon_2, epsilon_2_epsilon_4, epsilon_4	paper/stages/stage_096.tex:13	internal_consistency	provenance_built
096	fit_stage_096_numbering_script_output_band_plan NUMBERING_SCRIPT_OUTPUT_BAND_PLAN	redteam/pass2/reports/stage_096.md:125	no_constraint_record	provenance_built

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
096	fit_stage_096_yhat_cons Yhat_cons, Yhat_expected, c_geom, c_pole, eps_2, eps_4, rho_alpha, zeta_req	redteam/pass2/reports/ stage_096.md:84	internal_ consistency	provenance_built
097	fit_stage097_even_ledger_ fixed Kbar_0, Omega_Q	notes/stages/moving_ throat_pde_stage097_ single_normalization_ defect.md:38	internal_ consistency	provenance_built
097	fit_stage097_gamma5_fixed_ algebraically Gamma_5	notes/stages/moving_ throat_pde_stage097_ single_normalization_ defect.md:44	internal_ consistency	provenance_built
097	fit_stage097_yhatQcons_ forced_carrier Yhat_Q_cons	paper/stages/stage_097. tex:13	internal_ consistency	audited
097	fit_stage_097_epsilon_2 epsilon_2, epsilon_2_epsilon_ 4, epsilon_4	paper/stages/stage_097. tex:13	internal_ consistency	provenance_built
097	fit_stage_097_kbar_0 Kbar_0	notes/stages/moving_ throat_pde_stage097_ single_normalization_ defect.md:53	published_target	audited
097	fit_stage_097_r_i R_i	redteam/pass2/reports/ stage_097.md:103	published_target	audited
098	fit_stage098_forced_carrier_ card Yhat_Q_cons	paper/stages/stage_098. tex:13	internal_ consistency	audited
098	fit_stage098_rho_alpha_ selected_ratio rho_alpha	notes/stages/moving_ throat_pde_stage098_ family1_support_is_ automatic.md:16	internal_ consistency	provenance_built
098	fit_stage098_support_demand_ exact support_ratio, zeta_max	notes/stages/moving_ throat_pde_stage098_ family1_support_is_ automatic.md:18	internal_ consistency	audited
098	fit_stage098_zeta_max_f1_ imported_literal gap_F1, zeta_edge_F1, zeta_ max_F1	scripts/moving_throat_ pde_stage098_family1_ support_is_automatic_ sympy_audit.py:26	internal_ consistency	provenance_built
098	fit_stage098_zeta_max_f1_ opaque_carried_ceiling zeta_max_F1	notes/stages/moving_ throat_pde_stage098_ family1_support_is_ automatic.md:22	internal_ consistency	provenance_built
098	fit_stage_098_c_pole c_pole, rho_alpha, zeta_req, zmax_F1	redteam/pass2/reports/ stage_098.md:88	internal_ consistency	provenance_built
098	fit_stage_098_epsilon_2 epsilon_2, epsilon_2_epsilon_ 4, epsilon_4	paper/stages/stage_098. tex:13	internal_ consistency	provenance_built
098	fit_stage_098_gap_f1 gap_F1, zeta_edge, zeta_max, zeta_req, zmax_F1	redteam/pass2/reports/ stage_098.md:39(2 manifest anchors)	internal_ consistency	provenance_built
098	fit_stage_098_gap_f1_target gap_F1_target, zeta_edge_F1_ target	redteam/pass2/reports/ stage_098.md:105	internal_ consistency	provenance_built
098	fit_stage_098_zeta_edge_f1 zeta_edge_F1, zmax_F1	redteam/pass2/reports/ stage_098.md:102	internal_ consistency	provenance_built
098	fit_stage_098_zeta_max zeta_max	redteam/pass2/reports/ stage_098.md:100	internal_ consistency	provenance_built
099	fit_stage099_conservative_ split_exact Yhat_Q_cons, conservative_ split_3_4_1_4	notes/stages/moving_ throat_pde_stage099_ reduced_finish_line.md:18	internal_ consistency	provenance_built
099	fit_stage099_forced_by_ yhatQcons_script Gamma_5, K_0	scripts/moving_throat_ pde_stage099_reduced_ finish_line_sympy_audit. py:63	internal_ consistency	audited

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
099	fit_stage_099_epsilon_2 epsilon_2, epsilon_2_epsilon_4, epsilon_4	paper/stages/stage_099.tex:13	internal_consistency	provenance_built
099	fit_stage_099_k0_sym K0_sym, K0_target	redteam/pass2/reports/stage_099.md:89	free_choice	audited
099	fit_stage_099_xi_t Xi_T, Xi_v, r_F1, r_c, r_geom	notes/CHECKPOINT_CONSTANT_PROVENANCE.md:25	no_constraint_record	provenance_built
100	fit_stage100_chiQ_fixed_downstream_card chi_Q	paper/stages/stage_100.tex:13	free_choice	audited
100	fit_stage100_factorization_forced_script N_Q, chi_Q, mhat_0	scripts/moving_throat_pde_stage100_outgoing_normalization_factorization_sympy_audit.py:75	internal_consistency	audited
100	fit_stage_100_chi_q chi_Q, matched_fingerprint_value, mhat_0	paper/stages/stage_100.tex:13(4 manifest anchors)	free_choice	audited
100	fit_stage_100_gamma5_t Gamma5_t, Gammabar_5, K0_t, K2_t, K4_t, Kbar_0, Kbar_2, Kbar_4, Yhat_Q, chi_Q, closure_ratio, matched_fingerprint_value, mhat_0	redteam/pass2/reports/stage_100.md:65(2 manifest anchors)	published_target	audited
100	fit_stage_100_matched_fingerprint_value matched_fingerprint_value	paper/stages/stage_100.tex:24	internal_consistency	provenance_built
101	fit_stage101_stage105_fixes_by_matching chi_Q, mhat_0	notes/stages/moving_throat_pde_stage101_natural_source_map_reduction.md:49	internal_consistency	provenance_built
101	fit_stage_101_chi_q chi_Q, matched_fingerprint_value, mhat_0, to	paper/stages/stage_101.tex:13(5 manifest anchors)	internal_consistency	provenance_built
101	fit_stage_101_delta_q Delta_Q, chi_Q, matched_fingerprint_value, mhat_0, to	redteam/pass2/reports/stage_101.md:35(2 manifest anchors)	internal_consistency	provenance_built
101	fit_stage_101_matched_fingerprint_value matched_fingerprint_value	paper/stages/stage_101.tex:24(4 manifest anchors)	internal_consistency	provenance_built
102	fit_stage102_everything_else_fixed chi_Q, higher_odd_coeff_omega7	notes/stages/moving_throat_pde_stage102_higher_odd_irrelevance.md:58	internal_consistency	provenance_built
102	fit_stage_102_chi_q chi_Q	paper/stages/stage_102.tex:13(2 manifest anchors)	internal_consistency	provenance_built
102	fit_stage_102_gamma_5 Gamma_5, K_0, Yhat_Q, chi_Q, sigma_Q, stage_102, tau_Q	redteam/pass2/reports/stage_102.md:35	no_constraint_record	provenance_built
102	fit_stage_102_matched_fingerprint_value matched_fingerprint_value	paper/stages/stage_102.tex:24	internal_consistency	provenance_built
103	fit_stage103_split_exact_status conservative_split_3_4_1_4	notes/stages/moving_throat_pde_stage103_reduced_25pn_conditional_closure.md:8	internal_consistency	provenance_built
103	fit_stage_103_chi_q chi_Q	paper/stages/stage_103.tex:13	internal_consistency	provenance_built
103	fit_stage_103_matched_fingerprint_value matched_fingerprint_value	paper/stages/stage_103.tex:24	internal_consistency	provenance_built
104	fit_stage104_chiQ_canonical_condition_card chi_Q	paper/stages/stage_104.tex:13	internal_consistency	provenance_built

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
104	fit_stage104_gamma5_not_free Gamma_5_can, xi_Q	notes/stages/moving_throat_pde_stage104_outgoing_dtn_fingerprint.md:87	internal_consistency	audited
104	fit_stage104_unit_product_m0hat2_chiQ_NQ N_Q, chi_Q, m0_hat	research/pde_ledger/paper/stages/stage_104.tex:13	internal_consistency	provenance_built
104	fit_stage_104_chi_q chi_Q, matched_fingerprint_value	paper/stages/stage_104.tex:13(3 manifest anchors)	internal_consistency	provenance_built
104	fit_stage_104_gamma_5 Gamma_5, chi_Q, matched_fingerprint_value	redteam/pass2/reports/stage_104.md:39(2 manifest anchors)	internal_consistency	provenance_built
104	fit_stage_104_matched_fingerprint_value matched_fingerprint_value	paper/stages/stage_104.tex:24(17 manifest anchors)	internal_consistency	provenance_built
105	fit_stage105_carry_in_stale_multi_stage_attribution Lambda_out_fingerprint_coefficients, sigma_Q_can	notes/CHECKPOINT_CONSTANT_PROVENANCE.md:84	internal_consistency	provenance_built
105	fit_stage105_chiQ_eq_1_by_matching_card chi_Q	paper/stages/stage_105.tex:16	internal_consistency alias offit_stage_105_chi_q	scanned; alias folded to fit_stage_105_chi_q
105	fit_stage105_chi_q_canonical_fix chi_Q	research/pde_ledger/paper/stages/stage_105.tex:16	internal_consistency alias offit_stage_105_chi_q	scanned; alias folded to fit_stage_105_chi_q
105	fit_stage105_chi_q_canonical_unity chi_Q	notes/stages/moving_throat_pde_stage105_chiQ_fix_from_outgoing_dtn.md:69	internal_consistency	provenance_built
105	fit_stage105_chi_q_provenance_record_contradiction chi_Q	notes/CHECKPOINT_CONSTANT_PROVENANCE.md:75	internal_consistency	provenance_built
105	fit_stage105_last_scalar_fixed_exactly chi_Q, xi_Q	notes/stages/moving_throat_pde_stage105_chiQ_fix_from_outgoing_dtn.md:72	internal_consistency	audited
105	fit_stage105_omega_q_pole_frequency Omega_Q	notes/stages/moving_throat_pde_stage105_chiQ_fix_from_outgoing_dtn.md:29	internal_consistency	provenance_built
105	fit_stage_105_chi_q chi_Q, matched_fingerprint_value, sigma_Q, xi_Q	paper/stages/stage_105.tex:13(13 manifest anchors)	internal_consistency	provenance_built
105	fit_stage_105_matched_fingerprint_value matched_fingerprint_value, xi_Q	paper/stages/stage_105.tex:24(6 manifest anchors)	internal_consistency	provenance_built
105	fit_stage_105_t_dtn T_dtn, chi_Q, matched_fingerprint_value, outgoing_dtn_fingerprint	redteam/pass2/reports/stage_105.md:114	no_constraint_record	provenance_built
106	fit_stage106_coeffs_fixed_to_targets Gamma_5, K_2, K_4, N_Q	notes/stages/moving_throat_pde_stage106_canonical_outgoing_reduced_closure.md:40	published_target	audited
106	fit_stage106_odd_exactly_gr_target Gamma_5_normalized	notes/stages/moving_throat_pde_stage106_canonical_outgoing_reduced_closure.md:53	published_target	audited
106	fit_stage_106_chi_q chi_Q	paper/stages/stage_106.tex:13(3 manifest anchors)	internal_consistency	provenance_built

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
106	fit_stage_106_delta_q Delta_Q, Gamma_5, chi_Q, gamma_eff, overline	redteam/pass2/reports/ stage_106.md:108	internal_ consistency	provenance_built
106	fit_stage_106_gamma_5 Gamma_5, Gamma_5_2G_5c, K_0, K_2, K_4, chi_Q, gamma_rm, matched_fingerprint_value, overline	redteam/pass2/reports/ stage_106.md:35	published_target	audited
106	fit_stage_106_matched_ fingerprint_value matched_fingerprint_value	paper/stages/stage_106. tex:24(3 manifest anchors)	internal_ consistency	provenance_built
107	fit_stage107_chiQ_canonical_ normalization chi_Q	notes/stages/moving_ throat_pde_stage107_ general_dtn_deformation. md:22	internal_ consistency	provenance_built
107	fit_stage_107_lambda_def Lambda_def, chi_Q, matched_ fingerprint_value	redteam/pass2/reports/ stage_107.md:97(2 manifest anchors)	free_choice	audited
108	fit_stage108_beta_forced_ positive_branch beta	notes/stages/moving_ throat_pde_stage108_ robustness_classes.md:41	internal_ consistency	audited
108	fit_stage108_chiQ_robust_ claim chi_Q	notes/stages/moving_ throat_pde_stage108_ robustness_classes.md:102	internal_ consistency	provenance_built
108	fit_stage_108_chi_q chi_Q, chi_arg, matched_ fingerprint_value	redteam/pass2/reports/ stage_108.md:37(2 manifest anchors)	internal_ consistency	provenance_built
108	fit_stage_108_matched_ fingerprint_value matched_fingerprint_value, pde_ledger, stage_108, stage_ appendix_part04, toy_physics	notes/stages/moving_ throat_pde_stage108_ robustness_classes. md:105(6 manifest anchors)	internal_ consistency	provenance_built
108	fit_stage_108_y_can_literal Y_can_literal, Y_scale, matched_fingerprint_value	redteam/pass2/reports/ stage_108.md:86	internal_ consistency	provenance_built
109	fit_stage109_linearized_ condition_must_satisfy a_0, a_5, b	notes/stages/moving_ throat_pde_stage109_ linearized_branch_ selection.md:62	free_choice	audited
110	fit_stage110_chiQR_exact_ factor chi_Q_R, rho_R	notes/stages/moving_ throat_pde_stage110_ robin_outlet_model.md:52	internal_ consistency	provenance_built
110	fit_stage_110_a_0 a_0, a_5, chi_Q, rho_R	redteam/pass2/reports/ stage_110.md:134	internal_ consistency	provenance_built
110	fit_stage_110_chi_q chi_Q, rho_R	redteam/pass2/reports/ stage_110.md:51	internal_ consistency	provenance_built
110	fit_stage_110_matched_ fingerprint_value matched_fingerprint_value, pde_ledger, rho_R, stage_ appendix_part04, toy_physics	notes/stages/moving_ throat_pde_stage110_ robin_outlet_model.md:48(3 manifest anchors)	internal_ consistency	provenance_built
110	fit_stage_110_p_2 P_2	notes/stages/moving_ throat_pde_stage110_ robin_outlet_model.md:78	internal_ consistency	provenance_built
111	fit_stage111_sigmaW_forced_ to_zero kappa_W, sigma_W	notes/stages/moving_ throat_pde_stage111_ mixed_sidechannel_pole. md:55	internal_ consistency	audited
111	fit_stage_111_a_0 a_0, a_5, chi_Q, gamma_W, kappa_W, matched_fingerprint_ value, sigma_W	redteam/pass2/reports/ stage_111.md:35	internal_ consistency	provenance_built
111	fit_stage_111_a_w A_w, mu	notes/stages/moving_ throat_pde_stage111_ mixed_sidechannel_pole. md:82	free_choice	audited

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
111	fit_stage_111_chi_q chi_Q, chi_mix	redteam/pass2/reports/ stage_111.md:69	internal_ consistency	provenance_built
111	fit_stage_111_lambda_mix Lambda_mix, chi_mix, chi_mix_ lin	redteam/pass2/reports/ stage_111.md:110	free_choice	audited
111	fit_stage_111_matched_ fingerprint_value matched_fingerprint_value	notes/stages/moving_ throat_pde_stage111_ mixed_sidechannel_pole. md:5(2 manifest anchors)	internal_ consistency	provenance_built
112	fit_stage112_chiQhyb_exact_ factor chi_Q_hyb, gamma_W, sigma_W	notes/stages/moving_ throat_pde_stage112_ hybrid_robin_mixed_ compensation.md:74	internal_ consistency	provenance_built
112	fit_stage112_compensated_ branch_coefficients gamma_W, kappa_W, rho_R, sigma_W	notes/stages/moving_ throat_pde_stage112_ hybrid_robin_mixed_ compensation.md:47	internal_ consistency	provenance_built
112	fit_stage112_exactly_two_ branches kappa_W, rho_R, sigma_W	notes/stages/moving_ throat_pde_stage112_ hybrid_robin_mixed_ compensation.md:41	internal_ consistency	provenance_built
112	fit_stage_112_gamma_w gamma_W, matched_fingerprint_ value	redteam/pass2/reports/ stage_112.md:50	internal_ consistency	provenance_built
112	fit_stage_112_matched_ fingerprint_value matched_fingerprint_value	redteam/pass2/reports/ stage_112.md:39(3 manifest anchors)	internal_ consistency	provenance_built
113	fit_stage113_preserved_ exactly_when chi_Q, rho_R, sigma_W	notes/stages/moving_ throat_pde_stage113_ outlet_model_status.md:37	internal_ consistency	provenance_built
114	fit_stage114_bare_mixed_ denominator_ansatz gamma_0, kappa_0	notes/stages/moving_ throat_pde_stage114_ concrete_core_schur.md:29	free_choice	audited
114	fit_stage114_exact_core_ correction delta_Lambda_core, lambda, r_c	notes/stages/moving_ throat_pde_stage114_ concrete_core_schur.md:43	internal_ consistency	provenance_built
114	fit_stage_114_k_s K_s, g_q, g_s	redteam/pass2/reports/ stage_114.md:86	free_choice	audited
115	fit_stage115_gamma_0_pure_ scale_ansatz gamma_0	notes/stages/moving_ throat_pde_stage115_core_ balance_compensation. md:69	internal_ consistency	provenance_built
115	fit_stage115_gq_exact_two_ branch_law K_s, g_q, g_s	notes/stages/moving_ throat_pde_stage115_core_ balance_compensation. md:49	internal_ consistency	audited
115	fit_stage_115_k_q K_q, K_s, K_sK_q, K_sg_q, Lambda_rm, g_q, g_s, gamma_0, gamma_0_1_r_c_9, gamma_c, gamma_c_1_9, kappa_0, kappa_0_1_r_c_3, kappa_c, kappa_c_1_3, lambda, matched_fingerprint_value, mathfrak, pm, r_c, rho_c, rho_c_4sigma_c, sigma_c, sigma_g_s, sigma_sigma_1_z, tfrac	redteam/pass2/reports/ stage_115.md:35(2 manifest anchors)	free_choice	audited
115	fit_stage_115_k_s K_s, K_sK_q, K_sg_q, Lambda_ out, Y_target, balance_eq, g_q, g_s, lambda, matched_ fingerprint_value, r_c, rho_c, rho_c_g_s, sigma_c, sigma_c_ K_sg_q_lambda, target_delta	redteam/pass2/reports/ stage_115.md:39	free_choice	audited

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
115	fit_stage_115_matched_fingerprint_value matched_fingerprint_value	redteam/pass2/reports/stage_115.md:50(3 manifest anchors)	internal_consistency	provenance_built
115	fit_stage_115_sigma_c sigma_c	redteam/pass2/reports/stage_115.md:80	internal_consistency	provenance_built
115	fit_stage_115_y_eff Y_eff, Y_target, matched_fingerprint_value	redteam/pass2/reports/stage_115.md:61	internal_consistency	provenance_built
115	fit_stage_115_y_target Y_target, matched_fingerprint_value	redteam/pass2/reports/stage_115.md:121	internal_consistency	provenance_built
116	fit_stage116_LW_fixed_card L_W, gamma_0_bare	paper/stages/stage_116.tex:16	internal_consistency	provenance_built
116	fit_stage116_LW_fixed_note L_W	notes/stages/moving_throat_pde_stage116_dn_mixed_tube_realization.md:46	internal_consistency	provenance_built
116	fit_stage116_gamma0_compensation_backfill gamma_0	notes/stages/moving_throat_pde_stage116_dn_mixed_tube_realization.md:60	free_choice	audited
116	fit_stage116_lw_compensation_selected_length L_W	notes/stages/moving_throat_pde_stage116_dn_mixed_tube_realization.md:51	internal_consistency	provenance_built
116	fit_stage116_pure_scale_removed_exactly gamma_0_bare, r_c	notes/stages/moving_throat_pde_stage116_dn_mixed_tube_realization.md:70	free_choice	audited
116	fit_stage_116_l_w L_W, r_c	redteam/pass2/reports/stage_116.md:127	internal_consistency	provenance_built
117	fit_stage117_formulas_derived_claim L_W, gamma_0_bare, r_c	notes/stages/moving_throat_pde_stage117_outlet_core_status.md:26	internal_consistency	provenance_built
117	fit_stage_117_l_w L_W, matched_fingerprint_value, r_c	redteam/pass2/reports/stage_117.md:41	internal_consistency	provenance_built
117	fit_stage_117_matched_fingerprint_value matched_fingerprint_value	redteam/pass2/reports/stage_117.md:152	internal_consistency	provenance_built
118	fit_stage118_Kq_fixed_not_arbitrary K_q	notes/stages/moving_throat_pde_stage118_parent_core_extraction.md:149	internal_consistency	provenance_built
118	fit_stage118_frozen_n5_eos n_eos	notes/stages/moving_throat_pde_stage118_parent_core_extraction.md:18	free_choice	audited
118	fit_stage_118_a_q A_q, H_w, I_f, I_g, I_q, I_sq, K_q, K_s, L_W, T_m, Z_q, g_q, g_s	redteam/pass2/reports/stage_118.md:35	free_choice	audited
119	fit_stage119_g_fixed_exactly_once_r mathfrak_g, mathfrak_r	notes/stages/moving_throat_pde_stage119_parent_balance_family.md:52	internal_consistency	audited
119	fit_stage119_one_parameter_family_LW_fixed L_W, mathfrak_g	notes/stages/moving_throat_pde_stage119_parent_balance_family.md:162	internal_consistency	provenance_built
119	fit_stage_119_mathfrak mathfrak	notes/stages/moving_throat_pde_stage119_parent_balance_family.md:143	internal_consistency	provenance_built
120	fit_stage120_branch_exists_iff mathfrak_g, mathfrak_r	notes/stages/moving_throat_pde_stage120_core_parameter_status.md:24	internal_consistency	provenance_built

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
121	fit_stage121_aspect_ratio_37_20_reference_freeze L_over_a	notes/stages/moving_throat_pde_stage121_geometric_r_selection.md:60	free_choice	audited
121	fit_stage121_aspect_ratio_37_over_20 L_over_a	notes/stages/moving_throat_pde_stage121_geometric_r_selection.md:60	free_choice alias offit_stage121_aspect_ratio_37_20_reference_freeze	scanned; alias folded to fit_stage121_aspect_ratio_37_20_refere
121	fit_stage121_lw_equals_l_identification L_W	notes/stages/moving_throat_pde_stage121_geometric_r_selection.md:26	free_choice	audited
121	fit_stage121_r_geom_exact_branch L, a, mathfrak_r_geom	notes/stages/moving_throat_pde_stage121_geometric_r_selection.md:37	free_choice	audited
121	fit_stage121_r_not_tunable mathfrak_r	notes/stages/moving_throat_pde_stage121_geometric_r_selection.md:81	free_choice	audited
122	fit_stage122_g_nat_equal_normalized_unity g_nat	notes/stages/moving_throat_pde_stage122_mouth_source_compensation_test.md:26	free_choice	audited
122	fit_stage122_g_nat_unity_branch_ansatz g_nat	notes/stages/moving_throat_pde_stage122_mouth_source_compensation_test.md:26	free_choice	audited
122	fit_stage122_g_pm_exact_values mathfrak_g_minus_F1, mathfrak_g_plus_F1	notes/stages/moving_throat_pde_stage122_mouth_source_compensation_test.md:51	internal_consistency	provenance_built
122	fit_stage122_rF1_geometry_fixes mathfrak_r_F1	notes/stages/moving_throat_pde_stage122_mouth_source_compensation_test.md:148	free_choice	audited
122	fit_stage_122_t_m T_m, comp_def, expect_nonzero, g_nat, needs_user_resolution, r_F1	redteam/pass2/reports/stage_122.md:150	internal_consistency	provenance_built
123	fit_stage123_exact_branch_law Xi_T, Xi_v	notes/stages/moving_throat_pde_stage123_parent_normalized_branch_values.md:25	internal_consistency	provenance_built
123	fit_stage123_geometry_selected_hybridization mathfrak_r_F1	notes/stages/moving_throat_pde_stage123_parent_normalized_branch_values.md:109	free_choice	audited
123	fit_stage_123_mathfrak mathfrak, stage121_geometric_r_selection, stage_123	redteam/pass2/reports/stage_123.md:93	internal_consistency	provenance_built
124	fit_stage124_what_is_now_fixed mathfrak_g_pm_F1, mathfrak_r_F1	notes/stages/moving_throat_pde_stage124_core_branch_status.md:3	internal_consistency	provenance_built
124	fit_stage_124_g_nat g_nat, r_F1	redteam/pass2/reports/stage_124.md:64	free_choice	audited
124	fit_stage_124_r_f1 r_F1	redteam/pass2/reports/stage_124.md:110	free_choice	audited
124	fit_stage_124_stage_141 stage_141, stage_appendix_part04	redteam/pass2/reports/stage_124.md:89	no_constraint_record	provenance_built
125	fit_stage125_lower_branch_unique branch_sign, mathfrak_g_minus_F1	notes/stages/moving_throat_pde_stage125_positive_source_theorem.md:79	internal_consistency	audited

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
126	fit_stage126_g_match_pi_over_4_card mathfrak_g_match	paper/stages/stage_126.tex:16	internal_consistency	provenance_built
126	fit_stage126_not_a_coefficient_fit mathfrak_g_minus_F1, xi_star	notes/stages/moving_throat_pde_stage126_positive_source_families.md:115	internal_consistency	provenance_built
126	fit_stage126_xi_star_admixture_backsolve xi_star	notes/stages/moving_throat_pde_stage126_positive_source_families.md:101	internal_consistency	audited
126	fit_stage126_xi_star_exact_solution xi_star	notes/stages/moving_throat_pde_stage126_positive_source_families.md:93	internal_consistency	provenance_built
126	fit_stage_126_g_xi g_xi, input	redteam/pass2/reports/stage_126.md:113	internal_consistency	provenance_built
126	fit_stage_126_mathfrak mathfrak, pi	paper/stages/stage_126.tex:16(3 manifest anchors)	internal_consistency	provenance_built
127	fit_stage127_exp_depth_exact_branch x_star_exp	notes/stages/moving_throat_pde_stage127_penetration_families.md:82	internal_consistency	provenance_built
127	fit_stage127_slab_depth_unique x_star_slab	notes/stages/moving_throat_pde_stage127_penetration_families.md:37	internal_consistency	audited
127	fit_stage127_x_star_exp_backsolve x_star_exp	notes/stages/moving_throat_pde_stage127_penetration_families.md:78	internal_consistency	audited
127	fit_stage127_x_star_slab_backsolve x_star_slab	notes/stages/moving_throat_pde_stage127_penetration_families.md:40	internal_consistency	audited
128	fit_stage128_branch_sign_fixed_card branch_sign	paper/stages/stage_128.tex:16	internal_consistency	provenance_built
128	fit_stage128_lower_branch_uniquely_selected mathfrak_g_minus_F1	notes/stages/moving_throat_pde_stage128_mouth_source_bias_status.md:75	internal_consistency	audited
128	fit_stage128_rF1_fixed_by_geometry mathfrak_r_F1	notes/stages/moving_throat_pde_stage128_mouth_source_bias_status.md:12	internal_consistency	audited
128	fit_stage128_traction_gap_downstream_attribution traction_gap_percent	redteam/pass2/reports/stage_128.md:105	internal_consistency	provenance_built
129	fit_stage129_sigma_pi_derived_boundary_law sigma_Pi	notes/stages/moving_throat_pde_stage129_mouth_boundary_layer.md:7	internal_consistency	provenance_built
129	fit_stage_129_stop_cold stop_cold	redteam/pass2/reports/stage_129.md:128	no_constraint_record	provenance_built
130	fit_stage130_g_minus_compensation_target g_minus_F1, g_star	scripts/moving_throat_pde_stage130_mouth_bias_map_sympy_audit.py:10	internal_consistency	provenance_built
130	fit_stage130_pi_star_exact_unique Pi_star, g_Pi	paper/stages/stage_130.tex:16	internal_consistency	audited
130	fit_stage130_x_star_penetration_depth_match x_star	notes/stages/moving_throat_pde_stage130_mouth_bias_map.md:118	internal_consistency	provenance_built
131	fit_stage131_g_minus_closed_form_aspect_ratio L_over_a, g_minus_F1	scripts/moving_throat_pde_stage131_parent_mouth_threshold_sympy_audit.py:15	no_constraint_record	provenance_built

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
131	fit_stage131_parent_threshold_canonical A0_prime, Pi_star, T_m, Theta_sigma, q_star	paper/stages/stage_131.tex:16	free_choice	audited
131	fit_stage_131_a_0 A_0, Delta, Pi, T_m, Theta_sigma_L, pi	redteam/pass2/reports/stage_131.md:39	free_choice	audited
131	fit_stage_131_pi_star Pi_star, g_minus	redteam/pass2/reports/stage_131.md:71	internal_consistency	provenance_built
131	fit_stage_131_pi_theta_sigma_l Pi_Theta_sigma_L, T_m, g_minus, g_nat, input	redteam/pass2/reports/stage_131.md:106	internal_consistency	provenance_built
132	fit_stage132_source_family_fixed_claim sigma_Pi	notes/stages/moving_throat_pde_stage132_mouth_boundary_layer_status.md:8	internal_consistency	provenance_built
132	fit_stage_132_pi Pi	redteam/pass2/reports/stage_132.md:83	internal_consistency	provenance_built
133	fit_stage133_residual_freedom_fixed M_minus, M_plus, kappa_minus, kappa_plus	notes/stages/moving_throat_pde_stage133_coupled_mouth_fixedpoint.md:197	free_choice	audited
133	fit_stage_133_s_target S_target	redteam/pass2/reports/stage_133.md:39(2 manifest anchors)	internal_consistency	provenance_built
134	fit_stage134_bias_determined_by_fixedpoint M_q, M_s, Pi	notes/stages/moving_throat_pde_stage134_family1_mouth_fixedpoint.md:131	free_choice	audited
134	fit_stage134_kappa_q_first_mode_choice kappa_q	scripts/moving_throat_pde_stage134_family1_mouth_fixedpoint_sympy_audit.py:30	free_choice	audited
134	fit_stage134_pi_star_carried_literal Pi_star, S_q_star	scripts/moving_throat_pde_stage134_family1_mouth_fixedpoint_sympy_audit.py:39	internal_consistency	provenance_built
134	fit_stage134_sq_spotcheck_anchor_literals S_q_at_half, S_q_at_one, S_q_at_two	research/pde_ledger/scripts/moving_throat_pde_stage134_family1_mouth_fixedpoint_sympy_audit.py:61	internal_consistency	provenance_built
134	fit_stage_134_m_q M_q, M_qS_q, M_s, Pi, Pi_approx0_658075937605428, Pi_approx1_50882951349316, Pi_star, S_q, S_shell, approx, kappa_q, kappa_q_pi_2, kappa_s, kappa_s_0, mathcal, pi, tanh, tfrac	redteam/pass2/reports/stage_134.md:35(2 manifest anchors)	free_choice	audited
134	fit_stage_134_mathematica_mirror_policy MATHEMATICA_MIRROR_POLICY, MIRROR_POLICY, S_q, batch_IV4_paper_alignment	redteam/pass2/reports/stage_134.md:83	no_constraint_record	provenance_built
134	fit_stage_134_s_q S_q, stop_cold	redteam/pass2/reports/stage_134.md:114	internal_consistency	provenance_built
135	fit_stage135_exact_outlet_consistent_gain Sigma_0	notes/stages/moving_throat_pde_stage135_outlet_consistent_mouth_closure.md:83	internal_consistency	provenance_built
135	fit_stage135_outlet_gain_ratio M_q_over_Sigma_m, M_s_over_Sigma_m	scripts/moving_throat_pde_stage135_outlet_consistent_mouth_closure_sympy_audit.py:36	free_choice	audited

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
135	fit_stage135_pi_star_import_sigma_m_star Pi_star, Sigma_m_star	scripts/moving_throat_pde_stage135_outlet_consistent_mouth_closure_sympy_audit.py:46	internal_consistency	provenance_built
135	fit_stage_135_m_q M_q, M_s, S_q, Sigma_m, approx	redteam/pass2/reports/stage_135.md:35	internal_consistency	provenance_built
137	fit_stage137_gains_from_derived_core_response M_q, M_s	notes/stages/moving_throat_pde_stage137_core_to_mouth_gain_map.md:9	internal_consistency	provenance_built
137	fit_stage_137_d_w_bare D_W_bare, K_s, K_sK_q, K_sK_qD, K_sg_q, Lambda_schur, M_core, M_q, M_s, S_q, g_q, g_s, gamma_c, infty, kappa_c, kappa_c_gamma_c_r_c, lambda, neq, r_c, rho_c, rho_c_K_sg_q_lambda, rho_c_sigma_c, sigma_c, to	redteam/pass2/reports/stage_137.md:95	free_choice	audited
137	fit_stage_137_k_s K_s, K_sK_q, K_sg_q, Lg_s, M_q, M_s, Mq_paper, Ms_paper, Pi, S_q, Theta_sigma, Theta_sigma_rho_c, Theta_sigma_sigma_c, g_s, gamma_c, kappa_c, lambda, mathcal, rho_c, rho_c_g_s, rho_c_schur, rho_c_sigma_c_1_kappa_c, sigma_c, sigma_c_K_sg_q_lambda, sigma_c_schur	redteam/pass2/reports/stage_137.md:35(3 manifest anchors)	free_choice	audited
138	fit_stage138_compensation_quarter R_q_comp	scripts/moving_throat_pde_stage138_normalized_mouth_gain_family_sympy_audit.py:31	internal_consistency	provenance_built
138	fit_stage138_rq_quarter_derived_not_assumed R_q	notes/stages/moving_throat_pde_stage138_normalized_mouth_gain_family.md:106	internal_consistency	audited
138	fit_stage_138_k_s K_s, K_sK_q, M_q, M_s, Ms_norm, R_q, g_c, g_q, g_s	redteam/pass2/reports/stage_138.md:90	free_choice	audited
139	fit_stage139_aspect_ratio_literal L_over_a, r_F1	scripts/moving_throat_pde_stage139_family1_actual_mouth_gains_sympy_audit.py:6	free_choice	audited
139	fit_stage139_gains_land_exactly_on_canonical M_q_comp_star, M_s_comp_star	notes/stages/moving_throat_pde_stage139_family1_actual_mouth_gains.md:120	internal_consistency	audited
139	fit_stage_139_m_q M_q, M_s, R_q, S_q, Sq_star	redteam/pass2/reports/stage_139.md:122	internal_consistency	provenance_built
140	fit_stage140_carried_gain_magnitudes M_s_comp, M_s_nat, Sigma_0	scripts/moving_throat_pde_stage140_selfmatched_mouth_susceptibility_sympy_audit.py:18	internal_consistency	provenance_built
140	fit_stage140_sigma0_prefactor_fixed Sigma_0	notes/stages/moving_throat_pde_stage140_selfmatched_mouth_susceptibility.md:59	internal_consistency	provenance_built
140	fit_stage140_theta_sigma_not_fitted_by_hand Theta_sigma	notes/stages/moving_throat_pde_stage140_selfmatched_mouth_susceptibility.md:32	internal_consistency	provenance_built
140	fit_stage_140_j_s J_s, Sigma_0, T_m, ell, hbar, mathcal, rho	redteam/pass2/reports/stage_140.md:116	internal_consistency	provenance_built
140	fit_stage_140_m_s M_s, R_q, Sigma0_hat, Sigma_0, Sigma_0_M_s_20_9_widehat, T_m, ell, hbar, mathcal, rho	redteam/pass2/reports/stage_140.md:112	internal_consistency	provenance_built

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
140	fit_stage_140_theta_sigma Theta_sigma	notes/stages/moving_throat_pde_stage140_selfmatched_mouth_susceptibility.md:6	internal_consistency	provenance_built
141	fit_stage141_gain_pair_no_longer_free M_q, M_s	notes/stages/moving_throat_pde_stage141_mouth_gain_status.md:7	internal_consistency	audited
142	fit_stage142_three_quantities_no_longer_free R_q, T_m_hat, sigma_shape	notes/stages/moving_throat_pde_stage142_selfconsistent_mouth_branch.md:134	internal_consistency	audited
142	fit_stage_142_m_q M_q, M_s, Pi, R_q, S_q, Sigma_0, Sigma_0_1_R_q_Pi_mathcal, g_c, mathfrak, r_F1, stage_142	redteam/pass2/reports/stage_142.md:35	internal_consistency	provenance_built
142	fit_stage_142_r_q R_q, r_F1, stale_output	redteam/pass2/reports/stage_142.md:106	internal_consistency	provenance_built
143	fit_stage_143_m_q M_q, M_s	redteam/pass2/reports/stage_143.md:110	internal_consistency	provenance_built
144	fit_stage144_unique_regular_canonical_branch Pi_star	notes/stages/moving_throat_pde_stage144_unique_regular_canonical_branch.md:105	internal_consistency	audited
144	fit_stage_144_mathfrak mathfrak, pi	notes/stages/moving_throat_pde_stage144_unique_regular_canonical_branch.md:82	internal_consistency	provenance_built
144	fit_stage_144_r_q R_q, S_q, g_c, r_F1, stage_144	redteam/pass2/reports/stage_144.md:35(2 manifest anchors)	internal_consistency	provenance_built
145	fit_stage_145_m_q M_q, M_s, Pi_widehat, stage_145	redteam/pass2/reports/stage_145.md:35	internal_consistency	provenance_built
145	fit_stage_145_r_q R_q, S_q, g_c, r_F1, stage_appendix_part04	redteam/pass2/reports/stage_145.md:37	internal_consistency	provenance_built
146	fit_stage146_aspect_ratio_37_20_primitive L_over_a, r_F1	research/pde_ledger/scripts/moving_throat_pde_stage146_positive_deformation_expansion_sympy_audit.py:71	free_choice	audited
147	fit_stage147_AT_BT_circular_paper_anchor A_T, B_T	research/pde_ledger/paper/appendices/stage_appendix_part04.tex:846	internal_consistency	provenance_built
147	fit_stage147_AT_BT_paper_anchor A_T, B_T	scripts/moving_throat_pde_stage147_first_order_rigidity_kernel_sympy_audit.py:53	internal_consistency	provenance_built
147	fit_stage147_AT_BT_ratio_31_6785_mislabel_history abs_A_T_over_B_T	research/pde_ledger/notes/stages/moving_throat_pde_stage147_first_order_rigidity_kernel.md:77	internal_consistency	provenance_built
147	fit_stage_147_a_t A_T, B_T, K_q, Pi, Sigma, Sigma_T, Sigma_W_dx_0, Sigma_dx, T_m, bundle_inversion_four_drifts, codex_reviews, core_mouth_coevolution_status, dT_m, full_profile_residual, int, int_0, is_checkpoint, pi, representative_positive_families, stage_appendix_part04, varsigma	redteam/pass2/reports/stage_147.md:48(2 manifest anchors)	internal_consistency	provenance_built

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
148	fit_stage148_canonical_first_order_shifts_exact delta_Pi_lambda, delta_T_m_lambda	notes/stages/moving_throat_pde_stage148_representative_positive_families.md:96	internal_consistency	audited
148	fit_stage148_radical_constant_forced_by_rf1 r_F1, xi_star	scripts/moving_throat_pde_stage148_representative_positive_families_sympy_audit.py:91	internal_consistency	audited
148	fit_stage_148_a_t A_T, B_T, Pi_star, S_star, T_star, r_F1, stage_appendix_part04	redteam/pass2/reports/stage_148.md:70(3 manifest anchors)	internal_consistency	provenance_built
150	fit_stage150_residual_tangent_matched Pi_star, R_star	paper/stages/stage_150.tex:16	internal_consistency	provenance_built
150	fit_stage_150_phi_x_pi_x Phi_x_Pi_x	paper/stages/stage_150.tex:16	internal_consistency	provenance_built
150	fit_stage_150_s_q S_q, T_q	redteam/pass2/reports/stage_150.md:114	internal_consistency	provenance_built
151	fit_stage151_correction_fixed_by_covariances delta_Pi_act, delta_S_act	paper/stages/stage_151.tex:16	internal_consistency	audited
151	fit_stage_151_a_t A_T, B_T, Pi_star, r_1, r_2	redteam/pass2/reports/stage_151.md:103	internal_consistency	provenance_built
152	fit_stage152_affine_law_endpoints lambda_eff_Pi, lambda_eff_T	scripts/moving_throat_pde_stage152_family1_actual_correction_sympy_audit.py:112	internal_consistency	provenance_built
152	fit_stage152_carried_point_block A_T, B_T, Pi_star, S_star, Sigma_m_star, T_star, g_star, gprime_star	scripts/moving_throat_pde_stage152_family1_actual_correction_sympy_audit.py:28	internal_consistency	provenance_built
152	fit_stage152_corrected_mouth_state_pi_corr_t_corr Pi_corr, T_hat_m_corr, delta_S_act, delta_g_act	research/pde_ledger/notes/stages/moving_throat_pde_stage152_family1_actual_correction.md:78	internal_consistency	provenance_built
152	fit_stage152_profile_correction_samples_expected_pins cov_KR_expected, cov_cR_expected, deltaPi_expected, deltaT_expected	research/pde_ledger/scripts/numerical/stage150_152_profile_correction_samples.json:21	internal_consistency	provenance_built
152	fit_stage152_retuned_canonical_point A_T, B_T, delta_Pi_act, g_star_prime	notes/stages/moving_throat_pde_stage152_family1_actual_correction.md:48	internal_consistency	provenance_built
152	fit_stage_152_a_t A_T, B_T, Cov_cR, K_q, S_1, T_1, T_corr, T_q, T_s, expect_close, g_1, g_star	redteam/pass2/reports/stage_152.md:39	internal_consistency	provenance_built
154	fit_stage154_coevolving_branch_determined Sigma_0, Sigma_profile	notes/stages/moving_throat_pde_stage154_coevolving_core_mouth_map.md:170	internal_consistency	provenance_built
154	fit_stage_154_g_star g_star	redteam/pass2/reports/stage_154.md:85	internal_consistency	provenance_built
154	fit_stage_154_r_f1 r_F1	redteam/pass2/reports/stage_154.md:39(2 manifest anchors)	internal_consistency	provenance_built
154	fit_stage_154_t_q T_q, T_s, r_F1, stage_154	redteam/pass2/reports/stage_154.md:35	internal_consistency	provenance_built

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
155	fit_stage155_carried_constants_block Pi_star, S_star, Sigma_0_star, g_star, r_F1	scripts/moving_throat_pde_stage155_frozen_traction_fixedpoint_sympy_audit.py:25	internal_consistency	provenance_built
155	fit_stage155_fixedpoint_expected_targets_json Pi_can_expected, S_can_expected, Sigma0_can_expected, T_hat_can_expected	research/pde_ledger/scripts/numerical/stage155_156_fixedpoint_samples.json:16	internal_consistency	provenance_built
155	fit_stage155_frozen_traction_fixed_point_pins R_fp, S_fp, g_fp	research/pde_ledger/scripts/moving_throat_pde_stage155_frozen_traction_fixedpoint_sympy_audit.py:94	internal_consistency	provenance_built
155	fit_stage_155_g_star g_star	redteam/pass2/reports/stage_155.md:162	internal_consistency	provenance_built
156	fit_stage156_carried_constants_block Pi_star, Sigma_0_star, T_hat_star, g_star, r_F1	scripts/moving_throat_pde_stage156_renormalized_canonical_branch_sympy_audit.py:28	free_choice + internal_consistency	audited
156	fit_stage156_mouth_bias_increase_224_59_pct mouth_bias_increase_pct	research/pde_ledger/notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:136	internal_consistency	provenance_built
156	fit_stage156_unique_traction_renormalization Sigma_0_can, T_m_can	paper/stages/stage_156.tex:16	internal_consistency	audited
156	fit_stage_156_pi_can Pi_can, R_can, S_can, Sigma0_can, Sigma_0, StatusNumerical, T_hat, T_hat_m, g_can, g_fp	redteam/pass2/reports/stage_156.md:35	internal_consistency	provenance_built
156	fit_stage_156_statusnumerical StatusNumerical, T_hat, stale_output	redteam/pass2/reports/stage_156.md:105	internal_consistency	provenance_built
157	fit_stage157_expected_values_sidecar_json Pi_can_expected, S_can_expected, Sigma_0_can_expected, T_hat_can_expected	scripts/moving_throat_pde_stage157_core_mouth_coevolution_status_sympy_audit.py:49	internal_consistency	provenance_built
157	fit_stage_157_delta_kappa_w delta_kappa_W	redteam/pass2/reports/stage_157.md:59(2 manifest anchors)	internal_consistency	provenance_built
157	fit_stage_157_statusopen StatusOpen	redteam/pass2/reports/stage_157.md:170	internal_consistency	provenance_built
158	fit_stage158_canonical_quartet_literals S_can, Sigma_0_can, T_can, r_F1	scripts/moving_throat_pde_stage158_linear_defect_transport_sympy_audit.py:100	free_choice + internal_consistency	audited
158	fit_stage158_linear_defect_transport_coefficients slippage_coefficient_eps_perp, transport_coefficient_dR_dg	research/pde_ledger/notes/stages/moving_throat_pde_stage158_linear_defect_transport.md:103	internal_consistency	provenance_built
158	fit_stage_158_delta_q Delta_Q, M_q, Pi, Sigma_0, Sigma_0_R_dS_S_dR, Sigma_0_dS_dg, Sigma_0_dSigma_0_1_R_dR_S_dS, Sigma_0_dSigma_0_R_dR, Sigma_0_dg, Sigma_5, Sigma_5_3S_Sigma_0, a_0, chi, eps	redteam/pass2/reports/stage_158.md:40	internal_consistency	provenance_built
158	fit_stage_158_matched_fingerprint_value matched_fingerprint_value	notes/stages/moving_throat_pde_stage158_linear_defect_transport.md:410	internal_consistency	provenance_built

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
159	fit_stage159_canonical_even_baseline gamma_W, kappa_W	scripts/moving_throat_pde_stage159_hybrid_outlet_projection_sympy_audit.py:50	internal_consistency	provenance_built
159	fit_stage159_notes_only_numeric_illustrations Delta_Q_dg_coeff, delta_C_dg_coeff	research/pde_ledger/notes/stages/moving_throat_pde_stage159_hybrid_outlet_projection.md:172	internal_consistency	provenance_built
159	fit_stage_159_matched_fingerprint_value matched_fingerprint_value	notes/stages/moving_throat_pde_stage159_hybrid_outlet_projection.md:15(4 manifest anchors)	internal_consistency	provenance_built
160	fit_stage160_canonical_branch_characterized_kappa_gamma gamma_0, kappa_0, r_c	notes/stages/moving_throat_pde_stage160_bare_mixed_port_slippage.md:73	internal_consistency	provenance_built
160	fit_stage160_notes_only_slippage_numerics Delta_Q_dS_coeff, Delta_Q_dSigma0_coeff, Delta_Q_dThat_coeff, delta_BW_slip_coeff	research/pde_ledger/notes/stages/moving_throat_pde_stage160_bare_mixed_port_slippage.md:187	internal_consistency	provenance_built
160	fit_stage160_tangent_susceptibility_coefficients dPi_tan_coeff_S, dPi_tan_coeff_Sigma0	scripts/moving_throat_pde_stage160_bare_mixed_port_slippage_sympy_audit.py:66	internal_consistency	provenance_built
160	fit_stage_160_matched_fingerprint_value matched_fingerprint_value	notes/stages/moving_throat_pde_stage160_bare_mixed_port_slippage.md:140	internal_consistency	provenance_built
161	fit_stage161_delta_q_sigma_star_closure Xi_slip, sigma_star	paper/stages/stage_161.tex:16	internal_consistency	provenance_built
161	fit_stage161_dsigma0_dthat_coefficient dSigma0_per_dThat	research/pde_ledger/scripts/moving_throat_pde_stage161_dn_similarity_slippage_sympy_audit.py:122	internal_consistency	provenance_built
161	fit_stage161_dsigma0_dthat_conversion_ratio dSigma0_over_dThat	research/pde_ledger/scripts/moving_throat_pde_stage161_dn_similarity_slippage_sympy_audit.py:122	internal_consistency	provenance_built
161	fit_stage161_gamma0_star_branch_value gamma_0, kappa_0	research/pde_ledger/notes/stages/moving_throat_pde_stage161_dn_similarity_slippage.md:65	internal_consistency	provenance_built
161	fit_stage161_rF1_family1_radius r_F1	research/pde_ledger/scripts/moving_throat_pde_stage161_dn_similarity_slippage_sympy_audit.py:135	free_choice	audited
161	fit_stage161_stage159_tangential_transport_coeffs dPi_tan_coeff_dS, dPi_tan_coeff_dSigma0	research/pde_ledger/scripts/moving_throat_pde_stage161_dn_similarity_slippage_sympy_audit.py:117	internal_consistency	provenance_built
161	fit_stage161_upsilon_pi_one_ninth_coefficient Upsilon_Pi, r_c_star	notes/stages/moving_throat_pde_stage161_dn_similarity_slippage.md:25	internal_consistency	provenance_built

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
162	fit_stage162_gamma0_one_ninth_r_form gamma_0, r_frak	notes/stages/moving_throat_pde_stage162_parent_compensation_rigidity.md:61	free_choice	verdict_logged: PARTIAL non-fatal (CONVERGED Claude+Codex): card-badge/notes-wording over-scope, punch-list A8 (= A3 class), CONTINGENT on user ANSATZ_LEDGER 5-1 gate; NOT FIND_STANDS, no user gate
162	fit_stage162_gamma0_similarity_law gamma_0	notes/stages/moving_throat_pde_stage162_parent_compensation_rigidity.md:15	internal_consistency	provenance_built
162	fit_stage162_rF1_family1_radius r_F1	research/pde_ledger/scripts/moving_throat_pde_stage162_parent_compensation_rigidity_sympy_audit.py:71	free_choice	audited
162	fit_stage162_r_f1_family1_value r_F1	notes/stages/moving_throat_pde_stage162_parent_compensation_rigidity.md:155	free_choice	audited
162	fit_stage_162_g_lower g_lower	redteam/pass2/reports/stage_162.md:99	internal_consistency	provenance_built
163	fit_stage163_Rstar_quarter R_star	research/pde_ledger/scripts/moving_throat_pde_stage163_off_family_normal_coordinate_sympy_audit.py:112	internal_consistency	audited
163	fit_stage163_delta_perp_exact_parent_expression delta_perp, g_minus_prime_r_star	notes/stages/moving_throat_pde_stage163_off_family_normal_coordinate.md:21	internal_consistency	provenance_built
163	fit_stage163_family1_canonical_point_block S_can, Sigma0_can, g_F1, r_F1	research/pde_ledger/scripts/moving_throat_pde_stage163_off_family_normal_coordinate_sympy_audit.py:123	free_choice + internal_consistency	audited
163	fit_stage163_family1_readback_misattribution S_can, Sigma0_can, g_star, r_star	research/pde_ledger/scripts/moving_throat_pde_stage163_off_family_normal_coordinate_sympy_audit.py:18	internal_consistency	provenance_built
163	fit_stage163_sigma0_can_digit_variant Sigma0_can	research/pde_ledger/scripts/moving_throat_pde_stage163_off_family_normal_coordinate_sympy_audit.py:125	internal_consistency	provenance_built
164	fit_stage164_family1_point_injection g_star, r_star	research/pde_ledger/scripts/moving_throat_pde_stage164_microscopic_log_channels_sympy_audit.py:141	internal_consistency	audited
164	fit_stage164_log_channel_prefactors_27_40_27_320 first_log_channel_prefactor, second_log_channel_prefactor	research/pde_ledger/scripts/moving_throat_pde_stage164_microscopic_log_channels_sympy_audit.py:98	internal_consistency	provenance_built
164	fit_stage164_quarter_sqrt_weight delta_perp_channel_weight, r_star	notes/stages/moving_throat_pde_stage164_microscopic_log_channels.md:12	internal_consistency	provenance_built

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
164	fit_stage_164_l_w L_W, c_sw, v_w0	redteam/pass2/reports/stage_164.md:39	internal_consistency	provenance_built
165	fit_stage165_frozen_n5_wall_eos n_wall_eos	notes/stages/moving_throat_pde_stage165_exact_branch_drifts.md:27	free_choice	audited
165	fit_stage165_gstar_literal g_star	research/pde_ledger/scripts/moving_throat_pde_stage165_exact_branch_drifts_sympy_audit.py:110	internal_consistency	provenance_built
165	fit_stage165_pinned_parent_ratios g_frak, r_frak	notes/stages/moving_throat_pde_stage165_exact_branch_drifts.md:44	internal_consistency	audited
165	fit_stage165_rstar_closed_form_4107 r_star	research/pde_ledger/scripts/moving_throat_pde_stage165_exact_branch_drifts_sympy_audit.py:109	free_choice	audited
165	fit_stage_165_l_w L_W, T_m, Z_q, mathcal, rho_w, rho_w_deltaln, v_w0	redteam/pass2/reports/stage_165.md:35	internal_consistency	provenance_built
165	fit_stage_165_t_m_pref T_m_pref, expect_close, prod_pref, ratio_pref, v_pref	redteam/pass2/reports/stage_165.md:125	internal_consistency	provenance_built
166	fit_stage166_four_observables_determine_drifts K_q, K_s, P_0, Theta_w	notes/stages/moving_throat_pde_stage166_bundle_inversion_four_drifts.md:16	internal_consistency	provenance_built
166	fit_stage166_theta_chi_wall_datum Theta_w_chi, rho_w_chi	research/pde_ledger/scripts/moving_throat_pde_stage166_bundle_inversion_four_drifts_sympy_audit.py:94	internal_consistency	provenance_built
166	fit_stage166_theta_chi_wall_temperature Theta_chi, rho_w_chi	research/pde_ledger/scripts/moving_throat_pde_stage166_bundle_inversion_four_drifts_sympy_audit.py:94	internal_consistency	provenance_built
166	fit_stage_166_k_q K_q, K_s, P_0, Theta_w, Theta_w_K_s_K_q_P_0	paper/stages/stage_166.tex:15	internal_consistency	provenance_built
167	fit_stage167_parent_ratios_first_order_invariants g_frak, r_c, r_frak	notes/stages/moving_throat_pde_stage167_bundle_transport_tangent_compensation.md:26	internal_consistency	provenance_built
168	fit_stage168_canonical_mouth_block S_can, Sigma0_can, Sigma0_num, g_star	research/pde_ledger/scripts/moving_throat_pde_stage168_off_bundle_slippage_sympy_audit.py:119	internal_consistency	provenance_built
168	fit_stage168_eps_perp_exact_weights eps_perp, g_star, r_star	notes/stages/moving_throat_pde_stage168_off_bundle_slippage.md:171	internal_consistency	provenance_built
168	fit_stage168_rexact_closed_form_4107 r_F1	research/pde_ledger/mathematica/moving_throat_pde_stage168_off_bundle_slippage_mathematica_audit.wl:105	free_choice	audited
168	fit_stage168_sigma0_num_digit_variant Sigma0_num	research/pde_ledger/scripts/moving_throat_pde_stage168_off_bundle_slippage_sympy_audit.py:119	internal_consistency alias of fit_stage168_canonical_mouth_block	scanned; alias folded to fit_stage168_canonical_mouth_block
168	fit_stage_168_matched_fingerprint_value matched_fingerprint_value	notes/stages/moving_throat_pde_stage168_off_bundle_slippage.md:374	internal_consistency	provenance_built

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
168	fit_stage_168_s_can S_can, eps_perp	redteam/pass2/reports/ stage_168.md:132	internal_ consistency	provenance_built
169	fit_stage169_family1_point_ injection g_star, r_star	research/pde_ledger/ scripts/moving_throat_ pde_stage169_no_linear_ p2_scalar_slippage_sympy_ audit.py:127	no_constraint_ record	provenance_built
169	fit_stage169_seven_tenths_ invariant A_x_squared_coefficient, b_x_over_a_x	notes/stages/moving_ throat_pde_stage169_ no_linear_p2_scalar_ slippage.md:93	internal_ consistency	provenance_built
169	fit_stage169_xi_perp_paper_ targets Xi_perp_coeff_XiL, Xi_perp_ coeff_Xiv	research/pde_ledger/ scripts/moving_throat_ pde_stage169_no_linear_ p2_scalar_slippage_sympy_ audit.py:138	internal_ consistency	provenance_built
169	fit_stage_169_a_x a_x, b_x, r_num	redteam/pass2/reports/ stage_169.md:128	internal_ consistency	provenance_built
169	fit_stage_169_engine_ disagreement engine_disagreement	redteam/pass2/reports/ stage_169.md:124	no_constraint_ record	provenance_built
169	fit_stage_169_tautological_ check tautological_check	redteam/pass2/reports/ stage_169.md:66	no_constraint_ record	provenance_built
170	fit_stage170_canonical_ isotropic_loading_u2_u4 D2_over_D0, D4_over_D0, u2, u4	research/pde_ledger/ scripts/moving_throat_ pde_stage170_linear_ grouped_outlet_map_sympy_ audit.py:39	internal_ consistency	provenance_built
170	fit_stage170_canonical_u2_u4 u_2, u_4	notes/stages/moving_ throat_pde_stage170_ linear_grouped_outlet_ map.md:90	internal_ consistency	provenance_built
170	fit_stage170_k_a_one_ninth_ combination K_A_weight	notes/stages/moving_ throat_pde_stage170_ linear_grouped_outlet_ map.md:25	internal_ consistency	provenance_built
171	fit_stage171_outlet_feed_ coefficients delta_gamma_W_coefficient, delta_kappa_W_coefficient, sigma_star	notes/stages/moving_ throat_pde_stage171_ microscopic_grouped_ obstructions.md:16	internal_ consistency	provenance_built
172	fit_stage172_canonical_ branch_collapse G1_frak, K1_frak, u_2, u_4	notes/stages/moving_ throat_pde_stage172_ physical_slope_collapse. md:238	internal_ consistency	provenance_built
172	fit_stage172_canonical_even_ collapse_u2_pin u2	research/pde_ledger/ scripts/moving_throat_ pde_stage172_physical_ slope_collapse_sympy_ audit.py:71	internal_ consistency	provenance_built
172	fit_stage172_wl_canonical_ banner u_2	mathematica/moving_ throat_pde_stage172_ physical_slope_collapse_ mathematica_audit.wl:61	internal_ consistency	provenance_built
172	fit_stage_172_matched_ fingerprint_value matched_fingerprint_value	notes/stages/moving_ throat_pde_stage172_ physical_slope_collapse. md:396	internal_ consistency	provenance_built
173	fit_stage173_canonical_ loading_substitution D2_over_D0, D4_over_D0	research/pde_ledger/ scripts/moving_ throat_pde_stage173_ axisymmetric_loading_ mismatch_sympy_audit. py:60	internal_ consistency	provenance_built

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
173	fit_stage173_slopes_fixed_by_d01 D_01, u_2_slope, u_4_slope	notes/stages/moving_throat_pde_stage173_axisymmetric_loading_mismatch.md:226	free_choice	audited
173	fit_stage173_stage024_signature_triple lambda_20, lambda_21, lambda_22	notes/stages/moving_throat_pde_stage173_axisymmetric_loading_mismatch.md:23	internal_consistency	provenance_built
173	fit_stage_173_xi_load Xi_load, u21_zero_D21	redteam/pass2/reports/stage_173.md:176	internal_consistency	provenance_built
174	fit_stage174_self_similarity_weights omega_B, omega_Z	notes/stages/moving_throat_pde_stage174_static_self_similarity.md:29	internal_consistency	provenance_built
175	fit_stage175_wall_normalized_factorization B_0_alpha, K_wall_baseline, Z_0_r	notes/stages/moving_throat_pde_stage175_wall_normalized_load_shape.md:29	internal_consistency	provenance_built
175	fit_stage_175_sigma_n Sigma_N, subs_hat	redteam/pass2/reports/stage_175.md:110	internal_consistency	provenance_built
176	fit_stage176_square_root_mixed_leg_exponent G_W, K_wall_baseline, Omega_W, mixed_leg_exponent_half	notes/stages/moving_throat_pde_stage176_outgoing_load_factorization.md:21	internal_consistency	provenance_built
177	fit_stage177_xi1_exact_collapse P_0, P_1, Xi_1	notes/stages/moving_throat_pde_stage177_weak_axisymmetric_outgoing_slippage.md:353	internal_consistency	provenance_built
177	fit_stage177_xi1_prefactor_slope_no_derives_edge P_0, P_1, Xi_1, lambda_A	research/pde_ledger/notes/stages/moving_throat_pde_stage177_weak_axisymmetric_outgoing_slippage.md:67	internal_consistency	provenance_built
177	fit_stage_177_h_r H_r, I_r, M_r, P_0, P_1, Sigma_A_r, Xi, Xi_1, Xi_1Rightarrow, Xi_1_P_1_P_0, Xi_rm, bar, epsilon, h_r, i_r, kappa_1, lambda_20, lambda_21, lambda_22, lambda_A, lambda_Asigma_r, m_r, mathcal, mathfrak, r_r, sigma_r, sigma_r_2mathfrak, tfrac	redteam/pass2/reports/stage_177.md:35	internal_consistency	provenance_built
177	fit_stage_177_p_0 P_0, P_1, Xi_1, Xi_1_Xi_rm, Xi_rm, lambda_20_21_22, sigma_r, tfrac	redteam/pass2/reports/stage_177.md:94	no_constraint_record	provenance_built
177	fit_stage_177_xi_1 Xi_1	redteam/pass2/reports/stage_177.md:73	internal_consistency	provenance_built
178	fit_stage178_doubled_slope_factors kappa_1, nu_r_factor_2	notes/stages/moving_throat_pde_stage178_outgoing_port_coloading_theorem.md:97	free_choice	audited
178	fit_stage_178_d_from_series d_from_series, nu_bar, nu_from_series, p_from_series	redteam/pass2/reports/stage_178.md:122	internal_consistency	provenance_built
178	fit_stage_178_d_r d_r, nu_r, p_r, sigma_r	redteam/pass2/reports/stage_178.md:88	free_choice	audited
178	fit_stage_178_delta Delta, epsilon, neq, nu_r, rho	redteam/pass2/reports/stage_178.md:96	internal_consistency	provenance_built
179	fit_stage179_xi1_factor_two_transfer_slopes Xi_1_prefactor_2, rho_r_N, tau_r	notes/stages/moving_throat_pde_stage179_transfer_shape_theorem.md:43	internal_consistency	provenance_built
179	fit_stage_179_nu_r nu_r, rho_r, sum_r, tau_r	redteam/pass2/reports/stage_179.md:104	internal_consistency	provenance_built

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
180	fit_stage180_beta0_continuum_import K_0, Z_W, beta_0, epsilon_W, rho	notes/stages/moving_throat_pde_stage180_effective_transfer_shape_collapse.md:21	internal_consistency	provenance_built
180	fit_stage_180_pi_star Pi_star, gamma_0, pi	redteam/pass2/reports/stage_180.md:132	no_constraint_record	provenance_built
181	fit_stage181_eps_split_2_11_coefficient eps_split_coefficient	research/pde_ledger/scripts/moving_throat_pde_stage181_coherent_tracking_defect_sympy_audit.py:41	internal_consistency	provenance_built
181	fit_stage181_lambda_prefactor_27pi2_20 Lambda_prefactor	research/pde_ledger/scripts/moving_throat_pde_stage181_coherent_tracking_defect_sympy_audit.py:40	internal_consistency	provenance_built
181	fit_stage181_split_blocking_two_elevenths eps_W_split_coefficient	research/pde_ledger/notes/stages/moving_throat_pde_stage181_coherent_tracking_defect.md:29	internal_consistency	provenance_built
181	fit_stage181_transfer_shape_27pi2_over_20 R_target, T_A_squared_prefactor	notes/stages/moving_throat_pde_stage181_coherent_tracking_defect.md:14	internal_consistency	provenance_built
181	fit_stage181_two_elevenths_split_coefficient delta_U, epsilon_W_split_coefficient_2_11	notes/stages/moving_throat_pde_stage181_coherent_tracking_defect.md:29	internal_consistency	provenance_built
182	fit_stage182_eps_split_2_11_coefficient eps_split_coefficient	research/pde_ledger/scripts/moving_throat_pde_stage182_microscopic_coherent_slippage_sympy_audit.py:52	internal_consistency	provenance_built
182	fit_stage182_two_elevenths_carried_forward epsilon_W_split_coefficient_2_11	notes/stages/moving_throat_pde_stage182_microscopic_coherent_slippage.md:14	internal_consistency	provenance_built
182	fit_stage_182_sigma_z Sigma_Z, Sigma_Z_1, Sigma_chi, Sigma_delta, Sigma_epsilon, Sigma_eta_0, delta_U, epsilon, epsilon_W, value_mismatch	redteam/pass2/reports/stage_182.md:82	internal_consistency	provenance_built
183	fit_stage183_canonical_sigma_nt Sigma_nt	mathematica/moving_throat_pde_stage183_triangular_normal_form_mathematica_audit.wl:152	internal_consistency	provenance_built
183	fit_stage183_eps_split_2_11_coefficient eps_split_coefficient	research/pde_ledger/scripts/moving_throat_pde_stage183_triangular_normal_form_sympy_audit.py:46	internal_consistency	provenance_built
184	fit_stage184_b_star_exponent B_star, chi_0_star, delta_U_star	notes/stages/moving_throat_pde_stage184_branch_invariant_coordinates.md:36	internal_consistency	provenance_built
184	fit_stage184_eps_split_2_11_coefficient eps_split_coefficient	research/pde_ledger/notes/stages/moving_throat_pde_stage184_branch_invariant_coordinates.md:94	internal_consistency	provenance_built
184	fit_stage_184_eps_eta_var eps_eta_var	redteam/pass2/reports/stage_184.md:188	internal_consistency	provenance_built

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
184	fit_stage_184_lambda_0 Lambda_0, Lambda_0_1_epsilon_eta, Sigma_rm, eps_eta_var, epsilon_eta_Sigma_eta, mathcal, mathfrak	redteam/pass2/reports/stage_184.md:39	internal_consistency	provenance_built
185	fit_stage185_frozen_reference_coefficients_E_star_F_star E_star, F_star, chi_0_star, delta_U_star, epsilon_W_star, epsilon_star	research/pde_ledger/notes/stages/moving_throat_pde_stage185_microscopic_monomials.md:225	internal_consistency	provenance_built
185	fit_stage185_monomial_exponents C_tr_star_exponents, chi_0_star, delta_U_star	notes/stages/moving_throat_pde_stage185_microscopic_monomials.md:35	internal_consistency	provenance_built
185	fit_stage_185_a_tr_star A_tr_star, C_tr_star, F_star, Sigma_tr_compiled, Theta_1, engines_agree, outputs_fresh	redteam/pass2/reports/stage_185.md:92(2 manifest anchors)	internal_consistency	provenance_built
186	fit_stage186_not_fine_tuning_claim Sigma_eta, Sigma_nt, Sigma_tr	notes/stages/moving_throat_pde_stage186_similarity_orbit_closure.md:393	internal_consistency	provenance_built
186	fit_stage186_rank3_codimension_status_claim chi_0_star, rank_M_star	research/pde_ledger/notes/stages/moving_throat_pde_stage186_similarity_orbit_closure.md:194	internal_consistency	provenance_built
186	fit_stage186_wl_matches_reference_matrix M_star_matrix	mathematica/moving_throat_pde_stage186_similarity_orbit_closure_mathematica_audit.wl:83	internal_consistency	provenance_built
186	fit_stage_186_mathcal mathcal	notes/stages/moving_throat_pde_stage186_similarity_orbit_closure.md:442	internal_consistency	provenance_built
187	fit_stage187_exact_level_set_orbit_identity C_nt_star, C_tr_star, epsilon_eta	notes/stages/moving_throat_pde_stage187_orbit_quotient_closure.md:33	internal_consistency	audited
187	fit_stage187_g2_common_tangent_lambda_1 Lambda_1	notes/g2/g2_full_output.md:757	free_choice	audited
187	fit_stage_187_chi_0 chi_0	redteam/pass2/reports/stage_187.md:119	internal_consistency	provenance_built
187	fit_stage_187_ctr_ratio Ctr_ratio, chi_0, delta_U, expand_log, row_tr	redteam/pass2/reports/stage_187.md:115	internal_consistency	provenance_built
189	fit_stage189_eps_split_2_11_coefficient eps_split_coefficient	research/pde_ledger/scripts/moving_throat_pde_stage189_transfer_shape_prefactor_compiler_sympy_audit.py:110	internal_consistency	provenance_built
189	fit_stage189_lambda0_prefactor_27_20 Lambda_0	research/pde_ledger/scripts/moving_throat_pde_stage189_transfer_shape_prefactor_compiler_sympy_audit.py:98	internal_consistency	provenance_built
189	fit_stage189_mhat0_quadrupole_normalization_target Gamma_5, P_0, mhat_0	research/pde_ledger/notes/stages/moving_throat_pde_stage189_transfer_shape_prefactor_compiler.md:84	published_target	audited

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
189	`fit_stage189_normalization_target_2g_5c5` `Gamma_5_coefficient_1_27`, `m_hat0_squared_Gamma_5`, `normalization_54_5`	`notes/stages/moving_throat_pde_stage189_transfer_shape_prefactor_compiler.md`:515	`published_target`	audited
189	`fit_stage_189_k_bl` `K_bl`, `matched_fingerprint_value`	`redteam/pass2/reports/stage_189.md`:39	`internal_consistency`	provenance_built
189	`fit_stage_189_matched_fingerprint_value` `matched_fingerprint_value`	`notes/stages/moving_throat_pde_stage189_transfer_shape_prefactor_compiler.md`:79(6 manifest anchors)	`internal_consistency`	provenance_built
189	`fit_stage_189_p_0` `P_0`, `P_2`, `P_4`	`notes/stages/moving_throat_pde_stage189_transfer_shape_prefactor_compiler.md`:502	`published_target`	audited
190	`fit_stage_190_a_tr` `A_tr`, `C_tr`, `M_mix`, `M_supp`, `M_tr`, `R_target`, `b_x`, `stage_190`, `stage_appendix_part05`	`redteam/pass2/reports/stage_190.md`:35(2 manifest anchors)	`internal_consistency`	provenance_built
190	`fit_stage_190_b_x` `b_x`	`redteam/pass2/reports/stage_190.md`:146	`internal_consistency`	provenance_built
190	`fit_stage_190_m_mix` `M_mix`, `M_supp`	`redteam/pass2/reports/stage_190.md`:130	`internal_consistency`	provenance_built
191	`fit_stage191_carried_coefficients_in_packet_interconversion` `B_star`, `F_star`, `chi_0_star`	`research/pde_ledger/notes/stages/moving_throat_pde_stage191_minimal_pde_data_packet.md`:253	`free_choice+internal_consistency`	audited
191	`fit_stage191_fixed_by_previous_stages` `Delta_branch`, `Delta_orbit`	`notes/stages/moving_throat_pde_stage191_minimal_pde_data_packet.md`:341	`internal_consistency`	audited
191	`fit_stage_191_delta_branch` `Delta_branch`, `Delta_norm`, `Delta_pole`, `P0_target`	`redteam/pass2/reports/stage_191.md`:103	`internal_consistency`	provenance_built
191	`fit_stage_191_pi_star` `Pi_star`	`redteam/pass2/reports/stage_191.md`:115	`no_constraint_record`	provenance_built
192	`fit_stage192_canonical_quotient_section` `K_eta_eff`, `T_U`, `mu_W`	`notes/stages/moving_throat_pde_stage192_orbit_quotient_projectors.md`:176	`internal_consistency`	provenance_built
192	`fit_stage192_chi_q_unit_relations` `Delta_Q`, `N_Q`, `chi_Q`	`paper/stages/stage_192.tex`:21	`internal_consistency`	verdict_logged: batch_c1 YES paper_card_overclaim -> adjudication PARTIAL (non-fatal, Step-6 card fix); verdicts/batch_c1_adjudication.md
192	`fit_stage_192_delta_q` `Delta_Q`, `chi_Q`, `matched_fingerprint_value`	`paper/stages/stage_192.tex`:13	`internal_consistency`	audited
192	`fit_stage_192_i_3` `I_3`, `stale_output`	`redteam/pass2/reports/stage_192.md`:122	`internal_consistency`	provenance_built
192	`fit_stage_192_p_2` `P_2`, `matched_fingerprint_value`	`paper/stages/stage_192.tex`:9	`free_choice`	audited
192	`fit_stage_192_q_eta` `q_eta`, `q_nt`, `q_tr`	`redteam/pass2/reports/stage_192.md`:150	`internal_consistency`	provenance_built
193	`fit_stage193_grouped_p2_target_surface_coefficients` `Omega_Q`, `Y_asymptote_3_4`, `Y_pole_weight_1_4`	`scripts/moving_throat_pde_stage193_isotropic_grouped_p2_target_surface_sympy_audit.py`:112	`internal_consistency`	provenance_built

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
193	fit_stage193_one_pole_carrier_forced Delta_pole, Omega_Q_sq, a_2, a_4, b_2, b_4	notes/stages/moving_throat_pde_stage193_isotropic_grouped_p2_target_surface.md:217	internal_consistency	audited
193	fit_stage193_p0_target_surface_frozen_without_derivation_edge P0_target, mhat_0	research/pde_ledger/paper/stages/stage_193.tex:15	published_target	audited
193	fit_stage_193_delta_q Delta_Q, chi_Q, matched_fingerprint_value	paper/stages/stage_193.tex:13	internal_consistency	audited
193	fit_stage_193_p_2 P_2, matched_fingerprint_value	paper/stages/stage_193.tex:9	free_choice	audited
194	fit_stage194_canonical_even_moment_ratios L2_over_L0_1_9, L4_combination_4_81, odd_scale_1_27	scripts/moving_throat_pde_stage194_outgoing_l2_fingerprint_and_deformation_algebra_sympy_audit.py:158	internal_consistency	provenance_built
194	fit_stage194_chi_Q_canonical_fixing chi_Q	notes/stages/moving_throat_pde_stage194_outgoing_l2_fingerprint_and_deformation_algebra.md:184	internal_consistency	provenance_built
194	fit_stage194_chi_q_fixed_on_canonical_branch chi_Q	notes/stages/moving_throat_pde_stage194_outgoing_l2_fingerprint_and_deformation_algebra.md:333	internal_consistency	audited
194	fit_stage194_even_deformations_not_free Sigma_2, Sigma_4	notes/stages/moving_throat_pde_stage194_outgoing_l2_fingerprint_and_deformation_algebra.md:272	internal_consistency	audited
194	fit_stage_194_chi_q chi_Q, matched_fingerprint_value	paper/stages/stage_194.tex:15(3 manifest anchors)	internal_consistency	provenance_built
194	fit_stage_194_delta_q Delta_Q, chi_Q, matched_fingerprint_value	paper/stages/stage_194.tex:13	internal_consistency	audited
194	fit_stage_194_matched_fingerprint_value matched_fingerprint_value	notes/stages/moving_throat_pde_stage194_outgoing_l2_fingerprint_and_deformation_algebra.md:36(9 manifest anchors)	internal_consistency	provenance_built
194	fit_stage_194_p_2 P_2, matched_fingerprint_value	paper/stages/stage_194.tex:9	free_choice	audited
195	fit_stage195_mhat0_natural_source_map mhat_0	research/pde_ledger/scripts/moving_throat_pde_stage195_source_map_reduction_of_canonical_outgoing_branch_sympy_audit.py:108	free_choice	audited
195	fit_stage195_natural_source_map_mhat0 N_Q, m_hat_0	notes/stages/moving_throat_pde_stage195_source_map_reduction_of_canonical_outgoing_branch.md:196	free_choice	audited
195	fit_stage195_omega_Q_pole_frequency Omega_Q	scripts/moving_throat_pde_stage195_source_map_reduction_of_canonical_outgoing_branch_sympy_audit.py:45	internal_consistency	provenance_built

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
195	fit_stage195_p0_target_canonical_value P0_target	scripts/moving_throat_pde_stage195_source_map_reduction_of_canonical_outgoing_branch_sympy_audit.py:46	published_target	audited
195	fit_stage_195_chi_q chi_Q, matched_fingerprint_value	notes/stages/moving_throat_pde_stage195_source_map_reduction_of_canonical_outgoing_branch.md:38	internal_consistency	provenance_built
195	fit_stage_195_delta_q Delta_Q, chi_Q, matched_fingerprint_value	paper/stages/stage_195.tex:13	internal_consistency	provenance_built
195	fit_stage_195_matched_fingerprint_value matched_fingerprint_value	notes/stages/moving_throat_pde_stage195_source_map_reduction_of_canonical_outgoing_branch.md:5(4 manifest anchors)	internal_consistency	provenance_built
195	fit_stage_195_nq_from_def NQ_from_def	redteam/pass2/reports/stage_195.md:126	internal_consistency	provenance_built
195	fit_stage_195_p_2 P_2, matched_fingerprint_value	paper/stages/stage_195.tex:9	free_choice	audited
195	fit_stage_195_stale_output stale_output	redteam/pass2/reports/stage_195.md:122	no_constraint_record	provenance_built
196	fit_stage196_sigma_Q_canonical_outgoing_scale sigma_Q_can	scripts/moving_throat_pde_stage196_higher_odd_irrelevance_theorem_sympy_audit.py:45	internal_consistency	provenance_built
196	fit_stage_196_chi_q chi_Q	notes/stages/moving_throat_pde_stage196_higher_odd_irrelevance_theorem.md:368	internal_consistency	provenance_built
196	fit_stage_196_delta_q Delta_Q, chi_Q, matched_fingerprint_value	paper/stages/stage_196.tex:13	internal_consistency	provenance_built
196	fit_stage_196_l7_coeff_in_y L7_coeff_in_Y, assert_nonzero, matched_fingerprint_value	redteam/pass2/reports/stage_196.md:149	internal_consistency	provenance_built
196	fit_stage_196_matched_fingerprint_value matched_fingerprint_value	notes/stages/moving_throat_pde_stage196_higher_odd_irrelevance_theorem.md:5(2 manifest anchors)	internal_consistency	provenance_built
196	fit_stage_196_p_2 P_2, matched_fingerprint_value	paper/stages/stage_196.tex:9	internal_consistency	provenance_built
196	fit_stage_196_yhat_2 Yhat_2, matched_fingerprint_value	redteam/pass2/reports/stage_196.md:35	internal_consistency	provenance_built
197	fit_stage197_chi_q_unity_packet_a_target Delta_Q, chi_Q	research/pde_ledger/paper/stages/stage_197.tex:15	internal_consistency	provenance_built
197	fit_stage197_p0_target_quadrupole_normalization Gamma5_target, P0_target	research/pde_ledger/scripts/moving_throat_pde_stage197_conditional_packetA_closure_theorem_sympy_audit.py:76	published_target	audited
197	fit_stage_197_delta_q Delta_Q, chi_Q, matched_fingerprint_value	paper/stages/stage_197.tex:13	internal_consistency	provenance_built

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
197	fit_stage_197_matched_fingerprint_value matched_fingerprint_value	notes/stages/moving_throat_pde_stage197_conditional_packetA_closure_theorem.md:5(2 manifest anchors)	internal_consistency	provenance_built
197	fit_stage_197_p_2 P_2, matched_fingerprint_value	paper/stages/stage_197.tex:9	internal_consistency	provenance_built
197	fit_stage_197_y_stage194_hi Y_stage194_hi, matched_fingerprint_value, stale_output	redteam/pass2/reports/stage_197.md:127	internal_consistency	provenance_built
198	fit_stage_198_pi_star Pi_star, gamma_0	redteam/pass2/reports/stage_198.md:105	no_constraint_record	provenance_built
199	fit_stage_199_m_k m_K, m_T, missing_verification_script, r_K, r_T	redteam/pass2/reports/stage_199.md:170	internal_consistency	provenance_built
200	fit_stage200_mismatch_chart_rederived_not_posited M_star, q_eta, q_nt, q_tr	scripts/moving_throat_pde_stage200_reference_free_home_stretch_theorem_sympy_audit.py:446	internal_consistency	audited
200	fit_stage_200_eps_beta eps_beta, expand_log, expand_power_exp	redteam/pass2/reports/stage_200.md:98	free_choice	audited
200	fit_stage_200_stale_output stale_output	redteam/pass2/reports/stage_200.md:106	no_constraint_record	provenance_built
201	fit_stage201_canonical_repair_vector Delta_x_rep, Pi_can_orbit_projection	paper/stages/stage_201.tex:7	internal_consistency	provenance_built
201	fit_stage_201_delta Delta, Delta_rm, I_3, Pi, Pi_mathcal, T_U, chi_0, chi_Q, iff, m_K, m_T, mathbf, mathcal, mathfrak, mu, mu_W	notes/stages/moving_throat_pde_stage201_explicit_realization_compiler_and_canonical_orbit_projection.md:420(2 manifest anchors)	internal_consistency	provenance_built
201	fit_stage_201_i_3 I_3, K_eta, chi0_star, dx_dep, dx_rep, dx_rep_lin, m_K, m_T, m_mu	redteam/pass2/reports/stage_201.md:39	internal_consistency	provenance_built
202	fit_stage202_split_u_ratio_uniquely_fixed C_tr_target, delta_U_star_graph	notes/stages/moving_throat_pde_stage202_free_quintuple_target_graph.md:197	free_choice	audited
203	fit_stage_203_beta_path beta_path, cetaU_bar, cetaU_path, chi_from_stage197, closure_num_stage197, gamma_path	redteam/pass2/reports/stage_203.md:143	free_choice	audited
203	fit_stage_203_kw_t KW_t, expect_positive, hat_delta, lam_t, q_eta, q_nt, q_tr	redteam/pass2/reports/stage_203.md:116	free_choice	audited
204	fit_stage_204_delta Delta, mathbf, partial_tauln_cdot, s_i, sigma_deltatau, tau, tau_log, tau_rm, y_i	redteam/pass2/reports/stage_204.md:39	internal_consistency	provenance_built
206	fit_stage_206_mathbf mathbf, mathcal	notes/stages/moving_throat_pde_stage206_certified_ray_ranking_and_local_bracketing_theorem.md:467	no_constraint_record	provenance_built

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
207	fit_stage_207_q_i Q_i, mathcal	notes/stages/moving_throat_pde_stage207_primitive_ray_hessian_envelopes_and_certified_ray_table.md:375	internal_consistency	provenance_built
207	fit_stage_207_stale_output stale_output	redteam/pass2/reports/stage_207.md:116	no_constraint_record	provenance_built
207	fit_stage_207_tau_lo tau_lo	redteam/pass2/reports/stage_207.md:104	internal_consistency	provenance_built
208	fit_stage208_canonical_two_ray_screen a_i_equal_mix, a_j_equal_mix	paper/stages/stage_208.tex:15	internal_consistency	provenance_built
209	fit_stage_209_k_i k_i, k_j	redteam/pass2/reports/stage_209.md:39	internal_consistency	provenance_built
210	fit_stage210_canonical_triple_screens triple_screen_directions	paper/stages/stage_210.tex:15	internal_consistency	provenance_built
211	fit_stage_211_c_cross C_cross, Delta_iso, N_r, S_r, tau_iso	redteam/pass2/reports/stage_211.md:150	internal_consistency	provenance_built
211	fit_stage_211_l_r L_r, M_r, N_r	redteam/pass2/reports/stage_211.md:146	internal_consistency	provenance_built
211	fit_stage_211_s_r S_r, feedback_claude_reviews_codex_codes	redteam/pass2/reports/stage_211.md:119	internal_consistency	provenance_built
212	fit_stage212_evaluation_budget_600 evaluation_budget_600	paper/stages/stage_212.tex:15	internal_consistency	provenance_built
213	fit_stage213_four_way_canonical_screens four_way_screen_directions	paper/stages/stage_213.tex:7	internal_consistency	provenance_built
213	fit_stage_213_s_quad S_quad, beta_hi, beta_lo	redteam/pass2/reports/stage_213.md:53(2 manifest anchors)	internal_consistency	provenance_built
214	fit_stage214_preferred_bound_54 candidate_bound_54	paper/stages/stage_214.tex:13	internal_consistency	provenance_built
214	fit_stage_214_l_r L_r, L_s, M_r, M_s	redteam/pass2/reports/stage_214.md:177	internal_consistency	provenance_built
215	fit_stage215_cumulative_budget_1140 cumulative_budget_1140	paper/stages/stage_215.tex:15	internal_consistency	provenance_built
217	fit_stage217_preferred_bound_162_budgets candidate_bound_162, fallback_budget_1500, preferred_budget_324	paper/stages/stage_217.tex:13	internal_consistency	provenance_built
217	fit_stage_217_k_lin K_lin, M_r, M_x, k_L, k_c, stale_output	redteam/pass2/reports/stage_217.md:166	internal_consistency	provenance_built
218	fit_stage218_total_budgets_1464_2640 fallback_budget_2640, preferred_total_budget_1464	paper/stages/stage_218.tex:15	internal_consistency	provenance_built
219	fit_stage219_delta_v_mix_fixed_by_theorem alpha_2, alpha_6, delta_V_mix	notes/stages/moving_throat_pde_stage219_one_port_mixed_bundle_static_kernel_and_square_law_suppression_test_sympy_audit.md:386	internal_consistency	audited
219	fit_stage_219_c_2 C_2, C_6, kappa, to	redteam/pass2/reports/stage_219.md:113	internal_consistency	provenance_built
220	fit_stage_220_d_pi D_Pi, Delta_Pi, K_B, K_dyn, N_s, P_abs, Q_Pi, T_J, V_mix, chi_s	redteam/pass2/reports/stage_220.md:35	internal_consistency	provenance_built

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
220	fit_stage_220_delta_pi Delta_Pi, Pi, det, mathcal, times	paper/stages/stage_220.tex:13	internal_consistency	provenance_built
220	fit_stage_220_im Im, mathfrak	notes/stages/moving_throat_pde_stage220_dynamic_mixed_port_kernel_phase_lag_no_go_and_resonant_survival_gate_sympy_audit.md:219	internal_consistency	provenance_built
220	fit_stage_220_matched_fingerprint_value matched_fingerprint_value	notes/stages/moving_throat_pde_stage220_dynamic_mixed_port_kernel_phase_lag_no_go_and_resonant_survival_gate_sympy_audit.md:114	internal_consistency	provenance_built
220	fit_stage_220_p_abs P_abs, T_J, T_J0, V_mix	redteam/pass2/reports/stage_220.md:150	internal_consistency	provenance_built
220	fit_stage_220_re Re, mathfrak	notes/stages/moving_throat_pde_stage220_dynamic_mixed_port_kernel_phase_lag_no_go_and_resonant_survival_gate_sympy_audit.md:218	internal_consistency	provenance_built
221	fit_stage_221_f_0 F_0, dD_Pi	redteam/pass2/reports/stage_221.md:111	internal_consistency	provenance_built
222	fit_stage222_exact_overlap_constant_kappa kappa	notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:81	internal_consistency	audited
222	fit_stage222_illustrative_barrier_benchmark DeltaV_req, V_known, V_known_x1, epsilon, epsilon_barrier	scripts/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.py:213(2 manifest anchors)	free_choice	audited
222	fit_stage222_numerical_primitive_slice K, M, Omega_U, Omega_W, lambda_B, lambda_R, lambda_U, lambda_W, varpi	scripts/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.py:137	free_choice	audited
222	fit_stage222_primitive_sample_slice K, M, Omega_U, Omega_W, a, c_s, lam_B, lam_R, lam_U, lam_W, varpi	research/pde_ledger/scripts/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.py:135	free_choice	audited
222	fit_stage222_vknown_barrier_benchmark DeltaV_req, V_known, epsilon	research/pde_ledger/scripts/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.py:213	free_choice alias offit_stage222_illustrative_barrier_benchmark	scanned; alias folded to fit_stage222_illustrative_barrier_benchm

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
222	fit_stage_222_delta Delta	notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:436	internal_consistency	provenance_built
222	fit_stage_222_deltav_req DeltaV_req, V_known	redteam/pass2/reports/stage_222.md:131	free_choice	audited
222	fit_stage_222_lambda_b lambda_B, lambda_B_lambda_U_lambda_W_lambda_R, lambda_R, lambda_U, lambda_W	notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:434	free_choice	audited
222	fit_stage_222_n_star N_star, deg_y, lambda_W	redteam/pass2/reports/stage_222.md:39	internal_consistency	provenance_built
222	fit_stage_222_p_0 P_0, lambda_W, mathcal	notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:391	internal_consistency	provenance_built
223	fit_stage223_barrier_benchmark_and_eta_window V_known_x1, barrier_floor, eta_linewidth_window	scripts/moving_throat_pde_stage223_5pn_isotropic_target_surface_primitive_branch_compatibility_and_dynamic_survival_window_sympy_audit.py:296	free_choice	audited
223	fit_stage223_lambda02_rq_dual_engine_overclaim R_Q_lower_wall_lambdaW_0p2, R_Q_upper_wall_lambdaW_0p2	research/pde_ledger/notes/CHECKPOINT_CONSTANT_PROVENANCE.md:21	internal_consistency	audited
223	fit_stage223_p0_target_calibration_coefficient P0_target, coefficient_54_over_5, mhat_0	notes/stages/moving_throat_pde_stage223_5pn_isotropic_target_surface_primitive_branch_compatibility_and_dynamic_survival_window_sympy_audit.md:127	published_target	audited
223	fit_stage223_p0_target_import_unprovenanced P0_target	research/pde_ledger/paper/stages/stage_223.tex:9	published_target	audited
223	fit_stage223_universal_p0_target P_0_target	notes/stages/moving_throat_pde_stage223_5pn_isotropic_target_surface_primitive_branch_compatibility_and_dynamic_survival_window_sympy_audit.md:127	published_target	audited
223	fit_stage_223_d0_compat D0_compat, K_compat, P0_target, P0_target_compat	redteam/pass2/reports/stage_223.md:139	internal_consistency	provenance_built
223	fit_stage_223_k_norm K_norm, K_pole, P0_target, R_Q, assert_close	redteam/pass2/reports/stage_223.md:39	internal_consistency	provenance_built
223	fit_stage_223_mathrm mathrm	notes/stages/moving_throat_pde_stage223_5pn_isotropic_target_surface_primitive_branch_compatibility_and_dynamic_survival_window_sympy_audit.md:130	published_target	audited

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
224	fit_stage224_grouped_signature_exact b_P0_slope, lambda_20, lambda_21, lambda_22	notes/stages/moving_throat_pde_stage224_pde_branch_packet_compiler_weak_axisymmetric_ceiling_transport_and_first_actual_branch_kill_test_sympy_audit.md:195	internal_consistency	audited
224	fit_stage224_kill_test_budgets_slice_anchored P_crit_both_10, P_crit_both_30, P_crit_one_10, P_crit_one_30, barP0_compat	research/pde_ledger/scripts/moving_throat_pde_stage224_pde_branch_packet_compiler_weak_axisymmetric_ceiling_transport_and_first_actual_branch_kill_test_sympy_audit.py:134	internal_consistency	audited
224	fit_stage224_stage223_carryover_ceilings Pcrit_both_10, Pcrit_both_30, Pcrit_one_10, Pcrit_one_30, barP0_compat	scripts/moving_throat_pde_stage224_pde_branch_packet_compiler_weak_axisymmetric_ceiling_transport_and_first_actual_branch_kill_test_sympy_audit.py:134	internal_consistency	audited
224	fit_stage224_t_quad_target_scale T_quad	paper/stages/stage_224.tex:9	published_target	audited
224	fit_stage224_weak_axisymmetric_lane_signature lambda_20, lambda_21, lambda_22	scripts/moving_throat_pde_stage224_pde_branch_packet_compiler_weak_axisymmetric_ceiling_transport_and_first_actual_branch_kill_test_sympy_audit.py:66	free_choice	audited
224	fit_stage_224_delta_rm Delta_rm, P_0, P_A, StatusNumerical, Xi_1, Xi_1_4, Xi_1_P_rm, Xi_1_bar, Xi_1_hat, approx, bar, epsilon, ge, hat, lambda, lambda_A, lambda_AXi_1, le, mapsto, max, tfrac, varepsilon	redteam/pass2/reports/stage_224.md:35	free_choice	audited
224	fit_stage_224_hat hat	notes/stages/moving_throat_pde_stage224_pde_branch_packet_compiler_weak_axisymmetric_ceiling_transport_and_first_actual_branch_kill_test_sympy_audit.md:175(2 manifest anchors)	free_choice	audited
225	fit_stage225_K_compat_P0_target_pin D_0, D_2, D_4, K_compat, P_0_target	research/pde_ledger/scripts/moving_throat_pde_stage225_microscopic_xi1_compiler_first_order_conservative_compensation_surface_and_mixed_sector_survival_sieve_sympy_audit.py:239	internal_consistency	provenance_built
225	fit_stage225_stage223_sample_point OmU, OmW, kappa, lamB, lamR, lamU, lamW, varpi	scripts/moving_throat_pde_stage225_microscopic_xi1_compiler_first_order_conservative_compensation_surface_and_mixed_sector_survival_sieve_sympy_audit.py:202	free_choice	audited

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
225	fit_stage225_stage224_xi_budgets B_xi_both_10, B_xi_both_30, B_xi_one_10, B_xi_one_30	scripts/moving_throat_pde_stage225_microscopic_xi1_compiler_first_order_conservative_compensation_surface_and_mixed_sector_survival_sieve_sympy_audit.py:399	internal_consistency	provenance_built
225	fit_stage225_xk_xm_forced_zero x_K, x_M	notes/stages/moving_throat_pde_stage225_microscopic_xi1_compiler_first_order_conservative_compensation_surface_and_mixed_sector_survival_sieve_sympy_audit.md:404	internal_consistency	audited
225	fit_stage_225_d_4 D_4, Xi_1, assert_close, one_pole_D41, sigma_1, u2_1, u4_1, u_2	redteam/pass2/reports/stage_225.md:39(2 manifest anchors)	internal_consistency	provenance_built
225	fit_stage_225_xi_1 Xi_1	notes/stages/moving_throat_pde_stage225_microscopic_xi1_compiler_first_order_conservative_compensation_surface_and_mixed_sector_survival_sieve_sympy_audit.md:25	internal_consistency	provenance_built
226	fit_stage226_canonical_hidden_even_relation H_even	paper/stages/stage_226.tex:9	internal_consistency	provenance_built
226	fit_stage226_d01_forced_zero D_01	notes/stages/moving_throat_pde_stage226_strict_5pn_even_gate_package_surviving_mixed_corridor_and_pure_transfer_subcorridor_sympy_audit.md:207	internal_consistency	audited
226	fit_stage226_k1_h_even_open_gates H_even, K_1, Xi_load	research/pde_ledger/paper/stages/stage_226.tex:9	internal_consistency	provenance_built
226	fit_stage226_sigma_even_canonical_gain_scale sigma_even	notes/stages/moving_throat_pde_stage226_strict_5pn_even_gate_package_surviving_mixed_corridor_and_pure_transfer_subcorridor_sympy_audit.md:278	internal_consistency	provenance_built
226	fit_stage_226_k_1 K_1, Xi_1, Xi_1_N_01, Xi_load, sigma_even, sigma_transfer	redteam/pass2/reports/stage_226.md:39	internal_consistency	provenance_built
226	fit_stage_226_k_compat K_compat, N01_coeff, Xi_coeff	redteam/pass2/reports/stage_226.md:168	internal_consistency	provenance_built
226	fit_stage_226_xi_1 Xi_1, Xi_1_sigma_rm	redteam/pass2/reports/stage_226.md:49	internal_consistency	provenance_built
227	fit_stage227_kappa_exact_overlap_constant kappa	notes/stages/moving_throat_pde_stage227_pure_transfer_load_factor_outgoing_rigidity_sieve_and_first_co_loading_no_go_sympy_audit.md:86	internal_consistency	provenance_built
227	fit_stage227_sample_branch_coefficients_exact H_coeff, I_coeff	notes/stages/moving_throat_pde_stage227_pure_transfer_load_factor_outgoing_rigidity_sieve_and_first_co_loading_no_go_sympy_audit.md:256	internal_consistency	provenance_built

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
227	fit_stage227_xi1_sample_law_coefficients c_h, c_i, c_m	scripts/moving_throat_pde_stage227_pure_transfer_load_factor_outgoing_rigidity_sieve_and_first_co_loading_no_go_sympy_audit.py:201	internal_consistency	provenance_built
227	fit_stage_227_m_transfer M_transfer	redteam/pass2/reports/stage_227.md:95	internal_consistency	provenance_built
227	fit_stage_227_n_0 N_0, Xi_1, assert_close, pi	redteam/pass2/reports/stage_227.md:112	internal_consistency	provenance_built
227	fit_stage_227_xi_1 Xi_1, pi	redteam/pass2/reports/stage_227.md:39	internal_consistency	provenance_built
228	fit_stage228_k_compat_exact_stiffness K_compat	notes/stages/moving_throat_pde_stage228_numerator_denominator_split_of_the_pure_transfer_corridor_and_first_actual_dynamic_window_test_sympy_audit.md:77	internal_consistency	audited
228	fit_stage228_primitive_parameter_tuple M, Omega_U, Omega_W, lambda_B, lambda_R, lambda_U, lambda_W, varpi	notes/stages/moving_throat_pde_stage228_numerator_denominator_split_of_the_pure_transfer_corridor_and_first_actual_dynamic_window_test_sympy_audit.md:72	free_choice	audited
228	fit_stage228_rq_requirement_threshold RQ_req, threshold_eta_01	scripts/moving_throat_pde_stage228_numerator_denominator_split_of_the_pure_transfer_corridor_and_first_actual_dynamic_window_test_sympy_audit.py:432	published_target	audited
228	fit_stage_228_delta_1 delta_1, pi	redteam/pass2/reports/stage_228.md:102	internal_consistency	provenance_built
229	fit_stage229_delta_crossover_threshold delta, xi_star	paper/stages/stage_229.tex:15	internal_consistency	provenance_built
229	fit_stage_229_kappa_0 kappa_0, pi	notes/stages/moving_throat_pde_stage229_selected_branch_numerator_denominator_signature_and_softening_depth_crossover_theorem_sympy_audit.md:143	internal_consistency	provenance_built
230	fit_stage230_r_star_exact_threshold R_star	notes/stages/moving_throat_pde_stage230_selected_branch_classifier_to_dynamic_window_compiler_and_static_first_theorem_sympy_audit.md:224	internal_consistency	provenance_built
230	fit_stage230_stage228_rigid_slope_carry s_minus_den, s_minus_num, s_plus_den, s_plus_num	scripts/moving_throat_pde_stage230_selected_branch_classifier_to_dynamic_window_compiler_and_static_first_theorem_sympy_audit.py:77	internal_consistency	audited
231	fit_stage231_dynamic_budget_carry B_dyn_both_inf, B_dyn_nonempty_inf	scripts/moving_throat_pde_stage231_continuum_placement_pullback_of_the_selected_branch_dynamic_class_map_sympy_audit.py:272	internal_consistency	provenance_built

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
232	fit_stage232_family1_geometry_normalization Lambda_ell, Xi_drive_prefactor, kappa	scripts/moving_throat_pde_stage232_known_5pn_data_injection_and_current_branch_verdict_sympy_audit.py:50	free_choice	audited
232	fit_stage232_fom_prefactor_100 Xi_prefactor_100	research/pde_ledger/scripts/moving_throat_pde_stage232_known_5pn_data_injection_and_current_branch_verdict_sympy_audit.py:149	free_choice	audited
232	fit_stage232_injected_wall_depth_theta_w Theta_w_J, Theta_w_chi, Xi_J, Xi_chi	research/pde_ledger/scripts/moving_throat_pde_stage232_known_5pn_data_injection_and_current_branch_verdict_sympy_audit.py:146	internal_consistency	provenance_built
232	fit_stage232_known_5pn_data_injection rho_alpha_max, zeta_phys	notes/stages/moving_throat_pde_stage232_known_5pn_data_injection_and_current_branch_verdict_sympy_audit.md:23	internal_consistency	audited
232	fit_stage232_lambda_ell_geometry_literals Lambda_ell, chi_s, kappa	research/pde_ledger/scripts/moving_throat_pde_stage232_known_5pn_data_injection_and_current_branch_verdict_sympy_audit.py:50	free_choice	audited
232	fit_stage232_nq_canonical_normalization N_Q	notes/stages/moving_throat_pde_stage232_known_5pn_data_injection_and_current_branch_verdict_sympy_audit.md:314	free_choice	audited
232	fit_stage232_theta_w_5pn_injection Theta_w_J, Theta_w_chi	research/pde_ledger/scripts/moving_throat_pde_stage232_known_5pn_data_injection_and_current_branch_verdict_sympy_audit.py:146	internal_consistency	audited
232	fit_stage232_wall_depth_thetas Theta_w_J, Theta_w_chi	scripts/moving_throat_pde_stage232_known_5pn_data_injection_and_current_branch_verdict_sympy_audit.py:146	internal_consistency	provenance_built
232	fit_stage_232_a_k A_K, J_0	redteam/pass2/reports/stage_232.md:35(2 manifest anchors)	internal_consistency	provenance_built
232	fit_stage_232_lambda_mu_1 lambda_mu_1	notes/stages/moving_throat_pde_stage232_known_5pn_data_injection_and_current_branch_verdict_sympy_audit.md:149	free_choice	audited
232	fit_stage_232_theta_w_j Theta_w_J, Theta_w_chi	redteam/pass2/reports/stage_232.md:96	internal_consistency	provenance_built
233	fit_stage233_calibrated_branch_delta_norm Delta_norm	notes/stages/moving_throat_pde_stage233_rigid_mouth_orbit_lock_compiler_and_the_static_turbulence_gate_sympy_audit.md:203	free_choice	audited
233	fit_stage233_stage224_ceiling_carry Pbar, Pcrit_nonempty, Pcrit_robust	scripts/moving_throat_pde_stage233_rigid_mouth_orbit_lock_compiler_and_the_static_turbulence_gate_sympy_audit.py:115	internal_consistency	provenance_built

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
234	fit_stage234_canonical_least_deformation_family Xi_1	notes/stages/moving_throat_pde_stage234_direct_branch_observable_static_gate_and_the_two_observable_kill_test_sympy_audit.md:372	free_choice	audited
235	fit_stage_235_q_eta q_eta	notes/stages/moving_throat_pde_stage235_rigid_mouth_packet_projectors_static_blind_dressing_line_and_codimension_two_orbit_lock_point_sympy_audit.md:389	internal_consistency	provenance_built
236	fit_stage236_equal_drift_ray_forced K_eta, mu, q_eta	notes/stages/moving_throat_pde_stage236_rigid_mouth_microscopic_dependent_plane_projectors_equal_drift_dressing_ray_and_static_only_restoration_gap_sympy_audit.md:369	internal_consistency	audited
236	fit_stage_236_epsilon_eta epsilon_eta	notes/stages/moving_throat_pde_stage236_rigid_mouth_microscopic_dependent_plane_projectors_equal_drift_dressing_ray_and_static_only_restoration_gap_sympy_audit.md:452	internal_consistency	provenance_built
236	fit_stage_236_mu mu	notes/stages/moving_throat_pde_stage236_rigid_mouth_microscopic_dependent_plane_projectors_equal_drift_dressing_ray_and_static_only_restoration_gap_sympy_audit.md:448	internal_consistency	provenance_built
236	fit_stage_236_t_u T_U, mu	notes/stages/moving_throat_pde_stage236_rigid_mouth_microscopic_dependent_plane_projectors_equal_drift_dressing_ray_and_static_only_restoration_gap_sympy_audit.md:476	free_choice	audited
237	fit_stage237_branch_sample_point Rratio_base, Rtarget_base, eps_base, qeta_param_base, ref_base	scripts/moving_throat_pde_stage237_actual_branch_dressing_compiler_finite_static_blind_curve_and_support_blind_post_static_orbit_lock_theorem_sympy_audit.py:17	free_choice	audited
237	fit_stage_237_epsilon_eta epsilon_eta	notes/stages/moving_throat_pde_stage237_actual_branch_dressing_compiler_finite_static_blind_curve_and_support_blind_post_static_orbit_lock_theorem_sympy_audit.md:39	internal_consistency	provenance_built

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
237	fit_stage_237_q_eta q_eta	notes/stages/moving_throat_pde_stage237_actual_branch_dressing_compiler_finite_static_blind_curve_and_support_blind_post_static_orbit_lock_theorem_sympy_audit.md:10(3 manifest anchors)	internal_consistency	provenance_built
237	fit_stage_237_q_nt q_nt	notes/stages/moving_throat_pde_stage237_actual_branch_dressing_compiler_finite_static_blind_curve_and_support_blind_post_static_orbit_lock_theorem_sympy_audit.md:13	internal_consistency	provenance_built
238	fit_stage_238_pi_star Pi_star, gamma_0	redteam/pass2/reports/stage_238.md:99	no_constraint_record	provenance_built
238	fit_stage_238_q_eta q_eta	notes/stages/moving_throat_pde_stage238_physical_branch_transfer_shape_compiler_packet_factorization_and_post_static_dressing_invariance_theorem_sympy_audit.md:62	internal_consistency	provenance_built
239	fit_stage239_exact_microscopic_correction Delta_K_eta, Delta_T, Delta_mu	paper/stages/stage_239.tex:13	internal_consistency	provenance_built
239	fit_stage_239_mu mu	notes/stages/moving_throat_pde_stage239_rigid_mouth_physical_normal_form_exact_physical_to_microscopic_correction_compiler_and_cartesian_orbit_lock_theorem_sympy_audit.md:71	internal_consistency	provenance_built
240	fit_stage240_loading_ratio_from_minimal_precursor Pi_tr, rho_alpha, zeta_req	research/pde_ledger/paper/stages/stage_240.tex:15	internal_consistency	provenance_built
240	fit_stage240_loading_ratio_not_free C_mix, Pi_tr, rho_alpha	notes/stages/moving_throat_pde_stage240_selected_branch_loading_ratio_from_the_minimal_isotropic_quadrupole_precursor_sympy_audit.md:104	internal_consistency	audited
240	fit_stage240_minimal_isotropic_precursor c0_min, c1_min, rho_alpha, zeta_req	scripts/moving_throat_pde_stage240_selected_branch_loading_ratio_from_the_minimal_isotropic_quadrupole_precursor_sympy_audit.py:137	free_choice	audited
240	fit_stage240_rho_alpha_selected_demand Pi_tr_over_C_mix, rho_alpha	research/pde_ledger/notes/CHECKPOINT_CONSTANT_PROVENANCE.md:782	internal_consistency	audited
240	fit_stage240_zeta_req_fixed_exactly Pi_tr, zeta_req	notes/stages/moving_throat_pde_stage240_selected_branch_loading_ratio_from_the_minimal_isotropic_quadrupole_precursor_sympy_audit.md:277	internal_consistency	audited

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
240	fit_stage_240_pi_rm Pi_rm, Y_support, alpha_rm, c0_static, c1_static, epsilon_3, rho_alpha_4_3, to, varrho, zeta_rm	redteam/pass2/reports/stage_240.md:103	no_constraint_record	provenance_built
241	fit_stage241_beta_window_endpoint beta_max	scripts/moving_throat_pde_stage241_exact_primitive_ranking_on_the_selected_twin_support_branch_sympy_audit.py:180	internal_consistency	provenance_built
241	fit_stage241_varrho_no_longer_free epsilon_star, varrho	notes/stages/moving_throat_pde_stage241_exact_primitive_ranking_on_the_selected_twin_support_branch_sympy_audit.md:83	internal_consistency	audited
241	fit_stage_241_epsilon epsilon, varrho	notes/stages/moving_throat_pde_stage241_exact_primitive_ranking_on_the_selected_twin_support_branch_sympy_audit.md:707	internal_consistency	provenance_built
242	fit_stage242_placement_coeff_2_11 placement_coeff_2_11	research/pde_ledger/notes/CHECKPOINT_CONSTANT_PROVENANCE.md:779	internal_consistency	provenance_built
242	fit_stage242_varrho_phys_fixed sigma_phys, varrho_phys	notes/stages/moving_throat_pde_stage242_actual_twin_support_placement_and_coherent_orbit_lock_compiler_sympy_audit.md:671	internal_consistency	provenance_built
242	fit_stage_242_c_mix C_mix, Pi_tr	redteam/pass2/reports/stage_242.md:95	internal_consistency	provenance_built
242	fit_stage_242_epsilon epsilon	notes/stages/moving_throat_pde_stage242_actual_twin_support_placement_and_coherent_orbit_lock_compiler_sympy_audit.md:214	internal_consistency	provenance_built
242	fit_stage_242_lambda_0 Lambda_0	notes/stages/moving_throat_pde_stage242_actual_twin_support_placement_and_coherent_orbit_lock_compiler_sympy_audit.md:375	internal_consistency	provenance_built
243	fit_stage_243_e_w E_w, S_leak	redteam/pass2/reports/stage_243.md:87	free_choice	audited
243	fit_stage_243_k_turn K_turn, a_int, chi_lam, k_V	notes/CHECKPOINT_CONSTANT_PROVENANCE.md:17	no_constraint_record	provenance_built
243	fit_stage_243_xi_1 Xi_1	notes/stages/moving_throat_pde_stage243_relaxed_constraint_branch_declaration_and_short_range_open_system_compiler_sympy_audit.md:347	internal_consistency	provenance_built
244	fit_stage244_leakage_forced_through_fixed_demand Pi_tr	notes/stages/moving_throat_pde_stage244_selected_branch_leakage_and_scalar_photon_work_compiler_sympy_audit.md:557	internal_consistency	audited

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
245	fit_stage245_eps_eta_session1_match epsilon_eta	notes/stages/moving_throat_pde_stage245_nonrigid_mouth_dressing_packet_and_uv_drain_compiler_sympy_audit.md:516	free_choice	audited
245	fit_stage245_session1_readback U_obs, V_obs, eps_ref	research/pde_ledger/scripts/moving_throat_pde_stage245_nonrigid_mouth_dressing_packet_and_uv_drain_compiler_sympy_audit.py:209	free_choice	audited
245	fit_stage245_session1_readback_inputs U_obs, V_obs, a_U, a_V, eps_eta_ref, g_UV	scripts/moving_throat_pde_stage245_nonrigid_mouth_dressing_packet_and_uv_drain_compiler_sympy_audit.py:209	free_choice	audited
245	fit_stage_245_xi_1 Xi_1	notes/stages/moving_throat_pde_stage245_nonrigid_mouth_dressing_packet_and_uv_drain_compiler_sympy_audit.md:436	internal_consistency	provenance_built
246	fit_stage246_lower_branch_forced sigma_min	notes/stages/moving_throat_pde_stage246_compensated_multimode_source_compiler_beyond_positive_family1_sympy_audit.md:387	internal_consistency	audited
246	fit_stage246_rF1_coefficient rF1	scripts/moving_throat_pde_stage246_compensated_multimode_source_compiler_beyond_positive_family1_sympy_audit.py:145	internal_consistency	provenance_built
246	fit_stage246_session1_inputs a_0, b_0, r_eval, r_sigma	research/pde_ledger/scripts/moving_throat_pde_stage246_compensated_multimode_source_compiler_beyond_positive_family1_sympy_audit.py:230	free_choice	audited
246	fit_stage246_session1_readback_point a0, b0, r, r_sigma	scripts/moving_throat_pde_stage246_compensated_multimode_source_compiler_beyond_positive_family1_sympy_audit.py:230	free_choice	audited
246	fit_stage246_two_moment_source_matrix M_src	scripts/moving_throat_pde_stage246_compensated_multimode_source_compiler_beyond_positive_family1_sympy_audit.py:123	internal_consistency	provenance_built
247	fit_stage247_benchmark_pinned_to_paper_figures Lvar_soft, lambda_L	scripts/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.py:239	published_target	audited

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
247	fit_stage247_lambda_L_backsolve lambda_L	research/pde_ledger/scripts/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.py:238	published_target	audited
247	fit_stage247_lambda_L_closure lambda_L	scripts/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.py:238	published_target	audited
247	fit_stage247_lambda_l_fixed_by_recorded_point lambda_L	notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:497	published_target	audited
247	fit_stage247_lambda_w_natural_specialization lambda_W	notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:489	free_choice	audited
247	fit_stage247_session1_packet G_W, Rmix, UVdrop_obs, Veff_obs, Wsess_obs, beta_Q, beta_U, beta_W, eta_leak, mu_w, xi_R	research/pde_ledger/scripts/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.py:213	free_choice + published_target	audited
247	fit_stage247_session1_softening_point a0, b0, betaW, eta_leak, kappa, lam, mu_w, rF1, r_sigma, r_soft, xi_R	scripts/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.py:204	free_choice	audited
247	fit_stage_247_a_2 A_2, A_4, A_6, C_2, C_4, C_6, D_UV, DeltaE_UV, M_sigma, R_inf, S_leak, V_eff, V_short, W_w, alpha_2, alpha_6, eta_UV, g_inf, lambda_L, lambda_W, r_soft, xi_R	redteam/pass2/reports/stage_247.md:35	internal_consistency	provenance_built

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
247	fit_stage_247_alpha_2 alpha_2, alpha_6, alpha_6_alpha_2_0	notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:404	free_choice	audited
247	fit_stage_247_d_uv D_UV, K_red, Lvar_soft, M_sigma, S_leak, S_soft, V_eff_session, V_short, W_sess, g_inf, g_soft, lambda_L, lambda_L_soft	redteam/pass2/reports/stage_247.md:39	internal_consistency	provenance_built
247	fit_stage_247_delta Delta, stage_247	redteam/pass2/reports/stage_247.md:77	internal_consistency	provenance_built
247	fit_stage_247_lambda_l_soft lambda_L_soft	redteam/pass2/reports/stage_247.md:85	published_target	audited
247	fit_stage_247_lvar_from_w Lvar_from_W, lambda_L	redteam/pass2/reports/stage_247.md:93	internal_consistency	provenance_built
247	fit_stage_247_m_sigma M_sigma, S_soft, V_short, lambda_L_soft	redteam/pass2/reports/stage_247.md:68	internal_consistency	provenance_built
247	fit_stage_247_v_eff V_eff, V_short	redteam/pass2/reports/stage_247.md:51	published_target	audited
247	fit_stage_247_vrebuild_soft Vrebuild_soft, lambda_L, lambda_L_paper	redteam/pass2/reports/stage_247.md:79	published_target	audited
248	fit_stage248_chosen_contact_scale_benchmark r_contact	notes/stages/moving_throat_pde_stage248_dynamic_event_chain_compiler_from_relaxed_stationary_barrier_front_end_turning_point_threshold_speed_and_wkb_sympy_audit.md:538	free_choice	audited
248	fit_stage248_session2_launch_parameters E_sub, V0, Vpeak, m_s, r0, r_c	scripts/moving_throat_pde_stage248_dynamic_event_chain_compiler_from_relaxed_stationary_barrier_front_end_turning_point_threshold_speed_and_wkb_sympy_audit.py:237	free_choice	audited
248	fit_stage248_session2_readback E_sub, I_Coul_report, I_new, V_0, V_peak, r_0, r_c, r_coul_turn_report, r_inner, r_peak, r_turn_new, v_cross	research/pde_ledger/notes/CHECKPOINT_CONSTANT_PROVENANCE.md:892	free_choice	audited
248	fit_stage248_v0_cross_benchmark_speed v0_cross	notes/stages/moving_throat_pde_stage248_dynamic_event_chain_compiler_from_relaxed_stationary_barrier_front_end_turning_point_threshold_speed_and_wkb_sympy_audit.md:513	published_target	audited

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
248	fit_stage248_xi_turn_lambda_th_carried_hardcodes Xi_turn, lambda_th	research/pde_ledger/scripts/moving_throat_pde_stage248_dynamic_event_chain_compiler_from_relaxed_stationary_barrier_front_end_turning_point_threshold_speed_and_wkb_sympy_audit.py:247	published_target	audited
248	fit_stage_248_f_coul F_coul, I_Coul, T_Coul, T_new, r_turn, v_0, v_contact, v_crit	redteam/pass2/reports/stage_248.md:39	internal_consistency	provenance_built
248	fit_stage_248_i_coul I_Coul, I_coul, I_new, I_top, K_peak, T_Coul, T_new, V_eff, V_peak, engine_disagreement, m_s, r_contact, r_turn, v_contact, v_crit	redteam/pass2/reports/stage_248.md:35(2 manifest anchors)	internal_consistency	provenance_built
249	fit_stage249_helicity_ratio_matched H_sub_minus_final, H_sub_plus_final	notes/stages/moving_throat_pde_stage249_conditional_helicity_export_diagnostic_compiler_on_the_dynamic_event_chain_and_aligned_vs_anti_aligned_mixed_sector_closure_sympy_audit.md:486	free_choice	audited
249	fit_stage249_session2_helicity_readbacks H_int_aligned, H_int_antialigned, H_peak_aligned, H_peak_antialigned, alpha_h	scripts/moving_throat_pde_stage249_conditional_helicity_export_diagnostic_compiler_on_the_dynamic_event_chain_and_aligned_vs_anti_aligned_mixed_sector_closure_sympy_audit.py:167	free_choice	audited
249	fit_stage249_session3_peak_readback hint_aligned, hint_antialigned, peak_aligned, peak_antialigned	research/pde_ledger/scripts/moving_throat_pde_stage249_conditional_helicity_export_diagnostic_compiler_on_the_dynamic_event_chain_and_aligned_vs_anti_aligned_mixed_sector_closure_sympy_audit.py:167	free_choice	audited
249	fit_stage_249_alpha_rm alpha_rm	notes/stages/moving_throat_pde_stage249_conditional_helicity_export_diagnostic_compiler_on_the_dynamic_event_chain_and_aligned_vs_anti_aligned_mixed_sector_closure_sympy_audit.md:456	internal_consistency	provenance_built
249	fit_stage_249_c_sigma C_sigma, Hdot_sigma, Phi_sigma, alpha_h, eq_sub, eq_sub_expected	redteam/pass2/reports/stage_249.md:95	free_choice	audited
249	fit_stage_249_gamma_0 Gamma_0, R_pk, S_cov, alpha_h, alpha_pk, bar, eta_h, free_symbols	redteam/pass2/reports/stage_249.md:39	internal_consistency	provenance_built
249	fit_stage_249_r_int R_int, alpha_int, alpha_pk, engine_disagreement	redteam/pass2/reports/stage_249.md:99	free_choice	audited

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
250	fit_stage250_benchmark_masses_and_window E_max, chi_peak, chi_raw, g_UV, m_s, mu_eta, t_transit_max, t_transit_min	scripts/moving_throat_pde_stage250_crossing_vs_collapse_goldilocks_window_compiler_from_the_stage248_event_chain_and_relaxed_wall_timescale_closure_sympy_audit.py:177	free_choice+published_target	audited
250	fit_stage250_goldilocks_edge_forced E_edge, V_peak	notes/stages/moving_throat_pde_stage250_crossing_vs_collapse_goldilocks_window_compiler_from_the_stage248_event_chain_and_relaxed_wall_timescale_closure_sympy_audit.md:585	internal_consistency	audited
250	fit_stage250_goldilocks_inputs chi_peak, chi_raw, g_UV, m_s_ratio_1836	research/pde_ledger/scripts/moving_throat_pde_stage250_crossing_vs_collapse_goldilocks_window_compiler_from_the_stage248_event_chain_and_relaxed_wall_timescale_closure_sympy_audit.py:176	free_choice+published_target	audited
250	fit_stage_250_chi_rm chi_rm, lambda_rm, mu_eta	notes/stages/moving_throat_pde_stage250_crossing_vs_collapse_goldilocks_window_compiler_from_the_stage248_event_chain_and_relaxed_wall_timescale_closure_sympy_audit.md:622	free_choice	audited
250	fit_stage_250_e_edge E_edge, E_safe, dt_cross, m_s, min_raw, t_collapse, t_collapse_raw, v_safe	redteam/pass2/reports/stage_250.md:112(2 manifest anchors)	internal_consistency	provenance_built
250	fit_stage_250_t_cross t_cross	redteam/pass2/reports/stage_250.md:39	internal_consistency	provenance_built
251	fit_stage251_gamma3_exact_cubic_coefficient Gamma_3	notes/stages/moving_throat_pde_stage251_microscopic_damping_export_kernel_replacing_the_phenomenological_v_leg_envelope_law_sympy_audit.md:58	internal_consistency	provenance_built
251	fit_stage251_gamma5_exact_quintic_coefficient Gamma_5	notes/stages/moving_throat_pde_stage251_microscopic_damping_export_kernel_replacing_the_phenomenological_v_leg_envelope_law_sympy_audit.md:52	internal_consistency	provenance_built
251	fit_stage251_gamma_safe_exact_replacement gamma_safe	paper/stages/stage_251.tex:97	internal_consistency	audited
251	fit_stage251_session4_rate_constants gamma_crit, t_collapse0, t_cross	scripts/moving_throat_pde_stage251_microscopic_damping_export_kernel_replacing_the_phenomenological_v_leg_envelope_law_sympy_audit.py:176	free_choice	audited

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
251	fit_stage251_session4_time_inputs gamma_crit, s_c, t_collapse_0, t_cross	research/pde_ledger/scripts/moving_throat_pde_stage251_microscopic_damping_export_kernel_replacing_the_phenomenological_v_leg_envelope_law_sympy_audit.py:176	free_choice+internal_consistency	audited
251	fit_stage_251_cref cref, s_c	paper/stages/stage_251.tex:92	free_choice	audited
251	fit_stage_251_delta_0 Delta_0, G5hat_safe, Gamma_3, Gamma_5, K_exp, Omega_U0, P0_minus, P_exp, Pi_V, Pi_V0, beta_0, eta_0, gamma_1, gamma_crit, gamma_tot, kappa_V, mu_eta, s_0, s_c	redteam/pass2/reports/stage_251.md:35(2 manifest anchors)	internal_consistency	provenance_built
251	fit_stage_251_s_0 s_0, s_c	redteam/pass2/reports/stage_251.md:128	internal_consistency	provenance_built
251	fit_stage_251_s_c s_c	redteam/pass2/reports/stage_251.md:97(2 manifest anchors)	internal_consistency	provenance_built
252	fit_stage252_e_exp_min_completely_fixed E_exp_min_safe, V_in, mu_eta	notes/stages/moving_throat_pde_stage252_vacuum_vs_lattice_heat_partition_and_cold_survival_compiler_from_the_microscopic_export_kernel_sympy_audit.md:330	internal_consistency	audited
252	fit_stage252_three_to_one_split f_lat, lattice_vacuum_split_3, phi	research/pde_ledger/scripts/moving_throat_pde_stage252_vacuum_vs_lattice_heat_partition_and_cold_survival_compiler_from_the_microscopic_export_kernel_sympy_audit.py:155(2 manifest anchors)	free_choice	audited
252	fit_stage252_vin_benchmark_calibration_match V_in_match	notes/stages/moving_throat_pde_stage252_vacuum_vs_lattice_heat_partition_and_cold_survival_compiler_from_the_microscopic_export_kernel_sympy_audit.md:482	internal_consistency	audited
252	fit_stage_252_engine_disagreement engine_disagreement, f_lat, f_vac, mu_eta	redteam/pass2/reports/stage_252.md:97	no_constraint_record	provenance_built
252	fit_stage_252_moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_mathematica_audit moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_mathematica_audit	notes/CHECKPOINT_CONSTANT_PROVENANCE.md:942	no_constraint_record	provenance_built

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
252	fit_stage_252_moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit	notes/CHECKPOINT_CONSTANT_PROVENANCE.md:940	no_constraint_record	provenance_built
252	fit_stage_252_mu_eta mu_eta, safe_combo	redteam/pass2/reports/stage_252.md:129	free_choice	audited
252	fit_stage_252_r_v r_V	redteam/pass2/reports/stage_252.md:39	internal_consistency	provenance_built
253	fit_stage253_K_turn_force_match K_turn	research/pde_ledger/scripts/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.py:166	free_choice	audited
253	fit_stage253_calibration_slice_nominal_constants K_turn, f_lat, gamma_lattice_legacy, lambda_ref, mu_eta, r_turn, s_0, s_c	research/pde_ledger/scripts/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.py:161	free_choice + internal_consistency	audited
253	fit_stage253_gamma_lattice_red gamma_lattice_red	research/pde_ledger/scripts/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.py:163	free_choice	audited
253	fit_stage253_k_eff_force_matched k_eff_req	paper/stages/stage_253.tex:64	internal_consistency	audited
253	fit_stage253_legacy_lattice_rate_and_upsilon Upsilon_lat_sess, f_lat, gamma_lattice_legacy	scripts/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.py:163	internal_consistency	provenance_built
253	fit_stage253_session5_calibration_slice_recovery Upsilon_lat_sess, gamma_lattice_red	notes/stages/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.md:224	free_choice + internal_consistency	audited
253	fit_stage253_stage252_slice_inputs f_lat, mu_eta	research/pde_ledger/notes/CHECKPOINT_CONSTANT_PROVENANCE.md:954	free_choice	audited
253	fit_stage253_upsilon_lat_calibration Upsilon_lat	research/pde_ledger/notes/CHECKPOINT_CONSTANT_PROVENANCE.md:950	free_choice	audited

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
253	fit_stage253_upsilon_lat_calibration_factor Upsilon_lat	notes/stages/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.md:185	free_choice	audited
253	fit_stage_253_chi_lambda chi_lambda	notes/stages/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.md:62(3 manifest anchors)	internal_consistency	provenance_built
253	fit_stage_253_f_lat f_lat, mu_eta, s_0, stage_253	redteam/pass2/reports/stage_253.md:117	free_choice	audited
253	fit_stage_253_k_corr K_corr, K_turn, Pi_T, Pi_chi, Pi_ep, Pi_k, T_max, Upsilon_lat, a_int, chi_lambda, f_lat, gamma_lat, gamma_lattice, k_eff, lambda_ep, lambda_phys, lambda_ref, mu_eta, omega_D, r_turn, s_0, s_c, zeta_ep	redteam/pass2/reports/stage_253.md:35	free_choice	audited
253	fit_stage_253_k_turn K_turn, Pi_T, Pi_ep, Pi_k, a_int, chi_lambda, f_lat, gamma_lat, gamma_lattice_legacy, k_eff, lambda_ref, mu_eta, r_turn, s_0, s_c	redteam/pass2/reports/stage_253.md:39	free_choice	audited
253	fit_stage_253_mathcal mathcal	notes/stages/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.md:559	no_constraint_record	provenance_built
253	fit_stage_253_max max	notes/stages/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.md:54	no_constraint_record	provenance_built
253	fit_stage_253_moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_mathematica_audit moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_mathematica_audit	notes/stages/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.md:74	no_constraint_record	provenance_built

Stage	Candidate / parameter	Source anchor	constraint_kind	Audit disposition
253	`fit_stage_253_moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit` `moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit`	`notes/stages/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.md`:69	`no_constraint_record`	provenance_built
253	`fit_stage_253_pi_ep` `Pi_ep`, `T_max`, `a_int`, `f_lat`, `gamma_lat`, `mu_eta`, `s_0`, `s_c`	`redteam/pass2/reports/stage_253.md`:93	`internal_consistency`	provenance_built
253	`fit_stage_253_subsection` `subsection`	`paper/stages/stage_253.tex`:6	`no_constraint_record`	provenance_built

L.3 Visible Grouping and Exclusion Accounting

No Phase-A candidate is silently dropped. Alias-folded scanned candidates remain listed in the main table and are also enumerated here. Records whose provenance YAML has no constraints block are listed with constraint_kind no_constraint_record rather than being forced into a fit-vs-derive class.

Table L.3: Alias accounting from the dedup proposal.

Alias candidate	Canonical candidate
`fit_stage022_source_map_branch_mhat0_unity`	`fit_stage022_mhat0_unity_branch`
`fit_stage073_wall_fraction_epsilon_r`	`fit_stage073_epsilon_r_posited`
`fit_stage105_chiQ_eq_1_by_matching_card`	`fit_stage_105_chi_q`
`fit_stage105_chi_q_canonical_fix`	`fit_stage_105_chi_q`
`fit_stage121_aspect_ratio_37_over_20`	`fit_stage121_aspect_ratio_37_20_reference_freeze`
`fit_stage168_sigma0_num_digit_variant`	`fit_stage168_canonical_mouth_block`
`fit_stage222_vknown_barrier_benchmark`	`fit_stage222_illustrative_barrier_benchmark`

Appendix M

Layer-2 Parameter-Provenance Audit

M.1 Generated Coverage Summary

This generated appendix exposes the parameter-level genealogy used by the layer-2 fit-vs-derive audit. It covers 1017 top-level provenance records. The audit result is operational: no surviving fatal flaw was found in this toy-analog ledger; this is not a truth claim about the model.

Table M.1: Parameter-provenance generated coverage totals.

Item	Count
Top-level provenance parameter records	1017
Primary canonical families	563
Canonical candidates	915
Alias candidates excluded from canonical count but surfaced below	7
constraint_kind: internal_consistency	685
constraint_kind: free_choice	219
constraint_kind: published_target	57
constraint_kind: no_constraint_record	56

M.2 Published-Target Benchmark Sources

The rows below are the benchmark citations loaded from benchmarks.yaml. Published-target provenance rows cite these entries where their family map joins to one of these family identifiers; if no join exists, the row states that absence explicitly.

Table M.2: Benchmark citations available to published-target provenance rows.

Family	Type	Citation
`fam_0093_gamma_5`	textbook	bench_a187d7865e33: P. C. Peters, 'Gravitational Radiation and the Motion of Two Point Masses', Phys. Rev. 136, B1224 (1964), DOI 10.1103/PhysRev.136.B1224 (ADS bibcode 1964PhRv..136.1224P); see also M. Maggiore, 'Gravitational Waves, Vol. 1: Theory and Experiments', Oxford Univ. Press (2007), ISBN 978-0-19-857074-5, Ch. 3 (quadrupole formula L = (G/5c^5)<dddQ_ij dddQ_ij>).

Family	Type	Citation
fam_0174_n_q	textbook	bench_62250fff83d4: P. C. Peters, 'Gravitational Radiation and the Motion of Two Point Masses', Phys. Rev. 136, B1224 (1964), DOI 10.1103/PhysRev.136.B1224 (ADS bibcode 1964PhRv..136.1224P); see also M. Maggiore, 'Gravitational Waves, Vol. 1: Theory and Experiments', Oxford Univ. Press (2007), ISBN 978-0-19-857074-5, Ch. 3 (quadrupole formula L = (G/5c^5)<dddQ__ij dddQ__ij>).
fam_0187_p0_target	textbook	bench_2e5805a359c4: P. C. Peters, 'Gravitational Radiation and the Motion of Two Point Masses', Phys. Rev. 136, B1224 (1964), DOI 10.1103/PhysRev.136.B1224 (ADS bibcode 1964PhRv..136.1224P); see also M. Maggiore, 'Gravitational Waves, Vol. 1: Theory and Experiments', Oxford Univ. Press (2007), ISBN 978-0-19-857074-5, Ch. 3 (quadrupole formula L = (G/5c^5)<dddQ__ij dddQ__ij>).
fam_0457_m_s	CODATA	bench_051666ee239b: NIST CODATA, proton-electron mass ratio (m__p/m__e), https://physics.nist.gov/cgi-bin/cuu/Value?mpsme ; CODATA Internationally recommended values of the fundamental physical constants (2018 and 2022 adjustments).
fam_0458_m_s_ratio_1836	CODATA	bench_5d8d048b10f9: NIST CODATA, proton-electron mass ratio (m__p/m__e), https://physics.nist.gov/cgi-bin/cuu/Value?mpsme ; CODATA Internationally recommended values of the fundamental physical constants (2018 and 2022 adjustments).

M.3 Parameter Genealogy Rows

Table M.3: Parameter-level provenance records with basis and downstream dependents.

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
001	Kbar_0 fit_stage001_canonical_invariant_pair_determined fam_0141_kbar_0	origin stage 1 notes/stages/moving_throat_pde_stage001_geometry_lift.md:473	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage001_geometry_lift.md:471	022, 023
001	K_eta fit_stage001_wall_action_constitutive_coefficients fam_0129_k_eta, fam_0249_t_omega, fam_0260_t_w, fam_0478_mu_eta; overlays:chain_calibration_245_253, chain_wall_action	origin stage 1 notes/stages/moving_throat_pde_stage001_geometry_lift.md:204	free_choice	posited choice:notes/stages/moving_throat_pde_stage001_geometry_lift.md:214	002, 003
001	T_Omega fit_stage001_wall_action_constitutive_coefficients fam_0129_k_eta, fam_0249_t_omega, fam_0260_t_w, fam_0478_mu_eta; overlays:chain_calibration_245_253, chain_wall_action	origin stage 1 notes/stages/moving_throat_pde_stage001_geometry_lift.md:203	free_choice	posited choice:notes/stages/moving_throat_pde_stage001_geometry_lift.md:214	002, 003

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
001	T_w fit_stage001_wall_action_constitutive_coefficients fam_0129_k_eta, fam_0249_t_omega, fam_0260_t_w, fam_0478_mu_eta; overlays:chain_calibration_245_253, chain_wall_action	origin stage 1 notes/stages/moving_throat_pde_stage001_geometry_lift.md:202	free_choice	posited choice:notes/stages/moving_throat_pde_stage001_geometry_lift.md:214	002, 003
001	mu_eta fit_stage001_wall_action_constitutive_coefficients fam_0129_k_eta, fam_0249_t_omega, fam_0260_t_w, fam_0478_mu_eta; overlays:chain_calibration_245_253, chain_wall_action	origin stage 1 notes/stages/moving_throat_pde_stage001_geometry_lift.md:201	free_choice	posited choice:notes/stages/moving_throat_pde_stage001_geometry_lift.md:214	002, 003
001	K_l fit_stage_001_k_l fam_0133_k_l	origin stage 1 notes/stages/moving_throat_pde_stage001_geometry_lift.md:239	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage001_geometry_lift.md:239	002
002	K_2 fit_stage002_grouped_p2_stiffness_gate fam_0121_k_2	origin stage 2 notes/stages/moving_throat_pde_stage002_breathing_reduction.md:142	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage002_breathing_reduction.md:147	003
002	overlap_factor_4pi fit_stage002_overlap_factor_4pi_derived fam_0487_overlap_factor_4pi	origin stage 2 notes/stages/moving_throat_pde_stage002_breathing_reduction.md:83	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage002_breathing_reduction.md:21	003
002	Gamma_AB fit_stage_002_gamma_ab fam_0096_gamma_ab	origin stage 2 notes/stages/moving_throat_pde_stage002_breathing_reduction.md:108	free_choice	posited choice:notes/stages/moving_throat_pde_stage002_breathing_reduction.md:112	none recorded
003	wall_self_energy fit_stage003_wall_self_energy_no_fitted_data fam_0539_wall_self_energy	origin stage 3 notes/stages/moving_throat_pde_stage003_bdg_coupling.md:121	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage003_bdg_coupling.md:87	none recorded
004	I_WS_delta fit_stage_004_i_ws_delta fam_0108_i_ws_delta	origin stage 4 notes/em_projected/step_04_projection_reduction_comparison_notes.md:106	internal_consistency	located derivation step:notes/em_projected/step_04_projection_reduction_comparison_notes.md:100	005, 007
004	StatusExact fit_stage_004_statusexact fam_0246_statusexact	origin stage 004 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
005	assert_nonzero fit_stage_005_assert_nonzero fam_0317_assert_nonzero	origin stage 005 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
006	ampere_sign fit_stage006_projected_vector_signs_fixed_by_audit fam_0316_ampere_sign	origin stage 6 notes/em_projected/step_03_projected_maxwell_vector_notes.md:60	internal_consistency	located derivation step:notes/em_projected/step_03_projected_maxwell_vector_notes.md:68	008
006	J_i fit_stage_006_j_i fam_0113_j_i	origin stage 6 notes/em_projected/step_03_projected_maxwell_vector_notes.md:68	internal_consistency	located derivation step:notes/em_projected/step_03_projected_maxwell_vector_notes.md:109	008
008	S_w fit_stage008_source_profile_must_match_weight fam_0235_s_w	origin stage 8 notes/em_projected/step_05_projected_maxwell_extension_notes.md:220	internal_consistency	located derivation step:notes/em_projected/step_05_projected_maxwell_extension_notes.md:213	009, 010
008	inv_xi fit_stage_008_inv_xi fam_0428_inv_xi	origin stage 8 notes/em_projected/step_05_projected_maxwell_extension_notes.md:137	internal_consistency	located derivation step:notes/em_projected/step_05_projected_maxwell_extension_notes.md:139	none recorded
008	Z2_int fit_stage_008_z2_int fam_0300_z2_int	origin stage 8 notes/em_projected/step_05_projected_maxwell_extension_notes.md:222	internal_consistency	located derivation step:notes/em_projected/step_05_projected_maxwell_extension_notes.md:220	009, 010
009	mu_1 fit_stage009_mu1_kernel_specialization fam_0477_mu_1	origin stage 9 notes/em_projected/step_07_projected_maxwell_near_throat_notes.md:202	internal_consistency	located derivation step:notes/em_projected/step_07_projected_maxwell_near_throat_notes.md:194	none recorded
009	matched_fingerprint_value fit_stage_009_matched_fingerprint_value fam_0459_matched_fingerprint_value	origin stage 009 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
010	C_fixed fit_stage_010_c_fixed fam_0027_c_fixed	origin stage 10 notes/em_projected/step_08_projected_maxwell_push_bundle_master_notes.md:37	internal_consistency	located derivation step:notes/em_projected/step_08_projected_maxwell_push_bundle_master_notes.md:26	011, 012
010	dK_norm fit_stage_010_dk_norm fam_0356_dk_norm	origin stage 10 notes/em_projected/step_08_projected_maxwell_push_bundle_master_notes.md:53	internal_consistency	located derivation step:notes/em_projected/step_08_projected_maxwell_push_bundle_master_notes.md:60	011, 012
010	R_fixed fit_stage_010_r_fixed fam_0215_r_fixed	origin stage 10 notes/em_projected/step_08_projected_maxwell_push_bundle_master_notes.md:37	internal_consistency	located derivation step:notes/em_projected/step_08_projected_maxwell_push_bundle_master_notes.md:32	011, 012

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
011	B_0 fit_stage_011_b_0 fam_0014_b_0	origin stage 10 notes/em_projected/step_08_projected_maxwell_push_bundle_master_notes.md:53	free_choice	posited choice:notes/em_projected/step_09_projected_maxwell_p2_bridge_notes.md:50	012
012	assert_nonzero fit_stage_012_assert_nonzero fam_0317_assert_nonzero	origin stage 012 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
012	Delta fit_stage_012_delta fam_0051_delta	origin stage 12 notes/em_projected/step_10_projected_maxwell_stage4_primitive_bridge_notes.md:21	internal_consistency	located derivation step:notes/em_projected/step_10_projected_maxwell_stage4_primitive_bridge_notes.md:19	013, 014
012	K_norm fit_stage_012_k_norm fam_0136_k_norm	origin stage 10 notes/em_projected/step_08_projected_maxwell_push_bundle_master_notes.md:53	internal_consistency	located derivation step:notes/em_projected/step_10_projected_maxwell_stage4_primitive_bridge_notes.md:58	013
012	n_n fit_stage_012_n_n fam_0482_n_n	origin stage 11 notes/em_projected/step_09_projected_maxwell_p2_bridge_notes.md:94	internal_consistency	located derivation step:notes/em_projected/step_10_projected_maxwell_stage4_primitive_bridge_notes.md:74	013, 014
014	He_bundle_coefficients fit_stage014_even_gate_constants_2_3_1_27 fam_0103_he_bundle_coefficients	origin stage 14 notes/em_projected/step_12_projected_maxwell_mouth_taylor_gate_bridge_notes.md:135	internal_consistency	located derivation step:notes/em_projected/step_12_projected_maxwell_mouth_taylor_gate_bridge_notes.md:131	017
015	S_EM fit_stage_015_s_em fam_0227_s_em	origin stage 15 notes/em_projected/step_13_parent_throat_action_master_notes.md:35	free_choice	posited choice:notes/em_projected/step_13_parent_throat_action_master_notes.md:35	016, 017
016	boundary_value fit_stage_016_boundary_value fam_0331_boundary_value	origin stage 016 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
017	wall_gate_determinant fit_stage017_even_gate_determinant_1_27 fam_0538_wall_gate_determinant	origin stage 17 notes/em_projected/step_15_parent_throat_action_weak_axisym_notes.md:153	internal_consistency	located derivation step:notes/em_projected/step_15_parent_throat_action_weak_axisym_notes.md:154	none recorded
017	paragraph fit_stage_017_paragraph fam_0488_paragraph	origin stage 017 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
018	coeff_11 fit_stage018_stale_provenance_anchor_9_11_8 fam_0351_coeff_11	origin stage 018 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	030

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream

019	`K_Sigma`
	`fit_stage019_K_Sigma_fixed_by_one_pole`
	`fam_0123_k_sigma`;
	overlays:`chain_wall_action`
		origin stage 019
		`notes/stages/moving_throat_pde_stage023_full_grouped_bundle.md`:210
			`internal_consistency`	located derivation step:`notes/stages/moving_throat_pde_stage023_full_grouped_bundle.md`:296	016, 018, 019

019	`K_Sigma`
	`fit_stage019_K_Sigma_fixed_by_outgoing_normalization`
	`fam_0123_k_sigma`;
	overlays:`chain_wall_action`
		origin stage 019
		`notes/stages/moving_throat_pde_stage023_full_grouped_bundle.md`:210
			`published_target`	published_target; no benchmarks.yaml entry joined for listed family; evidence:`notes/stages/moving_throat_pde_stage023_full_grouped_bundle.md`:308	016, 018, 019

019	`P0_target`
	`fit_stage019_P0_target_upstream_pin`
	`fam_0187_p0_target`;
	overlays:`chain_quad_54_5`, `chain_wall_action`
		origin stage 18
		`notes/stages/moving_throat_pde_stage022_grouped_p2_normalization_bridge.md`:270
			`published_target`	benchmark: bench_2e5805a359c4: textbook - P. C. Peters, 'Gravitational Radiation and the Motion of Two Point Masses', Phys. Rev. 136, B1224 (1964), DOI 10.1103/PhysRev.136.B1224 (ADS bibcode 1964PhRv..136.1224P); see also M. Maggiore, 'Gravitational Waves, Vol. 1: Theory and Experiments', Oxford Univ. Press (2007), ISBN 978-0-19-857074-5, Ch. 3 (quadrupole formula L = (G/5c^5)<dddQ_ij dddQ_ij>).	016, 019, 025, 191

019	`P0_target`
	`fit_stage019_p0target_54_5_raw_retype`
	`fam_0187_p0_target`;
	overlays:`chain_quad_54_5`, `chain_wall_action`
		origin stage 18
		`notes/stages/moving_throat_pde_stage022_grouped_p2_normalization_bridge.md`:270
			`published_target`	benchmark: bench_2e5805a359c4: textbook - P. C. Peters, 'Gravitational Radiation and the Motion of Two Point Masses', Phys. Rev. 136, B1224 (1964), DOI 10.1103/PhysRev.136.B1224 (ADS bibcode 1964PhRv..136.1224P); see also M. Maggiore, 'Gravitational Waves, Vol. 1: Theory and Experiments', Oxford Univ. Press (2007), ISBN 978-0-19-857074-5, Ch. 3 (quadrupole formula L = (G/5c^5)<dddQ_ij dddQ_ij>).	016, 019, 025, 191

019	`P0_target`
	`fit_stage_019_p0_target`
	`fam_0187_p0_target`;
	overlays:`chain_quad_54_5`, `chain_wall_action`
		origin stage 18
		`notes/stages/moving_throat_pde_stage022_grouped_p2_normalization_bridge.md`:270
			`published_target`	benchmark: bench_2e5805a359c4: textbook - P. C. Peters, 'Gravitational Radiation and the Motion of Two Point Masses', Phys. Rev. 136, B1224 (1964), DOI 10.1103/PhysRev.136.B1224 (ADS bibcode 1964PhRv..136.1224P); see also M. Maggiore, 'Gravitational Waves, Vol. 1: Theory and Experiments', Oxford Univ. Press (2007), ISBN 978-0-19-857074-5, Ch. 3 (quadrupole formula L = (G/5c^5)<dddQ_ij dddQ_ij>).	016, 019, 025, 191

020	`even_gate_determinant`
	`fit_stage020_even_gate_determinant_1_27`
	`fam_0398_even_gate_determinant`
		origin stage 020
		not recorded
			`no_constraint_record`	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded

020	`Xi1`
	`fit_stage020_wall_slopes_fixed_uniquely`
	`fam_0282_xi1`
		origin stage 020
		not recorded
			`no_constraint_record`	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	020, 177, 189

020	`B_0`
	`fit_stage_020_b_0`
	`fam_0014_b_0`
		origin stage 19
		`notes/stages/moving_throat_pde_stage023_full_grouped_bundle.md`:144
			`free_choice`	posited choice:`notes/stages/moving_throat_pde_stage023_full_grouped_bundle.md`:150	019, 020

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
021	a fit_stage021_outgoing_fingerprint_omega5_coefficient fam_0301_a	origin stage 17 notes/stages/moving_throat_pde_stage021_reduced_one_port_normal_form.md:219	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage021_reduced_one_port_normal_form.md:214	018, 019
021	Omega_A_l fit_stage021_retained_reduced_adapter_couplings fam_0181_omega_a_l	origin stage 021 notes/stages/moving_throat_pde_stage021_reduced_one_port_normal_form.md:89	free_choice	posited choice:notes/stages/moving_throat_pde_stage021_reduced_one_port_normal_form.md:82	018, 019
021	C_a fit_stage_021_c_a fam_0025_c_a	origin stage 17 notes/stages/moving_throat_pde_stage021_reduced_one_port_normal_form.md:72	free_choice	posited choice:notes/stages/moving_throat_pde_stage021_reduced_one_port_normal_form.md:74	none recorded
021	EQ_COMPACT_L2_FINGERPRINT fit_stage_021_eq_compact_l2_fingerprint fam_0068_eq_compact_l2_fingerprint	origin stage 17 notes/stages/moving_throat_pde_stage021_reduced_one_port_normal_form.md:219	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage021_reduced_one_port_normal_form.md:214	018, 019
021	Gamma_5 fit_stage_021_gamma5_port fam_0093_gamma_5; overlays:chain_quad_54_5	origin stage 17 notes/stages/moving_throat_pde_stage021_reduced_one_port_normal_form.md:223	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage021_reduced_one_port_normal_form.md:214	018, 019
021	matched_fingerprint_value fit_stage_021_matched_fingerprint_value fam_0459_matched_fingerprint_value	origin stage 17 notes/stages/moving_throat_pde_stage021_reduced_one_port_normal_form.md:219	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage021_reduced_one_port_normal_form.md:225	018, 019
021	paragraph fit_stage_021_paragraph fam_0488_paragraph	origin stage 021 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
022	Gamma_5 fit_stage022_gamma_GR_quadrupole_target fam_0093_gamma_5; overlays:chain_quad_54_5	origin stage 18 notes/stages/moving_throat_pde_stage022_grouped_p2_normalization_bridge.md:262	published_target	benchmark: bench__a187d7865e33: textbook - P. C. Peters, 'Gravitational Radiation and the Motion of Two Point Masses', Phys. Rev. 136, B1224 (1964), DOI 10.1103/PhysRev.136.B1224 (ADS bibcode 1964PhRv..136.1224P); see also M. Maggiore, 'Gravitational Waves, Vol. 1: Theory and Experiments', Oxford Univ. Press (2007), ISBN 978-0-19-857074-5, Ch. 3 (quadrupole formula L = (G/5c^5)<dddQ__ij dddQ__ij>).	019
022	N_Q_universal fit_stage022_invariant_product_must_match_universal fam_0175_n_q_universal	origin stage 18 notes/stages/moving_throat_pde_stage022_grouped_p2_normalization_bridge.md:270	published_target	published__target; no benchmarks.yaml entry joined for listed family; evidence:notes/stages/moving_throat_pde_stage022_grouped_p2_normalization_bridge.md:256	019

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
022	mhat_0 fit_stage022_mhat0_unity_branch fam_0469_mhat_0; overlays:chain_chi_Q_norm, chain_quad_54_5, chain_wall_action	origin stage 18 notes/stages/moving_throat_pde_stage022_grouped_p2_normalization_bridge.md:279	free_choice	posited choice:notes/stages/moving_throat_pde_stage022_grouped_p2_normalization_bridge.md:279	019, 020
022	P0_target fit_stage022_p0_quadrupole_normalization_target fam_0187_p0_target; overlays:chain_quad_54_5, chain_wall_action	origin stage 18 notes/stages/moving_throat_pde_stage022_grouped_p2_normalization_bridge.md:270	published_target	benchmark: bench_2e5805a359c4: textbook - P. C. Peters, 'Gravitational Radiation and the Motion of Two Point Masses', Phys. Rev. 136, B1224 (1964), DOI 10.1103/PhysRev.136.B1224 (ADS bibcode 1964PhRv..136.1224P); see also M. Maggiore, 'Gravitational Waves, Vol. 1: Theory and Experiments', Oxford Univ. Press (2007), ISBN 978-0-19-857074-5, Ch. 3 (quadrupole formula L = (G/5c^5)<dddQ_ij dddQ_ij>).	016, 019, 025, 191
022	D_0 fit_stage_022_d_0 fam_0040_d_0	origin stage 18 notes/stages/moving_throat_pde_stage022_grouped_p2_normalization_bridge.md:30	free_choice	posited choice:notes/stages/moving_throat_pde_stage022_grouped_p2_normalization_bridge.md:74	019, 020
022	D_A fit_stage_022_d_a fam_0043_d_a	origin stage 18 notes/stages/moving_throat_pde_stage022_grouped_p2_normalization_bridge.md:30	free_choice	posited choice:notes/stages/moving_throat_pde_stage022_grouped_p2_normalization_bridge.md:47	019, 020
022	G_5 fit_stage_022_g_5 fam_0085_g_5	origin stage 17 notes/stages/moving_throat_pde_stage021_reduced_one_port_normal_form.md:219	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage021_reduced_one_port_normal_form.md:214	019
022	matched_fingerprint_value fit_stage_022_matched_fingerprint_value fam_0459_matched_fingerprint_value	origin stage 17 notes/stages/moving_throat_pde_stage021_reduced_one_port_normal_form.md:219	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage022_grouped_p2_normalization_bridge.md:151	019
022	P0_target fit_stage_022_p_0 fam_0187_p0_target; overlays:chain_quad_54_5, chain_wall_action	origin stage 18 notes/stages/moving_throat_pde_stage022_grouped_p2_normalization_bridge.md:270	published_target	benchmark: bench_2e5805a359c4: textbook - P. C. Peters, 'Gravitational Radiation and the Motion of Two Point Masses', Phys. Rev. 136, B1224 (1964), DOI 10.1103/PhysRev.136.B1224 (ADS bibcode 1964PhRv..136.1224P); see also M. Maggiore, 'Gravitational Waves, Vol. 1: Theory and Experiments', Oxford Univ. Press (2007), ISBN 978-0-19-857074-5, Ch. 3 (quadrupole formula L = (G/5c^5)<dddQ_ij dddQ_ij>).	016, 019, 025, 191

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
023	Gamma_5 fit_stage023_gamma5_port_27_coefficient fam_0093_gamma_5; overlays:chain_quad_54_5	origin stage 17 notes/stages/moving_throat_pde_stage021_reduced_one_port_normal_form.md:223	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage021_reduced_one_port_normal_form.md:214	020
023	N_Q fit_stage023_nq_target_gr_quadrupole_match fam_0174_n_q; overlays:chain_chi_Q_norm, chain_quad_54_5	origin stage 18 notes/stages/moving_throat_pde_stage022_grouped_p2_normalization_bridge.md:270	published_target	benchmark: bench__62250fff83d4: textbook - P. C. Peters, 'Gravitational Radiation and the Motion of Two Point Masses', Phys. Rev. 136, B1224 (1964), DOI 10.1103/PhysRev.136.B1224 (ADS bibcode 1964PhRv..136.1224P); see also M. Maggiore, 'Gravitational Waves, Vol. 1: Theory and Experiments', Oxford Univ. Press (2007), ISBN 978-0-19-857074-5, Ch. 3 (quadrupole formula L = (G/5c^5)<dddQ__ij dddQ__ij>).	020, 025, 191
023	B_0 fit_stage_023_b_0 fam_0014_b_0	origin stage 19 notes/stages/moving_throat_pde_stage023_full_grouped_bundle.md:144	free_choice	posited choice:notes/stages/moving_throat_pde_stage023_full_grouped_bundle.md:150	020
023	N_n fit_stage_023_n_n fam_0177_n_n	origin stage 19 notes/stages/moving_throat_pde_stage023_full_grouped_bundle.md:198	free_choice	posited choice:notes/stages/moving_throat_pde_stage023_full_grouped_bundle.md:288	020
024	b_over_a fit_stage024_b3a_signature_follows_directly fam_0320_b_over_a	origin stage 20 notes/stages/moving_throat_pde_stage024_overlap_isotropy.md:255	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage024_overlap_isotropy.md:249	164, 200
024	grouped_lane_coefficients_20_21_22 fit_stage024_grouped_bundle_forced_isotropic fam_0425_grouped_lane_coefficients_20_21_22	origin stage 20 notes/stages/moving_throat_pde_stage024_overlap_isotropy.md:183	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage024_overlap_isotropy.md:193	021, 022
024	mhat_ang fit_stage024_mhat_ang_exactly_one fam_0470_mhat_ang	origin stage 20 notes/stages/moving_throat_pde_stage024_overlap_isotropy.md:78	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage024_overlap_isotropy.md:68	021, 022
024	b_over_a fit_stage024_quadrupole_splitting_signature fam_0320_b_over_a	origin stage 20 notes/stages/moving_throat_pde_stage024_overlap_isotropy.md:255	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage024_overlap_isotropy.md:211	164, 200
024	kappa fit_stage_024_kappa fam_0431_kappa	origin stage 20 notes/stages/moving_throat_pde_stage024_overlap_isotropy.md:221	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage024_overlap_isotropy.md:211	022

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
024	`matched_fingerprint_value` `fit_stage_024_matched_fingerprint_value` `fam_0459_matched_fingerprint_value`	origin stage 20 `notes/stages/moving_throat_pde_stage024_overlap_isotropy.md`:183	`internal_consistency`	located derivation step:`notes/stages/moving_throat_pde_stage024_overlap_isotropy.md`:70	021, 022
025	`mhat_ang` `fit_stage025_mhat_ang_already_fixed` `fam_0470_mhat_ang`	origin stage 24 `notes/stages/moving_throat_pde_stage024_overlap_isotropy.md`:78	`internal_consistency`	located derivation step:`notes/stages/moving_throat_pde_stage025_minimal_isotropic_normalization.md`:95	026, 030, 031, 033
025	`mhat_rad` `fit_stage025_mhat_rad_tuned_to_target` `fam_0471_mhat_rad`	origin stage 25 `notes/stages/moving_throat_pde_stage025_minimal_isotropic_normalization.md`:97	`published_target`	published_target; no benchmarks.yaml entry joined for listed family; evidence:`notes/stages/moving_throat_pde_stage025_minimal_isotropic_normalization.md`:101	026, 030, 031, 033
025	`P0_target` `fit_stage025_one_mode_normalization_target` `fam_0187_p0_target`; overlays:`chain_quad_54_5`, `chain_wall_action`	origin stage 22 `notes/stages/moving_throat_pde_stage022_grouped_p2_normalization_bridge.md`:270	`published_target`	benchmark: bench__2e5805a359c4: textbook - P. C. Peters, 'Gravitational Radiation and the Motion of Two Point Masses', Phys. Rev. 136, B1224 (1964), DOI 10.1103/PhysRev.136.B1224 (ADS bibcode 1964PhRv..136.1224P); see also M. Maggiore, 'Gravitational Waves, Vol. 1: Theory and Experiments', Oxford Univ. Press (2007), ISBN 978-0-19-857074-5, Ch. 3 (quadrupole formula L = (G/5c^5)<dddQ__ij dddQ__ij>).	026, 030, 031, 033
025	`N_Q` `fit_stage025_radial_source_map_normalization` `fam_0174_n_q`; overlays:`chain_chi_Q_norm`, `chain_quad_54_5`	origin stage 22 `notes/stages/moving_throat_pde_stage022_grouped_p2_normalization_bridge.md`:270	`published_target`	benchmark: bench__62250fff83d4: textbook - P. C. Peters, 'Gravitational Radiation and the Motion of Two Point Masses', Phys. Rev. 136, B1224 (1964), DOI 10.1103/PhysRev.136.B1224 (ADS bibcode 1964PhRv..136.1224P); see also M. Maggiore, 'Gravitational Waves, Vol. 1: Theory and Experiments', Oxford Univ. Press (2007), ISBN 978-0-19-857074-5, Ch. 3 (quadrupole formula L = (G/5c^5)<dddQ__ij dddQ__ij>).	026, 030
025	`P0_target` `fit_stage025_target_coefficient_54_5` `fam_0187_p0_target`; overlays:`chain_quad_54_5`, `chain_wall_action`	origin stage 22 `notes/stages/moving_throat_pde_stage022_grouped_p2_normalization_bridge.md`:270	`published_target`	benchmark: bench__2e5805a359c4: textbook - P. C. Peters, 'Gravitational Radiation and the Motion of Two Point Masses', Phys. Rev. 136, B1224 (1964), DOI 10.1103/PhysRev.136.B1224 (ADS bibcode 1964PhRv..136.1224P); see also M. Maggiore, 'Gravitational Waves, Vol. 1: Theory and Experiments', Oxford Univ. Press (2007), ISBN 978-0-19-857074-5, Ch. 3 (quadrupole formula L = (G/5c^5)<dddQ__ij dddQ__ij>).	026, 030, 031, 033
025	`D_0` `fit_stage_025_d_0` `fam_0040_d_0`	origin stage 25 `notes/stages/moving_throat_pde_stage025_minimal_isotropic_normalization.md`:77	`internal_consistency`	located derivation step:`notes/stages/moving_throat_pde_stage025_minimal_isotropic_normalization.md`:77	026

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
025	P0_target fit_stage_025_p0_compact fam_0187_p0_target; overlays:chain_quad_54_5, chain_wall_action	origin stage 22 notes/stages/moving_throat_pde_stage022_grouped_p2_normalization_bridge.md:270	published_target	benchmark: bench_2e5805a359c4: textbook - P. C. Peters, 'Gravitational Radiation and the Motion of Two Point Masses', Phys. Rev. 136, B1224 (1964), DOI 10.1103/PhysRev.136.B1224 (ADS bibcode 1964PhRv..136.1224P); see also M. Maggiore, 'Gravitational Waves, Vol. 1: Theory and Experiments', Oxford Univ. Press (2007), ISBN 978-0-19-857074-5, Ch. 3 (quadrupole formula L = (G/5c^5)<dddQ_ij dddQ_ij>).	026, 030, 031, 033
026	K_req fit_stage026_K_req_solved_exactly_from_target fam_0138_k_req	origin stage 26 notes/stages/moving_throat_pde_stage026_concrete_axial_overlaps.md:214	published_target	published_target; no benchmarks.yaml entry joined for listed family; evidence:notes/stages/moving_throat_pde_stage026_concrete_axial_overlaps.md:218	027
026	K_req fit_stage026_Kreq_solved_backward fam_0138_k_req	origin stage 26 notes/stages/moving_throat_pde_stage026_concrete_axial_overlaps.md:214	published_target	published_target; no benchmarks.yaml entry joined for listed family; evidence:notes/stages/moving_throat_pde_stage026_concrete_axial_overlaps.md:212	027
026	dn_branch_constants fit_stage026_dn_constants_exact_derived_downstream fam_0378_dn_branch_constants	origin stage 26 notes/stages/moving_throat_pde_stage026_concrete_axial_overlaps.md:166	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage026_concrete_axial_overlaps.md:252	027, 035, 036
026	kappa_0 fit_stage026_dn_mode_basis_kappa0 fam_0433_kappa_0	origin stage 26 notes/stages/moving_throat_pde_stage026_concrete_axial_overlaps.md:126	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage026_concrete_axial_overlaps.md:124	027, 032, 035, 036
026	kappa fit_stage026_kappa_exact_overlap fam_0431_kappa	origin stage 26 notes/stages/moving_throat_pde_stage026_concrete_axial_overlaps.md:126	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage026_concrete_axial_overlaps.md:122	027, 032, 035, 036
026	kappa fit_stage026_kappa_overlap_2sqrt2_over_pi fam_0431_kappa	origin stage 26 notes/stages/moving_throat_pde_stage026_concrete_axial_overlaps.md:122	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage026_concrete_axial_overlaps.md:257	027, 032, 035, 036
026	A_w fit_stage_026_a_w fam_0012_a_w	origin stage 026 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
026	G_U fit_stage_026_g_u fam_0086_g_u	origin stage 26 notes/stages/moving_throat_pde_stage026_concrete_axial_overlaps.md:170	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage026_concrete_axial_overlaps.md:160	027
026	K_req fit_stage_026_k_req fam_0138_k_req	origin stage 26 notes/stages/moving_throat_pde_stage026_concrete_axial_overlaps.md:214	published_target	published_target; no benchmarks.yaml entry joined for listed family; evidence:notes/stages/moving_throat_pde_stage026_concrete_axial_overlaps.md:212	027

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
027	kappa_theta fit_stage027_max_coupling_branch_not_free fam_0439_kappa_theta	origin stage 27 notes/stages/moving_throat_pde_stage027_nonconstant_axial_family.md:100	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage027_nonconstant_axial_family.md:99	028
027	kappa_max fit_stage027_sin2_theta_max_2_11 fam_0437_kappa_max	origin stage 27 notes/stages/moving_throat_pde_stage027_nonconstant_axial_family.md:115	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage027_nonconstant_axial_family.md:111	028
027	sin2_theta_max fit_stage027_sin2_theta_max_exact_value fam_0514_sin2_theta_max	origin stage 27 notes/stages/moving_throat_pde_stage027_nonconstant_axial_family.md:151	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage027_nonconstant_axial_family.md:143	028
028	alpha fit_stage028_alpha_loading_coefficient fam_0309_alpha	origin stage 28 notes/stages/moving_throat_pde_stage028_loaded_profile_selection.md:94	free_choice	posited choice:notes/stages/moving_throat_pde_stage028_loaded_profile_selection.md:96	029, 030, 031
028	alpha_crit fit_stage028_eigenvalues_exact fam_0312_alpha_crit	origin stage 28 notes/stages/moving_throat_pde_stage028_loaded_profile_selection.md:240	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage028_loaded_profile_selection.md:236	029, 030, 031
028	theta_minus fit_stage028_theta_minus_sign_fixed fam_0527_theta_minus	origin stage 28 notes/stages/moving_throat_pde_stage028_loaded_profile_selection.md:132	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage028_loaded_profile_selection.md:147	029, 030, 031
028	det_eff fit_stage_028_det_eff fam_0376_det_eff	origin stage 28 notes/stages/moving_throat_pde_stage028_loaded_profile_selection.md:119	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage028_loaded_profile_selection.md:234	029
029	alpha fit_stage029_alpha_derived_dynamically fam_0309_alpha	origin stage 29 notes/stages/moving_throat_pde_stage029_dynamic_loading.md:115	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage029_dynamic_loading.md:133	030, 033
029	DeltaK_ax fit_stage029_deltaK_algebraically_forced fam_0053_deltak_ax	origin stage 28 notes/stages/moving_throat_pde_stage028_loaded_profile_selection.md:70	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage029_dynamic_loading.md:64	030, 031
029	alpha_0 fit_stage029_mixed_sign_convention_match fam_0310_alpha_0	origin stage 29 notes/stages/moving_throat_pde_stage029_dynamic_loading.md:149	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage029_dynamic_loading.md:150	030, 033

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
030	lambda_req fit_stage030_lambda_req_target_ratio fam_0455_lambda_req	origin stage 30 notes/stages/moving_throat_pde_stage030_selected_mode_normalization.md:187	published_target	published_target; no benchmarks.yaml entry joined for listed family; evidence:notes/stages/moving_throat_pde_stage030_selected_mode_normalization.md:185	031, 033
030	N_Q fit_stage030_nq_target_retype_54_5 fam_0174_n_q; overlays:chain_chi_Q_norm, chain_quad_54_5	origin stage 22 notes/stages/moving_throat_pde_stage022_grouped_p2_normalization_bridge.md:270	published_target	benchmark: bench_62250fff83d4: textbook - P. C. Peters, 'Gravitational Radiation and the Motion of Two Point Masses', Phys. Rev. 136, B1224 (1964), DOI 10.1103/PhysRev.136.B1224 (ADS bibcode 1964PhRv..136.1224P); see also M. Maggiore, 'Gravitational Waves, Vol. 1: Theory and Experiments', Oxford Univ. Press (2007), ISBN 978-0-19-857074-5, Ch. 3 (quadrupole formula L = (G/5c^5)<dddQ_ij dddQ_ij>).	031, 033
030	N_Q fit_stage030_selected_branch_target_equation fam_0174_n_q; overlays:chain_chi_Q_norm, chain_quad_54_5	origin stage 22 notes/stages/moving_throat_pde_stage022_grouped_p2_normalization_bridge.md:270	published_target	benchmark: bench_62250fff83d4: textbook - P. C. Peters, 'Gravitational Radiation and the Motion of Two Point Masses', Phys. Rev. 136, B1224 (1964), DOI 10.1103/PhysRev.136.B1224 (ADS bibcode 1964PhRv..136.1224P); see also M. Maggiore, 'Gravitational Waves, Vol. 1: Theory and Experiments', Oxford Univ. Press (2007), ISBN 978-0-19-857074-5, Ch. 3 (quadrupole formula L = (G/5c^5)<dddQ_ij dddQ_ij>).	031, 033
030	beta_0s_lambda fit_stage_030_beta_0s_lambda fam_0325_beta_0s_lambda	origin stage 30 notes/stages/moving_throat_pde_stage030_selected_mode_normalization.md:152	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage030_selected_mode_normalization.md:154	031, 033
030	C_5 fit_stage_030_c_5 fam_0024_c_5	origin stage 30 notes/stages/moving_throat_pde_stage030_selected_mode_normalization.md:116	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage030_selected_mode_normalization.md:114	031
030	M_eff fit_stage_030_m_eff fam_0158_m_eff	origin stage 30 notes/stages/moving_throat_pde_stage030_selected_mode_normalization.md:43	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage030_selected_mode_normalization.md:43	031
031	alpha_crit fit_stage031_alpha_crit_derived_directly fam_0312_alpha_crit	origin stage 28 notes/stages/moving_throat_pde_stage028_loaded_profile_selection.md:240	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage031_selected_branch_reachability.md:112	033
031	dlambda_minus_dalpha fit_stage031_hf_identity_derived_not_assumed fam_0377_dlambda_minus_dalpha	origin stage 30 notes/stages/moving_throat_pde_stage030_selected_mode_normalization.md:84	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage031_selected_branch_reachability.md:63	033

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
031	alpha_star fit_stage031_unique_alpha_star_crossing fam_0315_alpha_star	origin stage 31 notes/stages/moving_throat_pde_stage031_selected_branch_reachability.md:147	published_target	published_target; no benchmarks.yaml entry joined for listed family; evidence:notes/stages/moving_throat_pde_stage031_selected_branch_reachability.md:151	033
031	delta_kappa fit_stage_031_delta_kappa fam_0371_delta_kappa	origin stage 28 notes/stages/moving_throat_pde_stage028_loaded_profile_selection.md:137	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage030_selected_mode_normalization.md:55	033
031	stale_output fit_stage_031_stale_output fam_0517_stale_output	origin stage 031 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
032	kappa_0 fit_stage032_kappa_values_exact fam_0433_kappa_0	origin stage 26 notes/stages/moving_throat_pde_stage032_source_map_from_mode_integrals.md:67	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage032_source_map_from_mode_integrals.md:63	027, 028, 029, 030, 031, 032, 033, 034, 038, 039
032	mhat_0 fit_stage032_mhat_minus_completely_fixed fam_0469_mhat_0; overlays:chain_chi_Q_norm, chain_quad_54_5, chain_wall_action	origin stage 26 notes/stages/moving_throat_pde_stage032_source_map_from_mode_integrals.md:201	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage032_source_map_from_mode_integrals.md:208	027
032	rank_one_loading_pattern fit_stage032_pattern_now_derived fam_0499_rank_one_loading_pattern	origin stage 029 notes/stages/moving_throat_pde_stage032_source_map_from_mode_integrals.md:130	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage032_source_map_from_mode_integrals.md:130	027, 031
032	sigma fit_stage032_sigma_88_9pi2_and_mhat_limit_11_9 fam_0505_sigma	origin stage 26 notes/stages/moving_throat_pde_stage032_source_map_from_mode_integrals.md:75	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage032_source_map_from_mode_integrals.md:75	027, 031, 038, 039
032	I_2 fit_stage_032_i_2 fam_0104_i_2	origin stage 26 notes/stages/moving_throat_pde_stage032_source_map_from_mode_integrals.md:146	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage032_source_map_from_mode_integrals.md:146	027, 031
032	mhat_sq fit_stage_032_mhat_sq fam_0472_mhat_sq	origin stage 26 notes/stages/moving_throat_pde_stage032_source_map_from_mode_integrals.md:205	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage032_source_map_from_mode_integrals.md:36	027
033	kappa_0_sq fit_stage033_dn_constants_exact fam_0434_kappa_0_sq	origin stage 26 notes/stages/moving_throat_pde_stage032_source_map_from_mode_integrals.md:79	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage033_microscopic_normalization_equation.md:63	028, 029, 030, 031, 038

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
033	kappa_0_sq fit_stage033_kappa_overlap_literals fam_0434_kappa_0_sq	origin stage 26 notes/stages/moving_throat_pde_stage032_source_map_from_mode_integrals.md:79	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage033_microscopic_normalization_equation.md:135	028, 029, 030, 031, 038
033	K0_onset fit_stage_033_k0_onset fam_0117_k0_onset	origin stage 27 notes/stages/moving_throat_pde_stage033_microscopic_normalization_equation.md:185	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage033_microscopic_normalization_equation.md:175	028, 029
034	N_minus fit_stage034_normal_form_maps_fixed fam_0176_n_minus	origin stage 26 notes/stages/moving_throat_pde_stage032_source_map_from_mode_integrals.md:34	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage034_softening_depth_normal_form.md:185	029, 030
034	kappa0_sq fit_stage_034_kappa0_sq fam_0432_kappa0_sq	origin stage 26 notes/stages/moving_throat_pde_stage032_source_map_from_mode_integrals.md:79	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage034_softening_depth_normal_form.md:70	029, 030
034	Pi_star fit_stage_034_pi_star fam_0209_pi_star	origin stage 034 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
035	xi_req fit_stage035_xi_req_unique_locus fam_0548_xi_req	origin stage 29 notes/stages/moving_throat_pde_stage035_dimensionless_normalization_locus.md:110	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage035_dimensionless_normalization_locus.md:112	030, 032, 033, 034
035	F_target fit_stage_035_f_target fam_0082_f_target	origin stage 29 notes/stages/moving_throat_pde_stage035_dimensionless_normalization_locus.md:61	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage035_dimensionless_normalization_locus.md:59	030, 032
035	R_target fit_stage_035_r_target fam_0222_r_target	origin stage 29 notes/stages/moving_throat_pde_stage035_dimensionless_normalization_locus.md:71	published_target	published_target; no benchmarks.yaml entry joined for listed family; evidence:notes/stages/moving_throat_pde_stage035_dimensionless_normalization_locus.md:72	030, 038, 039
035	to fit_stage_035_to fam_0528_to	origin stage 035 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
036	dn_coeff_11 fit_stage036_dn_coeffs_stale_stage18_anchor fam_0379_dn_coeff_11	origin stage 036 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
036	g_B_req fit_stage036_no_free_hidden_eigenvector_param fam_0405_g_b_req	origin stage 28 notes/stages/moving_throat_pde_stage034_softening_depth_normal_form.md:165	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage036_support_feasibility_frontier.md:66	031, 038

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
036	M_mix fit_stage_036_m_mix fam_0160_m_mix	origin stage 30 notes/stages/moving_throat_pde_stage036_support_feasibility_frontier.md:56	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage036_support_feasibility_frontier.md:54	031, 038, 039
037	A fit_stage037_branch_data_exact_kernel_functionals fam_0001_a	origin stage 27 notes/stages/moving_throat_pde_stage033_microscopic_normalization_equation.md:44	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage037_continuum_kernel_extraction.md:233	032, 033, 034, 038, 039
037	DeltaK_ax fit_stage_037_deltak_ax fam_0053_deltak_ax	origin stage 31 notes/stages/moving_throat_pde_stage037_continuum_kernel_extraction.md:139	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage037_continuum_kernel_extraction.md:235	032, 033, 034, 038, 039
037	fix_loop fit_stage_037_fix_loop fam_0403_fix_loop	origin stage 037 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
037	M_mix fit_stage_037_m_mix fam_0160_m_mix	origin stage 30 notes/stages/moving_throat_pde_stage036_support_feasibility_frontier.md:56	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage037_continuum_kernel_extraction.md:257	038, 039
038	delta fit_stage038_delta_fixed_by_eps_eta fam_0363_delta	origin stage 29 notes/stages/moving_throat_pde_stage035_dimensionless_normalization_locus.md:39	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage038_dimensionless_continuum_placement.md:87	032, 033, 034, 039
038	Lambda fit_stage038_lambda_external_target_coefficient fam_0148_lambda	origin stage 038 notes/stages/moving_throat_pde_stage038_dimensionless_continuum_placement.md:69	published_target	published_target; no benchmarks.yaml entry joined for listed family; evidence:notes/stages/moving_throat_pde_stage038_dimensionless_continuum_placement.md:67	032, 039
038	P0_target fit_stage038_mhat_p0_fitted_scales fam_0187_p0_target; overlays:chain_quad_54_5, chain_wall_action	origin stage 038 notes/stages/moving_throat_pde_stage038_dimensionless_continuum_placement.md:91	published_target	benchmark: bench_2e5805a359c4: textbook - P. C. Peters, 'Gravitational Radiation and the Motion of Two Point Masses', Phys. Rev. 136, B1224 (1964), DOI 10.1103/PhysRev.136.B1224 (ADS bibcode 1964PhRv..136.1224P); see also M. Maggiore, 'Gravitational Waves, Vol. 1: Theory and Experiments', Oxford Univ. Press (2007), ISBN 978-0-19-857074-5, Ch. 3 (quadrupole formula L = (G/5c^5)<dddQ_ij dddQ_ij>).	032, 039
039	delta_split fit_stage039_delta_split_exact_value fam_0375_delta_split	origin stage 039 notes/stages/moving_throat_pde_stage039_split_u_sector.md:97	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage039_split_u_sector.md:89	032, 033, 034

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
039	D_dir fit_stage_039_d_dir fam_0048_d_dir	origin stage 039 notes/stages/moving_throat_pde_stage039_split_u_sector.md:170	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage039_split_u_sector.md:168	032, 033, 034
040	lambda_minus fit_stage040_lambda_exact_closed_form fam_0451_lambda_minus	origin stage 32 notes/stages/moving_throat_pde_stage040_generalized_selected_branch.md:45	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage040_generalized_selected_branch.md:65	033, 034
041	n_req fit_stage041_n_req_exact_support_loading fam_0483_n_req	origin stage 33 notes/stages/moving_throat_pde_stage041_rank2_support_completion.md:82	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage041_rank2_support_completion.md:80	034
041	D_sel fit_stage_041_d_sel fam_0050_d_sel	origin stage 33 notes/stages/moving_throat_pde_stage041_rank2_support_completion.md:64	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage041_rank2_support_completion.md:62	034
042	q fit_stage042_overlaps_exact fam_0490_q	origin stage 32 notes/stages/moving_throat_pde_stage040_generalized_selected_branch.md:57	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage042_rank2_selected_mode_normalization.md:77	034
042	F_src fit_stage_042_f_src fam_0080_f_src	origin stage 34 notes/stages/moving_throat_pde_stage042_rank2_selected_mode_normalization.md:145	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage042_rank2_selected_mode_normalization.md:143	034
043	B_baseline fit_stage043_baseline_8_over_pi2_circular_derivation fam_0017_b_baseline	origin stage 43 notes/stages/moving_throat_pde_stage043_support_direction_extraction.md:149	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage043_support_direction_extraction.md:149	044
043	R_phi fit_stage043_support_direction_extracted_not_guessed fam_0219_r_phi	origin stage 43 notes/stages/moving_throat_pde_stage043_support_direction_extraction.md:103	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage043_support_direction_extraction.md:13	044, 045
043	D_phi fit_stage_043_d_phi fam_0049_d_phi	origin stage 43 notes/stages/moving_throat_pde_stage043_support_direction_extraction.md:119	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage043_support_direction_extraction.md:125	044

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
043	D_U fit_stage_043_d_u fam_0045_d_u	origin stage 43 notes/stages/moving_throat_pde_stage043_support_direction_extraction.md:75	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage043_support_direction_extraction.md:71	044
043	M_mix fit_stage_043_m_mix fam_0160_m_mix	origin stage 39 notes/stages/moving_throat_pde_stage043_support_direction_extraction.md:173	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage043_support_direction_extraction.md:105	044, 045, 047
044	M_supp fit_stage044_rank2_inserted_direction_data fam_0170_m_supp	origin stage 43 notes/stages/moving_throat_pde_stage044_continuum_selected_rank2_closure.md:8	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage044_continuum_selected_rank2_closure.md:10	044, 045, 047
044	xi fit_stage044_xi_no_longer_free fam_0545_xi	origin stage 44 notes/stages/moving_throat_pde_stage044_continuum_selected_rank2_closure.md:101	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage044_continuum_selected_rank2_closure.md:14	045, 046, 048
044	B_cont fit_stage_044_b_cont fam_0018_b_cont	origin stage 44 notes/stages/moving_throat_pde_stage044_continuum_selected_rank2_closure.md:105	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage044_continuum_selected_rank2_closure.md:97	045
044	R_U fit_stage_044_r_u fam_0214_r_u	origin stage 39 notes/stages/moving_throat_pde_stage044_continuum_selected_rank2_closure.md:35	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage044_continuum_selected_rank2_closure.md:27	044, 045
045	R_tr fit_stage045_tracking_derived_not_postulated fam_0223_r_tr	origin stage 45 notes/stages/moving_throat_pde_stage045_coherent_local_tracking.md:125	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage045_coherent_local_tracking.md:206	046, 047, 048
045	R_U fit_stage045_tracking_forced fam_0214_r_u	origin stage 39 notes/stages/moving_throat_pde_stage045_coherent_local_tracking.md:23	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage045_coherent_local_tracking.md:89	046, 047, 048
045	F_tr fit_stage_045_f_tr fam_0083_f_tr	origin stage 45 notes/stages/moving_throat_pde_stage045_coherent_local_tracking.md:218	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage045_coherent_local_tracking.md:212	046, 048, 051
045	R_tr fit_stage_045_r_tr fam_0223_r_tr	origin stage 45 notes/stages/moving_throat_pde_stage045_coherent_local_tracking.md:125	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage045_coherent_local_tracking.md:126	046, 047, 048

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
046	N_tr fit_stage046_residual_comparison_exact fam_0179_n_tr	origin stage 046 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
046	E_flat fit_stage_046_e_flat fam_0071_e_flat	origin stage 46 notes/stages/moving_throat_pde_stage046_tracking_branch_bounds.md:181	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage046_tracking_branch_bounds.md:185	none recorded
046	F_flat fit_stage_046_f_flat fam_0079_f_flat	origin stage 46 notes/stages/moving_throat_pde_stage046_tracking_branch_bounds.md:54	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage046_tracking_branch_bounds.md:46	048, 051
047	R_target fit_stage047_R_target_zeta_invariance fam_0222_r_target	origin stage 47 notes/stages/moving_throat_pde_stage047_coherent_kernel_map.md:37	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage047_coherent_kernel_map.md:191	048, 051
047	epsilon fit_stage047_epsilon_support_saturation fam_0390_epsilon	origin stage 47 notes/stages/moving_throat_pde_stage047_coherent_kernel_map.md:136	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage047_coherent_kernel_map.md:137	048, 050, 051, 052
048	zeta_req fit_stage048_zeta_req_threshold fam_0555_zeta_req	origin stage 48 notes/stages/moving_throat_pde_stage048_support_compensation_theorem.md:99	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage048_support_compensation_theorem.md:97	049, 050, 051, 052
048	zeta_req fit_stage048_zeta_req_unique fam_0555_zeta_req	origin stage 48 notes/stages/moving_throat_pde_stage048_support_compensation_theorem.md:99	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage048_support_compensation_theorem.md:85	049, 050, 051, 052
048	F_tr fit_stage_048_f_tr fam_0083_f_tr	origin stage 45 notes/stages/moving_throat_pde_stage048_support_compensation_theorem.md:114	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage048_support_compensation_theorem.md:138	051
048	zeta_rm fit_stage_048_zeta_rm fam_0556_zeta_rm	origin stage 048 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
049	sigma_source_density fit_stage049_uniform_source_density fam_0512_sigma_source_density	origin stage 49 notes/stages/moving_throat_pde_stage049_dn_overlap_zeta.md:93	free_choice	posited choice:notes/stages/moving_throat_pde_stage049_dn_overlap_zeta.md:91	050, 053
049	zeta_n_phys fit_stage049_zeta_phys_no_longer_free fam_0554_zeta_n_phys	origin stage 49 notes/stages/moving_throat_pde_stage049_dn_overlap_zeta.md:82	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage049_dn_overlap_zeta.md:13	050

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
050	S_zeta_n fit_stage050_enhancement_factor_exact fam_0236_s_zeta_n	origin stage 50 notes/stages/moving_throat_pde_stage050_zeta_threshold_comparison.md:168	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage050_zeta_threshold_comparison.md:89	051, 052
051	xi_2x fit_stage051_xi_2x_closed_root fam_0546_xi_2x	origin stage 51 notes/stages/moving_throat_pde_stage051_lowest_twin_criterion.md:161	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage051_lowest_twin_criterion.md:159	none recorded
051	C_mix fit_stage_051_c_mix fam_0028_c_mix	origin stage 51 notes/stages/moving_throat_pde_stage051_lowest_twin_criterion.md:77	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage051_lowest_twin_criterion.md:75	052
052	Omega_asym fit_stage052_asymmetry_forced fam_0186_omega_asym	origin stage 52 notes/stages/moving_throat_pde_stage052_nontwin_asymmetry_threshold.md:109	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage052_nontwin_asymmetry_threshold.md:163	053, 055
053	A_I fit_stage053_overlap_window_exact fam_0006_a_i	origin stage 53 notes/stages/moving_throat_pde_stage053_overlap_boost_window.md:66	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage053_overlap_boost_window.md:90	055
054	A_K fit_stage054_robin_branch_determined fam_0007_a_k	origin stage 54 notes/stages/moving_throat_pde_stage054_robin_softening_support_lane.md:105	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage054_robin_softening_support_lane.md:122	055
055	x fit_stage055_reachability_window_exact fam_0540_x	origin stage 49 notes/stages/moving_throat_pde_stage049_dn_overlap_zeta.md:131	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage054_robin_softening_support_lane.md:109	055
055	stale_output fit_stage_055_stale_output fam_0517_stale_output	origin stage 055 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
056	Pe fit_stage056_pe_identification fam_0197_pe	origin stage 56 notes/stages/moving_throat_pde_stage056_transport_source_asymmetry.md:84	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage056_transport_source_asymmetry.md:86	057
056	I_W fit_stage_056_i_w fam_0107_i_w	origin stage 53 notes/stages/moving_throat_pde_stage053_overlap_boost_window.md:48	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage056_transport_source_asymmetry.md:106	none recorded

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
057	x fit_stage057_x_identified_as_stiffness_ratio fam_0540_x	origin stage 57 notes/stages/moving_throat_pde_stage057_physical_parameter_map.md:78	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage057_physical_parameter_map.md:81	058, 059, 079, 080, 081, 082, 083
058	Delta_0 fit_stage058_bracket_derived_not_guessed fam_0055_delta_0	origin stage 58 notes/stages/moving_throat_pde_stage058_coupled_support_source_operator.md:231	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage058_coupled_support_source_operator.md:234	059, 061, 063, 066
058	Delta_0 fit_stage_058_delta_0 fam_0055_delta_0	origin stage 58 notes/stages/moving_throat_pde_stage058_coupled_support_source_operator.md:231	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage058_coupled_support_source_operator.md:232	059, 061, 063, 066
059	Pe_req fit_stage059_pe_req_misattributed_carry fam_0200_pe_req	origin stage 59 notes/stages/moving_throat_pde_stage059_operator_branch_residual_bounds.md:175	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage059_operator_branch_residual_bounds.md:177	060, 061, 062, 063, 064, 065, 066
059	Pe_req fit_stage059_thresholds_exact fam_0200_pe_req	origin stage 59 notes/stages/moving_throat_pde_stage059_operator_branch_residual_bounds.md:175	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage059_operator_branch_residual_bounds.md:179	060, 061, 062, 063, 064, 065, 066
060	gamma fit_stage060_gamma_derived_rate fam_0414_gamma	origin stage 60 notes/stages/moving_throat_pde_stage060_entropic_microclosure.md:167	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage060_entropic_microclosure.md:180	061
060	Xi_micro fit_stage060_xi_micro_drop_prefactor_choice fam_0287_xi_micro	origin stage 60 notes/stages/moving_throat_pde_stage060_entropic_microclosure.md:203	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage060_entropic_microclosure.md:217	061, 062
061	G_fail fit_stage061_gain_thresholds_exact fam_0089_g_fail	origin stage 61 notes/stages/moving_throat_pde_stage061_microscopic_gain_thresholds.md:124	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage061_microscopic_gain_thresholds.md:122	062, 063, 065, 066
061	Delta_0 fit_stage_061_delta_0 fam_0055_delta_0	origin stage 58 notes/stages/moving_throat_pde_stage058_coupled_support_source_operator.md:231	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage061_microscopic_gain_thresholds.md:172	062, 063, 065, 066

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
061	`G_fail` `fit_stage_061_g_fail` `fam_0089_g_fail`	origin stage 61 `notes/stages/moving_throat_pde_stage061_microscopic_gain_thresholds.md`:124	`internal_consistency`	located derivation step:`notes/stages/moving_throat_pde_stage061_microscopic_gain_thresholds.md`:108	062, 063, 065, 066
061	`Pe_req` `fit_stage_061_pe_req` `fam_0200_pe_req`	origin stage 59 `notes/stages/moving_throat_pde_stage059_operator_branch_residual_bounds.md`:175	`internal_consistency`	located derivation step:`notes/stages/moving_throat_pde_stage061_microscopic_gain_thresholds.md`:124	062, 063, 065, 066
062	`K_eos` `fit_stage062_n5_eos_exponent` `fam_0128_k_eos`	origin stage 062 `notes/stages/moving_throat_pde_phase0_spec.md`:12	`free_choice`	posited choice:`notes/stages/moving_throat_pde_phase0_spec.md`:3	063, 064, 065, 076
062	`Theta` `fit_stage062_theta_not_free_entropy_constant` `fam_0261_theta`	origin stage 62 `notes/stages/moving_throat_pde_stage062_parent_action_gain.md`:153	`internal_consistency`	located derivation step:`notes/stages/moving_throat_pde_stage062_parent_action_gain.md`:176	063, 064
062	`tautological_check` `fit_stage_062_tautological_check` `fam_0526_tautological_check`	origin stage 062 not recorded	`no_constraint_record`	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
062	`Xi_micro` `fit_stage_062_xi_micro` `fam_0287_xi_micro`	origin stage 60 `notes/stages/moving_throat_pde_stage060_entropic_microclosure.md`:203	`internal_consistency`	located derivation step:`notes/stages/moving_throat_pde_stage062_parent_action_gain.md`:245	063, 064, 065, 066
063	`g_phi_fail` `fit_stage063_amplitude_thresholds_exact` `fam_0410_g_phi_fail`	origin stage 63 `notes/stages/moving_throat_pde_stage063_parent_thresholds.md`:104	`internal_consistency`	located derivation step:`notes/stages/moving_throat_pde_stage063_parent_thresholds.md`:105	064, 065, 066
063	`G_fail` `fit_stage_063_g_fail` `fam_0089_g_fail`	origin stage 61 `notes/stages/moving_throat_pde_stage061_microscopic_gain_thresholds.md`:124	`internal_consistency`	located derivation step:`notes/stages/moving_throat_pde_stage063_parent_thresholds.md`:184	064, 065, 066
064	`C` `fit_stage064_coherence_factor_derived` `fam_0022_c`	origin stage 64 `notes/stages/moving_throat_pde_stage064_equilibrium_alignment.md`:115	`internal_consistency`	located derivation step:`notes/stages/moving_throat_pde_stage064_equilibrium_alignment.md`:118	065, 066, 067
064	`C_sigma_phi_sq` `fit_stage064_matched_layer_coherence` `fam_0034_c_sigma_phi_sq`	origin stage 64 `notes/stages/moving_throat_pde_stage064_equilibrium_alignment.md`:146	`internal_consistency`	located derivation step:`notes/stages/moving_throat_pde_stage064_equilibrium_alignment.md`:132	065, 066, 067

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
064	F_eff fit_stage_064_f_eff fam_0078_f_eff	origin stage 64 notes/stages/moving_throat_pde_stage064_equilibrium_alignment.md:164	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage064_equilibrium_alignment.md:160	065, 066
064	G_eq fit_stage_064_g_eq fam_0088_g_eq	origin stage 64 notes/stages/moving_throat_pde_stage064_equilibrium_alignment.md:176	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage064_equilibrium_alignment.md:172	065, 066
064	H_w fit_stage_064_h_w fam_0102_h_w	origin stage 64 notes/stages/moving_throat_pde_stage064_equilibrium_alignment.md:136	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage064_equilibrium_alignment.md:194	065, 066
065	f_prime_0 fit_stage065_canonical_normalization_fprime fam_0400_f_prime_0	origin stage 65 notes/stages/moving_throat_pde_stage065_thin_wall_confinement.md:24	free_choice	posited choice:notes/stages/moving_throat_pde_stage065_thin_wall_confinement.md:108	066
065	V_0 fit_stage_065_v_0 fam_0270_v_0	origin stage 65 notes/stages/moving_throat_pde_stage065_thin_wall_confinement.md:23	free_choice	posited choice:notes/stages/moving_throat_pde_stage065_thin_wall_confinement.md:89	066
065	V_0f fit_stage_065_v_0f fam_0271_v_0f	origin stage 65 notes/stages/moving_throat_pde_stage065_thin_wall_confinement.md:23	free_choice	posited choice:notes/stages/moving_throat_pde_stage065_thin_wall_confinement.md:89	066
066	W_fail fit_stage066_exact_matched_verdict fam_0278_w_fail	origin stage 66 notes/stages/moving_throat_pde_stage066_wall_figure_of_merit.md:69	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage066_wall_figure_of_merit.md:67	067, 068, 069
066	Delta_0 fit_stage_066_delta_0 fam_0055_delta_0	origin stage 58 notes/stages/moving_throat_pde_stage058_coupled_support_source_operator.md:231	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage066_wall_figure_of_merit.md:71	067, 068, 069
066	W_wall fit_stage_066_w_wall fam_0280_w_wall	origin stage 66 notes/stages/moving_throat_pde_stage066_wall_figure_of_merit.md:17	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage066_wall_figure_of_merit.md:52	067, 068, 069
067	C_res_squared fit_stage067_C_res_unique_global_max fam_0032_c_res_squared	origin stage 67 notes/stages/moving_throat_pde_stage067_sech_gaussian_resonance.md:116	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage067_sech_gaussian_resonance.md:108	068, 069
067	C_res_sq fit_stage067_sech_gaussian_benchmark_constants fam_0031_c_res_sq	origin stage 67 notes/stages/moving_throat_pde_stage067_sech_gaussian_resonance.md:116	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage067_sech_gaussian_resonance.md:117	068, 069

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
067	w_g_over_w_f fit_stage067_w_ratio_self_duality_forced fam_0537_w_g_over_w_f	origin stage 67 notes/stages/moving_throat_pde_stage067_sech_gaussian_resonance.md:96	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage067_sech_gaussian_resonance.md:106	068, 069
067	C_res fit_stage_067_c_res fam_0030_c_res	origin stage 67 notes/stages/moving_throat_pde_stage067_sech_gaussian_resonance.md:116	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage067_sech_gaussian_resonance.md:117	068, 069
067	Iprime_dual fit_stage_067_iprime_dual fam_0111_iprime_dual	origin stage 67 notes/stages/moving_throat_pde_stage067_sech_gaussian_resonance.md:82	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage067_sech_gaussian_resonance.md:100	068, 069
068	W_res fit_stage068_W_res_derived_label fam_0279_w_res	origin stage 068 notes/stages/moving_throat_pde_stage068_resonance_thresholds.md:55	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage068_resonance_thresholds.md:73	069
068	P_res fit_stage068_threshold_translation_not_postulated fam_0194_p_res	origin stage 55 notes/stages/moving_throat_pde_stage067_sech_gaussian_resonance.md:127	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage067_sech_gaussian_resonance.md:116	069
068	C_res fit_stage_068_c_res fam_0030_c_res	origin stage 55 notes/stages/moving_throat_pde_stage067_sech_gaussian_resonance.md:116	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage067_sech_gaussian_resonance.md:106	068, 069
068	Delta_0 fit_stage_068_delta_0 fam_0055_delta_0	origin stage 49 notes/stages/moving_throat_pde_stage061_microscopic_gain_thresholds.md:162	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage061_microscopic_gain_thresholds.md:154	054, 068, 069
068	Delta_inf fit_stage_068_delta_inf fam_0063_delta_inf	origin stage 49 notes/stages/moving_throat_pde_stage061_microscopic_gain_thresholds.md:166	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage061_microscopic_gain_thresholds.md:154	054, 068, 069
068	P_res fit_stage_068_p_res fam_0194_p_res	origin stage 55 notes/stages/moving_throat_pde_stage067_sech_gaussian_resonance.md:127	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage067_sech_gaussian_resonance.md:116	068, 069
069	P_res fit_stage069_matched_window_exactly fam_0194_p_res	origin stage 55 notes/stages/moving_throat_pde_stage067_sech_gaussian_resonance.md:127	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage069_final_reduced_verdict.md:92	070, 071, 072, 073, 074, 075, 076, 077, 078

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
069	C_res fit_stage_069_c_res fam_0030_c_res	origin stage 55 notes/stages/moving_throat_pde_stage067_sech_gaussian_resonance.md:116	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage069_final_reduced_verdict.md:147	070, 071, 072, 073, 074, 075, 076, 077, 078
069	Delta_0 fit_stage_069_delta_0 fam_0055_delta_0	origin stage 49 notes/stages/moving_throat_pde_stage061_microscopic_gain_thresholds.md:162	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage069_final_reduced_verdict.md:29	070, 071, 072, 073, 074, 075, 076, 077, 078
069	Delta_eff fit_stage_069_delta_eff fam_0062_delta_eff	origin stage 069 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
069	P_res fit_stage_069_p_res fam_0194_p_res	origin stage 55 notes/stages/moving_throat_pde_stage067_sech_gaussian_resonance.md:127	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage069_final_reduced_verdict.md:21	070, 071, 072, 073, 074, 075, 076, 077, 078
070	n fit_stage070_n5_eos_exponent fam_0480_n	origin stage 50 notes/stages/moving_throat_pde_stage062_parent_action_gain.md:92	free_choice	posited choice:notes/stages/moving_throat_pde_stage062_parent_action_gain.md:88	070, 071, 073, 074, 167
070	J_1 fit_stage070_wall_shell_fixes_quintuple fam_0112_j_1, fam_0124_k_x, fam_0251_t_x, fam_0280_w_wall, fam_0431_kappa	origin stage 53 notes/stages/moving_throat_pde_stage065_thin_wall_confinement.md:69	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage070_gnls_wall_shell.md:120	071, 073, 074
070	K_X fit_stage070_wall_shell_fixes_quintuple fam_0112_j_1, fam_0124_k_x, fam_0251_t_x, fam_0280_w_wall, fam_0431_kappa	origin stage 56 notes/stages/moving_throat_pde_stage070_gnls_wall_shell.md:55	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage070_gnls_wall_shell.md:18	071, 073, 074
070	T_X fit_stage070_wall_shell_fixes_quintuple fam_0112_j_1, fam_0124_k_x, fam_0251_t_x, fam_0280_w_wall, fam_0431_kappa	origin stage 56 notes/stages/moving_throat_pde_stage070_gnls_wall_shell.md:53	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage070_gnls_wall_shell.md:17	071, 073, 074
070	W_wall fit_stage070_wall_shell_fixes_quintuple fam_0112_j_1, fam_0124_k_x, fam_0251_t_x, fam_0280_w_wall, fam_0431_kappa	origin stage 54 notes/stages/moving_throat_pde_stage066_wall_figure_of_merit.md:17	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage070_gnls_wall_shell.md:128	071, 073, 074
070	kappa fit_stage070_wall_shell_fixes_quintuple fam_0112_j_1, fam_0124_k_x, fam_0251_t_x, fam_0280_w_wall, fam_0431_kappa	origin stage 49 notes/stages/moving_throat_pde_stage061_microscopic_gain_thresholds.md:96	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage070_gnls_wall_shell.md:112	071, 073, 074
070	I_f fit_stage_070_i_f fam_0109_i_f	origin stage 53 notes/stages/moving_throat_pde_stage065_thin_wall_confinement.md:73	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage070_gnls_wall_shell.md:91	071, 073, 074

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
070	J_1 fit_stage_070_j_1 fam_0112_j_1	origin stage 53 notes/stages/moving_throat_pde_stage065_thin_wall_confinement.md:69	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage070_gnls_wall_shell.md:120	071, 073, 074
071	K_m fit_stage071_canonical_tanh_and_natural_closure fam_0135_k_m	origin stage 57 notes/stages/moving_throat_pde_stage071_tanh_wall_branch.md:93	free_choice	posited choice:notes/stages/moving_throat_pde_stage071_tanh_wall_branch.md:95	073, 074
071	K_m fit_stage071_mouth_closure_Km fam_0135_k_m	origin stage 57 notes/stages/moving_throat_pde_stage071_tanh_wall_branch.md:93	free_choice	posited choice:notes/stages/moving_throat_pde_stage071_tanh_wall_branch.md:95	073, 074
071	I_f fit_stage_071_i_f fam_0109_i_f	origin stage 57 notes/stages/moving_throat_pde_stage071_tanh_wall_branch.md:21	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage071_tanh_wall_branch.md:53	073, 074
072	L_over_a fit_stage072_canonical_controls_exact_thresholds fam_0146_l_over_a; overlays:chain_aspect_37_20	origin stage 072 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
073	L_over_a fit_stage073_Lambda_ell_fixed_37 fam_0146_l_over_a; overlays:chain_aspect_37_20	origin stage 59 notes/stages/moving_throat_pde_stage073_family1_geometry_map.md:54	free_choice	posited choice:notes/stages/moving_throat_pde_stage073_family1_geometry_map.md:56	074
073	L_over_a fit_stage073_aspect_ratio_L_over_a fam_0146_l_over_a; overlays:chain_aspect_37_20	origin stage 59 notes/stages/moving_throat_pde_stage073_family1_geometry_map.md:54	free_choice	posited choice:notes/stages/moving_throat_pde_stage073_family1_geometry_map.md:56	074
073	L_over_a fit_stage073_aspect_ratio_frozen fam_0146_l_over_a; overlays:chain_aspect_37_20	origin stage 59 notes/stages/moving_throat_pde_stage073_family1_geometry_map.md:54	free_choice	posited choice:notes/stages/moving_throat_pde_stage073_family1_geometry_map.md:56	074
073	epsilon_r fit_stage073_epsilon_r_posited fam_0394_epsilon_r; overlays:chain_aspect_37_20	origin stage 59 notes/stages/moving_throat_pde_stage073_family1_geometry_map.md:30	free_choice	posited choice:notes/stages/moving_throat_pde_stage073_family1_geometry_map.md:18	074
073	eta fit_stage073_eta_pinned_37 fam_0395_eta; overlays:chain_aspect_37_20	origin stage 59 notes/stages/moving_throat_pde_stage073_family1_geometry_map.md:92	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage073_family1_geometry_map.md:88	074

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
073	L_over_a fit_stage073_paper_fixes_geometry_ratios fam_0146_l_over_a; overlays:chain_aspect_37_20	origin stage 59 notes/stages/moving_throat_pde_stage073_family1_geometry_map.md:54	free_choice	posited choice:notes/stages/moving_throat_pde_stage073_family1_geometry_map.md:56	074
073	K_m fit_stage_073_k_m fam_0135_k_m	origin stage 57 notes/stages/moving_throat_pde_stage071_tanh_wall_branch.md:93	free_choice	posited choice:notes/stages/moving_throat_pde_stage071_tanh_wall_branch.md:95	074
073	T_X fit_stage_073_t_x fam_0251_t_x	origin stage 56 notes/stages/moving_throat_pde_stage070_gnls_wall_shell.md:53	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage070_gnls_wall_shell.md:17	074
074	chi_s fit_stage074_chi_s_now_fixes fam_0350_chi_s	origin stage 60 notes/stages/moving_throat_pde_stage074_family1_healing_lock.md:75	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage074_family1_healing_lock.md:71	075
074	ell fit_stage074_healing_lock_ell fam_0380_ell	origin stage 60 notes/stages/moving_throat_pde_stage074_family1_healing_lock.md:53	free_choice	posited choice:notes/stages/moving_throat_pde_stage074_family1_healing_lock.md:55	075
074	kappa fit_stage074_kappa_locked_reduction fam_0431_kappa	origin stage 60 notes/stages/moving_throat_pde_stage074_family1_healing_lock.md:95	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage074_family1_healing_lock.md:91	075
074	ell fit_stage074_natural_healing_closure fam_0380_ell	origin stage 60 notes/stages/moving_throat_pde_stage074_family1_healing_lock.md:53	free_choice	posited choice:notes/stages/moving_throat_pde_stage074_family1_healing_lock.md:55	075
075	alpha_r fit_stage075_alpha_r_wall_depth fam_0313_alpha_r	origin stage 61 notes/stages/moving_throat_pde_stage075_family1_threshold_window.md:91	free_choice	posited choice:notes/stages/moving_throat_pde_stage075_family1_threshold_window.md:89	076, 077, 078
075	L_over_a fit_stage075_triple_fixed_summary fam_0146_l_over_a; overlays:chain_aspect_37_20	origin stage 075 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
075	Delta_0 fit_stage_075_delta_0 fam_0055_delta_0	origin stage 61 notes/stages/moving_throat_pde_stage075_family1_threshold_window.md:39	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage075_family1_threshold_window.md:37	078

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
075	Pe_req fit_stage_075_pe_req fam_0200_pe_req	origin stage 059 notes/stages/moving_throat_pde_stage059_operator_branch_residual_bounds.md:52	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage059_operator_branch_residual_bounds.md:54	078, 079
076	lambda_mu fit_stage076_lambda_mu_normalization fam_0452_lambda_mu	origin stage 62 notes/stages/moving_throat_pde_stage076_n5_wall_depth_lock.md:60	free_choice	posited choice:notes/stages/moving_throat_pde_stage076_n5_wall_depth_lock.md:66	077, 078
076	Theta_w fit_stage076_target_25_derived fam_0264_theta_w	origin stage 61 notes/stages/moving_throat_pde_stage075_family1_threshold_window.md:104	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage076_n5_wall_depth_lock.md:17	077, 078
076	Theta_w fit_stage076_wall_depth_exactly_square fam_0264_theta_w	origin stage 61 notes/stages/moving_throat_pde_stage075_family1_threshold_window.md:104	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage076_n5_wall_depth_lock.md:93	077, 078
077	I_f fit_stage077_If_exact_canonical fam_0109_i_f	origin stage 57 notes/stages/moving_throat_pde_stage071_tanh_wall_branch.md:21	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage077_family1_theta_extraction.md:59	063
077	Theta_w fit_stage077_Theta_w_chi_derived_directly fam_0264_theta_w	origin stage 63 notes/stages/moving_throat_pde_stage077_family1_theta_extraction.md:93	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage077_family1_theta_extraction.md:96	078
077	I_f fit_stage_077_i_f fam_0109_i_f	origin stage 57 notes/stages/moving_throat_pde_stage071_tanh_wall_branch.md:21	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage077_family1_theta_extraction.md:59	063
077	Theta_w fit_stage_077_theta_w fam_0264_theta_w	origin stage 63 notes/stages/moving_throat_pde_stage077_family1_theta_extraction.md:93	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage077_family1_theta_extraction.md:96	078
078	L_over_a fit_stage078_verdict_fixed_pair fam_0146_l_over_a; overlays:chain_aspect_37_20	origin stage 078 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
078	Pe_req fit_stage_078_pe_req fam_0200_pe_req	origin stage 059 notes/stages/moving_throat_pde_stage059_operator_branch_residual_bounds.md:52	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage059_operator_branch_residual_bounds.md:54	079

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
079	Pe_req fit_stage079_unique_constructive_Pe_req fam_0200_pe_req	origin stage 059 notes/stages/moving_throat_pde_stage059_operator_branch_residual_bounds.md:52	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage079_family1_quadrupole_pe_map.md:104	080, 081, 082, 083, 084, 085, 087
079	A_F1 fit_stage_079_a_f1 fam_0005_a_f1	origin stage 079 notes/stages/moving_throat_pde_stage079_family1_quadrupole_pe_map.md:57	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage079_family1_quadrupole_pe_map.md:55	080, 081, 082, 083, 084, 085
080	A_F1 fit_stage_080_a_f1 fam_0005_a_f1	origin stage 079 notes/stages/moving_throat_pde_stage079_family1_quadrupole_pe_map.md:57	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage080_family1_zeta_thresholds.md:13	081, 082, 083, 084, 085
081	C_mix fit_stage_081_c_mix fam_0028_c_mix	origin stage 42 notes/stages/moving_throat_pde_stage052_nontwin_asymmetry_threshold.md:18	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage052_nontwin_asymmetry_threshold.md:47	082, 083, 084, 085, 086, 087, 088, 089
081	eps_blk fit_stage_081_eps_blk fam_0385_eps_blk	origin stage 081 notes/stages/moving_throat_pde_stage081_family1_pi_thresholds.md:41	free_choice	posited choice:notes/stages/moving_throat_pde_stage081_family1_pi_thresholds.md:23	082, 083, 084, 085, 086
082	Pi_fail fit_stage082_first_exact_product_window fam_0207_pi_fail	origin stage 082 notes/stages/moving_throat_pde_stage082_master_quadrupole_residual.md:202	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage082_master_quadrupole_residual.md:204	083, 084, 085, 086, 087
082	Pe_req fit_stage_082_pe_req fam_0200_pe_req	origin stage 059 notes/stages/moving_throat_pde_stage059_operator_branch_residual_bounds.md:52	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage059_operator_branch_residual_bounds.md:54	none recorded
082	zeta_max fit_stage_082_zeta_max fam_0552_zeta_max	origin stage 082 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
083	zeta_fail fit_stage083_windows_now_derived_directly fam_0551_zeta_fail	origin stage 080 notes/stages/moving_throat_pde_stage080_family1_zeta_thresholds.md:95	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage083_family1_direct_operator_window.md:91	084, 085, 086, 087
083	A_F1 fit_stage_083_a_f1 fam_0005_a_f1	origin stage 079 notes/stages/moving_throat_pde_stage079_family1_quadrupole_pe_map.md:57	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage079_family1_quadrupole_pe_map.md:55	084, 085

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
083	C_mix fit_stage_083_c_mix fam_0028_c_mix	origin stage 42 notes/stages/moving_throat_pde_stage052_nontwin_asymmetry_threshold.md:18	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage052_nontwin_asymmetry_threshold.md:47	084, 085, 086, 087, 088, 089
083	Delta0_F1 fit_stage_083_delta0_f1 fam_0052_delta0_f1	origin stage 61 notes/stages/moving_throat_pde_stage075_family1_threshold_window.md:39	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage083_family1_direct_operator_window.md:81	084, 085
083	Delta_0 fit_stage_083_delta_0 fam_0055_delta_0	origin stage 61 notes/stages/moving_throat_pde_stage075_family1_threshold_window.md:39	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage083_family1_direct_operator_window.md:81	084, 085
083	Pe_minus_J fit_stage_083_pe_minus_j fam_0199_pe_minus_j	origin stage 078 notes/stages/moving_throat_pde_stage078_family1_branch_verdict.md:54	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage083_family1_direct_operator_window.md:115	084
084	C_mix fit_stage_084_c_mix fam_0028_c_mix	origin stage 42 notes/stages/moving_throat_pde_stage052_nontwin_asymmetry_threshold.md:18	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage052_nontwin_asymmetry_threshold.md:47	085, 086, 087, 088, 089
084	Omega_Pe fit_stage_084_omega_pe fam_0182_omega_pe	origin stage 46 notes/stages/moving_throat_pde_stage056_transport_source_asymmetry.md:119	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage084_full_reduced_pde_writeup.md:58	085
084	Theta_w fit_stage_084_theta_w fam_0264_theta_w	origin stage 61 notes/stages/moving_throat_pde_stage075_family1_threshold_window.md:104	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage084_full_reduced_pde_writeup.md:121	085
084	zeta_max fit_stage_084_zeta_max fam_0552_zeta_max	origin stage 079 notes/stages/moving_throat_pde_stage079_family1_quadrupole_pe_map.md:35	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage084_full_reduced_pde_writeup.md:142	085, 086
085	eps_blk fit_stage085_unblocked_literal_law fam_0385_eps_blk, fam_0555_zeta_req	origin stage 081 notes/stages/moving_throat_pde_stage081_family1_pi_thresholds.md:41	free_choice	posited choice:notes/stages/moving_throat_pde_stage085_quadrupole_demand_cancellation.md:132	086

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
085	zeta_req fit_stage085_unblocked_literal_law fam_0385_eps_blk, fam_0555_zeta_req	origin stage 42 notes/stages/moving_throat_pde_stage052_nontwin_asymmetry_threshold.md:34	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage085_quadrupole_demand_cancellation.md:116	086, 087
085	rho_alpha fit_stage_085_rho_alpha fam_0500_rho_alpha	origin stage 085 notes/stages/moving_throat_pde_stage085_quadrupole_demand_cancellation.md:130	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage085_quadrupole_demand_cancellation.md:78	086, 087, 088
086	eps_blk fit_stage086_natural_lambda_mu_unblocked fam_0385_eps_blk, fam_0452_lambda_mu	origin stage 86 notes/stages/moving_throat_pde_stage086_family1_loading_ratio_window.md:81	free_choice	posited choice:notes/stages/moving_throat_pde_stage086_family1_loading_ratio_window.md:24	087, 088, 089
086	lambda_mu fit_stage086_natural_lambda_mu_unblocked fam_0385_eps_blk, fam_0452_lambda_mu	origin stage 86 notes/stages/moving_throat_pde_stage086_family1_loading_ratio_window.md:95	free_choice	posited choice:notes/stages/moving_throat_pde_stage086_family1_loading_ratio_window.md:152	087, 088, 089, 090
087	eps_blk fit_stage087_natural_lambda_mu_unblocked fam_0385_eps_blk, fam_0452_lambda_mu	origin stage 87 notes/stages/moving_throat_pde_stage087_outgoing_branch_loading_ratio_finish.md:57	free_choice	posited choice:notes/stages/moving_throat_pde_stage087_outgoing_branch_loading_ratio_finish.md:55	088, 089
087	lambda_mu fit_stage087_natural_lambda_mu_unblocked fam_0385_eps_blk, fam_0452_lambda_mu	origin stage 87 notes/stages/moving_throat_pde_stage087_outgoing_branch_loading_ratio_finish.md:57	free_choice	posited choice:notes/stages/moving_throat_pde_stage087_outgoing_branch_loading_ratio_finish.md:57	088, 089, 090
088	c0 fit_stage088_c0_c1_derived_not_defined fam_0333_c0	origin stage 88 notes/stages/moving_throat_pde_stage088_loading_ratio_from_minimal_module.md:17	free_choice	posited choice:notes/stages/moving_throat_pde_stage088_loading_ratio_from_minimal_module.md:81	089, 090, 091, 096
088	c_0 fit_stage088_minimal_module_c0_c1 fam_0335_c_0	origin stage 88 notes/stages/moving_throat_pde_stage088_loading_ratio_from_minimal_module.md:17	free_choice	posited choice:notes/stages/moving_throat_pde_stage088_loading_ratio_from_minimal_module.md:83	089, 090, 091, 096
088	c0 fit_stage088_minimal_module_coefficients_paper_quoted fam_0333_c0	origin stage 88 notes/stages/moving_throat_pde_stage088_loading_ratio_from_minimal_module.md:17	free_choice	posited choice:notes/stages/moving_throat_pde_stage088_loading_ratio_from_minimal_module.md:81	089, 090, 091, 096

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
088	c_contact fit_stage088_natural_identification_fixes_exactly fam_0336_c_contact	origin stage 88 notes/stages/moving_throat_pde_stage088_loading_ratio_from_minimal_module.md:60	free_choice	posited choice:notes/stages/moving_throat_pde_stage088_loading_ratio_from_minimal_module.md:83	089, 090, 091, 096
088	Omega_Q fit_stage088_pole_frequency_OmegaQ fam_0183_omega_q	origin stage 88 notes/stages/moving_throat_pde_stage088_loading_ratio_from_minimal_module.md:19	free_choice	posited choice:notes/stages/moving_throat_pde_stage088_loading_ratio_from_minimal_module.md:13	089, 091, 092, 105
089	Pe_req fit_stage089_Pe_req_forced_not_assumed fam_0200_pe_req	origin stage 89 notes/stages/moving_throat_pde_stage089_family1_minimal_isotropic_verdict.md:107	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage089_family1_minimal_isotropic_verdict.md:103	090
089	Pe_fail_chi fit_stage089_pe_window_upstream_misattribution fam_0198_pe_fail_chi	origin stage 78 notes/stages/moving_throat_pde_stage078_family1_branch_verdict.md:42	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage080_family1_zeta_thresholds.md:47	089
089	Pe_fail_chi fit_stage089_provenance_record_misattribution fam_0198_pe_fail_chi	origin stage 78 notes/stages/moving_throat_pde_stage078_family1_branch_verdict.md:42	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage080_family1_zeta_thresholds.md:47	089
089	A_F1 fit_stage_089_a_f1 fam_0005_a_f1	origin stage 89 notes/stages/moving_throat_pde_stage089_family1_minimal_isotropic_verdict.md:94	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage089_family1_minimal_isotropic_verdict.md:90	090
090	Pe_req fit_stage090_locked_triple_Pe_req fam_0200_pe_req	origin stage 89 notes/stages/moving_throat_pde_stage089_family1_minimal_isotropic_verdict.md:107	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage090_updated_reduced_status.md:19	none recorded
090	c_contact fit_stage090_minimal_module_upstream_contradiction fam_0336_c_contact	origin stage 88 notes/stages/moving_throat_pde_stage088_loading_ratio_from_minimal_module.md:17	free_choice	posited choice:notes/stages/moving_throat_pde_stage090_updated_reduced_status.md:15	none recorded
090	rho_alpha fit_stage_090_rho_alpha fam_0500_rho_alpha	origin stage 88 notes/stages/moving_throat_pde_stage088_loading_ratio_from_minimal_module.md:88	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage090_updated_reduced_status.md:17	091, 096

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
091	K_geom fit_stage091_K_geom_forced_by_identity fam_0132_k_geom	origin stage 91 notes/stages/moving_throat_pde_stage091_grouped_p2_static_geometry_derivation.md:70	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage091_grouped_p2_static_geometry_derivation.md:107	092, 093, 094, 096
091	K0_K4_over_K2sq fit_stage091_branch_identity_factor4 fam_0116_k0_k4_over_k2sq	origin stage 91 notes/stages/moving_throat_pde_stage091_grouped_p2_static_geometry_derivation.md:52	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage091_grouped_p2_static_geometry_derivation.md:50	092, 093, 094
091	Y_Q_cons fit_stage091_module_forced_paper fam_0291_y_q_cons	origin stage 91 notes/stages/moving_throat_pde_stage091_grouped_p2_static_geometry_derivation.md:32	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage091_grouped_p2_static_geometry_derivation.md:115	092, 093, 094, 096
091	epsilon_2 fit_stage_091_epsilon_2 fam_0391_epsilon_2	origin stage 92 notes/stages/moving_throat_pde_stage092_dynamic_geometry_obstruction.md:23	free_choice	posited choice:notes/stages/moving_throat_pde_stage091_grouped_p2_static_geometry_derivation.md:22	092, 093, 094
091	K_geom fit_stage_091_k_geom fam_0132_k_geom	origin stage 91 notes/stages/moving_throat_pde_stage091_grouped_p2_static_geometry_derivation.md:70	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage091_grouped_p2_static_geometry_derivation.md:113	092, 093, 094, 096
092	Y_Q_cons fit_stage092_forced_carrier_ledger_label fam_0291_y_q_cons	origin stage 91 notes/stages/moving_throat_pde_stage091_grouped_p2_static_geometry_derivation.md:32	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage092_dynamic_geometry_obstruction.md:27	093, 094, 096
092	K_geom fit_stage092_no_longer_forced_unless fam_0132_k_geom	origin stage 91 notes/stages/moving_throat_pde_stage091_grouped_p2_static_geometry_derivation.md:70	free_choice	posited choice:notes/stages/moving_throat_pde_stage092_dynamic_geometry_obstruction.md:68	093, 094
092	epsilon_2 fit_stage_092_epsilon_2 fam_0391_epsilon_2	origin stage 92 notes/stages/moving_throat_pde_stage092_dynamic_geometry_obstruction.md:23	free_choice	posited choice:notes/stages/moving_throat_pde_stage092_dynamic_geometry_obstruction.md:84	093, 094
092	Yhat_Q fit_stage_092_yhat_q fam_0297_yhat_q	origin stage 91 notes/stages/moving_throat_pde_stage091_grouped_p2_static_geometry_derivation.md:32	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage092_dynamic_geometry_obstruction.md:107	093, 094, 096

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
093	Y_Q_cons fit_stage093_module_forced_statement fam_0291_y_q_cons	origin stage 91 notes/stages/moving_throat_pde_stage091_grouped_p2_static_geometry_derivation.md:32	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage093_grouped_p2_status_update.md:25	094, 096
093	eps_2 fit_stage093_static_limit_hardcoded fam_0382_eps_2	origin stage 92 notes/stages/moving_throat_pde_stage092_dynamic_geometry_obstruction.md:23	free_choice	posited choice:notes/stages/moving_throat_pde_stage093_grouped_p2_status_update.md:21	094
093	c_pole fit_stage_093_c_pole fam_0339_c_pole	origin stage 92 notes/stages/moving_throat_pde_stage092_dynamic_geometry_obstruction.md:19	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage093_grouped_p2_status_update.md:53	094, 096
093	eps_2 fit_stage_093_eps_2 fam_0382_eps_2	origin stage 92 notes/stages/moving_throat_pde_stage092_dynamic_geometry_obstruction.md:23	free_choice	posited choice:notes/stages/moving_throat_pde_stage093_grouped_p2_status_update.md:55	094
093	epsilon_2 fit_stage_093_epsilon_2 fam_0391_epsilon_2	origin stage 92 notes/stages/moving_throat_pde_stage092_dynamic_geometry_obstruction.md:23	free_choice	posited choice:notes/stages/moving_throat_pde_stage093_grouped_p2_status_update.md:55	094
094	K_g2 fit_stage094_decoupling_kills_contamination fam_0131_k_g2	origin stage 94 notes/stages/moving_throat_pde_stage094_isotropic_geometry_decoupling.md:25	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage094_isotropic_geometry_decoupling.md:71	096
094	c_geom fit_stage094_static_split_assigned_not_derived fam_0337_c_geom	origin stage 92 notes/stages/moving_throat_pde_stage092_dynamic_geometry_obstruction.md:99	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage094_isotropic_geometry_decoupling.md:122	096
094	c_geom fit_stage_094_c_geom fam_0337_c_geom	origin stage 92 notes/stages/moving_throat_pde_stage092_dynamic_geometry_obstruction.md:99	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage094_isotropic_geometry_decoupling.md:122	096
094	epsilon_2 fit_stage_094_epsilon_2 fam_0391_epsilon_2	origin stage 92 notes/stages/moving_throat_pde_stage092_dynamic_geometry_obstruction.md:23	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage094_isotropic_geometry_decoupling.md:114	096

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
095	Y_Q_cons fit_stage095_forced_carrier_ledger_label fam_0291_y_q_cons	origin stage 95 notes/stages/moving_throat_pde_stage095_second_order_geometry_contamination.md:19	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage095_second_order_geometry_contamination.md:127	096, 097, 099, 100
095	c_pole fit_stage_095_c_pole fam_0339_c_pole	origin stage 95 notes/stages/moving_throat_pde_stage095_second_order_geometry_contamination.md:98	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage095_second_order_geometry_contamination.md:104	096, 097, 099
095	epsilon_2 fit_stage_095_epsilon_2 fam_0391_epsilon_2	origin stage 95 notes/stages/moving_throat_pde_stage095_second_order_geometry_contamination.md:80	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage095_second_order_geometry_contamination.md:86	096, 097, 099
096	c0 fit_stage096_constants_derived_from_eps_zero fam_0333_c0	origin stage 096 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	097, 098, 099, 100
096	c_geom fit_stage_096_c_geom fam_0337_c_geom	origin stage 96 notes/stages/moving_throat_pde_stage096_geometry_lane_check_verdict.md:43	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage096_geometry_lane_check_verdict.md:39	097, 098, 099, 100
096	eps_2 fit_stage_096_eps_2 fam_0382_eps_2	origin stage 96 notes/stages/moving_throat_pde_stage096_geometry_lane_check_verdict.md:31	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage096_geometry_lane_check_verdict.md:27	097, 099
096	epsilon_2 fit_stage_096_epsilon_2 fam_0391_epsilon_2	origin stage 96 notes/stages/moving_throat_pde_stage096_geometry_lane_check_verdict.md:31	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage096_geometry_lane_check_verdict.md:27	097, 099
096	NUMBERING_SCRIPT_OUTPUT_BAND_PLAN fit_stage_096_numbering_script_output_band_plan fam_0172_numbering_script_output_band_plan	origin stage 096 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
096	Yhat_cons fit_stage_096_yhat_cons fam_0299_yhat_cons	origin stage 96 notes/stages/moving_throat_pde_stage096_geometry_lane_check_verdict.md:47	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage096_geometry_lane_check_verdict.md:39	097, 098, 099, 100

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
097	Kbar_0 fit_stage097_even_ledger_fixed fam_0141_kbar_0	origin stage 97 notes/stages/moving_throat_pde_stage097_single_normalization_defect.md:30	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage097_single_normalization_defect.md:38	099, 100
097	Gamma_5 fit_stage097_gamma5_fixed_algebraically fam_0093_gamma_5; overlays:chain_quad_54_5	origin stage 97 notes/stages/moving_throat_pde_stage097_single_normalization_defect.md:46	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage097_single_normalization_defect.md:53	099, 100
097	Yhat_Q_cons fit_stage097_yhatQcons_forced_carrier fam_0298_yhat_q_cons	origin stage 97 notes/stages/moving_throat_pde_stage097_single_normalization_defect.md:11	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage097_single_normalization_defect.md:9	098, 099, 100
097	epsilon_2 fit_stage_097_epsilon_2 fam_0391_epsilon_2	origin stage 96 notes/stages/moving_throat_pde_stage096_geometry_lane_check_verdict.md:31	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage096_geometry_lane_check_verdict.md:27	099
097	Kbar_0 fit_stage_097_kbar_0 fam_0141_kbar_0	origin stage 97 notes/stages/moving_throat_pde_stage097_single_normalization_defect.md:30	published_target	published_target; no benchmarks.yaml entry joined for listed family; evidence:notes/stages/moving_throat_pde_stage097_single_normalization_defect.md:61	099, 100
097	R_i fit_stage_097_r_i fam_0217_r_i	origin stage 97 notes/stages/moving_throat_pde_stage097_single_normalization_defect.md:104	published_target	published_target; no benchmarks.yaml entry joined for listed family; evidence:notes/stages/moving_throat_pde_stage097_single_normalization_defect.md:114	099, 100
098	Yhat_Q_cons fit_stage098_forced_carrier_card fam_0298_yhat_q_cons	origin stage 96 notes/stages/moving_throat_pde_stage096_geometry_lane_check_verdict.md:47	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage098_family1_support_is_automatic.md:14	099, 100
098	rho_alpha fit_stage098_rho_alpha_selected_ratio fam_0500_rho_alpha	origin stage 96 notes/stages/moving_throat_pde_stage096_geometry_lane_check_verdict.md:51	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage096_geometry_lane_check_verdict.md:43	099, 100, 101, 102, 103, 104, 105, 106
098	support_ratio fit_stage098_support_demand_exact fam_0520_support_ratio	origin stage 98 notes/stages/moving_throat_pde_stage098_family1_support_is_automatic.md:32	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage098_family1_support_is_automatic.md:40	099, 100

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
098	zeta_max_F1 fit_stage098_zeta_max_f1_imported_literal fam_0553_zeta_max_f1	origin stage 79 notes/stages/moving_throat_pde_stage079_family1_quadrupole_pe_map.md:35	internal_consistency	located derivation step:notes/CHECKPOINT_CONSTANT_PROVENANCE.md:451	099
098	zeta_max_F1 fit_stage098_zeta_max_f1_opaque_carried_ceiling fam_0553_zeta_max_f1	origin stage 79 notes/stages/moving_throat_pde_stage079_family1_quadrupole_pe_map.md:35	internal_consistency	located derivation step:notes/CHECKPOINT_CONSTANT_PROVENANCE.md:451	099
098	c_pole fit_stage_098_c_pole fam_0339_c_pole	origin stage 95 notes/stages/moving_throat_pde_stage095_second_order_geometry_contamination.md:98	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage096_geometry_lane_check_verdict.md:41	099
098	epsilon_2 fit_stage_098_epsilon_2 fam_0391_epsilon_2	origin stage 96 notes/stages/moving_throat_pde_stage096_geometry_lane_check_verdict.md:31	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage096_geometry_lane_check_verdict.md:27	099
098	gap_F1 fit_stage_098_gap_f1 fam_0422_gap_f1	origin stage 98 notes/stages/moving_throat_pde_stage098_family1_support_is_automatic.md:92	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage098_family1_support_is_automatic.md:62	none recorded
098	gap_F1_target fit_stage_098_gap_f1_target fam_0423_gap_f1_target	origin stage 98 notes/stages/moving_throat_pde_stage098_family1_support_is_automatic.md:92	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage098_family1_support_is_automatic.md:64	none recorded
098	zeta_edge_F1 fit_stage_098_zeta_edge_f1 fam_0550_zeta_edge_f1	origin stage 98 notes/stages/moving_throat_pde_stage098_family1_support_is_automatic.md:96	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage098_family1_support_is_automatic.md:88	none recorded
098	zeta_max fit_stage_098_zeta_max fam_0552_zeta_max	origin stage 79 notes/stages/moving_throat_pde_stage079_family1_quadrupole_pe_map.md:35	internal_consistency	located derivation step:notes/CHECKPOINT_CONSTANT_PROVENANCE.md:451	099
099	Yhat_Q_cons fit_stage099_conservative_split_exact fam_0298_yhat_q_cons	origin stage 96 notes/stages/moving_throat_pde_stage096_geometry_lane_check_verdict.md:47	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage099_reduced_finish_line.md:18	100

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
099	Gamma_5 fit_stage099_forced_by_yhatQcons_script fam_0093_gamma_5; overlays:chain_quad_54_5	origin stage 97 notes/stages/moving_throat_pde_stage097_single_normalization_defect.md:46	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage099_reduced_finish_line.md:76	100
099	epsilon_2 fit_stage_099_epsilon_2 fam_0391_epsilon_2	origin stage 96 notes/stages/moving_throat_pde_stage096_geometry_lane_check_verdict.md:31	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage099_reduced_finish_line.md:17	100
099	K0_sym fit_stage_099_k0_sym fam_0118_k0_sym	origin stage 99 notes/stages/moving_throat_pde_stage099_reduced_finish_line.md:43	free_choice	posited choice:notes/stages/moving_throat_pde_stage099_reduced_finish_line.md:75	100
099	Xi_T fit_stage_099_xi_t fam_0285_xi_t	origin stage 099 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
100	chi_Q fit_stage100_chiQ_fixed_downstream_card fam_0345_chi_q; overlays:chain_chi_Q_norm	origin stage 100 notes/stages/moving_throat_pde_stage100_outgoing_normalization_factorization.md:15	free_choice	posited choice:notes/stages/moving_throat_pde_stage100_outgoing_normalization_factorization.md:111	105
100	N_Q fit_stage100_factorization_forced_script fam_0174_n_q; overlays:chain_chi_Q_norm, chain_quad_54_5	origin stage 97 notes/stages/moving_throat_pde_stage097_single_normalization_defect.md:88	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage100_outgoing_normalization_factorization.md:91	105, 106
100	chi_Q fit_stage_100_chi_q fam_0345_chi_q; overlays:chain_chi_Q_norm	origin stage 100 notes/stages/moving_throat_pde_stage100_outgoing_normalization_factorization.md:15	free_choice	posited choice:notes/stages/moving_throat_pde_stage100_outgoing_normalization_factorization.md:79	105
100	Gamma5_t fit_stage_100_gamma5_t fam_0090_gamma5_t	origin stage 100 notes/stages/moving_throat_pde_stage100_outgoing_normalization_factorization.md:69	published_target	published_target; no benchmarks.yaml entry joined for listed family; evidence:notes/stages/moving_throat_pde_stage100_outgoing_normalization_factorization.md:68	105, 106
100	matched_fingerprint_value fit_stage_100_matched_fingerprint_value fam_0459_matched_fingerprint_value	origin stage 100 notes/stages/moving_throat_pde_stage100_outgoing_normalization_factorization.md:20	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage100_outgoing_normalization_factorization.md:79	104, 105

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
101	chi_Q fit_stage101_stage105_fixes_by_matching fam_0345_chi_q; overlays:chain_chi_Q_norm	origin stage 101 notes/stages/moving_throat_pde_stage101_natural_source_map_reduction.md:13	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage101_natural_source_map_reduction.md:49	102, 103, 104, 105, 106, 107
101	chi_Q fit_stage_101_chi_q fam_0345_chi_q; overlays:chain_chi_Q_norm	origin stage 101 notes/stages/moving_throat_pde_stage101_natural_source_map_reduction.md:13	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage101_natural_source_map_reduction.md:49	102, 103, 104, 105, 106, 107
101	Delta_Q fit_stage_101_delta_q fam_0058_delta_q	origin stage 101 notes/stages/moving_throat_pde_stage101_natural_source_map_reduction.md:65	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage101_natural_source_map_reduction.md:65	102, 103, 106
101	matched_fingerprint_value fit_stage_101_matched_fingerprint_value fam_0459_matched_fingerprint_value	origin stage 104 notes/stages/moving_throat_pde_stage104_outgoing_dtn_fingerprint.md:89	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage101_natural_source_map_reduction.md:40	102, 103, 105, 106, 107
102	chi_Q fit_stage102_everything_else_fixed fam_0345_chi_q; overlays:chain_chi_Q_norm	origin stage 101 notes/stages/moving_throat_pde_stage101_natural_source_map_reduction.md:13	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage102_higher_odd_irrelevance.md:32	103, 104, 105, 106, 107
102	chi_Q fit_stage_102_chi_q fam_0345_chi_q; overlays:chain_chi_Q_norm	origin stage 101 notes/stages/moving_throat_pde_stage101_natural_source_map_reduction.md:13	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage102_higher_odd_irrelevance.md:28	103, 104, 105, 106, 107
102	Gamma_5 fit_stage_102_gamma_5 fam_0093_gamma_5; overlays:chain_quad_54_5	origin stage 102 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
102	matched_fingerprint_value fit_stage_102_matched_fingerprint_value fam_0459_matched_fingerprint_value	origin stage 104 notes/stages/moving_throat_pde_stage104_outgoing_dtn_fingerprint.md:89	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage102_higher_odd_irrelevance.md:28	103, 105, 106, 107
103	conservative_split_3_4_1_4 fit_stage103_split_exact_status fam_0352_conservative_split_3_4_1_4	origin stage 91 notes/stages/moving_throat_pde_stage091_grouped_p2_static_geometry_derivation.md:34	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage091_grouped_p2_static_geometry_derivation.md:90	103, 105, 106, 107
103	chi_Q fit_stage_103_chi_q fam_0345_chi_q; overlays:chain_chi_Q_norm	origin stage 101 notes/stages/moving_throat_pde_stage101_natural_source_map_reduction.md:13	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage103_reduced_25pn_conditional_closure.md:20	104, 105, 106, 107

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
103	matched_fingerprint_value fit_stage_103_matched_fingerprint_value fam_0459_matched_fingerprint_value	origin stage 104 notes/stages/moving_throat_pde_stage104_outgoing_dtn_fingerprint.md:89	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage103_reduced_25pn_conditional_closure.md:29	105, 106, 107
104	chi_Q fit_stage104_chiQ_canonical_condition_card fam_0345_chi_q; overlays:chain_chi_Q_norm	origin stage 101 notes/stages/moving_throat_pde_stage101_natural_source_map_reduction.md:13	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage104_outgoing_dtn_fingerprint.md:5	105, 106, 107
104	Gamma_5_can fit_stage104_gamma5_not_free fam_0094_gamma_5_can	origin stage 104 notes/stages/moving_throat_pde_stage104_outgoing_dtn_fingerprint.md:89	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage104_outgoing_dtn_fingerprint.md:87	105, 106, 107
104	chi_Q fit_stage104_unit_product_m0hat2_chiQ_NQ fam_0345_chi_q; overlays:chain_chi_Q_norm	origin stage 101 notes/stages/moving_throat_pde_stage101_natural_source_map_reduction.md:13	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage104_outgoing_dtn_fingerprint.md:92	105, 106, 107
104	chi_Q fit_stage_104_chi_q fam_0345_chi_q; overlays:chain_chi_Q_norm	origin stage 101 notes/stages/moving_throat_pde_stage101_natural_source_map_reduction.md:13	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage104_outgoing_dtn_fingerprint.md:87	105, 106, 107
104	Gamma_5 fit_stage_104_gamma_5 fam_0093_gamma_5; overlays:chain_quad_54_5	origin stage 104 notes/stages/moving_throat_pde_stage104_outgoing_dtn_fingerprint.md:89	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage104_outgoing_dtn_fingerprint.md:83	105, 106, 107
104	matched_fingerprint_value fit_stage_104_matched_fingerprint_value fam_0459_matched_fingerprint_value	origin stage 104 notes/stages/moving_throat_pde_stage104_outgoing_dtn_fingerprint.md:89	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage104_outgoing_dtn_fingerprint.md:83	105, 106, 107
105	Lambda_out_fingerprint_coefficients fit_stage105_carry_in_stale_multi_stage_attribution fam_0153_lambda_out_fingerprint_coefficients	origin stage 104 notes/stages/moving_throat_pde_stage104_outgoing_dtn_fingerprint.md:33	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage104_outgoing_dtn_fingerprint.md:83	105, 106, 107
105	chi_Q fit_stage105_chi_q_canonical_unity fam_0345_chi_q; overlays:chain_chi_Q_norm	origin stage 105 notes/stages/moving_throat_pde_stage105_chiQ_fix_from_outgoing_dtn.md:69	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage105_chiQ_fix_from_outgoing_dtn.md:67	106, 107
105	chi_Q fit_stage105_chi_q_provenance_record_contradiction fam_0345_chi_q; overlays:chain_chi_Q_norm	origin stage 105 notes/stages/moving_throat_pde_stage105_chiQ_fix_from_outgoing_dtn.md:69	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage105_chiQ_fix_from_outgoing_dtn.md:67	106, 107

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
105	chi_Q fit_stage105_last_scalar_fixed_exactly fam_0345_chi_q; overlays:chain_chi_Q_norm	origin stage 105 notes/stages/moving_throat_pde_stage105_chiQ_fix_from_outgoing_dtn.md:69	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage105_chiQ_fix_from_outgoing_dtn.md:72	106, 107
105	Omega_Q fit_stage105_omega_q_pole_frequency fam_0183_omega_q	origin stage 88 notes/stages/moving_throat_pde_stage088_loading_ratio_from_minimal_module.md:19	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage105_chiQ_fix_from_outgoing_dtn.md:29	106, 107
105	chi_Q fit_stage_105_chi_q fam_0345_chi_q; overlays:chain_chi_Q_norm	origin stage 105 notes/stages/moving_throat_pde_stage105_chiQ_fix_from_outgoing_dtn.md:69	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage105_chiQ_fix_from_outgoing_dtn.md:67	106, 107
105	matched_fingerprint_value fit_stage_105_matched_fingerprint_value fam_0459_matched_fingerprint_value	origin stage 104 notes/stages/moving_throat_pde_stage104_outgoing_dtn_fingerprint.md:89	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage105_chiQ_fix_from_outgoing_dtn.md:67	106, 107
105	T_dtn fit_stage_105_t_dtn fam_0253_t_dtn	origin stage 105 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
106	Gamma_5 fit_stage106_coeffs_fixed_to_targets fam_0093_gamma_5; overlays:chain_quad_54_5	origin stage 106 notes/stages/moving_throat_pde_stage106_canonical_outgoing_reduced_closure.md:49	published_target	benchmark: bench__a187d7865e33: textbook - P. C. Peters, 'Gravitational Radiation and the Motion of Two Point Masses', Phys. Rev. 136, B1224 (1964), DOI 10.1103/PhysRev.136.B1224 (ADS bibcode 1964PhRv..136.1224P); see also M. Maggiore, 'Gravitational Waves, Vol. 1: Theory and Experiments', Oxford Univ. Press (2007), ISBN 978-0-19-857074-5, Ch. 3 (quadrupole formula L = (G/5c^5)<dddQ__ij dddQ__ij>).	107
106	Gamma_5_normalized fit_stage106_odd_exactly_gr_target fam_0095_gamma_5_normalized	origin stage 106 notes/stages/moving_throat_pde_stage106_canonical_outgoing_reduced_closure.md:53	published_target	published__target; no benchmarks.yaml entry joined for listed family; evidence:notes/stages/moving_throat_pde_stage106_canonical_outgoing_reduced_closure.md:53	107
106	chi_Q fit_stage_106_chi_q fam_0345_chi_q; overlays:chain_chi_Q_norm	origin stage 105 notes/stages/moving_throat_pde_stage105_chiQ_fix_from_outgoing_dtn.md:69	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage106_canonical_outgoing_reduced_closure.md:73	107
106	Delta_Q fit_stage_106_delta_q fam_0058_delta_q	origin stage 101 notes/stages/moving_throat_pde_stage101_natural_source_map_reduction.md:65	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage101_natural_source_map_reduction.md:65	107

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
106	Gamma_5 fit_stage_106_gamma_5 fam_0093_gamma_5; overlays:chain_quad_54_5	origin stage 106 notes/stages/moving_throat_pde_stage106_canonical_outgoing_reduced_closure.md:49	published_target	benchmark: bench__a187d7865e33: textbook - P. C. Peters, 'Gravitational Radiation and the Motion of Two Point Masses', Phys. Rev. 136, B1224 (1964), DOI 10.1103/Phys-Rev.136.B1224 (ADS bibcode 1964PhRv..136.1224P); see also M. Maggiore, 'Gravitational Waves, Vol. 1: Theory and Experiments', Oxford Univ. Press (2007), ISBN 978-0-19-857074-5, Ch. 3 (quadrupole formula L = (G/5c^5)<dddQ__ij dddQ__ij>).	107
106	matched_fingerprint_value fit_stage_106_matched_fingerprint_value fam_0459_matched_fingerprint_value	origin stage 104 notes/stages/moving_throat_pde_stage104_outgoing_dtn_fingerprint.md:89	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage106_canonical_outgoing_reduced_closure.md:73	107
107	chi_Q fit_stage107_chiQ_canonical_normalization fam_0345_chi_q; overlays:chain_chi_Q_norm	origin stage 105 notes/stages/moving_throat_pde_stage105_chiQ_fix_from_outgoing_dtn.md:69	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage107_general_dtn_deformation.md:22	none recorded
107	Lambda_def fit_stage_107_lambda_def fam_0151_lambda_def	origin stage 107 notes/stages/moving_throat_pde_stage107_general_dtn_deformation.md:31	free_choice	posited choice:notes/stages/moving_throat_pde_stage107_general_dtn_deformation.md:26	none recorded
108	beta fit_stage108_beta_forced_positive_branch fam_0323_beta	origin stage 107 notes/stages/moving_throat_pde_stage107_general_dtn_deformation.md:38	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage108_robustness_classes.md:37	109, 110, 111, 112, 113, 114, 115
108	chi_Q fit_stage108_chiQ_robust_claim fam_0345_chi_q; overlays:chain_chi_Q_norm	origin stage 105 notes/stages/moving_throat_pde_stage105_chiQ_fix_from_outgoing_dtn.md:69	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage108_robustness_classes.md:102	109, 110, 111, 112, 113, 114, 115, 141
108	chi_Q fit_stage_108_chi_q fam_0345_chi_q; overlays:chain_chi_Q_norm	origin stage 105 notes/stages/moving_throat_pde_stage105_chiQ_fix_from_outgoing_dtn.md:69	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage105_chiQ_fix_from_outgoing_dtn.md:67	109, 110, 111, 112, 113, 114, 115, 141
108	matched_fingerprint_value fit_stage_108_matched_fingerprint_value fam_0459_matched_fingerprint_value	origin stage 107 notes/stages/moving_throat_pde_stage107_general_dtn_deformation.md:20	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage108_robustness_classes.md:21	109, 110, 111, 112, 113, 114, 115
108	Y_can_literal fit_stage_108_y_can_literal fam_0292_y_can_literal	origin stage 107 notes/stages/moving_throat_pde_stage107_general_dtn_deformation.md:20	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage108_robustness_classes.md:35	109, 110, 111, 112, 113, 114, 115

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
109	a_0 fit_stage109_linearized_condition_must_satisfy fam_0303_a_0	origin stage 109 notes/stages/moving_throat_pde_stage109_linearized_branch_selection.md:15	free_choice	posited choice:notes/stages/moving_throat_pde_stage109_linearized_branch_selection.md:57	110, 111, 112
110	chi_Q fit_stage110_chiQR_exact_factor fam_0345_chi_q; overlays:chain_chi_Q_norm	origin stage 110 notes/stages/moving_throat_pde_stage110_robin_outlet_model.md:55	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage110_robin_outlet_model.md:52	111, 112, 113
110	a_0 fit_stage_110_a_0 fam_0303_a_0	origin stage 109 notes/stages/moving_throat_pde_stage109_linearized_branch_selection.md:15	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage110_robin_outlet_model.md:68	111, 112
110	chi_Q fit_stage_110_chi_q fam_0345_chi_q; overlays:chain_chi_Q_norm	origin stage 105 notes/stages/moving_throat_pde_stage105_chiQ_fix_from_outgoing_dtn.md:69	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage110_robin_outlet_model.md:55	111, 112, 113
110	matched_fingerprint_value fit_stage_110_matched_fingerprint_value fam_0459_matched_fingerprint_value	origin stage 107 notes/stages/moving_throat_pde_stage107_general_dtn_deformation.md:20	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage110_robin_outlet_model.md:48	111, 112, 113
110	P_2 fit_stage_110_p_2 fam_0188_p_2	origin stage 91 notes/stages/moving_throat_pde_stage091_grouped_p2_static_geometry_derivation.md:1	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage110_robin_outlet_model.md:78	111, 112, 113
111	sigma_W fit_stage111_sigmaW_forced_to_zero fam_0508_sigma_w	origin stage 111 notes/stages/moving_throat_pde_stage111_mixed_sidechannel_pole.md:21	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage111_mixed_sidechannel_pole.md:53	112
111	a_0 fit_stage_111_a_0 fam_0303_a_0	origin stage 109 notes/stages/moving_throat_pde_stage109_linearized_branch_selection.md:15	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage111_mixed_sidechannel_pole.md:77	112
111	A_w fit_stage_111_a_w fam_0012_a_w	origin stage 111 notes/stages/moving_throat_pde_stage111_mixed_sidechannel_pole.md:9	free_choice	posited choice:notes/stages/moving_throat_pde_stage111_mixed_sidechannel_pole.md:5	112

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
111	chi_Q fit_stage_111_chi_q fam_0345_chi_q; overlays:chain_chi_Q_norm	origin stage 105 notes/stages/moving_throat_pde_stage105_chiQ_fix_from_outgoing_dtn.md:69	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage111_mixed_sidechannel_pole.md:64	112, 113
111	Lambda_mix fit_stage_111_lambda_mix fam_0152_lambda_mix	origin stage 111 notes/stages/moving_throat_pde_stage111_mixed_sidechannel_pole.md:12	free_choice	posited choice:notes/stages/moving_throat_pde_stage111_mixed_sidechannel_pole.md:9	112
111	matched_fingerprint_value fit_stage_111_matched_fingerprint_value fam_0459_matched_fingerprint_value	origin stage 107 notes/stages/moving_throat_pde_stage107_general_dtn_deformation.md:20	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage111_mixed_sidechannel_pole.md:41	112
112	chi_Q fit_stage112_chiQhyb_exact_factor fam_0345_chi_q; overlays:chain_chi_Q_norm	origin stage 112 notes/stages/moving_throat_pde_stage112_hybrid_robin_mixed_compensation.md:79	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage112_hybrid_robin_mixed_compensation.md:77	113, 114, 115
112	gamma_W fit_stage112_compensated_branch_coefficients fam_0417_gamma_w	origin stage 111 notes/stages/moving_throat_pde_stage111_mixed_sidechannel_pole.md:23	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage112_hybrid_robin_mixed_compensation.md:84	113, 114, 115
112	kappa_W fit_stage112_exactly_two_branches fam_0436_kappa_w	origin stage 111 notes/stages/moving_throat_pde_stage111_mixed_sidechannel_pole.md:22	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage112_hybrid_robin_mixed_compensation.md:47	113, 114, 115
112	gamma_W fit_stage_112_gamma_w fam_0417_gamma_w	origin stage 111 notes/stages/moving_throat_pde_stage111_mixed_sidechannel_pole.md:23	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage112_hybrid_robin_mixed_compensation.md:106	113, 114, 115
112	matched_fingerprint_value fit_stage_112_matched_fingerprint_value fam_0459_matched_fingerprint_value	origin stage 107 notes/stages/moving_throat_pde_stage107_general_dtn_deformation.md:20	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage112_hybrid_robin_mixed_compensation.md:35	113, 114, 115
113	chi_Q fit_stage113_preserved_exactly_when fam_0345_chi_q; overlays:chain_chi_Q_norm	origin stage 105 notes/stages/moving_throat_pde_stage105_chiQ_fix_from_outgoing_dtn.md:69	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage113_outlet_model_status.md:37	114, 115

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
114	gamma_0 fit_stage114_bare_mixed_denominator_ansatz fam_0415_gamma_0	origin stage 114 notes/stages/moving_throat_pde_stage114_concrete_core_schur.md:29	free_choice	posited choice:notes/stages/moving_throat_pde_stage114_concrete_core_schur.md:101	115
114	delta_Lambda_core fit_stage114_exact_core_correction fam_0365_delta_lambda_core	origin stage 114 notes/stages/moving_throat_pde_stage114_concrete_core_schur.md:33	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage114_concrete_core_schur.md:43	115
114	K_s fit_stage_114_k_s fam_0139_k_s	origin stage 114 notes/stages/moving_throat_pde_stage114_concrete_core_schur.md:18	free_choice	posited choice:notes/stages/moving_throat_pde_stage114_concrete_core_schur.md:97	115
115	gamma_0 fit_stage115_gamma_0_pure_scale_ansatz fam_0415_gamma_0	origin stage 114 notes/stages/moving_throat_pde_stage114_concrete_core_schur.md:29	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage115_core_balance_compensation.md:69	125, 139
115	g_q fit_stage115_gq_exact_two_branch_law fam_0411_g_q	origin stage 114 notes/stages/moving_throat_pde_stage114_concrete_core_schur.md:33	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage115_core_balance_compensation.md:52	125, 139
115	K_q fit_stage_115_k_q fam_0137_k_q	origin stage 114 notes/stages/moving_throat_pde_stage114_concrete_core_schur.md:19	free_choice	posited choice:notes/stages/moving_throat_pde_stage114_concrete_core_schur.md:98	125, 139
115	K_s fit_stage_115_k_s fam_0139_k_s	origin stage 114 notes/stages/moving_throat_pde_stage114_concrete_core_schur.md:18	free_choice	posited choice:notes/stages/moving_throat_pde_stage115_core_balance_compensation.md:33	125, 139
115	matched_fingerprint_value fit_stage_115_matched_fingerprint_value fam_0459_matched_fingerprint_value	origin stage 107 notes/stages/moving_throat_pde_stage107_general_dtn_deformation.md:20	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage115_core_balance_compensation.md:16	125, 139
115	sigma_c fit_stage_115_sigma_c fam_0509_sigma_c	origin stage 114 notes/stages/moving_throat_pde_stage114_concrete_core_schur.md:78	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage115_core_balance_compensation.md:35	125, 139
115	Y_eff fit_stage_115_y_eff fam_0293_y_eff	origin stage 115 notes/stages/moving_throat_pde_stage115_core_balance_compensation.md:92	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage115_core_balance_compensation.md:94	125, 139
115	Y_target fit_stage_115_y_target fam_0295_y_target	origin stage 107 notes/stages/moving_throat_pde_stage107_general_dtn_deformation.md:20	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage115_core_balance_compensation.md:94	125, 139

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
116	L_W fit_stage116_LW_fixed_card fam_0145_l_w	origin stage 116 notes/stages/moving_throat_pde_stage116_dn_mixed_tube_realization.md:46	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage116_dn_mixed_tube_realization.md:40	117, 118, 119, 120, 121, 122, 123, 124, 125
116	L_W fit_stage116_LW_fixed_note fam_0145_l_w	origin stage 116 notes/stages/moving_throat_pde_stage116_dn_mixed_tube_realization.md:46	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage116_dn_mixed_tube_realization.md:40	117, 118, 119, 120, 121, 122, 123, 124, 125
116	gamma_0 fit_stage116_gamma0_compensation_backfill fam_0415_gamma_0	origin stage 115 notes/stages/moving_throat_pde_stage115_core_balance_compensation.md:69	free_choice	posited choice:notes/stages/moving_throat_pde_stage116_dn_mixed_tube_realization.md:62	117, 118, 119, 120, 121, 122, 123, 124, 125
116	L_W fit_stage116_lw_compensation_selected_length fam_0145_l_w	origin stage 116 notes/stages/moving_throat_pde_stage116_dn_mixed_tube_realization.md:46	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage116_dn_mixed_tube_realization.md:40	117, 118, 119, 120, 121, 122, 123, 124, 125
116	gamma_0_bare fit_stage116_pure_scale_removed_exactly fam_0416_gamma_0_bare	origin stage 116 notes/stages/moving_throat_pde_stage116_dn_mixed_tube_realization.md:62	free_choice	posited choice:notes/stages/moving_throat_pde_stage116_dn_mixed_tube_realization.md:62	117, 118, 119, 120, 121
116	L_W fit_stage_116_l_w fam_0145_l_w	origin stage 116 notes/stages/moving_throat_pde_stage116_dn_mixed_tube_realization.md:46	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage116_dn_mixed_tube_realization.md:40	117, 118, 119, 120, 121, 122, 123, 124, 125
117	L_W fit_stage117_formulas_derived_claim fam_0145_l_w	origin stage 116 notes/stages/moving_throat_pde_stage116_dn_mixed_tube_realization.md:46	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage117_outlet_core_status.md:26	118, 119, 120, 121, 122, 123, 124, 125
117	L_W fit_stage_117_l_w fam_0145_l_w	origin stage 116 notes/stages/moving_throat_pde_stage116_dn_mixed_tube_realization.md:46	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage117_outlet_core_status.md:26	118, 119, 120, 121, 122, 123, 124, 125
117	matched_fingerprint_value fit_stage_117_matched_fingerprint_value fam_0459_matched_fingerprint_value	origin stage 104 notes/stages/moving_throat_pde_stage104_outgoing_dtn_fingerprint.md:46	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage117_outlet_core_status.md:53	118, 119, 120, 121, 122, 123, 124, 125
118	K_q fit_stage118_Kq_fixed_not_arbitrary fam_0137_k_q	origin stage 118 notes/stages/moving_throat_pde_stage118_parent_core_extraction.md:114	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage118_parent_core_extraction.md:149	119, 120, 121, 122, 123, 124, 125
118	n_eos fit_stage118_frozen_n5_eos fam_0481_n_eos	origin stage 118 notes/stages/moving_throat_pde_phase0_spec.md:12	free_choice	posited choice:notes/stages/moving_throat_pde_phase0_spec.md:12	119, 120, 121, 122, 123, 124, 125

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
118	`A_q` `fit_stage_118_a_q` `fam_0009_a_q`	origin stage 118 `notes/stages/moving_throat_pde_stage118_parent_core_extraction.md`:54	`free_choice`	posited choice:`notes/stages/moving_throat_pde_stage118_parent_core_extraction.md`:52	119, 120, 121, 122, 123, 124, 125
119	`mathfrak_g` `fit_stage119_g_fixed_exactly_once_r` `fam_0463_mathfrak_g`	origin stage 119 `notes/stages/moving_throat_pde_stage119_parent_balance_family.md`:15	`internal_consistency`	located derivation step:`notes/stages/moving_throat_pde_stage119_parent_balance_family.md`:52	120, 121, 122, 123, 124, 125
119	`L_W` `fit_stage119_one_parameter_family_LW_fixed` `fam_0145_l_w`	origin stage 116 `notes/stages/moving_throat_pde_stage116_dn_mixed_tube_realization.md`:46	`internal_consistency`	located derivation step:`notes/stages/moving_throat_pde_stage119_parent_balance_family.md`:162	120, 121, 122, 123, 124, 125
119	`mathfrak` `fit_stage_119_mathfrak` `fam_0462_mathfrak`	origin stage 119 `notes/stages/moving_throat_pde_stage119_parent_balance_family.md`:11	`internal_consistency`	located derivation step:`notes/stages/moving_throat_pde_stage119_parent_balance_family.md`:52	120, 121, 122, 123, 124, 125
120	`mathfrak_g` `fit_stage120_branch_exists_iff` `fam_0463_mathfrak_g`	origin stage 119 `notes/stages/moving_throat_pde_stage119_parent_balance_family.md`:15	`internal_consistency`	located derivation step:`notes/stages/moving_throat_pde_stage120_core_parameter_status.md`:24	121, 122, 123, 124, 125
121	`L_over_a` `fit_stage121_aspect_ratio_37_20_reference_freeze` `fam_0146_l_over_a`; overlays:`chain_aspect_37_20`	origin stage 73 `notes/stages/moving_throat_pde_stage073_family1_geometry_map.md`:54	`free_choice`	posited choice:`notes/stages/moving_throat_pde_stage073_family1_geometry_map.md`:56	122, 123, 124, 125, 139
121	`L_W` `fit_stage121_lw_equals_l_identification` `fam_0145_l_w`	origin stage 116 `notes/stages/moving_throat_pde_stage116_dn_mixed_tube_realization.md`:46	`free_choice`	posited choice:`notes/stages/moving_throat_pde_stage121_geometric_r_selection.md`:26	122, 123, 124, 125
121	`L` `fit_stage121_r_geom_exact_branch` `fam_0142_l`	origin stage 121 `notes/stages/moving_throat_pde_stage121_geometric_r_selection.md`:26	`free_choice`	posited choice:`notes/stages/moving_throat_pde_stage121_geometric_r_selection.md`:24	122, 123, 124, 125
121	`mathfrak_r` `fit_stage121_r_not_tunable` `fam_0466_mathfrak_r`	origin stage 119 `notes/stages/moving_throat_pde_stage119_parent_balance_family.md`:13	`free_choice`	posited choice:`notes/stages/moving_throat_pde_stage121_geometric_r_selection.md`:98	122, 123, 124, 125
122	`g_nat` `fit_stage122_g_nat_equal_normalized_unity` `fam_0409_g_nat`	origin stage 122 `notes/stages/moving_throat_pde_stage122_mouth_source_compensation_test.md`:26	`free_choice`	posited choice:`notes/stages/moving_throat_pde_stage122_mouth_source_compensation_test.md`:30	123, 124, 125

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
122	g_nat fit_stage122_g_nat_unity_branch_ansatz fam_0409_g_nat	origin stage 122 notes/stages/moving_throat_pde_stage122_mouth_source_compensation_test.md:26	free_choice	posited choice:notes/stages/moving_throat_pde_stage122_mouth_source_compensation_test.md:30	123, 124, 125
122	g_star fit_stage122_g_pm_exact_values fam_0412_g_star; overlays:chain_aspect_37_20	origin stage 122 notes/stages/moving_throat_pde_stage122_mouth_source_compensation_test.md:53	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage122_mouth_source_compensation_test.md:47	123, 124, 125
122	r_F1 fit_stage122_rF1_geometry_fixes fam_0493_r_f1; overlays:chain_aspect_37_20	origin stage 121 notes/stages/moving_throat_pde_stage121_geometric_r_selection.md:65	free_choice	posited choice:notes/stages/moving_throat_pde_stage122_mouth_source_compensation_test.md:49	123, 124, 125
122	T_m fit_stage_122_t_m fam_0255_t_m	origin stage 118 notes/stages/moving_throat_pde_stage118_parent_core_extraction.md:212	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage122_mouth_source_compensation_test.md:122	123, 124, 125
123	Xi_T fit_stage123_exact_branch_law fam_0285_xi_t	origin stage 123 notes/stages/moving_throat_pde_stage123_parent_normalized_branch_values.md:57	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage123_parent_normalized_branch_values.md:66	124, 125
123	r_F1 fit_stage123_geometry_selected_hybridization fam_0493_r_f1; overlays:chain_aspect_37_20	origin stage 121 notes/stages/moving_throat_pde_stage121_geometric_r_selection.md:65	free_choice	posited choice:notes/stages/moving_throat_pde_stage123_parent_normalized_branch_values.md:34	124, 125
123	mathfrak fit_stage_123_mathfrak fam_0462_mathfrak	origin stage 119 notes/stages/moving_throat_pde_stage119_parent_balance_family.md:11	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage123_parent_normalized_branch_values.md:25	124, 125
124	mathfrak_g_pm_F1 fit_stage124_what_is_now_fixed fam_0465_mathfrak_g_pm_f1	origin stage 122 notes/stages/moving_throat_pde_stage122_mouth_source_compensation_test.md:53	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage124_core_branch_status.md:23	125
124	g_nat fit_stage_124_g_nat fam_0409_g_nat	origin stage 122 notes/stages/moving_throat_pde_stage122_mouth_source_compensation_test.md:26	free_choice	posited choice:notes/stages/moving_throat_pde_stage122_mouth_source_compensation_test.md:30	125
124	r_F1 fit_stage_124_r_f1 fam_0493_r_f1; overlays:chain_aspect_37_20	origin stage 121 notes/stages/moving_throat_pde_stage121_geometric_r_selection.md:65	free_choice	posited choice:notes/stages/moving_throat_pde_stage124_core_branch_status.md:13	125

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
124	stage_141 fit_stage_124_stage_141 fam_0516_stage_141	origin stage 124 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
125	branch_sign fit_stage125_lower_branch_unique fam_0332_branch_sign	origin stage 125 notes/stages/moving_throat_pde_stage125_positive_source_theorem.md:76	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage125_positive_source_theorem.md:52	none recorded
126	mathfrak_g_match fit_stage126_g_match_pi_over_4_card fam_0464_mathfrak_g_match	origin stage 126 notes/stages/moving_throat_pde_stage126_positive_source_families.md:36	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage126_positive_source_families.md:27	133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145
126	g_star fit_stage126_not_a_coefficient_fit fam_0412_g_star; overlays:chain_aspect_37_20	origin stage 122 notes/stages/moving_throat_pde_stage122_mouth_source_compensation_test.md:63	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage126_positive_source_families.md:115	133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145
126	xi_star fit_stage126_xi_star_admixture_backsolve fam_0549_xi_star	origin stage 126 notes/stages/moving_throat_pde_stage126_positive_source_families.md:101	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage126_positive_source_families.md:91	126
126	xi_star fit_stage126_xi_star_exact_solution fam_0549_xi_star	origin stage 126 notes/stages/moving_throat_pde_stage126_positive_source_families.md:100	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage126_positive_source_families.md:98	126
126	g_xi fit_stage_126_g_xi fam_0413_g_xi	origin stage 126 notes/stages/moving_throat_pde_stage126_positive_source_families.md:82	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage126_positive_source_families.md:80	126
126	mathfrak fit_stage_126_mathfrak fam_0462_mathfrak	origin stage 126 notes/stages/moving_throat_pde_stage126_positive_source_families.md:36	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage126_positive_source_families.md:115	133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145
127	x_star_exp fit_stage127_exp_depth_exact_branch fam_0543_x_star_exp	origin stage 127 notes/stages/moving_throat_pde_stage127_penetration_families.md:78	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage127_penetration_families.md:73	133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145
127	x_star_slab fit_stage127_slab_depth_unique fam_0544_x_star_slab	origin stage 127 notes/stages/moving_throat_pde_stage127_penetration_families.md:40	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage127_penetration_families.md:35	133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
127	x_star_exp fit_stage127_x_star_exp_backsolve fam_0543_x_star_exp	origin stage 127 notes/stages/moving_throat_pde_stage127_penetration_families.md:78	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage127_penetration_families.md:73	133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145
127	x_star_slab fit_stage127_x_star_slab_backsolve fam_0544_x_star_slab	origin stage 127 notes/stages/moving_throat_pde_stage127_penetration_families.md:40	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage127_penetration_families.md:35	133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145
128	branch_sign fit_stage128_branch_sign_fixed_card fam_0332_branch_sign	origin stage 128 notes/stages/moving_throat_pde_stage128_mouth_source_bias_status.md:75	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage128_mouth_source_bias_status.md:29	133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145
128	g_star fit_stage128_lower_branch_uniquely_selected fam_0412_g_star; overlays:chain_aspect_37_20	origin stage 122 notes/stages/moving_throat_pde_stage122_mouth_source_compensation_test.md:63	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage128_mouth_source_bias_status.md:35	133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145
128	r_F1 fit_stage128_rF1_fixed_by_geometry fam_0493_r_f1; overlays:chain_aspect_37_20	origin stage 121 notes/stages/moving_throat_pde_stage121_geometric_r_selection.md:70	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage128_mouth_source_bias_status.md:12	122, 126, 127, 128, 133, 134
128	traction_gap_percent fit_stage128_traction_gap_downstream_attribution fam_0529_traction_gap_percent	origin stage 126 notes/stages/moving_throat_pde_stage126_positive_source_families.md:65	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage126_positive_source_families.md:61	128
129	sigma_Pi fit_stage129_sigma_pi_derived_boundary_law fam_0506_sigma_pi	origin stage 129 notes/stages/moving_throat_pde_stage129_mouth_boundary_layer.md:114	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage129_mouth_boundary_layer.md:122	132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145
129	stop_cold fit_stage_129_stop_cold fam_0518_stop_cold	origin stage 129 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
130	g_star fit_stage130_g_minus_compensation_target fam_0412_g_star; overlays:chain_aspect_37_20	origin stage 122 notes/stages/moving_throat_pde_stage122_mouth_source_compensation_test.md:63	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage130_mouth_bias_map.md:95	130, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145
130	Pi_star fit_stage130_pi_star_exact_unique fam_0209_pi_star	origin stage 130 notes/stages/moving_throat_pde_stage130_mouth_bias_map.md:108	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage130_mouth_bias_map.md:103	131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
130	x_star fit_stage130_x_star_penetration_depth_match fam_0542_x_star	origin stage 127 notes/stages/moving_throat_pde_stage127_penetration_families.md:78	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage130_mouth_bias_map.md:118	130, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145
131	L_over_a fit_stage131_g_minus_closed_form_aspect_ratio fam_0146_l_over_a; overlays:chain_aspect_37_20	origin stage 131 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
131	A0_prime fit_stage131_parent_threshold_canonical fam_0002_a0_prime	origin stage 131 notes/stages/moving_throat_pde_stage131_parent_mouth_threshold.md:57	free_choice	posited choice:notes/stages/moving_throat_pde_stage131_parent_mouth_threshold.md:70	133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145
131	A_0 fit_stage_131_a_0 fam_0003_a_0	origin stage 129 notes/stages/moving_throat_pde_stage129_mouth_boundary_layer.md:14	free_choice	posited choice:notes/stages/moving_throat_pde_stage131_parent_mouth_threshold.md:70	129, 131, 133, 134
131	Pi_star fit_stage_131_pi_star fam_0209_pi_star	origin stage 130 notes/stages/moving_throat_pde_stage130_mouth_bias_map.md:108	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage131_parent_mouth_threshold.md:30	132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145
131	Pi_Theta_sigma_L fit_stage_131_pi_theta_sigma_l fam_0203_pi_theta_sigma_l	origin stage 131 notes/stages/moving_throat_pde_stage131_parent_mouth_threshold.md:39	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage131_parent_mouth_threshold.md:27	133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145
132	sigma_Pi fit_stage132_source_family_fixed_claim fam_0506_sigma_pi	origin stage 129 notes/stages/moving_throat_pde_stage129_mouth_boundary_layer.md:114	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage132_mouth_boundary_layer_status.md:8	133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145
132	Pi fit_stage_132_pi fam_0202_pi	origin stage 130 notes/stages/moving_throat_pde_stage130_mouth_bias_map.md:108	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage132_mouth_boundary_layer_status.md:28	133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145
133	M_minus fit_stage133_residual_freedom_fixed fam_0159_m_minus	origin stage 133 notes/stages/moving_throat_pde_stage133_coupled_mouth_fixedpoint.md:185	free_choice	posited choice:notes/stages/moving_throat_pde_stage133_coupled_mouth_fixedpoint.md:216	146, 147, 148, 149, 150, 151, 152, 153
133	S_target fit_stage_133_s_target fam_0234_s_target	origin stage 133 notes/stages/moving_throat_pde_stage133_coupled_mouth_fixedpoint.md:147	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage133_coupled_mouth_fixedpoint.md:154	134, 146, 147, 148, 149, 150, 151, 152, 153

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
134	M_q fit_stage134_bias_determined_by_fixedpoint fam_0161_m_q	origin stage 134 notes/stages/moving_throat_pde_stage134_family1_mouth_fixedpoint.md:37	free_choice	posited choice:notes/stages/moving_throat_pde_stage134_family1_mouth_fixedpoint.md:145	135, 137, 140, 146, 147, 148, 149, 150, 151, 152, 153
134	kappa_q fit_stage134_kappa_q_first_mode_choice fam_0438_kappa_q	origin stage 134 notes/stages/moving_throat_pde_stage134_family1_mouth_fixedpoint.md:29	free_choice	posited choice:notes/stages/moving_throat_pde_stage134_family1_mouth_fixedpoint.md:31	135, 137, 140, 146, 147, 148, 149, 150, 151, 152, 153
134	Pi_star fit_stage134_pi_star_carried_literal fam_0209_pi_star	origin stage 130 notes/stages/moving_throat_pde_stage130_mouth_bias_map.md:108	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage134_family1_mouth_fixedpoint.md:70	135, 137, 140, 146, 147, 148, 149, 150, 151, 152, 153
134	S_q_at_half fit_stage134_sq_spotcheck_anchor_literals fam_0230_s_q_at_half	origin stage 134 notes/stages/moving_throat_pde_stage134_family1_mouth_fixedpoint.md:52	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage134_family1_mouth_fixedpoint.md:49	134
134	M_q fit_stage_134_m_q fam_0161_m_q	origin stage 134 notes/stages/moving_throat_pde_stage134_family1_mouth_fixedpoint.md:37	free_choice	posited choice:notes/stages/moving_throat_pde_stage134_family1_mouth_fixedpoint.md:145	135, 137, 140, 146, 147, 148, 149, 150, 151, 152, 153
134	MATHEMATICA_MIRROR_POLICY fit_stage_134_mathematica_mirror_policy fam_0157_mathematica_mirror_policy	origin stage 134 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
134	S_q fit_stage_134_s_q fam_0229_s_q	origin stage 134 notes/stages/moving_throat_pde_stage134_family1_mouth_fixedpoint.md:45	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage134_family1_mouth_fixedpoint.md:49	135, 137, 140, 146, 147, 148, 149, 150, 151, 152, 153
135	Sigma_0 fit_stage135_exact_outlet_consistent_gain fam_0239_sigma_0	origin stage 135 notes/stages/moving_throat_pde_stage135_outlet_consistent_mouth_closure.md:89	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage135_outlet_consistent_mouth_closure.md:88	136, 139, 140, 146, 152, 153
135	M_q_over_Sigma_m fit_stage135_outlet_gain_ratio fam_0163_m_q_over_sigma_m	origin stage 135 notes/stages/moving_throat_pde_stage135_outlet_consistent_mouth_closure.md:36	free_choice	posited choice:notes/stages/moving_throat_pde_stage135_outlet_consistent_mouth_closure.md:30	136, 139, 140, 141, 142
135	Pi_star fit_stage135_pi_star_import_sigma_m_star fam_0209_pi_star	origin stage 130 notes/stages/moving_throat_pde_stage130_mouth_bias_map.md:108	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage130_mouth_bias_map.md:103	131, 132, 134, 135, 136, 139, 142, 144, 145, 146, 152, 156

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
135	M_q fit_stage_135_m_q fam_0161_m_q	origin stage 135 notes/stages/moving_throat_pde_stage135_outlet_consistent_mouth_closure.md:36	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage135_outlet_consistent_mouth_closure.md:99	136, 139, 140, 141, 142, 143
137	M_q fit_stage137_gains_from_derived_core_response fam_0161_m_q	origin stage 137 notes/stages/moving_throat_pde_stage137_core_to_mouth_gain_map.md:65	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage137_core_to_mouth_gain_map.md:47	138, 139, 140, 141, 142, 146
137	D_W_bare fit_stage_137_d_w_bare fam_0047_d_w_bare	origin stage 114 notes/stages/moving_throat_pde_stage114_concrete_core_schur.md:29	free_choice	posited choice:notes/stages/moving_throat_pde_stage114_concrete_core_schur.md:36	115, 116, 137
137	K_s fit_stage_137_k_s fam_0139_k_s	origin stage 114 notes/stages/moving_throat_pde_stage114_concrete_core_schur.md:18	free_choice	posited choice:notes/stages/moving_throat_pde_stage114_concrete_core_schur.md:97	115, 116, 117, 119, 120, 121, 122, 123, 137, 138
138	R_q_comp fit_stage138_compensation_quarter fam_0221_r_q_comp	origin stage 138 notes/stages/moving_throat_pde_stage138_normalized_mouth_gain_family.md:69	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage138_normalized_mouth_gain_family.md:60	139, 140, 141, 142, 144, 145, 146
138	R_q fit_stage138_rq_quarter_derived_not_assumed fam_0220_r_q	origin stage 138 notes/stages/moving_throat_pde_stage138_normalized_mouth_gain_family.md:45	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage138_normalized_mouth_gain_family.md:60	139, 140, 141, 142, 144, 145, 146
138	K_s fit_stage_138_k_s fam_0139_k_s	origin stage 114 notes/stages/moving_throat_pde_stage114_concrete_core_schur.md:18	free_choice	posited choice:notes/stages/moving_throat_pde_stage114_concrete_core_schur.md:97	115, 116, 117, 119, 120, 121, 122, 123, 137, 138
139	L_over_a fit_stage139_aspect_ratio_literal fam_0146_l_over_a; overlays:chain_aspect_37_20	origin stage 73 notes/stages/moving_throat_pde_stage073_family1_geometry_map.md:54	free_choice	posited choice:notes/stages/moving_throat_pde_stage073_family1_geometry_map.md:48	121, 124, 139, 146
139	M_q_comp_star fit_stage139_gains_land_exactly_on_canonical fam_0162_m_q_comp_star	origin stage 139 notes/stages/moving_throat_pde_stage139_family1_actual_mouth_gains.md:116	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage139_family1_actual_mouth_gains.md:93	140, 141, 142, 144, 145, 146
139	M_q fit_stage_139_m_q fam_0161_m_q	origin stage 137 notes/stages/moving_throat_pde_stage137_core_to_mouth_gain_map.md:65	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage139_family1_actual_mouth_gains.md:93	140, 141, 142, 143, 144, 145, 146

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
140	M_s_comp fit_stage140_carried_gain_magnitudes fam_0165_m_s_comp	origin stage 139 notes/stages/moving_throat_pde_stage139_family1_actual_mouth_gains.md:110	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage139_family1_actual_mouth_gains.md:93	141, 142, 146, 152, 153
140	Sigma_0 fit_stage140_sigma0_prefactor_fixed fam_0239_sigma_0	origin stage 140 notes/stages/moving_throat_pde_stage140_selfmatched_mouth_susceptibility.md:75	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage140_selfmatched_mouth_susceptibility.md:42	141, 142, 146, 152, 153
140	Theta_sigma fit_stage140_theta_sigma_not_fitted_by_hand fam_0263_theta_sigma	origin stage 140 notes/stages/moving_throat_pde_stage140_selfmatched_mouth_susceptibility.md:29	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage140_selfmatched_mouth_susceptibility.md:32	146, 152, 153
140	J_s fit_stage_140_j_s fam_0114_j_s	origin stage 140 notes/stages/moving_throat_pde_stage140_selfmatched_mouth_susceptibility.md:22	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage140_selfmatched_mouth_susceptibility.md:15	146, 152, 153
140	M_s fit_stage_140_m_s fam_0164_m_s	origin stage 138 notes/stages/moving_throat_pde_stage138_normalized_mouth_gain_family.md:25	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage140_selfmatched_mouth_susceptibility.md:139	141, 142, 146, 152, 153
140	Theta_sigma fit_stage_140_theta_sigma fam_0263_theta_sigma	origin stage 140 notes/stages/moving_throat_pde_stage140_selfmatched_mouth_susceptibility.md:29	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage140_selfmatched_mouth_susceptibility.md:32	146, 152, 153
141	M_q fit_stage141_gain_pair_no_longer_free fam_0161_m_q	origin stage 137 notes/stages/moving_throat_pde_stage137_core_to_mouth_gain_map.md:65	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage141_mouth_gain_status.md:7	142, 143, 144, 145, 146
142	R_q fit_stage142_three_quantities_no_longer_free fam_0220_r_q	origin stage 138 notes/stages/moving_throat_pde_stage138_normalized_mouth_gain_family.md:45	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage142_selfconsistent_mouth_branch.md:112	143, 144, 145, 146
142	M_q fit_stage_142_m_q fam_0161_m_q	origin stage 137 notes/stages/moving_throat_pde_stage137_core_to_mouth_gain_map.md:65	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage142_selfconsistent_mouth_branch.md:16	143, 144, 145, 146

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
142	R_q fit_stage_142_r_q fam_0220_r_q	origin stage 138 notes/stages/moving_throat_pde_stage138_normalized_mouth_gain_family.md:69	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage142_selfconsistent_mouth_branch.md:112	143, 144, 145, 146
143	M_q fit_stage_143_m_q fam_0161_m_q	origin stage 137 notes/stages/moving_throat_pde_stage137_core_to_mouth_gain_map.md:65	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage143_equal_normalized_singular_limit.md:83	144, 145, 146
144	Pi_star fit_stage144_unique_regular_canonical_branch fam_0209_pi_star	origin stage 130 notes/stages/moving_throat_pde_stage130_mouth_bias_map.md:108	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage144_unique_regular_canonical_branch.md:49	145, 146, 152, 156
144	mathfrak fit_stage_144_mathfrak fam_0462_mathfrak	origin stage 144 notes/stages/moving_throat_pde_stage144_unique_regular_canonical_branch.md:17	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage144_unique_regular_canonical_branch.md:21	145, 146
144	R_q fit_stage_144_r_q fam_0220_r_q	origin stage 138 notes/stages/moving_throat_pde_stage138_normalized_mouth_gain_family.md:45	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage144_unique_regular_canonical_branch.md:15	145, 146
145	M_q fit_stage_145_m_q fam_0161_m_q	origin stage 137 notes/stages/moving_throat_pde_stage137_core_to_mouth_gain_map.md:65	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage145_mouth_branch_selection_status.md:26	146, 152, 153
145	R_q fit_stage_145_r_q fam_0220_r_q	origin stage 138 notes/stages/moving_throat_pde_stage138_normalized_mouth_gain_family.md:45	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage145_mouth_branch_selection_status.md:26	146, 152, 153
146	L_over_a fit_stage146_aspect_ratio_37_20_primitive fam_0146_l_over_a; overlays:chain_aspect_37_20	origin stage 73 notes/stages/moving_throat_pde_stage073_family1_geometry_map.md:54	free_choice	posited choice:notes/stages/moving_throat_pde_stage073_family1_geometry_map.md:48	121, 124, 139, 146
147	A_T fit_stage147_AT_BT_circular_paper_anchor fam_0008_a_t	origin stage 147 notes/stages/moving_throat_pde_stage147_first_order_rigidity_kernel.md:67	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage147_first_order_rigidity_kernel.md:50	148, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163
147	A_T fit_stage147_AT_BT_paper_anchor fam_0008_a_t	origin stage 147 notes/stages/moving_throat_pde_stage147_first_order_rigidity_kernel.md:67	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage147_first_order_rigidity_kernel.md:48	148, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
147	abs_A_T_over_B_T fit_stage147_AT_BT_ratio_31_6785_mislabel_history fam_0308_abs_a_t_over_b_t	origin stage 147 notes/stages/moving_throat_pde_stage147_first_order_rigidity_kernel.md:77	internal_consistency	located derivation step:notes/CHECKPOINT_CONSTANT_PROVENANCE.md:35	148
147	A_T fit_stage_147_a_t fam_0008_a_t	origin stage 147 notes/stages/moving_throat_pde_stage147_first_order_rigidity_kernel.md:67	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage147_first_order_rigidity_kernel.md:48	148, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163
148	delta_Pi_lambda fit_stage148_canonical_first_order_shifts_exact fam_0367_delta_pi_lambda	origin stage 148 notes/stages/moving_throat_pde_stage148_representative_positive_families.md:99	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage148_representative_positive_families.md:95	154, 155, 156, 157, 158, 159, 160, 161, 162, 163
148	r_F1 fit_stage148_radical_constant_forced_by_rf1 fam_0493_r_f1; overlays:chain_aspect_37_20	origin stage 121 notes/stages/moving_throat_pde_stage121_geometric_r_selection.md:65	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage121_geometric_r_selection.md:37	122, 123, 126, 127, 142, 143, 148, 154, 155
148	A_T fit_stage_148_a_t fam_0008_a_t	origin stage 147 notes/stages/moving_throat_pde_stage147_first_order_rigidity_kernel.md:48	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage147_first_order_rigidity_kernel.md:34	148, 150, 151, 152, 154, 155
150	Pi_star fit_stage150_residual_tangent_matched fam_0209_pi_star	origin stage 144 notes/stages/moving_throat_pde_stage144_unique_regular_canonical_branch.md:62	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage144_unique_regular_canonical_branch.md:58	146, 147, 148, 150, 151, 152, 154, 155
150	Phi_x_Pi_x fit_stage_150_phi_x_pi_x fam_0201_phi_x_pi_x	origin stage 150 notes/stages/moving_throat_pde_stage150_full_profile_residual.md:12	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage150_full_profile_residual.md:10	151, 152, 154, 155
150	S_q fit_stage_150_s_q fam_0229_s_q	origin stage 150 notes/stages/moving_throat_pde_stage150_full_profile_residual.md:92	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage150_full_profile_residual.md:98	151, 152, 154, 155
151	delta_Pi_act fit_stage151_correction_fixed_by_covariances fam_0366_delta_pi_act	origin stage 151 notes/stages/moving_throat_pde_stage151_first_order_selected_correction.md:96	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage151_first_order_selected_correction.md:42	152, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163
151	A_T fit_stage_151_a_t fam_0008_a_t	origin stage 147 notes/stages/moving_throat_pde_stage147_first_order_rigidity_kernel.md:48	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage147_first_order_rigidity_kernel.md:34	148, 150, 151, 152, 154, 155

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
152	lambda_eff_Pi fit_stage152_affine_law_endpoints fam_0450_lambda_eff_pi	origin stage 152 notes/stages/moving_throat_pde_stage152_family1_actual_correction.md:137	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage152_family1_actual_correction.md:134	154, 155
152	A_T fit_stage152_carried_point_block fam_0008_a_t, fam_0016_b_t, fam_0209_pi_star, fam_0233_s_star, fam_0244_sigma_m_star, fam_0259_t_star, fam_0412_g_star, fam_0424_gprime_star; overlays:chain_aspect_37_20	origin stage 147 notes/stages/moving_throat_pde_stage147_first_order_rigidity_kernel.md:48	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage152_family1_actual_correction.md:50	154, 155, 156, 157, 158, 159, 160, 161, 162, 163
152	B_T fit_stage152_carried_point_block fam_0008_a_t, fam_0016_b_t, fam_0209_pi_star, fam_0233_s_star, fam_0244_sigma_m_star, fam_0259_t_star, fam_0412_g_star, fam_0424_gprime_star; overlays:chain_aspect_37_20	origin stage 147 notes/stages/moving_throat_pde_stage147_first_order_rigidity_kernel.md:58	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage147_first_order_rigidity_kernel.md:34	154, 155, 156, 157, 158, 159, 160, 161, 162, 163
152	Pi_star fit_stage152_carried_point_block fam_0008_a_t, fam_0016_b_t, fam_0209_pi_star, fam_0233_s_star, fam_0244_sigma_m_star, fam_0259_t_star, fam_0412_g_star, fam_0424_gprime_star; overlays:chain_aspect_37_20	origin stage 144 notes/stages/moving_throat_pde_stage144_unique_regular_canonical_branch.md:62	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage144_unique_regular_canonical_branch.md:58	146, 147, 148, 150, 151, 152, 154, 155
152	S_star fit_stage152_carried_point_block fam_0008_a_t, fam_0016_b_t, fam_0209_pi_star, fam_0233_s_star, fam_0244_sigma_m_star, fam_0259_t_star, fam_0412_g_star, fam_0424_gprime_star; overlays:chain_aspect_37_20	origin stage 142 notes/stages/moving_throat_pde_stage142_selfconsistent_mouth_branch.md:114	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage142_selfconsistent_mouth_branch.md:47	147, 148, 150, 151, 152, 154, 155
152	Sigma_m_star fit_stage152_carried_point_block fam_0008_a_t, fam_0016_b_t, fam_0209_pi_star, fam_0233_s_star, fam_0244_sigma_m_star, fam_0259_t_star, fam_0412_g_star, fam_0424_gprime_star; overlays:chain_aspect_37_20	origin stage 135 notes/stages/moving_throat_pde_stage135_outlet_consistent_mouth_closure.md:89	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage135_outlet_consistent_mouth_closure.md:78	142, 144, 147, 148, 150, 151, 152, 154, 155
152	T_star fit_stage152_carried_point_block fam_0008_a_t, fam_0016_b_t, fam_0209_pi_star, fam_0233_s_star, fam_0244_sigma_m_star, fam_0259_t_star, fam_0412_g_star, fam_0424_gprime_star; overlays:chain_aspect_37_20	origin stage 142 notes/stages/moving_throat_pde_stage142_selfconsistent_mouth_branch.md:121	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage142_selfconsistent_mouth_branch.md:74	144, 147, 148, 152, 154, 155

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
152	g_star fit_stage152_carried_point_block fam_0008_a_t, fam_0016_b_t, fam_0209_pi_star, fam_0233_s_star, fam_0244_sigma_m_star, fam_0259_t_star, fam_0412_g_star, fam_0424_gprime_star; overlays:chain_aspect_37_20	origin stage 142 notes/stages/moving_throat_pde_stage142_selfconsistent_mouth_branch.md:97	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage144_unique_regular_canonical_branch.md:58	144, 146, 147, 148, 150, 151, 152, 154, 155
152	gprime_star fit_stage152_carried_point_block fam_0008_a_t, fam_0016_b_t, fam_0209_pi_star, fam_0233_s_star, fam_0244_sigma_m_star, fam_0259_t_star, fam_0412_g_star, fam_0424_gprime_star; overlays:chain_aspect_37_20	origin stage 131 notes/stages/moving_throat_pde_stage131_parent_mouth_threshold.md:92	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage146_positive_deformation_expansion.md:125	146, 147, 148, 151, 152, 154, 155
152	Pi_corr fit_stage152_corrected_mouth_state_pi_corr_t_corr fam_0205_pi_corr	origin stage 152 notes/stages/moving_throat_pde_stage152_family1_actual_correction.md:75	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage152_family1_actual_correction.md:77	154, 155, 156, 157
152	cov_KR_expected fit_stage152_profile_correction_samples_expected_pins fam_0353_cov_kr_expected	origin stage 152 notes/stages/moving_throat_pde_stage152_family1_actual_correction.md:30	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage152_family1_actual_correction.md:22	154, 155
152	A_T fit_stage152_retuned_canonical_point fam_0008_a_t	origin stage 147 notes/stages/moving_throat_pde_stage147_first_order_rigidity_kernel.md:48	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage152_family1_actual_correction.md:50	154, 155, 156, 157, 158, 159, 160, 161, 162, 163
152	A_T fit_stage_152_a_t fam_0008_a_t	origin stage 147 notes/stages/moving_throat_pde_stage147_first_order_rigidity_kernel.md:48	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage152_family1_actual_correction.md:50	154, 155, 156, 157, 158, 159, 160, 161, 162, 163
154	Sigma_0 fit_stage154_coevolving_branch_determined fam_0239_sigma_0	origin stage 140 notes/stages/moving_throat_pde_stage140_selfmatched_mouth_susceptibility.md:75	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage140_selfmatched_mouth_susceptibility.md:32	155, 156, 157
154	g_star fit_stage_154_g_star fam_0412_g_star; overlays:chain_aspect_37_20	origin stage 142 notes/stages/moving_throat_pde_stage142_selfconsistent_mouth_branch.md:97	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage154_coevolving_core_mouth_map.md:100	155, 156, 157
154	r_F1 fit_stage_154_r_f1 fam_0493_r_f1; overlays:chain_aspect_37_20	origin stage 121 notes/stages/moving_throat_pde_stage121_geometric_r_selection.md:65	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage121_geometric_r_selection.md:37	122, 123, 126, 127, 142, 143, 148, 154, 155

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
154	T_q fit_stage_154_t_q fam_0257_t_q	origin stage 154 notes/stages/moving_throat_pde_stage154_coevolving_core_mouth_map.md:51	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage154_coevolving_core_mouth_map.md:49	155, 156, 157
155	Pi_star fit_stage155_carried_constants_block fam_0209_pi_star, fam_0233_s_star, fam_0238_sigma0_star, fam_0412_g_star, fam_0493_r_f1; overlays:chain_aspect_37_20	origin stage 144 notes/stages/moving_throat_pde_stage144_unique_regular_canonical_branch.md:62	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage144_unique_regular_canonical_branch.md:58	156, 157, 158
155	S_star fit_stage155_carried_constants_block fam_0209_pi_star, fam_0233_s_star, fam_0238_sigma0_star, fam_0412_g_star, fam_0493_r_f1; overlays:chain_aspect_37_20	origin stage 142 notes/stages/moving_throat_pde_stage142_selfconsistent_mouth_branch.md:114	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage142_selfconsistent_mouth_branch.md:47	156, 157, 158
155	Sigma0_star fit_stage155_carried_constants_block fam_0209_pi_star, fam_0233_s_star, fam_0238_sigma0_star, fam_0412_g_star, fam_0493_r_f1; overlays:chain_aspect_37_20	origin stage 142 notes/stages/moving_throat_pde_stage142_selfconsistent_mouth_branch.md:119	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage142_selfconsistent_mouth_branch.md:74	156, 157, 158
155	g_star fit_stage155_carried_constants_block fam_0209_pi_star, fam_0233_s_star, fam_0238_sigma0_star, fam_0412_g_star, fam_0493_r_f1; overlays:chain_aspect_37_20	origin stage 142 notes/stages/moving_throat_pde_stage142_selfconsistent_mouth_branch.md:97	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage144_unique_regular_canonical_branch.md:58	156, 157, 158
155	r_F1 fit_stage155_carried_constants_block fam_0209_pi_star, fam_0233_s_star, fam_0238_sigma0_star, fam_0412_g_star, fam_0493_r_f1; overlays:chain_aspect_37_20	origin stage 121 notes/stages/moving_throat_pde_stage121_geometric_r_selection.md:65	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage121_geometric_r_selection.md:37	156, 157, 158
155	Pi_can fit_stage155_fixedpoint_expected_targets_json fam_0204_pi_can, fam_0228_s_can, fam_0237_sigma0_can, fam_0252_t_can; overlays:chain_sigma0_transport	origin stage 156 notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:72	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:55	156, 157, 158, 163
155	S_can fit_stage155_fixedpoint_expected_targets_json fam_0204_pi_can, fam_0228_s_can, fam_0237_sigma0_can, fam_0252_t_can; overlays:chain_sigma0_transport	origin stage 156 notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:69	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:55	156, 157, 158, 163

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
155	Sigma0_can fit_stage155_fixedpoint_expected_targets_json fam_0204_pi_can, fam_0228_s_can, fam_0237_sigma0_can, fam_0252_t_can; overlays:chain_sigma0_transport	origin stage 156 notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:31	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:23	156, 157, 158, 163
155	T_can fit_stage155_fixedpoint_expected_targets_json fam_0204_pi_can, fam_0228_s_can, fam_0237_sigma0_can, fam_0252_t_can; overlays:chain_sigma0_transport	origin stage 156 notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:43	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:37	156, 157, 158, 163
155	R_fp fit_stage155_frozen_traction_fixed_point_pins fam_0216_r_fp	origin stage 155 notes/stages/moving_throat_pde_stage155_frozen_traction_fixedpoint.md:69	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage155_frozen_traction_fixedpoint.md:66	156, 157
155	g_star fit_stage_155_g_star fam_0412_g_star; overlays:chain_aspect_37_20	origin stage 142 notes/stages/moving_throat_pde_stage142_selfconsistent_mouth_branch.md:97	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage144_unique_regular_canonical_branch.md:58	156, 157, 158
156	Pi_star fit_stage156_carried_constants_block fam_0209_pi_star, fam_0238_sigma0_star, fam_0254_t_hat_star, fam_0412_g_star, fam_0493_r_f1; overlays:chain_aspect_37_20	origin stage 156 notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:106	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:104	157, 158
156	Sigma0_star fit_stage156_carried_constants_block fam_0209_pi_star, fam_0238_sigma0_star, fam_0254_t_hat_star, fam_0412_g_star, fam_0493_r_f1; overlays:chain_aspect_37_20	origin stage 156 notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:113	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:104	157, 158
156	T_hat_star fit_stage156_carried_constants_block fam_0209_pi_star, fam_0238_sigma0_star, fam_0254_t_hat_star, fam_0412_g_star, fam_0493_r_f1; overlays:chain_aspect_37_20	origin stage 156 notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:127	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:104	157, 158
156	g_star fit_stage156_carried_constants_block fam_0209_pi_star, fam_0238_sigma0_star, fam_0254_t_hat_star, fam_0412_g_star, fam_0493_r_f1; overlays:chain_aspect_37_20	origin stage 122 notes/stages/moving_throat_pde_stage122_mouth_source_compensation_test.md:63	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage122_mouth_source_compensation_test.md:51	156, 157, 158, 159

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
156	r_F1 fit_stage156_carried_constants_block fam_0209_pi_star, fam_0238_sigma0_star, fam_0254_t_hat_star, fam_0412_g_star, fam_0493_r_f1; overlays:chain_aspect_37_20	origin stage 121 notes/stages/moving_throat_pde_stage121_geometric_r_selection.md:70	free_choice	posited choice:notes/stages/moving_throat_pde_stage121_geometric_r_selection.md:60	156, 157, 158, 159, 160, 161
156	mouth_bias_increase_pct fit_stage156_mouth_bias_increase_224_59_pct fam_0473_mouth_bias_increase_pct	origin stage 156 notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:136	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:120	none recorded
156	Sigma0_can fit_stage156_unique_traction_renormalization fam_0237_sigma0_can; overlays:chain_sigma0_transport	origin stage 156 notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:31	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:23	157, 158, 159
156	Pi_can fit_stage_156_pi_can fam_0204_pi_can; overlays:chain_sigma0_transport	origin stage 156 notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:71	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:55	157, 158
156	StatusNumerical fit_stage_156_statusnumerical fam_0247_statusnumerical	origin stage 156 notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:27	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:19	none recorded
157	Pi_can fit_stage157_expected_values_sidecar_json fam_0204_pi_can; overlays:chain_sigma0_transport	origin stage 156 notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:71	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage157_core_mouth_coevolution_status.md:36	158
157	delta_kappa_W fit_stage_157_delta_kappa_w fam_0372_delta_kappa_w	origin stage 159 notes/stages/moving_throat_pde_stage159_hybrid_outlet_projection.md:33	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage159_hybrid_outlet_projection.md:348	159, 160
157	StatusOpen fit_stage_157_statusopen fam_0248_statusopen	origin stage 157 notes/stages/moving_throat_pde_stage157_core_mouth_coevolution_status.md:79	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage157_core_mouth_coevolution_status.md:80	none recorded
158	S_can fit_stage158_canonical_quartet_literals fam_0228_s_can, fam_0237_sigma0_can, fam_0252_t_can, fam_0493_r_f1; overlays:chain_aspect_37_20, chain_sigma0_transport	origin stage 156 notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:68	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:65	159

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
158	Sigma0_can fit_stage158_canonical_quartet_literals fam_0228_s_can, fam_0237_sigma0_can, fam_0252_t_can, fam_0493_r_f1; overlays:chain_aspect_37_20, chain_sigma0_transport	origin stage 156 notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:31	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage158_linear_defect_transport.md:27	159
158	T_can fit_stage158_canonical_quartet_literals fam_0228_s_can, fam_0237_sigma0_can, fam_0252_t_can, fam_0493_r_f1; overlays:chain_aspect_37_20, chain_sigma0_transport	origin stage 156 notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:42	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:35	159
158	r_F1 fit_stage158_canonical_quartet_literals fam_0228_s_can, fam_0237_sigma0_can, fam_0252_t_can, fam_0493_r_f1; overlays:chain_aspect_37_20, chain_sigma0_transport	origin stage 121 notes/stages/moving_throat_pde_stage121_geometric_r_selection.md:70	free_choice	posited choice:notes/stages/moving_throat_pde_stage121_geometric_r_selection.md:60	159, 160, 161
158	slippage_coefficient_eps_perp fit_stage158_linear_defect_transport_coefficients fam_0515_slippage_coefficient_eps_perp	origin stage 158 notes/stages/moving_throat_pde_stage158_linear_defect_transport.md:151	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage158_linear_defect_transport.md:137	159, 160, 161
158	Delta_Q fit_stage_158_delta_q fam_0058_delta_q	origin stage 158 notes/stages/moving_throat_pde_stage158_linear_defect_transport.md:271	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage158_linear_defect_transport.md:289	159, 160, 161
158	matched_fingerprint_value fit_stage_158_matched_fingerprint_value fam_0459_matched_fingerprint_value	origin stage 115 notes/stages/moving_throat_pde_stage115_core_balance_compensation.md:25	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage158_linear_defect_transport.md:410	159, 160, 161
159	gamma_W fit_stage159_canonical_even_baseline fam_0417_gamma_w	origin stage 115 notes/stages/moving_throat_pde_stage115_core_balance_compensation.md:25	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage159_hybrid_outlet_projection.md:199	160, 161
159	Delta_Q_dg_coeff fit_stage159_notes_only_numeric_illustrations fam_0060_delta_q_dg_coeff	origin stage 159 notes/stages/moving_throat_pde_stage159_hybrid_outlet_projection.md:306	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage159_hybrid_outlet_projection.md:294	160, 161
159	matched_fingerprint_value fit_stage_159_matched_fingerprint_value fam_0459_matched_fingerprint_value	origin stage 115 notes/stages/moving_throat_pde_stage115_core_balance_compensation.md:25	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage159_hybrid_outlet_projection.md:202	160, 161

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
160	gamma_0 fit_stage160_canonical_branch_characterized_kappa_gamma fam_0415_gamma_0	origin stage 115 notes/stages/moving_throat_pde_stage115_core_balance_compensation.md:25	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage160_bare_mixed_port_slippage.md:73	161
160	Delta_Q_dS_coeff fit_stage160_notes_only_slippage_numerics fam_0059_delta_q_ds_coeff	origin stage 160 notes/stages/moving_throat_pde_stage160_bare_mixed_port_slippage.md:344	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage160_bare_mixed_port_slippage.md:335	161
160	dPi_tan_coeff_S fit_stage160_tangent_susceptibility_coefficients fam_0357_dpi_tan_coeff_s	origin stage 158 notes/stages/moving_throat_pde_stage158_linear_defect_transport.md:236	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage160_bare_mixed_port_slippage.md:250	161
160	matched_fingerprint_value fit_stage_160_matched_fingerprint_value fam_0459_matched_fingerprint_value	origin stage 115 notes/stages/moving_throat_pde_stage115_core_balance_compensation.md:25	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage160_bare_mixed_port_slippage.md:77	161
161	sigma_star fit_stage161_delta_q_sigma_star_closure fam_0513_sigma_star	origin stage 159 notes/stages/moving_throat_pde_stage159_hybrid_outlet_projection.md:83	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage159_hybrid_outlet_projection.md:74	160, 161
161	dSigma0_per_dThat fit_stage161_dsigma0_dthat_coefficient fam_0360_dsigma0_per_dthat	origin stage 158 notes/stages/moving_throat_pde_stage158_linear_defect_transport.md:178	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage158_linear_defect_transport.md:169	161
161	dSigma0_over_dThat fit_stage161_dsigma0_dthat_conversion_ratio fam_0359_dsigma0_over_dthat	origin stage 158 notes/stages/moving_throat_pde_stage158_linear_defect_transport.md:178	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage161_dn_similarity_slippage.md:363	161
161	gamma_0 fit_stage161_gamma0_star_branch_value fam_0415_gamma_0	origin stage 115 notes/stages/moving_throat_pde_stage115_core_balance_compensation.md:25	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage161_dn_similarity_slippage.md:65	none recorded
161	r_F1 fit_stage161_rF1_family1_radius fam_0493_r_f1; overlays:chain_aspect_37_20	origin stage 121 notes/stages/moving_throat_pde_stage121_geometric_r_selection.md:70	free_choice	posited choice:notes/stages/moving_throat_pde_stage121_geometric_r_selection.md:60	none recorded
161	dPi_tan_coeff_dS fit_stage161_stage159_tangential_transport_coeffs fam_0358_dpi_tan_coeff_ds	origin stage 158 notes/stages/moving_throat_pde_stage158_linear_defect_transport.md:236	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage161_dn_similarity_slippage.md:329	none recorded

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
161	Upsilon_Pi fit_stage161_upsilon_pi_one_ninth_coefficient fam_0267_upsilon_pi	origin stage 160 notes/stages/moving_throat_pde_stage160_bare_mixed_port_slippage.md:245	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage161_dn_similarity_slippage.md:239	none recorded
162	gamma_0 fit_stage162_gamma0_one_ninth_r_form fam_0415_gamma_0	origin stage 162 notes/stages/moving_throat_pde_stage162_parent_compensation_rigidity.md:61	free_choice	posited choice:notes/stages/moving_throat_pde_stage162_parent_compensation_rigidity.md:58	163
162	gamma_0 fit_stage162_gamma0_similarity_law fam_0415_gamma_0	origin stage 162 notes/stages/moving_throat_pde_stage162_parent_compensation_rigidity.md:61	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage162_parent_compensation_rigidity.md:100	163
162	r_F1 fit_stage162_rF1_family1_radius fam_0493_r_f1; overlays:chain_aspect_37_20	origin stage 121 notes/stages/moving_throat_pde_stage121_geometric_r_selection.md:65	free_choice	posited choice:notes/stages/moving_throat_pde_stage121_geometric_r_selection.md:60	163, 164, 165, 166, 167, 168
162	r_F1 fit_stage162_r_f1_family1_value fam_0493_r_f1; overlays:chain_aspect_37_20	origin stage 121 notes/stages/moving_throat_pde_stage121_geometric_r_selection.md:65	free_choice	posited choice:notes/stages/moving_throat_pde_stage121_geometric_r_selection.md:60	163, 164, 165, 166, 167, 168
162	g_lower fit_stage_162_g_lower fam_0408_g_lower	origin stage 162 notes/stages/moving_throat_pde_stage162_parent_compensation_rigidity.md:50	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage162_parent_compensation_rigidity.md:48	163
163	r_F1 fit_stage163_Rstar_quarter fam_0493_r_f1; overlays:chain_aspect_37_20	origin stage 163 notes/stages/moving_throat_pde_stage163_off_family_normal_coordinate.md:94	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage163_off_family_normal_coordinate.md:90	164, 168
163	delta_perp fit_stage163_delta_perp_exact_parent_expression fam_0373_delta_perp	origin stage 163 notes/stages/moving_throat_pde_stage163_off_family_normal_coordinate.md:65	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage163_off_family_normal_coordinate.md:163	164, 165, 167, 168
163	S_can fit_stage163_family1_canonical_point_block fam_0228_s_can, fam_0237_sigma0_can, fam_0412_g_star, fam_0493_r_f1; overlays:chain_aspect_37_20, chain_sigma0_transport	origin stage 156 notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:69	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:53	164, 168

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
163	Sigma0_can fit_stage163_family1_canonical_point_block fam_0228_s_can, fam_0237_sigma0_can, fam_0412_g_star, fam_0493_r_f1; overlays:chain_aspect_37_20, chain_sigma0_transport	origin stage 156 notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:53	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage158_linear_defect_transport.md:28	164, 168
163	g_star fit_stage163_family1_canonical_point_block fam_0228_s_can, fam_0237_sigma0_can, fam_0412_g_star, fam_0493_r_f1; overlays:chain_aspect_37_20, chain_sigma0_transport	origin stage 122 notes/stages/moving_throat_pde_stage122_mouth_source_compensation_test.md:63	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage162_parent_compensation_rigidity.md:50	164, 168
163	r_F1 fit_stage163_family1_canonical_point_block fam_0228_s_can, fam_0237_sigma0_can, fam_0412_g_star, fam_0493_r_f1; overlays:chain_aspect_37_20, chain_sigma0_transport	origin stage 121 notes/stages/moving_throat_pde_stage121_geometric_r_selection.md:65	free_choice	posited choice:notes/stages/moving_throat_pde_stage121_geometric_r_selection.md:60	164, 168
163	S_can fit_stage163_family1_readback_misattribution fam_0228_s_can; overlays:chain_sigma0_transport	origin stage 156 notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:69	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:53	164, 168
163	Sigma0_can fit_stage163_sigma0_can_digit_variant fam_0237_sigma0_can; overlays:chain_sigma0_transport	origin stage 156 notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:53	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage158_linear_defect_transport.md:28	164, 168
164	g_star fit_stage164_family1_point_injection fam_0412_g_star; overlays:chain_aspect_37_20	origin stage 122 notes/stages/moving_throat_pde_stage122_mouth_source_compensation_test.md:63	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage164_microscopic_log_channels.md:317	165, 168
164	first_log_channel_prefactor fit_stage164_log_channel_prefactors_27_40_27_320 fam_0402_first_log_channel_prefactor	origin stage 164 notes/stages/moving_throat_pde_stage164_microscopic_log_channels.md:226	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage164_microscopic_log_channels.md:214	165
164	delta_perp_channel_weight fit_stage164_quarter_sqrt_weight fam_0374_delta_perp_channel_weight	origin stage 163 notes/stages/moving_throat_pde_stage163_off_family_normal_coordinate.md:171	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage163_off_family_normal_coordinate.md:163	165, 168
164	L_W fit_stage_164_l_w fam_0145_l_w	origin stage 164 notes/stages/moving_throat_pde_stage164_microscopic_log_channels.md:31	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage164_microscopic_log_channels.md:101	165, 166, 167, 168

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
165	n_wall_eos fit_stage165_frozen_n5_wall_eos fam_0484_n_wall_eos	origin stage 118 notes/stages/moving_throat_pde_stage118_parent_core_extraction.md:18	free_choice	posited choice:notes/stages/moving_throat_pde_stage165_exact_branch_drifts.md:330	167
165	g_star fit_stage165_gstar_literal fam_0412_g_star; overlays:chain_aspect_37_20	origin stage 122 notes/stages/moving_throat_pde_stage122_mouth_source_compensation_test.md:63	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage165_exact_branch_drifts.md:200	167, 168
165	g_frak fit_stage165_pinned_parent_ratios fam_0407_g_frak	origin stage 165 notes/stages/moving_throat_pde_stage165_exact_branch_drifts.md:42	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage165_exact_branch_drifts.md:46	166, 167, 168
165	r_F1 fit_stage165_rstar_closed_form_4107 fam_0493_r_f1; overlays:chain_aspect_37_20	origin stage 121 notes/stages/moving_throat_pde_stage121_geometric_r_selection.md:69	free_choice	posited choice:notes/stages/moving_throat_pde_stage121_geometric_r_selection.md:60	167, 168
165	L_W fit_stage_165_l_w fam_0145_l_w	origin stage 165 notes/stages/moving_throat_pde_stage165_exact_branch_drifts.md:67	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage165_exact_branch_drifts.md:73	166, 167, 168
165	T_m_pref fit_stage_165_t_m_pref fam_0256_t_m_pref	origin stage 165 notes/stages/moving_throat_pde_stage165_exact_branch_drifts.md:212	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage165_exact_branch_drifts.md:207	167
166	K_q fit_stage166_four_observables_determine_drifts fam_0137_k_q	origin stage 166 notes/stages/moving_throat_pde_stage166_bundle_inversion_four_drifts.md:72	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage166_bundle_inversion_four_drifts.md:79	167
166	Theta_w fit_stage166_theta_chi_wall_datum fam_0264_theta_w	origin stage 77 notes/stages/moving_throat_pde_stage077_family1_theta_extraction.md:7	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage166_bundle_inversion_four_drifts.md:50	167
166	Theta_chi fit_stage166_theta_chi_wall_temperature fam_0262_theta_chi	origin stage 77 notes/stages/moving_throat_pde_stage077_family1_theta_extraction.md:94	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage077_family1_theta_extraction.md:93	167
166	K_q fit_stage_166_k_q fam_0137_k_q	origin stage 166 notes/stages/moving_throat_pde_stage166_bundle_inversion_four_drifts.md:72	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage166_bundle_inversion_four_drifts.md:79	167

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
167	g_frak fit_stage167_parent_ratios_first_order_invariants fam_0407_g_frak	origin stage 167 notes/stages/moving_throat_pde_stage167_bundle_transport_tangent_compensation.md:32	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage167_bundle_transport_tangent_compensation.md:322	168
168	S_can fit_stage168_canonical_mouth_block fam_0228_s_can, fam_0237_sigma0_can, fam_0412_g_star; overlays:chain_aspect_37_20, chain_sigma0_transport	origin stage 156 notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:69	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:53	none recorded
168	Sigma0_can fit_stage168_canonical_mouth_block fam_0228_s_can, fam_0237_sigma0_can, fam_0412_g_star; overlays:chain_aspect_37_20, chain_sigma0_transport	origin stage 156 notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:53	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage158_linear_defect_transport.md:28	none recorded
168	g_star fit_stage168_canonical_mouth_block fam_0228_s_can, fam_0237_sigma0_can, fam_0412_g_star; overlays:chain_aspect_37_20, chain_sigma0_transport	origin stage 122 notes/stages/moving_throat_pde_stage122_mouth_source_compensation_test.md:63	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage168_off_bundle_slippage.md:190	none recorded
168	eps_perp fit_stage168_eps_perp_exact_weights fam_0388_eps_perp	origin stage 168 notes/stages/moving_throat_pde_stage168_off_bundle_slippage.md:166	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage168_off_bundle_slippage.md:151	none recorded
168	r_F1 fit_stage168_rexact_closed_form_4107 fam_0493_r_f1; overlays:chain_aspect_37_20	origin stage 121 notes/stages/moving_throat_pde_stage121_geometric_r_selection.md:69	free_choice	posited choice:notes/stages/moving_throat_pde_stage121_geometric_r_selection.md:60	none recorded
168	matched_fingerprint_value fit_stage_168_matched_fingerprint_value fam_0459_matched_fingerprint_value	origin stage 168 notes/stages/moving_throat_pde_stage168_off_bundle_slippage.md:374	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage168_off_bundle_slippage.md:381	none recorded
168	S_can fit_stage_168_s_can fam_0228_s_can; overlays:chain_sigma0_transport	origin stage 156 notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:69	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage156_renormalized_canonical_branch.md:53	none recorded
169	g_star fit_stage169_family1_point_injection fam_0412_g_star; overlays:chain_aspect_37_20	origin stage 169 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
169	A_x_squared_coefficient fit_stage169_seven_tenths_invariant fam_0013_a_x_squared_coefficient	origin stage 169 notes/stages/moving_throat_pde_stage169_no_linear_p2_scalar_slippage.md:93	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage169_no_linear_p2_scalar_slippage.md:76	none recorded
169	Xi_perp_coeff_XiL fit_stage169_xi_perp_paper_targets fam_0288_xi_perp_coeff_xil	origin stage 169 notes/stages/moving_throat_pde_stage169_no_linear_p2_scalar_slippage.md:295	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage169_no_linear_p2_scalar_slippage.md:284	none recorded
169	a_x fit_stage_169_a_x fam_0307_a_x	origin stage 169 notes/stages/moving_throat_pde_stage169_no_linear_p2_scalar_slippage.md:54	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage169_no_linear_p2_scalar_slippage.md:54	none recorded
169	engine_disagreement fit_stage_169_engine_disagreement fam_0381_engine_disagreement	origin stage 169 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
169	tautological_check fit_stage_169_tautological_check fam_0526_tautological_check	origin stage 169 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
170	D2_over_D0 fit_stage170_canonical_isotropic_loading_u2_u4 fam_0039_d2_over_d0	origin stage 104 notes/stages/moving_throat_pde_stage104_outgoing_dtn_fingerprint.md:58	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage170_linear_grouped_outlet_map.md:90	171, 172, 173
170	u_2 fit_stage170_canonical_u2_u4 fam_0532_u_2	origin stage 104 notes/stages/moving_throat_pde_stage104_outgoing_dtn_fingerprint.md:58	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage104_outgoing_dtn_fingerprint.md:83	171, 172, 173
170	K_A_weight fit_stage170_k_a_one_ninth_combination fam_0122_k_a_weight	origin stage 170 notes/stages/moving_throat_pde_stage170_linear_grouped_outlet_map.md:25	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage170_linear_grouped_outlet_map.md:25	171, 172
171	delta_gamma_W_coefficient fit_stage171_outlet_feed_coefficients fam_0370_delta_gamma_w_coefficient	origin stage 170 notes/stages/moving_throat_pde_stage170_linear_grouped_outlet_map.md:46	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage171_microscopic_grouped_obstructions.md:18	172
172	G1_frak fit_stage172_canonical_branch_collapse fam_0084_g1_frak	origin stage 172 notes/stages/moving_throat_pde_stage172_physical_slope_collapse.md:34	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage172_physical_slope_collapse.md:166	173
172	u2 fit_stage172_canonical_even_collapse_u2_pin fam_0531_u2	origin stage 104 notes/stages/moving_throat_pde_stage104_outgoing_dtn_fingerprint.md:58	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage172_physical_slope_collapse.md:73	173

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
172	u_2 fit_stage172_wl_canonical_banner fam_0532_u_2	origin stage 104 notes/stages/moving_throat_pde_stage104_outgoing_dtn_fingerprint.md:58	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage172_physical_slope_collapse.md:73	173
172	matched_fingerprint_value fit_stage_172_matched_fingerprint_value fam_0459_matched_fingerprint_value	origin stage 104 notes/stages/moving_throat_pde_stage104_outgoing_dtn_fingerprint.md:58	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage172_physical_slope_collapse.md:322	173
173	D2_over_D0 fit_stage173_canonical_loading_substitution fam_0039_d2_over_d0	origin stage 104 notes/stages/moving_throat_pde_stage104_outgoing_dtn_fingerprint.md:58	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage173_axisymmetric_loading_mismatch.md:178	177
173	D_01 fit_stage173_slopes_fixed_by_d01 fam_0041_d_01	origin stage 173 notes/stages/moving_throat_pde_stage173_axisymmetric_loading_mismatch.md:27	free_choice	posited choice:notes/stages/moving_throat_pde_stage173_axisymmetric_loading_mismatch.md:226	177
173	lambda_20 fit_stage173_stage024_signature_triple fam_0444_lambda_20; overlays:chain_barrier_222_224	origin stage 24 notes/stages/moving_throat_pde_stage024_overlap_isotropy.md:231	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage024_overlap_isotropy.md:217	173, 177
173	Xi_load fit_stage_173_xi_load fam_0286_xi_load	origin stage 173 notes/stages/moving_throat_pde_stage173_axisymmetric_loading_mismatch.md:278	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage173_axisymmetric_loading_mismatch.md:441	174, 177
174	omega_B fit_stage174_self_similarity_weights fam_0486_omega_b	origin stage 174 notes/stages/moving_throat_pde_stage174_static_self_similarity.md:88	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage174_static_self_similarity.md:111	175
175	B_0_alpha fit_stage175_wall_normalized_factorization fam_0015_b_0_alpha	origin stage 175 notes/stages/moving_throat_pde_stage175_wall_normalized_load_shape.md:68	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage175_wall_normalized_load_shape.md:117	176, 177
175	Sigma_N fit_stage_175_sigma_n fam_0241_sigma_n	origin stage 175 notes/stages/moving_throat_pde_stage175_wall_normalized_load_shape.md:41	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage175_wall_normalized_load_shape.md:223	177, 178
176	G_W fit_stage176_square_root_mixed_leg_exponent fam_0087_g_w	origin stage 176 notes/stages/moving_throat_pde_stage176_outgoing_load_factorization.md:56	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage176_outgoing_load_factorization.md:234	177, 178

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
177	Xi_1 fit_stage177_xi1_exact_collapse fam_0283_xi_1	origin stage 177 notes/stages/moving_throat_pde_stage177_weak_axisymmetric_outgoing_slippage.md:191	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage177_weak_axisymmetric_outgoing_slippage.md:262	178, 179
177	Xi_1 fit_stage177_xi1_prefactor_slope_no_derives_edge fam_0283_xi_1	origin stage 177 notes/stages/moving_throat_pde_stage177_weak_axisymmetric_outgoing_slippage.md:191	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage177_weak_axisymmetric_outgoing_slippage.md:269	178, 179
177	H_r fit_stage_177_h_r fam_0100_h_r	origin stage 177 notes/stages/moving_throat_pde_stage177_weak_axisymmetric_outgoing_slippage.md:11	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage177_weak_axisymmetric_outgoing_slippage.md:117	178
177	P0_target fit_stage_177_p_0 fam_0187_p0_target; overlays:chain_quad_54_5, chain_wall_action	origin stage 177 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	178
177	Xi_1 fit_stage_177_xi_1 fam_0283_xi_1	origin stage 177 notes/stages/moving_throat_pde_stage177_weak_axisymmetric_outgoing_slippage.md:191	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage177_weak_axisymmetric_outgoing_slippage.md:262	178, 179
178	kappa_1 fit_stage178_doubled_slope_factors fam_0435_kappa_1	origin stage 178 notes/stages/moving_throat_pde_stage178_outgoing_port_coloading_theorem.md:70	free_choice	posited choice:notes/stages/moving_throat_pde_stage178_outgoing_port_coloading_theorem.md:37	179
178	d_from_series fit_stage_178_d_from_series fam_0361_d_from_series	origin stage 178 notes/stages/moving_throat_pde_stage178_outgoing_port_coloading_theorem.md:79	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage178_outgoing_port_coloading_theorem.md:238	179
178	d_r fit_stage_178_d_r fam_0362_d_r	origin stage 178 notes/stages/moving_throat_pde_stage178_outgoing_port_coloading_theorem.md:79	free_choice	posited choice:notes/stages/moving_throat_pde_stage178_outgoing_port_coloading_theorem.md:231	179
178	Delta fit_stage_178_delta fam_0051_delta	origin stage 178 notes/stages/moving_throat_pde_stage178_outgoing_port_coloading_theorem.md:178	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage178_outgoing_port_coloading_theorem.md:178	179

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
179	Xi_1_prefactor_2 fit_stage179_xi1_factor_two_transfer_slopes fam_0284_xi_1_prefactor_2	origin stage 179 notes/stages/moving_throat_pde_stage179_transfer_shape_theorem.md:251	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage179_transfer_shape_theorem.md:227	none recorded
179	nu_r fit_stage_179_nu_r fam_0485_nu_r	origin stage 178 notes/stages/moving_throat_pde_stage178_outgoing_port_coloading_theorem.md:90	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage179_transfer_shape_theorem.md:227	none recorded
180	beta_0 fit_stage180_beta0_continuum_import fam_0324_beta_0	origin stage 38 notes/stages/moving_throat_pde_stage038_dimensionless_continuum_placement.md:97	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage180_effective_transfer_shape_collapse.md:146	181, 182, 183, 184, 185, 189
180	Pi_star fit_stage_180_pi_star fam_0209_pi_star	origin stage 180 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
181	eps_split_coefficient fit_stage181_eps_split_2_11_coefficient fam_0389_eps_split_coefficient	origin stage 27 notes/stages/moving_throat_pde_stage027_nonconstant_axial_family.md:151	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage039_split_u_sector.md:123	039, 047, 182, 183, 184, 185, 186, 187, 189
181	Lambda_prefactor fit_stage181_lambda_prefactor_27pi2_20 fam_0154_lambda_prefactor	origin stage 38 notes/stages/moving_throat_pde_stage038_dimensionless_continuum_placement.md:32	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage181_coherent_tracking_defect.md:132	180, 182, 183, 184, 189
181	eps_W_split_coefficient fit_stage181_split_blocking_two_elevenths fam_0383_eps_w_split_coefficient	origin stage 39 notes/stages/moving_throat_pde_stage039_split_u_sector.md:123	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage039_split_u_sector.md:119	047, 181, 182, 183, 184, 185, 186, 187, 189
181	R_target fit_stage181_transfer_shape_27pi2_over_20 fam_0222_r_target	origin stage 38 notes/stages/moving_throat_pde_stage038_dimensionless_continuum_placement.md:28	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage181_coherent_tracking_defect.md:116	180, 182, 183, 184, 189
181	delta_U fit_stage181_two_elevenths_split_coefficient fam_0368_delta_u	origin stage 38 notes/stages/moving_throat_pde_stage038_dimensionless_continuum_placement.md:20	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage182_microscopic_coherent_slippage.md:94	039, 047, 181, 182, 183, 184, 185, 186, 187
182	eps_split_coefficient fit_stage182_eps_split_2_11_coefficient fam_0389_eps_split_coefficient	origin stage 27 notes/stages/moving_throat_pde_stage027_nonconstant_axial_family.md:151	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage182_microscopic_coherent_slippage.md:14	183, 184, 185, 186, 187, 189

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
182	epsilon_W_split_coefficient_2_11 fit_stage182_two_elevenths_carried_forward fam_0392_epsilon_w_split_coefficient_2_11	origin stage 27 notes/stages/moving_throat_pde_stage027_nonconstant_axial_family.md:151	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage182_microscopic_coherent_slippage.md:20	183, 184, 185, 186, 187, 189
182	Sigma_Z fit_stage_182_sigma_z fam_0242_sigma_z	origin stage 182 notes/stages/moving_throat_pde_stage182_microscopic_coherent_slippage.md:241	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage182_microscopic_coherent_slippage.md:241	183, 184, 185, 186, 187
183	Sigma_nt fit_stage183_canonical_sigma_nt fam_0245_sigma_nt	origin stage 183 notes/stages/moving_throat_pde_stage183_triangular_normal_form.md:99	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage183_triangular_normal_form.md:97	184, 185, 186, 187
183	eps_split_coefficient fit_stage183_eps_split_2_11_coefficient fam_0389_eps_split_coefficient	origin stage 27 notes/stages/moving_throat_pde_stage027_nonconstant_axial_family.md:151	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage183_triangular_normal_form.md:103	184, 185, 186, 187, 189
184	B_star fit_stage184_b_star_exponent fam_0020_b_star	origin stage 184 notes/stages/moving_throat_pde_stage184_branch_invariant_coordinates.md:218	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage184_branch_invariant_coordinates.md:213	185, 188, 189
184	eps_split_coefficient fit_stage184_eps_split_2_11_coefficient fam_0389_eps_split_coefficient	origin stage 27 notes/stages/moving_throat_pde_stage027_nonconstant_axial_family.md:151	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage184_branch_invariant_coordinates.md:94	185, 186, 187, 189
184	eps_eta_var fit_stage_184_eps_eta_var fam_0387_eps_eta_var	origin stage 38 notes/stages/moving_throat_pde_stage038_dimensionless_continuum_placement.md:12	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage182_microscopic_coherent_slippage.md:177	180, 181, 182, 183, 185, 186, 187, 189
184	Lambda_0 fit_stage_184_lambda_0 fam_0149_lambda_0	origin stage 38 notes/stages/moving_throat_pde_stage038_dimensionless_continuum_placement.md:32	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage184_branch_invariant_coordinates.md:114	180, 181, 189
185	E_star fit_stage185_frozen_reference_coefficients_E_star_F_star fam_0073_e_star	origin stage 185 notes/stages/moving_throat_pde_stage185_microscopic_monomials.md:218	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage185_microscopic_monomials.md:215	186, 187

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
185	C_tr_star_exponents fit_stage185_monomial_exponents fam_0035_c_tr_star_exponents	origin stage 185 notes/stages/moving_throat_pde_stage185_microscopic_monomials.md:159	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage185_microscopic_monomials.md:150	186, 187
185	A_tr_star fit_stage_185_a_tr_star fam_0011_a_tr_star	origin stage 183 notes/stages/moving_throat_pde_stage183_triangular_normal_form.md:118	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage185_microscopic_monomials.md:340	186, 187
186	Sigma_eta fit_stage186_not_fine_tuning_claim fam_0243_sigma_eta	origin stage 182 notes/stages/moving_throat_pde_stage182_microscopic_coherent_slippage.md:251	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage186_similarity_orbit_closure.md:66	187
186	chi_0_star fit_stage186_rank3_codimension_status_claim fam_0344_chi_0_star	origin stage 47 notes/stages/moving_throat_pde_stage182_microscopic_coherent_slippage.md:87	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage186_similarity_orbit_closure.md:191	187
186	M_star_matrix fit_stage186_wl_matches_reference_matrix fam_0169_m_star_matrix	origin stage 186 notes/stages/moving_throat_pde_stage186_similarity_orbit_closure.md:171	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage186_similarity_orbit_closure.md:181	187
186	mathcal fit_stage_186_mathcal fam_0461_mathcal	origin stage 186 notes/stages/moving_throat_pde_stage186_similarity_orbit_closure.md:262	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage186_similarity_orbit_closure.md:301	187
187	C_nt_star fit_stage187_exact_level_set_orbit_identity fam_0029_c_nt_star	origin stage 185 notes/stages/moving_throat_pde_stage185_microscopic_monomials.md:235	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage187_orbit_quotient_closure.md:89	none recorded
187	Lambda_1 fit_stage187_g2_common_tangent_lambda_1 fam_0150_lambda_1	origin stage 187 notes/stages/moving_throat_pde_stage187_orbit_quotient_closure.md:217	free_choice	posited choice:notes/stages/moving_throat_pde_stage187_orbit_quotient_closure.md:218	none recorded
187	chi_0 fit_stage_187_chi_0 fam_0343_chi_0	origin stage 47 notes/stages/moving_throat_pde_stage182_microscopic_coherent_slippage.md:87	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage187_orbit_quotient_closure.md:95	none recorded
187	Ctr_ratio fit_stage_187_ctr_ratio fam_0037_ctr_ratio	origin stage 185 notes/stages/moving_throat_pde_stage185_microscopic_monomials.md:159	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage187_orbit_quotient_closure.md:83	none recorded

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
189	eps_split_coefficient fit_stage189_eps_split_2_11_coefficient fam_0389_eps_split_coefficient	origin stage 27 notes/stages/moving_throat_pde_stage027_nonconstant_axial_family.md:151	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage189_transfer_shape_prefactor_compiler.md:273	none recorded
189	Lambda_0 fit_stage189_lambda0_prefactor_27_20 fam_0149_lambda_0	origin stage 38 notes/stages/moving_throat_pde_stage038_dimensionless_continuum_placement.md:32	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage189_transfer_shape_prefactor_compiler.md:144	none recorded
189	mhat_0 fit_stage189_mhat0_quadrupole_normalization_target fam_0469_mhat_0; overlays:chain_chi_Q_norm, chain_quad_54_5, chain_wall_action	origin stage 22 notes/stages/moving_throat_pde_stage022_grouped_p2_normalization_bridge.md:264	published_target	published_target; no benchmarks.yaml entry joined for listed family; evidence:notes/stages/moving_throat_pde_stage189_transfer_shape_prefactor_compiler.md:515	none recorded
189	P0_target fit_stage189_normalization_target_2g_5c5 fam_0187_p0_target; overlays:chain_quad_54_5, chain_wall_action	origin stage 22 notes/stages/moving_throat_pde_stage022_grouped_p2_normalization_bridge.md:256	published_target	benchmark: bench_2e5805a359c4: textbook - P. C. Peters, 'Gravitational Radiation and the Motion of Two Point Masses', Phys. Rev. 136, B1224 (1964), DOI 10.1103/PhysRev.136.B1224 (ADS bibcode 1964PhRv..136.1224P); see also M. Maggiore, 'Gravitational Waves, Vol. 1: Theory and Experiments', Oxford Univ. Press (2007), ISBN 978-0-19-857074-5, Ch. 3 (quadrupole formula L = (G/5c^5)<dddQ_ij dddQ_ij>).	none recorded
189	K_bl fit_stage_189_k_bl fam_0125_k_bl	origin stage 180 notes/stages/moving_throat_pde_stage180_effective_transfer_shape_collapse.md:80	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage189_transfer_shape_prefactor_compiler.md:345	none recorded
189	matched_fingerprint_value fit_stage_189_matched_fingerprint_value fam_0459_matched_fingerprint_value	origin stage 22 notes/stages/moving_throat_pde_stage022_grouped_p2_normalization_bridge.md:262	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage189_transfer_shape_prefactor_compiler.md:471	none recorded
189	P0_target fit_stage_189_p_0 fam_0187_p0_target; overlays:chain_quad_54_5, chain_wall_action	origin stage 22 notes/stages/moving_throat_pde_stage022_grouped_p2_normalization_bridge.md:256	published_target	benchmark: bench_2e5805a359c4: textbook - P. C. Peters, 'Gravitational Radiation and the Motion of Two Point Masses', Phys. Rev. 136, B1224 (1964), DOI 10.1103/PhysRev.136.B1224 (ADS bibcode 1964PhRv..136.1224P); see also M. Maggiore, 'Gravitational Waves, Vol. 1: Theory and Experiments', Oxford Univ. Press (2007), ISBN 978-0-19-857074-5, Ch. 3 (quadrupole formula L = (G/5c^5)<dddQ_ij dddQ_ij>).	none recorded

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
190	A_tr fit_stage_190_a_tr fam_0010_a_tr	origin stage 190 notes/stages/moving_throat_pde_stage190_direct_defect_vs_dressing_split.md:335	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage190_direct_defect_vs_dressing_split.md:333	191
190	b_x fit_stage_190_b_x fam_0321_b_x	origin stage 190 notes/stages/moving_throat_pde_stage190_direct_defect_vs_dressing_split.md:547	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage190_direct_defect_vs_dressing_split.md:574	191, 193
190	M_mix fit_stage_190_m_mix fam_0160_m_mix	origin stage 190 notes/stages/moving_throat_pde_stage190_direct_defect_vs_dressing_split.md:138	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage190_direct_defect_vs_dressing_split.md:140	none recorded
191	B_star fit_stage191_carried_coefficients_in_packet_interconversion fam_0020_b_star, fam_0081_f_star, fam_0344_chi_0_star	origin stage 190 notes/stages/moving_throat_pde_stage190_direct_defect_vs_dressing_split.md:78	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage190_direct_defect_vs_dressing_split.md:78	191
191	F_star fit_stage191_carried_coefficients_in_packet_interconversion fam_0020_b_star, fam_0081_f_star, fam_0344_chi_0_star	origin stage 191 notes/stages/moving_throat_pde_stage191_minimal_pde_data_packet.md:253	free_choice	posited choice:notes/stages/moving_throat_pde_stage191_minimal_pde_data_packet.md:253	192
191	chi_0_star fit_stage191_carried_coefficients_in_packet_interconversion fam_0020_b_star, fam_0081_f_star, fam_0344_chi_0_star	origin stage 191 notes/stages/moving_throat_pde_stage191_minimal_pde_data_packet.md:253	free_choice	posited choice:notes/stages/moving_throat_pde_stage191_minimal_pde_data_packet.md:253	192
191	Delta_branch fit_stage191_fixed_by_previous_stages fam_0061_delta_branch	origin stage 191 notes/stages/moving_throat_pde_stage191_minimal_pde_data_packet.md:200	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage191_minimal_pde_data_packet.md:197	192, 193, 195
191	Delta_branch fit_stage_191_delta_branch fam_0061_delta_branch	origin stage 191 notes/stages/moving_throat_pde_stage191_minimal_pde_data_packet.md:200	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage191_minimal_pde_data_packet.md:203	192, 193, 195
191	Pi_star fit_stage_191_pi_star fam_0209_pi_star	origin stage 191 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
192	K_eta_eff fit_stage192_canonical_quotient_section fam_0130_k_eta_eff	origin stage 192 notes/stages/moving_throat_pde_stage192_orbit_quotient_projectors.md:81	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage192_orbit_quotient_projectors.md:314	none recorded

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
192	chi_Q fit_stage192_chi_q_unit_relations fam_0345_chi_q; overlays:chain_chi_Q_norm	origin stage 194 notes/stages/moving_throat_pde_stage194_outgoing_l2_fingerprint_and_deformation_algebra.md:184	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage194_outgoing_l2_fingerprint_and_deformation_algebra.md:182	193, 194, 195
192	Delta_Q fit_stage_192_delta_q fam_0058_delta_q	origin stage 195 notes/stages/moving_throat_pde_stage195_source_map_reduction_of_canonical_outgoing_branch.md:185	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage195_source_map_reduction_of_canonical_outgoing_branch.md:185	196, 197
192	I_3 fit_stage_192_i_3 fam_0105_i_3	origin stage 192 notes/stages/moving_throat_pde_stage192_orbit_quotient_projectors.md:200	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage192_orbit_quotient_projectors.md:202	none recorded
192	P_2 fit_stage_192_p_2 fam_0188_p_2	origin stage 192 not recorded	free_choice	posited choice:notes/stages/moving_throat_pde_stage192_orbit_quotient_projectors.md:5	193, 194, 195
192	q_eta fit_stage_192_q_eta fam_0491_q_eta	origin stage 191 notes/stages/moving_throat_pde_stage191_minimal_pde_data_packet.md:268	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage192_orbit_quotient_projectors.md:93	192
193	Omega_Q fit_stage193_grouped_p2_target_surface_coefficients fam_0183_omega_q	origin stage 193 notes/stages/moving_throat_pde_stage193_isotropic_grouped_p2_target_surface.md:222	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage193_isotropic_grouped_p2_target_surface.md:227	194, 195
193	Delta_pole fit_stage193_one_pole_carrier_forced fam_0065_delta_pole	origin stage 191 notes/stages/moving_throat_pde_stage191_minimal_pde_data_packet.md:180	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage193_isotropic_grouped_p2_target_surface.md:161	194
193	P0_target fit_stage193_p0_target_surface_frozen_without_derivation_edge fam_0187_p0_target; overlays:chain_quad_54_5, chain_wall_action	origin stage 191 notes/stages/moving_throat_pde_stage191_minimal_pde_data_packet.md:190	published_target	benchmark: bench_2e5805a359c4: textbook - P. C. Peters, 'Gravitational Radiation and the Motion of Two Point Masses', Phys. Rev. 136, B1224 (1964), DOI 10.1103/PhysRev.136.B1224 (ADS bibcode 1964PhRv..136.1224P); see also M. Maggiore, 'Gravitational Waves, Vol. 1: Theory and Experiments', Oxford Univ. Press (2007), ISBN 978-0-19-857074-5, Ch. 3 (quadrupole formula L = (G/5c^5)<dddQ_ij dddQ_ij>).	195

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
193	Delta_Q fit_stage_193_delta_q fam_0058_delta_q	origin stage 195 notes/stages/moving_throat_pde_stage195_source_map_reduction_of_canonical_outgoing_branch.md:185	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage195_source_map_reduction_of_canonical_outgoing_branch.md:185	196, 197
193	P_2 fit_stage_193_p_2 fam_0188_p_2	origin stage 193 not recorded	free_choice	posited choice:notes/stages/moving_throat_pde_stage193_isotropic_grouped_p2_target_surface.md:5	194, 195
194	L2_over_L0_1_9 fit_stage194_canonical_even_moment_ratios fam_0143_l2_over_l0_1_9	origin stage 194 notes/stages/moving_throat_pde_stage194_outgoing_l2_fingerprint_and_deformation_algebra.md:260	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage194_outgoing_l2_fingerprint_and_deformation_algebra.md:252	195
194	chi_Q fit_stage194_chi_Q_canonical_fixing fam_0345_chi_q; overlays:chain_chi_Q_norm	origin stage 194 notes/stages/moving_throat_pde_stage194_outgoing_l2_fingerprint_and_deformation_algebra.md:184	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage194_outgoing_l2_fingerprint_and_deformation_algebra.md:182	195, 196, 197
194	chi_Q fit_stage194_chi_q_fixed_on_canonical_branch fam_0345_chi_q; overlays:chain_chi_Q_norm	origin stage 194 notes/stages/moving_throat_pde_stage194_outgoing_l2_fingerprint_and_deformation_algebra.md:184	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage194_outgoing_l2_fingerprint_and_deformation_algebra.md:186	195, 196, 197
194	Sigma_2 fit_stage194_even_deformations_not_free fam_0240_sigma_2	origin stage 194 notes/stages/moving_throat_pde_stage194_outgoing_l2_fingerprint_and_deformation_algebra.md:210	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage194_outgoing_l2_fingerprint_and_deformation_algebra.md:272	195
194	chi_Q fit_stage_194_chi_q fam_0345_chi_q; overlays:chain_chi_Q_norm	origin stage 194 notes/stages/moving_throat_pde_stage194_outgoing_l2_fingerprint_and_deformation_algebra.md:184	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage194_outgoing_l2_fingerprint_and_deformation_algebra.md:182	195, 196, 197
194	Delta_Q fit_stage_194_delta_q fam_0058_delta_q	origin stage 195 notes/stages/moving_throat_pde_stage195_source_map_reduction_of_canonical_outgoing_branch.md:185	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage195_source_map_reduction_of_canonical_outgoing_branch.md:185	196, 197

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
194	matched_fingerprint_value fit_stage_194_matched_fingerprint_value fam_0459_matched_fingerprint_value	origin stage 194 notes/stages/moving_throat_pde_stage194_outgoing_l2_fingerprint_and_deformation_algebra.md:182	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage194_outgoing_l2_fingerprint_and_deformation_algebra.md:158	195
194	P_2 fit_stage_194_p_2 fam_0188_p_2	origin stage 194 not recorded	free_choice	posited choice:notes/stages/moving_throat_pde_stage194_outgoing_l2_fingerprint_and_deformation_algebra.md:5	195
195	mhat_0 fit_stage195_mhat0_natural_source_map fam_0469_mhat_0; overlays:chain_chi_Q_norm, chain_quad_54_5, chain_wall_action	origin stage 195 notes/stages/moving_throat_pde_stage195_source_map_reduction_of_canonical_outgoing_branch.md:205	free_choice	posited choice:notes/stages/moving_throat_pde_stage195_source_map_reduction_of_canonical_outgoing_branch.md:202	196, 197
195	mhat_0 fit_stage195_natural_source_map_mhat0 fam_0469_mhat_0; overlays:chain_chi_Q_norm, chain_quad_54_5, chain_wall_action	origin stage 195 notes/stages/moving_throat_pde_stage195_source_map_reduction_of_canonical_outgoing_branch.md:205	free_choice	posited choice:notes/stages/moving_throat_pde_stage195_source_map_reduction_of_canonical_outgoing_branch.md:210	196, 197
195	Omega_Q fit_stage195_omega_Q_pole_frequency fam_0183_omega_q	origin stage 193 notes/stages/moving_throat_pde_stage193_isotropic_grouped_p2_target_surface.md:222	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage195_source_map_reduction_of_canonical_outgoing_branch.md:85	196, 197
195	P0_target fit_stage195_p0_target_canonical_value fam_0187_p0_target; overlays:chain_quad_54_5, chain_wall_action	origin stage 191 notes/stages/moving_throat_pde_stage191_minimal_pde_data_packet.md:190	published_target	benchmark: bench__2e5805a359c4: textbook - P. C. Peters, 'Gravitational Radiation and the Motion of Two Point Masses', Phys. Rev. 136, B1224 (1964), DOI 10.1103/PhysRev.136.B1224 (ADS bibcode 1964PhRv..136.1224P); see also M. Maggiore, 'Gravitational Waves, Vol. 1: Theory and Experiments', Oxford Univ. Press (2007), ISBN 978-0-19-857074-5, Ch. 3 (quadrupole formula L = (G/5c^5)<dddQ__ij dddQ__ij>).	196, 197
195	chi_Q fit_stage_195_chi_q fam_0345_chi_q; overlays:chain_chi_Q_norm	origin stage 194 notes/stages/moving_throat_pde_stage194_outgoing_l2_fingerprint_and_deformation_algebra.md:184	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage195_source_map_reduction_of_canonical_outgoing_branch.md:22	196, 197

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
195	Delta_Q fit_stage_195_delta_q fam_0058_delta_q	origin stage 195 notes/stages/moving_throat_pde_stage195_source_map_reduction_of_canonical_outgoing_branch.md:185	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage195_source_map_reduction_of_canonical_outgoing_branch.md:192	196, 197
195	matched_fingerprint_value fit_stage_195_matched_fingerprint_value fam_0459_matched_fingerprint_value	origin stage 194 notes/stages/moving_throat_pde_stage194_outgoing_l2_fingerprint_and_deformation_algebra.md:182	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage195_source_map_reduction_of_canonical_outgoing_branch.md:388	196
195	N_Q fit_stage_195_nq_from_def fam_0174_n_q; overlays:chain_chi_Q_norm, chain_quad_54_5	origin stage 195 notes/stages/moving_throat_pde_stage195_source_map_reduction_of_canonical_outgoing_branch.md:103	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage195_source_map_reduction_of_canonical_outgoing_branch.md:139	196, 197
195	P_2 fit_stage_195_p_2 fam_0188_p_2	origin stage 195 not recorded	free_choice	posited choice:notes/stages/moving_throat_pde_stage195_source_map_reduction_of_canonical_outgoing_branch.md:5	196, 197
195	stale_output fit_stage_195_stale_output fam_0517_stale_output	origin stage 195 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
196	sigma_Q_can fit_stage196_sigma_Q_canonical_outgoing_scale fam_0507_sigma_q_can; overlays:chain_chi_Q_norm	origin stage 194 notes/stages/moving_throat_pde_stage194_outgoing_l2_fingerprint_and_deformation_algebra.md:74	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage194_outgoing_l2_fingerprint_and_deformation_algebra.md:69	197
196	chi_Q fit_stage_196_chi_q fam_0345_chi_q; overlays:chain_chi_Q_norm	origin stage 194 notes/stages/moving_throat_pde_stage194_outgoing_l2_fingerprint_and_deformation_algebra.md:184	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage196_higher_odd_irrelevance_theorem.md:273	197
196	Delta_Q fit_stage_196_delta_q fam_0058_delta_q	origin stage 195 notes/stages/moving_throat_pde_stage195_source_map_reduction_of_canonical_outgoing_branch.md:185	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage196_higher_odd_irrelevance_theorem.md:54	197
196	L7_coeff_in_Y fit_stage_196_l7_coeff_in_y fam_0144_l7_coeff_in_y	origin stage 196 notes/stages/moving_throat_pde_stage196_higher_odd_irrelevance_theorem.md:178	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage196_higher_odd_irrelevance_theorem.md:268	none recorded

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
196	matched_fingerprint_value fit_stage_196_matched_fingerprint_value fam_0459_matched_fingerprint_value	origin stage 115 notes/stages/moving_throat_pde_stage115_core_balance_compensation.md:25	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage196_higher_odd_irrelevance_theorem.md:254	197
196	P_2 fit_stage_196_p_2 fam_0188_p_2	origin stage 91 notes/stages/moving_throat_pde_stage091_grouped_p2_static_geometry_derivation.md:1	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage196_higher_odd_irrelevance_theorem.md:22	197
196	Yhat_2 fit_stage_196_yhat_2 fam_0296_yhat_2	origin stage 194 notes/stages/moving_throat_pde_stage194_outgoing_l2_fingerprint_and_deformation_algebra.md:236	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage196_higher_odd_irrelevance_theorem.md:192	197
197	Delta_Q fit_stage197_chi_q_unity_packet_a_target fam_0058_delta_q, fam_0345_chi_q; overlays:chain_chi_Q_norm	origin stage 195 notes/stages/moving_throat_pde_stage195_source_map_reduction_of_canonical_outgoing_branch.md:185	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage197_conditional_packetA_closure_theorem.md:217	198
197	chi_Q fit_stage197_chi_q_unity_packet_a_target fam_0058_delta_q, fam_0345_chi_q; overlays:chain_chi_Q_norm	origin stage 194 notes/stages/moving_throat_pde_stage194_outgoing_l2_fingerprint_and_deformation_algebra.md:184	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage197_conditional_packetA_closure_theorem.md:194	198
197	P0_target fit_stage197_p0_target_quadrupole_normalization fam_0187_p0_target; overlays:chain_quad_54_5, chain_wall_action	origin stage 22 notes/stages/moving_throat_pde_stage022_grouped_p2_normalization_bridge.md:270	published_target	benchmark: bench__2e5805a359c4: textbook - P. C. Peters, 'Gravitational Radiation and the Motion of Two Point Masses', Phys. Rev. 136, B1224 (1964), DOI 10.1103/PhysRev.136.B1224 (ADS bibcode 1964PhRv..136.1224P); see also M. Maggiore, 'Gravitational Waves, Vol. 1: Theory and Experiments', Oxford Univ. Press (2007), ISBN 978-0-19-857074-5, Ch. 3 (quadrupole formula L = (G/5c^5)<dddQ__ij dddQ__ij>).	198
197	Delta_Q fit_stage_197_delta_q fam_0058_delta_q	origin stage 195 notes/stages/moving_throat_pde_stage195_source_map_reduction_of_canonical_outgoing_branch.md:185	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage197_conditional_packetA_closure_theorem.md:217	198

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
197	matched_fingerprint_value fit_stage_197_matched_fingerprint_value fam_0459_matched_fingerprint_value	origin stage 115 notes/stages/moving_throat_pde_stage115_core_balance_compensation.md:25	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage197_conditional_packetA_closure_theorem.md:221	198
197	P_2 fit_stage_197_p_2 fam_0188_p_2	origin stage 91 notes/stages/moving_throat_pde_stage091_grouped_p2_static_geometry_derivation.md:1	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage197_conditional_packetA_closure_theorem.md:72	198
197	Y_stage194_hi fit_stage_197_y_stage194_hi fam_0294_y_stage194_hi	origin stage 194 notes/stages/moving_throat_pde_stage194_outgoing_l2_fingerprint_and_deformation_algebra.md:236	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage196_higher_odd_irrelevance_theorem.md:202	198
198	Pi_star fit_stage_198_pi_star fam_0209_pi_star	origin stage 198 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
199	m_K fit_stage_199_m_k fam_0456_m_k	origin stage 199 notes/stages/moving_throat_pde_stage199_pairwise_orbit_transport_law.md:288	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage199_pairwise_orbit_transport_law.md:59	200
200	M_star fit_stage200_mismatch_chart_rederived_not_posited fam_0168_m_star	origin stage 192 notes/stages/moving_throat_pde_stage192_orbit_quotient_projectors.md:99	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage198_exact_finite_orbit_law.md:127	201
200	eps_beta fit_stage_200_eps_beta fam_0384_eps_beta	origin stage 197 notes/stages/moving_throat_pde_stage197_conditional_packetA_closure_theorem.md:265	free_choice	posited choice:notes/stages/moving_throat_pde_stage197_conditional_packetA_closure_theorem.md:271	none recorded
200	stale_output fit_stage_200_stale_output fam_0517_stale_output	origin stage 200 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
201	Delta_x_rep fit_stage201_canonical_repair_vector fam_0067_delta_x_rep	origin stage 201 notes/stages/moving_throat_pde_stage201_explicit_realization_compiler_and_canonical_orbit_projection.md:274	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage201_explicit_realization_compiler_and_canonical_orbit_projection.md:305	202
201	Delta fit_stage_201_delta fam_0051_delta	origin stage 195 notes/stages/moving_throat_pde_stage195_source_map_reduction_of_canonical_outgoing_branch.md:185	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage201_explicit_realization_compiler_and_canonical_orbit_projection.md:151	202

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
201	I_3 fit_stage_201_i_3 fam_0105_i_3	origin stage 192 notes/stages/moving_throat_pde_stage192_orbit_quotient_projectors.md:200	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage201_explicit_realization_compiler_and_canonical_orbit_projection.md:305	202
202	C_tr_target fit_stage202_split_u_ratio_uniquely_fixed fam_0036_c_tr_target	origin stage 202 notes/stages/moving_throat_pde_stage202_free_quintuple_target_graph.md:137	free_choice	posited choice:notes/stages/moving_throat_pde_stage202_free_quintuple_target_graph.md:134	203
203	beta_path fit_stage_203_beta_path fam_0330_beta_path	origin stage 203 notes/stages/moving_throat_pde_stage203_free_quintuple_scalar_closure_slice_and_crossing_theorem.md:456	free_choice	posited choice:notes/stages/moving_throat_pde_stage203_free_quintuple_scalar_closure_slice_and_crossing_theorem.md:334	none recorded
203	KW_t fit_stage_203_kw_t fam_0119_kw_t	origin stage 203 notes/stages/moving_throat_pde_stage203_free_quintuple_scalar_closure_slice_and_crossing_theorem.md:456	free_choice	posited choice:notes/stages/moving_throat_pde_stage203_free_quintuple_scalar_closure_slice_and_crossing_theorem.md:493	none recorded
204	Delta fit_stage_204_delta fam_0051_delta	origin stage 195 notes/stages/moving_throat_pde_stage195_source_map_reduction_of_canonical_outgoing_branch.md:185	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage204_explicit_log_ray_sweep_and_scalar_root_predictor.md:365	205
206	mathbf fit_stage_206_mathbf fam_0460_mathbf	origin stage 206 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
207	Q_i fit_stage_207_q_i fam_0211_q_i	origin stage 207 notes/stages/moving_throat_pde_stage207_primitive_ray_hessian_envelopes_and_certified_ray_table.md:367	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage207_primitive_ray_hessian_envelopes_and_certified_ray_table.md:375	208
207	stale_output fit_stage_207_stale_output fam_0517_stale_output	origin stage 207 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
207	tau_lo fit_stage_207_tau_lo fam_0525_tau_lo	origin stage 207 notes/stages/moving_throat_pde_stage207_primitive_ray_hessian_envelopes_and_certified_ray_table.md:261	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage207_primitive_ray_hessian_envelopes_and_certified_ray_table.md:272	208

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
208	a_i_equal_mix fit_stage208_canonical_two_ray_screen fam_0306_a_i_equal_mix	origin stage 208 notes/stages/moving_throat_pde_stage208_pairwise_mixed_rays_and_off_diagonal_hessian_synergy.md:221	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage208_pairwise_mixed_rays_and_off_diagonal_hessian_synergy.md:224	209, 210
209	k_i fit_stage_209_k_i fam_0430_k_i	origin stage 207 notes/stages/moving_throat_pde_stage207_primitive_ray_hessian_envelopes_and_certified_ray_table.md:248	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage209_pairwise_ratio_optimizer_and_mixed_ray_winner_theorem.md:62	210
210	triple_screen_directions fit_stage210_canonical_triple_screens fam_0530_triple_screen_directions	origin stage 210 notes/stages/moving_throat_pde_stage210_three_coordinate_mixed_simplex_audit.md:432	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage210_three_coordinate_mixed_simplex_audit.md:296	none recorded
211	C_cross fit_stage_211_c_cross fam_0026_c_cross	origin stage 211 notes/stages/moving_throat_pde_stage211_full_interior_simplex_optimizer_and_finite_candidate_reduction.md:238	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage211_full_interior_simplex_optimizer_and_finite_candidate_reduction.md:232	212
211	L_r fit_stage_211_l_r fam_0147_l_r	origin stage 211 notes/stages/moving_throat_pde_stage211_full_interior_simplex_optimizer_and_finite_candidate_reduction.md:167	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage211_full_interior_simplex_optimizer_and_finite_candidate_reduction.md:162	212
211	S_r fit_stage_211_s_r fam_0232_s_r	origin stage 211 notes/stages/moving_throat_pde_stage211_full_interior_simplex_optimizer_and_finite_candidate_reduction.md:262	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage211_full_interior_simplex_optimizer_and_finite_candidate_reduction.md:324	212
212	evaluation_budget_600 fit_stage212_evaluation_budget_600 fam_0397_evaluation_budget_600	origin stage 212 notes/stages/moving_throat_pde_stage212_full_primitive_triple_ranking_theorem.md:460	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage212_full_primitive_triple_ranking_theorem.md:457	215

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
213	four_way_screen_directions fit_stage213_four_way_canonical_screens fam_0404_four_way_screen_directions	origin stage 213 notes/stages/moving_throat_pde_stage213_four_coordinate_mixed_simplex_and_support_cardinality_4_gate.md:538	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage213_four_coordinate_mixed_simplex_and_support_cardinality_4_gate.md:517	214
213	S_quad fit_stage_213_s_quad fam_0231_s_quad	origin stage 213 notes/stages/moving_throat_pde_stage213_four_coordinate_mixed_simplex_and_support_cardinality_4_gate.md:538	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage213_four_coordinate_mixed_simplex_and_support_cardinality_4_gate.md:30	214
214	candidate_bound_54 fit_stage214_preferred_bound_54 fam_0342_candidate_bound_54	origin stage 214 notes/stages/moving_throat_pde_stage214_full_interior_four_coordinate_simplex_optimizer_and_finite_candidate_reduction.md:317	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage214_full_interior_four_coordinate_simplex_optimizer_and_finite_candidate_reduction.md:315	215
214	L_r fit_stage_214_l_r fam_0147_l_r	origin stage 214 notes/stages/moving_throat_pde_stage214_full_interior_four_coordinate_simplex_optimizer_and_finite_candidate_reduction.md:199	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage214_full_interior_four_coordinate_simplex_optimizer_and_finite_candidate_reduction.md:194	215
215	cumulative_budget_1140 fit_stage215_cumulative_budget_1140 fam_0355_cumulative_budget_1140	origin stage 215 notes/stages/moving_throat_pde_stage215_full_primitive_quadruple_ranking_theorem.md:440	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage215_full_primitive_quadruple_ranking_theorem.md:438	217, 218
217	candidate_bound_162 fit_stage217_preferred_bound_162_budgets fam_0341_candidate_bound_162	origin stage 217 notes/stages/moving_throat_pde_stage217_full_interior_five_coordinate_simplex_optimizer_and_finite_candidate_reduction.md:409	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage217_full_interior_five_coordinate_simplex_optimizer_and_finite_candidate_reduction.md:407	218

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
217	K_lin fit_stage_217_k_lin fam_0134_k_lin	origin stage 217 notes/stages/moving_throat_pde_stage217_full_interior_five_coordinate_simplex_optimizer_and_finite_candidate_reduction.md:286	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage217_full_interior_five_coordinate_simplex_optimizer_and_finite_candidate_reduction.md:289	218
218	fallback_budget_2640 fit_stage218_total_budgets_1464_2640 fam_0401_fallback_budget_2640	origin stage 218 notes/stages/moving_throat_pde_stage218_full_support_cardinality_5_completion_and_local_mixed_ray_search_closure.md:354	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage218_full_support_cardinality_5_completion_and_local_mixed_ray_search_closure.md:351	none recorded
219	delta_V_mix fit_stage219_delta_v_mix_fixed_by_theorem fam_0369_delta_v_mix	origin stage 219 notes/stages/moving_throat_pde_stage219_one_port_mixed_bundle_static_kernel_and_square_law_suppression_test_sympy_audit.md:158	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage219_one_port_mixed_bundle_static_kernel_and_square_law_suppression_test_sympy_audit.md:149	220
219	C_2 fit_stage_219_c_2 fam_0023_c_2	origin stage 219 notes/stages/moving_throat_pde_stage219_one_port_mixed_bundle_static_kernel_and_square_law_suppression_test_sympy_audit.md:330	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage219_one_port_mixed_bundle_static_kernel_and_square_law_suppression_test_sympy_audit.md:303	none recorded
220	D_Pi fit_stage_220_d_pi fam_0044_d_pi	origin stage 220 notes/stages/moving_throat_pde_stage220_dynamic_mixed_port_kernel_phase_lag_no_go_and_resonant_survival_gate_sympy_audit.md:154	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage220_dynamic_mixed_port_kernel_phase_lag_no_go_and_resonant_survival_gate_sympy_audit.md:247	none recorded
220	Delta_Pi fit_stage_220_delta_pi fam_0057_delta_pi	origin stage 220 notes/stages/moving_throat_pde_stage220_dynamic_mixed_port_kernel_phase_lag_no_go_and_resonant_survival_gate_sympy_audit.md:147	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage220_dynamic_mixed_port_kernel_phase_lag_no_go_and_resonant_survival_gate_sympy_audit.md:176	none recorded

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
220	Im fit_stage_220_im fam_0110_im	origin stage 220 notes/stages/moving_throat_pde_stage220_dynamic_mixed_port_kernel_phase_lag_no_go_and_resonant_survival_gate_sympy_audit.md:422	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage220_dynamic_mixed_port_kernel_phase_lag_no_go_and_resonant_survival_gate_sympy_audit.md:408	none recorded
220	matched_fingerprint_value fit_stage_220_matched_fingerprint_value fam_0459_matched_fingerprint_value	origin stage 220 notes/stages/moving_throat_pde_stage220_dynamic_mixed_port_kernel_phase_lag_no_go_and_resonant_survival_gate_sympy_audit.md:114	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage220_dynamic_mixed_port_kernel_phase_lag_no_go_and_resonant_survival_gate_sympy_audit.md:115	none recorded
220	P_abs fit_stage_220_p_abs fam_0189_p_abs	origin stage 220 notes/stages/moving_throat_pde_stage220_dynamic_mixed_port_kernel_phase_lag_no_go_and_resonant_survival_gate_sympy_audit.md:433	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage220_dynamic_mixed_port_kernel_phase_lag_no_go_and_resonant_survival_gate_sympy_audit.md:441	none recorded
220	Re fit_stage_220_re fam_0224_re	origin stage 220 notes/stages/moving_throat_pde_stage220_dynamic_mixed_port_kernel_phase_lag_no_go_and_resonant_survival_gate_sympy_audit.md:420	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage220_dynamic_mixed_port_kernel_phase_lag_no_go_and_resonant_survival_gate_sympy_audit.md:425	none recorded
221	F_0 fit_stage_221_f_0 fam_0076_f_0	origin stage 221 notes/stages/moving_throat_pde_stage221_resonance_linewidth_tradeoff_dispersive_no_free_lunch_theorem_and_linear_survival_window_sympy_audit.md:149	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage221_resonance_linewidth_tradeoff_dispersive_no_free_lunch_theorem_and_linear_survival_window_sympy_audit.md:143	222

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
222	kappa fit_stage222_exact_overlap_constant_kappa fam_0431_kappa	origin stage 222 notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:83	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:81	223
222	V_known fit_stage222_illustrative_barrier_benchmark fam_0274_v_known; overlays:chain_barrier_222_224	origin stage 222 notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:352	free_choice	posited choice:notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:350	223
222	K fit_stage222_numerical_primitive_slice fam_0115_k, fam_0156_m, fam_0184_omega_u, fam_0185_omega_w, fam_0445_lambda_b, fam_0447_lambda_r, fam_0448_lambda_u, fam_0449_lambda_w, fam_0534_varpi	origin stage 222 notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:289	free_choice	posited choice:notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:434	223
222	M fit_stage222_numerical_primitive_slice fam_0115_k, fam_0156_m, fam_0184_omega_u, fam_0185_omega_w, fam_0445_lambda_b, fam_0447_lambda_r, fam_0448_lambda_u, fam_0449_lambda_w, fam_0534_varpi	origin stage 222 notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:289	free_choice	posited choice:notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:434	223

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
222	Omega_U fit_stage222_numerical_primitive_slice fam_0115_k, fam_0156_m, fam_0184_omega_u, fam_0185_omega_w, fam_0445_lambda_b, fam_0447_lambda_r, fam_0448_lambda_u, fam_0449_lambda_w, fam_0534_varpi	origin stage 222 notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:289	free_choice	posited choice:notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:434	223
222	Omega_W fit_stage222_numerical_primitive_slice fam_0115_k, fam_0156_m, fam_0184_omega_u, fam_0185_omega_w, fam_0445_lambda_b, fam_0447_lambda_r, fam_0448_lambda_u, fam_0449_lambda_w, fam_0534_varpi	origin stage 222 notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:289	free_choice	posited choice:notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:434	223
222	lambda_B fit_stage222_numerical_primitive_slice fam_0115_k, fam_0156_m, fam_0184_omega_u, fam_0185_omega_w, fam_0445_lambda_b, fam_0447_lambda_r, fam_0448_lambda_u, fam_0449_lambda_w, fam_0534_varpi	origin stage 222 notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:289	free_choice	posited choice:notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:434	223
222	lambda_R fit_stage222_numerical_primitive_slice fam_0115_k, fam_0156_m, fam_0184_omega_u, fam_0185_omega_w, fam_0445_lambda_b, fam_0447_lambda_r, fam_0448_lambda_u, fam_0449_lambda_w, fam_0534_varpi	origin stage 222 notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:289	free_choice	posited choice:notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:434	223

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
222	lambda_U fit_stage222_numerical_primitive_slice fam_0115_k, fam_0156_m, fam_0184_omega_u, fam_0185_omega_w, fam_0445_lambda_b, fam_0447_lambda_r, fam_0448_lambda_u, fam_0449_lambda_w, fam_0534_varpi	origin stage 222 notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:289	free_choice	posited choice:notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:434	223
222	lambda_W fit_stage222_numerical_primitive_slice fam_0115_k, fam_0156_m, fam_0184_omega_u, fam_0185_omega_w, fam_0445_lambda_b, fam_0447_lambda_r, fam_0448_lambda_u, fam_0449_lambda_w, fam_0534_varpi	origin stage 222 notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:289	free_choice	posited choice:notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:434	223
222	varpi fit_stage222_numerical_primitive_slice fam_0115_k, fam_0156_m, fam_0184_omega_u, fam_0185_omega_w, fam_0445_lambda_b, fam_0447_lambda_r, fam_0448_lambda_u, fam_0449_lambda_w, fam_0534_varpi	origin stage 222 notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:289	free_choice	posited choice:notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:434	223
222	K fit_stage222_primitive_sample_slice fam_0115_k, fam_0156_m, fam_0184_omega_u, fam_0185_omega_w, fam_0301_a, fam_0340_c_s, fam_0440_lam_b, fam_0441_lam_r, fam_0442_lam_u, fam_0443_lam_w, fam_0534_varpi	origin stage 222 notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:289	free_choice	posited choice:notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:434	223

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
222	M fit_stage222_primitive_sample_slice fam_0115_k, fam_0156_m, fam_0184_omega_u, fam_0185_omega_w, fam_0301_a, fam_0340_c_s, fam_0440_lam_b, fam_0441_lam_r, fam_0442_lam_u, fam_0443_lam_w, fam_0534_varpi	origin stage 222 notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:289	free_choice	posited choice:notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:434	223
222	Omega_U fit_stage222_primitive_sample_slice fam_0115_k, fam_0156_m, fam_0184_omega_u, fam_0185_omega_w, fam_0301_a, fam_0340_c_s, fam_0440_lam_b, fam_0441_lam_r, fam_0442_lam_u, fam_0443_lam_w, fam_0534_varpi	origin stage 222 notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:289	free_choice	posited choice:notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:434	223
222	Omega_W fit_stage222_primitive_sample_slice fam_0115_k, fam_0156_m, fam_0184_omega_u, fam_0185_omega_w, fam_0301_a, fam_0340_c_s, fam_0440_lam_b, fam_0441_lam_r, fam_0442_lam_u, fam_0443_lam_w, fam_0534_varpi	origin stage 222 notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:289	free_choice	posited choice:notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:434	223
222	a fit_stage222_primitive_sample_slice fam_0115_k, fam_0156_m, fam_0184_omega_u, fam_0185_omega_w, fam_0301_a, fam_0340_c_s, fam_0440_lam_b, fam_0441_lam_r, fam_0442_lam_u, fam_0443_lam_w, fam_0534_varpi	origin stage 222 notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:289	free_choice	posited choice:notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:434	223

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
222	c_s fit_stage222_primitive_sample_slice fam_0115_k, fam_0156_m, fam_0184_omega_u, fam_0185_omega_w, fam_0301_a, fam_0340_c_s, fam_0440_lam_b, fam_0441_lam_r, fam_0442_lam_u, fam_0443_lam_w, fam_0534_varpi	origin stage 222 notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:289	free_choice	posited choice:notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:434	223
222	lam_B fit_stage222_primitive_sample_slice fam_0115_k, fam_0156_m, fam_0184_omega_u, fam_0185_omega_w, fam_0301_a, fam_0340_c_s, fam_0440_lam_b, fam_0441_lam_r, fam_0442_lam_u, fam_0443_lam_w, fam_0534_varpi	origin stage 222 notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:289	free_choice	posited choice:notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:434	223
222	lam_R fit_stage222_primitive_sample_slice fam_0115_k, fam_0156_m, fam_0184_omega_u, fam_0185_omega_w, fam_0301_a, fam_0340_c_s, fam_0440_lam_b, fam_0441_lam_r, fam_0442_lam_u, fam_0443_lam_w, fam_0534_varpi	origin stage 222 notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:289	free_choice	posited choice:notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:434	223
222	lam_U fit_stage222_primitive_sample_slice fam_0115_k, fam_0156_m, fam_0184_omega_u, fam_0185_omega_w, fam_0301_a, fam_0340_c_s, fam_0440_lam_b, fam_0441_lam_r, fam_0442_lam_u, fam_0443_lam_w, fam_0534_varpi	origin stage 222 notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:289	free_choice	posited choice:notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:434	223

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
222	lam_W fit_stage222_primitive_sample_slice fam_0115_k, fam_0156_m, fam_0184_omega_u, fam_0185_omega_w, fam_0301_a, fam_0340_c_s, fam_0440_lam_b, fam_0441_lam_r, fam_0442_lam_u, fam_0443_lam_w, fam_0534_varpi	origin stage 222 notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:289	free_choice	posited choice:notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:434	223
222	varpi fit_stage222_primitive_sample_slice fam_0115_k, fam_0156_m, fam_0184_omega_u, fam_0185_omega_w, fam_0301_a, fam_0340_c_s, fam_0440_lam_b, fam_0441_lam_r, fam_0442_lam_u, fam_0443_lam_w, fam_0534_varpi	origin stage 222 notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:289	free_choice	posited choice:notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:434	223
222	Delta fit_stage_222_delta fam_0051_delta	origin stage 222 notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:107	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:113	223
222	DeltaV_req fit_stage_222_deltav_req fam_0054_deltav_req; overlays:chain_barrier_222_224	origin stage 222 notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:356	free_choice	posited choice:notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:436	223

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
222	lambda_B fit_stage_222_lambda_b fam_0445_lambda_b	origin stage 222 notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:289	free_choice	posited choice:notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:434	223
222	N_star fit_stage_222_n_star fam_0178_n_star	origin stage 222 notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:180	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:178	223
222	P0_target fit_stage_222_p_0 fam_0187_p0_target; overlays:chain_quad_54_5, chain_wall_action	origin stage 222 notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:314	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:391	223
223	V_known fit_stage223_barrier_benchmark_and_eta_window fam_0274_v_known; overlays:chain_barrier_222_224	origin stage 222 notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:352	free_choice	posited choice:notes/stages/moving_throat_pde_stage223_5pn_isotropic_target_surface_primitive_branch_compatibility_and_dynamic_survival_window_sympy_audit.md:329	224, 225, 226, 227

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
223	R_Q_lower_wall_lambdaW_0p2 fit_stage223_lambda02_rq_dual_engine_overclaim fam_0213_r_q_lower_wall_lambdaw_0p2	origin stage 223 notes/stages/moving_throat_pde_stage223_5pn_isotropic_target_surface_primitive_branch_compatibility_and_dynamic_survival_window_sympy_audit.md:351	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage223_5pn_isotropic_target_surface_primitive_branch_compatibility_and_dynamic_survival_window_sympy_audit.md:8	224
223	P0_target fit_stage223_p0_target_calibration_coefficient fam_0187_p0_target; overlays:chain_quad_54_5, chain_wall_action	origin stage 22 notes/stages/moving_throat_pde_stage022_grouped_p2_normalization_bridge.md:270	published_target	benchmark: bench__2e5805a359c4: textbook - P. C. Peters, 'Gravitational Radiation and the Motion of Two Point Masses', Phys. Rev. 136, B1224 (1964), DOI 10.1103/PhysRev.136.B1224 (ADS bibcode 1964PhRv..136.1224P); see also M. Maggiore, 'Gravitational Waves, Vol. 1: Theory and Experiments', Oxford Univ. Press (2007), ISBN 978-0-19-857074-5, Ch. 3 (quadrupole formula L = (G/5c^5)<dddQ__ij dddQ__ij>).	224, 225
223	P0_target fit_stage223_p0_target_import_unprovenanced fam_0187_p0_target; overlays:chain_quad_54_5, chain_wall_action	origin stage 22 notes/stages/moving_throat_pde_stage022_grouped_p2_normalization_bridge.md:270	published_target	benchmark: bench__2e5805a359c4: textbook - P. C. Peters, 'Gravitational Radiation and the Motion of Two Point Masses', Phys. Rev. 136, B1224 (1964), DOI 10.1103/PhysRev.136.B1224 (ADS bibcode 1964PhRv..136.1224P); see also M. Maggiore, 'Gravitational Waves, Vol. 1: Theory and Experiments', Oxford Univ. Press (2007), ISBN 978-0-19-857074-5, Ch. 3 (quadrupole formula L = (G/5c^5)<dddQ__ij dddQ__ij>).	224, 225
223	P0_target fit_stage223_universal_p0_target fam_0187_p0_target; overlays:chain_quad_54_5, chain_wall_action	origin stage 22 notes/stages/moving_throat_pde_stage022_grouped_p2_normalization_bridge.md:270	published_target	benchmark: bench__2e5805a359c4: textbook - P. C. Peters, 'Gravitational Radiation and the Motion of Two Point Masses', Phys. Rev. 136, B1224 (1964), DOI 10.1103/PhysRev.136.B1224 (ADS bibcode 1964PhRv..136.1224P); see also M. Maggiore, 'Gravitational Waves, Vol. 1: Theory and Experiments', Oxford Univ. Press (2007), ISBN 978-0-19-857074-5, Ch. 3 (quadrupole formula L = (G/5c^5)<dddQ__ij dddQ__ij>).	224, 225
223	D0_compat fit_stage_223_d0_compat fam_0038_d0_compat	origin stage 223 notes/stages/moving_throat_pde_stage223_5pn_isotropic_target_surface_primitive_branch_compatibility_and_dynamic_survival_window_sympy_audit.md:268	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage223_5pn_isotropic_target_surface_primitive_branch_compatibility_and_dynamic_survival_window_sympy_audit.md:266	225, 226

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
223	K_norm fit_stage_223_k_norm fam_0136_k_norm	origin stage 223 notes/stages/moving_throat_pde_stage223_5pn_isotropic_target_surface_primitive_branch_compatibility_and_dynamic_survival_window_sympy_audit.md:144	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage223_5pn_isotropic_target_surface_primitive_branch_compatibility_and_dynamic_survival_window_sympy_audit.md:142	225
223	mathrm fit_stage_223_mathrm fam_0467_mathrm	origin stage 22 notes/stages/moving_throat_pde_stage022_grouped_p2_normalization_bridge.md:270	published_target	published_target; no benchmarks.yaml entry joined for listed family; evidence:notes/stages/moving_throat_pde_stage223_5pn_isotropic_target_surface_primitive_branch_compatibility_and_dynamic_survival_window_sympy_audit.md:127	224, 225
224	b_P0_slope fit_stage224_grouped_signature_exact fam_0319_b_p0_slope	origin stage 224 notes/stages/moving_throat_pde_stage224_pde_branch_packet_compiler_weak_axisymmetric_ceiling_transport_and_first_actual_branch_kill_test_sympy_audit.md:224	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage224_pde_branch_packet_compiler_weak_axisymmetric_ceiling_transport_and_first_actual_branch_kill_test_sympy_audit.md:228	225
224	P_crit_both_10 fit_stage224_kill_test_budgets_slice_anchored fam_0190_p_crit_both_10, fam_0191_p_crit_both_30, fam_0192_p_crit_one_10, fam_0193_p_crit_one_30, fam_0322_barp0_compat; overlays:chain_barrier_222_224	origin stage 223 notes/stages/moving_throat_pde_stage223_5pn_isotropic_target_surface_primitive_branch_compatibility_and_dynamic_survival_window_sympy_audit.md:373	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage224_pde_branch_packet_compiler_weak_axisymmetric_ceiling_transport_and_first_actual_branch_kill_test_sympy_audit.md:115	225, 226, 227
224	P_crit_both_30 fit_stage224_kill_test_budgets_slice_anchored fam_0190_p_crit_both_10, fam_0191_p_crit_both_30, fam_0192_p_crit_one_10, fam_0193_p_crit_one_30, fam_0322_barp0_compat; overlays:chain_barrier_222_224	origin stage 223 notes/stages/moving_throat_pde_stage223_5pn_isotropic_target_surface_primitive_branch_compatibility_and_dynamic_survival_window_sympy_audit.md:386	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage224_pde_branch_packet_compiler_weak_axisymmetric_ceiling_transport_and_first_actual_branch_kill_test_sympy_audit.md:108	225, 226, 227

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
224	P_crit_one_10 fit_stage224_kill_test_budgets_slice_anchored fam_0190_p_crit_both_10, fam_0191_p_crit_both_30, fam_0192_p_crit_one_10, fam_0193_p_crit_one_30, fam_0322_barp0_compat; overlays:chain_barrier_222_224	origin stage 223 notes/stages/moving_throat_pde_stage223_5pn_isotropic_target_surface_primitive_branch_compatibility_and_dynamic_survival_window_sympy_audit.md:377	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage224_pde_branch_packet_compiler_weak_axisymmetric_ceiling_transport_and_first_actual_branch_kill_test_sympy_audit.md:105	225, 226, 227
224	P_crit_one_30 fit_stage224_kill_test_budgets_slice_anchored fam_0190_p_crit_both_10, fam_0191_p_crit_both_30, fam_0192_p_crit_one_10, fam_0193_p_crit_one_30, fam_0322_barp0_compat; overlays:chain_barrier_222_224	origin stage 223 notes/stages/moving_throat_pde_stage223_5pn_isotropic_target_surface_primitive_branch_compatibility_and_dynamic_survival_window_sympy_audit.md:390	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage224_pde_branch_packet_compiler_weak_axisymmetric_ceiling_transport_and_first_actual_branch_kill_test_sympy_audit.md:112	225, 226, 227
224	barP0_compat fit_stage224_kill_test_budgets_slice_anchored fam_0190_p_crit_both_10, fam_0191_p_crit_both_30, fam_0192_p_crit_one_10, fam_0193_p_crit_one_30, fam_0322_barp0_compat; overlays:chain_barrier_222_224	origin stage 223 notes/stages/moving_throat_pde_stage223_5pn_isotropic_target_surface_primitive_branch_compatibility_and_dynamic_survival_window_sympy_audit.md:253	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage224_pde_branch_packet_compiler_weak_axisymmetric_ceiling_transport_and_first_actual_branch_kill_test_sympy_audit.md:13	225, 226, 227
224	Pcrit_both_10 fit_stage224_stage223_carryover_ceilings fam_0196_pcrit_both_10	origin stage 223 notes/stages/moving_throat_pde_stage223_5pn_isotropic_target_surface_primitive_branch_compatibility_and_dynamic_survival_window_sympy_audit.md:373	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage224_pde_branch_packet_compiler_weak_axisymmetric_ceiling_transport_and_first_actual_branch_kill_test_sympy_audit.md:103	225, 226, 227
224	T_quad fit_stage224_t_quad_target_scale fam_0258_t_quad; overlays:chain_barrier_222_224	origin stage 22 notes/stages/moving_throat_pde_stage022_grouped_p2_normalization_bridge.md:270	published_target	published_target; no benchmarks.yaml entry joined for listed family; evidence:notes/stages/moving_throat_pde_stage224_pde_branch_packet_compiler_weak_axisymmetric_ceiling_transport_and_first_actual_branch_kill_test_sympy_audit.md:79	225

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
224	lambda_20 fit_stage224_weak_axisymmetric_lane_signature fam_0444_lambda_20; overlays:chain_barrier_222_224	origin stage 224 notes/stages/moving_throat_pde_stage224_pde_branch_packet_compiler_weak_axisymmetric_ceiling_transport_and_first_actual_branch_kill_test_sympy_audit.md:200	free_choice	posited choice:notes/stages/moving_throat_pde_stage224_pde_branch_packet_compiler_weak_axisymmetric_ceiling_transport_and_first_actual_branch_kill_test_sympy_audit.md:195	225
224	Delta_rm fit_stage_224_delta_rm fam_0066_delta_rm	origin stage 224 notes/stages/moving_throat_pde_stage224_pde_branch_packet_compiler_weak_axisymmetric_ceiling_transport_and_first_actual_branch_kill_test_sympy_audit.md:68	free_choice	posited choice:notes/stages/moving_throat_pde_stage224_pde_branch_packet_compiler_weak_axisymmetric_ceiling_transport_and_first_actual_branch_kill_test_sympy_audit.md:177	225, 226
224	hat fit_stage_224_hat fam_0426_hat	origin stage 224 notes/stages/moving_throat_pde_stage224_pde_branch_packet_compiler_weak_axisymmetric_ceiling_transport_and_first_actual_branch_kill_test_sympy_audit.md:70	free_choice	posited choice:notes/stages/moving_throat_pde_stage224_pde_branch_packet_compiler_weak_axisymmetric_ceiling_transport_and_first_actual_branch_kill_test_sympy_audit.md:189	225
225	P0_target fit_stage225_K_compat_P0_target_pin fam_0187_p0_target; overlays:chain_quad_54_5, chain_wall_action	origin stage 223 notes/stages/moving_throat_pde_stage223_5pn_isotropic_target_surface_primitive_branch_compatibility_and_dynamic_survival_window_sympy_audit.md:253	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage225_microscopic_xi1_compiler_first_order_conservative_compensation_surface_and_mixed_sector_survival_sieve_sympy_audit.md:144	226, 227

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
225	`OmU` `fit_stage225_stage223_sample_point` `fam_0180_omu`	origin stage 223 `notes/stages/moving_throat_pde_stage223_5pn_isotropic_target_surface_primitive_branch_compatibility_and_dynamic_survival_window_sympy_audit.md`:209	`free_choice`	posited choice:`notes/stages/moving_throat_pde_stage225_microscopic_xi1_compiler_first_order_conservative_compensation_surface_and_mixed_sector_survival_sieve_sympy_audit.md`:119	226, 227
225	`B_xi_both_10` `fit_stage225_stage224_xi_budgets` `fam_0021_b_xi_both_10`	origin stage 224 `notes/stages/moving_throat_pde_stage224_pde_branch_packet_compiler_weak_axisymmetric_ceiling_transport_and_first_actual_branch_kill_test_sympy_audit.md`:318	`internal_consistency`	located derivation step:`notes/stages/moving_throat_pde_stage225_microscopic_xi1_compiler_first_order_conservative_compensation_surface_and_mixed_sector_survival_sieve_sympy_audit.md`:154	226, 227
225	`x_K` `fit_stage225_xk_xm_forced_zero` `fam_0541_x_k`	origin stage 225 `notes/stages/moving_throat_pde_stage225_microscopic_xi1_compiler_first_order_conservative_compensation_surface_and_mixed_sector_survival_sieve_sympy_audit.md`:407	`internal_consistency`	located derivation step:`notes/stages/moving_throat_pde_stage225_microscopic_xi1_compiler_first_order_conservative_compensation_surface_and_mixed_sector_survival_sieve_sympy_audit.md`:400	none recorded
225	`D_4` `fit_stage_225_d_4` `fam_0042_d_4`	origin stage 225 `notes/stages/moving_throat_pde_stage225_microscopic_xi1_compiler_first_order_conservative_compensation_surface_and_mixed_sector_survival_sieve_sympy_audit.md`:260	`internal_consistency`	located derivation step:`notes/stages/moving_throat_pde_stage225_microscopic_xi1_compiler_first_order_conservative_compensation_surface_and_mixed_sector_survival_sieve_sympy_audit.md`:254	226

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
225	Xi_1 fit_stage_225_xi_1 fam_0283_xi_1	origin stage 225 notes/stages/moving_throat_pde_stage225_microscopic_xi1_compiler_first_order_conservative_compensation_surface_and_mixed_sector_survival_sieve_sympy_audit.md:220	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage225_microscopic_xi1_compiler_first_order_conservative_compensation_surface_and_mixed_sector_survival_sieve_sympy_audit.md:224	226, 227
226	H_even fit_stage226_canonical_hidden_even_relation fam_0098_h_even	origin stage 226 notes/stages/moving_throat_pde_stage226_strict_5pn_even_gate_package_surviving_mixed_corridor_and_pure_transfer_subcorridor_sympy_audit.md:29	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage226_strict_5pn_even_gate_package_surviving_mixed_corridor_and_pure_transfer_subcorridor_sympy_audit.md:175	227
226	D_01 fit_stage226_d01_forced_zero fam_0041_d_01	origin stage 226 notes/stages/moving_throat_pde_stage226_strict_5pn_even_gate_package_surviving_mixed_corridor_and_pure_transfer_subcorridor_sympy_audit.md:214	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage226_strict_5pn_even_gate_package_surviving_mixed_corridor_and_pure_transfer_subcorridor_sympy_audit.md:205	227
226	H_even fit_stage226_k1_h_even_open_gates fam_0098_h_even	origin stage 226 notes/stages/moving_throat_pde_stage226_strict_5pn_even_gate_package_surviving_mixed_corridor_and_pure_transfer_subcorridor_sympy_audit.md:182	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage226_strict_5pn_even_gate_package_surviving_mixed_corridor_and_pure_transfer_subcorridor_sympy_audit.md:201	227
226	sigma_even fit_stage226_sigma_even_canonical_gain_scale fam_0510_sigma_even	origin stage 226 notes/stages/moving_throat_pde_stage226_strict_5pn_even_gate_package_surviving_mixed_corridor_and_pure_transfer_subcorridor_sympy_audit.md:276	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage226_strict_5pn_even_gate_package_surviving_mixed_corridor_and_pure_transfer_subcorridor_sympy_audit.md:270	227

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
226	K_1 fit_stage_226_k_1 fam_0120_k_1	origin stage 226 notes/stages/moving_throat_pde_stage226_strict_5pn_even_gate_package_surviving_mixed_corridor_and_pure_transfer_subcorridor_sympy_audit.md:26	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage226_strict_5pn_even_gate_package_surviving_mixed_corridor_and_pure_transfer_subcorridor_sympy_audit.md:179	227
226	K_compat fit_stage_226_k_compat fam_0126_k_compat	origin stage 223 notes/stages/moving_throat_pde_stage223_5pn_isotropic_target_surface_primitive_branch_compatibility_and_dynamic_survival_window_sympy_audit.md:260	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage223_5pn_isotropic_target_surface_primitive_branch_compatibility_and_dynamic_survival_window_sympy_audit.md:256	225, 226, 227
226	Xi_1 fit_stage_226_xi_1 fam_0283_xi_1	origin stage 225 notes/stages/moving_throat_pde_stage225_microscopic_xi1_compiler_first_order_conservative_compensation_surface_and_mixed_sector_survival_sieve_sympy_audit.md:220	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage226_strict_5pn_even_gate_package_surviving_mixed_corridor_and_pure_transfer_subcorridor_sympy_audit.md:154	227
227	kappa fit_stage227_kappa_exact_overlap_constant fam_0431_kappa	origin stage 26 notes/stages/moving_throat_pde_stage026_concrete_axial_overlaps.md:126	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage026_concrete_axial_overlaps.md:122	222, 223, 224, 225, 226, 227, 228, 229
227	H_coeff fit_stage227_sample_branch_coefficients_exact fam_0097_h_coeff	origin stage 227 notes/stages/moving_throat_pde_stage227_pure_transfer_load_factor_outgoing_rigidity_sieve_and_first_co_loading_no_go_sympy_audit.md:262	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage227_pure_transfer_load_factor_outgoing_rigidity_sieve_and_first_co_loading_no_go_sympy_audit.md:258	228

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
227	c_h fit_stage227_xi1_sample_law_coefficients fam_0338_c_h	origin stage 227 notes/stages/moving_throat_pde_stage227_pure_transfer_load_factor_outgoing_rigidity_sieve_and_first_co_loading_no_go_sympy_audit.md:269	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage227_pure_transfer_load_factor_outgoing_rigidity_sieve_and_first_co_loading_no_go_sympy_audit.md:264	228
227	M_transfer fit_stage_227_m_transfer fam_0171_m_transfer	origin stage 226 notes/stages/moving_throat_pde_stage226_strict_5pn_even_gate_package_surviving_mixed_corridor_and_pure_transfer_subcorridor_sympy_audit.md:305	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage227_pure_transfer_load_factor_outgoing_rigidity_sieve_and_first_co_loading_no_go_sympy_audit.md:122	228
227	N_0 fit_stage_227_n_0 fam_0173_n_0	origin stage 227 notes/stages/moving_throat_pde_stage227_pure_transfer_load_factor_outgoing_rigidity_sieve_and_first_co_loading_no_go_sympy_audit.md:111	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage227_pure_transfer_load_factor_outgoing_rigidity_sieve_and_first_co_loading_no_go_sympy_audit.md:145	228
227	Xi_1 fit_stage_227_xi_1 fam_0283_xi_1	origin stage 225 notes/stages/moving_throat_pde_stage225_microscopic_xi1_compiler_first_order_conservative_compensation_surface_and_mixed_sector_survival_sieve_sympy_audit.md:220	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage227_pure_transfer_load_factor_outgoing_rigidity_sieve_and_first_co_loading_no_go_sympy_audit.md:172	228
228	K_compat fit_stage228_k_compat_exact_stiffness fam_0126_k_compat	origin stage 228 notes/stages/moving_throat_pde_stage228_numerator_denominator_split_of_the_pure_transfer_corridor_and_first_actual_dynamic_window_test_sympy_audit.md:79	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage228_numerator_denominator_split_of_the_pure_transfer_corridor_and_first_actual_dynamic_window_test_sympy_audit.md:77	229, 230, 231, 232, 233, 234

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
228	M fit_stage228_primitive_parameter_tuple fam_0156_m	origin stage 228 notes/stages/moving_throat_pde_stage228_numerator_denominator_split_of_the_pure_transfer_corridor_and_first_actual_dynamic_window_test_sympy_audit.md:74	free_choice	posited choice:notes/stages/moving_throat_pde_stage228_numerator_denominator_split_of_the_pure_transfer_corridor_and_first_actual_dynamic_window_test_sympy_audit.md:73	229, 230, 231, 232, 233, 234
228	RQ_req fit_stage228_rq_requirement_threshold fam_0212_rq_req	origin stage 222 notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:360	published_target	published_target; no benchmarks.yaml entry joined for listed family; evidence:notes/stages/moving_throat_pde_stage222_concrete_finite_throat_primitive_branch_pole_census_and_residue_linewidth_survival_test_sympy_audit.md:352	230
228	delta_1 fit_stage_228_delta_1 fam_0364_delta_1	origin stage 228 notes/stages/moving_throat_pde_stage228_numerator_denominator_split_of_the_pure_transfer_corridor_and_first_actual_dynamic_window_test_sympy_audit.md:121	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage228_numerator_denominator_split_of_the_pure_transfer_corridor_and_first_actual_dynamic_window_test_sympy_audit.md:118	229, 230
229	delta fit_stage229_delta_crossover_threshold fam_0363_delta	origin stage 229 notes/stages/moving_throat_pde_stage229_selected_branch_numerator_denominator_signature_and_softening_depth_crossover_theorem_sympy_audit.md:62	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage229_selected_branch_numerator_denominator_signature_and_softening_depth_crossover_theorem_sympy_audit.md:239	230
229	kappa_0 fit_stage_229_kappa_0 fam_0433_kappa_0	origin stage 229 notes/stages/moving_throat_pde_stage229_selected_branch_numerator_denominator_signature_and_softening_depth_crossover_theorem_sympy_audit.md:101	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage229_selected_branch_numerator_denominator_signature_and_softening_depth_crossover_theorem_sympy_audit.md:143	230

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
230	r_F1 fit_stage230_r_star_exact_threshold fam_0493_r_f1; overlays:chain_aspect_37_20	origin stage 230 notes/stages/moving_throat_pde_stage230_selected_branch_classifier_to_dynamic_window_compiler_and_static_first_theorem_sympy_audit.md:227	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage230_selected_branch_classifier_to_dynamic_window_compiler_and_static_first_theorem_sympy_audit.md:220	231, 233, 234
230	s_minus_den fit_stage230_stage228_rigid_slope_carry fam_0504_s_minus_den	origin stage 230 notes/stages/moving_throat_pde_stage230_selected_branch_classifier_to_dynamic_window_compiler_and_static_first_theorem_sympy_audit.md:89	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage230_selected_branch_classifier_to_dynamic_window_compiler_and_static_first_theorem_sympy_audit.md:84	231, 233, 234
231	B_dyn_both_inf fit_stage231_dynamic_budget_carry fam_0019_b_dyn_both_inf	origin stage 230 notes/stages/moving_throat_pde_stage230_selected_branch_classifier_to_dynamic_window_compiler_and_static_first_theorem_sympy_audit.md:447	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage230_selected_branch_classifier_to_dynamic_window_compiler_and_static_first_theorem_sympy_audit.md:446	232, 233, 234
232	L_over_a fit_stage232_family1_geometry_normalization fam_0146_l_over_a; overlays:chain_aspect_37_20	origin stage 73 notes/stages/moving_throat_pde_stage073_family1_geometry_map.md:30	free_choice	posited choice:notes/stages/moving_throat_pde_stage073_family1_geometry_map.md:56	233, 234
232	Xi_prefactor_100 fit_stage232_fom_prefactor_100 fam_0289_xi_prefactor_100	origin stage 232 not recorded	free_choice	posited choice:notes/stages/moving_throat_pde_stage232_known_5pn_data_injection_and_current_branch_verdict_sympy_audit.md:153	233, 234
232	Theta_w fit_stage232_injected_wall_depth_theta_w fam_0264_theta_w	origin stage 77 notes/stages/moving_throat_pde_stage077_family1_theta_extraction.md:94	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage077_family1_theta_extraction.md:96	233, 234
232	rho_alpha_max fit_stage232_known_5pn_data_injection fam_0501_rho_alpha_max	origin stage 232 notes/stages/moving_throat_pde_stage232_known_5pn_data_injection_and_current_branch_verdict_sympy_audit.md:174	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage232_known_5pn_data_injection_and_current_branch_verdict_sympy_audit.md:176	233, 234
232	L_over_a fit_stage232_lambda_ell_geometry_literals fam_0146_l_over_a; overlays:chain_aspect_37_20	origin stage 73 notes/stages/moving_throat_pde_stage073_family1_geometry_map.md:30	free_choice	posited choice:notes/stages/moving_throat_pde_stage073_family1_geometry_map.md:56	233, 234

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
232	N_Q fit_stage232_nq_canonical_normalization fam_0174_n_q; overlays:chain_chi_Q_norm, chain_quad_54_5	origin stage 232 notes/stages/moving_throat_pde_stage232_known_5pn_data_injection_and_current_branch_verdict_sympy_audit.md:316	free_choice	posited choice:notes/stages/moving_throat_pde_stage232_known_5pn_data_injection_and_current_branch_verdict_sympy_audit.md:314	233, 234
232	Theta_w fit_stage232_theta_w_5pn_injection fam_0264_theta_w	origin stage 77 notes/stages/moving_throat_pde_stage077_family1_theta_extraction.md:94	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage077_family1_theta_extraction.md:96	233, 234
232	Theta_w fit_stage232_wall_depth_thetas fam_0264_theta_w	origin stage 77 notes/stages/moving_throat_pde_stage077_family1_theta_extraction.md:94	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage077_family1_theta_extraction.md:96	233, 234
232	A_K fit_stage_232_a_k fam_0007_a_k	origin stage 232 notes/stages/moving_throat_pde_stage232_known_5pn_data_injection_and_current_branch_verdict_sympy_audit.md:75	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage232_known_5pn_data_injection_and_current_branch_verdict_sympy_audit.md:67	233, 234
232	lambda_mu_1 fit_stage_232_lambda_mu_1 fam_0453_lambda_mu_1	origin stage 232 notes/stages/moving_throat_pde_stage232_known_5pn_data_injection_and_current_branch_verdict_sympy_audit.md:149	free_choice	posited choice:notes/stages/moving_throat_pde_stage232_known_5pn_data_injection_and_current_branch_verdict_sympy_audit.md:149	233, 234
232	Theta_w fit_stage_232_theta_w_j fam_0264_theta_w	origin stage 77 notes/stages/moving_throat_pde_stage077_family1_theta_extraction.md:104	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage077_family1_theta_extraction.md:115	233, 234
233	Delta_norm fit_stage233_calibrated_branch_delta_norm fam_0064_delta_norm	origin stage 233 notes/stages/moving_throat_pde_stage233_rigid_mouth_orbit_lock_compiler_and_the_static_turbulence_gate_sympy_audit.md:205	free_choice	posited choice:notes/stages/moving_throat_pde_stage233_rigid_mouth_orbit_lock_compiler_and_the_static_turbulence_gate_sympy_audit.md:203	234

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
233	Pbar fit_stage233_stage224_ceiling_carry fam_0195_pbar	origin stage 228 notes/stages/moving_throat_pde_stage228_numerator_denominator_split_of_the_pure_transfer_corridor_and_first_actual_dynamic_window_test_sympy_audit.md:307	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage233_rigid_mouth_orbit_lock_compiler_and_the_static_turbulence_gate_sympy_audit.md:248	234
234	Xi_1 fit_stage234_canonical_least_deformation_family fam_0283_xi_1	origin stage 234 notes/stages/moving_throat_pde_stage234_direct_branch_observable_static_gate_and_the_two_observable_kill_test_sympy_audit.md:328	free_choice	posited choice:notes/stages/moving_throat_pde_stage234_direct_branch_observable_static_gate_and_the_two_observable_kill_test_sympy_audit.md:372	none recorded
235	q_eta fit_stage_235_q_eta fam_0491_q_eta	origin stage 235 notes/stages/moving_throat_pde_stage235_rigid_mouth_packet_projectors_static_blind_dressing_line_and_codimension_two_orbit_lock_point_sympy_audit.md:92	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage235_rigid_mouth_packet_projectors_static_blind_dressing_line_and_codimension_two_orbit_lock_point_sympy_audit.md:226	236, 237, 238
236	K_eta fit_stage236_equal_drift_ray_forced fam_0129_k_eta; overlays:chain_wall_action	origin stage 236 notes/stages/moving_throat_pde_stage236_rigid_mouth_microscopic_dependent_plane_projectors_equal_drift_dressing_ray_and_static_only_restoration_gap_sympy_audit.md:102	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage236_rigid_mouth_microscopic_dependent_plane_projectors_equal_drift_dressing_ray_and_static_only_restoration_gap_sympy_audit.md:184	237, 238
236	epsilon_eta fit_stage_236_epsilon_eta fam_0393_epsilon_eta; overlays:chain_calibration_245_253	origin stage 236 notes/stages/moving_throat_pde_stage236_rigid_mouth_microscopic_dependent_plane_projectors_equal_drift_dressing_ray_and_static_only_restoration_gap_sympy_audit.md:46	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage236_rigid_mouth_microscopic_dependent_plane_projectors_equal_drift_dressing_ray_and_static_only_restoration_gap_sympy_audit.md:89	237, 238

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
236	mu fit_stage_236_mu fam_0476_mu	origin stage 236 notes/stages/moving_throat_pde_stage236_rigid_mouth_microscopic_dependent_plane_projectors_equal_drift_dressing_ray_and_static_only_restoration_gap_sympy_audit.md:104	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage236_rigid_mouth_microscopic_dependent_plane_projectors_equal_drift_dressing_ray_and_static_only_restoration_gap_sympy_audit.md:141	237, 238
236	T_U fit_stage_236_t_u fam_0250_t_u	origin stage 236 notes/stages/moving_throat_pde_stage236_rigid_mouth_microscopic_dependent_plane_projectors_equal_drift_dressing_ray_and_static_only_restoration_gap_sympy_audit.md:200	free_choice	posited choice:notes/stages/moving_throat_pde_stage236_rigid_mouth_microscopic_dependent_plane_projectors_equal_drift_dressing_ray_and_static_only_restoration_gap_sympy_audit.md:22	237, 238
237	Rratio_base fit_stage237_branch_sample_point fam_0226_rratio_base	origin stage 237 not recorded	free_choice	posited choice:notes/stages/moving_throat_pde_stage237_actual_branch_dressing_compiler_finite_static_blind_curve_and_support_blind_post_static_orbit_lock_theorem_sympy_audit.md:108	238
237	epsilon_eta fit_stage_237_epsilon_eta fam_0393_epsilon_eta; overlays:chain_calibration_245_253	origin stage 237 notes/stages/moving_throat_pde_stage237_actual_branch_dressing_compiler_finite_static_blind_curve_and_support_blind_post_static_orbit_lock_theorem_sympy_audit.md:15	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage237_actual_branch_dressing_compiler_finite_static_blind_curve_and_support_blind_post_static_orbit_lock_theorem_sympy_audit.md:117	238
237	q_eta fit_stage_237_q_eta fam_0491_q_eta	origin stage 235 notes/stages/moving_throat_pde_stage235_rigid_mouth_packet_projectors_static_blind_dressing_line_and_codimension_two_orbit_lock_point_sympy_audit.md:92	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage237_actual_branch_dressing_compiler_finite_static_blind_curve_and_support_blind_post_static_orbit_lock_theorem_sympy_audit.md:15	238

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
237	q_nt fit_stage_237_q_nt fam_0492_q_nt	origin stage 237 notes/stages/moving_throat_pde_stage237_actual_branch_dressing_compiler_finite_static_blind_curve_and_support_blind_post_static_orbit_lock_theorem_sympy_audit.md:177	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage237_actual_branch_dressing_compiler_finite_static_blind_curve_and_support_blind_post_static_orbit_lock_theorem_sympy_audit.md:13	238
238	Pi_star fit_stage_238_pi_star fam_0209_pi_star	origin stage 238 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
238	q_eta fit_stage_238_q_eta fam_0491_q_eta	origin stage 235 notes/stages/moving_throat_pde_stage235_rigid_mouth_packet_projectors_static_blind_dressing_line_and_codimension_two_orbit_lock_point_sympy_audit.md:92	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage238_physical_branch_transfer_shape_compiler_packet_factorization_and_post_static_dressing_invariance_theorem_sympy_audit.md:205	none recorded
239	Delta_K_eta fit_stage239_exact_microscopic_correction fam_0056_delta_k_eta	origin stage 239 notes/stages/moving_throat_pde_stage239_rigid_mouth_physical_normal_form_exact_physical_to_microscopic_correction_compiler_and_cartesian_orbit_lock_theorem_sympy_audit.md:522	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage239_rigid_mouth_physical_normal_form_exact_physical_to_microscopic_correction_compiler_and_cartesian_orbit_lock_theorem_sympy_audit.md:237	240, 245
239	mu fit_stage_239_mu fam_0476_mu	origin stage 239 notes/stages/moving_throat_pde_stage239_rigid_mouth_physical_normal_form_exact_physical_to_microscopic_correction_compiler_and_cartesian_orbit_lock_theorem_sympy_audit.md:522	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage239_rigid_mouth_physical_normal_form_exact_physical_to_microscopic_correction_compiler_and_cartesian_orbit_lock_theorem_sympy_audit.md:286	240, 245

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
240	Pi_tr fit_stage240_loading_ratio_from_minimal_precursor fam_0210_pi_tr	origin stage 240 notes/stages/moving_throat_pde_stage240_selected_branch_loading_ratio_from_the_minimal_isotropic_quadrupole_precursor_sympy_audit.md:285	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage240_selected_branch_loading_ratio_from_the_minimal_isotropic_quadrupole_precursor_sympy_audit.md:283	241, 242, 244
240	C_mix fit_stage240_loading_ratio_not_free fam_0028_c_mix	origin stage 240 notes/stages/moving_throat_pde_stage240_selected_branch_loading_ratio_from_the_minimal_isotropic_quadrupole_precursor_sympy_audit.md:301	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage240_selected_branch_loading_ratio_from_the_minimal_isotropic_quadrupole_precursor_sympy_audit.md:117	241, 242, 244
240	c0_min fit_stage240_minimal_isotropic_precursor fam_0334_c0_min	origin stage 240 notes/stages/moving_throat_pde_stage240_selected_branch_loading_ratio_from_the_minimal_isotropic_quadrupole_precursor_sympy_audit.md:252	free_choice	posited choice:notes/stages/moving_throat_pde_stage240_selected_branch_loading_ratio_from_the_minimal_isotropic_quadrupole_precursor_sympy_audit.md:249	241, 242
240	rho_alpha fit_stage240_rho_alpha_selected_demand fam_0500_rho_alpha	origin stage 240 notes/stages/moving_throat_pde_stage240_selected_branch_loading_ratio_from_the_minimal_isotropic_quadrupole_precursor_sympy_audit.md:265	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage240_selected_branch_loading_ratio_from_the_minimal_isotropic_quadrupole_precursor_sympy_audit.md:224	241, 242, 244
240	zeta_req fit_stage240_zeta_req_fixed_exactly fam_0555_zeta_req	origin stage 240 notes/stages/moving_throat_pde_stage240_selected_branch_loading_ratio_from_the_minimal_isotropic_quadrupole_precursor_sympy_audit.md:273	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage240_selected_branch_loading_ratio_from_the_minimal_isotropic_quadrupole_precursor_sympy_audit.md:232	241, 242
240	Pi_rm fit_stage_240_pi_rm fam_0208_pi_rm	origin stage 240 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
241	beta_max fit_stage241_beta_window_endpoint fam_0329_beta_max	origin stage 181 notes/stages/moving_throat_pde_stage181_coherent_tracking_defect.md:29	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage241_exact_primitive_ranking_on_the_selected_twin_support_branch_sympy_audit.md:360	242
241	varrho fit_stage241_varrho_no_longer_free fam_0535_varrho	origin stage 240 notes/stages/moving_throat_pde_stage240_selected_branch_loading_ratio_from_the_minimal_isotropic_quadrupole_precursor_sympy_audit.md:311	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage241_exact_primitive_ranking_on_the_selected_twin_support_branch_sympy_audit.md:158	242, 244
241	epsilon fit_stage_241_epsilon fam_0390_epsilon	origin stage 241 notes/stages/moving_throat_pde_stage241_exact_primitive_ranking_on_the_selected_twin_support_branch_sympy_audit.md:171	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage241_exact_primitive_ranking_on_the_selected_twin_support_branch_sympy_audit.md:164	242
242	placement_coeff_2_11 fit_stage242_placement_coeff_2_11 fam_0489_placement_coeff_2_11	origin stage 181 notes/stages/moving_throat_pde_stage181_coherent_tracking_defect.md:29	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage242_actual_twin_support_placement_and_coherent_orbit_lock_compiler_sympy_audit.md:117	244, 245
242	varrho_phys fit_stage242_varrho_phys_fixed fam_0536_varrho_phys	origin stage 242 notes/stages/moving_throat_pde_stage242_actual_twin_support_placement_and_coherent_orbit_lock_compiler_sympy_audit.md:143	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage242_actual_twin_support_placement_and_coherent_orbit_lock_compiler_sympy_audit.md:137	244
242	C_mix fit_stage_242_c_mix fam_0028_c_mix	origin stage 240 notes/stages/moving_throat_pde_stage240_selected_branch_loading_ratio_from_the_minimal_isotropic_quadrupole_precursor_sympy_audit.md:301	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage242_actual_twin_support_placement_and_coherent_orbit_lock_compiler_sympy_audit.md:129	244
242	epsilon fit_stage_242_epsilon fam_0390_epsilon	origin stage 181 notes/stages/moving_throat_pde_stage181_coherent_tracking_defect.md:29	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage242_actual_twin_support_placement_and_coherent_orbit_lock_compiler_sympy_audit.md:117	244, 245

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
242	Lambda_0 fit_stage_242_lambda_0 fam_0149_lambda_0	origin stage 188 notes/stages/moving_throat_pde_stage188_branch_observables_completion.md:144	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage242_actual_twin_support_placement_and_coherent_orbit_lock_compiler_sympy_audit.md:295	245
243	E_w fit_stage_243_e_w fam_0075_e_w	origin stage 243 notes/stages/moving_throat_pde_stage243_relaxed_constraint_branch_declaration_and_short_range_open_system_compiler_sympy_audit.md:220	free_choice	posited choice:notes/stages/moving_throat_pde_stage243_relaxed_constraint_branch_declaration_and_short_range_open_system_compiler_sympy_audit.md:217	244
243	K_turn fit_stage_243_k_turn fam_0140_k_turn; overlays:chain_calibration_245_253	origin stage 243 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
243	Xi_1 fit_stage_243_xi_1 fam_0283_xi_1	origin stage 243 notes/stages/moving_throat_pde_stage243_relaxed_constraint_branch_declaration_and_short_range_open_system_compiler_sympy_audit.md:344	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage243_relaxed_constraint_branch_declaration_and_short_range_open_system_compiler_sympy_audit.md:341	245
244	Pi_tr fit_stage244_leakage_forced_through_fixed_demand fam_0210_pi_tr	origin stage 240 notes/stages/moving_throat_pde_stage240_selected_branch_loading_ratio_from_the_minimal_isotropic_quadrupole_precursor_sympy_audit.md:285	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage244_selected_branch_leakage_and_scalar_photon_work_compiler_sympy_audit.md:282	none recorded
245	epsilon_eta fit_stage245_eps_eta_session1_match fam_0393_epsilon_eta; overlays:chain_calibration_245_253	origin stage 245 notes/stages/moving_throat_pde_stage245_nonrigid_mouth_dressing_packet_and_uv_drain_compiler_sympy_audit.md:510	free_choice	posited choice:notes/stages/moving_throat_pde_stage245_nonrigid_mouth_dressing_packet_and_uv_drain_compiler_sympy_audit.md:497	none recorded
245	U_obs fit_stage245_session1_readback fam_0266_u_obs	origin stage 245 notes/stages/moving_throat_pde_stage245_nonrigid_mouth_dressing_packet_and_uv_drain_compiler_sympy_audit.md:500	free_choice	posited choice:notes/stages/moving_throat_pde_stage245_nonrigid_mouth_dressing_packet_and_uv_drain_compiler_sympy_audit.md:497	none recorded

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
245	U_obs fit_stage245_session1_readback_inputs fam_0266_u_obs, fam_0275_v_obs, fam_0304_a_u, fam_0305_a_v, fam_0386_eps_eta_ref, fam_0406_g_uv	origin stage 245 notes/stages/moving_throat_pde_stage245_nonrigid_mouth_dressing_packet_and_uv_drain_compiler_sympy_audit.md:500	free_choice	posited choice:notes/stages/moving_throat_pde_stage245_nonrigid_mouth_dressing_packet_and_uv_drain_compiler_sympy_audit.md:497	none recorded
245	V_obs fit_stage245_session1_readback_inputs fam_0266_u_obs, fam_0275_v_obs, fam_0304_a_u, fam_0305_a_v, fam_0386_eps_eta_ref, fam_0406_g_uv	origin stage 245 notes/stages/moving_throat_pde_stage245_nonrigid_mouth_dressing_packet_and_uv_drain_compiler_sympy_audit.md:502	free_choice	posited choice:notes/stages/moving_throat_pde_stage245_nonrigid_mouth_dressing_packet_and_uv_drain_compiler_sympy_audit.md:497	none recorded
245	a_U fit_stage245_session1_readback_inputs fam_0266_u_obs, fam_0275_v_obs, fam_0304_a_u, fam_0305_a_v, fam_0386_eps_eta_ref, fam_0406_g_uv	origin stage 245 notes/stages/moving_throat_pde_stage245_nonrigid_mouth_dressing_packet_and_uv_drain_compiler_sympy_audit.md:155	free_choice	posited choice:notes/stages/moving_throat_pde_stage245_nonrigid_mouth_dressing_packet_and_uv_drain_compiler_sympy_audit.md:155	none recorded
245	a_V fit_stage245_session1_readback_inputs fam_0266_u_obs, fam_0275_v_obs, fam_0304_a_u, fam_0305_a_v, fam_0386_eps_eta_ref, fam_0406_g_uv	origin stage 245 notes/stages/moving_throat_pde_stage245_nonrigid_mouth_dressing_packet_and_uv_drain_compiler_sympy_audit.md:156	free_choice	posited choice:notes/stages/moving_throat_pde_stage245_nonrigid_mouth_dressing_packet_and_uv_drain_compiler_sympy_audit.md:156	none recorded
245	eps_eta_ref fit_stage245_session1_readback_inputs fam_0266_u_obs, fam_0275_v_obs, fam_0304_a_u, fam_0305_a_v, fam_0386_eps_eta_ref, fam_0406_g_uv	origin stage 245 notes/stages/moving_throat_pde_stage245_nonrigid_mouth_dressing_packet_and_uv_drain_compiler_sympy_audit.md:504	free_choice	posited choice:notes/stages/moving_throat_pde_stage245_nonrigid_mouth_dressing_packet_and_uv_drain_compiler_sympy_audit.md:497	none recorded
245	g_UV fit_stage245_session1_readback_inputs fam_0266_u_obs, fam_0275_v_obs, fam_0304_a_u, fam_0305_a_v, fam_0386_eps_eta_ref, fam_0406_g_uv	origin stage 245 notes/stages/moving_throat_pde_stage245_nonrigid_mouth_dressing_packet_and_uv_drain_compiler_sympy_audit.md:134	free_choice	posited choice:notes/stages/moving_throat_pde_stage245_nonrigid_mouth_dressing_packet_and_uv_drain_compiler_sympy_audit.md:157	none recorded
245	Xi_1 fit_stage_245_xi_1 fam_0283_xi_1	origin stage 245 notes/stages/moving_throat_pde_stage245_nonrigid_mouth_dressing_packet_and_uv_drain_compiler_sympy_audit.md:409	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage245_nonrigid_mouth_dressing_packet_and_uv_drain_compiler_sympy_audit.md:389	none recorded

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
246	sigma_min fit_stage246_lower_branch_forced fam_0511_sigma_min	origin stage 246 notes/stages/moving_throat_pde_stage246_compensated_multimode_source_compiler_beyond_positive_family1_sympy_audit.md:202	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage246_compensated_multimode_source_compiler_beyond_positive_family1_sympy_audit.md:199	none recorded
246	r_F1 fit_stage246_rF1_coefficient fam_0493_r_f1; overlays:chain_aspect_37_20	origin stage 121 notes/stages/moving_throat_pde_stage121_geometric_r_selection.md:65	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage121_geometric_r_selection.md:37	none recorded
246	a_0 fit_stage246_session1_inputs fam_0303_a_0, fam_0318_b_0, fam_0496_r_eval, fam_0497_r_sigma	origin stage 246 notes/stages/moving_throat_pde_stage246_compensated_multimode_source_compiler_beyond_positive_family1_sympy_audit.md:474	free_choice	posited choice:notes/stages/moving_throat_pde_stage246_compensated_multimode_source_compiler_beyond_positive_family1_sympy_audit.md:472	none recorded
246	b_0 fit_stage246_session1_inputs fam_0303_a_0, fam_0318_b_0, fam_0496_r_eval, fam_0497_r_sigma	origin stage 246 notes/stages/moving_throat_pde_stage246_compensated_multimode_source_compiler_beyond_positive_family1_sympy_audit.md:476	free_choice	posited choice:notes/stages/moving_throat_pde_stage246_compensated_multimode_source_compiler_beyond_positive_family1_sympy_audit.md:472	none recorded
246	r_eval fit_stage246_session1_inputs fam_0303_a_0, fam_0318_b_0, fam_0496_r_eval, fam_0497_r_sigma	origin stage 246 notes/stages/moving_throat_pde_stage246_compensated_multimode_source_compiler_beyond_positive_family1_sympy_audit.md:480	free_choice	posited choice:notes/stages/moving_throat_pde_stage246_compensated_multimode_source_compiler_beyond_positive_family1_sympy_audit.md:472	none recorded
246	r_sigma fit_stage246_session1_inputs fam_0303_a_0, fam_0318_b_0, fam_0496_r_eval, fam_0497_r_sigma	origin stage 246 notes/stages/moving_throat_pde_stage246_compensated_multimode_source_compiler_beyond_positive_family1_sympy_audit.md:478	free_choice	posited choice:notes/stages/moving_throat_pde_stage246_compensated_multimode_source_compiler_beyond_positive_family1_sympy_audit.md:472	none recorded
246	a0 fit_stage246_session1_readback_point fam_0302_a0	origin stage 246 notes/stages/moving_throat_pde_stage246_compensated_multimode_source_compiler_beyond_positive_family1_sympy_audit.md:474	free_choice	posited choice:notes/stages/moving_throat_pde_stage246_compensated_multimode_source_compiler_beyond_positive_family1_sympy_audit.md:472	none recorded

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
246	M_src fit_stage246_two_moment_source_matrix fam_0167_m_src	origin stage 246 notes/stages/moving_throat_pde_stage246_compensated_multimode_source_compiler_beyond_positive_family1_sympy_audit.md:282	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage246_compensated_multimode_source_compiler_beyond_positive_family1_sympy_audit.md:294	none recorded
247	lambda_L fit_stage247_benchmark_pinned_to_paper_figures fam_0446_lambda_l; overlays:chain_calibration_245_253	origin stage 247 notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:518	published_target	published_target; no benchmarks.yaml entry joined for listed family; evidence:notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:497	248
247	lambda_L fit_stage247_lambda_L_backsolve fam_0446_lambda_l; overlays:chain_calibration_245_253	origin stage 247 notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:493	published_target	published_target; no benchmarks.yaml entry joined for listed family; evidence:notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:497	248
247	lambda_L fit_stage247_lambda_L_closure fam_0446_lambda_l; overlays:chain_calibration_245_253	origin stage 247 notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:518	published_target	published_target; no benchmarks.yaml entry joined for listed family; evidence:notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:569	248

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
247	lambda_L fit_stage247_lambda_l_fixed_by_recorded_point fam_0446_lambda_l; overlays:chain_calibration_245_253	origin stage 247 notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:22	published_target	published_target; no benchmarks.yaml entry joined for listed family; evidence:notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:497	248
247	lambda_W fit_stage247_lambda_w_natural_specialization fam_0449_lambda_w	origin stage 247 notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:491	free_choice	posited choice:notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:489	248
247	G_W fit_stage247_session1_packet fam_0087_g_w, fam_0225_rmix, fam_0265_uvdrop_obs, fam_0276_veff_obs, fam_0281_wsess_obs, fam_0326_beta_q, fam_0327_beta_u, fam_0328_beta_w, fam_0396_eta_leak, fam_0479_mu_w, fam_0547_xi_r	origin stage 247 notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:391	free_choice	posited choice:notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:381	248
247	Rmix fit_stage247_session1_packet fam_0087_g_w, fam_0225_rmix, fam_0265_uvdrop_obs, fam_0276_veff_obs, fam_0281_wsess_obs, fam_0326_beta_q, fam_0327_beta_u, fam_0328_beta_w, fam_0396_eta_leak, fam_0479_mu_w, fam_0547_xi_r	origin stage 247 notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:393	free_choice	posited choice:notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:381	248

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
247	UVdrop_obs fit_stage247_session1_packet fam_0087_g_w, fam_0225_rmix, fam_0265_uvdrop_obs, fam_0276_veff_obs, fam_0281_wsess_obs, fam_0326_beta_q, fam_0327_beta_u, fam_0328_beta_w, fam_0396_eta_leak, fam_0479_mu_w, fam_0547_xi_r	origin stage 247 notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:431	published_target	published_target; no benchmarks.yaml entry joined for listed family; evidence:notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:427	248
247	Veff_obs fit_stage247_session1_packet fam_0087_g_w, fam_0225_rmix, fam_0265_uvdrop_obs, fam_0276_veff_obs, fam_0281_wsess_obs, fam_0326_beta_q, fam_0327_beta_u, fam_0328_beta_w, fam_0396_eta_leak, fam_0479_mu_w, fam_0547_xi_r	origin stage 247 notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:372	published_target	published_target; no benchmarks.yaml entry joined for listed family; evidence:notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:367	248
247	Wsess_obs fit_stage247_session1_packet fam_0087_g_w, fam_0225_rmix, fam_0265_uvdrop_obs, fam_0276_veff_obs, fam_0281_wsess_obs, fam_0326_beta_q, fam_0327_beta_u, fam_0328_beta_w, fam_0396_eta_leak, fam_0479_mu_w, fam_0547_xi_r	origin stage 247 notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:429	published_target	published_target; no benchmarks.yaml entry joined for listed family; evidence:notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:427	248
247	beta_Q fit_stage247_session1_packet fam_0087_g_w, fam_0225_rmix, fam_0265_uvdrop_obs, fam_0276_veff_obs, fam_0281_wsess_obs, fam_0326_beta_q, fam_0327_beta_u, fam_0328_beta_w, fam_0396_eta_leak, fam_0479_mu_w, fam_0547_xi_r	origin stage 247 notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:396	free_choice	posited choice:notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:381	248

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
247	beta_U fit_stage247_session1_packet fam_0087_g_w, fam_0225_rmix, fam_0265_uvdrop_obs, fam_0276_veff_obs, fam_0281_wsess_obs, fam_0326_beta_q, fam_0327_beta_u, fam_0328_beta_w, fam_0396_eta_leak, fam_0479_mu_w, fam_0547_xi_r	origin stage 247 notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:398	free_choice	posited choice:notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:381	248
247	beta_W fit_stage247_session1_packet fam_0087_g_w, fam_0225_rmix, fam_0265_uvdrop_obs, fam_0276_veff_obs, fam_0281_wsess_obs, fam_0326_beta_q, fam_0327_beta_u, fam_0328_beta_w, fam_0396_eta_leak, fam_0479_mu_w, fam_0547_xi_r	origin stage 247 notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:400	free_choice	posited choice:notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:381	248
247	eta_leak fit_stage247_session1_packet fam_0087_g_w, fam_0225_rmix, fam_0265_uvdrop_obs, fam_0276_veff_obs, fam_0281_wsess_obs, fam_0326_beta_q, fam_0327_beta_u, fam_0328_beta_w, fam_0396_eta_leak, fam_0479_mu_w, fam_0547_xi_r	origin stage 247 notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:445	free_choice	posited choice:notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:441	248
247	mu_w fit_stage247_session1_packet fam_0087_g_w, fam_0225_rmix, fam_0265_uvdrop_obs, fam_0276_veff_obs, fam_0281_wsess_obs, fam_0326_beta_q, fam_0327_beta_u, fam_0328_beta_w, fam_0396_eta_leak, fam_0479_mu_w, fam_0547_xi_r	origin stage 247 notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:447	free_choice	posited choice:notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:441	248

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
247	xi_R fit_stage247_session1_packet fam_0087_g_w, fam_0225_rmix, fam_0265_uvdrop_obs, fam_0276_veff_obs, fam_0281_wsess_obs, fam_0326_beta_q, fam_0327_beta_u, fam_0328_beta_w, fam_0396_eta_leak, fam_0479_mu_w, fam_0547_xi_r	origin stage 247 notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:476	free_choice	posited choice:notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:467	248
247	a0 fit_stage247_session1_softening_point fam_0302_a0	origin stage 246 notes/stages/moving_throat_pde_stage246_compensated_multimode_source_compiler_beyond_positive_family1_sympy_audit.md:474	free_choice	posited choice:notes/stages/moving_throat_pde_stage246_compensated_multimode_source_compiler_beyond_positive_family1_sympy_audit.md:472	247, 248
247	A_2 fit_stage_247_a_2 fam_0004_a_2	origin stage 247 notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:154	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:154	248
247	alpha_2 fit_stage_247_alpha_2 fam_0311_alpha_2	origin stage 247 notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:404	free_choice	posited choice:notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:404	248
247	D_UV fit_stage_247_d_uv fam_0046_d_uv	origin stage 245 notes/stages/moving_throat_pde_stage245_nonrigid_mouth_dressing_packet_and_uv_drain_compiler_sympy_audit.md:350	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:217	248

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
247	Delta fit_stage_247_delta fam_0051_delta	origin stage 247 notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:95	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:95	248
247	lambda_L fit_stage_247_lambda_l_soft fam_0446_lambda_l; overlays:chain_calibration_245_253	origin stage 247 notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:518	published_target	published_target; no benchmarks.yaml entry joined for listed family; evidence:notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:497	248
247	Lvar_from_W fit_stage_247_lvar_from_w fam_0155_lvar_from_w	origin stage 247 notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:204	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:454	248
247	M_sigma fit_stage_247_m_sigma fam_0166_m_sigma	origin stage 247 notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:247	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:247	248

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
247	V_eff fit_stage_247_v_eff fam_0272_v_eff	origin stage 247 notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:368	published_target	published_target; no benchmarks.yaml entry joined for listed family; evidence:notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:367	248
247	Vrebuild_soft fit_stage_247_vrebuild_soft fam_0277_vrebuild_soft	origin stage 247 notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:522	published_target	published_target; no benchmarks.yaml entry joined for listed family; evidence:notes/stages/moving_throat_pde_stage247_relaxed_stationary_barrier_compiler_from_one_port_short_range_leakage_uv_and_compensated_source_packets_sympy_audit.md:620	248
248	r_contact fit_stage248_chosen_contact_scale_benchmark fam_0495_r_contact	origin stage 248 notes/stages/moving_throat_pde_stage248_dynamic_event_chain_compiler_from_relaxed_stationary_barrier_front_end_turning_point_threshold_speed_and_wkb_sympy_audit.md:415	free_choice	posited choice:notes/stages/moving_throat_pde_stage248_dynamic_event_chain_compiler_from_relaxed_stationary_barrier_front_end_turning_point_threshold_speed_and_wkb_sympy_audit.md:165	none recorded
248	E_sub fit_stage248_session2_launch_parameters fam_0074_e_sub	origin stage 248 notes/stages/moving_throat_pde_stage248_dynamic_event_chain_compiler_from_relaxed_stationary_barrier_front_end_turning_point_threshold_speed_and_wkb_sympy_audit.md:417	free_choice	posited choice:notes/stages/moving_throat_pde_stage248_dynamic_event_chain_compiler_from_relaxed_stationary_barrier_front_end_turning_point_threshold_speed_and_wkb_sympy_audit.md:234	none recorded

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
248	E_sub fit_stage248_session2_readback fam_0074_e_sub	origin stage 248 notes/stages/moving_throat_pde_stage248_dynamic_event_chain_compiler_from_relaxed_stationary_barrier_front_end_turning_point_threshold_speed_and_wkb_sympy_audit.md:417	free_choice	posited choice:notes/stages/moving_throat_pde_stage248_dynamic_event_chain_compiler_from_relaxed_stationary_barrier_front_end_turning_point_threshold_speed_and_wkb_sympy_audit.md:470	none recorded
248	v0_cross fit_stage248_v0_cross_benchmark_speed fam_0533_v0_cross	origin stage 248 notes/stages/moving_throat_pde_stage248_dynamic_event_chain_compiler_from_relaxed_stationary_barrier_front_end_turning_point_threshold_speed_and_wkb_sympy_audit.md:515	published_target	published_target; no benchmarks.yaml entry joined for listed family; evidence:notes/stages/moving_throat_pde_stage248_dynamic_event_chain_compiler_from_relaxed_stationary_barrier_front_end_turning_point_threshold_speed_and_wkb_sympy_audit.md:513	none recorded
248	Xi_turn fit_stage248_xi_turn_lambda_th_carried_hardcodes fam_0290_xi_turn	origin stage 248 notes/stages/moving_throat_pde_stage248_dynamic_event_chain_compiler_from_relaxed_stationary_barrier_front_end_turning_point_threshold_speed_and_wkb_sympy_audit.md:436	published_target	published_target; no benchmarks.yaml entry joined for listed family; evidence:notes/stages/moving_throat_pde_stage248_dynamic_event_chain_compiler_from_relaxed_stationary_barrier_front_end_turning_point_threshold_speed_and_wkb_sympy_audit.md:419	none recorded
248	F_coul fit_stage_248_f_coul fam_0077_f_coul	origin stage 248 notes/stages/moving_throat_pde_stage248_dynamic_event_chain_compiler_from_relaxed_stationary_barrier_front_end_turning_point_threshold_speed_and_wkb_sympy_audit.md:290	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage248_dynamic_event_chain_compiler_from_relaxed_stationary_barrier_front_end_turning_point_threshold_speed_and_wkb_sympy_audit.md:297	none recorded

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
248	I_Coul fit_stage_248_i_coul fam_0106_i_coul	origin stage 248 notes/stages/moving_throat_pde_stage248_dynamic_event_chain_compiler_from_relaxed_stationary_barrier_front_end_turning_point_threshold_speed_and_wkb_sympy_audit.md:290	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage248_dynamic_event_chain_compiler_from_relaxed_stationary_barrier_front_end_turning_point_threshold_speed_and_wkb_sympy_audit.md:297	none recorded
249	H_sub_minus_final fit_stage249_helicity_ratio_matched fam_0101_h_sub_minus_final	origin stage 249 notes/stages/moving_throat_pde_stage249_conditional_helicity_export_diagnostic_compiler_on_the_dynamic_event_chain_and_aligned_vs_anti_aligned_mixed_sector_closure_sympy_audit.md:431	free_choice	posited choice:notes/stages/moving_throat_pde_stage249_conditional_helicity_export_diagnostic_compiler_on_the_dynamic_event_chain_and_aligned_vs_anti_aligned_mixed_sector_closure_sympy_audit.md:416	250
249	H_int_aligned fit_stage249_session2_helicity_readbacks fam_0099_h_int_aligned	origin stage 249 notes/stages/moving_throat_pde_stage249_conditional_helicity_export_diagnostic_compiler_on_the_dynamic_event_chain_and_aligned_vs_anti_aligned_mixed_sector_closure_sympy_audit.md:428	free_choice	posited choice:notes/stages/moving_throat_pde_stage249_conditional_helicity_export_diagnostic_compiler_on_the_dynamic_event_chain_and_aligned_vs_anti_aligned_mixed_sector_closure_sympy_audit.md:416	250
249	hint_aligned fit_stage249_session3_peak_readback fam_0427_hint_aligned	origin stage 249 notes/stages/moving_throat_pde_stage249_conditional_helicity_export_diagnostic_compiler_on_the_dynamic_event_chain_and_aligned_vs_anti_aligned_mixed_sector_closure_sympy_audit.md:419	free_choice	posited choice:notes/stages/moving_throat_pde_stage249_conditional_helicity_export_diagnostic_compiler_on_the_dynamic_event_chain_and_aligned_vs_anti_aligned_mixed_sector_closure_sympy_audit.md:416	250

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
249	alpha_rm fit_stage_249_alpha_rm fam_0314_alpha_rm	origin stage 249 notes/stages/moving_throat_pde_stage249_conditional_helicity_export_diagnostic_compiler_on_the_dynamic_event_chain_and_aligned_vs_anti_aligned_mixed_sector_closure_sympy_audit.md:229	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage249_conditional_helicity_export_diagnostic_compiler_on_the_dynamic_event_chain_and_aligned_vs_anti_aligned_mixed_sector_closure_sympy_audit.md:372	250
249	C_sigma fit_stage_249_c_sigma fam_0033_c_sigma	origin stage 249 notes/stages/moving_throat_pde_stage249_conditional_helicity_export_diagnostic_compiler_on_the_dynamic_event_chain_and_aligned_vs_anti_aligned_mixed_sector_closure_sympy_audit.md:204	free_choice	posited choice:notes/stages/moving_throat_pde_stage249_conditional_helicity_export_diagnostic_compiler_on_the_dynamic_event_chain_and_aligned_vs_anti_aligned_mixed_sector_closure_sympy_audit.md:198	250
249	Gamma_0 fit_stage_249_gamma_0 fam_0091_gamma_0	origin stage 249 notes/stages/moving_throat_pde_stage249_conditional_helicity_export_diagnostic_compiler_on_the_dynamic_event_chain_and_aligned_vs_anti_aligned_mixed_sector_closure_sympy_audit.md:220	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage249_conditional_helicity_export_diagnostic_compiler_on_the_dynamic_event_chain_and_aligned_vs_anti_aligned_mixed_sector_closure_sympy_audit.md:220	250
249	R_int fit_stage_249_r_int fam_0218_r_int	origin stage 249 notes/stages/moving_throat_pde_stage249_conditional_helicity_export_diagnostic_compiler_on_the_dynamic_event_chain_and_aligned_vs_anti_aligned_mixed_sector_closure_sympy_audit.md:463	free_choice	posited choice:notes/stages/moving_throat_pde_stage249_conditional_helicity_export_diagnostic_compiler_on_the_dynamic_event_chain_and_aligned_vs_anti_aligned_mixed_sector_closure_sympy_audit.md:460	250

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
250	E_max fit_stage250_benchmark_masses_and_window fam_0072_e_max, fam_0347_chi_peak, fam_0348_chi_raw, fam_0406_g_uv, fam_0457_m_s, fam_0478_mu_eta, fam_0523_t_transit_max, fam_0524_t_transit_min; overlays:chain_calibration_245_253, chain_wall_action	origin stage 250 notes/stages/moving_throat_pde_stage250_crossing_vs_collapse_goldilocks_window_compiler_from_the_stage248_event_chain_and_relaxed_wall_timescale_closure_sympy_audit.md:486	free_choice	posited choice:notes/stages/moving_throat_pde_stage250_crossing_vs_collapse_goldilocks_window_compiler_from_the_stage248_event_chain_and_relaxed_wall_timescale_closure_sympy_audit.md:483	none recorded
250	chi_peak fit_stage250_benchmark_masses_and_window fam_0072_e_max, fam_0347_chi_peak, fam_0348_chi_raw, fam_0406_g_uv, fam_0457_m_s, fam_0478_mu_eta, fam_0523_t_transit_max, fam_0524_t_transit_min; overlays:chain_calibration_245_253, chain_wall_action	origin stage 250 notes/stages/moving_throat_pde_stage250_crossing_vs_collapse_goldilocks_window_compiler_from_the_stage248_event_chain_and_relaxed_wall_timescale_closure_sympy_audit.md:417	free_choice	posited choice:notes/stages/moving_throat_pde_stage250_crossing_vs_collapse_goldilocks_window_compiler_from_the_stage248_event_chain_and_relaxed_wall_timescale_closure_sympy_audit.md:388	251
250	chi_raw fit_stage250_benchmark_masses_and_window fam_0072_e_max, fam_0347_chi_peak, fam_0348_chi_raw, fam_0406_g_uv, fam_0457_m_s, fam_0478_mu_eta, fam_0523_t_transit_max, fam_0524_t_transit_min; overlays:chain_calibration_245_253, chain_wall_action	origin stage 250 notes/stages/moving_throat_pde_stage250_crossing_vs_collapse_goldilocks_window_compiler_from_the_stage248_event_chain_and_relaxed_wall_timescale_closure_sympy_audit.md:526	free_choice	posited choice:notes/stages/moving_throat_pde_stage250_crossing_vs_collapse_goldilocks_window_compiler_from_the_stage248_event_chain_and_relaxed_wall_timescale_closure_sympy_audit.md:518	none recorded
250	g_UV fit_stage250_benchmark_masses_and_window fam_0072_e_max, fam_0347_chi_peak, fam_0348_chi_raw, fam_0406_g_uv, fam_0457_m_s, fam_0478_mu_eta, fam_0523_t_transit_max, fam_0524_t_transit_min; overlays:chain_calibration_245_253, chain_wall_action	origin stage 250 notes/stages/moving_throat_pde_stage250_crossing_vs_collapse_goldilocks_window_compiler_from_the_stage248_event_chain_and_relaxed_wall_timescale_closure_sympy_audit.md:415	free_choice	posited choice:notes/stages/moving_throat_pde_stage250_crossing_vs_collapse_goldilocks_window_compiler_from_the_stage248_event_chain_and_relaxed_wall_timescale_closure_sympy_audit.md:172	251, 253

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
250	m_s fit_stage250_benchmark_masses_and_window fam_0072_e_max, fam_0347_chi_peak, fam_0348_chi_raw, fam_0406_g_uv, fam_0457_m_s, fam_0478_mu_eta, fam_0523_t_transit_max, fam_0524_t_transit_min; overlays:chain_calibration_245_253, chain_wall_action	origin stage 250 notes/stages/moving_throat_pde_stage250_crossing_vs_collapse_goldilocks_window_compiler_from_the_stage248_event_chain_and_relaxed_wall_timescale_closure_sympy_audit.md:419	published_target	benchmark: bench__051666ee239b: CODATA - NIST CODATA, proton-electron mass ratio (m__p/m__e), https://physics.nist.gov/cgi-bin/cuu/Value?mpsme ; CODATA Internationally recommended values of the fundamental physical constants (2018 and 2022 adjustments).	251, 253
250	mu_eta fit_stage250_benchmark_masses_and_window fam_0072_e_max, fam_0347_chi_peak, fam_0348_chi_raw, fam_0406_g_uv, fam_0457_m_s, fam_0478_mu_eta, fam_0523_t_transit_max, fam_0524_t_transit_min; overlays:chain_calibration_245_253, chain_wall_action	origin stage 250 notes/stages/moving_throat_pde_stage250_crossing_vs_collapse_goldilocks_window_compiler_from_the_stage248_event_chain_and_relaxed_wall_timescale_closure_sympy_audit.md:419	published_target	benchmark: bench__051666ee239b: CODATA - NIST CODATA, proton-electron mass ratio (m__p/m__e), https://physics.nist.gov/cgi-bin/cuu/Value?mpsme ; CODATA Internationally recommended values of the fundamental physical constants (2018 and 2022 adjustments).	251, 253
250	t_transit_max fit_stage250_benchmark_masses_and_window fam_0072_e_max, fam_0347_chi_peak, fam_0348_chi_raw, fam_0406_g_uv, fam_0457_m_s, fam_0478_mu_eta, fam_0523_t_transit_max, fam_0524_t_transit_min; overlays:chain_calibration_245_253, chain_wall_action	origin stage 250 notes/stages/moving_throat_pde_stage250_crossing_vs_collapse_goldilocks_window_compiler_from_the_stage248_event_chain_and_relaxed_wall_timescale_closure_sympy_audit.md:506	free_choice	posited choice:notes/stages/moving_throat_pde_stage250_crossing_vs_collapse_goldilocks_window_compiler_from_the_stage248_event_chain_and_relaxed_wall_timescale_closure_sympy_audit.md:502	none recorded
250	t_transit_min fit_stage250_benchmark_masses_and_window fam_0072_e_max, fam_0347_chi_peak, fam_0348_chi_raw, fam_0406_g_uv, fam_0457_m_s, fam_0478_mu_eta, fam_0523_t_transit_max, fam_0524_t_transit_min; overlays:chain_calibration_245_253, chain_wall_action	origin stage 250 notes/stages/moving_throat_pde_stage250_crossing_vs_collapse_goldilocks_window_compiler_from_the_stage248_event_chain_and_relaxed_wall_timescale_closure_sympy_audit.md:504	free_choice	posited choice:notes/stages/moving_throat_pde_stage250_crossing_vs_collapse_goldilocks_window_compiler_from_the_stage248_event_chain_and_relaxed_wall_timescale_closure_sympy_audit.md:502	none recorded

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
250	E_edge fit_stage250_goldilocks_edge_forced fam_0069_e_edge	origin stage 250 notes/stages/moving_throat_pde_stage250_crossing_vs_collapse_goldilocks_window_compiler_from_the_stage248_event_chain_and_relaxed_wall_timescale_closure_sympy_audit.md:229	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage250_crossing_vs_collapse_goldilocks_window_compiler_from_the_stage248_event_chain_and_relaxed_wall_timescale_closure_sympy_audit.md:224	251
250	chi_peak fit_stage250_goldilocks_inputs fam_0347_chi_peak, fam_0348_chi_raw, fam_0406_g_uv, fam_0458_m_s_ratio_1836	origin stage 250 notes/stages/moving_throat_pde_stage250_crossing_vs_collapse_goldilocks_window_compiler_from_the_stage248_event_chain_and_relaxed_wall_timescale_closure_sympy_audit.md:417	free_choice	posited choice:notes/stages/moving_throat_pde_stage250_crossing_vs_collapse_goldilocks_window_compiler_from_the_stage248_event_chain_and_relaxed_wall_timescale_closure_sympy_audit.md:388	251
250	chi_raw fit_stage250_goldilocks_inputs fam_0347_chi_peak, fam_0348_chi_raw, fam_0406_g_uv, fam_0458_m_s_ratio_1836	origin stage 250 notes/stages/moving_throat_pde_stage250_crossing_vs_collapse_goldilocks_window_compiler_from_the_stage248_event_chain_and_relaxed_wall_timescale_closure_sympy_audit.md:526	free_choice	posited choice:notes/stages/moving_throat_pde_stage250_crossing_vs_collapse_goldilocks_window_compiler_from_the_stage248_event_chain_and_relaxed_wall_timescale_closure_sympy_audit.md:518	none recorded
250	g_UV fit_stage250_goldilocks_inputs fam_0347_chi_peak, fam_0348_chi_raw, fam_0406_g_uv, fam_0458_m_s_ratio_1836	origin stage 250 notes/stages/moving_throat_pde_stage250_crossing_vs_collapse_goldilocks_window_compiler_from_the_stage248_event_chain_and_relaxed_wall_timescale_closure_sympy_audit.md:415	free_choice	posited choice:notes/stages/moving_throat_pde_stage250_crossing_vs_collapse_goldilocks_window_compiler_from_the_stage248_event_chain_and_relaxed_wall_timescale_closure_sympy_audit.md:172	251, 253

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
250	m_s_ratio_1836 fit_stage250_goldilocks_inputs fam_0347_chi_peak, fam_0348_chi_raw, fam_0406_g_uv, fam_0458_m_s_ratio_1836	origin stage 250 notes/stages/moving_throat_pde_stage250_crossing_vs_collapse_goldilocks_window_compiler_from_the_stage248_event_chain_and_relaxed_wall_timescale_closure_sympy_audit.md:419	published_target	benchmark: bench__5d8d048b10f9: CODATA - NIST CODATA, proton-electron mass ratio (m__p/m__e), https://physics.nist.gov/cgi-bin/cuu/Value?mpsme ; CODATA Internationally recommended values of the fundamental physical constants (2018 and 2022 adjustments).	251, 253
250	chi_rm fit_stage_250_chi_rm fam_0349_chi_rm	origin stage 250 notes/stages/moving_throat_pde_stage250_crossing_vs_collapse_goldilocks_window_compiler_from_the_stage248_event_chain_and_relaxed_wall_timescale_closure_sympy_audit.md:417	free_choice	posited choice:notes/stages/moving_throat_pde_stage250_crossing_vs_collapse_goldilocks_window_compiler_from_the_stage248_event_chain_and_relaxed_wall_timescale_closure_sympy_audit.md:388	251
250	E_edge fit_stage_250_e_edge fam_0069_e_edge	origin stage 250 notes/stages/moving_throat_pde_stage250_crossing_vs_collapse_goldilocks_window_compiler_from_the_stage248_event_chain_and_relaxed_wall_timescale_closure_sympy_audit.md:229	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage250_crossing_vs_collapse_goldilocks_window_compiler_from_the_stage248_event_chain_and_relaxed_wall_timescale_closure_sympy_audit.md:224	251
250	t_cross fit_stage_250_t_cross fam_0522_t_cross	origin stage 250 notes/stages/moving_throat_pde_stage250_crossing_vs_collapse_goldilocks_window_compiler_from_the_stage248_event_chain_and_relaxed_wall_timescale_closure_sympy_audit.md:140	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage250_crossing_vs_collapse_goldilocks_window_compiler_from_the_stage248_event_chain_and_relaxed_wall_timescale_closure_sympy_audit.md:144	251

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
251	Gamma_3 fit_stage251_gamma3_exact_cubic_coefficient fam_0092_gamma_3	origin stage 251 notes/stages/moving_throat_pde_stage251_microscopic_damping_export_kernel_replacing_the_phenomenological_v_leg_envelope_law_sympy_audit.md:164	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage251_microscopic_damping_export_kernel_replacing_the_phenomenological_v_leg_envelope_law_sympy_audit.md:158	252
251	Gamma_5 fit_stage251_gamma5_exact_quintic_coefficient fam_0093_gamma_5; overlays:chain_quad_54_5	origin stage 251 notes/stages/moving_throat_pde_stage251_microscopic_damping_export_kernel_replacing_the_phenomenological_v_leg_envelope_law_sympy_audit.md:189	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage251_microscopic_damping_export_kernel_replacing_the_phenomenological_v_leg_envelope_law_sympy_audit.md:181	252
251	gamma_safe fit_stage251_gamma_safe_exact_replacement fam_0421_gamma_safe	origin stage 251 notes/stages/moving_throat_pde_stage251_microscopic_damping_export_kernel_replacing_the_phenomenological_v_leg_envelope_law_sympy_audit.md:393	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage251_microscopic_damping_export_kernel_replacing_the_phenomenological_v_leg_envelope_law_sympy_audit.md:385	252
251	gamma_crit fit_stage251_session4_rate_constants fam_0418_gamma_crit	origin stage 251 notes/stages/moving_throat_pde_stage251_microscopic_damping_export_kernel_replacing_the_phenomenological_v_leg_envelope_law_sympy_audit.md:455	free_choice	posited choice:notes/stages/moving_throat_pde_stage251_microscopic_damping_export_kernel_replacing_the_phenomenological_v_leg_envelope_law_sympy_audit.md:562	none recorded
251	gamma_crit fit_stage251_session4_time_inputs fam_0418_gamma_crit, fam_0503_s_c, fam_0521_t_collapse_0, fam_0522_t_cross	origin stage 251 notes/stages/moving_throat_pde_stage251_microscopic_damping_export_kernel_replacing_the_phenomenological_v_leg_envelope_law_sympy_audit.md:455	free_choice	posited choice:notes/stages/moving_throat_pde_stage251_microscopic_damping_export_kernel_replacing_the_phenomenological_v_leg_envelope_law_sympy_audit.md:562	none recorded

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
251	s_c fit_stage251_session4_time_inputs fam_0418_gamma_crit, fam_0503_s_c, fam_0521_t_collapse_0, fam_0522_t_cross	origin stage 251 notes/stages/moving_throat_pde_stage251_microscopic_damping_export_kernel_replacing_the_phenomenological_v_leg_envelope_law_sympy_audit.md:382	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage251_microscopic_damping_export_kernel_replacing_the_phenomenological_v_leg_envelope_law_sympy_audit.md:382	252
251	t_collapse_0 fit_stage251_session4_time_inputs fam_0418_gamma_crit, fam_0503_s_c, fam_0521_t_collapse_0, fam_0522_t_cross	origin stage 251 notes/stages/moving_throat_pde_stage251_microscopic_damping_export_kernel_replacing_the_phenomenological_v_leg_envelope_law_sympy_audit.md:453	free_choice	posited choice:notes/stages/moving_throat_pde_stage251_microscopic_damping_export_kernel_replacing_the_phenomenological_v_leg_envelope_law_sympy_audit.md:449	none recorded
251	t_cross fit_stage251_session4_time_inputs fam_0418_gamma_crit, fam_0503_s_c, fam_0521_t_collapse_0, fam_0522_t_cross	origin stage 251 notes/stages/moving_throat_pde_stage251_microscopic_damping_export_kernel_replacing_the_phenomenological_v_leg_envelope_law_sympy_audit.md:451	free_choice	posited choice:notes/stages/moving_throat_pde_stage251_microscopic_damping_export_kernel_replacing_the_phenomenological_v_leg_envelope_law_sympy_audit.md:449	252
251	cref fit_stage_251_cref fam_0354_cref	origin stage 251 notes/stages/moving_throat_pde_stage251_microscopic_damping_export_kernel_replacing_the_phenomenological_v_leg_envelope_law_sympy_audit.md:181	free_choice	posited choice:notes/stages/moving_throat_pde_stage251_microscopic_damping_export_kernel_replacing_the_phenomenological_v_leg_envelope_law_sympy_audit.md:181	252
251	Delta_0 fit_stage_251_delta_0 fam_0055_delta_0	origin stage 251 notes/stages/moving_throat_pde_stage251_microscopic_damping_export_kernel_replacing_the_phenomenological_v_leg_envelope_law_sympy_audit.md:141	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage251_microscopic_damping_export_kernel_replacing_the_phenomenological_v_leg_envelope_law_sympy_audit.md:141	252

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
251	s_0 fit_stage_251_s_0 fam_0502_s_0	origin stage 251 notes/stages/moving_throat_pde_stage251_microscopic_damping_export_kernel_replacing_the_phenomenological_v_leg_envelope_law_sympy_audit.md:220	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage251_microscopic_damping_export_kernel_replacing_the_phenomenological_v_leg_envelope_law_sympy_audit.md:220	252
251	s_c fit_stage_251_s_c fam_0503_s_c	origin stage 251 notes/stages/moving_throat_pde_stage251_microscopic_damping_export_kernel_replacing_the_phenomenological_v_leg_envelope_law_sympy_audit.md:382	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage251_microscopic_damping_export_kernel_replacing_the_phenomenological_v_leg_envelope_law_sympy_audit.md:382	252
252	E_exp_min_safe fit_stage252_e_exp_min_completely_fixed fam_0070_e_exp_min_safe	origin stage 252 notes/stages/moving_throat_pde_stage252_vacuum_vs_lattice_heat_partition_and_cold_survival_compiler_from_the_microscopic_export_kernel_sympy_audit.md:323	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage252_vacuum_vs_lattice_heat_partition_and_cold_survival_compiler_from_the_microscopic_export_kernel_sympy_audit.md:330	253
252	f_lat fit_stage252_three_to_one_split fam_0399_f_lat; overlays:chain_calibration_245_253	origin stage 252 notes/stages/moving_throat_pde_stage252_vacuum_vs_lattice_heat_partition_and_cold_survival_compiler_from_the_microscopic_export_kernel_sympy_audit.md:385	free_choice	posited choice:notes/stages/moving_throat_pde_stage252_vacuum_vs_lattice_heat_partition_and_cold_survival_compiler_from_the_microscopic_export_kernel_sympy_audit.md:78	253
252	V_in_match fit_stage252_vin_benchmark_calibration_match fam_0273_v_in_match	origin stage 252 notes/stages/moving_throat_pde_stage252_vacuum_vs_lattice_heat_partition_and_cold_survival_compiler_from_the_microscopic_export_kernel_sympy_audit.md:476	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage252_vacuum_vs_lattice_heat_partition_and_cold_survival_compiler_from_the_microscopic_export_kernel_sympy_audit.md:474	none recorded

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
252	engine_disagreement fit_stage_252_engine_disagreement fam_0381_engine_disagreement	origin stage 252 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
252	moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_mathematica_audit fit_stage_252_moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_mathematica_audit fam_0474_moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_mathematica_audit	origin stage 252 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
252	moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit fit_stage_252_moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit fam_0475_moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit	origin stage 252 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
252	mu_eta fit_stage_252_mu_eta fam_0478_mu_eta; overlays:chain_calibration_245_253, chain_wall_action	origin stage 252 notes/stages/moving_throat_pde_stage252_vacuum_vs_lattice_heat_partition_and_cold_survival_compiler_from_the_microscopic_export_kernel_sympy_audit.md:447	free_choice	posited choice:notes/stages/moving_throat_pde_stage252_vacuum_vs_lattice_heat_partition_and_cold_survival_compiler_from_the_microscopic_export_kernel_sympy_audit.md:434	253

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
252	r_V fit_stage_252_r_v fam_0494_r_v	origin stage 252 notes/stages/moving_throat_pde_stage252_vacuum_vs_lattice_heat_partition_and_cold_survival_compiler_from_the_microscopic_export_kernel_sympy_audit.md:152	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage252_vacuum_vs_lattice_heat_partition_and_cold_survival_compiler_from_the_microscopic_export_kernel_sympy_audit.md:240	253
253	K_turn fit_stage253_K_turn_force_match fam_0140_k_turn; overlays:chain_calibration_245_253	origin stage 253 notes/stages/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.md:403	free_choice	posited choice:notes/stages/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.md:401	none recorded
253	K_turn fit_stage253_calibration_slice_nominal_constants fam_0140_k_turn, fam_0399_f_lat, fam_0419_gamma_lattice_legacy, fam_0454_lambda_ref, fam_0478_mu_eta, fam_0498_r_turn, fam_0502_s_0, fam_0503_s_c; overlays:chain_calibration_245_253, chain_wall_action	origin stage 253 notes/stages/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.md:403	free_choice	posited choice:notes/CHECKPOINT_CONSTANT_PROVENANCE.md:956	none recorded
253	f_lat fit_stage253_calibration_slice_nominal_constants fam_0140_k_turn, fam_0399_f_lat, fam_0419_gamma_lattice_legacy, fam_0454_lambda_ref, fam_0478_mu_eta, fam_0498_r_turn, fam_0502_s_0, fam_0503_s_c; overlays:chain_calibration_245_253, chain_wall_action	origin stage 253 notes/stages/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.md:255	free_choice	posited choice:notes/CHECKPOINT_CONSTANT_PROVENANCE.md:954	none recorded

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
253	gamma_lattice_legacy fit_stage253_calibration_slice_nominal_constants fam_0140_k_turn, fam_0399_f_lat, fam_0419_gamma_lattice_legacy, fam_0454_lambda_ref, fam_0478_mu_eta, fam_0498_r_turn, fam_0502_s_0, fam_0503_s_c; overlays:chain_calibration_245_253, chain_wall_action	origin stage 253 notes/stages/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.md:222	free_choice	posited choice:notes/CHECKPOINT_CONSTANT_PROVENANCE.md:952	none recorded
253	lambda_ref fit_stage253_calibration_slice_nominal_constants fam_0140_k_turn, fam_0399_f_lat, fam_0419_gamma_lattice_legacy, fam_0454_lambda_ref, fam_0478_mu_eta, fam_0498_r_turn, fam_0502_s_0, fam_0503_s_c; overlays:chain_calibration_245_253, chain_wall_action	origin stage 248 notes/stages/moving_throat_pde_stage248_dynamic_event_chain_compiler_from_relaxed_stationary_barrier_front_end_turning_point_threshold_speed_and_wkb_sympy_audit.md:438	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.md:384	none recorded
253	mu_eta fit_stage253_calibration_slice_nominal_constants fam_0140_k_turn, fam_0399_f_lat, fam_0419_gamma_lattice_legacy, fam_0454_lambda_ref, fam_0478_mu_eta, fam_0498_r_turn, fam_0502_s_0, fam_0503_s_c; overlays:chain_calibration_245_253, chain_wall_action	origin stage 253 notes/stages/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.md:260	free_choice	posited choice:notes/CHECKPOINT_CONSTANT_PROVENANCE.md:954	none recorded
253	r_turn fit_stage253_calibration_slice_nominal_constants fam_0140_k_turn, fam_0399_f_lat, fam_0419_gamma_lattice_legacy, fam_0454_lambda_ref, fam_0478_mu_eta, fam_0498_r_turn, fam_0502_s_0, fam_0503_s_c; overlays:chain_calibration_245_253, chain_wall_action	origin stage 248 notes/stages/moving_throat_pde_stage248_dynamic_event_chain_compiler_from_relaxed_stationary_barrier_front_end_turning_point_threshold_speed_and_wkb_sympy_audit.md:428	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.md:382	none recorded

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
253	s_0 fit_stage253_calibration_slice_nominal_constants fam_0140_k_turn, fam_0399_f_lat, fam_0419_gamma_lattice_legacy, fam_0454_lambda_ref, fam_0478_mu_eta, fam_0498_r_turn, fam_0502_s_0, fam_0503_s_c; overlays:chain_calibration_245_253, chain_wall_action	origin stage 251 notes/stages/moving_throat_pde_stage251_microscopic_damping_export_kernel_replacing_the_phenomenological_v_leg_envelope_law_sympy_audit.md:461	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage252_vacuum_vs_lattice_heat_partition_and_cold_survival_compiler_from_the_microscopic_export_kernel_sympy_audit.md:300	none recorded
253	s_c fit_stage253_calibration_slice_nominal_constants fam_0140_k_turn, fam_0399_f_lat, fam_0419_gamma_lattice_legacy, fam_0454_lambda_ref, fam_0478_mu_eta, fam_0498_r_turn, fam_0502_s_0, fam_0503_s_c; overlays:chain_calibration_245_253, chain_wall_action	origin stage 251 notes/stages/moving_throat_pde_stage251_microscopic_damping_export_kernel_replacing_the_phenomenological_v_leg_envelope_law_sympy_audit.md:459	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage252_vacuum_vs_lattice_heat_partition_and_cold_survival_compiler_from_the_microscopic_export_kernel_sympy_audit.md:298	none recorded
253	gamma_lattice_red fit_stage253_gamma_lattice_red fam_0420_gamma_lattice_red; overlays:chain_calibration_245_253	origin stage 253 notes/stages/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.md:222	free_choice	posited choice:notes/CHECKPOINT_CONSTANT_PROVENANCE.md:952	none recorded
253	k_eff_req fit_stage253_k_eff_force_matched fam_0429_k_eff_req	origin stage 253 notes/stages/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.md:344	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.md:338	none recorded

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
253	Upsilon_lat_sess fit_stage253_legacy_lattice_rate_and_upsilon fam_0269_upsilon_lat_sess	origin stage 253 notes/stages/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.md:227	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.md:224	none recorded
253	Upsilon_lat_sess fit_stage253_session5_calibration_slice_recovery fam_0269_upsilon_lat_sess, fam_0420_gamma_lattice_red, fam_0478_mu_eta; overlays:chain_calibration_245_253, chain_wall_action	origin stage 253 notes/stages/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.md:227	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.md:224	none recorded
253	gamma_lattice_red fit_stage253_session5_calibration_slice_recovery fam_0269_upsilon_lat_sess, fam_0420_gamma_lattice_red, fam_0478_mu_eta; overlays:chain_calibration_245_253, chain_wall_action	origin stage 253 notes/stages/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.md:222	free_choice	posited choice:notes/CHECKPOINT_CONSTANT_PROVENANCE.md:952	none recorded
253	mu_eta fit_stage253_session5_calibration_slice_recovery fam_0269_upsilon_lat_sess, fam_0420_gamma_lattice_red, fam_0478_mu_eta; overlays:chain_calibration_245_253, chain_wall_action	origin stage 253 notes/stages/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.md:260	free_choice	posited choice:notes/CHECKPOINT_CONSTANT_PROVENANCE.md:954	none recorded

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
253	f_lat fit_stage253_stage252_slice_inputs fam_0399_f_lat, fam_0478_mu_eta; overlays:chain_calibration_245_253, chain_wall_action	origin stage 253 notes/stages/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.md:255	free_choice	posited choice:notes/CHECKPOINT_CONSTANT_PROVENANCE.md:954	none recorded
253	mu_eta fit_stage253_stage252_slice_inputs fam_0399_f_lat, fam_0478_mu_eta; overlays:chain_calibration_245_253, chain_wall_action	origin stage 253 notes/stages/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.md:260	free_choice	posited choice:notes/CHECKPOINT_CONSTANT_PROVENANCE.md:954	none recorded
253	Upsilon_lat fit_stage253_upsilon_lat_calibration fam_0268_upsilon_lat; overlays:chain_calibration_245_253	origin stage 253 notes/stages/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.md:161	free_choice	posited choice:notes/stages/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.md:185	none recorded
253	Upsilon_lat fit_stage253_upsilon_lat_calibration_factor fam_0268_upsilon_lat; overlays:chain_calibration_245_253	origin stage 253 notes/stages/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.md:161	free_choice	posited choice:notes/stages/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.md:185	none recorded

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
253	chi_lambda fit_stage_253_chi_lambda fam_0346_chi_lambda	origin stage 253 notes/stages/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.md:312	internal_consistency	located derivation step:notes/stages/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.md:316	none recorded
253	f_lat fit_stage_253_f_lat fam_0399_f_lat; overlays:chain_calibration_245_253	origin stage 253 notes/stages/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.md:255	free_choice	posited choice:notes/CHECKPOINT_CONSTANT_PROVENANCE.md:954	none recorded
253	K_corr fit_stage_253_k_corr fam_0127_k_corr	origin stage 253 notes/stages/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.md:432	free_choice	posited choice:notes/stages/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.md:576	none recorded
253	K_turn fit_stage_253_k_turn fam_0140_k_turn; overlays:chain_calibration_245_253	origin stage 253 notes/stages/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.md:403	free_choice	posited choice:notes/CHECKPOINT_CONSTANT_PROVENANCE.md:956	none recorded
253	mathcal fit_stage_253_mathcal fam_0461_mathcal	origin stage 253 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
253	max fit_stage_253_max fam_0468_max	origin stage 253 not recorded	no_constraint_record	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded

Stage	Parameter / candidate / family	Origin	constraint_kind	Basis	Downstream
253	`moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_mathematica_audit` `fit_stage_253_moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_mathematica_audit` `fam_0474_moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_mathematica_audit`	origin stage 253 not recorded	`no_constraint_record`	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
253	`moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit` `fit_stage_253_moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit` `fam_0475_moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit`	origin stage 253 not recorded	`no_constraint_record`	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded
253	`Pi_ep` `fit_stage_253_pi_ep` `fam_0206_pi_ep`	origin stage 253 `notes/stages/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.md`:471	`internal_consistency`	located derivation step:`notes/stages/moving_throat_pde_stage253_physical_calibration_and_material_threshold_companion_from_the_stage252_export_and_cold_survival_compiler_sympy_audit.md`:475	none recorded
253	`subsection` `fit_stage_253_subsection` `fam_0519_subsection`	origin stage 253 not recorded	`no_constraint_record`	no constraints block in provenance record; surfaced as scanner/non-parameter accounting	none recorded

M.4 Visible Alias Accounting

The family map canonicalizes 915 candidates and excludes 7 alias candidates from the canonical count. Those aliases are visible here so the appendix does not silently drop them.

Table M.4: Alias accounting used by the parameter-provenance family map.

Alias candidate	Canonical candidate
`fit_stage022_source_map_branch_mhat0_unity`	`fit_stage022_mhat0_unity_branch`
`fit_stage073_wall_fraction_epsilon_r`	`fit_stage073_epsilon_r_posited`
`fit_stage105_chiQ_eq_1_by_matching_card`	`fit_stage_105_chi_q`
`fit_stage105_chi_q_canonical_fix`	`fit_stage_105_chi_q`
`fit_stage121_aspect_ratio_37_over_20`	`fit_stage121_aspect_ratio_37_20_reference_freeze`
`fit_stage168_sigma0_num_digit_variant`	`fit_stage168_canonical_mouth_block`
`fit_stage222_vknown_barrier_benchmark`	`fit_stage222_illustrative_barrier_benchmark`

Appendix N

Fill Workflow and Editorial Rules

N.1 Purpose of this appendix

This appendix is the operating manual for completing the derivation companion. The companion is intentionally not a normal narrative paper. It is a verification ledger: a document whose job is to let a reader trace every reusable moving-throat formula from its source stage, through its assumptions, to its status label, to its final citation anchor. The workflow below is therefore stricter than ordinary drafting workflow. It is designed to prevent three common failures:

1. turning an open branch-realization target into a completed theorem by accident;
2. losing the provenance of a formula when it is promoted from a stage note to a theorem block;
3. creating future papers that cite a convenient equation without carrying the hypotheses, status label, or realization caveat that make the equation true.

The appendix should be used whenever a new section, stage entry, theorem block, source card, reproducibility card, or downstream citation is added. It is also the maintenance checklist for freezing a stable release of the companion.

The workflow assumes the document structure already fixed in the front matter: the theorem-block chapters in Parts I–VIII give the readable derivation path; the stage appendices preserve the 253-stage verification trail; the result-anchor map gives citation handles; the source-file index records provenance; and the reproducibility map records what kind of evidence supports each claim. The central rule is that no fill operation is complete until all of those layers agree.

N.2 What "filled" means

A section is filled only when it is usable in three different ways. A reader should be able to read it for understanding, audit it for correctness, and cite it safely. These are different requirements.

Readable. The section states what problem it solves, which stages it covers, which assumptions are active, and what the result means in the larger derivation.

Auditable. The section gives enough displayed algebra, references to detailed stage entries, and checks for a reader to verify the result without relying on hidden reasoning.

Citable. The section carries a stable result anchor, a claim-status label, explicit hypotheses, and a clear boundary between framework success, realization success, and closure success.

A stage appendix entry is filled only when it contains purpose, inputs, status, derivation, output, checks, and downstream use. A theorem-block subsection is filled only when it states the result in reusable form and points to the stage entries where the detailed derivation is audited. A result anchor is filled only when the anchor map, stage ledger, theorem block, and reproducibility map all point to the same stage range and status ceiling.

The companion should not try to hide open work. A stage or theorem block may be complete while still carrying Open or Numerically Located status. Completeness means the claim is honestly bounded and auditable, not that every branch has been realized.

N.3 Governing maps used during filling

The workflow relies on the governance layers listed in table N.1. These maps should be treated as active editorial controls rather than as front-matter decoration.

Table N.1: Governance layers used during fill work.

Governance layer	What it controls	Fill-workflow use
Claim-status firewall	The six permitted claim levels and the weakest-essential-input rule.	Assign the strongest safe status for each result, then demote if any essential input has weaker status.
Notation firewall	Charge ontology, grouped-lane labels, mixed-sector variables, response symbols, quotient coordinates, orbit-lock variables, and relaxed-branch notation.	Check every filled section for symbol drift, especially around charge, circulation, grouped labels, and prefactors.
Result anchor map	Public citation handles such as **MTDC-T1–MTDC-T12**.	Decide which results deserve theorem-level citation anchors and which should remain stage-local provenance.
Dependency map	Upstream and downstream theorem dependencies.	Prevent silent use of formula gates and keep the fill order aligned with actual dependency order.
Stage ledger	Consolidated Stage 001–253 map.	Verify that every filled theorem block points to the right stages and that every stage entry exports the correct result.

Governance layer	What it controls	Fill-workflow use
Reproducibility map	Script, search, symbolic, numerical, and repository-freeze evidence.	Decide whether a derivation is hand-checkable, script-backed, numerically located, or still open.
Source-file index	Canonical source hierarchy and precedence rules.	Resolve conflicts between summaries, full notes, compact masters, generated TEX files, and scripts.

This appendix should be read as the procedural layer on top of those maps. It does not reassign the mathematical status of any theorem. It tells the author how to fill, check, revise, and eventually freeze the theorem and stage entries without accidentally overstating them.

N.4 The four-layer fill model

Every reusable result should pass through four layers. This is the safest way to preserve both readability and provenance.

Table N.2: Four-layer fill model.

Layer	Fill target	Required content	Main file type
Source layer	Identify where the result came from.	Source file, stage number, original local claim, local assumptions, and any script/output artifact mentioned by the source.	Source index; stage ledger.
Stage-audit layer	Preserve the derivation in verifiable order.	Inputs, status, derivation, boxed output, checks, downstream use, and provenance notes.	Stage appendices.
Synthesis layer	Turn many stages into a readable theorem block.	Definitions, hypotheses, theorem/proposition statement, proof sketch, interpretation, failure modes, and citation anchor.	Parts I–VIII.
Citation layer	Make the result safe for future papers.	Result anchor, equation labels, status wording, citation guardrails, and downstream crosswalk entry.	Result-anchor map; future papers.

The fill order should normally be source layer, then stage-audit layer, then synthesis layer, then citation layer. The only exception is when a theorem block is needed as a planning device. In that case, write the theorem block as a provisional synthesis, then immediately backfill the corresponding stage-audit entries before the result is treated as stable.

N.5 Repository roles

The generated companion is split into files so that each fill operation has a natural target. The roles in table N.3 should be preserved when the repository is maintained.

Table N.3: File roles during fill work.

File family	Role	Fill cadence
`frontmatter/00_purpose_scope.tex` through `frontmatter/05_dependency_map.tex`	Defines purpose, reading rules, status firewall, notation firewall, result anchors, and dependency map. These files set the rules for the rest of the document.	Stabilize early; edit rarely after Part I is filled.
`parts/part01_*.tex` through `parts/part08_*.tex`	Readable theorem-block chapters. These files explain the derivation by result families rather than by raw stage order.	Fill after corresponding stage range has been extracted.
`appendices/stage_ledger.tex`	One-line registry of every stage, status, output, anchor, and source range.	Update whenever a stage status, title, output, or anchor changes.
`appendices/stage_appendix_part*.tex`	Detailed verification trail for Stages 001–253. This is the audit spine of the companion.	Fill stage by stage; revise when theorem-block wording exposes missing assumptions.
`appendices/reproducibility_map.tex`	Evidence and audit registry: symbolic scripts, hand checks, numerical searches, branch-realization gates, and freeze protocol.	Update whenever a result becomes script-backed, numerically located, or branch-realization dependent.
`appendices/source_file_index.tex`	Provenance index for source files and generated companion files.	Update when a source file is added, superseded, or demoted.
`appendices/fill_workflow.tex`	Editorial workflow and maintenance protocol.	Edit only when the ledger process itself changes.

The front matter is allowed to define rules. The theorem-block chapters are allowed to synthesize. The stage appendices are allowed to be long. The source and reproducibility appendices are allowed to be operational. Mixing these roles is the main way a derivation ledger becomes hard to audit.

N.6 Recommended global fill order

The safest fill order is deliberately conservative. It avoids polishing a theorem block before the associated stage entries are stable, but it also avoids burying the reader in 253 raw entries before the citation structure exists.

1. **Freeze the front-matter rules.** Confirm that the purpose, reading rules, status firewall, notation firewall, result anchors, and dependency map are internally consistent.
2. **Fill one complete pilot block.** Use Part I and Stages 001–023 as the pilot because those stages define the parent lift, reduced wall variables, first BdG coupling, Maxwell/mixed bridge, grouped normalization bridge, and full grouped bundle.
3. **Audit the pilot block end to end.** Check that the theorem statements, stage entries, equation labels, result anchors, source entries, and reproducibility notes agree.
4. **Fill the remaining parts by stage range.** Work through Parts II–VIII in order, always keeping the corresponding stage appendix ahead of or tied with the theorem-block prose.
5. **Update the consolidated stage ledger after each part.** Do not wait until the end to reconcile stage statuses and outputs.
6. **Promote stable formulas to equation labels.** Only formulas that downstream papers will cite should receive stable equation labels. Intermediate scratch algebra should remain unpromoted.
7. **Run the status audit.** Every theorem, proposition, boxed formula, and result anchor must inherit the weakest essential status among its inputs.
8. **Run the notation audit.** Check high-risk symbols against the notation firewall, especially charge labels, grouped real lanes, quotient coordinates, support variables, and relaxed-branch symbols.
9. **Run the reproducibility audit.** Confirm that every script-backed, numerical, search-based, or branch-realization claim has the required artifact card or open-gate entry.
10. **Freeze a release manifest.** Only after the ledger builds cleanly and the status audit passes should downstream framework and application papers be revised to cite the companion.

This order is not meant to slow down writing. It is meant to keep later corrections local. If a status demotion is discovered in Stage 171, the demotion should affect the corresponding theorem block, result anchor, and downstream wording immediately rather than being discovered after future papers have already cited the stronger claim.

N.7 Unit of work

The most useful unit of work is not a page count. It is a result packet. A result packet is the smallest derivation unit that can be stated, audited, and cited without importing unrelated stages. Typical result packets are:

1. a single exact algebraic identity, such as a projector inverse map;
2. a reduced compiler, such as the bridge from operator moments to response moments;

3. a branch-test packet, such as an outgoing normalization ratio;
4. a finite search verdict, such as a support-cardinality sieve;
5. an open gate, such as actual branch realization of a target packet;
6. a relaxed-branch extension, such as a barrier or export kernel companion.

A good writing session fills one or a few result packets completely. It should not partially polish every chapter at once.

N.8 Single-section workflow

When filling one section or subsection of a theorem-block chapter, use the following workflow.

N.8.1 Step 1: identify the source packet

Before writing prose, identify the source material that controls the section. The minimum source packet is:

$$\mathcal{S} = (\text{source file},\ \text{stage range},\ \text{prior anchors},\ \text{script artifacts},\ \text{status ceiling}).$$

For example, a section in Part V may draw from Stages 164–200, from the weak-axisymmetric transport results, from the exact quotient/orbit theorem, and from the compact program status summary. The section should not begin by restating every stage. It should begin by deciding which result it is trying to make citable.

N.8.2 Step 2: write the hypothesis box first

Every theorem-level section should have an explicit hypothesis paragraph before any final result is stated. The hypothesis paragraph should answer four questions:

1. Which branch, closure, or reduction is being used?
2. Which quantities are treated as already fixed inputs?
3. Which quantities are being solved, transported, or tested?
4. Which realization question remains outside the claim?

A good hypothesis paragraph prevents later citation misuse. It should be written in plain prose and should not hide assumptions inside notation.

N.8.3 Step 3: state the result at the right strength

State the result with its status in the title line or immediately before the statement. Suitable forms include:

$$\text{Theorem: Exact within the reduced one-port branch.}$$

$$\text{Proposition: Controlled weak-axisymmetric transport law.}$$

$$\text{Open branch gate: Actual realization of } \Xi_1.$$

Avoid status-neutral wording such as “we have shown” unless the exact scope is obvious from the same paragraph.

N.8.4 Step 4: separate proof sketch from audit trail

The main text should give a proof sketch. The appendix should give the audit trail. The distinction is:

Table N.4: Division of labor between main text and stage appendix.

Layer	Main-text theorem block	Stage appendix
Purpose	Explain why this result matters and what bottleneck it resolves.	Record the stage-local purpose and how it connects to adjacent stages.
Inputs	List only the essential inputs.	List all local inputs, inherited assumptions, source equations, and prior stage dependencies.
Derivation	Show the core transformation or two to five decisive equations.	Show the full algebraic chain, including sign conventions, substitutions, and reductions.
Checks	Mention the decisive checks.	Record all available checks: dimensional, sign, limiting, isotropy, script-backed, numerical residual, and branch-domain checks.
Output	Name the citable theorem or formula packet.	Box the exact stage output and state downstream uses.
Status	Give the final status tag.	Explain why that tag is justified and what would be needed to upgrade it.

N.8.5 Step 5: close the packet with a downstream-use sentence

Every filled subsection should end, or nearly end, with a sentence of the form:

This result is used downstream to ...

or

The remaining branch-realization gate is ...

That sentence is not filler. It is the local citation map. It tells future papers whether they are allowed to cite the result as a completed derivation, a reduced compiler, a numerical placement, or an open test condition.

N.9 Stage appendix fill protocol

The stage appendices are the verification backbone of the companion. Each stage entry should preserve the stage number and local result even when the theorem-block chapters reorganize the narrative.

N.9.1 Canonical stage-entry template

Use the following template unless a stage is purely tabular or purely numerical.

1. **Stage title and source.** Preserve the original stage number and title. Name the controlling source file.
2. **Purpose.** State the stage-local problem in one paragraph.
3. **Inputs.** List prior stage outputs, branch assumptions, variables, and status ceilings.
4. **Claim status.** Assign one of the companion status tags and explain the limiting assumption.
5. **Derivation.** Give the algebra in enough detail that a reader can reproduce it.
6. **Output.** Box only the result that survives downstream.
7. **Checks.** Record symbolic scripts, limiting cases, sign checks, dimensional checks, or numerical residuals.
8. **Downstream use.** State where the result is used later.

N.9.2 Stage-entry completion checklist

A stage entry is complete only when the following checklist is satisfied.

- ☐ The stage number and title match the stage ledger.
- ☐ The source file is identified.
- ☐ The status tag is present and justified.
- ☐ All nonlocal assumptions are named rather than implied.
- ☐ Every symbol that changes meaning from earlier sections is redefined or explicitly inherited.
- ☐ The final output is visually distinguishable from intermediate algebra.
- ☐ Script-backed claims name the script or output artifact.
- ☐ Numerical claims include the target equation, branch domain, and residual or tolerance.
- ☐ Open branch-realization targets are not written as achieved results.
- ☐ The downstream-use line points to a part, theorem anchor, later stage, or future-paper citation use.

N.10 Theorem-block fill protocol

Theorem-block chapters are not chronological transcripts. They are the readable synthesis of the stage ledger. Each theorem block should be organized around what a future paper needs to cite.

N.10.1 Theorem-block skeleton

A filled theorem block should normally contain:

1. a short orientation paragraph;
2. a branch or closure declaration;
3. definitions of the local variables and observable packet;
4. one or more named lemmas or propositions;
5. a main theorem or result statement;
6. a proof sketch with stage-appendix pointers;
7. a status and non-claim paragraph;
8. a downstream citation paragraph.

The orientation paragraph should not retell the whole program. It should answer the local question: what bottleneck does this theorem block remove?

N.10.2 When to promote a result to a theorem

Promote a result to a theorem only when it satisfies all of the following conditions:

1. It is reused by later stages, later parts, or future papers.
2. Its assumptions can be stated in a compact hypothesis list.
3. Its status can be assigned without ambiguity.
4. Its proof can be located in one or more stage appendices.
5. Its conclusion can be stated without referencing the chronological accident of how it was discovered.

If a result is important but not yet stable, use a proposition, lemma, or open gate instead. The label "theorem" should be reserved for results whose citation behavior is clear.

N.11 Result-anchor promotion protocol

Result anchors are citation tools. They should not be created merely because a formula looks important. The correct promotion chain is:

$$\begin{aligned} &\text{stage output} \longrightarrow \text{appendix boxed formula} \longrightarrow \text{main-text theorem} \\ &\longrightarrow \text{result anchor} \longrightarrow \text{downstream citation.} \end{aligned}$$

N.11.1 Anchor-card requirements

Each promoted result should have a result card containing:

Table N.5: Minimum contents of a result-anchor card.

Field	Required content
Anchor ID	Stable identifier, for example **MTDC-T8**.
Short name	Human-readable result name.
Status	One of the companion status tags.
Stage range	The source stage interval.
Main location	The theorem-block section where the result is synthesized.
Appendix location	The stage appendix entry or entries containing the detailed derivation.
Hypotheses	Branch, closure, reduction, search class, or numerical-domain assumptions.
Exported formulas	The exact equations future papers are allowed to cite.
Non-claims	Nearby stronger statements that are not supported.
Downstream use	Papers, sections, or result packets that should cite the anchor.

N.11.2 Anchor stability rule

Once a result anchor is introduced, its meaning should not drift. If a later pass strengthens, weakens, or changes the result, do not silently reuse the same anchor. Either revise the anchor card explicitly or create a subanchor. Stage numbers are provenance markers; result anchors are citation commitments.

N.12 Boxing and equation-label rules

Boxed equations should be scarce. A boxed equation tells the reader, "this is an exported result." It should therefore be used only for:

1. final stage outputs;
2. theorem conclusions;
3. branch-test packets;
4. exact realization or obstruction conditions;
5. open target equations that future branch work must hit.

Intermediate algebra should receive ordinary equation labels only when it is referenced later. If a formula is used only in the local derivation, leave it unboxed and usually unlabeled.

N.12.1 Equation-label naming convention

Use stable semantic labels rather than line-number-like labels. Examples:

`eq:stage022-p0-normalization-target`,

`eq:weak-axisymmetric-b-equals-3a`,

`eq:rigid-mouth-uv-lock`.

Avoid labels such as `eq:important`, `eq:main`, or `eq:formula7`. They will become meaningless once the document grows.

N.13 Source-handling protocol

The source-file index defines the provenance hierarchy. During filling, use the following precedence rules.

Table N.6: Source precedence during the fill process.

Question	Preferred source	Use rule
What did a stage derive?	Canonical stage files plus stage appendices	Treat the per-stage TEX files and their assembled appendices as the canonical chronological stage source.
What is the current global theorem-status reading?	Frontmatter firewall plus theorem-block chapters	Use for present bottleneck, status firewall, strict/relaxed split, and final packet language.
How is the framework meant to be used?	`pde.tex`	Use for operational branch-test language and future-paper citation style.
What exact parent equations are inherited?	Parent action and Maxwell/plasma summaries.	Use for exact field equations, projection/reduction distinction, and charge/mixed-sector ontology.
What lower-order target is being matched?	PN summary and full PN files.	Use for target ledgers, carry-forward constants, and PN-specific status.
How should application papers cite this material?	Reduced-sector companion files.	Use only after the derivation companion status is fixed.

When two sources appear to disagree, do not harmonize by prose smoothing. Add an explicit reconciliation note that says which file controls which layer. For example, the full stage archive may control the chronological derivation, while the compact master controls the current status interpretation after later stages changed the endgame.

N.14 Notation audit protocol

Notation drift is one of the main risks in this ledger. The notation firewall should be rechecked whenever a packet is filled.

N.14.1 Symbols that require special care

The following symbols and symbol families require explicit checking:

Table N.7: High-risk notation during filling.

Symbol family	Risk	Fill rule
q, Q, $q_\star$, q_{eff}	Confusion between electric charge, scalar coefficients, grouped coordinates, and quotient drifts.	Electric charge must use $\eta_Q, q_\star, q_{\mathrm{eff}}$. Gravity-side mass dressing is κ_ρ, not electric charge.
20/21/22	Mistaken as spacetime indices.	Always state that these are grouped real P_2 lanes when first used in a section.
$A_w, J^w, F_{\mu w}$, E_w, C_a	Accidental deletion by zero-mode Maxwell language.	Say they are suppressed only in strict brane reductions, not removed from the microscopic ontology.
$P_0, P_1, \Xi_1, N_Q, \chi_Q$	Confusion between normalization, prefactor, and transported slope.	Define the local packet before using any of these quantities in a theorem statement.
U, V	Confusion between potentials and rigid-mouth log chart.	In Part VII and later, declare the physical logarithmic chart explicitly.
$\epsilon_\eta, \epsilon_W, \zeta, \rho_\alpha$	Different branch-placement roles in strict and relaxed sectors.	State whether the packet is strict selected support, weak-axisymmetric, rigid-mouth, or relaxed/open-system.
Δ packets	Many unrelated deficits and verdict packets.	Give the full subscripted packet name and avoid bare Δ in exported results.

N.14.2 Notation-audit checklist

- ☐ No electric-charge statement uses circulation, throat radius, or breathing variables as the charge definition.
- ☐ No gravity-side scalar coefficient is written as bare electric charge.
- ☐ Grouped real P_2 lanes are distinguished from tensor or spacetime indices.
- ☐ Mixed-sector variables are not erased by a brane-limit statement.
- ☐ Strict branch, rigid-mouth branch, selected support branch, and relaxed branch variables are not interchanged.
- ☐ Reused letters such as K, G, P, N, R, and Δ are locally disambiguated.

N.15 Status-audit protocol

After each packet is filled, perform a status audit. The status audit asks whether the written section accidentally upgraded any result.

N.15.1 Status-upgrade tests

Use the following tests.

Table N.8: Tests for accidental status upgrades.

Question	If yes	Required action
Does the proof assume a branch ansatz, finite-dimensional reduction, or low-frequency expansion?	It is not unrestricted EXACT.	Mark as EXACT WITHIN CLOSURE or REDUCED / CONTROLLED REDUCTION.
Does the result depend on an actual branch landing on a target value?	The branch realization is not automatic.	Mark the target or landing as OPEN unless realized.
Does the result use a numerical root, ranking, or placement?	The location is not closed form.	Mark as NUMERICALLY LOCATED and record tolerances/residuals.
Does the result use a physically motivated but not uniquely derived constitutive law?	It is a closure choice.	Mark as EFFECTIVE CLOSURE or EXACT WITHIN CLOSURE inside that closure.
Does a script verify only a finite symbolic model?	The verification scope is finite.	State the model and search class.
Does a theorem require a later relaxation or open-system extension?	Strict and relaxed claims differ.	Split the statement into strict-front-end and relaxed-branch pieces.

N.16 Reproducibility fill protocol

Whenever a stage or theorem depends on a script, numerical solve, finite search, or generated output, the reproducibility map must contain enough information for a future reader to reconstruct the evidence trail.

N.16.1 Artifact-card template

Use this template for script-backed or numerical packets:

Table N.9: Artifact-card template for reproducibility entries.

Field	Content
Artifact name	Script, notebook, output transcript, or generated table basename.
Stage range	Stages certified by the artifact.
Mathematical object	Projector, Schur complement, polynomial identity, search sieve, branch solve, Hessian, etc.
Inputs	Symbolic assumptions, numerical parameters, branch constraints, search class, tolerances.
Outputs	Equations, candidate list, roots, residuals, rankings, or theorem gates certified.
Status ceiling	Exact symbolic, exact within closure, numerical location, or open branch test.
Independent checks	Limiting cases, dimension checks, duplicate script, residual evaluation, hand derivation, or sanity check.
Failure modes	What the artifact does not prove.

N.16.2 Script-backed result wording

Use precise wording for script-backed results. Prefer:

> The accompanying symbolic audit verifies the stated identity inside the declared finite model.

Avoid:

> The script proves the full PDE theorem.

unless the script really implements the full PDE theorem, including all branch-realization hypotheses.

N.17 Numerical-location protocol

Numerically located results must be written so that a future reader can tell the difference between an exact target equation and the numerical point currently found on it.

A numerical-location paragraph should contain:

1. the exact equation being solved;
2. the branch domain or admissible interval;
3. the numerical value reported;
4. the residual or tolerance;
5. the script/output artifact;
6. whether uniqueness in the domain was checked;
7. whether the result is used as an input or only as a diagnostic.

If uniqueness is not checked, do not write "the" branch point without qualification. Write "the located branch point in the searched domain" or give the domain explicitly.

N.18 Open-gate protocol

Open gates are valuable results. They turn vague missing work into explicit branch tests. They should be written with the same care as completed theorems.

N.18.1 Open-gate template

An open gate should contain:

1. the exact observable packet to be returned by the actual branch;
2. the target value or target condition;
3. the branch assumptions under which the target is meaningful;
4. the formulas that reduce the branch data to the observable packet;
5. the reason the previous algebra is complete but branch realization remains open;
6. the downstream consequence of success or failure.

For example, the outgoing quadrupole normalization gate should be written as an explicit condition on the relevant prefactor or normalization packet, not as a vague instruction to "solve radiation reaction."

N.19 Strict versus relaxed branch workflow

The strict branch and relaxed/open-system branch should be filled as related but distinct ledgers.

Table N.10: Strict and relaxed branch separation during filling.

Layer	Strict branch	Relaxed/open-system branch
Purpose	Tests whether the selected support/orbit/outgoing packet is realized without added leakage/work/source repair.	Studies the codimension-three extension around the strict front end.
Recovery map	The strict packet itself.	Must state the recovery map back to the strict front end.
Same-charge verdict	Frozen by the strict one-port audit unless explicitly reopened.	Should not create a new long-range same-charge attraction by wording drift.
Barrier/export quantities	Usually absent or only target-side.	Leakage, work, non-rigid mouth, compensated source, event chain, export kernel, and heat partition are active.
Citation style	Cite as strict branch theorem or strict branch open gate.	Cite as relaxed branch companion, not as proof of strict branch realization.

When a relaxed result reduces to the strict branch under a recovery map, state the recovery map explicitly. Do not let the relaxed branch silently backfill a missing strict-branch theorem.

N.20 Local consistency pass

After filling any packet, perform a local consistency pass before moving on.

- ☐ Stage numbers match the stage ledger.
- ☐ Result anchors match the result-anchor map.
- ☐ All referenced sections, equations, and appendices exist.
- ☐ No equation label is duplicated.
- ☐ No status tag appears without a reason.
- ☐ No boxed equation lacks downstream use.
- ☐ Every open target is labeled as open.
- ☐ Every branch-dependent result states the branch.
- ☐ The notation firewall is obeyed.
- ☐ The source-file index supports the provenance claim.

N.21 Global consistency pass

The global consistency pass should be done after all packets in a part are filled, and again after the full document is filled.

N.21.1 Global pass A: status consistency

Search the whole document for every instance of Exact, Exact Within Closure, Reduced / Controlled Reduction, Effective Closure, Numerically Located, and Open. Confirm that each use is consistent with the claim-status firewall. If a result appears with two different statuses in different chapters, either harmonize the status or explain that one is a local subclaim and the other is a global branch-realization claim.

N.21.2 Global pass B: anchor consistency

For each **MTDC-T** result, verify that:

1. the anchor appears in the result-anchor map;
2. the theorem-block location exists;
3. the stage appendix location exists;
4. the stage ledger agrees with the stage range;
5. the status tag agrees across all appearances;
6. downstream papers would know exactly what can be cited.

N.21.3 Global pass C: formula-packet consistency

Some quantities appear in multiple parts of the companion. For those, verify that the definitions are identical or that the change of meaning is explicit. High-priority packets include:

$$P_0, \quad P_1, \quad \Xi_1, \quad N_Q, \quad \chi_Q, \quad \Delta_{\mathrm{full}}, \quad q_{\mathrm{tr}}, \quad q_{\mathrm{nt}}, \quad q_\eta, \quad U, V.$$

N.22 Suggested packet sizes for future writing sessions

Because the document is large, future writing sessions should use packet sizes that are small enough to audit. The recommended sizes are:

Table N.11: Recommended packet sizes for filling sessions.

Task type	Recommended size	Reason
Main-text theorem block	One section or subsection.	Keeps assumptions and proof sketch focused.
Stage appendix	One stage for long stages, or three to five short consecutive stages.	Preserves chronological provenance without losing local algebra.
Ledger table update	One part at a time.	Avoids mismatched stage ranges and anchor IDs.
Result-anchor pass	One anchor family at a time.	Prevents anchor drift.
Reproducibility pass	One script family or theorem block.	Keeps evidence trails complete.
Downstream-paper retrofit	One paper and one result family.	Prevents broad citation rewrites before the ledger is stable.

This is also the recommended collaboration protocol: fill one packet, audit it, then move to the next. Large unreviewed rewrites should be avoided.

N.23 Downstream-paper retrofit protocol

After the ledger is stable, future papers should be revised to cite this companion rather than raw working notes. The retrofit should happen in this order:

1. identify claims currently citing raw notes, compact masters, old framework text, or informal derivations;
2. map each claim to a companion result anchor and, when needed, a stage appendix range;
3. check whether the downstream wording matches the companion status;
4. replace raw-note citations with theorem/result-anchor citations;
5. preserve raw source references only when the paper is discussing archive provenance rather than using a theorem.

A downstream paper should not cite the derivation companion as a blanket proof of the whole program. It should cite the specific anchor that proves or audits the specific claim being used.

N.23.1 Safe downstream wording

Use status-aware language. Examples:

> The grouped outgoing-normalization compiler is derived in **MTDC-T3**; actual branch realization is tested by the scalar packet stated there.
>
> The weak-axisymmetric quotient reduction to $\Xi_1 = P_1/P_0$ is given in **MTDC-T8** under the declared grouped-lane assumptions.
>
> The relaxed barrier/export companion uses the recovery map stated in **MTDC-T12**; it is not part of the strict conservative branch proof.

Unsafe wording would remove the status condition, for example by saying that the actual PDE has been solved merely because the compiler is exact.

N.24 Release-freeze workflow

A release freeze is a repository event, not just a typesetting event. A frozen release should include:

1. a clean full-project build;
2. a source-file manifest;
3. script and output artifact manifest;
4. status-audit notes for each result anchor;
5. list of open branch-realization gates;
6. list of superseded or failed branches preserved in the ledger;
7. hash or version identifier for the release;
8. downstream papers whose citations have been updated to the release.

The release manifest should not change the mathematical status of a result. It only makes the status reproducible.

N.25 Maintenance rules

The following rules should govern future edits.

1. **Do not weaken provenance for readability.** A theorem block may be readable, but the stage appendix must remain detailed enough to audit.
2. **Do not strengthen status for convenience.** A useful target equation remains a target equation until branch realization is proven.

3. **Do not merge strict and relaxed branches.** Use the recovery map when moving between them.

4. **Do not cite a source file as a theorem.** Cite result anchors and stage appendices; use source files for provenance.

5. **Do not hide failed routes.** Failed branches should be preserved with their domain and obstruction.

6. **Do not change notation silently.** If a symbol changes meaning, update the notation firewall and the local definitions.

7. **Do not promote equations casually.** Stable labels and boxes create future citation obligations.

8. **Do not leave a status change local.** Update theorem block, stage ledger, anchor map, and reproducibility map together.

N.26 Minimal templates

The following templates are not placeholders in the document; they are forms to copy when adding new material.

N.26.1 Stage-entry template

Stage NNN: local title.

Purpose. State the local problem and why it matters.

Inputs. List prior equations, assumptions, branch restrictions, and source-stage provenance.

Claim status. State the local status and the reason for that status.

Derivation. Show the algebra, reduction, or search logic.

Output. Box only the final reusable formulas or verdicts.

Checks. State symbolic, numerical, limiting, sign, dimensional, or symmetry checks.

Downstream use. Name later stages, result anchors, and future citation uses.

N.26.2 Theorem-block template

Local setup. Define variables and active branch class.

Status. State exact, exact-within-closure, reduced, effective, numerical, or open status.

Theorem or proposition. State the reusable result with hypotheses.

Proof sketch. Identify the stage outputs used and show the essential algebra.

Interpretation. Explain what the result means for the moving-throat program.

Failure modes. State when the result does not apply.

Citation target. Identify result anchor, equation labels, and stage appendix range.

N.26.3 Open-gate template

Open branch gate. The compiler reduces the problem to observable packet X. The remaining realization question is whether the actual moving-throat branch in class B returns X satisfying target condition Y. Until that branch construction is supplied, the compiler result is Exact Within Closure while the realized-branch claim is Open.

N.27 Minimal build and static-check workflow

A useful local build workflow is:

1. Run a static scan for duplicate labels, unmatched braces, unmatched environments, literal control-character corruption, and accidental placeholder markers.
2. Compile the edited file in a minimal harness if the full project is large.
3. Compile the full project at least once after each major packet group.
4. Compile twice before a release candidate so cross-references stabilize.
5. Inspect warnings about undefined references, overfull boxes in long equations, and missing citations or anchors.

The build check is not a mathematical proof. It is a source-integrity check. A file can compile and still overstate a theorem; conversely, a mathematically correct packet can fail to compile because of a label or macro issue. Both kinds of checks are required.

N.28 Recommended handoff note after each filled packet

After a packet is filled, leave a short handoff note in the working discussion or project log with the following fields:

1. file edited;
2. stage or result range covered;
3. strongest new citable result;
4. status tag;
5. open gates left behind;
6. checks performed;
7. files that should be updated next.

The handoff note is not part of the final paper, but it prevents future sessions from losing track of what was completed and what merely looked complete.

N.29 Closing rule

The derivation companion is allowed to be large, but it should never be vague. Every filled section should answer four questions: what is being proved or audited, under what assumptions, with what evidence, and how future papers may cite it. If those four questions are answered locally and consistently across the stage ledger, theorem blocks, result anchors, source index, and reproducibility map, then the ledger is doing its job.